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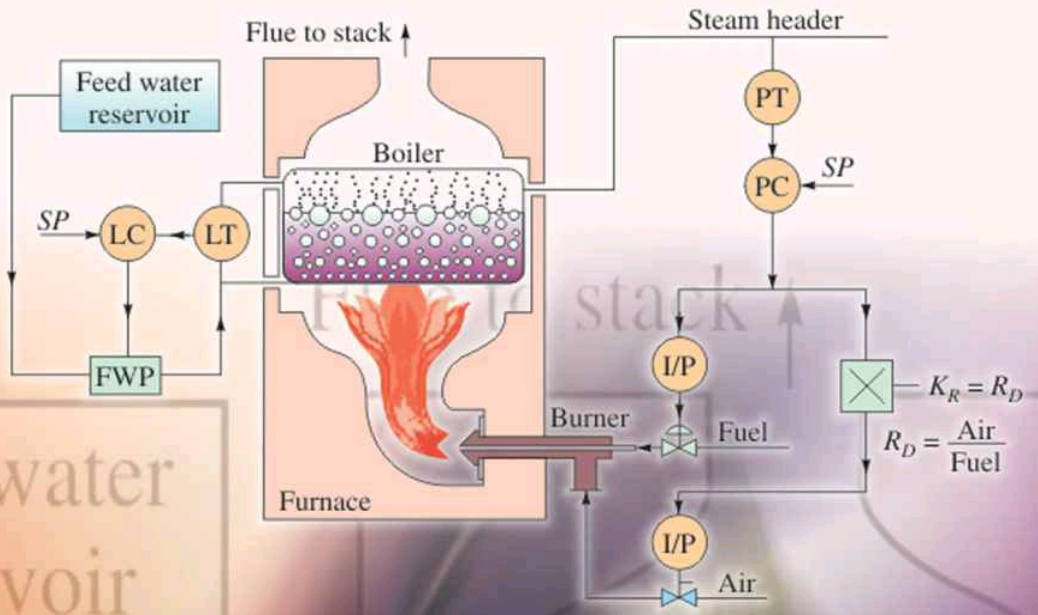

PROCESS DYNAMICS AND CONTROL

Prabir Kumar Sarkar

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PROCESS DYNAMICS AND CONTROL

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Delhi-110092
2014

PROCESS DYNAMICS AND CONTROL (with CD-ROM)
Prabir Kumar Sarkar

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To

My parents, wife and students

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Preface

The text fully covers undergraduate syllabus of process dynamics and control offered in Indian universities and technical institutes, including some institutions abroad. Extensive use of descriptive and worked-out examples are cited for conception development. A number of exercise problems are given at the end of every chapter for self testing of the readers thereby increasing their level of assimilation and aptitude.

The reason for completing this project is not only to launch a text, and add another member in the long list of excellent books written in this area, probably starting by the pioneering work of Ceagleske. There are two main objectives that I have tried to fulfill:

1. Process Dynamics and Control is not a subject to be feared off, but may become one of your favourites.

This subject is offered in some of the latter semesters of undergraduate course of chemical engineering to an audience who has practically no exposure of dynamic manifestation of process systems. In the earlier semesters, principal subjects of the course are taught, where all equipment systems are presented by their steady state functional relation. Introduction to the discipline of dynamic analysis exposes a new kind of treatment for hitherto known systems like a heat exchanger or an absorption column.

I have experienced that a large body of chemical engineering students have a mixed feeling toward this area of learning. In the first place, they think that this subject is nothing but scores of exercises through a lot of tougher variety of mathematical procedures like analysis involving complex variable, operational methods of solution of differential equations, matrix representation and solution of differential equations, frequency domain analysis that reminds vector plots used in AC circuit analysis, Fourier integral series and transform, etc. Also an undercurrent of notion exists that the dynamics and control should not be considered as one of the principal or core subjects of chemical engineering like stoichiometry, thermodynamics, unit operations, and processes because in process equipment analysis and design that is taught under unit operations, no dynamic model is ever used. These are probably the reasons, for which, during this conception jump from steady to transient state, a kind of soft landing is necessary.

Another unfortunate fact should be mentioned that though the significance of control systems for successful operation of equipment are included in many textbooks of unit operations and processes, however, the teacher concerned while treating the

subject matter, hardly mentions their existence or importance. This omission is probably due to the volume of syllabus to be covered within the time frame of a semester. These are the reasons why the discipline is generally received with an amount of unfamiliarity and no loss of love in between. Even students of upper tiers also approach this subject as a collection of dry mathematical muscle show that has to be tackled with cramming power, and treated as a necessary evil.

This came as the primary challenge to me, when I started teaching in this area. I addressed this issue by stating that the familiar steady state relations are still there in the dynamic model, only, they have been contained in the steady state gain term of the transfer function. The extra term that involves the Laplace operator '*s*', spells a dimensionless time function and tells how much of the gain term would be recovered with the passing of time (Section 2.4). I know that the gap could not be completely bridged just by this statement and establishing it with some examples, nevertheless, the students become aware that the teacher appreciates their *misery* and will do something positive about it in future classes.

2. Assimilation of Dynamic Behaviour by Drawing Simile from Examples

I have seen weakness in identifying dynamic parameters, their inter-relations and disturbance types from their physical description. The answer that came up is to think as many number of physical examples, not only from process area but also from areas like physiology, sociology, economics, and even political systems. The criterion of selection was universal that those respond to the environmental changes and demonstrate dynamic feature comparable to an industrial process. I found that the key word is physical, and its meaning extends from the expanse of learner's everyday experience, to the knowledge base he has acquired since the early schooling days through the walk of life. Acceptance or appeal of an example to different persons within a target audience is different, but examples developed by computer programs with animated graphic output had the maximum acceptance. Through the years I have developed more than twenty of such examples; eighteen of them are introduced in the accompanying CD. I have named them as demonstration (DM01.EXE to DM18.EXE) programs.

I had to face many awkward questions like—'Why so much song-and-dance with differential equations? Is it impossible to build a dynamic model with only algebraic equations? I have formulated the answer in DM01.EXE.

There was another query: "The suddenness of putting '*iw*' in place of '*s*' in the transfer function is not easily digestible, and in response the long fangled complex variable theorem by which this substitution has been justified, also hardly caries any physical significance." To address this question I have tried to draw a simile between dynamic representation in complex and time domain (Section 11.2.2).

Certain aspects about treatment of a particular topic gradually became apparent that by slightly altering the approach, the ease of assimilation could be improved.

(a) In linearization, if the slope evaluation point is shifted from the beginning of the sweep of input to the midpoint, the accuracy will increase (Section 5.8).

(b) To operate with a complex input comprising several ideal inputs, use of superposition may be more acceptable to a greater body of students than the analytical approach. For implementation, only two simple dynamic results are necessary as prerequisites (Section 5.5.2).

I have extensively used worked-out examples for examining a result or a conception from a number of areas of process applications. Also, in the solution procedure, the maximum detail of the algebraic and arithmetic steps has been exposed. This had been done as a reciprocation to the demand of a large number of students. In placing the topics and designing these examples, I have maintained a sequence of conception development, so that the subject matter may be presented as an unbroken chain that gradually unfolds before a beginner.

I have emphasised the importance of digital simulation for checking dynamic response shapes and control system design and optimisation. An indigenously developed simulator Process Control System Analyser (PCSA) has been employed for this purpose (Chapter 14). The time domain part of this chapter starts with a simple linear interpolation procedure and ends with a simulation program of multiple loop control. Through the chapter, I have exposed the usage of about 20-subroutines through about 34-worked out example programs through which complexity of program writing is gradually exposed. The chapter ends with four programs for solving frequency response problems. A 3-D plotting subroutine and an optimization subroutine are also developed, these are not included in Chapter 14, however, their usage are shown in the two demonstration programs included in the accompanying CD.

My ambition had been to nurture and develop a sort of knowledge about the dynamic behaviour of systems, which would become part of the learner's thought process. Thus, an analytical result or conclusion no longer remain as a mathematical jugglery, instead becomes a physical tangible reality that is corroborated by this knowledge about how any system should behave in a similar dynamic situation. I have always preached that first priority is your feeling about how it should react, if the result is contrary to it, doubt the mathematics not your feeling. I guaranteed them that no other subject could offer a more logical and sequential development of conceptions that is required to learn the basics with least amount data required to be crammed within.

ORGANISATION OF THE BOOK

Chapter 1 introduces preliminary conceptions, e.g, process, disturbance, control systems, linear and non-linear processes, lumping and distribution of parameters, development of block diagram from physical description of a process, degree of freedom, order of a system, stability of a dynamic system and self-regulation.

Chapter 2 starts by showing the dynamic manifestation of a process with examples from assorted areas, then introduces the building blocks (dynamic elements) necessary for construction of the process dynamic model.

Chapter 3 introduces the process model building techniques for lumped parameter, multiple stage lumped parameter and distributed parameter systems. The chapter introduces different analytical descriptions of process models and linearization technique.

Chapter 4 deals with transfer function development of first order systems for a number of example systems, also emphasises the linearisation technique of nonlinear systems.

Chapter 5 discusses the response derivation of first order system to, ideal and non-ideal input functions, certain related properties and the some further development of linearisation technique.

Chapter 6 is about the transfer function development of second and higher order systems and derivation of their response to ideal inputs and certain related properties. The transfer and transportation lags are also introduced in this chapter.

Chapter 7 discusses the dynamic models of three hardware devices, Sensor-transmitters, Signal transmission line, Final control elements related to a control system.

Chapter 8 is about the parameters of an ideal automatic controller and dynamic model for different modes of control. Discussion of controller response to ideal inputs in error function and transfer functions for certain industrial configuration of controllers are included.

Chapter 9 discusses about the principle of feedback logic, two basic kinds of disturbances to a control loop, thus introduces the closed loop transfer function. A number of closed loop responses are demonstrated to show the effect of process parameters for different modes of control action.

Chapter 10 introduces the stability of a dynamic system with special reference to a control loop in complex and time domain by the help of Root locus method and Routh–Hurwitz stability criterion.

Chapter 11 introduces the fundamentals of frequency response of a dynamic system, depiction of frequency response data by Bode', Nuyquist and Nichol's plots and the information accrued from these plots, also stability analysis in frequency domain.

Chapter 12 discusses about the experimental method of process construction. The step response methods for parameter evaluation of FO, FOPDT, SO and SOPDT model structure.

Chapter 13 covers the controller tuning methods by Ziegler–Nichol, Cohen–Coon and Lopez by the integral error criterion minimization.

In Chapter 14 an effort has been made to introduce and establish a digital simulation platform 'Process Control System Analyzer (PCSA)' based on the language QBASIC. Starting from very simple programs the level is gradually enhanced to include complex open loop and closed loop processes. At the end of the chapter, programs for frequency response plot generation are also presented.

Chapter 15 discusses the necessity of control strategies extra to feedback for systems featuring several kinds of control difficulties. Thus, following advanced control strategies are presented: Feedforward, Ratio, Cascade, Smith's predictor algorithm, Dynamic compensator for processes with inverse response, Inferential, Selective control and adaptive control strategies.

Chapter 16 has two parts. The first part is devoted to the methods of state variable analysis of dynamic system. This includes time domain solution, state transition matrix, conversion of state variable to transfer function and vice-versa, stability aspect, eigen-values and matrix diagonalization.

The second part of the chapter discusses the treatment of multiple input multiple output (MIMO) systems. The topics include measure of interaction, relative gain matrix, and Bristol's method.

Appendix 1A, contains some cgs to fps conversion factor of common physical quantities and constants required in numerical problem solution.

Appendix 1B, displays the symbol of analog and digital instrument blocks used in making Process and Instrument Diagrams in this text.

Appendix 2, contains the theory and application of Laplace' Transform in dynamic analysis and control systems.

Appendix 3A, discusses the characteristics of control valves, certain related properties and sizing procedures in a given application. This part may be used as a follow up of the control valve section of Chapter 7.

Appendix 3B, discusses dynamic aspect of two important flow path configurations around a process: Recycle and Bypass. This part may be used as a follow up of Chapter 9.

Appendix 4, gives transform chart, response equations and controller tuning formulae.

ACCOMPANYING CD CONTENT

The CD contains a FOLDER of the name: APP-4. Insert the CD in the CD-ROM drive of your computer and COPY separately all the contents of APP-4 in one of the DRIVES, say D-DRIVE. There are following FILE and sub-FOLDERs in APP-4:

(A) APP4.doc—a WORD file containing formulae chart, to be printed for use as support for solving problems.

(B) The Q-BASIC FOLDER of the name qbO. This folder contains the following qbasic programs:

1. C10-01.BAS – A program to draw root locus corresponding to seven (7) built-in problems.

2. C14-01.BAS to C14-26.BAS and C14-F1.BAS to C14-F4.BAS, in total thirty-seven (37) nos. of programs. These are amply discussed in Chapter 14 of the text.

3. Eighteen (18) nos. of demonstration programs:

(i) DM01.EXE − Shows the performance difference of 'Algebraic equation' and 'Differential equation' models of a dynamic system.

(ii) DM02.EXE − In this program the concept of superposition of two or more parallel phenomena occurring in a dynamic process to establish its linear character has been shown by three example programs plus an introductory example that reveals the notion of superposition.

(iii) DM03.EXE − The program demonstrates the dynamic manifestation of a 'lumped parameter' process by means of two examples, (1) Heat transfer, and (2) Liquid level.

(iv) DM04.EXE − The program demonstrates the dynamic manifestation of a 'distributed parameter' process by means of two examples, (1) Heat transfer, and (2) Liquid level.

(v) DM05.EXE − The program demonstrates the types of stable and unstable conditions of a process as shown under transient condition. The time domain response together with 'root location' of characteristic equation on complex plane has been shown.

(vi) DM06.EXE − The visual quality of the Step responses of four common example systems are demonstrated.

(vii) DM07.EXE − The dynamics of Non-interacting and Interacting type two tank liquid level system are shown corresponding to sequential pulse and step change of inflow rate.

(viii) DM08.EXE − This program shows the under-damped step response of two classical examples of inherently second order systems, (1) U-tube manometer, and (2) Damped vibrator.

(ix) DM09.EXE − This program can be used to obtain dynamic responses of a

first or second order systems, with or without a dead time element. The user can define the input function by putting the input, x, values in a data array. The instructions provided in the opening screen should be carefully read. The minimum value of time constant should not be less than 20 time units to avoid numerical integration error.

(x) DM10.EXE − This program depicts pulse responses of a first order system. Progressively, each of the pulses are of decreasing width and increasing height, but their magnitude remains same. In limiting situation with an infinitesimally small width, the pulse becomes an impulse and what we obtain is an impulse response of the same magnitude.

(xi) DM11.EXE − Corresponding to a sinusoidal input concentration variation of a solution, this program demonstrates the exit concentration dynamics of a single stirred tank for, (1) Only mixing, and (2) Mixing with dilution with a pure solvent stream.

(xii) DM12.EXE − This program examines the liquid level dynamics of a cylindrical tank with a non-linear resistance in its bottom exit line. The dynamic response has been studied for different descriptions of non-linearity.

(xiii) DM13.EXE − This program examines the exit liquid temperature response of a stirred liquid heater. Heat is transferred from the circulating hot liquid in the jacket of the vessel with the help of an electrical immersion heater situated at the vessel bottom. Input disturbances in (1) Liquid flow rate, (2) Inlet temperature of the liquid, and (3) Heat input rate, can be made by describing the input functions as data sets.

(xiv) DM14.EXE− The program is designed to reveal different aspects of the integrating character of a dynamic system by citing certain example systems.

(xv) DM15.BAS − The program simulates a closed loop FOPDT system and generates the response corresponding to user provided parameter values like process dead time, controller mode, controller parameter values, and type of disturbance (in load or set point).

(xvi) DM16.EXE − The program simulates the closed loop FOPDT response of an FOPDT process, when the final control element acts in PWM mode. For a given set of parameter values, response for +ve step change in load, and –ve step change in set point are demonstrated.

(xvii) DM17.EXE − This program uses a gradient search optimization algorithm for minimization of Integral Time Absolute Error (ITAE) by searching through the 3-D parameter space (K_C, t_I, t_D).

(xviii) DM18.EXE − The program displays the cascade scheme, kinetic and heat transfer model for an exothermic batch reactor, and generates the response

of reacting species concentration, reactor and jacket liquid temperatures for total run time.

To run an -.EXE program file, open the qbO folder where all the contained files are listed. Select the chosen -.EXE file from the list, then 'double click' the 'mouse arrow' on the selected file. The program will run.

To run a -.BAS program file, open the qbO folder where all the contained files are listed. Select the QB.EXE file from the list, then 'double click' the 'mouse arrow' on the selected file. An active screen on the full extent of your monitor will appear. Drop down 'File Menu' from the extreme top left of the screen, and click on the program list. The program list will appear in a separate box. Select the chosen program and click on <OK> option. After the program appears it can run:

1. By pressing F5;
2. By dropping the Run menu and clicking on the item Start.

To develop your programming skills read any textbook on BASIC programming; you can also print the Help items from the Help menu, which will reveal the usage of different command items of QBASIC programming. The QBASIC 4.5 compiler and support files are available from following sites:

(a) www.programmersheaven.com.

(b) cn.softonic.com/qbasic 4.5

3. A DOS based software may be used for transfer of graphic outputs. This is called DOSBOX. Install DOSBOX on the DESKTOP. Use of this software is primarily for generation of graphic output of QBASIC programs on a temporary drive created by calling the QB.EXE from the DOSBOX. The obtained output will be transferable to any of the MS.PAINT files. Assuming that qbO folder is within the D-drive, you may follow the following steps:

(a) Double click on the DOSBOX icon on the desktop. On the DOSBOX screen, the following lines will appear:

Z:\>_

print: **mount d d:**, then press ENTER. Next line will appear as:

Drive D is mounted as local directory D:

Z:\>_

Print D: then ENTER.

Next line will appear as:

Z:\>D:_

Print **cd\qbO** then ENTER.

Next line will appear as:

D:\>cd>QBO>_

Print **cd qb**_then ENTER

Again, the active screen on the DOSBOX screen of your monitor will appear. Drop down 'File Menu' from the extreme top left of the screen, and click on the program list. The program list will appear in a separate box. Select the chosen program and click on <OK> option. After the program appears it can run: (i) By pressing F5, and (ii) By dropping the Run menu and clicking on the item Start.

After the graphic output appears on the screen, press ALT + PRINT SCREEN. Then open any chosen file of MS-PAINT. Drop the EDIT menu and click the Paste item to get the graphic output from DOSBOX screen on to the PAINT screen. The picture will be drawn in white on a black background. Drop the Image menu and click on the item- Invert Colors to obtain the figure drawn in black on a white background. This figure may be edited to add scale divisions, captions, etc., and pasted in a word file for report generation.

The DOSBOX package can also be downloaded from the following site:

www.dosbox.com/download.php? Main = 1

Prabir Kumar Sarkar

Acknowledgements

In the first place this opportunity reminds me the person who urged to take up this task and rendered continuous boost and morale leverage in times of set backs, frustrations during long bad periods of intellectual impasse. In a nutshell without whom this project could not even be started.

I am indebted to the following of my students who also urged me not only to write this book but gave their active help by checking the procedural detail of analytical results, solution of examples and in preparing the simulation platform 'Process Control System Analyzer (PCSA)':

Mr. Abhirup Banerjee, Mr. Devdutta Mukherjee, Mr. Biswajit Rakshit, Dr. Tushar Sen, Sumantra Gupta, Mr. Dhurbajyojit Sinha, Mr. Pritam Das, Miss. Indrani Basu, Mr. Arjun Chakraborty, Mr. Soumyajit Sengupta, Mr. Sudip De Sarkar, Mr. Abhishek Das, Mr. Somenath Dutta, Rahul Agarwalla.

A special word of thanks goes to Mr. Somak Jyoti Sahoo for his continued help through the years for preparing the text.

At last, for all the failures, shortcomings and incompleteness of the project I may take refuge behind the following words spoken by the famous philosopher, Paul Newman:

A man would do nothing if he waited until he could it so well that no one could find fault with what he has done.

Prabir Kumar Sarkar

1

Introduction

1.1 Historical

History of Control probably dates back to the history of human civilization when man first learnt to harness (control) fire in his cave corner. In the temple of Greek Gods, cleverly designed mechanical and acoustic systems were used to show miracles to common people. That was the first social abuse of Control Technology. During the **14th** and **15th** century, Scottish coalmines used hydro-turbines called 'water wheels' to drive a lift like device for lowering and lifting of workers in the shaft. In the period of Industrial Renaissance, Steam Governors came up for speed control of steam driven machinery and in 1868 Maxwell[1] published its formal analysis. Inspired by this work, Hurwitz (1875), Liapunov (1893) and Routh (1905) forwarded important contributions in the area of stability analysis of dynamic systems. In a separate study carried in Bell Laboratories, Nuyquist[2] proposed a stability analysis based on frequency response in 1932. Within a few decades, Servomechanism was introduced in industrial applications and their analysis by Hazen (1943)[3] was available only after a short period from their appearance. Thus, Control Engineering started its long saga. The period of World War-II witnessed an explosion in Control Technology and probably a step jump of several decades was completed in three to four years.

The first application of control in Chemical Process Technology may be recognized by the work of Mason in 1904, where he introduced the conception of Dead Time and its significance in a control system's dynamics. In 1930, Grebe[4], Boundy and Cermak, discussed using derivative mode in some difficult pH control problem in their publication. Further, in 1958 Amundson and Aris[5] published their works on Exothermic Reactors and introduced State Variable Technique for the process system that probably marked the entry of classical analysis into the area of process dynamics and control. The last few decades have witnessed a colossal amount of development in this field. With respect to hardware, we may mention the phasing out of modular instruments, introduction of digital micro-controllers, which gradually gave way for PLC (Programmable Logic Controller) based DCS (Distributed Control System) systems. Largely, this hardware development has made it possible to develop and implement the most advanced process control strategies like Adaptive Schemes,

Expert Systems, Self Learning type Neural Network controllers, and Statistical Process Control (SPC) systems.

In the recent years, it has been felt from many quarters that a more critical examination of control schemes at the design stage can also reduce the investment in instruments, which may be equal to a small percent of the total plant cost. Many existing plants are over-instrumented and even DCS based systems show questionable performance, particularly in face of a critical combination of external disturbance. In many cases this is due to the use of a 'package technology' or firmware with a generally defined objective without permitting any leeway for customization required for a specific application.

Dynamic analysis and carefully designed compensation systems can show which controllers are really needed and may lead to better system performance with respect to operational ease and economic achievement.

With such an objective we shall begin our journey, first by defining certain terms and ideas related to a process and then their control aspects that will be extensively used in future discussions.

1.2 Preliminary Definitions

The term **Process** has a different meaning when used in process dynamics and control literature than in chemical engineering.

In the latter, a **unit process** signifies a conversion mechanism within an equipment system that involves chemical reaction, where as a **unit operation** is only a physical conversion, e.g. separation of one entity from a mixture, reduction of size of solid material feed, mixing of more than one raw material, etc.

In the present context, we may define a **Process** as a conversion mechanism by which a low value raw material is converted to a high value product. Or, a **Process** may be conceived as associated with an equipment in which, a material and/or energy stream occurs from input (entry point) to the output (delivery) end of the equipment, which embodies the desired conversion. We can pictorially define a process as (refer to Figure 1.1):

Figure 1.1 Definition of a process.

An industrial process is usually a combination of recognizable small elemental process units. Each element is identified by a single kind of conversion. For the purpose of analysis it is a well-known technique to first understand or analyze these 'bits-n-pieces', and subsequently try to get the 'big picture' of the total process by combining them.

To carry out the desired conversion, an assortment of electro-mechanical equipment

is used in a well thought-out (designed) manner. These equipments are also called **Systems**. Often processes are identified with systems because in engineering you cannot carry out a process without a particular equipment system. Thus, these two terms have an overlapping meaning.

The physical boundary or envelope within which a process occurs is called the **system boundary**. Everything outside this is called the **environment** or **surrounding**.

A process equipment is designed while conforming to a list of required performance specification. An optimum design method determines the physical details of the equipment, as well as finds the parameter values of the process to be maintained during operation to obtain best performance in terms of conversion or profit. Such parameter values may be temperature, pressure, pH, composition and a few other process variables that are key conditions to be maintained at the recommended design value during the operation. This **optimum state of operation**, in terms of parameter values, remain constant in case of a continuous (steady state) process, but becomes a function of time for a batch (unsteady state) process.

Unavoidably, certain disturbances in the form of material and/or energy streams from the environment enters as inputs into the process. These inputs are called **excitation**, **disturbance** or **load**. Such inputs may also originate from a disturbed upstream sub-process that is within the system boundary. These external inputs may be further subdivided into two classes as follows:

- The class which affects all processes nesting a similar type of conversion in a similar manner, in spite of differences in application and physical location, is called **macro** or **global** disturbances.
- The other class of inputs whose effect varies due to physico-chemical description of the process, or is dependent on any other operational parameter of an otherwise similar process, is called **micro** or **local** disturbances.

The load inputs tend to drive the process away from the designed optimum state of operation. To bring such a disturbed process back to the desired state, a separate set of equipment or instruments are employed. By being employed in a proper combination, they constitute what is known as the **control** or **instrumentation system**. A control system employed for maintaining a chemical process is known as **process control system**. Practically all processes are run with the help of a control system to achieve certain objectives.

The primary objective of implementing control system is for the *Product Quality* assurance. The other objectives are: to obtain the highest possible *Energy and Mass Transfer/conversion Efficiency*; enhancement of *Safety* features; and *Eco-Friendly*

aspect.

Fundamentally, in a control system there is a hardware, called **measuring element**, which measures the state of the process in terms of **controlled variable**, and transmits the information to the **controller**. In the controller, this information is compared with the desired or **set value** of the variable in a **comparator** section, thus an **error** is computed, and processed to produce the controller output as a function of error. The energy content of the controller output signal is essentially low. The low power controller output is transmitted to the **final control element**, where it is sufficiently amplified to regulate the flow rate of the **manipulated variable** to the process. This is called the **control action**. The manipulated variable enters into the process parallely with the **load or disturbance**. Load is the uncontrollable external disturbance that drives the process away from the set point. A manipulated variable is the manifestation of control action that neutralizes the disturbance and brings the controlled variable back to its desired or set value. The flow of signals inside such a control system through its elements has been presented in Figure 1.2.

The control scheme depicted in Figure 1.2 is an example of the schemes used in the process control system. This scheme is by far the most widely used, and is called **feedback control** because the information used to control and the resulting control action, flows back or goes reverse to normal flow of material and/or energy through the process. Though we are mainly

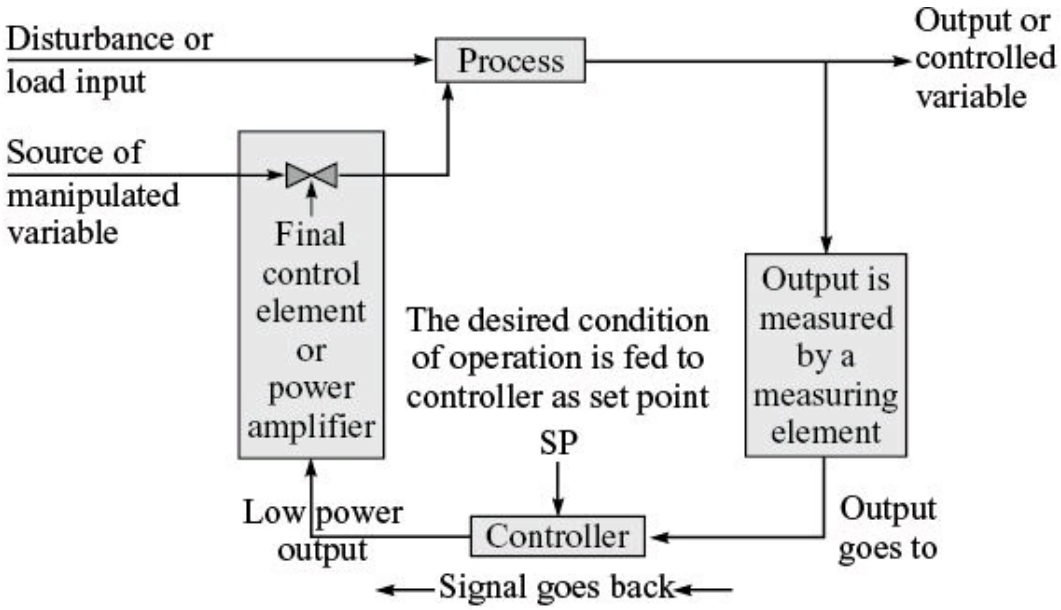


Figure 1.2 Information flow inside a process control system.

concerned with chemical processes, but any other controlled process bears a similar relationship with their control system, under the feedback scheme. Such a control system is also called a **closed loop system** because the flow of information occurs in a path from the process output round through the control system elements and appears again as control action at the entry point of the process. A process without its control

system is called an **open loop process**.

1.3 The Block Diagram

The scheme of controlling a process that we have just exposed may be efficiently expressed by a graphical representation. This pictorial description of a control system is called its **Block Diagram**. Such a block diagram contains information like:

(a) *Material*, *Energy* and *Signal Flows* within the system.

(b) The *Mathematical Processing* done for conversion of a signal to another (as happens when a temperature sensor-transmitter converts the process temperature to an electrical current).

(c) *Branching* and *Algebraic Addition* of the signal streams.

A Block Diagram generally consists of four basic elements:

(i) ***Lines*** represent a signal, material or energy stream path and the associated *Arrow-Head* expresses the direction of flow.

(ii) The ***summing points*** represent the algebraic summation of signals coming through the incoming arrows with the +ve or –ve signs as their algebraic attributes. The resulting signal flows out through the outgoing arrow.

(iii) A ***branch point*** is the location on a line at which the signal branches out and goes paralelly to their individual destination.

(iv) A ***block*** represents the mathematical operation done on an incoming signal to convert that to the next one. Thus, a block is where the input signal to the block terminates and the output signal from the block is generated. In a block diagram the blocks generally represent the hardware used in a control system.

Any block diagram can be handled or manipulated algebraically. Figure 1.3 shows the essential rules of block diagram algebra.

The general feedback control scheme of Figure 1.2 may be expressed in the form of block diagram. But before that, we should introduce notations or symbols for representing the different operational blocks used in Figure 1.2. These are:

- The Process: G_P;
- The Measuring Element: G_M;
- The Controller: G_C;
- The FCE: G_V.

(*Throughout the text we will be using the same nomenclature for these blocks of an Instrumentation System.*)

The block diagram for the system of Figure 1.2 is shown in Figure 1.4. The first

summing point before the controller block is called the **Comparator** because this performs the following algebraic comparison between Set Point and Measured Variable (mv) to find the error as:

$$e = R - \text{mv} \qquad (1.1)$$

The second summing point before the process block is where the Load (L) and Manipulated Variables (MV) are coupled to form the Input to the process as:

$$X = \text{L} + \text{MV} \qquad (1.2)$$

It has become customary to represent the control system elements in a block diagram by the following notations or symbols:

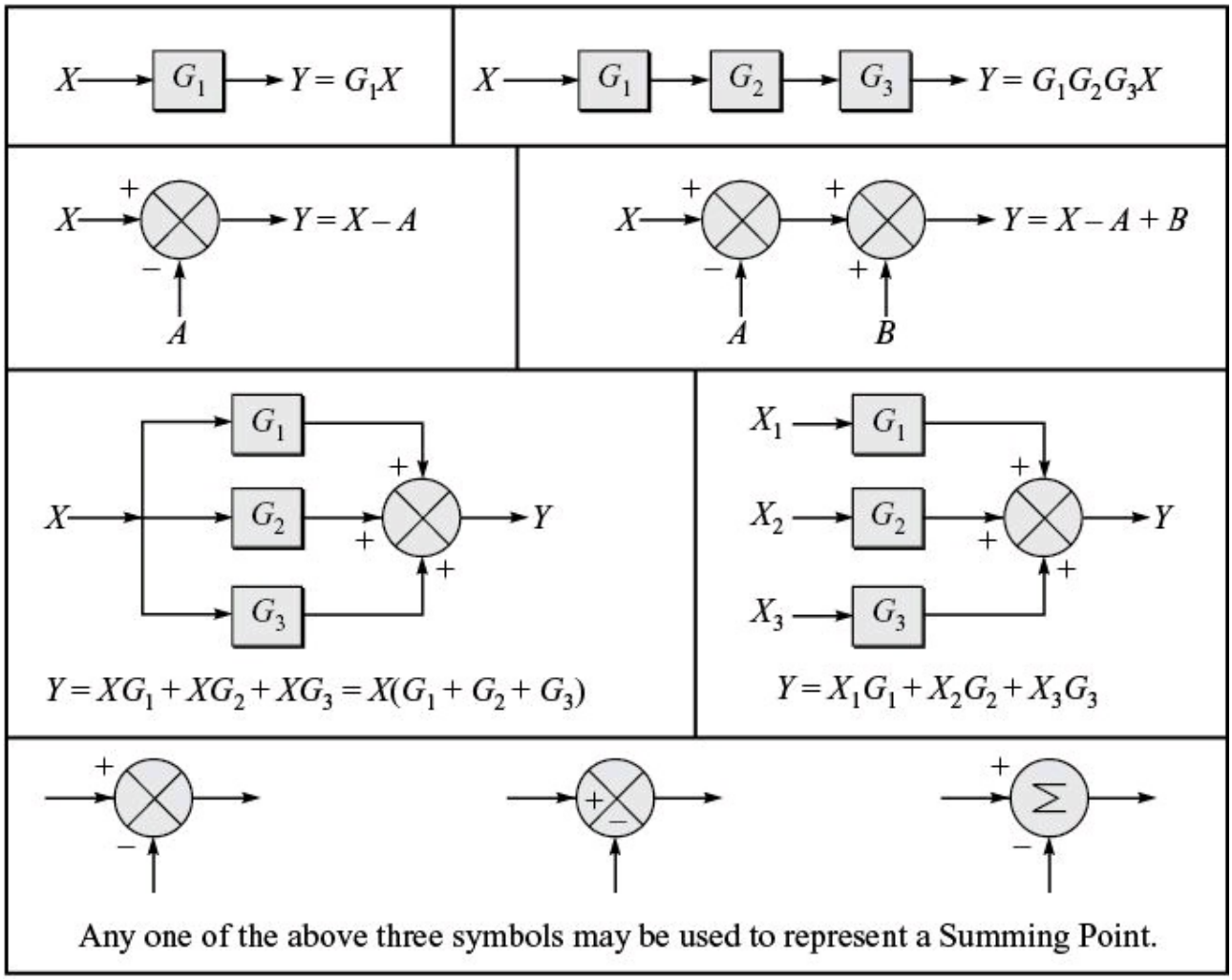


Figure 1.3 The Rules of block diagram algebra.

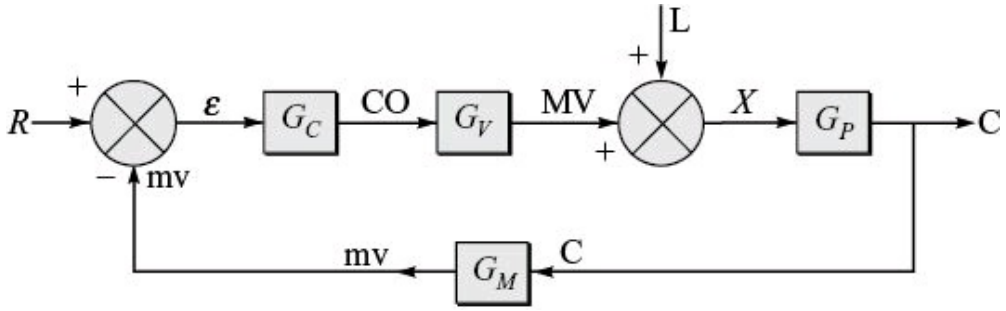


Figure 1.4 Block Diagram of a Control System.

We will use the block diagram to represent process dynamic models as well as in

dealing dynamics of closed loop systems.

The units or elements of a control system are passive pieces of machineries. They require some form of energy to operate. Depending on the type of energy used, a classification of the instrumentation system may be done. A pneumatic system operates within an input/output variable range of 3–15 psig air pressure, and an electronic system operates within 4–20 mA DC. The input–output compatibility in these systems may be appreciated more thoroughly from the following signal trains.

For a pneumatic control system:

$$G_P \xrightarrow[\text{From process}]{\text{Contr. Var.}} G_M \xrightarrow[\text{O/P} \rightarrow 3-15 \text{ psig} \rightarrow \text{I/P}]{\text{Measrd. Var. (mv)}} G_C \xrightarrow[\text{O/P} \rightarrow 3-15 \text{ psig} \rightarrow \text{I/P}]{\text{Contr. O/P (CO)}} G_C \xrightarrow[\text{to process}]{\text{Mnpltd. Var. (MV)}} G_p$$

In an electronic control system:

$$G_P \xrightarrow[\text{From process}]{\text{Contr. Var. (C)}} G_M \xrightarrow[\text{O/P} \rightarrow 4-20 \text{ mA} \rightarrow \text{I/P}]{\text{Measrd. Var. (mv)}} G_C \xrightarrow[\text{O/P} \rightarrow 4-20 \text{ mA} \rightarrow \text{I/P}]{\text{Contr. O/P (CO)}} G_C \xrightarrow[\text{to process}]{\text{Mnpltd. Var. (MV)}} G_p$$

If the instruments in a loop need more than one type of energy source, we may call such an arrangement as the **mixed** o r **heterogeneous** instrumentation. For many practical considerations, a mixed instrumentation is often recommended for realization of certain control systems. A typical example system comprising of electronic measuring element and controller, may employ a pneumatic final control element, e.g. a Pneumatic Control Valve. The electrical output from the controller of this loop needs to be proportionally converted to pneumatic signal to actuate the Final Control Element (FCE). A signal transducer known as Electro-Pneumatic-Converter (EPC) is available for this job. In an EPC 4–20 mA DC signal is proportionally converted to 3–15 psig air pressure.

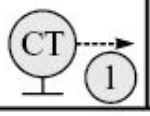
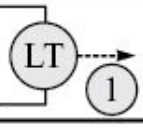
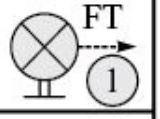
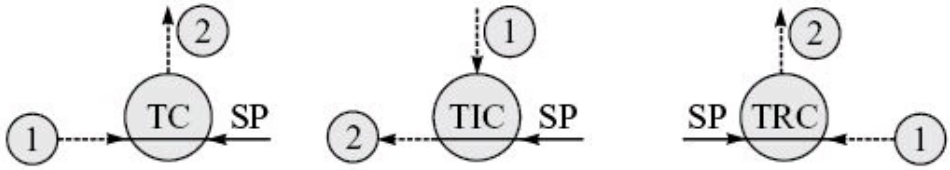
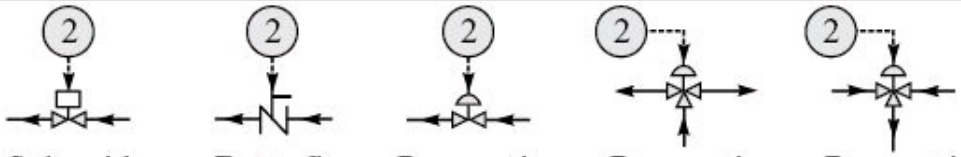
To finish this discussion, we should mention that there are two more types of instruments that require energy sources other than we have already mentioned. These are **hydraulic** and **fluidic** instruments systems. As these types are rarely or not used at all in process control applications, we would refrain from discussing any further about them at present. However, these instrumentation would be discussed a little further at the beginning of Chapter 8. If you are interested, you may go through the references mentioned at the end of Chapter 8.

1.4 The Symbol of Instruments

We have already presented one method of describing a control system, i.e. the block diagram method, which does not contain any implementation detail, and is primarily used for the purpose of analysis. For practical and professional presentation, standard symbols for instruments and process equipments are used to describe the system. As the control system alone cannot be expressed without the associated process, a diagram to represent the close loop system by the symbols is called a **Process and**

Instrumentation (P&I) diagram. In Table 1.1, we present a comprehensive collection of instrument symbols that have been used in this text and are easy to follow; a more detail list is included in Appendix-IB. However, a detailed list of instruments, recommended by Instrument Society of America (ISA) is compiled by Liptak[6].

TABLE 1.1 Common Instrumentation Symbols for Control Loop Presentation

Sensor-transmitter	For pressure, temperature composition, by changing the first letter in the circle 	for level 	for flow 
Controller	 The first letter in the circle denotes the controlled variable type The second letter if any denotes indicating (I) or recording (R), facility. SP is the set point. Arrow (1) brings information from transmitter. Arrow (2) carries the controller output to FCE.		
Final control element (control valves)	 Solenoid valve for flow switching	Butterfly type for vapour or gas	Pneumatic two-way for liquid	Pneumatic three-way (diverting) for liquid	Pneumatic three-way (mixing) for liquid
Signal converter	Electro-pneumatic-transducer (EPT): 4–20 mA 3–15 psig				

To demonstrate the manifestation or existence of control action that are necessary for driving a process, we have selected some examples from different application areas. These examples have been chosen not only from the process industry but also from other facets of experience to emphasize that control action is embedded in many areas of our society and life. In fact, the road to civilization has been paved by a continuous struggle to win over or control the elemental forces of nature. Hence, the first lighting of fire might be an accident, but the really smart thing has been to harness it for mankind. However, the really marvellous control systems are functioning inside you. Do you know how many muscle movements your brain controls when you advance your foot just one step? About thirty-eight.

1.5 Examples of Control Systems

EXAMPLE 1.1 Before the advent of instrumentation science, control action was brought manually in many industrial applications and still it is done in our daily life

and also in a fair number of industries. The study of such a system may give us an opportunity to compare part by part with that of an instrumental control. Figure 1.5 depicts a human being riding a bicycle.

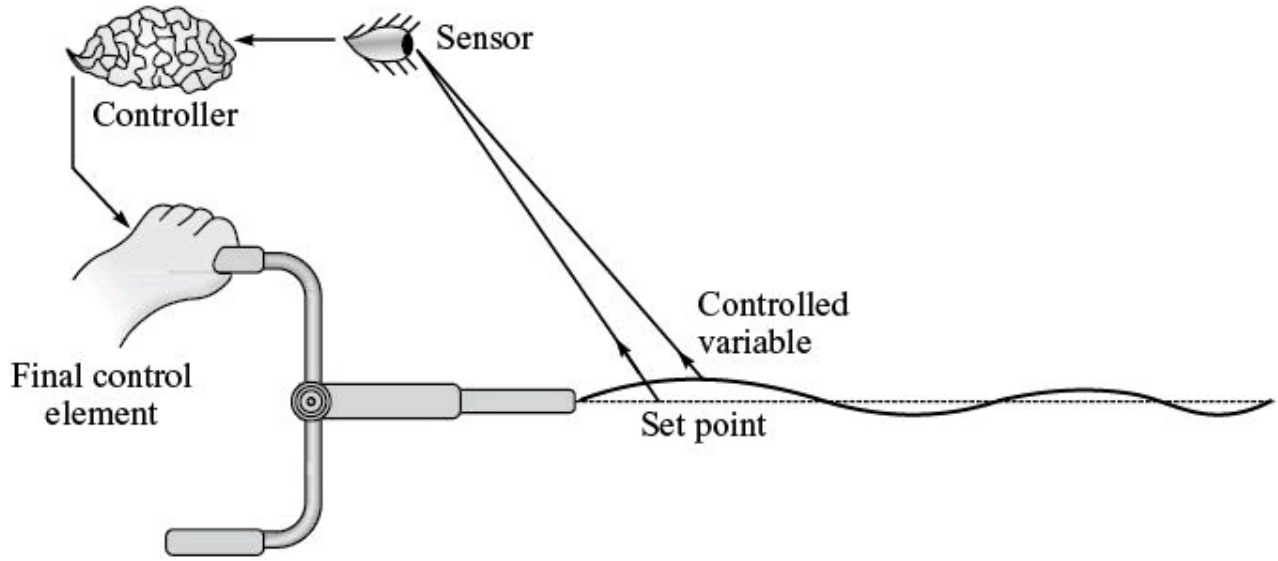


Figure 1.5 A man riding a bicycle and trying to advance on an imaginary straight line.

The rider looks ahead and receives information about his current position (**Controlled Variable**) and also about the imaginary dotted line (**Set Point**). So obviously the eye acts as the **Sensor**, that sends the information to the human brain (**the Controller**), and after processing the data, the decision or control action is achieved by steering the bicycle by the rider's hand (**Final Control Element**).

Note that a person, who has just learnt how to ride a cycle, gradually increases his skill as he continues the practice. From his initial failures and incompetence, he picks up what are the *dos* and *do-nots* about cycle riding. The controller stores this information, putting more importance (weight) on the 'do-list'. Thus, an efficient control strategy grows up.

Is it possible to artificially build up such a self-learning intelligent controller?

EXAMPLE 1.2 The control system, called **Fly-ball Governor**, was probably developed as one of the first of its kind, which was successfully used in industrial application for more than a century. The principal purpose of a Governor is to maintain rotational speed of a steam turbine.

The operation of this control system is simple; the assembly of the twin mass and linkage mechanism rotates with the turbine through a gear and pulley and rides on a free sleeve over the vertical stem. The steady rotational speed of the turbine may change due to load change as well as inlet steam pressure variation. This will cause the spherical masses to come closer or move away as the centrifugal forces decreases or increases until a new balance point is reached with the compressed spring. Thus the vertical stem goes upward or downward. The valve-plug attached with the stem will move away or come closer to the valve-seat, causing the steam flow to vary in a

manner so that the turbine speed comes back to its desirable value.
Figure 1.6 illustrates the action of such a Governor. Adjusting the spring compression may change the desired speed or set point of this control system. A higher compression will raise the set point to higher speed.

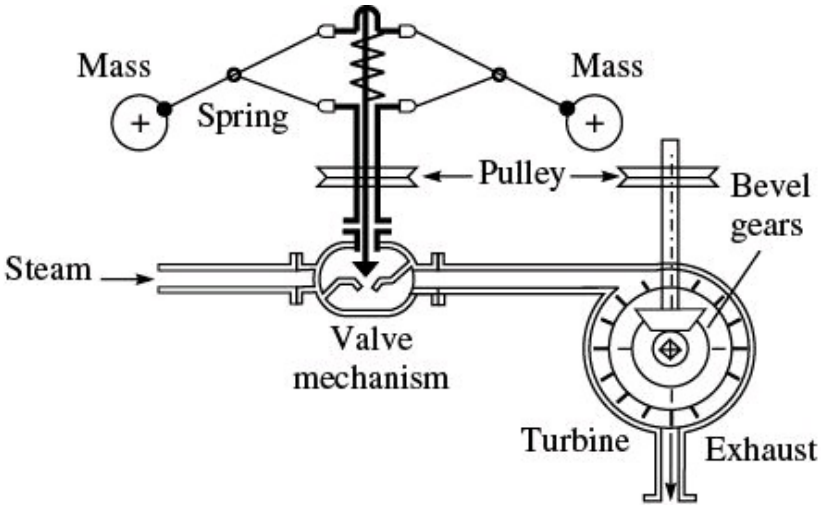


Figure 1.6 The steam governor probably the oldest controller.

There is no historical information available about the inventor of this machine; it was used in the industry for a very long time probably from the period when the first development of steam engine was going on during industrial revolution. There was no analytical or design data available and a small bunch of senior mechanics could build Governors by experience only. Then in 1868, Maxwell published a paper, in which he put forward the complete analysis of this machine, and probably started the era of an analytical way of control system analysis and design.

EXAMPLE 1.3 Probably not so historical but much more popular and in current use is the **Float type Liquid Level Controller** (Figure 1.7).

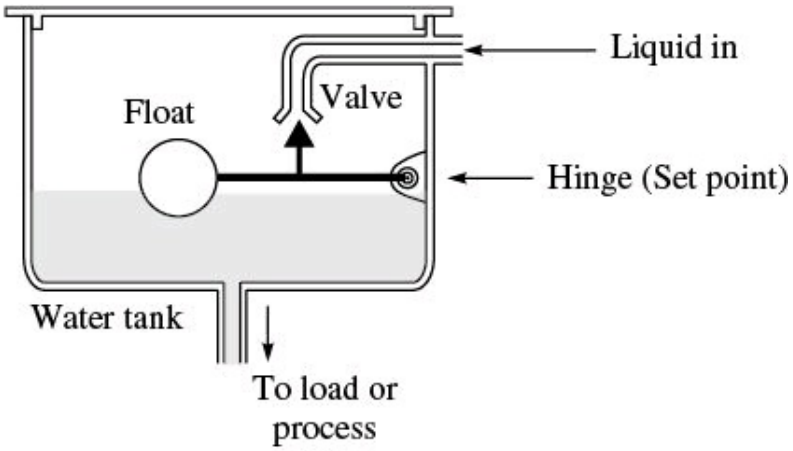


Figure 1.7 The float type level controller.

A rise in liquid level causes the hollow Ball–Float to lift, and the attached 'plug' of a valve mechanism comes closer to its 'seat' (built on the end of the liquid inlet) to reduce the liquid inflow. This system has a similarity with the Governor of Example 1.2, in the fact that inception of this level control system also has been lost from

annuls of technology development. In the system, the set point is the location of the hinge on the tank wall which is usually fixed because in major number of applications it is required that the tank should be filled to the fullest before use.

EXAMPLE 1.4 This is an example from a typical mechanical operation: rolling of a thick metal sheet as feed into a thinner sheet as a product of controlled thickness. Product thickness is measured by measuring the intensity of the *g*-ray beam coming through the product from a radioactive source.

The source (a typical source may be the radio-isotope of Cobalt with an atomic number–60), detector and the associated electronics that interprets the beam intensity in terms of sheet thickness, constitute the measuring system. The measured value of the product sheet as a current signal is transmitted to the thickness controller that compares the thickness with the set value, finds the error, and computes corrective action as its output as a current signal. A typical set of electronic control hardware may operate within an input/output range specification of
4–20 mA DC. Controller output, as current signal is converted to hydraulic fluid pressure in a Signal Transducer, is symbolically represented by 'I/P inside a circle' as standard P&I drawing symbol, as illustrated in Figure 1.8.

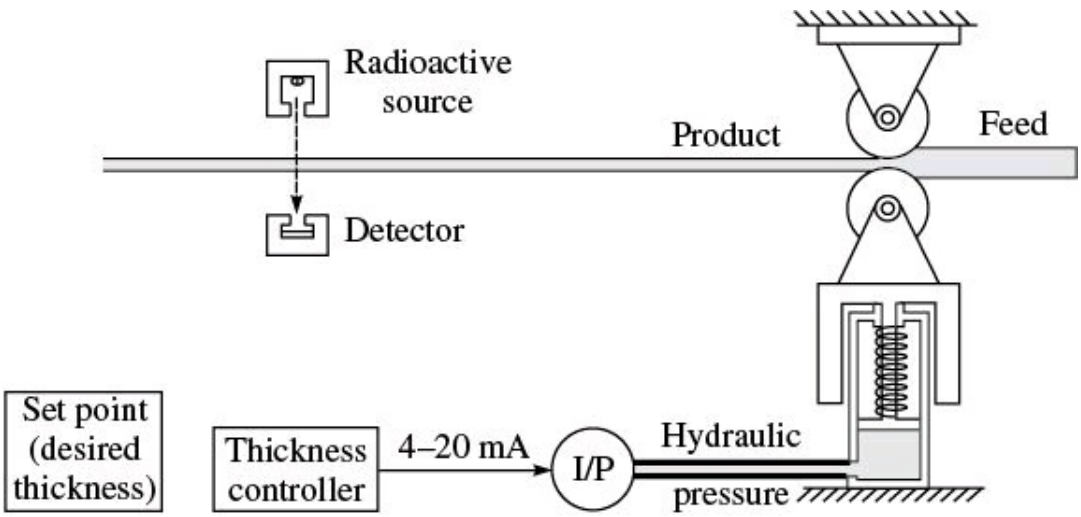


Figure 1.8 The thickness control of a metal sheet rolling system.

In case the manufactured sheet is of incorrect thickness, the change in fluid pressure is amplified in a hydraulic actuator. We get a vertical displacement of the lower roller as the piston works and comes in balance with the spring force, and the gap between two rollers is suitably adjusted to obtain a metal sheet of correct thickness. This arrangement may be called the Final Control Element (FCE) of the control system.

The Thickness Control System looks apparently fine. But an interesting question crops up. Any defective portion of the sheet has to travel the distance from the Roller assembly to the detecting point before the defect is sensed, and at each of these instance the smallest length of defective thickness is at least equal to the distance between roller and thickness detector. Note that we have no information about the

thickness of the sheet during the 'time' when it travels between the production and detection point. Let us call this phenomenon **Dead time** because we are completely unaware of what is happening at the process output during this time. The information currently available from the sensor has actually happened in the past or earlier by an amount equal to the dead time of the system. Please remember that this delay or dead time poses a serious problem in control system design as we have seen here. We shall have to talk about these delays in proper time, in detail, in further chapters.

EXAMPLE 1.5 In this example we demonstrate the control of the exit liquid stream temperature from a constant hold up mixer cum heater (Figure 1.9). This is one of the most popular example systems that have been used in numerous texts to introduce many important conceptions of control engineering. We shall also do so in our discussions.

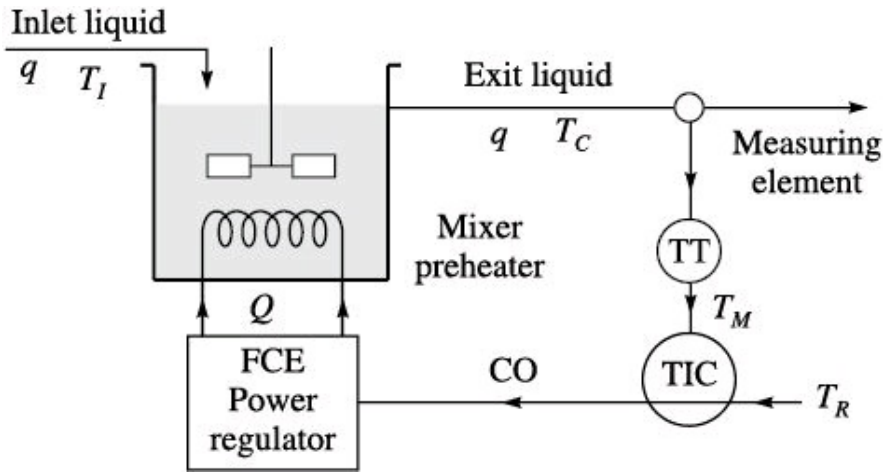


Figure 1.9 The temperature control system of a mixer pre-heater.

A Thermocouple Sensor, TC, measures the exit stream temperature, T_C. The TC output is electronically conditioned in the Temperature Transmitter, TT, to 4–20 mA DC current signal and transmitted as the measured variable, T_M, to the Temperature Indicator Controller, TIC. Such a Controller, beside its control capability has the facility of displaying the Measured Variable, T_M, and Set value, T_R. Error is found as, $e = T_R - T_M$, and the Controller Output, CO, is produced as a function of error, CO = $f[e(t)]$. The low energy output from the Controller is amplified through the FCE (which may be a Thyristor type power regulator) that changes the power mains (230 V) current to the resistance heater in the tank.

There are many industrial applications of this system. For instance, furnace oil (FO) is a commercially available liquid fuel used in furnaces. At normal temperature, FO is a thick liquid with waxy and tarry particulate deposits that reduce its flow-ability. This oil has to be heated to about 90°C–100°C before it is sent to the burner. On the other hand, over heating of FO may raise some of its lighter components to their ignition temperature and cause fire hazards. Hence, temperature control of the exit liquid from

the pre-heater is needed to have an uninterrupted flow of FO and furnace operation.

The thermal quality of feed stream to a distillation column needs to be uniform and as close to the design specification of the feed as possible, because otherwise the column may become unbalanced. A temperature controlled feed pre-heating system is again the answer to such an application.

The next three examples have been chosen to introduce some new aspects of process operation problems, and also because they have been addressed by realization of control system.

EXAMPLE 1.6 A tank acts as a constant head reservoir for supplying liquid to a process. The process has variable demands. A Constant head in the tank is required so that exit flow rate is only dependent on the resistance in the outflow line. The resistance may be the Control Valve (FCE) of the control system of the down stream process. The liquid level of the tank is measured by a level sensor-transmitter, and sent to the level Controller. The Controller Output manipulates inflow through the Control Valve. The open loop process, and the process with its control system has been shown in Figure 1.10.

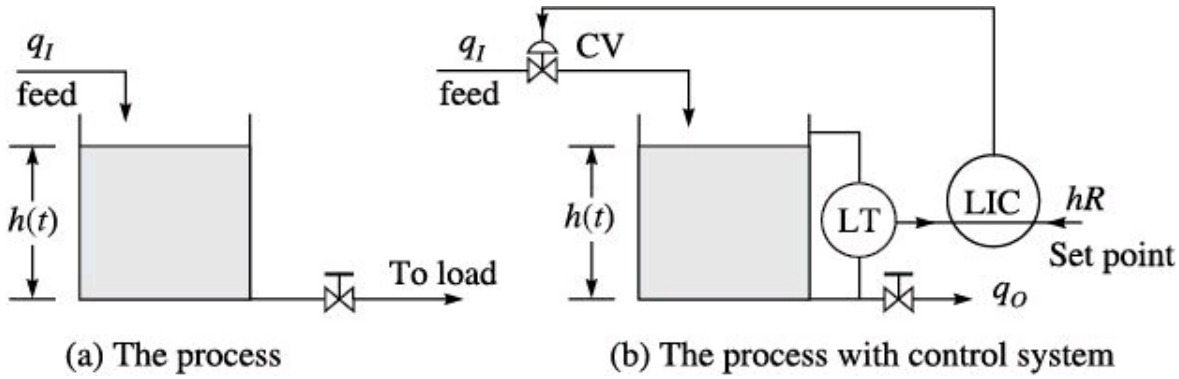


Figure 1.10 The dynamic level system and its control.

EXAMPLE 1.7 In a distillation column, the reflux ratio is one of most important parameters to be maintained to keep up the top product or distillate composition to the desired value; and in many instances top product is the desired product of this operation. Reflux ratio, as the name suggests, is the ratio of molal or mass flow rates of two streams R and D.

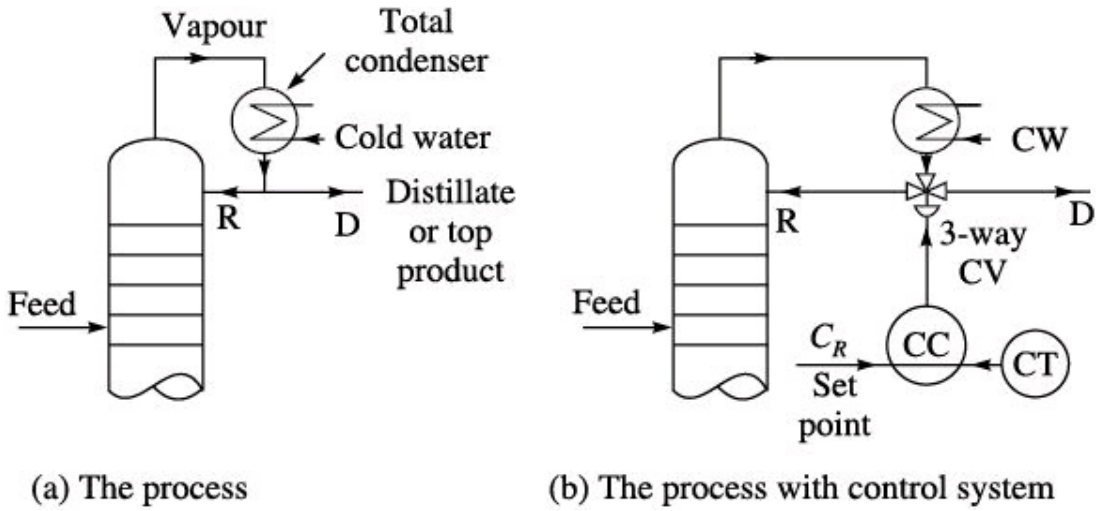


Figure 1.11 The reflux ratio control of a distillation column.

The Total Condenser condenses all the vapours coming from the top plate of the column. The condensed liquid is diverted into two streams, R and D. R is fed back to the top plate, and D is the product stream. A composition sensor-transmitter on the product line measures the product composition and sends the information to the composition Controller, which manipulates the reflux ratio through the three-way control valve. Figure 1.11 depicts the control system.

[Three way Control Valves are a variety of control valves and acts as special final control elements. They come in two basic configurations: (1) **Diverting type** has one inlet and two outlet ports. Liquid coming through inlet goes out through the outlet ports subdivided into two streams at a definite volume ratio. This ratio is dependent on the valve stem position. (2) **Mixing type** has one outlet and two inlet ports. Liquids coming through inlet ports are mixed in a definite volume ratio inside the valve body, and go out through the outlet port. Again this ratio is dependent on the valve stem position.]

The same control action may be achieved by using two sets of two-way Control Valves. But discussion on this topic should be made after the **Action** of a Control Valve is introduced in Chapter 7.

EXAMPLE 1.8 In many process industries, including production of heavy chemicals, steel, aluminum, and fertilizers, where high pressure steam is used as a reaction component or utility at certain stages of manufacture, high temperature saturated water or aqueous solution streams containing mineral solutes are produced as a by-product.

Such a stream contains a large amount of energy. Unless this energy is recovered, the production economics becomes very poor. Recovery through heat exchangers by heating an incoming cold liquid is less efficient than 'Flashing' operation. Flashing of a high temperature high-pressure liquid means suddenly expanding it in a low-pressure chamber where the stream will separate into saturated vapour (relatively low pressure pure steam) and saturated liquid fractions.

Then the generated steam may be put into the medium pressure steam supply line and the liquid sent either to the next flashing stage or to a heat exchanger. Successful operation of a flashing stage requires that the chamber is maintained at a slightly higher pressure than the medium pressure steam header, and a positive liquid head has to be kept so that full flow of liquid occurs through the liquid outlet of the flashing unit.

A control system has been illustrated in Figure 1.12, for this purpose. Note that in this system there are two Controlled Variables involved in the same process. If physical laws relate these variables to each other, a situation may arise when manipulation in one loop may affect the second loop so that the second Controlled Variable is disturbed from its set value. This is one of the classical problems associated with multivariable control systems, known as **variable interaction** or simply interaction. We have many things to talk about interaction but only when we start our discussion on multivariable system. It is sufficient to comment here that in this example the loops have very negligible interaction and cause no operational problem as such.

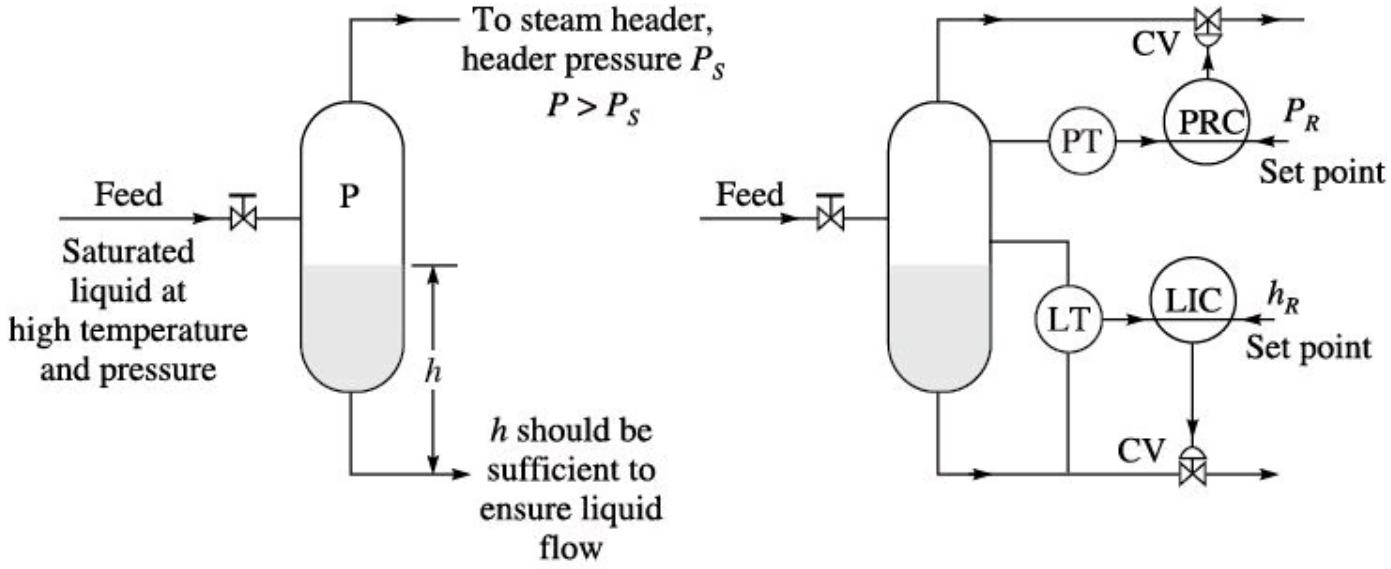


Figure 1.12 Multi-loop control system of a flashing stage.

Still, there is a possibility of another kind of control difficulty about which you should be aware of. It is very usual in a plant that there are few load or demand points connected in a parallel fashion to the steam line, and they have their own on-off schedule. This will cause fluctuation of pressure in the steam line. Now you may appreciate the difficulty that such a pressure fluctuation will bring into the pressure control loop of the Flashing Stage. This problem will be tackled when we discuss about **Cascade Control**.

To appreciate the existence of control action in areas other than technology and science, we cite the following two examples.

EXAMPLE 1.9 An advanced diabetic patient needs to be injected with insulin regularly round the clock depending on his fluctuating blood sugar level. A special needle like glass electrode (marketed by Pyrex Corpn.) inserted in an artery of the patient continuously measures the blood sugar level, and transmits the information to the Controller. A microcomputer based concentration controller computes the amount of insulin to be injected, with respect to current sugar level.

The final control element of this control system is a special pump known as Peristaltic

Pump through which Insulin is injected. A portable version of this machine is also available. Figure 1.13 presents the system.

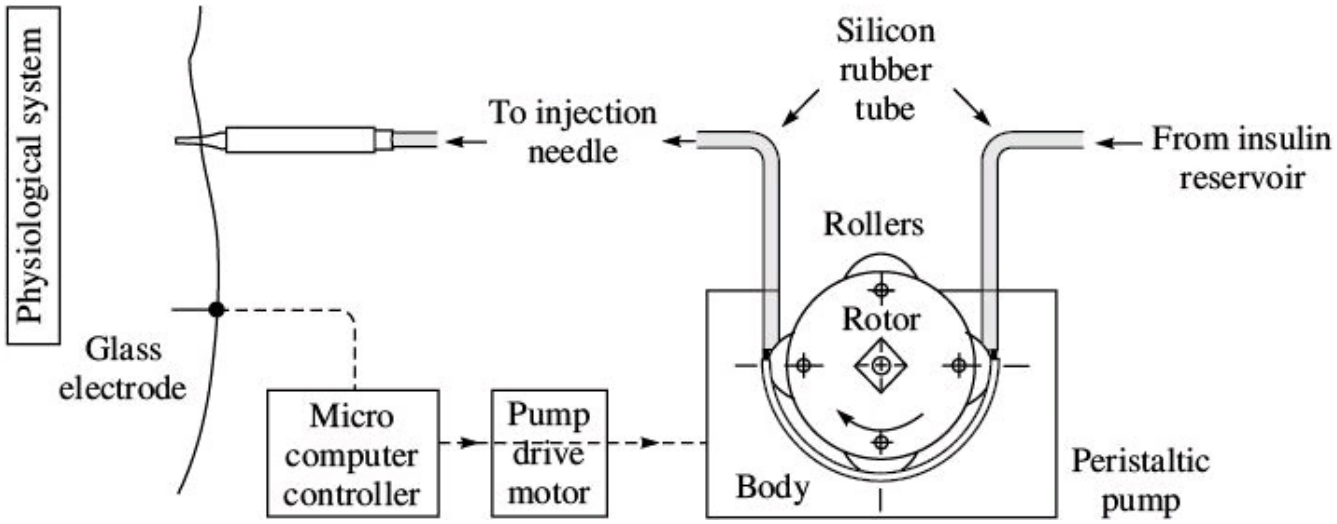


Figure 1.13 The blood sugar level control system of an advanced diabetic patient.

EXAMPLE 1.10 An example from a socio-political system should be a proper ending of this section. An army General has become the Dictator (Controller) of the state. He wants to maintain a particular type of condition (Set Point) in the country. He has no real communication with his countrymen.

His only way to know about the real state of affairs (Controlled Variable) is the information received from certain gentlemen (sensor-transmitter) who generally perform night duties under his employment (Figure 1.14). They wear low hats, black goggle, long coats with upturned collars and are known only by numerals starting with double zeros.

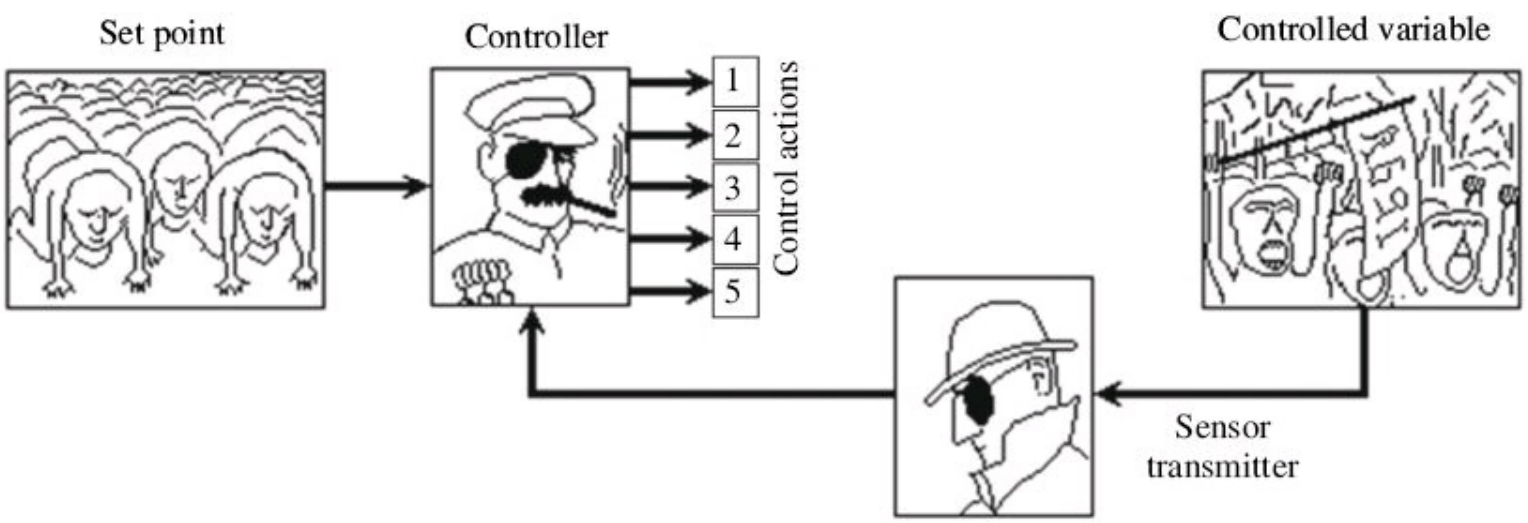


Figure 1.14 Control of a state under a dictator.

But currently (for $t \geq 0$) the information received about the Controlled Variable value has been very disturbing, and quite deviant from the set point value. The Controller has five alternative control actions:

1. A statewide civil action by employing police, armed police, militia forces (FCE is the Home Ministry).
2. A statewide military action by employing the State Guards, and regular Army divisions with an air cover (FCE is the Defense Ministry).
3. One specially chosen gentleman he is equipped with a long range target rifle mounted with a high power scope and made to stay in front of the rebel's house (FCE is the Department of Covert Operations).
4. Breaking out a war with a neighbouring country, and trying to convince the people that all their hardships are due to the aggressive attitude of the neighbour (FCE is the Information and Cultural Ministry).
5. The dictator, using a disguise, mixes with his countrymen to know about their grievances like the famous emperor Harun-all-Rashid, and removes the socio-political dirt. Well, this control action is an idealistic utopia that never happens.

1.6 Block Diagram Development from the Description of Control Systems

The control system examples given above have been presented as physical or narrative descriptions of the application. In Section 1.3, the closed loop block diagram has been introduced as a common platform for describing control systems (Figure 1.4). Presentation of a control loop in this form develops a generalized view towards any physical variety of such system, and trains to think in terms of hardware blocks in the loop. Not only for that, this technique reduces the overlapping and abstractness in the physical description, and configures all the sub-systems in the loop, and the overall system in terms of input–output variables, and internal parameters. In fact, the

block diagram presentation is a step toward analysis and design of a given control problem.

To render a preliminary training of this exercise, a few of the given examples would be taken up to develop the descriptive figures in a way from which block diagram construction will be a natural step forward. That is, all the blocks with their labels as in Figure 1.4, will be attributed to the properly identified hardware in descriptive figures of the systems. Remember that at present we can attempt to address this task only in a qualitative way. This topic will be again taken up in Chapter 9 on Closed Loop Dynamics and there we shall workout a rigorous way to develop block diagrams containing quantitative information.

EXAMPLE 1.11 Let us begin with the Stirred Tank Liquid Heater of Example 1.5. The development of the block diagram of this system should not be difficult because from Figure 1.9, the blocks, relevant variables and the feedback path are easily recognizable. In the following Figure 1.15, we have redrawn Figure 1.9, renaming the system parts and variables in terms of block diagram nomenclature.

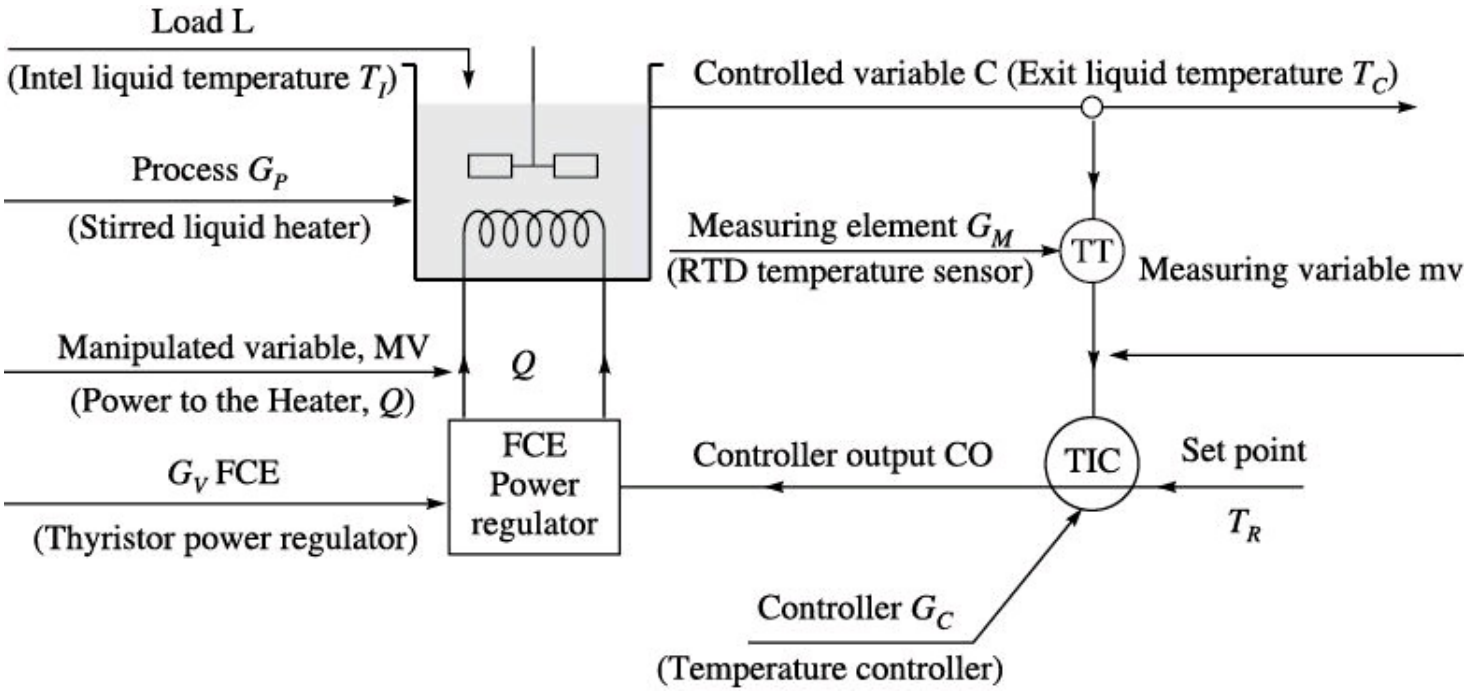


Figure 1.15 Example 1.5 with block diagram nomenclature.

The reader should note that there is no separate module for the Comparator in Figure 1.15, as that unit is imbedded in a commercial Controller. Error e, and Controller Output, CO, are computed within the Controller module.

EXAMPLE 1.12 Next we consider Example 1.7, the Reflux Ratio Control of a Distillation Column, depicted in Figure 1.11. Similar to the above example, Figure 1.16 is the version of Figure 1.11 with block diagram notations.

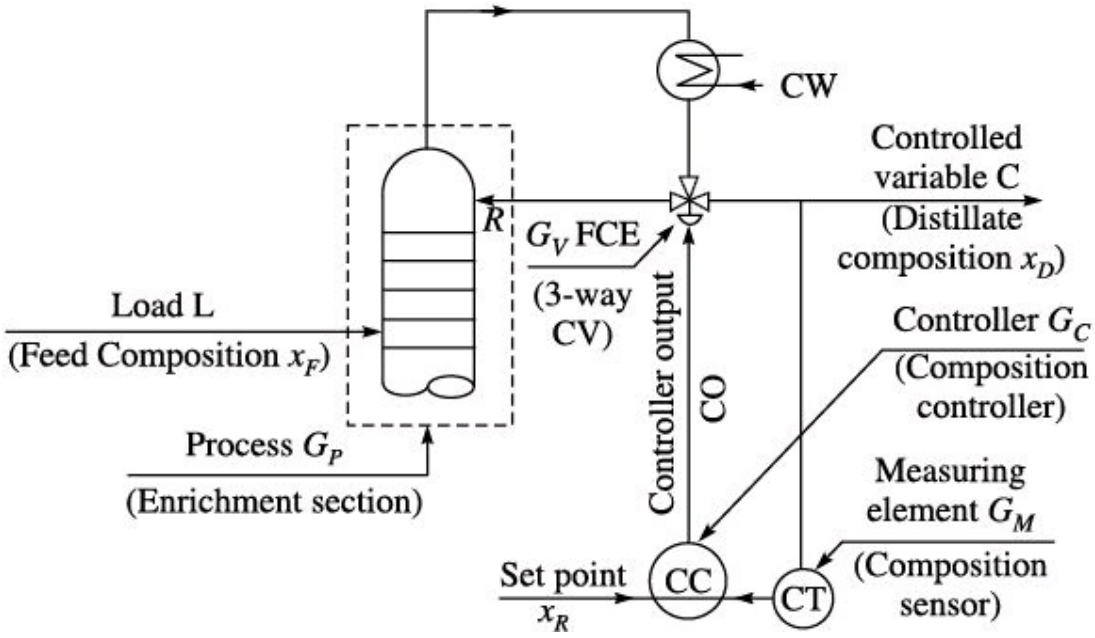


Figure 1.16 Example 1.7 with block diagram nomenclature.

A few things should be noted in the presentation of Figure 1.16:

1. The Feed Composition x_F has been considered to be the only Load Variable, although in a practical distillation system, other recognized load variables are Flow rate and Total enthalpy of the feed stream.
2. The controller output CO is actually the ratio of the two exit stream flow rates from the three-way control valve.
3. The distillate composition x_D depends also on the dynamics of the bottom part of the Column, although this influence is minor in comparison to the contribution of the top part.

The exercise of identifying the blocks and variables in terms of block diagram notation becomes harder for systems in which more than one block resides in one or more system hardware. There are instances where these blocks overlap in one or more system elements. Two such examples are shown as under.

EXAMPLE 1.13 Let us begin with the Governor of Example 1.2. The physical description of the system in Figure 1.9 has been developed to the required form in the following Figure 1.17.

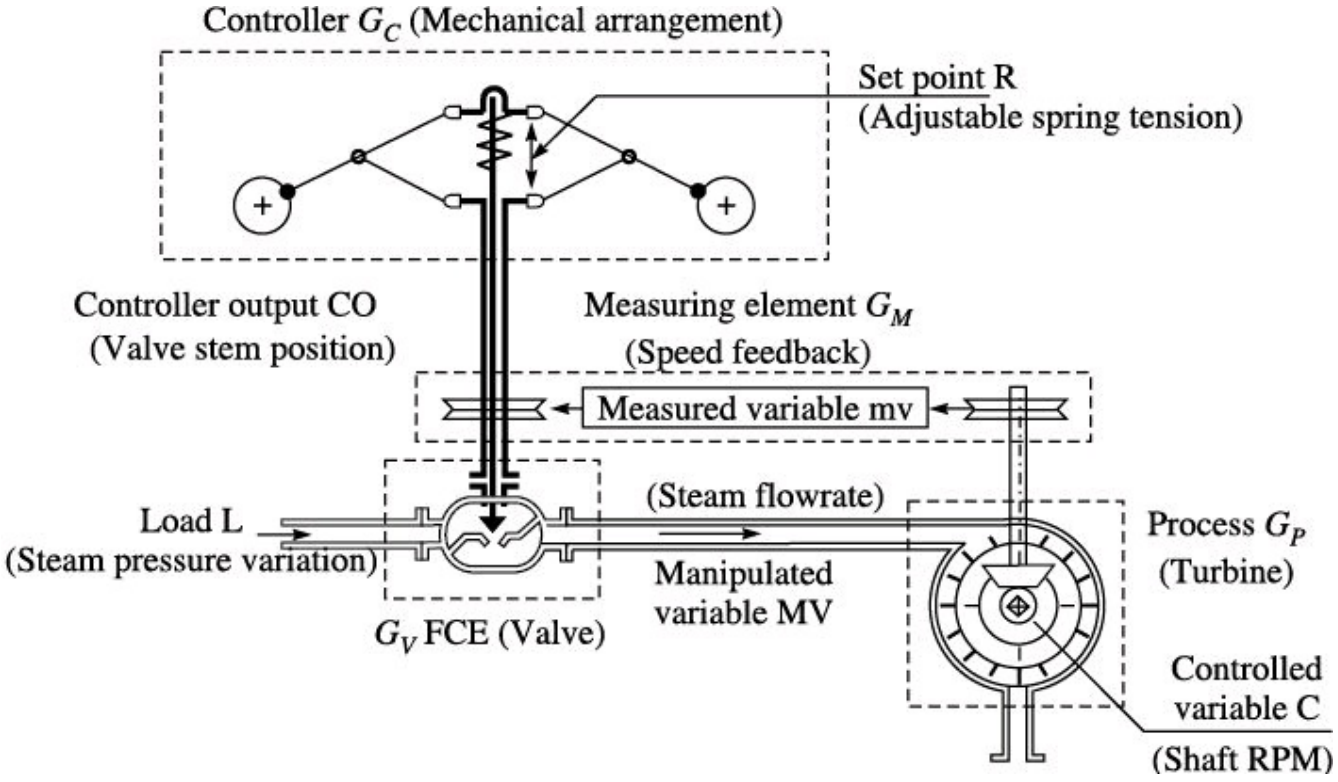


Figure 1.17 Example 1.2 with block diagram nomenclature.

Note the presence of set point and controller mechanism in the same system element.

EXAMPLE 1.14 In the next example, we shall consider Example 1.3, in which the phenomenon of overlapping blocks also exists. The corresponding development has been shown in Figure 1.18.

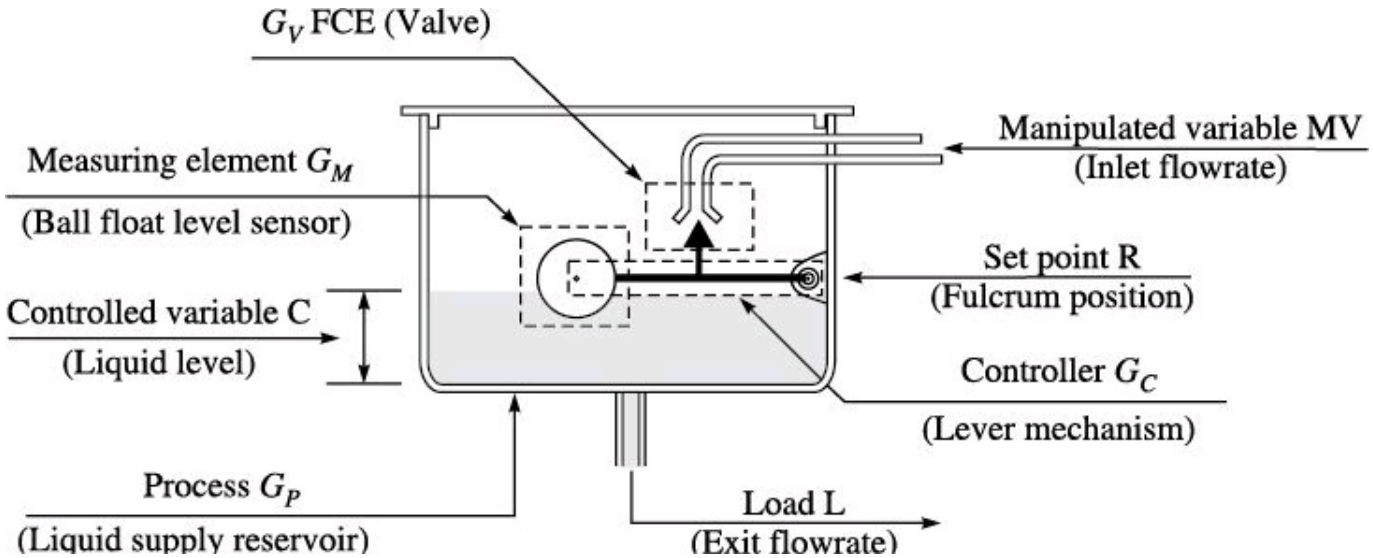


Figure 1.18 Example 1.3 with block diagram nomenclature.

EXAMPLE 1.15 The last example in this discussion would be the Flashing System of Example 1.8, described in Figure 1.19. The possibility of occurrence of more than one control loop in the same process has been demonstrated in this example.

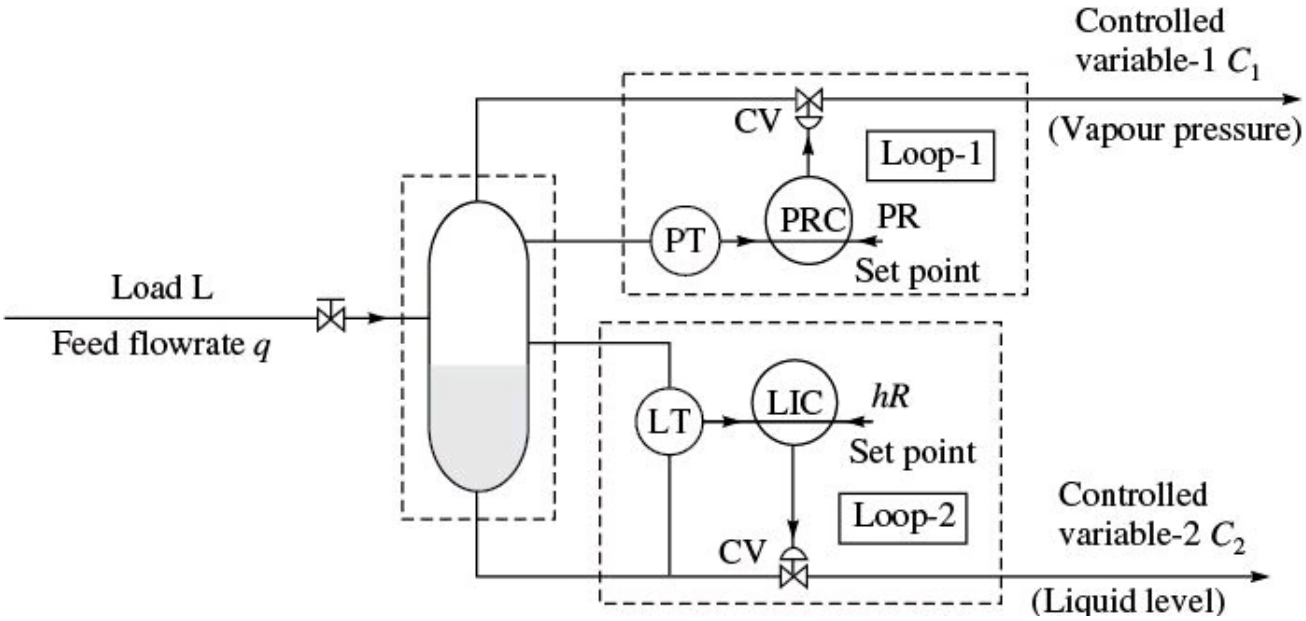


Figure 1.19 Example 1.8 with block diagram nomenclature.

Physical description of control systems have been labelled by proper block diagram nomenclature in the above discussion. One can easily express this information in the standard form, by putting the identified parts of the system into the proper blocks of Figure 1.4.

However, in the last example system, existence of a pair of control loops has been identified. The Load affects both these loops simultaneously. We have drawn the relevant block diagram in Figure 1.20 with the information shown in Figure 1.19. Though, there is a possibility of interaction between the two loops, i.e. the manipulation of one loop may affect the other in such systems. However, for the sake of simplicity, the block diagram has been drawn considering them as separate loops.

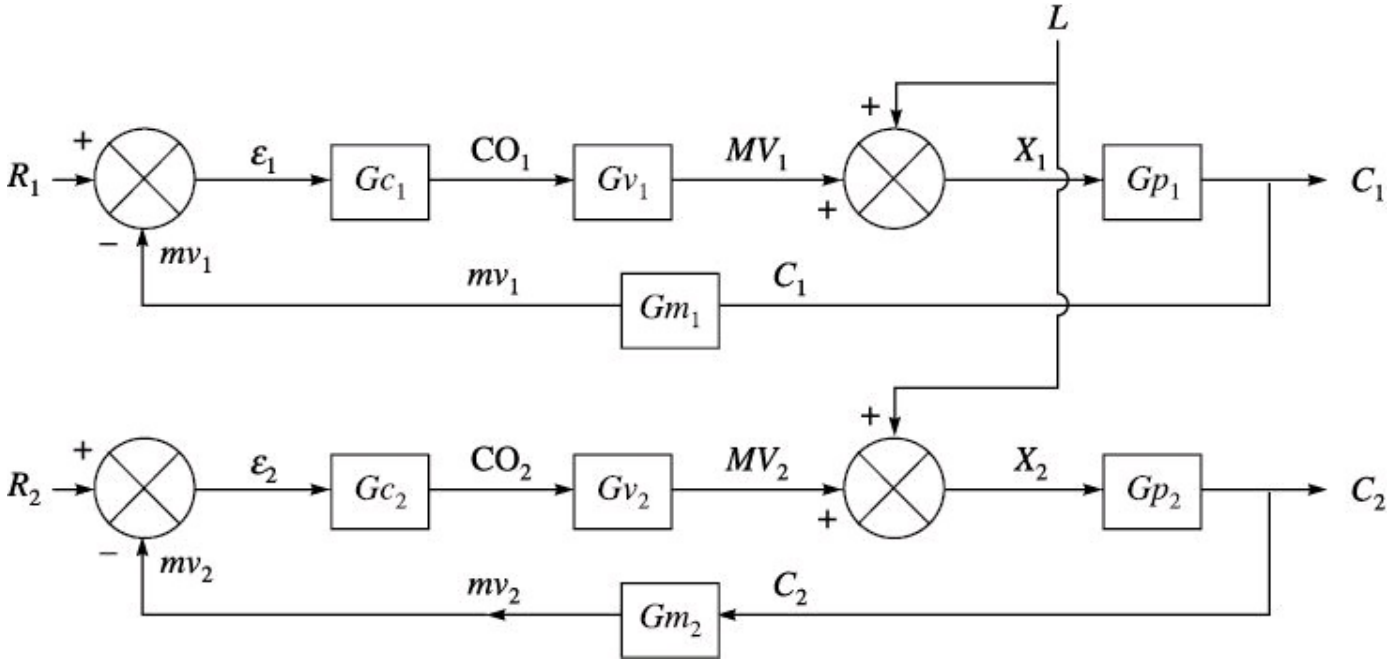


Figure 1.20 Block diagram of two control loops in a flashing stage.

Block diagrams have been used in Process Dynamic Analysis by putting the process dynamic model as a mathematical function inside the block representing the process.

This function expresses how the process is going to convert any input variable to the corresponding response. This approach is more extensively used for the development of closed loop dynamic model of a control system. The exercise starts from the block diagram of the system. But such an analysis calls for the block diagram containing quantitative information about the signal and process streams, as well as the exact configuration of mathematical functions residing within the blocks. We will discuss this procedure in the chapter on closed loop dynamics.

The qualitative treatment on the subject that we have just described should be considered as a preliminary training to build up block diagrams from the physical description of a system. It is hoped that this exposure will support the reader as a conceptual help during the more rigorous treatment of the subject.

We have already presented a logical basis for the existence of control system around a process. This is probably the right place to demonstrate that the necessity of a control system also could be analytically established. Let us introduce the idea of **Degree of Freedom** of a dynamic system.

1.7 Concept of Degree Of Freedom (DOF)

A railway train, being always mounted on rails, can only move forward and backward as if it has only that much of liberty to move in one direction. An automobile can move in right-and-left as well as backward and forward, so it has got a double amount of movement liberty by adding another coordinate to its capability than the railway train. Similarly, for an aircraft, this liberty of movement will be trebled. In the language of analysis this 'liberty of movement' is called Degree Of Freedom (DOF). In mechanically moving systems, where more maneuverability is required, a higher DOF is more desirable.

Let us examine this issue with respect to a process application. In an industry, a process usually renders service by supplying a stream of a substance or species at a desired potential in a stable and trouble-free manner. Any fluctuations in the flow or potential of the stream, are undesirable. We shall take up two cases of a hot water supply system, to introduce the conception of Degree Of Freedom (DOF).

Case–1

Let us consider a metal pot containing water being heated by a gas burner. The water temperature, $T_W(t)$, gradually rises, and so does the dissipated heat through the sides of the pot to surrounding atmosphere. The atmospheric air temperature, T_A, is constant. If the heat input rate, W, is constant, the water temperature eventually reaches a constant value.

If we are interested in the water heater as the process and concerned about T_W as

the process variable that is required to remain constant, we need to fix the heating rate. The required heating rate for a particular value of T_W, may be worked out from the heat balance equation,

$$W = h \cdot A(T_W - T_A) \qquad (1.3)$$

where h = Outside film coefficient of the pot surface, and A = Heat transfer area.

In this condition, the process is said to be at 'zero degree of freedom' because the process variable T_W has no leeway or allowance to change (move), it is fixed. But if we have no regulation over W, the above heat balance takes the following form,

$$W(t) = h \cdot A[T_W(t) - T_A] \qquad (1.4)$$

Time t within parentheses of the quantities, indicates that they are no longer constants but may vary with time, i.e. the process variable T_W can move about depending on the value of W, and in this condition as the number of variables are two [$W(t)$ and $T_W(t)$] but number of equation is still one (the heat balance), the system has one (1) degree of freedom.

Case–2

Now let us sufficiently increase the time variant heating rate so that the water temperature increases to reach the boiling point (T_{Bp}), and boiling starts. Thus, so long as the water temperature remains at its boiling point ($T_W = T_{Bp}$), it will remain constant despite any changes of the heating rate. The variable $T_W(t)$ has become constant because we have introduced another relation, the phase equilibrium, so that variation of heating rate is taken care of by the amount of vapour generated keeping $T_W = T_{Bp}$.

Up till now we have forgotten to mention one important thing in the above discussion that constancy of boiling point is possible only when the atmospheric or superincumbent pressure P on the boiling liquid would remain invariant. If we consider a variable pressure $P(t)$ situation, again the $T_W(t) = T_{Bp}(t)$ become variable. That is, $T_W(t)$ moves with one degree of freedom when number of equations still remain two (2), but number of variables has become three (3).

These conclusions of our discussion for Case–1 and Case–2 have been presented in Table 1.2.

TABLE 1.2 Degree of Freedom Analysis for a Water Heating System

Case	Situation	No. of Variables	No. of Equations	Degree of Freedom (DOF)
Case – 1	W constant	1	1	0
	$W(t)$ variable	2	1	1
	$W(t)$ variable			

Case – 2	P constant Liq. is boiling	2	2	0
	$W(t)$ variable $P(t)$ variable Liq. is boiling	3	2	1

By this time, from the information in the above table, we should generalize the result in the following equation form:

Degree Of Freedom (DOF) = No. of Variables (NV) – No. of Equations (NE) (1.5)

In the above table, Case 1 refers to sensible heat transfer, and Case 2 refers to sensible plus latent heat transfer required for phase change. Under each case, the temperature T_W may vary with one degree of freedom (DOF = 1), when NV is greater than NE by one (1). If we still want that T_W should remain constant, another dependency of W with respect to T_W in the form of an equation has to be brought within the system to render DOF = 0. ***Well if such a relation does not naturally exists in the system, we can buy the required relation from the instrument vendor.*** This is probably the analytical way of allocating and introducing the control system into a dynamic process, because the control equation will furnish the missing relation so that DOF = 0. In both the cases, the control system will measure T_W and manipulate W. [It should be mentioned that in Case 2 situation, the system somewhat resembles a steam generator if the pot is closed on the top with an outlet for the generated steam. In many of such applications, measuring the system pressure would be more adequate instead of temperature.]

We shall close the discussion on this topic at present. In Chapter 3 on Process Modeling, the DOF analysis will be demonstrated for certain important process examples.

As we have established the necessity of control system for process and what good it can do, we should also mention the disadvantageous features of a controlled system. The controller is also called a **dynamic compensator** because it compensates (controls) the process aberrations in the transient situation. Hence, the possibility of over compensation appears. In any of the examples in Section 1.5, if the controller is over sensitive or misinformed about the process (as in Example 1.4), it may take a control action that is more than necessary or altogether in the wrong direction. This may cause the process to go beyond its physical limits of operation, resulting in various kinds of failure. Such a crisis may be described as the process that has gone beyond its **Stability** limit. Let's talk about stability.

1.8 Stability of Dynamic Systems

All dynamic systems come under the influence of disturbance that drives the process away from their steady state condition. The question is whether such a disturbance

may incur some sort of damage to the process. This damage may cause the failure of any part of the process equipment, or may crop up as the unacceptable quality deterioration of the product. Before we go further with this topic, one thing should be made clear that if the disturbance magnitude is too large, any system might fail. Hence, we are talking about disturbances which are smaller than the design limits of the process system. After facing such a disturbance, whether a process has got the capability of coming to the next steady state condition, unharmed in any way, needs to be evaluated. To address this problem, we have to consider the **Stability Analysis** of a dynamic system. From the above discussion a preliminary definition of a stable system may be proposed as:

"*A process is said to be stable if its response is finite (bounded) for a finite (bounded) input*."

Stability is an important aspect of process dynamics and its control system design. We shall offer a few analytical approaches for determination of stability in later chapters; at present let us qualitatively examine the issues involved.

Certain processes may be unstable by themselves, that is, due to their dynamic character, cannot remain at a particular state. This character may appear from physical shape or physical and chemical description. Such systems are called–**open loop unstable** [Nuclear spontaneous chain reactions are good examples of such systems.].

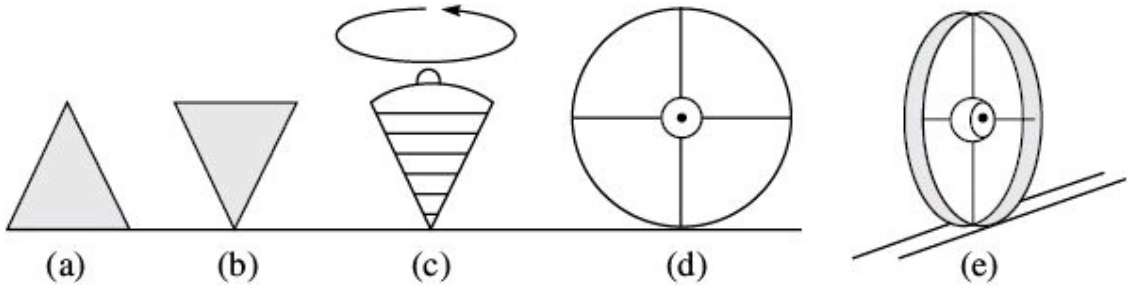


Figure 1.21 Examples of open loop unstable systems.

In Figure 1.21, you can put the cone base-wise on a flat surface (a), but cannot place on its apex (b). Similarly, a wheel at rest would fall sidewise if placed vertically (d). But a spinning top (c) or a rotating wheel (e) will retain their vertical position as long as the rotational speed is above a minimum value. A reactor operation carrying an exothermic reaction may be thermally unstable, but if a cooling fluid is put through the external reactor jacket for heat extraction, a stable operating temperature may be obtained. This external excitation may come from a control system as in the case of the bicycle rider in Example 1.1, and the manipulated flow of cooling fluid in reactor temperature control system.

Thus, an open loop system may be made stable by adding compensation or control. But there is an alarming possibility that a closed loop process may become unstable due to **Over Compensation** (Figure 1.22).

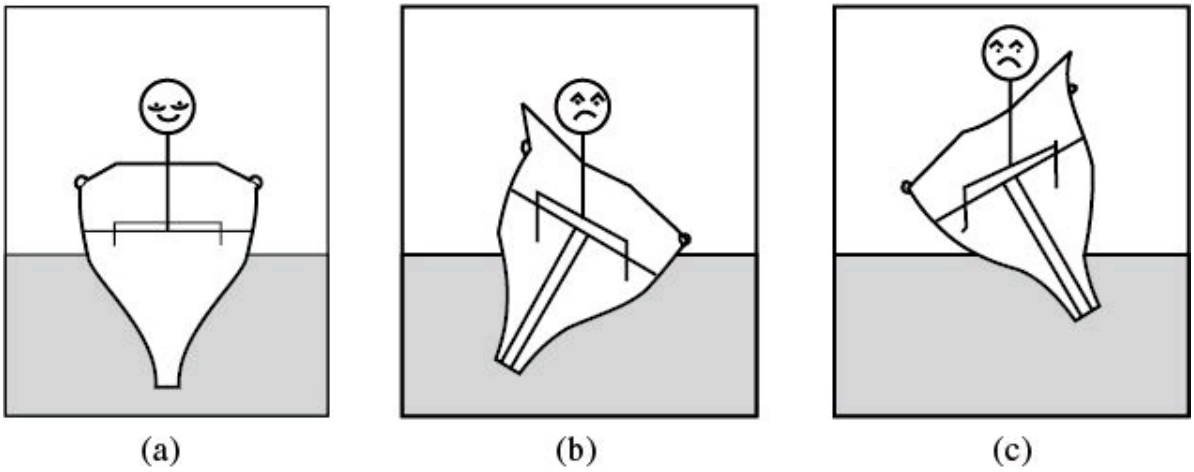


Figure 1.22 A guy riding a boat during tide.

A person feels tranquil while riding a boat in calm and placid waters. This tranquility is harshly removed when the tidewater hits the boat side-on, and it tilts at an angle q from the vertical even keel position. An experienced rider's reaction would be to shift his body weight in a way so that q reduces when the boat swings in reverse direction. We may consider the boat as our system, the rider being the Controller, and q the Controlled Variable. In the desired situation, the amplitude of q vs. time locus will be reduced in each oscillation [Figure 1.23(a)]. But what will happen if the rider is a novice or nervous (the Controller is over sensitive) and takes over compensation by shifting his weight too much, thereby increasing the amplitude of q in each oscillation – The Boat will Capsize as the closed loop system will become unstable [Figure 1.23(b)].

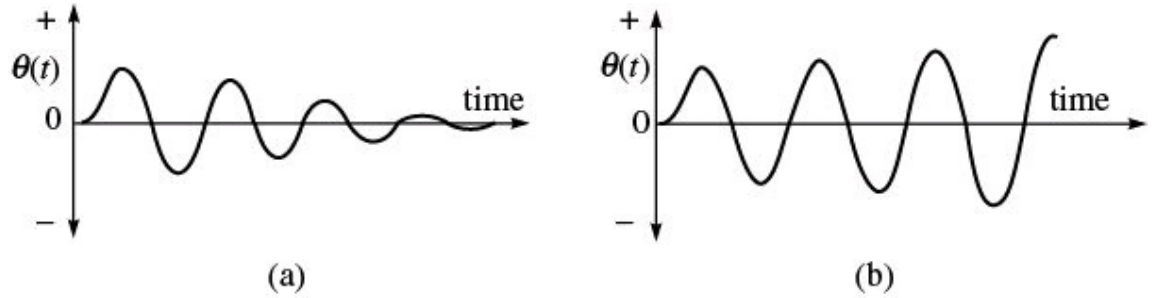


Figure 1.23 Closed loop stability (a) A boat rider with experience (b) A nervous rider.

It should be mentioned here that, besides Controller Sensitivity, the stability of a closed loop system also depends on the existence and amount of 'dead time', if any, present in the system. You must have remembered the dead time as a dynamic phenomenon that has been mentioned in Example 1.4 in the context of the thickness control system in a rolling mill. We shall analytically prove this statement in Chapter 10 on Stability Analysis. This exposure brings up the importance of stability consideration of a closed loop system during its Controller design.

We shall see later that the measure of stability and speed of response of a dynamic system are sort of complementary phenomena. Speed of response is a measure of how fast the Controller can drive out the bad influence of disturbance, as out of quality

product from the system, and stability renders the measure of safety with which the above operation would be carried out. Generally speaking, stability of a system can be increased at the expense of its speed of response. Thus, the issue of Optimum Controller Design crops up, and presently it will be sufficient to say that control system design is a trade-off between stability and response speed of a given closed loop system.

We have expressed the scope and the overall problem that this text is intending to address. The sequence of subsequent development should be to analyze each of the elements of the feedback control loop. After that, the knowledge of individual dynamics of the blocks may be integrated to obtain the overall dynamics of the loop. Only then we will posses the necessary background for stability analysis and Controller design.

EXERCISE PROBLEMS

1.1 A person gets his pay packet on the first day of the month and his spending follows the well known exponential decay pattern. On the twenty-fifth, he has to receive debt (from a citizen of Afghanistan, presently in exile) but maintains a similar spending pattern. Draw an approximate graph of his financial index (the amount of money) against time during a month.

1.2 *Claustridium botulinum* is a dangerous bacteria, causes virulent stomach disorder if ingested, as I have heard from my Bio-technologist friends. One of them is trying to control the strain's toxicity to a lesser degree by under feeding them and hoping to discover a miracle drug for constipation patients. Draw a figure to illustrate the control scheme, where the controlled variable is the amount of toxin secretion per unit time, and the manipulated variable is amount of nutrient supplied.

1.3 Develop the block diagram of the control systems shown in Examples 1.4–1.6.

1.4 Draw the block diagrams in standard form of Figure 1.4 for the systems discussed in Section 1.6.

1.5 Analyze the DOF of the level control system of Example 1.6, and allocation of its control equation.

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2

Preliminary Concepts for Process Dynamic Model Development

The most important and offering the maximum variety of dynamic character is the Process block GP. For the development of dynamic model a process should be looked from a slightly different view-point, than we did while proposing their steady state model. That is to consider the start and continuation of a process in terms of a **Cause–Effect** relation. This will help us to introduce the **Dynamic Elements** of a process, which primarily influence its dynamic character.

2.1 Process Looked upon as a Cause–Effect Phenomenon

Recalling the notion given in Figure 2.1 about a process, that a conversion or flow of certain entity may constitute a process, we should more closely examine this species called as entity.

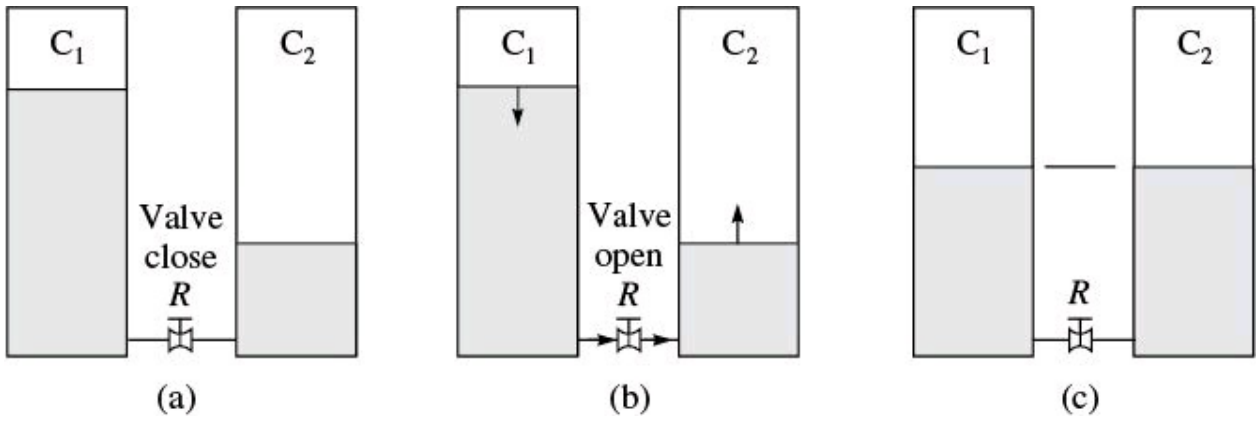


Figure 2.1 The minimum requirement for a process to happen. (a) There is no process when the valve is close ($R = \infty$). (b) The process starts as the connecting path is made by opening the valve. (c) The process is complete as the potential of the two reservoirs, C_1 and C_2 become equal, and there is no flow through the connecting path.

The entity may be in various form, e.g. Mass, Momentum or many other kinds of material or energy. Generally, the operational state of a process is a measure of the potential of this **principal entity (PE)**, and we shall call the potential of PE as **principal variable (PV)**.

For the realization of a process through the flow of PE, the minimum requirements are: (1) Two reservoirs where PE is initially stored up to different levels; and (2) A connecting path through which PE flows from high to low level reservoir, if it is allowed to. The flow rate depends upon the driving potential and the resistance

offered by the connecting path. Thus the process starts from the instant when the connection is made, say by opening the valve in Figure 2.1(b). Initially the rate of the process, i.e. the flow rate of PE through the valve and connecting line is highest, because the level (potential) difference between the two reservoirs is maximum. As the process continues, the level difference decreases and the flow rate of PE slows down, which completely stops when the level difference becomes zero, or an equilibrium is established. We may, at this stage, say that the process is complete. But for achieving this zero potential, theoretically one has to wait for infinite time, or at least long enough so that the potential difference reduces to a negligible magnitude. Practically such an investment of time in a process operation may seriously hamper the economic aspect or profitability by loosing the capacity utilization of the plant. Let us examine the industrial process operation from this outlook.

2.2 Industrial Process Operation

Industrial Processes are carried out after a careful analysis and design effort, to obtain an optimum performance. The ultimate criterion of optimization of a process operation is economic, where the goal is to maximize profit. Hence, in industrial practice, such an effort is realized by making a compromise between the potential which is still unused, and utilization of the efficiency of plant capacity. Thus, one does not wait for unlimited time to achieve equilibrium and the conversions are carried out up to a certain extent less than the equilibrium conversion. Following these considerations, three main types of industrial process operations have been developed during many years of practice.

(*There are several ways to utilize the part of the potential still remaining in the product mix, e.g. after separating the desired product, the balance is recycled with the feed, or it is sold as a low grade product which may be successfully used in certain other processes as feed material. See Appendix-IIIB.*)

1. **Unsteady state or Batch Operation:** Where all the raw materials are brought together and the process is allowed to continue under carefully maintained conditions up to a definite extent. It is evident that all concerned variables in such a system will be non- stationary, that is, their time derivatives have finite values.
2. **Steady state or Continuous Operation:** Where a continuous flow of raw materials and other utilities occurs from entry. The potential difference is exploited up to an extent depending on the time period that the inflowing material is allowed to stay for (residence time) within the system. Hence, the time derivative of the concerned variables are zero within the system.

3. **Semi-Batch Operation:** Where some raw materials are brought together in a batch in the plant, but a few of them are fed continuously during continuation of the process. Hence, the time derivative of the concerned variables are non zero. Figure 2.2 may help you to have a better idea about the above topics.

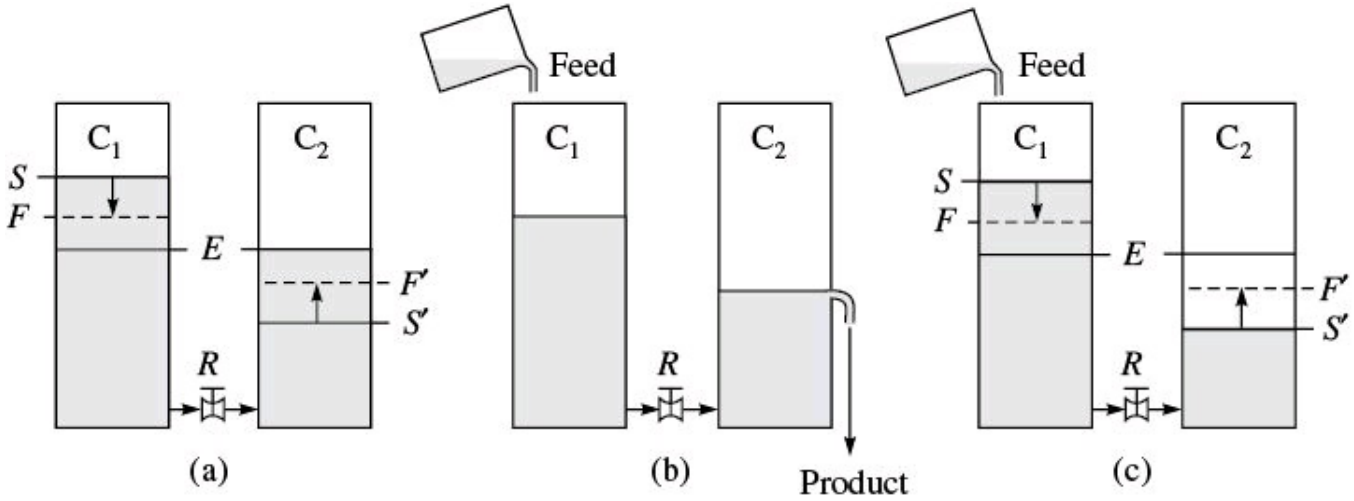


Figure 2.2 Types of industrial process operation. (a) Starting from S, S□ process is finished at F, F□ before the equilibrium point E, in a batch operation. (b) In a continuous operation a constant potential is maintained during the plant operation. (c) In semi-batch operation a few raw or feed material are fed into a plant that is otherwise operating in batch fashion.

We would like to clarify an important consideration at this point regarding the selection of principal variable. There are other associated variables beside the principal variable in an ongoing process, the choice of PV is not unique. That a particular variable will be identified as the PV depends entirely on the application or performance specification of the process. Physically identical processes may have different sets of PV depending upon the application and instrumentation system. We shall discuss this interesting topic with examples in a latter part of our discussion.

In the following few chapters we will restrict our discussion mainly on continuous systems, because a large number of preliminary ideas may be introduced with lesser mathematical rigours. With such a preliminary exposure on why a process happens, let us examine how the process manifests its dynamics.

2.3 Dynamic Manifestation of a Process

Everyday we are witnessing a number of phenomena (process) happening around us. They are of many categories; physical, social, economic, political, etc. Continuation of these processes are demonstrated by continuous changes of certain quantities associated with the process, as for example, the temperature increase of a pot full of water put over a flame, a person in exile picking up the language and other facets of culture of the people he is living with, thus increasing his communication power with his neighbours, or the discovery of a new resource (say an oil reserve) in a country increases its economic strength. We may list the time dependent quantities in these

examples as, water temperature, the person's communication power and the currency value of the country. They are called the process variables. A process shows or manifests its dynamics through the type of time dependency of these variables.

The above discussion about the conception of process dynamic behaviour should be complemented with the practical aspect of what is meant by when a process demonstrates its dynamics or how you will recognize the dynamics of a process in real physical terms. Not only to know the true sense of term 'dynamic behaviour', but also to examine the possibility that some important conclusions may be drawn from the observation, we cite an example—**Cooling of a hot piece of metal**.

A piece of metal was in a furnace for sufficiently long time, and has eventually attained the furnace temperature. It was withdrawn at a time instant $t = 0$, kept for a little time in the open air, and then put into a pool of water which was at room temperature. A sample record of temperatures of the piece of metal has been made:

TABLE 2.1 Temperature of the Object vs. Time During Different Stages of Cooling

Temperature,°C	Time in seconds	Remarks (Room temp. 25°C)
1260	0	Withdrawn from the furnace
1100	10	Put into water pool
700	20	Still in the water pool
50	120	Cooling operation complete

This data has been plotted on a graph in Figure 2.3. It should be noted that this is only a partial description of the Cooling Dynamics of the metal piece. Unless we have the time-temperature locus through the recorded points, the dynamic description of the process of cooling would not be complete. Still, an observation is made that while in air the temperature drop has been 160°C in 10 sec, whereas in water the drop has been 400°C for equal time duration. We will restrain from drawing any conclusion until we go a little further in our discussion.

In the absence of recorded temperature data between these points, let us consider an example of different physical descriptions having dynamic equivalence with the present system. The observations may give us an insight into their dynamic character. Also, such a process may offer a facility for recording the dynamic locus.

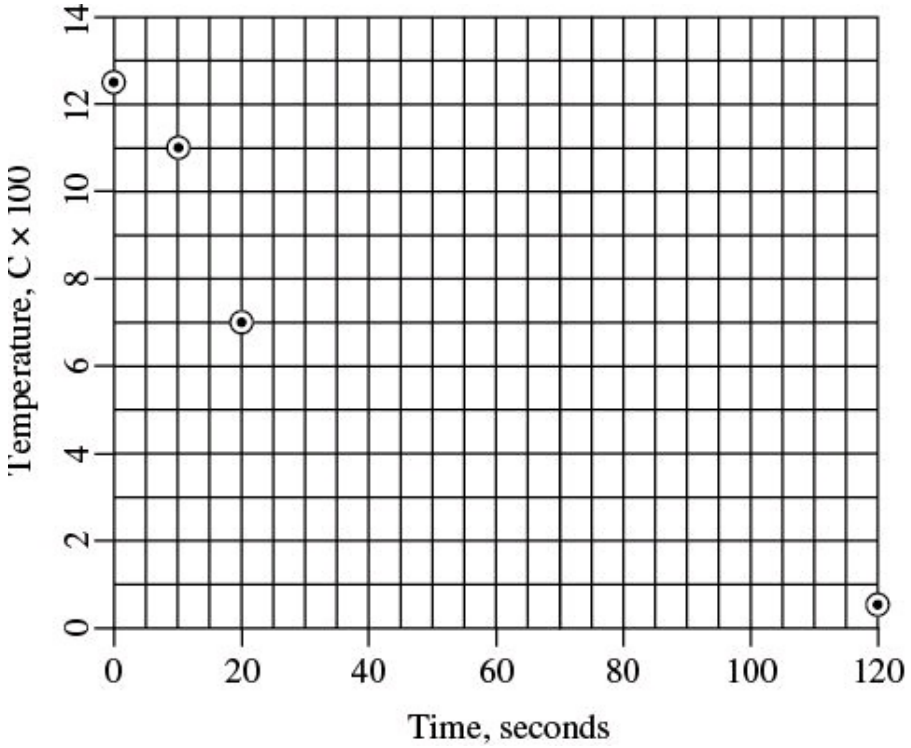


Figure 2.3 Plot of temperature data of a hot metal body during cooling.

A similar system may be a tall glass cylinder with level graduation in mm, and a bottom discharge line with a valve, as in Figure 2.4. Initially the cylinder was filled up to a height of 1260 mm. Now the bottom valve is opened slightly at time $t = 0$. Knowing the inner cross sectional area of the cylinder, the valve opening may be adjusted in such a way that the time required by the liquid height to fall up to 1100 mm is 10 seconds. At this instant the valve is further opened to a value so that liquid height falls to 700 mm at 20th second and to 50 mm at 120th second. The liquid is discharged to a sump of extremely large area so that the discharged liquid has a negligible effect on the sump liquid level that remains practically constant at 25 mm.

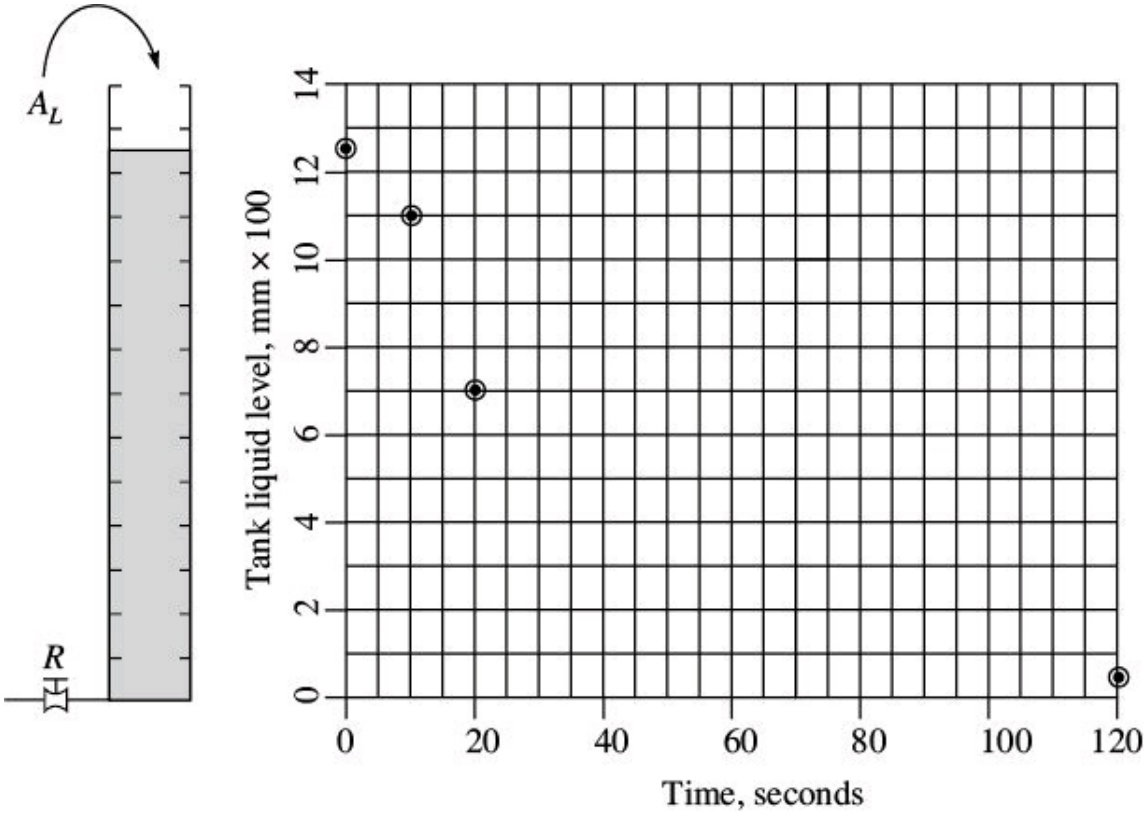


Figure 2.4 Liquid discharging from a cylindrical tank through a linear resistance.

Thus we have found another physical system that has reproduced a similar dynamic data as the shown by the hot body during its cooling. We may now put the necessary instruments—a level sensor transmitter to the bottom of the cylindrical tank, and a strip chart recorder that records the transmitter output to obtain a hard copy of the complete level dynamics during discharge. The recorded data is shown in Figure 2.5.

The shape of the tank draining dynamics through points BCD, is characterized by the highest initial rate, which gradually and asymptotically reduces to zero value as the process progresses. Such behaviour tells us two things—that at any instant the rate of discharge is proportional to the remaining value of the system potential, and secondly it reminds that we have come across similar time dependency in two other kinds of physical systems: (1) In electrical circuit theory, the time vs. voltage [E(t)] locus of a charged capacitor that is discharging through a constant resistance R, and (2) the time vs. concentration [C(t)] locus of reacting species in a first order irreversible reaction system. The equations that describe the dynamics are given in the Table 2.2. They confirm that the discharge process starts at highest rate (putting $t = 0$ in the equations) from the initial potential value of the capacitor, and asymptotically approach zero potential at equilibrium as time increases (putting $t = \infty$). Processes manifesting such dynamics are said to obey **Exponential Potential Depletion or Decay**.

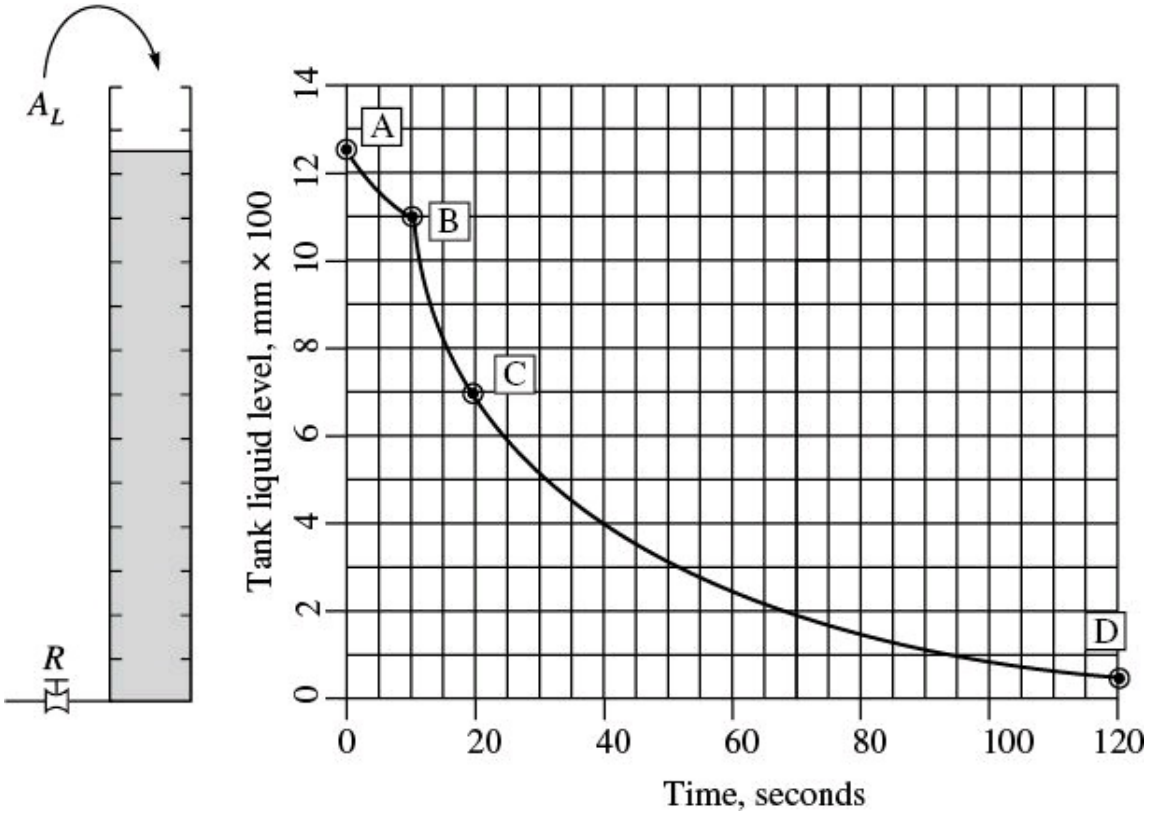


Figure 2.5 The complete dynamic locus of a liquid discharging from a cylindrical tank through a linear resistance.

TABLE 2.2 Equations Describing the Dynamics of Draining	
First order Reaction	**Discharge of a Capacitor**
$C(t) = C_I \exp(-kt)$	$E(t) = E_I \exp(-t/CR)$

The suffix I refers to starting condition of the variable.

Let us state the basic definitions of Capacitance and Resistance elements of a process. The **Capacitance** of a capacitor is the amount of PE required to raise the capacitor potential or PV by unit amount, and the **Resistance** of a resistor element is the flow rate of PE per unit potential difference (unit PV) across the resistor. By comparison, we may write the expressions of capacitance and resistance for the thermal and level dynamic systems. The results are compiled in Table 2.3.

TABLE 2.3 Comparison between the parameters of temperature and Level dynamics	
Temperature (*T*)	**Liquid level (*h*)**
Thermal Capacitance (mC_P)	Tank area (A_L)
Heat transfer resistance ($1/UA_T$)	Valve resistance (R)

The parameter names used in the above table are as follows. Tank area A_L, and thermal capacity mC_P(m = mass of the metal piece, C_P = heat capacity of metal) are property of the reservoirs. Valve resistance R and the thermal resistance ($1/UA_T$) are the properties of the connecting path, where U is the heat transfer coefficient between the metal and surrounding, and A_T is the area available for heat transfer.

Let us take a stock about what we have gained by this discussion—a hot body discharging heat to a low temperature surrounding and the tall tank discharging liquid

to a low-level sump. The atmosphere or sump has such a large thermal capacity or area, that discharging PE into them hardly changes their potential. The two systems are dynamically similar. This fact may be confirmed by the comparable parameters of the two systems listed in the above table.

The significance of two valve openings should become apparent to you by this time, i.e. the difference of heat transfer resistance from metal to air is equivalent to a small valve opening (greater resistance), and from metal to water is equivalent to a large valve opening (smaller resistance).

We may conclude that heat dissipation and liquid drainage processes could be bracketed under the same heading of *Depletion of potential from a reservoir against a resistance that has two successive values.*

Not only for these physical systems, such responses may result from many processes that have apparently different descriptions, and may originate from social, psychological or natural phenomena beside physical or technical area. We present some more examples from other application areas.

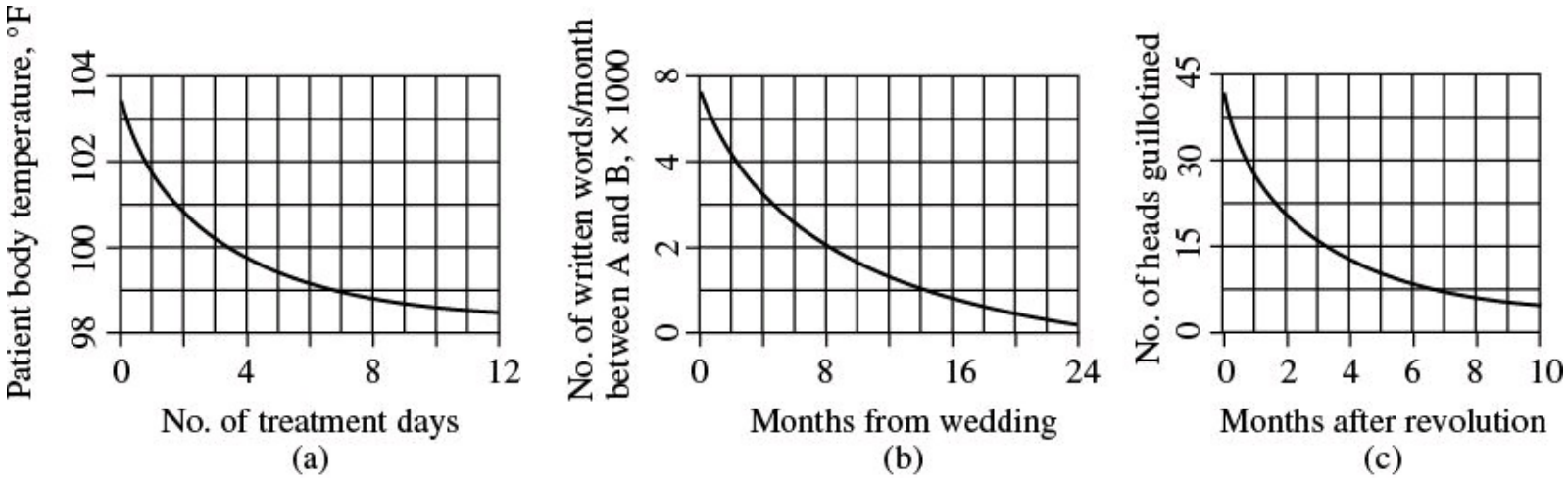


Figure 2.6 Similar potential depletion response from: (a) Physiological system–body temperature of a person during ailment after an illness. (b) Psychological system–number of written words/month between A and B after their wedding date. (c) Socio-Political system–number of heads of the aristocrats guillotined per day from the day of French revolution.

Though their general nature has a resemblance with the heat and liquid discharge dynamics, the graphs contain small local departures—these are called noise or micro-effects of miscellaneous system variables. There may be quite a few of them, whose number and quantitative amount of influences over the principal variable are often impossible to estimate. However, these effects in general do not clutter the main picture that the PV shows a similar exponential dynamics when a resistive flow path connects reservoirs having dissimilar potential.

After this exposure you should be able to stretch the logic a little further to realize that the charging dynamics of an empty reservoir (capacitor) from a constant potential source should be complementary to the decay dynamics. We would discuss this topic

in detail in future. At present, the functional expression may be arbitrarily stated as M [1 – exp(– t/k)] where, M = Ultimate value of the potential in the receiver = Source Reservoir potential, and k = system constant. The charging and discharging locus is shown in Figure 2.7.

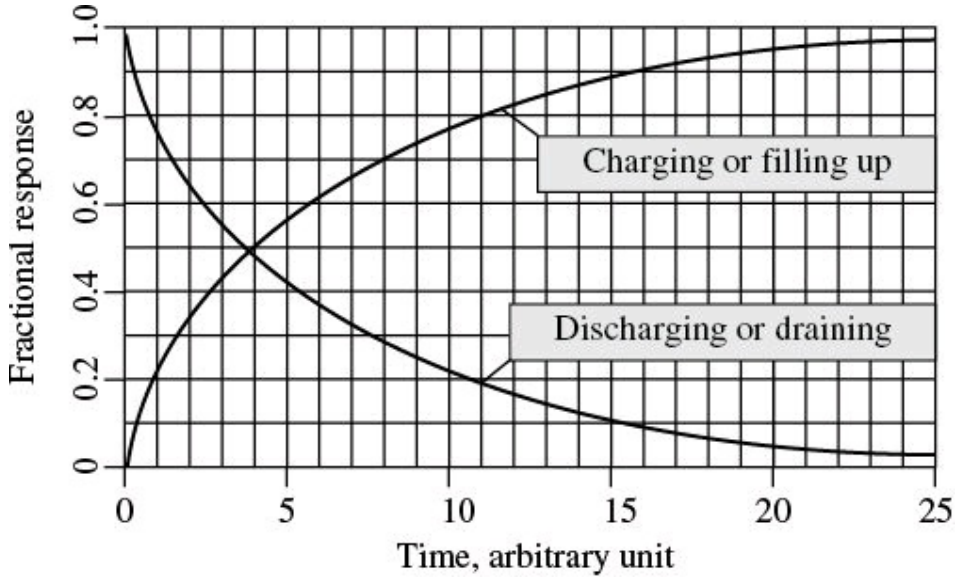


Figure 2.7 The charging dynamics should be complementary to the drainage.

Before proceeding further along this line of discussion, we would like to introduce an approach to bridge the gap between the steady state and dynamic model of a process, because this is the most opportune moment to do so.

2.4 Steady State to Dynamic Model—A Step Forward

If you are a Chemical Engineering student then you must be familiar with the steady state relations of process systems. The equations that you have learnt to use for heat and mass transfer equipment design in Unit Operation classes are steady state relations, and they represent the steady state model of the process, i.e. given the input variable values and physical data of the equipment, you can calculate the output variable magnitude. (*For certain applications the procedure may be implicit but eventually you can arrive at the required result.*)

Let us consider a shell and tube heat exchanger. All the physical data of the equipment, material properties of the streams handled and the variable values are available while the system is operating at steady state. Now if the inlet tube fluid temperature is suddenly and permanently changed to a new value, keeping all other things constant, it is possible to calculate the new steady state value of fluid temperature at the tube outlet by using the steady state relations.

If the input and output variables are recognized as the tube fluid temperatures T_{TI}, and T_{TO} respectively at inlet and outlet of the heat exchanger, the steady state change may be expressed in the following ratio form:

$$\frac{T_{TO2} - T_{TO1}}{T_{TI2} - T_{TI1}} = \frac{\Delta T_{TO}}{\Delta T_{TI}} \qquad (2.1)$$

where, suffix 1 and 2 refer to the old and new steady state conditions respectively.

This ratio of input and output variable magnitude differences at two steady state conditions of a process is called the **Steady State Process Gain** and denoted as K_{PSS}, or simply K_P. The general expression for K_P for any process is,

$$K_P = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y}{\Delta x} \qquad (2.2)$$

where,

x = The general notation of input variable,

y = The general notation of output variable,

Suffix 1 and 2 refer to the old and new steady state conditions respectively.

(In process dynamics vocabulary, the sudden and permanent change of any variable, is called a **step change.**)

We have already established in the previous section that after an input step change has been made the process output cannot change immediately (instantaneously) to its new steady state value. The total change takes some time for completion. However, we know to what ultimate value the output variable would arrive from the steady state analysis, but if you have only the steady state model at your disposal, it is impossible to decide about the locus or path through which it will go from old to the new steady state. Let us examine the situation with respect to a continuous system. The conclusions are illustrated in the following Figure 2.8 for a continuously operating process that has experienced a step change in the input from x_1 to x_2, at time $t = t_1 = 0$, and the complete change of output variable y, from y_1 to the next steady state value y_2 has occurred at $t = t_2$. In the figure, the two possible directions of input change (+ve and –ve), and the corresponding output change have been included.

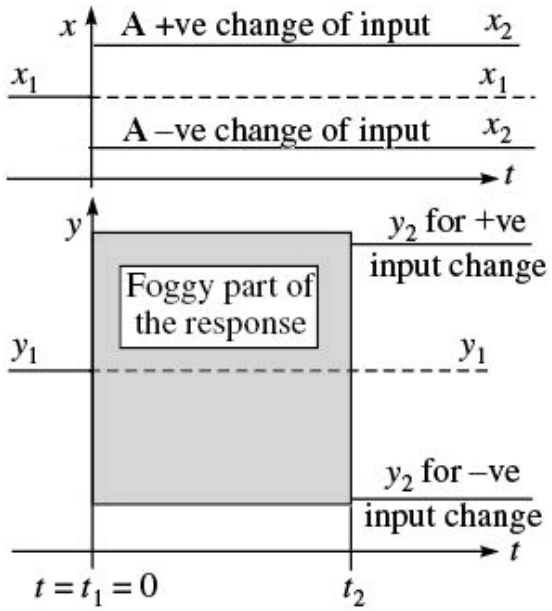


Figure 2.8 Input and corresponding output change from old to the new steady state.

The output goes through a dynamic region, as it travels from old to the new steady state. We have made this part 'Foggy' to emphasize that at present we have no knowledge about the output value when it is travelling through this region. This foggy region is also called the **Transient** part of the response.

(*There are certain processes that react instantaneously after an input change. As for example, a mechanical lever, linkages and drives, very fast chemical reactions like blasting of conventional and nuclear warheads. These processes may be considered as* ***Pure Gain Elements****, having incredibly fast dynamics or practically no lag in response. We shall treat this topic in Section 5.5.*)

Now recalling the discussion in Section 2.3 on 'depletion dynamics' and conclusions depicted in Figure 2.7 about a process moving from one steady state to the next, we can at least qualitatively draw the nature of dynamic locus of the output variable in the 'foggy' part of Figure 2.8. This has been done in Figure 2.9.

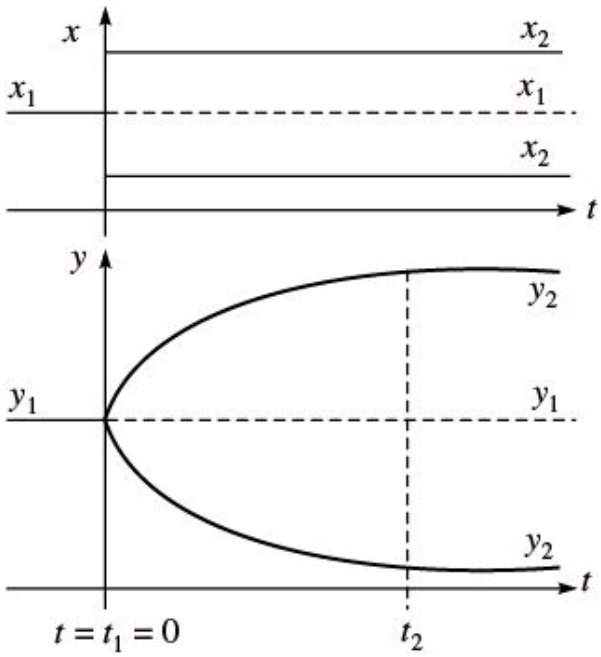


Figure 2.9 The complete process response.

What does this discussion amount to? From the knowledge of K_P, which may be obtained by steady state relations alone, we can have the ultimate value of output or response for a given input change. Not only that, the process gain, K_P, contains the dimensional conversion between input to output variable.

Now if we consider that the process input and output were at zero initial value, and a unit +ve step change of input has been made, the output may increase up to maximum value
$y_M = y_2 = K_P$, and the response dynamics may be expressed as:

$$y(t)\big|_{t_1<t<t_2} = K_P f(t) \qquad (2.3)$$

where, $f(t)$ = A dimensionless time variant function that increases from 0 to 1 as the process moves from old to the new steady state.

In fact Equation 2.3 expresses the fractional recovery of K_P in the response during its transient part. At steady state the recovery becomes complete. So the dynamic response can be divided into two parts K_P and $f(t)$. The first part may be calculated by the steady state model equation that we already know and the second part—a dimensionless time function about which we have no quantitative knowledge, but qualitatively we may guess about its shape. That we have done in Figure 2.10. The process dynamic analysis is the subject that will teach us to quantitatively determine the response shape. Thus, dynamic characteristic is one step forward from the steady state consideration about a process. If we think about the information content about the process, its dynamic model contains much more than the steady state one. It is some- what like looking through a slit in the window to a landscape with the steady state knowledge, and that the dynamic equation has opened up the window to get a full view.

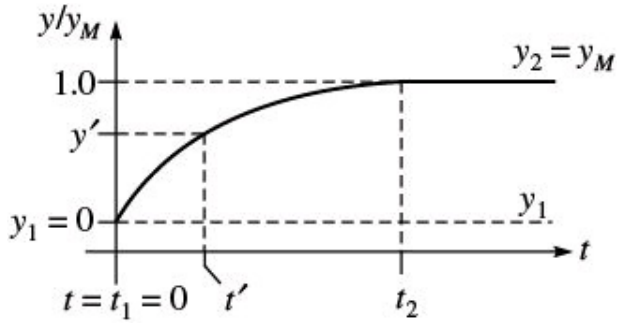


Figure 2.10 Transient response in the normalized form.

Another idea may be introduced at this stage as a corollary of this discussion. The definition of a new term **Dynamic Gain**, denoted as $K_{P\square}$ may be obtained from Equation 2.3 by considering the response magnitude of a unit step at any time $t = t\square$. During the transient part, Equation 2.3 may be written as:

$$K_{P'} = \frac{y'}{x_2} = \frac{y'}{1.0} = y' = K_P f(t') \qquad (2.4)$$

If we consider the deviation of the variables from their old steady state, rather than their absolute values, the assumption of a zero value of input and output of a process at the initial steady state does not by any means undermine the significance of the result. To increase the generality of this result we may consider the fractional or normalized response (y/y_M) instead of its absolute value y. Figure 2.10 expresses these ideas.

Just for the completeness of this discussion we should mention that there are two other possible shapes of response that you will come across in future discussions. They have been mentioned here in Figure 2.11 as type I, II and III. At present you need not bother about type II, and III; we will continue with type I only, but the ideas developed are equally applicable to the other two types of response.

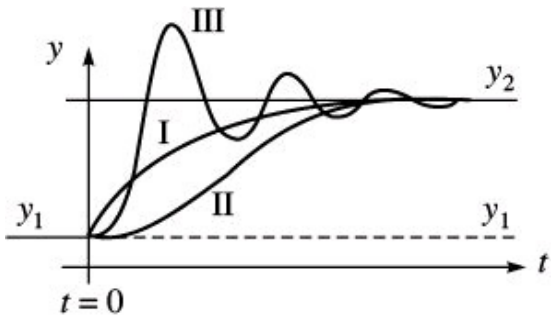


Figure 2.11 Qualitative nature of basic three types of process responses.

The idea of process gain K_P is an important parameter even in the process dynamic models. It will re-enter our future discussions, e.g. stability analysis, Controller design, etc.

Another important feature of a process that will be used during model development is the consideration about the **Linearity** of a process character. Let us define the linearity of a dynamic process.

2.5 Linearity of a Process

The classical definition of linearity of a system may be given as "If the relation between input x and output y, given by, $y = f(x)$, contains any nonlinear functions (e.g. powers, indices, trigonometric or any other types of transcendental function) in the argument, the system is a **Non-Linear** system".

However, from the gain calculation, a more practical notion of gain may be proposed. If in the working range (i.e. maximum possible variation of input and output variables) of a process, the gain remains constant, the process is said to possess **Linear Characteristics** and it is termed a **Linear Process**.

Another feature of a linear process is that, if there are two phenomena occurring

simultaneously in the process, and we know the dynamic description of both, the overall dynamics may be obtained by superimposition of the two processes. An example of this superimposition may be the treatment of a CSTR where the reactor performance may be obtained by combining the hydro-dynamics of mixing and the reaction kinetics.

In our foregoing discussion of Section 2.3, all the examples of dynamic responses are of processes after they have experienced a step change at their input condition. But input variable changes of any type may be described as an arbitrary shape on a graph against time. It may be associated with or without noise. We present the following examples and hope these will help you to develop a bias-free notion about the dynamic manifestation of processes.

2.6 More Examples of Dynamics of a Process

EXAMPLE 2.1 (Share Value of a Private Company): Trustworthy Corporation is a well known company covering various sectors of manufacturing, trading and import/export. It has been single handedly created, nurtured and brought up by the present Managing Director Mr. C.M. The share value of this company as reported by the Index of Bombay Stock Exchange (BSE Index) has shown dynamic response corresponding to some events during the last year.

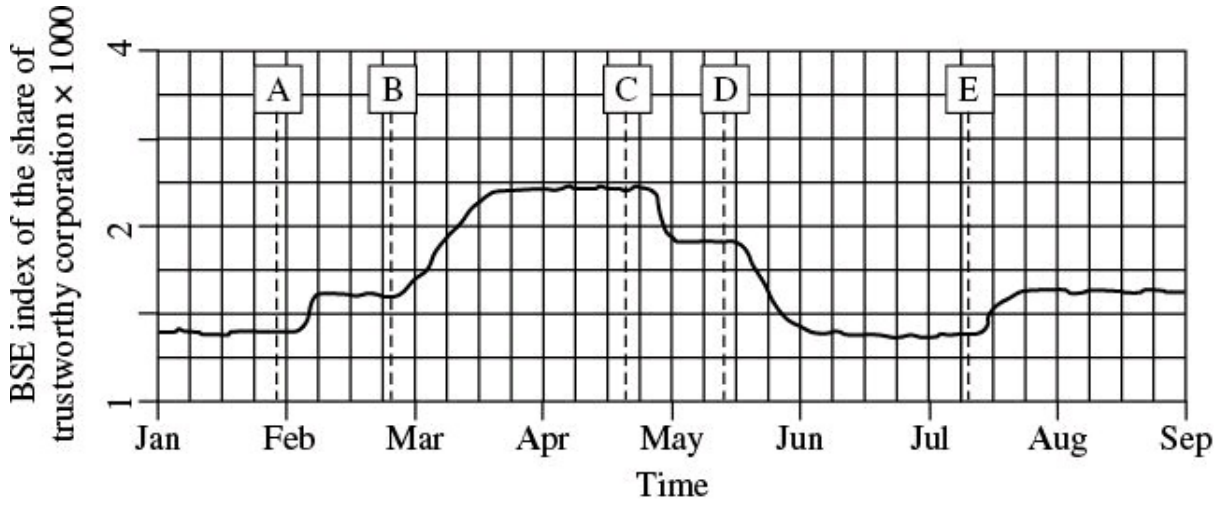


Figure 2.12 The dynamics of BSE Index of the share value of Trustworthy Corporation in last year. At (A) there has been a rumor that the Company is going to have a merger with a large multinational company. At (B) the merger actually takes place with African Green Hammery Inc. At (C) Mr. C.M. has fallen ill—there is a nation wide concern! At (D) Mr. C.M. has a massive stroke and gone into 'coma, doctor's panel report states that there is hardly any hope – the nation is stricken! At (E) The Miracle has happened—Mr. C.M. is out of danger and slowly gaining health.

EXAMPLE 2.2 (Pressure in the Steam Header of a Chemical Plant): In this example we present the dynamics of steam pressure in the steam delivery line from the boiler of a process plant (say IOC, Haldia). The pressure fluctuation occur due to the starting and stopping of an assortment of parallely connected loads.

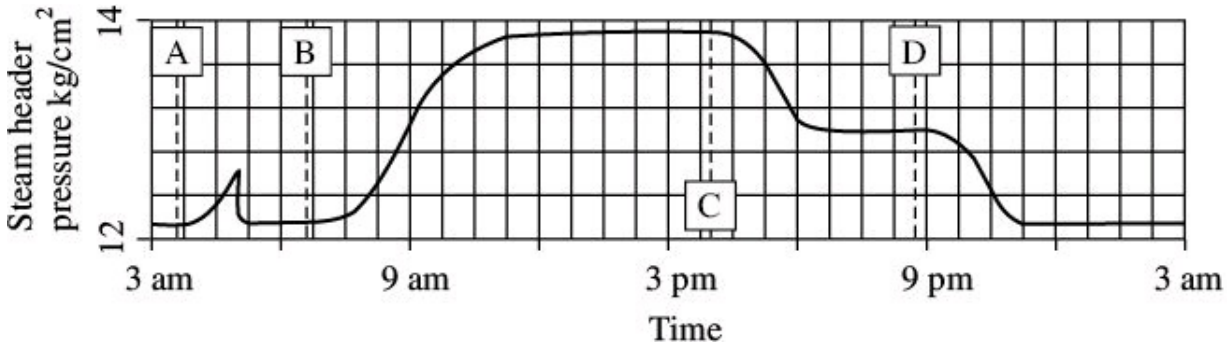


Figure 2.13 In a large process plant, the dynamics of steam pressure in the header recorded during a 24 hr. period. At (A) there has been an attempt to close the second turbine valve, unsuccessful due to pneumatic line failure. At (B) this valve is closed successfully. At (C) the First auxiliary compressor is turned on. At (D) the second auxiliary compressor is turned on.

EXAMPLE 2.3 (A Physiological Variable) : The next example shows the blood-pressure dynamics of a person who has come across a particular piece of very bad news—that his only son has been held in custody following an 'Anti-drug' mop-up organized by the special branch of the enforcement department.

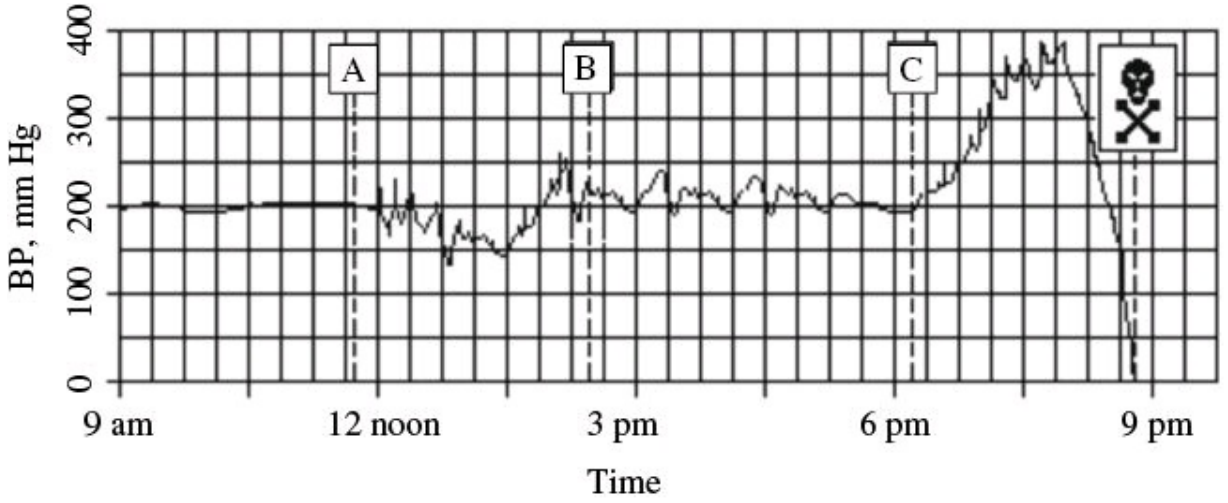


Figure 2.14 An ordinary day in a simple person's life but only up to point (A) when a very bad piece of news hits him. At (B) he is admitted to a nursing home and is in the ICU. At (C) – there is a power failure, and as there is no emergency power facility, the heartlung m/c. stops.

In all the above examples, the processes generally are not under some kind of tight control that has been designed with an adequate knowledge of dynamics. Whatever amount of internal control is with the last two example systems, may be considered as the part of the process. The bio-feed-back loops existing in the physiological system (Example 2.3) of the person or the financial management system (Example 2.1) built up inside the company organization act like a 'cushion' or certain amount of extra capacity. In face of a high magnitude of disturbance many times more than this cushion capacity, the system will behave just like it has no control. In Example 2.2, the boiler has an internal control to assure only the output steam quality within a range of throughput, but generally the control system is provided with dynamic knowledge to cope with any arbitrary type of loads. However, this has also got its capacity limitations beyond which such closed loop systems would also crash. The process

behaviour without control, in face of disturbance, is known as **open loop dynamics**. But what happens when there is a control action associated with the process and disturbance regulation is within the capacity of control system?

EXAMPLE 2.4 (Composition of a Component in a Mixing Process): In the next example we present a mixing system for two inlet component streams A and B.

The controlled variable is the composition of A measured at the outlet. Load may enter as flow fluctuation of stream A. Control action is achieved by manipulating the flow rate of B. The time history of load variable has been shown in Figure 2.15(a), and the controlled variable in Figure 2.15(b). Note that following each disturbance the control action comes after a little delay. Also the dynamic response is a little oscillatory. In future study of control system analysis and design, these phenomena will become very significant. In this example we observe a process with control action and it demonstrates what is called as **closed loop dynamics**.

By now you have been given an ample opportunity to understand the idea about the dynamics of a process. Now we must attempt to learn about constructing a dynamic model for processes that will give us the knowledge base of developing quantitative information of the process response corresponding to an input. The first step toward this objective is to learn about the dynamic elements of a process because these elements fundamentally determine the process dynamic character. Although, in Section 2.3 we have already introduced two of the most important dynamic elements—Capacitance and Resistance in a little informal manner, in the following section we shall have a proper discussion on this topic.

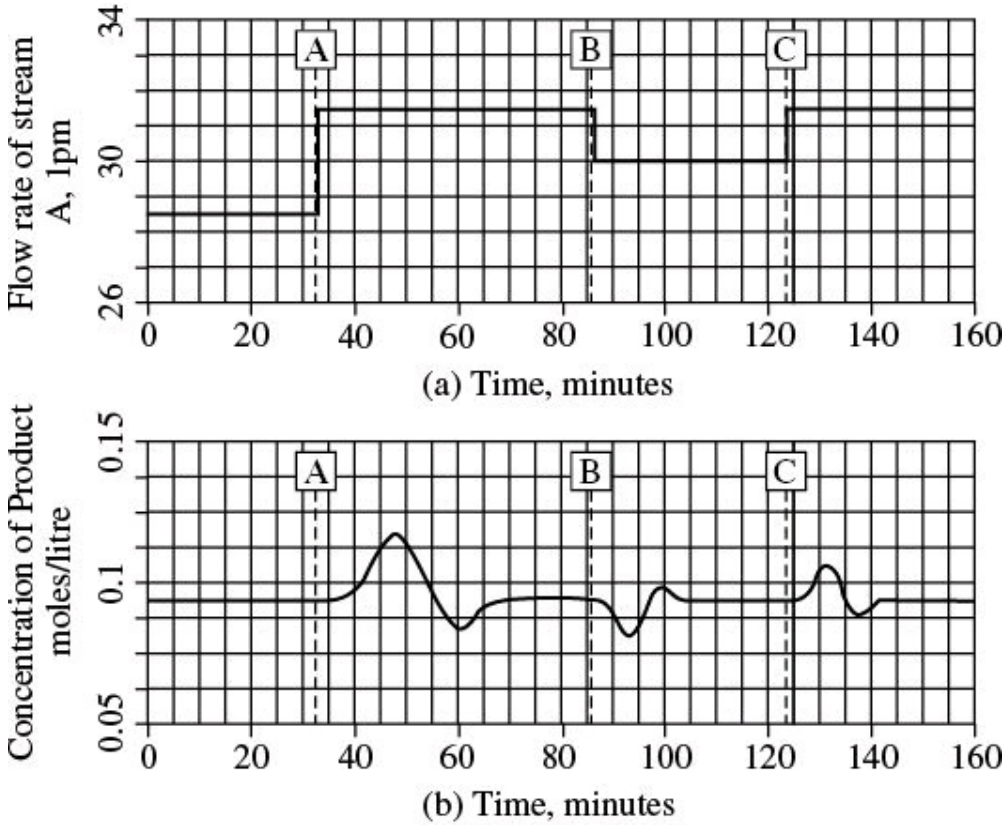


Figure 2.15 (a) The time history of the flow fluctuation of stream A as the load variable. At (A) flow suddenly increases from 28.5 lpm to 31.5 lpm. At (B) flow reduces from 31.5 lpm to 30 lpm. At (C) flow increases from 30 lpm to 31.5 lpm. (b) The dynamics of mixing process with control.

2.7 Dynamic Elements of Processes

The similarity of dynamic behaviour of the two systems (heat dissipation, and tank level depletion) in Section 2.3 is not surprising. If Process systems from different application areas contain parts in them that produce similar dynamic characteristics in the response should be bracketed under similar nomenclature. These parts are called dynamic elements of a process.

The reservoir part is called **capacity element** as this can store PE, and the connecting path may be called **resistive element** due to its restrictive effect on the flow of PE. The potential of PE in any capacity element is the principle variable PV, as its measure at any time will tell us the present state of the process—whether it is following the desired operational condition or whether any corrective action is necessary.

There is also a third category of dynamic element, in processes in which transfer of PE is associated with inertia. Thus, such a process does not stop at zero potential condition; as in the case of a mass sliding down an inclined plane, this effect will not allow the mass to stop at the point where the inclined planes join the horizontal plane. Hence, the inertial momentum drives it further; the low level reservoir temporarily becomes high level relative to the other one; and the flow of PE occurs in the reverse direction. Dynamically such a behaviour is known as oscillation. The part of the process that brings in the inertia in a process is called the **inductive element**. To complete this discussion we should state that an oscillatory behaviour may be identified with the presence of inductive element, but the vice versa is not true, i.e. processes containing inductive element may not oscillate due to high values of resistance and capacitance.

Due to the inherent mechanism of certain types of flow of PE during the continuation of a process, there is no possibility of existence of an inductive element. Examples are:
(1) Diffusional mass transfer where the PV is concentration, and (2) Conductive or Radiation heat transfer where PV is temperature. Dynamics of such processes are non-oscillatory.

Process engineers should confess that the concept of dynamic element has originated from electrical technology, as their name suggests. In latter period, not only from process analysis, but from other areas also, like Rheology, Solid State Physics, Mechanical Engineering, identification and establishment of another set of dynamic elements have been made. This has been possible by defining fundamental model

building blocks with configurations selected from system elements familiar in that branch of physical science and engineering discipline. They are conceptual ideal mechanical bodies and possess similar dynamic character with their electrical counterparts, capacitance, resistance, and inductance elements. Let us introduce them.

2.7.1 Mechanical and other Equivalents of Dynamic Elements

In the following we propose the input-output relations of mechanical and other equivalent of dynamic elements.

(1) **Spring:** If the applied force ranges within elastic limit of the spring, Hooke's law is applicable between force and spring displacement.

$$y(t) = K_S f(t) \qquad (2.5)$$

where K_S is the 'Spring Constant'. We obtain a linear relation, that is comparable to the relation between amount of electron, Q, stored in a charged capacitor when connected to voltage source of volts V.

A similar linearity also exists in case of the liquid level in a cylindrical tank, where the volume, V_L, and mass, M, of stored liquid bears a linear relation with liquid level, h , in the tank,

$$V_L = A_L h \qquad (2.6)$$

and,

$$M = rA_L h \qquad (2.7)$$

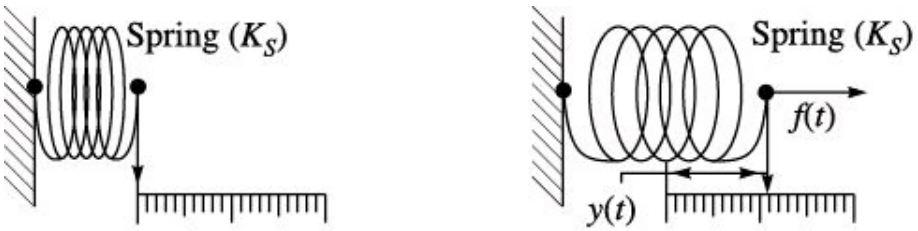


Figure 2.16 (a) The spring at rest. (b) The spring when stretched under an applied force $F(t)$ an elongation of $y(t)$ is obtained. The spring constant is K_S.

where, A_L is the tank area, and r is the liquid density. Such a capacitance element is also identifiable in heat transfer system,

$$H = mC_P(T - T_D) \qquad (2.8)$$

or,

$$H = mC_P\text{DT} \qquad (2.9)$$

where, m and C_P are mass and heat capacity of the substance raised through the temperature from T_D (datum temp.) to T, by introducing H amount of heat. The

product mC_P is known as thermal capacitance of the substance. You must agree that the name has understandable significance.

In a mixing system involving solute transfer in solution phase, the mass m of solute transferred into or from volume V of solution to raise or lower the concentration from Ca_I to Ca_O, the relation between m and DCa:

$$m = V(Ca_O - Ca_I) \qquad (2.10)$$

or,

$$m = V\mathrm{D}C_A \qquad (2.11)$$

where, volume V may be called as the capacitance and Ca, the potential (the solute concentration) in a mixing system. In all the examples of capacitance elements from different areas we may conceive that it is measured as the amount of principal entity required to be introduced into the element to raise its potential by unity.

In a solution system involving chemical reaction, the conception of capacitance and resistance elements are a bit difficult to establish. We shall discuss and work on them after we have covered certain important prerequisites.

Next we shall discuss the resistance element, and the various forms of this element identified in different application areas.

(2) **Dashpot:** This is a conceptual body comprising of a frictionless piston-cylinder assembly with a connecting capillary between upper and lower part of the cylinder. The cylinder including the capillary is filled with an incompressible liquid. The piston stem comes out of the cylinder body through a leak-proof seal. When force is applied to the piston stem, the piston will move either into or out of the cylinder causing the contained liquid to flow through the capillary.

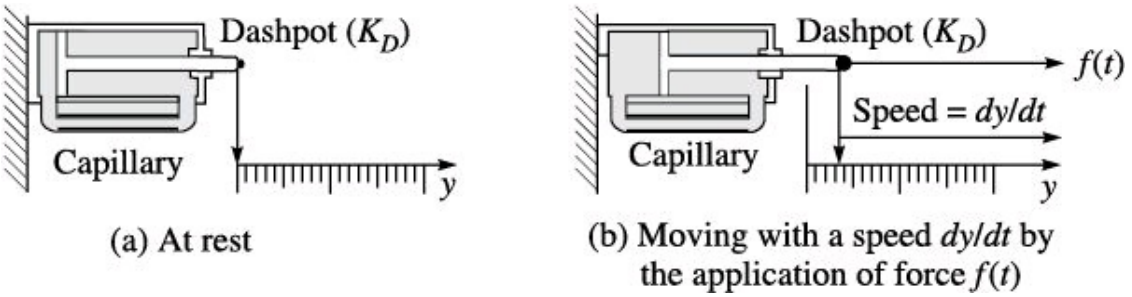


Figure 2.17 (a) The dashpot at rest. (b) The dashpot stem is moving with a speed, dy/dt under an applied force $f(t)$. The dashpot constant is K_D.

If the flow through the capillary always remains within laminar regime ($N_{RE} < 2100$) through the range of applied force, the Hagen-Poisseuli equation will be applicable for the system. Assuming a negligible flow resistance in the piston movement, and considering that all the resistance comes from the flow inside the capillary, a linear relation will exist between pressure difference DP across the capillary and liquid volume flow rate, F, through the capillary. Writing the relation in

terms of the applied force [$f(t)$] instead of DP and linear velocity of the fluid, u, instead of F, we obtain,

$$u = \left(\frac{d^3 g_C}{32 D^2 \mu L} \right) f(t) \qquad (2.12)$$

where, u = Linear velocity of fluid through the capillary; D = Cylinder internal diameter;
d = Capillary internal diameter; g_C = Gravitational constant; μ = Liquid viscosity; L = Length of the capillary, $f(t)$ = Applied force.

The above equation may be further simplified by uniting the constant terms to define the dashpot constant K_D,

$$\text{or,} u = K_D f(t) \qquad (2.13)$$

where, $K_D = \dfrac{d^3 g_C}{32 D^2 \mu L}$

In equation 2.13, K_D contains all physical parameters of the system except μ which is a fluid property. We call K_D as $1/\mu^*$, to recognize the fact that the constant μ^* is similar to viscosity μ described by the 'flow curve' of the dashpot liquid, i.e. the shear stress versus shear rate relation of Newtonian fluids.

Also, in equation 2.13 the velocity term u may be replaced by the derivative of displacement (dy/dt), and we obtain the $dQ/dt = I = (1/R).V$ type description of the mechanical resistance element, and that μ^* has become the mechanical equivalent of Resistance. Finally we may write,

$$\frac{dy}{dt} = u = \frac{1}{\mu^*} f(t) \qquad (2.14)$$

If we assume fluid flow occurs through a linear resistance (restriction), that is liquid level h and flow rate ($q = dV_L/dt$) bears a 'Ohms-law' like relation.

$$q = \frac{dV_L}{dt} = \frac{1}{R} h \qquad (2.15)$$

In a thermal transfer system if the total heat transfer area is A_T and heat transfer coefficient is U, the thermal resistance is $1/(U.A_T)$. The $I = V/R$ type relation for heat transfer become,

$$\text{Heat flux} = \frac{dH}{dt} = \left(\frac{1}{UA_T} \right) \Delta T \qquad (2.16)$$

In a mixing system involving solute transfer in solution phase, if we assume a

perfectly stirred condition in the mixer so that any solution, as soon as it reaches the mixer interior, becomes equal in concentration with that of the bulk liquid. Under such conditions, as there is no resistance of dissolution due to diffusion or any other mechanism, the effective resistance to such a transfer becomes the reciprocal of solution flow rate in to the mixer, that is, $1/q$.

(3) **Mass:** This is a conceptual system comprising of a block of mass with weight W at rest on a smooth plane over frictionless wheels (Figure 2.18).

When force $f(t)$ is applied to the block, if there is no friction, it will move with an acceleration d^2y/dt^2 along the axis of force application. According to Newton's law:

$$f(t) = \left(\frac{W}{g_C}\right)\frac{d^2y}{dt^2} \qquad (2.17)$$

The relation is readily comparable with voltage drop expression across an inductor:

$$V_L = L\left(\frac{dI}{dt}\right) = L\frac{d^2Q}{dt^2} \qquad (2.18)$$

Thus, a mass may be considered as a mechanical equivalent of inductance. With conductive heat transfer and solute transfer in a mixer, with or without reaction under diffusion and/or dispersion mechanism, there is no contribution of inertia effect, and hence, no possibility of existence of an inductance element.

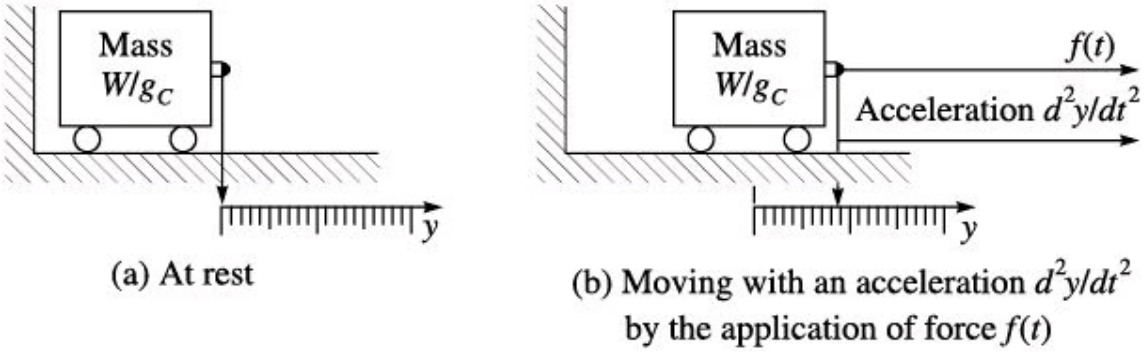


Figure 2.18 (a) The mass is at rest. (b) The block of mass is moving with an acceleration d^2y/dt^2 under an applied force $f(t)$.

Thus, a mass may be considered as a mechanical equivalent of inductance. For the liquid level systems with large tank diameter and a relatively small discharge tube diameter (high resistance), the effect of inductance is negligible, and only the resistance and capacitance elements are to be considered. But inertial effect due to the momentum of a flowing body of liquid becomes worth to be accounted for, if the tank and line diameters are comparable, as in a liquid filled Manometer. We shall discuss this later in the context of second order system dynamics.

In Table 2.4, we present a comparison chart for the dynamic elements as they exist in different disciplines. Hope this will help you for a quick check during process

model building.

TABLE 2.4 Comparison of dynamic element configurations

System Items	Thermal	Liquid level	Electrical	Mechanical
Principal Entity (PE)	Enthalpy H (cal)	Volume/Mass, V_L l/M (kg)	Charged particle (Electron) Q	Displacement y (cm)
Principal Variable (PV)	Temperature (°C)	Level h (m)	Electric potential V (Volt)	Force f (kg-force Dynes)
Current	Heat flux dH/dt (cal/s)	Flow rate q q or rq (m^3/s or kg/s)	I (Ampere or Coulombs/s)	Rate of displacement speed, velocity, $dy/dt = u$ (m/s)
Capacitance	Thermal capacitance MC_P (cal/°C) $DT = H/(MC_P)$	Tank area, A_L $(pD^2/4)$ m^2 $Dh = V_L/A_L$	Capacitance, C (Farad) $V = Q/C$	Spring (Figure 2.16) □ Ideal elastic body $f = K_S y$ □ □ □ (V) $(1/C)$ (Q)
Resistance	Thermal resistance $1/(UA_T)$, °C/(cal/s) $T = [1/(UA_T)]$ (dH/dt)	Fluid flow resistance R [(m/(m^3/s)] $h = qR$	Electrical resistance, R (Ohms) $V = R.I$ $= R(dQ/dt)$	Dashpot (Figure 2.17) □ Ideal plastic body $f = \mu*(dy/dt)$ □ □ □ (V) (R) $(I) = dQ/dt$
Inductance	—	—	Electrical inductance L (Henry) $V = L(dI/dt)$ $= L(d^2Q/dt^2)$	Mass (Figure 2.18) □ Ideal inertial body $f = (W/g_C)(du/dt)$ or, $f = (W/g_C)(d^2y/dt^2)$ □ □ □ (V) (L) (dI/dt)

After introducing the idea of dynamic elements of a process, it should be mentioned that there is an important classification of process in relation to the location of these elements within the process.

2.8 Lumping and Distribution of Dynamic Elements within a System

This spatial allocation of dynamic elements inside a process equipment is largely conceptual, i.e. what is our hypothesis or conception about how a process is performing the conversion of input variable to obtain the output? Within the system boundary, whether the processing or conversion is happening: (1) at a particular location in a concentrated manner, or (2) it is occurring in a scattered way through out the equipment volume? It is obvious that the presence of dynamic elements is necessary at the conversion points, hence for option: (1) the dynamic elements are concentrated or **Lumped**, and for option, (2) they are scattered or **Distributed**.

In fact, the classification of process systems under lumped or distributed parameter heading comes solely from our idea about how we think about the mechanism of conversion. In lumped parameter description, the process model is given by ordinary differential equations (ODEs) and distributed parameter description involves partial differential equations (PDEs). As we have said this classification originates directly

from modeling conceptions, it is obvious that initially we should attempt a lumped parameter model that is analytically easier to treat, provided the lumping of parameters is not far from the description of conversion mechanism in the process.

However, if we draw examples from reaction engineering, typical examples of lumped and distributed parameter systems are stirred tank and plug flow reactors respectively.

Also, a process may be considered lumped for an approximate or crude model description at first, but later for higher accuracy demands, may be modified to a partially distributed configuration and for a still higher accuracy, may be considered as a fully distributed parameter system.

2.9 Order of a System

A classification of processes associated with lumped parameter definition is the **Order** of a process. Order of a process is defined as "order of the ODE required to describe its dynamics". We will define this term in another way in the next chapter.

2.10 Self Regulation

Self regulation is a dynamic property of processes by virtue of which, when disturbed from a steady state operating condition, they can achieve to attain a subsequent steady state. Two example processes are shown in Figure 2.19. In Figure 2.19(a), a simple liquid level dynamic system has been shown in which, in transient condition, the outflow rate differs from inflow rate and tank level varies. As the system attains steady state, both the flow rates will equalize and the tank level would become and remain constant. Now if the inflow rate is changed to a new value, the outflow will begin to change and gradually become equal to the new value of inflow rate and the tank level will again attain a new constant value at the second steady state. This phenomenon will be repeated if the concerned variables do not cross the physical limits of the system (say if the tank does not overflow).

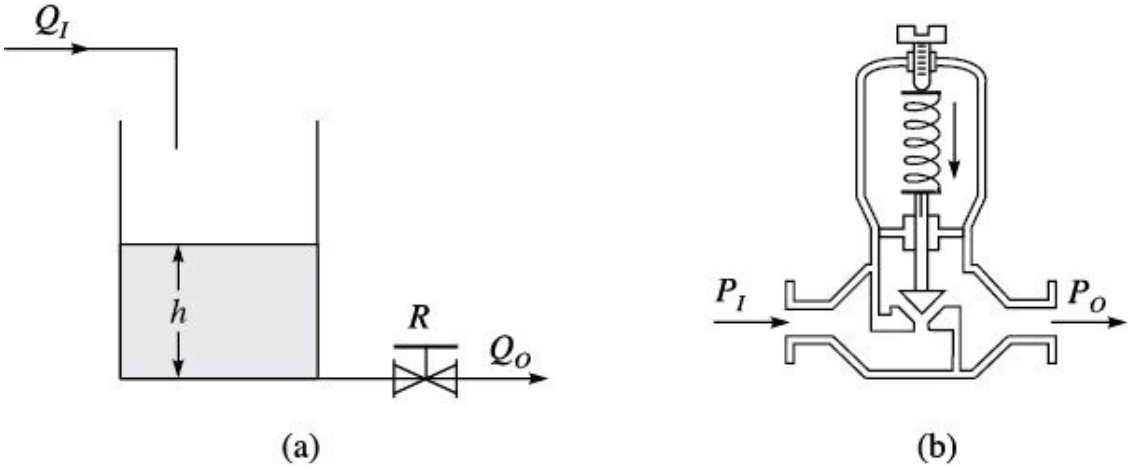


Figure 2.19 Two example processes that possess self regulation.

The reason of this self regulation may be attributed to that of an inflow increase. The potential (the tank level) will start to increase, and this change of potential will play on the fixed resistance, the valve on the bottom exit line, to accordingly change the outflow so that the system eventually attains a new steady state. It seems as if a negative feedback action lies in the mechanism of the process. After a change of inflow, if we define an error as the difference between inflow and outflow rates, this feedback action will drive the process in a direction toward reducing this error, which eventually would become zero as the system attains steady state.

On the other hand if the exit line valve is replaced by a constant through the output pump, the tank would continually increase or decrease, depending on whether the inflow rate is lower or higher than pump output. Thus self regulation property of the system will be lost by changing the R–C system to a pure integrator.

The second example in Figure 2.19(b) is that of pressure regulating valve, which has many uses in gas or vapour handling processes. The inlet pressure plays on the valve plug and hence onto the spring and the balance between the upward force (= projected area of the plug x inlet pressure) and downward spring force produces a clearance between the plug and seat of the valve. This is the effective valve resistance that determines the outlet pressure. If the inlet pressure decreases, the clearance between valve seat and plug would increase to decrease the valve resistance so that outlet pressure remains as close to its previous value as possible. This is a feedback control action that has gained the name ‘pressure regulating valve’. In case this valve is replaced by a fixed resistance valve, the system becomes comparable to the liquid level system of Figure 2.19(a). Thus, adding the control action in the form of a variable resistance has increased the amount of self regulation.

Consideration of the amount of self regulation of a given process may be considered as a measure of stability, thus, often becomes important in designing its control system.

In the next four chapters we shall be discussing the dynamic model development of process systems because for designing a control system, the primary need is to know the dynamics of the process concerned. In fact the type, required features and specification of the required control system is justified by the transient need of the process. We have redrawn Figure 1.4, the fundamental block diagram of the feedback control system as Figure 2.20, in which, to emphasize our immediate objective, the block labeled G_P has been made bold.

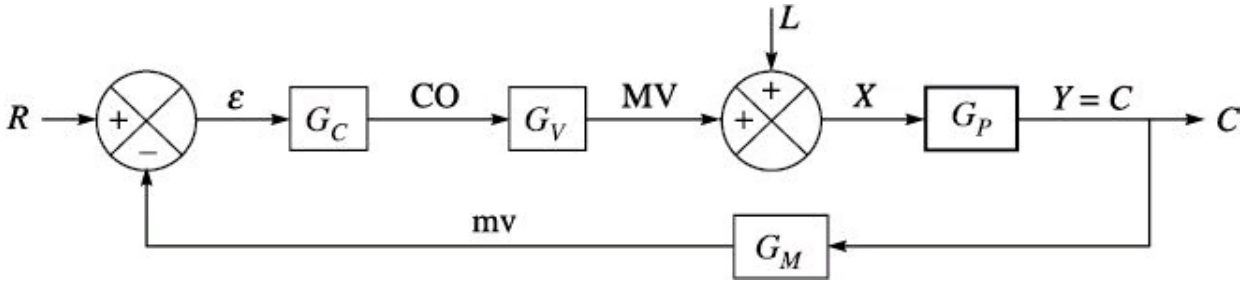


Figure 2.20 The process block in bold to begin the discussion on process modeling.

The objectives of constructing the process dynamic model are: (1) Designing the control system, (2) Better design possibilities of process equipment through its model, and (3) Diagnosis of a malfunctioning plant to identify the bottle-neck point.

EXERCISE PROBLEMS

2.1 With the mechanical dynamic elements presented in Table 1.3, think up of a combination by which you can measure the amount of acceleration of a moving body (a) along a horizontal plane, (b) along a vertical path under gravity.

2.2 It has been commented in the foregoing discussion that there is no effect of inductance in heat transfer by conduction and radiation, but what about convection, where heat transfer occurs due to mass flow of a relatively hot fluid? Can you cite an example to show the inertial or inductance effect in convection?

(As the density difference due to relative heating of part of fluid mass is very small, specially if the fluid is gas, a large amount is required to get a demonstrable effect, say when sun rays strike a large body of air for some time. Hot air goes up through the atmosphere and relatively cold air rushes into the low pressure region. But the rushing air misses its target because in the mean time, the low pressure spot has shifted slightly, due to Earth's rotation.

After missing its target, carried away by its momentum, the rushing air turns to follow the spot in a circular motion, but cannot reach it, because of the angular momentum of its rotational motion (like the proverbial dog trying catch its tail). If the low pressure spot happens to be on a large water surface, say somewhere in South Pacific, the circular motion become self sustained and grows up in size as it shifts over water surface, due to the ready availability of low density water vapour which keeps up the density difference. Another factor for its sustained existence is the very low friction between moving air and water surface.

The whole thing becomes a huge spinning top, few hundred kilometers in diameter, but relatively dwarf in size – one to two kilometers in height – and we get a …….. in our hand. The total energy content of such phenomena may be equivalent to the released energy from a blasted 'nuke-warhead' of say a few hundred mega-ton in rating. The phenomenon has many names in different

countries. (Write at least two of them.)

2.3 Dimension of the product of capacitance and resistance elements of all example systems in Table 1.3 has an interesting similarity—verify yourself. Can you make a comment on this similarity?

2.4 A very rich and fat man while walking in the garden of his residence suddenly felt a sting like pain on the side of his torso. Initially he ignored the pain but finding a small trickle of blood he called the Doctor. The X-ray examination revealed that a 'bullet' had entered and travelled 204 mm inside his body without touching any vital organ! Define the dynamic process by specifying the input and output variables. What are the dynamic elements involved in this process?

2.5 The level dynamics of water draining through a constant resistance from a tall cylindrical tank is known. The same experiment is carried with glycerin as the liquid. Draw your best guesses about the level dynamics in the two situations.

2.6 A metal block is riding a conveyor as it travels through a tunnel type furnace and is subjected to a gradually increasing temperature. With respect to speed of travel, the surrounding temperature of the metal block (i.e. the furnace interior) increases by:

$T = 50°C + 10°C/min.$

Where, T = furnace temperature in °C. 50°C is the atmospheric temperature at the furnace entry. Draw your best guess of the metal temperature vs. time on a graph for 60 min. from start.

2.7 In FPS unit, the numerical value of μ^* of a dashpot is 15. Calculate the speed of travel of the piston stem if a force of 10 lb is applied there. Convert your result to MKS and SI systems.

2.8 Rheology is a branch of fluid mechanics which deals with the mechanism of flow behaviour under stress, of substances that have structure ranging from purely plastic bodies to a combination of plastic and elastic bodies. It has been an age-old practice of rheologists to represent an elastic body by a spring and plastic body by a dashpot. They have come up with two fundamental combinations of these two elements: (1) The spring and dashpot are combined in series, called a Maxwell element; (2) The spring and dashpot are combined in parallel, called a Voigt element. If these bodies are suddenly put under a finite force, try to guess their responses as displacement. Draw the two graphs on a common time base.

2.9 The reaction rate constant of a 1st order irreversible reaction is $0.01 sec^{-1}$. Calculate the 'half life period' of the reaction, or the time required for lowering the reactant concentration to 50% of its starting value.

2.10 Comment on whether the configurations of Spring–Dashpot combination shown in
Figure P2.1, would behave like a 'solid (elastic) or a 'liquid' (plastic) under a stress [force $f(t)$].

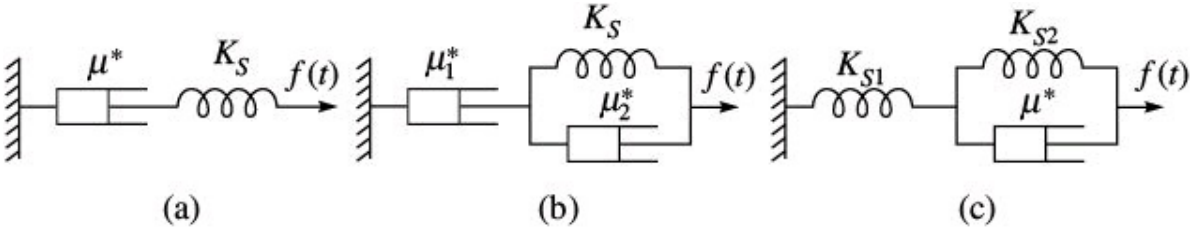


Figure P2.1 Systems for different configurations of spring–dashpot elements.

2.11 In an absorption column, an NH_3 air mixture rising from the bottom of the column is contacted in counter current fashion with water as the absorbent flowing down from the top. Data of the NH_3 concentration of the inlet and exit streams are collected for three steady states of column operation.

Steady state-1 (SS1): Liquid flow rate, Q_W = 250 lpm, liquid inlet concentration, C_{LI} = 0 gm/lit, liquid outlet concentration C_{LO} gm/lit, gas flow rate, Q_W = 85 m^3/min, inlet gas concentration, C_{GI} = 6.5% by volume, gas outlet concentration, C_{GO} = 0.8% by volume. If only C_{GI} is increased to 7% by volume and steady state-2 (SS2) is brought, it was found that the total mass transfer rate M gm/min has decreased by 2.8%. Again starting from SS1 if only Q_W is raised to 260 lpm, and steady state-3 (SS3) is brought,
M increases by 3.2%. Considering C_{GI} to be the load variable, Q_W, the manipulated variable, and C_{GO}, the output variable, calculate the system gains K_{PL} = DC_{GO}/DC_{GI}, and K_{PM} = DC_{GO}/DQ_W. Assume that there is no volume change of the liquid and gas throughout the column, and the column internal temperature remain constant at 30ºC, that is equal to the gas and liquid stream temperatures.

3

Basic Modelling Principles (Time Domain Analysis)

The terms **Process Analysis** and **Process Modelling** have approximately synonymous meanings—that is to develop a mathematical statement of the transfer or flow phenomena by which raw material conversion occurs inside a process system. Alternatively, it is an exercise of developing a hypothesis about the conversion mechanism that exists inside a process system, and then to convert this hypothesis or notion into a mathematical description for obtaining process output corresponding to any input changes. The mathematical description is achieved by developing a set of equations that contain all concerned variable and system parameters, and truly reflect the process behaviour at dynamic and steady state conditions. This means, if the real physical input to the process is interpreted as a mathematical function and then introduced into the process model, its output will replicate the actual process response.

At this point we shall discard the terms 'Raw material' and 'Product' and instead use 'Input' and 'Output' or 'Response' in concern to a process, because they are more compatible to the analytical vocabulary. These models are basically of two categories with reference to the condition of process operation.

3.1 Steady State Model

This describes the process output/s under steady state or stationary condition of input variable/s. In unit operation courses, the performance describing equations for heat and mass transfer equipment that you have learnt are all based on steady state models, often supported by experimental data from pilot set-ups.

3.2 Dynamic Model

This describes the process reaction or response when there has been a change in one more of its inputs from the steady state value, or, it is under unsteady state condition, and the solution of such model equations in time domain is often called Process Dynamics. The unsteady condition is also called 'Transient' state.

A dynamic model contains more information about the process than the steady state model, and is more difficult to construct. The steady state solution may be derived

from a complete dynamic model by stretching the time domain solution to an infinite value of time. The objectives of developing the dynamic model for a given process may be listed as:

1. Equipment Design by using the model equations has proved to obtain better performance than could be achieved by the steady state design relations
2. To Diagnose an Operational Fault in a running plant. The model may pin-point the source of malfunction
3. The exercise of Control System Design for a process starts from the dynamic model building, and the accuracy of the developed model determines the performance of the control system.

As stated above, for control system design the first step is to develop the dynamic model of the process because a control system fundamentally tries to negotiate a disturbed process under unsteady state condition and brings it back to an optimum steady state. For this reason the alternative name, *dynamic compensation*, is often used to denote the control action.

Generally for the real physical systems, the developed model equations are differential equations. But is it impossible to obtain algebraic equations from such an exercise? We shall treat this topic a little bit seriously to expose certain relevant conceptions.

Consider the mixing of two solution streams transported by two pipes. The two pipes are joined at two ends of a 'Tee', and at the junction point, the streams get mixed instantaneously. The mixed stream flows through the third or branch port of the Tee joint. Figure 3.1(a) illustrates the arrangement. Alternatively, a centrifugal pump may be also used for the mixing purpose as in the Figure 3.1(b). The principal assumptions or dynamic features of these systems are: (1) The mixing happens in *negligible time*; (2) There is only a negligible '*space*' or capacity provided for the process to happen. In the figures the 'Q's denote the solution mass flow rates and 'x's, the solute concentrations.

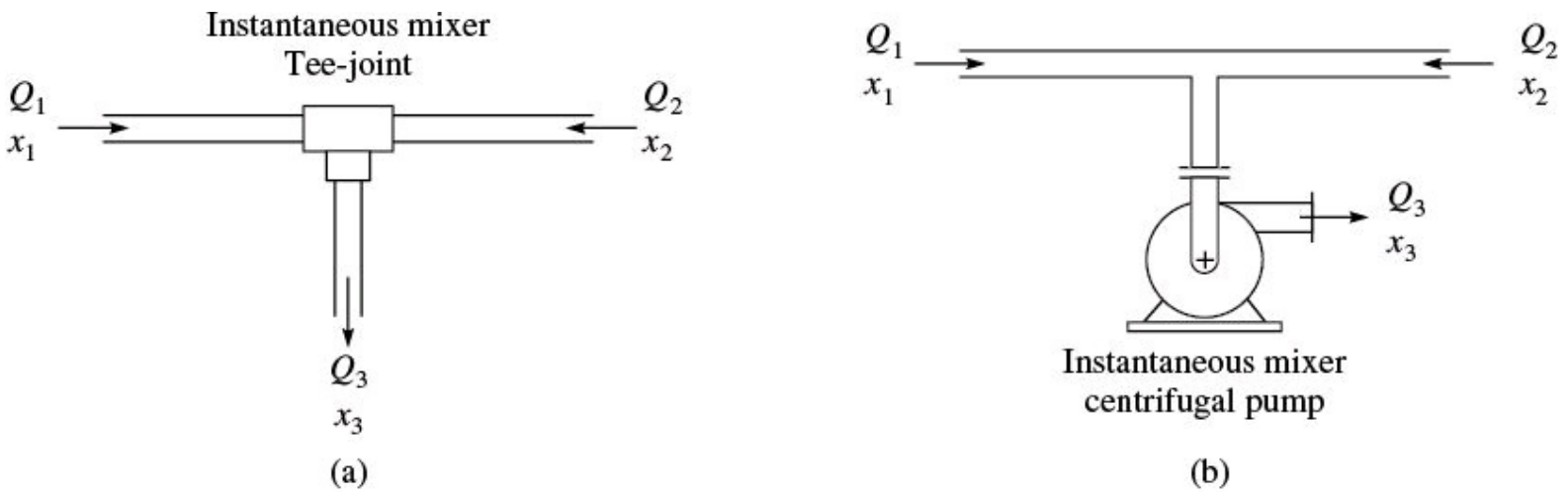


Figure 3.1 Two alternative ways of instantaneous mixing of two solution streams.

First let us consider the steady state situation, i.e. the inflow rates and concentrations to the mixer are constant. It is sufficient to write the following two equations to describe the steady state behaviour of the system:

Overall mass balance:

$$Q_1 + Q_2 = Q_3 \qquad (3.1)$$

Solute mass balance:

$$Q_1x_1 + Q_2x_2 = Q_3x_3 \qquad (3.2)$$

Equations 3.1 and 3.2 are nothing but the classical Law of Conservation of Mass and may be stated in language as:

$$\text{Inlet Rate of Mass} - \text{Exit Rate of Mass} = 0 \qquad (3.3)$$

Now think if any of the inlet stream's flow rate or concentration change from their steady state value, in the dynamic condition of the mixing system, the corresponding responses would be instantaneous and equations 3.1 and 3.2 are still sufficient to describe the system behaviour in terms of exit concentration and flow rate.

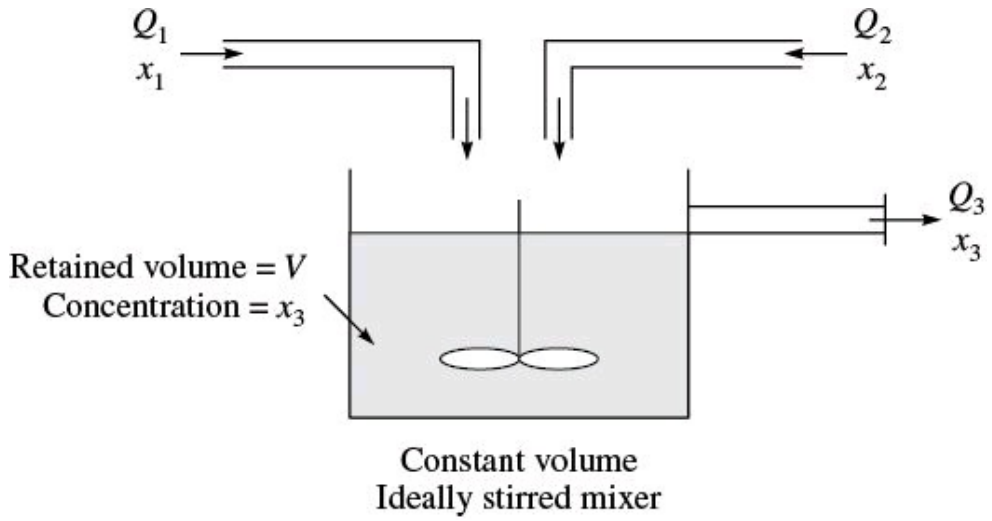


Figure 3.2 Constant hold-up ideally stirred mixer.

Now suppose we are not satisfied with the performance of this arrangement due to the presence of too much concentration fluctuations in the exit stream, and it has been

proposed to use an ideally or perfectly stirred cylindrical tank with a side outlet, as the mixer. As the outlet is located at a height from the tank bottom, there will always be a definite volume of liquid retained within. The two incoming streams are pouring down in the tank and the mixed stream goes out from the side outlet(Refer to Figure 3.2).

At the steady state operational condition, equations 3.1 and 3.2 are still valid and sufficient for describing the input-output dependency of this mixing system. What if any of the inflow rate changes? Assuming that the solution density is independent of concentration, as it has been stated that retained liquid volume or hold-up volume always remain constant, there would be no accumulation or depletion of total liquid mass within the tank, hence, Equation 3.1 would still remain valid. We may say that flow response of the system is instantaneous. But what if any of the inflowing stream concentration changes?

Before discussing this topic, let us first understand what property or performance of a mixer is referred by the adjectives: **Ideal** or **Perfect Mixing**. If you have already gone through the course of Reaction Engineering, the term **Back-mixing** is equivalent of these terms expressing that there is no concentration gradient and perfect homogeneity remain within the mixer, i.e. bulk and exit concentrations are equal.

Physically thinking about the dynamic response for an input concentration change, as the tank is filled up with liquid of steady state concentration, corresponding to any inflowing stream's concentration change, the total change in exit stream's concentration should require a definite amount of time to become complete (or to reach the next steady state), because at least that much amount of mass of solute has to be carried into the mixer that is required to bring up the hold-up volume of liquid to the next steady state concentration value.

(In fact more time will be required for this task, if we realize that a certain amount of solution of excess concentration will leave the mixer in the mean time, because bulk liquid concentration inside the tank is changing gradually.)

This gradual change of liquid bulk concentration in the mixer and that of the exit stream may be analytically accounted for by introducing a differential amount of accumulation (or depletion) term in equation 3.2:

$$Q_1 x_1 + Q_2 x_2 - Q_3 x_3 = V \frac{dx_3}{dt} \qquad (3.4)$$

The liquid density, *r*, need not be put with flow rate or mixer holdup volume, because it has been assumed that the mixing is a constant density operation. In fact, the time derivative of bulk liquid concentration really reflects the gradual change as interpreted as 'Rate of accumulation (or depletion)' of mass.

If the 'so-called' instantaneous mixers of Figures 3.1(a) and (b) are critically

considered, it should be appreciated that both, the Tee joint and the Centrifugal Pump, have finite inside volumes (however small) for mixing and Equation 3.4 may still be required to account for their dynamic character. On the other hand, the hold-up volume of the stirred mixer may be reduced only to a limit to assume the character of an instantaneous mixer. So by an effort to propose a more realistic model for the mixing system, the *Algebraic Equation Model* has developed into a *Differential Equation Model*.

(Examples of such digression of model configuration may be also cited in other physical systems.)

Consider a billiard ball driven at an initial speed in a particular direction on the board surface. If we consider the surface to be perfectly smooth, the travelling speed would not change until the ball strikes any of the side walls. If we consider the collision to be perfectly elastic, the travelling speed would not change; only the ball would change direction according to the Law of Reflection of Light. The ball will keep on moving endlessly and bouncing on the side walls. The model describing the Motion or Displacement of the ball is a set of algebraic equations.

To propose a more realistic model, let us say the ball is initially driven with a force F in X-direction. To obtain the velocity V_X in X-direction, we have to divide the force by the mass of the ball m. Then to recognize the roughness of the board surface, we introduce a friction coefficient m^*. This frictional drag force, due to this roughness, is proportional to the velocity and acts in reverse to the direction of motion. According to Newton's law, the difference between the applied force and drag force will be the rate of change of momentum of the ball or the product of ball mass and its acceleration. Finally, the force balance becomes:

$$F - \mu^* \frac{dX}{dt} = m\frac{d^2X}{dt^2} \quad \text{or} \quad m\frac{d^2X}{dt^2} + \mu^* \frac{dX}{dt} = F \tag{3.5}$$

Thus, we have arrived at a model described by a second order differential equation starting from an algebraic equation model by trying to propose a more accurate dynamic description of a given system.

Let us again return to the instant mixing systems and their model Equations 3.1 and 3.2. In fact, these equations signify that the input-output relation of the system is a *constant* or a multiplier that instantly and continuously multiplies the input to form the output or response. We have already introduced the notion of Gain of a Process in the previous chapter. The constant multiplier type I/O relation that has been just mentioned is the character of a Gain element. We may summarize the conclusions from the above discussion as:

1. The systems with similar dynamic and steady state models (Equation 3.3) are algebraic in nature. They posses a very fast dynamic response, and may be considered as a Pure Gain element. (We shall discuss more about gain elements in Chapter 5.)
2. The systems with finite lag in response should have an accumulation term in their unsteady state model equation (as in Equation 3.4), and results in a single or a set of differential equations. Steady state model for such systems may be obtained by omitting the accumulation term.

It may be concluded that for a given process, we want to develop an unsteady or dynamic balance that would take the form of Equation 3.4. But balance of what? We have already mentioned the concept of **Principal Entity** in the introductory chapter, which may be defined in a round-about way as: *A process may be considered as the manifestation of movement or flow of the principal entity within the system boundary*. Principal Entity takes the form of Mass, Heat, Momentum, etc. for different physical applications.

It should be mentioned at this point that selection or identification of a particular species as the principal entity is not unique. There may be others also whose movement can be attributed to the continuation of a process. Generally, the principal entity is selected so that its *potential* or *level* may describe the state of the process to an extent sufficient for the intended application of the developed model.

To develop the fundamental differential equation, the following dynamic balance for the principal entity, PE, is constructed from the statement of the problem.

$$\left\{\begin{matrix}\text{Inlet Rate}\\ \text{of PE}\end{matrix}\right\} - \left\{\begin{matrix}\text{Outlet Rate}\\ \text{of PE}\end{matrix}\right\} = \left\{\begin{matrix}\text{Accumulation Rate}\\ \text{of PE}\end{matrix}\right\} \qquad (3.6)$$

As an example, for the temperature control of a heat exchanger, we will build up a model for the process (heat exchanger) by selecting enthalpy as the principal entity and develop equation in terms of terminal temperatures of the associated fluid streams as process variables.

We can now proceed to a more down-to-earth discussion about the procedures of model building. It should be mentioned that at present, our efforts will be restricted to the analytical ways of model development, i.e. only ideal physical and chemical laws shall be used for the exercise. However, there are methods that use dynamic experimental data obtained from the actual plant for model development. These procedures would be taken up in the subsequent Chapter 12 on '**Process Identification**'.

The mathematical prerequisites for the ensuing discussion are the Theory of Laplace

Transformation, and Theory of Matrix and some related topics. It is expected that the reader has been already exposed to these techniques in some earlier semester. For refreshing the memory, a short comprehensive form of Laplace' Transform has been included in Appendix-II, and that of Matrix in the beginning of Chapter 16.

3.3 Different Descriptions of Process Model

The exercise of modelling begins from establishing an unsteady state balance of principle entity, PE, around the process expressed by Equation 3.6. From this balance, the fundamental differential equation could be developed and may be expressed by certain alternative forms. We express the more important of these configurations in context of a single input single output (SISO) process system.

Figure 3.3 A SISO process.

For a SISO process, the dynamic model may be expressed by the following type of differential equation:

$$a_n \frac{d^N y}{dt^N} + a_{n-1} \frac{d^{N-1} y}{dt^{N-1}} + \cdots + a_1 \frac{dy}{dt} + a_0 y = b_0 x(t) \qquad (3.7)$$

It has been already mentioned in the previous chapter that the order of the above Ordinary Differential Equation (ODE) that completely describes the process dynamic character is called the **Order** of the process.

3.3.1 State-Variable

State-Variable form of the model is developed by writing n number of first order ODEs from Equation 3.7, thus,

$$\left.\begin{aligned} dy/dt &= y_1 \\ dy_1/dt &= y_2 \\ &\vdots \\ dy_{N-2}/dt &= y_{N-1} \\ dy_{N-1}/dt &= (b_0/a_N)x(t) - (a_{N-1}/a_N)y_{N-1} - \cdots - (a_1/a_N)y_1 - (a_0/a_N)y \end{aligned}\right\} \qquad (3.8)$$

And expressing the set of equations in the following matrix form:

$$x = AX + BU \qquad (3.9)$$

$$y = CX \qquad (3.10)$$

where,

$\dot{x} = \begin{bmatrix} \dot{y} \\ \dot{y}_1 \\ . \\ . \\ \dot{y}_{N-2} \\ \dot{y}_{N-1} \end{bmatrix}$ is the vector representing the derivative of state variable vector,

$\mathbf{A} = \begin{bmatrix} 0 & 1 & . & . & 0 & 0 \\ 0 & 0 & 1 & . & 0 & 0 \\ . & . & . & . & . & . \\ . & . & . & . & . & . \\ 0 & 0 & . & . & 1 & 0 \\ -\frac{a_0}{a_N} & -\frac{a_1}{a_N} & . & . & -\frac{a_{N-1}}{a_N} & 0 \end{bmatrix}$ is the process matrix, $\mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ . \\ . \\ 0 \\ \frac{b_0}{a_N} \end{bmatrix}$ is the input vector,

u = [0 0 . . 0 1] is the input disturbance,

C = [1 0 . . 0 0] is the output matrix, and $\boldsymbol{x} = \begin{bmatrix} y \\ y_1 \\ . \\ . \\ y_{N-2} \\ y_{N-1} \end{bmatrix}$ is the state vector.

3.3.2 The Transfer Function

The transfer function corresponding to Equation 3.7 may be simply defined as the "*ratio of Laplace transformed output is to input variable*". But we know that transformation of derivatives $[d^N y/dt^N]$ will contain initial values of the variable $[y_0]$ and its derivatives $[d^{N-1}y_0/dt^{N-1}, d^{N-2}y_0/dt^{N-2}, \ldots, dy_0/dt]$. These terms may cause additional and unnecessary algebraic effort in the subsequent dynamic response analysis. Elimination of these terms to obtain the final form of the model is done either by: (1) first assuming that process variables and their derivative have zero initial values, or (2) translating the absolute variable values in Equation 3.7 to their corresponding **Deviation Variables,** then taking Laplace transform, and finally

expressing the result as a ratio of the transformed output to input variables:

$$G_P(s) = \frac{Y(s)}{X(s)} = \frac{b_0}{a_N s^N + a_{N-1} s^{N-1} + \cdots + a_1 s + a_0} \qquad (3.11)$$

This ratio $GP(s)$ is called the **Transfer Function** of the given system. The deviation form of a variable is defined as the difference of its absolute value from the corresponding steady state value, thus,

$$\left.\begin{aligned} X &= x(t) - x_S \\ Y &= y(t) - y_S \end{aligned}\right\} \qquad (3.12)$$

Using the deviation form of the variables is equivalent to the first assumption, if we consider that the process was at steady state during all values of time previous to the beginning of our observation, i.e. for $t < 0$, and dynamic observation and analysis are performed for $t \geq 0$. This is a logical assumption that the process was at steady state before our observation begins, because the dynamic study of a process is only relevant after it has been disturbed.

A more generalized form of the transfer function may be written as:

$$G_P(s) = \frac{Y(s)}{X(s)} = \frac{b_M s^M + b_{M-1} s^{M-1} + \cdots + b_1 s + b_0}{a_N s^N + a_{N-1} s^{N-1} + \cdots + a_1 s + a_0} \qquad (3.11a)$$

Equation 3.11(a) recognizes the possibility of differential expression existing in numerator for input variable processing within the process after which the integration in the denominator with respect to output variable would produce the response. Considering separately the polynomials in s in the numerator and denominator of R.H.S. of Equation 3.11(a), they may be equated to zero to obtain the following two equations:

$$b_M s^M + b_{M-1} s^{M-1} + \cdots + b_1 s + b_0 = 0 \qquad (3.11b)$$

$$a_N s^N + a_{N-1} s^{N-1} + \cdots + a_1 s + a_0 = 0 \qquad (3.11c)$$

The roots of the above equations may be found as, $z_1, z_2, \ldots, z_M$, for Equation 3.11(b) and $p_1, p_2, \ldots, p_N$, for 3.11(c). The numerator roots $(z - s)$ are locations on the complex s-plane at which Equation 3.11(a) would have zero value, hence these roots are called '*zeros*' of the transfer function, and denominator roots $(p - s)$ are locations on the complex s-plane at which Equation 3.11(a) would result an infinite ∞ value, hence these roots are called '*poles*' of the transfer function. It is obvious that the number of poles is equal to the order of the system. In factored form, Equation 3.11(a) assumes the following configuration:

$$G_P(s) = \frac{(s - z_1)(s - z_2)\cdots(s - z_{M-1})(s - z_M)}{(s - p_1)(s - p_2)\cdots(s - p_{N-1})(s - p_N)} \qquad (3.11d)$$

An Equation like 3.11(c), derived from a transfer function, is called the **Characteristic Equation** of the process. It spells the fundamental dynamic character of an undisturbed process, along with a number of conclusions regarding its stability and speed of response. A frequency domain description of the process can be also derived from its transfer function.

Formation of state variable and the transfer function description from the basic differential equation form of a process model is shown in the following example.

EXAMPLE 3.1 The following differential equation has been obtained as the dynamic characteristic of a given process. It is required to develop the corresponding state and transfer function model.

$$a_0 \frac{d^2y(t)}{dt^2} + a_1 \frac{dy(t)}{dt} + a_2 y(t) = b_0 x(t) \qquad (3.13)$$

This 2nd order ordinary differential equation (ODE) describes the process dynamics. The output variable $y(t)$ is related to the input $x(t)$ by the above Equation 3.13, where, a_0, a_1, a_2, b_0 are the process parameters

To develop the state variable model first from the given 2nd order equation, two 1st order differential equations may be written as:

$$\left.\begin{aligned} dy/dt &= y_1 \\ dy_1/dt &= (b_0/a_0)x(t) - (a_1/a_0)y_1 - (a_2/a_0)y \end{aligned}\right\} \qquad (3.14)$$

In matrix notation, Equation 3.14 may be written as:

$$[\dot{\mathbf{y}}] = \begin{bmatrix} 0 & 1 \\ -a_2/a_0 & -a_1/a_0 \end{bmatrix} \mathbf{y} + \begin{bmatrix} 0 \\ b_0/a_0 \end{bmatrix} \mathbf{x} \qquad (3.15)$$

The transfer function corresponding to Equation 3.13 may be readily written by first converting Equation 3.13 in terms of the deviation variables Y and X, and taking Laplace transform of the resulting equation:

$$\frac{Y(s)}{X(s)} = \frac{b_0}{a_0 s^2 + a_1 s + a_2} \qquad (3.16)$$

In dynamic and control system applications, Equation 3.16 is expressed in another alternative way called '*Standard Form*'. This is given as in the Equation 3.17. We shall have a detailed discussion about this form in a later section.

$$\frac{Y(s)}{X(s)} = \frac{b_0/a_2}{(a_0/a_2)s^2 + (a_1/a_2)s + 1} \qquad (3.17)$$

At the beginning of this chapter, a guideline has been presented for process model development. After introducing the alternative descriptions of the dynamic model, it

should be proper to introduce the development procedure of process models of certain physical systems that occur commonly in applications. We will start with continuously operating systems. The lumped and distributed parameter kind of systems have been already introduced, and we know that the later variety involves a lot more analytical rigour in model development than the former kind. We shall begin with the simplest kind of processes, i.e. continuously operating single stage lumped parameter systems.

3.4 Modelling for Continuous Systems

3.4.1 Lumped Parameter Single Stage Systems

EXAMPLE 3.2 (Liquid Level Dynamic System): A large tank placed in a process liquid flow path may serve as a surge or capacitance to smoothen out any upstream flow fluctuation. Such a tank with cylindrical cross-section (area = A_L) has one liquid inlet from top and an outlet line with a valve at the bottom of the tank.

The exit flow occurs due to the potential of liquid level inside the tank. At steady state, the inlet (q_I) and outlet (q_O) flow rates are constant and of course the height of the liquid (h) in the tank is also constant. Disturbance or load may enter into the system as variation of inlet flow rate $q_I(t)$, and we need to build the model that will tell us the resulting variation of either $h(t)$ or $q_O(t)$.

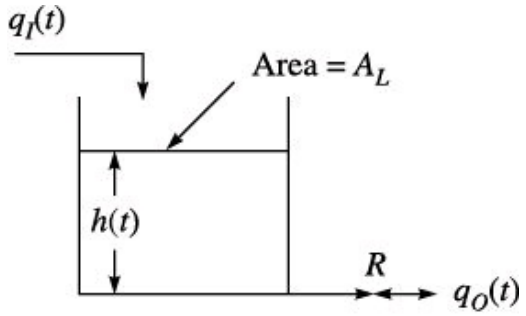


Figure 3.4 The Liquid level dynamics of a cylindrical tank.

Before attempting the analysis, we must state the assumptions made as the basis of the model.

1. All the liquid storage facility is only in the tank
2. All the resistance to liquid flow resides in the valve

The above two assumptions are an alternative way of expressing that the two relevant dynamic elements (capacitance and resistance) are concentrated or **lumped** at certain location within the system boundary

The model based on such assumptions is called a **lumped parameter model** and it should be emphasized that the character of a model largely depends on the statement

of the assumptions. If we want to be closer to reality, it should be recognized that in reality, the outlet pipe can also store a relatively small amount of liquid and the tank wall does offer a small amount of resistance to the flowing liquid.

1. The resistance, R, of the valve has a linear characteristic, that is, it follows an ohm laws like relationship: $R = h/q_O$
2. The tank is a cylinder with constant diameter at all heights. This assumption states that the capacitance of the system (internal area of the tank) is independent of the potential and the classical relation valid for electrical capacitance ($Q = CV$), may be applied, i.e. $V_L = A_L h$
3. The system operates at constant temperature and fluid properties like density, r, and viscosity, m, are constant

Now the unsteady state principle entity (mass) balance may be proposed with reference to Figure 3.4:

(Rate of mass input) – (Rate of mass output) = (Rate of mass accumulation) (3.18)

Writing the proper parameters and variables, Equation 3.18 becomes:

$$\rho q_I - \rho q_O = \rho A \frac{dh}{dt} \qquad (3.19)$$

Cancelling r on each side of Equation 3.19,

or $$q_I - q_O = A \frac{dh}{dt}$$

In the above equation, though there is an single input variable, q_I, it could effect either the tank level, h, or out flow rate, q_O, simultaneously, that is, the system has two possible output variables and fortunately they are mutually combined by a linear relation. To obtain a single input single output model, we have to choose one of the outputs by converting the other by the ohms law like linear relation, $h = q_O R$. The two possible model equations would be:

$$A \frac{dh}{dt} + \frac{h}{R} = q_I \qquad (3.20)$$

and

$$AR \frac{dq_O}{dt} + q_O = q_I \qquad (3.21)$$

Equations 3.20 and 3.21 are the required forms of the process model. They are linear 1st order ODEs.

We have already defined the concept of **Degree of Freedom** (DOF) in the first

(introductory) chapter. Let us apply the concept to find the DOF of this system.

No. of variables, NV = 2 (q_I and h, or q_I and q_O)

No. of equations, NE = 1 (Equation 3.20 or Equation 3.21)

That is, if q_I is constant, Equation 3.20 may be used to find h by making $dh/dt = 0$. But if q_I is variable, h cannot be made stationary at any desired value because the system

DOF = 2 – 1 = 1. Another equation is needed to accomplish this objective, say a relation between h and the exit flow rate q_O.

$$q_O = f(h) \qquad (3.22)$$

This is in fact the control equation and is implemented by making the exit line resistance R a function of the tank level h, which in turn realizes the Equation 3.22.

EXAMPLE 3.3 (Temperature Dynamics in a Mercury in Glass Thermometer): Mercury in a glass thermometer is placed totally immersed in an atmosphere (Figure 3.5) so that heat transfer between the thermometer and surrounding atmosphere always occurs in one direction. While the changing surrounding temperature, t_I, is the input variable, the temperature recorded in the thermometer, t_O, is the output variable. It is required to construct a dynamic model relating these two variables.

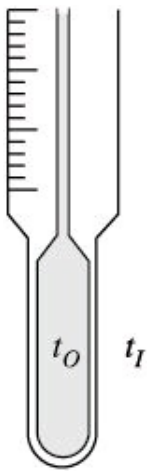


Figure 3.5 A mercury in glass thermometer.

As in the previous example, we list our assumptions for this system as follows,

1. All the thermal storage capacity is only in the mercury
2. All the resistance to heat transfer is in the heat transfer coefficient residing on the film between glass bulb and surrounding atmosphere.

Again we have lumped the resistance and capacitance elements. We should recognize in reality that the glass bulb also has a relatively small amount of thermal capacitance and a small amount of heat transfer resistance resides in the stagnant

mercury film inside the bulb. An important thing should be mentioned that the amount of lumping may be reduced in phases to bring the model closer to reality. Thus, we can derive the same system model with partial lumping. We shall do so in a later part of our discussion in Chapter 6. But for now, let us carry on the task of listing rest of the assumptions.

1. The heat transfer resistance ($1/UA_T$) is of linear character
2. The thermal capacity remains unchanged with respect to temperature
3. The material properties like thermal conductivity, k, heat capacity, C_P, etc. are independent of temperature.

Now we can write the unsteady state heat balance around the thermometer as:

(rate of heat input) – (rate of heat output) = (rate of heat accumulation) (3.23)

As the thermometer is totally immersed in the surrounding, there would be no heat output term and Equation 3.23 becomes:

(3.24)

where m and C_P are respectively mass and heat capacity of mercury. U is the heat transfer coefficient between surrounding atmosphere and thermometer, and A_T is heat transfer area of the glass bulb.

$$\frac{dt_O}{dt} = \frac{UA_T}{mC_P}(t_I - t_O) \qquad (3.25)$$

Equation 3.25 is the required form of the process model. This is also a linear 1st order ODE.

EXAMPLE 3.4 (Temperature Dynamics in a Perfectly Stirred Constant Holdup Liquid Heater): A tank of working volume V_L with a mechanical agitator serves the purpose of mixing or homogenization of a flowing liquid stream that is moving with a flow rate q through the mixer (Figure 3.6). A heat input device in the form of either a resistance type electric element, a heat transfer immersed coil or a jacket carrying a hot fluid is associated with the mixer so that the exit liquid leaves at an elevated temperature. The mixing tank is otherwise thermally insulated.

The inlet temperature of the inlet liquid, $T_I(t)$, and the heat input rate from the heating system, $W(t)$, may both vary. Here we have a system that has two inputs and one output variable, and two possible input-output relations could be obtained. It is required to find the dynamics of exit liquid temperature, $T_O(t)$, with respect to $T_I(t)$

and $W(t)$. The bulk liquid temperature may be assumed as equal to that of the exit liquid, due to perfect mixing.

Other relevant assumptions are: (1) There is no heat transfer between liquid and the vessel wall; (2) The liquid density and heat capacity are independent of temperature: (3) The heat transfer efficiency of the heating system is 100%. A schematic diagram of the system is shown in Figure 3.6.

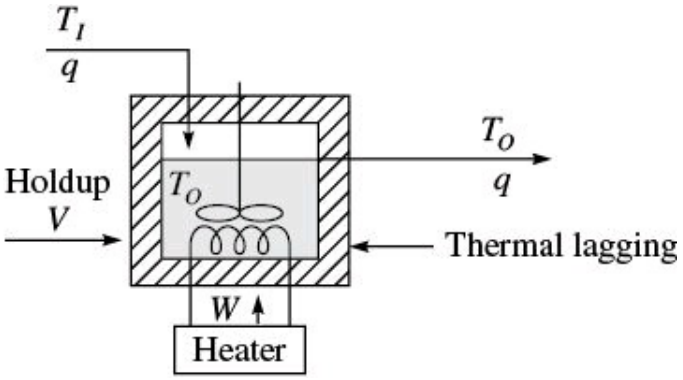


Figure 3.6 Stirred tank constant holdup liquid heater.

In the constant hold situation, there will be no accumulation of mass, and we need not write the dynamic mass balance. The heat balance equation for the system is:

$$\rho q C_P (T_I - T_D) + W - \rho q C_P (T_O - T_D) = \rho C_P V \frac{dT_O}{dt} \qquad (3.26)$$

where,

T_D = Datum temp., ºC; r = Liq. density, g/ml; C_P = Liq. Ht. Capacity, cal/(g.ºC)

In this system, we have two input variables, $T_I(t)$ and $W(t)$, and hence by way of linear analysis we can analyze the response with respect to only one input, assuming the other input to be constant during the transient period. However, the last equation may be further simplified by considering T_D = 0ºC,

$$\rho q C_P T_I + W - \rho q C_P T_O = \rho C_P V \frac{dT_O}{dt} \qquad (3.27)$$

$$\frac{dT_O}{dt} = \frac{q}{V} T_I - \frac{q}{V} T_O + \frac{1}{\rho C_P V} W \qquad (3.28)$$

Consideration of the system DOF reveals that if T_I varies while W remains constant,

$$\text{DOF} = \text{NV}\,[T_I, T_O] - \text{NE}\,[\text{Equation (3.28)}] = 2 - 1 = 1.$$

That is corresponding to any change in T_I if we want to render T_O at a desired value, an extra equation has to be implemented relating T_O and the heater power, W:

$$W = f(T_O) \qquad (3.29)$$

Equation 3.29 may be called the control equation whose appearance will reduce DOF = 0. You should compare this result with that has been obtained in the example

of liquid level dynamics. The two control Equations, 3.22 and 3.29 have similar form and the manipulated variables q_O and W are comparable in these applications.

EXAMPLE 3.5 (Concentration Dynamics in a Perfectly Stirred Constant Holdup Isothermal Reactor): A tank of working volume V_L with a mechanical agitator serves the purpose of mixing or homogenization of a flowing liquid stream that is moving with a flow rate q through the mixer (Figure 3.7). The inlet liquid is a solution of variable solute concentration that enters the tank in the vicinity of moving impellers to obtain maximum and immediate dissolution.

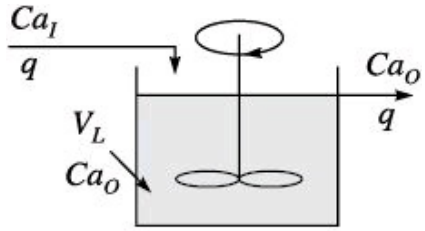


Figure 3.7 The concentration dynamics in a perfectly stirred mixer.

The exit stream issues as overflow somewhere from the top of the vessel near its rim. Thus the tank always remains filled up to the exit point, hence, it's working or effective volume for material processing remains constant. We call such an equipment to be operating with a **constant hold-up** volume of V_L. The actual volume of such equipment is always greater than the working volume.

The exit flow rate does not depend on the tank liquid level, but is equal to the inlet flow rate q, so that if q remains constant, the theoretical average residence time, t, defined as "the length of time each entering liquid particle stays in the mixer", will also be of constant value, $t = V_L/q$.

We have already defined the term 'Perfect Mixing' at the beginning of this chapter, that by this ideal phenomenon we mean, the inlet liquid as soon as reaches the mixer interior abruptly changes its concentration and assumes the bulk liquid concentration.

In such a system, a first order isothermal and irreversible reaction without any heat effect occurs, which may be given by:

$$A \rightarrow \text{Product} \qquad (3.30)$$

The reaction rate may be expressed by:

$$r = -\frac{dCa}{dt} = kCa \qquad (3.31)$$

where, k is the specific reaction rate constant.

The unsteady state reactant mass balance will result in an equation very similar to the mass balance equation of mixing system, but with one extra term to account for reactant mass disappearance due to consumption in reaction. This term will be

$kV_L Ca_O$. Now we may write the unsteady state balance as:

$$qCa_I - qCa_O - kV_L Ca_O = V_L \frac{dCa_C}{dt} \qquad (3.32)$$

$$\frac{dCa_O}{dt} = \frac{q}{V_L} Ca_I - \frac{q}{V_L} Ca_O - kCa_O \qquad (3.33)$$

A single stage lumped parameter system may contain more than one state equation depending on the number of output variables associated in the system. A simple example may be cited of the stirred liquid heater with slightly different description, where the heat addition occurs through the common surface of the stirred vessel body and the jacket through which a hot fluid is flowing.

EXAMPLE 3.6 (Constant Holdup Jacketed Stirred Liquid Heater): This example retains all the essential features of Example 3.4, except that the heat transfer occurs through the tank wall surface outside of which a jacket on the stirred tank body carries the heat transfer fluid.

Both the stirred heater and the jacket are considered to be lumped parameter units, i.e. both are assumed to be operating in perfectly stirred condition. With reference to Figure 3.8, the variable and parameter nomenclature is as follows:

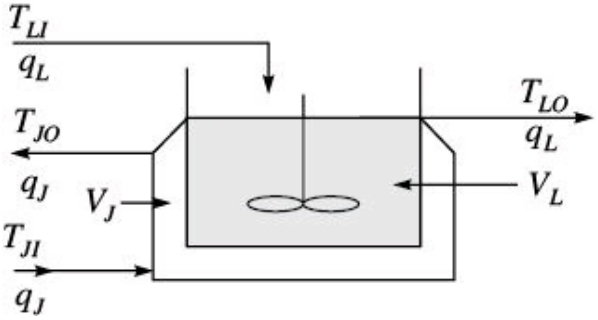


Figure 3.8 Liquid heating through heat transfer surface in a constant holdup mixer.

V_L = Stirred tank volume; q_L = Process liquid flow rate; r_L = Process liquid density; Cp_L = Process liquid heat capacity; V_J = Jacket volume; q_J = Jacket liquid flow rate; r_J = Jacket liquid density; Cp_J = Jacket liquid heat capacity; U = Overall heat transfer coefficient between stirred tank and jacket; A = Heat transfer area.

Again the mass balance need not be written, because the tank operating volume is constant, thus, we can directly write the energy balances.

For the process stream,

$$V_L \rho_L Cp_L \frac{dT_{LO}}{dt} = q_L \rho_L Cp_L (T_{LI} - T_{LO}) + UA(T_{JO} - T_{LO}) \tag{3.34}$$

For the jacket liquid,

$$V_J \rho_J Cp_J \frac{dT_{JO}}{dt} = q_J \rho_J Cp_J (T_{JI} - T_{JO}) - UA(T_{JO} - T_{LO}) \tag{3.35}$$

The corresponding state equations are:

$$\frac{dT_{LO}}{dt} = \frac{q_L}{V_L}(T_{LI} - T_{LO}) + \frac{UA}{V_L \rho_L Cp_L}(T_{JO} - T_{LO}) \tag{3.36}$$

$$\frac{dT_{JO}}{dt} = \frac{q_J}{V_J}(T_{JI} - T_{JO}) - \frac{UA}{V_J \rho_J Cp_J}(T_{JO} - T_{LO}) \tag{3.37}$$

The corresponding matrix form of Equations 3.36 and 3.37 may be written as:

$$\begin{bmatrix} \dot{T}_{LO} \\ \dot{T}_{JO} \end{bmatrix} = \begin{bmatrix} -a_{11}T_{LO} & a_{12}T_{JO} \\ a_{21}T_{LO} & a_{22}T_{JO} \end{bmatrix} \begin{bmatrix} T_{LO} \\ T_{JO} \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \begin{bmatrix} T_{LI} & T_{JI} \end{bmatrix} \tag{3.38}$$

where, $a_{11} = \frac{q_L}{V_L} + \frac{UA}{V_L \rho_L Cp_L}$; $a_{12} = \frac{UA}{V_L \rho_L Cp_L}$; $b_1 = \frac{q_L}{V_L}$

and, $a_{21} = -\frac{UA}{V_J \rho_J Cp_J}$; $a_{22} = \frac{UA}{V_J \rho_J Cp_J} - \frac{q_J}{V_J}$; $b_{21} = \frac{q_J}{V_J}$

This is your first introduction to a system that has more than one input and output variables, the so-called multiple input–multiple output (MIMO) process. This multiplicity of I/O of a system may feature that change in one input will not only change the corresponding output but may effect the remaining outputs also, a phenomenon called *Interaction of Variables*. We shall present another example of this class in the next chapter.

In the following example of lumped parameter single stage system, a process has been discussed which involves slightly more of algebra, the flashing system.

EXAMPLE 3.7 (A Single Stage Flashing Unit): Flashing systems are used for energy recovery from a high temperature saturated liquid stream coming out as exit stream from a process, or for component enrichment and separation from a multi-component high temperature saturated liquid mixture with *N*-number of components. In any of these applications, the high-energy liquid is passed through a resistance to escape into a relatively low-pressure region. Thus, if we assume adiabatic operation, i.e. total energy of the incoming and outgoing stream around the restriction remain constant and the pressure in which the incoming stream escapes is lower than its vapour pressure at the prevailing temperature, the stream splits into two phases: (1)

Saturated liquid, and (2) Saturated vapour. This phenomenon is called 'Flashing' operation.

In the following we shall propose the modelling equations for the flashing operation of a multi-component liquid mixture shown as a schematic in Figure 3.9, and determine the degree of freedom of such a model. Assuming constant density and heat capacity of incoming liquid and outgoing vapour and liquid, the dynamic balance equations may be written as:

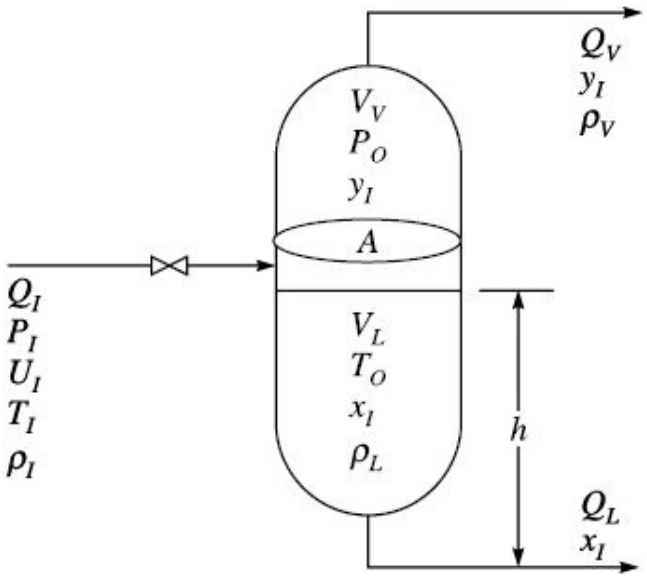


Figure 3.9 The single stage flashing unit.

Total mass balance,

$$A\rho_L \frac{dh}{dt} = Q_I - Q_V - Q_L \tag{3.39}$$

Component mass balance,

$$A\rho_L \frac{d(hx_I)}{dt} = Q_I u_I - Q_V y_I - Q_L x_I \tag{3.40}$$

Heat balance,

$$Cp_L A \frac{d(hT_O)}{dt} = Cp_I Q_I T_I - Cp_V Q_V T_O - Cp_L Q_L T_O \tag{3.41}$$

Beside these, the following relations are also to be used for formulating the model:

1. Vapour-liquid equilibrium:

$$y_I = f(T, P)x_I \tag{3.42}$$

2. Composition constraints:

$$\sum_{i=1}^{N} x_I = 1 \tag{3.43}$$

and

$$\sum_{i=1}^{N} y_I = 1 \tag{3.44}$$

Hence the system is described by:

Total mass balance:	1 equation,
Component mass balance:	$N - 1$ equations,
Total energy balance:	1 equation,
Vapour-liquid equilibrium:	N equations,
Composition constraints:	2 equations
Total number of equations:	$2N + 3$

(3.45)

In this system, the concerned variables are: P_I, T_I, Q_I and u_I for $i = 1, 2, 3, \ldots, (N - 1)$, P_O, T_O, h, Q_V, Q_L, P_O, T_O, and x_I, y_I for $i = 1, 2, 3, \ldots, N$. (3.46)

In which the disturbance variables are: P_I, T_I and u_I for $i = 1, 2, 3 \ldots, (N - 1)$.

Thus, the floating variables are: P_O, T_O, h, Q_V, Q_L, Q_I and x_I, y_I for $i = 1, 2, 3 \ldots, N$.

In total: $2N + 6$. (3.47)

Hence, the system has a DOF = NOV – NOE = (2N + 6) – (2N + 3) = 3. (3.48)

Similarly in a flashing system for a binary mixture, the DOF = (4 + 6) – (4 + 3) = 3.

(3.49)

3.4.2 Non-Linear Process Characteristics—Linearization

In all the examples of Section 3.3, so far we have deduced process models for linear systems. In certain cases, the possibility for non-linearity has been avoided by restricting the problem statement as in examples 3.5, 3.6, and in some applications, as in Example 3.1, we have assumed linear character of a part of the process (the linear flow-head relation of the exit line valve) that is in fact non-linear.

In this section, we shall introduce the non-linear configuration of process models in a preliminary way, and the linearization procedure by Taylor's expansion. Let us consider the CSTR in Example 3.5, with second order kinetics.

EXAMPLE 3.8 (Concentration Dynamics of a 2nd Order Irreversible Reaction in a Perfectly Stirred Constant Holdup Isothermal Reactor): All the statements of this problem are similar to those we made in Example 3.5 except that the reaction kinetics for the 2nd order system is given as:

$$r = -\frac{dCa}{dt} = kCa^2 \qquad (3.50)$$

where, k is the specific reaction rate constant.

The unsteady state reactant mass balance will result in an equation very similar to 3.32. Only the reactant mass disappearance term due to reaction would be $kV_L Ca_O{}^2$ for the 2nd order kinetics. Now we may write the unsteady state balance as:

$$qCa_I - qCa_O - kV_L Ca_O^2 = V_L \frac{dCa_O}{dt} \qquad (3.51)$$

$$\frac{dCa_O}{dt} = \frac{q}{V_L} Ca_I - \frac{q}{V_L} Ca_O - kCa_O^2 \qquad (3.52)$$

Equation 3.52 is non-linear 1st order ODE due to existence of second degree concentration term in its argument and all the linear methods of analysis fail to deal with such systems.

In the next section a method to obtain the linear form of such non-linear equations will be discussed. The procedure would be easy to implement if the variables of Equation 3.52 are expressed in their deviation form. The idea of deviation variable has been already explained in Section 3.2.2. Thus, Equation 3.52 becomes:

$$\frac{d(Ca_O - Ca_{OS})}{dt} = \frac{q}{V_L}(Ca_I - Ca_{IS}) - \frac{q}{V_L}(Ca_O - Ca_{OS}) - (kCa_O^2 - kCa_{OS}^2) \qquad (3.53)$$

3.4.3 Method of Linearization by Taylor's Expansion

"Whatever be the curvature of the locus of a function, it may be assumed to be linear for an arbitrarily small range of increment of the independent variable"—just like the earth's surface seems to be flat as we stand at the centre of a football ground. This is the philosophy of linearization procedure, and also the linear interpolation techniques. In this context, Taylor's series for any given function may be written around its steady state as:

$$y = f(x) = y_S + \left.\frac{dy}{dx}\right|_{x_S} (x - x_S) + \left.\frac{1}{2!}\frac{d^2y}{dx^2}\right|_{x_S} (x - x_S)^2 + \left.\frac{1}{3!}\frac{d^3y}{dx^3}\right|_{x_S} (x - x_S)^3 + \ldots \qquad (3.54)$$

The suffix s indicates the steady state values of the variables, and the suffix x_S on the derivative terms signify that they are also evaluated at steady state condition. If only up to the first derivative term of Equation 3.54 is considered to represent the function, any departure from the steady state value may be expressed by:

$$y = f(x) = y_S + \left.\frac{dy}{dx}\right|_{x_S} (x - x_S) \qquad (3.55)$$

Equation 3.55 is called the linear approximation of the non-linear function $y = f(x)$ around any steady state operating condition, and the implementation of this procedure has been illustrated in Figure 3.10. You must note the vertical distance between intersections of the tangent and the actual function with the given value of independent variable, x. This is the *error* in linear representation of the function $f(x)$.

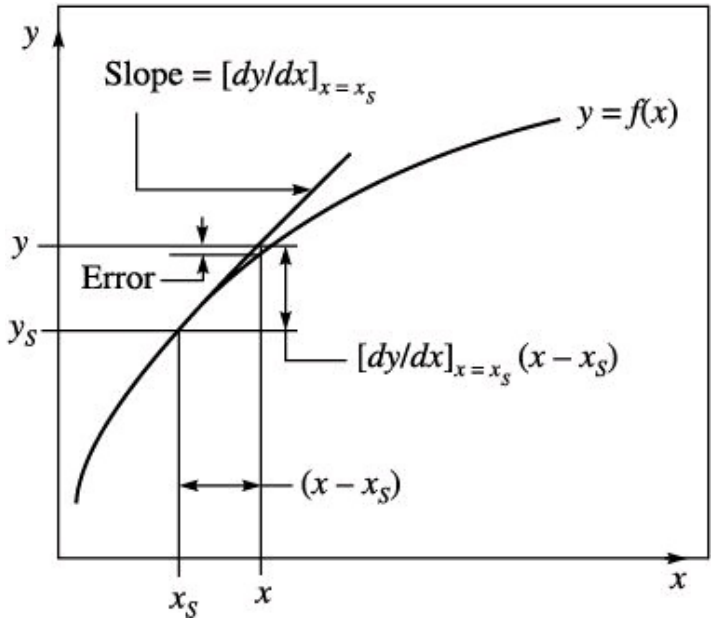


Figure 3.10 Linearization by Taylor's expansion.

In the present Example 3.8, it is sufficient to treat the nonlinear, that is the 3rd term in the right hand side of Equation 3.52 by the above procedure, thus,

$$y = f(Ca_O) = kCa_O^2 = kCa_{OS}^2 + k\frac{dCa_O^2}{dCa_O}\bigg|_{Ca_O = Ca_{OS}} (Ca_O - Ca_{OS})$$

or,

$$kCa_O^2 - kCa_{OS}^2 = 2kCa_{OS}(Ca_O - Ca_{OS}) \qquad (3.56)$$

By inserting Equation 3.56 in 3.53 and introducing the deviation variables $[\bar{C}a_O = Ca_O - Ca_{OS}, \text{ and } \bar{C}a_I = Ca_I - Ca_{IS}]$, we obtain,

$$\frac{d\bar{C}a_O}{dt} = \frac{q}{V_L}\bar{C}a_I - \frac{q}{V_L}\bar{C}a_O - 2kCa_{OS}\bar{C}a_O \quad (3.57)$$

This is the linear approximation of Equation 3.52.

Non-linearity may also appear due to existence of terms which are the product of two or more number of variables in the argument of model equation. To illustrate this class of non-linearity, Example 3.4 has been reconsidered with a slight alteration of the problem statement. Example 3.4 dealt with the temperature dynamics in a perfectly stirred constant holdup liquid heater with constant inflow of the liquid. Only the variability of inlet liquid temperature was considered. In the present case, the liquid flow rate, liquid inlet temperature and the heater power, all three have been considered as variables.

EXAMPLE 3.9 (Temperature Dynamics in a Perfectly Stirred Constant Holdup Liquid Heater): In this example, all other descriptions remain as they were for Example 3.4, except that not only T_I, but q and W are also time dependent inputs to the system. In fact, in a process application, say as a Pre-Heater of a liquid fuel (FO) stream to the burner of a furnace, the time dependency of the mentioned quantities are realistic and should be considered. To recognize this multiple variability, these quantities along with the system response, T_O, appear with a t in the parenthesis in the heat balance equation of the system.

$$\rho q(t)CpT_I(t) + W(t) - \rho q(t)CpT_O(t) = \rho CpV\frac{dT_O(t)}{dt}$$

or,

$$\frac{dT_O(t)}{dt} = \frac{1}{V}q(t)T_I(t) - \frac{1}{V}q(t)T_O(t) + \frac{1}{\rho.CpV}W(t) \qquad (3.58)$$

The first and second term in the right hand side of Equation 3.58 contain variable products, hence these terms have to be linearized. As in the previous example, let us write Equation 3.58 in a difference form between absolute and steady state values of the variables to facilitate the incorporation of deviation variables.

$$\frac{d(T_O - T_{OS})}{dt} = \frac{1}{V}[qT_I - q_S T_{IS}] - \frac{1}{V}[qT_O - q_S T_{OS}] + \frac{1}{\rho.CpV}[W - W_S] \qquad (3.59)$$

Note that the parenthesis t has been removed because they have served their purpose. But here the non-linearity contains two variables, hence they have to be expanded according to Taylor's expansion of a function of two independent variables, i.e. $y = f(x_1, x_2)$.

The expansion up to the first derivative term may be written as:

$$y = f(x_1, x_2) = y_S + \left.\frac{\partial y}{\partial x_1}\right|_{\substack{x_1 = x_{1S} \\ x_2 = x_{2S}}} (x_1 - x_{1S}) + \left.\frac{\partial y}{\partial x_2}\right|_{\substack{x_2 = x_{2S} \\ x_1 = x_{1S}}} (x_2 - x_{2S}) \qquad (3.60)$$

We write first term in the R.H.S. of Equation 3.58 according to Equation 3.60 as,

$$qT_I = q_S T_{IS} + \left.\frac{\partial (qT_I)}{\partial q}\right|_{\substack{q = q_S \\ T_I = T_{IS}}} (q - q_S) + \left.\frac{\partial (qT_I)}{\partial T_I}\right|_{\substack{T_I = T_{IS} \\ q = q_S}} (T_I - T_{IS})$$

or, $$qT_I - q_S T_{IS} = T_{IS}(q - q_S) + q_S(T_I - T_{IS}) \qquad (3.61)$$

Similarly, the second term in the R.H.S. of Equation 3.58 may be expanded as:

or, $$qT_O - q_S T_{OS} = T_{OS}(q - q_S) + q_S(T_O - T_{OS}) \qquad (3.62)$$

Equations 3.61 and 3.62 may be put into Equation 3.59 to obtain the required result as:

$$\frac{d(T_O - T_{OS})}{dt} = \frac{T_{IS}}{V}(q - q_S) + \frac{q_S}{V}(T_I - T_{IS}) - \frac{T_{OS}}{V}(q - q_S) - \frac{q_S}{V}(T_O - T_{OS}) + \frac{1}{\rho CpV}(W - W_S)$$

Introducing the following deviation variables:

$$\bar{T}_O = T_O - T_{OS}; \quad \bar{T}_I = T_I - T_{IS}; \quad Q = q - q_S; \quad \text{and} \quad \bar{W} = W - W_S$$

in the above equation and slightly rearranging, we obtain the linear model of the system, which is as follows:

$$\frac{d\bar{T}_O}{dt} = \left[\frac{T_{IS} - T_{OS}}{V}\right] Q + \frac{q_S}{V}\bar{T}_I - \frac{q_S}{V}\bar{T}_O + \frac{1}{\rho CpV}\bar{W} \qquad (3.63)$$

Equations 3.57 and 3.63 are the linearized dynamic model that we obtained for Examples 3.8 and 3.9. Presently we have introduced two kinds of non-linearity that appear in process model formulation.

In Example 3.8, non-linearity is in some parts of the process, to be more specific it is due to the characteristics of a dynamic element of the process. The non-linearity of exit flow line resistance in dynamic liquid level system is another example of this kind.

The second category of non-linearity appear due to existence of terms in the

argument of the model that contains the product of more than one variable. This happens according to the condition of variability in process description, as in Example 3.9. In a process description, both categories of non-linearity may appear simultaneously. Fortunately, they may be treated in the same way as of Taylor's expansion.

A number of non-linear process examples from a variety of physical systems have been compiled in Chapter 4, on first order transfer function analysis. However, before closing this preliminary exposure on linearization, you should be aware of a few things while using this procedure.

1. The linearized model is only valid within an arbitrarily small range around the steady state operating condition of the process. The error in linear approximation has been mentioned pictorially when we presented Figure 3.10. The error definitely depends on the amount of departure of the disturbed system from the steady state point where function derivative is calculated and definitely on the amount of non-linearity of process characteristic. We will talk a little more about choosing the derivative evaluation point in Chapter 5.
2. The quantitative result of the linear model response analysis depends on the given steady state condition of the process, because this determines the parameter values of the linear model. Hence, at different steady states, the dynamic nature of the process differs.
3. For process models containing non-linearity of the second kind—involving variable product terms—the linear model allows you to analyze the process behaviour corresponding to only one of the inputs, considering others to be constant. By linear method, response analysis for more than one input is not possible. However for such a linear model "*response for simultaneous input changes may be obtained by super-imposition of responses obtained for two independent input changes*"—this point would be more understandable in the next chapter during the discussion of transfer function models.

(However, for more accurate results, variation of more than one input is very much allowed in the analysis by resorting to Digital Simulation of Process Models.)

3.4.4 Lumped Parameter Multiple Stage Systems

Often discrete number of differential (lumped) stages are used for facilitating certain kinds of material conversion. A few examples will establish this notion.

EXAMPLE 3.10 (First Order Irreversible Reaction in a Three CSTR Battery): In this system, all the descriptions and statements remain same as they are in the

example for the single CSTR with the only exception that there are three CSTRs in a battery operating in series.

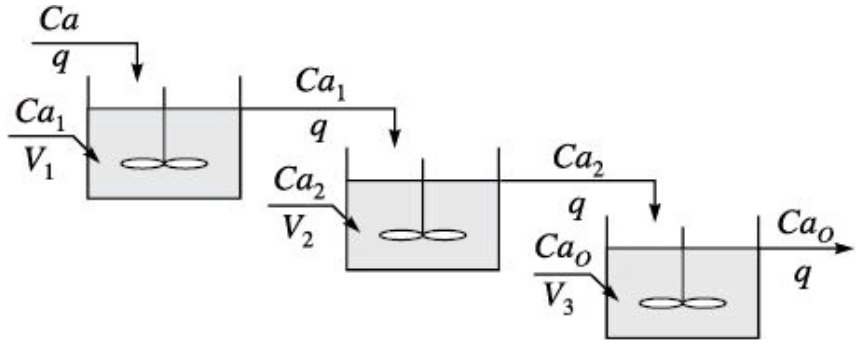


Figure 3.11 Three CSTR battery.

The reactor system is employed to carry out an irreversible first order reaction in solution phase given by:

A □ Product (3.64)

The reaction occurs under isothermal conditions without any heat effect, and its rate may be expressed by:

$$r = -\frac{dCa}{dt} = kCa \qquad (3.65)$$

The mass balance for the reacting component in each of the reactors yields:

$$\frac{dCa_1}{dt} = \frac{q}{V_1}Ca_I - \frac{q}{V_1}Ca_1 - kCa_1 \qquad (3.66)$$

$$\frac{dCa_2}{dt} = \frac{q}{V_2}Ca_1 - \frac{q}{V_2}Ca_2 - kCa_2 \qquad (3.67)$$

$$\frac{dCa_O}{dt} = \frac{q}{V_3}Ca_2 - \frac{q}{V_3}Ca_O - kCa_O \qquad (3.68)$$

The above three equations are written in state equation form. Introducing the definition of residence times of the three reactors, $t_1 = V_1/q$; $t_2 = V_2/q$; $t_3 = V_3/q$, we may write the vector form of the state variables as:

$$\begin{bmatrix} \dot{C}a_1 \\ \dot{C}a_2 \\ \dot{C}a_O \end{bmatrix} = \begin{bmatrix} -(1/\tau_1 + k) & 0 & 0 \\ 1/\tau_2 & -(1/\tau_2 + k) & 0 \\ 0 & 1/\tau_3 & -(1/\tau_3 + k) \end{bmatrix} \begin{bmatrix} Ca_1 \\ Ca_2 \\ Ca_O \end{bmatrix} + \begin{bmatrix} 1/\tau_1 \\ 0 \\ 0 \end{bmatrix} [Ca_I] \qquad (3.69)$$

and the output equation becomes:

$$y = [0 \quad 0 \quad 1] \begin{bmatrix} Ca_1 \\ Ca_2 \\ Ca_O \end{bmatrix} \qquad (3.70)$$

EXAMPLE 3.11 (Plate Type Gas Absorber): In process applications, gas absorbers are widely used to separate a component in the gas stream by contacting with a liquid in which the component has much higher solubility. Removal of NH_3 or HCl vapours from a gas stream, where the other component is air or some inert gas, by using water as the absorbent is an example of such an application. According to the natural flow of gases and liquids under gravity in a two phase system, the streams are contacted in counter current fashion in the absorber. Gas goes upward and liquid falls downwards. Many designs of such absorption systems are available. Figure 3.12 illustrates the Plate type column, which is one of the most popular configurations. They feature multiple plates installed in a tall cylindrical column providing stagewise gas-liquid contact. Each of the plates is equipped with some sort of vapour riser (say Bubble cap type) and liquid down comer (say weir type).

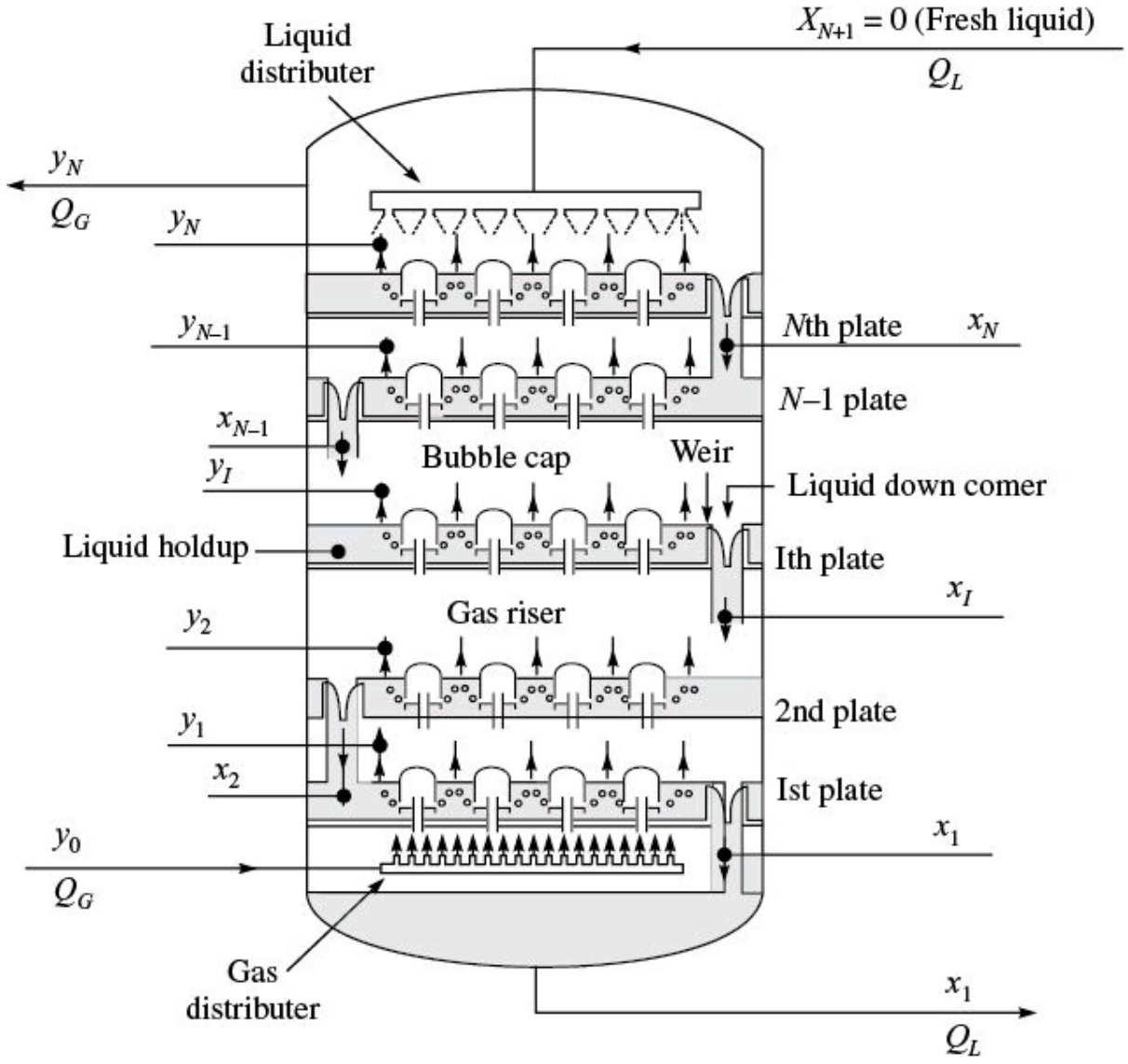


Figure 3.12 Plate type gas absorber.

The model for an Absorber, Extraction systems, Rectifying and Stripping section of a distillation column are fundamentally similar. Two immiscible phases containing some transferable component are brought into intimate contact, usually by dispersing

one of the phases in the form of small droplets or bubbles. The concentration difference of the transferable component is reduced as the liquid and vapour move toward equilibrium on a particular plate. The reduction of concentration difference depends on the residence time and mixing efficiency available in the hold up volume over a plate.

At present, we assume ideal mixing on each plate so that bulk as well as exit concentration of the phases are equal, and the plates are 100% efficient, i.e. the phases leaving any of the plates are in equilibrium.

The equilibrium relation for low concentration range may be expressed as:

$$y_N = mx_N \qquad (3.71)$$

where, m is a constant, and y and x are mole fractions of the transferable component in gas and liquid phase respectively. The suffix N refers to the plate number. To further simplify the modelling effort, it has been assumed so that (1) molal flow changes of gas Q_G and liquid Q_L streams are negligible through the column; (2) the molar liquid hold up V on each plate are constant.

For the Ith plate of the column the dynamic component balance may be written as:

$$V\frac{dx_I}{dt} = Q_G y_{I-1} - Q_G y_I + Q_L x_{I+1} - Q_L x_I \qquad (3.72)$$

In the last equation, the gas composition may be expressed in terms of liquid composition by using the equilibrium relation given by Equation 3.71 as:

$$V\frac{dx_I}{dt} = mQ_L x_{I+1} - (Q_L + mQ_G)x_I + mQ_G x_{I-1} \qquad (3.73)$$

Considering a three plate column, we may write the dynamic balance equations. Assuming that pure liquid absorbent is coming at the top plate, $x_4 = 0$, the gas composition from the top plate, y_3, is the output variable, and the gas composition, y_0, coming from below the bottom plate is the input variable to the system. We write the dynamic balance in state equation form.

$$\frac{dx_1}{dt} = Ax_2 - Bx_1 + Cx_0 \qquad (3.74)$$

$$\frac{dx_2}{dt} = Ax_3 - Bx_2 + Cx_1 \qquad (3.75)$$

$$\frac{dx_3}{dt} = -Bx_3 + Cx_2 \qquad (3.76)$$

where, $A = \frac{mQ_L}{V}$; $B = \frac{Q_L + mQ_G}{V}$; $C = \frac{mQ_G}{V}$

In the input-output matrix notation, the above three state equations become:

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix}=\begin{bmatrix}-B & A & 0\\ C & -B & A\\ 0 & C & -B\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}+\begin{bmatrix}C\\ 0\\ 0\end{bmatrix}[x_0] \qquad (3.77)$$

and the output equation is: $y=[0 \quad 0 \quad 1]\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$ (3.78)

3.4.5 Distributed Parameter Systems[1],[2],[3]

A distributed parameter system may be defined as an equipment in which the transport of principal entity takes place not at a particular location but throughout a portion of the volume of system boundary. Alternatively, these are systems in which the dynamic elements are not concentrated in one location but scattered or distributed through the equipment. By parameter lumping assumption, the model would be so far removed from reality that its accuracy may become questionable. In such cases, the process variables have non-zero partial derivatives with respect to spatial coordinates inside the system boundary. If stirring inside a mixer could not be assumed as ideal, which happens in paste or high Viscosity liquid mixing, we have to resort either to Finite Element Mixing Models or, Non-ideal Mixing Models to represent the distributed parameter hydrodynamics in such mixers.

Shell and tube heat exchangers, packed absorption columns, tubular reactors are all examples of this class.

Double pipe heat exchanger

Heat exchangers are widely used process equipment for energy transfer in process plants. Analysis of double pipe heat exchanger is important because this exposes a numerical method that may be also applied to packed absorption columns, tubular reactors and other dynamically similar systems. Another fact is that the modelling approach may be extended to more complex exchangers having multiple passes.

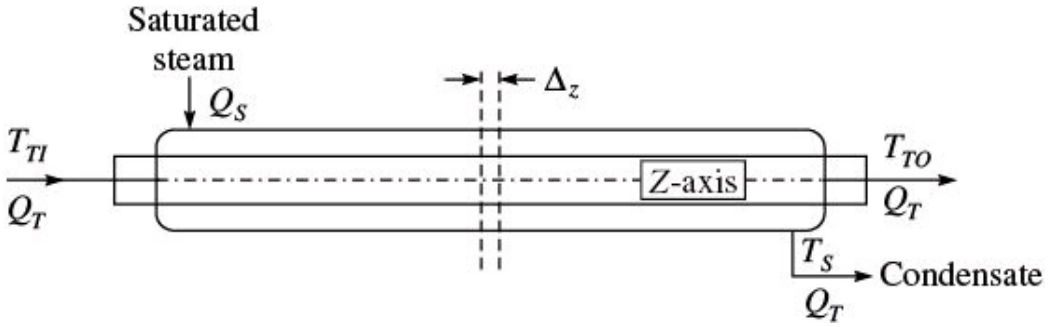


Figure 3.13 Double pipe heat exchanger with steam condensing in the shell side.

Let steam be used as the heating fluid in the shell side of a concentric tube double

pipe heat exchanger. The fluid that moves through the tube side has to be heated. The steam condenses on the outside of the tube. The condensate temperature depends only on the steam pressure but is independent of the position along z-axis.

Based on the following assumptions: (1) The only resistance to heat transfer exists in the inside film of the tube; (2) Fluid density, viscosity and heat capacity are independent of temperature variation inside the exchanger; (3) Thermal capacitance resides only in the flowing fluids; (4) The thermal capacitance of the metal wall is neglected; and (5) The fluid velocity profile is flat, so that there is no radial temperature gradient. The energy balance over a differential length Δz obtains:

$$\underset{\text{accumulation}}{\rho A Cp \Delta z \left[T\big|_{z+\Delta z} - T\big|_{z} \right]} = \underset{\text{flow in}}{ACp\rho v \Delta t \left[T\big|_{z} - T\big|_{z+\Delta z} \right]} + \underset{\text{through wall}}{hA_S \Delta z \Delta t (T_S - T)} \qquad (3.79)$$

where,

Cp is the fluid heat capacity, cal.gm^{-1}.°C^{-1};

r is the fluid density, gm.cm^{-3};

A is the tube cross sectional area, cm^2;

v is the Fluid velocity, cm.sec^{-1};

A_S is the tube heat transfer area per unit length, cm^2.cm;

h is the inside film coefficient of heat transfer, cal.sec^{-1}.cm^{-2}.°C^{-1};

T is the fluid temperature, °C;

T_S is the steam temperature, °C.

Dividing Equation 3.79 by Δz and taking the limit as $\Delta z \to 0$, $\Delta t \to 0$,

$$A\rho Cp \frac{\partial T}{\partial t} + A\rho Cpv \frac{\partial T}{\partial z} = A_S h (T_S - T) \qquad (3.80)$$

This is a first order linear, first order partial differential equation, which may have variable coefficients. Analytical solution of Equation 3.80 is possible for constant coefficient situation. Also, for the same system considering the thermal capacitance of metal wall analytical solution of $[\partial T_M/\partial z]$ and $[\partial T_M/\partial t]$ along with $[\partial T/\partial z]$ and $[\partial T/\partial t]$, have been proposed.

An alternative representation is popular in digital simulation of similar heat transfer systems, which subdivides a distributed part of the equipment into a number of lumped parameter units. Probably the dead time element or a plug flow reactor represented by a serially connected, ideally stirred, lumped parameter pots, are the most easy to understand examples of this approach.

EXAMPLE 3.12 (Sensible Heat Transfer in a Co-Current Exchanger): We first develop the multiple lumped parameter stage model by considering a concentric two tube exchanger where the shell and tube are divided into N-number of equal volume segments. The fluid flow through shell and tube side happen in a co-current fashion and only sensible heat transfer occurs in the exchanger. The nomenclature of process variables for shell and tube sides have been illustrated in Figure 3.14. The following nomenclature for system parameters holds through the analysis:

Q_S—Flow rate of shell fluid, $cm^3.sec^{-1}$; Q_T — Flow rate of tube fluid, $cm^3.sec^{-1}$; Cp_S—Ht. Cap. of shell fluid, $cal.gm^{-1}.^oC^{-1}$; Cp_T—Ht. Cap. of shell fluid, $cal.gm^{-1}.^oC^{-1}$; r_S—Shell fluid density, $gm.cm^{-3}$; r_T—Tube fluid density, $gm.cm^{-3}$; N—Number of segments; L—Length of exchanger, cm; V_S—Shell side volume, cm^3; V_T—Tube volume, cm^3; A_S—Cross sectional area of shell, cm^2; A_T—Cross sectional area of tube, cm^2; A—Heat transfer area, cm^2; U—Heat transfer coefficient, $cal.sec^{-1}.cm^{-2}.^oC^{-1}$; $V_{SN} = V_S/N$; $V_{TN} = V_T/N$; $A_N = A/N$; Suffix 'S' refers to shell side; Suffix 'T' refers to tube side.

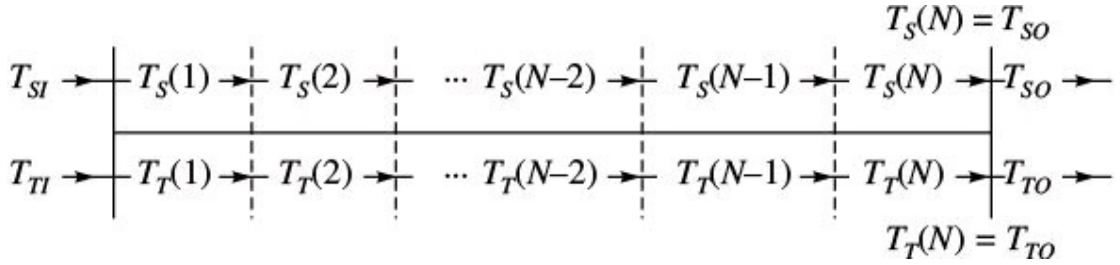


Figure 3.14 Lumped parameter segmental model of sensible heat transfer in a double co-current pipe heat exchanger.

The heat balance at the entry may be written as:

For shell:
$$Cp_S\rho_S V_{SN}\frac{dT_S(1)}{dt} = Q_S\rho_S Cp_S T_{SI} - Q_S\rho_S Cp_S T_S(1) - UA_N[T_S(1) - T_T(1)]$$

For tube:
$$Cp_T\rho_T V_{TN}\frac{dT_T(1)}{dt} = Q_T\rho_T Cp_T T_{TI} - Q_T\rho_T Cp_T T_T(1) + UA_N[T_S(1) - T_T(1)]$$

The heat balance for a general Ith segment:

For shell:
$$Cp_S\rho_S V_{SN}\frac{dT_S(I)}{dt} = Q_S\rho_S Cp_S T_S(I-1) - Q_S\rho_S Cp_S T_S(I) - UA_N[T_S(I) - T_T(I)]$$

For tube:
$$Cp_T\rho_T V_{TN}\frac{dT_T(I)}{dt} = Q_T\rho_T Cp_T T_T(I-1) - Q_T\rho_T Cp_T T_T(I) + UA_N[T_S(I) - T_T(I)]$$

The heat balance at the exit segment:

For shell: $$Cp_S\rho_S V_{SN}\frac{dT_S(N)}{dt} = Q_S\rho_S Cp_S T_S(N-1) - Q_S\rho_S Cp_S T_S(N) - UA_N[T_S(N) - T_T(N)]$$

For tube: $$Cp_T\rho_T V_{TN}\frac{dT_T(N)}{dt} = Q_T\rho_T Cp_T T_T(N-1) - Q_T\rho_T Cp_T T_T(N) + UA_N[T_S(N) - T_T(N)]$$

For simplification of the above equations, the following group of parameters have been introduced:

$$BS = \frac{Q_S}{V_{SN}};\quad BT = \frac{Q_T}{V_{TN}};\quad CS = \frac{UA_N}{Cp_S\rho_S V_{SN}};\quad CT = \frac{UA_N}{Cp_T\rho_T V_{TN}}$$

The resulting state equations are presented as follows:

At the entry segment:

For shell, $$\frac{d}{dt}T_S(1) = BS[T_{SI} - T_S(1)] - CS[[T_S(1) - T_T(1)] \tag{3.81}$$

For tube, $$\frac{d}{dt}T_T(1) = BT[T_{TI} - T_T(1)] + CT[[T_S(1) - T_T(1)] \tag{3.82}$$

At a general Ith segment:

For shell, $$\frac{d}{dt}T_S(I) = BS[T_S(I-1) - T_S(I)] - CS[[T_S(I) - T_T(I)] \tag{3.83}$$

For tube, $$\frac{d}{dt}T_T(I) = BT[T_T(I-1) - T_T(I)] + CT[[T_S(I) - T_T(I)] \tag{3.84}$$

At the exit segment:

For shell, $$\frac{d}{dt}T_S(N) = BS[T_S(N-1) - T_S(N)] - CS[[T_S(N) - T_T(N)] \tag{3.85}$$

For tube, $$\frac{d}{dt}T_T(N) = BT[T_T(N-1) - T_T(N)] + CT[[T_S(N) - T_T(N)] \tag{3.86}$$

The above set of equations may be solved in time domain by numerical procedure.

EXAMPLE 3.13 (Sensible Heat Transfer in a Counter Current Exchanger): The nomenclature of process variables for shell and tube sides have been illustrated in the Figure 3.15.

The nomenclature for system parameters that has been proposed in the co-current situation will hold through this analysis. After writing the dynamic heat balance equations, the identical group of parameters have been applied to simplify them. These state equations are presented as follows:

At the tube and shell entry segment:

For shell, $$\frac{d}{dt}T_S(N) = BS[T_{SI} - T_S(N)] - CS[T_S(N) - T_T(N)] \tag{3.87}$$

For tube, $$\frac{d}{dt}T_T(1) = BT[T_{TI} - T_T(1)] + CT[T_S(1) - T_T(1)] \tag{3.88}$$

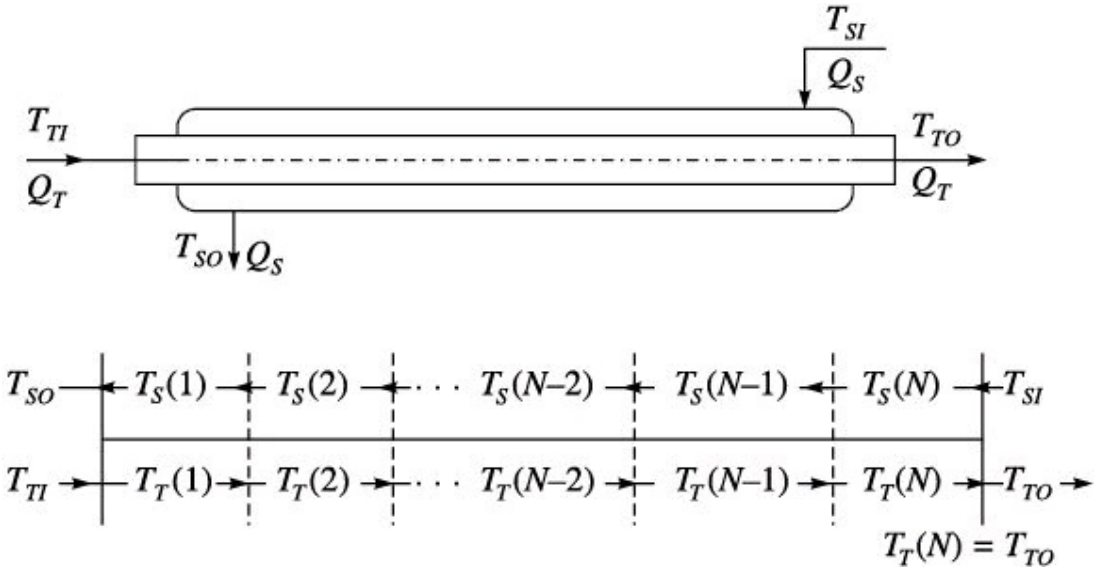


Figure 3.15 Lumped parameter segmental model of sensible heat transfer in a double counter-current pipe heat exchanger.

At a general Ith segment: [in shell side|2 □ I □ (N–2); in tube side|2 □ I □ (N–2)]

For shell, $$\frac{d}{dt}T_S(I) = BS[T_S(I+1) - T_S(I)] - CS[T_S(I) - T_T(I)] \tag{3.89}$$

For tube, $$\frac{d}{dt}T_T(I) = BT[T_T(I-1) - T_T(I)] + CT[T_S(I) - T_T(I)] \tag{3.90}$$

At the tube and shell exit segment:

For shell, $$\frac{d}{dt}T_S(1) = BS[T_S(2) - T_S(1)] - CS[T_S(1) - T_T(1)] \tag{3.91}$$

For tube, $$\frac{d}{dt}T_T(N) = BT[T_T(N-1) - T_T(N)] + CT[T_S(N) - T_T(N)] \tag{3.92}$$

EXAMPLE 3.14 (Steam Condensing in The Shell Side Tube Fluid is Heated): In the shell side, steam is condensing at constant pressure, the saturation pressure of the steam, and we can assume that throughout shell volume the temperature remains constant. There is no spatial temperature distribution in the shell side, whereas, in the tube side there is, because sensible heat transfer occurs in that part of the system. Thus, the system features a lumped parameter part coupled with a distributed parameter part. We shall write multiple segmental models for the tube and a lumped

model for the shell. The modelling principle has been illustrated in Figure 3.16. We have to go for numerical solution method for such problems. Thus, for an integration time step, Δt, the heat transferred from constant temperature shell to each of the tube segments will be summed up and the result subtracted from the total shell side enthalpy while the tube side model will be treated according to the method described in Example 3.12.

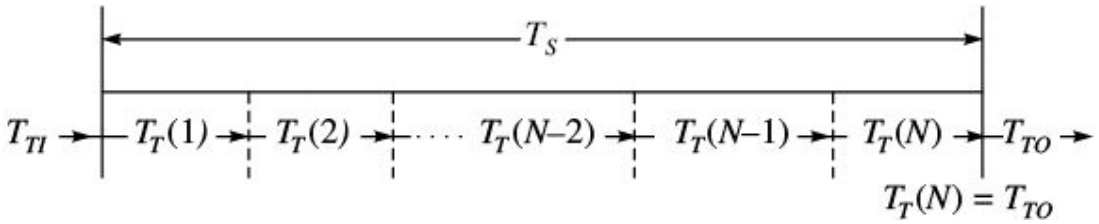


Figure 3.16 Segmental model of a double pipe heat exchanger with steam condensing in the shell side.

For tube,
$$\frac{d}{dt}T_T(1) = BT[T_{TI} - T_T(1)] + CT[[T_S - T_T(1)] \tag{3.93}$$

At a general Ith segment: [in tube side$|2 \square I \square (N-2)$]

For tube,
$$\frac{d}{dt}T_T(I) = BT[T_T(I-1) - T_T(I)] + CT[[T_S - T_T(I)] \tag{3.94}$$

At the exit segment:

For tube,
$$\frac{d}{dt}T_T(N) = BT[T_T(N-1) - T_T(N)] + CT[[T_S - T_T(N)] \tag{3.95}$$

The total heat transfer may be obtained as the sum of the heat transfer terms from all tube segments:

$$H_{TR} = \sum_{I=1}^{N} CT[T_S - T_T(I)] \tag{3.96}$$

The shell side temperature remain constant at T_S, the saturation temperature of steam. The shell side thermal dynamics may be expressed in terms of inlet mass flow rate of steam. Let the mass of steam consumed in time Δt be m_S, and its latent heat l. Assuming only latent heat transfer occurs in an ideal heat transfer condition, and the condensate leaves at T_S, we may write:

$$m_S = \frac{H_{TR}}{\lambda}$$

and the rate of steam consumption,

$$\frac{d}{dt}m_S = \underbrace{Lt}_{\Delta t \to 0} \frac{H_{TR}/\lambda}{\Delta t} \tag{3.97}$$

Not only for the present system, this analysis would be applicable to jacketed

reactors with mechanically stirred inside and the jacket is the helically wound half pipe section or has transverse inside baffles. Heat transfer through jacket is essential to control an exothermic reaction in the reactor. The heat transfer may be effected by a tank coil immersed in the reactor interior. In all these situations, the fluid inside the reactor may be assumed to be in thermally homogeneous conditions so that we may consider a lumped parameter model, but the jacket should still be a distributed parameter unit. A similar analysis may be exercised for such system as we have done in the present example.

Packed absorption column

As in the plate column, a similar kind of mass transfer operation may be carried out in packed columns. The packing materials are essentially inert to the gas and liquid that are contacted within the column in counter current fashion. The packing also provides intimate contact between gas and liquid phases. A packed type absorption column has been illustrated in Figure 3.17 to demonstrate the stream arrangement with blown up view to understand the gas-liquid mass transfer on the packing surface. The following assumptions are applicable for the analysis.

1. There is a linear equilibrium relation between the concentration of the transferable component in gas and liquid phases as:

$$y_E = mx \qquad (3.98)$$

2. Gas and liquid phases move through the column in plug flow (flat velocity profile) fashion. Hence, concentration gradient in r-axis and q-axis does not exist.
3. The concentration change of the transferable component is negligible, so that Mass or molar flow of any of the phases remain constant through the column.
4. The mass transfer occurring inside the column is controlled by gas film only.
5. Gas and liquid hold up in the packed bed are constant.
6. The column operates in isothermal condition.

The following nomenclature for the variables and parameters hold through the analysis:

Q_G — Molar flow rate of gas, g-mol/sec;

Q_L — Molar flow rate of liquid, g-mol/sec;

x — Mole fraction of the transferable component in liquid;

y — Mole fraction of the transferable component in gas;

V_G — Hold up volume of gas in packing per unit packing volume;

V_L — Hold up volume of liquid in packing per unit packing volume;

k_G — Mass transfer coefficient, g-mol.sec^{-1}cm^{-2}$(y - y_E)^{-1}$;

a — Mass transfer area per unit packing volume, cm^2.cm^{-3};

A — Cross-sectional area of the column, cm^2 .

A total dynamic mass balance for the element Δz gives:

$$V_G \frac{\partial x}{\partial t} + V_L \frac{\partial y}{\partial t} = -Q_G \frac{\partial y}{\partial z} + Q_L \frac{\partial x}{\partial z} \qquad (3.99)$$

Similarly, a mass balance for the gas phase alone is:

$$V_G \frac{\partial y}{\partial t} = -Q_G \frac{\partial y}{\partial z} - k_G a(y - y_E) \qquad (3.100)$$

Putting the equilibrium relation of Equation 3.98 in the last equation to eliminate y_E,

$$V_G \frac{\partial y}{\partial t} = -Q_G \frac{\partial y}{\partial z} - k_G a(y - m_X) \qquad (3.101)$$

Subtracting Equation 3.101 from 3.99, the mass balance for just the liquid phase may be obtained as:

$$V_G \left[\frac{\partial x}{\partial t} - \frac{\partial y}{\partial t}\right] + V_L \frac{\partial y}{\partial t} = Q_L \frac{\partial x}{\partial z} + k_G a(y - y_E) \qquad (3.102)$$

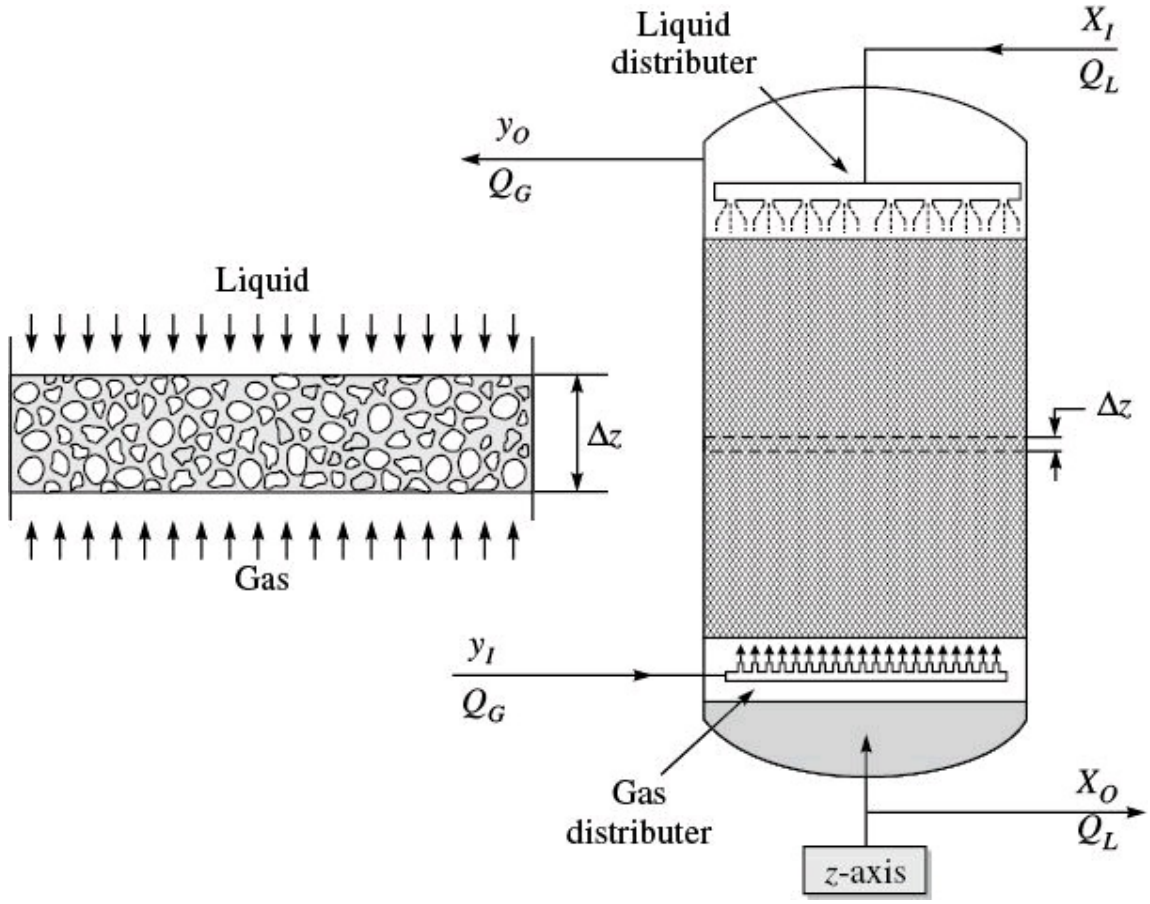


Figure 3.17 The packed type absorption column.

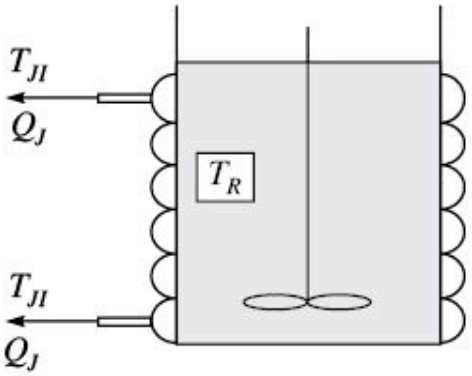
An analytical solution of Equation 3.100 and 3.102, or Equation 3.101 and 3.102 is possible. Also, the solution based on first or second degree difference equations, e.g.

Crank–Nicholson method, is possible. But numerical solution of packed absorption systems assuming lumped parameter segments is probably more suitable for process control system simulation exercises. The same procedure that has been done for counter current heat exchanger analysis for sensible heat transfer may be followed.

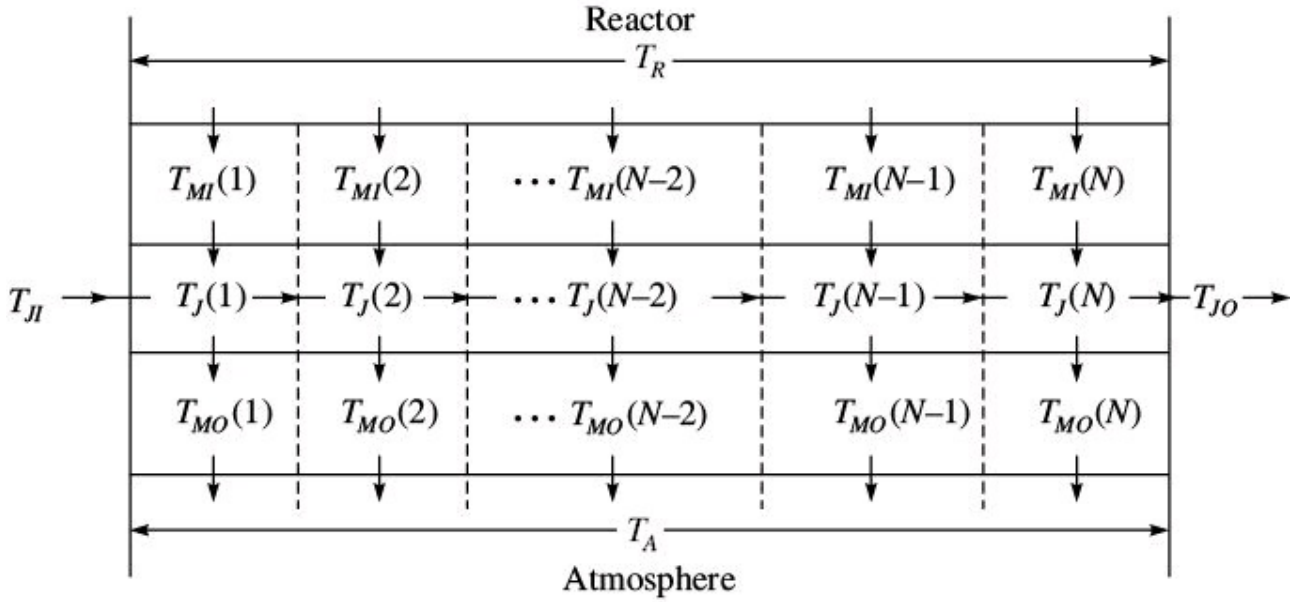
3.4.6 Batch Process

Such systems have special dynamic feature that from start to finish they never achieve steady state condition. The related variables always have non-zero derivatives during continuation and even at the completion point of the process.

EXAMPLE 3.15 (Exothermic Batch Reactor): Batch reactors are important industrial units that are used for value added fine chemical production. Many of such reactions evolve heat during the raw material conversion. In this example we discuss an exothermic reaction carried in a batch reactor. The generated heat is taken away by a cooling liquid passing through the jacket. The jacket is a longitudinally cut pipe section helically formed and fabricated over the reactor outer surface. This configuration of the jacket is a more recent trend in design that practically removes any possibility of jacket fluid short circuit, and guarantees high heat transfer efficiency in the equipment. A schematic of the reactor with its helical tubular jacket is illustrated in
Figure 3.18. A thermal lagging exists over the jacket surface and reactor bottom.



(a) Schematic diagram of the batch reactor.



(b) The heat transfer system model.

Figure 3.18 A batch reactor with helical jacket for heat removal.

We have assumed that the reactor is an ideally stirred lumped parameter unit connected by the thermal flux to the continuously operating distributed parameter jacket. This idea has been already discussed in the Example 3.14, where the shell side containing condensing saturated steam is treated as a lumped parameter section connected to the distributed parameter tube. Also, we have assumed that the reactor metal wall and jacket metal wall have capacitance and associated resistances. These considerations have been expressed in the Figure 3.18(b).

Let us first consider the jacket:

At the entry section,

$$\frac{dT_{MI}(1)}{dt} = [T_R - T_{MI}(1)]B_{MI} - [T_{MI}(1) - T_J(1)]C_{MI} \tag{3.103}$$

$$\frac{dT_J(1)}{dt} = [T_{MI}(N) - T_J(1)]B_J - [T_J(1) - T_{MO}(1)]C_J + [T_{JI} - T_J(1)]D_J \tag{3.104}$$

$$\frac{dT_{MO}(1)}{dt} = [T_J(1) - T_{MO}(1)]B_{MO} - [T_{MO}(1) - T_A]C_{MO} \tag{3.105}$$

In the sections for I increasing 2 to (N–1)

$$\frac{dT_{MI}(I)}{dt} = [T_R - T_{MI}(I)]B_{MI} - [T_{MI}(I) - T_J(I)]C_{MI} \tag{3.106}$$

$$\frac{dT_J(I)}{dt} = [T_{MI}(N) - T_J(I)]B_J - [T_J(I) - T_{MO}(I)]C_J + [T_J(I-1) - T_J(I)]D_J \tag{3.107}$$

$$\frac{dT_{MO}(I)}{dt} = [T_J(I) - T_{MO}(I)]B_{MO} - [T_{MO}(I) - T_A]C_{MO} \tag{3.108}$$

At the exit section,

$$\frac{dT_{MI}(N)}{dt} = [T_R - T_{MI}(N)]B_{MI} - [T_{MI}(N) - T_{JO}]C_{MI} \tag{3.109}$$

$$\frac{dT_J(N)}{dt} = [T_{MI}(N) - T_{JO}]B_J - [T_{JO} - T_{MO}(N)]C_J + [T_J(N-1) - T_{JO}]D_J \tag{3.110}$$

$$\frac{dT_{MO}(N)}{dt} = [T_{JO} - T_{MO}(N)]B_{MO} - [T_{MO}(N) - T_A]C_{MO} \tag{3.111}$$

The amount of heat transferred from the reaction mass in one integral time step Δt,

$$\begin{aligned} H_{TRF} &= [T_R - T_{MI}(1)]U_{RI}AR_{IN} + \sum_{I=2}^{N-1}[T_R - T_{MI}(I)]U_{RI}AR_{IN} + [T_R - T_{MI}(N)]U_{RI}AR_{IN} \\ &= U_{RI}AR_{IN}\left[\left\{\{T_R - T_{MI}(1)\} + \sum_{I=2}^{N-1}\{(T_R - T_{MI}(I)\} + \{(T_R - T_{MI}(N)\}\right\}\right] \end{aligned} \tag{3.112}$$

In the reactor the concentration and heat generation dynamics:

Initial concentration = C_{AI}, and Initial temperature = T_{RI}.

Also at any time instant t, let the concentration and temperature be C_{AI} and T_{RI}, and

they become C_{AF} and T_{RF} after the time increment Δt. The reaction kinetics and heat effect terms for a first order exothermic and irreversible reaction may be written as:

$$k_T = k_0 e^{-B/T_R} \tag{3.113}$$

$$\frac{dC_A}{dt} = -k_T C_A \tag{3.114}$$

For any time instant t, $C_A = C_{AI}$, $T_R = T_{RI}$; at $t = t + \Delta t$, $C_A = C_{AF}$, $T_R = T_{RF}$. Thus,

$$C_{AF} = C_{AI} + \int_t^{t+\Delta t} \frac{dC_A}{dt} dt = C_{AI} + \int_t^{t+\Delta t} -k_{T_{RI}} C_{AI} dt \tag{3.115}$$

The reaction mass temperature at $t + \Delta t$ may be obtained by subtracting the transferred heat, H_{TRF}, from the generated H_{GEN} and dividing the result by the thermal capacitance of the reaction mass.

$$T_{RF} = T_{RI} + \frac{H_{GEN} - H_{TRF}}{C_{PR}\rho_R V_R} \tag{3.116}$$

where,

$$H_{GEN} = (C_{AI} - C_{AF}) V_R H_R \tag{3.117}$$

In the analysis, we have used the following group of parameters:

$$B_{MI} = \frac{UR_I AR_{IN}}{Cp_{RW} RW_{MN}}; \quad C_{MI} = \frac{UR_O AR_{ON}}{Cp_{RW} RW_{MN}}; \quad B_{MO} = \frac{UJ_I AJ_{IN}}{Cp_{JW} JW_{MN}}; \quad C_{MO} = \frac{UJ_O AJ_{ON}}{Cp_{JW} JW_{MN}}$$

$$B_J = \frac{UR_O AR_{ON}}{Cp_{JF} RO_{JF} V_{JN}}; \quad C_J = \frac{UJ_I AJ_{IN}}{Cp_{JF} RO_{JF} V_{JN}}; \quad D_J = \frac{V_{JN} Cp_{JF} RO_{JF}}{Q_J Cp_{JF} RO_{JF}}$$

where,

AR_I = Heat transfer area on the inside of the reactor shell.

AR_O = Heat transfer area on the outside of the reactor shell.

AJ_I = Heat transfer area on the inside of the jacket wall.

AJ_O = Heat transfer area on the outside of the jacket wall.

UR_I = Overall heat transfer coefficient on inside of reactor shell.

UR_O = Overall heat transfer coefficient on outside of reactor shell.

UJ_I = Overall heat transfer coefficient on inside of jacket wall.

UJ_O = Overall heat transfer coefficient on outside of jacket shell.

RW_M = Mass of the reactor shell.

JW_M = Mass of the jacket wall.

Cp_{RM} = Heat capacity of the reactor shell material.

Cp_{JM} = Heat capacity of the jacket wall material.

Cp_{RL} = Heat capacity of the reactor liquid.

Cp_{JL} = Heat capacity of the shell liquid.

V_R = Reactor volume.

V_J = Jacket volume.

RO_{RL} = Density of the reactor liquid.

RO_{JL} = Density of the jacket liquid.

N = Number of assumed segments lumped parameter segments to represent the jacket, and,

$AR_{IN} = AR_I/N$; $AR_{ON} = AR_O/N$; $AJ_{IN} = AJ_I/N$; $AJ_{ON} = AJ_O/N$

$RW_{MN} = RW_M/N$; $JWR_{MN} = JW_M/N$; $V_{RN} = V_R/N$; $V_{JN} = V_J/N$

All the examples in which distributed parameter systems have been represented by multiple number of lumped parameter segments, may be simulated by numerical methods through computer programming. The principles of digital simulation have been discussed with a number of worked out examples are presented in Chapter 14. Many of the process models discussed in this chapter plus a few more are presented in Chapter 14 with their simulation programs.

EXERCISE PROBLEMS

3.1 A variable volume cylindrically shaped perfectly stirred mixer through which a solution stream is being processed. The exit line is located at the bottom of the cylindrical tank and contains a linear resistance R in the form of a valve. The inlet solution concentration C_I gm/lit, and flow rate q lit/min, are time dependent, that is the tank liquid level is also variable. Write the model equations resulting from total and component mass balance. You may assume that solution density does not depend on concentration and the system is working isothermally. If the resulting equations are non-linear, try to linearize them.

3.2 Develop the state model for a plate type absorber having five bubble-cap plates.

3.3 Develop a multiple segment lumped parameter model for a shell and tube heat exchanger that has one shell and two tube passes.

3.4 Develop a multiple segment lumped parameter model for a packed absorption

column.

3.5 A single effect short tube type evaporator shown in Figure P3.1, concentrates a dilute NaOH solution from initial concentration C_I gms/lit, to the final concentration C_F gms/lit. Saturated steam at pressure P_S kg/cm^2 is used as the heating fluid that supplies only the latent heat and condensate issues as saturated liquid at same temperature. A pressure of P_E less than atmospheric pressure is maintained in the evaporator drum. The mass flow rate of feed is Q_M kg/min. The available heat transfer area is A_T m^2. The overall heat transfer coefficient from steam callendria to the drum liquid is U kcal/(°C.min.m^2). The boiling point rise of NaOH solution, as a function of concentration, is expressed by During Plot[4].

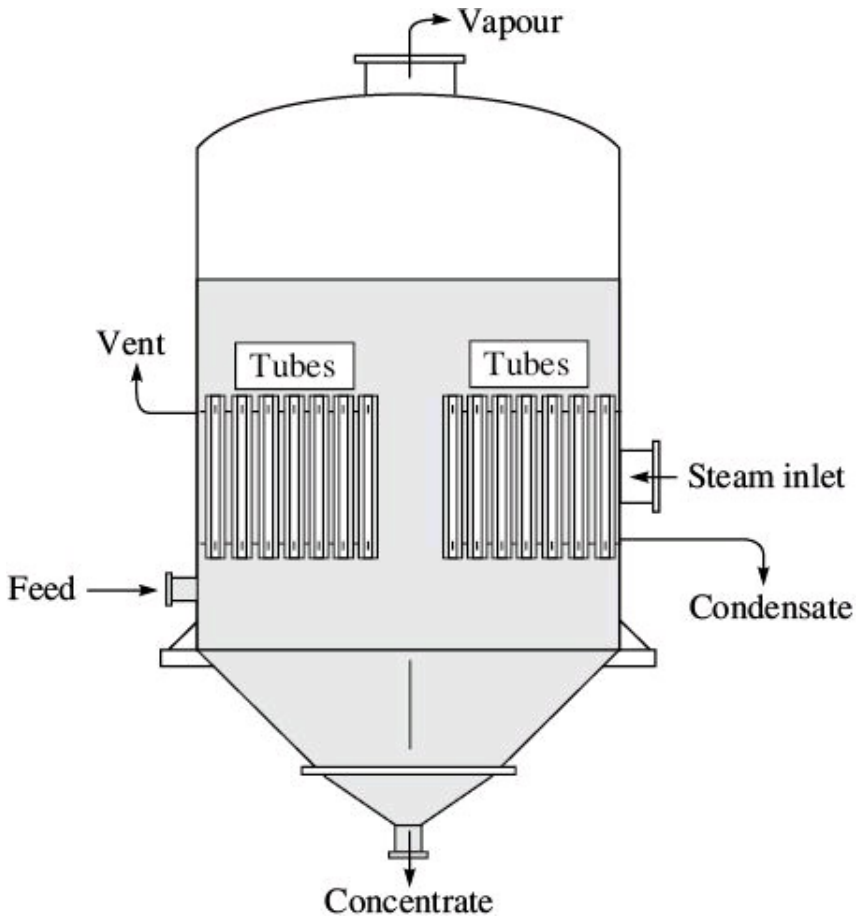


Figure P3.1 A short tube single effect evaporator.

Develop a model for the system identifying C_F as the output and C_I as the time dependent input variable. All other quantities are constants.

3.6 A wet cat, as you know all wet cats are a morose looking unhappy bunch of guys, was very timidly walking along a narrow deserted alley. However, from his front, a refreshing cool breeze was blowing, and the wet cat was gradually becoming a dry cat. If the initial moisture in his furs was 7% by wt., and the drying operation was carried out under a constant rate of 0.05 mg/(cm^2.min), Weight of cat = 3.45 kg on

the dry basis. But there is a little twist, as the cat was gradually loosing the moisture his footsteps were becoming more spry, and his speed was increasing according to:

$$S = 2.5 + 0.5 \text{ m}^2/\text{min}$$

where, M = % moisture content in the cat's fur by wt. Assuming the available drying surface area to be 0.75 m^2, prepare a model relating the moisture in the fur and the distance travelled by the cat in meters down the alley. Also calculate the length of the alley that the cat has to walk to become a dry cat having final moisture content of 1.5% by wt. Air-water vapour equilibrium properties are available in literature[4].

3.7 A tunnel type air drier schematically shown in Figure P3.2 is used to dry cakes of a wet solid riding over a conveyor. The cakes are placed D (= 180 mm) distance apart over the conveyor moving with a linear speed of V (5 cm/sec) through the drier chamber of length L (=250 cm).

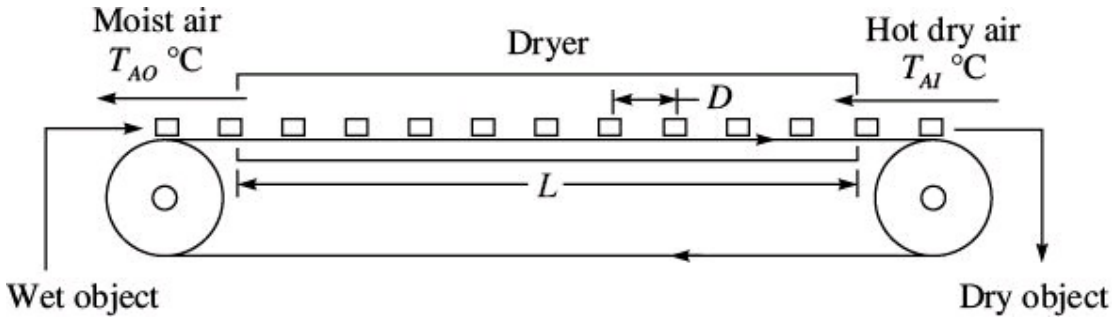


Figure P3.2 Operation of a tunnel type continuous drier.

Hot air of DBT = T_{AI} °C and RH = R_I % is available. This air leaves at DBT = T_{AO} °C and RH = R_O%. The drying operation is done within the constant drying rate period of the solid that is = 0.012 gm moisture/(cm^2.sec). Cake size: 76 mm(L) × 50(W) × 30 mm(H). Each cake weighs 115 gm. Data: T_{AI} = 76°C, R_I = 18%, R_O = 32%. To calculate T_{AO}, assume only latent heat transfer under adiabatic condition.

Develop the dynamic model to relate R_O as the output variable with R_I as the input disturbance.

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[3] Luyben, W.L., *Process Modelling Simulation and Control for Chemical Engineers*, McGraw Hill Pub. Co, NY, 1973.

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4

Transfer Function of First Order Systems (Complex Domain Analysis)

In this chapter we are going to discuss Transfer Function (TF) type of process model building procedure. The general form of a process transfer function has already been defined in Section 3.2.2, as an alternative description of the basic differential equation that represents the process dynamic models. This variety of model description is more popular and more widely used in process control literature. The more important reason is that, except for processes that feature multiple lumped parameter stages, major number of process systems offer a fewer number of state equations to be handled than in electrical or mechanical systems. The transfer function models are easy to handle compared to their corresponding state equation representation.

Secondly, transfer functions are a compact and general expression of process model without requiring any specification of the input function. Ideally for any input function, the TF may be used to obtain the process response.

Moreover, the TF model is an expression of ratio of response to the input of a process, and by examining the expression in its denominator we may have a good idea about the dynamic character of the process.

We have already mentioned the importance of making certain assumptions regarding the physical and chemical phenomena that constitutes the fundamentals of the model and dictates its quality (accuracy). The assumptions made would be listed during the model building procedure. The order of a system has been already defined. We shall begin with first order systems.

(*For the reader, who are not familiar with the theory and application of Laplace transform procedure, we have included a short exposition of this operational method in Appendix-II.*)

4.1 First Order Systems

The first order model is the simplest dynamic description of a process. A first order linear ODE is sufficient to express the process dynamics. Only two parameter values, K_P (Gain) and t (Time Constant) are sufficient to describe a first order system. The

standard form of first order transfer function may be written as,

$$G_P(s) = \frac{K_P}{\tau s + 1} \qquad (4.1)$$

If the order of the given process is one, the modelling effort should produce a single ordinary first order differential equation. In many examples it is required that the character of the dynamic element/s has to be approximated by linearity and lumping assumptions to realize a first order description. For more complex applications, the first order model is constructed as a first attempt, which may be modified later to build up complex or higher order models of greater accuracy for more sophisticated applications.

To demonstrate the first order model building procedure, we have cited a few examples. Some of the discussed examples are follow-ups from the previous chapter; for them we will not rewrite the problem statement and the assumptions. The proposed time domain model will be carried over for further processing.

EXAMPLE 4.1 (Liquid Level Dynamic System): This system has been taken up as Example 3.3 in the previous chapter, where the statement of the problem has been presented together with the associated assumptions for model development. We rewrite Equations 3.20 and 3.21 as the starting point of this discussion

$$A\frac{dh}{dt} + \frac{h}{R} = q_I \qquad (3.20)$$

and

$$AR\frac{dq_O}{dt} + q_O = q_I \qquad (3.21)$$

At steady state, Equation 3.20 becomes:

$$q_{IS} - \frac{h_S}{R} = A\frac{dh_S}{dt} = 0 \qquad (4.2)$$

where, subscript '*S*' is used to denote steady state values of the respective variables. Now subtracting Equation 4.2 from Equation 3.20,

$$(q_I - q_{IS}) - \frac{h - h_s}{R} = A\left[\frac{d(h - h_S)}{dt}\right] \qquad (4.3)$$

Introducing deviation variables, $Q_I = q_I - q_{IS}$, and $H = h - h_S$, in Equation 4.3,

$$Q_I - \frac{H}{R} = A\frac{dH}{dt} \qquad (4.4)$$

Taking Laplace Transform of both sides of Equation 4.4, we get,

$$Q_I(s) - \frac{H(s)}{R} = AsH(s)$$

Recalling the definition of Transfer function, the above equation may be rearranged as:

$$\frac{H(s)}{Q_I(s)} = \frac{R}{ARs+1} \qquad (4.5)$$

Note the combination of parameters as in the coefficient of the Laplace operator '*s*' and the numerator of transfer function expressions.

You must have covered the solution of the linear ordinary differential equations in time domain. If we consider the general time domain solution method, the transfer function may be equated with complementary function and the input or disturbance to the particular integral. The complete solution is available only when these two are specified.

To complete this discussion we should mention, Laplace transform is an operational method and the analysis procedure is strictly for linear systems, i.e. systems having dynamic elements of linear character. We do not have to bother about the tank (capacitance = cross sectional area of the tank) because even fabricated cylindrical tanks are quite good cylinders in the sense that tank diameter is independent of height, but what about the flow resistance offered by a valve restriction or orifice, particularly, if you have covered fluid mechanics in some earlier semester. The familiar relation between flow, q_O, through an orifice or restriction, due to a fluid head, h, is given by:

$$q_O = C_D\sqrt{h} \qquad (4.6)$$

where, C_D is known as the orifice discharge coefficient. This non-linear relation is quite different from the Ohm's law like linear relation, $q_O = h/R$ that has been used to develop the model. The linear version of the process model is obviously easier to construct and involves a lesser analytical effort for response analysis. The linear approximation is justified only when the response of the linear model is within the warranted accuracy limit of the application. Otherwise more rigorous nonlinear models should be developed. Then linearization of the rigorous model is performed to obtain a linearized approximate version of the model. Response analysis and system design may proceed using this approximate model.

(*We have already introduced the procedure of linearization in the previous chapter. In the latter part of the present chapter, more examples of non-linear systems and their linearization procedure have been included.*)

EXAMPLE 4.2 (Temperature Dynamics in a Mercury in Glass Thermometer): This system was taken up in Example 3.4, where the statements of the problem have been presented together with the associated assumptions for model development. We

rewrite Equation 3.25 as the starting point of this discussion.

$$\frac{dt_O}{dt} = \frac{UA_T}{mC_P}(t_I - t_O) \qquad (3.25)$$

where m and C_P are respectively mass and heat capacity of mercury, U is the overall heat transfer coefficient between surrounding atmosphere and the thermometer, and A_T is the heat transfer area. At steady state the Equation 3.25 takes the following form:

$$0 = \frac{dt_{OS}}{dt} = \frac{UA_T}{mC_P}(t_{IS} - t_{OS}) \qquad (4.7)$$

Now subtracting Equation 4.7 from Equation 3.25,

$$\frac{d(t_O - t_{OS})}{dt} = \frac{UA}{mC_P}(t_i - t_{IS}) - \frac{UA}{mC_P}(t_O - t_{OS}) \qquad (4.8)$$

Introducing deviation variable $T_I = t_I - t_{IS}$, and $T_O = t_O - t_{OS}$, in Equation 4.8,

$$\frac{dT_O}{dt} = \frac{UA}{mC_P}T_I - \frac{UA}{mC_P}T_O \qquad (4.9)$$

Taking Laplace Transform of both sides of Equation 4.9, we get,

$$sT_O(s) = \frac{UA}{mC_P}T_I(s) - \frac{UA}{mC_P}T_O(s) \qquad (4.10)$$

Again recalling the definition of Transfer function, the above Equation 4.10 may be rearranged as:

$$\frac{T_O(s)}{T_I(s)} = \frac{1}{\frac{mC_P}{UA}s + 1} \qquad (4.11)$$

Note the combination of parameters as in the coefficient of the Laplace operator 's' and the numerator of transfer function expressions. Let us try a few more examples from an assortment of application areas.

EXAMPLE 4.3 (Electrical Potential Dynamics in an R–C Circuit):
At the input of the circuit shown in Figure 4.1, a variable voltage, $E_I(t)$, constitutes the input disturbance, and the output is the voltage $E_O(t)$. It is required to develop the TF between E_O and E_I.

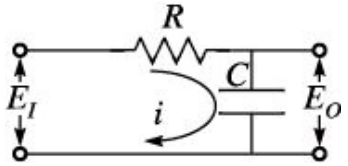

Figure 4.1 An electrical circuit with resistance and capacitance elements.

Again we list the assumptions that are fundamental to the model building:

1. All the electrical charge storage capacity is only in the capacitance element
2. All the resistance to electron or charged particle transfer is in the resistance element

In this example, the lumping of parameters is more justified because from the knowledge of electricity–magnetism, one is ready to accept that a capacitance and resistance element really contains the storing and resisting capability, respectively. Manifestations of connecting wire or the energy source in this respect are so small that similar assumptions have been made even in the textbooks on pure physics during circuit analysis calculations.

The same notion of ideality is also valid for the next two assumptions:

3. The resistance (R ohms) has a linear character
4. The capacitance value (C farad) is independent of applied voltage (V volts).

The following unsteady state charge balance is not done in the way of the above two examples, because there is another more familiar way of treating electrical circuit. We apply Kirchhoff's law, i.e. in a single loop circuit, the sum of the voltage drops across all components of the circuit is equal to the source voltage. If i be the current flowing through the loop, let the voltage drops across C be V_C, and across R, V_R, then according to Kirchhoff's law:

$$E_I(t) = V_C + V_R \qquad (4.12)$$

We may write individual expressions for V_C, and V_R as:

$$V_C = \frac{1}{C}\int_0^t i\,dt \qquad (4.13)$$

and,

$$V_R = R_I \qquad (4.14)$$

Assuming that all initial variable values are zero, we may directly take Laplace Transform of equations 4.13 and 4.14. Combining the results, and recognizing that V_C is in fact the output voltage $E_O(t)$, we may write,

$$E_I(s) = I(s)R + \frac{I(s)}{Cs} = I(s)\left(R + \frac{1}{Cs}\right) = E_O(s)Cs\left(R + \frac{1}{Cs}\right) = E_O(s)[RCs + 1] \qquad (4.15)$$

Rearranging Equation 4.15 we obtain the required TF,

$$\frac{E_O(s)}{E_I(s)} = \frac{1}{RCs + 1} \qquad (4.16)$$

Again note the combination of parameters as in the coefficient of the Laplace operator '*s*' and the numerator of transfer function expression.

By this time you must have realized that if the associated variables of a system have a threshold or datum value equal to zero, we need not convert the absolute variables into their deviation form. This has been mentioned in the beginning of the previous chapter where we introduced the deviation variables. We have followed the same principle in this example.

EXAMPLE 4.4 (Displacement Dynamics of a Spring–Dashpot Combination): A combination of an ideal spring and dashpot elements has been shown in Figure 4.2. Let us assume that the resulting system does not contain any mass so that there would be no contribution of inertia or inductive element during displacement or motion. The characteristics of spring and dashpot elements have been presented in Chapter 2, where the mechanical equivalents of dynamic element are introduced.

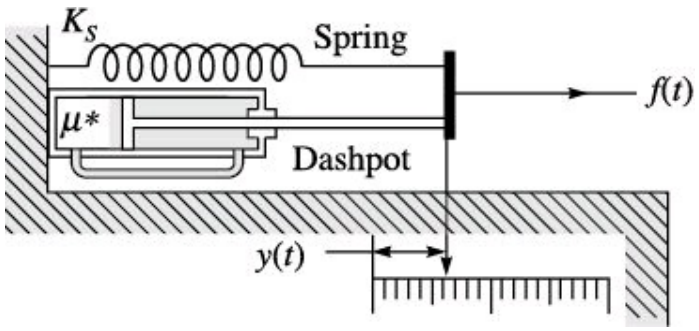

Figure 4.2 Displacement dynamics of a spring–dashpot assembly.

The system is at rest until $t = 0$, when force $f(t)$ is applied at the end where spring and dashpot joins together. We attempt to find the transfer function between the applied force and displacement as, $Y(s)/F(s)$. The assumptions regarding parameter lumping and linearity have been covered where we defined these mechanical dynamic elements in Chapter 2. Hence, we may directly go into the model building effort.

The mechanical equivalent of Kirchhoff's law is the so-called 'force balance' for the given system that may be expressed by the following free-body diagram of Figure 4.3.

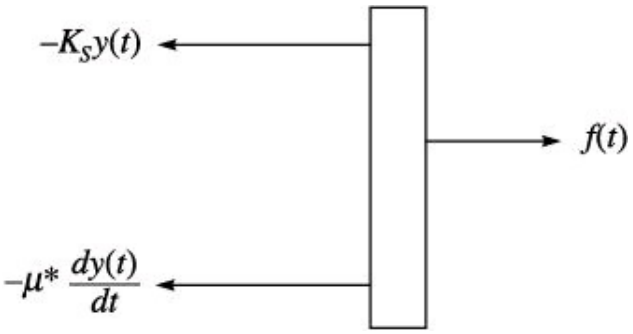

Figure 4.3 The free-body diagram of the mechanical system.

We put –ve signs on the drag force of dashpot and elastic force of spring to recognize that they act in the reverse direction to the applied force $f(t)$ and the direction of displacement $y(t)$. If these forces are summed together the result remains

always zero at any point of motion. Thus,

$$f(t) - K_S y(t) - \mu * \frac{dy(t)}{dt} = 0$$

or,

$$f(t) = K_S y(t) + \mu * \frac{dy(t)}{dt} \qquad (4.17)$$

This is an equation of motion of the spring-dashpot system. The drag force of dashpot and the elastic force of spring acting reverse to the applied force, causing the system to stop at a finite distance from the starting point. Following the reasoning given in case of Example 4.3, Equation 4.17 may be Laplace transformed by introducing deviation form of the variables $F(t)$ and $Y(t)$.

$$F(s) = K_S Y(s) + \mu * sY(s) \qquad (4.18)$$

Rearranging Equation 4.18, the standard form of the TF may be obtained as:

$$\frac{Y(s)}{F(s)} = \frac{(1/K_S)}{(\mu */K_S)s + 1} \qquad (4.19)$$

Note the combination of parameters as in the coefficient of the Laplace operator 's' and the numerator of transfer function expressions. If the spring constant term K_S is brought in the L.H.S of Equation 4.19,

$$\frac{Y(s)}{F(s)/K_S} = \frac{1}{(\mu */K_S)s + 1} \qquad (4.20)$$

The term $[f(t)/K_S]$ has a dimension of length, and this length is the ultimate displacement at $t = \infty$, if the force $f(t)$ applied at $t = 0$ remains constant during the time period. Hence, at any time t, within the interval $f(t)/K_S$ may be considered as the displacement equivalent of force, and if we introduce $x(t)$ to replace $f(t)/K_S$, as the input variable, so that the TF of Equation 4.20 becomes:

$$\frac{Y(s)}{X(s)} = \frac{1}{(\mu */K_S)s + 1} \qquad (4.21)$$

Note that this TF has a dimensionless unity in its numerator.

EXAMPLE 4.5 (Concentration Dynamics in a Perfectly Stirred Constant Holdup Mixer): A tank of working volume V_L with a mechanical agitator serves the purpose of mixing or homogenization of a flowing liquid stream that is moving with a flow rate q through the mixer (Figure 4.4). The mixing in the tank happens ideally.

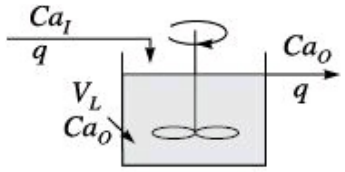

Figure 4.4 The concentration dynamics in a perfectly stirred mixer.

The exit stream issues as overflow somewhere from the top of the vessel near its rim. Thus the tank always remains filled up to the exit point, that is its working or effective volume for material processing, and we call such an equipment to be operating with a **constant hold-up** volume of V_L, which is less than the actual volume of the tank.

The exit flow rate does not depend on the tank liquid level, but is equal to q, so that if q remain constant, the theoretical average residence time, t, defined as "the length of time each entering liquid particle stays in the mixer" will be $t = V_L/q$.

The conception of perfect- or back-mixing has been introduced in Section 3.2 in rather detail. This means that the inlet liquid as soon as reaches the mixer interior abruptly changes its concentration and assumes the bulk liquid concentration. This is another way of saying that the solute transfer due to concentration difference does not depend on diffusion or any such mechanisms, but only on how soon the inlet stream could be brought into the tank, hence, reciprocal of flow rate ($1/q$) may be conceived as the resistance element.

As the bulk liquid is the storing device into which the solute has to enter, its capacitance may be defined as "the amount of solute required to raise the concentration by unit amount", which is numerically equal to the volume V_L, if expressed in consistent engineering unit system. (*Verify the comment considering V_L in litres and concentration in gms/lit., and also in moles/lit.*)

For simulation and modelling purpose many process equipment have been represented by a single or combination of such **Ideal Mixed Stage**. In fact, this is one of the fundamental building blocks (a first order combination of resistance and capacitance elements) for model development, of more complex processes. There is another such conceptual body called **Dead Time**, **Delay**, or, **Plug-flow** element. On the contrary to the ideally mixed stage, in a delay or plug flow element, there occurs no mixing or homogenization. The variable profile (concentration, temperature, etc.) with the entering fluid remains intact during its passage through the element and comes out from the exit without any loss of shape. We shall have a more detail discussion about this dynamic element in Chapter 6.

Thus, for the present system, the basic assumption is that the mixing happens in an ideal manner that automatically defines and quantifies the capacitance and resistance elements. However, to complete the list we should also mention the following ones:

1. There is no density or viscosity change due to mixing
2. The mixing occurs in isothermal manner

The unsteady state solute mass balance around the mixer may be written as:
(Rate of mass input) – (Rate of mass output) = (Rate of mass accumulation) (4.22)
Putting the variable and parameter names in Equation 4.22,

$$qCa_I - qCa_O = V_L \frac{dCa_O}{dt} \qquad (4.23)$$

In the above equation q, Ca, and V_L should be in any consistent engineering unit system. Introducing deviation variables $\bar{C}a_I = Ca_I - Ca_{IS}$; $\bar{C}a_O = Ca_O - Ca_{OS}$; and rearranging, we get:

$$\bar{C}a_I - \bar{C}a_O = \frac{V_L}{q}\frac{d\bar{C}a_O}{dt} \qquad (4.24)$$

Taking Laplace transform of Equation 4.24, and rearranging to obtain the TF as:

$$\frac{\bar{C}a_O(s)}{\bar{C}a_I(s)} = \frac{1}{(V_L/q)s+1} \qquad (4.25)$$

Again the TF has a dimensionless unity in its numerator, and the product of capacitance and resistance has been recognized as a characteristic time concerning the system, the residence time. Not only the product in this example, but all such parameter combinations as coefficient of the Laplace operator '*s*', mentioned in previous example systems, have dimension of time. In process dynamics literature they are called **Time Constant** of the respective systems, designated by the Greek letter *t*. Time constant is one of the significant dynamic parameters of a system. This obtains a measure of how quick a system will respond after it has been disturbed from steady state.

The other important parameter is the numerator of TF, which may be conceived as the ratio of ultimate magnitude of response at steady state, relative to the magnitude of input disturbance. This parameter is called as the **Steady state Gain** or **Process Gain** or simply **Gain** (K_P) of the system. Its manifestation will appear in numerous important considerations of analysis and design of control systems. Gain also embodies the dimensional conversion of input to output variable.

EXAMPLE 4.6 (Temperature Dynamics in a Perfectly Stirred Constant Holdup Liquid Heater): A tank of working volume V in which a perfect mechanical agitator serves the purpose of mixing or homogenization of a flowing liquid stream that is moving with a flow rate q through the mixer. A heat input device in the form of a resistance type electric heater element is associated with the mixer so that the exit

liquid leaves at an elevated temperature. The mixing tank is otherwise thermally insulated.

The inlet temperature of the inlet liquid T_I (t), and the heat input rate from the heating system $W(t)$ may both vary. It is required to find the dynamics of exit liquid temperature T_O (t) with respect to T_I (t) and $W(t)$. The bulk liquid temperature may be assumed as equal to that of the exit liquid, due to perfect mixing.

Other relevant assumptions are that the liquid density and heat capacity are not dependent on temperature. A schematic diagram of the system is shown in Figure 4.5.

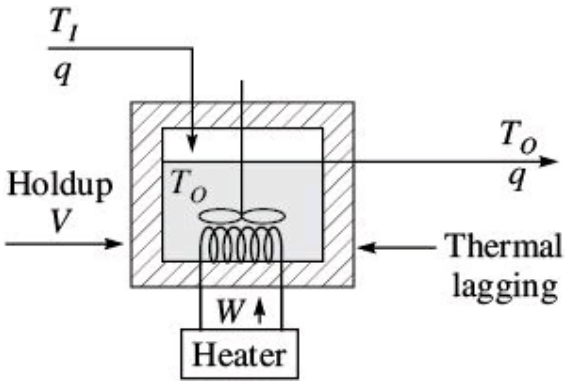

Figure 4.5 Stirred tank constant holdup liquid heater.

There will be no accumulation of mass, and we need not write the dynamic mass balance.

We may write the heat balance equation for the system as:

$$\rho q C_P (T_I - T_D) + W - \rho q C_P (T_O - T_D) = \rho C_P V \frac{dT_O}{dt} \qquad (4.26)$$

where,

T_D = Datum temp.,°C; r = Liq. density, g/ml; C_P = Liq. Ht. Capacity, cal/(g°C)

In this system we have two input variables T_I (t) and $W(t)$ and hence by way of linear analysis, we should determine transfer functions T_O $(s)/T_I$ (s) considering W to be constant, and T_O $(s)/W$ (s) while T_I remain constant.

1. To determine $T_O(s)/T_I(s)$ Equation 4.26 is rewritten for $W = W_S$, as:

$$\rho q C_P (T_I - T_D) + W_S - \rho q C_P (T_O - T_D) = \rho C_P V \frac{dT_O}{dt} \qquad (4.27)$$

Next we write Equation 4.27 in the steady state condition of the variables as:

$$\rho q C_P (T_{IS} - T_D) + W_S - \rho q C_P (T_{OS} - T_D) = \rho C_P V \frac{dT_{OS}}{dt} \qquad (4.28)$$

Subtracting Equation 4.28 from Equation 4.27:

$$\rho q C_P (T_I - T_{IS}) - \rho q C_P (T_O - T_{OS}) = \rho C_P V \frac{d(T_O - T_{OS})}{dt}$$

Introducing the deviation variables, $\bar{T}_I = T_I - T_{IS}$; $\bar{T}_O = T_O - T_{OS}$, into the last equation,

$$\rho q C_P \bar{T}_I - \rho q C_P \bar{T}_O = \rho C_P V \frac{d\bar{T}_O}{dt} \qquad (4.29)$$

Taking Laplace Transform of Equation 4.29, and rearranging we obtain the Transfer function:

$$\frac{\bar{T}_O(s)}{\bar{T}_I(s)} = \frac{1}{(V/q)s + 1} \qquad (4.30)$$

2. To determine $T_O(s)/W(s)$, Equation 4.26 is rewritten for $T_I = T_{IS}$, as:

$$\rho q C_P (T_{IS} - T_D) + W - \rho q C_P (T_O - T_D) = \rho C_P V \frac{dT_O}{dt} \qquad (4.31)$$

As above, we write Equation 4.31 in the steady state condition of the variables as:

$$\rho q C_P (T_{IS} - T_D) + W_S - \rho q C_P (T_{OS} - T_D) = \rho C_P V \frac{dT_{OS}}{dt} \qquad (4.32)$$

Subtracting Equation 4.32 from Equation 4.31:

$$(W - W_S) - \rho q C_P (T_O - T_{OS}) = \rho C_P V \frac{d(T_O - T_{OS})}{dt}$$

Introducing the deviation variables, $\bar{W} = W - W_S$; $\bar{T}_O = T_O - T_{OS}$, into the last equation,

$$\bar{W} - \rho q C_P \bar{T}_O = \rho C_P V \frac{d\bar{T}_O}{dt} \qquad (4.33)$$

Taking Laplace Transform of Equation 4.33, and rearranging we obtain the Transfer function:

$$\frac{\bar{T}_O(s)}{\bar{W}(s)} = \frac{1/\rho C_P q}{(V/q)s + 1} \qquad (4.34)$$

Such dependency on two input variables may be expressed in Laplace domain in the following form:

$$\bar{T}_O(s) = \bar{T}_I(s)\left[\frac{1}{(V/q)s + 1}\right] + \bar{W}(s)\left[\frac{1/\rho C_P q}{(V/q)s + 1}\right] \qquad (4.35)$$

Equation 4.35 could be also expressed in the form of block diagram as in the Figure 4.6.

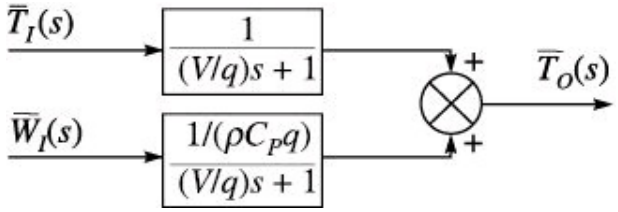

Figure 4.6 The block diagram representation of Equation 4.35.

Recall the comment at the end of Section 3.4 that "*response for simultaneous inputs may be obtained by super-imposition of responses obtained for two independent input changes"*– the block diagram of Figure 4.6 illustrates this additive procedure. We shall include a numerical example to demonstrate the procedure in the next chapter.

EXAMPLE 4.7 (Concentration Dynamics in a Perfectly Stirred Constant Holdup Isothermal Reactor): In Example 4.5, the Figure 4.4, may well describe this system. A perfectly stirred tank having a constant holdup volume V_L with all the comment and observations made for the system in Example 4.5 are valid, we need not repeat them. Also, you should note that this system has been taken up as Example 3.4, in the previous chapter where the statement of the problem has been presented together with the associated assumptions for model development. We rewrite Equation 3.33 as the starting point of this discussion.

$$\frac{dCa_O}{dt} = \frac{q}{V_L} Ca_I - \frac{q}{V_L} Ca_O - kCa_O \qquad (3.33)$$

After showing the subsequent procedure so many times for deriving the transfer function in earlier examples, we may skip those steps and directly write the transfer function in standard form as:

$$\frac{\bar{C}a_O(s)}{\bar{C}a_I(s)} = \frac{\frac{q}{q+kV_L}}{\frac{V_L}{q+kV_L}s+1} \qquad (4.36)$$

Note the expressions for gain and time constant in the above TF (Equation 4.36), and their difference with the TF (Equation 4.25) found in case of the mixing system of Example 4.5. The gain for the mixing system has been unity and for a reactor it is $q/(q + kV_L)$, i.e. less than 1, this tallies with the fact that a sudden change of inlet concentration of a finite magnitude M will produce an ultimate change of output concentration less than M because an amount of reactant mass would be consumed in reaction. By similar reasoning, it follows that reactor time constant $V_L/(q + kV_L)$, is less than the mixer time constant V_L/q. In these examples the system dynamic character is expressed by linear first order ODE, the fundamental requirement for a

first order process. Also, in all the cases, the product of capacitance and resistance term has dimension of time and is called the time constant of the respective systems, designated by t.

The gain terms in Examples 4.2, 4.3 and 4.5 are dimensionless with magnitude 1.0. This tells us that input and output variables in these examples have same dimension and there is no internal mechanism by which certain portion of principle entity would be consumed (go out) or generated (come in) except the inlet or outlet streams. In Examples 4.1 and 4.4, the gain terms in the TF has dimension. This is for the fact that input and output variables are dimensionally different, and the gain term acts as the dimensional conversion factor between input and output. This notion may be confirmed by the TF of Example 4.4 (Equation 4.19 and Equation 4.21). Thus, it has been established that the general expression of first order transfer function may be given by Equation 4.1.

$$\frac{Y(s)}{X(s)} = \frac{K_P}{\tau s + 1} \qquad (4.1)$$

The fundamental dynamic part of Equation 2.40, i.e. the term $[1/(t.s + 1)]$, is called in some literature as **single exponential stage**, because this term when reverted to time domain obtains the exponential time decay function $e^{-t/t}$.

A dimensionless gain with non-unity magnitude ($KP \neq 1$) indicates an internal depletion or generation mechanism that either amplifies or attenuates the magnitude of input change into output change.

In the next section we will discuss a few examples which feature a little more complexity of analysis.

4.2 More Complex Examples of First Order Systems[2]

A system involving storage and flow of compressible fluids has an inherent difficulty that volume flow rate depends not only the local pressure but if the flow occurs across a resistance then it is also dependent on the pressures on both sides of the resistance.

One way of negotiating this difficulty is to perform a dynamic mass balance for the gas flow system.

EXAMPLE 4.8 (Develop the transfer function between *P*(*s*)/*Pi*(*s*) for the following surge tank in a compressible fluid flow line):

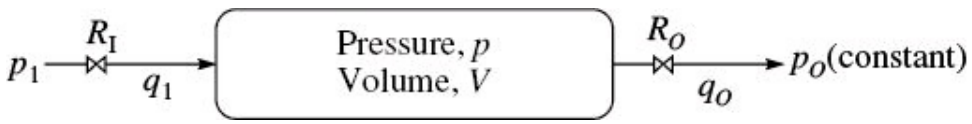

Figure 4.7 The pressure dynamics of a compressible fluid in a constant volume vessel.

Solution: Assuming inlet and outlet gases have equal densities, the unsteady state

mass balance may be written as:

$$\rho(q_I - q_O) = \frac{dm}{dt} \qquad (4.37)$$

From the gas-law, we may write:

$$m = Mn = \frac{MV}{RT}P$$

or

$$\frac{dm}{dt} = \frac{MV}{RT}\frac{dP}{dt} \qquad (4.38)$$

where, R = is the universal gas constant. Putting Equation 4.37 in Equation 4.38,

$$\rho\left(\frac{p_I - p}{R_I}\right) - \rho\left(\frac{p - p_O}{R_O}\right) = \frac{MV}{RT}\cdot\frac{dP}{dt} \qquad (4.39)$$

At steady state the above equation becomes:

$$\rho\left(\frac{p_{IS} - p_S}{R_I}\right) - \rho\left(\frac{p_S - p_O}{R_O}\right) = \frac{MV}{RT}\cdot\frac{dp_S}{dt} \qquad (4.40)$$

Subtracting Equation 4.40 from Equation 4.39:

$$\rho\left[\frac{(p_I - p_{IS}) - (p - p_S)}{R_I} - \frac{(p - p_S) - (p_O - p_O)}{R_O}\right] = \frac{MV}{RT}\cdot\frac{d(p - p_S)}{dt}$$

Introducing the deviation variables:

$$\frac{P_I}{R_I} - \frac{P}{R_I} - \frac{P}{R_O} = \frac{MV}{\rho RT}\cdot\frac{dP}{dT} \qquad (4.41)$$

taking Laplace transform of Equation 4.41;

$$\frac{P_I(s)}{R_I} = P(s)\left[\frac{1}{R_I} + \frac{1}{R_O} + \frac{MV}{\rho RT}s\right]$$

Simplifying the above equation:

$$\frac{P(s)}{P_I(s)} = \frac{\dfrac{R_O}{(R_I + R_O)}}{\dfrac{MVR_IR_O}{\rho RT(R_I + R_O)}s + 1} = \frac{K_P}{\tau s + 1} \qquad (4.42)$$

where, $K_P = \dfrac{R_O}{R_I + R_O}$; and $\tau = \dfrac{MVR_IR_O}{\rho.RT(R_I + R_O)}$

EXAMPLE 4.9 A cylindrical tank as in Figure 4.8, has an internal area A and height H, and a normal liquid depth of h. The flow of a liquid to the tank is q_I, and liquid from the tank is discharged through a valve to a process at atmospheric pressure. Develop the transfer function between $Q_I(s)$ and $H(s)$, if the tank is closed with a constant pressure P.

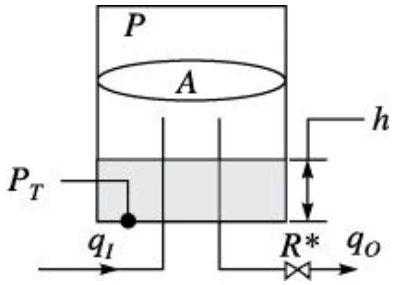

Figure 4.8 Pressure dynamics of a confined volume of compressible fluid.

Solution: The unsteady state mass balance may be written as:

$$\rho(q_I - q_O) = \rho A \frac{dh}{dt} \qquad (4.43)$$

Differentiating Equation (a) again with respect to t:

$$\frac{dq_I}{dt} - \frac{dq_O}{dt} = A\frac{d^2h}{dt^2} \qquad (4.44)$$

The flow–head relation may be given as: $q_O = \dfrac{P_T}{R^*}$

Applying the flow–head relation into Equation 4.44:

$$\frac{dq_I}{dt} - \frac{1}{R^*}\frac{dP_T}{dt} = A\frac{d^2h}{dt^2} \qquad (4.45)$$

Under constant temperature the gas law may be written as: $PV = K$
Differentiating the above equation with respect to t,

$$\frac{dP}{dt} = -\frac{P}{V}\cdot\frac{dV}{dT} = -\frac{PA}{V}\left(\frac{dh}{dt}\right) = -\frac{PA}{V}\cdot\frac{dh}{dt}$$

Applying the relation: $P_T = P + hrg$, in the above equation,

$$\frac{dP_T}{dt} = \frac{dP}{dt} + \rho g\frac{dh}{dt} = \frac{PA}{V}\cdot\frac{dh}{dt} + \rho g\frac{dh}{dt} = \left(\frac{PA}{V} + \rho g\right)\frac{dh}{dt} \qquad (4.46)$$

Putting the expression of dP_T/dt from Equation 4.46 into Equation 4.45:

$$\frac{dq_I}{dt} - \frac{1}{R^*}\left(\frac{PA}{V} + \rho g\right)\frac{dh}{dt} = A\frac{d^2h}{dt^2} \qquad (4.47)$$

Let us skip the steps of writing Equation 4.47 at steady state, subtracting that from

Equation 4.47, and we directly write Equation 4.47 in deviation variables:

$$\frac{dQ_I}{dt} - \frac{1}{R^*}\left(\frac{PA}{V} + \rho g\right)\frac{dH}{dt} = A\frac{d^2H}{dt^2} \qquad (4.48)$$

$$sQ_I(s) - \frac{1}{R^*}\left(\frac{PA}{V} + \rho g\right)sH(s) = As^2H(s) \qquad (4.49)$$

Rearranging Equation 4.49 and putting, $K^* = \frac{PA}{V} + \rho g$, we obtain the transfer function:

$$\frac{H(s)}{Q_I(s)} = \frac{R^*/K^*}{\{(AR^*)/K^*\}s + 1} = \frac{K_P}{\tau s + 1} \qquad (4.50)$$

where the system gain, $K_P = R^*/K^*$; and, the time constant, $\tau = \frac{AR^*}{K^*}$.

4.3 Non-Linear System Examples

In the last chapter we have introduced process non-linearity of two categories. In the following section some more examples have been discussed that feature individual classes and also both classes of non-linearity coexisting within a system.

EXAMPLE 4.10 A fluid of constant density is pumped into a cone shaped tank of total volume equal to $(pR^2H)/3$. The flow out of the bottom of the tank is proportional to the square root of the level h of liquid in tank. Derive the TF relating the tank level to inlet flow rate.

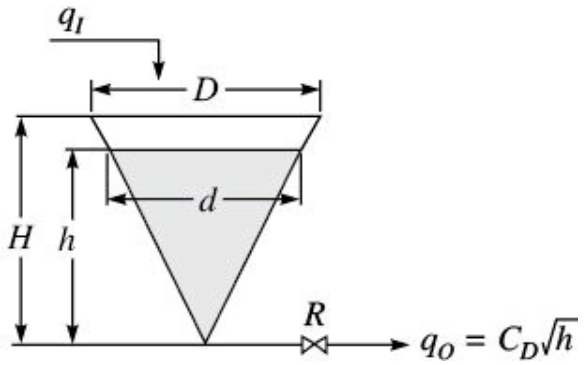

Figure 4.9 System featuring non-linear capacitance and resistance.

Solution:

Unsteady state mass balance
(Assuming R to be linear)

$$\rho q_I - \rho q_O = \rho \frac{dV}{dt}$$

$$q_I - \frac{h}{R} = K'' \frac{d}{dt}(h^3) \qquad (4.51)$$

Expressing tank volume V as a function of level h

$$\frac{h}{H} = \frac{d}{D}; \; d = \frac{D}{H} h$$

Area a at height h,

$$a = \frac{\pi d^2}{4} = \frac{\pi}{4}\left(\frac{Dh}{H}\right)^2 = K' h^2$$

At steady state the above Equation becomes:

$$q_{IS} - \frac{h_S}{R} = K'' \frac{d}{dt}(h_S^3) \qquad (4.52)$$

Vol. at level h

$$v = \frac{h\pi d^2}{12} = \frac{\pi h}{12}\left(\frac{Dh}{H}\right)^2 = K'' h^3$$

Subtracting Equation 4.52 from Equation 4.51:

$$(q_I - q_{IS}) - (q_O - q_{OS}) = K'' \frac{d}{dt}(h^3 - h_S^3) \qquad (4.53)$$

$$\frac{d}{dt}(h^3) = \frac{dh}{dt} \cdot \frac{d}{dh}(h^3) = \frac{dh}{dt} \cdot \frac{d}{dh}\left[f(h)\right] = \frac{dh}{dt} \cdot \frac{d}{dh}\left[f(h_S) + \left\{\frac{df(h)}{dh}\right\}_{h=h_S} (h - h_S) \right]$$

$$= \frac{dh}{dt} \cdot \frac{d}{dh}\left[h_S^3 + 3h_S^2 h - 3h_S^3 \right] = \frac{dh}{dt} \cdot 3h_S^2 = \frac{d(h - h_S)}{dt} \cdot 3h_S^2 \qquad (4.54)$$

Putting Equation 4.54 in Equation 4.53:

$$(q_I - q_{IS}) - (q_O - q_{OS}) = 3h_S^2 K'' \frac{d}{dt}(h - h_S) \tag{4.55}$$

Introducing the non-linearity in resistance and again using the Taylor's expansion,

$$(q_I - q_{IS}) - \frac{C_D}{2\sqrt{h_S}}(h - h_S) = 3h_S^2 K'' \frac{d}{dt}(h - h_S) \tag{4.56}$$

Introducing deviation variables:

$$Q_I - \frac{C_D}{2\sqrt{h_S}} H = 3h_S^2 K'' \frac{dH}{dt}$$

Taking Laplace transform and rearranging the TF, we obtain:

$$\frac{H(s)}{Q_I(s)} = \frac{\dfrac{2\sqrt{h_S}}{C_D}}{\left(\dfrac{h_S^2 \pi D^2}{4H^2}\right)\left(\dfrac{2\sqrt{h_S}}{C_D}\right)s + 1} = \frac{K}{\tau s + 1} \tag{4.57}$$

EXAMPLE 4.11 Two vessel profiles to present two different non-linear capacitances have been shown in Figure 4.10. In (a) the vessel has a spherical shape; in (b) it is a parabolloid.

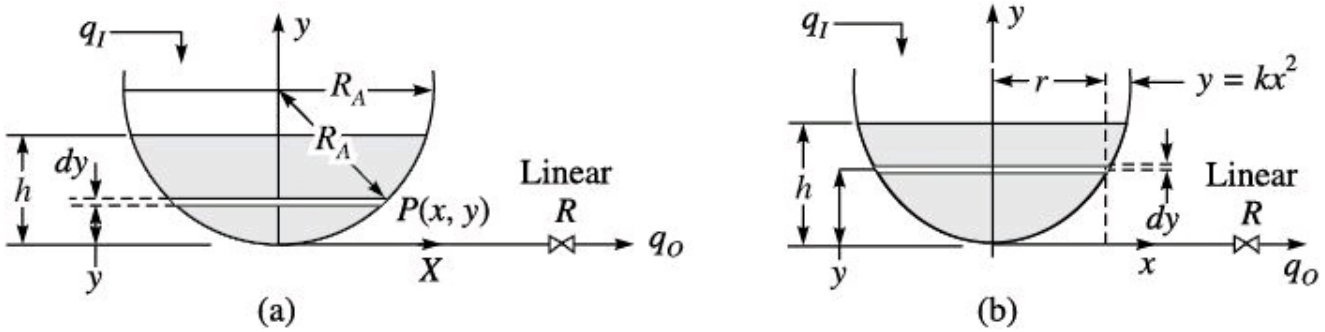

Figure 4.10 Two more varieties of non-linear capacitances.

It is required to find the transfer function $H(s)/Q_I(s)$ for the systems in Figures 4.10(a) and (b).

Solution: The unsteady state mass equation would have the same familiar form for the two vessels:

$$\rho q_I - \rho q_O = \rho \frac{dv}{dt}$$

To find the linearized time derivative of volume in the above equation, first let us relate liquid volume, v, with the liquid level, h, within the vessels as $v = f(h)$,

(a)

At any point P, on a circle,

$$x^2 = R_A^2 - (y - R_A)^2$$

Differential volume of the strip:

$$dv = \pi x^2 dy = \pi[R_A^2 - (y - R_A)^2]dy$$

As the ordinate y represent the level h, to obtain volume up to a level h, we integrate the above expression from $y = 0$ to h.

Thus,

$$v = \pi \int_0^h (2R_A y - y^2)dy = \pi(R_A h^2 - h^3/3)$$

(b)

At any height y, $y = kx^2$;

or, $r^2 = x^2 = y/k$.

The differential volume of the strip:

$$dv = \pi r^2 dy$$

Hence, total tank volume up to a height h may be found by integrating the differential volume expression from $y = 0$ to h.

Thus,

$$v = \frac{\pi}{k} \int_0^h y\, dy = \frac{\pi h^2}{2k}$$

$$v = f(h) = f(h_S) + \left.\frac{df(h)}{dh}\right|_{h=h_S} (h - h_S)$$

$$f(h) = \pi\left[R_A h_S^2 - \frac{h_S^3}{3}\right] + \pi\left[2R_A h_S - h_S^2\right](h - h_S)$$

$$\frac{dv}{dt} = \frac{dh}{dt} \cdot \frac{d[f(h)]}{dh} = \frac{dh}{dt}\pi\left[2R_A h_S - h_S^2\right]$$

$$f(h) = \frac{\pi}{2k} h_S^2 + \frac{\pi h_S}{k}(h - h_S)$$

$$\frac{dv}{dt} = \frac{dh}{dt} \cdot \frac{d[f(h)]}{dh} = \frac{dh}{dt} \cdot \frac{\pi h_S}{k}$$

Putting the time derivative of volume in the mass balance equation:

$$q_I - q_O = \frac{dh}{dt}\pi[2R_A h_S - h_S^2]$$

$$q_I - q_O = \frac{dh}{dt} \cdot \frac{\pi h_S}{k}$$

Subtracting the mass balance equation at steady state condition from the above:

$$(q_I - q_{IS}) - (q_O - q_{OS}) = \frac{d(h - h_S)}{dt}\pi[2R_A h_S - h_S^2]$$

$$(q_I - q_{IS}) - (q_O - q_{OS}) = \frac{d(h - h_S)}{dt} \cdot \frac{\pi h_S}{k}$$

Introducing deviation variables, eliminating the $q_O(t)$ by using the flow–head relation, and taking Laplace transform of the above equations:

$$Q_I(s) - \frac{H(s)}{R} = \pi[2R_A h_S - h_S^2]sH(s)$$

$$Q_I(s) - \frac{H(s)}{R} = \frac{\pi h_S}{k} sH(s)$$

By rearranging we obtain the transfer functions:

$$\frac{H(s)}{Q_I(s)} = \frac{R}{R\pi[2R_A h_S - h_S^2]s + 1} \qquad (4.58)$$

$$\frac{H(s)}{Q_I(s)} = \frac{R}{\frac{R\pi h_S}{k}s + 1} \qquad (4.59)$$

Non linearity may come from Principal Entity transfer characteristics as in radiation heat transfer.

EXAMPLE 4.12 A thermocouple junction of area A, mass m, heat capacity C_P, and emissivity, e, is located in a furnace, normally which is at T_I °C. The thermocouple output is proportional to its junction temperature T_O °C. At these temperatures only radioactive heat transfer prevails. Determine the linearized TF between junction temperature, T_O, and furnace temperature, T_I. Figure 4.11, illustrates the system.

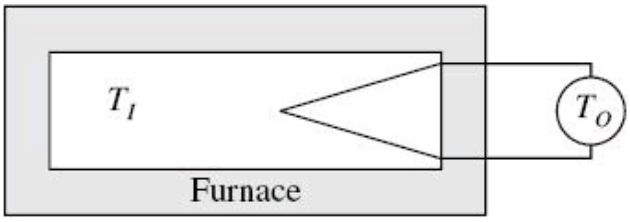

Figure 4.11 Thermocouple temperature sensor in a furnace.

Solution:

Input rate of heat to the Thermocouple junction:

$$W_I = A\sigma\varepsilon(T_I^4 - T_O^4) \text{ erg/sec}$$
$$= KA\sigma\varepsilon(T_I^4 - T_O^4) \text{ Cal/sec}$$

Output rate of heat from Thermocouple junction:

$$W_O = 0 \text{ Cal/sec}$$

Rate of Heat accumulation:

$$mC_P \frac{dT_O}{dt} \text{ Cal/sec}$$

The system parameters are:

ε = Emissivity; σ =Stefan-Boltzmann K

C_P = Heat capacity of junction

m = Mass of the Thermocouple

A = Heat transfer area

The unsteady state heat balance becomes:

$$KA\sigma\varepsilon(T_I^4 - T_O^4) = mC_P \frac{dT_O}{dt}$$

The above equation may be simplified as:

$$T_I^4 - T_O^4 = C\frac{dT_O}{dt} \tag{4.60}$$

where, $C = \dfrac{mC_P}{KA\sigma\varepsilon}$

At steady state Equation (a) becomes:

$$T_{IS}^4 - T_{OS}^4 = C\frac{dT_{OS}}{dt} \tag{4.61}$$

Subtracting Equation 4.61 from Equation 4.60

$$(T_I^4 - T_{IS}^4) - (T_O^4 - T_{OS}^4) = C\frac{d(T_O - T_{OS})}{dt} \tag{4.62}$$

Expanding the non-linear functions:

$$Y_I = f(T_I) = T_I^4 \text{ ; Hence, } T_I^4 = T_{IS}^4 + 4T_{IS}^3(T_I - T_{IS}) \text{ ; and } T_O^4 = T_{OS}^4 + 4T_{OS}^3(T_O - T_{OS})$$

Putting the above results in Equation 4.62

$$4T_{IS}^3(T_I - T_{IS}) - 4T_{OS}^3(T_O - T_{OS}) = C\frac{d(T_O - T_{OS})}{dt} \qquad (4.63)$$

Introducing deviation variables, and recognizing that at steady stare, $T_{IS} = T_{OS}$, Equation 4.63 becomes:

$$\bar{T}_I - \bar{T}_O = \frac{C}{4T_{IS}^3} \cdot \frac{d\bar{T}_O}{dt} \qquad (4.64)$$

Taking L.T and rearranging

$$\frac{\bar{T}_O(s)}{\bar{T}_I(s)} = \frac{1}{\tau * s + 1} \qquad (4.65)$$

where, $\tau^* = \dfrac{mC_P}{KA\sigma\varepsilon 4T_{IS}^3}$

4.4 Generalized Models for Stirred Tank Mixer Systems

In process applications, **Stirred Pots** are frequently used for *Heat* and/or *Mass Transfer* operations. A single liquid stream may be heated or cooled by adding or subtracting energy from the stream. The blending of two or more number of incoming liquid streams carrying miscible species may happen adiabatically or by supplying energy (non-adiabatically). In certain applications, a solution stream may require to be diluted by adding pure solvent or has to be concentrated by adding a thicker solution, before it is sent to the next processing stage. Practically, in all such applications, a vessel (capacity element) is used in which mixing takes place either by a mechanical stirrer or by a strategically placed baffle arrangement to obtain a fairly homogeneous exit stream. We may consider the flowing fluid through the pot as a *vehicle fluid* that caries the primary variable, e.g. temperature, and/or concentration.

The stirred reactors are further examples in this line that feature a rate process along with the hydrodynamics.

Generally, the stirred pots may be of two operational categories:

(a) **Constant Holdup**—Where the exit stream is taken out at a certain height from the tank so that the working volume or the tank holdup volume always remains constant. In this condition the exit flow rate is equal to that of the inlet flow rate of the liquid,

(b) **Variable Volume** —In such a system the exit line is located at the bottom of the tank, and this line contains an adjustable restriction or resistance represented by a

valve. Thus, the outlet flow q_O will be proportional to the tank level h, according to which $q_O = h/R$, where, R is the linear resistance of the valve. Except in certain systems where supply dynamics of a liquid from its constantly fed storage is required to be evaluated, the latter variety has a very limited use in process applications. These two configurations of the stirred pots have been shown in Figure 4.12.

Analysis of such systems assuming perfect or ideal stirring condition may be done with little effect to the accuracy of the results obtained. We should remember that in the Examples 4.5, 4.6 and 4.7, certain systems of this class have been discussed for specific applications. In the following analysis, we have tried to forward a more general solution of the representative example systems identified in Figure 4.12.

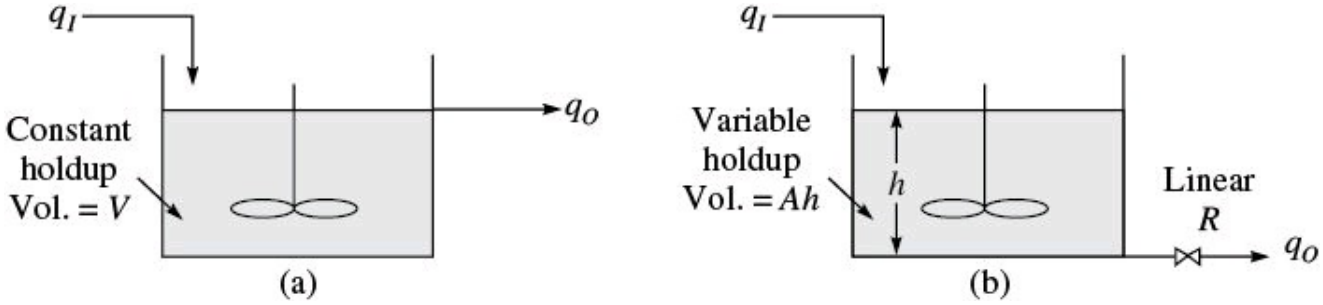

Figure 4.12 The configurations of stirred pot operation (a) constant holdup, (b) variable holdup.

During the analysis of the following example systems, we did not always spell all the assumptions made and the formal way of introducing the deviation variables. Because some simpler configurations of these systems have been already taken up in early part of the previous and present chapter where we forwarded these details. It has been assumed that by this time the reader can fill up these missing steps.

4.4.1 Enthalpy Transfer Systems

Energy transfer in situ with material flow in a system increases the model complexity by giving forth more number of differential equation to be solved. Let us establish the idea with the help of following examples.

EXAMPLE 4.13 (Non-Adiabatic Mixing in a Variable Holdup Mixer): [$q_I(t)$, $T_I(t)$, $h(t)$, $W(t)$ □ $T_O(t)$].

In this example we build up the linear model for the dependence of $T_O(t)$, exit liquid temperature, on:

1. Inflow rate $q_I(t)$,
2. Tank level $h(t)$,
3. Inlet liquid temperature $T_I(t)$ and
4. Heat input $W(t)$.

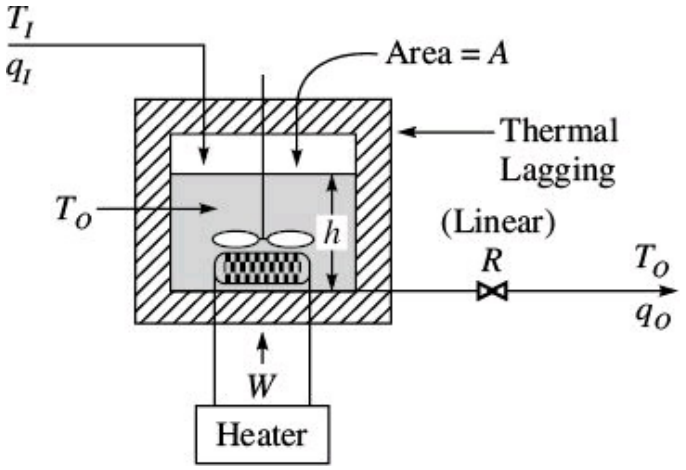

Figure 4.13 Non-Adiabatic mixing in a variable holdup mixer.

With reference to the Figure 4.13, the mass balance would be,

$$\rho q_I - \rho q_O = \rho A \frac{dh}{dt} \qquad \text{or} \qquad \frac{dh}{dt} = \frac{q_I}{A} - \frac{q_O}{A} \qquad (4.66)$$

From Equation 4.66 the transfer function between $H(s)$ and $Q_I(s)$, may be derived as:

$$\frac{Q_O(s)}{Q_I(s)} = \frac{1}{ARs+1} \qquad \text{and} \qquad \frac{H(s)}{Q_I(s)} = \frac{R}{ARs+1} \qquad (4.67)$$

The corresponding heat balance after expanding the derivative term becomes:

$$\rho q_I C_P (T_I - T_D) + W - \rho q_O C_P (T_O - T_D) = \rho C_P \frac{d(VT_O)}{dt} = \rho C_P Ah \frac{dT_O}{dt} + \rho C_P A T_O \frac{dh}{dt} \qquad (4.68)$$

(It should be mentioned that beside electrical heating, the heat exchange facility also may occur through the vessel wall from a jacket or an immersed tube coil bank carrying the heat transfer fluid, and heat addition term, W, may take the following form:

$$W = UA(T_J - T_O)$$

where, T_J = Jacket fluid temperature.

In such a description, the time dependency of $W(t)$ may be attributed to $T_J(t)$.)

Assuming the Datum temperature, T_D = 0, and putting expression of dh/dt into Equation 4.68 from Equation 4.66 and simplifying the result, we obtain:

$$q_I T_I - q_O T_O + \frac{W}{\rho Cp} = Ah \frac{dT_O}{dt} + AT_O \frac{q_I}{A} - AT_O \frac{q_O}{A}$$

or,

$$\frac{dT_O}{dt} = \frac{1}{A}\left[q_I T_I h^{-1}\right] - \frac{1}{A}\left[q_I T_O h^{-1}\right] + \frac{1}{A\rho C_P}\left[W h^{-1}\right] \qquad (4.69)$$

The R.H.S. of Equation 4.69 contains product of time dependent variables that

requires to be treated as non-linear functions. Hence, expanding each of these terms by Taylor's series, we obtain,

$$q_I T_I h^{-1} = q_{IS} T_{IS} h_S^{-1} + q_{IS} h_S^{-1}(T_I - T_{IS}) + T_{IS} h_S^{-1}(q_I - q_{IS}) - q_{IS} T_{IS} h_S^{-2}(h - h_S) \quad (4.70)$$

$$q_I T_O h^{-1} = q_{IS} T_{OS} h_S^{-1} + q_{IS} h_S^{-1}(T_O - T_{OS}) + T_{OS} h_S^{-1}(q_I - q_{IS}) - q_{IS} T_{OS} h_S^{-2}(h - h_S) \quad (4.71)$$

$$W h^{-1} = W_S h_S^{-1} + h_S^{-1}(W - W_S) - W_S h_S^{-2}(h - h_S) \quad (4.72)$$

Writing Equation 4.69 in the difference form between absolute and steady state value of the variables, and putting the expressions of differences from the Equations 4.70, 4.71 and 4.72:

$$\frac{d(T_O - T_{OS})}{dt} = \frac{1}{A}\left[\frac{q_{IS}}{h_S}(T_I - T_{IS}) + \frac{T_{IS}}{h_S}(q_I - q_{IS}) - \frac{q_{IS}T_{IS}}{h_S^2}(h - h_S)\right] -$$

$$\frac{1}{A}\left[\frac{q_{IS}}{h_S}(T_O - T_{OS}) + \frac{T_{OS}}{h_S}(q_I - q_{IS}) - \frac{q_{IS}T_{OS}}{h_S^2}(h - h_S)\right] + \frac{1}{A\rho Cp}\left[\frac{W - W_S}{h_S} - \frac{W_S(h - h_S)}{h_S^2}\right]$$

Putting the corresponding deviation variables in the above equation:

$$\frac{d\bar{T}_O}{dt} = \frac{1}{A}\left[\frac{q_{IS}}{h_S}\bar{T}_I + \frac{T_{IS}}{h_S}Q_I - \frac{q_{IS}T_{IS}}{h_S^2}H\right] - \frac{1}{A}\left[\frac{q_{IS}}{h_S}\bar{T}_O + \frac{T_{OS}}{h_S}Q_I - \frac{q_{IS}T_{OS}}{h_S^2}H\right] +$$

$$\frac{1}{A\rho C_P}\left[\frac{\bar{W}}{h_S} - \frac{W_S}{h_S^2}H\right] \quad (4.73)$$

By taking Laplace transform of both sides of Equation 4.73 and rearranging, we get:

$$\bar{T}_O(s)\left[s + \frac{q_{IS}}{Ah_S}\right] = \bar{T}_I(s)\left[\frac{q_{IS}}{Ah_S}\right] + Q_I(s)\left[\frac{T_{IS} - T_{OS}}{Ah_S}\right] + \bar{W}(s)\frac{1}{A\rho C_P h_S} +$$

$$H(s)\left[\frac{q_{IS}T_{OS}}{Ah_S^2} - \frac{q_{IS}T_{IS}}{Ah_S^2} - \frac{W_S}{A\rho C_P h_S^2}\right]$$

or,

$$\bar{T}_O(s) = \bar{T}_I(s)\left[\frac{1}{(Ah_S/q_{IS})s + 1}\right] + Q_I(s)\left[\frac{(T_{IS} - T_{OS})/q_{IS}}{(Ah_S/q_{IS})s + 1}\right] + \bar{W}(s)\left[\frac{1/(\rho C_P q_{IS})}{(Ah_S/q_{IS})s + 1}\right] +$$

$$H(s)\left[\frac{(T_{OS} - T_{IS})/h_S - W_S/(\rho C_P h_S q_{IS})}{(Ah_S/q_{IS})s + 1}\right] \quad (4.74)$$

Equations 4.66 and 4.69 are the state equations for this system. Equations 4.67 and

4.74 are the corresponding transfer functions. The system features two output variables H and $\bar{T}_O$. Equation 4.67 bears the relation of H with Q_I and Equation 4.74 describe the relation of $\bar{T}_O$ with the inputs $\bar{T}_I$, Q_I, and $\bar{W}$. Note, that Q_I affects $\bar{T}_O$ in two ways:

1. through the 2nd term in R.H.S. of Equation 4.74, and
2. through Equations 4.66 and the 4th term in the R.H.S. of Equation 4.74. These two are parallel paths and the total change of $\bar{T}_O$ may be obtained by summing up the two effects by way of supper-imposition. The significance of these two paths is that path:

   1. describes the change in the rate of enthalpy input to the system due to variation of inlet flow, and path,
   2. is the effect of change of system thermal capacity due to change in level (h is a measure of the retained mass), of liquid within the mixer. Path 2. contains two serially connected 1st order TFs constituting a 2nd order dynamics. In Chapter 6 we shall discuss about the 2nd order systems obtained by serially connected 1st order systems.

Equation 4.74 is a linear relation and we should remember that response study is allowable corresponding to only one input variable considering others to be constant. However, additive super-imposition of response change for multiple input variations is possible, but that may lead to a sacrifice of accuracy due to linear approximation.

To show how this generalized relation may be used in a specific application, let us consider the dependence of $\bar{T}_O$ on inlet liquid temperature $\bar{T}_I$ and heat input heat addition rate $\bar{W}$. As the inlet liquid flow rate Q_I and the tank level H must be considered as constant the two resulting transfer functions may be written as:

$$\frac{\bar{T}_O(s)}{\bar{T}_I(s)} = \left[\frac{1}{(Ah_S/q_{IS})s+1}\right] = \frac{1}{(V/q_I)s+1} \quad \text{and} \quad \frac{\bar{T}_O(s)}{\bar{W}(s)} = \left[\frac{1/(\rho C p q_{IS})}{(Ah_S/q_{IS})s+1}\right] = \frac{1/(\rho C p q_I)}{(V/q_I)s+1}$$

where, V = The operating tank volume at constant q_I. These are the results that we obtained in the Example 4.6. Hence, it is apparent that from the generalized model represented by the transfer function of Equation 4.74, the solution for any of the special cases may be obtained by considering variable terms to be constants specified by the problem statement. To demonstrate this notion, in the next example the corresponding constant hold up condition considered in the previous example has been analyzed. The result would show that we could have obtained the same by assuming that $h = h_S$, and $V = A.h_S$ = Hold up volume of the stirred heater, into the

final transfer function of the above example.

Equations 4.67 and 4.74 may be coupled in the block diagram of Figure 4.14, to describe the multiple input, two output relation.

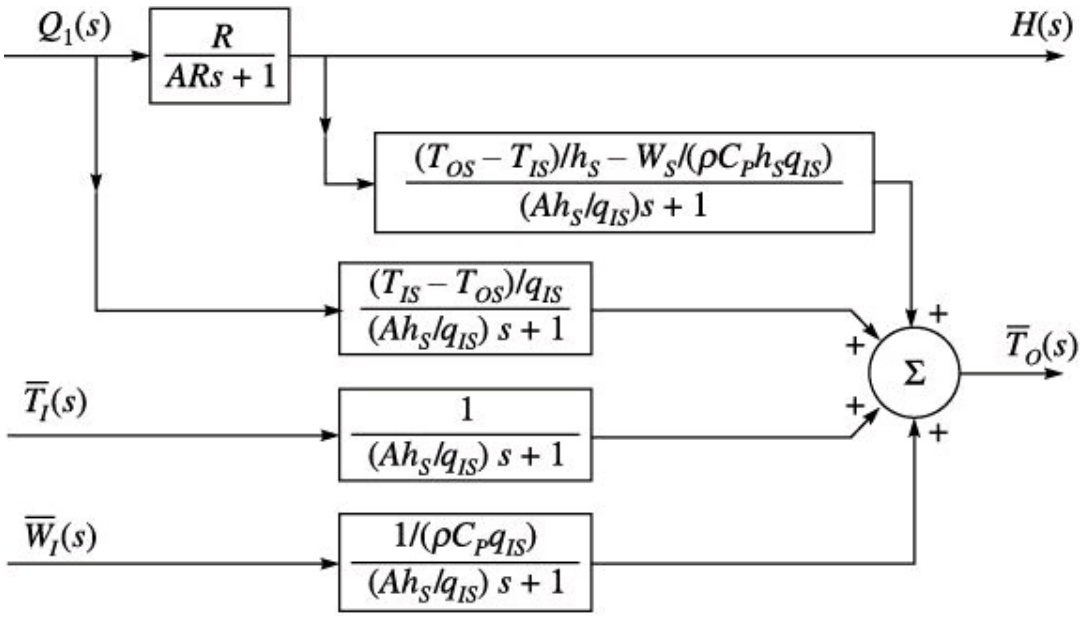

Figure 4.14 The linearized model of thermal dynamics in a stirred tank.

The corresponding constant hold up model may be readily developed by slightly modifying the assumptions as in the following example.

EXAMPLE 4.14 (Non-Adiabatic Mixing in a Constant Holdup Mixer): [q_I (t), T_I (t),
W (t) □ T_O (t)].

In this example, we build up the linear model for the dependence of T_O (t), exit liquid temperature $T_O(t)$, on the:

1. Inflow rate q_I (t),
2. Inlet liquid temperature T_I (t) and
3. Heat input $W(t)$.

Solution: With a constant hold up condition it follows that the input and exit flow rates are equal:

i.e. $$q_1 = q_0 = q \qquad (4.75)$$

There will be no accumulation of mass, and we need not write the dynamic mass balance.

Also $Ah = V$ (the hold up volume). The corresponding heat balance equation would be:

$$\rho q C_P T_I + W - \rho q C_P T_O = \rho C_P V \frac{dT_O}{dt} \qquad (4.76)$$

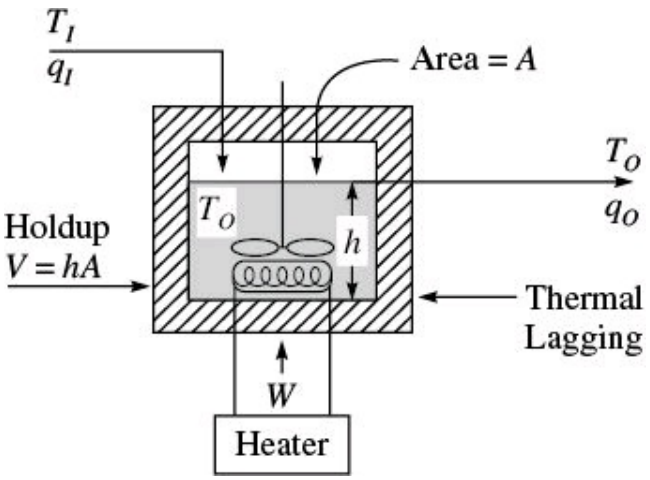

Figure 4.15 Non-Adiabatic mixing in a constant holdup mixer.

Following the procedure of expanding of variable product terms according to Taylor's series, introducing deviation variables as we did for the previous example:

$$\frac{d\bar{T}_O}{dt} = \left[\frac{q_S}{V}\bar{T}_I + \frac{T_{IS}}{V}Q\right] - \left[\frac{q_S}{V}\bar{T}_O + \frac{T_{OS}}{V}Q\right] + \frac{1}{\rho CpV}\bar{W} \qquad (4.77)$$

By taking Laplace transform of Equation 4.77 and rearranging, the composite transfer function becomes:

$$\bar{T}_O(s) = \bar{T}_I(s)\left[\frac{1}{(V/q_S)s+1}\right] + Q(s)\left[\frac{(T_{IS}-T_{OS})/q_S}{(V/q_S)s+1}\right] + \bar{W}(s)\left[\frac{1/(\rho C p q_S)}{(V/q_S)s+1}\right] \qquad (4.78)$$

Equation 4.78 may be expressed in block diagram as in Figure 4.16, to describe the multiple input, single output relation.

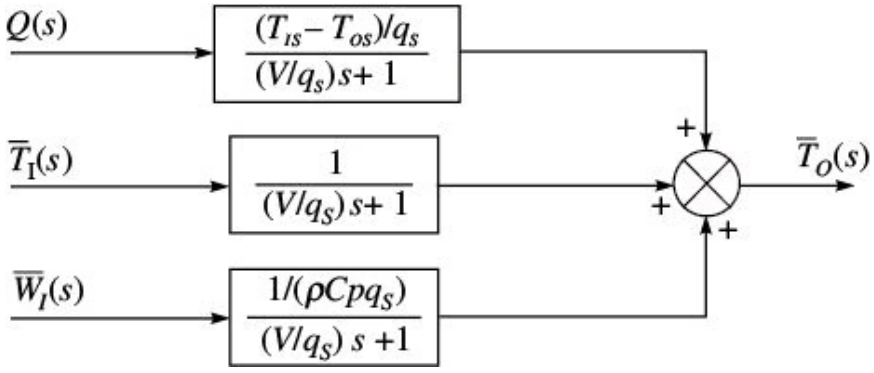

Figure 4.16 The linearized model of thermal dynamics in a constant holdup stirred tank.

4.4.2 Mass Transfer Systems

The problem of mass transfer (mixing and associated concentration change) may be modeled in the similar system description of stirred pots with or without a kinetic rate process. However, to enhance the generality of the developed model we shall include an *n*-th order irreversible reaction as the kinetic process. If there is no reaction, it is

only a mixing or blending system. For the sake of simplicity we assume in either case the system operates in isothermal condition, i.e. if there is any reaction, it runs without any heat effect, $\Delta H_R = 0$.

EXAMPLE 4.15 (Isothermal n-th Order Irreversible Reaction in a Variable Holdup Stirred Reactor): [$q_I(t)$, $C_I(t)$, $h(t)$ □ $C_O(t)$].

In this example, we build up the linear model for the dependence of exit concentration of the reacting species $C_O(t)$, on the:

1. Inflow rate $q_I(t)$;
2. Concentration of the reacting species in the Inlet liquid stream $C_I(t)$; and
3. Tank level $h(t)$.

There is an n-th order irreversible reaction: A □ Product, occurring in the reactor. Its kinetics may be expressed as:

$$r = -\frac{dC_O}{dt} = kC_O^n$$

It is further assumed that no density change occurs due to concentration variation during reaction. Such a system may be represented by the following Figure 4.17.

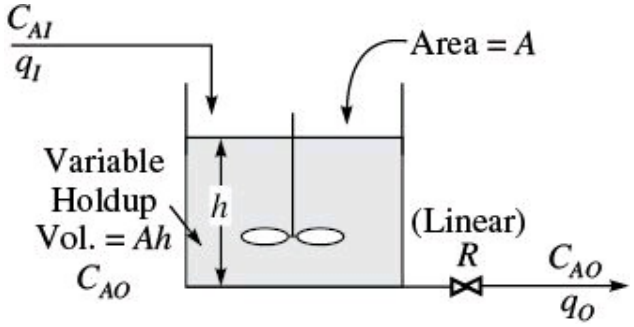

Figure 4.17 Isothermal reaction in a variable holdup stirred reactor.

With reference to Figure 4.17, and the previous example making a similar mass balance,

$$\rho q_I - \rho q_O = \rho A \frac{dh}{dt} \qquad \text{or} \qquad \frac{dh}{dt} = \frac{q_I}{A} - \frac{q_O}{A} \qquad (4.79)$$

The transfer function between $H(s)$ and $Q_I(s)$, may be directly written as:

$$\frac{Q_O(s)}{Q_I(s)} = \frac{1}{ARs+1} \qquad \text{and} \qquad \frac{H(s)}{Q_I(s)} = \frac{R}{ARs+1} \qquad (4.80)$$

In the following mass balance of the reacting component, we can omit the density term, because the concentrations either expressed on the basis of volume or mass need not carry the density term. After expanding the derivative we obtain:

$$q_I C_I - kAhC_O^n - q_O C_O = \frac{d(VC_O)}{dt} = Ah\frac{dC_O}{dt} + AC_O\frac{dh}{dt} \qquad (4.81)$$

Putting Equation 4.79 into Equation 4.81,

$$q_I C_I - q_O C_O - kAhC_O^n = Ah\frac{dC_O}{dt} + AC_O\frac{q_I}{A} - AC_O\frac{q_O}{A}$$

or,

$$\frac{dC_O}{dt} = \frac{1}{A}\left[q_I C_I h^{-1}\right] - \frac{1}{A}\left[q_I C_O h^{-1}\right] - kC_O^n \qquad (4.82)$$

The R.H.S. of Equation 4.82 contains product of time dependent variables and the kinetic disappearance term that need to be expanded by Taylor's series we obtain,

$$q_I C_I h^{-1} = q_{IS} C_{IS} h_S^{-1} + q_{IS} h_S^{-1}(C_I - C_{IS}) + C_{IS} h_S^{-1}(q_I - q_{IS}) - q_{IS} C_{IS} h_S^{-2}(h - h_S) \qquad (4.83)$$

$$q_I C_O h^{-1} = q_{IS} C_{OS} h_S^{-1} + q_{IS} h_S^{-1}(C_O - C_{OS}) + C_{OS} h_S^{-1}(q_I - q_{IS}) - q_{IS} C_{OS} h_S^{-2}(h - h_S) \qquad (4.84)$$

$$C_O^n = C_{OS}^n + nC_{OS}^{n-1}(C_O - C_{OS}) \qquad (4.85)$$

Writing Equation 4.82 in the difference form between absolute and steady state value of the variables:

$$\frac{d(C_O - C_{OS})}{dt} = \frac{1}{A}\left[\frac{q_{IS}}{h_S}(C_I - C_{IS}) + \frac{C_{IS}}{h_S}(q_I - q_{IS})\frac{q_{IS}C_{IS}}{h_S^2}(h - h_S)\right] -$$

$$\frac{1}{A}\left[\frac{q_{IS}}{h_S}(C_O - C_{OS}) + \frac{C_{OS}}{h_S}(q_I - q_{IS}) - \frac{q_{IS}C_{OS}}{h_S^2}(h - h_S)\right] - k(C_O^n - C_{OS}^n)$$

Now putting the expressions of differences from the Equations 4.83, 4.84 and 4.85:

$$\frac{d(C_O - C_{OS})}{dt} = \frac{1}{A}\left[\frac{q_{IS}}{h_S}(C_I - C_{IS}) + \frac{C_{IS}}{h_S}(q_I - q_{IS}) - \frac{q_{IS}C_{IS}}{h_S^2}(h - h_S)\right]$$

$$-\frac{1}{A}\left[\frac{q_{IS}}{h_S}(C_O - C_{OS}) + \frac{C_{OS}}{h_S}(q_I - q_{IS}) - \frac{q_{IS}C_{OS}}{h_S^2}(h - h_S)\right] - nkC_{OS}^{n-1}(C_O - C_{OS})$$

Putting the corresponding deviation variables in the above equation:

$$\frac{d\bar{C}_O}{dt} = \frac{1}{A}\left[\frac{q_{IS}}{h_S}\bar{C}_I + \frac{C_{IS}}{h_S}Q_I - \frac{q_{IS}C_{IS}}{h_S^2}H\right] - \frac{1}{A}\left[\frac{q_{IS}}{h_S}\bar{C}_O + \frac{C_{OS}}{h_S}Q_I - \frac{q_{IS}C_{OS}}{h_S^2}H\right] - nkC_{OS}^{n-1}\bar{C}_O \qquad (4.86)$$

By taking Laplace transform of both sides of Equation 4.86 and rearranging,

$$\bar{C}_O(s)\left[s+\frac{q_{IS}}{Ah_S}+nkC_{OS}^{n-1}\right]=\bar{C}_I(s)\left[\frac{q_{IS}}{Ah_S}\right]+Q_I(s)\left[\frac{C_{IS}-C_{OS}}{Ah_S}\right]+H(s)\left[\frac{q_{IS}C_{OS}}{Ah_S^2}-\frac{q_{IS}C_{IS}}{Ah_S^2}\right]$$

or,

$$\bar{C}_O(s)=\bar{C}_I(s)\left[\frac{q_{IS}/(q_{IS}+Ah_SnkC_{OS}^{n-1})}{[Ah_S/(q_{IS}+Ah_SnkC_{OS}^{n-1})]s+1}\right]+Q_I(s)\left[\frac{(C_{IS}-C_{OS})/(q_{IS}+Ah_SnkC_{OS}^{n-1})}{[Ah_S/(q_{IS}+Ah_SnkC_{OS}^{n-1})]s+1}\right]+$$

$$H(s)\left[\frac{q_{IS}(C_{OS}-C_{IS})/[h_S(q_{IS}+Ah_SnkC_{OS}^{n-1})]}{[Ah_S/(q_{IS}+Ah_SnkC_{OS}^{n-1})]s+1}\right] \qquad (4.87)$$

Equation 4.87 may be expressed in block diagram as in Figure 4.18, to describe the multiple input, single output relation.

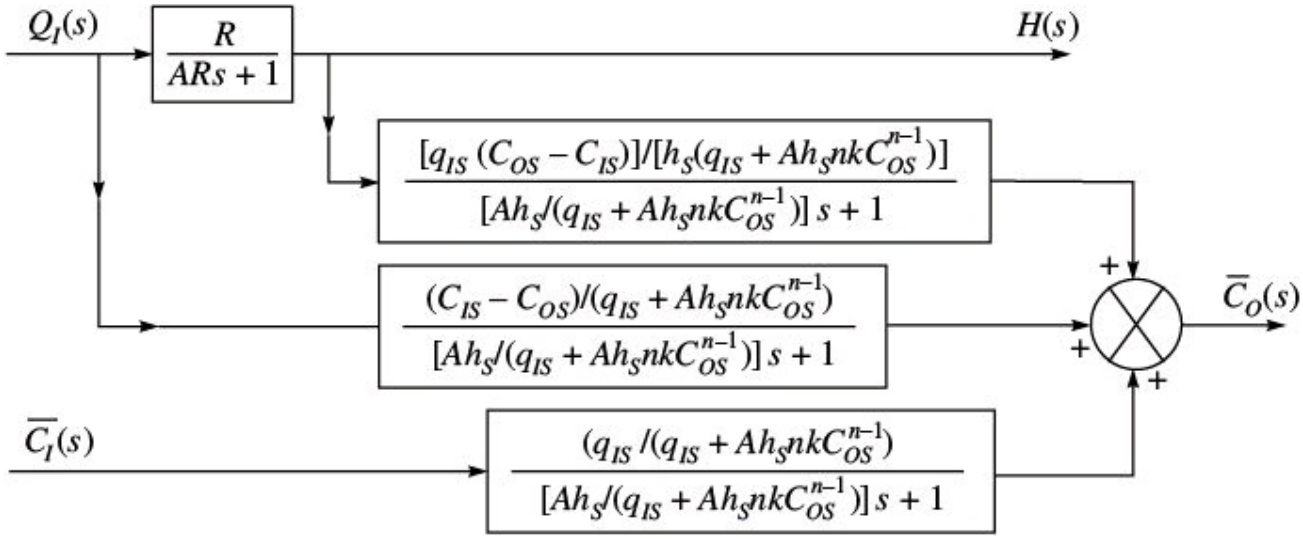

Figure 4.18 The linearized model of solute concentration dynamics in a variable holdup stirred tank reactor.

The model for the corresponding constant hold up condition may be developed as we have done in Example 4.15, following a similar approach.

EXAMPLE 4.16 (Isothermal n-th Order Irreversible Reaction in a Constant Holdup Stirred Reactor): $[q_I(t), C_I(t) \rightarrow C_O(t)]$.

In this example we present the linear model for the dependence of $C_O(t)$, the Exit concentration of the reacting species, on the inflow rate $q_I(t)$, and the Concentration of the reacting species in the Inlet liquid stream $C_I(t)$.

The preliminary assumptions made in Example 4.15 are again valid in this problem, that is, for a constant hold up condition, as the input and exit flow rates are equal:

$$q_1 = q_0 = q \qquad (4.88)$$

and, $Ah = V$ (the hold up volume) (4.89)

We may proceed either from the component mass balance and following the steps of Example 4.16, or by reducing the obtained model of Example 4.16 to fit in the present problem by making the adequate changes that follows from the stated conditions for Equations 4.88 and 4.89.

The final form of the transfer function may be given as:

$$\text{or,} \quad \bar{C}_O(s) = \bar{C}_I(s)\left[\frac{q_S/(q_S + nkVC_{OS}^{n-1})}{[V/(q_S + nkVC_{OS}^{n-1})]s + 1}\right] + Q(s)\left[\frac{(C_{IS} - C_{OS})/(q_S + nkVC_{OS}^{n-1})}{[V/(q_S + nkVC_{OS}^{n-1})]s + 1}\right] \qquad (4.90)$$

Development of the corresponding block diagram has been left to the reader as an exercise.

To demonstrate how the generalized model could be used for developing TFs for specific problems we consider the following cases.

Let us work out some examples with specific system descriptions to demonstrate how these above may be used for these applications.

EXAMPLE 4.17 1. Develop the TF for an Isothermal 1st Order Irreversible Reaction in a Constant Holdup Stirred Reactor: [$q_I(t)$, $C_I(t)$ □ $C_O(t)$].

For the given problem, it is sufficient to put $n = 1$ in the kinetic disappearance term of Equation 4.90. This term becomes kV for 1st order reaction and the required TF is obtained as:

$$\bar{C}_O(s) = \bar{C}_I(s)\left[\frac{q_S/(q_S + kV)}{[V/(q_S + kV)]s + 1}\right] + Q(s)\left[\frac{(C_{IS} - C_{OS})/(q_S + kV)}{[V/(q_S + kV)]s + 1}\right] \qquad (4.91)$$

with the same general conditions for constant reactant inflow rate, the 2nd term in the R.H.S. of Equation 4.91 is discarded, that describe the flow rate dependency, and $q = q_S$ is put to get the required TF as:

$$\bar{C}_O(s) = \bar{C}_I(s)\left[\frac{q/(q + kV)}{[V/(q + kV)]s + 1}\right] \qquad (4.92)$$

and for only mixing without any reaction, the TF is obtained by making the reaction disappearance term $kV = 0$, in Equation 4.92,

$$\bar{C}_O(s) = \bar{C}_I(s)\left[\frac{1}{(V/q)s + 1}\right] \qquad (4.93)$$

2. Develop the TF for an Isothermal 2nd Order Irreversible Reaction in a Constant Holdup Stirred Reactor: [$q_I(t)$, $C_I(t)$ □ $C_O(t)$].

For the given problem it is sufficient to put $n = 2$ in the kinetic disappearance term of Equation 4.90. This term becomes $2kVC_{OS}$ for 2nd order reaction and the required

TF is obtained as:

$$\bar{C}_O(s) = \bar{C}_I(s)\left[\frac{q_S/(q_S + 2kVC_{OS})}{[V/(q_S + 2kVC_{OS})]s + 1}\right] + Q(s)\left[\frac{(C_{IS} - C_{OS})/(q_S + 2kVC_{OS})}{[V/(q_S + 2kVC_{OS})]s + 1}\right] \qquad (4.94)$$

with the same general conditions for constant reactant inflow rate the 2nd term in the R.H.S of Equation 4.91, is discarded that describe the flow rate dependency, and $q = q_S$ is put to get the required TF as:

$$\bar{C}_O(s) = \bar{C}_I(s)\left[\frac{q/(q + 2kVC_{OS})}{[V/(q + 2kVC_{OS})]s + 1}\right] \qquad (4.95)$$

We have already obtained many of these results in the examples discussed in Chapter 2 and in the early part of this chapter, while treating individual problems of this kind. The generalized models in the last four examples have been included to develop an overview on the enthalpy and mass transfer systems in stirred pots.

The reader should be able to compare the situations of adiabatic ($W = 0$) thermal dynamics with only mixing no reaction[$nkVC_{OS}^{n-1} = 0$] type concentration dynamics situations. It follows that process gain in the TFs relating output to input temperature or concentration would be unity ($K_P = 1$) in adiabatic or no reaction conditions.

Also, the fact that the time constant, as the coefficient of Laplace operator 's' in the denominator of the TFs is only dependent on the inflow rate q_I , indicates that for variable flow the system has a **Non-Stationary** character.

But what if enthalpy and mass transfer simultaneously occur associated with an exothermic reaction in a perfectly stirred pot? This is the classical problem of *Exothermic CSTR Dynamics and Control* probably first formally investigated by Amundson and Aris in 1958[1] and since then has attracted a lot of research interest. At present, only the linear approximation of a simpler version of the system model is presented.

EXAMPLE 4.18 (Exothermic Jacketed CSTR): A reactant solution stream is fed to a constant holdup ideally stirred reactor. The feed temperature and concentration are C_I, T_I respectively. The exothermic, irreversible, constant density, liquid phase reaction proceeds according to:

$$A \rightarrow \text{Product};$$

The kinetics of the reaction is: $r = -\frac{dC_O}{dt} = kC_O$; The heat of reaction = $\Delta H_R \frac{\text{cal}}{\text{mol}}$;

The temperature effect on rate constant may be expressed as: $k = k_O.e^{-E/RT}\,\text{time}^{-1}$.

The reactor has an outside jacket that carries the cooling fluid. The feed flow rate and jacket liquid temperature may be assumed constant at q and T_J. The heat transfer occurs through an area $= A$ and the overall heat transfer coefficient is U. The exit liquid concentration C_O and temperature T_O, are similar to that in the bulk liquid in the reactor. It is required to develop dynamic relations between C_O and T_O with C_I and T_I.

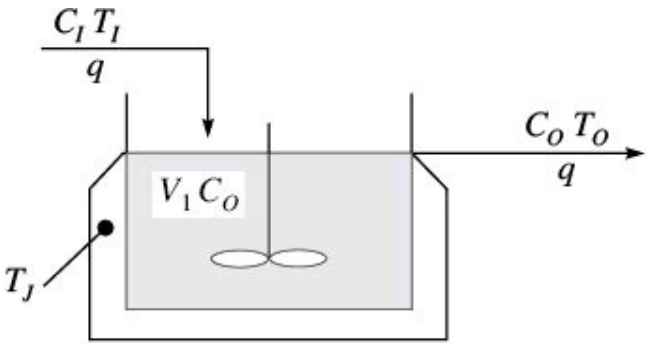

Figure 4.19 The constant holdup exothermic jacketed CSTR.

Solution:

The Mass Balance:

$$qC_I - qC_O - kVC_O = V\frac{dC_O}{dt} \tag{4.96}$$

The Heat Balance:

$$q\rho C_P T_I - q\rho C_P T_O - UA(T_O - T_J) + VkC_O\Delta H_R = V\rho C_P\frac{dT_O}{dt} \tag{4.97}$$

Only the 4th term in the L.H.S of Equation 4.97 is a product two variables due to the temperature dependency of reaction rate constant (Arhenius equation). To linearize, we expand this term according to Taylor's Series:

$$kC_O = k_O e^{-E/RT_O(t)}C_O(t) = (kC_O)_{SS} + \left.\frac{\partial(kC_O)}{\partial T_O}\right|_{SS}(T_O - T_{OS}) + \left.\frac{\partial(kC_O)}{\partial C_O}\right|_{SS}(C_O - C_{OS})$$

or

$$kC_O - (kC_O)_{SS} = \frac{k_O E C_{OS}}{RT_{OS}^2}e^{-E/RT_{OS}}(T_O - T_{OS}) + k_O e^{-E/RT_{OS}}(C_O - C_{OS})$$

or

$$kC_O - (kC_O)_{SS} = K_1(T_O - T_{OS}) + K_2(C_O - C_{OS}) \tag{4.98}$$

Writing Equations 4.96 and 4.97 in the difference form between absolute and steady state values of the variables, and putting Equation 4.98 into the difference:

$$q(C_I - C_{IS}) - q(C_O - C_{OS}) - V[(kC_O) - (kC_O)_{SS}] = V\frac{d(C_O - C_{OS})}{dt} \quad (4.99)$$

and,

$$q\rho C_P(T_I - T_{IS}) - q\rho C_P(T_O - T_{OS}) - UA[(T_O - T_{OS}) - (T_J - T_J)] +$$

$$V\Delta H_R[(kC_O) - (kC_O)_{SS}] = V\rho C_P\frac{d(T_O - T_{OS})}{dt} \quad (4.100)$$

Introducing the deviation variables:

$$\bar{C}_I = C_I - C_{IS};\quad \bar{C}_O = C_O - C_{OS};\quad \bar{T}_I = T_I - T_{IS};\quad \bar{T}_O = T_O - T_{OS}$$

into equations 4.99 and 4.100:

we obtain:

$$q\bar{C}_I - q\bar{C}_O - VK_1\bar{T}_O + VK_2\bar{C}_O = V\frac{d\bar{C}_O}{dt} \quad (4.101)$$

and

$$q\rho C_P\bar{T}_I - q\rho C_P\bar{T}_O - UA\bar{T}_O + V\Delta H_R K_1\bar{T}_O - V\Delta H_R K_2\bar{C}_O = V\rho C_P\frac{d\bar{T}_O}{dt} \quad (4.102)$$

Rearranging equations 4.101 and 4.102:

$$\frac{q}{q + VK_2}\bar{C}_I - \frac{VK_1}{q + VK_2}\bar{T}_O = \frac{V}{q + CK_2}\cdot\frac{d\bar{C}_O}{dt} + \bar{C}_O \quad (4.103)$$

and,

$$\frac{q\rho.Cp}{q\rho C_P + UA - V\Delta H_R K_1}\bar{T}_I + \frac{V\Delta H_R K_2}{q\rho C_P + UA - V\Delta H_R K_1}\bar{C}_O =$$

$$\frac{V\rho Cp}{q\rho C_P + UA - V\Delta H_R K_1}\cdot\frac{d\bar{T}_O}{dt} + \bar{T}_O \quad (4.104)$$

To simplify the TFs to standard form, introduce the following constants:

$$\frac{q}{q + VK_2} = K_{11};\quad \frac{VK_1}{q + VK_2} = K_{12};\quad \frac{V}{q + CK_2} = \tau_1$$

$$\frac{q\rho C_P}{q\rho C_P + UA - V\Delta H_R K_1} = K_{21};\quad \frac{V\Delta H_R K_2}{q\rho C_P + UA - V\Delta H_R K_1} = K_{22};\quad \frac{V\rho C_P}{q\rho C_P + UA - V\Delta H_R K_1} = \tau_2$$

Introducing these constants into the last pair of equations 4.103 and 4.104, and taking Laplace Transform of the resulting equations:

$$K_{11}\bar{C}_I(s) - K_{12}\bar{T}_O(s) = \tau_1 s\bar{C}_O(s) + \bar{C}_O(s)$$

and,

$$K_{21}\bar{T}_I(s) + K_{22}\bar{C}_O(s) = \tau_2 s\bar{T}_O(s) + \bar{T}_O(s)$$

Rearranging:

$$\bar{C}_O(s) = \frac{K_{11}}{\tau_1 s + 1}\bar{C}_I(s) - \frac{K_{12}}{\tau_1 s + 1}\bar{T}_O(s) \tag{4.105}$$

$$\bar{T}_O(s) = \frac{K_{21}}{\tau_2 s + 1}\bar{T}_I(s) - \frac{K_{22}}{\tau_2 s + 1}\bar{C}_O(s) \tag{4.106}$$

So this is your second introduction to a system that has more than one input and output variables. The first one has been presented as Example 3.6 in the previous chapter. Such a multiple input – multiple output process is called a MIMO system in literature. This multiplicity of I/O of a system may feature that change in one input, will not only change the corresponding output, but may effect the remaining outputs also. The exothermic CSTR is a classical example where this **Interaction of Variables** exits.

To understand this phenomenon, let us write equations 4.105 and 4.106 in splitted form in time domain in terms of intermediate variables C_{O1}, C_{O2} and T_{O1}, T_{O2}:

$$\frac{dC_{O1}}{dt} = \frac{K_{11}}{\tau_1}C_I - \frac{1}{\tau_1}C_{O1} \tag{4.107a}$$

$$\frac{dC_{O2}}{dt} = \frac{K_{12}}{\tau_1}T_O - \frac{1}{\tau_1}C_{O2} \tag{4.107b}$$

$$\frac{dT_{O1}}{dt} = \frac{K_{21}}{\tau_2}T_I - \frac{1}{\tau_2}T_{O1} \tag{4.107c}$$

$$\frac{dT_{O2}}{dt} = \frac{K_{22}}{\tau_2}C_O - \frac{1}{\tau_2}T_{O2} \tag{4.107d}$$

These equations have been written by assuming, $C_O = C_{O1} - C_{O2}$ and $T_O = T_{O1} - T_{O2}$. Let us introduce the following set of coefficients in the above equations:

$$A_{11} = K_{11}/t_1;\ A_{12} = K_{12}/t_1;\ T_1 = 1/t_1;\ A_{21} = K_{21}/t_2;\ A_{22} = K_{22}/t_2;\ T_2 = 1/t_2$$

Now we may construct a variable flow diagram in terms of inputs, outputs and intermediates with the coefficients that are to be multiplied with them to complete the

solution. This is illustrated in Figure 4.20(c).

Meaning of the triangles containing an integration symbol used in Figure 4.20(c) is explained in part (a) and (b) of the figure that this stands for an integrator and can produce the result of the time derivative of a dynamic variable as its time description, and in part (b) the solution of a 1st order system written in time domain form is obtained by using one integrator and variable flow paths in which the coefficient or constants are written as multipliers in circles. The reactor model in part (c) is nothing but solution of four 1st order ODEs, equations 4.107a, b, c, d. Figure 4.20 reveals that any one of the output variables contain effect of the remaining output via feedback paths, that is, the feedback paths that connect C_O to T_O through T_{O2}, and T_O to C_O through C_{O2}.

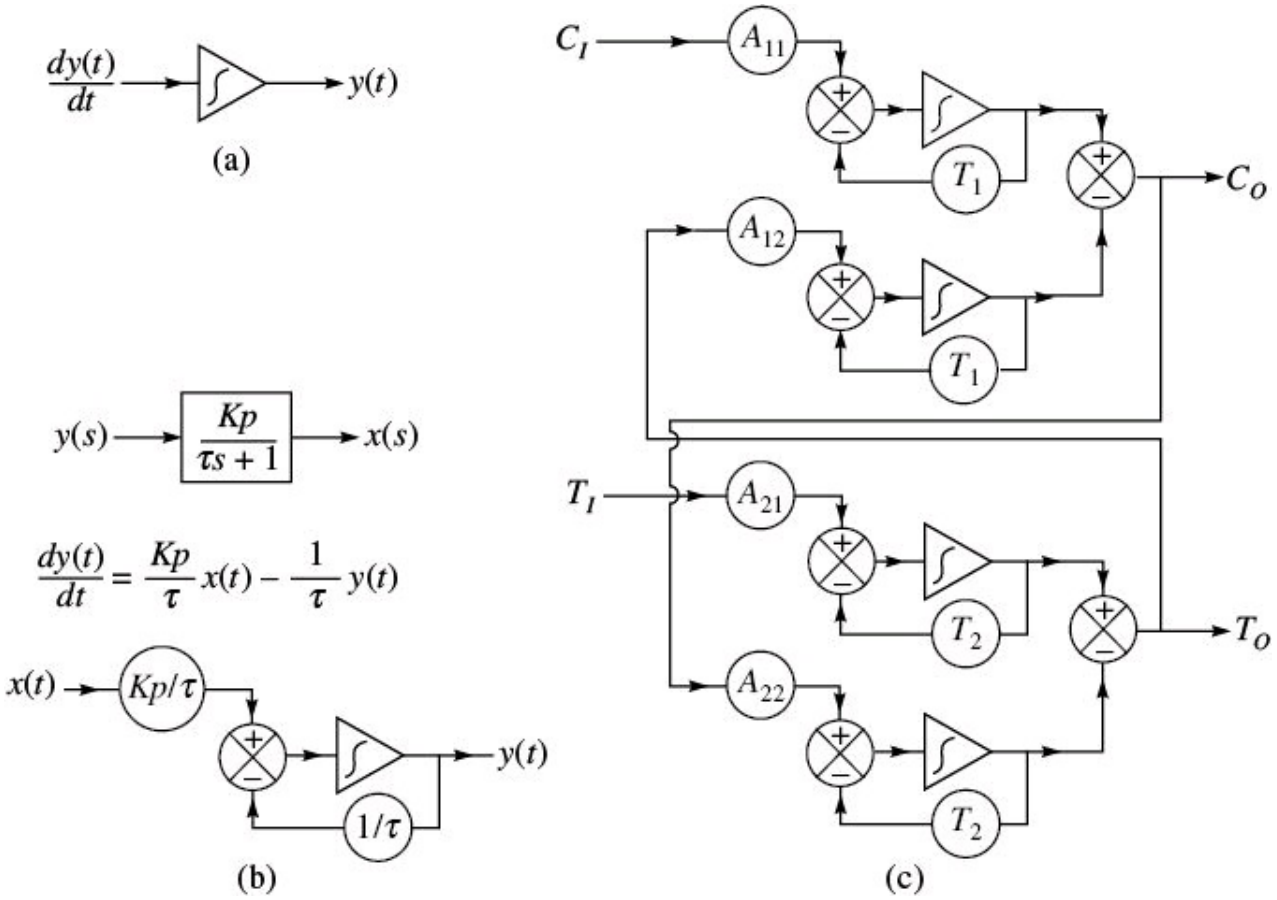

Figure 4.20 (a) The analog integrator. (b) Solution of a 1st order ODE. (c) The exothermic reactor model.

In terms of conventional block diagram representation that have been used in some of the previous examples, Equations 4.105 and 4.106 describing the linear model of this example may also be used to construct Figure 4.21.

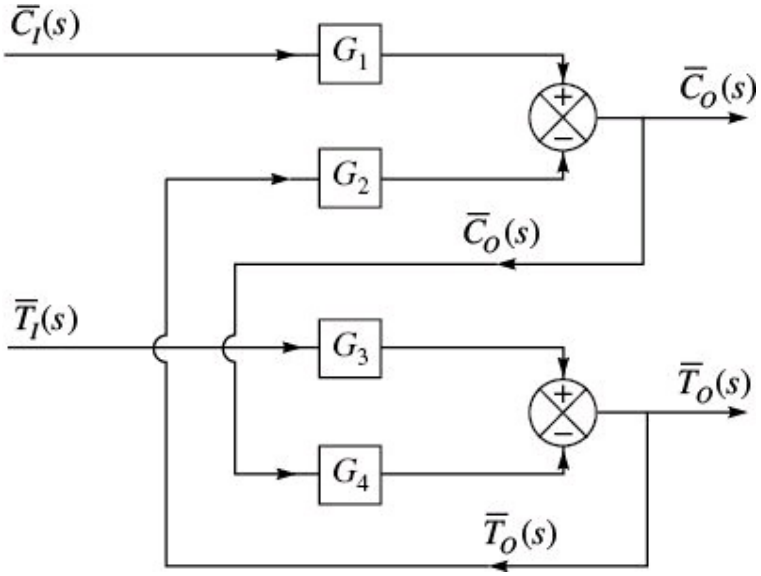

Figure 4.21 The block diagram description of the exothermic CSTR.

The 1st order TF blocks G_1, G_2, G_3 and G_4 used in Figure 4.21 can be interpreted from Equations 4.105 and 4.106 as:

$$G_1 = \frac{\bar{C}_O(s)}{\bar{C}_I(s)} = \frac{K_{11}}{\tau_1 s + 1}; \quad G_2 = \frac{\bar{C}_O(s)}{\bar{T}_O(s)} = \frac{K_{12}}{\tau_1 s + 1}; \quad G_3 = \frac{\bar{T}_O(s)}{\bar{T}_I(s)} = \frac{K_{21}}{\tau_2 s + 1}; \quad G_4 = \frac{\bar{T}_O(s)}{\bar{C}_O(s)} = \frac{K_{22}}{\tau_2 s + 1}$$

This illustration also displays the feedback paths, from $\bar{T}_O$ to $\bar{C}_O$ via G_2, and from $\bar{C}_O$ to $\bar{T}_O$ via G_4. All the efforts have been made to show that any of the input changes would effect both of the output variables through these feedback paths. Such influences of one input on more than one output in a multiple input-output system, is called interaction and create a special class of control difficulty that we shall address in Chapter 18.

The existence of non-linear character of a process involving two or more number of independent variables may not be always due to the presence of variable product terms. That could be the manifestation of the total process characteristic or, a part there of.

EXAMPLE 4.19 A level control system is shown in Figure 4.22. The controller output,
CO, may be 3–15 psig air pressure that is applied to the pneumatic control valve and produces a proportional displacement of valve stem. The valve has a parabolic characteristic (see Appendix-III), i.e. the stem position, *x*, to port opening, AP, has a relation: $AP = k.x^2$, where *k* is a constant. The manipulated variable, flow through the valve q_O will depend on:

1. The controller output, which determines the port opening AP; and

2. The tank level, h, which is spelled as, the flow-head relation through any orifice or opening: $q_O = C_D h^{0.5}$. For a valve, we use valve coefficient, C_V, instead of the orifice discharge coefficient, C_D. The complete dependency of tank exit flow on controller output and tank level is expressed as:

$$q_O = COC_V\sqrt{h} = xC_V\sqrt{h} = K'\sqrt{A_P}C_V\sqrt{h} = K_1(A_P)^{0.5}K_2(h)^{0.5}$$

It is required to find a linearized relation that will describe the dependency of exit flow, within a short departure of the independent variables from their steady values A_{PS} and h_S. Express the result as the TF of the liquid reservoir $H(s)/Q_I(s)$. This result may be conveniently used in the level control system analysis.

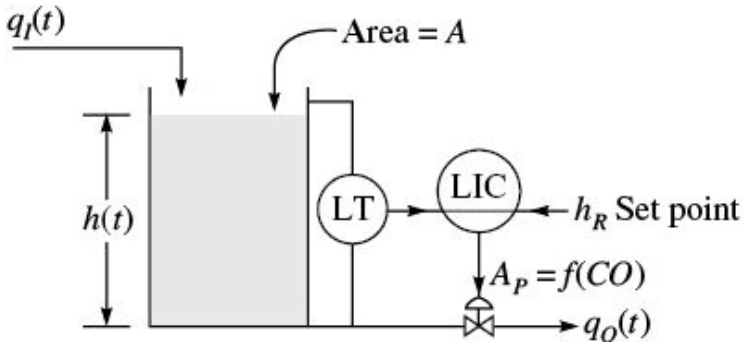

Figure 4.22 The level control system involving non-linearity in two variables.

Solution: The above equation relating A_P and h to could be readily linearized around the steady states A_{PS} and h_S as:

$$q_O = q_{OS} + \left.\frac{\partial q_O}{\partial A_P}\right|_{\substack{A_P=A_{PS}\\h=h_S}}(A_P - A_{PS}) + \left.\frac{\partial q_O}{\partial h}\right|_{\substack{h=h_S\\A_P=A_{PS}}}(h - h_S)$$

or,
$$q_O - q_{OS} = \frac{K_1K_2}{2}\sqrt{\frac{h_S}{A_{PS}}}(A_P - A_{PS}) + \frac{K_1K_2}{2}\sqrt{\frac{A_{PS}}{h_S}}(h - h_S) \qquad (4.108)$$

The dynamics of the liquid level system in the form of difference between absolute and steady state values of concerned variables is:

$$(q_I - q_{IS}) - (q_O - q_{OS}) = A\frac{d(h - h_S)}{dt} \qquad .(4.109)$$

Putting Equation (4.108) in Equation (4.109),

$$(q_I - q_{IS}) - \frac{K_1K_2}{2}\sqrt{\frac{h_S}{A_{PS}}}(A_P - A_{PS}) - \frac{K_1K_2}{2}\sqrt{\frac{A_{PS}}{h_S}}(h - h_S) = A\frac{d(h - h_S)}{dt}$$

Introducing deviation variables and taking Laplace transform:

$$Q_I(s) - C_1\bar{A}_P(s) - C_2H(s) = AsH(s)$$

or

$$Q_I(s) - C_1\bar{A}_P(s) = H(s)[C_2 + As]$$

or

$$\frac{1}{C_2}Q_I(s) - \frac{C_1}{C_2}\bar{A}_P(s) = H(s)\left[1 + \frac{A}{C_2}s\right]$$

or

$$K_{P1}Q_I(s) - K_{P2}\bar{A}_P(s) = [1 + \tau_P s]$$

The dependency of H on Q and A can be expressed by the following two TFs:

$$\frac{H(s)}{Q_I(s)} = \frac{K_{P1}}{1 + \tau_P s} \tag{4.110}$$

and

$$\frac{H(s)}{A_P(s)} = \frac{-K_{P2}}{1 + \tau_P s} \tag{4.111}$$

In the block diagram form, the dynamic relation between output and the two input variables may be expressed as in Figure 4.23.

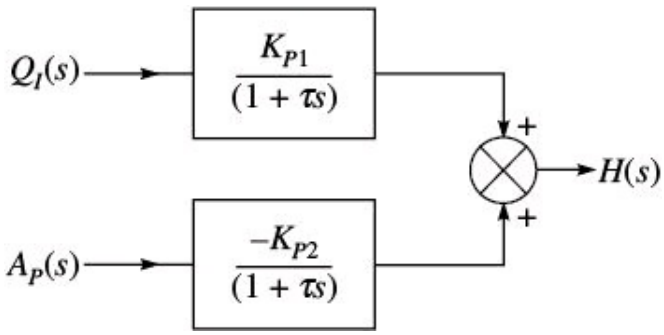

Figure 4.23 The transfer function representation of the system of Figure 4.22.

It is interesting to note that AP has a –ve Gain in its transfer function with H. Can you explain?

EXAMPLE 4.20 A colligative property μ, of the solution of a fluorescent dye in a non-polar organic solvent is dependent on:

1. Solute concentration, x gm-mole/lit;
2. Pressure, p mmHg; and
3. Temperature T°K. This dependency is expressed as:

$$\mu = (A_O + A_1x + A_2x^2)(B/p)(D_1e^{-D_2T})$$

It is required to find a linearized equation for expressing this dependency for small

departures of concentration, pressure and temperature from their steady state values x_S, p_S and T_S.

Solution: To obtain the required solution, the above equation is expanded according Taylor's series in three independent variables:

$$\mu = \mu_S + \left.\frac{\partial \mu}{\partial x}\right|_{\substack{x=x_S \\ p=p_S \\ T=T_S}} (x - x_S) + \left.\frac{\partial \mu}{\partial p}\right|_{\substack{p=p_S \\ T=T_S \\ x=x_S}} (p - p_S) + \left.\frac{\partial \mu}{\partial T}\right|_{\substack{T=T_S \\ x=x_S \\ p=p_S}} (T - T_S) \tag{A}$$

Evaluating the partial derivatives:

$$\mu = \mu_S + [A_1 + 2A_2 x_S](x - x_S) - \frac{B}{p_S^2}(p - p_S) - D_1 D_2 e^{-D_2 T_S}(T - T_S)$$

or,
$$\mu - \mu_S = [A_1 + 2A_2 x_S](x - x_S) - \frac{B}{p_S^2}(p - p_S) - D_1 D_2 e^{-D_2 T_S}(T - T_S) \tag{B}$$

Introducing the deviation variables, and taking Laplace transform of the resulting equation:

$$\bar{\mu}(s) = K_1 X(s) - K_2 P(s) - K_3 \vec{T}(s) \tag{C}$$

Equation C can be expressed in block diagram form as shown in Figure 4.24.

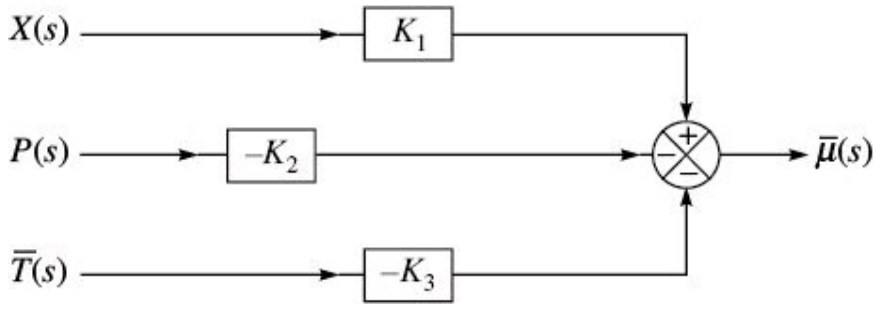

Figure 4.24 The linearized model in block diagram form.

EXERCISE PROBLEMS

4.1 Find the transfer function $H(s)/Q_I(s)$ for the systems depicted in Figures P4.1(a), (b), and (c).

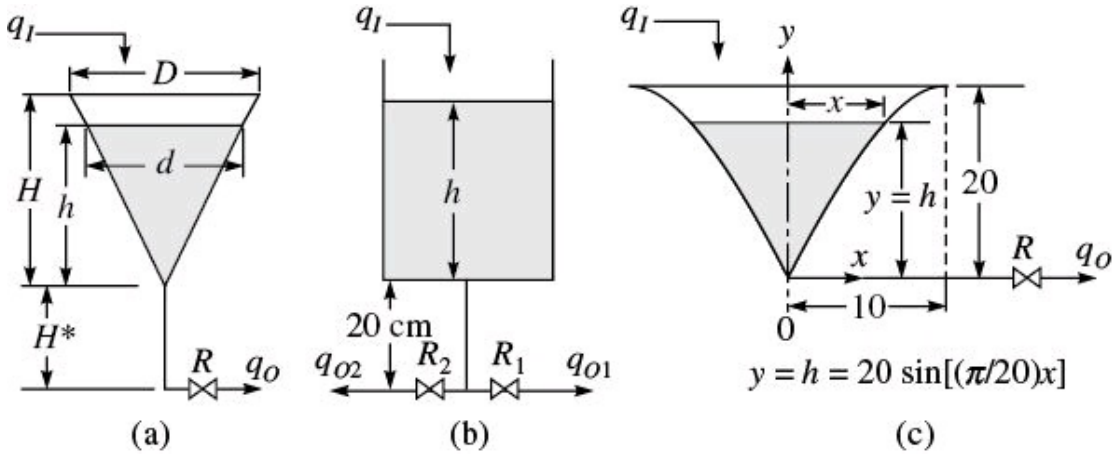

Figure P4.1 Liquid level systems of different configurations.

4.2 The system shown in Figure P4.2, has been disturbed as:

1. at $t = 0$, only T_I is step increased;
2. at $t = t_1$, only q_I is step increased.

(a) Prove that by increasing q_I to whatever magnitude T_O could not be returned to its original value that was at $t < 0$.

(b) T_I should be changed how much to bring T_O back to its original value.

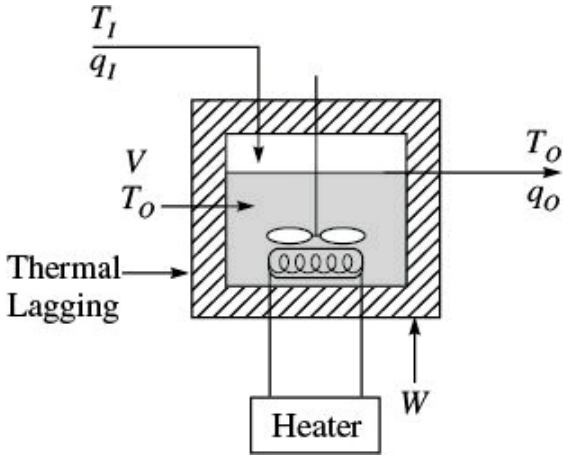

Figure P4.2 Constant holdup stirred heater.

4.3 For the transfer function: $\dfrac{Y(s)}{X(s)} = \dfrac{K_1}{\tau_1 s + 1} + \dfrac{K_2}{\tau_2 s + 1}$; what is the steady state gain.

4.4 Exit flow through a Weir (Figure P4.3) fitted at the end of a tank follows a nonlinear flow-head relationship:

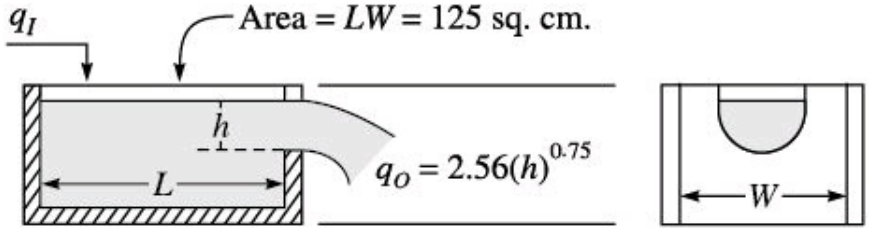

Figure P4.3 Discharge from the weir.

where h is in cm and q_O in litre per min. Deduce the linearized transfer function for the tank, if the steady state liquid height is 15 cm.

4.5 Consider a single plate gas-liquid absorption system depicted in Figure P4.4.

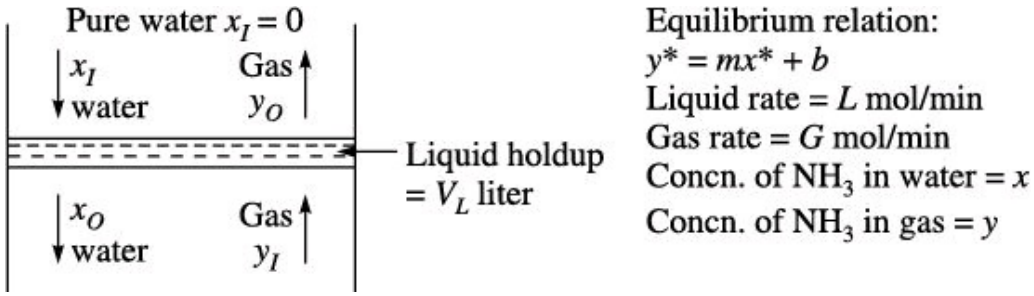

Figure P4.4 Streams contacting on a plate of an absorber.

From an upgoing NH_3 + Air mixture, NH_3 is absorbed by a down going water stream over the plate by intermixing. Assuming complete mixing over the plate and a liquid holdup of V_L litre, prepare an unsteady state component, NH_3, balance for the tray where it may be considered that all mass transfer takes place. You may convert the gas phase, NH_3 concentration y to equivalent equilibrium liquid concentration x^* by applying the equilibrium relation. However, from the balance derive the differential equation of the form:

$$dxo/dt = \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$

Assuming L and G are constant, and choosing y_I or x_I^* as the input and x_O as the output variable, try to develop the transfer function: $x_O(s)/y_I(s)$.

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[2] Smith, C.A and Corripio, A.B., *Principles and Practices of Automatic Process Control*, John Wiley & Sons, NY, 1985.

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5

Response of First Order Systems (Complex Domain Analysis)

Dynamic study of a Process, or the term process dynamics, signifies the examination of response of processes when they are in the transient state. Transient state of a system is defined as the state it belongs to when, after a disturbance, it has moved away from the steady state. The transient state may end if the system has sufficient self regulation to reach a subsequent steady state. We begin the study by first describing certain ideal type of disturbances or input functions, and their Laplace transform that we shall put as the $X(s)$ in Equation 4.1 and invert the resulting expression to time domain. The solution will give us the system's response corresponding to the respective input function. A short exposure on the analytical aspects of Laplace transform, transformation of the input functions, and the method of solving ordinary differential equations are covered in Appendix-II.

5.1 Ideal Input Function

5.1.1 The Step Function

In a mid-summer noon if you *suddenly* step out of the Computer Lab of your department, your body or the physiological system will experience a jump of surrounding temperature of the order of 20^{o} C or more.

At the top of an empty tank if there is a water tap and you *suddenly* turn it on to the full flow value, the empty tank system will experience an instantaneous change of inflow from 0 to the full flow rate.

When you *suddenly* step on a weighing machine, the weighing system is subjected to an instantaneous input weight change, equal to your body weight.

In the above descriptions, the word 'suddenly' has been underlined to emphasize that in all the cases, the changes have been made instantaneously, i.e. in no time or in a very small time interval that may be considered as zero in engineering sense. Another fact to be noted is that after the change has occurred, the input magnitude has remained constant at its new value at least for a reasonably long period of time. The name *step* has been coined for such changes, because pictorially such an input function on a time abscissa would look like a step. We may say a step change has

been made as an input disturbance to the system. (To express the practical notion about the length of this time, two alternative descriptions may be proposed as:

1. *The longest conceivable time during which the process would be under observation*
2. *The time should be large enough so that it spans long after the process has attained its steady state after the disturbance has occurred.*)

Thus, a **'step function'** may be defined as an instantaneous change of the function value at a point in time after which the function remains at this new value anywhere in +ve time. The vertical height of the step in each case is the magnitude of the function. The general time description of a step function should consider that the input, $x(t)$, was at its steady state value, x_S, before the sudden change has occurred at an arbitrary value of time, t_1. The absolute function may be expressed in the deviation form so that the origin of y-axis would be shifted to the zero value of the function. The zero of the time scale may be shifted in this way by defining a new time variable $t^* = t - t_1$. But this is not necessary if we consider that our observation and time scale have started from the instance of the input change. Thus, t_1 becomes the new origin of the time axis. The idea may be understood from Figures 5.1(a) and (b). Note that its shape really resembles the step of a staircase. We present the time domain description and the Laplace transform of a step function of magnitude M in the following Figure. A unit step has a magnitude equal to 1.0.

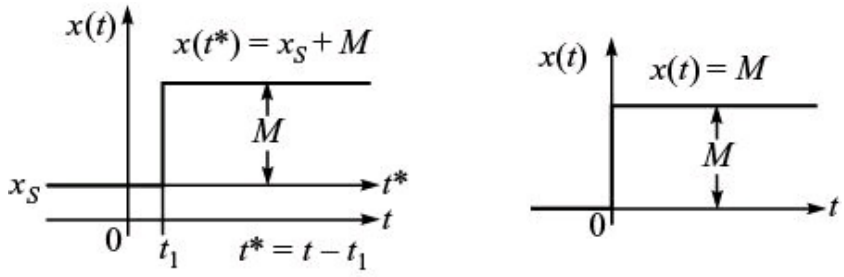


Figure 5.1 The step function.

A unit step may be defined in time domain as:

$X(t) = 0$, for $t < 0$, and, $X(t) = 1.0$, for $t \geq 0$.(5.1)

and, the Laplace Transform of Equation 5.1 will be:

$$X(s) = \frac{1}{s} \qquad (5.2)$$

A step of magnitude M may be defined in time domain as:

$X(t) = 0$, for $t < 0$ and $X(t) = M$, for $t \geq 0$ (5.3)

and, the Laplace Transform of Equation 5.3 will be:

$$X(s) = \frac{M}{s} \qquad (5.4)$$

Note that in the time domain definition by Equations 5.1 and 5.3, we have used the deviation form of the input variable, so that whatever be its threshold value at steady state condition, the initial condition for $t > 0$ remains valid. In the following discussion, we shall be using the same variable definition.

You should recognize that in our previous discussions in the introductory chapter, the expressions '+ve input change' and '–ve input change' refer to +ve and –ve step changes.

5.1.2 The Ramp Function

Consider an electrically heated furnace lying in 'off' condition for sufficiently long time so that it has attained, and is at steady state at the surrounding room temperature. At time $t = 0$, the input power is turned 'on'. Let us assume that the interior chamber of the furnace has a perfect thermal insulation from the surrounding, the furnace body material and the object inside the furnace have equal thermal capacity, C_P, and that its value is independent of the temperature. If the energy input rate is constant, the furnace interior and the object will be subjected to a linearly increasing temperature with respect to time, starting from the room temperature.

From a compressed Helium gas cylinder, a weather balloon is being filled up. We assume the pressure build up inside the balloon is low, and the pressure change of the supply cylinder may also be neglected, at any time during the filling process. The temperature and the atmospheric pressure could be justifiably assumed to be constant during the filling process. Thus, the mass as well as the volume of the gas inside the balloon, starting from zero value will linearly increase with time.

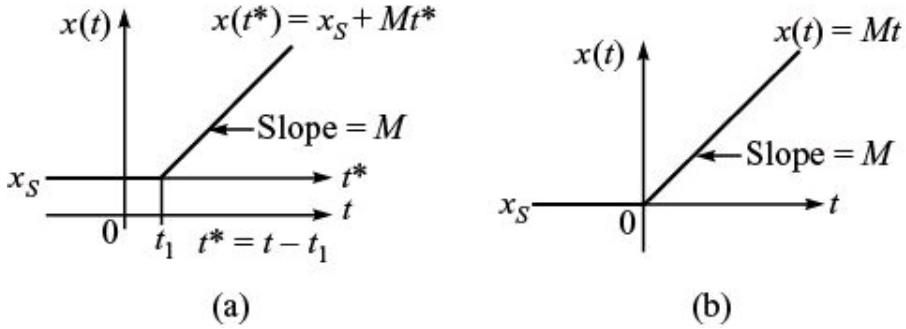


Figure 5.2 The ramp function.

In the above examples, the time description of the variables resemble the smooth rise that is built alongside the stairs of the front porch, as we see in many houses used for moving two-wheelers, or the smoothly rising approach to bridges or culverts. This gradually rising shaped constructions are called ramps, and from this usage, the linearly changing time function has coined its name as the '**ramp function**'.

The general time description of a ramp function is shown in Figure 5.2(a), and the shifting of axes follows the similar procedure indicated for step function, and obtains the necessary function configuration in deviation variable form shown in Figure 5.2(b). In time domain, the dynamic description of a ramp function has been presented in Figure 5.2. Analytically, a ramp function of slope M may be defined in time domain as:

$X(t) = 0$, for $t < 0$, and $X(t) = Mt$, for $t \geq 0$ (5.5)

and, the Laplace Transform of Equation 5.5 will be:

$$X(s) = \frac{M}{s^2} \qquad (5.6)$$

5.1.3 The Impulse Function

A cricket bat striking an incoming ball, a thunder bolt striking on a tree, or a mass of explosive blasting in a drilled hole in a coal mine; think about the time taken for the transfer of energy in each of these examples. This time must be very small, say of the order of micro or nano seconds. Let the transferred amount of energy in each case be M. If this transfer time could be reduced to an infinitesimally small value, the resulting input function may be called an **impulse function.** An impulse is a hypothetical function that attains an infinitely large value at a particular point of time, but before and after this point (instant) any where in time, the function has a zero value.

Magnitude of an impulse is the integrated area under the function. A unit impulse is denoted by $d(t)$, because its magnitude is unity. Hence, an impulse of magnitude M is denoted by $Md(t)$. We may derive the definition of impulse from a **Pulse Function.** A pulse function has a finite height, h, and occurs during a finite width or duration, w, of time. To define a pulse of unit magnitude, the width w should be $= 1/h$. The corresponding unit impulse is obtained in the limiting case when h approaches infinity ($h \to \infty$). These definitions and the shifting of the origin of time, as well as input axis (as we have shown in case of step and ramp inputs), have been illustrated in Figure 5.3(a), (b), (c). The function may be analytically presented as:

For an unit impulse:

$X(t) = 0$, for $t < 0$, and $t > 0$, and $X(t) = \underset{h \to \infty}{\text{Lt}} \left[h \frac{1}{h} \right] = \delta(t)$, for $t = 0$ (5.7)

and

Laplace transform of Equation 5.7 will be, $X(s) = 1$ (5.8)

Referring to Figure 5.3(a), as the limit is taken, the height h of the function rises to infinity, and its width $1/h$ approaches ‘zero’, thus the area within the function remains equal to unity.

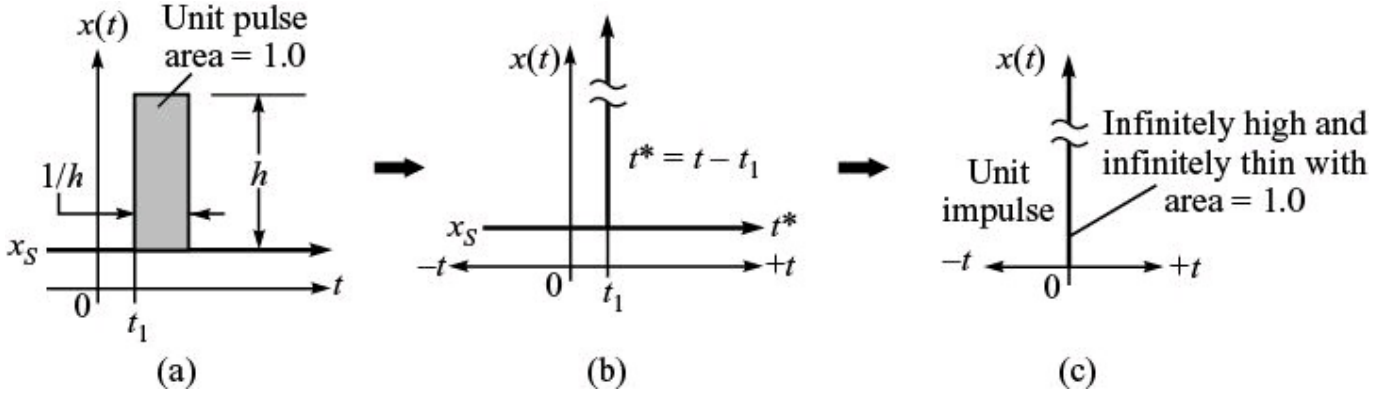


Figure 5.3 (a) A unit pulse function. (b) The unit impulse function in the limit. (c) Unit impulse in deviation variable form.

For an impulse of magnitude M, the above limit expression becomes,

$$X(t) = \lim_{h \to \infty} \left[Mh \frac{1}{h} \right] = M\delta(t), \text{ for } t = 0 \qquad (5.9)$$

and the Laplace transform will be, $X(s) = M$ (5.10)

5.1.4 The Sinusoidal Function

Many physical variables like the atmospheric temperature during day and night time, tidal variation of coastal sea level, or the Cardiac Potential that maintains our heartbeat, are periodic in nature. They may not be truly sinusoidal, but if analyzed, many of them show that they are a combination of several sinusoids of different frequency and amplitude. True sinusoidal functions appear in the voltage fluctuation of AC power supply, and the motion of crankshaft of an automobile moving at a constant speed.

A Sinusoidal function may be defined in time domain as:

$X(t) = 0$, for $t < 0$ and $X(t) = M \sin(wt)$, for $t \geq 0$ (5.11)

where, M is the amplitude, and w is the radian frequency of the function. Time period, T_P, that is the 'time required by the function to cover one cycle or $2.p$ radians', bears the following relation with w:

$$T_P = (2p)/w$$

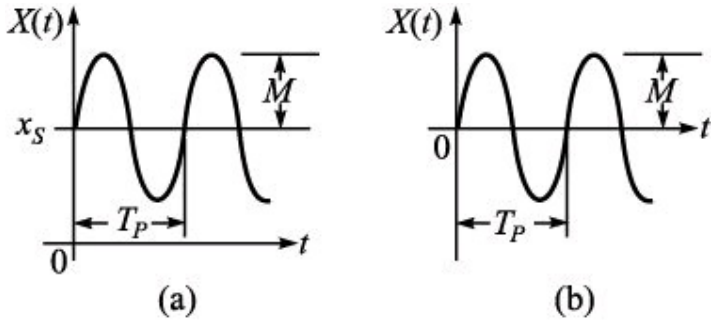


Figure 5.4 The sinusoidal function. (a) Absolute value of the variable. (b) Deviation form.

The Laplace Transform of Equation 5.11 will be,

$$X(s) = \frac{M\omega}{s^2 + \omega^2} \qquad (5.12)$$

We have drawn the absolute and the deviation form of a sinusoidal function in Figure 5.4. Please note the position of threshold or steady state value x_S of the function in the two forms of the function. The presentation of sinusoidal change in absolute form is necessary for scalar quantities as concentration, temperature, etc. Whereas variables that are vectors, e.g. force, velocity, voltage, may be treated in the deviation form.

5.2 First Order Process Response Corresponding to Ideal Input Functions

The response in time domain as $Y(t)$ would be found by first putting the proper input expression as $X(s)$ corresponding to a particular input, in the general transfer function of Equation 4.1 of a first order process, and inverting the resulting $Y(s)$ to time domain. We begin by finding the response for step input function.

5.2.1 The Step Response

In the first chapter we tried to establish the step response of the concerned first order systems with examples of charging or discharging of a liquid reservoir. We already have a good notion about their qualitative nature. Let us now develop the analytical expression of a first order step response.

Rewriting the transfer function of a first order process,

$$\frac{Y(s)}{X(s)} = \frac{K_P}{\tau s + 1} \qquad (4.1)$$

For a unit step, in Laplace domain, $X(s) = 1/s$, thus,

$$Y(s) = \frac{K_P}{s(\tau s + 1)} \qquad (5.13)$$

The above equation may be inverted by the way of partial fraction expansion, and the resulting time domain solution is given by:

$$Y(t) = K_P(1 - e^{-t/\tau}) \qquad (5.14)$$

Equation 5.14 expresses the step response of a first order process in time domain. The bracketed term $(1 - e^{-t/t})$ should be recognized as the 'time variant function' referring to our discussion in Section 2.4. This term may have values ranging from 0 to 1 as time increases from 0 to ∞. The other term in the LHS of Equation 5.14 is the process gain, K_P, thus the step response, $Y(t)$, may have values ranging from 0 to K_P,

as time increases from 0 to ∞. How does this response look like? We have plotted the normalized output, $Y(t)/Y_{MAX}$, as ordinate with respect to a reduced time scale as the abscissa, in Figure 5.5.

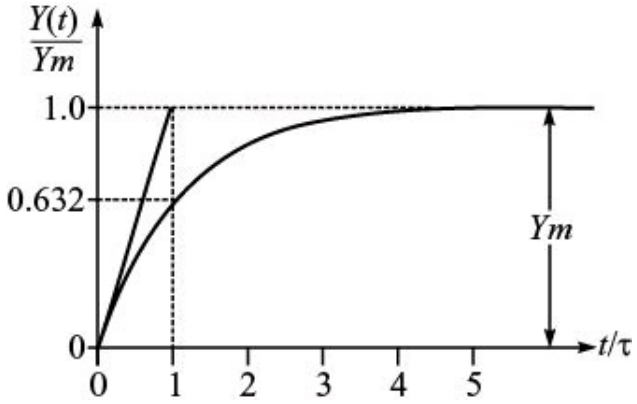


Figure 5.5 Step response of a first order system.

Certain properties of this response may be derived from Equation 5.14 as follows:

1. The initial slope of response, i.e. $dy(t)/dt|_{T=0}$ is maximum, and it has a finite value.
2. The response value, at $t = t$, the time constant of the system, is 63.2% of the total step height.
3. With initial speed of response, time required to cover the total step height is equal to one time constant t of the system.

These properties are important for identification of a first order process from other higher order processes. We shall discuss these after an exposure of the second and higher order processes. We should mention that observing Equation 5.14, the response for a step of magnitude, M, may be guessed.
For a step of magnitude M the response equation is:

$$Y(t) = MK_P(1 - e^{-t/\tau}) \qquad (5.15)$$

We shall see with every input function type, the response falls behind, or the effect of input takes a certain time to appear fully. Also, due to the internal dynamic processing of the system, the shape of the input function is distorted as it appears in the response. You should understand this 'dynamics processing' is indeed the dynamic character, or simply the dynamics of a process. The phenomenon of response falling behind the input is in fact the interaction of the resistive and capacity elements in the flow path of the principal entity. You may confirm that in case of step response after an input change, it takes $4t$ amount of time for the response to reach 98% of the final value, and for $5t$ the recovery is more than 99%.

EXAMPLE 5.1 Prove that if a first order system, after a step disturbance, is allowed

to travel at its initial speed, it would cover the entire step magnitude within a time equal to its time constant.

Solution: We shall start by differentiating the step response Equation 5.18;

$$\frac{dY(t)}{dt} = \frac{MK_P}{\tau} e^{-t/\tau} \qquad (5.16)$$

Initial value of this slope may be obtained by setting $t = 0$.

Thus, $$\left.\frac{dY(t)}{dt}\right|_{t=0} = \frac{MK_P}{\tau} \qquad (5.17)$$

This is the speed with which the system would cover a distance = MK_P. Hence, the required time = t, is the system time constant.

EXAMPLE 5.2 A manufacturer of a brand of a clinical Thermometer proudly claims his product is a 1 min thermometer that requires only 1 min contact time to register true temperature (say 99% recovery of the step magnitude). Find the time constant.

Solution: First let us find the time required by a step disturbed first order system to reach 99% of the step. From Equation 5.15 we may write:

$$0.99\,M = M(1 - e^{-X}); \text{ or } -X = \ln(0.01); \text{ or } X = 4.6052$$

Hence, $$\tau = \frac{60}{4.6052} \cong 13 \text{ sec}$$

5.2.2 The Ramp Response

Let us try to visualize what will really happen when the temperature of the exemplary furnace in Section 5.1.2 increases constantly from the initial room temperature and a thermometer is trying to record the temperature.

It is like a 'lazy-boy' athlete, who after a heavy and lengthy schedule on the previous day, tries to run along with his spry trainer in the morning. In the beginning, with a foggy head and sluggish body he will be falling behind his trainer who is running at a constant speed. Initially, the distance between them will increase, but gradually after overcoming the initial lethargy, when he attains the same speed as his trainer, this 'gap' will no longer increase. Let us assume he continues to maintain this speed and so does his trainer during the rest of their lap. Now, though they are running at the same speed, the trainee is at a constant distance behind his trainer. You may have by now got an idea about the thermometer response. To rationalize this idea quantitatively, let us find the response equation.

For a ramp input of slope, M, in Laplace domain, $X(s) = M/s^2$, for a unity gain TF,

$$Y(s) = \frac{M}{s^2(\tau s + 1)} \qquad (5.18)$$

The above equation may be inverted by the way of partial fraction expansion, but this time the L'Hospital theorem will be necessary due to the presence of repetitive poles, s^2 in the denominator, and the resulting time domain solution is obtained as:

$$Y(t) = M(t - \tau + \tau e^{-t/\tau}) \qquad (5.19)$$

Equation 5.19 expresses the ramp response of a unity gain first order process in the time domain. The input function and the response equation have been drawn on the same plot and are illustrated in Figure 5.6.

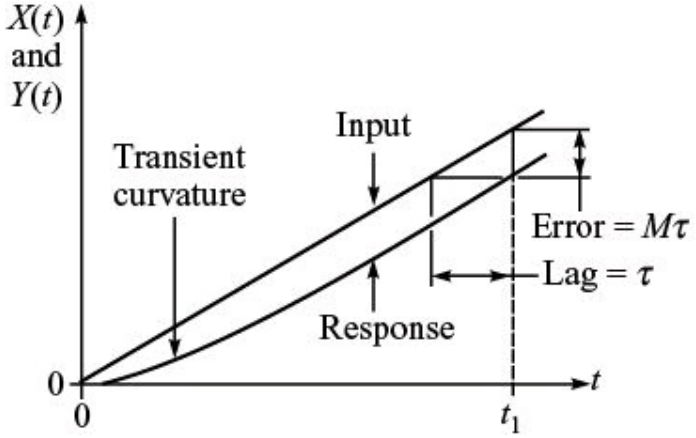


Figure 5.6 The ramp response of a first order system.

The bracketed terms ($t - t + te^{-t/t}$), describe the system's dynamics. Note that the third term, ($te^{-t/t}$), gradually diminishes in magnitude as time increases, and in fact is responsible for the initial rounded portion of the response.

Can you equate the slowly decreasing lethargy of our example athlete, and the response of the thermometer with this ($te^{-t/t}$) transient term, and when this term gradually vanishes, speed of the trainee athlete (or rate of increase of the indicated temperature by the thermometer), becomes equal to that of his trainer (or rate of rise of true furnace temperature)?

Thus, after the initial rounded portion, the input and response become parallel straight lines with equal slope, M. The input, $X(t) = Mt$, passes through origin, and if the straight portion of the response, $Y(t) = Mt - Mt$, is extended back to $t = 0$, it will cut the temperature (ordinate) axis at $-Mt$.

Two important properties of ramp response are 'Lag' and 'Error'. These are defined as follows:

Lag : Time difference between input and response to cross identical variable values.

Error : Difference between the true value (input) and output or indicated value(response).

Lag and error may be calculated at any +ve time instant by using the time domain definition of the input ramp function and the response Equation 5.20 or may be found by the graphical construction indicated in Figure 5.6.

For graphical evaluation indicated in Figure 5.6, a vertical line is drawn at the given time instant. The vertical distance between response and input function is the error. The horizontal distance of the input function from the crossing point of the time vertical and the response locus is the lag. At large values of time, when the input and response are parallel straight lines, the values of lag and error become independent of time and there magnitudes are:

$$\text{Lag} = t \qquad (5.20)$$

and

$$\text{Error} = Mt \qquad (5.21)$$

One thing should be mentioned. We have to use a unity gain first order transfer function (TF) *to obtain constant values of lag and error. Can you think up of the reason?*

EXAMPLE 5.3 You have been asked to ascertain the transfer function of a Thermocouple (TC) type temperature sensor–transmitter system by dynamic testing. The TC is to be used in a programmed temperature control system for a Heat Treatment Furnace. An optical type temperature indicator (OTI) is also available to determine the true furnace temperature. Data obtained on a Strip Chart Recorder as continuous traces of temperature readings from TC and OTI. To confirm the validity of the result, it has been proposed to obtain the data at two different heat input rates at two different settings of the Thyristor drive. Also, it is intended to use the power input data as scale up guidance if and when the furnace has to be driven at higher temperature ranges.

(a) Assuming a first order dynamic model for the TC as:

$$\frac{T_{\text{TC}}(s)}{T_{\text{OTI}}(s)} = \frac{K_P}{\tau s + 1}$$

determine the model parameters K_P and t, by using the experimental data.

(b) You have made a mistake—forgot to record the Thyristor output readings. But being a bright young engineer, have been later able to dig out the following information from reference and furnace supplier literature:

Heat capacity of furnace material, C_P = 1.5 cal/(g°C); Furnace mass, M = 15.4 kg.

How would you workout the missing data and report as W_1 and W_2 in kcal/sec or kW, assuming negligible dissipation of heat from the furnace exterior surface.

(*The main feature of a programmed control is that its set point is not constant, but changes with time according to a pre-selected* ***Time—Set point Schedule****. In a heat*

treatment furnace, a material is subjected to a sequence of gradually rising temperature ramps. The end temperature of the preceding ramp is the starting temperature of the next ramp and after the furnace temperature attains the desired highest value, comes a constant temperature period after which the temperature decreases. This operation is done for a material in the furnace to enhance its physical properties. Other examples of set point tracking may be controlling a temperature scheduled batch reactor, and guidance system of a moving object in a curved path – say a ship or an aircraft.)

Solution (a) The experimental data has been plotted in the following Figure 5.7(a). Note that in the initial part of the experimental run and around the point where change of ramp slope has occurred, the TC and OTI data is omitted to avoid the clutter of the transient part of response.

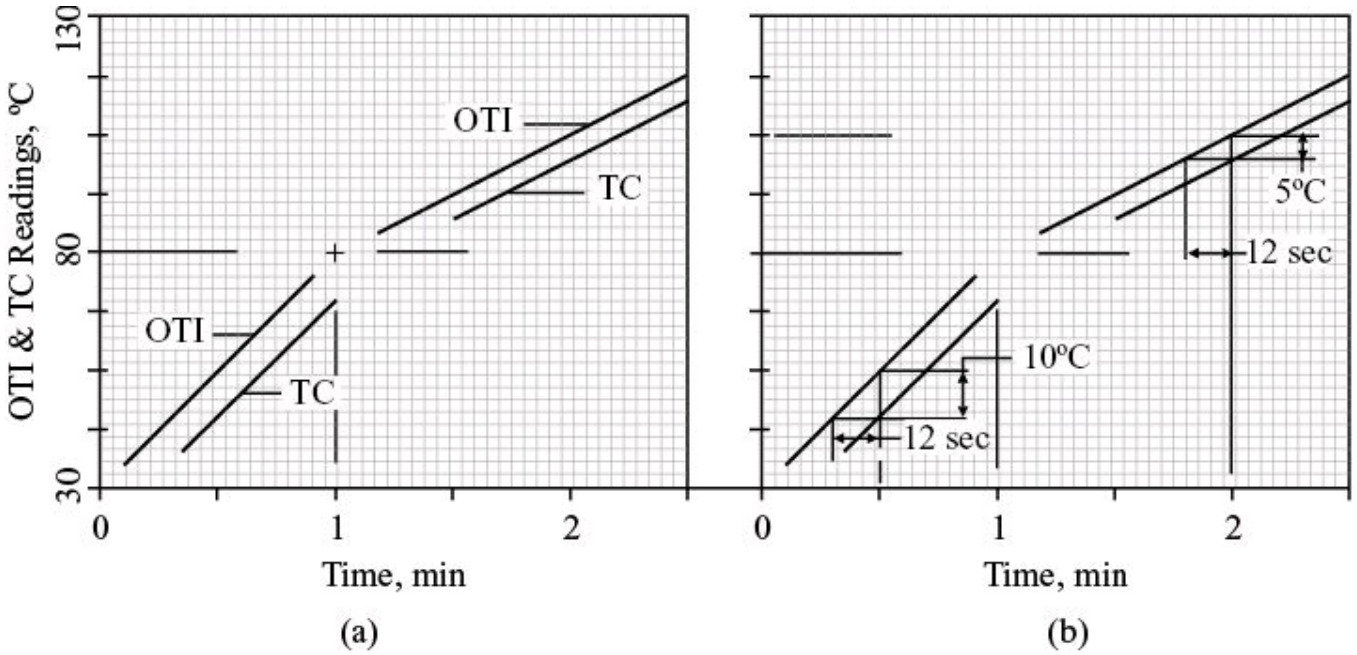


Figure 5.7 The OTI and TC-output on a common time base. (a) Raw data. (b) The geometrical construction on the response.

You should have recognized that we are dealing with ramp response of the TC corresponding to two distinct time slopes of input function. As the two traces are parallel at the stable part of the response, we may conclude that $KP = 1$.

In Figure 5.7(b), the geometrical construction has been done from which the lag of the TC may be determined, $lag = 12$ sec. and, $lag = t = 12$ sec. Thus,

$$\frac{T_{TC}(s)}{T_{OTI}(s)} = \frac{K_P}{\tau s + 1} = \frac{1}{12s + 1}$$

(b) From Figure 5.7, the slope of the true furnace temperature, (OTI-output) vs. time, are:

$S_1 = \frac{50°C}{60\,\text{sec}}$; and $S_2 = \frac{37.5°C}{90\,\text{sec}}$; W_1 and W_2 may be calculated as:

$W_1 = 15.4 \times 1.5 \times \frac{50}{60} = 19.25$ kcal/sec; and $W_1 = 15.4 \times 1.5 \times \frac{37.5}{90} = 9.625$ kcal/sec.

5.2.3 Impulse Response

We shall begin by analytically deriving the impulse response equation of a first order system. The Lapalce transform of a unit impulse function $d(t)$ is given by:

$X(s) = 1$, putting this in the TF of a unity gain first order system $\frac{Y(s)}{X(s)} = \frac{1}{\tau s + 1}$,

we obtain, $Y(s) = \frac{1}{\tau s + 1}$

Inverting to time domain, the required response equation is:

$$Y(t) = \frac{1}{\tau} e^{-t/\tau} \qquad (5.22)$$

It follows that the corresponding response for an impulse of magnitude, M, and a system with a gain K_P would be,

$$Y(t) = \frac{MK_P}{\tau} e^{-t/\tau} \qquad (5.23)$$

We may examine these response equations by considering a physical example. A set of experiments being carried out with the cylindrical tank liquid level system were discussed in the beginning of Chapter 2. The tank area is A_L. Let us assume, at the start of each run, the tank is empty and it has an open outlet pipe with a partially open valve constituting the linear outflow resistance R. Each time we have to pour M litre of a liquid from the top. The pouring occurs at a constant flow rate. Hence, at initial condition $t = 0$, the liquid begins to fall in the empty tank, and we are recording the tank level, $H(t)$, starting from $t = 0$, until the tank becomes empty again. *Note that within a finite pouring time, some amount of liquid will flow out from the tank through the bottom outlet line due to the liquid level build up in the tank.* Several experimental runs have been performed with same amount of the liquid, M, beginning with the pouring time, t_1. In the subsequent runs, the pouring time has been halved, with respect to the earlier run, i.e. in the 2nd run it is $0.5t_1$, in the 3rd it is $0.25t_1$, and so on.

What will happen in the limit when the same amount of liquid is poured in 'no-time' or instantly? No liquid will escape through the bottom outlet during the infinitesimally small charging time, and we will have an initial liquid level, $H^* = M/A_L$, at $t = 0$. To

have an idea about the discharge dynamics, we may recall Section 2.3, where we concluded that the discharge dynamics of a system, having an initial finite potential, follows a gradually diminishing discharge rate according to a decaying locus. By generalization with:

1. Discharging of capacitance through a resistance, and
2. Kinetics of a first order irreversible reaction, we concluded that the discharge dynamics follows an exponential decay function of the type e^{-Ct}. Applying Equation 5.23 for the tank level response for an impulse of M volume of water,

$$H(t) = \frac{MR}{\tau} e^{-t/\tau} = \frac{H * A_L R}{A_L R} e^{-t/\tau} = H * e^{-t/\tau} \qquad (5.24)$$

This result confirms the impulse response expressed as Equations 5.23 and 5.22, and determines the coefficient, C, on the exponential function as $C = 1/t$.

But what about the experimental runs that required finite time intervals for pouring same amount of water into the tank? We can at least say that longer the pouring time, smaller will be the maximum level attained within the tank at the end of pouring time. This is due to two reasons:

1. With longer pouring time, the input flow would be smaller, and
2. Longer input time would allow that much amount of water to flow out during the pouring interval.

The inputs that occur at $t = 0$ and remain at a finite constant value for a finite time interval are called Pulse input. At present, we may not derive the nature of the level build up locus, but it is obvious that the discharge dynamics, starting from the end of a pulse, should follow a similar exponential decay function as in case of the impulse response. The response would start from the level attained at the end of a pulse type water addition. The qualitative conclusions about the pulse responses and quantitative result of the impulse response are illustrated in Figure 5.8.

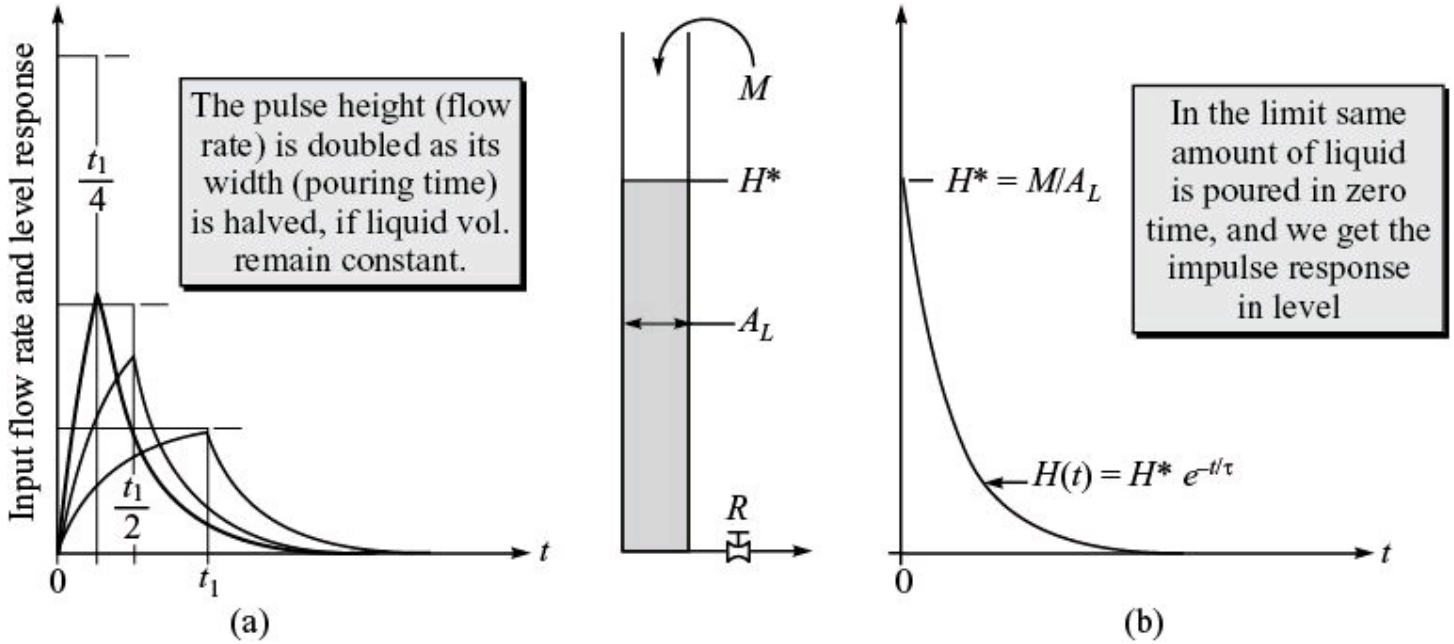


Figure 5.8 In the limit, a pulse response becomes an impulse response.

The response analysis of pulse and some other special input functions will be dealt after the treatment of ideal inputs. The following example will show the functional similarity of impulse response to another ideal input response.

EXAMPLE 5.4 You have been appointed as caterer for the reception of a wedding ceremony. On the day you are to serve a finely made soft drink as the preliminary appetizer to each of the guests. The drink mixture will be stored in a 10 litre vessel from which 100 ml drinks will be served per guest, through a bottom outlet valve, in glasses. After each serving, the server has to add an adequate amount water and soft drink concentrate into the tank, followed by an ample amount of stirring, so that the drink quality remains constant.

But to increase your profit unscrupulously, you have instructed the server (one of your employees) to pour only a glass of water per glass of drink served, and stir the storage tank content homogeneously before the next serving. Obviously, as the queue of the guest progresses, they will become increasingly annoyed about the drink quality.

(a) It is anticipated that when the drink concentration reduces to half of its initial value, the guests may go berserk and burst into a riot that could interrupt the ceremony. If the average time required for serving a single drink is 1 min, and the rituals for the ceremony will take 1 hr from start to finish, and the time required for dispatching the bride and groom to a safe house is 5 min, is it possible that the ceremony could be safely conducted? Assume that the ceremony rituals and drink serving start at the same time.

(b) If we gradually reduce the volume of drink per serving but still maintain the average outflow rate constant at 100 ml/min, the system will tend towards a constant exit flow of drink (of reducing concentration) and constant inflow of

fresh water, both at 100 ml/min. As the tank is ideally stirred, hence, we may assume that the drink concentrations inside the tank and in the exit flow are identical. In such a situation, the drink concentration in the exit flow describes the response to an ideal input function. Name the input.

Solution

(a) Storage tank volume, V = 10 lit. Volume of each drink, v = 100 ml.

Initial drink concentration, C_I^0

After serving the 1st drink, makeup water added and contents stirred,

Drink concentration in the tank, $C_I^1 = C_I^0\left(\frac{V-v}{V}\right)$

Similarly after serving the 2nd drink, makeup water added and contents stirred,

Drink concentration in the tank, $C_I^2 = C_I^1\left(\frac{V-v}{V}\right) = C_I^0\left(\frac{V-v}{V}\right)^2$

Proceeding in this manner, after serving the N no. of drink, makeup water added and contents stirred,

Drink concentration in the tank, $C_I^N = C_I^0\left(\frac{V-v}{V}\right)^N$

We want to know the value of N, when $C_I^N = C_I^0/2$; knowing that $\frac{V-v}{V} = 0.99$

Solving this, $0.5 = (0.99)^N$; or $\log(0.5) = N\log(0.99)$

or, $N = \frac{-0.30103}{-4.3648\times10^{-3}} = 68.9676 \cong 69$ glasses.

Hence the time required will be 69 min and you will be safely home.

(b) The continuous addition of fresh solvent to a solution at constant flow rate in a constant volume (holdup) perfectly mixed tank, while the exit flow is the overflow from the tank. In terms of response to ideal inputs the above description may be realized in two ways:

1. A pure solvent stream was running through the mixer. At time t = 0, an exact amount of solute required to obtain the initial concentration, C_I^0, has been put and mixed instantaneously in the mixer. In the outflow solution the concentration would show the response due to the impulse input, or
2. There are two input lines to the tank, one that carries the solution of concentration C_I^0, and the other caries only the pure solvent. Both inputs operate at the same flow rate of 100 ml/min. Previously, the system has been running with the

solution flowing through, while the pure solvent flow was zero and the holdup volume of the tank filled with solution of C_I^0 concentration. At time $t = 0$, the solvent flow is commenced and solution flow is stopped. The system receives a sudden input concentration change from C_I^0 to zero, and the exit solution would show a response to this –ve step input. The responses corresponding to the two descriptions should be identical because they describe a system behaviour under similar initial and dynamic condition. At present, the response analysis has been done for the impulse input. The –ve step response will be taken up a little later. It should be mentioned that for the analysis, we have assumed that the solution density is independent of concentration.

Let us check the previous result according to impulse response:

Tank volume = 10 l; Solvent flow rate = 100 ml/min. Hence, t = 10000/100 = 100 min.

The exit concentration response, $C_O(t) = C_I^0 e^{-t/\tau}$

Time, T for $C_O(t)$ to become half of C_I^0 may be found by putting $C_O(t) = C_I^0/2$.

Thus, $\ln(0.5) = -(T/\tau)$; or $T = 69.315$ min.

This value tallies with our previous result of 69 min. The slight discrepancy is due to the difference of discrete and continuous flows.

5.2.4 The Sinusoidal Response

Following the approach applied to the other three types of dynamic responses. We start with the transfer function of a unity gain first order system,

$$\frac{Y(s)}{X(s)} = \frac{1}{\tau s + 1} \qquad (5.25)$$

The Laplace transform of a sinusoidal function has been given by:

$$X(s) = \frac{M\omega}{s^2 + \omega^2} \qquad (5.26)$$

Putting Equation 5.26 into Equation 5.25,

$$Y(s) = \frac{M.\omega}{(s^2 + \omega^2)(\tau s + 1)} \qquad (5.27)$$

The above equation may be inverted by the way of partial fraction expansion, and the resulting time domain solution is given by:

$$Y(t) = \frac{M\omega\tau}{1 + \tau^2\omega^2} e^{-t/\tau} - \frac{M\omega\tau}{1 + \tau^2\omega^2} \cos(\omega t) + \frac{M}{1 + \tau^2\omega^2} \sin(\omega t) \qquad (5.28)$$

We shall use the following trigonometric identity to condense the two periodic functions in Equation 5.28.

$$p\cos A + q\sin A = r\sin(A+\varphi)\text{; where } r=\sqrt{(p^2+q^2)}\text{, and } \phi=\tan^{-1}\left(\frac{p}{q}\right)$$

Applying the above identity, Equation 5.28 becomes:

$$Y(t)=\frac{M\omega\tau}{1+\tau^2\omega^2}e^{-t/\tau}+\frac{M}{\sqrt{1+\tau^2\omega^2}}\sin\left[\omega t-\tan^{-1}(\omega\tau)\right] \qquad (5.29)$$

Similar to the ramp response equation, the first term of Equation 5.29 contains time the decay factor ($e^{-t/t}$), that will become negligible at large values of time. Ignoring this transient term, we have the stable part of the response as:

$$Y(t)=\frac{M}{\sqrt{1+\tau^2\omega^2}}\sin\left[\omega t-\tan^{-1}(\omega\tau)\right] \qquad (5.30)$$

Leaving the transient portion, the above expression for stable part of sinusoidal response is known as the 'frequency response' of the given dynamic system. Thus, Equation 5.30 is the frequency response of a first order system. The input and the response have been presented in Figure 5.9.

Let us examine the response.

The coefficient term in front of the trigonometric function is, in fact, the amplitude of the response or output. This term is only dependent on frequency, *w*, provided the other system parameters, e.g. *KP, t* are constant. Except for a zero value, for any finite value of *w*, the output amplitude must be less than the input amplitude, and it will gradually decrease with an increase of *w*.

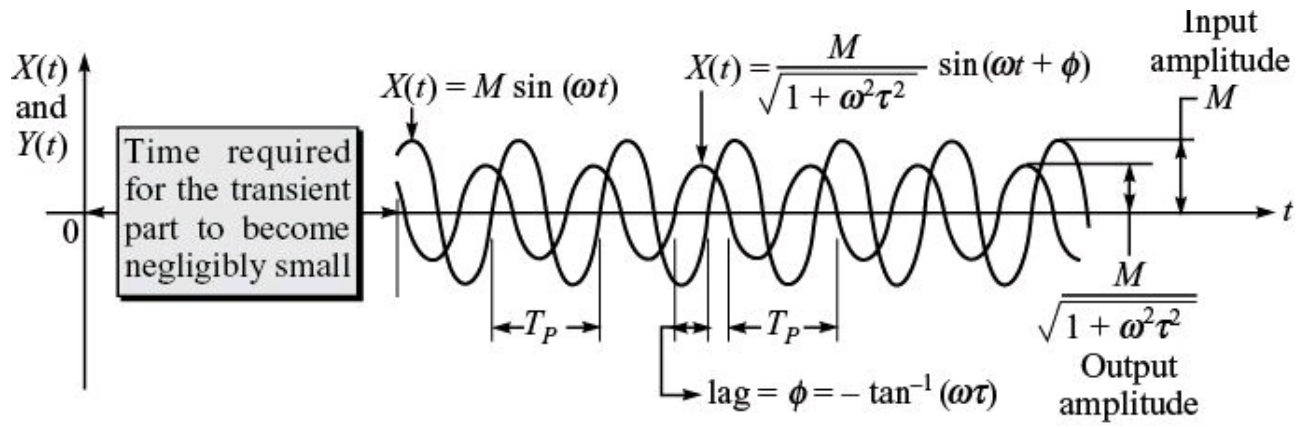


Figure 5.9 The frequency response of a first order system.

The Amplitude Ratio (AR) has been defined in relation to frequency response of a dynamic system as:

$$\text{AR} = \frac{\text{Output amplitude}}{\text{Input amplitude}} = \frac{M/[\sqrt{1+\omega^2\tau^2}]}{M} = \frac{1}{\sqrt{1+\omega^2\tau^2}} \qquad (5.31)$$

An alternative way for expressing AR is to convert it to 'Decibel' (*dB*), a term formerly used in electrical and electronic discipline and also in acoustic and audio systems to express the gain of an amplifier.

$$dB = 20[\text{Log}_{10}(\text{AR})] \qquad (5.32)$$

Thus, for a first order system,

$$dB = 20\,\text{Log}_{10}\left(\frac{1}{\sqrt{1+\tau^2\omega^2}}\right) \qquad (5.33)$$

The trigonometric sine function brings the oscillation in response and we observe that the response and input are at same frequency because the time variant part in both of them is *wt*. The response contains an extra term, $[\tan^{-1}(wt)]$. This term remains constant for a given frequency, *w*, and denotes the phase difference between input and response sinusoids. Phase difference in terms of angle is denoted by F. A 'phase-lead' signifies +ve value of F, and a 'phase-lag' signifies –ve value of F. A first order response will always lag behind its input as indicated by Equation 5.34.

$$\text{F} = -\tan^{-1}(wt) \qquad (5.34)$$

The input function types ramp and sinusoid are continuously changing with time whereas step and impulse only changes once during $0 < t < \infty$, i.e. at $t = 0$. The first order system poses an R–C type lag. Or it may be said that the full value of the response appears after some time due to the delaying effect of resistance-capacitance element couple.

EXAMPLE 5.5 A mixing process may be described as follows: A stream with solute concentration, C_I (gm/lit), is fed to a perfectly stirred tank at a constant flow rate of q (vol/time). The perfectly mixed product is withdrawn from the tank, also at the flow rate q, at the same concentration as the material in the tank C_O. The total volume of the solution in the tank is constant at V. Density may be considered to be independent of concentration. A trace of tank concentration is shown in Figure 5.10(a).

(a) Plot on this same figure your best guess of the quantitative behaviour of the inlet concentration vs. time. Label the graph with quantitative information regarding times and magnitudes of any other data which demonstrate your understanding of the situation.

(b) Write an equation for C_I as a function of time.

Data: Tank hold up volume, $V = 2830$ lit; Flow rate, $q = 100$ lit/min

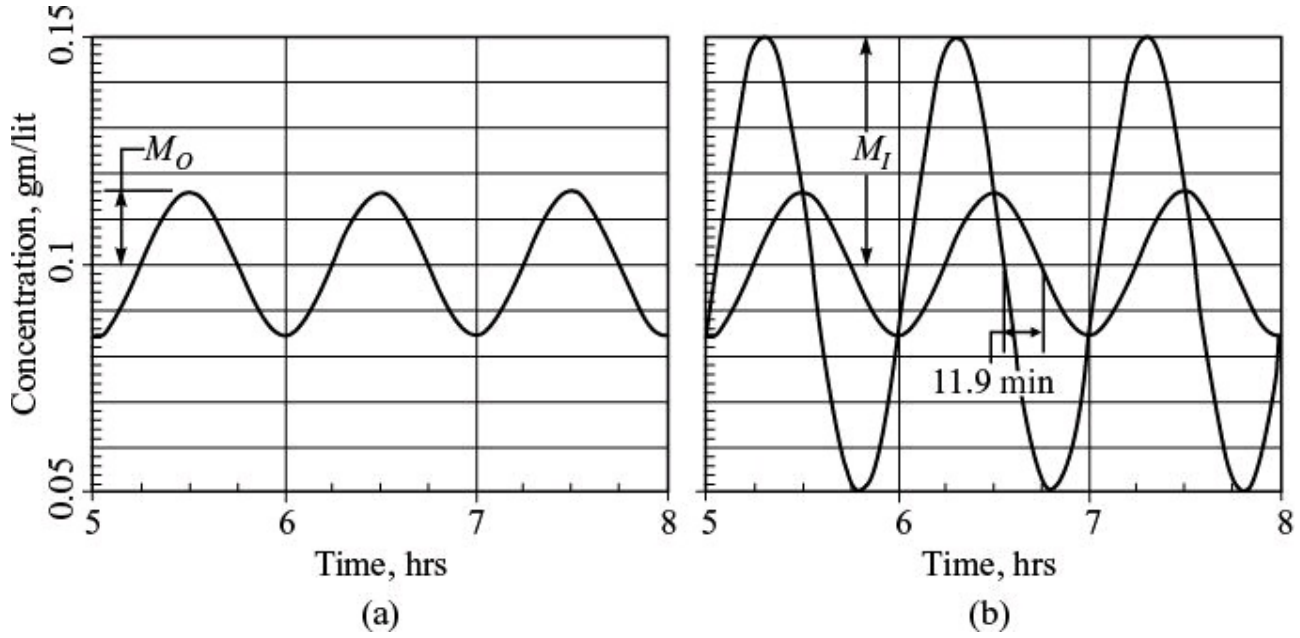


Figure 5.10 (a) The exit concentration, (b) The inlet and exit concentration of the mixer.

Solution From the trace of exit concentration oscillation, its time period TP = 1 hr = 60 min.

Hence, the angular frequency $w = 2p$ rad/hr = $(2p/60)$ rad/min= 0.10472 rad/min.

From the trace of exit concentration oscillation, the exit amplitude, M_O = 0.016 g/l.

From the given data the time constant, $t = V/q$ = 28.3 min.

By using Equation 5.30,

$$AR = \frac{M_O}{M_I} = \frac{1}{\sqrt{1+\tau^2\omega^2}}$$

Hence,

$$M_I = 0.016\sqrt{1+\{28.3\times(2\pi/60)\}^2} = 0.05 \text{ g/l}$$

By using the same Equation 5.30, the phase angle f may be calculated as:

$$\varphi = -\tan^{-1}(\omega\tau) = -71.34^\circ$$

Hence,

$$\text{lag} = \frac{60\times 71.34}{360} = 11.9 \text{ min}$$

With this information, the inlet concentration trace has been drawn in Figure 5.11(b), together with the exit concentration. Now the inlet concentration function may be proposed as:

$C_I(t) = 0.1 + 0.05 \sin(0.10472t)$ g/l., where t is in min.

Note that the threshold or average concentration around which the oscillation occurs remains unaffected by the mixing operation. This may be easily understood, because as there is no addition or depletion of mass in such a mixing system, the total and component mass balance will result in an identical average concentration of inlet and

exit streams.

Such mixers may be used in process applications where a small but concentrated solution is injected in a large stream of liquid by a reciprocating type +ve displacement pump (say plunger or diaphragm). A typical application may be the dosing of liquid chlorine in drinking water. Also, in continuous polymerization processes, addition of initiator, inhibitor, chain length modifier, and addition of different additives in petroleum product development are other examples of such pulsed dosing system. Also, ON–OFF control of any process produces an oscillatory character of the controlled variable. The average concentration may be what is required, but if the amplitude of oscillation in the exit stream is excessive, such mixing systems with designed holdups may be put on the exit stream to bring down the oscillation amplitude.

Note that the time base in Figure 5.10 starts not from 0, instead a finite value of 5 hrs. The reason being that the response is traced after a sufficiently large value of time, to allow the Transient part of the response (first term in the R.H.S. of Equation 5.29) to die down. Is there any possibility of obtaining any quantitative estimate of this 'sufficient time' as a general guide-line? At present, a couple of hints may be forwarded.

1. The transient envelope of the oscillations are due to the function ($e^{-t/t}$) in the response, and
2. The numerical value of the function (e^{-X}) becomes less than 0.01, for $X \geq 5$. The rest is left to the reader.

What happens when a pure solvent stream is added to dilute the inlet stream as well as to reduce the amplitude of concentration? The average concentration is definitely reduced but there is a question about the amplitude reduction relative to the case with no dilution.

EXAMPLE 5.6 (a) A solution is fed into an ideally stirred mixer at a constant rate of 10 lpm. The solute concentration at the mixer inlet varies according to a sinusoid, oscillating between 7.2 and 2.8 g/l with a time period of 64 sec. Another stream of pure solvent is also fed into the mixer at a constant rate of 5 lpm. The effective volume of the mixer is 12.5 lit. The outlet stream comes out from the mixer as an overflow. Find the concentration response in the exit stream. Plot the inlet and outlet concentrations on a common time base.

(b) For the same process described in part (a), if there is no addition of pure solvent stream, that is, there is no dilution, plot the inlet and outlet stream concentration on a

common time base and compare the amplitude ratio of concentration obtained part (a) and (b).

Solution: (a) The best way to treat such a problem is to conceive that the solution and the solvent streams are not being joined in the mixer, instead they are premixed (say coming from two sides and mixing at the meeting point of a 'Tee', and the joined stream goes out from the third limb into the mixing tank). This arrangement has been explained in Figure 5.11.

(*The same system may feature either a dilution operation of a solution by putting a pure solvent, or a solution of lower concentration as the second stream, or an enrichment operation by putting a concentrated solution as the second stream to the mixer. In the two situations, we shall obtain an exit stream of lower or higher average concentration as the case may be. The treatment for such cases would follow a similar approach as indicated in this example. In calculating the output amplitude, one should remember that the time constant has changed due to the inclusion of the second stream.*)

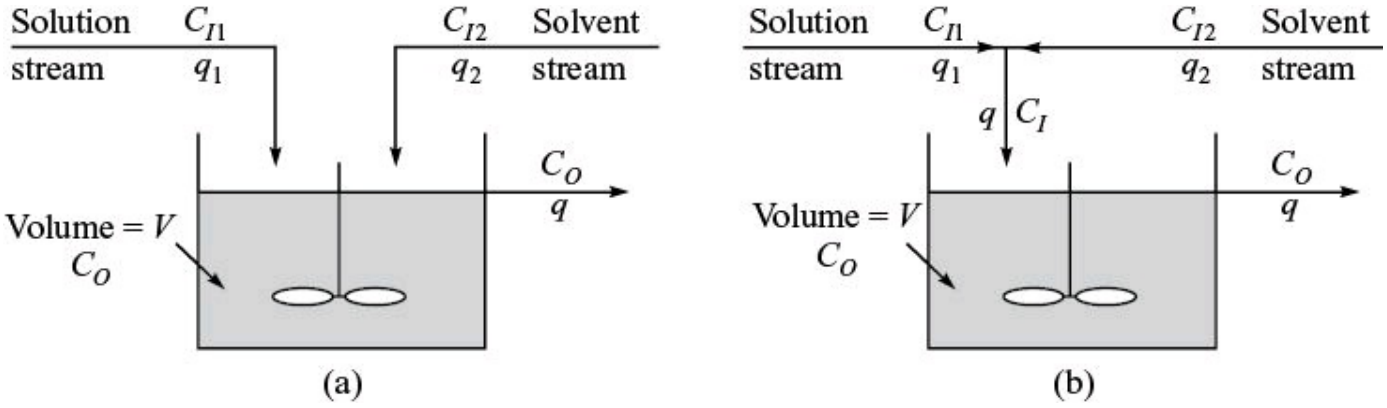


Figure 5.11 (a) The actual stream arrangement. (b) Conceptual presentation of premixing.

For the solute balance around the Tee mixer, consider the average concentrations of the inlet and outlet streams: $\bar{C}_{I1} = (7.2 + 2.8)/2 = 5$ g/l, $\bar{C}_{I2} = 0$; and we want to find $\bar{C}_I$. The stream flow rates are: q_1 = 10 lpm; q_2 = 5 lpm; q = 15 lpm. The solute balance for the two inlet and the outlet streams would be:

$$C_{I1}q_1 + C_{I2}q_2 = C_I q$$

Putting the known values in the above equation we get:

The average concentration in the mixed stream, $\bar{C}_I = \dfrac{5 \times 10}{15} = \dfrac{50}{15} = 3.33$ g/l

The input amplitude in the mixed stream will be proportionally reduced to new value of,

$$M_I = 2.2 \times \frac{10}{15} = 1.47 \text{ g/l.}$$

The time constant, $t = 12.5/15 = 0.833$ min = 50 sec.

But the angular frequency w, will remain unchanged.

Angular frequency of oscillation, $w = (2p)/TP = 0.0982$ rad/sec.

The sinusoidal response equation may be directly applied:

$$C_O(t) = \bar{C}_I + \frac{M_I}{\sqrt{1+\omega^2\tau^2}} \sin[\omega t - \tan^{-1}(\omega\tau)]$$

where, t = Time constant of the mixer = 50 sec.

The exit concentration amplitude,

$$M_O = \frac{M_I}{\sqrt{1+\omega^2\tau^2}} = \frac{1.47}{\sqrt{1+(0.0982\times 50)^2}} = \frac{1.47}{5.0108} = 0.29337 \text{ g/l}$$

and the phase lag,

$$\Phi = \tan^{-1}(\omega\tau) = \tan^{-1}(0.0982\times 50) = 78.48824^\circ = \frac{64\times 78.48824}{360} = 13.9535 \text{ sec.}$$

The above results, together with the mixing and dilution system, have been demonstrated in Figure 5.12.

(b) Without the pure solvent stream, we will analyze the mixing effect on the solution stream in which the solute concentration is varying according to a sine function.

$$C_I(t) = \bar{C}_{I1} + M_{I1} \sin(\omega t)$$

where, Average concentration, $\bar{C}_{I1} = (7.2 + 2.8)/2 = 5$ g/l;

Amplitude of the inlet concentration sinusoid, $M_I = 7.2 - \bar{C}_{I1} = \bar{C}_{I1} - 2.8 = 2.2$ g/l;

Time period of oscillation, TP = 64 sec;

Angular frequency of oscillation, $w = (2p)/TP = 0.0982$ rad/sec.

The sinusoidal response equation may be directly applied:

$$C_O(t) = \bar{C}_{I1} + \frac{M_I}{\sqrt{1+\omega^2\tau^2}} \sin[\omega.t - \tan^{-1}(\omega\tau)]$$

where, t = Time constant of the mixer = V/q = 12.5/10 = 1.25 min = 75 sec.

The exit concentration amplitude,

$$M_O = \frac{M_I}{\sqrt{1+\omega^2\tau^2}} = \frac{2.2}{\sqrt{1+(0.0982\times 75)^2}} = 0.296 \text{ g/l}$$

and the phase lag,

$$\Phi = \tan^{-1}(\omega\tau) = \tan^{-1}(0.0982 \times 75) = 82.2678^\circ = \frac{64 \times 82.2678}{360} = 14.6254 \text{ sec.}$$

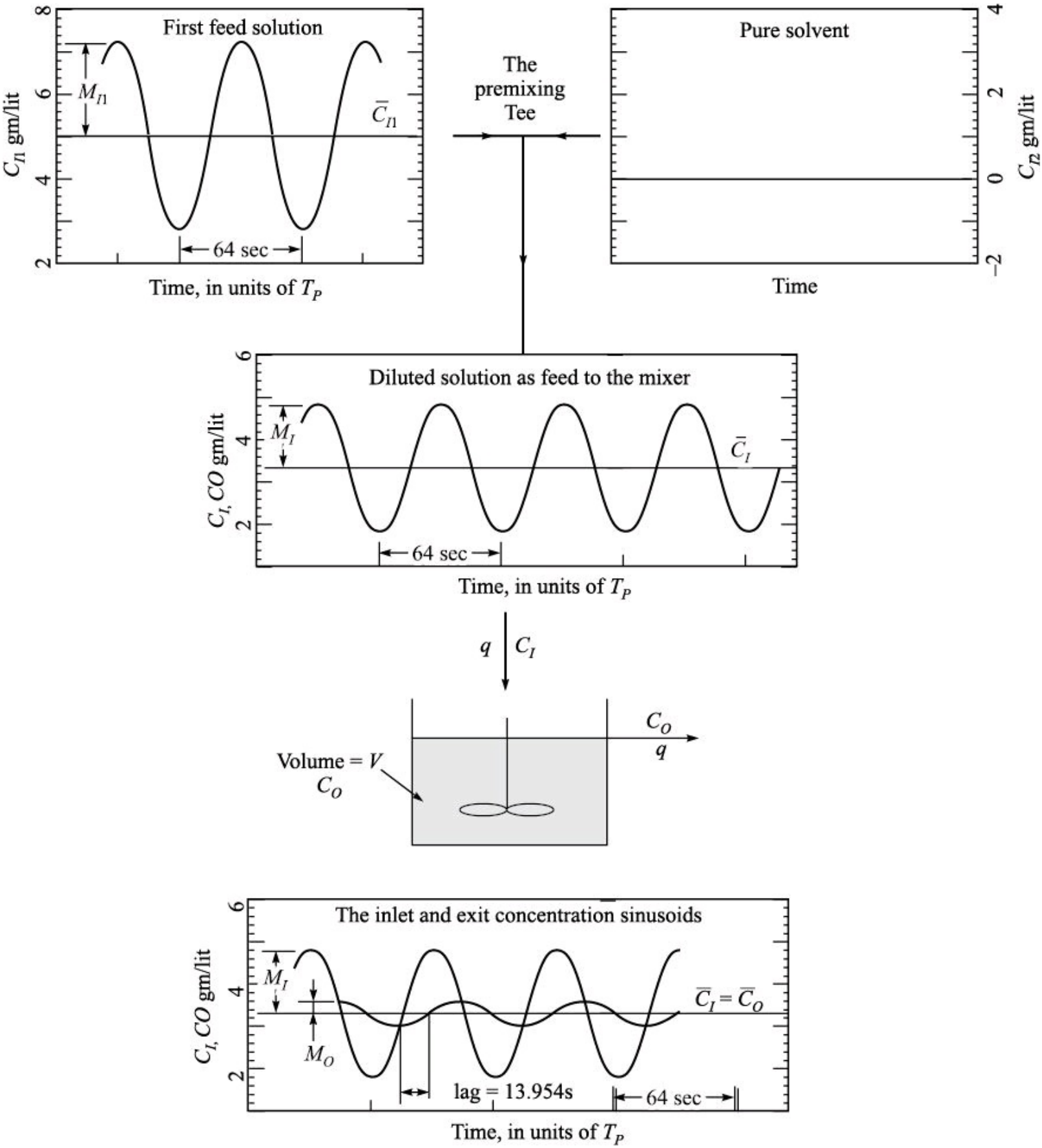


Figure 5.12 The conceptual premixing and shifting of the average concentration value.

The sinusoidal concentration and input and response have been plotted on a common time base in Figure 5.13. By designing the volume of such a mixer the inlet

amplitude of concentration, oscillation may be reduced to the desired extent. This is definitely a low-cost solution to obtain more uniform product quality, not only for sinusoidal variation, but may be applied for any arbitrary type of periodic fluctuation of the concerned process variable.

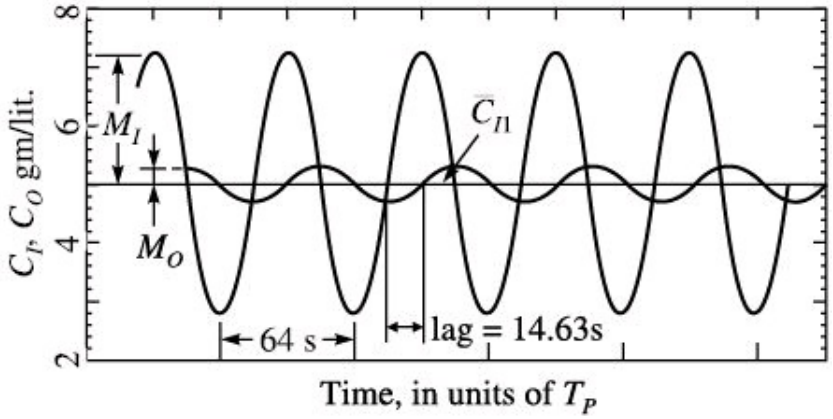


Figure 5.13 The input response without premixing and shifting of the average concentration value.

The Amplitude Ratio of concentration oscillation between inlet and final outlet streams found for part (a):

In part (a) diluted condition $[AR]_D = 0.29337/2.2 = 0.1335$

In part (b) Undiluted condition $[AR]_{UD} = 0.296/2.2 = 0.1345$

So the overall reduction in AR is marginal by dilution, but if we consider the AR* with respect to actual entering and leaving streams of the mixer,

$$[\text{AR}^*]_D = \frac{1}{\sqrt{1+\omega^2\tau_D^2}} = \frac{1}{\sqrt{1+(0.0982\times 50)^2}} = 0.19957$$

$$[\text{AR}^*]_{UD} = \frac{1}{\sqrt{1+\omega^2\tau_{UD}^2}} = \frac{1}{\sqrt{1+(0.0982\times 75)^2}} = 0.13454$$

where, t_D = Time constant of the mixer (with dilution) = 50 sec, and t_{UD} = Time constant of the mixer (undiluted condition) = 75 sec.

The higher value of $(\text{AR}^*)_D$ than $(\text{AR}^*)_{UD}$ is due to the lower value of t_D than t_{UD}.

5.3 More Examples of First Order Process Response Analysis

EXAMPLE 5.7 A cylindrical closed bottom liquid level tank is provided with a top liquid inlet line and a bottom exit line. The exit line has an adjustable resistance as a valve. The valve is initially closed. An incoming liquid flow of 22.5 l/min is suddenly started from the top inlet line. The level rises at a uniform rate up to a height of 150 cm in 5 min, when the exit line valve is partially opened. It is observed that subsequently the tank level remains at 100 cm even at large values of time. Find the

transfer function, $H(S)/Q_I(s)$, at this operating condition of the tank.

Solution Level rises through 150 cm in 5 min for an inflow rate of 22.5 lpm. From this data the tank internal area (A) may be calculated as:

$$\mathrm{A} = \frac{22.5 \times 5 \times 1000}{150} = 750 \text{ cm}^2$$

At large value of time, that is after steady state has been attained and $q_I = q_O = h/R$, the level remain at 150 cm. From this information the valve resistance may be calculated as:

$$R = \frac{150}{22.5 \times 1000} = 0.0067 \frac{\text{cm}}{\text{cc/min}}$$

Thus, TF of the liquid level system is obtained as:

$$\frac{H(s)}{Q_I(s)} = \frac{R}{\mathrm{AR}s + 1} = \frac{0.0067}{5s + 1} \frac{\text{cm}}{\text{cc/min}}$$

Of course, this transfer function is valid only when the exit valve is open to the extent as described in the problem statement. In the closed condition, the same system would behave like an integrator. We have devoted Section 5.5 on the dynamics of integrating processes. The reader is urged to go through the portion, to understand the following result. The transfer function for the closed valve condition may be given as:

$$\frac{H(s)}{Q_I(s)} = \frac{1}{As} = \frac{1}{750s} \frac{\text{cm}}{\text{cc/min}}$$

In the next example, model development from dynamic data for an energy transfer to a fixed mass has been proposed. There is no material flow in or out from the system—a batch of fluid is heated, hence, input to the system is energy from a constant potential source, and output variable is the temperature of the body of fluid. Also, there is a demonstration of how the initial speed of step response property has been used to calculate a required system parameter value.

EXAMPLE 5.8 During heating of a cylindrical batch type water heating tank after starting a constant rate of energy flow, the following data of time vs. temperature are obtained. The tank (1.0 m ID × 2.0 m H) is well stirred and full to its capacity.

Time, min	0	4	8	12	20	30	40	60	80	100	120	160	200	240
Temp. °C	35	40	44.8	49	55	61	66.2	73	78	81	83.7	86.2	87.5	87.5

Determine the time constant of the system by as many methods as you can. Can you explain, if these values do not agree? Find the overall heat transfer coefficient, U, in kcal/[(min)(°C)(m^2)], and the change in the heat input, W, in kcal/min. The ambient temperature is 35°C. The heating tank was full to its capacity.

Solution This thermal dynamic system may be represented by Figure 5.14.

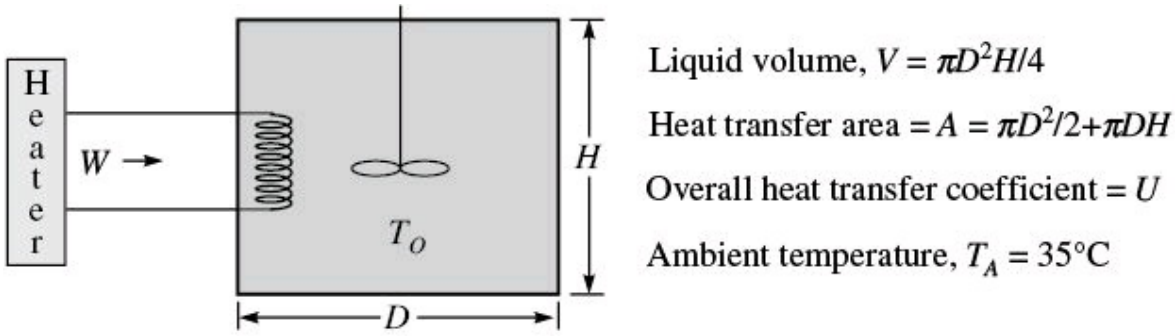


Figure 5.14 The thermal dynamic system of example 5.8.

A dynamic heat balance for this system may be proposed a:

$$W - UA(T_O - T_A) = V\rho C_p \frac{dT_O}{dt} \qquad \text{(A)}$$

In writing this heat balance equation, we have defined the electrical power input, W, as the input to the system and heat dissipation through the tank wall due to the difference of the tank liquid temperature, T_O, from the ambient temperature T_A = 35ºC.

Equation A at steady state condition,

$$W_S - UA(T_{OS} - T_A) = V\rho C_P \frac{dT_{OS}}{dt} \qquad \text{(B)}$$

Subtracting Equation B from A and introducing deviation variables, $\bar{W} = W - W_S$, and $\bar{T}_O = T_O - T_{OS}$

$$\bar{W} - UA\bar{T}_O = V\rho C_P \frac{d\bar{T}_O}{dt} \qquad \text{(C)}$$

By Laplace transformation of Equation C, the following transfer function is obtained:

$$\frac{\bar{T}_O(s)}{\bar{W}(s)} = \frac{(1/UA)}{(V\rho C_P/UA)s + 1} \qquad \text{(D)}$$

where process gain, $K_P = 1/UA$; and time constant, $\tau_P = V\rho C_P/UA = mC_P/UA$.

For a step input in W of magnitude M, the response equation in $\bar{T}_O$ will be:

$$\bar{T}_O(t) = K_P M(1 - e^{-t/\tau}) \qquad \text{(E)}$$

The initial rate of the response would be:

$$\left.\frac{d\bar{T}_O}{dt}\right|_{t=0} = \left.\frac{K_P M}{\tau} e^{-t/\tau}\right|_{t=0} = \frac{MUA}{UAmC_P} = \frac{M}{mC_P} \qquad \text{(F)}$$

From the tank dimensions, the mass of the liquid in the tank will be as follows:

$$m = \rho V = \frac{\pi \times 100^2 \times 200}{4 \times 1000} \text{S lit.} \times 1 \frac{\text{kg}}{\text{lit.}} = 1570.8 \text{ kg}$$; and for water, $C_P = 1 \frac{\text{kcal}}{(\text{kg})(°\text{C})}$

To obtain the initial value of the slope of response, we plot T_O vs. time data and construct the tangent at the beginning of the curve as in Figure 5.15.

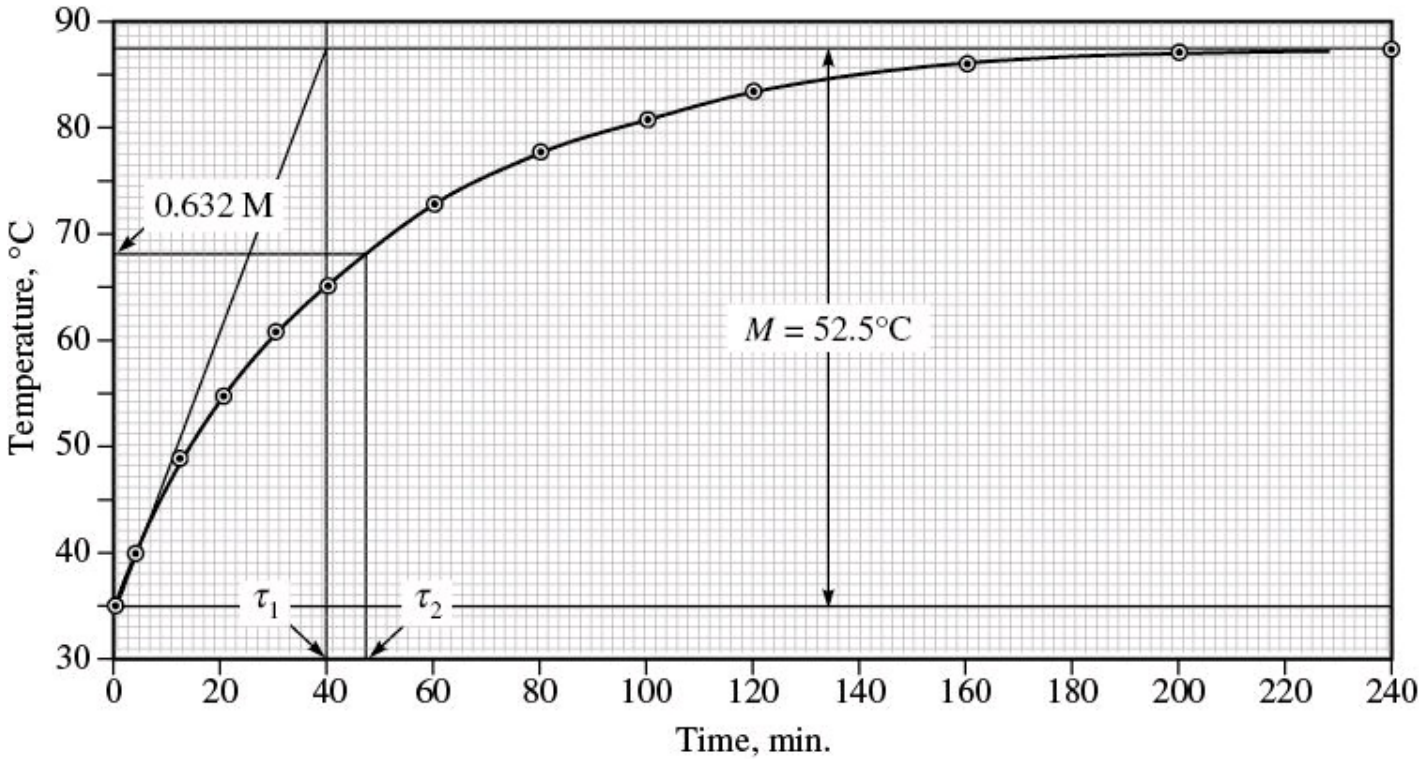


Figure 5.15 Construction of initial slope on the Experimental data.

From above the initial slope, $S_I = \frac{52.5}{40} = \frac{M}{mC_P} = \frac{M}{1570.8}$; □ M = 2061.673 kcal/min

This heat input rate equals the heat dissipation rate from the tank wall when the tank liquid temperature becomes constant at 87.5 ºC. Hence, we may write for this steady state condition:

$$M = UA(T_{OS} - T_A) \text{ or } U = \frac{2061.73}{6.28 \times 52.5} = 6.2532 \frac{\text{kcal}}{(\text{min})(°\text{C})(\text{m}^2)}$$

The three values of time constants, t_1 = 40 min, evaluated from initial slope, and t_2 = 46 min, equal to the time required for the recovery of 63.2% of step magnitude, and $t_3 = (mC_P)/(UA)$ = 40 min, do not differ much. However, this difference of time constant values shows the limitation of the assumption that the tank is ideally stirred, i.e. no temperature distribution in the tank liquid exists. As such, we may accept t_3 to be more accurate as this has been determined based on the steady state condition and thus, less possibility of non-homogeneous temperature distribution, and the fact remains that $t_3 = t_1$.

In the following Examples 5.9, 5.10, and 5.11, various developments from Section 4.4 have been used. The reader is urged to go through the referred material, given under Examples 4.16, 4.17 and 4.18.

EXAMPLE 5.9 In a constant volume isothermal CSTR (V = 2 m^3), a first order liquid phase reaction, $-r = kC_O$ (k = 0.05 min^{-1}) is taking place. Feed flow is constant at q = 0.1 m^3/min. At steady state, the inlet concentration, C_I = 1.0 mol/m^3. The exit and bulk concentrations are equal due to ideal mixing conditions, and is denoted by C_O. An alarm sounds if and when the reactant concentration in the reactor reaches 0.9 mol/m^3, because there is a risk of the onset of a second series reaction that will greatly reduce reaction yield.

(a) If C_I is doubled from its present steady state value, when will the alarm sound?

(b) Suppose an operator intervention is made by reducing C_I to its previous value at the time of alarm, derive the response C_O (t) after the initial and through the final change in C_I .

(c) What is the maximum permissible change in C_I from its steady state value, that would not require any intervention/control action?

Solution: (a) From the problem statement, the system parameters and variable value at steady state may be written as:

$$V = 2 \text{ m}^3; k = 0.05 \text{ min}^{-1}; q = 0.1 \text{ m}^3/\text{min}; C_{IS} = 0.9 \text{ mol/m}^3.$$

Referring to Section 4.4, we know that for the constant inlet flow rate, q, of the reactant solution and constant holdup volume condition, the transfer function between exit and inlet concentration may be written as:

$$\frac{\bar{C}_O(s)}{\bar{C}_I(s)} = \frac{q_S/(q_S + kV)}{[V/(q_S + kV)]s + 1} = \frac{0.1/(0.1 + 0.05 \times 2)}{2/(0.1 + 0.05 \times 2)s + 1} = \frac{0.5}{10s +}$$

where, the Gain, K_P = 0.5; and Time constant, t_P = 10 min, and the corresponding step response equation of exit concentration $C_O(t)$ in the deviation form may be given as:

$$\bar{C}_O(t) = M \times K_P[1 - e^{-t/\tau_P}] = 1 \times 0.5 \times [1 - e^{-t/\tau_P}] = 0.5 \times [1 - e^{-t/\tau_P}]$$

By the problem statement, the absolute and deviation form of the variables are equal, hence, we may use the absolute values and drop the bar over the variables. At steady state

$$C_{OS} = 0.5 \times (1 - 0) = 0.5 \text{ mol/m}^3$$

thus C_{OS} is within the alarm limit.

After the system has attained steady state with C_I = 1.0 mol/m^3, C_I is doubled from its previous value, i.e. a second step change in C_I from 1.0 to 2.0 mol/m^3, has been made. Hence, C_{OS} will be also doubled.

That is, the new steady state value of exit concentration will be, $C_{OS} = 2 \times 0.5 = 1.0$ mol/m^3.

This value is above the alarm limit. Let us denote the time to reach the limit as T_M. The dynamics in terms of deviation variables for the 2nd step change up to $t = T_M$ would be:

$$\frac{\bar{C}_O(T_M)}{\bar{C}_{OS}} = \frac{0.9 - 0.5}{1.0 - 0.5} = \frac{0.4}{0.5} = [1 - e^{-T_M/10.0}] \text{ or } T_M = 16.0944 \cong 16.1 \text{ min}$$

The alarm will sound after 16.1 min.

(b) The response equation when C_I was changed from 0.85 to 1.7 mol/m^3 would be, $\bar{C}_O(t) = 0.5 \times [1 - e^{-t/\tau_P}]$ mol/m^3; in terms of deviation variables, and in terms of absolute value of the variable, $C_O(t) = 0.5 + 0.5 \times [1 - e^{-t/\tau_P}]$ mol/m^3.

After the exit concentration has reached the alarm limit, i.e. C_O = 0.9 mol/m^3 and C_I has been brought down to its original value of 1.0 mol/m^3, C_O will show a negative step response that may be described as:

$$C_O(t) = 0.9 - 0.4 \times [1 - e^{-(t - T_M)/\tau_P}] \text{ mol/m}^3 \text{ for, } t > 16.1 \text{ min.}$$

(c) Maximum permissible input concentration change, C_M, from its steady state value of
0.85 mol/m^3 that would produce an exit concentration within the limit may be calculated as:

$$K_P = 0.5 = \frac{\Delta C_O}{\Delta C_I} = \frac{0.4}{C_M} \text{ or } C_M = \frac{0.4}{0.5} = 0.8 \text{ mol/m}^3$$

If the operator has an option of creating an additional flow of pure solvent as a dilution liquid in the alarm situation, then what should be its flow rate to keep the exit concentration just under the limit? Let q_S be the required flow rate in m^3/min, then,

$$q \times 1.0 = (q + q_S) \times 0.9 \text{ or } q_S = 0.0101 \text{ m}^3\text{/min.}$$

EXAMPLE 5.10 In a constant volume (2.0 m^3) isothermal CSTR, the reaction $A \rightarrow$ Products, takes place with 1st order kinetics. The feed flow rate remains constant (q = 0.005 m^3/s), but the feed concentration, C_I, may vary. The reactor dynamics is experimentally tested by putting two disturbances in C_I, one following the other, the corresponding exit concentration responses as deviation from their steady state values are recorded against time.

Response for Disturbance (1)									
Time sec	120	130	140	150	160	170	180	190	200
$C_O - C_{OS}$ mol/m^3	0	0.394	0.632	0.776	0.864	0.938	0.95	0.97	0.982

Response for Disturbance (2)									
Time sec	600	610	620	630	640	650	660	670	680
$C_O - C_{OS}$ mol/m^3	0	0.394	0.632	0.776	0.864	0.524	0.318	0.192	0.116

Prepare two separate plots of the response data and find the corresponding input functions. Also find the reaction rate constant, k.

Solution (a) From the problem statement, the system parameters may be written as:

$$V = 2.0 \text{ m}^3; q = 0.005 \text{ m}^3/\text{min}; C_{IS} = 0.1 \text{ mol/m}^3.$$

Referring to Section 4.4, we know that for the constant inlet flow rate, $q = q_S$ of the reactant solution and constant holdup volume condition, the transfer function between exit and inlet concentration may be written as:

$$\frac{\bar{C}_O(s)}{\bar{C}_I(s)} = \frac{q/(q+kV)}{[V/(q+kV)]s+1}$$

We have drawn the response plots from the given tabular data as deviation variables in concentration as ordinates and time from the introduction of inputs, in Figures 5.16(a) and (b) respectively for the 1st and 2nd disturbances:

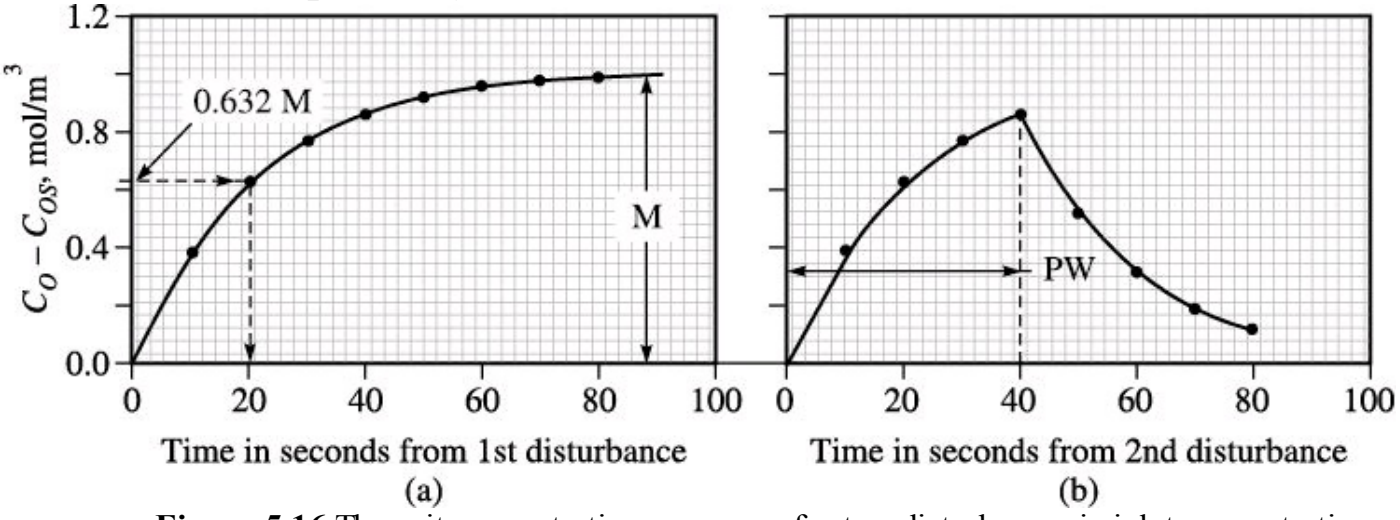


Figure 5.16 The exit concentration responses for two disturbances in inlet concentration.

In Figure 5.16(a), we have a step response and the time required for 63.2% step magnitude recovery has been found to be 20 sec, which is the time constant of the system. We equate this value to the expression of t in the transfer function to obtain the value of reaction rate constant, k:

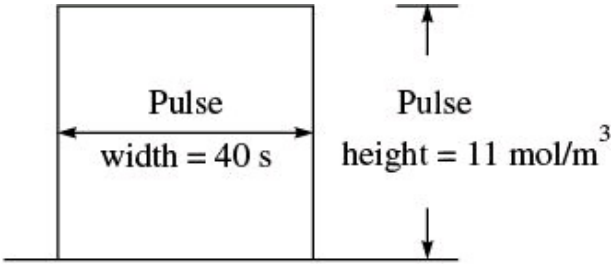
Thus, $\tau = \dfrac{V}{q + kV} = 20$ sec, putting values of q and V, $k = 0.025$ sec^{-1}

Also the second respond response in Figure 5.16(b) corresponds to a pulse input. As the first five points of response (b) are equal to that of (a), it may be assumed that the input pulse height was equal to that of the step magnitude of the first disturbance. For a linear system without any delay element, the response starts the decay dynamics at the end of pulse duration. Hence, input Pulse Width, PW = 40 sec.

To find the pulse height, we have to find the input step magnitude in case of the first disturbance (a), because pulse height and step height are equal as the rising part of the pulse response retraces the step response dynamics. First, the system gain is found from the transfer function expression as follows:

$$K_P = \frac{q}{q + kV} = \frac{0.005}{0.005 + 0.025 \times 2.0} = 0.090909, \text{ dimensionless.}$$

From Figure 5.17(a) the ultimate value of the response is 1.0 mol/m^3. Hence, input magnitude of the step must be:



$M = 1.0/0.0909 = 11.0$ mol/m^3.

This will be equal to the Pulse height.

EXAMPLE 5.11 In a constant volume (2 m^3) isothermal CSTR, the reaction $A \rightarrow$ Products, takes place with 1st order kinetics. The feed flow rate, q, as well as the inlet concentration of the reactant material C_I may both vary. Using the following inlet data, $C_I = 0.925$ mol/m^3,

$q = 0.085$ m^3/min, $k = 0.05$ min^{-1}, find the values of gain, time constant and steady state exit concentration, C_O, of the reactant. At this condition, the inlet reactant concentration, C_I, is increased by 0.0925 mol/m^3 from its previous value according to

a step change. Find the new steady state value of exit concentration, C_O, after the change. Find the step magnitude of flow rate, q, by which it should be changed to bring back C_O to the steady state value that was obtained for $C_I = 0.925$ mol/m^3 and $q = 0.085$ m^3/min.

Solution We could use the linear dynamic model developed in Section 4.4 for a CSTR, handling isothermal irreversible first order kinetics for:

1. variable concentration, constant inlet flow, and
2. variable q, but constant C_I.

But according to the problem statement, the changes have been made only under steady state conditions, hence, an alternative and easier way may be adopted.

The TF between C_O and C_I for constant q is:

$$\frac{C_O(s)}{C_I(s)} = \frac{q/(q+kV)}{[V/(q+kV)]s+1}$$

Hence, the steady state gain, $K_P = q/(q + kV) = 0.085/(0.085 + 0.05 \times 2) = 0.459459459$.

Thus, under steady state, the exit concentration value would be:

$$C_O = K_P C_I = 0.425 \text{ mol/m}^3.$$

The time constant in this condition is $t_1 = V/(q + kV) = 2/(0.085 + 0.05 \times 2) = 10.81$ min.

The inlet concentration after a step change of magnitude 0.0925 is $C_I = 0.925 + 0.0925 = 1.0175$ mol/m^3. The new steady state exit concentration may be calculated considering the variables in their deviation form, but the calculation may also proceed with their absolute values. For the increased value of C_I, the steady state value of exit concentration would be, $C_O = C_I \times K_P = 1.0175 \times 0.459459459 = 0.4675$ mol/m^3.

After the concentration change and the subsequent flow rate change, the system is allowed to come to steady state. For this situation, the above TF relating exit and inlet concentration is still applicable by considering that q is constant at its new value for a sufficiently long time. Let the new steady state gain be denoted by K_{PN} that will bring down the exit concentration to its previous value, $C_O = 0.425$ mol/m^3.

$$K_{PN} = \frac{0.425}{1.0175} = 0.417690417 = \frac{Q}{Q + kV}$$

where Q is the new value of flow rate.

From the above expression, the new value of flow rate is found as, Q = 0.07173 m^3/min. The new value of time constant will be, $t_2 = V/(Q + kV)$ = 11.65 min.

The flow rate has to be step changed by a –ve magnitude of M = –0.01327 m^3/min. This decrease of flow rate is necessary, because after the increase of inlet concentration, if the exit concentration has to remain as it was, the reactor has to run for a higher conversion, which is possible by increasing the residence time of the reactant stream or time constant of the reactor. This notion will be confirmed, if the ratios of conversions and time constants could be found equal with respect to the two conditions:

Ratio of conversions: $\frac{(0.925 - 0.425)/0.925}{(1.0175 - 0.425)/1.0175} = 0.92827$.

Ratio of residence times: $\frac{10.81081081}{11.64618878} = 0.92827$.

5.4 Lead-Lag Element

A special dynamic element called the **Lead-Lag** element is treated in the next example. It has the basic transfer function structure:

$$\frac{Y(s)}{X(s)} = \frac{\tau_D s + 1}{\tau_I s + 1}$$

The numerator and denominator of the TF represent the single **Zero** (at $-1/t_D$, 0 on the complex plane) and the single **Pole** (at $-1/t_I$, 0). The zero imparts a **Differentiating** and the pole imparts an **Integrating** character to the overall TF of lead-lag element. The following example will explain how the two properties are realized.

EXAMPLE 5.12 Given is a system with the following transfer function. Find $Y(t)$ if $X(t)$ is a unit step function, for (A) t_1/t_2 = 5, and (B) t_1/t_2 = 1/5, sketch $Y(t)$ vs. t/t_2. What are the maximum and minimum values of the response during the transient?

$$\frac{Y(s)}{X(s)} = \frac{\tau_1 s + 1}{\tau_2 s + 1}$$

Solution:

Putting $X(s) = \frac{1}{s}$, we obtain, $Y(s) = \frac{\tau_1 s + 1}{s(\tau_2 s + 1)} = \frac{\tau_1}{\tau_2 s + 1} + \frac{1}{s(\tau_2 s + 1)}$

Clearly, the corresponding time domain solution will contain two parts—an impulse and a step response. We may write the complete time domain solution as a sum of the solutions of these two parts:

$$Y(t) = Y_1(t) + Y_2(t) = \frac{\tau_1}{\tau_2} e^{-t/\tau_2} + [1 - e^{-t/\tau_2}]$$

We have obtained the solutions for two ratios of t_1/t_2 , for $Y_1(t)$, $Y_2(t)$ and $Y(t)$ as plots on a common reduced time base of t/t_2 in Figure 5.17. In part (a) for $t_1/t_2 = 5$, and in part (b) for $t_1/t_2 = 1/5$. From the output traces the maximum and minimum values of $Y(t)$ may be readily read as $Y_{MAX} = 10.0$ and $Y_{MIN} = 1.0$ for $t_1/t_2 = 5$, and (b) $Y_{MAX} = 1.0$ and $Y_{MIN} = 0.2$ for $t_1/t_2 = 1/5$.

(*These results could be also obtained by analytical way. Reader is asked to try to find these results by differentiating the response equation and analytically determining the maximum and minimum values by equating the derivative to zero.)*

Compared to part (a), an amplified scale has been used for plotting the response for part (b) to obtain a visual clarity of the response.

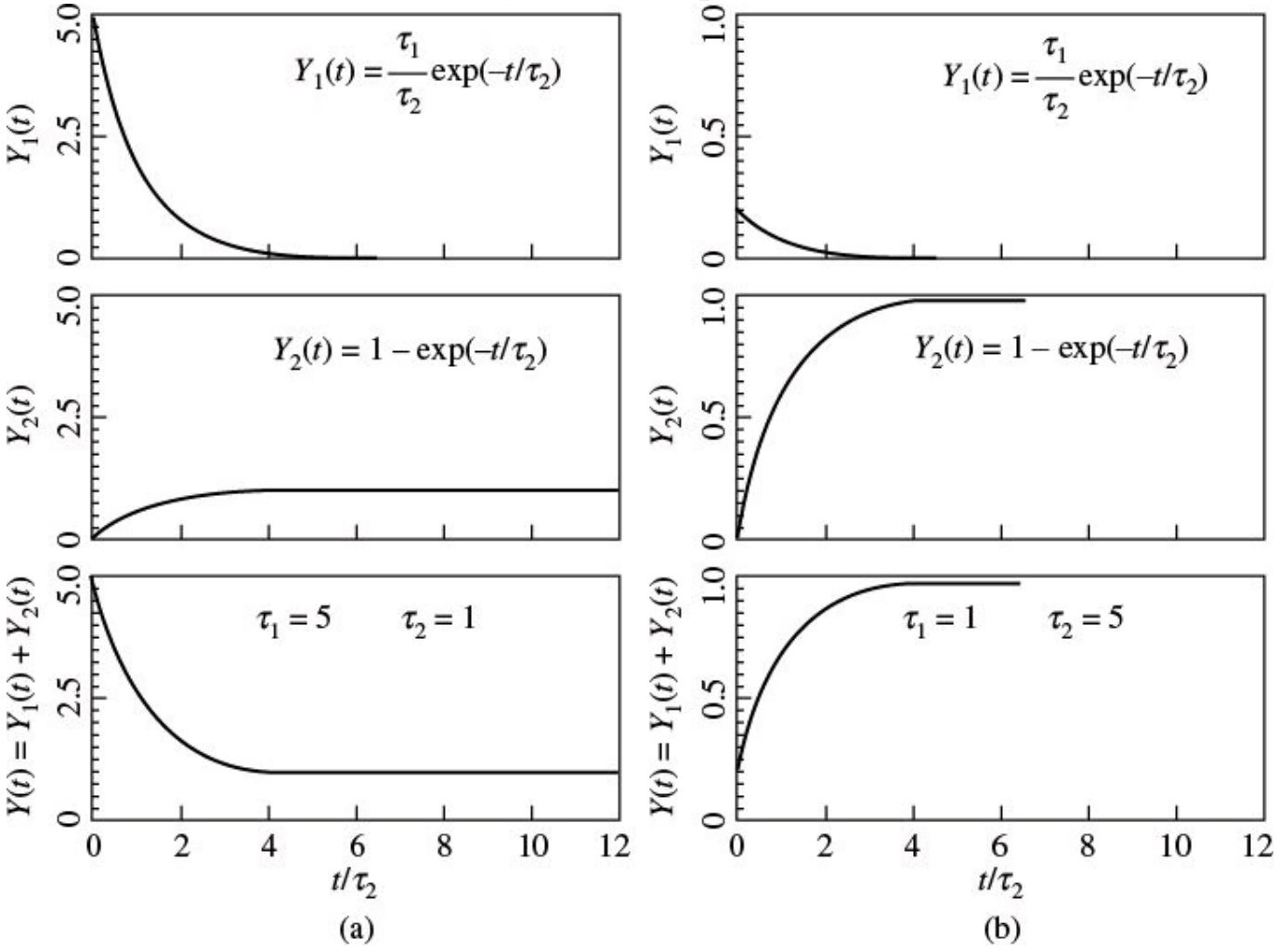


Figure 5.17 Unit step response of the system $G_P(s) = (t_1 s + 1)/(t_2 s + 1)$ for (a) $t_1/t_2 = 5$, for (b) $t_1/t_2 = 1/5$.

Let us examine these plots with a little more attention, specially the total response $Y(t)$ vs. t/τ_2, for the two ratios of τ_1/τ_2.

In part (a) for $\tau_1 > \tau_2$, the response looks somewhat like an impulse response.	In part (b) for $\tau_2 > \tau_1$, the response looks somewhat like a step response.

If we interpret this response character to the fact that:

In part (a), the s in the numerator has a greater emphasis as $\tau_1 > \tau_2$.	In part (b), the s in the denominator has a greater emphasis as $\tau_2 > \tau_1$.
Or, going a little step further,	
In part (a), the system moves closer to a differentiator, because of the larger coefficient of numerator s.	In part (b), the system moves closer to an integrator, because of the larger coefficient of denominator s.

The differentiating and integrating properties of **Lead-Lag** element is further enhanced by assuming more extreme values of the ratio, t_1/t_2. The differentiating property of this element is used in the realization of derivative mode of an automatic controller. To discuss about this we shall reopen this topic in Chapter 8.

5.5 Special Input Functions and Their Responses

During process operation, a dynamic system often faces an input that can be hardly defined as ideal. Combining the knowledge that we have acquired so far about ideal inputs, many of such disturbances may be treated. In Figure 5.18, four such disturbances have been illustrated. The response of a unit gain 1st order system for these input functions would be discussed.

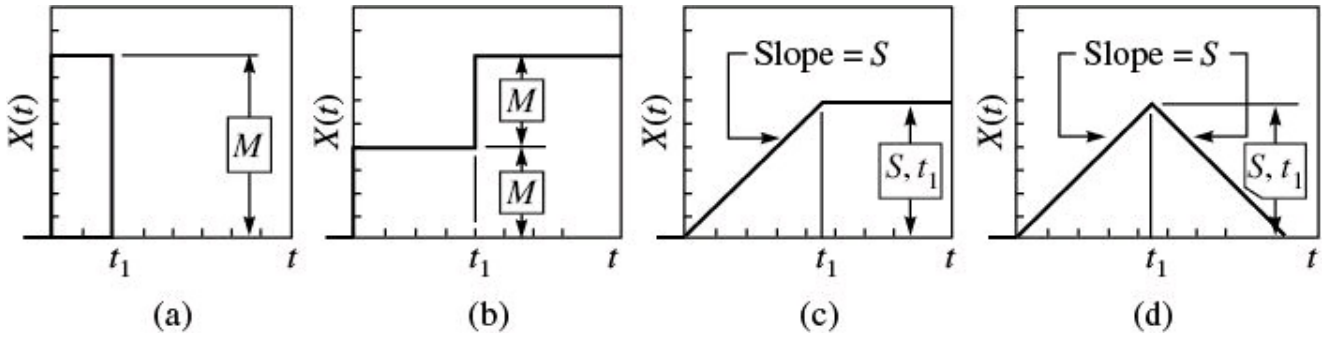


Figure 5.19 Non-ideal inputs (a) Pulse. (b) Double-step or stair-case. (c) Ramp with saturation. (d) Triangular.

5.5.1 The Analytical Approach

Generally such complex or non ideal input functions consists of ideal input functions occurring together of slightly time staggered fashion. Hence, treatment of ideal input functions coupled with time translation (time shift) theorem is required for their

response generation.

The Response of Pulse Input

The pulse function has been already defined and may be conceived as a combination of two steps. The 1st one is +ve of magnitude, M, occurring at $t = 0$, the 2nd one is of –ve magnitude $– M$, occurring at $t = t_1$. Hence, the pulse height is M, and its width is t_1. In Figure 5.20(a) these step functions have been shown. In time domain, the functions are:

$$X_1(t) = MU(t) \quad \text{and} \quad X_2(t) = -MU(t - t_1)$$

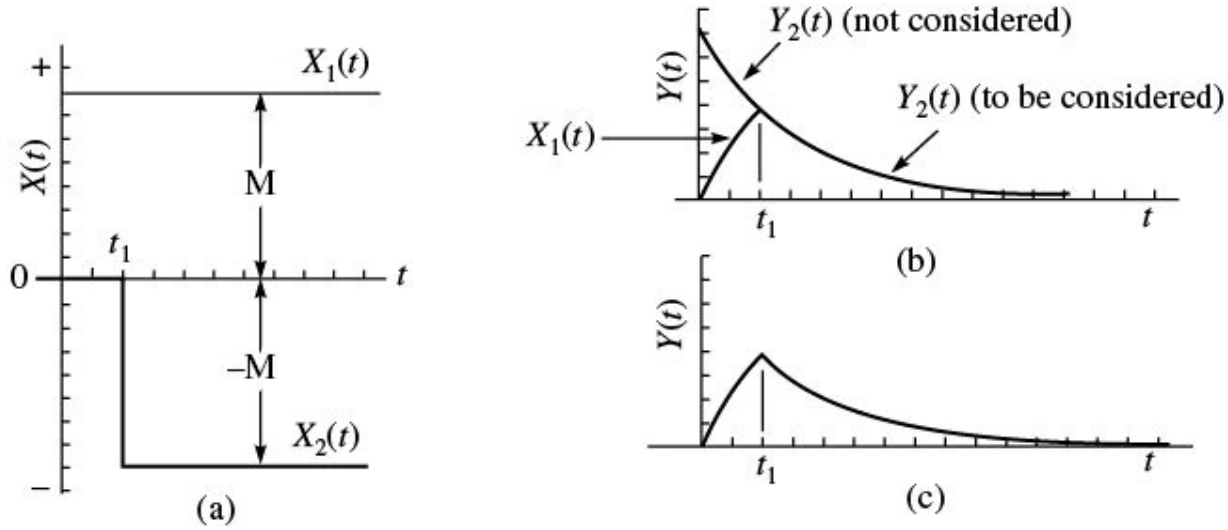


Figure 5.19 Analytical treatment to obtain the pulse response.

The +ve step occurring at $t = 0$ may be readily transformed, and the delayed –ve step occurring at $t = t_1$ may be transformed by the help of the Time Shift Theorem (see Appendix-II). The results are:

$$X_1(s) = \frac{M}{s} \quad \text{and} \quad X_2(s) = -\frac{M}{s}e^{-st_1}$$

The total input function, as a combination of these two, may be written as:

$$X(s) = X_1(s) + X_2(s) = \frac{M}{s}(1 - e^{-st_1}) \qquad (5.35)$$

When such an input enters into a unity gain the 1st order system, there will be two response equations valid for two time zones.

1. For $0 < t < t_1$, there is only one input to the system (+ve step), so the response is:

$$Y(t) = Y_1(t) = M(1 - e^{-t/\tau}) \qquad (5.36)$$

2. For $t > t_1$, the system is under the +ve step and the delayed –ve step, so the response will be a combination of these two inputs, thus, is:

$$Y(t) = Y_1(t) + Y_2(t) = M(1 - e^{-t/\tau}) - M(1 - e^{-(t-t_1)/\tau}) = Me^{-t/\tau}(e^{t_1/\tau} - 1) \qquad (5.37)$$

In Figure 5.19(b) results of equations 5.36 and 5.37 have been plotted with the

following parameter values: $Kp = 1$, $t = 40$, $M = 80$, $t_1 = 20$. The + ve step response $Y_1(t) = Y(t)$ valid for $0 < t < t_1$. Though Equation 5.37 is only valid for $t > t_1$, yet the time function $e^{-t/t}$ in this equation makes it possible to be evaluated for $t > 0$. We have plotted the function but denoted the portion by a dotted locus that is not to be considered. The total response has been plotted in Figure 5.19(c).

A Two Step Staircase Input

For such a function of equally spaced two steps of equal height, M, the time description is a combination of two equal magnitude steps, occurring after equal time intervals of t_1.

$$X(t) = MU(t) + MU(t - t_1) \qquad (5.38)$$

The corresponding transform is: $X(s) = \frac{M}{s}(1 + e^{-st_1})$ (5.39)

For a unit gain 1st order system, the response equation would be:

$$Y(t) = M(1 - e^{-t/\tau}) + M(1 - e^{-(t-t_1)/\tau}) \qquad (5.39a)$$

Note that the 2nd part of the response equation has a zero value for time values less than t_1.

Ramp Input With Saturation

The time description and the necessary splitting of the function into component functions have been shown respectively in Figures 5.20(a) (b) and (c).

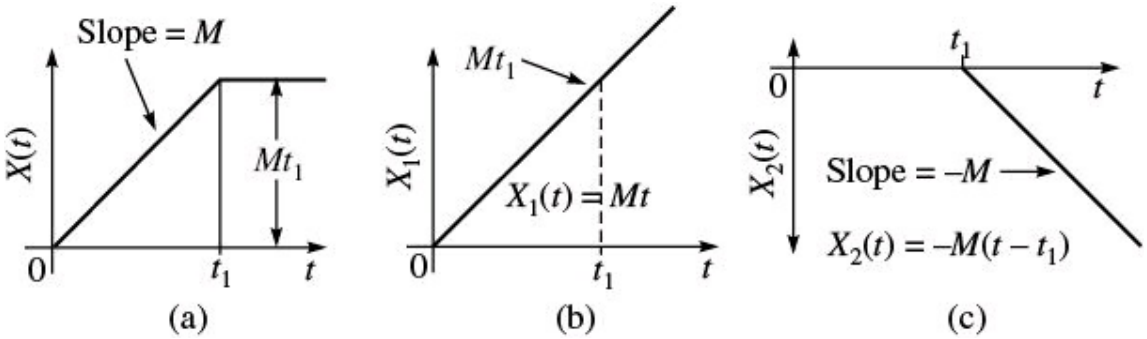


Figure 5.20 Splitting of a ramp with saturation into component function.

The time domain functions and their transform are given by:

$$X(t) = MtU(t) - MtU(t - t_1) \qquad (5.40)$$

and,

$$X(s) = \frac{M}{s^2} - \frac{M}{s^2}e^{-st_1} \qquad (5.41)$$

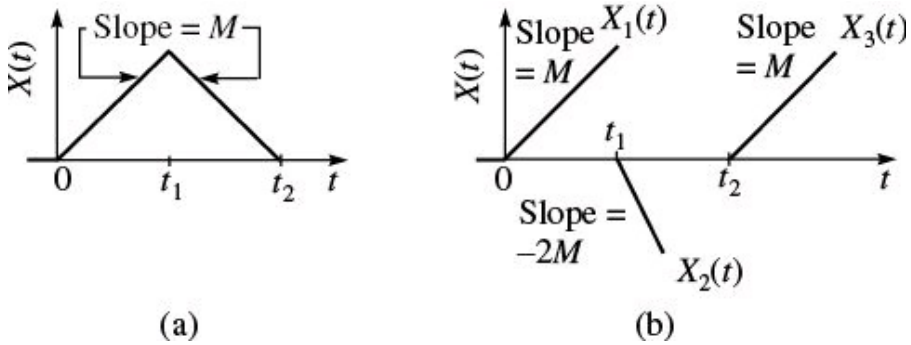
For a unit gain 1st order system, the response equation would be,

$$Y(t) = M(t - \tau + \tau e^{-t/\tau}) - M[(t - t_1) - \tau + \tau e^{-(t-t_1)}] \qquad (5.41a)$$

Note that the 2nd part of the response equation has a zero value for time values less than t_1.

Triangular Input

The time description and the necessary splitting of the function into component functions has been shown respectively in Figures 5.21(a) and (b).



(a) (b)

Figure 5.21 Splitting of a triangular input into component function.

The time domain functions and their transform are given by:

$$X(t) = MtU(t) - 2MtU(t-t_1) + MtU(t-t_2)$$
$$= MtU(t) - 2MtU(t-t_1) + MtU(t-2t_1) \quad (5.42)$$

and, $$X(s) = \frac{M}{s^2} - \frac{2M}{s^2}e^{-st_1} + \frac{M}{s^2}e^{-2st_1} = M\left(\frac{1-2e^{-st_1}+e^{-2st_1}}{s^2}\right) \quad (5.43)$$

Note that the time description of each component function for any of the complex function requires the introduction of the Unit Step $U(t)$ as a multiplier of a step of magnitude M or a ramp of slope M, and the history of occurrence of any of the component is contained by $U(t)$, as $M.U(t - t_1)$ means a step of magnitude M appearing at $t = t_1$ amount of delay. A delayed ramp is stated similarly as $Mt.U(t - t_1)$. The subsequent response analysis becomes lengthy and each delayed function response is obtained from origin, but the portion from the origin ($t = 0$), up to the appearance point ($t = t_1$) of the component function is not to be considered, as shown in Figure 5.19(b).

5.5.2 An Alternative Approach

Here we present another method for treatment of such special functions that has more relevancy with the physical significance of the functions related to their time description. The total function is split into parts that are ideal and continuous. A delayed part of the function is represented as $f(t - t_{DL})$ in time domain, where, t_{DL} is the time interval after which the function has appeared. If two ideal input functions are required to construct a part of the input, responses corresponding to those parts

are summed up to obtain the total response. For all the ideal parts of the input, responses are derived following the results of Section 5.2. The representation of complex functions by this approach may be easier to understand and practice, as it has been seen through student feedback. Before discussing the details, it is necessary to expose a few conceptions.

(A) Consider two dynamic liquid level systems made of two identical cylindrical tanks (Area = A) having equal resistances, R in their bottom outlet line are subjected to two kinds of input:

1. The first tank was previously empty when a volume V (= AH^*) of water at t = 0, is poured instantaneously into the tank as an impulse.
2. The second tank was previously operating at a steady state with constant inflow and outflow rates while the tank level was constant at H^*. At t = 0, the inflow, q_I, is suddenly and completely stopped. Thus, a –ve step input in inflow q_I has been introduced. Let us derive the level response for the two cases.

1. We have already calculated the level response while introducing the impulse response and Equation 5.24 may be directly written as:

$$H(t) = \frac{MR}{\tau} e^{-t/\tau} = \frac{H^* AR}{AR} e^{-t/\tau} = H^* e^{-t/\tau} = H^* e^{-t/(AR)}$$

2. We start from the unsteady state mass balance around the tank for the condition when inflow is stopped.

$$0 - \rho q_O = \rho A \frac{dh}{dt} \qquad (5.44)$$

As inflow term is zero, canceling r on each side of Equation 5.44,

$$-q_O = A \frac{dh}{dt}$$

Again, we apply the linear relation, $h = q_O.R$, to eliminate q_O and obtain,

$$-\frac{h}{R} = A \frac{dh}{dt} \qquad (5.45)$$

Then writing the above equation at steady state condition, subtracting the steady state form of Equation 5.45 from Equation 5.45, and finally introducing deviation variables,

$$-\frac{H}{R} = A \frac{dH}{dt} \qquad (5.46)$$

Note that we have to integrate the above Equation 5.46 between the limits H^* to

H, and 0 to t, to obtain the required response equation. Thus,

$$-\int_{H(t)}^{H^*} \frac{dH}{H} = \int_{t}^{0} dt$$

Performing the integration, we obtain:

$$H(t) = H * (e^{-t/\tau}) = H * e^{-t/(AR)} \qquad (5.47)$$

Thus, impulse and –ve step inputs yielded identical response, provided the initial value of the output variable H^* and parameter values A and R are equal. The result may be formally generalized as:

*When the principal entity (*PE*) from a capacitance element (C) initially stored up to a principal variable (potential) value Y_I, discharges through a resistance (R) into a sink at zero potential, the principal variable response (discharge dynamics of the potential) may be given as:*

$$Y(t) = Y_I e^{-t/\tau} = Y_I e^{-t/CR} \qquad (5.48)$$

Alternatively, a 1st order system starting from any initial value of the output variable, Y_I, would show the drainage or depletion dynamics given by Equation 5.48, where C and R, respectively, are the capacitance and resistance of the system.

(B) In a continuous tunnel drier, there are two zones maintained at different temperatures in increasing order. The first zone, which is shorter in length, is at 100ºC, and the second zone has a longer path length which is at 150ºC. The article to be dried initially at room temperature of 30ºC, is put on a conveyor outside the drier. The conveyor moves the article through the temperature zones. Heat enters into the article according to a first order dynamics. Article temperature is considered to be the output variable. Conveyor speed is such that the article passes through the first zone before it attains the furnace temperature (100ºC) of that zone. But in the second zone, the article attains the zone temperature (150ºC) because the residence time is long enough. The article is subjected to an input function depicted in Figure 5.22(a). This is what is called the Double Steps of a staircase function. At $t = t_1$, the article enters the second zone from the first zone.

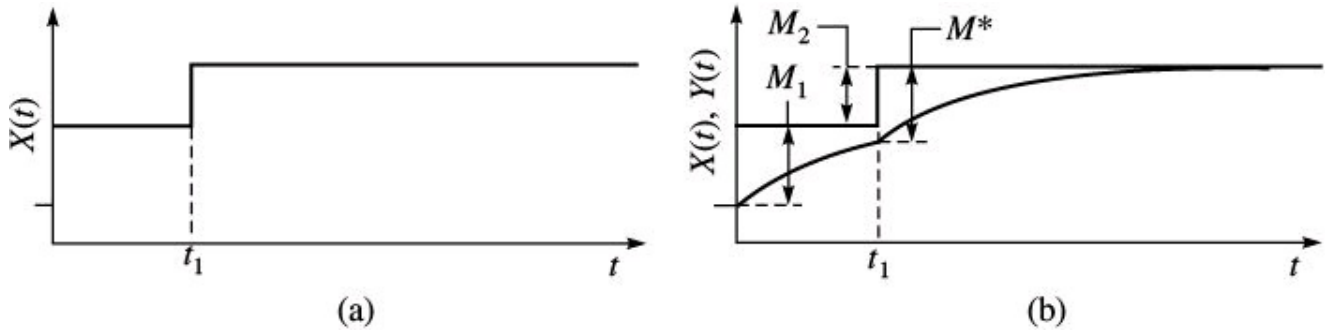


Figure 5.22 The double step input and the response.

The first step response equation is valid for $0 > t > t_1$, and may be written as:

$$T_A(t) = 30 + 70(1 - e^{t/t}) \text{ for } 0 < t < t_1$$

We are interested about the second step response. Apparently, the second step magnitude is 150ºC – 100ºC = 50ºC (M_2). But actually at t_1, the article temperature, $T_A(t)$, following the first step magnitude of 100ºC – 30ºC = 70ºC (M_1), is:

$$T_A(t_1) = 30 + 70(1 - e^{t_1/\tau})$$

Putting the parameter values $t = 40$, and $t_1 = 60$, in above equation, we get

$$T_A(t_1) = 84.4^{\circ}\text{C}.$$

This is less than 100ºC. Hence, the article just entering into the second zone of the furnace would face a temperature difference between itself and the furnace temperature = 150ºC – 84.4ºC = 65.6ºC. This is the second step magnitude and the system (article) would show the step response with respect to this corrected magnitude of 65.6ºC (M*). These observations have been presented in Figure 5.22(b). Recognizing the fact that the second step response is valid for $t > t_1$, the second step response equation may be written as:

$$T_A(t) = 84.4 + 65.6(1 - e^{-(t-t_1)/\tau}) \text{ for } t > t_1$$

5.5.3 Response Calculation Examples

Using the above procedure we may try to derive the response of a first order unity gain system to the inputs given in Figure 5.18.

EXAMPLE 5.13 (Pulse): The initial part of the pulse is treated as a step of magnitude M appearing at $t = 0$. The corresponding response is:

$$Y(t) = M(1 - e^{-t/\tau}) \text{ for } 0 > t > t_1$$

Following the +ve step at $t = 0$, the system responds up to $t = t_1$ and at t_1, the output is:

$$Y(t_1) = M(1 - e^{-t_1/\tau}) = M^*$$

After that, the input value is zero. Hence, after t_1, the system may be considered to be draining or dissipating the potential or response value it has acquired at t_1. This potential is expressed as $Y(t_1)$. Using the conclusion of Section 5.5.2(A), the 2nd part of the response (drainage dynamics) for $t > t_1$ may be directly written as:

$$Y(t) = Y(t_1)[e^{-(t-t_1)/\tau}] = M * e^{-(t-t_1)/\tau} \text{ for } t > t_1$$

The input function and details of the response derivation are illustrated in Figure 5.23.

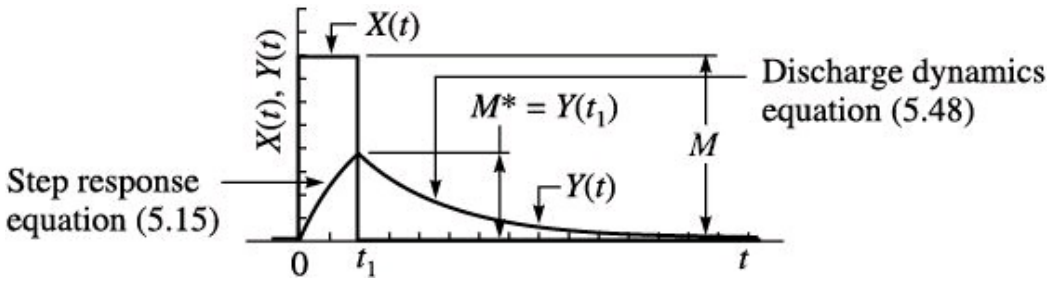


Figure 5.23 Pulse Input and the Response.

EXAMPLE 5.14 (Double-Step or Staircase): The method of deriving response to this function has been already discussed in Section 5.5.2(B). We may directly write the response equations for equal input step magnitude M, as in Figure 5.19(b).

$$Y(t) = M(1 - e^{-t/\tau}) \text{ for } 0 > t > t_1 \qquad (5.49)$$

Following the first +ve step at $t = 0$, the system responds up to $t = t_1$ and at t_1 the output is:

$$Y(t_1) = M(1 - e^{-t_1/\tau}) = M' \qquad (5.50)$$

The magnitude of the second step will be, $M^* = 2M - M'$. The response for $t > t_1$ is:

$$Y(t) = M_1 + M * (1 - e^{-(t-t_1)/\tau}) \text{ for } t > t_1 \qquad (5.51)$$

For response visualization, Figure 5.22 may be used with the modification $M_1 = M_2$. This approach may be applied for a staircase function with more than two steps.

EXAMPLE 5.15 (Ramp with Saturation): From $t = 0$ to $t = t_1$, there will be a ramp response as:

$$Y(t) = M(t - \tau + \tau e^{-t/\tau}) \text{ for } 0 < t < t_1 \qquad (5.52)$$

At $t = t_1$, the response value is:

$$Y(t_1) = M(t_1 - \tau + \tau e^{-t_1/\tau}) \qquad (5.53)$$

At the end of the ramp, the input does not increase with a fixed slope but becomes constant or remains at the fixed value of $M.t_1$. Hence, the potential (difference between the input and the system output value) at $t = t_1$ that the system has to recover is:

$$Mt_1 - Y(t_1) = M^* = M\tau(1 - e^{-t_1/\tau}) \qquad (5.54)$$

Recovery of this potential will be according to a step response of magnitude M^* for $t > t_1$ according to which:

$$Y(t) = Y(t_1) + M*(1 - e^{-(t-t_1)/\tau}) \text{ for } t > t_1 \qquad (5.55)$$

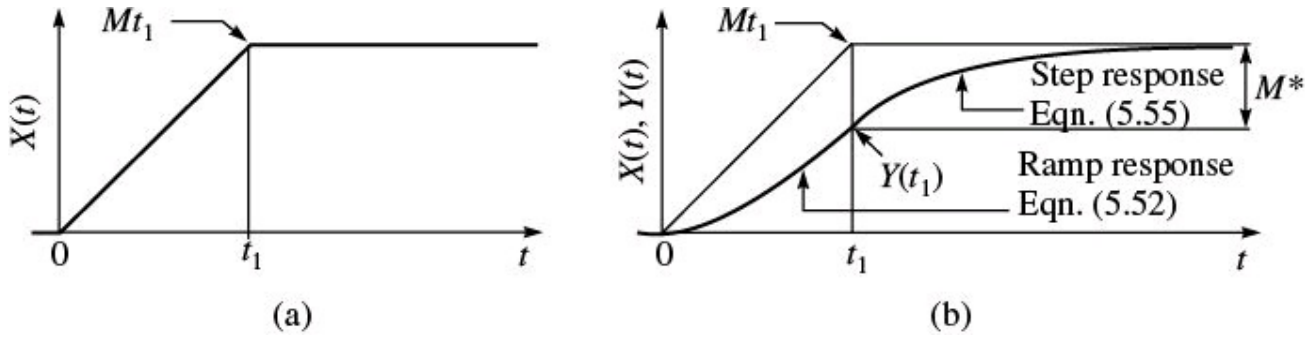


Figure 5.24 Response for a ramp input with saturation.

The derivation of response has been illustrated in Figure 5.24.

EXAMPLE 5.16 (Triangular): The input function has been illustrated in Figure 5.18. The first part of response from $t = 0$ to $t = t_1$, for the +ve ramp input is identical to Example 5.15 and Equation 5.52 is applicable. Also, at $t = t_1$, the response value may be expressed by Equation 5.53, and at the end of the ramp, the difference between the input and the system output value is given by Equation 5.54. The input and output for $0 > t > t_1$ is shown in Figure 5.25(a).

After $t = t_1$, the input proceeds according to a ramp of slope = – M, starting from a height of Mt_1 up to $t = t_2$. This input may be split into two ideal input functions acting parallely and additively on the system:

1. A ramp of slope – M starting from a height of $Y(t_1)$ found by Equation 5.53,

$$Y'(t) = [Y(t_1) - M(t - t_1 + \tau e^{-(t-t_1)/\tau})] \text{ for } t_1 < t < t_2 \qquad (5.56)$$

and,

2. A step of magnitude M^* starting from base line,

$$Y''(t) = [M*(1 - e^{-(t-t_1)/\tau})] \text{ for } t_1 < t < t_2 \qquad (5.57)$$

Both inputs act during $t_1 < t < t_2$.

The splitting of the –ve ramp input from Mt_1 into a step and a ramp has been shown in Figure 5.25(b). The total response $[Y(t) = Y'(t) + Y''(t)]$ may be given as:

$$Y(t) = \underbrace{[Y(t_1) - M(t - t_1 + \tau e^{-(t-t_1)/\tau})]}_{\text{The response part for –ve ramp}} + \underbrace{[M*(1 - e^{-(t-t_1)/\tau})]}_{\text{The response part for step}} \text{ for } t_1 < t < t_2 \qquad (5.58)$$

The responses for the two parts of the input have been shown in Figure 5.25(c), the total response for $t_1 < t < t_2$ is shown in Figure 5.25(d).

After $t = t_2$, the input value becomes zero, but the system output value is obtained by putting $t = t_2$ in Equation 5.58,

$$Y(t_2) = [Y(t_1) - M(t_2 - t_1 + \tau e^{-(t_2 - t_1)/\tau})] + [M * (1 - e^{-(t_2 - t_1)/\tau})] \qquad (5.59)$$

The response for the period $t > t_2$ is clearly the dissipation dynamics discussed in Section 5.5.2, and we may write the response as:

$$Y(t) = Y(t_2)e^{-(t - t_2)/\tau} \qquad (5.60)$$

This part of the dynamics is in Figure 5.25(e). The overall dynamics may be obtained by joining the part of responses in 5.25(a), (d) and (e). This has been shown in Figure 5.25(f) along with the input function.

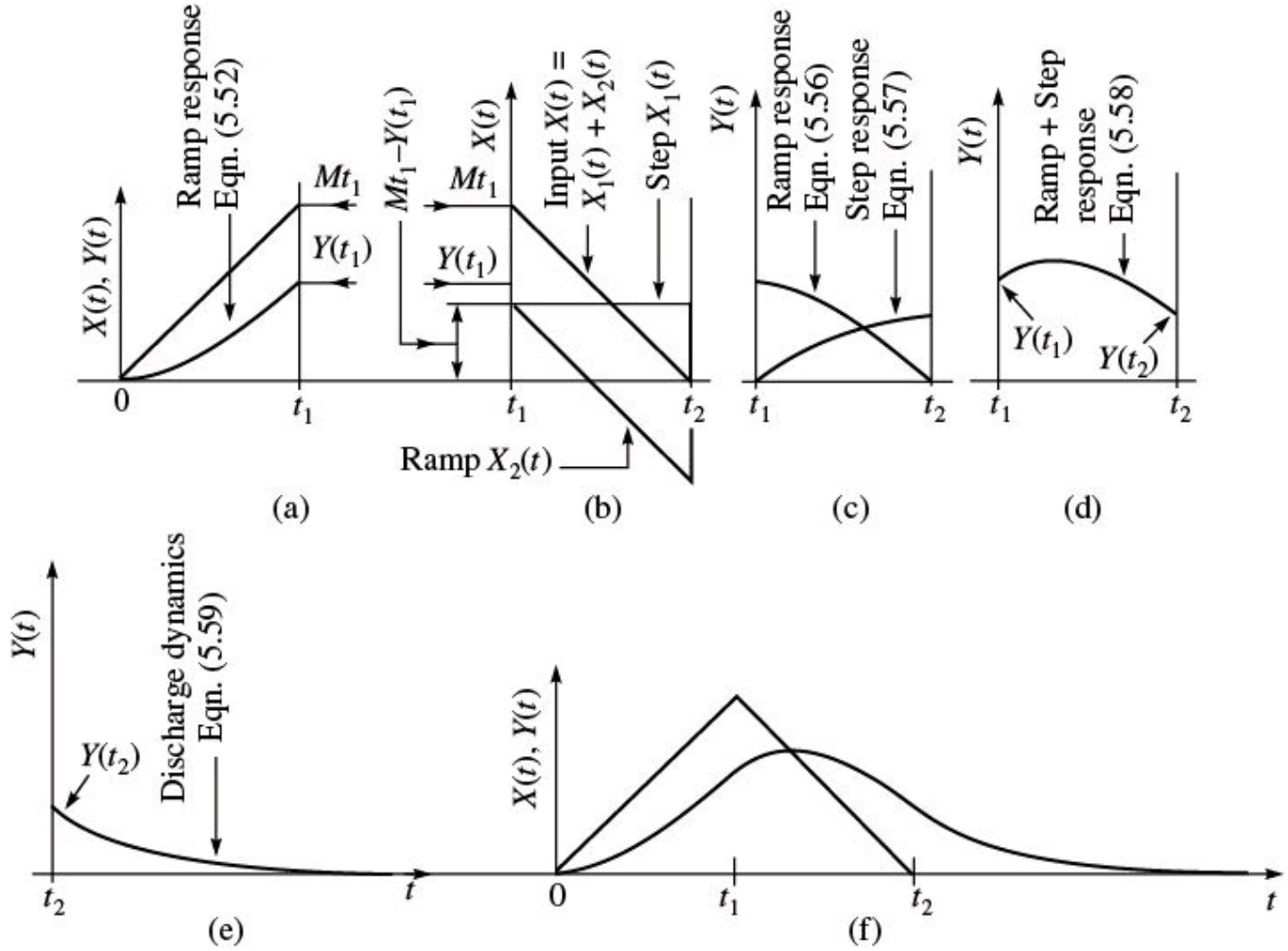


Figure 5.25 The derivation of response for a triangular input.

It is a proper place to introduce two process elements with special dynamic character, which may exist as a part of a process or individually. In a block diagram representation, they are often written in the transfer function of the closed loop element that is more significant for their existence.

5.6 Pure Gain Element

We have introduced the idea of Steady State Gain (K_P) of a process. There are self-contained processes, with closed system boundary or sub-processes as the part of a large process that show an instantaneous response corresponding to an input change. Such processes are called Pure Gain Elements and their TF may be written as $G_P(s) = K_P$. The steady state and dynamic gains are equal for pure gain elements. We explain the idea by the following example.

Assume that you are dynamically testing a first order process with an adjustable time constant. In each experimental run, you are putting the system to step change of equal magnitude from the same initial state, recording the response, and after each run decreasing the time constant in small steps from an initial large value. If we plot the responses on a graph, it looks like Figure 5.26.

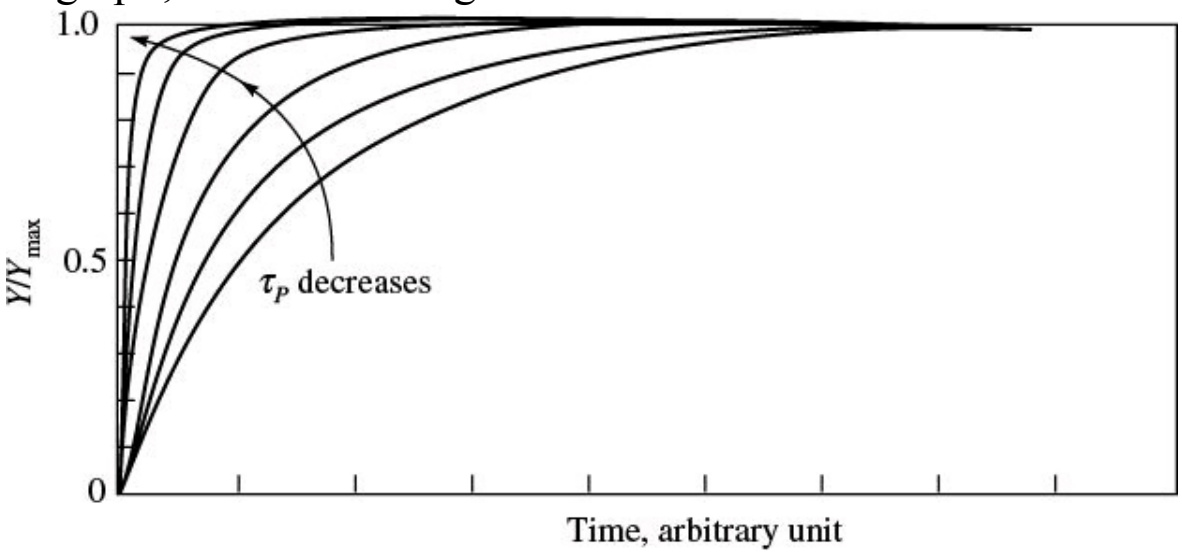


Figure 5.26 Speed of step response as a function of the time constant of a first order system.

As the time constant, t_P, of a first order system decreases, the transfer function tends to become a pure gain element, thus,

$$\underset{\tau \to 0}{\text{Lt}}\left[G_P(s)\right] = \underset{\tau \to 0}{\text{Lt}}\left[\frac{K_P}{\tau_P s + 1}\right] = K_P \qquad (5.61)$$

The above result is also true for higher order processes, and it tells us that if we increase the response speed of any process infinitely large, at the limit we will obtain a pure gain element, numerically and dimensionally equal to the steady state gain of the process. In other words, a process that reacts very fast (instantaneously) in response to an input may be called a pure gain element. Thus, ideally a pure gain element reacts with zero dynamic lag. A practical or engineering definition would be – such a process whose response lag is undetectable by any conventional instrument.

A mechanical lever function or motion transmission linkage with zero backlashes may be cited as examples of pure gain element. Figures 5.27(a) and (b) illustrates two

such examples.

The –ve sign of K_P in the first example comes, because the input and output displacement are in opposite direction. In Figure 5.27(b), K_P is again –ve, considering clockwise pinion rotation to be −ve, and downward rack movement to be +ve.

High-speed operational amplifiers, digital ICs, computer mother-board clocks, all have fabulous operational speeds and may be examples of pure gain in electronic applications.

In chemical systems, rate processes with very fast reaction kinetics, e.g. neutralization of strong alkali by mineral acids, combustion or detonation of explosives and spontaneous chain reaction of radioactive elements are examples of pure gain elements.

In process control applications, we should consider the speed of response of an instrument relative to the process for this definition. Process variable measuring instruments employing electrical, electromagnetic, optical and certain other property changes as the primary detection technique, offer very high speed of response. The transmitter part of these instruments that converts the primary detector output to electronic signal, conforming to some accepted standards of voltage or current range, basically operates through an electronic circuitry. These are also very high-speed systems.

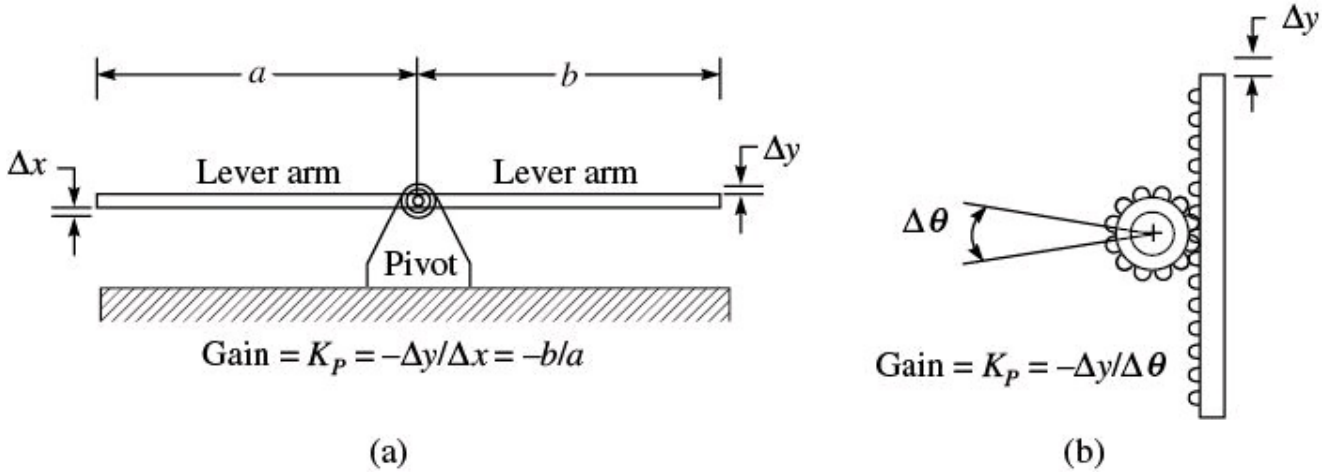


Figure 5.27 Examples of pure gain elements in mechanical applications.

As we have said earlier, whether a process element should be considered as pure gain, depends on the response speed of the associated blocks in the loop. If there is at least one block of much slower response speed in the loop, we may class the other blocks as gain elements because of their relative speed of response.

A control valve, regulating steam to the re-boiler of a distillation column, has a response speed in terms of seconds. The distillation column, re-boiler, and the condenser, all operate with a response speed in terms of hours. Thus, the control valve may be treated as a pure gain element. We shall reopen this topic in the Chapter 7. There, more examples of this kind would be presented.

5.7 Integrating Processes

There are process elements that possess an integrating character. That is, they keep on integrating or accumulating any PE that enters as input, inside the system boundary. Such dynamic character requires special attention for closed loop system design. At present let us understand this property by considering the following examples of dynamic liquid level systems shown in Figure 5.28.

For the initially empty tank of Figure 5.28(a), The dynamic mass balance for $t \geq 0$ will obtain:
balance:

$$\rho q_I - 0 = \rho A \frac{dh}{dt}$$

Introducing deviation variables:

$$Q_I = A \frac{dH}{dt}$$

Taking Laplace transform and rearranging,

$$\frac{H(s)}{Q_I(s)} = \frac{1}{As}$$

For the tank in Figure 5.28(b), with a liquid level sufficient to keep the pump suction filled, the dynamic mass

$$\rho q_I - \rho q_O = \rho A \frac{dh}{dt}$$

At steady state:

$$\rho q_{IS} - \rho q_O = A \frac{dh_S}{dt}$$

Introducing deviation variables,

$$Q_I = A \frac{dH}{dt}$$

Taking Laplace transform and rearranging,

$$\frac{H(s)}{Q_I(s)} = \frac{1}{As}$$

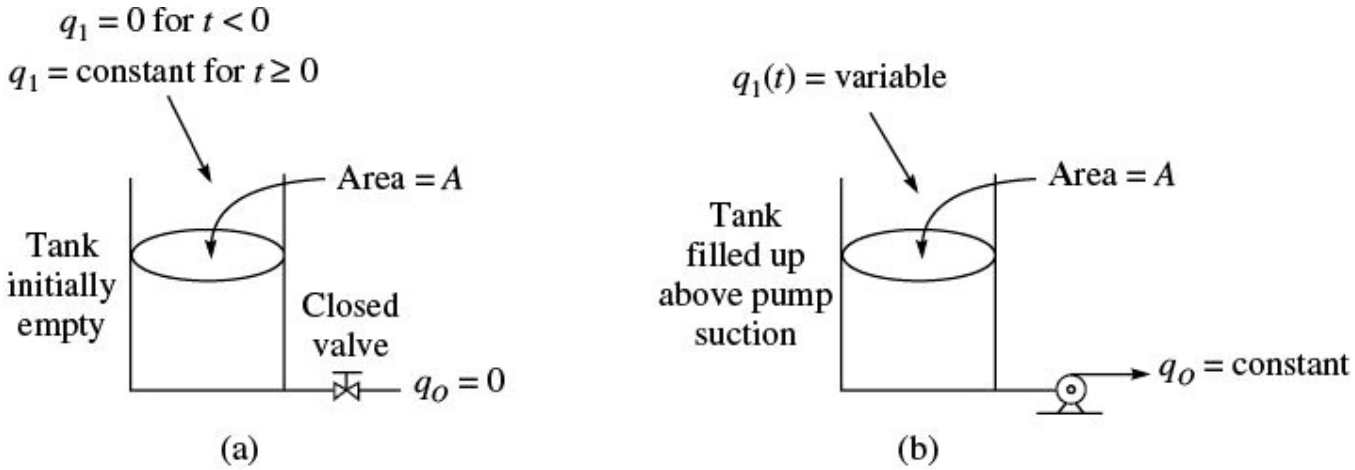


Figure 5.28 (a) An empty tank with zero exit flow. (b) With a constant output, both have an integrating character.

The similarity between the above two systems becomes more understandable, if we think a new inflow variable, $q_I^*(t)$ for the system of Figure 5.28(b), and define this as $q_I^*(t) = M + q_O$. Thus, if inflow suddenly starts with a constant magnitude of M in the system of Figure 5.28(a), and if the inflow to the system of Figure 5.28(b) was equal to q_O in the past, but suddenly has been increased to a new higher value of

$q_I^*(t) = M + q_O$, both of these inputs would produce a similar dynamic response in the respective systems, i.e. the tank levels will start rising at a constant rate according to a ramp function. The slope of their level rises would be:

Slope = $\frac{M}{A}$, cm/sec., where M is in cm^3/sec and A is in cm^2.

Let us subject these systems to a new kind of input. Returning to the initial conditions before the dynamic test described in the previous paragraph, if we pour a volume of M litres of liquid instantaneously into both the tanks according to an impulse function, at $t = 0$, without any other kind of input, what should happen? In both of the systems, the level will rise by
H cm, according to a step change, where,

$$H = \frac{M}{A} \text{ cm, if } M \text{ is expressed in cm}^3 \text{ and } A \text{ is in cm}^2.$$

We may summarize the above observations as:

Step Input: $X(s) = M/s$ → $G_P(s) = 1/As$ → $Y(s) = M/As^2$ Ramp response.

Impulse Input: $X(s) = M$ → $G_P(s) = 1/As$ → $Y(s) = M/As$ Step response.

The above result, that is, the inclusion of an s in the denominator of the input function, as it goes through process and appear as response, confirms the integrating characteristic. This is corroborated by the fact that in Laplace transform table, we find the concerned transform as:

$$L\left[\int f(t)dt\right] = \frac{f(s)}{s}$$

Recalling the discussion about self regulation of a process that we had in Section 2.10, it can be concluded that a purely integrating process posses no self regulation in its dynamic character.

Let us solve some related problems to further enrich ourselves about this topic.

EXAMPLE 5.17 Try to guess the input flow changes and any system parameter changes required to obtain the given level dynamics.

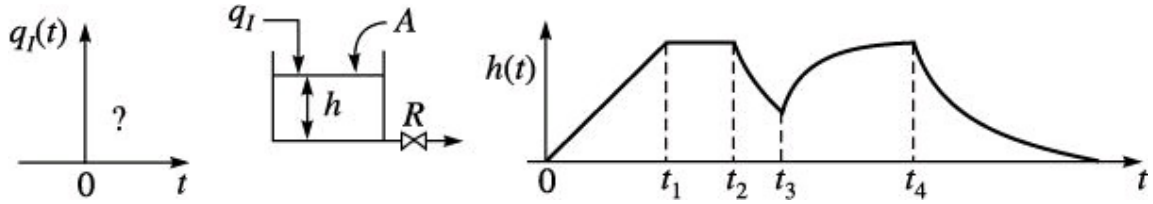


Figure 5.29 Response dynamics of the liquid level system.

Solution From the given Figure 5.29, we may guess that during the time interval $0 < t < t_1$, the exit valve was closed, and q_I started and remained at a constant magnitude, thus the tank was integrating, and its internal liquid level was rising at a constant slope (ramp response to a step input).

Between t_1 and t_2, there are two possibilities:

1. The inflow q_I has been turned off, or
2. While maintaining q_I at its previous value, the bottom exit valve has been opened to an extent so that q_O was exactly equal to q_I. In both of these options, the tank level would remain constant as in the figure.

At $t < t_2$, the bottom exit valve has been opened (if we choose option 1 in the previous time slot), or the bottom exit valve has been opened to a further extent (if we choose option 2 in the previous time slot) so that the tank drains according to a –ve step response of a first order system within the time slot t_2 to t_3.

For the time interval $t_3 < t < t_4$, clearly the response corresponds to a pulse in the inflow rate, q_I. The pulse width must be $t_3 - t_4$. The pulse height or magnitude increase in the inflow, q_I, may be worked out if we know the numerical values of process parameters and input function specification.

EXAMPLE 5.18 From the above discussion, we expect that ramp response equation of a first order system, if differentiated, should obtain the corresponding equation for step response.

Solution: We first write the ramp response equation for a first order system:

$$Y(t) = K_P M(t - \tau + \tau e^{-t/\tau})$$

where, t = Time constant, K_P = Process Gain, M = Slope of the Ramp,

Differentiating the above equation we obtain,

$$\frac{dY(t)}{dt} = Y'(t) = K_P M\left[1 - 0 + \tau\left(-\frac{1}{\tau}\right)e^{-t/\tau}\right] = K_P M(1 - e^{-t/\tau}),$$

the Step response, and, again differentiating, we obtain,

$$\frac{d^2Y(t)}{dt^2} = Y''(t) = \frac{K_P M}{\tau} e^{-t/\tau},$$

the Impulse response.

EXAMPLE 5.19 Derive the Transfer Function, $H(s)/Q_I(s)$ for the system shown in

Figure 5.30

(a) The tank level operates about the steady state value of $h_S = 30$ cm.

(b) The tank level operates about the steady state value of $h_S = 70$ cm.

The constant rate pump throughput is 25 lpm. This rate is independent of tank level.

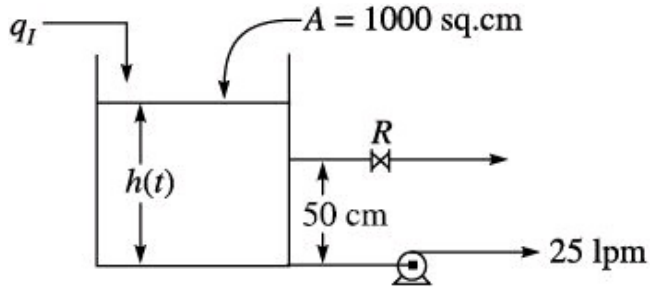


Figure 5.30 The liquid level system of Example 5.19.

The linear liquid flow resistance R is 0.002 cm/(cc/min).

Solution (a) If the operating tank level remains below the exit pipe with valve, the system would behave as an integrator. The transfer function in this condition becomes:

$$\frac{H(s)}{Q_I(s)} = \frac{1}{As} = \frac{1}{s} \qquad (5.62)$$

where, $A = 1000$ sq.cm.

(b) When the operating tank level is above the exit pipe with valve, the tank would behave as a first order system. To derive the transfer function in this condition, we may redefine the level as $H^*(t) = h(t) - 50$ cm, and the inflow $Q_I(t) = q_I(t) - 25000$ cc/min. Thus, the transfer function is:

$$\frac{H*(s)}{Q_I(s)} = \frac{0.002}{(0.002)(1000)s+1} = \frac{0.002}{2s+1} \qquad (5.63)$$

In chemical applications, a semi-batch reactor operation, that starts with some of the raw materials but needs certain other substances to be put into while the reaction is in progress, is an example of integrating process in the sense that once a material added into the reactor, stays within and never escapes. Such a process offers certain difficulty to a control system designer.

Before closing this chapter we would like to reveal certain observations on the procedure of linearization. The reason for the late appearance of this topic is that, this could be proposed only after the discussion on process dynamic responses.

5.8 Linearization Revisited

In Chapter 3, the technique of linearization was introduced as a procedure for deriving an approximate linear description of a complex non-linear system. Such an

approximate model would produce a reasonably accurate response compared to the actual process, provided in the transient period, the process remains within a close vicinity of the steady state from which it has been disturbed. From Section 3.4.3, we know that the technique of linearization, essentially is calculating the derivative or slope of the function at the steady state operating point and representing the process characteristic by equation of the slope. Could we shift the location of this slope evaluation point to improve the accuracy of the resulting linear model?

Let us demonstrate this notion with the help of the following example.

EXAMPLE 5.20 (A liquid level system has the following data): Tank area =300 cm^2; Tank height = 150 cm. The steady operating characteristics of the tank are given below:

q_{OS}, lpm	0	1.26	1.79	2.53	3.1	3.58	4.0
h_S, cms	0	10	20	40	60	80	100

Check the non-linearity of the data and find an empirical relation to express the Flow-Head character. In a process application, the q_I is held at 2530 cc/min for 1 hr and then suddenly raised to 3100 cc/min. Derive the non-linear transfer function $H(s)/Q_I(s)$, (a) By the method discussed in Section 3.4.3, (b) According to the method when slope evaluation point is shifted. Calculate Steady state level as the model response in part (a) and (b). Compare these values with the values given in the Table above and report the results as % error.

Solution To check the non-linearity of the flow vs. head data, we have plotted them on a linear graph in Figure 5.31(a). Recalling the knowledge of fluid mechanics, that in case of a flow through an orifice or restriction under turbulent condition, the relation between flow, Q vs. head in terms of pressure drop, ΔP, may be expressed by:

$$Q = C_D\sqrt{\Delta P} \qquad (5.64)$$

where, C_D is called the 'Discharge Coefficient' of the orifice or restriction. Following this notion, we have also prepared a Log-Log plot of the data in Figure 5.31(b). The resulting relation is:

$$q_{OS} = 400\,(h_S)^{0.5} \qquad (5.65)$$

corroborates Equation 5.64.

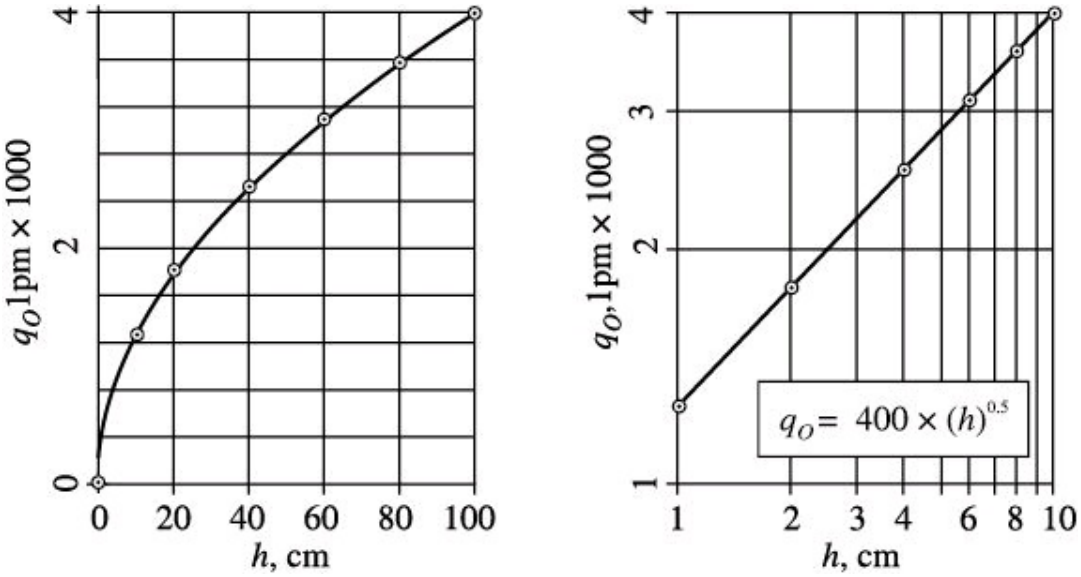


Figure 5.31 Linear and log-log plot of exit flow rate vs. tank level at steady state.

From the experimental data, the liquid level corresponding to the flow rate of 3100 cc/min is 60 cm, at steady state condition.

(a) Following the procedure discussed in Section 3.4, the linear TF for the system may be derived. The equation that is obtained just before introducing the deviation variables is:

$$(q_I - q_{IS}) - \frac{C_D}{2\sqrt{h_S}}(h - h_S) = A\frac{d(h - h_S)}{dt} \qquad (5.66)$$

Introducing the deviation variables, $Q_I - \frac{H}{\frac{2\sqrt{h_S}}{C_D}} = A\frac{dH}{dt}$

The term $\frac{2\sqrt{h_S}}{C_D}$ may be considered as the linearized resistance R^*.

Taking Laplace transform of Equation 5.66, the desired transfer function is obtained:

$$\frac{H(s)}{Q_I(s)} = \frac{R^*}{\tau^* s + 1} \qquad (5.67)$$

where the linearized Resistance, $R^* = \frac{2\sqrt{h_S}}{C_D}$,

and linearized Time Constant, $\tau^* = A\frac{2\sqrt{h_S}}{C_D}$

Using the numerical values from Equation 5.66,

$$R^* = \frac{1}{\left[dq_O/dh\right]_{h=40}} = \frac{1}{\frac{C_D}{2\sqrt{h_s}}} = \frac{2\sqrt{40}}{400} = 0.03162\,\frac{\text{cm}}{\text{cc/min}}$$

The steady state response after the step change of magnitude $M = 3100 - 2530 = 570$ cc/min, may be calculated as:

$$H(t)\big|_{t=\infty} = K_P M[1 - e^{-t/\tau^*}]_{t=\infty} = R.M = 0.03162 \times 570 = 18.025 \text{ cm}$$

In absolute values, the steady state level is:

$$h(t)|_{T=\infty} = h_S + H(t)|_{T=\infty} = 40 + 18.025 = 58.025 \text{ cm}$$

Thus, the % error is:

$$\%\ \text{Error} = \frac{(h_{S,\text{EXPTL}} - h_{S,\text{CALCD}})100}{h_{S,\text{EXPTL}}} = \frac{(60 - 58.025)100}{60} = 3.3\%$$

(b) In part (a) the linearization procedure amounts to finding the resistance R^* by taking reciprocal of the derivative of the function $q_0 = f(h)$ at a point the system is at the beginning of the step change. During the transient period, the operating point of the linearized system remains on the equation of the derivative. This point gradually moves away from the process characteristic locus. The departure is maximum at the end of response or at the steady state condition. This maximum departure is in fact the Error (E) calculated above and has the sole dependency on the derivative or slope evaluation point (P_{SE}). As the operating range of the point extends throughout the transient region, from previous to the next steady state, the rational choice of P_{SE} should be at the mid-point of the input step magnitude. These considerations have been illustrated in Figure 5.32.

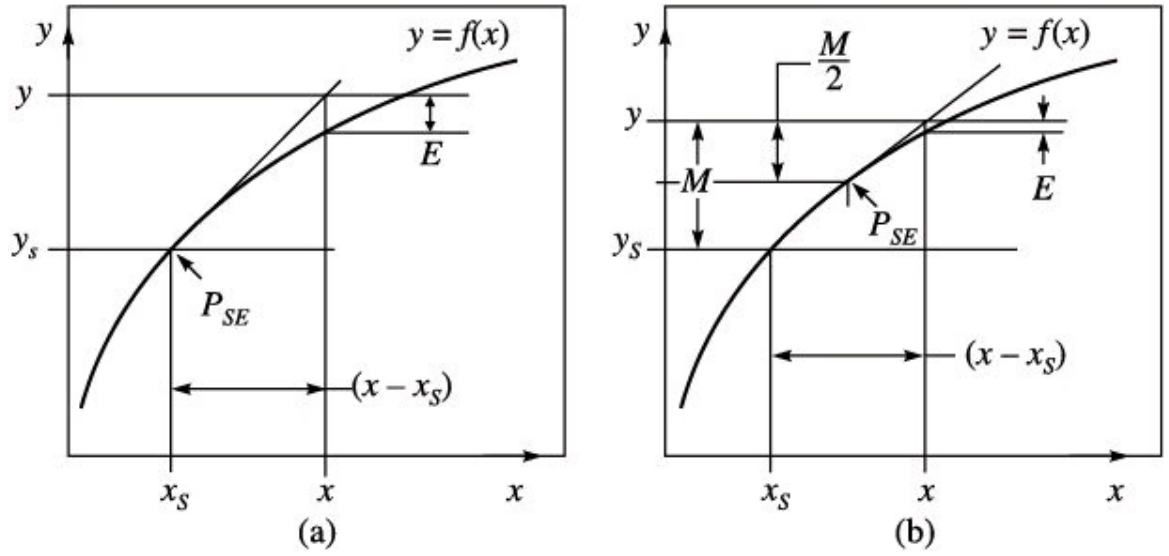


Figure 5.32 Consideration for the modified slope evaluation point.

Let us try this procedure for the same problem.

The value of q_O at the mid-point of step change is:

$$q_{OA} = (2530 + 3100)/2 = 2815 \text{ cc/min}$$

Putting this value of flow rate in the flow-head relation, the corresponding value of level may be found as:

$$h_A = \left(\frac{2815}{400}\right)^2 = 49.5264 \text{ cm}$$

The derivative evaluated at this point would obtain:

$$R^* = \frac{1}{\left[dq_O/dh\right]_{h=h_A}} = \frac{1}{\frac{C_D}{2\sqrt{h_A}}} = \frac{2\sqrt{49.5264}}{400} = 0.0351875 \ \frac{\text{cm}}{\text{cc/min}}$$

Again as above, $H(t)\big|_{t=\infty} = R.M = 0.0351875 \times 570 = 20.056875$ cm

And $h(t)|_{T=\infty} = h_S + H(t)|_{T=\infty} = 40 + 20.056875 = 60.056875$, we may calculate the % error as:

$$\% \text{ Error} = \frac{(h_{S,\text{EXPTL}} - h_{S,\text{CALCD}})100}{h_{S,\text{EXPTL}}} = \frac{(60 - 60.056875)100}{60} = -0.095\%$$

The only limitation of this technique would be for systems those are described by complex non-linear relations:

$$Y = f_{NL}(X)$$

where, calculation of X for a given Y could not be done as readily as in this example but requires implicit method. Nevertheless, for the sake of accuracy, one may go for any of such methods, e.g. Interval Halving or Newton-Raphson algorithm.

EXERCISE PROBLEMS

5.1 In the last periods of Third Reich (Adolf Hitler's Regime), secret documents were recovered by the American 3rd Army, led by Gen. George S. Patton, from an infamous concentration camp, containing detailed information about medical and physiological tests conducted on Jewish prisoners.

Nutrient transfer dynamics from blood vessels to starved tissues was studied in human subjects after they were put under different period of starvation ranging from 5 to 30 days. Concentrated glucose solution was injected into the subject's jugular vein and glucose concentration was measured from a special sensor (glass electrode type) put in a prominent blood vessel, usually adjacent to the ocular sensory center of human brain. (The casualty rate of the subjects was about 83%.

Those survived became blind.)

Sugar Concn. μ-gm/lit.	17.5	20	22.4	24.5	27.5	30.5	33.1
Time, min.	0	2	4	6	10	15	20
Sugar Concn. μ-gm/lit.	36.5	39	40.5	41.9	43.1	43.8	43.8
Time, min.	30	40	50	60	80	100	120

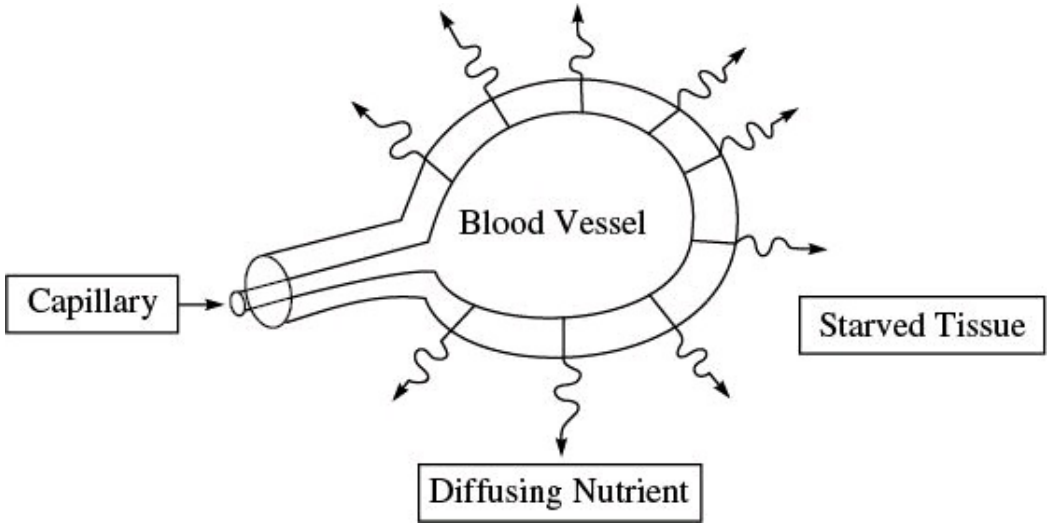


Figure P5.1 A simple representation of nutrient diffusion into starved tissue from a blood vessel.

Considering the simple pictorial representation in Figure P5.1, calculate,

(a) Mass transfer coefficient in $\frac{\mu g}{(\text{min})(\text{cm}^2)(\mu g/\text{cc})}$ for the nutrient diffusion from the blood vessel to the tissue.

(b) Magnitude of the step jump of the nutrient.

(c) Time constant of the nutrient transfer process assuming that this is a 1st order system.

Assume:

(i) The blood vessel is a 2 mm diameter sphere with all outer surface available for mass transfer;

(ii) During the experiment, the sugar concentration in the surrounding tissue remain constant at 17.5 μ-gm/cc. (The last assumption is physiologically justified by the fact that starved tissues require a considerable time to assimilate suddenly available nutrient.)

5.2 Aqueous solution streams A and B are fed continuously into a CSTR and react rapidly to completion with a release of heat = 250 kcal/kg of solution A. Cooling water is circulated through the jacket at such a rate that its temperature rise may be neglected. The following data may be used for analysis: Feed rate Q_A = 750 kg/h, Q_B = 750 kg/h, Cp_A = 0.9 cal/

(g °C), Cp_B = 1cal/(gm °C), hold up volume, V_R = 2500 l, Jacket, V_J = 750 litre, r_A

$= r_B = 1$ gm/cc, heat transfer area = 3.5 m^2, $U = 30$ BTU/(hr.m^2.°C). Inlet temperature of streams A and B = 40°C, Cooling water temperature = 32°C.

What is the exit temperature of the stream of reaction mixture? What is the exit temperature 5 min after the cooling water temperature suddenly drops to 25°C.

5.3 The process in this problem is constant volume overflow type stirred tank water heater. The density, r, and specific heat capacity, C_P, of the liquid, may be assumed constant, but the mass flow rate, m, inflow stream temperature, T_I, and power input, W, say through a steam coil may all vary. Physical properties of water may be assumed for the analysis.

(a) Obtain the transfer functions

(i) $T_O(s)/T_I(s)$

(ii) $T_O(s)/m(s)$

(iii) $T_O(s)/W(s)$, where all variables are in their deviation variable form.

(b) If the holdup volume of the tank is 100 litre water, find the steady state water exit stream temperature, if $T_I = 25^oC$, $m = 12.5$ kg/sec and $W = 5$ kW.

(c) After the system has achieved steady under conditions described in part (b), if m is decreased by 5% (at $t = 0$), find the required change in W to bring the exit temperature T_O to its previous value.

5.4 A stirred cylindrical tank with flat top and bottom cover is made of thin SS-sheet. It is used as a water heater by placing an electrical heater of negligible volume inside the tank. The water at 30°C is taken to the full capacity of the tank and the heater is switched 'ON'. It is switched 'OFF' when the tank water temperature has reached 85°C. The heat rejection (cooling) dynamics of tank water to atmosphere has been found to follow the exponential relation,

$$T(t) = 30 + 55.e^{-0.125t}$$

where T = Water temperature, in °C, and t, time in min.

Find the heat transfer coefficient, U, (cal.cm^{-2}.min^{-1}.°C^{-1}) of the tank surface, and the atmospheric temperature in °C. The tank height is 750 mm, and tank diameter is 500 mm.

5.5 In the introductory chapter, we cited few examples of 'Potential Depletion Dynamics'. One of these examples was about decrease of number of heads of aristocrats guillotined per day after French Revolution. Is this a first order process? If it is, the process is showing such a dynamic response under what kind of disturbance?

5.6 The liquid level dynamic system comprising a cylindrical tank (area = A) and a linear outflow resistance, R, is of first order in nature. The system parameters, Time Constant
($= t = A.R$), and Gain ($= K_P = R$) have been calculated by performing a dynamic test of the system. The experiment was done by creating a step change in the input flow rate, and recording the tank level dynamics against time. Water was used as the fluid during the experiment.

If the same experiment had been carried with Glycerin as the liquid, try to figure out what would be the values of the two system parameters. A qualitative answer will be sufficient.

5.7 A first order, irreversible, isothermal reaction is carried in a constant volume CSTR. The reaction, A □ Product proceeds with the following rate expression, $dCa_O/dt = kCa_O$. With the following steady state operating data:

$Ca_I = 1.25$ mol/m^3, $Ca_O = 0.25$ mol/m^3, $Q = 0.1$ m^3/min, $V = 2.5$ cu.m, calculate the numerical value and dimension of the reaction rate constant k. The variable names have their usual significance.

5.8 From the following response of a first order system (Figure P5.2), find a quantitative estimate about the nature of the input function.

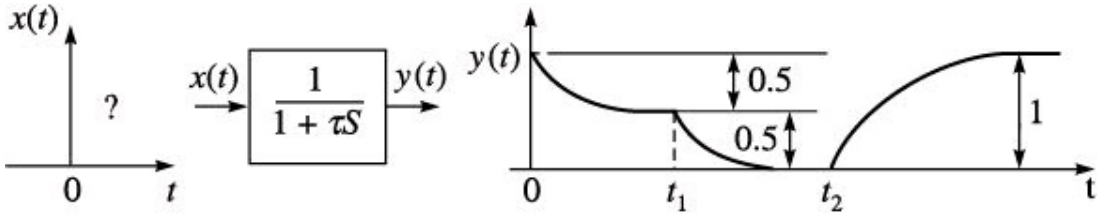


Figure P5.2.

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[1] Coughanowr, D.R., *Process System Analysis and Control*, 2nd Ed., McGraw-Hill Inc, NY, pp. 68, 1991.

[2] Smith, C.A. and Corripio, A.B., *Principles and Practices of Automatic Process Control*, John Wiley & Sons, NY, pp. 35, 1985.

[3] Perlmutter, D.D., *Introduction to Chemical Process Control*, John Wiley and Sons, Inc., 1965.

6

Transfer Function Development and Response Analysis of Second and Higher Order Systems (Complex Domain Analysis)

A class or variety of processes is identified as of second order, if a second order ordinary differential equation may describe its dynamics. At the beginning of process dynamic model development, we discussed about the importance and significance of modelling assumptions, and their all-important role in determining the final configuration of the derived dynamic characteristic equation. We shall begin this chapter by considering two physical examples of first order systems and demonstrate that by modifying the earlier modelling assumptions, we could end up with a second order model for the same systems. The objective of such revision and modification of assumptions is to bring the system description closer to reality and having a more accurate model with respect to the prediction performance of real system behaviour. Handling such a revised model involves lengthier and more complex computation. A trade-off between computation, speed and accuracy, or the selection of optimum model description will obviously depend on the accuracy demand of the given application.

EXAMPLE 6.1 In Example 3.2, the model of a Mercury in Glass Thermometer has been developed as a first order ODE and the corresponding TF is derived in Example 4.2. During model formulation, we assumed that the thermal resistances are lumped in the film coefficient on the glass bulb and the thermal capacitances are lumped in the mercury.

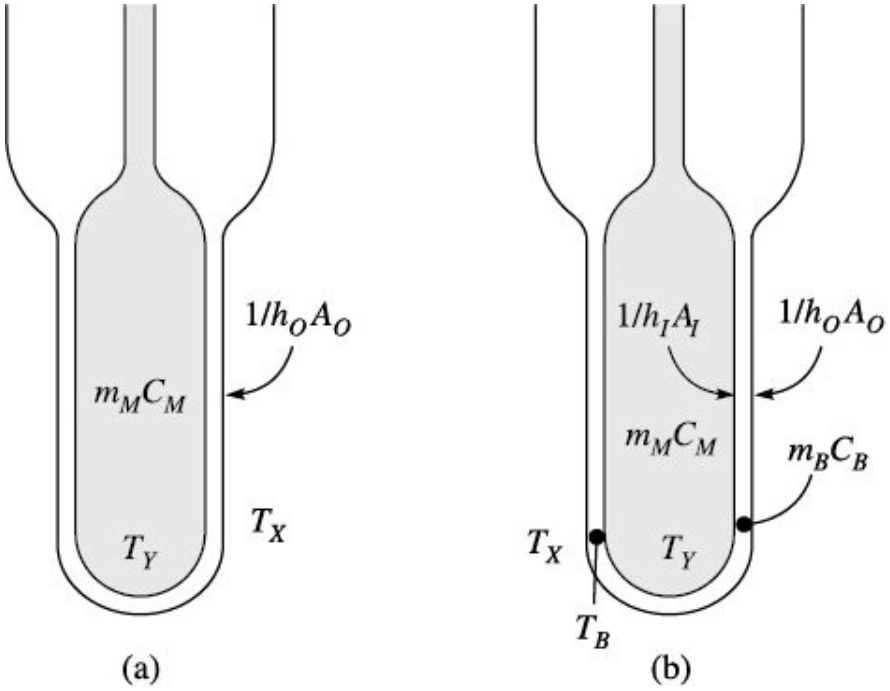


Figure 6.1 (a) First order and (b) Second order model configurations of Hg in glass thermometer.

In this example, trying to obtain a more exact description of the process, we introduce the following modification of assumptions: in addition to the mass of mercury, the glass bulb also has a thermal capacitance, expressed as $m_B C_B$. Also, we assume an extra heat flow resistance residing at the thin film of mercury on the inside wall of glass bulb, $1/(h_I A_I)$. That is, this dynamic description of the process recognizes two capacitance–resistance pairs instead of one as we did for the first order model consideration. In Figure 6.1, we present the two configurations for comparison.

The following nomenclature for the variable and parameters has been used in the above figure and following analysis:

T_X = Surrounding temperature

T_B = Glass Bulb temperature

T_Y = Mercury temperature that is equal to the indicated temperature by the thermometer

C_B = Heat capacity of glass

C_M = Heat capacity of mercury

h_O = Outside heat transfer film coefficient

h_I = Inside heat transfer film coefficient

A_O = Outside heat transfer area

A_I = Inside heat transfer area

m_B = Mass of the glass bulb

m_M = Mass of the mercury.

To develop the model, we write a dynamic heat balance around the glass bulb,

$$h_O A_O (T_X - T_B) - h_I A_I (T_B - T_Y) = m_B C_B \frac{dT_B}{dt} \qquad (6.1)$$

and around the mercury,

$$h_I A_I (T_B - T_Y) - 0 = m_M C_M \frac{dT_Y}{dt} \qquad (6.2)$$

Obviously, we want an equation relating T_Y and T_X. Elimination of T_B may be done in time domain, but the algebra involved will be a little clumsier than, if we do it in Laplace domain. Assuming zero initial value for all variables and taking Laplace transform of the above Equations 6.1 and 6.2,

$$h_O A_O T_X(s) - h_O A_O T_B(s) - h_I A_I T_B(s) + h_I A_I T_Y(s) = m_B C_B s T_Y(s) \quad (6.3)$$

$$h_I A_I T_B(s) - h_I A_I T_Y(s) = m_M C_M s T_Y(s) \quad (6.4)$$

From the last equation we can write,

$$T_B(s) = [m_M C_M s T_Y(s) + h_I A_I T_Y(s)] \frac{1}{h_I A_I} \quad (6.5)$$

Putting this in Equation 6.3,

$$h_O A_O T_X(s) + h_I A_I T_Y(s) - \frac{1}{h_I A_I}[m_M C_M h_O A_O s T_Y(s) + h_I A_I h_O A_O T_Y(s)$$

$$+ m_M C_M h_I A_I s T_Y(s) + h_I^2 A_I^2 T_y(s)] = m_B C_B s [m_M C_M s T_y(s) + h_I A_I T_Y(s)] \frac{1}{h_I A_I}$$

or, $$h_O A_O T_X(s) + h_I A_I T_Y(s) - \frac{m_M C_M h_O A_O}{h_I A_I} s T_Y(s) - h_O A_O T_Y(s)$$

$$- m_M C_M s T_Y(s) - h_I A_I T_Y(s) = \frac{m_B C_B m_M C_M}{h_I A_I} s^2 T_y(s) + m_B C_B s T_Y(s)$$

or, $$T_Y(s)\left[\frac{m_B C_B m_M C_M}{h_I A_I} s^2 + \left(\frac{m_M C_M h_O A_O}{h_I A_I} + m_M C_M + m_B C_B\right) s + h_O A_O\right] = h_O A_O T_X(s)$$

Writing $m_B C_B = A_B$ and $m_M C_M = A_M$, the thermal capacitances of glass bulb and mercury, and $1/h_I A_I = R_I = R_{MI}$ and $1/h_O A_O = R_O = R_B$, the thermal resistances of outside and inside surfaces of the glass bulb, the above equation becomes:

$$T_y(s)\left[A_B A_M R_M s^2 + \left(\frac{A_M R_M}{R_B} + A_M + A_B\right) s + \frac{1}{R_O}\right] = \frac{T_X(s)}{R_B}$$

or, $$T_Y(s)[A_B R_B A_M R_M s^2 + (A_M R_M + A_M R_B + A_B R_B) s + 1] = T_X(s)$$

As the product $A_B R_B$ involves heat transfer to the glass bulb, and $A_M R_M$ involves heat transfer to the mercury, we may define the following two time constants as: $t_B = A_B R_B$, and
$t_M = A_M B_M$. Introducing these definitions in the above equation and rearranging,

$$\frac{T_Y(s)}{T_X(s)} = \frac{1}{\tau_B \tau_M s^2 + (\tau_B + \tau_M + A_M R_B)s + 1} \qquad (6.6)$$

The corresponding time domain differential equation must be of the following form:

$$\tau_B \tau_M \frac{d^2 T_Y}{dt^2} + (\tau_B + \tau_M + A_M R_B)\frac{dT_Y}{dt} + T_Y = T_X \qquad (6.7)$$

This is a second order differential equation, and thus the same thermometer has become a second order system just by slightly changing the proposed assumptions that we made during the formulation of the first order model in Example 3.2.

The modification of assumption may be also looked upon as 'Instead of total lumping', we have made them partially distributed or 'splitted'. This modification of assumption is definitely a step towards recognizing the true mechanism of heat transfer in such systems. A similar consideration for Thermocouple, RTD, etc. placed inside a protecting tube or thermo-well or even for jacketed small volume reactors will produce a more accurate model response than we would obtain with a fully lumped parameter description.

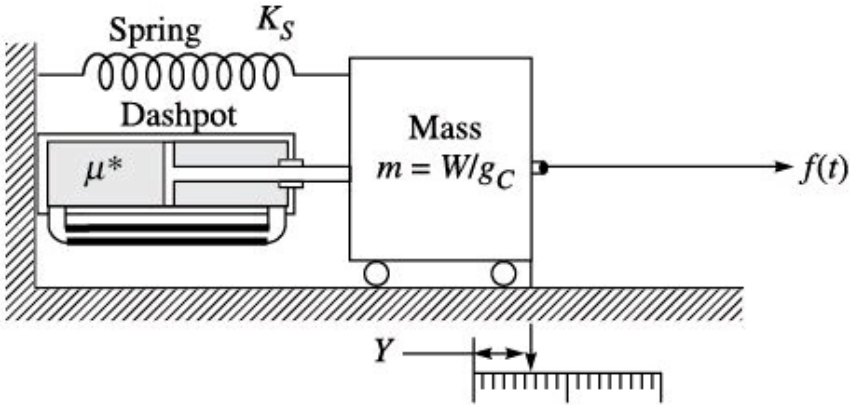
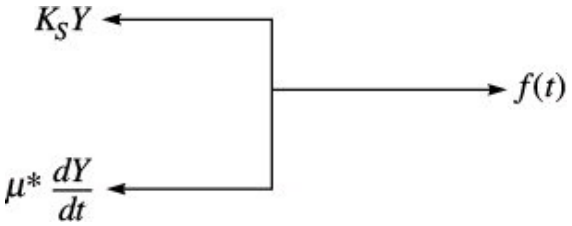


Figure 6.2 The Spring-dashpot system having a finite mass.

EXAMPLE 6.2 Next we choose the 'Spring-Dashpot' system that we discussed in Example 4.4, where we assumed that the mechanical system had no mass and developed a first order model. Here we modify the assumption by recognizing the reality that mechanical components of the system must have some finite mass. However, we aggregate all the bits and pieces of mass elements around the system into a block for convenience of analysis. Now the spring-dashpot assembly will look like the one in Figure 6.2.

A force balance may be developed based on the free body diagram:



The difference between $f(t)$ and $K_S Y + m^*(dY/dt)$, will be the rest of the force due to rate of change of momentum, i.e. acceleration times mass. Thus the complete force balance equation becomes:

$$m\frac{d^2Y}{dt^2} = f(t) - K_S Y - \mu * \frac{dY}{dt}$$

or

$$\frac{W}{g_C} \cdot \frac{d^2Y}{dt^2} + \mu * \frac{dY}{dt} + K_S Y = f(t) \tag{6.8}$$

This is an ordinary second order differential equation. Thus, we have again obtained a second order system in place of a first order for the same physical process by improving the modelling assumption to bring it closure to reality. The bottom line of the foregoing discussion is that a system should not be labeled by a particular order value based on its physical description, rather than the order, and complexity of the model significantly depends on the model building considerations for a given process to account for the physical and chemical phenomena occurring inside. The above two examples have been chosen that represent the two classes of second order system as we shall see in the next section.

Definition and Classification of Second Order Systems: Whatever be the cause of origin of a second order system, they come in two varieties.

6.1 Synthetic Type Second Order System

When a second order system is constructed by the physical combination of two first order system elements and these subsystem elements functionally operate in series, we may call such combination—a Synthetic Second Order System. In many texts they have been also termed as **First Order Systems in Series**. The following two kinds of series combination are possible with a pair of first order systems. These are shown in Figures 6.3(a) and (b), where two liquid level dynamic systems have been used as first order subsystem.

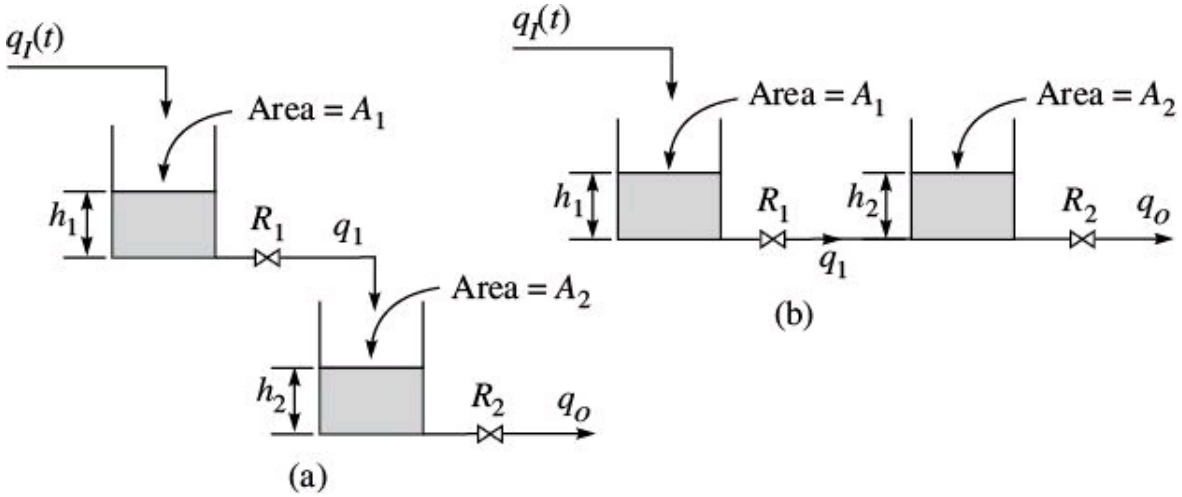


Figure 6.3 Two classes of synthetic second order systems. (a) Non-interacting type. (b) Interacting type.

Let us develop the model equations for theses two subclasses of synthetic second order systems.

6.1.1 Non-Interacting Combination [Figure 6.3(a)]

Following the analysis procedure for a single tank system, we may write straightway for first and second tanks in this system:

$$\frac{Q_1(s)}{Q_I(s)} = \frac{1}{A_1R_1s+1} = \frac{1}{\tau_1 s+1} \tag{a}$$

and,

$$\frac{H_2(s)}{Q_1(s)} = \frac{R_2}{A_2R_2s+1} = \frac{1}{\tau_2 s+1} \tag{b}$$

also,

$$\frac{Q_O(s)}{Q_1(s)} = \frac{1}{A_2R_2s+1} = \frac{1}{\tau_2 s+1} \tag{c}$$

From the above three subsystem transfer functions (a), (b) and (c), we may obtain the two possible transfer functions of the resulting system as:

$$\frac{H_2(s)}{Q_I(s)} = \frac{Q_1(s)}{Q_I(s)} \cdot \frac{H_2(s)}{Q_1(s)} = \frac{R_2}{(\tau_1 s+1)(\tau_2 s+1)} = \frac{R_2}{\tau_1\tau_2 s^2+(\tau_1+\tau_2)s+1} \tag{6.9}$$

$$\frac{Q_2(s)}{Q_I(s)} = \frac{Q_1(s)}{Q_I(s)} \cdot \frac{Q_2(s)}{Q_1(s)} = \frac{1}{(\tau_1 s+1)(\tau_2 s+1)} = \frac{1}{\tau_1\tau_2 s^2+(\tau_1+\tau_2)s+1} \tag{6.10}$$

This method could be applied to non-interacting systems comprising of any number of tanks. As in the following example of tree tank system, the transfer functions could be derived straightway.

EXAMPLE 6.3 Find the transfer functions $H_3(s)/Q_I(s)$ and $Q_O(s)/Q_I(s)$ for the system shown in Figure 6.4.

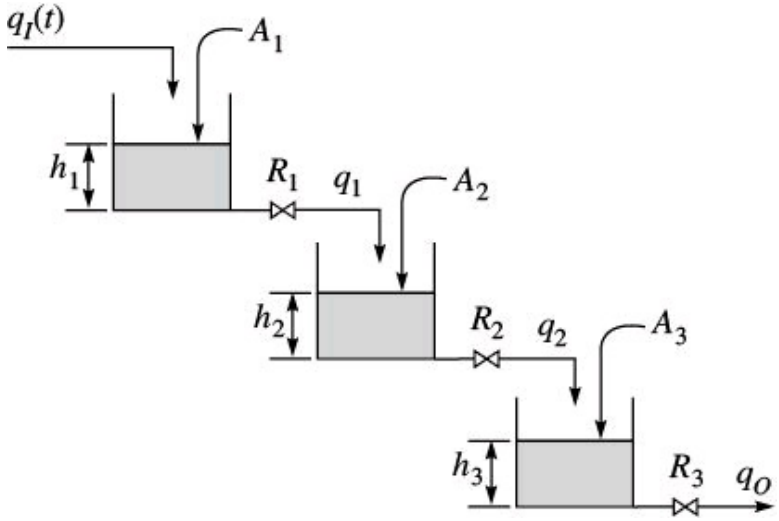


Figure 6.4 A Three-tank system connected in non-interacting combination.

Solution:

$$\frac{H_3(s)}{Q_I(s)} = \frac{Q_1(s)}{Q_I(s)} \cdot \frac{Q_2(s)}{Q_1(s)} \cdot \frac{H_3(s)}{Q_2(s)} = \frac{R_3}{(\tau_1 s + 1)(\tau_2 s + 1)(\tau_3 s + 1)} \tag{6.11}$$

$$\frac{Q_O(s)}{Q_I(s)} = \frac{Q_1(s)}{Q_I(s)} \cdot \frac{Q_2(s)}{Q_1(s)} \cdot \frac{Q_O(s)}{Q_2(s)} = \frac{1}{(\tau_1 s + 1)(\tau_2 s + 1)(\tau_3 s + 1)} \tag{6.12}$$

6.1.2 Interacting Combination [Figure 6.3(b)]

The two tanks in Figure 6.3(b) are said to be interacting because the exit flow from the first tank, q_1, is not dependent only on the first tank level, h_1, as it is in Figure 6.3(a), but on the difference between h_1 and h_2. The related flow-head relations may mathematically spell this statement as:

$q_1 = \frac{h_1}{R_1}$ (for non-interacting system) and $q_1 = \frac{h_1 - h_2}{R_1}$ (for interacting system)

Thus, the second tank level, h_2, is said to be 'Interacting' or 'Loading the exit flow, q_1, of the first tank. The effect of interaction dynamically hinders the process response as the second tank level (potential) tries to choke the first tank exit flow. In fact, this effect makes the overall system response more sluggish than the system with non-interacting combination. To find the dynamic model of the system in Figure 6.3(b), we cannot directly write the transfer functions of the two tanks, as we have done for non-interacting system and combine them to derive the overall system's dynamics. Why it is so, will be clear if we write the concerned mass balance and flow-head relations.

For the first tank:

Mass balance: $q_I - q_1 = A_1 \frac{dh_1}{dt}$, and flow-head relation: $q_1 = \frac{h_1 - h_2}{R_1}$

For the second tank:

Mass balance: $q_1 - q_O = A_2 \frac{dh_2}{dt}$, and flow-head relation: $q_O = \frac{h_2}{R_2}$

Omitting the steps for writing the variables in their deviation form, we directly write the above four equations in Laplace domain:

$$Q_I(s) - Q_1(s) = A_1 s H_1(s) \tag{6.13}$$

$$R_1 Q_I(s) = H_1(s) - H_2(s) \tag{6.14}$$

$$Q_I(s) - Q_O(s) = A_2 s H_2(s) \tag{6.15}$$

$$R_2 Q_O(s) = H_2(s) \tag{6.16}$$

To find the transfer function of the system between $H_2(s)$ and $Q_I(s)$, we have to eliminate the intermediate connecting variables, $Q_1(s)$, $Q_O(s)$ and $H_1(s)$, by using Equations 6.13 through 6.16. The result is:

$$\frac{H_2(s)}{Q_I(s)} = \frac{R_2}{A_1 R_1 A_2 R_2 s^2 + (A_1 R_1 + A_2 R_2 + A_1 R_2)s + 1} \tag{6.17a}$$

$$\frac{H_2(s)}{Q_I(s)} = \frac{R_2}{\tau_1 \tau_2 s^2 + (\tau_1 + \tau_2 + A_1 R_2)s + 1} \tag{6.17b}$$

and to find the transfer function of the system between $Q_O(s)$ and $Q_I(s)$, we may use Equation 6.14 to eliminate $H_2(s)$ from the above equation and obtain,

$$\frac{Q_O(s)}{Q_I(s)} = \frac{1}{\tau_1 \tau_2 s^2 + (\tau_1 + \tau_2 + A_1 R_2)s + 1} \tag{6.18}$$

The appearance of the extra $A_1 R_2$ term in the coefficient of s in the denominator of the transfer function is due to the presence of interaction, as we have already found in the dynamics of the modified thermometer model. Existence of this extra term is the reason for more sluggish response.

We shall discuss a few more examples of such systems to strengthen the notion required for their dynamic model development.

EXAMPLE 6.4 Redo the Mercury in glass thermometer system dynamics, but for the present analysis, derive the transfer function $T_Y(s)/T_X(s)$ from a Liquid level dynamic equivalent of the heat transfer process and the resulting mass balance and flow-head relations.

Solution: Recalling Figure 6.1, we prepare the equivalent liquid level dynamic system in Figure 6.5. For the heat transfer from the surrounding, there is no mass balance relation, because the surrounding may be considered as a reservoir (tank) with infinite capacitance (area, A), so that due to any finite flow, q, there would be no depletion of potential, T_X. However, we may write the flow-head relation for the connecting path between surrounding and the glass bulb as:

$$q_1 = \frac{T_X - T_B}{R_O} = h_O A_O (T_X - T_B)$$

Heat balance for the glass bulb: $q_1 - q_2 = A_O \dfrac{dT_B}{dt}$

Flow-head relation between bulb and mercury: $q_2 = \dfrac{T_B - T_Y}{R_I} = h_I A_I (T_B - T_Y)$

Heat balance for the mercury: $q_2 - 0 = A_I . \dfrac{dT_Y}{dt}$

The above four equations in Laplace domain will yield:

$R_O Q_I(s) = T_X(s) - T_B(s)$ (6.19)

$Q_I(s) - Q_2(s) = A_O s T_B(s)$ (6.20)

$R_I Q_2(s) = T_B(s) - T_Y(s)$ (6.21)

$Q_2(s) = A_I s T_B(s)$ (6.22)

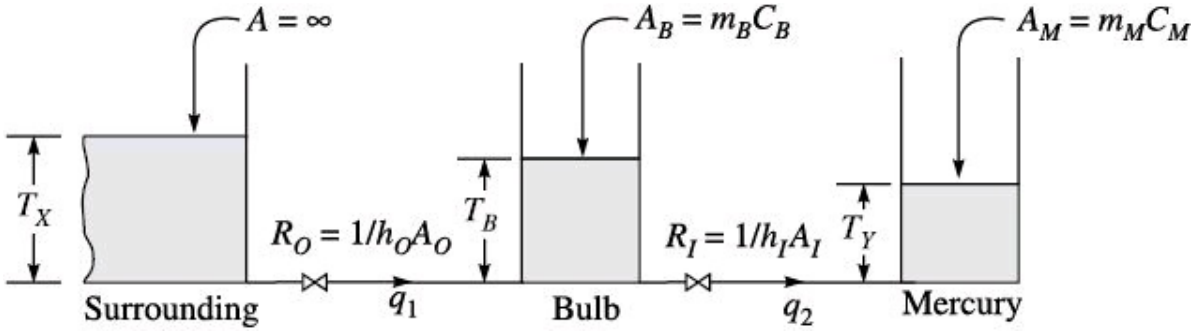


Figure 6.5 Liquid level dynamic equivalent of the heat transfer in a mercury in glass thermometer.

To obtained the desired transfer function, $T_Y(s)/T_X(s)$, from the four Equations 6.19 through 6.22, we have to eliminate the intermediate variables, $T_B(s)$, $Q_1(s)$ and $Q_2(s)$. By combining Equations 6.21 and 6.22,

$$T_B(s) = R_I A_I T_Y(s) + T_Y(s) \qquad (6.23)$$

Putting the expression for $Q_1(s)$ from Equation 6.19, and for $Q_2(s)$ from Equation 6.21 into Equation 6.20,

$$\left[\frac{T_X(s)}{R_O}-\frac{T_B(s)}{R_O}\right]-\left[\frac{T_B(s)}{R_I}-\frac{T_Y(s)}{R_I}\right]=A_O s T_B(s)$$

or,
$$\frac{T_X(s)}{R_O}+\frac{T_Y(s)}{R_I}=\left[\frac{1}{R_O}+\frac{1}{R_I}+A_O s\right]T_B(s) \qquad (6.24)$$

Putting the expression for $T_B(s)$ from Equation 6.23 into the above Equation 6.24,

$$\frac{T_X(s)}{R_O}+\frac{T_Y(s)}{R_I}=\left[\frac{1}{R_O}+\frac{1}{R_I}+A_O s\right]\left[R_I A_I s+1\right]T_Y(s)$$

On simplification, the above equation yields:

$$\frac{T_Y(s)}{T_X(s)}=\frac{1}{A_I R_I A_O R_O s^2+(A_I R_I+A_O R_O+A_I R_O)s+1} \qquad (6.25)$$

Equation 6.25 is similar to the transfer function that we obtained starting from the differential equations of the system in the earlier section.

For a given system of one physical description, drawing equivalent systems of other areas often increases the understanding and simplifies the analysis. This is a fundamental advantage of **System Approach** of analysis applied to processes that contain internal conversion of multiple types of physical variables.

EXAMPLE 6.5 Find the transfer functions $H_2(s)/Q_I(s)$ and $Q_O(s)/Q_I(s)$ for the system shown in Figure 6.6.

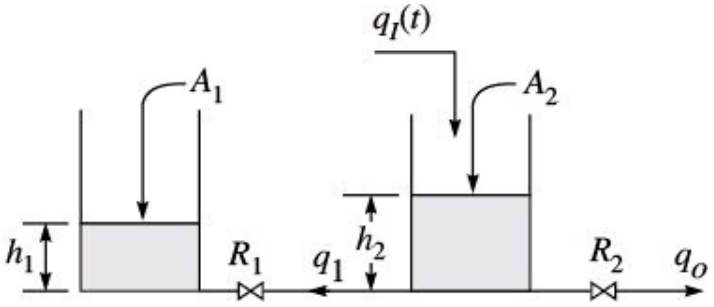


Figure 6.6 The interacting liquid level of Example 6.5.

Solution:

The mass balance equations are: $q_I - q_O - q_1 = A_2\frac{dh_2}{dt}$; and $q_1 = A_1\frac{dh_1}{dt}$;

And the Flow-head relations are: $q_O = \frac{h_2}{R_2}$; and $q_1 = \frac{h_2 - h_1}{R_1}$

Laplace transform of the equations yields:

$Q_I(s) - Q_O(s) - Q_1(s) = A_2 s H_2(s)$ (6.26)

$Q_1(s) = A_1 s H_1(s)$ (6.27)

$R_2 Q_O(s) = H_2(s)$ (6.28)

$R_1 Q_1(s) = H_2(s) - H_1(s)$ (6.29)

Rearranging Equation 6.26 and putting the expression of $Q_1(s)$ from Equation 6.27 into Equation 6.26:

$$Q_I(s) = Q_O(s) + A_1 s H_1(s) + A_2 s H_2(s) \qquad \text{(a)}$$

Putting expression for $Q_1(s)$ from Equation 6.27 into Equation 6.29:

$$R_1 A_1 s H_1(s) = H_2(s) - H_1(s)$$

or
$$H_1(s) = \frac{H_2(s)}{A_1 R_1 s + 1} \qquad (6.30)$$

Putting the expression for $H_1(s)$ from Equation 6.30 into (a):

$$Q_I(s) = Q_O(s) + \frac{A_1 s H_2(s)}{A_1 R_1 s + 1} + A_2 s H_2(s) \qquad (6.31)$$

Putting expression for $Q_O(s)$ from Equation 6.28 into Equation 6.31:

$$Q_I(s) = \frac{H_2(s)}{R_2} + \frac{H_2(s) A_1 s}{A_1 R_1 s + 1} + A_2 s H_2(s)$$

or,

$$Q_I(s) = H_2(s)\left[\frac{A_1 R_1 A_2 R_2 s^2 + (A_1 R_1 + A_2 R_2 + A_1 R_2)s + 1}{R_2(A_1 R_1 s + 1)}\right]$$

or,
$$\frac{H_2(s)}{Q_I(s)} = \frac{R_2(A_1 R_1 s + 1)}{A_1 R_1 A_2 R_2 s^2 + (A_1 R_1 + A_2 R_2 + A_1 R_2)s + 1}$$

using the expression for $H_2(s)$ from Equation 6.28 into above, we obtain:

$$\frac{Q_O(s)}{Q_I(s)} = \frac{A_1 R_1 s + 1}{A_1 R_1 A_2 R_2 s^2 + (A_1 R_1 + A_2 R_2 + A_1 R_2)s + 1} \qquad (6.32)$$

EXAMPLE 6.6 Find the transfer functions $H_3(s)/Q_I(s)$ and $Q_O(s)/Q_I(s)$ for the system shown in Figure 6.7.

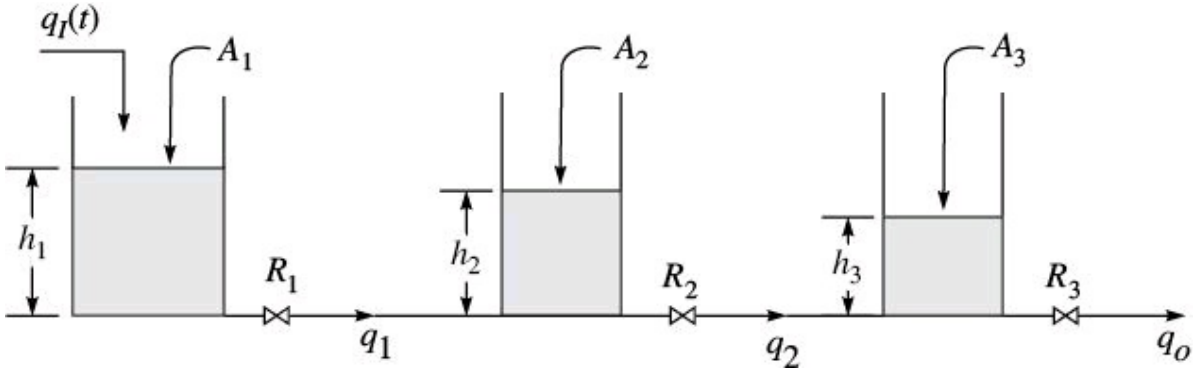


Figure 6.7 The interacting liquid level system of Example 6.6.

Solution: The mass balance equations are:

$$q_I - q_1 = A_2 \frac{dh_1}{dt};\ q_1 - q_2 = A_2 \frac{dh_2}{dt};\ \text{and}\ \ q_2 - q_O = A_3 \frac{dh_3}{dt}$$

and the Flow-head relations are:

$$q_1 = \frac{h_1 - h_2}{R_1};\ q_2 = \frac{h_2 - h_3}{R_2};\ \text{and}\ \ q_O = \frac{h_3}{R_3}$$

The corresponding equations in Laplace domain are:

$$Q_I(s) - Q_1(s) = A_1 s H_1(s) \quad (6.33)$$

$$Q_1(s) - Q_2(s) = A_2 s H_2(s) \quad (6.34)$$

$$Q_2(s) - Q_O(s) = A_3 s H_3(s) \quad (6.35)$$

$$R_1 Q_1(s) = H_1(s) - H_2(s) \quad (6.36)$$

$$R_2 Q_2(s) = H_2(s) - H_3(s) \quad (6.37)$$

$$R_3 Q_O(s) = H_3(s) \quad (6.38)$$

equations 6.33 + 6.34: $Q_I(s) - Q_2(s) = A_1 s H_1(s) + A_2 s H_2(s)$ (6.39)

equations 6.35 + 6.39: $Q_I(s) - Q_O(s) = A_1 s H_1(s) + A_2 s H_2(s) + A_3 s H_3(s)$ (6.40)

Again equations 6.37 + 6.38: $H_2(s) = R_2 Q_2(s) + R_3 Q_O(s)$ (6.41)

And equations 6.41 + 6.36: $H_1(s) = R_1 Q_1(s) + R_2 Q_2(s) + R_3 Q_O(s)$ (6.42)

and from equation 6.38: $H_3(s) = R_3 Q_O(s)$ (6.43)

Putting expressions for $H_1(s)$, $H_2(s)$ and $Q_O(s)$ from equations 6.42, 6.41 and 6.43 respectively into Equation 6.40:

$$Q_I(s) = \frac{H_3(s)}{R_3} + A_1 s[R_1 Q_1(s) + R_2 Q_2(s) + H_3(s)] + A_2 s[R_2 Q_2(s) + H_3(s)] + A_3 s H_3(s)$$

Rearranging the above equation,

or, $$Q_I(s) = \underbrace{Q_2(s)[A_2R_2s + A_1R_2s] + Q_1(s)A_1R_1s}_{\text{Term 1}} + \underbrace{H_3(s)\left[\frac{1}{R_3} + A_3s + A_1s + A_2s\right]}_{\text{Term 2}} \quad (6.44)$$

From Equation 6.34, putting expressions for $Q_1(s)$ into Term I of Equation 6.44,

$$\text{Term I} = Q_2(s)[A_2R_2s + A_1R_2s + A_1R_1s] + A_1R_1A_2s^2H_2(s) \quad (6.45)$$

From Equation 6.35:

$$Q_2(s) = A_3sH_3(s) + Q_O(s) \quad (6.46)$$

Putting the expression for $Q_O(s)$ from Equation 6.38 in Equation 6.46:

$$Q_2(s) = A_3sH_3(s) + \frac{H_3(s)}{R_3} \quad (6.47)$$

Applying Equation 6.47 into Equation 6.45,

$$\text{Term I} = \left[A_3s + \frac{1}{R_3}\right](A_2R_2s + A_1R_2s + A_1R_1s)H_3(s) + A_1R_1A_2s^2H_2(s) \quad (6.48)$$

From Equation 6.37, $$H_2(s) = R_2Q_2(s) + H_3(s) \quad (6.49)$$

From Equation 6.35, putting expression for $Q_2(s)$ into the above Equation 6.49:

$$H_2(s) = R_2[A_3sH_3(s) + Q_O(s)] + H_3(s) = H_3(s)\left[R_2A_3s + \frac{R_2}{R_3} + 1\right] \quad (6.50)$$

Putting $H_2(s)$ from Equation 6.50 into Equation 6.48:

$$\text{Term I} = \left[\left\{\frac{A_3R_3s + 1}{R_3}\right\}(A_2R_2s + A_1R_2s + A_1R_1s) + A_1R_1A_2s^2\left\{\frac{A_3R_3R_2s + R_2 + R_3}{R_3}\right\}\right]H_3(s)$$

putting the above expression for Term 1 in Equation 6.44, and rearranging, we obtain the transfer function:

$$\frac{H_3(s)}{Q_I(s)} = \frac{R_3}{A_1R_1A_2R_2A_3R_3s^3 + (A_1R_1A_2R_2 + A_1R_1A_3R_3 + A_2R_2A_3R_3 + A_1R_2A_3R_3 + A_1R_1A_2R_3)s^2 + (A_1R_1 + A_2R_2 + A_3R_3 + A_1R_2 + A_2R_3 + A_1R_3)s + 1} \quad (6.51)$$

Also applying Equation 6.38, we may obtain:

$$\frac{Q_O(s)}{Q_I(s)} = \frac{1}{A_1R_1A_2R_2A_3R_3s^3 + (A_1R_1A_2R_2 + A_1R_1A_3R_3 + A_2R_2A_3R_3 + A_1R_2A_3R_3 + A_1R_1A_2R_3)s^2 + (A_1R_1 + A_2R_2 + A_3R_3 + A_1R_2 + A_2R_3 + A_1R_3)s + 1} \quad (6.52)$$

We may apply the following time constant expression to further concise Equations 6.51 and 6.52:

$$t_1 = A_1R_1;\ t_2 = A_2R_2;\ t_3 = A_3R_3$$

(*You may note that in all the examples of synthetic second order system, the last term in the denominator of the transfer function without any s is always unity. In the time domain-differential equation form it is equivalent to, that the free output variable term has no coefficient. Only for the 'Spring-Dashpot' example it is not so. However, that can be remedied by a little algebraic manipulation. We shall discuss the significance of this observation a little later in concern with the* ***Standard form of Transfer Functions***.)

We have found out the dynamic model equations for non-interacting and interacting second and third order systems. Let us write down them for comparison:

Second order non-interacting:

$$\frac{H_2(s)}{Q_I(s)} = \frac{R_2}{\tau_1\tau_2 s^2 + (\tau_1 + \tau_2)s + 1} \qquad \text{(A1)}$$

Second order interacting:

$$\frac{H_2(s)}{Q_I(s)} = \frac{R_2}{\tau_1\tau_2 s^2 + (\tau_1 + \tau_2 + A_1R_2)s + 1} \qquad \text{(A2)}$$

Third order non-interacting:

$$\frac{H_3(s)}{Q_I(s)} = \frac{R_2}{\tau_1\tau_2\tau_3 s^3 + (\tau_1\tau_2 + \tau_2\tau_3 + \tau_1\tau_3)s^2 + (\tau_1 + \tau_2 + \tau_3)s + 1} \qquad \text{(B1)}$$

Third order interacting:

$$\frac{H_3(s)}{Q_I(s)} = \frac{R_3}{\tau_1\tau_2\tau_3 s^3 + (\tau_1\tau_2 + \tau_1\tau_3 + \tau_2\tau_3 + A_1R_2\tau_3 + \tau_1A_2R_3)s^2 + (\tau_1 + \tau_2 + \tau_3 + \\ + A_1R_2 + A_2R_3 + A_1R_3)s + 1} \qquad \text{(B2)}$$

Third order transfer functions are cited together with the second order, to establish the generality of the statement that we made earlier—For an Nth order system, in comparison to the non-interacting configurations [(A1) and (B1)], appearance of extra terms in the coefficients of s^{N-1}, s^{N-2}, …, s^0, in the denominator of the transfer functions [(A2) and (B2)] is due the presence of 'interaction' or 'loading' within the process. The dynamic manifestation of this phenomenon is a more sluggish or lethargic response, relative to the corresponding non-interacting configurations. This comment may be established by examining the step response of these systems on a common time base, as in Figure 6.8. A little later, we shall also propose an analytical

reason for this behaviour during the discussion of dynamic response of second order systems.

6.2 Transfer Lag

At this point, to introduce an important process response property of higher order systems, we have to concentrate on the initial part of their response, i.e. $Y(t)$ for t in the vicinity of 0.

An important observation regarding the initial part of the response in Figure 6.8 is that, just after the disturbance, the system moves with a zero slope,

or
$$\lim_{t \to 0} \left[\frac{dY}{dt} \right] = 0 \text{ ; for } N > 1$$

where N is the order of the system

This means if the order of the system is more than 1, the system response not only starts from a zero value, but its initial slope of response (speed) is also zero. That is, the response must stay at zero value for a finite time, however small. The zero response part indicates that after the instant of disturbance, there is a time lag or delay, after which the response shows a finite measurable value. Also, from Figure 6.8 as we go from second to third order, this phenomenon of response lag becomes greater, irrespective of the fact whether the system contains interaction. To confirm this dynamic character, we have drawn the step response of systems of order, $N = 1, 2, 3, 4, 5$, in Figure 6.9. The illustrated responses corroborate that except for $N = 1$, as we go for higher order, the Lag increases. This kind of lag has been termed as the **Transfer Lag**, to differentiate with another kind of lag, called **Transportation Lag** that occurs in process dynamic systems. Pictorially transfer lag (TRL) is demonstrated as the **'Foot'** of response during which the response grazes the time abscissa, shown as in the 'Inset' of Figure 6.9.

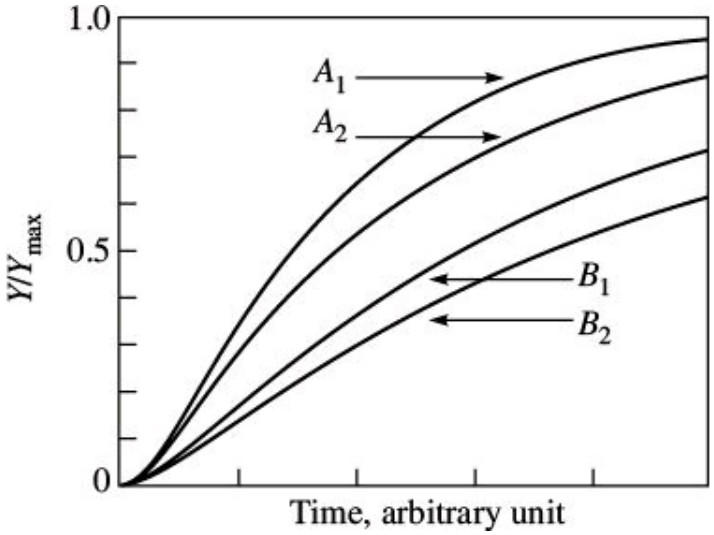


Figure 6.8 Presence of interaction slows down

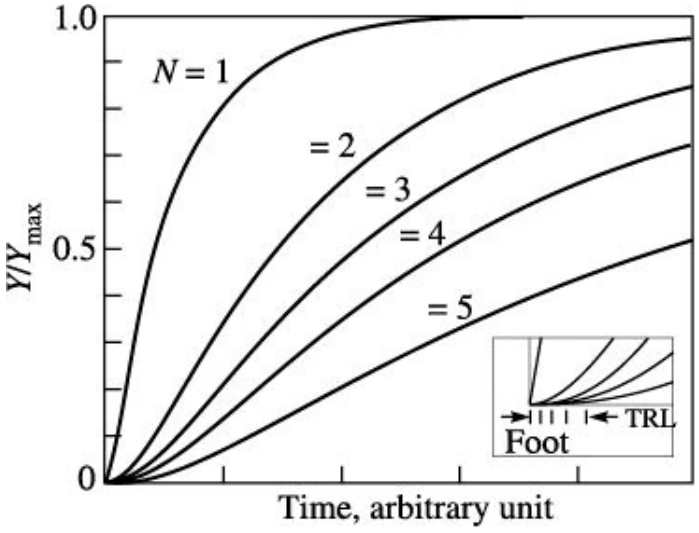


Figure 6.9 The initial zero response time a process irrespective of its order (foot) increases for higher order.

At this point, the following example may establish an analytical confirmation of the above comments.

EXAMPLE 6.7 From the following second order transfer function,

$$G_P(s) = \frac{Y(s)}{X(s)} = \frac{K_P}{\tau_1\tau_2 s^2 + (\tau_1 + \tau_2)s + 1} = \frac{K_P}{\tau_1\tau_2} \cdot \frac{1}{(s + 1/\tau_1)(s + 1/\tau_2)},$$

derive the step response equation and evaluate the initial rate (speed) of response.

Solution: Putting, $X(s) = M/s$, into $G_P(s)$ and effecting the partial fraction expansion, the final time domain solution comes as:

$$Y(t) = K_P M\left[1 - \frac{\tau_1}{\tau_1 - \tau_2}.\exp\left(-\frac{t}{\tau_1}\right) + \frac{\tau_2}{\tau_1 - \tau_2}.\exp\left(-\frac{t}{\tau_2}\right)\right] \qquad \text{(a)}$$

Putting $t = 0$, in Equation (a): $Y(t)|_{T=0} = K_P M(1 - 1) = 0$

Differentiating the above response equation with respect to time,

$$\frac{dY(t)}{dt} = \frac{1}{\tau_1 - \tau_2} e^{-\frac{t}{\tau_1}} - \frac{1}{\tau_1 - \tau_2} e^{-\frac{t}{\tau_2}}, \text{ setting } t = 0, \left.\frac{dY(t)}{dt}\right|_{t=0} = \frac{1}{\tau_1 - \tau_2} - \frac{1}{\tau_1 - \tau_2} = 0$$

Thus, it is proved that the initial value of the step response, and its time derivative, both have zero values.

(Only for a first order system the initial speed of step response is finite, and in fact the initial slope has the highest value throughout the range of response. You may recall the example in the earlier chapter regarding the first order dynamics in which we proved that ***If a first order system could move with the initial speed after a step change, the time required to cover the step magnitude would equal the system time constant,*** t.)

Let us try to obtain a physical interpretation of existence of transfer lag with

reference to a first order and a second order dynamic liquid level system with reference to the Figure 6.10.

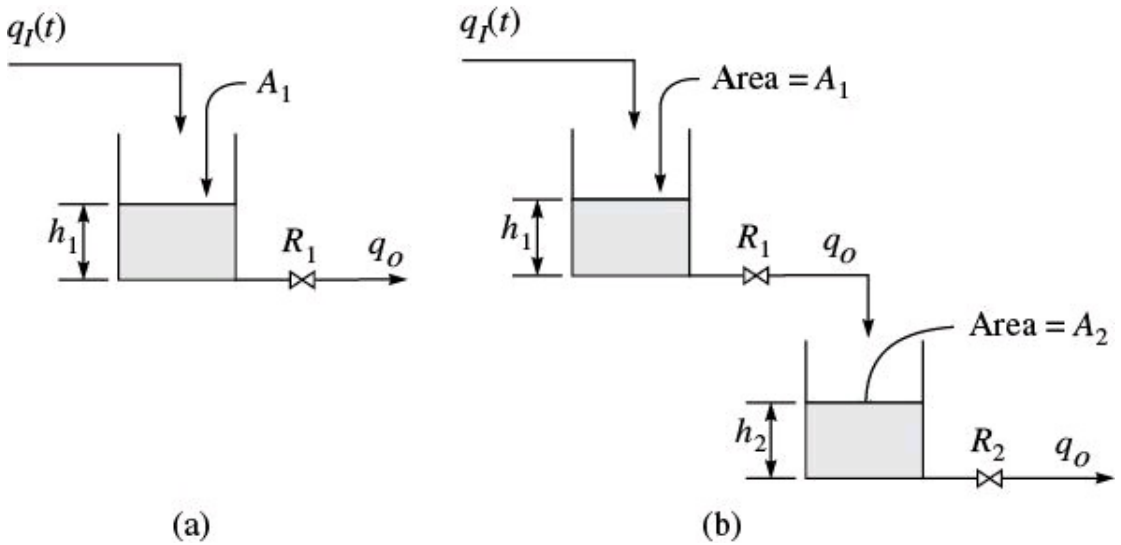


Figure 6.10 Liquid level systems of (a) First order (b) Second order dynamics.

Corresponding to a step change of q_I, level h_1 in systems shown in Figures 6.10(a) and (b) will start to rise with maximum rate because the exit flow, which is the only cause for depletion of h_1 can only increase corresponding to a higher value of h_1. That is unless h_1 attains a higher value, q_I cannot increase and in the initial stage of response there will be a temporary accumulation of liquid in the tank of Figure 6.10(a) and in the first tank of Figure 6.10(b). This is the reason for the maximum initial rate of response in a first order system.

In the second tank of Figure 6.10(b) the input q_1 would not increase with a step or sudden 'jump' because h_1, is only starting to rise, though with a maximum rate, however, in the initial stage ($t \approx 0$), it will be close to its previous steady state value. Also, the output q_O, operating at steady state value, will be able to drain the mass of liquid as the inflow is just increasing. Hence, very near to $t = 0$, h_2 will remain at or very close to its steady state value, may be for a very small time interval, and a 'foot' or 'transfer lag' will appear at the beginning of response. In a third order system configured in the similar way, the response of level h_3 may show a transfer lag that will be summation of the lags occurring in the second and third tank. Hence, the transfer lag will increase as we go for higher order systems.

The conclusions we just made is applicable also for interacting configurations of second and higher order systems. In fact, the transfer lag would be more pronounced due to the choking effect of the second tank level on the exit flow from the first tank.

The second order systems of the second category will also exhibit transfer lag in their dynamic response as we shall see in the next section, because they can be described by the similar analytical description.

The term transfer lag has a legacy from industrial instrument practice. In many of the process instruments, the primary detector first senses the process variable as a change of certain physical variable. In the primary detector for this detection at least one pair of *R–C* elements are required. This primary change is converted to proportional change of another suitable physical variable either for display or transmission to a destination. This conversion again involves at least one pair of *R–C* elements, as the minimum requirement. Thus for a practical instrument, we have at least a second order system that will exhibit the transfer lag in its dynamics. We exemplify this fact in Figure 6.11, by illustrating a Hydrostatic Head Type Level Indicator.

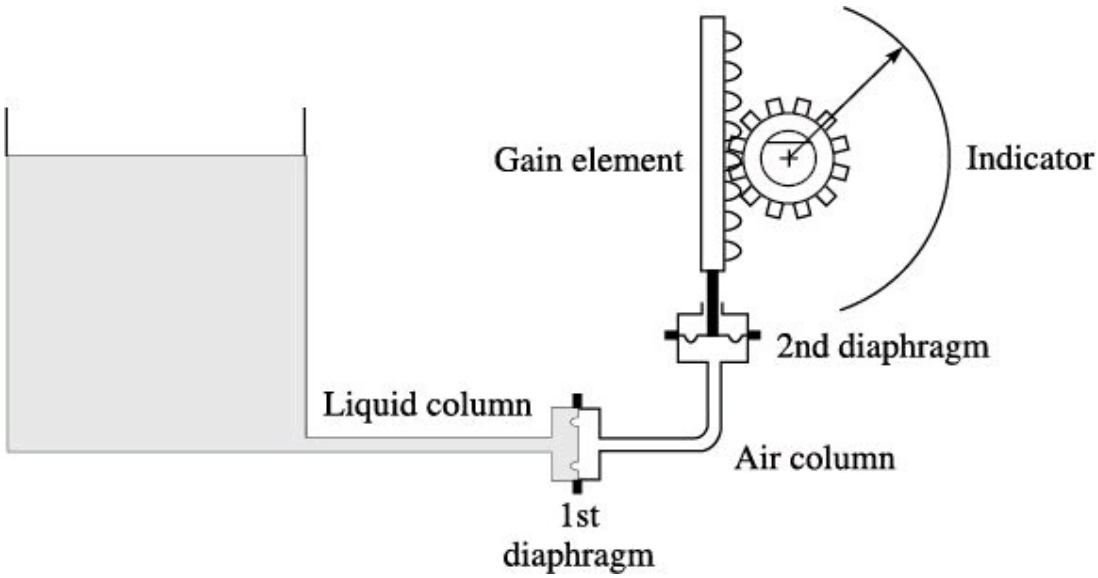


Figure 6.11 In a hydrostatic head type level indicator, a series combination of two first order elements will show transfer lag.

Transfer lag is also called '**Quadratic Lag**' because at least a second order transfer function that has a quadratic polynomial in *s* in the denominator, is necessary for its existence.

If we consider a second order system with a large time constant ratio, say $t_2/t_1 = 0.1$, it is obvious that the system dynamics would be close to that of a first order system with a time constant t_1, because the other first order system of t_2 time constant will move with a much faster speed (refer Figure 5.27), and may be considered as a pure gain element. Such lower order approximation considerably reduces computation with minimum sacrifice of accuracy level. This philosophy of approximation is also applicable for higher order or systems having complex dynamics. If there is one time constant much larger than the others, then it is called the **Significant Time Constant**, and an approximate first order model with the significant time constant value may describe the system dynamics with reasonable accuracy.

As we have started discussing on the topic of lag, another important category of lag called **Dead Time element** or **Transportation Lag** should be mentioned here. This

lag element appears frequently in industrial applications independently or as part of a process. It has individual response characteristics that need to be investigated to understand its difference with transfer lag. This lag element has many synonyms, e.g. **Distance Velocity Lag, Delay Element, Pipe Line Delay,** etc.

6.3 Dead Time Element

Consider a section of pipeline carrying a flowing liquid at constant rate, and also assume that the flow profile inside the pipe is essentially flat. Such a flow in terms of fluid mechanics is called **Plug flow**. The significance of plug flow is that the liquid layers move with equal velocity, independent of their position from the flow field boundary, hence, there will be no intermixing in any direction while the liquid is moving. If the flowing liquid carries the principal entity, so that its potential becomes the process variable, the liquid may be termed as the **carrier** or **vehicle** fluid. The process variable may be aconcentration of a solute, or the temperature of the liquid. Also let us assume that there is no density effect due to variation of concentration or temperature. Thus, any dynamic change of these variables associated with the carrier fluid, entering at the inlet end of the pipe, should come out from the exit end without any deformation with respect to its shape and magnitude; only it will be shifted in time to the extent that the liquid resides in the pipe, or the time required by the liquid to pass through the pipe.

In Figure 6.12, the features of a dead time element have been illustrated. Following the foregoing discussion:

Dead time, $$\tau_{DL} = \frac{V}{q} \qquad (6.53)$$

where V = Internal volume of the pipe, and q = Volumetric flow rate of the carrier fluid.

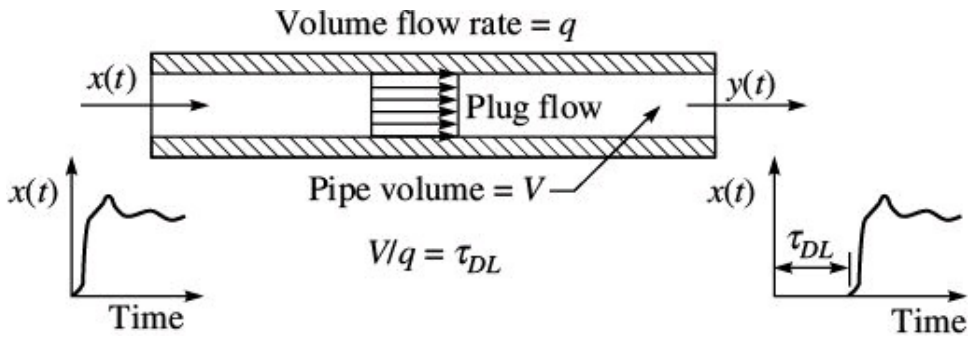


Figure 6.12 Definition of dead time element.

In reaction engineering studies, we have already come across an equivalent term. The **Residence time**, related to Continuous Reactor Systems, has a similar definition at least for the constant density systems.

(*We have assumed a flat velocity profile that guarantees zero intermixing and*

distortion free passage of the input variable profile through the flow field for proposing the following dynamic model. Surely, this assumption puts the system description away from reality, but not very far. In process industries, a major number of fluid flow occurs in highly turbulent conditions, and turbulent flow models predict a near flat velocity profile through a large part of the flow area. But such profiles are always associated with noise. If the noise is removed by ***'time-smoothening'****, or averaging over small time intervals, the result would prove the validity of plug flow assumption at least in the engineering sense. On the other hand, simplicity of this model warrants easy conceptual assimilation and lesser computational efforts during analysis, especially in case the lag element is a part of a complex process network.*)

From Figure 6.12, it follows that $y(t)$ is in fact the time shifted description of $x(t)$, that is in $x(t)$, if we replace t by $t - t_{DL}$, we will obtain $y(t)$, i.e.

$$y(t) = x(t - \tau_{DL}) \qquad (6.54)$$

Recalling the definition integral of Laplace transformation,

$$x(s) = L[x(t)] = \int_0^\infty x(t)e^{-st}dt \qquad (6.55)$$

It should be noted that t variable no longer exists in the result of the integration by applying the limits on t. In other words, t is integrated out, and for this feature, in some literature it is called 'the dummy variable of integration'. Hence, if we replace t by $t - t_{DL}$ in the above integral equation, and if t_{DL} is finite, the replacement does not hamper the significance of Laplace integral because t or $t - t_{DL}$ will be integrated out. Thus, we may write Equation 6.55 as:

$$x(s) = \int_0^\infty x(t - \tau_{DL})e^{-(t-\tau_{DL})s}d(t - \tau_{DL}) = e^{\tau_{DL}s}\int_0^\infty x(t - \tau_{DL})e^{-st}dt \qquad (6.56)$$

Using Equation 6.54 for the definition of $y(t)$, and Equation 6.55 for the definition of $y(s)$, in the above Equation 6.56 we obtain the transfer function of the dead time element:

$$x(s) = e^{\tau_{DL}s}\int_0^\infty y(t)e^{-st}dt = e^{\tau_{DL}s}y(s) \qquad (6.57)$$

or,
$$\frac{y(s)}{x(s)} = e^{-\tau_{DL}s} \qquad (6.58)$$

Dead time is a non-linear element. It brings in a discontinuity due to presence of the 'lag' in the response, i.e. we have no information of $y(t)$ during the time interval $t = 0$

to $t = t_{DL}$, for an input occurring at $t = 0$.

The primary concern would be to formulate the dynamic analysis of a process that contains dead time element. However, if a process under steady state receives a current disturbance, we may proceed in the following way. Suppose we have the complete process model given by:

$$G_P(s) = \frac{Y(s)}{X(s)} = \frac{K_P e^{-\tau_{DL}s}}{\tau_P s + 1} \qquad (6.59)$$

That is a first order element acting in series with a dead time. Assume that this system has been disturbed by a step input of magnitude, M at time $t = 0$.

If we consider the process transfer function without its dead time element, the response will be:

$$y(t) = MK_P(1 - e^{-t/\tau_P}) \qquad (6.60)$$

and the effect of dead time may be added to the solution as,

$$y(t - \tau_{DL}) = MK_P[1 - e^{-(t-\tau_{DL})/\tau_P}], \text{ for } t \geq t_{DL} \qquad (6.61)$$

That is, the above response equation is valid for time values greater than t_{DL}. But for the time interval $0 < t < t_{DL}$, there is no analytical solution for $y(t)$ possible by linear methods. However, by applying our common sense and the information that before being disturbed, the process was at steady state, we may decide that within the interval $0 < t < t_{DL}$, $y(t) = 0$, and the complete solution would be:

$$\begin{aligned} y(t) &= 0, \text{ for } 0 \leq t \leq \tau_{DL} \\ y(t) &= MK_P[1 - e^{-(t-\tau_{DL})/\tau_P}], \text{ for } t \geq \tau_{DL} \end{aligned} \qquad (6.62)$$

But the above procedure cannot be applied when the response is fed back or looped back with the incoming fresh disturbance, as it happens when the process is within a control loop. Certain approaches have been developed for approximation of e^{-tDLS} that may be used in the framework of linear analysis.

6.3.1 Bode's Formula

Bode's formula is based on the expansion of e^{-tDLS} as an exponential series. The fundamental expansion:

$$e^{-\tau_{DL}s} = 1 - \tau_{DL}s + \frac{\tau_{DL}^2 s^2}{2} - \frac{\tau_{DL}^3 s^3}{6} + \frac{\tau_{DL}^4 s^4}{24} \qquad (6.63)$$

is seldom used because such an expression in the numerator of a transfer function will

act as a differentiator of multiple order that may pose stability problem. However, the above series truncated to:

$$e^{-\tau_{DL}s} = 1 - \tau_{DL}s \qquad (6.64)$$

may be used for solving problems where low accuracy result is acceptable.

An alternative form of Equation 6.63 is:

$$e^{-\tau_{DL}s} = \frac{1}{e^{\tau_{DL}s}} = \frac{1}{1 + \tau_{DL}s + \frac{\tau_{DL}^2 s^2}{2} + \frac{\tau_{DL}^3 s^3}{6} + \frac{\tau_{DL}^4 s^4}{24} + \cdots} \qquad (6.65)$$

By using Equation 6.64, the stability problem is removed, but if higher order terms are considered for better accuracy, one has to resort to numerical analysis methods. This will cause another kind of error: the truncation error of numerical integration, and presence of higher degree terms of s would result in higher order integration. Thus, the truncation error generated at any stage of the procedure would also be raised to the power equal to the highest order of s in the denominator polynomial. This may create unacceptable error magnitude, and numerical stability problem.

6.3.2 Pade's Formula

It is a better approach to split the e^{-tDLS} term into numerator and denominator parts, so that the approximate transfer function, written up to fourth order of the exponential series, for a dead time element becomes:

$$e^{-\tau_{DL}s} = \frac{e^{-\frac{\tau_{DL}s}{2}}}{e^{\frac{\tau_{DL}s}{2}}} = \frac{1 - \frac{\tau_{DL}s}{2} + \frac{\tau_{DL}^2 s^2}{12} - \frac{\tau_{DL}^3 s^3}{48} + \frac{\tau_{DL}^4 s^4}{384}}{1 + \frac{\tau_{DL}s}{2} + \frac{\tau_{DL}^2 s^2}{12} + \frac{\tau_{DL}^3 s^3}{48} + \frac{\tau_{DL}^4 s^4}{384} + \cdots} \qquad (6.66)$$

Equation 6.66 is more popularly known as Pade's fourth order approximation formula, because the order of approximation is equal to the highest power of s in the transfer function up to which the series is considered. The equal order terms in the denominator and numerator of Equation 6.66 produces a better dynamic performance than obtained with Equations 6.63, 6.64 or 6.65.

6.3.3 Approximation in Frequency Domain

In frequency domain analysis, an exact representation of e^{-tDLS} is possible by using the Euler's identity,

$$e^{-i\theta} = \cos\theta - i\sin\theta \qquad (6.67)$$

In frequency domain, i.e. for sinusoidal inputs, process response may be obtained by putting *iw* for *s* in its transfer function. The analytical basis of this substitution has been covered at the beginning of the section on Frequency Response. Applying the substitution to $e^{-t_{DL}S}$, and expanding according to Equation 6.67:

$$e^{-\tau_{DL}s}\Big|_{i\omega \to s} \Rightarrow e^{-i\omega\tau_{DL}} = \cos\omega\tau_{DL} - i\sin\omega\tau_{DL} \qquad (6.68)$$

Equation 6.68 may be used to represent $e^{-t_{DL}S}$ in frequency domain, a useful expression for certain dynamic property evaluation, e.g. stability analysis of processes that contain dead time element.

Pade's approximation formulae were developed for dead time representation in analog simulation of dynamic processes. In later years, special Tape Recorders called Instrumentation Tape Recorder were developed that used frequency modulation (FM) technique for the recording of slowly changing DC voltages (process variables). These equipments used separate recording and retrieval heads with adjustable inter-distance and variable tape speed for simulating a wide range of lag magnitude. In such a device, the simulated lag is the product of inter-distance between recording and retrieval heads and the effective tape speed. For simulating a large magnitude of lag, recorders with multiple channel recording and retrieval facility were also developed

In Figure 6.13, we have illustrated the working of a multi-channel tape recorder. The input signal, $x(t)$, will be delayed to produce the output signal, $x(t - t_{DL})$. The magnitude of the lag, t_{DL}, for an N-channel system would be,

$$t_{DL} = NDs$$

where

D = distance between recording and retrieval heads, cm,

s = Tape speed, cm/s, and N = No. of channels.

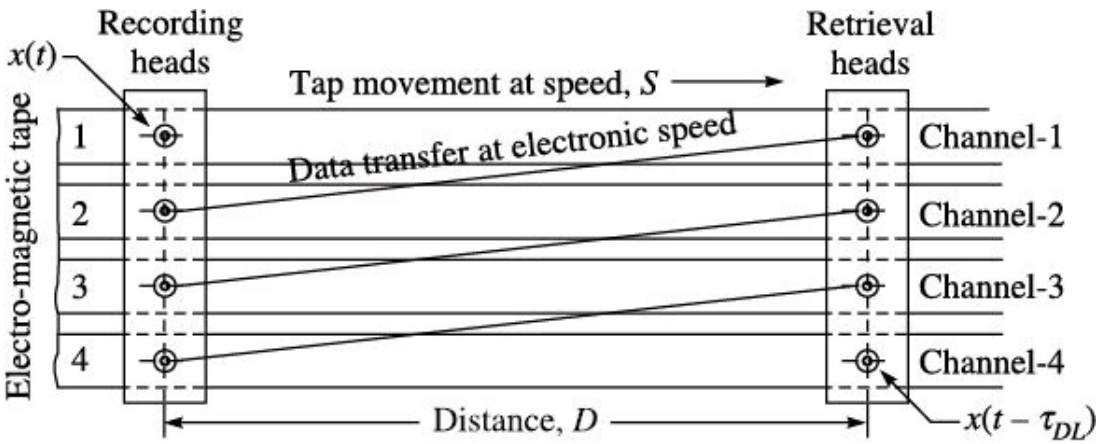


Figure 6.13 Principle of operation of a four-channel instrument tape recorder.

With the advent of digital computers and numerical analysis procedures, analog simulation suffered a degree of obsolescence, because among many other reasons,

simulation of a dead time element reduced to the effort of writing a few lines of computer programme.

You should be aware about the dynamic significance of the presence of a dead time element in control systems. We have already pointed out such a presence in the example of output sheet thickness control in a rolling mill cited in the introductory chapter. The product quality (sheet thickness) can only be measured at a finite distance from the roller heads at the location of the thickness detector. That is, the effective dead time, t_{DL}, equals the product of sheet travelling speed and the distance between the roller heads and the thickness detection point.

As a second example, consider a flow reactor where the exit concentration sensor is placed at a distance from the actual reactor exit point on the outlet pipeline. Such instrument location often occurs in process plants due to design and/or operational demands. Thus, a dead time, t_{DL}, is created as the time required for the reaction mass to reach the measurement point from the reactor exit. Thus, the sheet thickness controller of the rolling mill and reactor concentration controller are always receiving information about the product quality that is t_{DL} time older than its current value. This lack of current information may cause serious deterioration of control quality and the stability aspect of the loop. We shall discuss these topics in Chapters 9 and 10.

With this much excursion on the subject of 'lag', we move on to the second category of second order systems in the next section. In the section on first category of second order systems, we have refrained from deducing their dynamic response equations. The reason being that in the following section, we will propose the general standard form of second order transfer function that possesses the capability of representing all categories of second order systems. We shall use that standard form to deduce response equations corresponding to ideal input functions.

6.4 Second Order Systems of the Natural Variety or Inherently Second Order Systems

This kind of second order systems cannot be split into two first order systems. They contain three kinds of dynamic elements, e.g. Capacitance (C), Resistance (R) and Inductance (L). As these systems cannot be dissembled, hence, we call them natural. In certain literatures they are also called **Inherently Second Order Systems**. The second and first derivative terms of the output variable usually comes from the respective attributes of inductance, and resistance elements, and the free output variable from the capacitance. Whereas, the coupled variety of second order systems usually contains a pair of first order exponential lag elements, i.e. a pair of $R - C$ couples.

We shall discuss physical examples of this category to demonstrate the model building procedure in the following and also in Chapter 7, devoted to Dynamics of Measuring and Final Control Element. Let us begin with the example of Spring-Dashpot system with mass that we discussed up to the development of fundamental differential equation at the beginning of the discussion about second order systems.

Let us begin by rewriting the differential equation that we derived in Example 6.2:

or
$$\frac{W}{g_C}\cdot\frac{d^2Y}{dt^2}+\mu^*\frac{dY}{dt}+K_SY=f(t) \qquad (6.8)$$

where W = Mass of the block, l bm.; g_C = 32.2 [(l bm)(ft)]/[(lbf)(ft^2)]; μ^* = Dashpot constant, (lbf)/(ft/s); K_S = Spring constant, lbf/ft; $f(t)$ = Applied force, lbf.

Assuming that the initial values of all concerned variables and their derivatives are zero, we may directly take the Laplace transform of Equation 6.70 to obtain the transfer function:

$$\frac{Y(s)}{F(s)}=\frac{1/K_S}{\frac{W}{g_CK_S}s^2+\frac{\mu^*}{K_S}s+1} \qquad (6.69)$$

Knowing that at equilibrium, the total applied force will be borne by the spring only, because as $t \to \infty$, $y(t) \to Y_{ULTIMATE}$, hence at steady state, the drag force contribution proportional to (dy/dt), and the momentum contribution proportional to (d^2y/dt^2), are zero, and we may express the force in terms of the spring displacement, i.e. if we put $f(t)/K_S = X(t)$, the Equation 6.69 becomes:

$$\frac{Y(s)}{X(s)}=\frac{1}{\frac{W}{g_CK_S}s^2+\frac{\mu^*}{K_S}s+1} \qquad (6.70)$$

The rearrangement of Equations 6.69 to 6.70 has converted the input variable dimensionally equal to the output variable.

If we compare the transfer functions of Equation 6.69 or 6.70 with all the transfer functions of synthetic kind of second order systems, a common feature becomes apparent, that the last term in the denominator, or the term without any s is always unity. In the differential equation form, the free variable term has a unity coefficient. In the above spring-dashpot system, it was not so in Equation 6.8, hence, we divided the equation by K_S. The result as in Equation 6.70 is the so-called **standard form** of second order transfer function. We have already mentioned this aspect of transfer function at the end of Section 6.1. It should be noted that for the first order systems

also, this rearrangement was done to transfer functions that were not directly obtained with the derivative free output variable term with unity coefficient. In this connection, you may recall the examples of zero mass-spring dashpot and ideally stirred reactor systems.

We have already introduced the definition of **characteristic equation** and its roots in beginning of Chapter 3. If the characteristic equation is obtained from the standard form of transfer function, the root evaluation becomes easier. If we want to examine two systems that are physically almost similar, say, at least in a qualitative way, and we want to compare their dynamics, an examination of their transfer functions in the standard form may be a good starting point for this exercise.

Thus, this standard form is a sort of common platform for comparing the dynamic properties of a number of systems of similar order. As for example, for a first order transfer function, in the standard form, the coefficient of s (having dimension of T) is the time constant and the numerator is the steady state gain. Correspondingly, in a second order transfer function, the coefficient of s^2 (having dimension of T^2) is the square of time constant and the numerator is the steady state gain of the system. Extending this observation further, we find that the coefficient of the s, the middle term, has the dimension of time, but generally its numerical value is not equal to the square root of the coefficient of s^2 term. (However, this equality could be obtained for particular combinations of physical parameter values.) Hence, to have a general form of the transfer function, another dynamic parameter, z, has been introduced into the middle term along with the time constant, t; and for dimensional consistency, parameter z must be dimensionless. This parameter, z, is called the **'Damping Coefficient'** because its value determines the speed of response and whether a disturbed second order system will produce a monotonic or oscillatory response.

Finally, to recognize the quadratic polynomial in the denominator, the general form of the second order transfer function may be written as:

$$\frac{Y(s)}{X(s)} = \frac{K_P}{\tau^2 s^2 + 2\tau\zeta s + 1} \qquad (6.71)$$

The advantage of Equation 6.71 is that it could express the dynamics of both kinds of second order systems. To check this statement, let us find the roots of the quadratic type characteristic equation obtained from the above transfer function.

The characteristic equation:

$$\tau^2 s^2 + 2\tau\zeta s + 1 = 0 \qquad (6.72)$$

has the following roots:

$$s_1 = -\frac{\zeta}{\tau} + \frac{1}{\tau}\sqrt{\zeta^2 - 1} \quad \text{and} \quad s_2 = -\frac{\zeta}{\tau} - \frac{1}{\tau}\sqrt{\zeta^2 - 1} \qquad (6.73)$$

Observing the root expressions, we may conclude that for values of $z > 1$, the roots are real and unequal and the denominator of Equation 6.71 may be reduced to two factors. This is what was obtained for synthetic second order systems. If $z = 1$, a special case for which roots are real but equal,

$$s_1 = -\frac{\zeta}{\tau} \quad \text{and} \quad s_2 = -\frac{\zeta}{\tau} \tag{6.74}$$

as would happen for equal time constants of the two first order systems in non-interacting combination. If the two tanks are joined in interacting combination, the presence of the extra term, A_1R_2 in the coefficient of s will increase the value of z greater than 1, and render the roots unequal. If we relate this increase of z to the more sluggish response of interacting systems, it justifies the parameter name- damping coefficient.

You must remember that with the inherently second order systems (as the mass-spring-dashpot combination) all three root-categories are realizable. The third category for $z < 1$, the roots become complex conjugate pair,

$$s_1 = -\frac{\zeta}{\tau} + i\frac{1}{\tau}\sqrt{1-\zeta^2}, \quad \text{and} \quad s_2 = -\frac{\zeta}{\tau} - i\frac{1}{\tau}\sqrt{1-\zeta^2} \tag{6.75}$$

As we shall very shortly see that with $z < 1$, the response is oscillatory, and for $z \geq 1$, the response become non-oscillatory or monotonic, that includes the special case for $z = 1$, a damping condition corresponding to fastest non-oscillatory response. We may now introduce the names for these damping conditions and relate them with category of roots as in the following:

Range of z	Damping Category	Type of Roots of Characteristic eqn.	Nature of Dynamic Response
> 1	Over Damped	Real unequal	Non-oscillatory
= 1	Critically Damped	Real equal	Fastest non-oscillatory
< 1	Under Damped	Complex conjugate	Oscillatory

The second order transfer function has another form. This is obtained by introducing the natural frequency, w_N, as the reciprocal of time constant, t, i.e. $w_N = 1/t$. By this substitution for a unity gain system Equation 6.71 becomes,

$$\frac{Y(s)}{X(s)} = \frac{\omega_N^2}{s^2 + 2\omega_N\zeta s + \omega_N^2} \tag{6.76}$$

6.5 Response of Second Order Systems for Ideal Input Functions

To find the root expressions for different damping categories, the same substitution $w_N = 1/t$ may be applied to Equations 6.73, 6.74 and 6.75. In the following section, we present them in factorized form of the denominator and putting $X(s)$ in the numerator.

These will be used for the derivation of response equations, substituting the expressions of $X(s)$ for different input functions.

For $z < 1$;
$$Y(s) = \frac{X(s)\omega_N^2}{(s + \omega_N\zeta - i\omega_N\sqrt{1-\zeta^2})(s + \omega_N\zeta + i\omega_N\sqrt{1-\zeta^2})} \quad (6.77)$$

For $z = 1$;
$$Y(s) = \frac{X(s)\omega_N^2}{(s + \omega_N)(s + \omega_N)} \quad (6.78)$$

For $z > 1$;
$$Y(s) = \frac{X(s)\omega_N^2}{(s + \omega_N\zeta - \omega_N\sqrt{\zeta^2-1})(s + \omega_N\zeta + \omega_N\sqrt{\zeta^2-1})} \quad (6.79)$$

For deriving the response equations, we shall use the natural frequency version of transfer function Equations 6.77 through 6.79, because certain dynamic aspects, e.g. resonance, etc. will be more understandable in this form. If required, by using the reciprocal relation between time constant and natural frequency ($t = 1/w_N$), the results obtained in terms of w_N may be easily converted to the corresponding t form.

6.5.1 Step Response for an Input of Magnitude M

Step response equations are derived for three kinds of damping conditions.

Case I: Under Damped condition ($z < 1$)

Putting $X(s) = M/s$ into Equation 6.77, applying partial fraction expansion, and inverting to time domain the response equation is obtained as:

$$Y(t) = M\left[1 - e^{-\omega_N\zeta t}\left(\cos\omega_N\sqrt{1-\zeta^2}\,t + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\omega_N\sqrt{1-\zeta^2}\,t\right)\right] \quad (6.80)$$

Recalling the treatment we applied in sinusoidal response of first order system, we shall use the following trigonometric identity to condense the two periodic functions in Equation 6.80.

$$p\cos A + q\sin A = r\sin(A+\varphi)\text{; where } r = \sqrt{(p^2+q^2)}\text{, and } \phi = \tan^{-1}\left(\frac{p}{q}\right)$$

Applying the above identity in the Equation, Equation 6.80 becomes:

$$Y(t) = M\left[1 - \frac{1}{\sqrt{1-\zeta^2}}\cdot e^{-\omega_N\zeta t}\sin\left(\omega_N\sqrt{1-\zeta^2}\,t + \tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta}\right)\right] \quad (6.81)$$

A typical under damped step response in the normalized form has been shown in Figure 6.14. The oscillation happens because of the sine function, and amplitude

reduction in successive peaks is due to the –ve exponential time function in Equation 6.81.

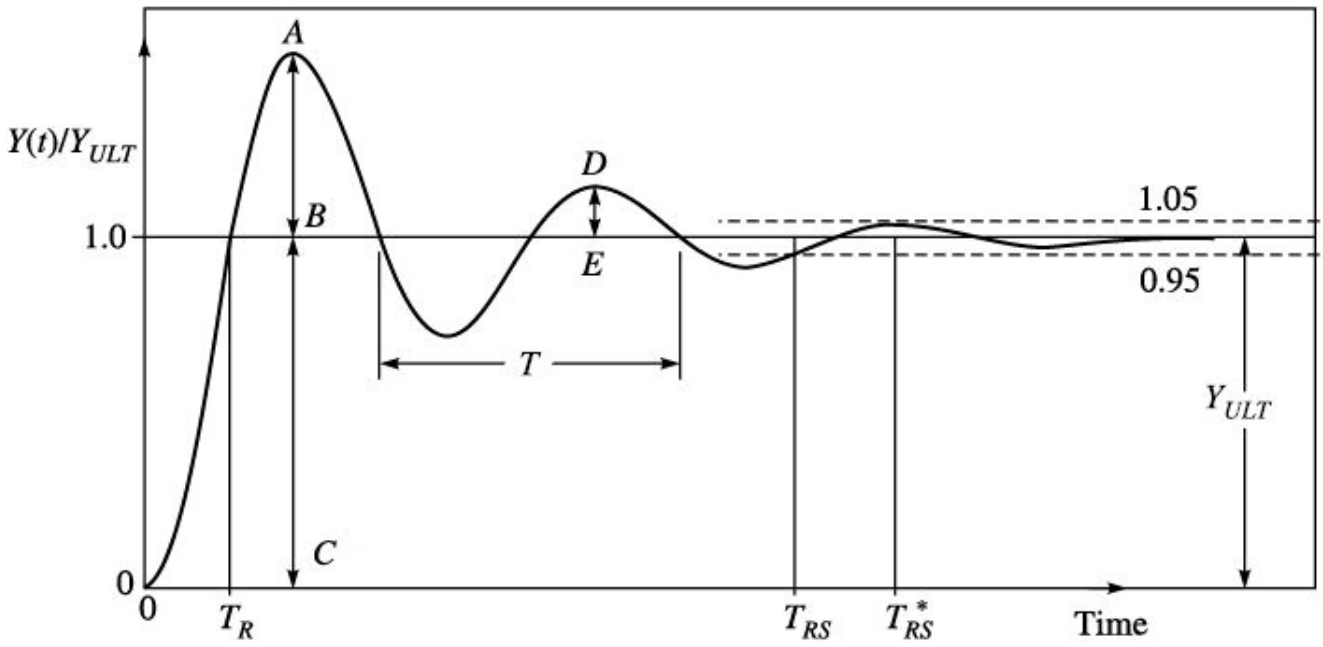


Figure 6.14 Definitions of the characteristics of under damped second order step response.

Before moving on to other kind of damping conditions, we would like to present certain characteristics of the under damped step response. The control systems and measuring instruments are often designed to produce responses that may be approximated by or have the exact form of an under damped kind of response. It is important therefore to have quantitative estimates of these dynamic properties for further reference. These characteristics are illustrated in Figure 6.14.

Overshoot

Overshoot is how much the response, $Y(t)$, exceeds the ultimate response value, Y_{ULT}, following a step change. Y_{ULT} is defined as $Y(t)|_{T\to\infty} = Y_{ULT}$. Overshoot is expressed as the ratio $(AB)/(BC)$ in Figure 6.14.

Overshoot is related to the damping coefficient by the following equation:

$$\text{Overshoot} = e^{-(\pi\zeta/\sqrt{1-\zeta^2})}\text{; and Percent Overshoot} = 100\,e^{-(\pi\zeta/\sqrt{1-\zeta^2})} \qquad (6.82)$$

For deriving Equation 6.82, differentiate Equation 6.81 with respect to time and set the derivative equal to zero for finding the extremes of the response.

$$\frac{dY}{dt} = \frac{M\omega_N e^{-\omega_N\zeta t}}{\sqrt{1-\zeta^2}} \cdot \sin(\omega_N\sqrt{1-\zeta^2}\,t)$$

setting the derivative expression to zero, we obtain two solutions:

1. $t = \infty$, *indicating that the response assumes ultimate value at infinite time*
2. $\sin(\omega_N\sqrt{1-\zeta^2}\,t) = 0$; *or,* $\omega_N\sqrt{1-\zeta^2}\,t = n\pi$; *where* $n = 0, 1, 2, 3, \ldots$ *up to* ∞.

The second solution is a periodic solution that may be used to determine the 'crests'

and 'troughs' of the response. The initial foot of response corresponds to $n = 0$, and the first peak to $n = 1$. Putting $n = 1$ in the periodic solution,

$$t = \frac{\pi}{\omega_N \sqrt{1-\zeta^2}}$$

Putting this value of t in Equation 6.81, we obtain:

$$Y = M\left[1 + e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}\right]$$

Hence, the required expression would be:

$$\text{Overshoot} = \frac{M\left[1 + e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}\right] - M}{M} = e^{-(\pi\zeta/\sqrt{1-\zeta^2})} \qquad (6.82)$$

Clearly the overshoot increases as z decreases.

Decay ratio

Decay ratio is the ratio of magnitudes of successive peaks of response and is given by (AB/DE) in Figure 6.14. Decay ratio is related to the damping coefficient by the following equation:

$$\text{Decay ratio} = e^{-(2\pi\zeta/\sqrt{1-\zeta^2})} \qquad (6.83)$$

Equation 6.83 may be derived by following the method for overshoot; the next peak or crest will occur corresponding to the value of t obtained by putting $n = 3$, in the periodic solution. Thus, height of the second peak is $M[1 + e^{-(3\pi\zeta/\sqrt{1-\zeta^2})}]$,

and $$\text{Decay ratio} = \frac{M[1 + e^{-(3\pi\zeta/\sqrt{1-\zeta^2})}] - M}{M[1 + e^{-(\pi\zeta/\sqrt{1-\zeta^2})}] - M} = e^{-(2\pi\zeta/\sqrt{1-\zeta^2})} \qquad (6.83)$$

Rise Time (T_R)

Rise time is the time required for the response to first reach the ultimate value Y_{ULT}. This has been labeled T_R in Figure 6.14. To develop a relation for T_R, Equation 6.81 may be set equal to the ultimate value to obtain the time solution of crossing points of $Y(t)$ and the horizontal line representing Y_{ULT}.

Again we obtain a first solution at infinite time due to an exponential –ve time function. The second periodic solution is obtained as:

$$\sin\left(\omega_N \sqrt{1-\zeta^2}\, t + \tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta}\right) = \sin(n\pi)$$

T_R is the minimum +ve value of time obtained by putting $n = 1$ in the above expression,

Thus, $$T_R = \frac{1}{\omega_N \sqrt{1-\zeta^2}} \left[\pi - \tan^{-1} \frac{\sqrt{1-\zeta^2}}{\zeta} \right] \qquad (6.84)$$

Response Time (T_{RS})

This is the time required by the response to come within ± 5% of its ultimate value and remain there. In Figure 6.14, this has been denoted by T_{RS}. One method of finding the value of T_{RS} is to first find the peak or trough that occurred within the range of normalized response value of 1.05 and 0.95, shown by the dotted horizontal lines in Figure 6.14. This can be done by following the approach of finding relations of overshoot and decay ratio. The general expressions for all the peaks and troughs of the normalized response may be expressed by:

$$Y/M = \left[1 + e^{-(n\pi\zeta)/(\sqrt{1-\zeta^2})} \right] \qquad (6.85)$$

For an odd value of n, it is a peak, and for an even value, it is a trough. It is required to find the minimum value of n for which the right hand side of Equation 6.85 is within a magnitude range of 1.05 and 0.95. However, this is a preliminary estimate of response time denoted as T_{RS}* in Figure 6.14.

To find the actual value of T_{RS}, a trial-and-error procedure may be followed between the time interval corresponding to n, as found above, and $n - 1$.

(*The trial error effort may be reduced if, within a few trials, two time values are obtained which are close to either side of the solution. Remembering that whatever be the curvature of a function, within a small range it can be assumed as to be linear, we may apply linear interpolation technique between the last two trial values to obtain the solution.*)

Period of Oscillation

From Equation 6.81, the angular frequency (rads/time) is the coefficient of t in the trigonometric function; thus,

Angular frequency, $$w = \omega_N \sqrt{1-\zeta^2} \qquad (6.86a),$$

and the cycle frequency, $$f = \frac{1}{T} = \frac{\omega_N \sqrt{1-\zeta^2}}{2\pi} \qquad (6.86b)$$

We can conclude that the time constant, $t = 1/w_N$, as well as the damping character, z, affects the time period, T. The dependence of overshoot and decay ratio on the

value of z has been illustrated in Figure 6.15.

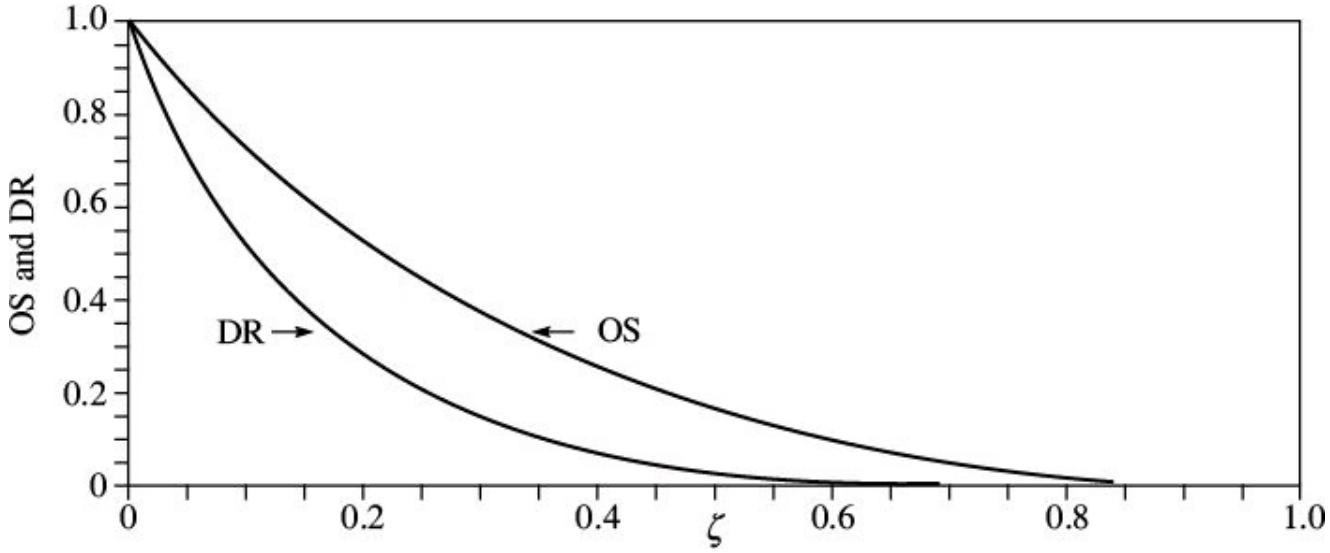


Figure 6.15 The dependence of overshoot and decay ratio on the damping coefficient.

Case II: Critically Damped condition ($z = 1$)

Putting $X(s) = M/s$ into Equation 6.78, applying partial fraction expansion, and inverting to time domain, the response equation is obtained as:

$$Y(t) = M\left[1 - e^{-\omega_N t}\left(1 + \omega_N t\right)\right] \qquad (6.87)$$

The time functions in Equation 6.87 suggest that the critically damped second order step response would be non-oscillatory or monotonic.

Case III: Over Damped condition ($z > 1$)

Putting $X(s) = M/s$ into Equation 6.79, applying partial fraction expansion, and inverting to time domain, the response equation is obtained as:

$$Y(t) = M\left[1 - e^{-\omega_N \zeta t}\left(\cos h\omega_N\sqrt{\zeta^2 - 1}t + \frac{\zeta}{\sqrt{\zeta^2 - 1}}\sin h\omega_N\sqrt{\zeta^2 - 1}t\right)\right] \qquad (6.88)$$

The hyperbolic functions of Equation 6.88 may be defined as:

$$\sin h(x) = (e^x - e^{-x})/2 \text{ and } \cos h(x) = (e^x + e^{-x})/2$$

EXAMPLE 6.8 We have already obtained the over damped response in Example 6.7, where we found the step response of a combination of two non-interacting first order systems. The time constants are t_1 and t_2 and gain K_P. The transfer function of this system is:

$$G_P(s) = \frac{Y(s)}{X(s)} = \frac{K_P}{\tau_1\tau_2 s^2 + (\tau_1 + \tau_2)s + 1} = K_P\frac{1/(\tau_1\tau_2)}{(s + 1/\tau_1)(s + 1/\tau_2)}$$

The corresponding response equation for a step of magnitude, M, has been obtained as:

$$Y(t) = K_P M\left[1 - \frac{\tau_1}{\tau_1 - \tau_2}\exp\left(-\frac{t}{\tau_1}\right) + \frac{\tau_2}{\tau_1 - \tau_2}\exp\left(-\frac{t}{\tau_2}\right)\right] \qquad (6.89)$$

with a little algebra, it could be proved that the time functions within the third bracket of Equations 6.88 and 6.89 are identical. As a first step, let us try to relate the roots of two time constants, t_1, t_2 of Equation 6.89 with the one time constant, t, and damping coefficient, z, of Equation 6.88.

Solution: From the two transfer functions:

$$\frac{-1}{\tau_1} = \frac{-\zeta}{\tau} + \frac{\sqrt{\zeta^2 - 1}}{\tau} \text{ (A)}$$

and $$\frac{-1}{\tau_2} = \frac{-\zeta}{\tau} - \frac{\sqrt{\zeta^2 - 1}}{\tau} \text{ (B)}$$

Now, (A) + (B): $$\frac{\tau_1 + \tau_2}{\tau_1\tau_2} = \frac{2\zeta}{\tau} \text{ (C)}$$

and (A) – (B): $$\frac{\tau_1 - \tau_2}{\tau_1\tau_2} = 2\frac{\sqrt{\zeta^2 - 1}}{\tau} \text{ (D)}$$

[(C)/(D)]^2: $$\frac{\tau_1^2 + 2\tau_1\tau_2 + \tau_2^2}{\tau_1^2 - 2\tau_1\tau_2 + \tau_2^2} = \frac{\zeta^2}{\zeta^2 - 1} \text{ or } \frac{\tau_1^2 + 2\tau_1\tau_2 + \tau_2^2}{\tau_1^2 - 2\tau_1\tau_2 + \tau_2^2} - 1 = \frac{\zeta^2}{\zeta^2 - 1} - 1$$

simplifying, we obtain,

$$\zeta = \frac{\tau_1 + \tau_2}{2\sqrt{\tau_1\tau_2}}$$

and putting this value of z in (C): $\tau = \sqrt{\tau_1\tau_2}$

The effect of damping coefficient through the range from under damped up to over damped condition, on the nature of the step response, can be understood most clearly by plotting Equations 6.81, 6.87 and 6.89 as shown in Figure 6.16. The normalized response value
($Y(t)/Y_{ULT}$) has been drawn against the dimensionless time (tw_N, or t/t). The Y_{ULT} or ultimate value of the response is the value of $Y(t)|_{T\to\infty} = M$, the step magnitude. For $z < 1$, all response curves are oscillatory in nature, and they become less oscillatory as z increases. When $z = 1$, the system is critically damped, the response becomes non-oscillatory or monotonic, and in fact, critically damped response shows fastest monotonic recovery. For increase of z beyond 1, the responses become more

sluggish.

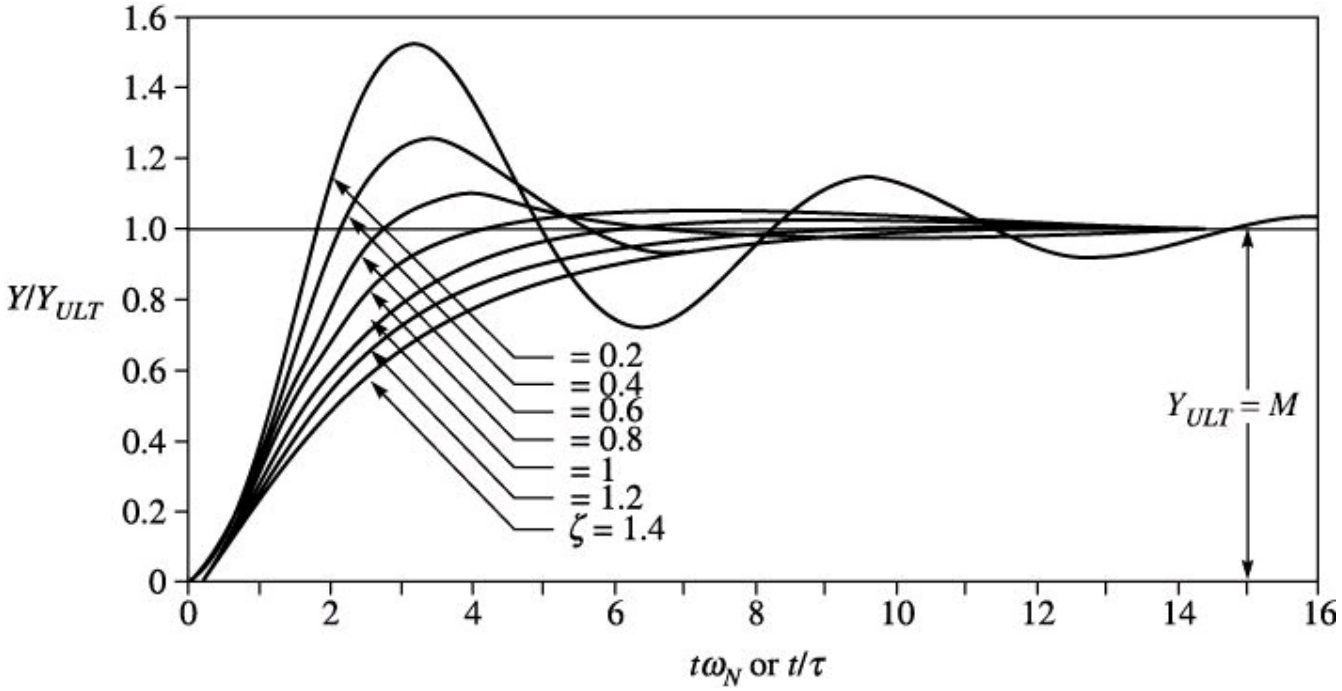


Figure 6.16 Step response of magnitude *M* of second order system.

EXAMPLE 6.9 The transfer function of a second order system is given by:

$$\frac{Y(s)}{X(s)} = \frac{2.3}{s^2 + 4s + 6}$$

Find the values of w_N, w, t, z, K_P, T_R, T_{RS}, overshoot and decay ratio. If this system is disturbed by a step input of magnitude 4, what would be value of Y_{ULT}?

Solution: A general quadratic expression of second order transfer function may be processed as follows:

$$\frac{Y(s)}{X(s)} = \frac{K}{As^2 + Bs + C} = \frac{K/C}{(A/C)s^2 + (B/C)s + 1} = \frac{K}{C} \cdot \frac{C/A}{s^2 + (B/A)s + (C/A)} \qquad \text{(a)}$$

Comparing the last expression of Equation (a) with Equation 6.76 we may use,

$$w_N = \sqrt{(C/A)},\ 2w_N z = B/A, \text{ and } K_P = K/C$$

putting the values of *K, A, B, C* from the given transfer function in the problem, we obtain the following system parameter values:

$w_N = \sqrt{(6/1)} = 2.45$; $t = 1/w_N = 0.4083$; $z = 4/(2 \times 2.45) = 0.8165$; and $K_P = 2.3/6 = 0.3833$.

Thus the under damped characteristics may be evaluated as,

1. Overshoot = $e^{-(\pi\zeta/\sqrt{1-\zeta^2})} = e^{-4.44065} = 0.01179$, or 1.18%
2. Decay ratio = $e^{-(2\pi\zeta/\sqrt{1-\zeta^2})}$ = (Overshoot)^2 = 0.000139

3. Rise time, $T_R = \frac{1}{2.45 \times 0.57735}\left[3.14 - \tan^{-1}\frac{0.57735}{0.8165}\right] = 1.785$ time unit

4. Period of oscillation, $T = \frac{2\pi}{\omega_N\sqrt{1-\zeta^2}} = \frac{2 \times 3.14}{2.45 \times 0.57735} = 4.8073$ time unit

5. Ultimate Response, $Y_{ULT} = MKP = 4 \times 0.3833 = 1.533$

The transfer function parameters may also be obtained if the response plot is available.

EXAMPLE 6.10 An open loop dynamic test has been performed on an air drier to relate the input power to the exit. Air temperature from the drier has been given in Figure 6.17. From the shape of the normalized response an under damped model may be proposed. An ultimate temperature difference of 7.5ºC is obtained from a base temperature of 72ºC, for an input power change of 22 kcal/min (calculated from the change of steam flow rate).

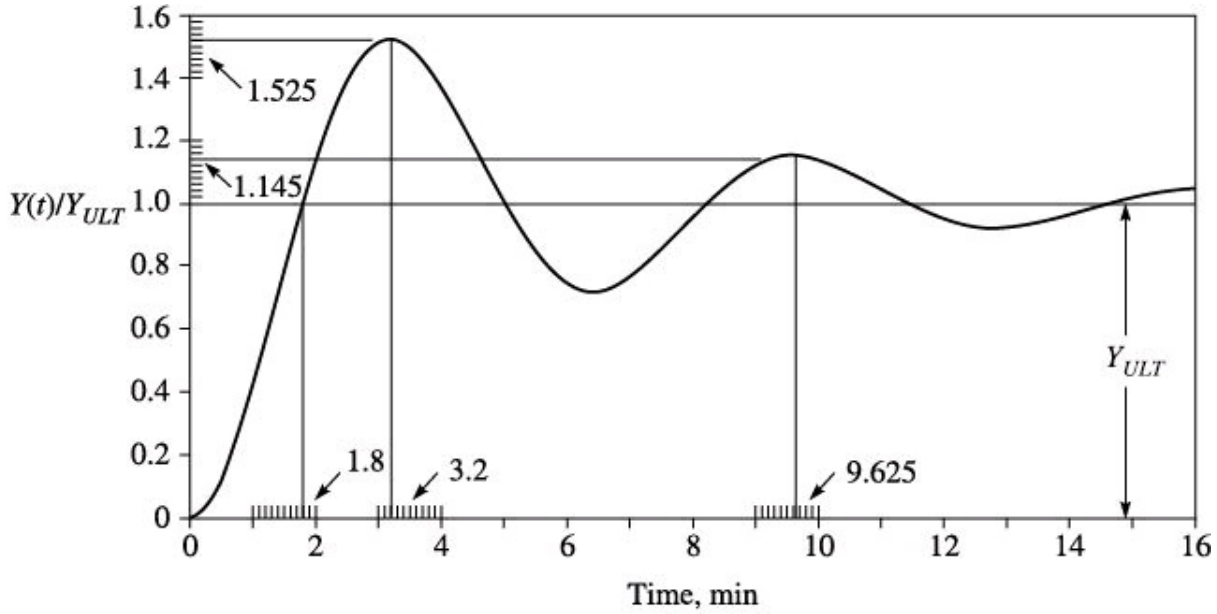


Figure 6.17 The normalized step response of a drier with air recycle.

Using the graphical parameters of Figure 6.17, find the values of w_N, w, t, z, K_P, T_R, overshoot and decay ratio for the proposed second order transfer function.

Solution: We have included the small divisions in the graph to improve the measurement accuracy from the graph. From the given data:

Process Gain, $K_P = \frac{\Delta \text{Output}}{\Delta \text{Input}} = \frac{7.5°\text{C}}{22 \text{ kcal/min}} = 0.341 \frac{°\text{C}}{\text{kcal/min}}$

From the graph: $\text{Overshoot} = \frac{0.525}{1.0} = 0.525$

Now the equation relating overshoot to z may be rearranged to obtain:

$$\zeta = \frac{[\ln(OS)]^2}{\pi^2 - [\ln(OS)]^2} = \frac{[\ln(0.525)]^2}{(3.14159)^2 - [\ln(0.525)]^2} = 0.10577$$

From the response the period of oscillation, $T = 9.625 - 3.2 = 6.425$ min.

Hence, $$\omega_N = \frac{2\pi}{T\sqrt{1-\zeta^2}} = \frac{6.28318}{6.425 \times 0.9944} = 0.98344 \text{ rads/min.}$$

and, $$\omega = \frac{2\pi}{T} = 0.978 \text{ rads/min}$$

The rise time may be found from the graph as:
$T_R = 1.8$ min. The reader should check this result by using Equation 6.84.

EXAMPLE 6.11 The transfer function parameters could also be expressed by the location of roots of the characteristics equation on complex s – plane. Let the roots be $s_1 = -3 + 2i$ and
$s_2 = -3 - 2i$. Find the second order transfer function.

Solution: The quadratic expression in the denominator of the transfer function may be found by product of the two roots as:

$$s_1 \times s_2 = (-3 + 2i)(-3 - 2i) = s^2 + 6s + 13.$$

Recognizing the last numerical term '13' to be equal to w^2_N, and the coefficient of s term, '6' to be equal to $2w_N z$, following Equation 6.76, we may write the transfer function as:

$$\frac{Y(s)}{X(s)} = \frac{13}{s^2 + 6s + 13} = \frac{K_P \omega_N^2}{s^2 + 2\omega_N \zeta s + \omega_N^2}$$

where $\omega_N = \sqrt{13} = 3.6056$ rads/time, and $\zeta = \frac{6}{2 \times 3.6056} = 0.832$

However, to find the gain, K_P, we need to know input vs. output magnitude at least at two steady state conditions. Let this additional information be provided as follows:
The above system represents an Electronic Pressure Transducer that records a final output change of 3 mA corresponding to an input change of 75 mm Hg.

Thus, $$K_P = \frac{\Delta \text{ Output}}{\Delta \text{ Input}} = \frac{3}{75} = 0.04 \frac{\text{mA}}{\text{mmHg}}$$

and the complete transfer becomes

$$\frac{Y(s)}{X(s)} = \frac{K_P \omega_N^2}{s^2 + 2\omega_N \zeta s + \omega_N^2} = \frac{0.52}{s^2 + 6s + 13} \frac{\text{mA}}{\text{mmHg}}$$

6.5.2 Ramp Response for an Input of Constant Slope M

Ramp response equations are derived for three types of damping conditions.

Case I: Under Damped condition ($z < 1$)

Putting $X(s) = M/s^2$ into Equation 6.77, applying partial fraction expansion, and inverting to time domain, the response equation is obtained as:

$$Y(t) = M\left[t - \frac{2\zeta}{\omega_N} + 2e^{-\omega_N \zeta t}\left(\frac{\zeta}{\omega_N^3}\cos \omega_N \sqrt{1-\zeta^2}\, t + \frac{2\zeta^2 - 1}{2\omega_N^3 \sqrt{1-\zeta^2}} \sin \omega_N \sqrt{1-\zeta^2}\, t\right)\right] \quad (6.90)$$

Recalling the treatment we applied in sinusoidal response of first order, and step response of under damped second order systems, we shall use the following trigonometric identity to condense the two periodic functions in Equation 6.90

$$p\cos A + q \sin A = r \sin(A + \varphi); \text{ where, } r = \sqrt{(p^2 + q^2)}, \text{ and } \phi = \tan^{-1}\left(\frac{p}{q}\right)$$

Applying the above identity in equation, Equation 6.90 becomes,

$$Y(t) = M\left[t - \frac{2\zeta}{\omega_N} + \frac{1}{\omega_N \sqrt{1-\zeta^2}} e^{-\omega_N \zeta t} \sin\left\{\omega_N \sqrt{1-\zeta^2}\, t + \tan^{-1}\left(\frac{\zeta\sqrt{1-\zeta^2}}{\zeta^2 - 0.5}\right)\right\}\right] \quad (6.91)$$

Case II: Critically Damped condition ($z = 1$)

Putting $X(s) = M/s^2$ into Equation 6.78, applying partial fraction expansion, and inverting to time domain, the response equation is obtained as:

$$Y(t) = M\left[\left(t - \frac{2}{\omega_N}\right) + \left(t + \frac{2}{\omega_N}\right)e^{-\omega_N t}\right] \quad (6.92)$$

Case III: Over Damped condition ($z > 1$)

Putting $X(s) = M/s^2$ into Equation 6.79, applying partial fraction expansion, and inverting to time domain, the response equation is obtained as:

$$Y(t) = M\left[t - \frac{2\zeta}{\omega_N} + e^{-\omega_N t}\left\{2\zeta \cos h\left(\omega_N \sqrt{\zeta^2 - 1}t\right) + \frac{2\zeta^2}{\sqrt{\zeta^2 - 1}} \sin h\left(\omega_N \sqrt{\zeta^2 - 1}t\right)\right\}\right] \quad (6.93)$$

As we have seen for first order systems, here also the ramp responses for any value of z have a part that gradually vanishes as time increases due to the associated

exponential –ve time function contributing the initial curved nature of the response near origin. The remaining part is a linear time function that becomes parallel to the input function at large values of time. Of course this observation is only valid for systems with dimensionless unity gain. From the stable part of the response, we may evaluate the constant steady state **'Error'** and **'Lag'**, as we had done for first order systems. From Equations 6.91, 6.92 and 6.93, the expressions for steady values of lag and error would be:

$$\text{Lag, } L = \frac{2\zeta}{\omega_N} = 2\zeta\tau \text{ and Error, } E = \frac{2M\zeta}{\omega_N} = 2M\zeta\tau$$

For $z < 1$, there should be oscillation due to the sine function that will diminish in amplitude for the presence of exponential –ve time term. But the critically and over-damped responses would not show any such oscillation but in an anticipated way the sluggishness of response will increase with increase of z.

To demonstrate these observations, we have plotted the ramp responses of a second order system for the three possible damping conditions in Figure 6.18.

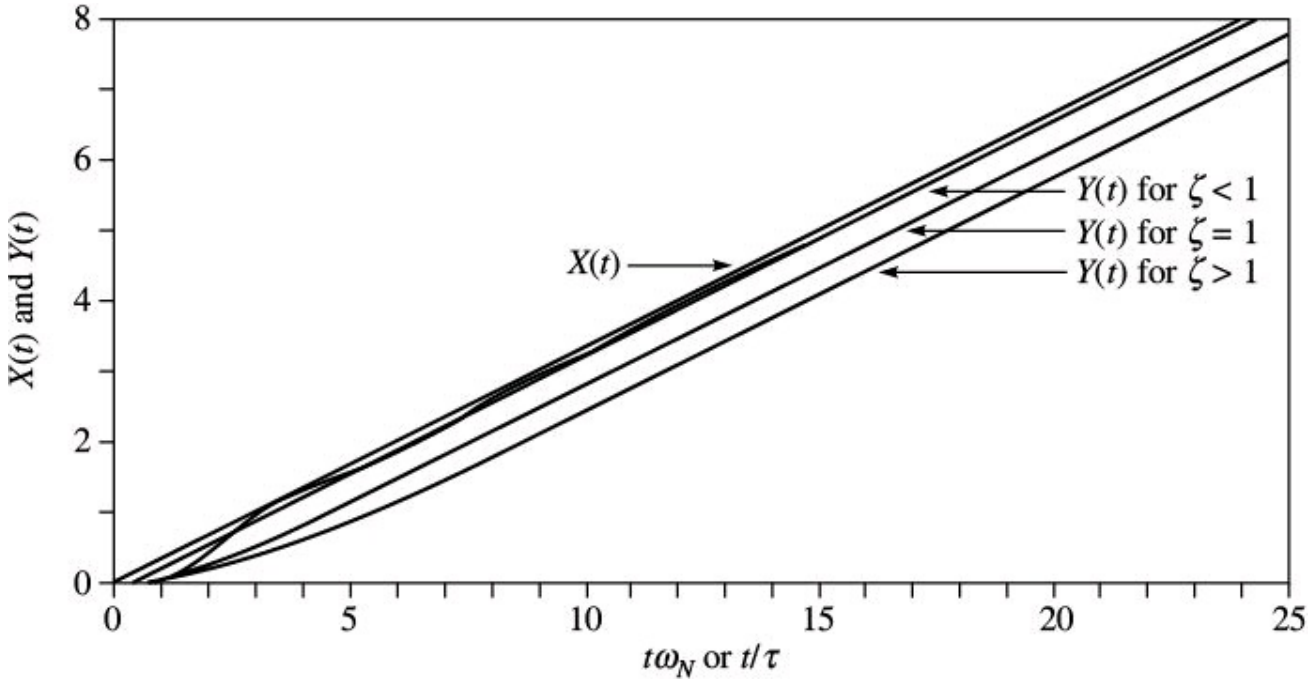


Figure 6.18 The ramp response of a second order system for three damping conditions.

EXAMPLE 6.12 In a small laboratory type heat treatment furnace, the temperature is linearly increasing according to: $T_X = 50°C + Mt$ °C, where M is the slope in °C/min and t is in min. The initial furnace temperature was 50°C. A non-ferrous object is inside the furnace.

After sufficient time from the start of the heat treatment run, the recorded steady 'Lag and 'Error' between the furnace and object temperature are 6 min and 12°C respectively.

From a separate step test, to observe the dynamics of heat transfer from furnace to

the object, it was found to be of over-damped second order in nature, with the time constant
t = 2 min. Find z and M.

Solution: From Equation 6.93, the non-time vanishing part of the response may be written as:

$$Y(t) = M\left[t - \frac{2\zeta}{\omega_N}\right]$$

Thus Lag, $$L = \frac{2\zeta}{\omega_N} = 2\tau\zeta = 2 \times 2 \times \zeta = 6\,\text{min}$$

Hence, z = 1.5, and Error, $E = M\tau 2\zeta = 12°\text{C}$. Hence M = 2°C/min.

6.5.3 Impulse Response for an Input of Magnitude M

Impulse response equations are derived for three types of damping conditions.

Case I: Under Damped condition ($z < 1$)

Putting $X(s) = M$ into Equation 6.77, applying partial fraction expansion, and inverting to time domain, the response equation is obtained as:

$$Y(t) = M\frac{\omega_N}{\sqrt{1-\zeta^2}}e^{-\omega_N\zeta t}[\sin(\omega_N\sqrt{1-\zeta^2}t)] \qquad (6.94)$$

Case II: Critically Damped condition ($z = 1$)

Putting $X(s) = M$ into Equation 6.78, applying partial fraction expansion, and inverting to time domain, the response equation is obtained as:

$$Y(t) = M\omega_N^2 te^{-\omega_N t} \qquad (6.95)$$

Case III: Over Damped condition ($z > 1$)

Putting $X(s) = M$ into Equation 6.79, applying partial fraction expansion, and inverting to time domain, the response equation is obtained as:

$$Y(t) = M\frac{\omega_N}{\sqrt{\zeta^2-1}}e^{-\omega_N\zeta t}\sin h(\omega_N\sqrt{\zeta^2-1}t) \qquad (6.96)$$

Unlike step and ramp, the impulse and sinusoidal inputs brings a temporary change to the process, because the process variable eventually returns to its threshold or previous steady state value.

To demonstrate the response character, we have plotted the impulse responses of a second order system for the three possible damping conditions in Figure 6.19.

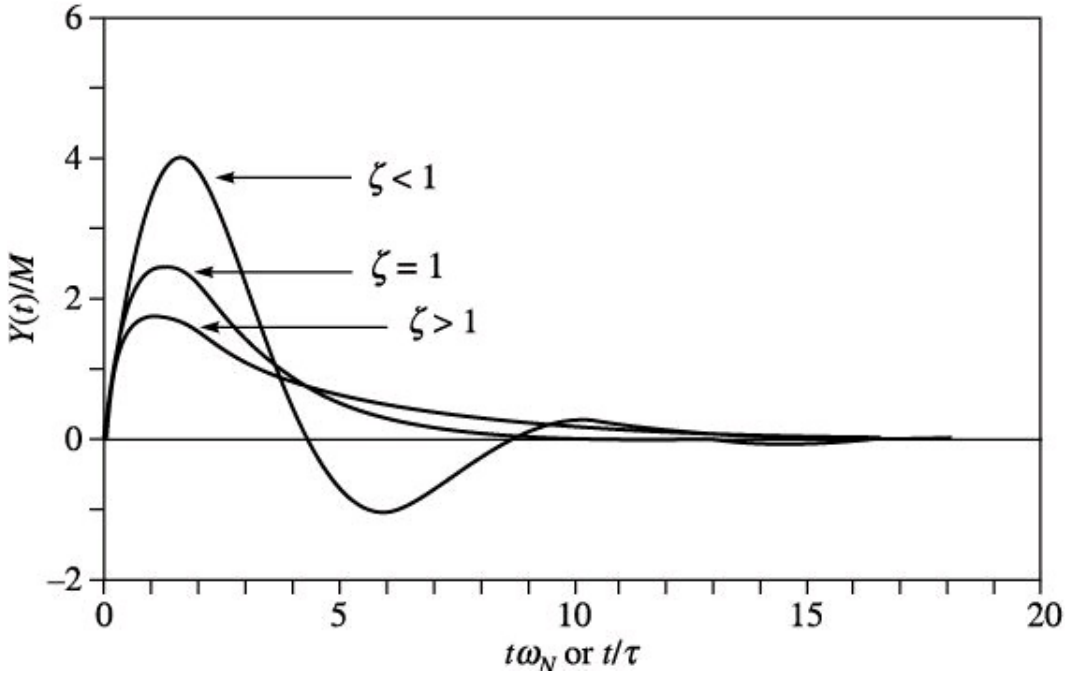


Figure 6.19 The normalized impulse response of a second order system for three damping conditions.

A property demonstrated by impulse response of a second order system is that its initial slope has a +ve finite value irrespective of damping condition. For all the other three kind of responses this value is zero. You must recall the discussion about the initial part of step response of first order system, where we introduced the conception and reason for the existence of *transfer lag*. Here also, before an analytical proof, we try to find the reason of this special property from the physical example of two-tank liquid level dynamic system illustrated in
Figure 6.10 (b).

What will happen when a bucket of water is poured into the first tank while the system was operating at steady state? If the pouring has been done in no time (pouring time approaching zero value), the input may be called an impulse of magnitude, M = the volume of water in the bucket. As the ideal pouring time of an impulse input is zero, there will be no outflow through the bottom exit line of the first tank during the pouring, and the corresponding instantaneous rise of level will be, $H_1 = M/A_1$, where, A_1 is the cross-sectional area of the first tank. Thus, as the cause for the first tank outflow q_1, is the level h_1 hence, q_1 will also increase with a 'sudden jump'. (This did not happen for a step input where q_1 initially increased with a finite slope.) But for an impulse input, the sudden increase in q_1 will cause the second tank level h_2 to rise with a finite slope, an almost similar phenomenon that occurred for the first tank level h_1 in step response situation.

EXAMPLE 6.13 Prove that the initial value of slope of the impulse response is always finite and independent of the system's damping character.

Solution: Differentiating Equations 6.94, 6.95 and 6.96, we respectively obtain A, B

and C.

$$\frac{dY(t)}{dt} = M\left[k_1(-k_2)e^{-k_2t}\sin k_3t + k_1e^{-k_2t}k_3\cos k_3t\right] \qquad \text{(A)}$$

where, $k_1 = \dfrac{\omega_N}{\sqrt{1-\zeta^2}}$; $k_2 = \omega_N\zeta$; $k_3 = \omega_N\sqrt{1-\zeta^2}$

$$\frac{dY(t)}{dt} = M\left[k_4e^{-k_5t} + k_4(-k_5)te^{-k_5t}\right] \qquad \text{(B)}$$

where, $k_4 = \omega_N^2$; $k_5 = \omega_N$

$$\frac{dY(t)}{dt} = M\left[k_6(-k_7)e^{-k_7t}\sin h(k_8t) + k_6e^{-k_7t}k_8\cos h(k_8t)\right]$$

$$= M\left[k_6(-k_7)e^{-k_7t}\left(\frac{e^{k_8t} - e^{-k_8t}}{2}\right) + k_6e^{-k_7t}k_8\left(\frac{e^{k_8t} + e^{-k_8t}}{2}\right)\right] \qquad \text{(C)}$$

where $k_6 = \dfrac{\omega_N}{\sqrt{\zeta^2-1}}$; $k_7 = \omega_N\zeta$; $k_8 = \omega_N\sqrt{\zeta^2-1}$

Putting $t = 0$ in equations A, B and C, we find the initial values of slope of responses are equal to:

$$\left.\frac{dY(t)}{dt}\right|_{t=0} = \omega_N^2$$

EXAMPLE 6.14 Find the time at which the critically damped second order system response under an impulse input reaches its maximum value. Is this value dependent on system time constant?

Solution: From the previous example, rewriting Equation B:

$$\frac{dY(t)}{dt} = M[k_4e^{-k_5t} + k_4(-k_5)te^{-k_5t}] \qquad \text{(B)}$$

where, $k_4 = w^2_N$; $k_5 = w_N$

If we set Equation B to zero and find the required time solution,

$$t = \frac{k_4}{k_4k_5} = \frac{1}{\omega_N} = \tau$$

EXAMPLE 6.15 The two tank liquid level system shown in Figyre 6.20 is operating

at steady state when a step change is made in the flow rate to the top tank.

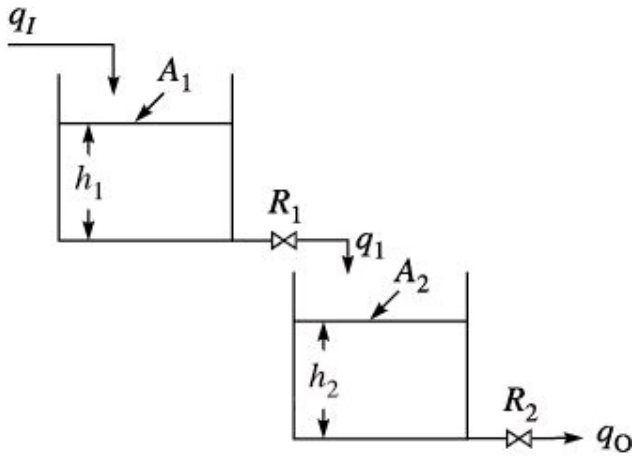


Figure 6.20 The liquid level system of Example 6.15.

The transient response is critically damped. Calculate the system time constant if it takes 2.0 min for the change in the level of the second tank to reach 72% of the total change.

Solution: Let us start by writing the related response Equation 6.87 in terms of time constant t:

$$Y(t) = M\left[1 - e^{-\frac{t}{\tau}}\left(1 + \frac{t}{\tau}\right)\right] \tag{6.87a}$$

Putting $t = 2$ min, $Y(t)/M = 0.72$, and $k_1 = w_N = 1/t$; and simplifying,

$$1 - e^{-2k_1}[1 + 2k_1] = 0.72 \text{ or } e^{-2k_1} = \frac{0.28}{1 + 2k_1}$$

It is apparent that an implicit solution for $k_1 = w_N = 1/t$ may be obtained as:

No. of Trials	1	2	3	4	5	6	7
k_1	1	2	1.5	1.4	1.3	1.2	1.25
$\text{Exp}(-2k_1)$	0.135	0.0183	0.056	0.061	0.0743	0.0907	0.0821
	0.093	0.056	0.07	0.074	0.078	0.08235	0.08

No. of Trials	8	9	10	11	12	13
k_1	1.24	1.26	1.27	1.265	1.266	1.268
$\text{Exp}(-2k_1)$	0.0837	0.0804	0.0788	0.07966	0.0795	0.07918
	0.0804	0.0795	0.079	0.07932	0.0793	0.07918

Thus, from the last trial, the system time constant, t = 1/1.268 = 0.789 min. The reader may attempt this example by the implicit convergence procedures presented in Chapter 14.

6.5.4 Sinusoidal Response for an Input of Amplitude M

and Angular frequency ~

We shall only discuss about the solution for under damped condition and indicate that there is a much simpler way of obtaining the responses for sinusoidal inputs for any damping conditions.

Case I: Under Damped condition ($z < 1$)

Putting $X(s) = [(Mw)/(s^2 + w^2)]$ into Equation 6.77, we obtain,

$$Y(s) = \left[\frac{M\omega\omega_N^2}{(s + i\omega)(s - i\omega)(s + \omega_N\zeta - i\omega_N\sqrt{1-\zeta^2})(s + \omega_N\zeta + i\omega_N\sqrt{1-\zeta^2})} \right] \tag{6.97}$$

Partial fraction expansion of Equation 6.97 is possible but that will involve a lot of tedious algebra. However, we may apply the knowledge of inverting real roots for the case of imaginary and complex roots and decide that the form of the complete time domain solution, $Y(t)$ would comprise of $Y_1(t)$ and $Y_2(t)$, the respective contribution of the forcing function (input) and process dynamic character:

$$Y_1(t) = A_1 \cos\omega t + B_1 \sin\omega t \tag{6.98a}$$

and,

$$Y_2(t) = e^{-\omega_N\zeta t}\left[A_2 \cos(\omega_N\sqrt{1-\zeta^2}\,t) + B_2 \sin(\omega_N\sqrt{1-\zeta^2}\,t) \right] \tag{6.98b}$$

where A_1, B_1, A_2, B_2, are constants that may be evaluated by partial fraction expansion. But the second part of the solution $Y_2(t)$ will not exist at large values of t due to the presence of the exponential –ve time function e^{-wNzt}. This is a similar situation that we came across while deriving the sinusoidal response of first order system. The stable part of the response may be worked out to obtain,

$$Y(t) = \frac{M}{\sqrt{[1 - (\omega/\omega_N)^2 + \{(2\zeta\omega)/\omega_N\}^2}} \sin(\omega t + \phi) \tag{6.99}$$

where,

$$\phi = -\tan^{-1}\frac{2\zeta\omega/\omega_N}{1 - (\omega/\omega_N)^2} \tag{6.100}$$

In the frequency response analysis of Chapter 11, you would be presented with an easy and much quicker method of obtaining sinusoidal response for a given transfer function without going through the rigors of partial fraction method.

EXAMPLE 6.16 A catalytic flow reactor has been operating in adiabatic condition, at 250ºC. A temperature sensor situated at the cylindrical reactor wall, records the temperature. This has a first order dynamics with unity gain and time constant of 0.5 min. The heat transfer dynamics from reactor interior to the temperature sensor may also be expressed by a first order model with unity gain and time constant of 2.0 min.

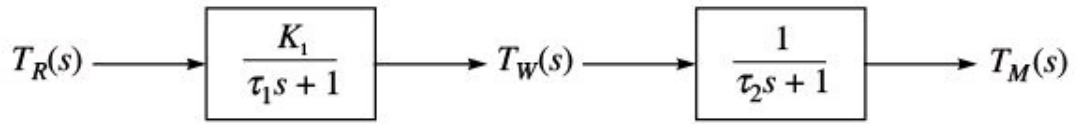
It was proposed to run the reactor with cyclic variation of temperature to increase the average production because the catalyst activity increases exponentially with temperature. The frequency of temperature sinusoid would be 0.1 cycle/min around an average temperature of 250ºC.

However, there is an upper limit of catalyst temperature, that is, if the catalyst bed is subjected to a temperature of more than 290ºC but within 300ºC for more than 5 min, it may be irreversibly damaged due to sintering.

If the measured temperature shows a sinusoidal variation around 250ºC with amplitude of 25ºC, confirm whether under such a temperature cycling, the catalyst would remain safe.

Solution: In the following block diagram, we depict the variable nomenclature and the dynamic relations between them. T_R, T_W and T_M are the temperature of reactor mass plus catalyst, temperature of the reactor wall, and the measured temperature respectively.

$T_R(t) = A_R \sin \omega t$ $\quad$ $T_W(t) = A_W \sin(\omega t - \varphi_1)$ $\quad$ $T_M(t) = A_M \sin(\omega t - \varphi_2)$



We assume that T_R and T_M are related by a synthetic second order system comprising of a non-interacting combination of two first order systems, but as the intermediate variable, T_W, is recognizable, we can treat the problem using the sinusoidal response equation of first order systems. If we denote T_{WA} and T_{MA} as amplitude of T_W and T_M around their average value then they are related as,

$$T_{MA} = \frac{T_{WA}}{\sqrt{1 + \tau_2^2 \omega^2}}$$

we know $T_{MA} = 25$ºC,

hence, $T_{WA} = 25 \times \sqrt{1 + (0.5 \times 0.1 \times 2 \times 3.14159)^2} = 26.2047$°C

A similar relation exists between T_{RA} and T_{WA}

$$T_{WA} = \frac{T_{RA}}{\sqrt{1 + \tau_1^2 \omega^2}}$$

using the calculated value of T_{WA} = 26.2047°C, we obtain

$$T_{RA} = 26.2047 \times \sqrt{1 + (2 \times 0.1 \times 2 \times 3.14159)^2} = 42.084°C$$

Instead of two first order if we consider the reactor to temperature indicator as a second order process, the second order amplitude ratio expression between T_{RA} and T_{MA} obtained from Equation 6.99:

$$AR = \frac{1}{\sqrt{(1 - (\omega/\omega_N)^2 + [(2\zeta\omega)/\omega_N]^2}}$$

may be used to find T_{RA}, where

$$\omega_N = \frac{1}{\tau} = \frac{1}{\sqrt{\tau_1 \tau_2}} = 1 \text{ and } \zeta = \frac{\tau_1 + \tau_2}{2\sqrt{\tau_1 \tau_2}} = 1.25$$

The reader may confirm that the resulting values are equal. However, we have established that the maximum temperature, 250 + 42.084 = 292.084°C, exceeds the upper allowable limit of catalyst operation. But does this excess temperature condition exist for more than 5 min?

The peak temperature 292.084°C corresponds to $wt = p/2$.

If at an angle x, the variable value is 40°C, then

$$\sin x = \frac{40}{42.084} \text{ and } x = \sin^{-1}(0.9505) = 72°C$$

Thus the over temperature condition will remain for an angle value of = 2 × (90 – 72) = 36°. As the sinusoid sweeps 36° per min (w = 0.1 cycles/min), hence the over temperature condition in the reactor will remain for 1 min. This value is small enough to confirm that the catalyst bed will be within the safety limit.

The dynamic analysis by the conventional method of partial fraction expansion and inversion to time domain becomes laborious for processes that contain zero/s in their TF. Certain useful information may be obtained for such processes by using properties of Laplace Transform theory with a lot of lesser amount of computational exercise.

EXAMPLE 6.17 Two processes are represented by the following TFs (a) and (b).

(i) It is required to find their steady state gain K_P.

(a) $\frac{Y(s)}{X(s)} = \frac{Ds + E}{As^2 + Bs + C}$; *and*

(b) $\frac{Y(s)}{X(s)} = \frac{Ds^2 + Es + F}{As^2 + Bs + C}$

(ii) If the systems (a) and (b) are disturbed by a unit step input, find the ultimate values of their response.

Solution: (i) A straightforward evaluation of gain becomes difficult for processes that contain zero/s, or a polynomial of s in numerator of the TF. Putting $s = 0$, in the TF is a shortcut method for finding K_P. Provided an undefined term (0/0) does not result by this substitution.

For system (a): the $K_P = E/C$, and for (b) $K_P = F/C$.

(ii) This may be found by applying Final Value Theorem, $F(t)\,|_{t\to\infty} = [sF(s)]_{s=0}$

For system (a): $Y(s) = \frac{Ds + E}{s(As^2 + Bs + C)}$; and for

(b): $Y(s) = \frac{Ds^2 + Es + F}{s(As^2 + Bs + C)}$

Hence, for (a): $Y(t)\,|_{t=\infty} = E/C$; and for (b): $Y(t)\,|_{t=\infty} = F/C$

Alternatively, the step response of system (b) may be written in the splitted form:

$$Y(s) = \frac{Ds^2}{s(As^2 + Bs + C)} + \frac{Es}{s(As^2 + Bs + C)} + \frac{F}{s(As^2 + Bs + C)}$$

Term I Term II Term III

In the above expansion, Term II would produce an impulse response, and Term I would be the derivative of an impulse response. Thus, contribution of Term I and II are temporary, which would disappear as time increases, and the ultimate response value would be due to the step response due to Term III. That the step response will produce an ultimate value of F/C may be confirmed by Equation (a) derived in Example 6.9. For convenience, we have rewritten the development of the referred equation:

$$\frac{Y(s)}{X(s)} = \frac{K}{As^2 + Bs + C} = \frac{K/C}{(A/C)s^2 + (B/C)s + 1} = \frac{K}{C} \cdot \frac{C/A}{s^2 + (B/A)s + (C/A)} \quad \text{(a)}$$

A similar treatment for system (a) would also confirm the obtained result. Also the result of part (i) could be checked for system (a) and (b) by using the same equation.

6.6 Higher Order Systems

Transfer function development of third order non-interacting and interacting systems are shown in Examples 6.3 and 6.6 respectively. In Section 6.2, step responses of certain higher order systems are depicted to establish the notion of transfer lag. To show the way of using Laplace transform to obtain higher order system corresponding to a given input, let us solve the following examples.

EXAMPLE 6.18 Derive the response equations in time domain for the following four process TFs:

I.A. $\frac{Y(s)}{X(s)} = \frac{1}{(s+a)(s+b)(s+c)}$ I.B. $\frac{Y(s)}{X(s)} = \frac{1}{(s+a)^3}$

II.A. $\frac{Y(s)}{X(s)} = \frac{1}{(s+a)(s+b)(s+c)(s+d)(s+f)}$ II.B. $\frac{Y(s)}{X(s)} = \frac{1}{(s+a)^5}$

The input functions are: (i) $X(s) = M/s$ (step of magnitude M), and (ii) $X(s) = M$ (impulse of magnitude M).

Solution: Time domain responses could be obtained by the way of partial fraction expansion. The following time domain responses are obtained for the eight cases.

1. Step response of the system I.A.

$$L^{-1}\left[\frac{M}{s}\cdot\frac{1}{(s+a)(s+b)(s+c)}\right]$$
$$= M\left[\frac{1}{abc} - \frac{e^{-at}}{a(b-a)(c-a)} - \frac{e^{-bt}}{b(a-b)(c-b)} - \frac{e^{-ct}}{c(a-c)(b-c)}\right]$$

2. Impulse response of the system I.A.

$$L^{-1}\left[\frac{M}{(s+a)(s+b)(s+c)}\right] = M\left[\frac{e^{-at}}{(b-a)(c-a)} + \frac{e^{-bt}}{(a-b)(c-b)} + \frac{e^{-ct}}{(a-c)(b-c)}\right]$$

3. Step response of the system I.B.

$$L^{-1}\left[\frac{M}{s}\cdot\frac{1}{(s+a)^3}\right] = M\left[\frac{1}{a^3} - \frac{t^2e^{-at}}{2a} - \frac{te^{-at}}{a^2} - \frac{e^{-at}}{a^3}\right]$$

4. Impulse response of the system I.B.

$$L^{-1}\left[\frac{M}{(s+a)^3}\right] = M\left[\frac{t^2e^{-at}}{2}\right]$$

5. Step response of the system II.A.

$$L^{-1}\left[\frac{M}{s}\cdot\frac{1}{(s+a)(s+b)(s+c)(s+d)(s+f)}\right]$$

$$= M\left[\frac{1}{abcdf} - \frac{e^{-at}}{a(b-a)(c-a)(d-a)(f-a)} - \frac{e^{-bt}}{b(a-b)(c-b)(d-b)(f-b)}\right.$$

$$- \frac{e^{-ct}}{c(a-c)(b-c)(d-c)(f-c)} - \frac{e^{-dt}}{d(a-d)(b-d)(c-d)(f-d)}$$

$$\left. - \frac{e^{-ft}}{f(a-f)(b-f)(c-f)(d-f)}\right]$$

6. Impulse response of the system II.A.

$$L^{-1}\left[\frac{M}{(s+a)(s+b)(s+c)(s+d)(s+f)}\right]$$

$$= M\left[\frac{e^{-at}}{(b-a)(c-a)(d-a)(f-a)} + \frac{e^{-bt}}{(a-b)(c-b)(d-b)(f-b)} + \right.$$

$$\frac{e^{-ct}}{(a-c)(b-c)(d-c)(f-c)} + \frac{e^{-dt}}{(a-d)(b-d)(c-d)(f-d)} +$$

$$\left. \frac{e^{-ft}}{(a-f)(b-f)(c-f)(d-f)}\right]$$

7. Step response of the system II.B.

$$L^{-1}\left[\frac{M}{s}\cdot\frac{1}{(s+a)^5}\right] = M\left[\frac{1}{a^5} - \frac{t^4e^{-at}}{24a} - \frac{t^3e^{-at}}{6a^2} - \frac{t^2e^{-at}}{2a^3} - \frac{te^{-at}}{a^4} - \frac{e^{-at}}{a^5}\right]$$

8. Impulse response of the system II.B.

$$L^{-1}\left[\frac{M}{(s+a)^5}\right] = M\left[\frac{t^4e^{-at}}{24}\right]$$

EXAMPLE 6.19 A concentrated solution of a solute is injected into a stream of a solvent by reciprocating pump action, so that the solute concentration varies sinusoidally according to

$C(t) = C_{AVG} + C_{AMP}.\text{Sin}(wt)$, where, C_{AVG} = Average concentration, 2 kg-moles/l; C_{AMP} = Amplitude of concentration varation, 0.5 kg-moles/l; w = Radian frequency, 0.1 rads/min. To reduce this cycling amplitude of concentration, it has been proposed to process the solution through five (5) no. of constant holdup perfectly stirred mixers operating in series. Two alternative mixer designs are forwarded:

(A) The mixer holdup volumes are, $V_1 = 0.6\ m^3$, $V_2 = 0.9\ m^3$, $V_3 = 1.2\ m^3$, V_4 =

1.5 m^3 and V_5 = 1.8 m^3,

(B) The mixers have identical volumes = 1.2 m^3. If the solution flow rate, Q is constant
at Q = 100 lpm, find which of the designs would obtain a better mixing performance.

Solution: This problem involves sinusoidal response of a higher order system. In Section 6.5.4, we have commented that for such systems, it is convenient to use frequency response properties presented in Chapter 11, than to go through the partial fraction route. However, in this problem, the system is a fifth order system comprising of serial combination of five first order elements, hence we have to use only the sinusoidal response properties of a first order system already given in Chapter 5. As this is a strictly mixing process without any reaction, the gain of the system is unity. Thus, K_P = 1.Q = 100 lpm

For part A, the parameter values we are going to use are:

K_P = 1, $t_1 = V_1/Q$ = 0.6 min, t_2 = 0.9 min, t_3 = 1.2 min, t_4 = 1.5 min, t_5 = 1.8 min.

Thus, the amplitude ratio

$$AR = \frac{1}{\sqrt{(1+\omega^2\tau_1^2)(1+\omega^2\tau_2^2)(1+\omega^2\tau_3^2)(1+\omega^2\tau_4^2)(1+\omega^2\tau_5^2)}} = \frac{1}{9.108} = 0.1098$$

For part B, the parameter values we are going to use are:

$$K_P = 1,\ t_1 = V/Q = 1.2 \text{ min} = t_2 = t_3 = t_4 = t_5$$

Thus, the amplitude ratio

$$AR = \frac{1}{(1+\omega^2\tau^2)^{2.5}} = \frac{1}{9.2998} = 0.10753$$

From the result, it appears that design A gives a slightly poorer result. The reader is urged to examine whether the result is dependent on the input frequency.

EXERCISE PROBLEMS

6.1 If we define transfer lag as the time required by a system's response to become greater than 1% of its ultimate value, after it is disturbed by a step input, find the transfer lag for a fourth order system with four equal time constants of 4 min and unity gain.

6.2 On a linear graph trace the response of a second order plus dead time system for a step disturbance of magnitude 4. The system parameters are: K_P = 0.5, t_1 = 2 min,

$t_2 = 4$ min, $t_{DL} = 1$ min.

6.3 Find the TF $E_O(s)/E_I(s)$ for the following electrical network shown in Figure P6.1:

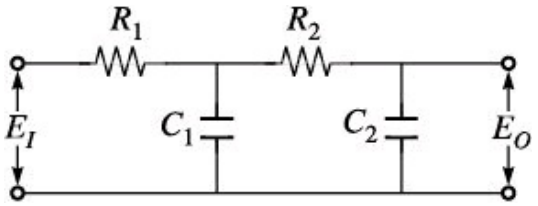


Figure P6.1

6.4 A step change of magnitude 4 is introduced into a system having the TF

$$\frac{Y(s)}{X(s)} = \frac{15.4}{14s^2 + 7s + 21}$$

Determine: (a) Percent overshoot; (b) Rise time; (c) Maximum value of $Y(t)$; (d) Ultimate value of $Y(t)$; (e) Period of oscillation; (f) Decay ratio.

6.5 The 2-tank system shown in Figure P6.2, is operating at steady state. At time $t = 0$, 300 litres of liquid is suddenly poured into the first tank. The parameter values of the two tank liquid level system are: $A_1 = A_2 = 10000$ cm^2, $R_1 = 0.001$ cm/(cc/min), $R_2 = 0.0035$ cm/(cc/min).

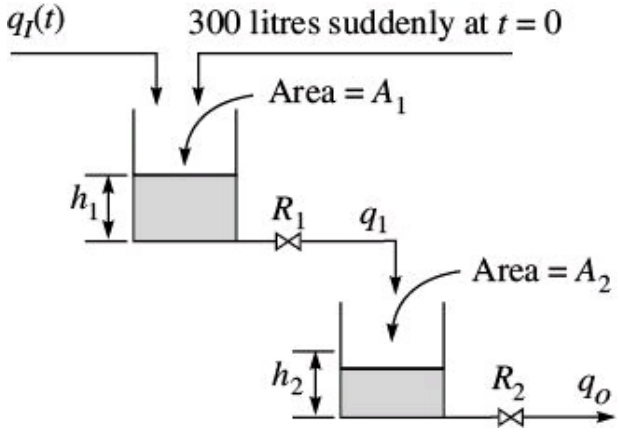


Figure P6.2

Determine the maximum deviation in level (in cm.) in both tanks from the ultimate steady state values and the time at which each maximum occurs,

6.6 For the manometer shown in Figure P6.3, determine the TF $H(s)/P_1(s)$.

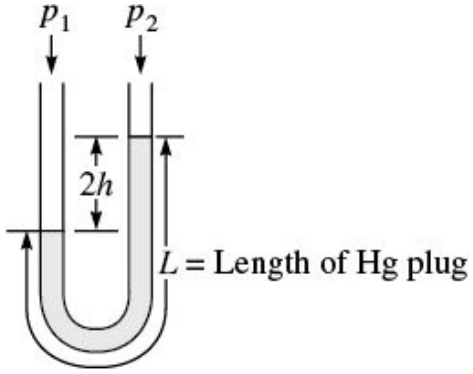


Figure P6.3

where, A = Cross-sectional area of the manometer tube; r = Liquid density; p_1, p_1 = Applied pressures; R = Frictional resistance, m = Fluid viscosity.

Assume the flow of fluid inside the manometer tube is laminar and the velocity profile being flat. From the TF written in standard second order form, list:

(a) The steady state gain

(b) T_P, w_N and z

Comment on these parameters as they are related to the physical nature of the problem.

6.7 Calculate the liquid viscosity and density of a U-tube manometer that will measure pressures up to 1.25 atm absolute having each limb made of 0.5 meter glass tube of 3 mm ID, and will give responses which are slightly under-damped with $z = 0.8$.

6.8 A unity gain system with second order dynamics is subjected to an input $x(t) = M_S + Mt$, (where $x = M_S$, for $t < 0$). After sufficient time has elapsed for the responses to move with constant velocity, find the relation between $y(t)$ and $x(t)$. Find the dependency of 'lag' and 'error' on the damping coefficient value of the system.

6.9 Find the transfer function $Y(s)/F(s)$ for the systems depicted in Figures P6.4(a), (b), and (c).

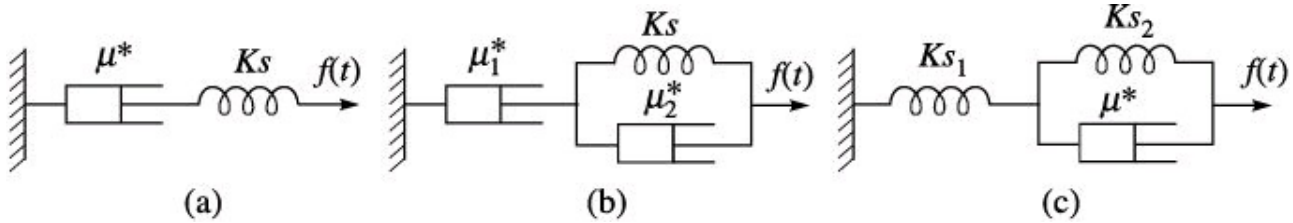


Figure P6.4 Systems for different configurations of spring-dashpot elements.

6.10 Two constant hold up stirred liquid heaters are in non-interacting combination. Only the first has the external heating facility. Each tank has a hold up volume of 50 litres through which a constant liquid flow rate of 12.5 l/min is maintained. Initially the inlet liquid temperature (T_I) is constant at 70°C and the electrical heater remains switched off.

At $t = 0$, T_I is suddenly reduced to 50°C, and the electrical heater is switched on so that

$W = 2.5$ kW. Assuming liquid properties identical to water, derive the response of exit liquid temperatures from the two tanks.

6.11 Find step and impulse response of the systems represented by the following two transfer functions, for three states of damping coefficients.

(a) $\dfrac{Y(s)}{X(s)} = \dfrac{Ds + E}{As^2 + Bs + C}$ (b) $\dfrac{Y(s)}{X(s)} = \dfrac{Ds^2 + Es + F}{As^2 + Bs + C}$

6.12 Find the response equation of a second order system for the three damping conditions and plot on a graph paper for the input functions depicted in Figures P6.5(a), (b), (c), (d), (e), (f) and (g).

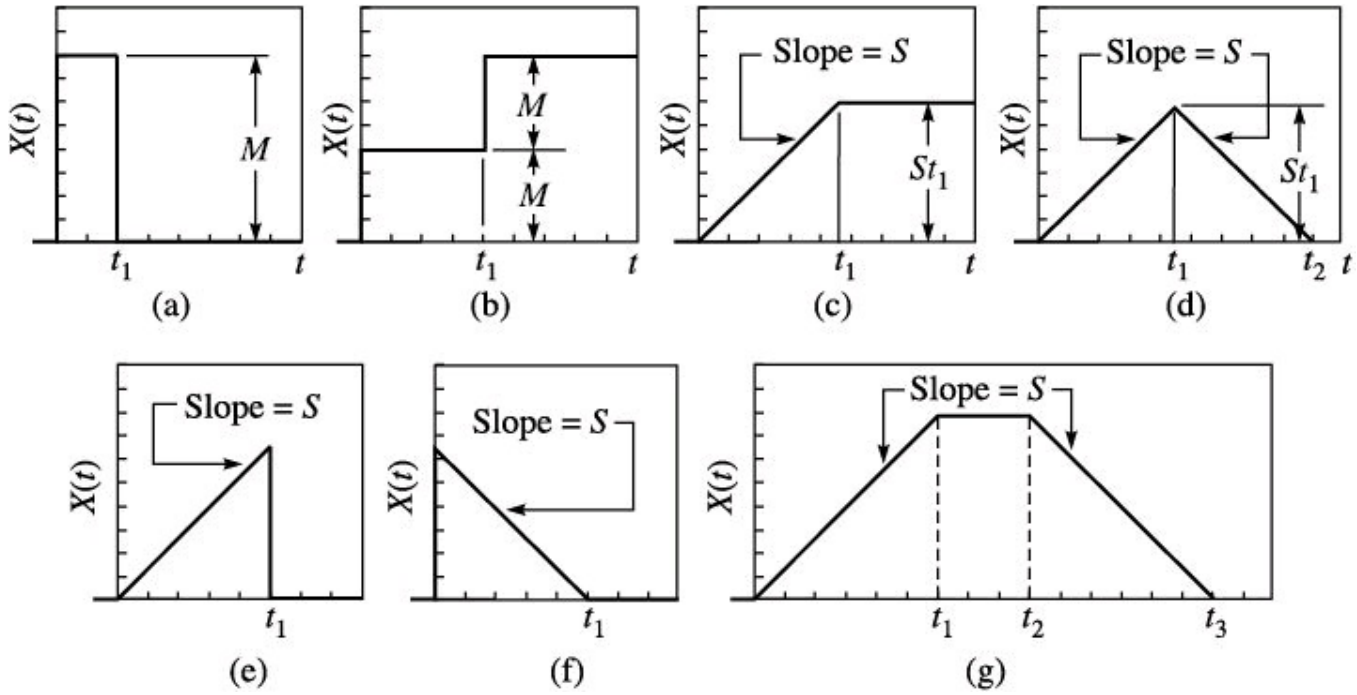


Figure P6.5 Different configurations of complex input functions.

6.13 We are intending to unfold a secret part of the incidents that was narrated by Mr. William Shakespeare in his immortal drama …,well, we will refrain from mentioning any name. Mr. Shakespeare was also daunted in including this part lest some violent public reaction may unhinge his reputation.

The three witches, as you know the three characters of the story, dwelt in the cave of the famous mountain. They had a vicious plan to take over world power by blowing away of human mind. There were three water holes in the cave through which a secret hot spring supplied water that cascaded through the three holes and finally came down the hillside as a small brook.

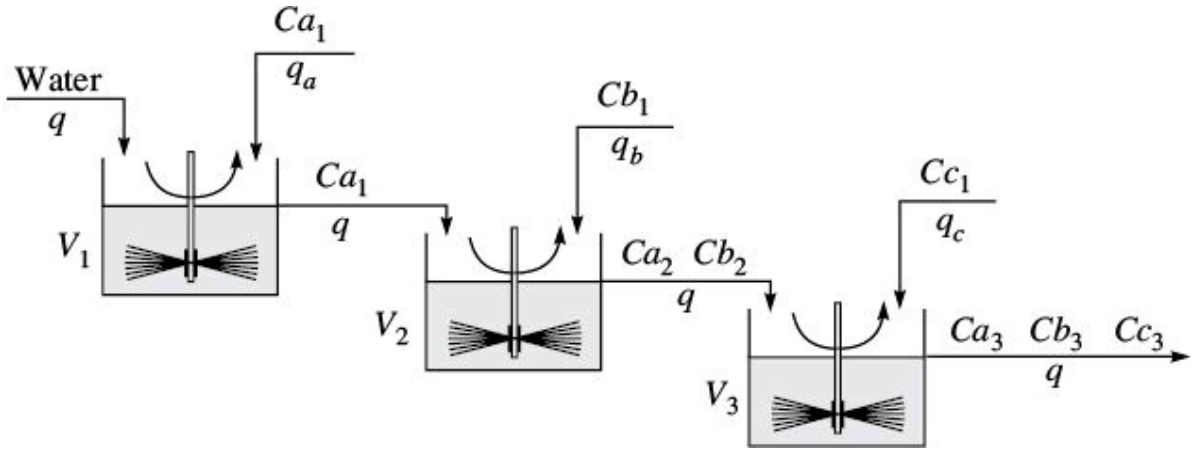


Figure P6.6 Three CSTR battery with parallel feeding of reactants.

This was the sole water supply for the community. The mind-blowing plan came from an ancient script that three separate potions (liquid reactants) A, B and C had to be parallely added to the three water holes (just like separately fed 3-CSTR battery) at constant rates, while the aqueous reaction mixture in the holes were kept under perfect mixed condition by the witches brooms rotating at high speed. The reaction rate constants dictate that different residence times had to be provided to obtain the three products at right concentration in exit stream as in Figure P6.6.

Investigative journalist Miss Lois Lane got some fishy smell in the wind as all journalists do, and reached there by simple means of time travel, but was immediately apprehended by the junior witch. Bound by hand and foot, she was left at the cave corner, and was informed with a gleeful wicked smile that she was in the grand stand seat to watch the nemesis of civilization.

The villains had taken away all of her effects. The only mistake that a small white box made of unknown shiny material hanging from her neck by a chain was left. In fact it is the state-of-art voice controlled cellular phone. By hanging her neck in apparent dejection she whispered to raise Kit Walker, Bruce Wayne, Peter Parker, Tintin, even Asterix, and all the other guys in the trade. But alas all of them were either busy in removing other threats to human race or could not be contacted. Then at last she called Clark Kent. Fortunately Superman was aerially roaming in the vicinity and came for the rescue. By the simple use of his ESP for a few nano-seconds he assimilated the following technical detail:

The inlet temperature (T_I) and flow rate (q) of hot water to the first CSTR are T_I = 70ºC, and q = 2000 l/min. The holdup volume of each reactors are, $V_1 = V_2 = V_3$ = 25000 l. All three reactors operate under adiabatic condition at a constant temperature of T_I = 70ºC. The inlet reactant concentrations are: $C_{AI} = C_{BI} = CI$ = 50 g-mol/1. All three reactants undergo first order irreversible reactions:

A □ Product; B □ Product; C □ Product;

The reaction rate constants are temperature dependent according to:

$$k_{AT} = k_{A0}e^{-K_A/T};\ \ k_{BT} = k_{B0}e^{-K_B/T};\ \ \text{and}\ \ k_{CT} = k_{C0}e^{-K_C/T}$$

But the reactions have no heat effect so that,

$$\Delta H_{RA} = \Delta H_{RB} = \Delta H_{RC} = 0$$

The only available kinetic data is: $k_{A0} = 0.050651602\ \text{min}^{-1}$; $K_A = 26.56449468°\text{C}$;

Superman took the decision and implemented it by his well-known frosty blow from his mouth, cooled the inlet stream to the first water hole so that reactions were only half complete and the calamity was prevented.

(a) Calculate the value of T_I for which the conversion is halved. If the air temperature in his exhaled breath was 0ºC, and heat transfer happened under adiabatic condition with 100% efficiency, calculate the mass and volume rate of air in the blow.

From the third chapter of Perry's Handbook, 7th ed, you can obtain the values of density and heat capacity of air.

(b) Calculate the values of k_{B0}, K_B, k_{C0} and K_C for reactants B and C for the similar conversion values as in part (a). (*Consider steady state expressions of the TF for the analysis*.)

6.14 The top view of a sliding door has been shown in Figure P6.7. A specially designed Spring–Dashpot system has been used for quick opening and retarded, soft closure.

Data: Mass of the door panel and mechanism = 15 kgf.

Dashpot constant, K_D = 2.8 kgf s/ft.

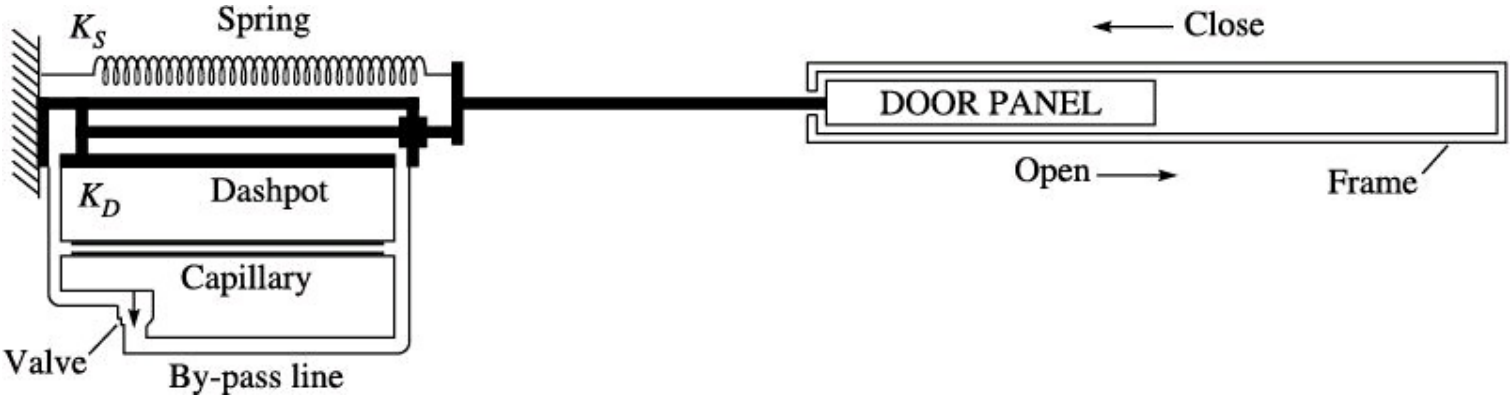


Figure P6.7 Damped door closing system.

(During the opening of the door, the valve on the by-pass line opens to allow the major portion of the contained fluid in the dashpot cylinder to flow through the large diameter by-pass line, thus featuring the quick opening facility. But during closur the valve remains closed, and the fluid has to pass only through the capillary.)

6.15 A step change in the steam pressure to the shell side of a heat exchanger is made

from 12 to 15 psig. The tube outlet temperature response recorded by a strip chart recorder given in Figure P6.8, appears to be of second order in nature. But the chart speed is not available.

A second test using a sinusoidal variation of steam pressure of 3 psig amplitude and a frequency of 2 cycles/min yields an output temperature variation with 5°C amplitude. Determine the transfer function for the system.

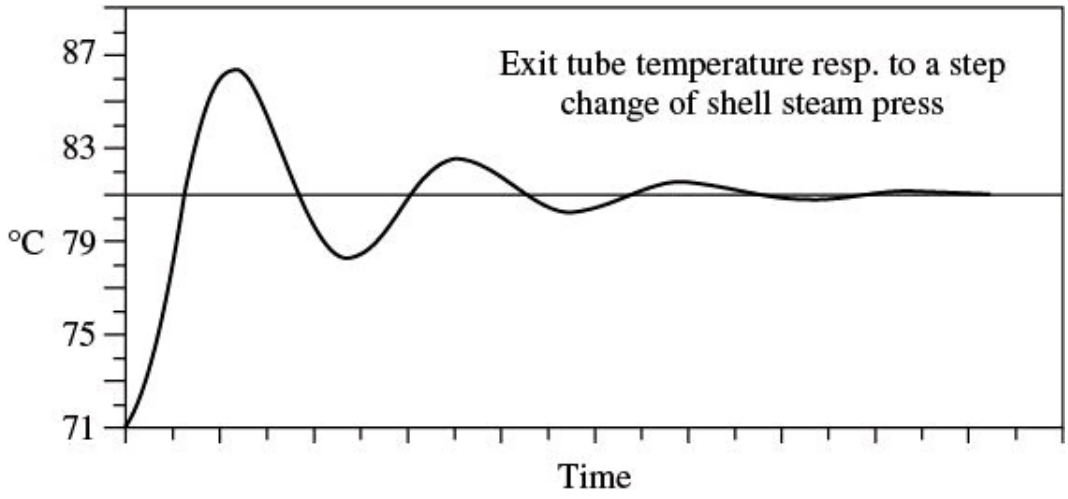


Figure P6.8

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[1] Coughanowr, D.R., *Process System Analysis and Control*, 2nd ed., McGraw-Hill Inc, NY, pp 102, 1991.

[2] Seborg, D.E., Edgar, T.F., and Mellichamp, D.A., *Process Dynamics and Control*, Ch-7, John Wiley and Sons, Inc. 1989.

7

Measuring Elements, Signal Transducers and Final Control Elements

Referring to the block diagram of Figure 1.4, of a feedback control system, we have dealt with the dynamics of the process block in the earlier Chapters 2 through 6. In this chapter we shall discuss about dynamic characteristics of rest of the hardware blocks of a feedback control loop, except the controller. In principle, these elements are signal transducers, i.e. they proportionally convert one kind of physical variable into another. Basically there are two such blocks in the block diagram of a feedback control loop. They are drawn in bold in Figure 7.1. The block labeled G_V denotes the final control element (FCE), and G_M, the measuring element in the control system. The FCE receives controller output (CO) as its input, and produces manipulated variable (MV) as the output, and the measuring element receives the process or controlled variable and produces measured variable (mv) as the output.

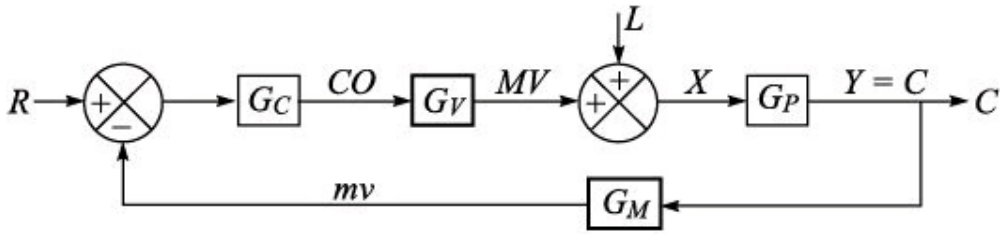


Figure 7.1 Location and I/O variables of G_V and G_M in a feedback loop.

In certain closed loop systems, instruments are selected that use more than one type of actuation energy or utility. A common example would be an electronic sensor-transmitter and an electronic controller with a pneumatic control valve. Such heterogeneity is intentionally brought due to safety aspects, speed of closed loop response, and may be for economic reasons. Hence, another instrument is needed to convert the DC current signal from the controller to the pneumatic or air pressure signal for driving the control valve. Such a signal transducer is called an Electro-Pneumatic-Converter (EPC) and denoted by [I/P] or G_{IP} in the block diagram in Figure 7.2.

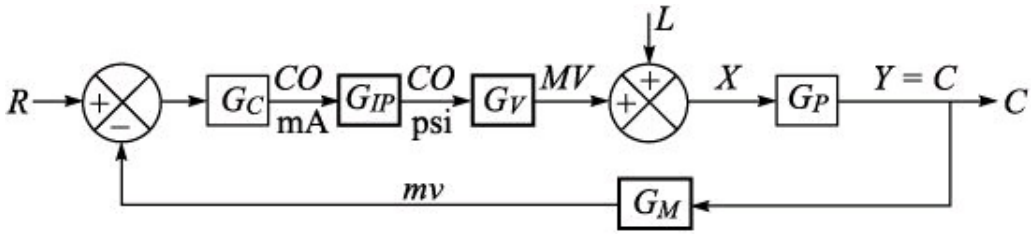


Figure 7.2 The G_{IP} block is required if Instruments in a loop use heterogeneous actuation.

The developers and manufacturers have designed these instruments in a way that their speed of response remains too fast compared to majority of industrial processes. Thus while considering the closed loop response, the instrument items may be assumed as pure gain elements, i.e.

$$G_M(s) = K_M;\ G_{IP}(s) = K_{IP};\ \text{and}\ G_V(s) = K_V.$$

However, before applying this simplification, the relative speed of response of the hardware and the process system should be carefully evaluated and compared. The idea should be clearer from the following two examples.

EXAMPLE 7.1 A temperature control system is employed to keep the tube outlet temperature of the heat exchanger at a desired value. There is an EPC in the loop because the sensor-transmitter and controller are electronic, but the FCE is a pneumatic control valve. The system has been illustrated in Figure 7.3.

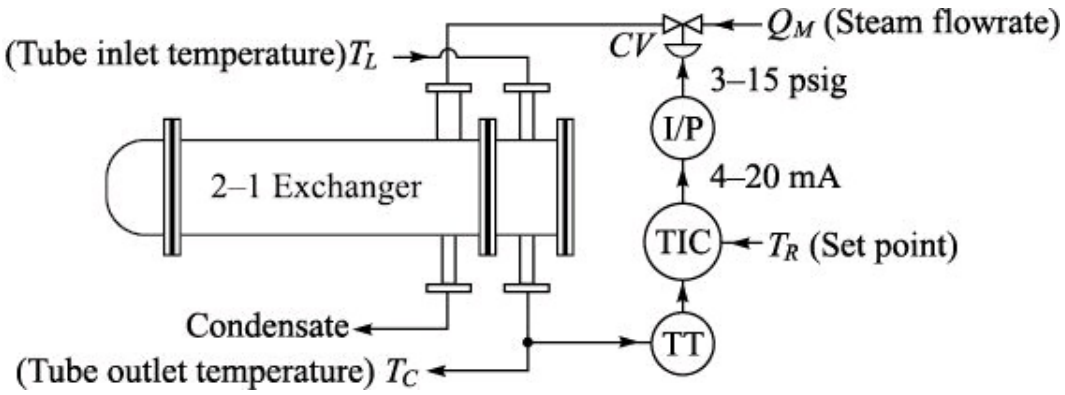


Figure 7.3 Temperature control system of a shell and tube heat exchanger.

Transfer functions of the process have been determined with respect to load, L and manipulated variable, MV, by step test method (see Chapter 12 for details):

$$G_{PL}(s) = \frac{Tc(s)}{T_L(s)} = \frac{0.43e^{-2.8\,\text{min}.S}}{12.5\,\text{min}\cdot s + 1};\ G_{PM}(s) = \frac{Tc(s)}{Q_M(s)} = \frac{0.3^{\circ}\text{C/lpm}\cdot e^{-2.1\,\text{min}.S}}{10.8\,\text{min}\cdot s + 1} \quad (7.1)$$

and the dynamics of the instruments may be expressed as:

For TT (RTD): $$G_M(s) = \frac{0.08\ \text{mA/°C}}{2.4\text{s}\cdot s + 1}; \tag{7.2}$$

For I/P: $$G_{IP}(s) = 0.75\ \text{psi/mA} \tag{7.3}$$

For *CV*(1-in two way equal %): $$G_V(s) = \frac{1.2\ \text{lpm/psi}}{4\text{s}\cdot s + 1} \tag{7.4}$$

That the dynamics of the instruments are much faster than the process, is confirmed by the time constants given in Equations 7.2, 7.3, 7.4 and Equation 7.1. Hence, we may assume the steady state gain of these instruments to represent their dynamics. Thus,

$$G_M(s) = 0.08\ \text{mA/°C};\ G_{IP}(s) = 0.75\ \text{psi/mA};\ \text{and}\ G_V(s) = 1.2\ \text{lpm/psi} \tag{7.5}$$

In Equation 7.3, the dynamics of EPC is expressed as a gain element before any simplification is done, because the hardware of this system involves mechanical lever and spring action with a very low mass and you shall see in a following section that justifiably, the dynamic model becomes a pure gain.

EXAMPLE 7.2 The next example system is about the temperature control of Continuous Sterilization Process of Molasses in a yeast production plant. The sterilization is done by injecting a medium pressure live steam through a control valve (1") into the molasses carrying pipe (4") at a point, and measuring the stream temperature by a temperature sensor transmitter installed at a distance downstream from the steam entry point. An electronic PID controller receives the temperature signal. The output is converted by an EPC and the pneumatic output goes to the control valve. The system has been illustrated in Figure 7.4.

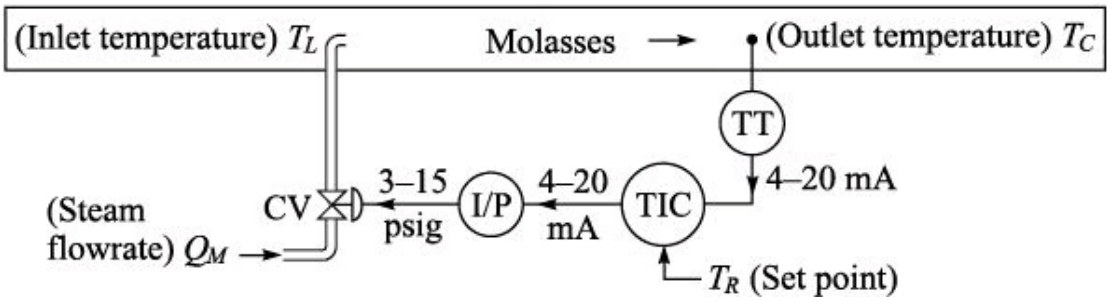


Figure 7.4 Temperature control of a continuous molasses sterilization system.

As in the previous example, the transfer functions of the process has been determined with respect to load *L* and manipulated variable, MV, by step test method:

$$G_{PL}(s) = \frac{Tc(s)}{T_L(s)} = \frac{0.89e^{-4.0\,\text{s}\cdot s}}{16.2\,\text{s}\cdot s + 1};\ G_{PM}(s) = \frac{Tc(s)}{Q_M(s)} = \frac{0.51°\text{C/lpm}\ e^{-3.2\,\text{s}\cdot s}}{12.3\,\text{s}\cdot s + 1} \tag{7.6}$$

If we assume that the same instruments of Example 7.1 have been used here, then except for the EPC, the time constants of Equation 7.6 become so close to those of the instruments, that they could not be considered as gain elements. Equations 7.2 and 7.4 should be used to describe their dynamics.

In this chapter we shall first consider the steady state input–output relation, then move on to develop the dynamic model of few representative examples of each class of hardware. These instruments often contain mechanical elements like spring, diaphragm, bellow, linkage, lever, gear, etc. coupled with primary sensors like mechanical elastic elements, electrical resistors, capacitors, optical devices, etc.

In a typical system the force is transmitted by the displacement of compressible or incompressible fluid, or rigid mechanical elements. This is often opposed by fluid frictional drag or sliding or bearing friction-drag force plus force due to deformation of any elastic element. After applying a finite input force to the system that is at steady state, we obtain a finite output displacement when the next steady state is attained. Hence, during the transient period, the system should have been moving with acceleration (may be of a –ve value). If we perform a force balance, just as we did for the spring-dashpot system in Chapter 6, between the input and the two opposing components (drag and deformation), the residue will be equal to acceleration times mass of the system. This equality is in fact the Newton's law [P (force) = m (mass) × f (acceleration)]. Thus the dynamic model involves the inductance element (rate of change of momentum term), Capacitance element (force of elastic deformation), and Resistance element (frictional or drag force).

The art of model development is not only to identify the dynamic elements in a system, but more important is to judge the significance or contribution of a particular element in the of system response. Such a consideration would justify whether an element should be included or excluded in the model.

7.1 The Input/Output Signal Level of Pneumatic and Electronic Instruments

Pneumatic instrumentation has evolved during the last sixty to seventy years. Even after so much progress and popularity of electronic and digital instruments, pneumatic systems are still in use, specifically in the front end of an automation system due to their inherent safety features, robustness and reliability. Pneumatic instruments operate within an input-output variable range of 3–15 psig air pressure. A pneumatic mains, capable of supplying dust free, oil free, dry air at 20–25 psig, is required as a utility service to the instrument system, and normally it should have a supply capacity of 1 Nm^3/h of air per instrument module.

Electronic instrument's manufacturers used different variable range of operation,

e.g. 1–5 V, 0–20 mA, 4–20 mA, etc. Although, different I/O ranges are still specified for certain instrument modules, but in recent times the range of 4–20 mA DC has been accepted universally. The output load capability of each of such unit is normally 500 □ (maximum), but in some critical application with larger 'fan-out requirement this may extend up to 650 □.

7.2 Measuring Elements

A measuring element supplies information regarding the state of the process to the controller so that it can act properly for up-keeping of the process. That is, it carries out the vital responsibility of communicating the process (controlled) variable value to the controller. In a feedback loop, this communication of information constitutes the feedback path. A measuring element essentially has two parts:

1. Process variable sensor, and
2. Transmitter.

Sensor or **Primary Detector** is a device that first converts the process variable into a suitable form of physical variable that would be easy to handle for further processing. These variables may be mechanical deformation of elastic elements, volume or pressure change of a fluid, change in electrical and magnetic properties like resistance, capacitance, reluctance, change of acoustic and certain optical properties. Also, electro-chemical properties like half-cell potential measurement have been used in sensors.

A sensor is rarely used without its associated transmitter, except only as a field mounted indicator at a location near the plant. Even in such applications, the system contains a secondary transducer to convert the process variable value into a readable format. For example, a simple Hg-in-steel thermometer may have an elastic element attached through a capillary to the mercury bulb. The elastic element may be a C-type bourdon element. Any pressure variation in the Hg bulb due to a change of process temperature would produce a deformation of the elastic element that in turn displays an angular change of the indication needle position on the circular dial calibrated in temperature unit.

The **Transmitter** is a system that converts the sensor output into a suitable form of physical variable, electrical or pneumatic, sufficiently strong in power so that its quality does not deteriorate even when the signal has to travel a long distance (say few hundred meters). Though the majority of sensors bear a non-linear relationship with the measured variable, when coupled to the associated transmitter the overall characteristic between measured variable and transmitter output, become linear due to

the presence of a linearization procedure (hardware based) inside the transmitter. Generally, the linearization hardware is specialized for a particular type of variable conversion. However, the magnitude of non-linearity varies for different sensors. The RTD type temperature sensor has the characteristic described by the temperature versus electrical resistance of a metal:

$$R_T = R_O(1 + \alpha T + \beta T^2 + \gamma T^3 + \cdots) \qquad (7.7)$$

The influence of quadratic and higher order terms in Equation 7.7 are so small that an RTD is considered to have a linear character within a temperature span of 100ºC to 150ºC, and for higher ranges, the compensation is brought about by using the temperature dependence of biasing resistors in the transmitter circuit (metal oxide film resistors have a +ve and the carbon resistors have a –ve coefficient).

However, for sensors having a pronounced non-linear character, specially designed linear compensators are used in their transmitter system. Two examples of this class are:

(1) **Flow measurement by Area Flow Meters,** e.g. Orifice, or Venturi. These sensors possess an input-output relation given by:

$$Q_O = C_D\sqrt{\Delta P} \qquad (7.8)$$

where Q_O = Average volumetric flow rate, and ΔP = Differential pressure across Orifice or Venturi, C_D = Discharge coefficient. Hence, the differential pressure transmitter (DPT) should have an in-built linear compensator for the above square root type character. Pneumatic and electronic DPT with square root extractor has been developed and are commercially available since mid-1960s. An interesting configuration of such a transmitter has been described in Example 7.5. There, a specially designed resistance type pressure transmitter performs the square root extraction.

(2) **In the pH measurement by Glass Electrode,** the dependence of the developed emf (E) in the electrode with the concentration of hydrogen ions follows the Nernst Equation:

$$E = E^O - 0.0591 \cdot \log H^+ \qquad (7.9)$$

where E = developed potential in the glass electrode, E^O = the potential at zero pH, and H^+ is the hydrogen ion concentration in the solution of unknown pH. Equation 7.9 contains a power law type non-linear character. But by using the definition of pH,

$$pH = -\log H^+ \qquad (7.10)$$

The non-linearity may be removed in the sensor-transmitter by putting Equation 7.10

into 7.9.

We may conclude that, generally, the steady state sensor-transmitter character is linear. The input-output relation of a pneumatic and an electronic measuring system may be considered for a level sensor-transmitter that is installed for a water reservoir in which the level is expected to vary between 0 and 3000 mm. The output signal with respect to the input variable water level has been illustrated in Figure 7.5.

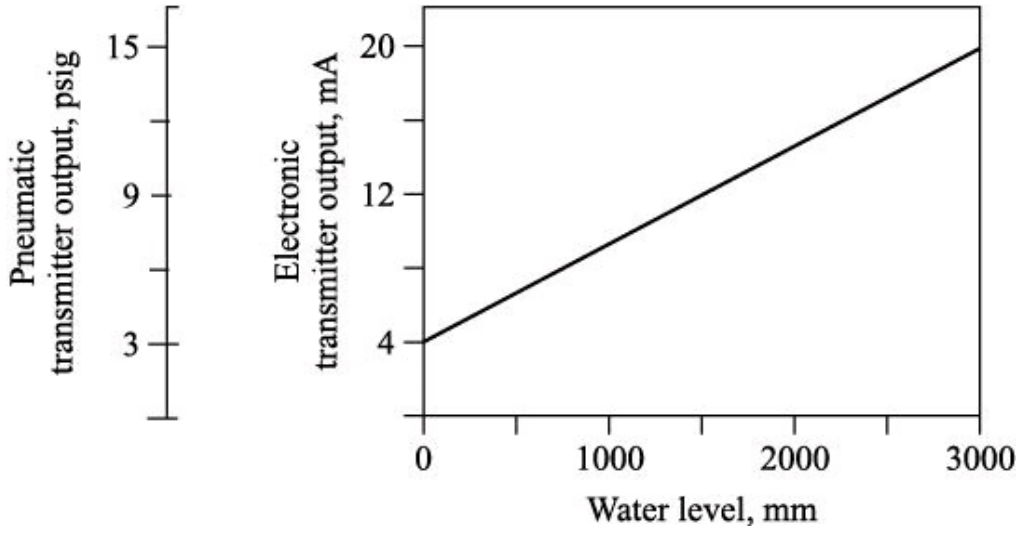


Figure 7.5 Linear character of a level sensor with pneumatic or electronic transmitter.

For any given level, L mm, the transmitter output may be found by the $Y = mX + c$ type characteristic equation as:

The pneumatic transmitter output: $P = \dfrac{(15-3)\,\text{psig}}{(3000-0)\,\text{mm}} L + 3\ \text{psig}$, and

The electronic transmitter output: $I = \dfrac{(20-4)\,\text{mA}}{(3000-0)\,\text{mm}} L + 4\ \text{mA}$

7.2.1 Instrument Terminology and Performance Measures

In this section, certain terms used frequently in practice of instruments have been defined. They are used to express the technical specification and performance measures of an instrument and some of them have an overlapping usage.

(1) **Range of an Instrument:** It is stated by mentioning the minimum and maximum value of the variable that the instrument can measure, receive, or transmit.

Example A: 0 to 150 psi,

Example B: –50 to 600ºC,

Example C: 0.5 to 10.5 gm-mole/lit.

The maximum and minimum variable values are also called the upper and lower range values.

(2) **Elevated Zero Range:** An instrument range in which the zero value of the

measured variable is greater than its lower range value, as in the above Example (B).

(3) **Suppressed Zero Range:** An instrument range in which the zero value of the measured variable is less than its lower range value, as in the above Example (C).

(4) **Span of an Instrument:** It is the algebraic difference between the upper and lower range values. Example:

A. Range: 0 to 150 psi, Span: 150 psi,
B. Range: –50 to 600ºC, Span: 650ºC,
C. Range: 0.5 to 10.5 gm-mole/lit, Span: 10.0 gm-mole/lit.

(5) **Error/Measurement Error:** It is the algebraic difference between the indication and the true value of the measured signal. It is the quantity which when algebraically subtracted from the indication, gives the true value, i.e. Error = Indicated Value – True value.

(6) **Accuracy:** It is the degree of conformity of an indicated value by the instrument to the accepted standard value or ideal value.

(7) **Accuracy Rating:** A number or quantity that defines a limit that the errors will not exceed when an instrument is used under specified operating condition. Accuracy rating is actually an expression of maximum possible errors that may appear in the instrument output and may be expressed in the following two ways:

A. In terms of the measured variable. A typical expression of such usage would be ±1ºC or, ±2ºF.
B. In terms of percent full scale. A typical expression of such usage would be ±0.5%.

(8) **Resolution:** The least interval between two closely adjacent input values that the instrument can distinguish. Alternatively, it is the variable value of the smallest division of instrument output scale.

(9) **Calibration:** It is to record output of an instrument corresponding to a series of values of the input variable, which the instrument is to measure, indicate, or transmit. Data so obtained are used to:

A. Determine the location at which the scale graduations are to be placed.
B. Adjust the output, to bring it to the desired value, within a specified tolerance.
C. Ascertain the error by comparing the instrument output reading against a standard.

(10) **Calibration Curve:** A tabular or graphical representation by drawing a smooth curve through the data points that expresses the relationship of the instrument over its range, as compared against a standard.

(11) **Calibration of an Instrument:** In the commercially available instruments there

are always adjustment facilities for 'Zero' and 'Span' settings. Two standard sources of measured variable, one near the minimum (LO) and the other, near the maximum (HI) of the instrument range, are set up. Then the following procedure is adopted:

A. The sensor is brought in equilibrium with the LO source, and the output reading is adjusted by manipulating the Zero-set.

B. The sensor is brought in equilibrium with the HI source, and the output reading is adjusted by manipulating the Span-set.

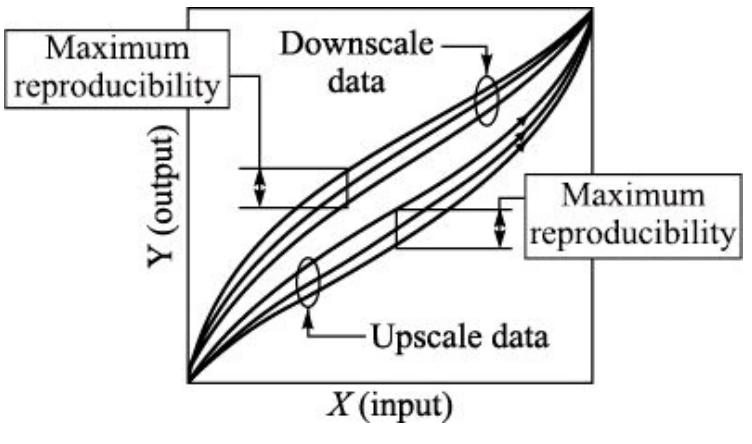
C. Steps A, and B are repeated until the deviation at the two extreme points of output scale become zero or very small. Usually a straight line between these two extreme points is assumed to represent the input-output relation.

(12) **Repeatability:** It is defined as the closeness of agreement among a number of consecutive measurements of the output for the same value of input under the same operating conditions while the observations are made during either **Up-scale** or **Down-scale** traverse. The data is obtained for the full range of the instrument. The repeatability is reported in percent of the instrument span.

(13) **Reliability:** This is defined as the probability that an instrument's output quality will remain within its reported performance measures, for the specified period of time, under specified operating conditions.

(14) **Gain (Steady State) of an Instrument:** Its value is the ratio of total output change after a step change in input, to the input step magnitude.

(15) **Gain (Dynamic) of an Instrument:** For a sinusoidal input to an instrument, it is the ratio of output to input magnitude, when the stable oscillatory part of response appears, after the transient part has decayed.



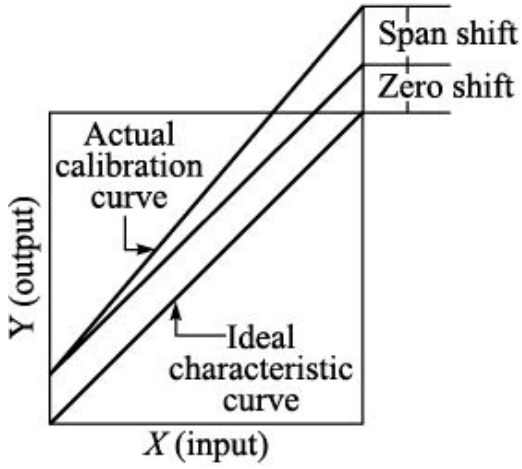


Figure 7.6 Determination of reproducibility **Figure 7.7** Zero and span shift.

(16) **Drift:** An undesirable change in the input-output relation of an instrument over a specified period of time.

(17) **Point Drift:** The change in output over a specified period of time for a constant input under specified (reference) operating conditions.

(18) **Shift:** A change in the input-output relation of an instrument that may come during a long usage of the device.

(19) **Span Shift:** Any change of slope of the input-output curve.

(20) **Zero Shift:** Any parallel shift of the input-output curve.

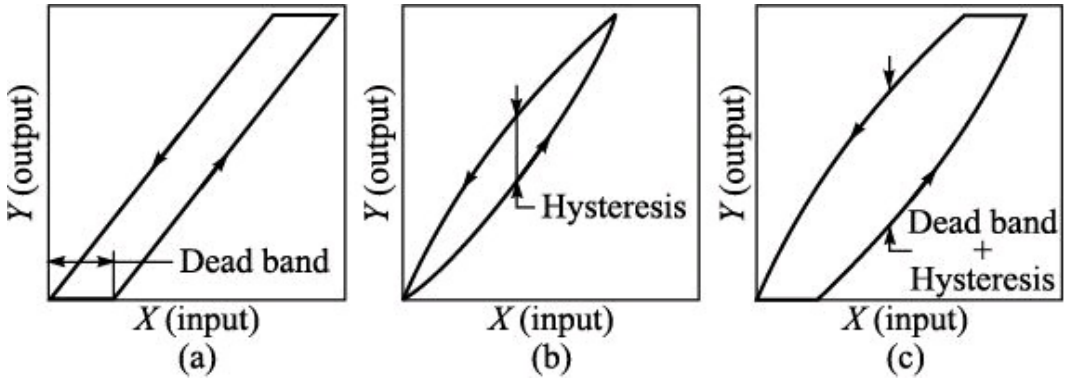


Figure 7.8 Occurrence of dead band and hysteresis in an instrument characteristic.

(21) **Dead Band:** The range of an instrument through which the input can be varied without producing any observable response. Dead band is expressed in percent of span.

(22) **Hysteresis:** The property of a measuring device evidenced by the dependence of the output value, for a given excursion of input, upon the history of prior excursions and the direction of the current traverse. It is usually determined by subtracting the value of dead band from the maximum measured separation between upscale going and downscale coming locus of the measured variable during a full-scale traverse after transients have decayed.

(23) **Hunting:** An undesirable oscillation of appreciable amplitude of measured variable that is prolonged after the external stimuli (disturbance) has disappeared.

(24) **Sensitivity:** This term expresses two slightly different definitions/meanings:

A. It is the minimum value of input variable required to produce the smallest detectable change of instrument output.

B. The ratio of output by input span of the instrument. Obviously, this definition of sensitivity is equivalent to the gain.

(25) **Linearity:** The closeness to which a curve approximates a straight line. The more practical definition of linearity in instrumentation practice may be stated as: The maximum deviation of the calibration curve from a straight line drawn in such a way that the deviation becomes minimum.

Or, if a straight line is drawn through the calibration curve in such a way that the deviation between them becomes minimum, then the maximum deviation at any point within the instrument span, expressed as %-full scale (span), is defined as the %-linearity.

(26) **Response Time:** After a step change in the input, it is the time required by the output or response to come within a specified band around the ultimate response value. For an oscillatory (under-damped) response, this band is specified as ± D% around the ultimate response value of 100%. For an over-damped response it is usually expressed as – D%. The value of D may be taken as 5, 2, or 1, depending on usage and practice in a particular application area. Typically for a first order type response we may use D = 1%, i.e. response time will be the time required for 99% step recovery, hence response time ≈ 5t (t = time constant of the system).

EXAMPLE 7.3 A transducer measures flow linearly with a gain of 33 m V/(l/min) and has a first order time constant =0.5 min. Find the transducer output 15 s after the input changes from 20 cc/min to 41 cc/min. Calculate the flow rate this signal represents.

Solution: The transfer function of the transducer may be written as:

$$G_M(s) = \frac{K_M}{\tau s + 1}$$

where $K_M = 33\ \dfrac{\text{mv}}{\text{lpm}}$ and $\tau = 30$ sec

$$\text{Input} = \text{step of } 11 \times 10^{-3} \text{ l pm.}$$

Hence, the response after 15 s: $Y|_{15S} = y + MK_M[1 - e^{-t/t}]$

$$= 20 \times 10^{-3} \times 33 + 21 \times 10^{-3} \times 33[1 - e^{-0.5}] = 0.66 + 0.2727 = 0.9327 \text{ mv.}$$

EXAMPLE 7.4 An instrument, after some time in service, was recalibrated. It was found that a zero shift of +2.5% (full scale), and a span shift expressed as –1.8% (full scale) has occurred in its characteristic. If it is a temperature measuring system with a range of 25 to 200ºC, calculate the true temperature values corresponding to the following indicated values: (a) 45ºC, 87ºC, and 190ºC.

Solution: Using the zero shift value, corresponding to a true temperature of 25ºC,

The instrument reading will be = 25 + 0.025 × 175 = 25 + 4.375 = 29.375ºC.

Using the zero and span shift value, corresponding to a true temperature of 200ºC,

The instrument reading will be = 200 + 0.025 × 175 – 0.018 × 175 = 201.225ºC.

Let the faulty characteristic line have an equation of the type: $y = mx + c$. To use this equation with the variables in ºC unit, it must be written as:

$$y = m(x - 25) + c$$

Slope of the line, $m = \dfrac{201.225 - 29.375}{200 - 25} = 0.982$; and, c = 25 + 4.375 = 29.375°C

Rearranging the above equation to find x for a given y, $x = (y - 4.825)/0.982$,

(a) For an indicated temperature, $y = 45°C$; the true temperature $x = (45 - 4.825)/0.982 = 40.911°C$.

(b) For $y = 87°C$, $x = (87 - 4.825)/0.982 = 83.68°C$.

(c) For $y = 190°C$, $x = (190 - 4.825)/0.982 = 188.57°C$.

7.2.2 The Dynamic Character of Measuring Element

We would again like to point out that the transmitters of both pneumatic and electronic type possess very fast dynamic response owing to their design features. The dynamics of a measuring element is solely due to its sensor parts. In this section, we shall discuss the dynamic responses of the more important types of process variable sensors.

The Temperature Sensors

Generally, the **Thermocouple, RTD, Filled System and Bimetal temperature sensors** are installed with a protective casing or Thermo-well. According to the analysis presented in Example 6.1, these systems also feature a couple $R-C$ pair in their model that would show an over-damped second order dynamics due to the presence of two first order elements in interactive combination, as given in Figure 7.9(b). In some special applications (Hg in glass thermometer), they may be used in bare form and we obtain a first order dynamic model, as shown in Figure 7.9(a).

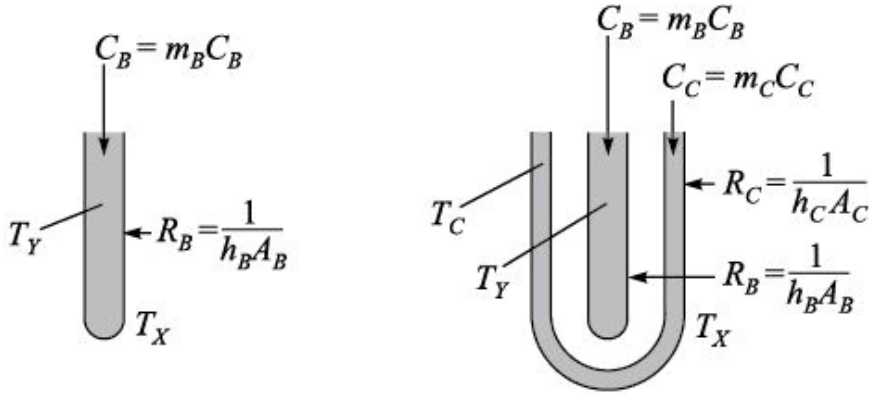


Figure 7.9 The location of dynamic elements in temperature sensor configurations. (a) Bare sensor. (b) Sensor within a thermowell.

From our former knowledge of model building, we can directly write the dynamics as transfer functions for these sensor configurations as:

(a) For bare sensor: $$\frac{T_Y(s)}{T_X(s)} = \frac{1}{\frac{m_B C_B}{h_B A_B} s + 1} \qquad (7.11)$$

(b) For sensor within thermowell:

$$\frac{T_Y(s)}{T_X(s)} = \frac{1}{\frac{m_B C_B}{h_B A_B} \cdot \frac{m_B C_B}{h_B A_B} s^2 + \left(\frac{m_B C_B}{h_B A_B} + \frac{m_C C_C}{h_C A_C} + \frac{m_B C_B}{h_C A_C} \right) + 1} \qquad (7.12)$$

Radiation Pyrometers use a primary detector called '**thermopile**' that is a series of combinations of a number of thermocouples with their hot junctions located at a point. The emitted radiation energy from the hot body (process) is concentrated by a lens system to fall on that point. The output of this instrument is thermo-electric emf generated by the thermopile that is in fact the emf of each thermocouple multiplied by their number, just like the output voltage of a battery comprising of a number of unit cells in series combination. Though the model for a thermopile may be described by a Resistance Capacitance combination as in case of a thermocouple, but the heat gain expression would differ due to the fact that heat transfer in this case is occurring through radiation instead of conduction/convection as in case of Equation 7.11. We derived the linearized TF as Equation 4.65 for the radiation heat transfer to a thermocouple placed inside a furnace in Example 4.12. This TF has a unity gain, which means that after a disturbance at steady state, the furnace and thermocouple junction temperatures will become equal ($T_{OS} = T_{IS}$).

In the present case, only a fraction of total energy enters into the thermopile because only the radiation part of the total energy is allowed as the input signal, and again a part of this energy is absorbed in the lens and other associated hardware in the instrument. This means that after a disturbance at steady state, thermocouple temperature will be less than thermocouple junction temperature ($T_{OS} < T_{IS}$). These considerations may be recognized by introducing a gain term K_M in the TF of Equation 4.65. K_M is, in fact, an attenuator because its value is always less than 1.0. (*Just like the reaction disappearance term in the CSTR model of Chapter 4 that leads to a less than 1.0 gain in the TF.*) Note that there will be no appreciable change of the time constant, t^*, the linearized time constant calculated in Example 4.12, because in this new description of the system, the values of resistance and capacitance elements practically remain unchanged. The resulting TF may be given as:

$$\frac{T_O(s)}{T_I(s)} = \frac{K_M}{\tau^* s + 1} \qquad (7.13)$$

where $\tau^* = \frac{mC_P}{KA\sigma\varepsilon 4T_{IS}^3}$; and, K_M = gain < 1.0

For the significance of the variable and parameter names please see Example 4.12.

In an **Optical Pyrometer,** the incident intensity of the visible part of the total emitted radiation from the hot body (process) is used as the input variable. In older designs of this instrument, the incident intensity was compared with the brightness of a current carrying Platinum filament. By manually adjusting the impressed voltage on the filament, its brightness is made equal to that of the hot body, and its temperature directly reads from the scale calibrated on the voltage adjustment knob. This configuration of the instrument is not a continuous monitor; hence building a dynamic model does not arise.

A modern version of an **Optical Pyrometer** uses an electronic optical sensor that measures the visible part of the radiation energy and sends the temperature information as a continuous electrical signal. The speed of response of such an instrument is extremely fast, hence may be considered as a pure gain element.

Pressure Sensors

The hydrostatic force balance type instruments or **Manometers,** have a number of configurations that are successfully used as pressure and differential pressure sensors in process control applications. As we shall see in the following example, due to existence of capacitance, resistance, and inductance elements, a second order dynamic model results for such systems.

EXAMPLE 7.5 A class of low range absolute or differential pressure sensor uses mercury manometer as the primary detector. A number of thin metal rods are placed in one the manometer tubes and the lower end of these rods are at progressively higher heights from the bottom of the tube. This tube is used as the measuring arm and is connected to the process for pressure or level measurement. The remaining arm of the U may remain open to atmosphere. For differential pressure monitoring as required for an Area flow meter, the measuring arm and the other tube are connected respectively to the higher and lower pressure taps of the flow sensor. The upper ends of these thin rods are joined by electrical resistance elements in series combination. Initially for zero pressure difference, all the metal rods remain dipped into or are in contact with mercury. In Figure 7.10, the schematic of such an instrument has been illustrated.

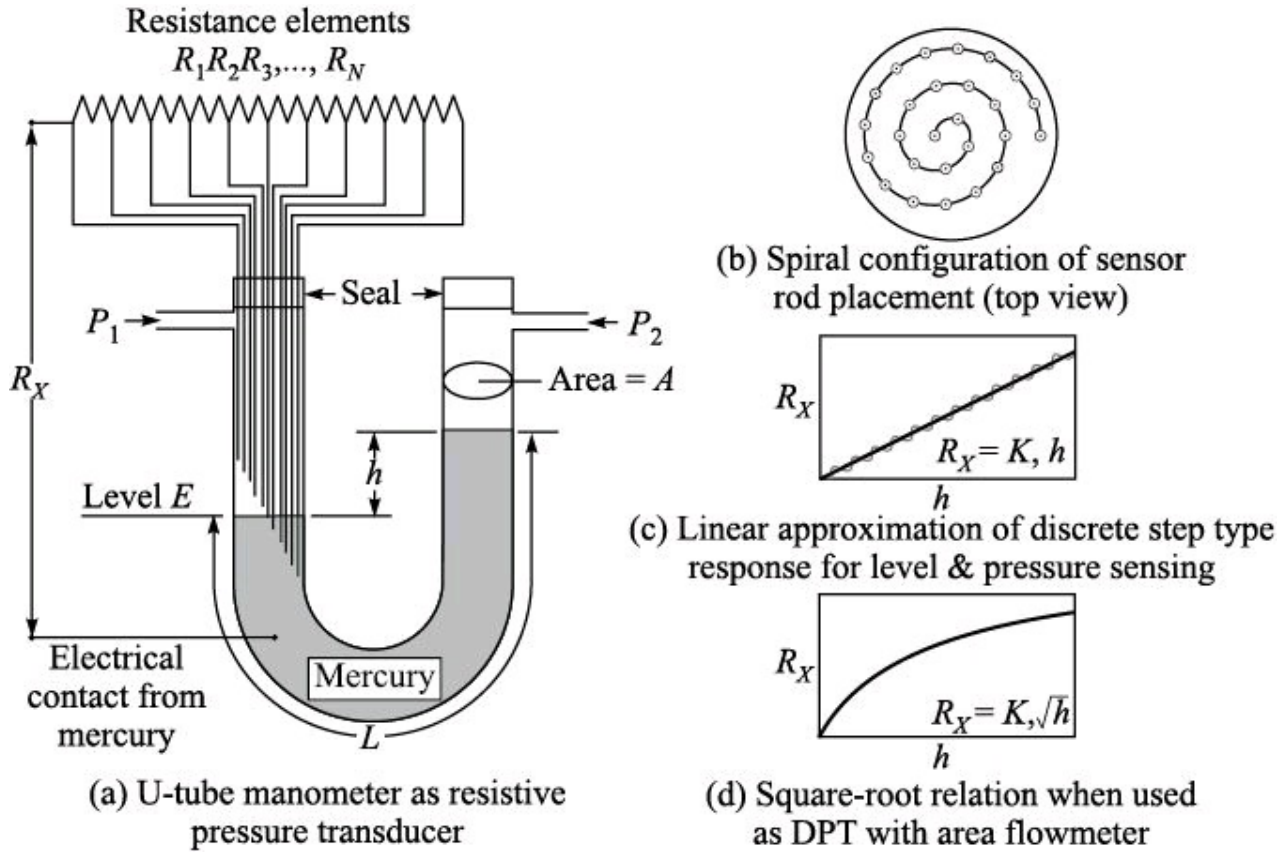


Figure 7.10 U-tube mercury manometer as absolute or differential pressure sensor and resistive transducer.

For any of the pressure difference between two tubes of the manometer, the mercury level goes down in the tube containing the metal rods, and gradually more metal rods loose contact with the mercury, and the resistance between the mercury contact and resistance, R_X, increases. Thus resistance, R_X, the instrument output for a given input pressure or pressure difference that is measured by a Wheatstone's bridge circuit, will be proportional to the applied pressure difference. Certain manufacturers design resistance element in such a way that when the instrument is used with an area flowmeter, the output resistance, R_X, bears a square root relation with input, ΔP. Thus, the required linearization due to the flow-head relation [flow rate $\propto \sqrt{(\Delta P)}$], takes place within the instrument.

Figure 7.10(b) shows that the metal rods are arranged in a spiral configuration to provide more number of rods in the measuring arm so that the steps in the discrete response of R_X as ΔP change become smaller and the characteristic come closer to the continuous smooth line of Figure 7.10(c). For the following analysis, we shall use this continuous linearity shown in Figure 7.10(c), and the square root type continuous nonlinear function in Figure 7.10(d).

Initially $P_1 = P_2$, there is no level difference in the two tubes as $\Delta P = 0$. Now P_1 has become greater than P_2 and the mercury slug is moving. At a certain time in the measuring arm, the mercury level is at Level E, and we make a force balance at this level.

Force in the L.H. tube at level E = Force in the R.H. tube at level E (7.14)

In Equation 7.14, the right hand side contains more than one force components.

Force due to pressure P_1	=	Force due to pressure P_2	+	Force due to Hydrostatic head of Mercury above E	+	Frictional drag force due to moving mercury slug

(7.15)

If any residual force remains after performing the above balance of Equation 7.15, it will be accounted by the rate of change of momentum of the moving mass of mercury. That is, by applying Newton's law, Equation 7.15 takes the form:

Force due to pressure P_1 I	–	Force due to pressure P_2 II	–	Hydrostatic force of the mercury column above E III	–	Frictional drag due to moving mercury slug IV	=	Mass of mercury V	×	Acceleration

(7.16)

Let us put the variable names and system parameters into each term of Equation 7.16:

Term I – Term II = $A(P_1 - P_2)$; Term III = $\rho \frac{g}{g_C} Ah$; Term V = $\frac{\rho AL}{g_C} \cdot \frac{d^2h}{dt^2}$

where,

A = Internal cross-sectional area of the manometer tube,

$\Delta P = P_1 - P_2$ = Input pressure differential,

h = Output mercury level difference in the two manometer tubes,

r = Density of mercury

g and g_C have their usual significance.

To express term IV, we assume the movement of mercury slug occurs under laminar condition, hence we may apply Poiseuille's equation to express the frictional drag force, thus

$$q = A\frac{dh}{dt} = \frac{\pi \Delta P_F r^4}{8\mu L} \qquad (7.17)$$

where,

ΔP_F = The pressure difference causing the mercury flow;

μ = Mercury viscosity;

r = Radius of the manometer tube;

L = Length of the mercury slug;

q = Volumetric flow rate of mercury.

We may also define the frictional drag force, F_D, as,

$$F_D = \frac{\Delta P_F \pi r^2}{g_C} \qquad (7.18)$$

By putting the value of ΔP_F from Equation 7.17 into 7.18, the expression of F_D becomes:

$$\text{Term IV} = F_D = \frac{8\mu LA}{r^2 g_C} \cdot \frac{dh}{dt} \qquad (7.19)$$

Now we may put expressions for all the terms of Equation 7.16 to obtain:

$$A\Delta P - \rho Ah\frac{g}{g_C} - \frac{8\mu LA}{r^2 g_C} \cdot \frac{dh}{dt} = \frac{\rho AL}{g_C} \cdot \frac{d^2h}{dt^2} \qquad (7.20)$$

To present this second order differential equation in standard form, we rearrange and divide each side by $[(rgA)/g_C]$;

$$\frac{L}{g} \cdot \frac{d^2h}{dt^2} + \frac{8\mu L}{\rho r^2 g} \cdot \frac{dh}{dt} + h = \frac{g_C}{\rho g}\Delta P \qquad (7.21)$$

Assuming zero initial values of all variables and their derivatives, Equation 7.21 may be Laplace transformed to obtain the transfer function:

$$\frac{H(s)}{\Delta P(s)} = \frac{K_{P'}}{\tau^2 s^2 + 2\tau\zeta s + 1} \qquad (7.22)$$

where $\tau = \sqrt{\frac{L}{g}}$; and, $\zeta = \frac{4\mu}{\rho r^2} \cdot \sqrt{\frac{L}{g}}$; and $K_{P'} = \frac{g_C}{\rho g}$

But we want to have the final transfer function of the instrument as $[R_X(s)/\Delta P(s)]$.

In applications like pressure or level monitoring, a linear relation is designed between R_X and h, that is $R_X = Kh$, and we have $R_X(s)/H(s) = K$, thus,

$$\frac{R_X(s)}{\Delta P(s)} = \frac{R_X(s)}{H(s)} \cdot \frac{H(s)}{\Delta P(s)} = \frac{K_P}{\tau^2 s^2 + 2\tau\zeta s + 1} \qquad (7.23)$$

where $K_P = KK_{P\Delta}$.

In applications where the instrument is used as a DPT in conjunction with an area flow-meter, a square root type non-linearity is designed in the h vs. R_X relation. Such a relation, e.g. $R_X = K\sqrt{h}$, would linearise the inherent flow-head non-linearity ($q = C_D\sqrt{h}$) associated with these instruments. We may obtain the linearised transfer function of $R_X(s)/H(s)$, as

$$R_X = R_X(s) + \left[\frac{dR_X}{dh}\right]_{h=hs}(h - h_S) \text{ or } \frac{R_X(s)}{H(s)} = \frac{K}{2\sqrt{h_S}} \qquad (7.24)$$

Similarly, the final transfer function may be obtained as:

$$\frac{R_X(s)}{\Delta P(s)} = \frac{R_X(s)}{H(s)} \cdot \frac{H(s)}{\Delta P(s)} = \frac{K_P}{\tau^2 s^2 + 2\tau\zeta s + 1} \qquad (7.25)$$

where, $K_P = \dfrac{KK_{P'}}{2\sqrt{h_S}}$

The **Elastic Elements** like **Bourdon tube, Diaphragm, Capsule,** and **Bellows,** are used as pressure sensors. They contain *capacitance* due to the elastic deformation of the element, *resistance* arising out of the frictional drag experienced by the flowing fluid in or out of the element through the connecting pipe due to any pressure variation on the process side, and *inductance* or inertia associated with the mass of the moving fluid and hardware. But all these elements do not have equal share in dictating the system dynamics. In many of the systems, the mass has a relatively small contribution in the force balance equation compared to frictional drag and elastic deformation, hence, the inductance element may be neglected. We have presented the approach in the following Example 7.6.

EXAMPLE 7.6 In Figure 7.11(a), a bellow type pressure sensor and the spring displacement as the output indication has been presented:

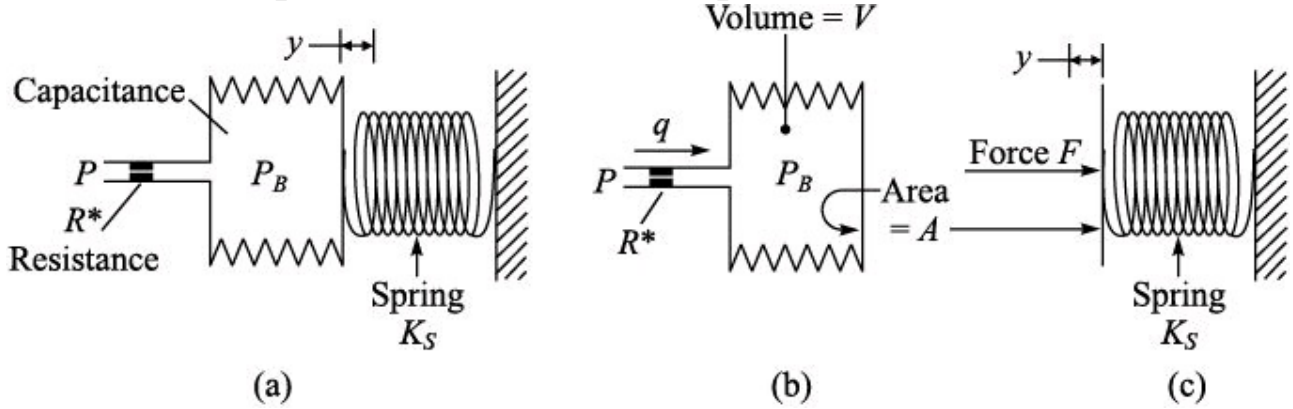


Figure 7.11 Bellow-spring assembly for pressure sensing and indication.

The above arrangement is a simplified representation of a pressure sensor indicator in which the unknown pressure causes an elastic deformation of the bellow that is balanced with the counteracting force of the compressed spring. The resulting bellow deformation may be displayed on an edgewise or circular scale by a suitable mechanical arrangement. We proceed for the analysis assuming that the total mass of the moving parts is negligible. We also assume that the gas density remains constant for any change in pressure.

From Figure 7.11(b), and (c)

$$P_B A = F = K_S y \text{ or } P_B = y\frac{K_S}{A}$$

This linear relation may be differentiated to obtain:

$$\frac{dP_B}{dt} = \frac{K_S}{A} \cdot \frac{dy}{dt} \qquad (32)$$

Volume at any instant, $V = Vs + (y - y_S)A$; The suffix s is for the steady state condition.

Mass of gas inside the bellow $m = nM = \frac{MP_B V}{RT}$

where, M = molecular wt. of the gas, R = Universal gas constant, T = Ambient temp. ºK.

Due to a pressure change from steady state, change in mass within the bellow is dm in time dt, thus,

$$\frac{dm}{dt} = \frac{M}{RT}\left(Pb_S \frac{dV}{dt} + V_S \frac{dP_P}{dt}\right) = \frac{M}{RT}\left(Pb_S A + \frac{V_S K_S}{A}\right)\frac{dy}{dt} \qquad (7.26)$$

With these preliminaries, we proceed for the dynamic mass balance.

$$\rho q - 0 = \frac{dm}{dt} \qquad (7.27)$$

The gas flow rate, q, through linear resistance, R^*, due a pressure difference, $P - P_B$, may be given by,

$$q = \frac{P - P_B}{R^*} \qquad (7.28)$$

Putting Equations 7.26 and 7.28 in 7.27, we may obtain,

$$\frac{P - P_B}{R^*} = \frac{M}{\rho RT}\left(Pb_S A + \frac{V_S K_S}{A}\right)\frac{dy}{dt}$$

or

$$P = \frac{K_S}{A} y + \frac{MR^*}{\rho RT}\left(Pb_S A + \frac{V_S K_S}{A}\right)\frac{dy}{dt} \qquad (7.29)$$

Assuming zero initial values of all variables, Equation 7.29 may be Laplace transformed to obtain:

$$\frac{Y(s)}{P(s)} = \frac{K_M}{\tau_M s + 1} \qquad (7.30)$$

Thus the instrument has a first order dynamics, with the following parameters:

The instrument gain, $K_M = \frac{A}{K_S}$, and the time constant, $\tau_M = \frac{AMR^*}{K_S \rho RT}\left(Pb_S A + \frac{V_S K_S}{A}\right)$

By considering a finite value of mass of the gas moving inside the bellow and the mass of the deforming bellow and spring, we could get a second order transfer function, but for the present system and certain other classes of instrument, the effective mass is negligibly small, and hence the effect of inertia or rate of change of momentum may be discarded from the analysis.

A variety of electrical transducers are used as secondary device with such sensors. These may be **Resistive (Strain Gage), Capacitive, Inductive (LVDT), Piezo-Electric,** and **Magneto-Strictive**. These transducers feature very fast response speed; hence, the dynamics of any combination of the elastic deformation type sensor, with any of these secondary transducers, is determined only by the sensor itself.

But if an auxiliary liquid (sealing fluid), separated by a secondary elastic diaphragm, is used in the instrument, to convey the pressure to the primary measuring elastic element, the mass of the moving fluid becomes significant. An example of this kind is the **Capacitive Type Differential Pressure** sensor used with area type flow meters. We present the dynamic model of this instrument in the next example.

EXAMPLE 7.7 Process fluid is isolated from the sensor interior by the two outer or isolation diaphragms. The differential pressure signal from the process is communicated by these diaphragms to a sealing liquid contained in the measuring chamber. The sealing liquid (e.g. Silicone oil) moves through the capillaries on each side of the diaphragm to subject it to differential pressure. The measuring chamber consists of two curved dielectric surfaces and the measuring diaphragm. The capacitances C_1 and C_2 are measured between the electrical leads connected to the diaphragm and each dielectric plate. Thus, any pressure difference between two ends will produce a deformation of measuring diaphragm, resulting in a difference of capacitance measured between the electrical leads due to the resulting proximity difference of the measuring diaphragm from the two dielectric plates.

Solution: From Figure 7.12, we may obtain the following parameters of the system, Length of each capillary = L; Cross-sectional area of the capillary = A_C; Effective area of the measuring diaphragm = A_D.

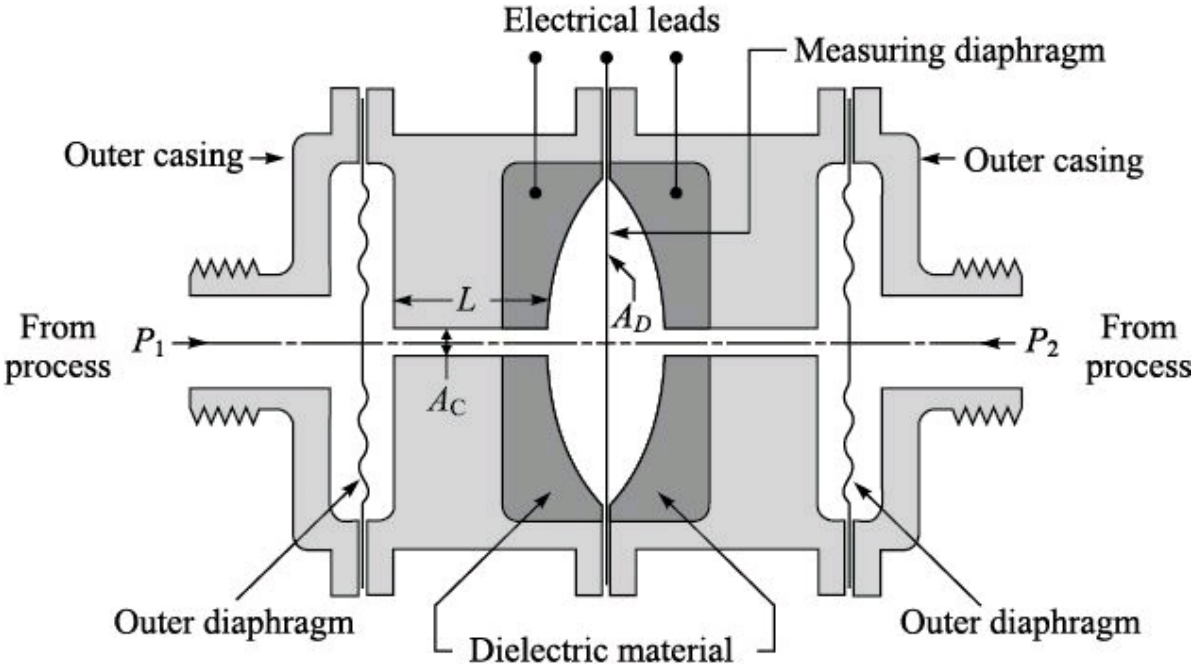


Figure 7.12 The schematic of a capacitive differential pressure sensor-transducer.

As we had done for the mercury manometer, we shall perform a force balance around the measuring diaphragm. The different force terms that will be appear in the balance are:

(1) The input force on the measuring diaphragm due the difference between P_1 and P_2:

$$P_1 - P_2 = \Delta P \qquad (7.31)$$

(2) The elastic force due to diaphragm deformation, y, acting opposite to the input:

$$-K_S y \qquad (7.32)$$

(3) The drag force due to sealing liquid velocity through the capillaries:

$$-\mu * \frac{dy}{dt} \cdot \frac{A_D}{A_C} \qquad (7.33)$$

After deducting Equations 7.32 and 7.33 from 7.31, the residue according to Newton's law will be the rate of momentum change:

$$\frac{2A_C L\rho}{g_C} \cdot \frac{d^2y}{dt^2} \qquad (7.34)$$

where

K = Elastic constant of the measuring diaphragm

μ = Viscous drag constant of the sealing liquid

r = Density of the sealing liquid.

y = Displacement of the measuring diaphragm.

Combining Equations 7.31, 7.32, 7.33 and 7.34 results in the following dynamic force balance equation:

$$P_1 A_D - P_2 A_D = \Delta P = -K_S y - \mu^* \frac{dy}{dt} \cdot \frac{A_D}{A_C} = \frac{2A_C L\rho}{g_C} \cdot \frac{d^2 y}{dt^2} \qquad (7.35)$$

From Equation 7.35, we may obtain the second order transfer function in standard form:

$$\frac{Y(s)}{\Delta P(s)} = \frac{K_{P'}}{\tau^2 s^2 + 2\tau\zeta s + 1} \qquad (7.36)$$

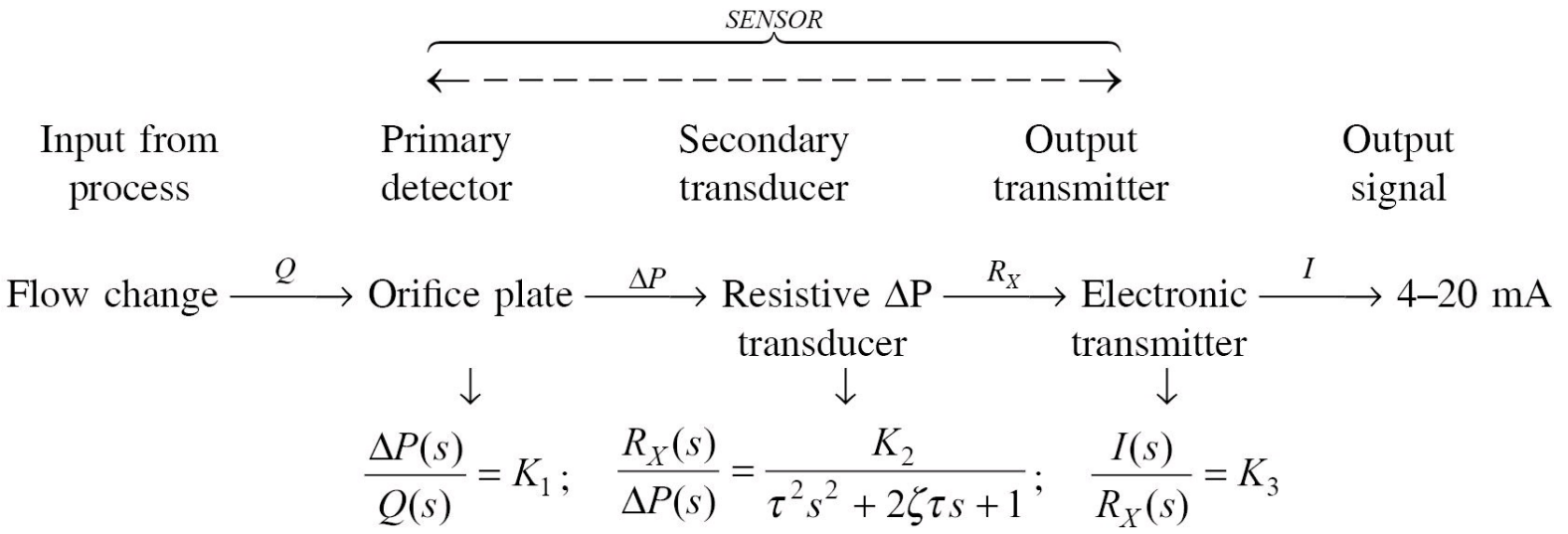
where $\tau = \sqrt{\frac{2L\rho}{K_S g_C}}$; and, $\zeta = \frac{\mu^*}{2A_C}\sqrt{\frac{g_C}{2K_S L\rho}}$; and $K_{P\square} = K_S$

There are a few categories of extremely low absolute pressure or High Vacuum Sensors. Among them **McLeod Gage, Pirani Gage, Hot and Cold Ionization Gages**, and **Thermocouple Gage** are important. McLeod Gage requires manual intervention for measurement, hence excluded from dynamic model building effort. The other instruments have electronic primary devices like Strain Gage, Thermionic Emission Detector, thermoelectric emf response of thermocouple of very small mass. Their speed of response is very fast and hence may be considered as pure gain elements.

Flow Sensors

Flow sensors of **Constant Area** category, e.g. **Orifice** and **Venturi** tube, and also fluid velocity sensors, e.g. **Pitot Tube**, **Target Flow Sensors**, **Annubar Flow Sensors,** feature a fast dynamics. But the associated secondary transducer that is a part of the sensor system, may have a considerable dynamic lag. That the U-tube Manometer type Resistive, and Diaphragm type Capacitive DPTs have second order dynamics, has been shown in Examples 7.5 and 7.6.

We have already mentioned that the final output signal within a standard range of variable is obtained by further conditioning of the measured signal by another transducer connected in series. This transducer, whether pneumatic or electronic, is called the transmitter and possesses a very fast dynamic response. Thus, the overall dynamics of such a sensor-transmitter combination may be appreciated from the following signal flow scheme and development of Equation 7.37. This also confirms our earlier notion that the dynamics of a measuring system is generally dominated by the sensor part of the instrument. The sensor may consist of only the primary detector, or it may be a combination of sensor and its secondary transducer, as it happens in the present system.



Hence, the overall dynamics of the measuring system will be:

$$\frac{I(s)}{Q(s)} = \frac{\Delta P(s)}{Q(s)} \cdot \frac{R_X(s)}{\Delta P(s)} \cdot \frac{I(s)}{R_X(s)} = \frac{K_1 K_2 K_3}{\tau^2 s^2 + 2\zeta\tau s + 1} = \frac{K_P}{\tau^2 s^2 + 2\zeta\tau s + 1} \qquad (7.37)$$

In the **Variable Area Flow Meter** or **Rotameter,** the fluid is made to flow upward through a transparent vertical tube with an upward flaring internal taper. A carefully designed solid object called *'Float'* is put inside the tube. For a given flow rate of the fluid, the float rides to a particular height and the flow rate is directly read on a scale corresponding to the float position. The scale is either etched on the tube or mounted externally. This arrangement is only possible when the flow tube is made of transparent material and the instrument is used as a flow indicator. However, the basic design may be adopted as a flow transmitter with an opaque tube and the float attached to a stem that has a magnetic core at the top end. With flow rate change, the core rides within an LVDT type variable reluctance transducer.

EXAMPLE 7.8 The Rotameter: The schematic diagram of a typical Rotameter is shown in Figure 7.13. The associated geometrical development in the figure relates the tapered tube diameter to the level of the float during a flow measurement.

To analyze this problem, let us identify the force components that are applied on the float with the fluid flowing in an upward direction.

The weight of the float (acting □) = $V\rho\frac{g}{g_C}$

The buoyancy of the float (acting □) = $V\rho_F\frac{g}{g_C}$

The frictional drag of the flowing fluid (acting □) = $F_D = \frac{C_D v^2 \rho A_P}{2g_C}$

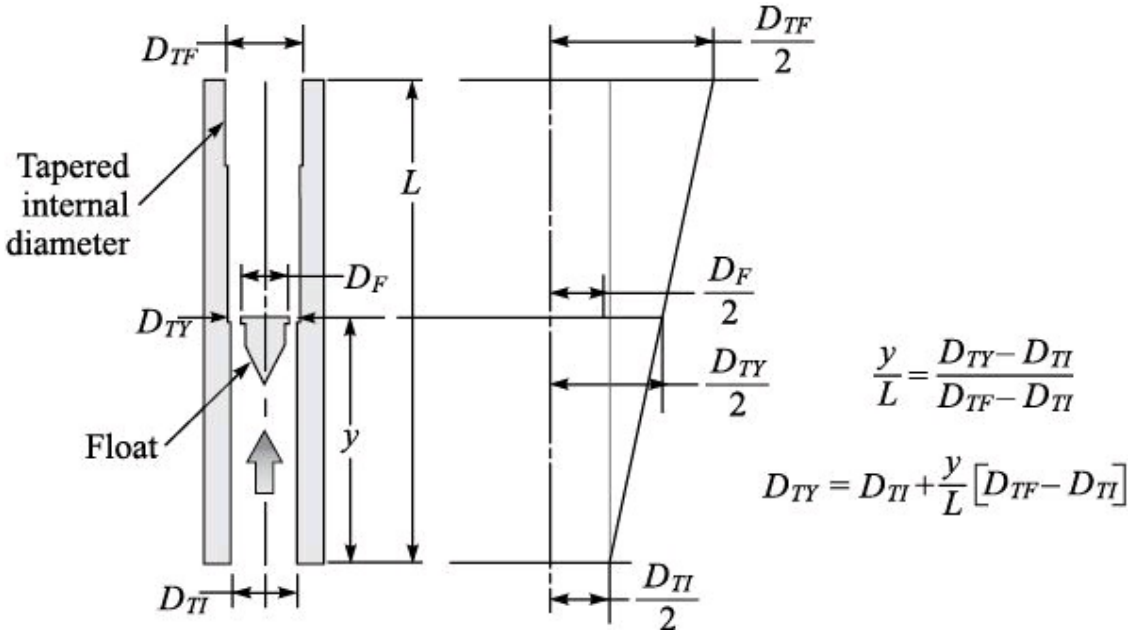


Figure 7.13 The schematic of a rotameter, and the tube diameter at a float height, y.
(The dimensions in the above figure have the following names: D_{TF} = Top diameter of the tapered Rotameter tube, D_{TI} = Bottom diameter of the tapered Rotameter tube, D_F = Diameter of the float, D_{TY} = Rotameter tube diameter at a height Y from the bottom.)

The balance of these forces would cause the rate of change of momentum of the float while it is motion due to flow rate change and yet to assume an equilibrium vertical height. Thus for a vertical height, $y(t)$, the expression for the momentum term is:

$$\frac{V\rho}{g_C}\cdot\frac{d^2y}{dt^2}$$

where,

V = Volume of the float,

r = Density of the float material,

Recognizing that upward be the +ve direction, now the force balance equation may be written as,

$$-V\rho\frac{g}{g_C}+V\rho_F\frac{g}{g_C}+\frac{C_D v^2\rho A_P}{2g_C}=\frac{V\rho}{g_C}\cdot\frac{d^2y}{dt^2} \qquad (7.38)$$

Where,

r_F = Density of the fluid,

C_D = Discharge coefficient for the flow through the annulus between float and tube,

v = Velocity of the fluid through the annulus between float and tube,

A_P = Projected area of the float on a plane perpendicular to the axis of flow.

The first two terms in the LHS of Equation 7.38 are constants and their sum will have –ve sign due to the fact that r_F is always less than r. We shall denote this sum by

C_1. If volumetric flow rate of the fluid is Q, the expression for the velocity through the annulus, v, in terms of the vertical float position, y, may be found by the relation developed in Figure 7.13, as:

$$v = \frac{Q}{A_P} = \frac{4Q}{\pi\left[\left\{D_{TI} + (y/L)(D_{TF} - D_{TI})\right\}^2 - {D_F}^2\right]} = K_1 Q \qquad (7.39)$$

Putting Equation 7.39 and C_1 in Equation 7.38, we may obtain the following non-linear differential equation as the description of the system dynamic response:

$$\frac{d^2y}{dt^2} = \frac{C_D A_P}{2V}\left[\frac{4Q}{\pi\left[\left\{D_{TI} + (y/L)(D_{TF} - D_{TI})\right\}^2 - {D_F}^2\right]}\right]^2 - \frac{C_1 g_C}{V\rho} \qquad (7.40)$$

Is it possible to find a linearized transfer function $\frac{Y(s)}{Q(s)}$ from Equation 7.40?

The task is left to the reader.

In the anemometer class **Hot Wire Anemometer** uses the resistance change due to dissipation of heat of a current carrying Pt-wire immersed in the flowing fluid, to sense its flow rate. As the mass of Platinum in the wire is very small, and heat transfer coefficient from hot wire to flowing fluid is quite high, the response of such a system would be fast enough to be considered as a pure gain element.

However, the **Moving Vane Anemometer** for air or gas flow measurement and **Turbine Flow Meter** for liquids are members of a separate class. These instruments convert a part of the momentum of the flowing fluid into the rotational motion of their vanes. The rotational speed is proportional to the fluid velocity and in fact, the output of the instrument. We should investigate their dynamics. But before that, one thing is worth mentioning. Two kinds of secondary transducers are used with such sensors that may produce Discrete or Analog outputs. First let us examine the design of the two kinds of speed sensors or **Pick-ups**:

1. The **Optical Pick-up** uses a visible light source and photo-electric receiver with a small gap in between, through which a small opaque object attached to the rotating shaft passes once per rotation. For each rotation, one electrical pulse is generated from the photo-electric device due to one interruption of the light beam as the opaque object passes through the gap between source and receiver. For low speed range of the shaft, by attaching multiple opaque objects, more than one interruption is created.

2. In a **Magnetic Pick-up,** small ferromagnetic substances are attached to the vane tips, and a variable reluctance pick-up is situated outside but at near proximity to the rotating vanes. Thus, a pulse train is generated as output from these devices that can be handled in two ways:

A. The pulses are counted in a digital counter for a selected time interval, after which they are displayed on a digital display. The display is held unchanged for the next pulse counting interval or Hold Time. This is a sampled data output system for which derivation of transfer function is impossible.

B. The frequency of the pulse train is electronically converted to a proportional voltage signal. This is an Analog output system and we may attempt to derive its transfer function.

EXAMPLE 7.9 The schematic of a Turbine type Flow Meter with a Magnetic Pick-up has been presented in Figure 7.14. In Figure 7.14(a), the Rotor has multiple number impellers (blades). Each of the blades experience a force as the fluid flows in the arrowed direction and the blades move in the anti-clockwise direction seen from down streamside. The force vectors on a single impeller have been given in Figure 7.14(b).

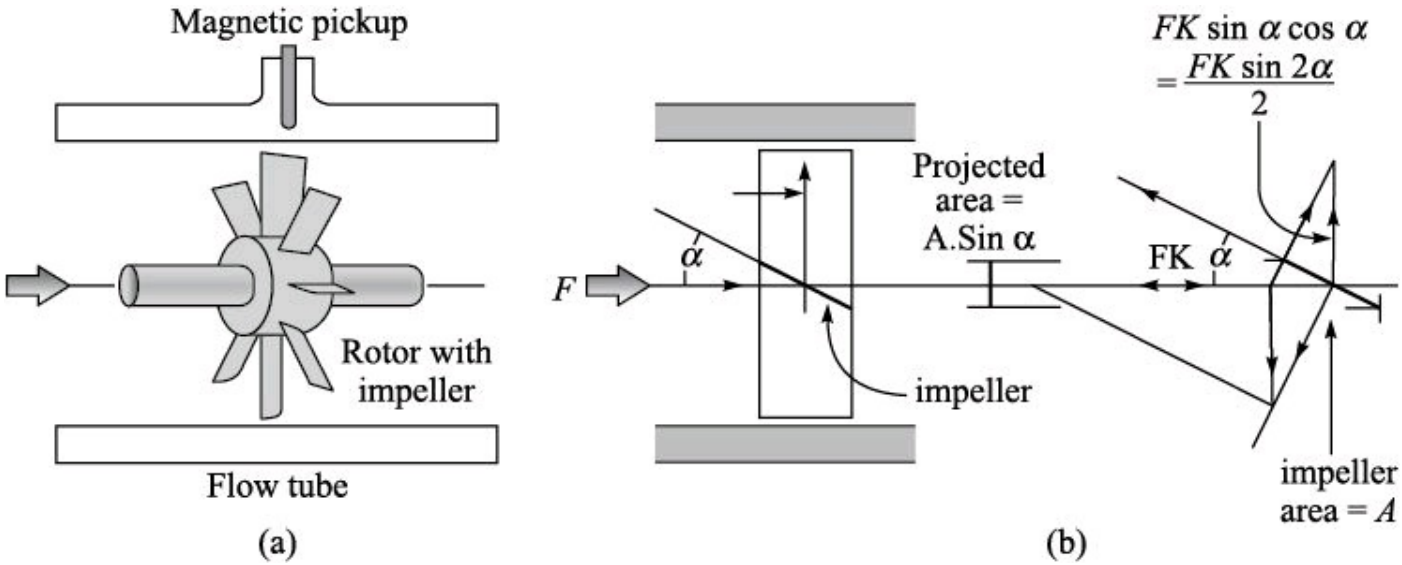


Figure 7.14 The turbine type flow meter. (a) Internal schematic. (b) The force vectors.

If we assume the fluid to be incompressible, the moving fluid before striking the impeller contains a total energy comprising of potential, velocity, and static pressure heads. Just after striking the impeller, the fluid velocity will be reduced, hence we may assume that the energy is transferred to the impeller at the expense of the velocity head.

(At a short distance away from the impeller in the down stream section, the velocity again becomes equal to the upstream velocity, because the fluid is incompressible and the flow is steady. This make-up energy comes from the total energy of the fluid. This

fact may be demonstrated by measuring the total pressure drop across the anemometer at two points where the up and downstream velocities are equal.)

However, the velocity head multiplied by the fluid density becomes pressure due to velocity head. A part of this energy will be transferred to the impellers. This fraction of pressure head multiplied by the projected area of the impellers, on a plane perpendicular to axis of flow, is the input force vector to the impeller. Thus,

Input force vector to the impellers: $$\frac{v^2 \rho K}{2 g_C} NA \sin \alpha \qquad (7.41)$$

where,

v = Upstream fluid velocity,

r = Fluid density,

a = Impeller angle with the axis of flow,

K = The Fractional constant relating available to the transferred energy from the fluid to the impeller,

N = Number of impeller on the rotor,

A = Area of an impeller.

From the construction of Figure 7.14(b), the resultant force vector that causes the rotational motion may be derived from Equation 7.41, and this would be:

$$\frac{v^2 \rho K}{4 g_C} NA \sin \alpha \sin 2\alpha \qquad (7.42)$$

This force will be opposed by the bearing friction of the rotor. This opposing force component may be written as:

$$-\mu * \frac{d\theta}{dt} \qquad (7.43)$$

Where

μ^* = Friction constant of the bearing,

q = Angular travel of the impellers.

The balance of these forces would cause the rate of change of momentum of the rotor while it is in motion due to flow rate change, and yet to achieve a constant speed condition. The momentum term would be:

$$\frac{m}{g_C} \cdot \frac{d^2\theta}{dt^2} \qquad (7.44)$$

where m = Mass of the rotor assembly.

By combining Equations 7.42, 7.43 and 7.44, the complete equation of motion for the system becomes:

$$\frac{m}{g_C}\cdot\frac{d^2\theta}{dt^2}=\frac{v^2\rho K}{4g_C}NA\sin\alpha\sin 2\alpha-\mu*\frac{d\theta}{dt}$$

or,

$$v^2C_1=\frac{m}{g_C}\cdot\frac{d^2\theta}{dt^2}+\mu*\frac{d\theta}{dt}$$

Recognizing that the output of the instrument is angular speed of the rotor, □, or, dq/dt and the input being the volumetric flow rate, $q = vA_P$, where, A_P is the cross-sectional area of the pipe. Introducing q, and denoting dq/dt = □, the above equation becomes:

$$\frac{q^2}{A_P}C_1=\frac{m}{g_C}\cdot\frac{d\omega}{dt}+\mu*\omega$$

or,

$$q^2C_2=\frac{m}{g_C}\cdot\frac{d\omega}{dt}+\mu*\omega \qquad (7.45)$$

To derive a transfer function from this dynamic relation, we need to linearize the input variable in squared form in Equation 7.45:

$$2q_SQ(s)C_2=\frac{m}{g_C}s\omega(s)+\mu*\omega(s)=\omega(s)\left[\frac{m}{g_C}s+\mu*\right]$$

Rearranging, the transfer function may be obtained as:

$$\frac{\omega(s)}{Q(s)}=\frac{K_M^*}{\tau*s+1} \qquad (7.46)$$

where, the linearized gain, $K_M^*=\frac{2q_SC_2}{\mu*}=\frac{q_S\rho K}{2g_CA_P\mu*}NA\sin\alpha\sin 2\alpha$

and, the linearized Time Constant, $\tau*=\frac{m}{g_C\mu*}$

For this type of flow measuring systems, often the mass of the rotor assembly is relatively very small, specially in case of anemometers for air flow measurement, and the instrument may be considered as a pure gain element.

Level Sensors

A class of level sensors have been developed for detecting level while it rises to reach, or diminishes to leave, a particular height (or level mark) inside a tank or reservoir. They change their output state from fully ON to fully OFF or vice versa, while the liquid level crosses this particular value. These devices may be called **'Level Switches'** rather than level sensors as they can produce only a binary (yes/no) type

information about the measured variable. These instruments find their use in many process applications, e.g. where information about filling of an overhead reservoir to a level slightly below its full capacity is required to prevent flooding. Obviously, it is not possible to propose the dynamic model of a measuring system that is not a continuous monitor but can only produce binary information.

Level is always measured in a capacitance-resistance type system. The capacity is realized by the area of the tank in which the level is measured, and resistance resides in the inlet and/or outlet pipes of the tank. The disturbance in such a system may only appear due to a change in the inflow or outflow changes. If the tank area is large, which is true in many industrial systems, such a disturbance would result in a considerably slow dynamic response in tank level. Thus, it becomes a little meaningless to bother about the speed of response of a sensor when the concerned process variable may only have a maximum achievable rate of change that is considerably slower even when compared to the slowest of the level sensors.

Probably the slowest level sensors employ the **Measurement of Hydrostatic Head** of the liquid column by a **Manometer**, and **Force Balance** of the liquid column with deformation of an **Elastic Element** such as a **Pressure Sensor**. We have already discussed their dynamics a little earlier in Section 7.2.2. The different configurations o f **Float and Riders**, and **Gage Glass** type level indicators have a high speed of response. Also, **Capacitive**, **Ultra-Sonic**, and **Nuclear or Radioactive** type level sensors possess very high response speed.

Chemical Composition Sensors

The more important categories of chemical composition sensors are mentioned in the following for the consideration of their dynamic properties.

1. Potentiometric Composition Sensors: Composition measurement performed by instruments of the following categories:

A. pH measurement by Glass Electrodes;

B. Reduction and Oxidation Potential measurement by Pt or Au Electrodes;

C. Measurement of Concentration of certain Chemical Species, e.g. dissolved O_2 concentration in a Bio-chemical reaction, sugar concentration in blood, etc. by ion selective electrodes;

is performed by potentiometric method. That is, depending on the concentration of the target species in the process solution an emf is produced in the electrode that is placed within the solution. This emf is measured relative to standard electrode as the cell potential where the complete cell consists of the measuring half-cell and the reference half-cell. Corresponding to the concentration change of the ionic species, the cell potential varies. This weak electronic signal is amplified and conditioned

through the transmitter circuit to produce the required output. The primary detector response together with the transmitter response is very fast, hence such instruments may be considered as pure gain element.

2. Optical Type Composition Sensors: Dynamically these instruments fall under two distinct classes of which the variety that may be used as coninuous sensors is indicated in the following:

A. Measurement done on the **Absorbance, Transmittance or Reflectance** *of the electromagnetic radiation at a selected wavelength, e.g.*

(a) Continuous Colorimeter—Dye concentration in a solution, by passing a filtered visible light beam of selected wavelength and measuring the solution absorbance.

(b) Long Path Length IR Cell—Flue gas composition is measured by a selected wavelength IR beam across the flowing gas and determining the absorbance or transmittance.

(c) X-ray Nephelometer—Concentration of suspended solid, living cell concentration in a biochemical reaction system is determined by passing an X-ray beam and measuring the *fluorescence* produced by the living cells, e.g. yeast cell concentration in the fermentation tank during production of commercial yeast.

In all these instruments, the basic measurement is done by electronic sensing of optical property change of a given process fluid, that happens very fast. These sensors may also be treated dynamically as pure gain elements.

B. **Measurement based on Spectrographic Analysis** *over a range of Wavelength: Whether it be an* **IR, Visible,** *or* **UV** *type analyzer, or even* **Mass, NMR, ESR,** *the measurement proceeds by taking a sample from a running process (say sample from a particular outlet stream of a petroleum refinery column), and performing the test that takes a definite amount of time to complete. These measuring systems fall in the category of sampled data devices and they have no transfer function in Laplace domain.*

3. Chromatographic Analyzers: Not only the systems mentioned in the previous Section, but certain other composition analyzers also have the sampled data character. Chromatographic analyzers and Automatic Titrators are important examples of this class. **Chromatographs** use the differential desorption characteristic of different chemical species to separate and analyze them from a mixture when a definite amount of the mixture as sample has been adsorbed in a matrix that is put within a stream of an ellutrient (inert fluid stream).

4. Automatic Titrators employ an automated replica of a sequence of procedural

steps to complete the analysis of a mixture of chemical species, as a chemist manually does in the laboratory. These systems also require a finite processing time, after that only the result of the last measurement could be obtained.

To demonstrate how large this sampling interval could be, we consider the following example. In an Aluminum extraction plant, the alumina production happens in a section of the plant where Bauxite ore is converted to Alumina-tri-hydrate crystal. The concentrated reaction mixture in solution form, from the bauxite digester after filtration is brought in the crystalliser, where $Al_2O_3 \square 3H_2O$ is crystallized and the lean liquor is recycled with fresh bauxite into the digester.

The reaction yield in the digester, as well as the recovery of alumina in the crystallizer, is expressed by a term **'A/C Ratio'**. This expresses the molar ratio of dissolved alumina, A, as Sodium Aluminate ($NaAlO_2$) is to that of dissolved Caustic Soda, C, as alkali in the solution. Thus, by determining A/C ratio of the thick liquor, the product from the digester,

$$Al_2O_3 \square 3H_2O + NaOH \square\square NaAlO_2 + 3H_2O,$$

and the lean liquor from the crystallizer,

$$NaAlO_2 + 3H_2O \square\square Al_2O_3 3H_2O + NaOH,$$

one could estimate the total performance of the alumina plant. But what would be the time required for the determination of A/C ratio? As late as 1985, the only available method was a laboratory based complex sequential processing that embodied several precision titrations done on a sample taken from the plant. This took 3 to 4 h to complete, and the plant had to run for an average time period of 3.5 h, practically without this vital information.

Later, a complex automatic titrator system was developed that took about 10 to 15 min for the complete procedure. This was a welcome news to the Aluminum industry personnel.

7.3 Signal Converters and Transmission Line

We have mentioned in the beginning of this chapter that in realization of certain control loops, instruments of dissimilar actuation energy or utility type are used. This happens due to:

1. Process design limitations,
2. Capacity limitation of the instrument,
3. Availability of hardware.

Examples 7.1 and 7.2, illustrated in Figures 7.3 and 7.4, demonstrate two closed loop processes where a special instrument has been used that converts the 4–20 mA

current input from electronic controller to 3–5 psig air pressure output acceptable to the pneumatic control valve. Such a signal transducer is called an Electro-Pneumatic-Converter (EPC) and denoted by (I/P) or G_{IP} in the block diagram of Figure 7.2. If the Control valve is physically situated away from the EPC the relatively long pneumatic signal transmission line would have a considerable dynamic lag. In this section we propose to develop the dynamic models for these two items.

7.3.1 Electro Pneumatic Signal Converter

The EPC proportionally converts 4–20 mA current input into 3–15 psig air pressure output. It requires only one actuation or utility, i.e. 20–25 psig instrument quality air supply for the output. The signal flow from input to output occurs in a straight-forward path. In the following example, we attempt to develop its transfer function.

EXAMPLE 7.11 In Figure 7.15 a simplified design of an EPC has been presented. The input current causes an electromagnetic induction in the solenoid coupled to the lever arm. The permanent magnet repeals this electromagnet (solenoid) upward with a force proportional to the coil current. Thus, the right hand part of the lever arm dips down. The downward force is balanced by spring tension and at equilibrium, a gap, x, between the lever arm acting as a 'Flapper' and the 'Nozzle' is created. The amount of 'bleed off' air from the nozzle depends on this gap, x, thus output pressure, P_O, from the branch would vary between 3 and 15 psig as input current changes between 4 and 20 mA.

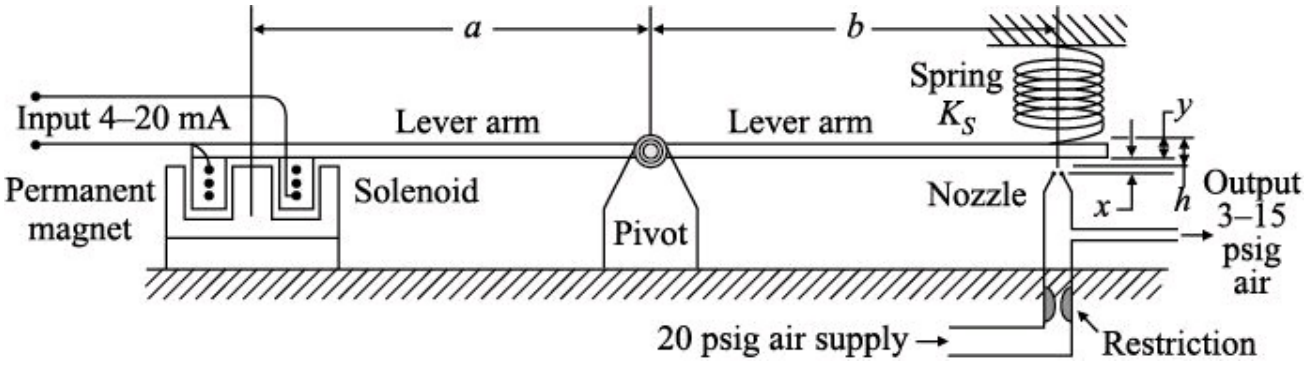


Figure 7.15 The schematic of an electro pneumatic transducer.

The upward force developed by the interaction of the permanent magnet and the current induced magnet, F_U, would be proportional to the product of Current, I, and number of turns of conductor in the solenoid, N_T; hence,

$$F_U = K_1 I N_T = K_2 I \qquad (7.47)$$

If the downward force generated in the right hand side of the lever arm at the top of the nozzle is F_D, then substituting for F_U from Equation 7.47, we obtain:

$$F_D = F_U K_2 I = \left(\frac{a}{b}\right) F_U = \left(\frac{a}{b}\right) K_2 I \qquad (7.48)$$

This force, F_D, will be balanced by the tension of the spring having constant, K_S, to produce a vertical displacement of the lever arm, y, from its initial position corresponding to 4 mA current input in the solenoid; hence,

$$F_D = yK_S \qquad (7.49)$$

The vertical distance of the lever arm from the nozzle mouth is denoted by x. This mechanical configuration comprising of the lever arm and nozzle actually constitutes a 'Baffle–Nozzle' assembly.

Before proceeding further, let us introduce the character of a Baffle–Nozzle assembly, which is one of the building blocks of pneumatic instruments. The arrangement of Baffle–Nozzle assembly has been illustrated in Figure 7.16.

The supply air at a pressure of 20 psig goes past a restriction and bleeds or leaks through the gap, x, between the nozzle mouth and flapper. The amount of bleeding air increases as this gap increases, as a result the output pressure, P, through the branch decreases. Let the lever arm which acts as the flapper is at a point A when there is no current in the solenoid, and is at B corresponding to a current 4 mA, and at C when the current is 20 mA. The flapper has a maximum vertical travel denoted by h as the input current to the solenoid varies between
4 and 20 mA. Note that when the baffle is at C, there is still a finite separation from the nozzle mouth.

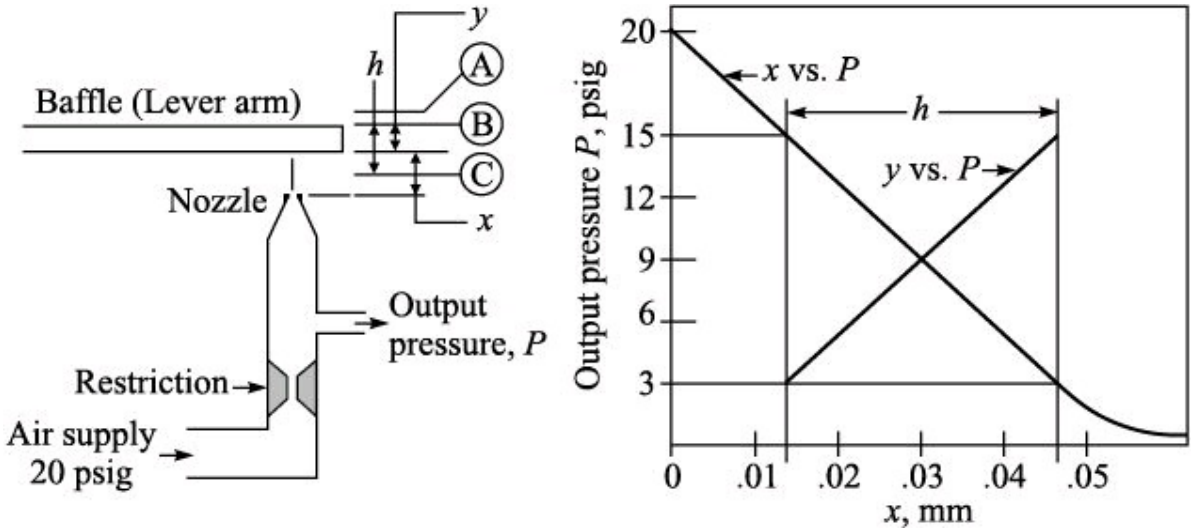


Figure 7.16 The baffle–nozzle, and its characteristics relating x vs. P and y vs. P.

Figure 7.16 shows the output branch line pressure as a function of x and y. Such a figure is known as Baffle–Nozzle characteristic in instrumentation literature, and this features a quite large linear portion together with a high gain, that is a high value of dp/dy. These properties are used in instrument design.

Equating Equations 7.48 and 7.49:

$$y = \left(\frac{a}{b} \cdot \frac{K_2}{K_S}\right) I \qquad (7.50)$$

Now let us assume the designed maximum gap between nozzle and flapper is h. Hence, as x varies from 0 to h. The relation between y and output pressure, P, may be given by:

$$P = 3 + \frac{12y}{h} \text{ ; or, } y = \frac{h(P-3)}{12} \qquad (7.51)$$

Equating Equation 7.50 and 7.51:

$$P = I\left(\frac{12aK_2}{hbK_S}\right) + 3 \qquad (7.52)$$

From Equation 7.52 we may obtain the transfer function:

$$\frac{P(s)}{I(s)} = K_{IP} \qquad (7.53)$$

7.3.2 Pneumatic Transmission Line

Drawn Cu or Nylon tubes are used to carry pneumatic signals between instruments. For a small distance (usually < 50 feet), the response is very fast and may be considered as a unity gain element. For longer distances above 100 feet, there would be a recognizable lag in signal transmission. A lumped parameter model, by assuming the total flow resistance, is concentrated at a particular point would be too inaccurate.

Harriot proposed a lumped parameter model for transmission lines recognizing all the three dynamic elements (R, C and L) in this system. However, the air density (causing the appearance of the inductance term) is small enough to discard inductance from the model. Also, the investigation of Bradner on the dynamic response of Pneumatic Transmission Line indicates that a distributed parameter model may represent the system.

In the following example we have attempted such an analysis.

EXAMPLE 7.12 The flow of a compressible fluid through a long circular conduit may be modeled by recognizing the flow resistance distributed throughout the length of the tube. Without going for an analytical solution by solving the partial differential equation for the flow system, we have attempted the multiple lumped parameter representation of a distributed parameter system. You may go through the Heat Exchanger models in Chapter 3.

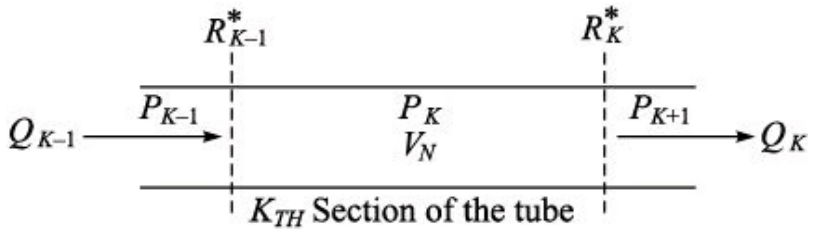


Figure 7.17 The variable and parameter configuration of the K_{TH} section of the transmission line.

Conceptually divide the tube into N number of equal length sections. Each of these sections would be treated as lumped parameter units. In Figure 7.17, a general K_{TH} section has been depicted with the input, output flow variables, and capacitance (volume) and resistance parameters.

Assuming inlet and exit gases are of equal density, r, we may write the unsteady state mass balance for the above section as:

$$\rho(Q_{K-1} - Q_K) = \frac{dm}{dt} \qquad (7.54)$$

Assuming ideal gas law:

$$PV = nRT \quad \text{or} \quad n = \frac{PV}{RT}$$

Now the mass of the gas inside the section may be expressed as: $m = Mn = \frac{MVP}{RT}$

where M = Molecular weight of the gas, R = Universal gas constant.

By differentiating the last expression with respect to time: $\frac{dm}{dt} = \frac{MV}{RT} \cdot \frac{dp}{dt}$

(7.55)

Putting Equation 7.55 into 7.54:

$$\rho\left(\frac{P_{K-1} - P_K}{R_{K-1}{}^*}\right) - \rho\left(\frac{P_K - P_{K+1}}{R_K{}^*}\right) = \frac{MV_N}{RT} \cdot \frac{dP_K}{dt}$$

where R^* = Flow resistance experienced by the fluid in a single section, V_N = Volume of a single section of the tube.

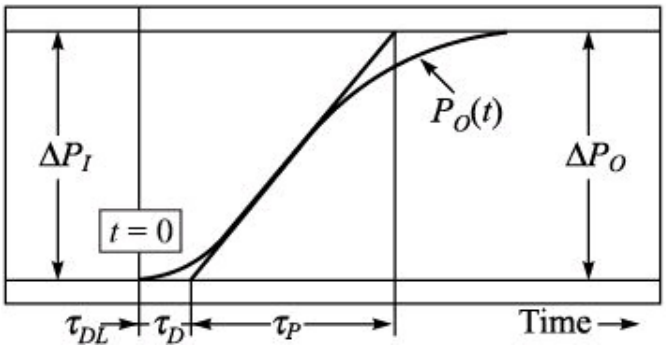
The above expression may be rearranged and simplified by recognizing that flow resistances in each section are equal:

or,

$$\frac{dP_K}{dt} = \frac{\rho RT}{MV}\left(\frac{P_{K-1}}{R^*} - \frac{2P_K}{R^*} + \frac{P_{K+1}}{R^*}\right) = \frac{\rho RT}{MVR^*}\left(P_{K-1} - 2P_K + P_{K+1}\right) \qquad (7.56)$$

For K = 1 to N, there will be N number of first order ODEs (Equation 7.56). Identifying the initial conditions and with proper values of system parameters, the set

of equations may be digitally simulated by writing a computer programme. The flow resistance may be evaluated assuming Laminar flow inside the tube and applying Hagen-Poiuseullie's equation. The programme may be used to obtain output pressure response for a step increase of input pressure. The response is found to be an S-shaped curve characteristic of a distributed parameter system. A typical response shown in Figure 7.18 agrees with the experimental findings of Bradner.[2]



$$K_P = \frac{\Delta P_O}{\Delta P_I} = 1$$

From the graphical construction, the FOPDT time constant ratio,

$$\frac{\tau_{DL}}{\tau_P} \simeq 0.25$$

Figure 7.18 Pressure response of pneumatic tubing for input pressure change ΔP_I.

The FOPDT type approximation of the step responses have shown a general feature that the ratio $t_{DL}/t_P \approx 0.25$.

Figure 7.19 may be used for design of pneumatic transmission line dynamics as an FOPDT approximate model. Development by Smith may be used for this purpose. We demonstrate the procedure by considering following example.

Let the 1/4" tubing length is 500 ft. The time required for 63.2% step recovery is 5.9 s. According to the Smith this time (5.9 s) is equal to $t_P + t_{DL}$. But in such a tubing system, it has been established that $t_P/t_{DL} \approx 4.0$. Thus, the two characteristic time values of may be solved to obtain $t_P = 4.72$ s and $t_{DL} = 1.18$ s.

Description of FO, FOPDT, SO and SOPDT type approximate Lumped Parameter model structures and their parameter evaluation methods are described in Chapter 12 on Process Identification.

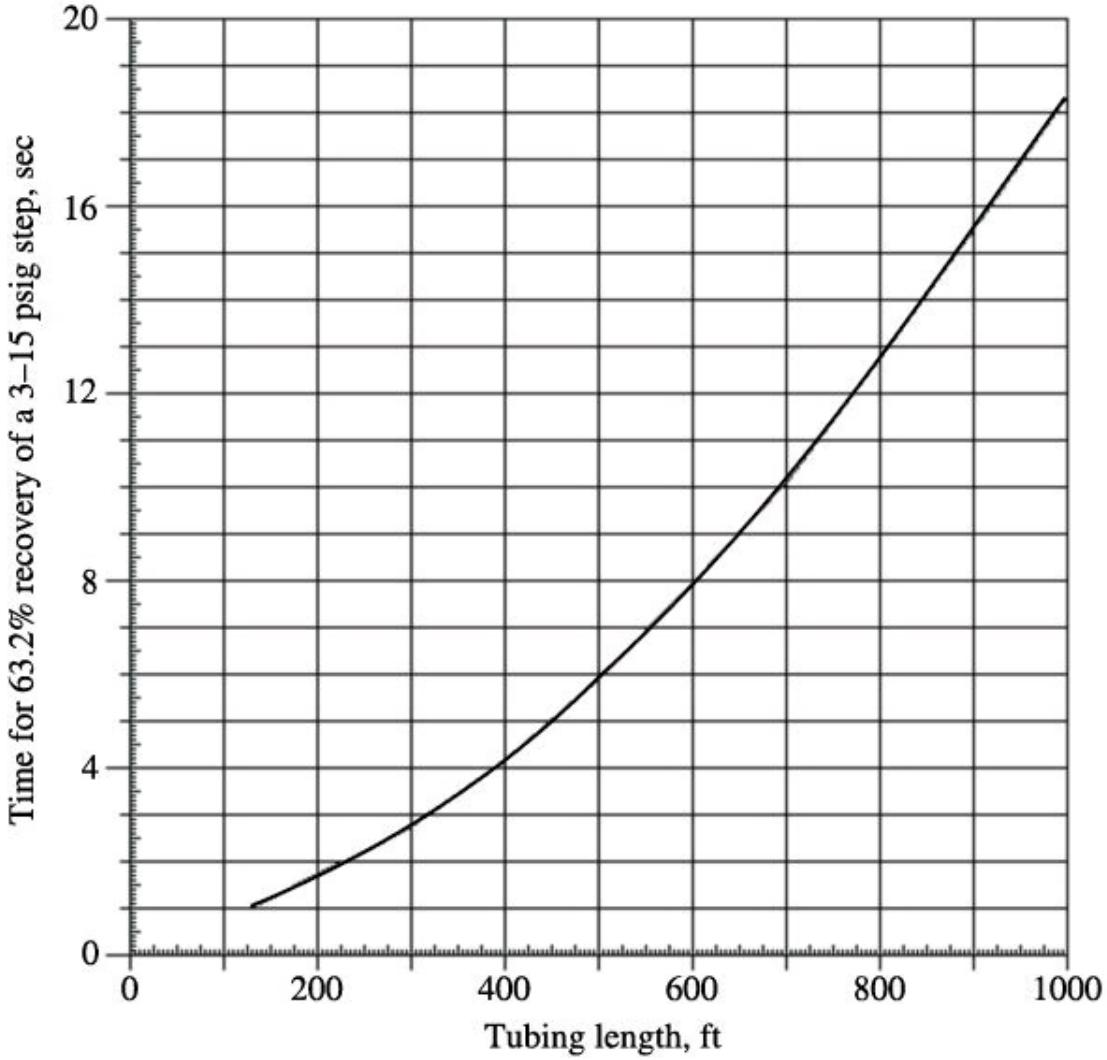


Figure 7.19 Response data from Bradner for a ¼" tubing (ID = 3/4") courtesy of Instrument Society of America.

7.3.3 Electronic and Digital Data Transmission

With the progress and popularity of electronic and subsequently digital control systems, many varieties of data transmission device have come up. From the older versions of Data-Links, e.g. RS-432, up to the modern Data Bus and Data Highways that have been designed for DCS applications. Fundamentally, all these systems use very high speed electronic data transmission, and may be safely considered as pure gain element.

7.4 Final Control Element (FCE)

Final control element (FCE) regulates the manipulated variable (MV) stream at the desired rate, to enter into the process as compensation. The manipulated variable (MV) is usually available as one of the utilities. In most cases, the MV stream has a high-energy content, and the regulation of such a stream requires considerable power, that is not available in the controller output (CO). The controller can perform rigorous calculations on error $e(t)$ to generate its output (CO) that is usually a low power signal in milli-watt, or micro-watt range. Hence, low power sophisticated output from the

'Brainy piece' or intelligent element of the loop has to be amplified by the next in line 'Muscle man' hardware—FCE. The high-energy manipulated stream from the FCE enters into the process for compensating dynamic error.

Practically, in all the FCEs there are two basic parts:

1. Actuator, where the power amplification takes place.
2. The mechanical/electro-mechanical system through which the manipulated variable is managed by receiving the high power actuation from the actuator. We shall discuss working of an FCE, in the context of Pneumatic control valves. This will be done because their number is largest among process control applications, and they contain practically all general features common to other classes of FCE.

7.4.1 Pneumatic Control Valve

In Figure 7.20, the actuator and the valve body has been demonstrated to explain the operation of a pneumatic control valve. Such a valve of 3□□ line size for medium pressure application will have a diaphragm of around 15□□ in diameter. Corresponding to an air pressure change from of 3 psig (minimum) to 15 psig (maximum), as the controller output, would produce a thrust of:

$$\frac{15^2 \times \pi \times 3}{4 \times 2.2 \times 1000} = 0.24 \text{ ton (minimum), to } \frac{15^2 \times \pi \times 15}{4 \times 2.2 \times 1000} = 1.2 \text{ ton (maximum)}$$

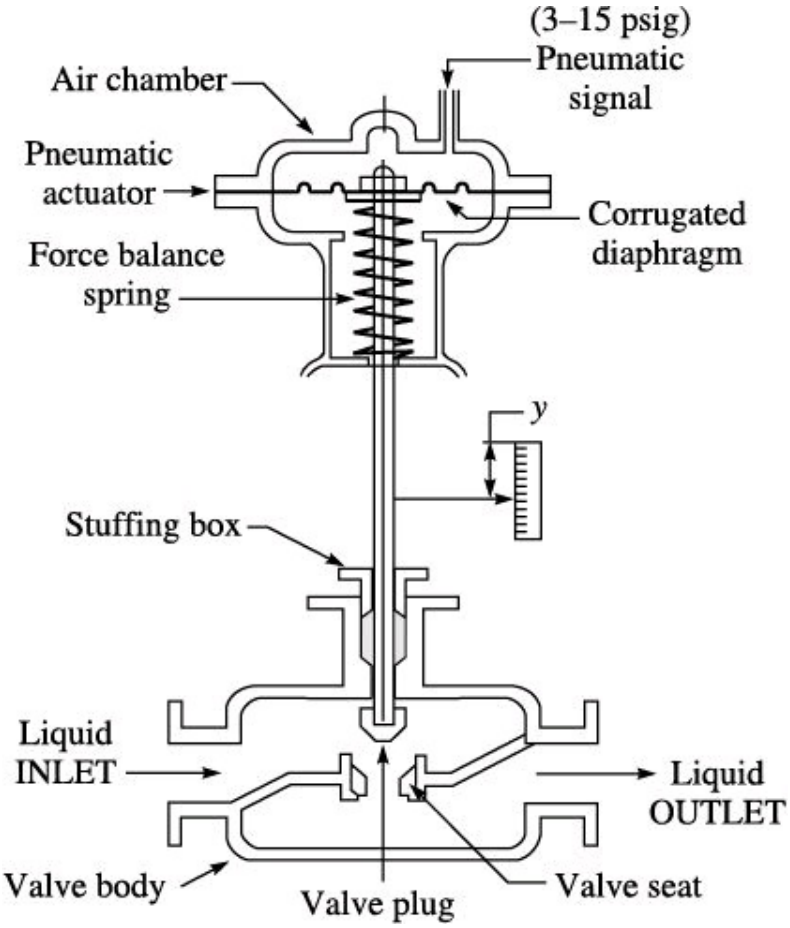


Figure 7.20 The parts of a control valve those determine its dynamic character.

The valve could produce this amount of variation of sealing or throttling pressure to regulate the flow of a fluid stream flowing through its body. For an applied pressure within this range, the final position of valve stem is the balance point between this thrust acting downward, and the compressed spring force acting upward.

The plug and seat profile of such a valve is very carefully engineered so that from fully-closed to fully-open position, the fraction of maximum flow through the valve (measured at constant pressure drop across the valve) bears a definite relation with valve port opening. This relation is called the characteristics of a control valve.

EXAMPLE 7.13 Dynamics of a Pneumatic Control Valve: Depending on the application, there are many variation of the basic design of a control valve and its actuation system. Usually, the design and selection of a control valve is optimized with respect to the property of the fluid handled, associated instrumentation system in the loop, and the type of atmosphere or plant environment in which it is to be installed. There are special designs (3-way valve configurations) also available for splitting of a single inlet stream into two exits streams (diverting type) or joining two inlet streams into a single exit stream (mixing type). The mixing or splitting ratio depends linearly on the stem position and that in turn on the valve top pressure or, controller output.

These topics have been thoroughly covered in any of the Instrumentation texts and a short coverage is included in Appendix-III. Here we would like to derive the dynamic model of such a valve preferably in transfer function form.

If displacement of valve stem from its fully closed position is Y, and the valve has a linear characteristic.

Then the fractional port opening, $Po = Y/Y_{MAX}$, and flow rate, Q, through this opening at constant pressure drop across the valve may be expressed as,

$$Q = CvPo$$

where Cv is a constant

Hence, $$Y = PoY_{MAX} = (Q/Cv)Y_{MAX} = K^*Q$$

In the dynamic form the above equation becomes:

$$Y(t) = K^*Q(t) \qquad (7.57)$$

The stem displacement, $Y(t)$ may be related to the applied pressure, $p(t)$, over the diaphragm of area, A, by making a dynamic force balance.

The relevant components of this force balance would be:

The applied force, $f(t) = Ap(t)$, will be balanced by,

1. The elastic force due to spring deformation, $K_S Y(t)$, and
2. The viscous drag force in the stuffing box due to the friction between the valve stem and the packing, $\mu^*(dY/dt)$.

According to Newton's law, if these two components are subtracted from the applied force, the residue is the inertial component that causes the change in momentum, thus,

$$p(t)A - K_S Y(t) - \mu^* \frac{dY(t)}{dt} = W\frac{d^2Y(t)}{dt^2} \qquad (7.58)$$

where,

K_S is Spring constant,

μ^* is frictional drag constant of the stuffing box or seal, and

m is the mass of the diaphragm, stem and plug assembly, thus $W = m/g_C$.

Inserting the result of Equation 7.57 in 7.58 and rearranging slightly,

$$\frac{W}{K_S}\cdot\frac{d^2Q(t)}{dt^2} + \frac{\mu^*}{K_S}\cdot\frac{dQ(t)}{dt} + Q(t) = p(t)\frac{A}{K^*} = p(t)Kv \qquad (7.59)$$

If the input and output variables are in deviation variable form, by taking Laplace transform,

$$\frac{Q(s)}{p(s)} = \frac{Kv}{\frac{W}{K_S}s^2 + \frac{\mu^*}{K_S}s + 1} \qquad (7.60)$$

In practical design of such a valve, the inertial term usually appears to have a very low value and hence may be neglected. With this simplification, the second order transfer function reduces to the following first order transfer function:

$$\frac{Q(s)}{p(s)} = \frac{K_V}{\frac{\mu^*}{K_S}s + 1} \qquad (7.61)$$

If the smallest time constant of other elements in the loop has a considerably larger value than the first order time constant (μ^*/K_S), Equation 7.61 may be further reduced to:

$$\frac{Q(s)}{p(s)} = K_V \qquad (7.62)$$

Equation 7.62 defines the control valve as a pure gain element and in majority of problems this description is quite satisfactory.

(In the following section we have used the principle of developing closed loop dynamics in the form of transfer function, from the open loop dynamic description of elements in the loop. The reader may bypass this section until he has acquired the prerequisites, or he may read the necessary materials: (1) Block diagram algebra, (2) Principles of Feedback, and (3) Determination of Closed loop transfer function, from Chapter 9.0)

7.4.2 Motorized Control Valve

The second most commonly used configuration of control valve is the Motorized Control Valve. In such valves, the actuator is an electric motor. Two basic types of motor actuation used are:

1. A DC-shunt or AC or AC-Synchronous-motor having bi-directional movement capacity is used with a Servo system,
2. A Stepper-motor is used with a sequenced pulse generator coupled to a power amplifier.

We would be discussing the motorized control valve with a DC-shunt Motor and servo system acting as the actuator. Thus, the actuator comprises of several parts. We first develop the dynamics of a DC-Shunt motor.

Dynamics of a DC-Shunt Motor

DC shunt motors are also called separately excited armature controlled DC motor, because they feature high field resistance and low armature resistance. The field current is usually very low, but the armature current produces the sole amount shaft

torque, and its direction through armature dictates the direction of shaft rotation. Figure 7.21 illustrates the schematic of such a motor.

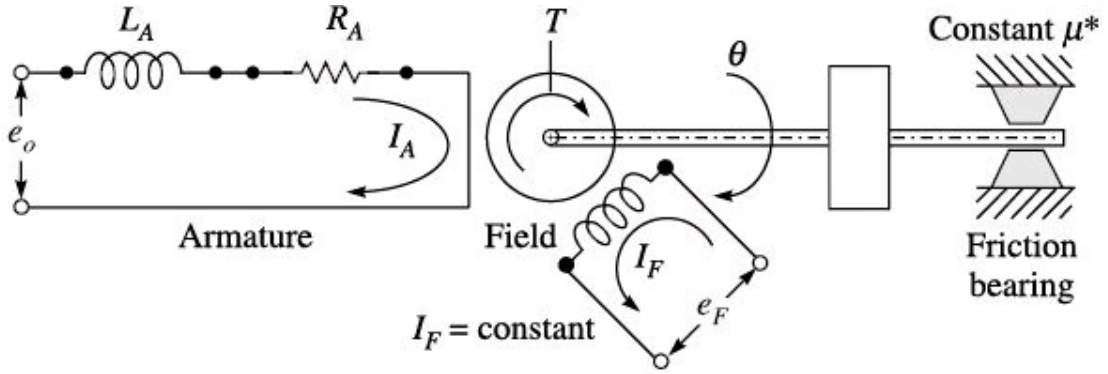


Figure 7.21 The schematic circuit of a DC shunt motor.

L_A = Armature winding inductance, henrys.

R_A = Armature winding resistance, ohms.

I_A = Armature current, Amp.

I_F = Field current, Amp.

e_O = Applied armature voltage, Volt.

e_B = Back emf, Volt.

q = Angular displacement of the armature shaft, radians.

T = Torque delivered by the motor, kg-m.

M = Equivalent moment of inertia of the motor and load referred to the motor shaft.

μ^* = Equivalent viscous friction at the bearings of motor and load referred to the motor shaft, (kg-m)/(rad/s).

Recognizing I_F will be constant in an armature controlled situation, the torque, T, can be related to the I_A in the following manner:

$$T = K_1 K_F I_F I_A = K_2 I_A \qquad (7.63)$$

where, K_1, K_F, K_2 are constants.

When in motion, a voltage (back emf) proportional to flux multiplied by the angular velocity of the shaft is induced in the armature. For a constant flux the induced voltage, e_B, would be directly proportional to angular velocity, dq/dt. Thus,

$$e_B = K_B \frac{d\theta}{dt} \qquad (7.64)$$

where K_B is the back emf constant.

The speed, dq/dt, is controlled by the armature voltage, e_O, that is supplied by an amplifier or a relay. The dynamic equation for the armature circuit may be written as:

$$L_A \frac{dI_A}{dt} + R_A I_A + e_B = e_O \qquad (7.65)$$

The armature current produces the torque that is applied to the inertia and friction, hence,

$$M\frac{d^2\theta}{dt^2} + \mu^* \frac{d\theta}{dt} = T = K_2 I_A \qquad (7.66)$$

Assuming all initial conditions are zero, by taking Laplace Transform of Equations 7.64, 7.65 and 7.66,

$$K_B s\theta(s) = E_B(s) \qquad (7.67)$$

$$\left(L_A s + R_A\right) I_A(s) + E_B(s) = E_A(s) \qquad (7.68)$$

$$\left(Ms^2 + \mu^* s\right)\theta(s) = T(s) = K_2 I_A(s) \qquad (7.69)$$

Recognizing $E_A(s)$ as the input and $q(s)$ as the output variable, we may develop the motor dynamics in block diagram form based on Equations 7.67, 7.68 and 7.69. This is given in
Figure 7.22.

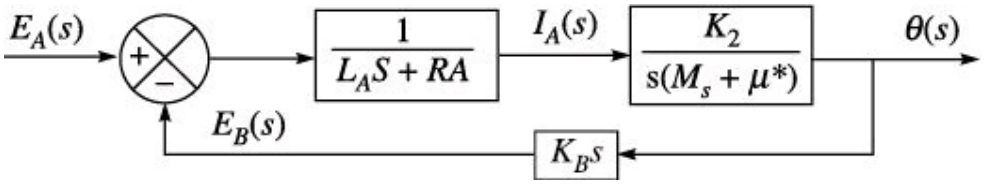


Figure 7.22 The block diagram representation of a DC-shunt motor dynamics.

The effect of back emf produces a damping character in the motor response. The closed loop transfer function of the DC motor may be obtained from the block diagram of Figure 7.22.

$$\frac{\theta(s)}{E_A(s)} = \frac{K_2}{s\left\{L_A M s^2 + \left(L_A \mu^* + R_A M\right)s + \mu^* R_A + K_2 K_B\right\}} \qquad (7.70)$$

The L in practical motor design is negligible. Thus, Equation 7.70 becomes:

$$\frac{\theta(s)}{E_A(s)} = \frac{K_M}{s(\tau_M s + 1)} \qquad (7.71)$$

where

$K_M = \dfrac{K_2}{R_A \mu^* + K_2 K_B}$, is the motor gain,

and, $\tau_M = \dfrac{R_A M}{R_A \mu^* + K_2 K_B}$, is the motor time constant.

(*You should note that there is an integrator in the motor transfer function. Try to explain this from your physical experience of the starting of a DC motor.*)

Electronic Servo System

An Electronic Servo System with its associated hardware has been shown in the following Figure 7.23. We shall see that the configuration shown in Figure 7.23 is rather theoretical but easy to understand. A more practical design of the same system is given in Figure 7.27.

EXAMPLE 7.14 Transfer Function of Motorized Control Valve: An Electronic Servo System comprising of a DC-shunt motor-coupled gear drive to a potentiometer, an operational amplifier based voltage comparator-current amplifier, and relay is part of the actuator. These are shown in Figure 7.23, together with the complete schematic of a control valve with a motor actuator used for driving the valve stem.

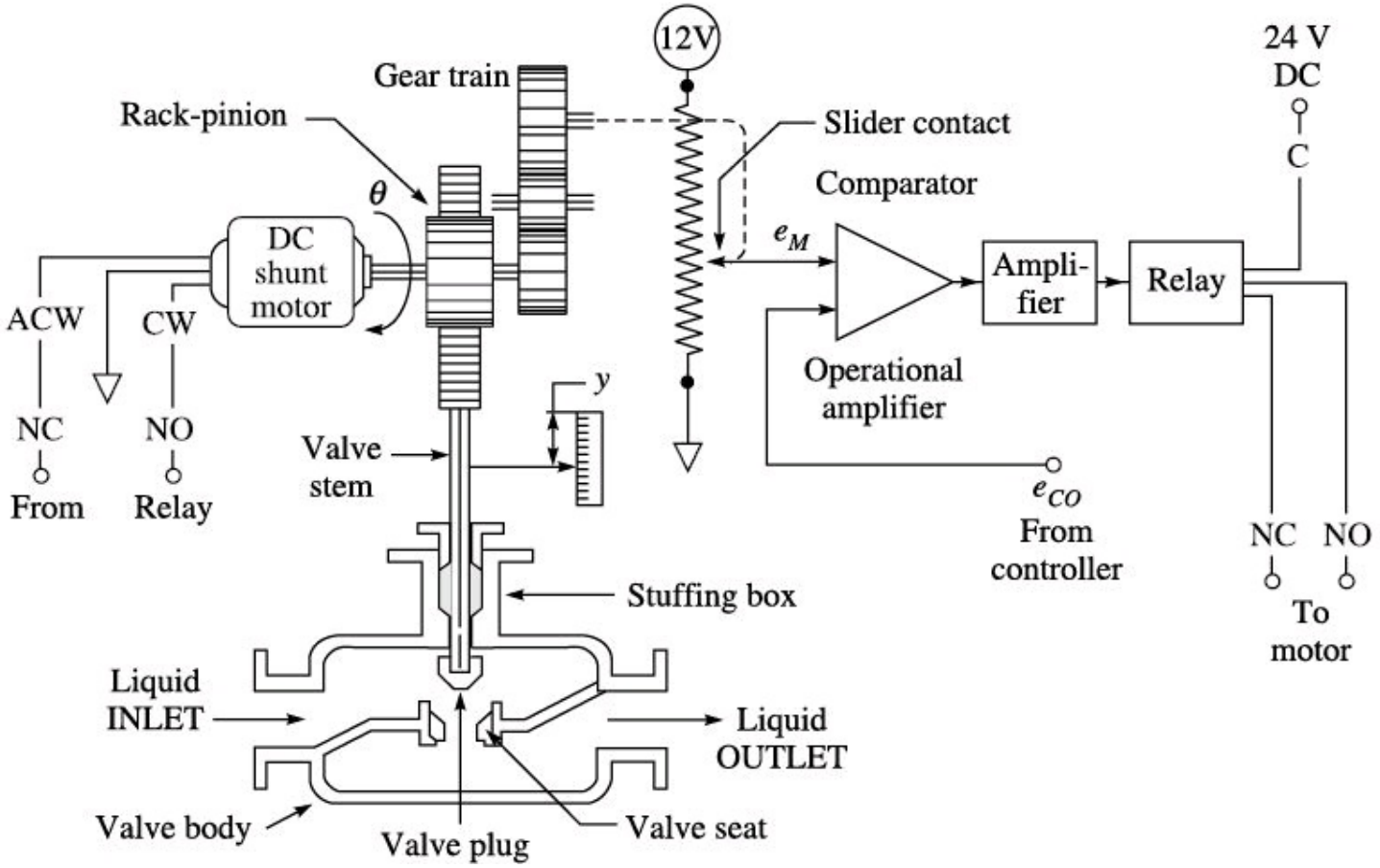


Figure 7.23 The schematic of a motorized control valve.

The servo system operates in the following manner. The bi-directional DC motor has three terminals; the middle one is grounded while the other two are connected to Normally Open (NO) and Normally Closed (NC) contacts of the Relay, and the Common (C) contact of the relay, is connected to the rated supply of the motor, 24 V DC.

As the relay can only have two states during operation, current can either go through the motor by NC to ground, causing an Anti-Clock Wise (ACW) motion, or by NO to the ground causing a Clock Wise (CW) motion of the motor. The motor shaft through

the Rack-Pinion arrangement not only moves the valve stem, it also drives the potentiometer through the gear train.

The potentiometer is actually a voltage divider that uses a fixed resistance with a slider coupled to the gear drive. Two ends of this resistance are connected respectively to +12 V and ground as shown, so the slider contact output voltage, e_M, may vary from 0 up to +12 V, depending on the slider position. An Operation Amplifier (OP-AMP) configured as a Comparator compares e_{CO} with e_M. An operational amplifier has two input terminals on the left vertical side of the triangle in the Figure 7.23. The higher one is called the Inverting Input (I), and the lower one is the Non-Inverting Input (NI) terminal. The property of such an amplifier is that when e_{CO} connected to NI is lower than e_M connected to Terminal I , the amplifier output voltage is zero, and the relay is in OFF state. Thus, the 24 V supply goes through the NC contact to move the motor in an ACW direction. This motion causes the valve stem as well as the slider to move downward so that e_M gradually becomes equal to e_{CO}.

On the other hand, if e_{CO} is higher than e_M, the comparator delivers full output voltage that after amplification, turns the relay ON, and the motor receives the 24 V supply through the NO contact and moves in a CW direction to lift the valve stem. Thus e_M becomes equal to e_{CO}. This action of the valve stem following the controller output voltage, and following each disturbed condition of control system, the difference between two voltages becoming equal, is called the Servo-Action. As the final action of this mechanism is position change or displacement of a device to a desired location, it is called a Positional Servo System. This much is sufficient about the operation of a Motorized control valve employing a positional servo system at this point. Now let us proceed to develop the dynamic model.

The variable nomenclature that we used for proposing the DC-shunt motor dynamics will remain almost similar with exception that the inertial term will be due to the combination of: Motor armature and the Load of gear train and Valve stem, i.e.

$$M = M_A + M_{GT} + M_{VS}$$

Also, the friction will be the sum of motor and gear bearing friction, plus the friction between the valve stem and the stuffing box packing:

$$m^* = m_B + m_P$$

The DC motor transfer function of Equations 7.70, and 7.71, and expressions for the motor gain and motor time constant would contain these new values of M and μ^*.

The required relation for the potentiometer type error detector:

$$E(s) = E_{CO}(s) - E_M(s) \qquad (7.72)$$

For the amplifier we may write:

$$E_O(s) = K_A E(s) \qquad (7.73)$$

where K_A = the amplifier gain. Note that E is the result of Equation 7.72. This subtraction operation happens within the integrated circuit based op-amp chip and E_O is the amplifier output. In fact K_A the amplifier gain of such an amplifier is extremely high ($\approx 10^6$), so that E_O may be either full value of the amplifier output or zero depending upon the algebraic sign of the result of Equation 7.72. The signal flow from the subtractor (comparator) on-word is shown in Figure 7.24. Such a binary kind of control action would be again evidenced while we discuss ON-OFF control in the next chapter.

The valve stem displacement y will bear the following relation with the angular displacement of the motor shaft, q, after transformation:

$$Y(s) = K_1 q(s) \qquad (7.74)$$

where K_1 = Gear ratio. Finally, the potentiometer output voltage from the slider, E_M, would have the following relation with valve stem linear displacement, y:

$$E_M(s) = \left(\frac{1}{K_1 K_2 K_A}\right) Y(s) \qquad (7.75)$$

where, K_2 = proportionality constant between angular displacement, q and potentiometer output voltage, E_M.

As we had done for the DC motor, Equations 7.71, 7.72, 7.73, 7.74 and 7.75 may be synthesized into the following closed loop block diagram of Figure 7.24.

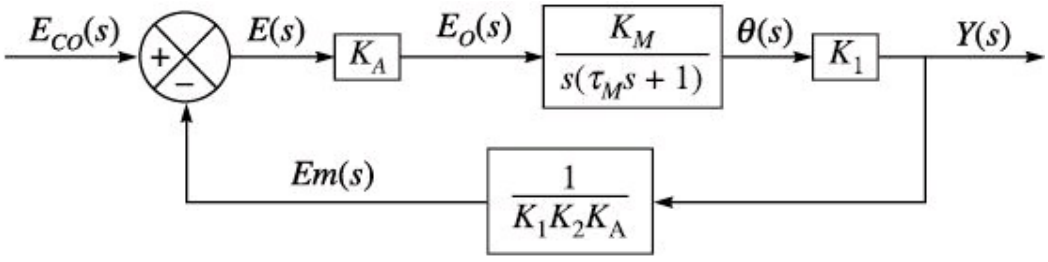


Figure 7.24 The block diagram representation of a motorized control valve dynamics.

From Figure 7.24, the required transfer function may be obtained as:

$$\frac{Y(s)}{E_{CO}(s)} = \frac{\dfrac{K_A K_M K_1}{s(\tau_M s + 1)}}{1 + \dfrac{K_A K_M K_1}{s(\tau_M s + 1)} \cdot \dfrac{1}{K_1 K_2 K_A}} = \frac{K_A K_M K_1 \cdot K_1 K_2 K_A}{\tau_M K_1 K_2 K_A s^2 + K_1 K_2 K_A s + K_A K_M K_1}$$

In the standard second order form, the above transfer function becomes:

$$\frac{Y(s)}{E_{CO}(s)} = \frac{K_1 K_2 K_A}{\left(\frac{\tau_M K_2}{K_M}\right)s^2 + \left(\frac{K_2}{K_M}\right)s + 1} = \frac{K_V}{\tau_V{}^2 s^2 + 2\tau_V \zeta s + 1} \qquad (7.76)$$

where the gain, $K_V = K_1 K_2 K_A$;

The Time Constant, $\tau_V = \sqrt{\frac{\tau_M K_2}{K_M}}$; and the Damping Coefficient, $\zeta = \sqrt{\frac{K_2}{4\tau_M K_M}}$

The configuration of the servo system of Figure 7.23 has some practical limitations. The most troublesome of these may be appreciated from Figure 7.25. Near the balance point of e_M and e_{CO}, when error ($e = e_{CO} - e_M$) changes from a very small +ve to very small –ve value, the relay will change state, and the motor will reverse its direction. Add this knowledge to the fact that the motor dynamics constitutes an appreciable amount of inertia due to its armature, gear mechanism and valve stem drive, which will not allow the motor to stop exactly at the balance point. The result will be a constant to-and-fro motion or oscillation. In the chapter on Automatic Controller, you will come across a similar phenomenon called **Chatter** while we discuss about ON-OFF controller in Chapter 8.

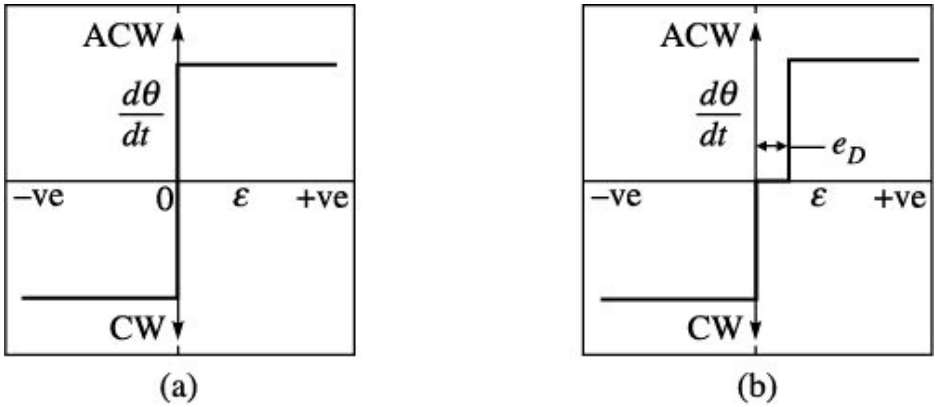


Figure 7.25 The error vs. direction of motion. (a) for zero differential voltage. (b) for a finite differential voltage.

We apply the same remedy here that has been applied for the ON-OFF controller, that is to add a small differential voltage e_D to e_M and apply two comparators. One will compare between e_{CO} and e_M and the other between e_{CO} and $e_M + e_D$. The comparator outputs would drive two relays, one for CW and the other for ACW motion. When the error (voltage) is less than e_D, the motor stops and we obtain a stable balance point. The operational difference of these two strategies has been illustrated in Figure 7.25. The indicated circuit modification is shown in Figure 7.26.

The other difficulty is that the full voltage range of the potentiometer should not be used by the slider output as e_M, because at the two ends there should be a provision for current limiting resistances. Also, at two extreme limits, there may be non-linear

characteristics of displacement vs. output voltage relation arising from mechanical erosion and failure and resistance heating due to high current.

The conventional output from an electronic controller is 4–20 mA, but in the above analysis, we have taken the input variable to be the voltage, e_{CO}. If the input current, I_{CO}, is sent through a grounded resistor of 500 □ and the voltage across the resistor is recognized as the e_{CO} , then the full range of e_{CO} will be 2 to 10 volts. Thus, e_M would be required to sweep the same voltage range. With a 12 V supply for the potentiometer, this could be achieved by putting two current limiting resistances so that 2 to 10 V remains as the full range for the potentiometer. We present in Figure 7.26 the circuit configuration required for these practical aspects, and in Figure 7.27, the complete diagram for motorized control valve operation.

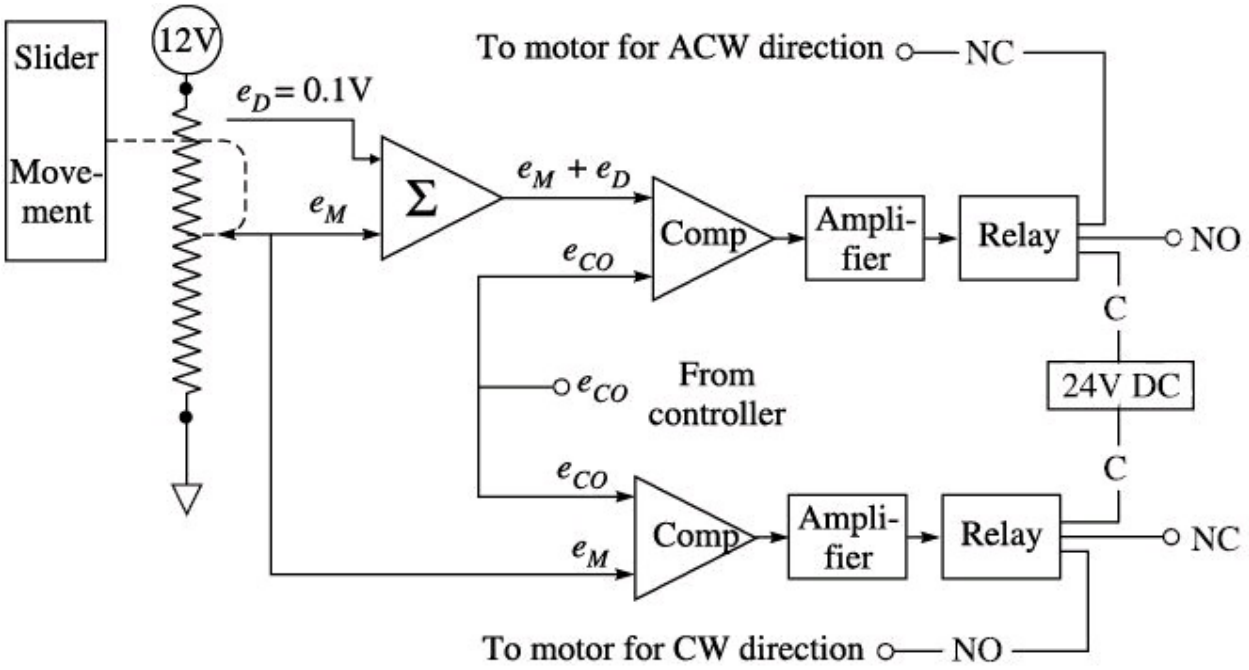


Figure 7.26 Addition of a differential voltage by a summer op-Amp and two comparators for ACW and CW motor rotation.

In this chapter, we have discussed about the dynamic characteristic of control valves. For specifying such a valve for a given control duty, there are certain other aspects of this hardware that we need to know. These topics are: **Sizing**, **Characteristics**, **Action** and **Rangeability** of a control valve. We have included this discussion in the Appendix-III.

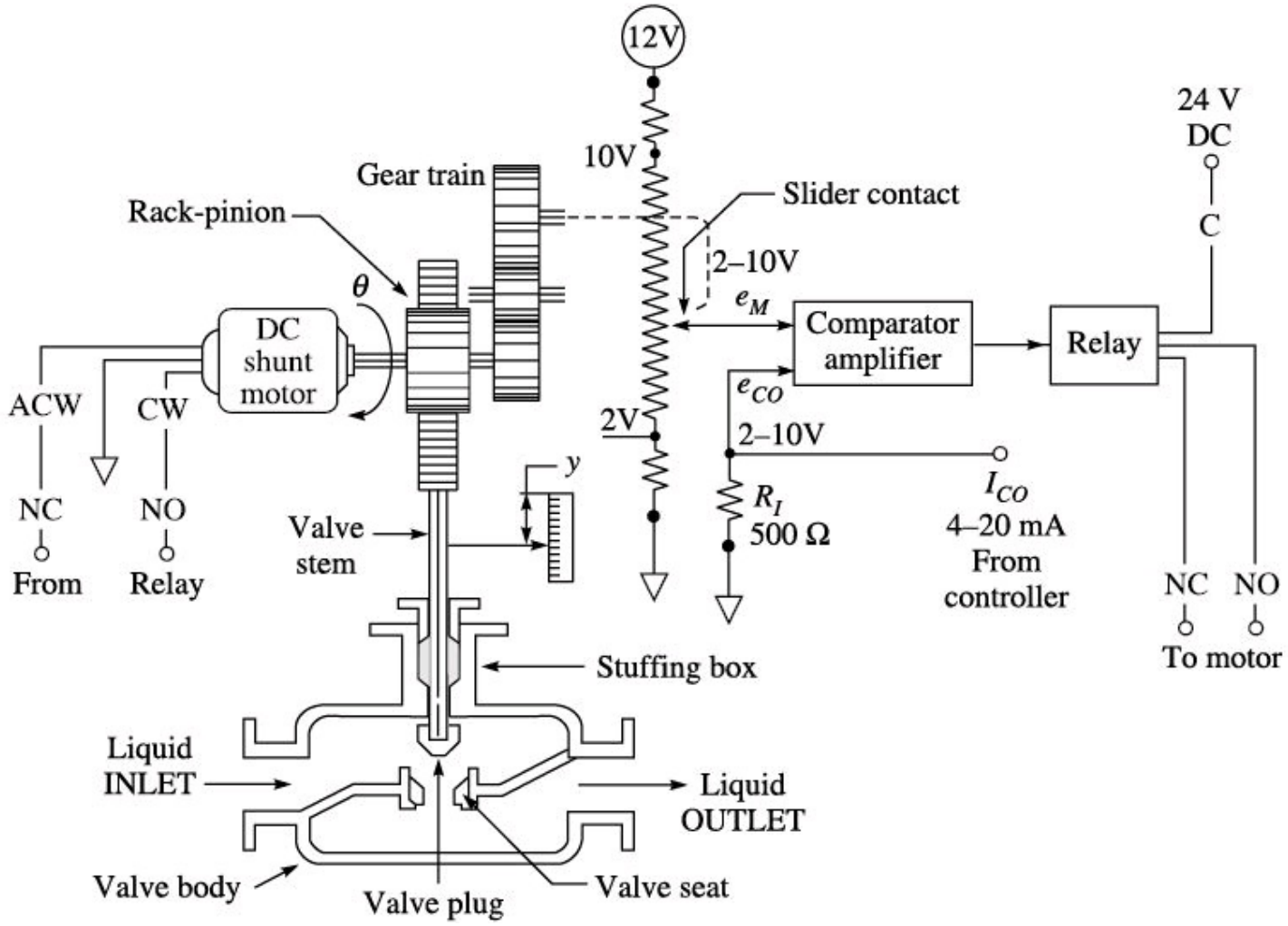


Figure 7.27 A practical configuration of the motorized control valve.

7.4.3 Pump Speed vs. Throttling

Consider a control loop where a pump is used for supplying the fluid, whose flow rate has to be regulated as the control action. The conventional way of doing this is to put a control valve on the pump delivery line. But an alternative way for this job is to vary the pump speed instead of using a valve. Variation of pump throughput by impeller speed variation of a centrifugal pump has some operational advantages over the technique of direct throttling. For a liquid driven by a centrifugal pump through the process equipment and associated pipeline may be thought as a resistive load, and its flow-head characteristic is considered as the load characteristic (LC). Similarly, the flow-head characteristic of the pump is known as the pump characteristic (PC). LC depends on the amount of throttling (increase of flow line resistance by the control valve action); PC is insensitive to this at constant pump shaft speed N. Whereas, reduction of speed from N_1 to N_2 means that PC(N_2) will be valid instead of PC(N_1) for the new operating condition of the pump.

In a typical installation, the intersection of LC and PC on a flow-head chart is the operating point of the flow system. Direct throttling by a control valve makes the LC steeper and shifts the operating point A to a lower efficiency zone C. By applying

speed variation there will be a new PC(N_2) on which the new operating point B will lye. The new operating point B obtained by pump speed reduction would be in a higher efficiency zone than C. This fact is valid for a considerable range of flow variation.

By this way of flow regulation, the pump is saved from over pressure conditions. Also, energy saving is achieved due to:

1. The pump does not have to supply the head required to negotiate the added resistance of valve throttling,
2. The pump operates at higher efficiency.

These ideas have been illustrated in Figures 7.28(a) and (b) for flow rate variation from
Q_1 to Q_2.

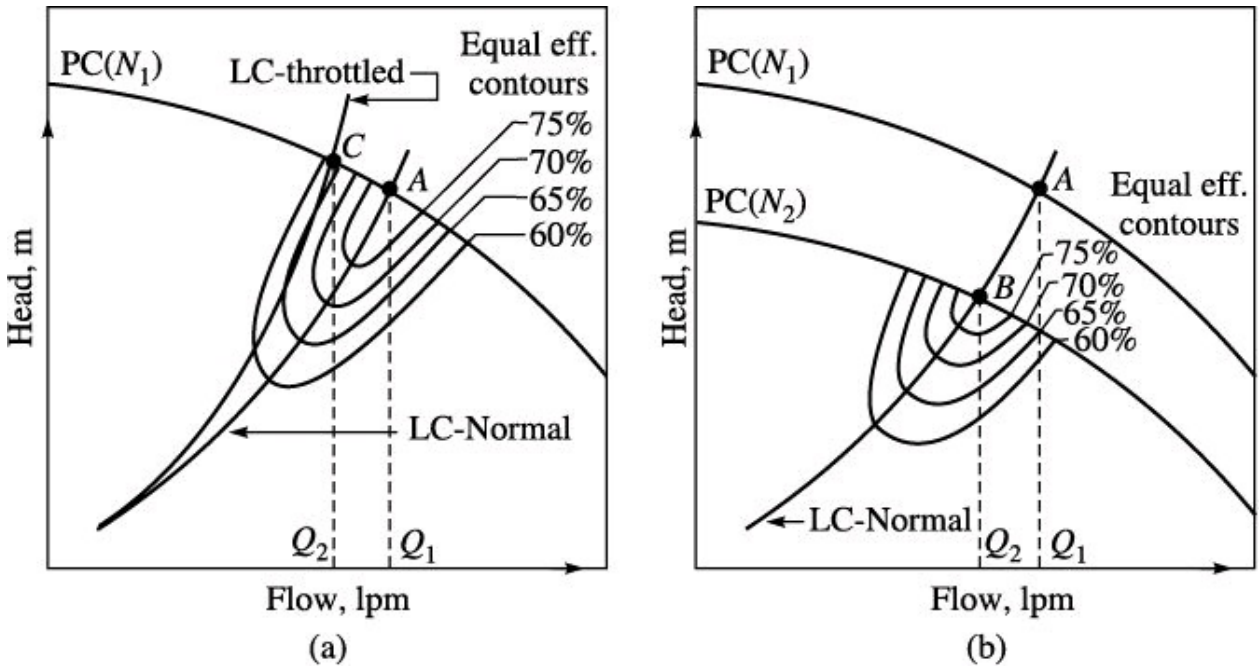


Figure 7.28 Pump flow regulation by throttling and shaft speed variation.

There are some added advantages of this method of flow regulation. The relation between throughput, Q, of a centrifugal pump and its shaft speed, N, may be given by:

$$\frac{Q_1}{Q_2} = \frac{N_1}{N_2} \qquad (7.77)$$

Thus, such a manipulation scheme would offer more linear behaviour. Also the reason for energy saving by this method of flow regulation becomes apparent from the relation between pump shaft power, P, and its shaft speed, N:

$$\sqrt[3]{\frac{P_1}{P_2}} = \frac{N_1}{N_2} \qquad (7.78)$$

The speed variation is mainly implemented by two methods:

1. PWM mode of power regulation, and
2. Variable Frequency Drives (VFD).

PWM Mode Power Regulators

It should be mentioned here that in many process control applications beside pump speed regulation, PWM mode is also used for low to moderate values (5 kW–50 kW) of electrical heater power regulation, as in the temperature control of electrical pre-heater for fuel oil (FO) supply to the burner of a furnace.

For the power regulation services, solid-state semi-conductor modules are widely used. These devices fall under three categories: Silicon Controlled Rectifiers (SCR), TRIACs, and Gate Turn-Off (GTO) devices. They are implemented to regulate the input power to a load from an alternating current type mains supply. For SCR application, an unfiltered full wave rectified DC is needed, whereas, TRIAC may directly operate with an AC supply, and a GTO may operate with both AC and DC variety of supply.

Principally the solid-state switching devices (SSSD) are electronic switches that can allow very fast (faster than power line frequency, > 50 Hz) switching cycles. In contrast to analog regulation, this discrete kind of regulation is called Pulse Width Modulation (PWM) Mode of regulation. A low voltage DC pulse is generated from the mains power and used to trigger the SSSD. Mains power is allowed to the load during the ON-time period of the pulse, and is blocked during the OFF-time period. The time period of the pulse train is kept constant, which is equal to ON + OFF time. In literature, this time period is also called the 'Window Time' or simply 'Window'. The input power to the load is regulated by varying the ratio of: ON-time/(ON-time + OFF-time). In a typical design, the trigger pulse frequency may be equal or less than mains power frequency. Thus, 35% analog regulation of the manipulated variable (MV) will be equivalent to:

$$\frac{\text{ON-time}}{\text{ON-time} + \text{OFF-time}} = \frac{\text{ON-time}}{\text{Window-time}} = 0.35$$

In Figure 7.29, the PWM mode of power regulation by an SCR, has been shown for a full wave rectified DC supply from the original AC.

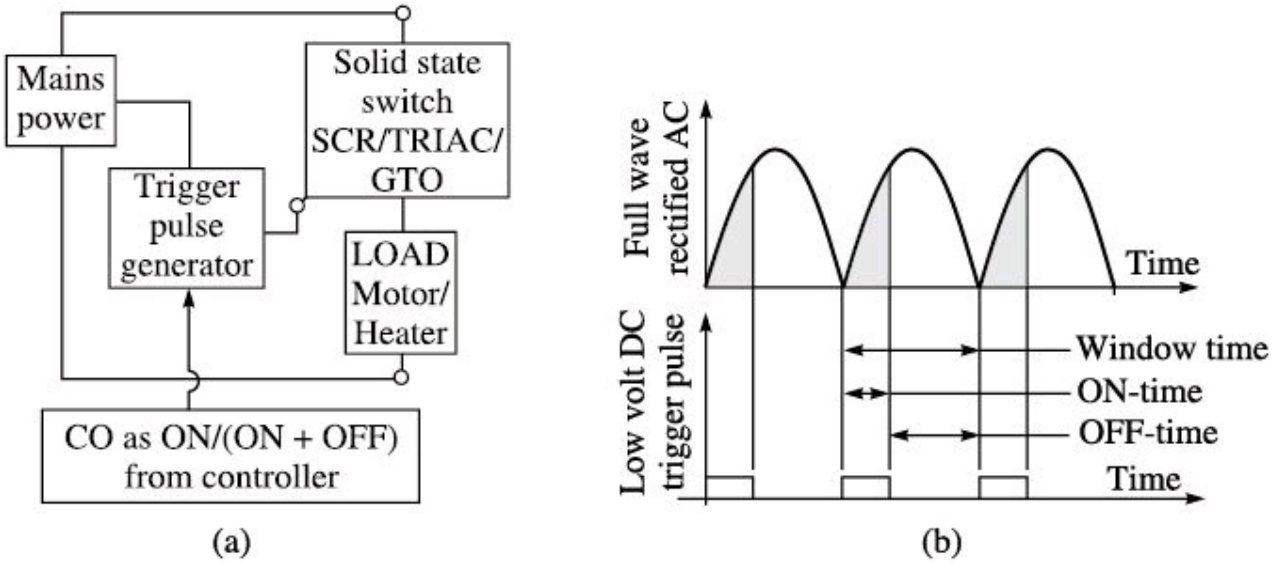


Figure 7.29 (a) Block diagram of a PWM mode regulator. (b) Regulation of power.

The solid-state switches are commercially available in modular packages of different operating voltage and current ratings from which the designer may select according to the requirement in a particular application.

One important consideration about the capacity selection of final control elements is that the maximum value of the manipulated variable stream that the FCE can deliver, determines the maximum amount of error the control system can cope with. When a critically tuned control system is operating not too far away from its stability limit, maximum capacity of FCE will determine the amount of overshoot that will appear for a given error magnitude.

Flow Regulation by Direct Application of PWM Mode

There is another way the PWM mode technology is used in implementing the final control action in a control loop. Instead of power, the flow of a stream of fluid is directly manipulated in pulsed fashion into the process. The controller output, as a fraction of full scale, is again made equal to the ratio of ON-time/Window-time. A pulse generator produces a symmetric pulse train at a designed frequency. The analog controller output is used for modulation of the pulses to achieve the desired ratio of ON is to OFF-time periods. The signal is power amplified to drive a Solenoid Valve, the FCE of the control loop. Determination of the pulse frequency is critical. The pulse frequency should be less than (1/10)th of the maximum allowable switching frequency of the solenoid valve. On the other hand, the frequency should be high enough so that the 'Ripple' in the process variable, caused by the introduction of pulse type modulation, remains sufficiently small to be neglected. Generally, this method is applied to regulate relatively small flow rates of manipulated stream.

An application of this technology that has found popularity is in regulating the fuel flow to moderate to large capacity furnaces. This method is also employed in dosing of small amount of costly or hazardous chemicals continuously into a large bulk of

stationary or flowing liquid.

Solenoid Valve: Solenoid valves are widely used in industry for implementation of PWM type flow regulation or where a liquid stream has to be suddenly shut or commenced, say for the adding a metered amount of liquid. These valves act like 'flow-switches' having very fast opening or closing speed (a typical maximum switching frequency of 635 ON-OFF operations/min has been reported by a manufacturer). A solenoid or coil, wound over a bobbin when energized, displaces a ferro-magnetic element by electromagnetic induction. This movement of the element, attached to a follower that carries the liquid seal on the inlet liquid nozzle, releases the process liquid to flow through the outlet port. When the energy input to the solenoid is stopped, by a mechanical spring action, the follower-element assembly returns to their initial position to stop the flow. Solenoid valves are available with 'normally-open' and 'normally-closed' action options. They have typical energization voltage specifications, e.g. 12 V DC, 24 V DC, 24 V AC, 230 V AC. Figure 7.30 depicts a typical solenoid valve in which the

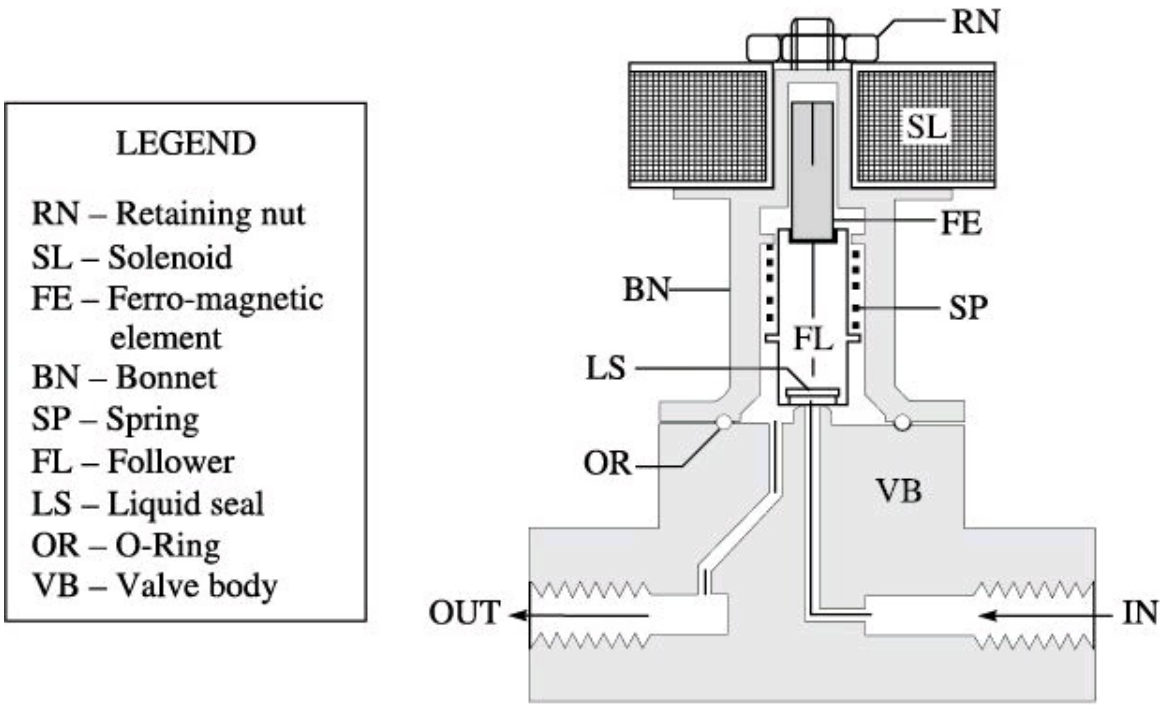


Figure 7.30 The essential parts of a typical two-way solenoid valve.

ferro-magnetic element, follower and spring are housed in a bonnet which are attached to the valve body by a standard fastener.

Implementation of PWM Mode in Digital Control with Solenoid Valve: In such an application, solenoid valves are suitable because the final control element, instead of being proportionally open to the controller output (CO), is rapidly switched between ON and OFF states, and the ratio of (ON-time period) is to the (ON + OFF time period) is made equal to the normalized controller output. The ON + OFF time period is called the window time, W. (The value of W is usually set equal to an integral

multiple of the sampling time of a digital control loop.) A schematic view of the valve operation corresponding to the controller output is shown in Figure 7.31.

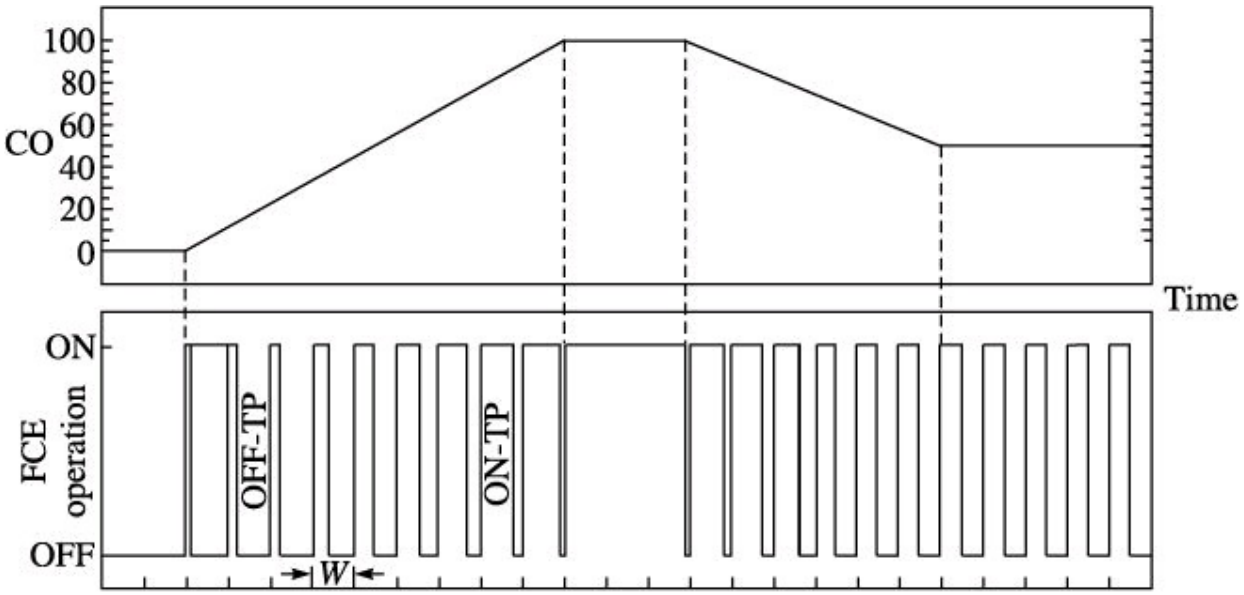


Figure 7.31 The controller output manipulates the on/off ratio of the FCE pulses.

It is expected that the controlled variable will show rapid oscillation of small amplitude around the set point at steady state, as a result of the pulsed nature of manipulated variable stream. By careful selection of window time, *W*, minimum pulse width (DT) and ratio of DT/WT, the oscillation amplitude or ripple could be reduced to an acceptable limit.

Application areas in Process Control:

1. Various dosing systems in water treatment and similar applications.
2. Additive incorporation in continuous and semi-batch systems in polymer synthesis and modification, petroleum processing and finishing.
3. Fuel injection to the burner of a furnace.

Advantages: Rapid pulsation of the manipulated variable has a self cleaning feature of the final control element that eliminates any clogging problem of the nozzle or orifice due to accumulation of minute amount of dirt or carried over solids in the fluid. This is specially important in furnace operation where nozzle clogging of the burner from soot formation is a well-known problem.

If the manipulated variable flow is from a constant head source, a better flow regulation results (you don't have to bother about 'inherent characteristics' and 'installed characteristics' that are of essential consideration while installing a control valve).

In computer control application, the ON-OFF pulses could be generated right from the control programme, only an optically isolated current amplifier will do the job instead of the more costly DAC card with data hold facility and data link for

communication.

The cost of a solenoid valve is dramatically less than a control valve and its accessories of the same port size.

AC Variable Frequency Drives (VFD)

The speed of an induction motor may be controlled by the frequency of the power supply to the motor as an alternative to the PWM mode of power regulation. Pump flow manipulation by the frequency variation of pump drive is a technique being practiced for the last few decades. VFDs convert incoming AC electrical power to DC, then convert the DC power into AC power of suitable frequency and voltage. Conversion of DC to AC of desirable frequency and voltage occurs in a solid-state circuit module called Inverter. VFDs are designed to take advantage of the fact that speed, torque and horsepower of an AC motor are all related to the frequency and voltage of the power supply:

$$\text{Speed} = \frac{2 \text{ Hz}. 60}{\text{No. of Poles}} \tag{7.79}$$

$$\text{Torque} = f\left(\frac{\text{Volt}}{\text{Hz}}\right) \tag{7.80}$$

$$\text{HP} = f(\text{Torque} \cdot \text{Speed}) \tag{7.81}$$

In practice, the VFD technology is implemented with an internal feedback to improve the speed regulation performance. The principle of operation of a VFD with such an internal feedback control is shown in Figure 7.32.

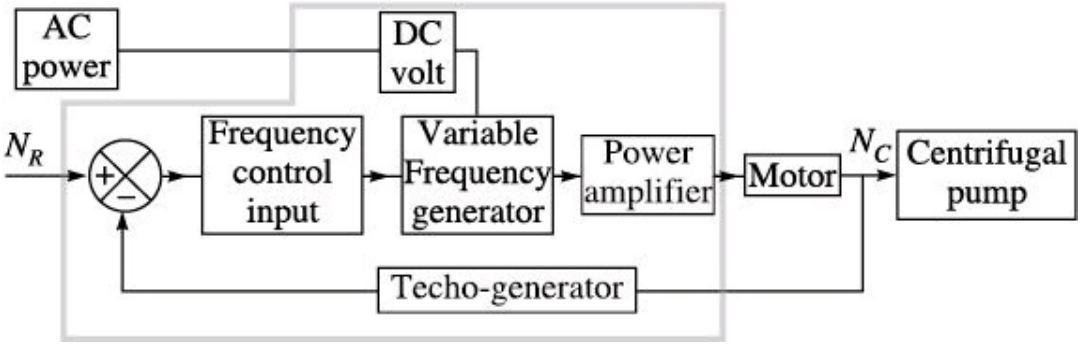


Figure 7.32 Operation of a VFD with an internal speed feedback.

We may consider the application of VFD in controlling the exit shell liquid temperature of a shell-and-tube heat exchanger in which a hot liquid is cooled by flowing a coolant stream through the tube side of the exchanger.

A VFD drives the centrifugal pump that transports the coolant. The internal nested speed control loop ensures that the manipulation in terms of coolant flow rate warranted by

the temperature controller is realized by the FCE. The control loop has been shown in Figure 7.33.

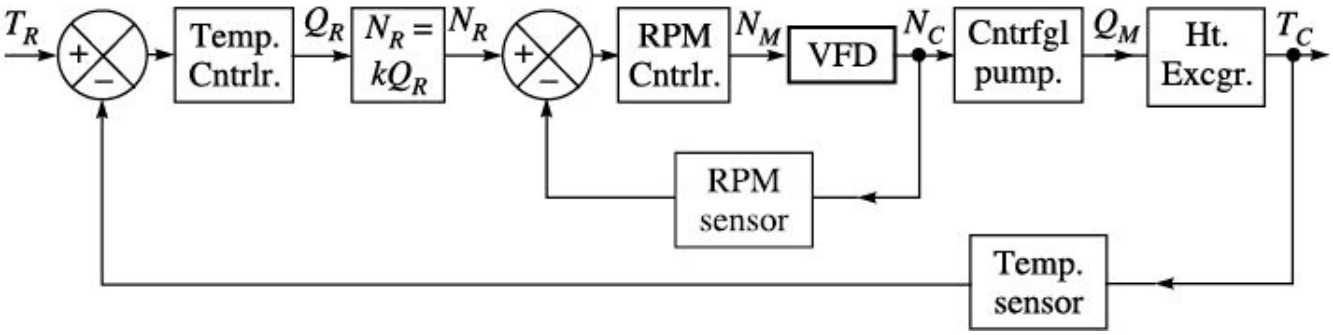


Figure 7.33 Temperature control of a heat exchanger. The nested loop implements a VFD for the coolant flow pump as FCE.

7.4.4 Monitoring/Dosing Pumps (Flow Manipulator and Integrating Pumps)

Positive displacement pumps transport a fluid by physically trapping a definite volume of the fluid at the low pressure suction end, then conveying and releasing the amount fluid at the high pressure delivery end. According to the mechanical design, they may be classed under two broad headings:

1. Reciprocating type, as for example, Piston-Cylinder, Diaphragm, etc.
2. Rotary type, examples are Gear, Twin-Screw, Lobe, Nutating-Disc, etc.

Whatever be the type, the delivered mass by these pumps may be slightly less than the trapped amount at suction, due to Leakage, a phenomenon that occurs because a small amount of fluid may flow from delivery to suction through the little clearance between mechanical parts in relative motion. Leakage happens due to the pressure difference against which the pump is operating, and its amount is dependent on the magnitude of ΔP. However, by increasing the design accuracy and suitable choice of construction material, this leakage could be made smaller than 0.1% of the total flow.

With such a performance measure, it may be considered that volume transferred per stroke (for reciprocating types) or per rotation (for rotary types) is independent of the operating pressure differential between delivery and suction. The throughput would be only dependent on pump speed. Not only that, the delivered amount of fluid may be obtained by counting the number of strokes or rotation within a time interval and multiplying the number, by the mass of fluid delivered per stroke or rotation.

Flow rate through these pumps may be controlled by speed variation of the drive. This may be achieved either by PWM or VFD technology.

Following these considerations, it should be appreciated that positive displacement pumps may be used for flow regulation as well as flow integration with a high

precision performance. Among a number of designs, the Piston and Diaphragm pumps from the reciprocating class and Gear and Twin-Screw pumps of the rotary class have found use as final control element applications. A special application in medical instrument has been cited in Example 1.9 featuring a 'Peristaltic Pump' used as a controlled volume insulin dispenser for blood sugar level control of an advanced diabetic patient.

EXERCISE PROBLEMS

7.1 During a calibration test following Input (X), and Output (Y) data has been obtained and expressed as % of full scale.

Run No.	X	0	10	20	30	40	50	60	70	80	90	100
1 (X □)	Y	0.5	9.3	21.1	30.2	40.4	49.8	60.1	70.2	80.9	90.4	99.5
2 (X □)	Y	2.1	10.2	19.89	31.0	42.1	49.8	59.3	70.5	79.2	89.7	99.6
3 (X □)	Y	0.95	9.5	20.5	28.9	41.5	50.1	58.8	71.3	78.9	89.6	98.1
4 (X □)	Y	1.7	11.1	19.9	29.5	39.5	51.9	59.9	70.8	80.4	90.1	99.2
5 (X □)	Y	0.9	8.9	18.7	30.4	39.1	49.2	60.6	69.7	81.3	91.5	99.7
6 (X □)	Y	1.65	11.2	19.8	30.9	38.5	50.9	61.9	71.8	81.4	90.9	98.2

Prepare a plot on a square graph. Draw the x-y diagonal on the graph to show the deviations, and calculate following steady state characteristics: Zero Error, Maximum error as % full scale, % Linearity, Repeatability as % full scale.

7.2 Ring balance manometer is a special configuration that measures differential pressure by the angular displacement of the pointer due to the shifting of mercury mass at one side of the ring that is free to rotate around the pivot. The pressure taps are connected to the manometer by flexible rubber tubing. The system is shown in Figure P7.1.

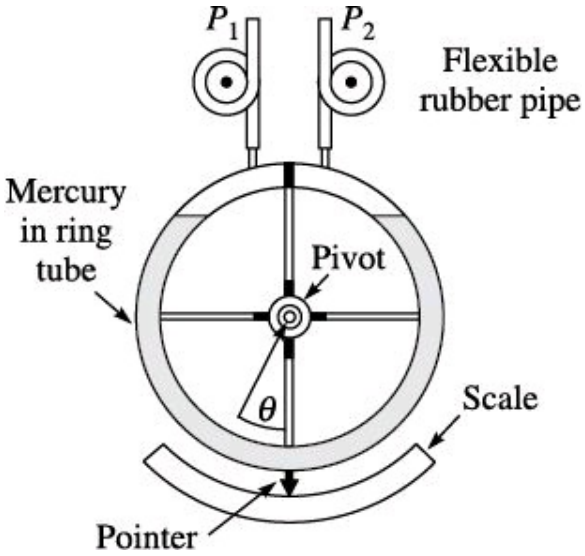


Figure P7.1 Ring balance manometer

It proposed to develop the dynamic model of this instrument relating the input variable- differential pressure to the output angular displacement. Assume the

required physical parameter and material property values for the analysis.

7.3 You have thought out a concentration measurement scheme based on dependence of RI on concentration of a soluble carbohydrate in water. The schematic arrangement of the system is illustrated in Figure P7.2. The Refractive Index of the aqueous solution (RI_S) has the following dependence with solution concentration:

$$RI_S = 1.32 + 0.01C_H$$

where C_H = Solution concentration by weight.

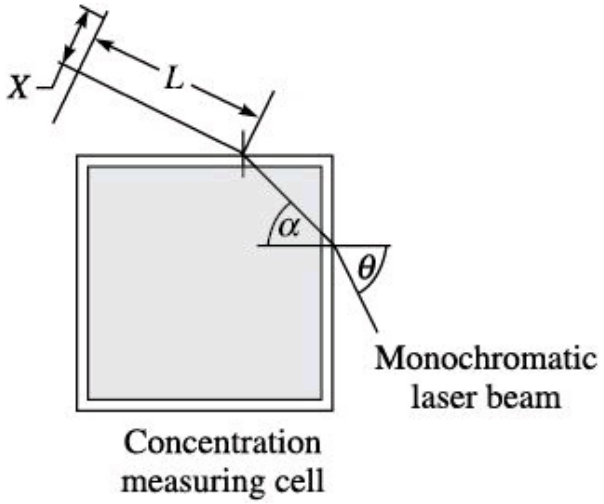


Figure P7.2 Concentration measurement based on refractive index.

Find the relation between solution concentration (C_H) and the lateral shift, x, of the Laser Beam. Assume q = 60º, and L = 10 cm. For a concentration change of 0 to 17.5% by weight, calculate the value of the lateral shift of Laser beam x.

7.4 The bulb of a Hg-in-Steel thermometer is inserted in a hot air flow line in bare condition. A lumped parameter first order TF may be assumed for the dynamics of the temperature sensor-indicator. Find the transfer function □(*s*)/T_I(*s*) for the system, where, $T_I(t)$ is the variable air temperature, and □(*t*) the angular travel of the pointer on the circular dial. The arrangement has been shown in Figure P7.3.

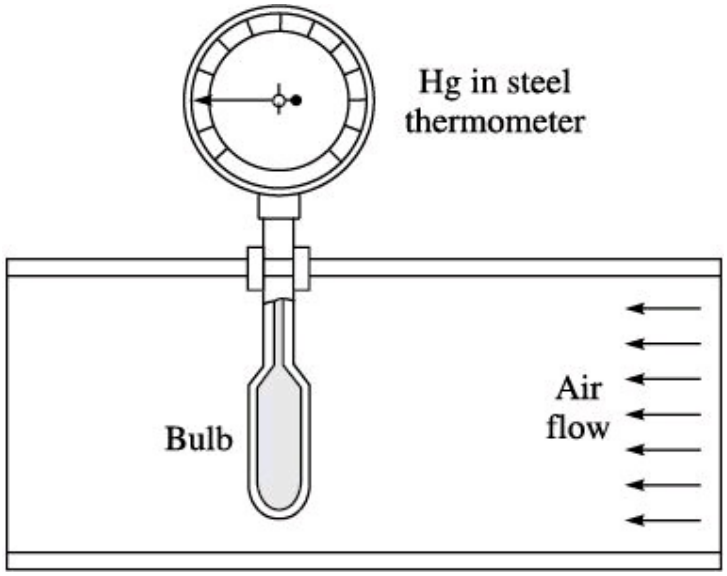


Figure P7.3 Temperature sensor indicator in a hot air pipe line.

Data: Constant air velocity, u = 100 cm/s, Mass of Hg, m_M = 10 gm, Air viscosity, μ_A = 0.01cp, Diameter of the bulb, D_B = 4 mm, Length of the bulb, L_B = 2 cm, Thermal conductivity of air, k_A = 0.0014 kcal/(cm² .s.°C/cm, Density of Air, r_A = 0.01 gm/cc,
0–150°C is marked on the dial through an angular span of 270°. The form of Dittus-Boelter equation for such a heat transfer situation may be taken as:

$$\frac{hD_B}{k_A} = 0.023\left(\frac{D_B u \mu_A}{\rho_A}\right)^{0.88}\left(\frac{Cp_A \mu_A}{k_A}\right)^{0.33}$$

We have neglected the viscosity correction term due to wall effect. Also, the value of N_{PR} may be taken as 1.0.

7.5 By using the optical refraction of a monochromatic laser beam, a liquid level sensor system has been depicted in Figure P7.4. Find out the value of incident angle a of the laser beam, if the working depth 600 cm of the tank. Level would be measured between a maximum level, L_2 and a minimum level, L_1 = 50 cm that will produce a lateral shift of the beam, D = 1.5 cm. Assume the stored liquid to be water having an RI, μ_W = 1.32. Find the value of L_2 in cm. Neglect the auxiliary refractive shift within the thin flat glass bottom of the liquid tank.

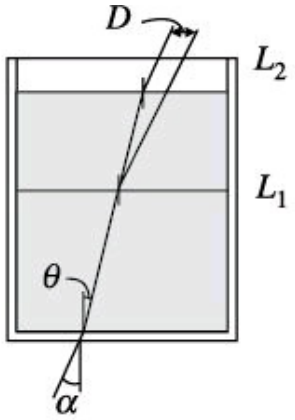


Figure P7.4 Level sensor based on optical refraction.

7.6 A pneumatic control valve (CV) is to be installed on a1/2" liquid fuel line. The effective diameter of its diaphragm is 150 mm. (a) Design the spring constant that will allow a 10 mm stem travel for a 3–15 psig pressure difference over the diaphragm. (b) Considering a first order dynamics of the valve (Equation 7.61), the process control loop design demands that valve response time ($5t_V$) should be 10 s. Calculate the frictional drag constant of the stuffing box (μ^*) in c.g.s. units.

7.7 A pneumatic pressure sensor has a first order dynamic response (K_M = 0.08 and

t_M = 8 s). The measured signal is sent to the control room through 1/4" pneumatic tubing at a distance of 600 ft. Calculate the response time of the sensor-transmitter for 99% step recovery. (Hint: Find the response time excluding the delay, by using second order response equation, then add the delay time.)

7.8 A constant holdup stirred liquid heater has an associated temperature control system featuring a Thyristor type FCE that uses 230 V, 1- □, AC power. The liquid, a petroleum derivative, flows at constant flow rate of 50 l/min. The inlet liquid temperature varies between 15ºC to 35ºC, and the desirable exit liquid temperature = 70ºC = Set Point of the controller. Calculate the Thyristor current and voltage rating, assuming the heat transfer process is 87% efficient and a design safety factor of 1.5. Data: Liquid density = 0.93 gm/cc.; Liquid *Cp* = 0.85 kcal/(kgºC). (Though the current calculation should be based on the line voltage, yet in a 230 V AC supply, a 400 V thyristor should be used due to the peak value consideration of an alternating voltage.)

REFERENCES

[1] Harriot, P., *Process Control*, McGraw-Hill Book Co., pp. 209, 1964.

[2] Bradner, M., *Pneumatic Transmission Lag*, Instruments, pp. 22, 7, 618, 1949.

[3] Ogata, K., *Modern Control Engineering*, Prentice-Hall of India, New Delhi, pp. 99, 1982.

8

Automatic Controllers

In this chapter we shall discuss about the dynamic characteristic of the control element of a control system. The basic block diagram of the closed loop system is again depicted in Figure 8.1 with the G_C labelled block drawn in bold to emphasize the topic of the present discussion. The controllers perform the duty of bringing a disturbed process back to the desired steady state condition by dynamically compensating the principal entity flow as the manipulation. In literature, controllers are often called the **Dynamic Compensators**.

Controllers are the brainy piece of instrumentation hardware that do all the calculation on information received from the process to figure out in what way it should be run to ensure optimum steady state operation condition.

Though external as well as certain internal features of the controllers have gone through several stages of change, still, certain basic computational and processing capabilities remain as they were about seventy years ago.

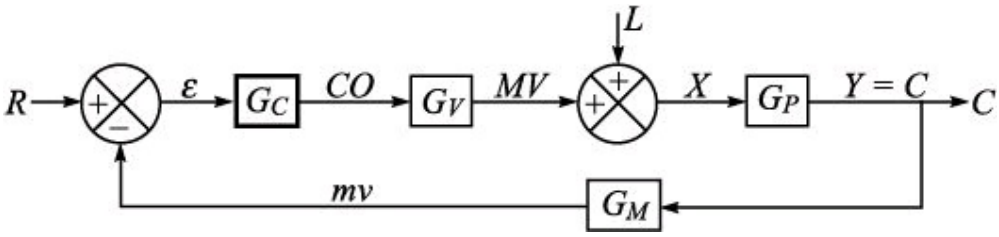


Figure 8.1 The location of the controller block within a feedback control loop.

During analysis, the following variables are used with the corresponding notations given in Figure 8.1:

Measured variable, mv; Set point, R; Error, e; and Controller Output, CO

In a feedback control loop, the controller is situated before the final control element and after the Comparator element, as shown in Figure 8.1. The duty of a controller is to receive the error variable, e, that is generated in the comparator according to:

$$e = R - mv \qquad (8.1)$$

and operate this error with respect to a particular analytic and/or logical function, $f(e)$. Controller output, CO, is the result of this operation.

$$CO = f(e) \qquad (8.2)$$

8.1 Fundamental Considerations

To appreciate the working automatic controller, certain basic feature of these

hardware should be exposed.

8.1.1 Controller Actuation

Controllers are the intelligent part of instrumentation systems with necessary amount of decision-making power that have been packaged after a careful design during their manufacture, and later fine-tuned prior to usage. But with all their intelligence, they can only produce an output signal that is basically weak in power. Hence, the feeble signal generated from a controller has to go through a power amplification stage for regulation of streams of high energy content, known as manipulated variables. The so called power amplifier is called **actuator** in control vocabulary. Thus, a final control element (FCE) comprises of an actuator and a throttling device at its front end for regulation of high energy streams. The actuators require some form of energy to perform their function. The energy source is usually supplied as a utility to the control system. Based on the form of energy the system utilizes, the controllers and the associated hardware may be classified into following categories.

1 . Pneumatic Systems: Probably the oldest variety, but still in use due to their inherent safety features and robustness. The utility is a dust, oil, and moisture free air supply at a constant pressure ranging from 20 to 25 psig. Generally all the connected hardware in a pneumatic control system has an input–output variable range of 3 to 15 psig. However, for higher power applications, auxiliary air supply at pressure in the range of 80 to 100 psig. are also used in the FCE of a control loop. For constructing pneumatic controllers and signal transmitters, designers use hardware units or basic building blocks like 'Flapper-nozzle' assembly, linkages of various configurations, levers, cam-followers, pulleys, gears, springs, cylinder-piston dashpots, bellows, diaphragms, etc.

2. Hydraulic Systems: These are used in systems where high power manipulation is required at the front end, e.g. in heavy duty crane, earth movers, material handling devices, industrial robotics, automobile and aircraft navigation. Special fluids having incompressible character and high boiling point, and suitable viscosity index are used for transmitting power in the final control element. Usually the power is available from hydraulic fluids stored at high pressure in storage tanks. The high pressure is created by +ve displacement pumps capable of producing high head. Supply conduits carry the fluid to the utilization points. The low-pressure return fluid goes back to another reservoir and subsequently goes through the pump for reuse. Usually the final control element produces displacement as the manipulated variable, and 'hydraulic servos' are applied to obtain output precision. For constructing hydraulic control systems, designers use hardware device units or basic building blocks like 'Piston-Cylinder' assembly, servos, various linkages, levers, cam-followers, pulleys, gears,

springs, bellows, diaphragms, etc. The pressure rating of such systems may range from few kg/cm^2, up to several hundred kg/cm^2.

3. Fluidic Systems: A fluid jet escaping at one end of a cavity has two possible outlets at the other end. Two auxiliary jets are constructed on two opposite sides of the cavity that may be used to issue any of the two auxiliary jets to impinge onto the main jet in perpendicular to its flow axis. When an auxiliary jet is applied, the main jet adheres to the opposite wall of the cavity and the outflow from the outlet on that side of the cavity carries the bulk of flow from the main nozzle, the other outlet become starved. A fluid jet has this special property that once it is adhered to a wall by an auxiliary jet application, it stays in that way until the opposite jet pushes it, and outflow shifts from the previous outlet to the other that was in the starved condition. This is called a 'Toggle' action. Such a fluidic assembly constitutes a by-stable or binary memory state and is used as the building block of fluidic systems to construct digital computing facilities required to develop control algorithms. (Ogata)**[2]**.

4. Electronic Systems: In recent times, they are probably the most widely used in process plant automation. Starting from the earliest vacuum tube devices to the modern integrated circuit based systems they have gone a number of evolutionary stages. The control algorithms are based on analog computation where an operational amplifier stands for the fundamental data processor that is configured to obtain addition, subtraction, multiplication, division, integration, differentiation, and comparator units to realize the desired control action. Other supporting components are resistors, capacitors, and inductor coils for properly biasing the amplifiers. The most widely used input and output variable range for electronic control system is 4 to 20 mA DC. Although other specifications are also used, e.g. 0 to 20 mA DC, 0 to 20 mV DC.

5. Digital Systems: This category of controllers basically falls within the class of electronic variety. However, they have grown a special class by themselves, because of the type of data processing that is carried within the hardware. Modular digital controller using certain commercial models of Micro-processors were available since mid-seventies, later came devices that used Micro-controllers—a system that is realized by custom built motherboards and limited I/O peripherals. The control programme is stored in memory by burning a ROM, thus making a dedicated device for the control problem to be addressed. The data communication is done through Analog to Digital Converter (ADC) for incoming process variable signal from measuring device, and by Digital to Analog Converter (DAC) modules for communicating CO to the FCE. The input and output signal range at the digital side of the hardware is 0 to 5 V DC. These controllers use a programme that is the

mathematical procedure written in sequential steps by using a language understandable to the operating system of the CPU. In earlier versions of the system, the programme had to be written in assembly language, even in mnemonic form, but later, modified versions became available that could operate in a PC platform using the environment of Real Time Operating System (RTOS). Programming language compatibility includes high level languages like FORTRAN, BASIC, C++, etc.

8.1.2 Saturation of Controller Output

In controller operation there is always a maximum and minimum value that the CO cannot cross in either way. This is called the output range of the controller. In pneumatic and electronic controllers, the typical values are 3–15 psig air pressure and 4–20 mA DC, respectively.

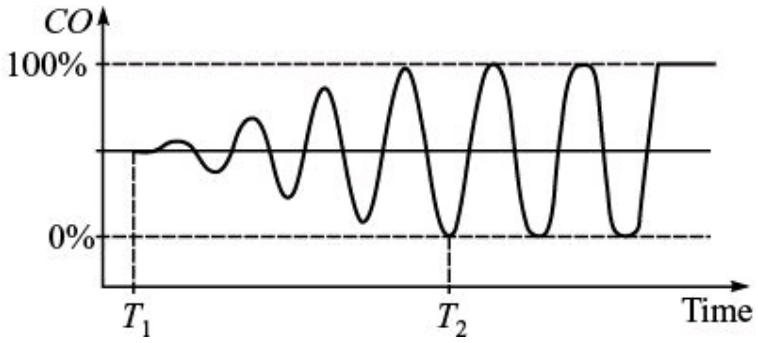


Figure 8.2 Controller output saturation.

We know that the controller performs a mathematical operation on errors defined by Equation 8.2, and if the error at some point of operation becomes too high or low, so that mathematically CO becomes more or less than the higher or lower bound of its output limits, the CO value will physically still remain at its limiting values while mathematically CO remains beyond the limits. During these periods, the controller output is said to be saturated. Figure 8.2 illustrates the saturation phenomena.

8.1.3 Control Functions and Mode

In a controller the type of mathematical operation or **function** that is performed on the error is known as **Mode** of the controller. Such a mode may consist of a single, or more than one function acting together. In the following section, we shall describe these basic **control functions**. In many texts, the control functions have been alternatively termed as action. However, to avoid any confusion with the terms **Direct** and **Reverse Action,** a very important controller parameter, we shall use the term 'control function' to describe these individual basic mathematical operations. In the next section, we are going to discuss these functions as they appear in certain ideal controller transfer functions to construe the controller mode.

1. Proportional (P) Function: The proportional function (or action) multiplies the

error, e, by a constant, usually called the **controller gain**, K_C, to produce the output. K_C is also known as the parameter of the (P) control function. Hence, Equation 8.2 becomes,

$$CO = K_C \varepsilon(t) \qquad (8.3)$$

2. Integral (I) Function: This function (or action) integrates the error variable with respect to time for generating the output, according to,

$$CO = \frac{1}{\tau_I} \int_0^t \varepsilon(t) dt \qquad (8.4)$$

The constant, t_I, that appears as reciprocal before the integral, has dimension of time and is called the **integral time.** t_I is also known as the parameter of the (I) control function.

3. Derivative (D) Function: The derivative function (or action) differentiates the error variable to produce the output according to the following relation:

$$CO = \tau_D \frac{d\varepsilon(t)}{dt} \qquad (8.5)$$

The constant, t_D, that appears before the derivative term has dimension of time and is called the **derivative time.** t_D is also known as the parameter of the (D) control function.

4. On-Off Action: This is a logical function that may be described as:
(i) If $mv < R$ (SP) or, $e > 0$, then CO = 100% (full value/fully open)
(ii) If $mv > R$ (SP) or, $e < 0$, then CO = 0% (zero value/completely closed)

To understand the dynamics of the first three control functions, their responses for ideal error input functions have been illustrated in Figure 8.3. The dynamic behaviour of On-Off control function is discussed latter separately in the Section 8.5.

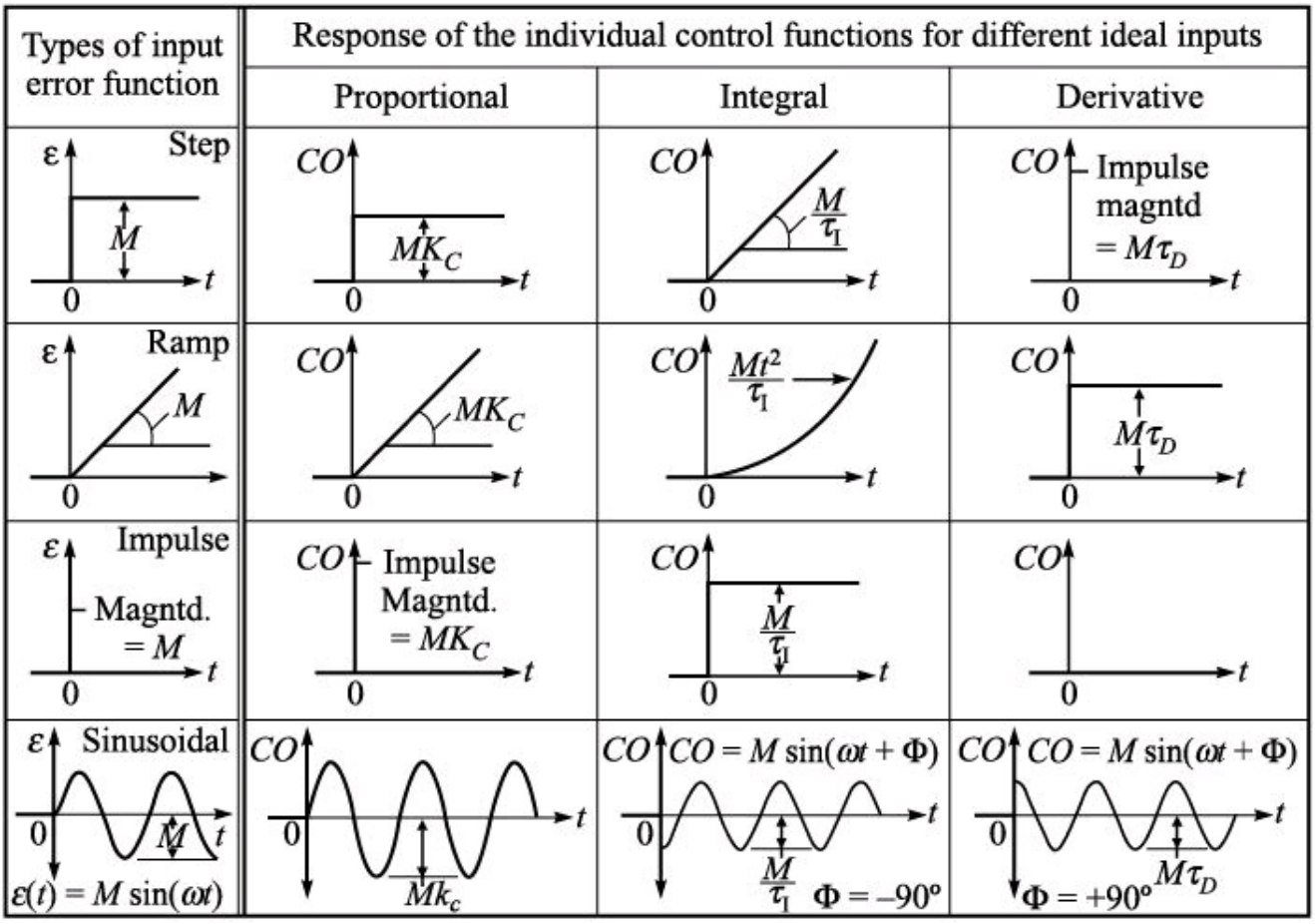


Figure 8.3 Response of controller function output CO(t) for different types of input error functions.

In commercial controllers often more than one control function exist. To have a qualitative idea of the fundamental dynamics of a multiple function/mode controller, we only need to select the individual responses given in the row for the given error type of Figure 8.3 and add them by superposition. This is possible because the control functions are linear and they act parallely. As an example, we may obtain the fundamental response quality of a proportional plus integral function controller, for step type error input, by graphically adding the second and third figures in the first row of Figure 8.3.

A controller with only proportional function acts as a multiplier of the input variable to produce the controller output. This property enables the operator of a plant, which is running close to the desired steady state, to bring the controlled variable exactly at the set point by adjusting the controller gain, K_C. But now in this new error free steady state of operation if the process is disturbed, the controller with the previously adjusted value of K_C, cannot bring the controlled variable back to the set point value. There will always be an amount of error after the process reaches the next steady state condition. This steady state error is called the '**Off-set**'. The analytical treatment of off-set has been dealt in Chapter 9. A physical explanation of the off-set is provided in the following section on Proportional Mode controller.

Proportional function is singly used in Proportional (P) Mode controller. This will be discussed in the next section.

After a disturbance has shifted the controlled variable from the set point, the integral function is used for bringing the controller output again back to the error free or zero offset steady state condition. From Figure 8.3, it is evident that for any magnitude of error remaining in the response, the integral action will keep on changing the controller output (CO), at a rate proportional to the error magnitude. That is, the control valve (final control element) stem will travel with a speed proportional to the amount of error present at that instant in a direction, determined by the algebraic sign of error, $e(t)$. Due to the –ve feedback logic of the loop, the stem travel will be in the direction required for reducing the error.

The coefficient, $1/t_I$, before the integral in Equation 8.4 determines the magnitude of the integral function. For a large value of $1/t_I$, the integral function may carry the controlled variable beyond the zero error point. Thus, error would appear again but with a reversed sign. Eventually, integral function will bend the response back towards the set point. This process will continue until the process output is brought back to zero error steady state condition. Thus, integral function may cause oscillatory response of the controlled variable. This is just like having an inductance element in the system. Too large a value of the coefficient may cause a highly oscillatory response, that requires a long time to settle, or the oscillations may even grow in amplitude and drive a control system towards an unstable condition. This problem will be addressed during the discussion on controller tuning.

In many literatures, the derivative function has been called as an '**anticipatory**' action. This differentiating property, in a slowly changing error situation, will produce a controller output proportional to the slope of the error variable. That is, at a certain positive value of time (in future) the error may become relatively large in magnitude, and the derivative action will take a preventive measure by producing a controller output proportional to the rate of error change. This is equivalent of taking action beforehand, because the slope of a variable is an indication to what it will become in magnitude in the future. Therefore, in (D) function, there is an essence of prophecy about what will happen in future, and hence, there is ample reason for calling the derivative function as '**The Prophet**'.

In comparison to this, the integral function 'sums up' or 'collects' the error values from the past up to the present, and produces the corrective action based on this sum of error values. In other words, integral action is realized by the past behaviour (history) of the error function and thus we may call integral function by the name '**The Historian**'.

We shall see in the following discussion that in real controllers the integral and derivative functions are present in combination with proportional function, because such combinations offer more compensation features required for a large variety of

process systems.

8.2 Controller Mode

With so much of an introduction, we may move on to the ideal version of controller as they exist in industrial application. These controllers have a couple of extra parameters that make it possible to convert the mathematical intention, of Equation 8.2 into practical reality. These parameters are **Output Bias** and **Action** of a controller.

8.2.1 Output Bias

This is the controller output value when it is running under zero error input. It is denoted in this literature as OPB. OPB is generally expressed as percent full-scale of controller output, and acts somewhat like an 'intercept' in the error versus controller output equation and plays an important role during controller tuning for a non-linear process. We shall discuss more about this topic later, through some examples. For the present, it is sufficient to consider this as a constant appearing in the controller output equation. This parameter may be adjusted from the controller front panel.

$$CO = \text{OPB} + f(e) \qquad (8.6)$$

We will discuss the significance of OPB selection by considering two closed loop processes. Both of them have a similar block diagram description as given in Figure 8.1. Among them, process A has a linear relation between manipulated variable (MV) and control variable (C), and a corresponding non-linear relation exist for process B. These process characteristics have been illustrated in Figures 8.4(a) and (b). In these figures, we have drawn the error scale adjacent to the *C*-scale, in which, at the zero error value, $C = R$ (the set point).

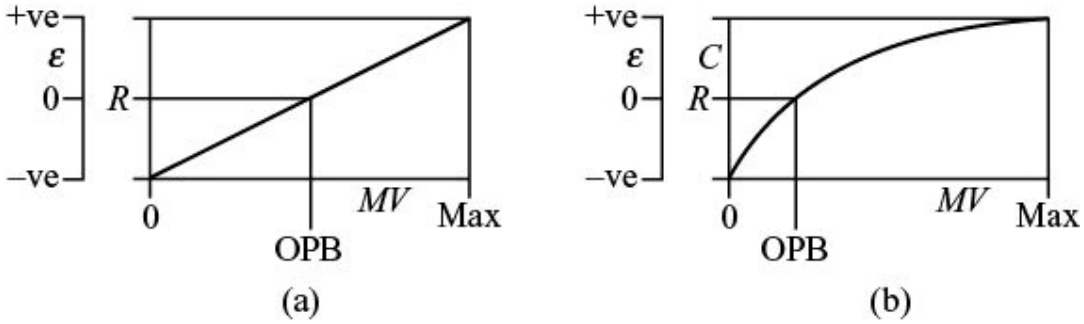


Figure 8.4 Selection of output bias. (a) for a linear process.(b) for a non-linear process.

A good design of the final control element with adequate energy or mass transfer capacity should be capable of manipulating *C* by equal magnitudes above and below the set point for the full range variation of *MV* for compensating equal magnitudes of +ve and –ve error. This fact has been indicated in the *e*-scale drawn parallel and adjacent to *C*-scale in Figures 8.4(a) and (b). If we know the relation between *C* and *MV*, the *MV* corresponding to mid-point of range of *C* should give us the OPB value.

This value of OPB would be expressed as % full scale of *CO*.

From Figures 8.4(a) and (b) it is understood, that for a linear process, the OPB will be 50%, but for a non-linear process of the type shown in Figure 8.4(b), the OPB value should be less than 50%.

8.2.2 Action

The idea about action of a controller may be understandable by considering the following examples of level control of a liquid reservoir illustrated in Figures 8.5(a) and (b).

In the control system of Figure 8.5(a), the load or disturbance is the variable demand of a down stream equipment hence, exit flow is the load variable. The level control is realized by the manipulation of the inflow to the reservoir.

While in the control system of Figure 8.5(b), the load or disturbance is the variable inlet liquid rate to the reservoir, the level control is realized by manipulation of the exit flow from the reservoir.

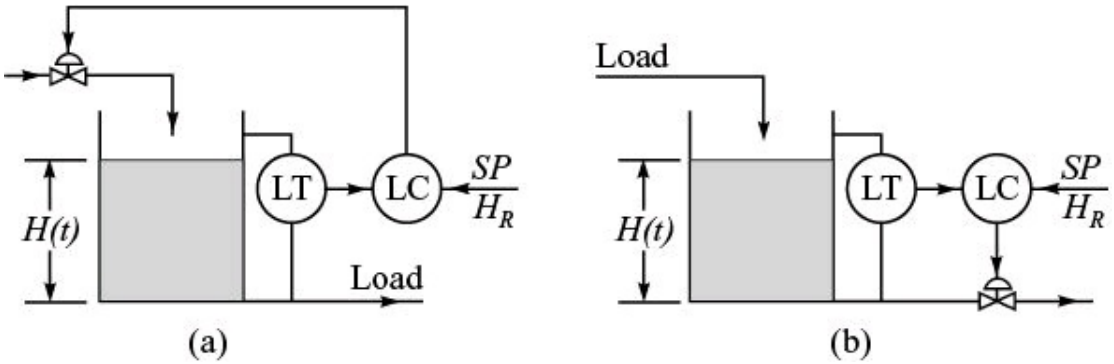


Figure 8.5 The two possible configurations of the level control of a liquid reservoir.

In the first system description of Figure 8.5(a), for a +ve value of error, that is, measured level being less than the set point, the manipulated variable (inlet flowrate) has to be increased or controller output has to move toward higher values (in +ve direction), to increase the control valve opening.

In the second system description of Figure 8.5(b), for a +ve value of error that is measured level being less than the set point, the manipulated variable (exit flowrate) has to be decreased or controller output has to move towards lower values (in –ve direction), to decrease the control valve opening.

If the same controller is employed for both of these two systems, it should be equipped with another parameter, i.e. a multiplier of the variable part of the controller output Equation 8.3, that may have values either +1 or –1. This parameter is called the **Action** of the controller and denoted by ACN. In a direct action controller, ACN = 1, and in a reverse action controller, ACN = –1. Thus, after addition of this parameter, Equation 8.6 becomes:

$$CO = OPB + (ACN)f(e) \qquad (8.7)$$

A direct action controller can be reversed by choosing a final control element, say, the control valve with reverse action. That is we select an 'Air-to-Close' action valve instead of an Air-to-Open'.

8.2.3 Modes of Practical Controllers

After this necessary detour, we recommence our discussion of mode of a controller. Mode is the most important characteristic of a controller that determines its dynamic response quality and suitability for a given control duty. In fact, the mode constitutes the type of dynamic compensation a controller is going to offer to a process in its transient state. We shall first examine the control action produced by each of these modes. In industrial controllers, these modes occur single and in combination, thus they are called single or mixed mode controllers. There are four basic modes available in commercial controllers.

Proportional (P) Mode

We already know the behaviour of proportional function. With introduction of action and output bias in Equation 8.3, the dynamics of a (P) mode controller may be expressed as:

$$CO = \text{OPB} + (\text{ACN})K_C e \qquad (8.8)$$

The corresponding transfer function would be:

$$\frac{CO(s)}{\varepsilon(s)} = (\text{ACN})K_C \qquad (8.9)$$

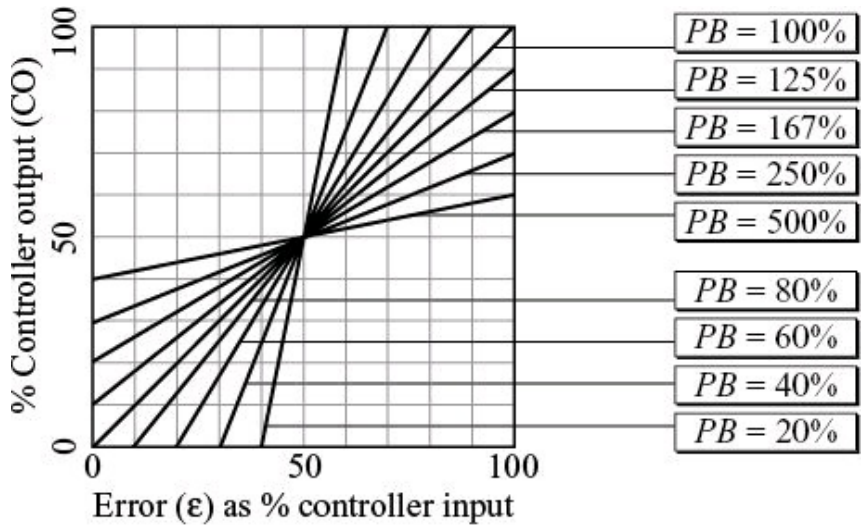


Figure 8.6 The definition of proportional band.

Controller gain, K_C, is the controller parameter, which may be adjusted in the field or off-line condition. In many commercially available controllers for realizing the (P) action, the proportional gain, K_C, is alternatively expressed as Proportional Band (PB). PB is usually expressed in percent, as % PB. Percent PB may be defined as "The value of error in % of its full scale that produces a 100% controller output change".

The idea of *PB* as a relation between error and controller output should become clearer from Figure 8.6. We know that K_C for any system is equal to the ratio of output change to the input change at two subsequent steady states. Thus, for a (P) controller,

$$K_C = \frac{\Delta CO}{\Delta \varepsilon} \qquad (8.10)$$

and from any curve of Figure 8.6, the interrelation between PB and K_C, may be readily obtained as:

$$\text{PB} = \frac{100}{K_C} \text{ ; and, } K_C = \frac{100}{\text{PB}} \qquad (8.11)$$

To demonstrate the definitions of PB and K_C in an application let us consider the following example.

EXAMPLE 8.1 A pneumatic proportional controller has input–output range of 3 to 15 psig air pressure. A temperature-measuring element installed in the process produces 0.75 psi output for a change of 1ºC. It has been observed that for 10ºC change in the process side, there is a
12 psi change of controller output. Calculate the K_C and % PB of the (P) controller.

Solution: The controller input (i.e. the sensor output) changes 0.75 psi/ºC.

Assuming constant set point, for a 10ºC change of measured value, the controller output changes full scale of 12 psi.

For 10ºC input change, the sensor output will change by 10 × 0.75 = 7.5 psi.

Thus, K_C = 12/7.5 = 1.6 (dimensionless).

Now 7.5 psi is (7.5 × 100)/12 = 62.5% of full scale. That is, 62.5% change in input produces 100% controller output change (12 psi = 3 to 15 psig), thus, PB = 62.5%. (You may check the result by applying Equation 8.11.)

Proportional + Integral (PI) Mode

The input–output relation of a (PI) controller may be obtained by simply adding integral function (Equation 8.4) to the proportional function (Equation 8.3). Adding action and output bias to this combination, as in Equation 8.8, the output equation of a (PI) mode controller becomes:

$$CO = \text{OPB} + \text{ACN}\left[K_C\varepsilon(t) + \frac{K_C}{\tau_I}\int_0^t \varepsilon(t)dt\right] \qquad (8.12)$$

In the transfer function form, Equation 8.12 will become:

$$\frac{CO(s)}{\varepsilon(s)} = (\text{ACN})\,K_C\left[1+\frac{1}{\tau_I s}\right] \qquad (8.13)$$

In a (PI) mode controller, the integral function is added to the proportional function primarily to have an off-set free or zero error steady state output. If a slightly sluggish closed loop response is allowable and the process does not contain any large lag elements (multiple *RC*s or dead time), a (PI) controller may be used to obtain error free response. To understand the dynamics of such a controller with direct action, we have presented the response equations for different error inputs in Table 8.1, and illustrated the response in Figure 8.7.

Table 8.1 Response Equations of a Direct Action (PI) Controller to Error Input Functions

Type of Error Input	**The Direct Action (PI) Controller Response, for $t > 0$**
Step: $e(t) = M$; for $t \geq 0$ $e(t) = 0$; for $t < 0$	$CO = \text{OPB} + \text{ACN}\left[MK_C + \frac{MK_C t}{\tau_I}\right]$ (8.14)
Ramp: $e(t) = Mt$; for $t \geq 0$ $e(t) = 0$; for $t < 0$	$CO = \text{OPB} + \text{ACN}\left[MK_C t + \frac{MK_C t^2}{2\tau_I}\right]$ (8.15)
Impulse: $e(t) = dM$; at $t = 0$ $e(t) = 0$; for $t < 0$, and $t > 0$	$CO = \text{OPB} + \text{ACN}\left[MK_C\delta(t) + \frac{MK_C}{\tau_I}\right]$ (8.16)
Sinusoidal: $e(t) = M \sin wt$; for $t \geq 0$ $e(t) = 0$; for $t < 0$	$CO = \text{OPB} + \text{ACN}\left[\frac{MK_C}{\omega\tau_I}\sqrt{1+\omega^2\tau_I^2}\sin\left\{\omega t - \tan^{-1}\left(\frac{1}{\omega\tau_I}\right)\right\}\right]$ (8.17)

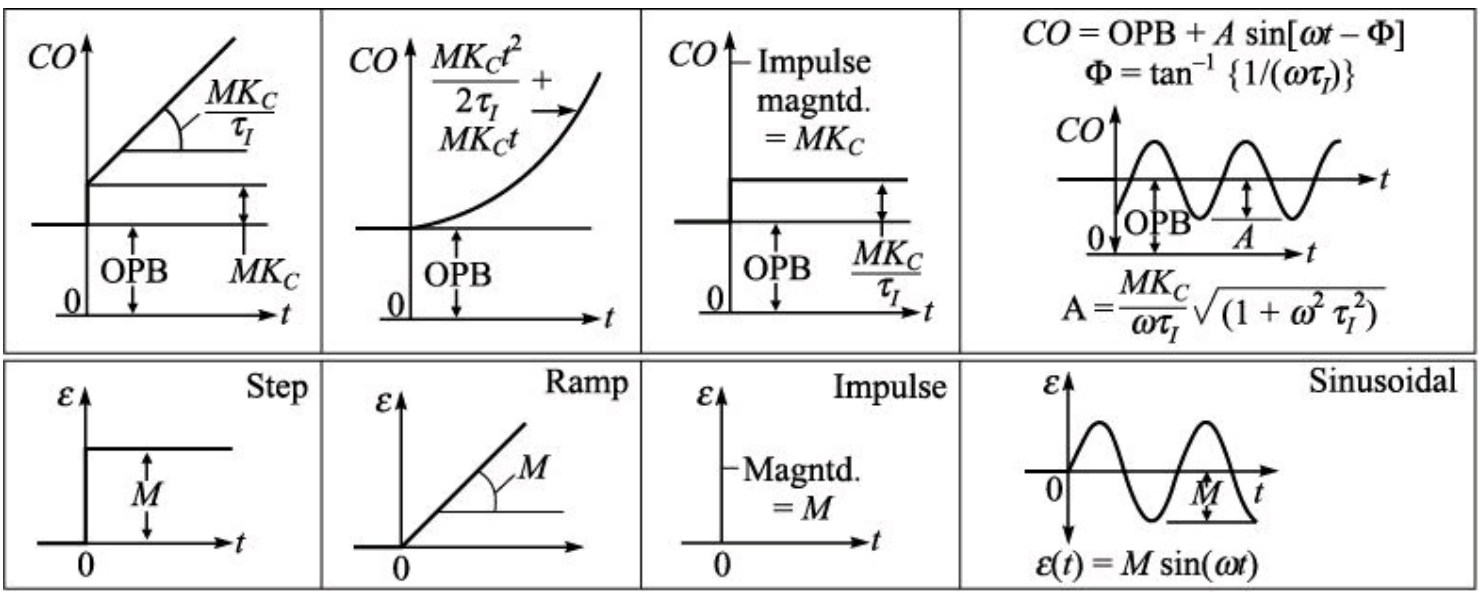


Figure 8.7 The response of an ideal PI controller for input error functions.

The **Step** Response shows a step rise due to (P) mode, followed by a linear change due to (I) mode, in the top sketch of the first column, Equation 8.14.

The **Ramp** Response shows a non-linear change comprising of a linear function due to (P) mode, and a parabolic (quadratic) function due to (I) mode, in the top sketch of the second column, Equation 8.15.

The **Impulse** Response shows an Impulse at $t = 0$, due to (P) mode, and for $t > 0$ a

step or constant change due to (I) mode, in the top sketch of the third column, Equation 8.16.

The **Sinusoidal** Response shows a sine wave of the same frequency *w*, but in phase it will 'lag' the input sinusoid due to (I) mode, the top sketch of the fourth column, Equation 8.17. The expressions for 'amplitude ratio' and 'phase lag' are given by:

$$\text{Amplitude Ratio} = \frac{K_C}{\omega\tau_I}\sqrt{1+\omega^2{\tau_I}^2} \qquad (8.18a)$$

and,

$$\text{Phase Angle, F} = -\tan^{-1}\left(\frac{1}{\omega\tau_I}\right) \qquad (8.18b)$$

Let us do an example.

EXAMPLE 8.2 An electronic (PI) controller output range of 4 to 20 mA DC. Initially the controller output (CO) was 16 mA, while the error, *e*, was zero. At $t = 0$, *e* suddenly became 4 mA, according to a step change. The controller output response was recorded and given in the following figure. Find: the action, % output bias, and parameter values of the controller.

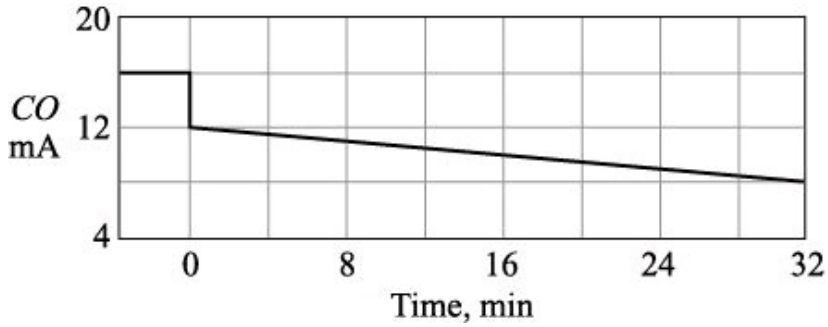


Figure 8.8 The open loop (PI) controller response for a step change in error.

Solution: For a +ve increase of error, controller output goes to lower values, thus this is a reverse action controller, i.e. ACN = –1.

Initially in the zero error input condition, the controller output was 16 mA. Hence, the output bias in % will be, OPB = [(16 – 4)/(20 – 4)] × 100 = 75%.

Applying –ve action to the step response Equation 8.14 of a PI-controller, the sudden change in output is due the proportional function only, i.e. (ACN) $MK_C = -4$ mA, where,

M = 4 mA = the step magnitude of error. Hence, $K_C = 1$. Also, the –ve slope of linear part of the response is (ACN MK_C)/t_I. Slope of this portion is –4/32 = –0.125 mA/min, and this equals to (ACN) MK_C/t_I.

or $MK_C/t_I = 0.125$ or (4 mA) × $1/t_I = 0.125$ mA/min

Thus, $t_I = 4/0.125 = 32$ min.

Proportional + Derivative (PD) Mode

The (PD) mode controller equation may be obtained by combining equations 8.3 and 8.5, and adding action and output bias to this combination.

$$CO = \text{OPB} + \text{ACN}\left[K_C\varepsilon(t) + K_C\tau_D\frac{d\varepsilon(t)}{dt}\right] \qquad (8.19)$$

In the transfer function form Equation 8.19 would become:

$$\frac{CO(s)}{\varepsilon(s)} = (\text{ACN})K_c\left[1 + \tau_D s\right] \qquad (8.20)$$

To understand the dynamics of such a controller we have presented the response equations for different error inputs in Table 8.2, and illustrated the responses in Figure 8.9.

Table 8.2 Response Equations of a Direct Action (PD) Controller to Error Input Functions

Type of Error Input	**The Direct Action (PD) Controller Response, for $t > 0$**
Step: $e(t) = M$; for $t \geq 0$ $e(t) = 0$; for $t < 0$	$CO = \text{OPB} + \text{ACN}[MK_C + MK_C t_D d(t)]$ (8.21)
Ramp: $e(t) = Mt$; for $t \geq 0$ $e(t) = 0$; for $t < 0$	$CO = \text{OPB} + \text{ACN}[MK_C t + MK_C t_D]$ (8.22)
Impulse: $e(t) = dM$; at $t = 0$ $E(t) = 0$; for $t < 0$, and $t > 0$	$CO = \text{OPB} + \text{ACN}[MK_C d(t)]$ (8.23)
Sinusoidal: $e(t) = M.\sin wt$; for $t \geq 0$ $e(t) = 0$; for $t < 0$	$CO = \text{OPB}+\text{ACN}\left[MK_C\sqrt{1+\omega^2\tau_D{}^2}\,\sin\left\{\omega t + \tan^{-1}(\omega\tau_D)\right\}\right]$ (8.24)

The **Step** Response shows an Impulse at $t = 0$, due to (D) mode, and for $t > 0$ a step or constant change due to (P) mode, in the top sketch of the first column, Equation 8.21.

The **Ramp** Response shows a step rise due to (D) mode, followed by a linear change due to (P) mode, in the top sketch of the second column, Equation 8.22.

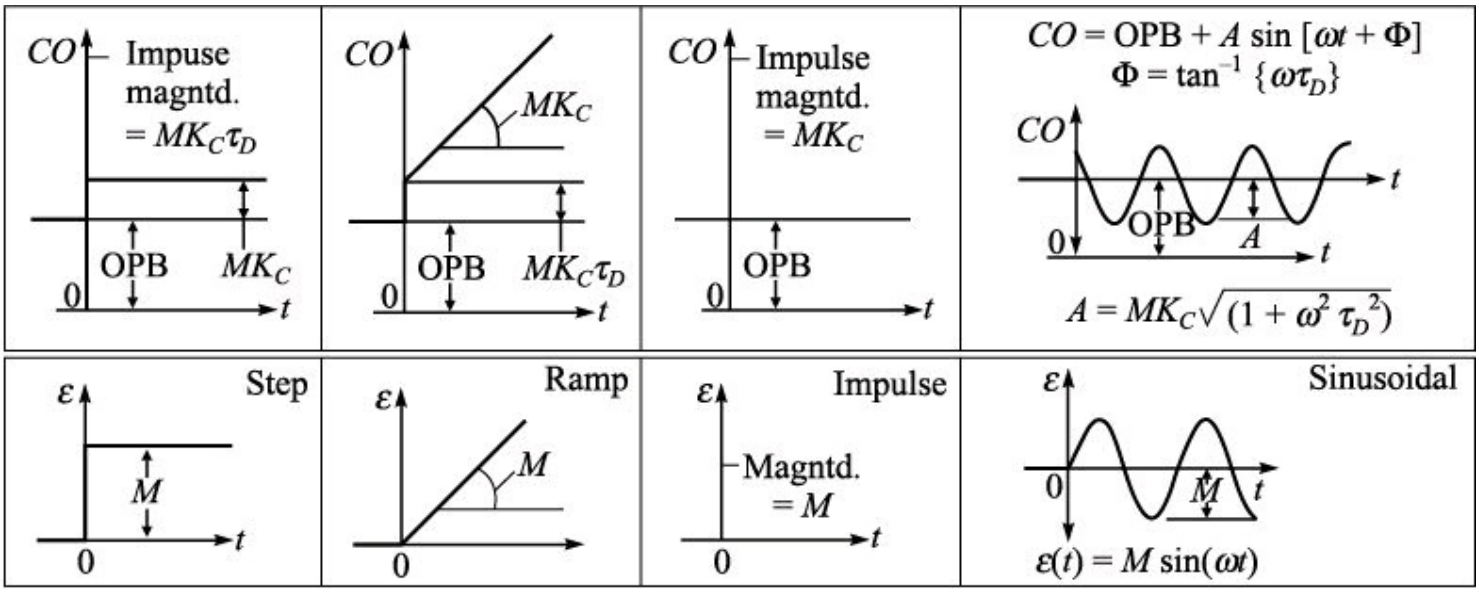


Figure 8.9 The response of an ideal PD controller for input error functions.

The **Impulse** Response shows only an Impulse at $t = 0$, due to (P) mode, but there will be no response for (D) mode because an impulse, when differentiated, obtains a zero output, as in the top sketch of the third column, Equation 8.23.

The **Sinusoidal** Response shows a sine wave of the same frequency, *w*, but in phase it will 'lead' the input sinusoid due to (D) mode, the top sketch of the fourth column, Equation 8.24. The expressions for 'amplitude ratio' and 'phase lag' are given by:

$$\text{Amplitude Ratio} = K_C\sqrt{1+\omega^2\tau_D{}^2} \tag{8.25a}$$

$$\text{Phase Angle, } \Phi = \tan^{-1}(\omega\tau_D) \tag{8.25b}$$

Application of (PD) mode is not very common in general process control practice. In closed loop systems where, in the feedback path, the measuring element block $G_M(s)$ can be represented by unity or by a gain K_M, this mode hardly improves the dynamic character of the loop response beyond a (P) mode application. However, when the $G_M(s)$ block has a recognizable dynamics, i.e.

$$G_M(s) = \frac{K_M}{\tau_M s + 1} \tag{8.26a}$$

or

$$G_M(s) = \frac{K_M e^{-\tau_M s}}{\tau_M s + 1} \tag{8.26b}$$

application of (PD) mode shows potential improvement of the loop dynamics. We shall discuss this topic in the next chapter.

[Equation 8.26a) becomes necessary to represent $G_M(s)$ for sensor-transmitter combinations that uses secondary or tertiary transducer stages for signal processing, and Equation 8.26b is required where combined to the above, the final transmitted variable, (that may be the pressure of a compressible fluid in case of pneumatic systems), has to travel a long distance between the plant and control station.]

EXAMPLE 8.3 A pneumatic controller has an input and output range of 3 to 15 psig air pressure. Initially the controller output (CO) was 6 psig, while the error, *e* was zero. From $t = 0$, *e* started to increase by 0.325 psi/min. The controller output response was recorded in Figure 8.6. Find: the mode, action, % output bias, and parameter values of the controller.

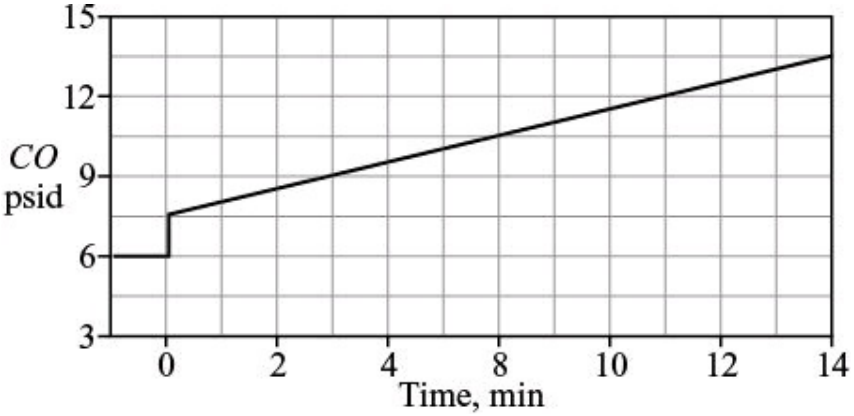


Figure 8.10 The open loop ramp response of a controller.

Solution For a +ve increase of error, the controller output goes to higher values, thus this is a direct action controller, i.e. ACN = +1.

Initially in the zero error input condition, the controller output was 6 psig. Hence, the output bias in % will be

$$\text{OPB} = \frac{6-3}{15-3} \times 100 = 25\%$$

The input is a ramp in error, with a slope, $M = 0.325$ psi/min. The response shows a step jump followed by a ramp. Examining Figure 8.3, we find that for a ramp input, the (D) function produces a step output, and the (P) function produces a ramp response. Thus, the controller mode is (PD).

The ramp part of the response has a slope = (13.5 – 7.5)/(14 – 0) = 0.42857143 psi/min. This value of the slope is equal to $M.K_C = 0.325K_C$. Hence, K_C = 1.318681319.

The magnitude of step part of response in Figure 8.10, $M.K_C.t_D = 1.5$ psi.

Hence, $t_D = 1.5/(MK_C) = 3.5$ min.

Proportional + Integral + Derivative (PID) Mode

The (PID) mode controller equation may be obtained by adding derivative function, i.e.

Equation 8.5 to Equation 8.1:

$$CO = \text{OPB} + \text{ACN}\left[K_C\varepsilon(t) + \frac{K_C}{\tau_I}\int_0^t \varepsilon(t)dt + K_C\tau_D\frac{d\varepsilon(t)}{dt}\right] \qquad (8.27)$$

In the transfer function form, Equation 8.2 will become:

$$\frac{CO(s)}{\varepsilon(s)} = (\text{ACN})K_C\left[1 + \frac{1}{\tau_I s} + \tau_D s\right] \qquad (8.28)$$

For a (PID) controller, the control action becomes faster than a (PI) controller, obviously due to the additional compensation of (D) function. To understand the

dynamics of such a controller, we have presented the response equations for different error inputs in Table 8.3, and illustrated the response in Figure 8.11.

Table 8.3 Response Equations of a Direct Action (PID) Controller to Error Input Functions

Type of Error Input	**The Direct Action (PID) Controller Response, for $t > 0$**
Step: $e(t) = M$; for $t \geq 0$ $e(t) = 0$; for $t < 0$	$CO = \text{OPB} + \text{ACN}\left[MK_C + \frac{MK_C t}{\tau_I} + MK_C \tau_D \delta(t) \right]$ (8.29)
Ramp: $e(t) = Mt$; for $t \geq 0$ $e(t) = 0$; for $t < 0$	$CO = \text{OPB} + \text{ACN}\left[MK_C t + \frac{MK_C t^2}{2\tau_I} + MK_C \tau_D \right]$ (8.30)
Impulse: $e(t) = dM$; at $t = 0$ $e(t) = 0$; for $t < 0$, and $t > 0$	$CO = \text{OPB} + \text{ACN}\left[MK_C \delta(t) + \frac{MK_C}{\tau_I} \right]$ (8.31)
Sinusoidal: $e(t) = M \sin wt$; for $t \geq 0$ $e(t) = 0$; for $t < 0$	$CO = \text{OPB} + \text{ACN}\begin{bmatrix} \frac{MK_C}{\omega\tau_I}\sqrt{1+\omega^4\tau_I^2\tau_D^2+\omega^2\tau_I^2-2\omega^2\tau_I\tau_D} \\ \sin\left\{\omega t + \tan^{-1}\left(\omega\tau_D - \frac{1}{\omega\tau_I}\right)\right\} \end{bmatrix}$ (8.32)

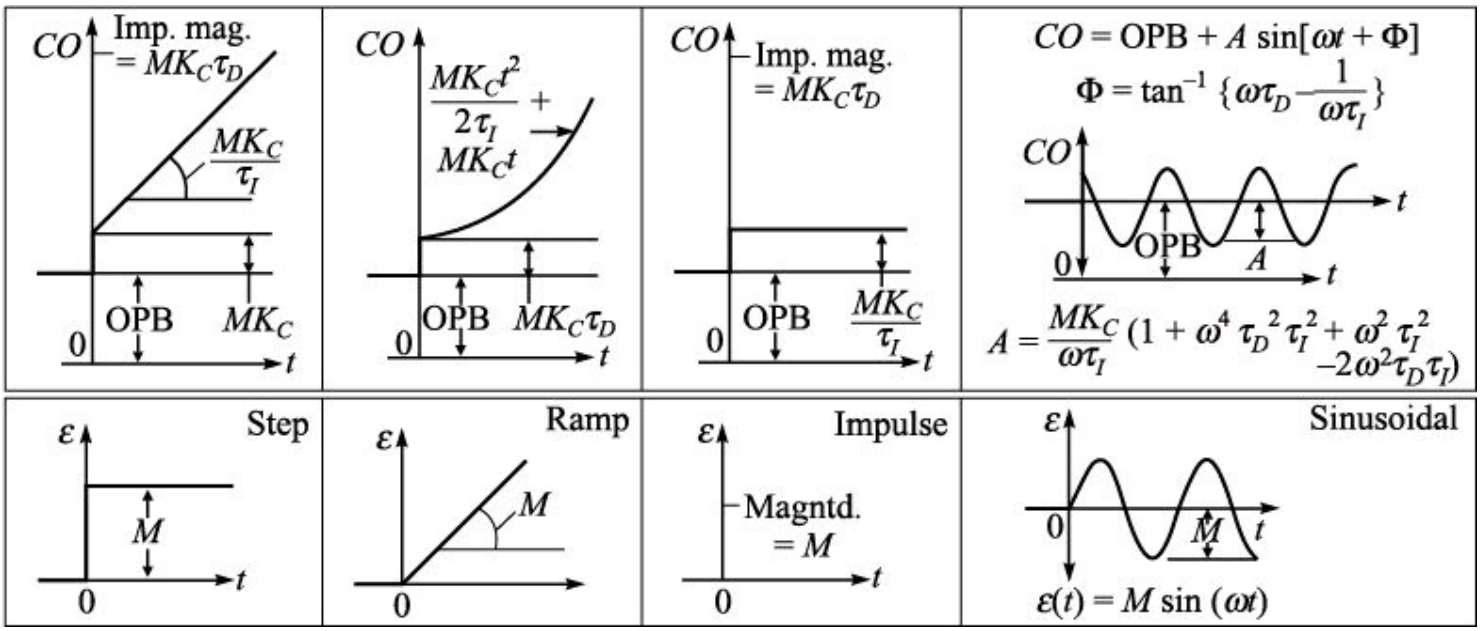


Figure 8.11 The response of an ideal PID controller for input error functions.

The **Step** Response shows an Impulse at $t = 0$, due to (D) mode, then a step rise due to (P) mode also at $t = 0$, followed by a linear change due to (I) mode, in the top sketch of the first column, Equation 8.29.

The **Ramp** Response shows a step rise due to (D) mode at $t = 0$, followed by a non-linear change, comprising a linear function due to (P) mode, and a parabolic (quadratic) function due to (I) mode, in the top sketch of the second column, Equation 8.30.

The **Impulse** Response shows an Impulse at $t = 0$, due to (P) mode, and a step rise also at $t = 0$, due to (I) mode, but there will be no response for (D) mode because an impulse when differentiated obtains a zero output, in the top sketch of the third column, Equation. 8.31.

The **Sinusoidal** Response shows a sine wave of the same frequency, *w*, but its phase

will depend on the difference between t_D and $1/t_I$. The phase will lead if this difference is a +ve quantity, and will lag if the difference becomes –ve. The three sinusoids due to (P), (I), and (D) modes may be coupled together to obtain Equation 8.32, and the response trace in the top sketch of the fourth column. The expressions for 'amplitude ratio' and 'phase lag' are given by:

$$\text{Amplitude Ratio} = \frac{K_C}{\omega\tau_I}\sqrt{1+\omega^4\tau_I^{\,2}\tau_D^{\,2}+\omega^2\tau_I^{\,2}-2\omega^2\tau_I\tau_D} \qquad (8.33a)$$

$$\text{Phase Angle, F} = \tan^{-1}\left(\omega\tau_D - \frac{1}{\omega\tau_I}\right) \qquad (8.33b)$$

EXAMPLE 8.4 If a sinusoidal change in error is impressed upon a (PID) controller, the proportional action will produce a sinusoid whose amplitude will only be affected but not the phase. The (I) and (D) functions will produce two sinusoids 'lagging' by 90º and 'leading' by 90º respectively from the input sine function. Find a relation between relevant controller parameters so that the response is a sinusoid with zero phase shift.

Solution: The (I) mode will produce a sine function with an amplitude = MK_C/t_I, with F = – 90º, and (D) mode will produce a sine function with an amplitude = MK_Ct_D, with F = +90º.

These should cancel each other, to obtain the response as a sine function due to proportional function only, that is, in phase with the input. Equating the above expressions for amplitude we obtain, $t_It_D = 1$, the required relation between controller parameters. The reader may confirm this result from Equation 8.33b.

8.3 Comments on the Closed Loop Characteristics of Controller Modes

In this section, we shall discuss the effect of mode of an existing controller in the loop, on certain general features of closed loop response. At present, we can only propose a simple qualitative exposure on this topic. More detail analysis on this matter is included in Chapter 9 on closed loop response of feedback control systems.

We have already mentioned that a proportional controller will always exhibit offset, or a steady state error in the closed loop response. A physical explanation of this comment may be put by considering a closed loop system with a (P) controller operating at steady state with zero error. At this condition by Equation 8.8, CO = OPB. Now if the set point is moved to a new value, and subsequently steady state condition is reached, the controller output, CO, cannot have the correct value to produce zero error condition in this new steady state, because OPB is a constant. Thus, a steady

state error (off-set) will result. The initial error free steady state may have been brought by manually adjusting the K_C, or OPB value.

With a (P) controller in the loop, the offset value reduces with increase of K_C or reduction of PB. However, by continuously increasing the K_C in this way, we cannot completely eliminate offset. Instead, at a certain value of K_C ($= K_{C3}$) initially the response will start with oscillation and then settle to produce a still lower value of offset. Further increase of K_C ($= K_{C5}$) may give forth an oscillatory response in which the oscillations grow in amplitude and drive the system to unstable condition. One important fact is that there is always a value of K_C ($= K_{C4}$) lying between K_{C3} and K_{C5} for which the response will oscillate with constant (undiminished) amplitude. K_{C4} is called the **Ultimate Gain, K_U**, of the control system. K_U has an important part in the tuning of control systems. These properties of a proportional controller have been illustrated in Figure 8.12, where the controlled variable response of a simple process has been considered under proportional control. The traces of the controlled variable have been obtained for five increasing values of K_C. The responses have been obtained for step change in Load (L) as well as Set Point (SP). The limiting value of controller gain ($K_U = K_{C4}$) remains almost unchanged both for load and set point excitations. For a practical design value of controller gain for a (P)-controller, we may use $K_C = K_{C3}$ to obtain a slightly oscillatory response and a large reduction in offset.

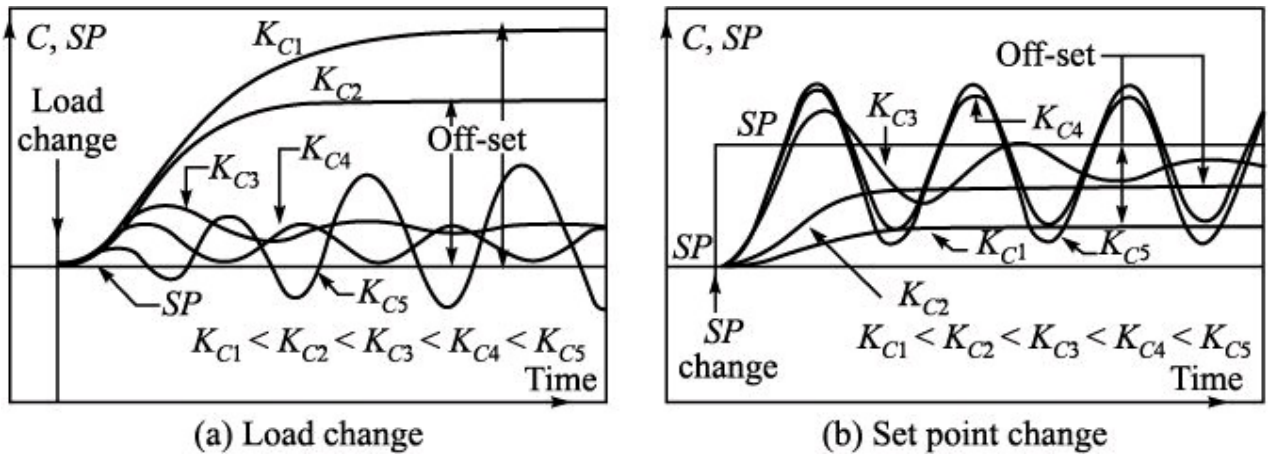
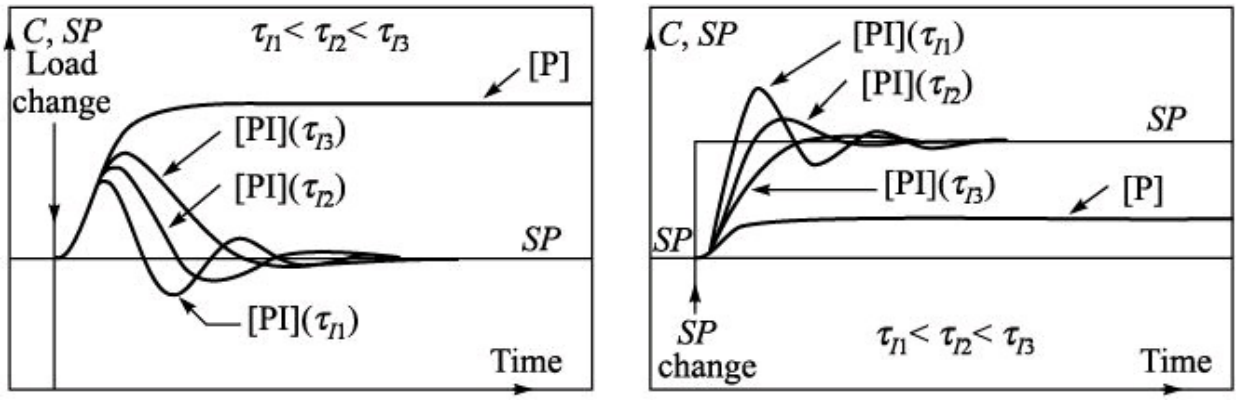


Figure 8.12 Effect of K_C value on closed loop response of a process with a (P) controller.

Note that in the dynamic equation of combined mode controllers, the coefficient term before the time integral, and the derivative term contain two independently adjustable parameters (K_C and t_I, and K_C and t_D). This facility is provided to have independent access to the individual actions of the controller, so that the weight or bias to individual actions becomes adjustable for obtaining an optimum controller performance.

The (I), and (D) modes are generally not used single in an individual controller. Because, integral function would only become effective when the error of a reasonable magnitude remains for some time within the loop, but proportional function will immediately take action as soon as the error signal is detected. On the other hand, for a very slowly changing error signal or when the error becomes steady at a constant value, the effect of a single existing derivative function will hardly take any effect, whatever be the error magnitude, but the proportional function will take an action corresponding to the magnitude of error.

We have also mentioned that Integral function is primarily used to have an error free steady state condition by reducing the offset to zero value. This is impossible with (P), or (D) functions. But integral function also tends to make the response oscillatory, which may pose stability problem to the loop. The (I) function generates a control action that is proportional to the magnitude of accumulated error at that instant. The proportionality constant being the coefficient K_C/t_I of the time integral of error in Equations 8.13 and 8.27. By this action, the process variable will move towards and eventually cross the set point value. At that instant, the controlled system will be at zero error state, but the accumulated error due to integral action may not be zero at the same time. In fact, due to the inherent 'lag' character of an integrator (pure capacity element), the '**error-integral-zero**' will always be behind in time relative to '**zero error point**'.



(a) Load change (b) Set point change

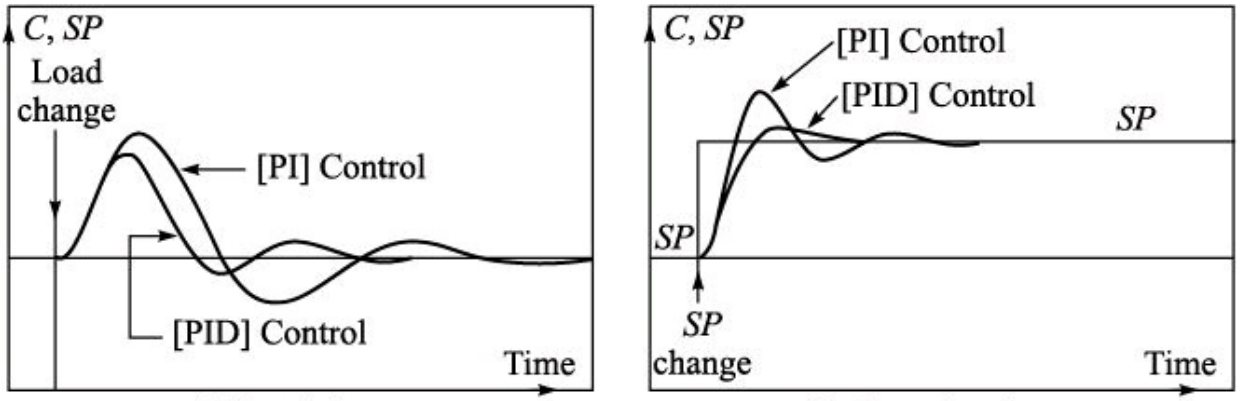
Figure 8.13 Comparison of closed loop response of a (PI) with a (P) controlled system, and the effect of integral time on (PI) response.

Hence, the coefficient, K_C/t_I of the integral term is tuned to a low value to make a compromise between speed and stability. Thus, with a tuned (PI) mode controller, we may obtain a slower loop response.

In Figure 8.13, the closed loop responses of a (PI) controller has been compared with a (P) controller, also the effect of increasing t_I of the (PI) controller on the loop response has been shown. From Figure 8.13, the integral function of the (P) controller may be tuned with a value of $t_I = t_{I2}$. This produces reasonably low overshoot,

smaller settling time, and zero offset. As the integral function imparts oscillatory character or decreases stability in the response, a (PI) controller is not suitable for systems having large lag in their dynamics.

Addition of (D) mode to a (P) controller simultaneously improves the response speed and dampens the oscillation of loop response. These features have been presented in Figure 8.14, for step change of load variable and set point. The increase of response speed may be attributed to the extra compensation due to (D) action while the process variable is changing. The reduction of oscillation is due to the phase lead character of derivative action, which counters the cause of its generation (just like the eddy current in a DC motor).



(a) Load change (b) Set point change

Figure 8.14 Effect of adding (D) function into a (PI) controller.

Let us look upon a (PD) controller as a (P) controller in which the (D) action has been added. An enhancement of loop stability will occur in such a (PD) controller when properly tuned due to the addition of a zero in the closed loop transfer function.

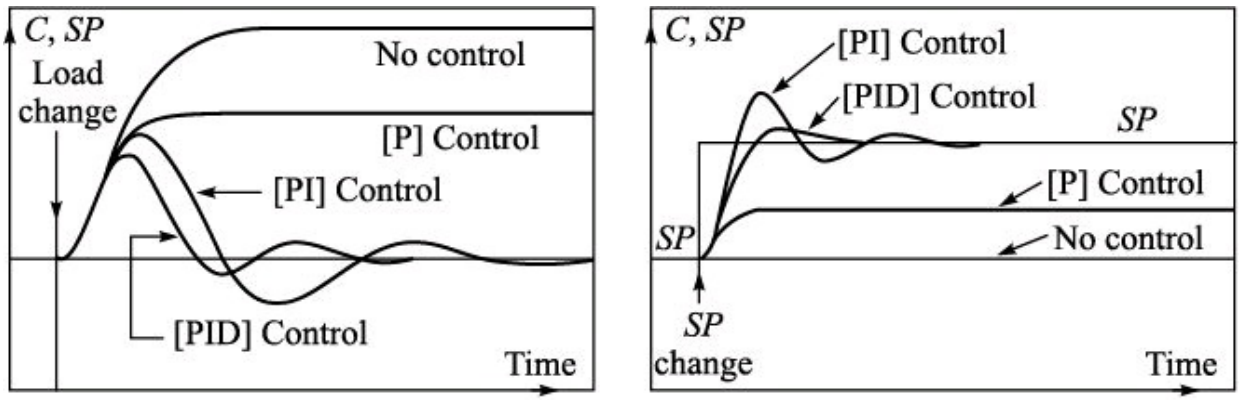
Compared to a (P) controller, the speed of response of a (PD) controller is higher due to the extra compensation of derivative action, but the magnitude of 'off-set' would not change. The contribution of derivative action only appears when the process output is changing, hence, at steady state or very near to the steady state, its effect will be zero or negligible. Thus, off-set, which is a steady state feature, will remain unchanged.

We may generalize the notions about the effect of adding (D) mode to a (P) and a (PI) controller in the following way. If simple proportional control gives a reasonably acceptable performance, we may try (PD) control. Similarly, we may try (PID) over (P). Obviously the additional stabilizing action due to addition of (D) function will allow us to use larger values of K_C and obtain a faster response.

One disadvantage about the (D) function is that if the measured process variable contains noise component, the derivative mode would amplify the small magnitude high frequency oscillations to saturate the controller output in both direction. That is, CO will have a variation from zero to maximum value in rapid succession, due to

alternative +ve and –ve high slope regions of the oscillatory component. Thus, in flow control loops, or in the loops where the measured variables are associated with random movement of particulate entities and/or high frequency electro-mechanical vibrations, use of (D) mode is normally prohibited. However, if such signals are 'filtered' or 'time-smoothened', the (D) mode may be used with proper precaution and care.

To conclude this discussion, we consider a process with feedback but initially without any controller, i.e. $G_C = 1$, then gradually add (P), (I), and (D) modes into the controller block. Improvement in the closed loop response for the 4-states of G_C block for load and set point change are recorded in Figures 8.15(a) and (b) respectively. The results agree with our foregoing discussion in this section.



(a) Load change (b) Set point change

Figure 8.15 Effect of addition of (P), (I), and (D) function to a no-control situation.

8.4 More Realistic Features of Industrial Controllers

At the beginning of this discussion, we should mention about a minor difference between the block labelled G_C that stand for the controller in the block diagram of Figure 8.1, and the hardware module sold commercially as controllers. This difference has been illustrated in
Figure 8.16.

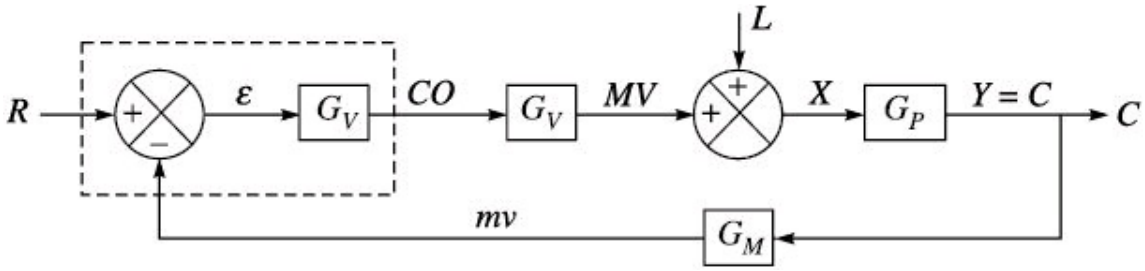


Figure 8.16 The difference in hardware of an industrial with a conceptual controller.

In an industrial controller, the controller block G_C, and the comparator unit are coupled together indicated by the dotted box in Figure 8.16. That is, such a unit would have two Inputs, e.g. R (set point), and mv (measured variable), unlike the block G_C, that has a single input of error, e. In industrial controllers, the error is calculated

within. However, they have a single output (CO).

In the following section, we shall discuss certain features of industrial controllers that are either alternative expressions of the controller parameters or additional facilities related to certain precautionary dynamic measures needed in an industrial application.

8.4.1 Reset Time, Time per Repeat, Repeats per Time

In commercial controllers, the integral and derivative functions are often called 'Reset' and 'Rate' actions, and the associated parameters are called 'Reset Time' and 'Rate Time'. Generally reset time is equivalent to integral time, t_I.

But some manufacturers use a term 'Times per Repeat' to express the reset time. This refers to a controller test procedure. A step input in error is impressed upon a (PI) or (PID) controller, the responses will have an initial step jump followed by a linear increase [we discard the initial derivative 'kick' due to (D) mode]. If the K_C value is adjusted to unity (PB = 100%), the time required by the integral function to produce an equal output to that of the proportional function is defined as the 'Repeat time' or 'Time per repeat'. With reference to Figure 8.17, T_{RP} is the time where the ramp part of the controller output due to (I) mode become same to the step part of the controller output due to (P) mode with unity controller gain (PB = 100%).

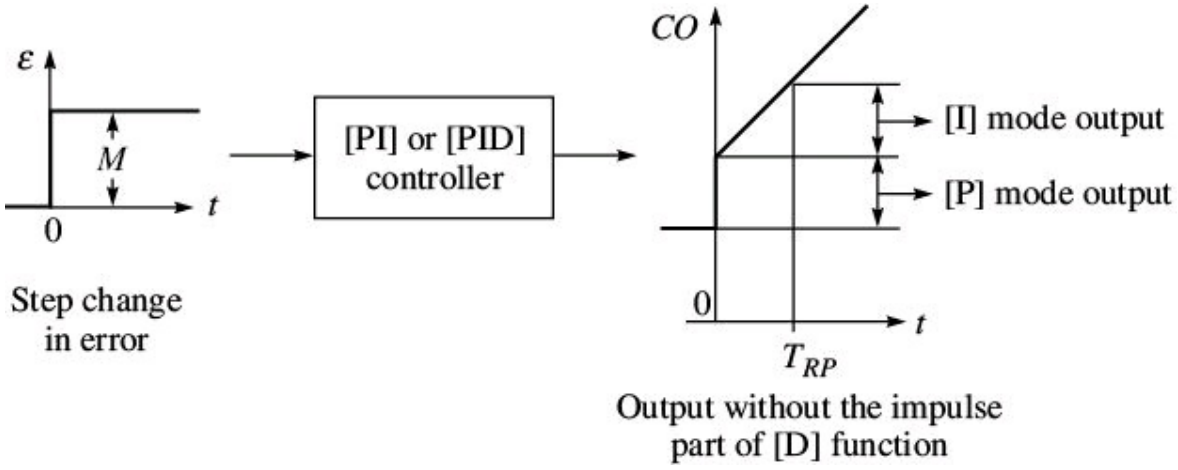


Figure 8.17 Definition of 'time per repeat'(T_{RP}).

A few of the manufacturers prefer to use another name for the same parameter 'Repeats per time', which is in fact reciprocal of time per repeat, and is more familiar because it proportionally expresses the part of the time integral multiplier, and directly reflects the weight or bias of the integral function on the controller output.

8.4.2 Rate Time

In common usage rate time is equivalent to derivative time, t_D, related to derivative function of a (PD) or a (PID) controller. In industrial practice, however, certain

manufacturers use the term Rate Time in somewhat parallel to the definition of time per repeat. This again refers to a controller test procedure. The test is done with an input error ramp function for a (PD) mode controller, and for the (P) and (D) modes together for a (PID) mode controller after making (I) mode inoperative by raising the integral time to a very high value. In such a condition, the response will initially show a step jump due to derivative function followed by a linear change due to proportional function. The K_C value is set to unity. In this condition, the time required by the proportional mode output to become equal to the derivative mode output is defined as the 'Rate Time'.

With reference to Figure 8.18, T_{RT} is the time where the ramp part of the controller output due to (P) mode become same to the step part of the controller output due to (D) mode with unity controller gain (PB = 100%).

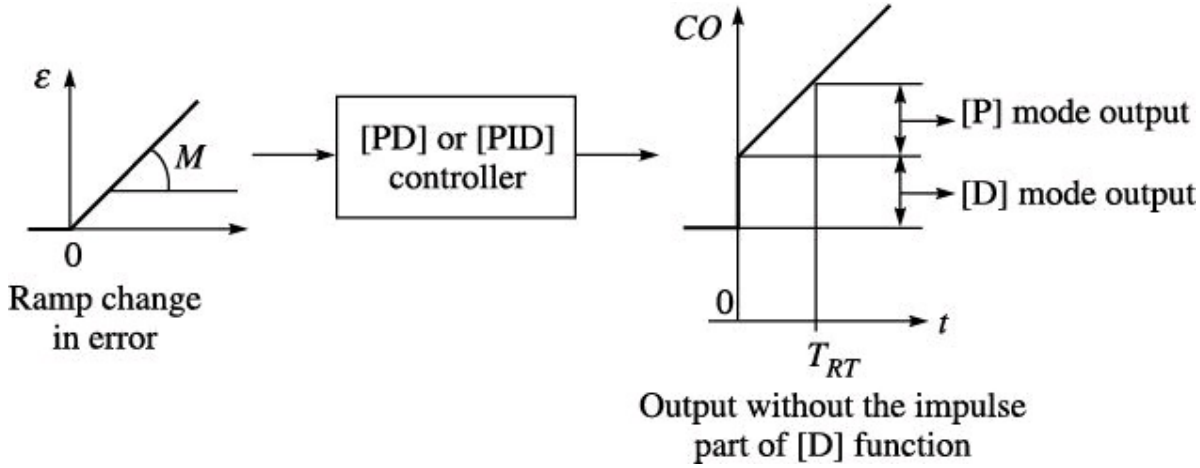


Figure 8.18 Definition of 'time per repeat'(T_{RT}).

8.4.3 Lead–Lag Element for Derivative Mode

Most of the industrial process controllers contain pneumatic or electronic hardware components. Although accurate (P) and (I) functions could be constructed using such building blocks, but realization of (D) mode is physically impossible with them. Due to this reason, the controllers that we have discussed so far are called as the ideal controllers, and the derived equations either in time or Laplace domain are referred to as Ideal Controller characteristics. Hence, with respect to (P) and (I) modes, the ideal and real controllers have no difference in construction, only the (D) function in an industrial controller is designed by the so-called 'Lead–Lag' element. This dynamic element is represented by the following transfer function:

$$\frac{Y(s)}{X(s)} = \frac{1+\tau_1 s}{1+\tau_2 s} \qquad (8.34)$$

The differentiating and integrating properties of a lead-lag element have been demonstrated in Example 5.12. The reader should revisit the referred matter. To

understand how this element can approximate the derivative function, we have illustrated the step response in the left and that of an ideal differentiator in the right part of Figure 8.19.

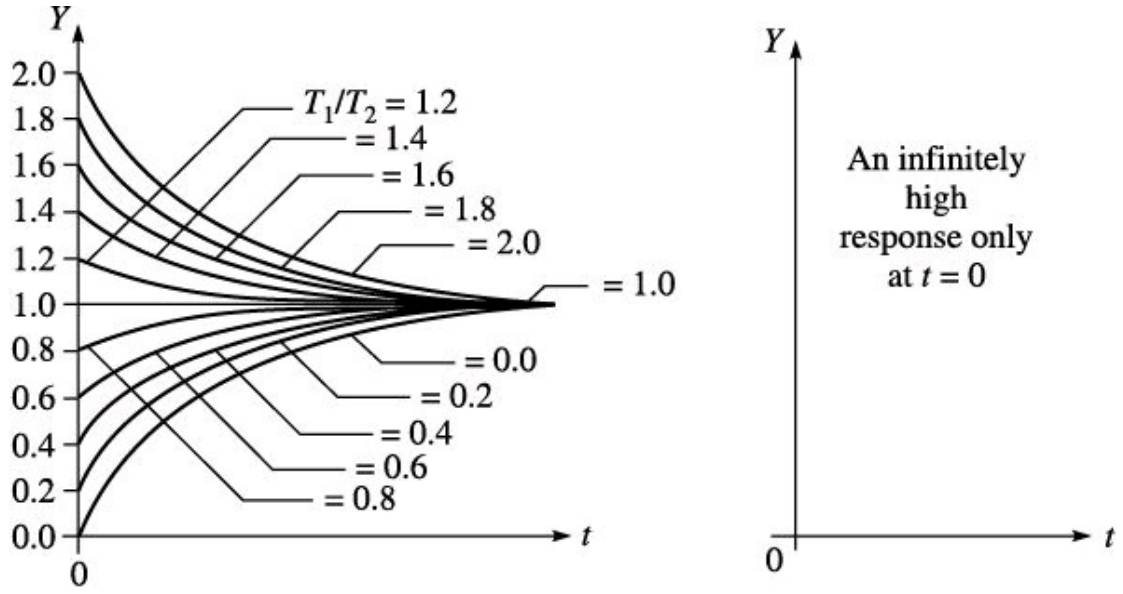


Figure 8.19 The unit step response of a 'lead–lag' element for different ratios of t_1/t_2 in equation 8.36, compared to that of an ideal differentiator.

In practical controllers, another form of lead–lag element is also used to represent the derivative function. This form may be expressed by:

$$\frac{Y(s)}{X(s)} = \frac{\tau_1 s}{1+\tau_2 s} \qquad (8.35)$$

For a step input, response of both the systems of Equations 8.34 and 8.35 starts from equal initial values as a function of the parameter t_1/t_2, but the ultimate response value of the system of Equation 8.35 approaches zero, whereas, that of the system of Equation 8.34 tends to a value of 1.0. Thus, the second variety of lead–lag element has a closer dynamic response to the ideal differentiator, and represents a physically realizable transfer function.

With the (D) function realized by the Equations 8.34 or 8.35, the (PD) and (PID) controller transfer function has certain alternative forms, mainly due to manufacturer's design difference. Two most widely used configurations are presented as follows:

1. In the first category, the modes act parallel to each other as we have seen they do in ideal controller transfer functions. With the lead–lag element of Equation 8.35, the (PD) and (PID) controller transfer functions take the following form:

(PD) $$\frac{CO(s)}{\varepsilon(s)} = K_C\left[1+\left(\frac{\tau_D s}{1+\alpha\tau_D s}\right)\right] = K_C\left[\frac{(\alpha+1)\tau_D s+1}{\alpha\tau_D s+1}\right] \qquad (8.36)$$

(PID): $$\frac{CO(s)}{\varepsilon(s)} = K_C\left[1 + \frac{1}{\tau_I s} + \frac{\tau_D s}{1+\alpha\tau_D s}\right] = K_C\left[\frac{1}{\tau_I s} + \frac{(\alpha+1)\tau_D s + 1}{\alpha\tau_D s + 1}\right] \quad (8.37)$$

2 . An alternative form is available by putting the control functions in series combination and using Equation 8.34 for lead–lag block:

(PD): $$\frac{CO(s)}{\varepsilon(s)} = K_C\left[\frac{\tau_D s + 1}{\alpha\tau_D s + 1}\right] \quad (8.38)$$

(PID): $$\frac{CO(s)}{\varepsilon(s)} = K_C\left[1 + \frac{1}{\tau_I s}\right]\left[\frac{\tau_D s + 1}{\alpha\tau_D s + 1}\right] \quad (8.39)$$

The controller configuration of Equation 8.39 is also called Interactive PID Algorithm'. K_C, t_I, t_D, and a are parameters, adjustable from the front panel of the controller. From Figure 8.10, you should note that as a goes lower, the 'Lead–Lag' block approaches the behaviour of a true differentiator. Typical values of a range from 0.05 to 0.1. We shall have further discussion about the dynamic character of 'Lead–Lag' block in frequency response analysis.

8.4.4 Derivative Mode on the Measurement Loop

In industrial process control application, often the final control element is an electro-mechanical device that regulates the flow of a high-energy stream. The manifestation of energy content may be high temp. and/or pressure of a fluid, which has to be regulated at a reasonably high flow rate. Suppose a control valve is throttling a 15 atm steam through an 8 inch line. Any sudden change of controller output would produce a 'jerk' in the valve assembly that may cause serious mechanical damage to the system operating under high temperature and pressure condition. To avoid a similar contingency, we have cautioned against the application of (D) mode controller in a flow loop where the signal measured may contain noise. A parallel may result, again due to (D) mode, in face of a sharp set point change (say step change). A sharp change of error will be produced resulting in the so-called jerk that should have been avoided.

In relatively modern designs of controllers, the (D) action is applied on the measurement path so that when the result of the sharp change in set point comes back through the loop to the final control element, becomes smooth enough (as a result of capacity elements and RC-lags present in the loop), and there would be no sudden changes. This also modifies the dynamic loop performance in certain ways. In time domain the control equation of a (PID) controller with this modification takes the following form:

$$CO(t) = K_C\left[\varepsilon(t) + \frac{1}{\tau_I}\int_0^t \varepsilon(t)dt + \tau_D \frac{d}{dt} mv(t)\right] \qquad (8.40)$$

An alternative procedure is to have a ramp generator not only for handling set point, but changes of any of the controller parameter values, so that whenever such a change is manually implemented, the ramp generator produces such a change linearly with predetermined variable vs. time slope to achieve the change smoothly.

8.4.5 Auto and Manual Modes, Bump-Less Transfer

The dynamic equations that have been forwarded for control functions describe the control performance while it is operating in the Automatic Mode. However, at certain periods of plant maintenance, human intervention is preferred or may become necessary. Such situations may arise during:

1. Start up or shut down of a sub-system or the total plant.
2. Failure of a system and subsequent duty switch over to a 'back-up' or stand by device.
3. Intentional closing of a part of the plant in case of a contingency or emergency situation, e.g. accidental damage, personnel safety, elemental calamity, etc.

There may be many other reasons for shifting the controller to manual mode. Usually in the front panel of the controller, a switch is offered to provide the AUTO□MANUAL change over facility.

In the manual mode, the controller is driven by a variable resistance element to regulate the *CO* value. The operator does the variation of *CO* by either turning a knob or pushing one of the two push-button switches to decrease or increase the *CO* value. Suppose in the immediately previous event of MANUAL operation, the setting was left at 50% *CO* value, when the mode had been changed over to AUTO. At the present instant when controller is sending an output value, *CO* = 30%, if the operator wishes to run the plant in manual mode and throws the mode change over switch to manual, *CO* will suddenly change to 50%, because as the manual regulation happens through a hardware, the last setting is retained before any human intervention. This will cause a jump of the signal to the final control element, and we know such sharp changes should be avoided lest they bring a potential damage to the system.

A special feature called **'Bump-less Transfer'** is added to the industrial controllers that allow the output value to approach the last manual setting from current automatic mode output value in a smooth fashion, without any sharp change. To realize this feature, manufacturers differ in their approach. However, there are certain general

categories of design and realization of this feature.

1. At the instant of change over, the output management shifts to an active type first order RC-lag system. This produces a step response dynamics with the difference between auto and manual mode output as the magnitude of step.
2. At the instant of change over, the output duty shifts to a preset timer that reduces the difference between auto and manual mode output linearly within the preset time interval.
3. In a digital control system, the control duty may shift to a part of the control software (a subroutine), in which any suitable algorithm may perform the task of bump-less transfer.

8.4.6 Reset Windup Problem

The fundamental control functions that have been introduced so far are linear algorithms, defined as if the process measurements and fixed control elements were capable of perfect operation to whatever range might be needed in any process condition. In fact, the sensor transmitter output, controller output and final control element, all have their input and output variable range. These variables may get saturated in case of over- or under-range variable states. We have already addressed this issue in the discussion of controller actuation and output variable saturation. Hence, the controller must be designed to accommodate temporary final control element output saturation and allow the system to be restarted smoothly after a long time failure or shutdown.

The most common of such problems is the 'Wind-up' that arise only when (I) function is present in the controller mode. Thus, the reset function continues to integrate the error, even after the final control element has gone to its limit (saturated) and is incapable of further change. This phenomenon is called wind-up and under this condition, the controller requires recovery time even after the error reverses its sign. The recovery time is necessary because the accumulated sum of error has to be neutralized before any effective control action could be obtained.

Remedy to this problem requires that the algorithm be provided some indication of the saturation and in that event the controller behaviour can be altered. One method is to use a 'flag' or a switch that changes state in the event of targeted variable reaches any of its range limits. The information regarding the state of the switch is propagated back to the affected controller and causes the integration process to stop, in some appropriate way.

A superior approach is called external feedback that senses the actual state of the manipulated variable and feeds it back in comparison to its intended value. By

working with the true state of the process, the algorithm can make much more refined limit of variable adjustment.

Another simple but effective method of achieving this feature is to realize a limiter logic function on the controller output as given below,

If $CO \geq C_{MX}$, then $CO = C_{MX}$,

and, if $CO \leq C_{MN}$, then $CO = C_{MN}$.

where, C_{MX} and C_{MN} are the desired maximum and minimum values of controller output. This limiter-logic is necessary for arresting the reset windup problem associated with the integral action, however, to obtain a better safety against limit crossing, in any part of the controller calculations, it should be logical to put the same limiter to other actions as well.

8.5 ON-OFF Mode Control

These are probably the oldest and still most widely used controllers in industry. They have two other names, e.g. **Two Position** and **Bang-Bang** controllers. Their action principle is quite simple. Ideally the controller acts as a switch. A switch can have only two states, e.g. fully-on, or, fully-off. Similarly an on-off controller output goes from 0 to 100% as the controlled variable crosses the set point. During operation, the *CO* will become 100% for the slightest amount of +ve error, and *CO* would be 0, if error has a –ve value however small.

Thus, for a very small amount of error magnitude, the response changes full scale. From this viewpoint one may consider that an on-off controller is, in fact, a proportional controller with very high value of K_C.

With a proportional controller, the closed loop process response becomes oscillatory, if the K_C value is greater than a limiting value. Reason for this oscillatory response is that for a relatively small error magnitude, the controller would always over-react or over-compensate. Due to the presence of dynamic lag in the process characteristics, the control action would take some time to affect the process, and in the mean time the controlled variable would have a little excursion away from the set point. This is a temporary phenomenon, eventually the manipulation would take effect and the controlled variable will come down toward the set point and cross it. Again, the cycle would start with taking the over compensation and its delayed action on controlled variable, everything similar but in the reverse direction from the set point. Such repeated sequence of events is reflected as the oscillatory process character.

We have commented that on-off controller may be considered as a (P) controller with a K_C of infinite value. It follows that with an on-off control, process variable

oscillations would be to the ultimate value. This has been illustrated in Figure 8.20. The oscillations may resemble a sine function for some processes but largely they are distorted repetitions of arbitrary nature, even they may not be symmetric with respect to the set point axis.

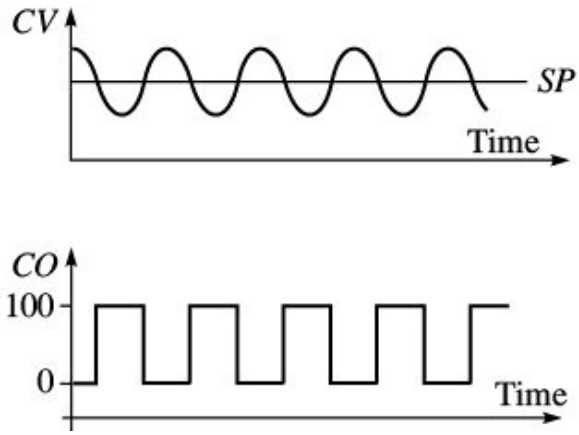


Figure 8.20 The ideal on-off control action.

Such an ideal on-off action has some disadvantages to be used in an industrial application. In real processes, the controlled variables often contain certain amount of noise, i.e. the process output shows small amplitude oscillations of high frequency around its average value. These oscillations may be due to electro-mechanical vibration or may originate from the hydrodynamic characteristics of the equipment systems in which the process is carried out. Whatever may be the cause, when such variables approach the vicinity of the set point, they may cross the set point several times before finally going across the zero error axis. For each of these crossings, there will two switch actions, that is, the controller output will go either from zero to maximum and back or vice versa. The resulting series of opening and closing of the final control element will happen in rapid succession because the noise frequency is high. This characteristic is called '**Chatter**' of the final control element, and may be quite harmful for electrical and mechanical systems associated with a final control element, say a pneumatic control valve, or a Thyristor drive for a resistance type heating load. The phenomenon of chatter has been illustrated in Figure 8.21.

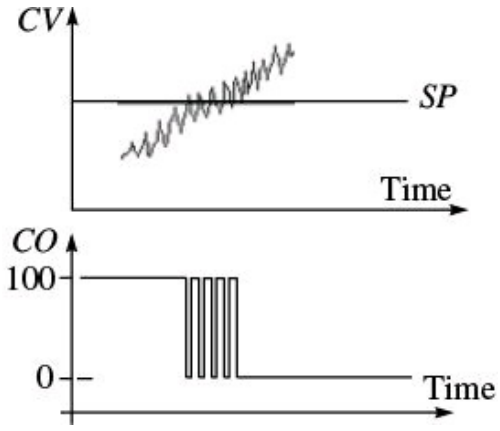


Figure 8.21 The phenomena of 'chatter'.

In an ON-OFF control system, the controlled variable (CV) without any appreciable noise content may also show *chatter*. This happens if the *lag* or *delay* between CO and the control action appearing as process variable change is very small. This delay may be in the form of transfer lag or dead time. There are several techniques for eliminating chatter from the closed loop dynamics of an ON-OFF control system. We shall describe three of the more important types within this category.

1. The first one is realized by creating a **Differential** (D_F) around the set point or splitting up the single set point into two set points SPH, and SPL, and implementing a control logic that allows the controller output (CO) to go from 0 to 100%, only when the lower set point (SPL) is crossed by the process variable from high to low value. Similarly, the controller output (CO) can go from 100% to 0, only when the higher set point (SPH) is crossed by the process variable from low to high value. The difference between two set points is called 'differential' or 'overlap' and expressed in the same unit as that of the process variable. In electrical engineering texts, the term differential is called **Hysteresis**. The differential gap between two set points is adjustable, and is set at a value that is greater than the maximum amplitude of noise carried by the process variable to eliminate chatter. This has been illustrated in Figure 8.22(a).
2. The second technique uses a **Dead-Zone** (D_Z) along with differential around the set point. A dead zone may be defined as the range of controlled variable (CV) with set point at the center of the range, for which there will be no response in the controller output (CO). This dead zone may be sandwiched between two halves of the differential. Working of such a control action is illustrated in Figure 8.22(b).
3. The third method uses **Time Differential** (D_T) instead of a differential in set point. In this technique, as soon as the process variable crosses the set point in either way, a time delay circuit is turned 'on'. After the delay time is over, the controller is allowed to operate with respect to the error present at that instant. The delay time is designed to an appropriate magnitude so that all the high frequency crossing points would occur in between, after which the process variable is permanently on the other side of set point. This control action has been illustrated in Figure 8.22(c).

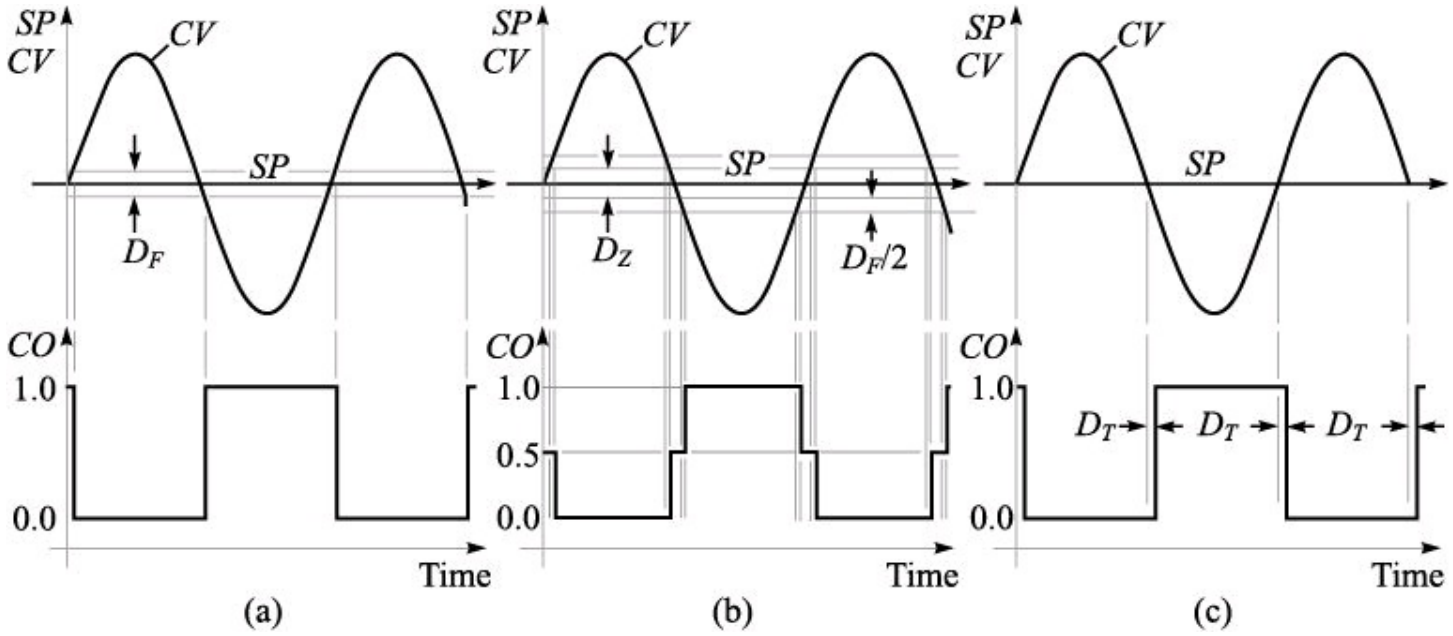


Figure 8.22 Modifications of ideal on-off control strategy to avoid chatter.

Main limitation of an on-off controller lies in its working principle, i.e. under such a controller the controlled variable oscillates. Unless the oscillation amplitude is within the tolerance limit around the set point, the control mode is unacceptable for the process under consideration. This problem has been addressed by the designers in the following way.

Suppose a furnace has to be controlled at constant temperature with a maximum tolerance of ± 5ºC. For different applications, the range of constant set temperature is from 250 to 1000ºC. The furnace is installed with an on-off controller and a 6.0 kW resistance heating element as the final control element to control the temperature. An on-off controller that operates on the total FCE load of 6.0 kW, the furnace shows a closed loop dynamics exhibiting ±10ºC oscillations around the set point and hence, is unacceptable. Now suppose the FCE load of resistance heater is subdivided into four parts of 1.5 kW segments, and the control is applied on only one of them. Thus, according to this '**Split Loading**' technique, instead of using a single 6.0 kW heating element, 4 nos. 1.5 kW elements are installed as parallel loads, three of which can be run by manual switch operation and remaining one is under on-off control action. As the maximum and minimum power under on-off action under this split load strategy has been reduced from 6.0 kW to 1.5 kW, it is expected that the oscillation amplitude of furnace temperature would come down to be within the tolerance limit of ±5ºC. Depending on set point temperature, the operator may use, one, two, three constant loads plus the controlled load for controlling the temperature. He may be using a guide chart somewhat like Table 8.4. This improvement in response has been possible because now the load under control is divided (or splitted) to fourth of its original value, and the resulting power that is made on or off by the controller is only sufficient to produce an amplitude much smaller than the warranted tolerance limit.

In the context split loading technique, two more system parameters should be mentioned. The first one is Threshold Value of MV, M_{TV}, signifying the value of heating load that always remains ON, while the remaining part is switched ON or OFF. The second one can be determined from an energy and/or material balance around the process to determine the theoretical requirement of MV to keep the CV at set point for a given disturbance entering into the process. This theoretical requirement of MV is termed as M_{TH}. For a physical process, determination of M_{TH} is often very difficult, if the process dynamics is not well known. But an experimental estimate may be made by gradually varying the M_{TV} and keeping the maximum value of MV, M_{MX}, a constant, until the oscillation of CV becomes symmetrical, i.e. a wave with crests and troughs of equal height. Now a rough estimate of M_{TH} may be made as: $M_{TH} = (M_{MX} + M_{TV})/2$.

Table 8.4 Guideline for Split Loading of the FCE

Set Point Range °C	Load – 1	Load – 2	Load – 3	Load – 4
250 – 500	OFF	OFF	ON	Under on-off control
500 – 750	OFF	ON	ON	Under on-off control
750 – 1000	ON	ON	ON	Under on-off control

This scheme has certain advantages in the way of reduced cost of installation and maintenance because of the reduced requirement of final control element capacity rating.

ON-OFF control in any of the forms those mentioned above are typical non-linear control actions. In this chapter, we have discussed the closed loop features of linear control modes, e.g. (P), (PI), (PD) and (PID) in a qualitative manner. In the next chapter (Chapter 9) there will be a fuller treatment of the topic.

8.6 Selection of Controller Mode

The design of a control system fundamentally depends on the dynamic character of the process under consideration. For a professional design exercise, the term process has a slightly larger scope of meaning than its expression and usage in the first few chapters of this text, that is, it includes the combined dynamics of process system $G_P(s)$, the final control element $G_V(s)$, and the measuring element. In a less formal analysis, the $G_M(s)$ and $G_V(s)$ may be considered to be of unity dynamics so that the design is carried out by only considering the $G_P(s)$ block. The total exercise involved in this procedure could be subdivided into two distinct activities:

1. Selection of the most suitable mode of control
2. Tuning or adjustment of the controller parameters of the selected mode

We shall discuss the mode selection in this section, and leave the discussion on controller tuning techniques for the subsequent Chapter 13 devoted on this topic, because you need to know a few prerequisites before going into that.

We shall present this problem of mode selection from two approaches which are:

- Ascertaining the suitable dynamic character of the process for a given controller mode (i.e. Mode □ Process Loop Character Approach)
- Selection of the controller mode for the common control loops encountered in typical process industries (i.e. Process Loop Character □ Mode, Approach)

It is believed that for a given problem, the assimilation of the results from the two approaches should produce a more optimum solution.

8.6.1 Mode □ Process Loop Character—Approach

For a given controller mode suitability of the process for its application may be judged from its close loop character.

1. On-Off Controller

Produces oscillation in the process output.

1. If the magnitude of the manipulated variable stream is large enough, e.g. regulation of a high value of electrical power by electro-mechanical contactor, or an incompressible fluid flow occurring in pipe sizes more than 2". The rapid state change of the final control element is undesirable.
2. The oscillations or 'ripple' may be reduced by calculated use of 'Load splitting' technique.

2. (P) Controller

In dynamic situation when steady state is reached there is a steady state error 'Offset' with (P) controller. That is for systems where controlled variable may remain anywhere between widely spaced HI and LO tolerance limits above and below the set point – a (P) controller may be used.

3. (PI) Controller

Except on-off controlled processes, about 75% of control loops are of PI category. Integral action completely eliminates 'off-set'. Low t_I brings faster response but also renders the loop more under-damped. The selection of K_C and t_I depends on process dynamic character and requires proper tuning.

4. (PID) Controller

The addition of (D) mode is required where the process output moves slowly and can produce a large error in future. This is due to existence of one more type of 'Lag' in the process (transfer and/or distance velocity). The (D) mode takes a preventive action depending upon the slope or rate of error rather than its magnitude. Hence it is called Anticipatory Control Action.

8.6.2 Process Loop Character Mode—Approach

For the common process control loops, selection of suitable controller mode may be done on the basis of process characteristics.

1. Flow

PI mode is used almost exclusively. Wide PB (150 – 250%) to reduce noise, if any (flow signals are usually associated with noise due to turbulence), with low t_I (0.1 to 0.25 min) may be used to get a fast response. (D) mode cannot be used due to the presence of noise in the measured flow signal.

2. Level

1. Level control for tanks used as surge/capacity to dampen any upstream flow fluctuation. Hence, level may be anywhere between widely spaced HI and LO marks. (P) controller may be used with a medium PB (80 – 120%) with level sensor measuring range fully covering the level range from LO to HI.
2. Heat transfer in submerged tube banks (Reboilers, vapourizers, steam generators, etc.). Requirement: – no tube should run dry, i.e. level must not go below the LO mark and above the HI mark. The set point may be defined as the mark where 85% of the shell is filled. A (P) controller may be used with similar recommendations as in the case 1.

3. Pressure

In tindustry pressure control systems have a wide variety of dynamic specification.

1. Fast Response Loop, [Figure 8.23(a)]: Pressure control in a vessel from which a vapour/gas is flowing. The manipulation is directly on the exit line. The response is fast, hence a (PI) controller may be used with same recommendation as was in the Flow Loop.
2. Slow Response Loop, [Figure 8.23(b)]: The vapour from the top of the vessel goes to a total condenser. Pressure in the vessel is controlled by cold water flow manipulation on the shell side of the condenser. Slow response due to:

(a) Heat transfer dynamics of distributed parameter character (Figure 8.24).

(b) Dead time associated with manipulated variable action because the cold water is flowing through a pipeline of definite length. Hence PID control action is necessary. *PB* is adjusted to make forward path Gain =1.0, t_I relatively large 5 to 10 mins t_D = 0.5 to 1 min.

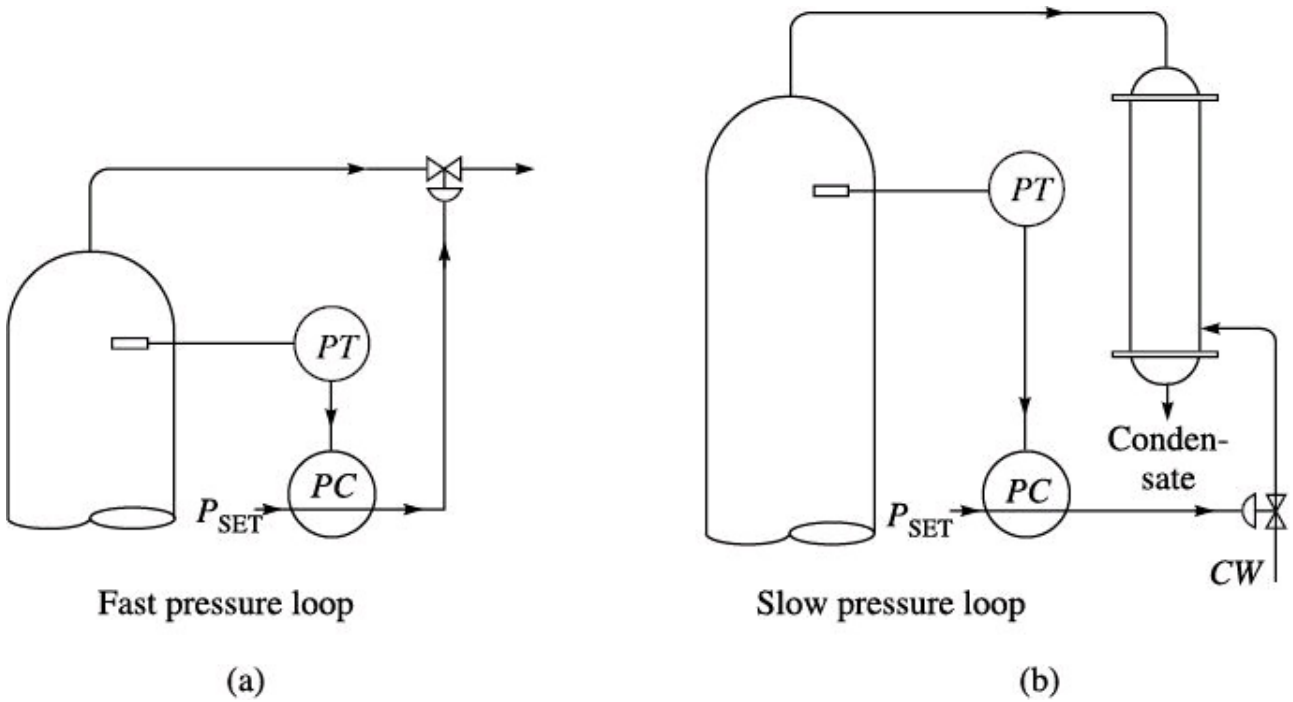


Figure 8.23 Demonstrating the fast and slow pressure loops in the two configurations of distillation column control.

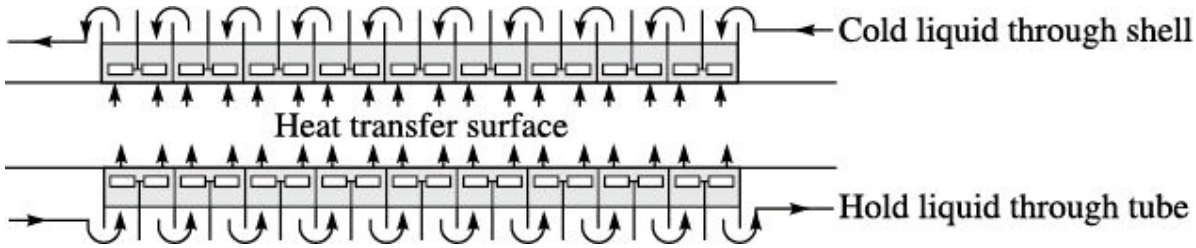


Figure 8.24 A commonly used model configuration for analysis of heat transfer dynamics explains the large lag associated with this kind of system representation.

4. Temperature

Temperature control loops are usually slow due to:

1. Heat transfer dynamics is distributed parameter type, which may be approximated by higher order process model resulting in large transfer lag (Figure 8.25).
2. Dead time due to fluid flow through pipes from manipulation point to the process, and from process to the measurement point.

PID mode is used generally, with similar consideration as in the case of slower pressure loop.

5. Chemical Composition

These are also slow loops like temperature due to:

1. Mass transfer is associated with distributed parameter system dynamics resulting a high value of transfer lag.
2. Dead time due to pipeline flow for manipulation as well as measurement.

PID mode is recommended with similar consideration as was in the case of slower pressure loops. For an offered process application if the answers obtained by the above two approaches are identical, you may apply the solution.

8.7 Digital Version of Analog Controller

So far in this chapter we have discussed about the control action as the mathematical function that operates on the input error to produce the controller output. This operation is done in a continuous manner, that is, the error and controller output may be described by a continuous graph with respect to time abscissa. This is the reason they are called Analog controllers. From the past two decades digital computers are increasingly being used in process applications for signal communication, data logging and control duties. To fulfill a controller's duty, a programme or code has to be written in real time platform that would numerically perform any of the mathematical operation of Equations 8.8, 8.12, 8.19 and 8.27 respectively as (P), (PI), (PD) and (PID) controller. Such a numerical representation of control action is called the digital version of its analog counterpart. This is an important tool of digital control systems.

Converting the analytical equations to their programmable numerical form may be also considered as the digital simulation of analog control functions. Discussion of this topic is included in the Chapter 14 on Digital Simulation.

EXERCISE PROBLEMS

8.1 A pneumatic (PI) controller has an outlet pressure of 10 psig when the set point and pen point are together. The set point is suddenly displaced by 0.5 inch (i.e. a step change in error is introduced) and the following data are obtained:

Time (sec)	0	0	20	60	90
Controller output (psig)	10	8	7	5	3.5

Determine the Gain, K_C (Psi/inch), the integral time t_I, percent output bias, OPB, and the Action (ACN) of the controller.

8.2 A proportional controller controls temperature in the range 50°–100°C. The controller is so adjusted that the output pressure goes from 3 psig to 15 psig as the measured temperature goes from 71°–75°C with the set point held constant. Find the controller gain, K_C, and the Proportional Band, PB.

8.3 A unit step change in error is introduced into a PID controller. If $K_C = 10$, $t_I = 1$, and,

$t_D = 0.5$, plot the response of the controller output CO(t).

8.4 A PID controller is at steady state with an output pressure of 9 psig. At time $t = 0$, the set point is moved away from the pen point at a rate of 0.5 inch/min. The motion of the set point is toward lower readings. If $K_C = 2$ psi/inch, $t_I = 1.25$ min, and $t_D = 0.4$ min, plot the response of the controller output pressure versus time.

8.5 A PID controller is at steady state with an output of 12 mA. At time $t = 0$, the error signal starts to change according to $e(t) = 12 + 1.4 \sin 2t$ mA., where, t is in minutes. The controller parameters are: $K_C = 2.0$, $t_I = 1.25$ min, and, $t_D = 0.4$ min. Derive the response equation CO(t) containing a single trigonometric function.

8.6 The input error signal to a controller was 12 mA and the output was 9 mA at steady zero error condition. From the instant $t = 0$, the error started to change according to:

$e(t) = 12$ mA + 0.1 mA/s. The response is shown in tabular form in the following table:

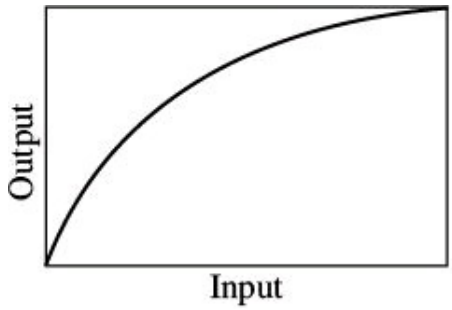
t, sec	–0	+0	40	80	120	160
CO, mA	9	12	13	14	15	16

Plot the functions $e(t)$ and CO(t) as two separate graphs (on the same graph paper), with a common time base and find, (a) Mode of the controller; (b) Controller parameter values; (c) Output bias in %; (d) Action of the controller.

8.7 The two actions of a controller and two varieties of control valve (viz. 'Air-to-open' and 'Air-to-close') has an operational similarity—Explain. (First go through the discussion on action of a control valve in Appendix III.)

8.8 For nonlinear processes often the output bias (OPB) value is set higher or lower than 50%. Explain why? For the following kinds of process non-linearity (Figure P8.1) comment whether the OPB value should be higher or lower than 50%.

1. Process non-linearity Type A
2. Process non-linearity Type B



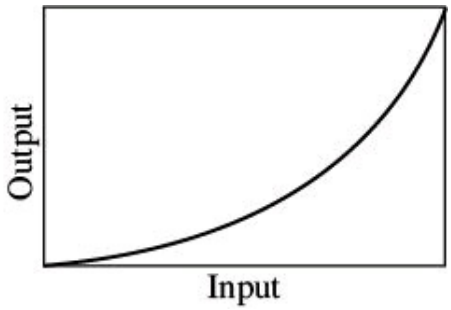


Figure P8.1 Two types of process non-linearity.

8.9 Find the responses for (P), (PD), (PI) and (PID) for the following error input functions shown in Figure P8.2:

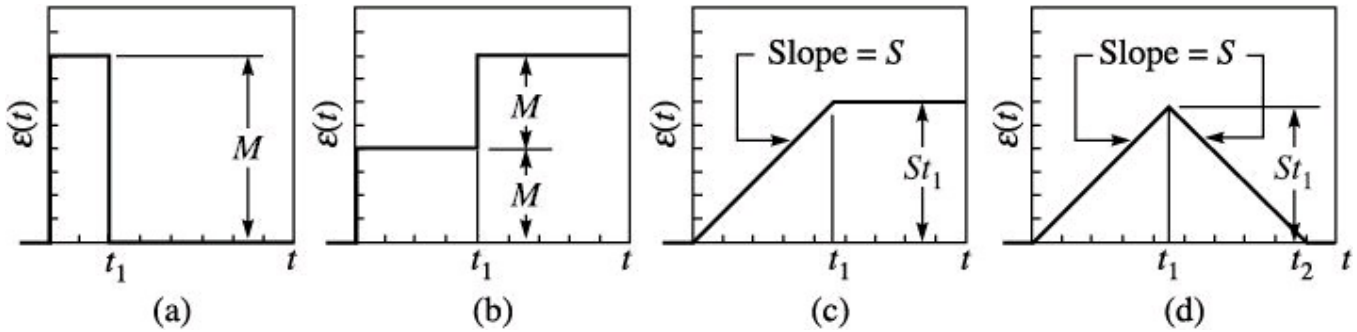


Figure P8.2 Special input functions.

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9

Dynamic Model and Response Analysis of Closed Loop Systems

After covering the preceding chapters, we should be capable of dynamically analyzing all the elements in the block diagram of a control loop. As a sign of this achievement, we present Figure 1.4 with all of its blocks drawn in **bold** in Figure 9.1(a). We have compiled the knowledge of the bits-n-pieces of the system and now want to attempt for the big-picture, that is, the dynamic model of the total control system. The new system boundary containing the closed loop control system with its input and output variables has been drawn in Figure 9.1(b).

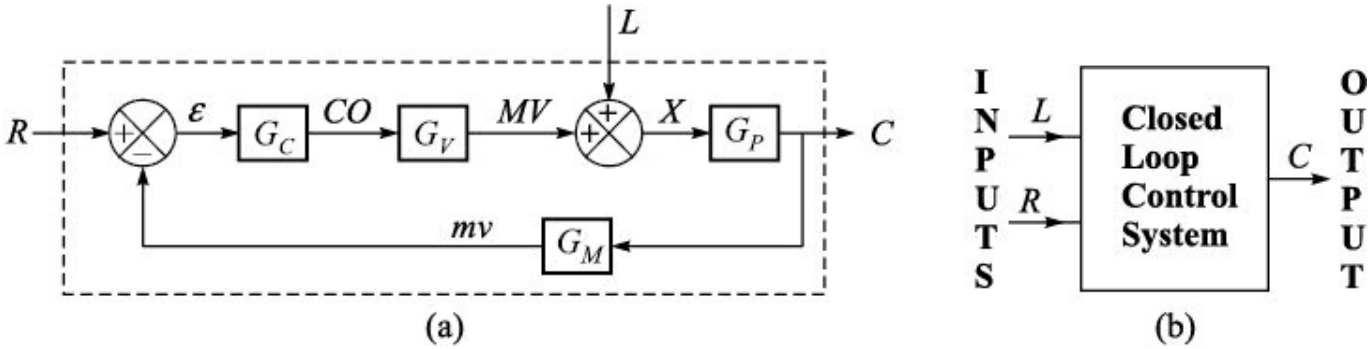

Figure 9.1 (a) The dotted box defines the system boundary. (b) The new system of our interest, and its input/output variable.

9.1 Principle of Feedback Control

By this time the idea of feedback control must have become familiar through many example systems that we treated earlier, but before proceeding further with the present discussion, it is the proper place to formally propose the strategy of feedback control. Figure 9.1(a) is the general presentation of a negative (–ve) feedback control system. The adjective –ve originates from the fact that process information, mv, obtained by the measuring element, G_M, is fedback to the controller with a –ve sign. This is the fundamental logic of controlling a process. Consider the example of the stirred liquid heater where load is the inlet liquid temperature, T_I, controlled variable is the exit liquid temperature, T_O, and manipulated variable is the heater power, W. Let the desirable value of $T_O = T_R = 70^\circ C$, the set point temperature. At any instant of operation, if T_O is 75°C, the error, e , would be – 5°C (e = 70°C – 75°C = –5°C) and any control mode with a direct action will tend to reduce W, which is the right thing to

be done. Similarly for a T_O value lower than T_R, the controller would produce an action to increase the magnitude of W. We may generalize as—**In a –ve feedback control system, the control action due to any error,** e (generated by the subtraction of mv from the set point R) **is such, that it cancels the error to bring the process back to the set point**.

The status of controlled variable C, set Point SP, deviation of C from SP, error e, and the resulting control action CO, for a direct action controller for +ve and –ve values of error are illustrated in Figure 9.2. We hope this depiction should help to understand the logic of feedback control.

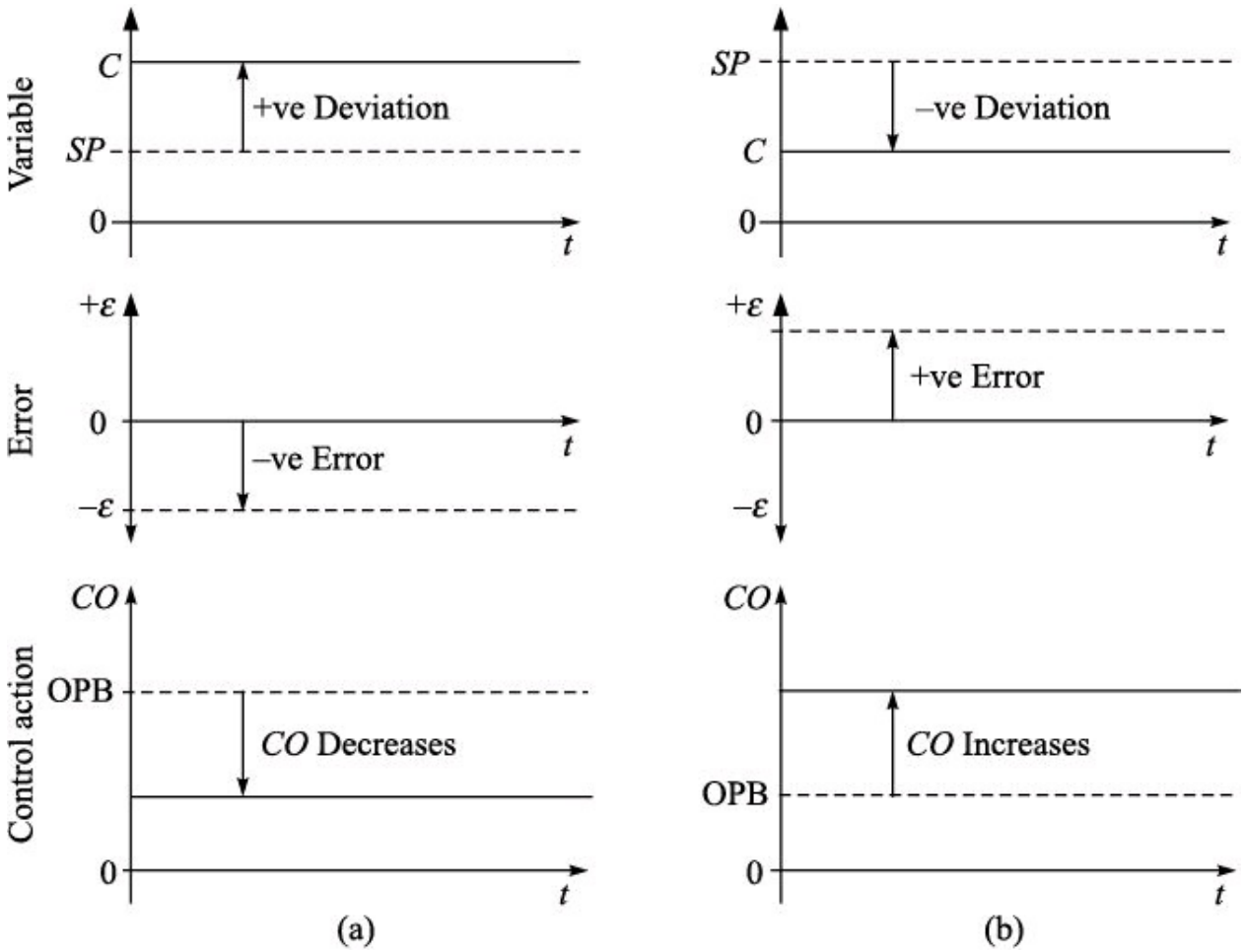

Figure 9.2 The control variable deviation from set point to generate the error and the resulting control action in a feedback control loop for (a) +ve and (b) –ve values of deviation.

A similar operation has been exemplified in the positional servo control system of DC Shunt Motor while discussing the TF of a Motorized Control Valve in Chapter 7. At the beginning of Chapter 8, introducing the idea of action of a Controller by two alternative examples of level control system, we again resorted to the logical basis of feedback control.

In place of –ve, if a control loop uses +ve feedback, i.e. the error, e, is generated according to, $e = R + mv$, you may examine by applying the above logic sequence that the control action would cause to increase the error instead of decreasing. The process variable would move further away from set point. So, shall we stamp ‘villain’ over

the term +ve feedback that no control engineer might wrongfully apply this logic. The answer is 'no'. There are special control loops where +ve feedback may be beneficially used for getting a better closed loop performance. Generally these are not feedback loops and +ve feedback is used only in a branch of the total loop. We may discuss some of the examples in Chapter 15.

It has been mentioned in the introductory chapter that there are few other control schemes other than feedback. But a majority of closed loop studies and dynamic results are based on feedback loops, because of their wide spread use and popularity. Generally the other schemes have been proposed and developed for processes that contain some form of dynamic anomalies. Such dynamic features restrict the application of feedback control, because the closed loop performance becomes too poor to be accepted. We will discuss these special control strategies in Chapter 15.

Let us continue the discussion from the first paragraph. The next obvious step should be to build up a model for the total control system. The dotted line box in Figure 9.1(a) represents the newly defined system boundary. Everything within the box is the system to be considered. This has been shown in Figure 9.1(b) with properly directed arrows for input and output variables. There are two possible inputs or disturbances:

(a) Load L, the excitation or wild variable on which we have no regulation or control,

(b) Set point R, which could be occasionally changed by the plant operator or a variable set point signal may come from a set point programming module.

The first kind of occasional change is brought due to various kinds of plant operation demands and contingencies. The second kind of set point change occurs according to a designed time schedule. This is the feature of programmed control systems.

As the objective of the analysis is to develop TF of the closed loop system, and it is already known that TF could only describe the relation between the output C, to only one input, either L or R, hence, two TFs may be derived to describe the dynamics of the loop.

These TFs define two kinds of control problem that a control system may come under.

I. Regulatory Control Problem: When load L varies but set point R remains constant. The closed loop dynamics is expressed by $C(s)/L(s)$ [R is constant, hence, $R(s) = 0$].

II. Servo Control Problem: When load R varies but load L remains constant. The closed loop dynamics is expressed by $C(s)/R(s)$ [L is constant, hence, $L(s) = 0$].

In continuously operating process plants, the major number of control applications are concerned about keeping the process at a fixed set value to obtain a good quality

product, hence regulatory control is the most commonly found feature of such applications. Set point variation is found in certain units operated in batch fashion, e.g. batch reactors operated according to a temperature schedule, heat treatment furnaces. As the name suggests, Servo Systems were introduced to address position control problems. We have seen this happen in case of Motorized Control Valve in Example 7.14. They are widely used in Robotics, Navigation of Ships and Aircrafts, and Artificial Satellite Positioning.

9.2 The Closed Loop Transfer Functions C(s)/L(s) and C(s)/R(s)

To find the TFs with respect to regulatory and servo control problems, we will resort to the rules of block diagram algebra that we introduced in Section 1.3. Considering Figure 9.1(a), a general relation between $C(s)$, $L(s)$ and $R(s)$ may be developed in the following manner.

$$C(s) = X(s)G_P(s) = [L(s) + MV(s)]G_P(s) = [L(s) + CO(s)G_V(s)]G_P(s)$$
$$= [L(s) + \varepsilon(s)G_C(s)G_V(s)]G_P(s)$$

or,
$$C(s) = [L(s) + \{R(s) - mv(s)\}G_C(s)G_V(s)]G_P(s) \tag{9.1}$$

For a regulatory problem, R is constant hence, putting $R(s) = 0$ in Equation 9.1, we obtain,

$$C(s) = [L(s) + \varepsilon(s)G_C(s)G_V(s)]G_P(s) = [L(s) - mv(s)G_C(s)G_V(s)]G_P(s)$$

The last unwanted variable, $mv(s)$, may be removed by using the relation:

$$mv(s) = C(s)G_M(s)$$

Thus,
$$C(s) = L(s)G_P(s) - C(s)G_M(s)G_C(s)G_V(s)G_P(s)$$

After a slight rearrangement, the required TF is obtained as:

$$\frac{C(s)}{L(s)} = \frac{G_P(s)}{1 + G_M(s)G_C(s)G_V(s)G_P(s)} \tag{9.2}$$

Similarly, the TF for the servo control problem $[C(s)/R(s)]$ may be determined by considering $L(s) = 0$. The result is:

$$\frac{C(s)}{R(s)} = \frac{G_C(s)G_V(s)G_P(s)}{1 + G_M(s)G_C(s)G_V(s)G_P(s)} \tag{9.3}$$

By closely observing Equations 9.2 and 9.3, we may arrive at the following general relation as a short cut for deriving closed loop TFs,

$$\frac{Y(s)}{X(s)} = \frac{[\text{Product of Block TFs in the path from } Y \text{ to } X\,]}{[1 \pm \text{Product of Block TFs in the signal path from } Y \text{ through the loop back to } Y]} \tag{9.4}$$

The ± sign in the denominator of the TF in Equation 9.4 signifies that the equation has to be used with the +ve sign for –ve feedback and with a –ve sign for +ve feedback systems. The reader should confirm the validity of Equation 9.4 by applying the same to obtain Equations 9.2 and 9.3. Use of Equation 9.4 may be demonstrated by considering an example.

In the following closed loop TF derivations and analyses, we will discard the Laplace operator *s* from transformed variable, and block TF parentheses, because we are adult enough to workout the block diagram algebra without this. However, remember that any block containing a *K* with or without a suffix is considered to be a gain element, and naturally do not have an *s*.

EXAMPLE 9.1 It is required to find the regulator and servo TFs for the system described by the block diagram shown in Figure 9.3(a).

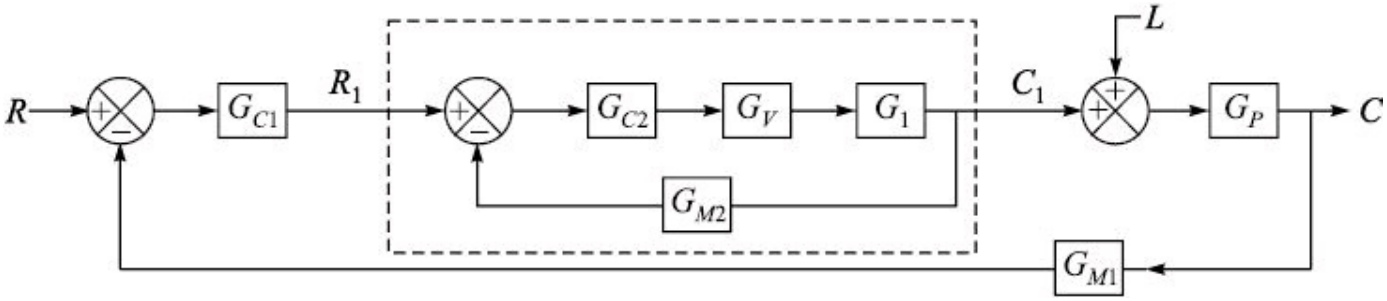

Figure 9.3(a) The control system block diagram featuring a nested loop.

Solution: The dotted box has been drawn on the given block diagram as a part of solution to indicate that reduction is possible by applying Equation 9.4 first to the part of the system included within the box. Let the nested loop TF be designated as G^*, where,

$$G^* = \frac{G_{C2}G_V G_1}{1 + G_{M2}G_{C2}G_V G_1}$$

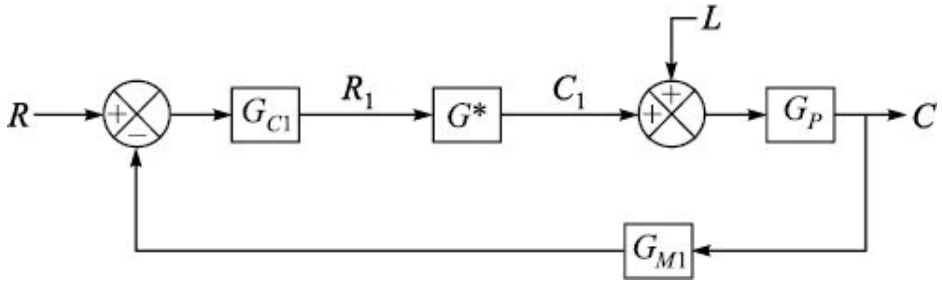

Figure 9.3(b) The reduced block diagram of the control system of Figure 9.3(a).

The reduced control loop takes the form as shown in Figure 9.3(b). Now we may again apply Equation 9.4 to obtain the two required TFs.

$$\frac{C}{L} = \frac{G_P}{1 + G_{M1}G_{C1}G^*G_P} \qquad \frac{C}{R} = \frac{G_{C1}G^*G_P}{1 + G_{M1}G_{C1}G^*G_P}$$

Putting the expression of G^* in these equations,

$$\frac{C}{L} = \frac{G_P[1 + G_{M2}G_{C2}G_V G_1]}{1 + G_{M2}G_{C2}G_V G_1 + G_{M1}G_{C1}G_{C2}G_V G_1 G_P}$$

and

$$\frac{C}{R} = \frac{G_{C1}G_{C2}G_V G_1 G_P}{1 + G_{M2}G_{C2}G_V G_1 + G_{M1}G_{C1}G_{C2}G_V G_1 G_P}$$

The given problem is a simplified version of the type encountered in cascade control systems that we shall learn in Chapter 15. The block diagram presented in Figure 9.2(a) is a simplified version of cascade control. In many such systems, the load enters into the inner loop and only the servo TF may be derived through block diagram reduction. The regulatory TF had to be derived directly.

As shown in proposing Equation 9.1, the direct derivation method may be applied to more involved block diagrams and without even applying the constancy conditions $R(s) = 0$ or
$L(s) = 0$. The result usually describes the relation between either of the two inputs $L(s)$ and $R(s)$ and the output variable, $C(s)$. After that the constancy conditions, $R(s) = 0$ and $L(s) = 0$ may be respectively used to obtain regulatory or servo TFs. This method is followed in solving the next example problem.

EXAMPLE 9.2 Find the closed loop TFs $C(s)/L(s)$, $C(s)/R(s)$ for the feed forward-feedback control system given in the block diagram of Figure 9.4. K_F and K_B are the confidence factors or weightage attributed respectively to the feed forward and feedback controllers.

Solution: The direct derivation proceeds as:

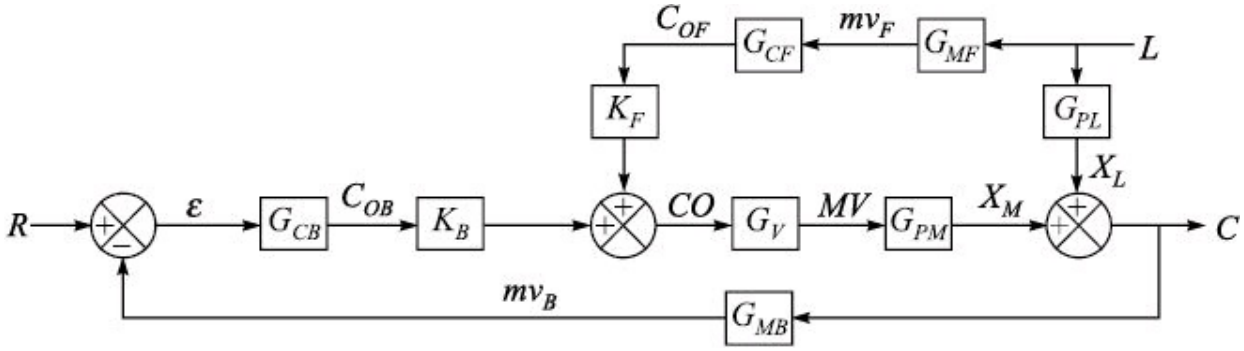

Figure 9.4 Block diagram of a feed forward-feedback control system.

$$C = (X_L + X_M) = LG_{PL} + MVG_{PM} = LG_{PL} + COG_V G_{PM}$$
$$= LG_{PL} + [LG_{MF}G_{CF}K_F + (R - mv_B)G_{CB}K_B]G_V G_{PM}$$
$$= LG_{PL} + LG_{MF}G_{CF}K_F G_V G_{PM} + RG_{CB}K_B G_V G_{PM} - CG_{MB}G_{CB}K_B G_V G_{PM}$$

Rearranging,

$$C(1 + G_{MB}G_{CB}K_B C_V G_{PM}) = L(G_{PL} + G_{MF}G_{CF}K_F G_V G_{PM}) +$$
$$RG_{CB}K_B G_V G_{PM}$$

By applying the constancy conditions, the required TFs are obtained as:

$$\frac{C}{L}=\frac{G_{PL}+G_{MF}G_{CF}K_FG_VG_{PM}}{1+G_{MB}G_{CB}K_BG_VG_{PM}} \quad \frac{C}{R}=\frac{G_{CB}K_BG_VG_{PM}}{1+G_{MB}G_{CB}K_BG_VG_{PM}}$$

EXAMPLE 9.3 Find the TF $C(s)/R(s)$ for the closed loop system given in the block diagram of Figure 9.5.

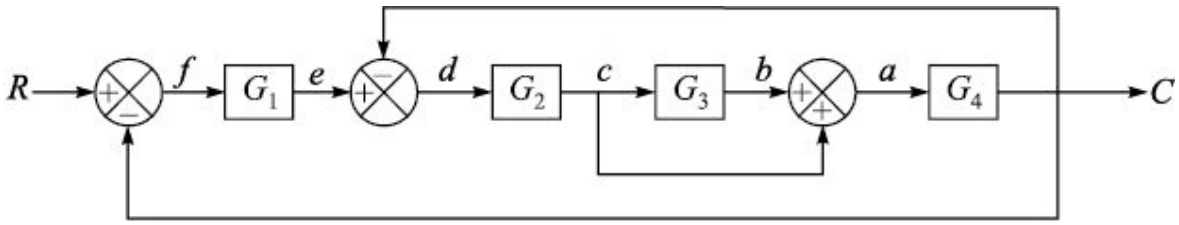

Figure 9.5 The closed loop system of example 9.3.

Solution:

$$C=aG_4=(b+c)G_4=(cG_3+c)G_4=c(G_3+1)G_4=dG_2(G_3+1)G_4$$
$$=(e-C)G_2G_4(G_3+1)=(fG_1-C)G_2G_4(G_3+1)=\{(R-C)G_1-C\}G_2G_4(G_3+1)$$

Rearranging,

$$C[1+G_1G_2G_4(G_3+1)+G_2G_4(G_3+1)]=RG_1G_2G_4(G_3+1)$$

Finally the closed loop TF takes the form: $\frac{C}{R}=\frac{G_1G_2G_4(G_3+1)}{1+(G_3+1)(G_1+1)G_2G_4}$

In certain closed loop systems, the existence of more than one feedback loop or branching of signal, increases the complication, and variable relations excess to those found by following the main loop are needed, for elimination of some of the intermediate variables.

EXAMPLE 9.4 Find the TFs $C(s)/L(s)$ and $C(s)/R(s)$ for the closed loop system in the block diagram of Figure 9.6, configured according to Internal Model Reference Control algorithm.

Solution:

$$C=XG_3=(M+L)G_3=(CoG_1+L)G_3=(\varepsilon_1G_CG_1+L)G_3=\{(RK_R-\varepsilon_2)G_CG_1+L\}G_3$$
$$=[\{RK_R-(C-CoG_2)K_M\}G_CG_1+L]G_3$$

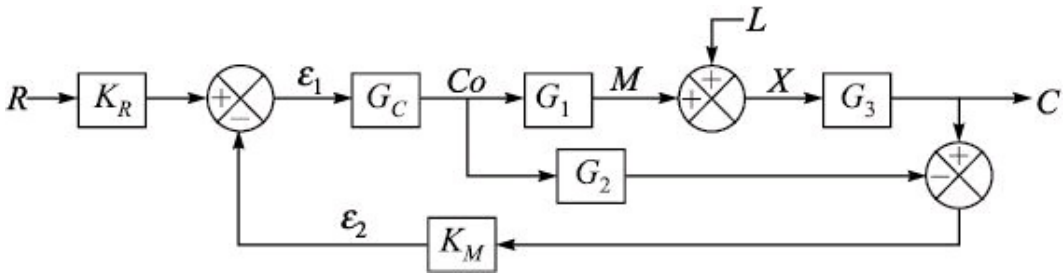

Figure 9.6 IMRC control loop.

At this point, the variable Co, existing in the main loop, reappears. This has to be

eliminated by developing a relation between *Co* and the variables that will remain in the final TF. Thus,

$(CoG_1 + L)G_3 = C$ or $Co = \dfrac{C - LG_3}{G_1 G_3}$, applying this equation into the last expression,

$$C = \left[\left\{RK_R - \left(C - \frac{C - LG_3}{G_1 G_3} G_2\right) K_M\right\} G_C G_1 + L\right] G_3$$

$$= \left[\left\{\frac{RK_R G_1 G_3 - CG_1 G_3 K_M + CG_2 K_M - LG_3 G_2 K_M}{G_1 G_3}\right\} G_C G_1 + L\right] G_3$$

$$= \left[\frac{RK_R G_1 G_3 G_C - CG_1 G_3 G_C K_M + CG_2 K_M G_C - LG_3 G_2 K_M G_C + LG_3}{G_3}\right] G_3$$

Rearranging right left sides of the equation:

$$C(1 + G_1 G_3 G_C K_M - G_2 K_M G_C) = RK_R G_1 G_3 G_C + L(G_3 - G_3 G_2 K_M G_C)$$

thus, $\dfrac{C}{L} = \dfrac{G_3(1 - G_2 K_M G_C)}{1 + G_C K_M (G_1 G_3 - G_2)}$ and $\dfrac{C}{R} = \dfrac{K_R G_1 G_3 G_C}{1 + G_C K_M (G_1 G_3 - G_2)}$

Once the block diagram is available, its reduction to obtain closed loop TFs is not very difficult. To develop the block diagram even from P&I diagram sometimes becomes critical. Often, equipment systems are integrated in a large plant due to several types of process operational restrains. The resulting configurations become difficult to unscramble and label the parts of the system by the known block element names of a feedback loop. In the introductory chapter the examples of steam governor and float type level controller may be cited as examples of this class of systems.

However, after deriving the closed loop dynamics in terms of TF, the next exercise should be the attempt to obtain the dynamic response of the loop for input disturbance. For such analysis, disturbances of both kinds, i.e. in Load and Set Point would be considered. The results will contain exact quantitative information of closed loop dynamics. To do this, we need to have the block diagram in which every block contains numerical values of the parameters. A detailed description of the process, along with its control system with quantitative information regarding the streams handled within the system, are necessary for construction of such, let's say, **Quantitative Block Diagrams**. As a first step we should remember that the input and output variables to and from the comparator element before the controller, and the summer element before the process must be of the same engineering unit. Certain hidden blocks may appear during this development. We shall discuss a few examples of different process systems to demonstrate the procedure.

EXAMPLE 9.5 Control of the Exit Liquid Temperature of a Stirred Liquid Heater: Assuming a constant liquid flow through the constant holdup mixer heater, the inlet liquid temperature, T_I, may be considered as the Load variable to the system. Liquid flows at a volumetric flow rate, q, and the tank volume is V. The liquid properties are: Density = r, Heat capacity = C_P. The desired set point temperature is T_R°C. The control system has been shown in Figure 9.7(a). During system operation, it has been found that the load variable, T_I may vary from $T_{I\text{MX}}$°C to $T_{I\text{MN}}$°C. From this set of information, we may draw the fundamental block diagram as in Figure 9.7 (b).

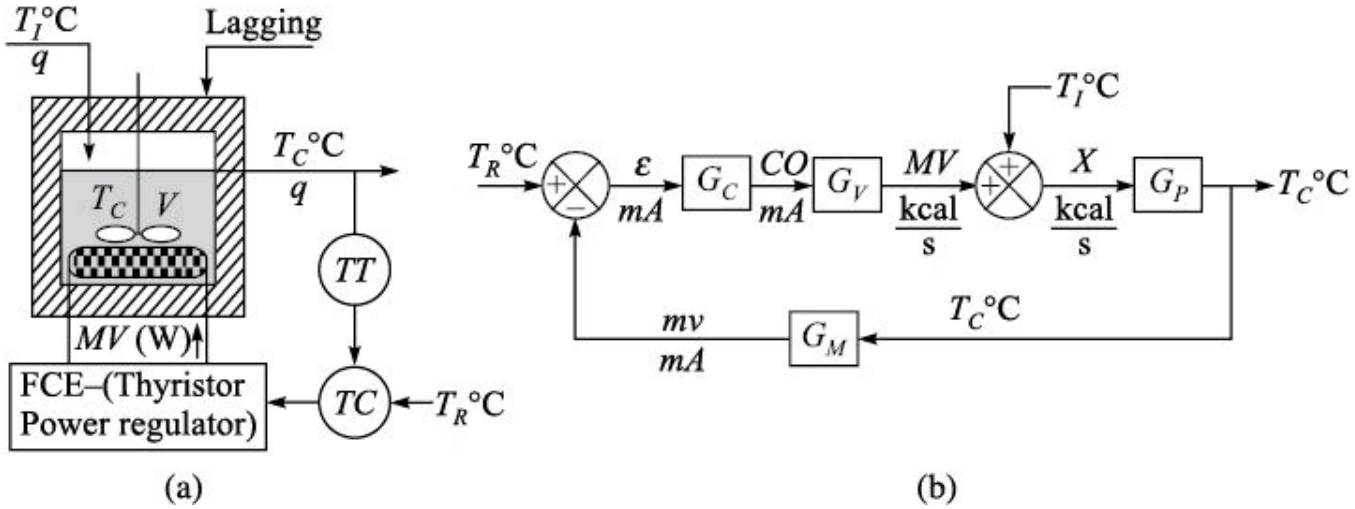

Figure 9.7 (a) The temperature control system. (b) The fundamental block diagram of the control system.

To indicate the dimensional discrepancy, we have put the units along each signal or material streams between the blocks. The comparator has two inputs of different dimensions (mv □ mA and R □ °C); also the summer before the process block has two inputs of different dimensions (MV □ kcal/s and T_I □ °C). Two converter blocks K_R and K_L, which are in-fact gain elements, may be introduced.

To express K_R, we require the additional information about the working range of the controller. Let's specify that the controller has a working range of 0°C to T_M °C. Thus, for an electronic controller $K_R = 16/T_M$, mA/°C.

The gain K_L should be such that when it is multiplied with the inlet stream temperature T_I, the product expresses the rate of enthalpy influx through the stream into the system, thus, $K_L = qrC_P$. The process TF has a first order time constant, $t_P = V/q$ but it should also contain a process gain, $K_P = 1/(qrC_P)$ in the TF numerator, which would convert the rate of enthalpy back to the temperature of the exit stream T_C in °C.

Further assuming that the electronic temperature-measuring element has the same working range of 0°C to T_M °C as that of the controller and has a negligible dynamics,

$G_M = K_M = 16/T_M$, mA/ºC.

To ascertain the FCE dynamics G_V, we first consider that it has a negligible dynamics so that $G_V = K_V$. The expression of K_V may be found in the following manner. From the problem statement the load variable, T_I may vary from $T_{I\text{MX}}$ºC to $T_{I\text{MN}}$ºC. The controller and the temperature-measuring element have the same working range of 0ºC to T_M ºC. By doing a heat balance and taking a safety factor, F_S , we may ascertain the gain, K_V, of the FCE as:

$$K_V = \frac{[q\rho C_p(T_R - T_{I\text{mn}})F_S]}{16} \text{ kcal/(s/mA)}$$

Now the quantitative form of the block diagram may be developed. This has been shown in Figure 9.8.

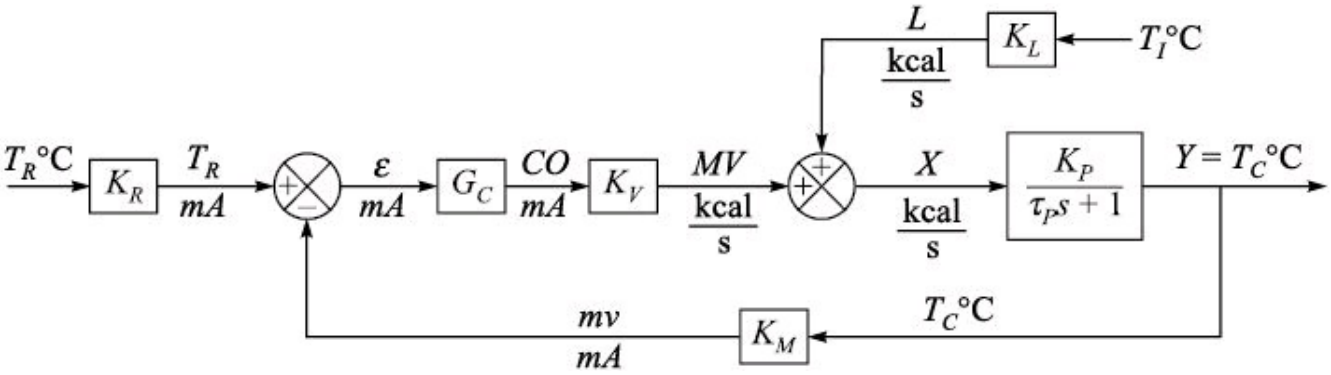

Figure 9.8 The quantitative block diagram of the stirred liquid heater.

From Figure 9.8, the closed loop TFs may be readily found as:

$$\frac{T_C}{T_I} = \frac{K_L K_P/(\tau_P s + 1)}{1 + G_C K_V K_P K_M/(\tau_P s + 1)} \text{ and } \frac{T_C}{T_R} = \frac{K_R G_C K_V K_P/(\tau_P s + 1)}{1 + G_C K_V K_M K_P/(\tau_P s + 1)}$$

The response analysis may proceed for any given input function with the above TFs.

EXAMPLE 9.6 Level Control in a Liquid Reservoir: In Chapter 8 we introduced the idea of action of a controller with two kinds of level control systems. In this chapter, the system with a direct action controller has been analyzed. The controller, measuring element and FCE have pneumatic actuation. The load variable is the exit flow stream to some downstream process. Definitely there is an exit valve, and its resistance (port opening) determines the outflow from the reservoir. The resistance R in the exit flow line varies between R_{MN} to R_{MX} during the normal operation of the system, so that the flow through this valve may be considered as the load variable.

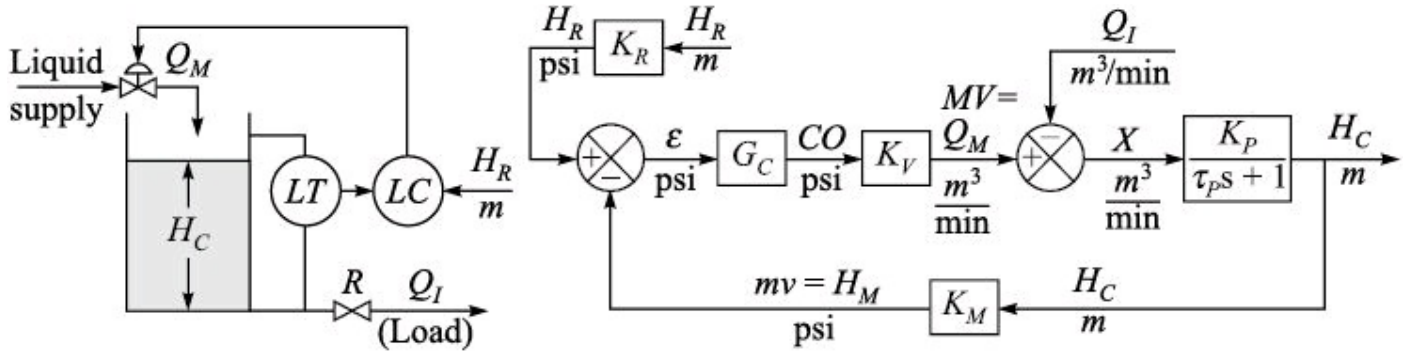

Figure 9.9 The level control system and its quantitative block diagram.

The quantitative block diagram that contains all the hidden blocks may be directly drawn. The gains have been depicted for removing dimensional inconsistency of the input variables meeting in the summer or comparator. One interesting thing is that load variable effects the controlled variable in a –ve way, thus the summer before the process block in fact is a subtractor. The process gain, K_P, must be the exit line resistance, R. Assuming that the pneumatic level sensor-transmitter and controller have the same working range of two meters and the dynamics of the level sensor-transmitter is negligible, the gains K_R and K_M may be calculated. Also assuming that the control valve may be represented by a gain element, the K_V could be evaluated. Let us write these expressions:

$$K_R = K_M = \frac{12}{2} = 6 \ \frac{\text{psi}}{\text{m}} \quad \text{and} \quad K_V = \frac{2 \text{ m}/[\ R_{\text{MN}} \text{ m}/(\text{m}^3/\text{min})]}{12 \text{ psi}} = \frac{R_{\text{MN}}}{6} \ \frac{\text{m}^3/\text{min}}{\text{psi}}$$

Before deriving the regulatory TF, one thing should be mentioned regarding the process TF, that load change could be affected only by changing the value of exit line resistance, R, which changes the dynamic parameter, t_P. This makes the process **Non-Stationary** and hence non-linear with respect to any load change. Thus, the regulatory TF can be written for only a small departure from the zero error steady state condition. However, in deriving the servo TF there is no such limitation,

$$\frac{H_C}{Q_I} = \frac{-K_P^*/(\tau_P^* s + 1)}{1 + G_C K_V K_M K_P^*/(\tau_P^* s + 1)} \quad \text{and} \quad \frac{H_C}{H_R} = \frac{K_R G_C K_V K_P/(\tau_P s + 1)}{1 + G_C K_V K_M K_P/(\tau_P s + 1)}$$

Where, K^*_P and t^*_P are the corresponding steady state values of the parameters. As shown in the regulatory TF, the –ve sign on the numerator of regulator TF indicates that the influence of an increase in load is to lower the controlled variable magnitude, because more liquid is being drained out. Thus, compensation requires that the manipulation must be stepped up to bring the controlled variable to the set point.

EXAMPLE 9.7 Concentration in an Ideally Stirred Mixer: Such concentration control systems (Figure 9.10) are common in process applications. They feature either

simple mixing or blending as in the preparation of petroleum products, lubricants and dyes or mixing with a kinetic rate process as in a CSTR.

In this example, a simple configuration of a mixing process has been considered. The load variable is the inlet solution concentration, $C_I(t)$, that under operating condition may vary between C_{IMN} and C_{IMX}. The inlet solution flow rate, q_L, is constant. The mixer has a constant holdup volume of V litres. The overflow or exit solution concentration sensor-transmitter is

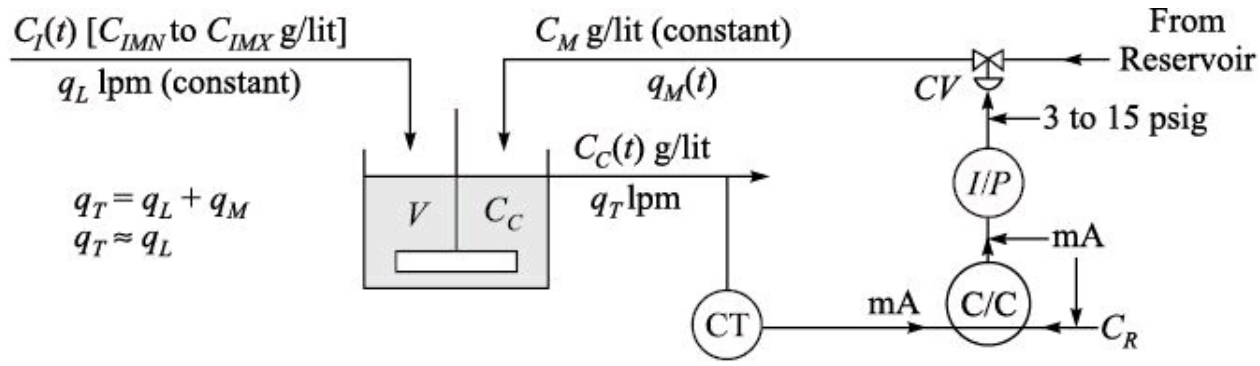

Figure 9.10 The concentration control system.

electronic but the FCE (control valve) is pneumatic, hence, an Electro-Pneumatic Transducer (EPT) converts the 4–20 mA from the concentration transmitter to 3–15 psig for the control valve. The manipulated variable is the flow rate, $q_M(t)$, of a concentrated solution stream having a concentration, C_M, of the same material. Let us assume that $q_M << q_L$ (because $C_M >> C_I$), then the exit flow rate, q_T, may be considered to be equal to q_L. The manipulated and the load variable needs to be converted to the same unit of g/min the mass input rate to the system. The process must contain a gain, K_P, that would convert mass/min to the concentration unit of mass/volume.

Let us write these parameters required for constructing the quantitative block diagram:

$$K_R = K_M = \frac{16}{C_{IMX} - C_{IMN}} \frac{\text{mA}}{\text{g/l}}; \; K_{IP} = 0.75 \frac{\text{psi}}{\text{mA}}; \; K_V = \frac{q_L(C_{IMN} - C_R)}{12C_M} \frac{\text{l/m}}{\text{psi}};$$

$$K_{MV} = C_M \frac{\text{g}}{\text{l}}; \; K_L = q_L \frac{1}{\text{min}}; \; K_P = \frac{1}{q_L} \frac{\text{min}}{\text{l}}; \; \tau_P = \frac{V}{q_L} = \frac{V}{q_T} \text{ min}$$

The block diagram is shown in Figure 9.11.

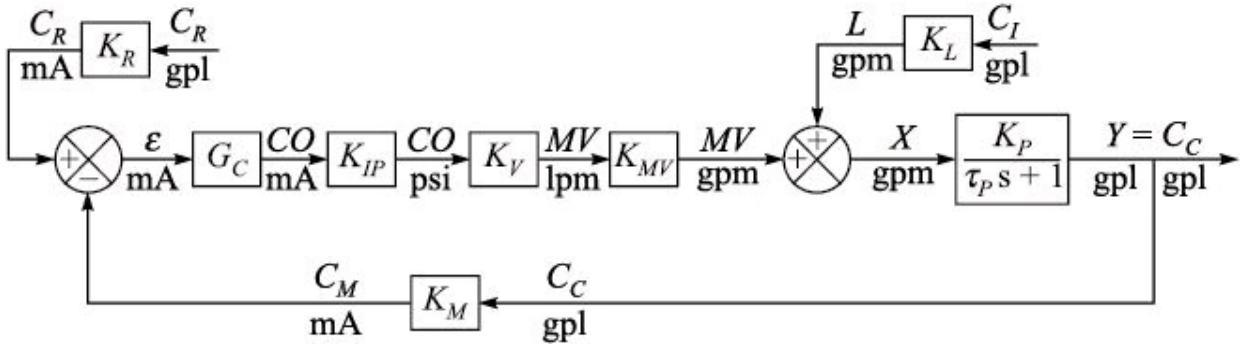

Figure 9.11 The quantitative block diagram of the concentration control system.

The closed loop Tfs are:

$$\frac{C_C}{C_I} = \frac{K_L K_P/(\tau_P s+1)}{1+G_C K_{IP} K_V K_{MV} K_P K_M (\tau_P s+1)} \quad \text{and} \quad \frac{C_C}{Q_M} = \frac{K_R G_C K_{IP} K_V K_{MV} K_P/(\tau_P s+1)}{1+G_C K_{IP} K_V K_{MV} K_P K_M (\tau_P s+1)} \ \frac{\text{g/l}}{\text{l/m}}$$

If q_M and q_L are of comparable magnitude, the mixer time constant would be variable and such systems are called **Non-Stationary Systems**. Linear analysis is only possible for an arbitrarily small departure from zero error steady state condition.

9.3 Control Loop Response

After discussing the procedure of quantitative block diagram development from the description of closed loop systems, the next step should be to attempt the response analysis of such systems. A simplified block diagram, shown in Figure 9.12, has been considered for this purpose, where the process has first order dynamics, the measuring and final control elements are pure gain elements, also a hidden gain block appears before the summer block in the load entry path. This configuration of a block diagram is very commonly found in many applications. The analysis will comprise of:

1. Determination of close loop response expressions for load and set point changes in case of four controller modes and,
2. Obtaining the close loop response shapes as quantitative results for getting an idea about the effect of parameter values on the response character.

Principally this study will be concerned about the effect of controller parameters, but in the certain pertinent cases, effect of process parameters would be also addressed.

To keep up the generality in this investigation, the variables are of arbitrary unit, the time variable has been expressed in reduced form as t/t_P. In every analysis, initially with the set point value, R = 50 and load variable value, L = 25, the system is running at zero error steady state. To achieve this, the FCE gain, K_V, has been set at 0.5. All the other gain blocks, K_L, K_R, K_M, have unity values. The process contains a unity

gain first order TF, i.e. KP is also = 1. A step disturbance of 10 arbitrary units is introduced at a finite time. For regulatory case, the L is changed from 25 to 35 units, for servo disturbance, R is changed from 50 to 60 units. The assumption of unity value for the gain blocks do not lessen the significance of results, but helps to quantitatively compare the responses with respect to variable magnitudes and time values. The responses are believed to develop a notion about the influence of controller parameter on the closed loop performance.

The responses are obtained by using computer programmes that digitally simulate the control loop. The responses, as results of the programme, are quantitative within the accuracy tolerance of numerical integration of differential equations, and hence may be used for obtaining performance measure of similar systems. The programme may incorporate any of the controller modes (P), (PD), (PI) and (PID) and has other features too. For any more detail of the scope and feature of such a programme, the reader may go through Chapter 14, dedicated on digital simulation.

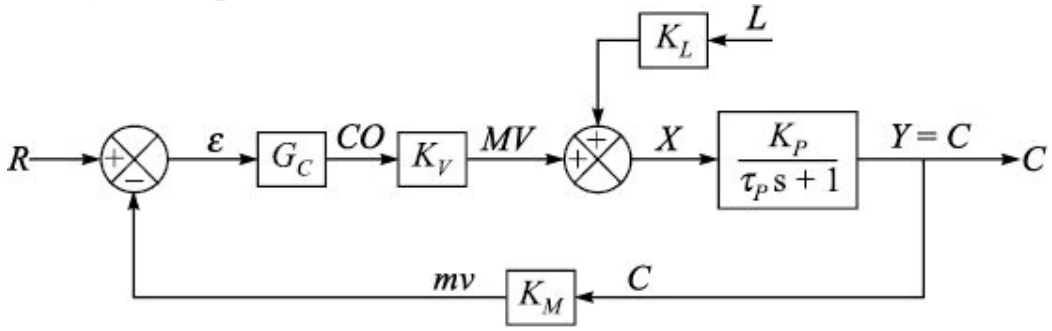

Figure 9.12 The general configuration of the block diagram for response analysis.

The closed loop TFs for the system in Figure 9.12 are:

$$\frac{C}{L}=\frac{1/(\tau_P s+1)}{1+0.5G_C/(\tau_P s+1)} \qquad (9.5)$$

and,

$$\frac{C}{R}=\frac{0.5G_C/(\tau_P s+1)}{1+0.5G_C/(\tau_P s+1)} \qquad (9.6)$$

Note that in all the above examples, we have represented the controller with the general block label G_C. The analysis of closed loop response would be done with different controller modes. The characteristics of response shapes with respect to the controller and certain process parameters are the other relevant aspects of closed loop performance.

An important performance measure of a closed loop system is the steady state error or **offset** of the system when its response has settled after a disturbance. Offset is defined as:

$$\text{Offset} = R - C(t)|_{T=\infty} \qquad (9.7)$$

That is, the difference between set point, R, and the ultimate value of the controlled variable, $C(t)$, of a disturbed process.

9.3.1 Proportional (P)-Control

The regulatory and servo TFs are derived for the controller shown in Figure 9.12 with a
[P] controller.
For the regulatory TF, putting $G_C = K_C$ in Equation 9.5:

$$\frac{C}{L} = \frac{K_L K_P/(\tau_P s+1)}{1+K_C K_{IP} K_V K_M K_P/(\tau_P s+1)} = \frac{K_1/(\tau_P s+1)}{1+K_C K_2/(\tau_P s+1)} = \frac{K_1}{\tau_P s+1+K_C K_2}$$

or,
$$\frac{C}{L} = \frac{K_1/(1+K_C K_2)}{[\tau_P/(1+K_C K_2)]s+1} \tag{9.8}$$

Similarly, for the servo TF, putting $G_C = K_C$ in Equation 9.6:

$$\frac{C}{R} = \frac{K_R K_C K_{IP} K_V K_P/(\tau_P s+1)}{1+K_C K_{IP} K_V K_M K_P/(\tau_P s+1)} = \frac{K_C K_3/(\tau_P s+1)}{1+K_C K_2/(\tau_P s+1)} = \frac{K_C K_3}{\tau_P s+1+K_C K_2}$$

or,
$$\frac{C}{R} = \frac{K_C K_3/(1+K_C K_2)}{[\tau_P/(1+K_C K_2)]s+1} \tag{9.9}$$

where, $K_1 = K_L K_P$, $K_2 = K_{IP} K_V K_M K_P$ and $K_3 = K_R K_{IP} K_V K_P$

Equations 9.8 and 9.9 are TFs of a first order system. By addition of a (P) controller, the closed loop system remain of the same order as of the open loop process. The closed loop response becomes faster than the open loop process. This is apparent from the following table, i.e. $t_{CL} < t_P$. The reader should verify that this conclusion is also true for any of the higher order systems.

For a unit step input, the offset may be determined from the ultimate response value =
$C(t)|_{T=\infty}$, which will be equal to the closed loop system gain K_{CL}. In Table 9.1, we have compiled the offset and closed loop time constant as a measure of speed of response.

Table 9.1 The Offset and Closed Loop Time Constant

Type of Disturbance	**Offset value**	**The closed Loop time constant, t_{CL}, the measure of Speed of Response**
$L(s) = 1/s$	Offset $= R - C(t)\|_{T=\infty}$ $= 0 - K_{CL} = 0 - K_1/(1+K_C K_2)$ $= -K_1/(1+K_C K_2)$	$t_{CL} = t_P/(1+K_C K_2)$
$R(s) = 1/s$	Offset $= R - C(t)\|_{T=\infty}$ $= 1 - K_{CL}$	$t_{CL} = t_P/(1+K_C K_2)$

$= (1 + K_C K_2 - K_C K_3)/(1 + K_C K_2)$

The closed loop responses with a proportionally controlled system have been shown in Figure 9.13. In part (a), the regulatory response and in part (b) the servo responses have been shown for increasing values of controller gain, K_C. The results confirm that:

1. The offset and closed loop time constant decreases as K_C increases.

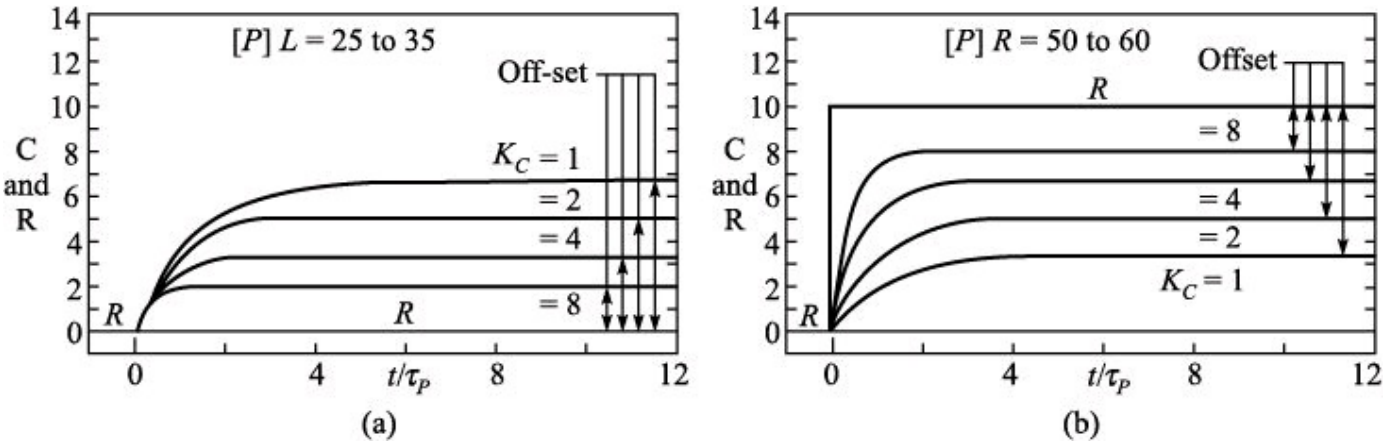

Figure 9.13 Closed loop responses for different K_C values of (P) controller.

However, the response shapes of Figure 8.12 in Section 8.3 for a (P) controlled process show an increasingly oscillatory behaviour as K_C goes higher and ultimately become unstable if K_C is higher than K_U, called the 'ultimate gain'. Such oscillatory response could not be obtained with first order processes. To analytically establish this comment, we need to go through the theory of closed loop stability analysis in complex and frequency domain included in Chapters 10 and 11. In fact, the responses of Figure 8.12 are obtained by simulating a closed loop system with a third order process.

9.3.2 Proportional Derivative [PD]-Control

The regulatory and servo TFs are derived for the control loop shown in Figure 9.12 with a [PD] controller.

For the regulatory TF, putting $G_C = K_C(1 + t_D s)$ in Equation 9.5:

$$\frac{C}{L} = \frac{K_L K_P/(\tau_P s + 1)}{1 + K_C[1 + \tau_D s]K_{IP}K_V K_M K_P/(\tau_P s + 1)} = \frac{K_1/(\tau_P s + 1)}{1 + K_C[1 + \tau_D s]K_2/(\tau_P s + 1)}$$

or,
$$\frac{C}{L} = \frac{K_1/(K_2 K_C + 1)}{[(\tau_P + K_2 K_C \tau_D)/(K_2 K_C + 1)]s + 1} \qquad (9.8)$$

Equation 9.8 shows the closed loop TF is that of a first order system. The time constant and gain of the response are respectively:

$$\tau_{CL} = \frac{\tau_P + K_2 K_C \tau_D}{K_2 K_C + 1} \text{ and } K_{CL} = \frac{K_1}{K_2 K_C + 1}$$

In the t_{CL} expression t_D is in the numerator. It is expected that the speed of response would increase for higher values of t_D. Also, the practical range of t_D values are considerably less than t_P. If we slightly rearrange the R.H.S. of t_{CL} expression,

$$\tau_{CL} = \frac{\tau_P + K_2 K_C \tau_D}{K_2 K_C + 1} = \frac{\tau_P [1 + K_2 K_C (\tau_D / \tau_P)]}{1 + K_2 K_C}$$

It becomes understandable that t_{CL} will be less than t_P.
For a unit step input, the ultimate response value is K_{CL}, hence,

$$\text{Offset} = 0 - \frac{K_1}{K_2 K_C + 1} = -\frac{K_1}{K_2 K_C + 1}$$

The offset value is equal to that we obtained with a proportional controller.

For the servo TF, putting $G_C = K_C(1 + t_D s)$ in Equation 9.6:

$$\frac{C}{R} = \frac{K_3 K_C (1 + \tau_D s)/(\tau_P s + 1)}{1 + K_2 K_C [1 + \tau_D s]/(\tau_P s + 1)} = \frac{K_3 K_C (1 + \tau_D s)}{(\tau_P s + 1)} x \frac{(\tau_P s + 1)}{\tau_P s + 1 + K_2 K_C (1 + \tau_D s)}$$

or,
$$\frac{C}{R} = \frac{K_3 K_C / (K_2 K_C + 1) + K_3 K_C \tau_D s / (K_2 K_C + 1)}{[(\tau_P + K_2 K_C \tau_D)/(K_2 K_C + 1)]s + 1} \qquad (9.9)$$

The servo TF of The R.H.S. of Equation 9.9 represents a Lead-Lag element and we have already treated such a TF in Example 5.12, by splitting into two terms. Thus, Equation 9.9 may be rearranged as:

$$\frac{C}{R} = \underbrace{\frac{K_3 K_C \tau_D s / (K_2 K_C + 1)}{[(\tau_P + K_2 K_C \tau_D)/(K_2 K_C + 1)]s + 1}}_{\text{Term I}} + \underbrace{\frac{K_3 K_C / (K_2 K_C + 1)}{[(\tau_P + K_2 K_C \tau_D)/(K_2 K_C + 1)]s + 1}}_{\text{Term II}} \qquad (9.9a)$$

For a unit step in R, Term I part of $C(t)$ would show an impulse response of magnitude of $K_3 K_C t_D/(K_2 K_C + 1)$ and Term II part of $C(t)$ would show a step response with an ultimate value of $K_{CL} = K_3 K_C/(K_2 K_C + 1)$. As the impulse response would vanish as time increases, the ultimate response would be $= K_{CL} = K_3 K_C/(K_2 K_C + 1)$.

Thus, $\text{the offset} = 1 - \frac{K_3 K_C}{K_2 K_C + 1} = \frac{1 + K_C K_2 - K_C K_3}{1 + K_C K_2}$ and $\tau_{CL} = \frac{\tau_P + K_2 K_C \tau_D}{K_2 K_C + 1}$

This offset value is again equal to that has been found for (P) control under servo disturbance, and the response moves faster than the open loop situation because:

1. $t_{CL} < t_P$, we have already stated the reason for this conclusion, while discussing the regulatory disturbance,
2. The impulse part of the response would also contribute in hastening the response during the transient period. The closed loop responses with a (PD) controller have been shown in Figure 9.14 for variation of K_C at constant t_D, and in Figure 9.15 for variation of t_D at constant K_C.

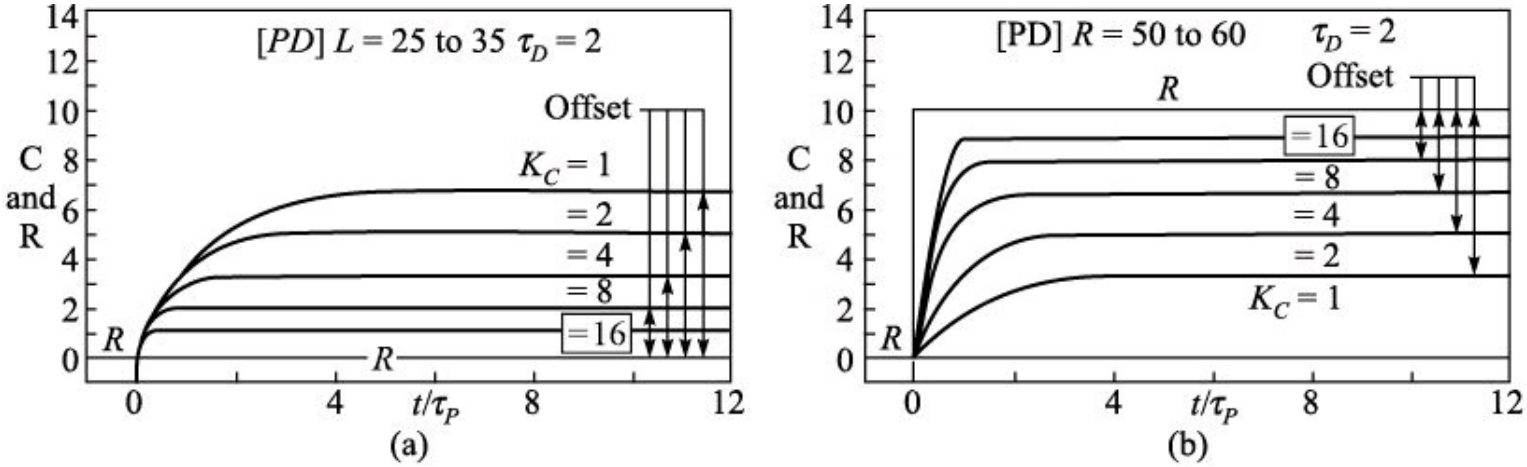

Figure 9.14 Closed loop responses for different K_C values at constant t_D for (PD) controlled system.

In part (a), the regulatory response and in part (b), the servo responses have been shown for increasing values of controller gain, K_C. The results confirm that the offset decreases as K_C goes higher and the offset values are equal to their corresponding values obtained with only a (P) controller in Figure 9.13. Also, it may be confirmed that the response speed has gone higher with the addition of (D) mode than with only a (P) controller.

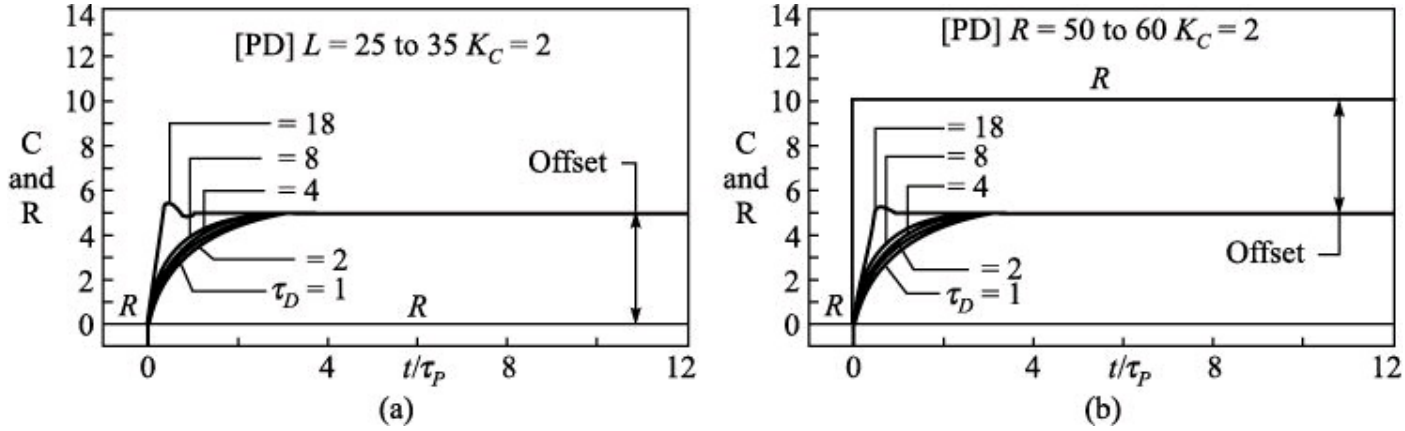

Figure 9.15 Closed loop responses for different t_D values at constant K_C for (PD) controller.

The last comment about the increase of response speed with addition of (D) mode and increment of t_D has been confirmed in Figure 9.15. During the closed loop TF derivation, we remarked that the (D) mode contributes an impulse part in the

response. This impulse part becomes recognizable for high values of either t_D or K_C increases, because the product of t_D and K_C appears as the coefficient of differentiator s in the controller TF. In Figures 9.13 and 9.14, the overshoot or 'kick' is obtained for $K_C = 16$ or $t_D = 18$. In Chapter 8, a caution about this derivative kick has been made and as a remedy to this problem, it is suggested to put the (D) mode to operate on measurement, mv instead of error, e.

We may generalize the results found as: By adding (D) mode to a (P) controller, whether it be for a regulatory or servo disturbance, the offset remains unchanged but the closed loop response moves with a faster speed.

9.3.3 Proportional Integral [PI]-Control

The regulatory and servo TFs are derived for the control loop shown in Figure 9.12 with a [PI] controller.

For the regulatory TF, putting $G_C = K_C\,[1 + 1/(t_I s)]$ in Equation 9.5:

$$\frac{C}{L} = \frac{K_L K_P/(\tau_P s + 1)}{1 + K_C[1 + 1/(\tau_I s)]K_{IP}K_V K_M K_P/(\tau_P s + 1)} = \frac{K_1/(\tau_P s + 1)}{1 + K_C[1 + 1/(\tau_I s)]K_2/(\tau_P s + 1)}$$

or,
$$\frac{C}{L} = \frac{K_1 \tau_I s}{\tau_I \tau_P s^2 + (\tau_I + K_C K_2 \tau_I)s + K_C K_2} \qquad (9.10)$$

The development regarding the general quadratic polynomial in the TF denominator to express as the parameter expressions of a second order system, that we have proposed as Equation a in Example 6.9, is used again. Thus, corresponding to standard form of second order TF,

$$\frac{C(s)}{L(s)} = \frac{K_{CL}\omega_{NCL}^2}{s^2 + 2\zeta_{CL}\omega_{NCL}s + \omega_{NCL}^2} \qquad (9.10a)$$

The parameter expressions for the TF of Equation 9.10 are:

$$\tau_{CL} = \frac{1}{\omega_{NCL}} = \sqrt{\frac{\tau_I \tau_P}{K_C K_2}}\;;\; \zeta_{CL} = \frac{1 + K_C K_2}{2\tau_P}\sqrt{\frac{K_C K_2}{\tau_I \tau_P}} \text{ and } K_{CL} = \frac{K_1 \tau_I}{K_C K_2}$$

The closed loop response will be the response of a second order system for an impulse input of magnitude, K_{CL}. The ultimate response value would be zero for an impulse response, hence, the offset value would be:

Offset = 0 – 0 = 0.

The damping character of the response could be determined only when the gain and

parameter values are available. However, it is a practice to tune the controller parameters to obtain a slightly under damped response.

Similarly, for the servo TF, putting $G_C = K_C\,[1 + 1/(t_I s)]$ in Equation 9.6:

$$\frac{C}{R} = \frac{K_R K_C(\tau_I s + 1)K_{IP}K_V K_P/[\tau_I s(\tau_P s + 1)]}{1 + K_C(\tau_I s + 1)K_{IP}K_V K_M K_P/[\tau_I s(\tau_P s + 1)]}$$

or, $$\frac{C}{R} = \frac{K_3 K_C \tau_I s + K_3 K_C}{\tau_I \tau_P s^2 + (1 + K_C K_2)\tau_I s + K_C K_2} \quad (9.11)$$

The above TF may be split into two terms as has been done in case of servo response of (P) control.

$$\frac{C}{R} = \underbrace{\frac{K_3 K_C \tau_I s}{\tau_I \tau_P s^2 + (1 + K_C K_2)\tau_I s + K_C K_2}}_{\text{Term I}} + \underbrace{\frac{K_3 K_C}{\tau_I \tau_P s^2 + (1 + K_C K_2)\tau_I s + K_C K_2}}_{\text{Term II}}$$

Comparing the denominator terms of Equations 9.10 and 9.11, the expressions for t_{CL} and z_{CL} would be similar to what has been found for the regulator TF. However, the closed gains would differ. These are:

$$K_{CLI} = \frac{K_3 \tau_I}{K_2} \text{ and } K_{CLII} = \frac{K_3}{K_2}$$

For a unit step in R, Term I part of $C(t)$ would show an impulse response of magnitude = $K_{CLI} = \frac{K_3 \tau_I}{K_2}$, and Term II part of $C(t)$ would show a step response with an ultimate value of $K_{CLII} = \frac{K_3}{K_2}$. As the impulse response would vanish as time increases, the ultimate response value would be $= \frac{K_3}{K_2}$.

In a major number of control system application, the input range of measuring element and the controller are equal, i.e. $K_M = K_R$. By the definition of K_2 and K_3 given at the end of the discussion regarding (P) control, $K_2 = K_3$. Hence, the ultimate response value would be = 1.0, and the

$$\text{Offset} = 1 - 1 = 0$$

The closed loop responses with a (PI) controller have been shown in Figure 9.16 for variation of K_C at constant t_I, and in Figure 9.17 for variation of t_I at constant K_C.

Considering the oscillatory response, it may be concluded that increase of K_C or reduction of x_I, produce an approximately similar effect, i.e. z_{CL} as well as t_{CL} are reduced to obtain more tight oscillation. In Figure 9.16(b), the extreme case for K_C = 8, where the overshoot increase has been shown with an insignificant change in w_{NCL}.

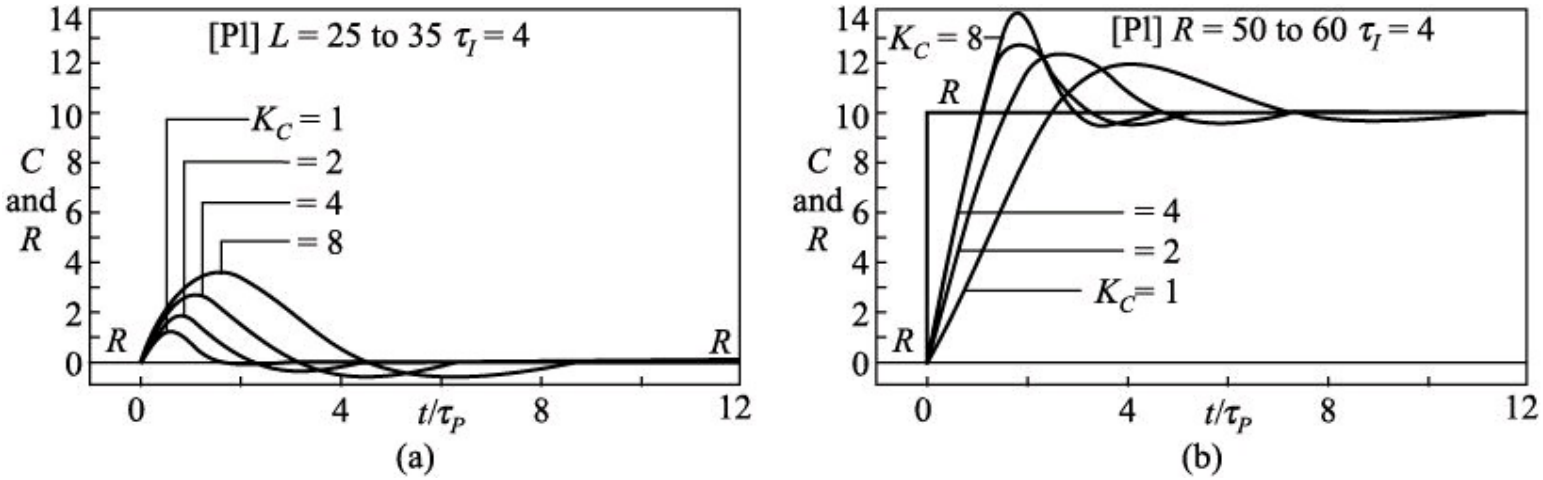

Figure 9.16 Closed loop responses for different K_C values at constant t_I for (PI) controlled system.

The response characters in Figure 9.17 are approximately similar to that we obtained in Figures 8.13(a) and (b) for a third order system, and again the only difference is that a closed loop system with a first order process cannot be made unstable by decreasing t_I. The reason for this has already been discussed.

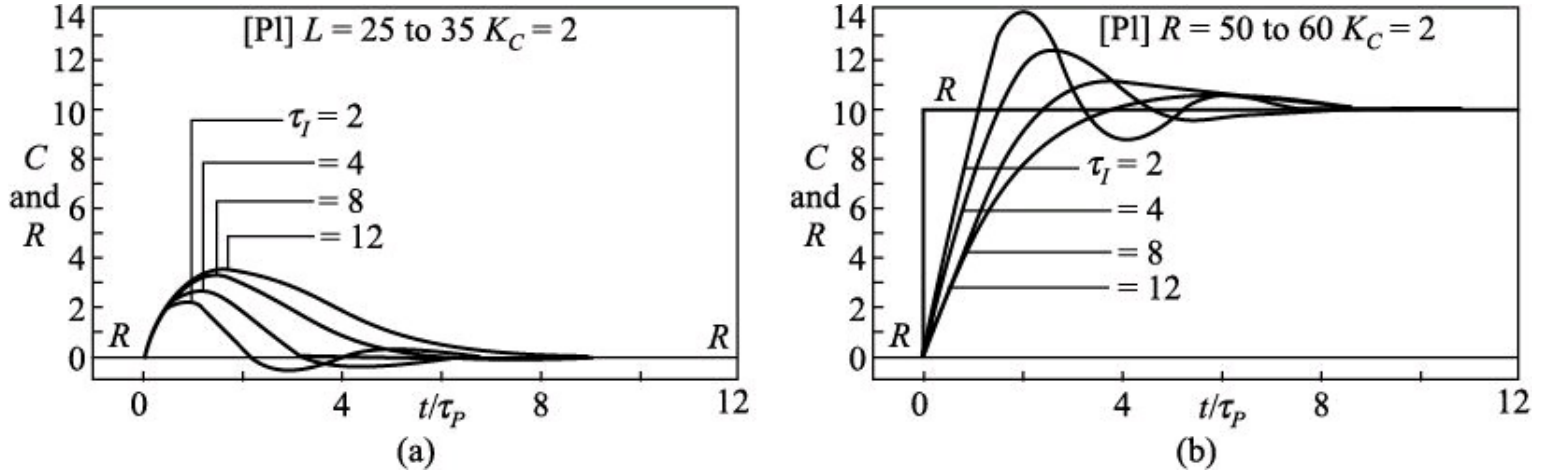

Figure 9.17 Closed loop responses for different t_I values at constant K_C for (PI) controller.

An important performance measure of closed loop response is the so-called 'quarter decay ratio response'. The oscillatory response may not be exactly that of an under damped second order system, as in the case of servo disturbance. Still the decay ratio expressed by Equation 6.83 may be applied to find the required value of close loop damping coefficient:

$$\text{Decay ratio} = e^{-(2\pi\zeta/\sqrt{1-\zeta^2})} = 0.25 \text{ or } \ln(0.25) = -1.386294361 = -\frac{2\pi\zeta}{\sqrt{1-\zeta^2}}$$

or

$$1.921812056(1 - z^2) = 39.4784176z^2 \text{ or } z = 0.215453762$$

If values of K_2 and t_P, are provided as process parameters, this value of $z = z_{CL}$ may be used to determine a relation between K_C and t_I, by applying the developed relation:

$$\zeta_{CL} = \frac{1 + K_C K_2}{2\tau_P}\sqrt{\frac{K_C K_2}{\tau_I \tau_P}} \qquad (9.12)$$

This is, in a sense, the stability measure of the closed loop system. The other important closed loop performance criterion is how fast the controller drives away the effect of disturbance from the system or, response time of the oscillatory response. This may be measured by applying the found relation between K_C and t_I into the developed equation for w_{NCL}:

$$\omega_{NCL} = \sqrt{\frac{K_C K_2}{\tau_I \tau_P}} \qquad (9.13)]$$

9.3.4 Proportional Integral Derivative [PID]-Control

The regulatory and servo TFs are derived for the control loop shown in Figure 9.12 with a [PID] controller.

For the regulatory TF, putting $G_C = K_C\,[1 + 1/(t_I s) + t_D s]$ in Equation 9.4:

$$\frac{C}{L} = \frac{K_1/(\tau_P s + 1)}{1 + K_C[1 + 1/(\tau_I s) + \tau_D s]K_2/(\tau_P s + 1)}$$

Rearranging the above equation,

or,
$$\frac{C}{L} = \frac{K_1 \tau_I s}{(\tau_I \tau_P + K_C K_2 \tau_I \tau_D)s^2 + (\tau_I + K_C K_2 \tau_I)s + K_C K_2} \qquad (9.14)$$

Again the procedure for determining the parameters of the standard second order TF has been used, and the following values are obtained with respect to standard form of Equation 9.14.

$$\tau_{CL} = \frac{1}{\omega_{NCL}} = \sqrt{\frac{\tau_I(\tau_P + K_C K_2 \tau_D)}{K_C K_2}}\,;\ \zeta_{CL} = \frac{1 + K_C K_2}{2}\sqrt{\frac{\tau_I}{(\tau_P + K_C K_2 \tau_D)K_C K_2}}\,;$$

and
$$K_{CL} = \frac{K_1 \tau_I}{K_C K_2}$$

The closed loop response will be the response of a second order system for an

impulse input of magnitude = K_{CL}. The ultimate response value would be zero for an impulse response, hence, the offset value would be:

$$\text{Offset} = 0 - 0 = 0.$$

The damping character of the response could be determined only when the parameter values are available. However, it is a practice to tune the controller parameters to obtain a slightly under damped response.

Similarly, for the servo TF, putting $G_C = K_C\,[1 + 1/(t_I s) + t_D s]$ in Equation 9.6, and rearranging:

$$\frac{C}{R} = \frac{K_C K_3 \tau_I \tau_D s^2 + K_C K_3 \tau_I s + K_C K_3}{(\tau_I \tau_P + K_C K_2 \tau_I \tau_D)s^2 + (\tau_I + K_C K_2 \tau_I)s + K_C K_2} \quad (9.15)$$

The above TF may be split into three terms as follows:

$$\frac{C}{R} = \underbrace{\frac{K_C K_3 \tau_I \tau_D s^2}{(\tau_I \tau_P + K_C K_2 \tau_I \tau_D)s^2 + (\tau_I + K_C K_2 \tau_I)s + K_C K_2}}_{\text{Term I}} +$$

$$\underbrace{\frac{K_C K_3 \tau_I s}{(\tau_I \tau_P + K_C K_2 \tau_I \tau_D)s^2 + (\tau_I + K_C K_2 \tau_I)s + K_C K_2}}_{\text{Term II}} +$$

$$\underbrace{\frac{K_C K_3}{(\tau_I \tau_P + K_C K_2 \tau_I \tau_D)s^2 + (\tau_I + K_C K_2 \tau_I)s + K_C K_2}}_{\text{Term III}} \quad (9.15)$$

Again the procedure for determining the parameters of the standard second order TF has been used, and the following values are obtained with respect to Equation 9.15.

$$\tau_{CL} = \frac{1}{\omega_{NCL}} = \sqrt{\frac{\tau_I(\tau_P + K_C K_2 \tau_D)}{K_C K_2}}; \qquad \zeta_{CL} = \frac{1 + K_C K_2}{2}\sqrt{\frac{\tau_I}{(\tau_P + K_C K_2 \tau_D)K_C K_2}};$$

These are similar expressions as have been found for regulator disturbance. But the closed loop gain expressions are different for the three terms; they are:

$$K_{CLI} = \frac{K_3 \tau_I \tau_D}{K_2}; \; K_{CLII} = \frac{K_3 \tau_I}{K_2} \text{ and } K_{CLIII} = \frac{K_3}{K_2}$$

Term III will contribute a step response, Term II would be an impulse response corresponding to a magnitude of K_{CLII}, and Term I is the time derivative of an impulse response of magnitude of K_{CLI}. The responses of Term I and II would vanish as time increases.

The closed loop responses with a (PID) controller have been shown in Figure 9.18 for variation of K_C at constant t_I and t_D, in Figure 9.19 for variation of t_I at constant K_C and t_D, and in Figure 9.20 for variation of t_D at constant K_C and t_I.

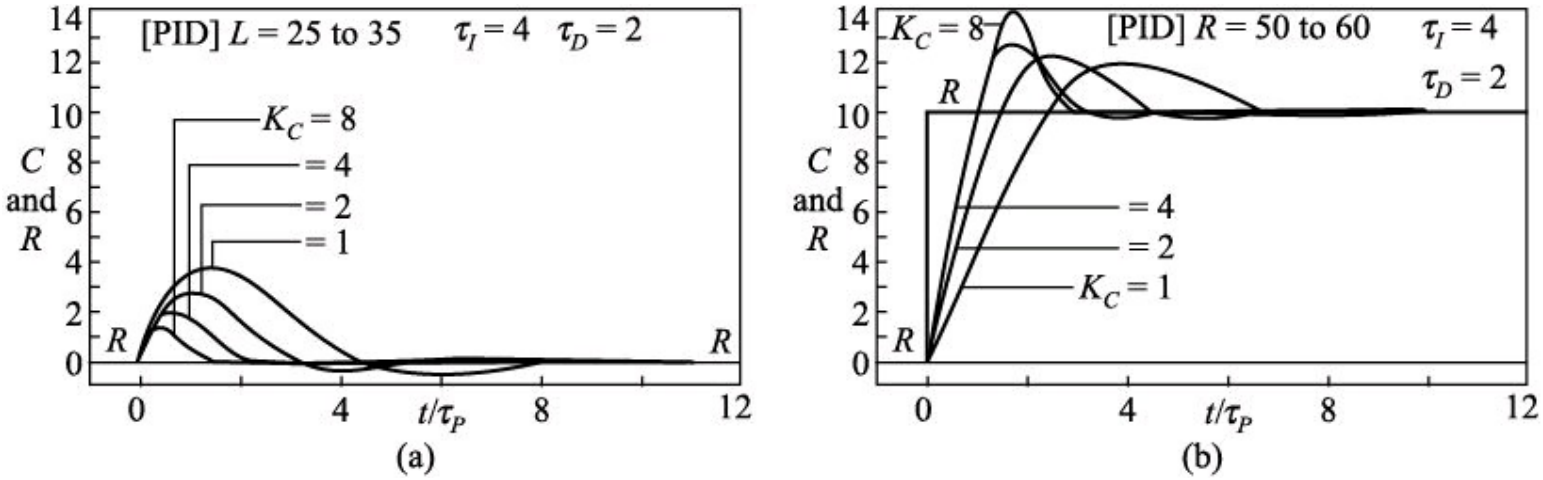

Figure 9.18 Closed loop responses for different K_C values at constant t_I and t_D for (PID) controlled system.

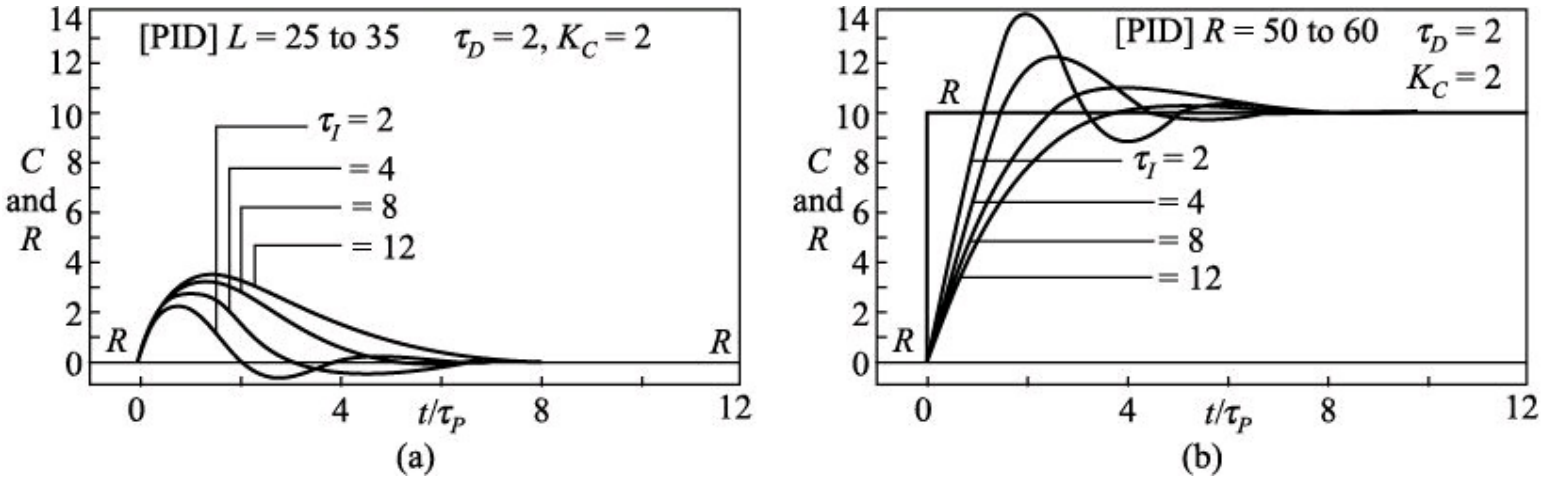

Figure 9.19 Closed loop responses for different t_I values at constant K_C and t_D for (PID) controlled system.

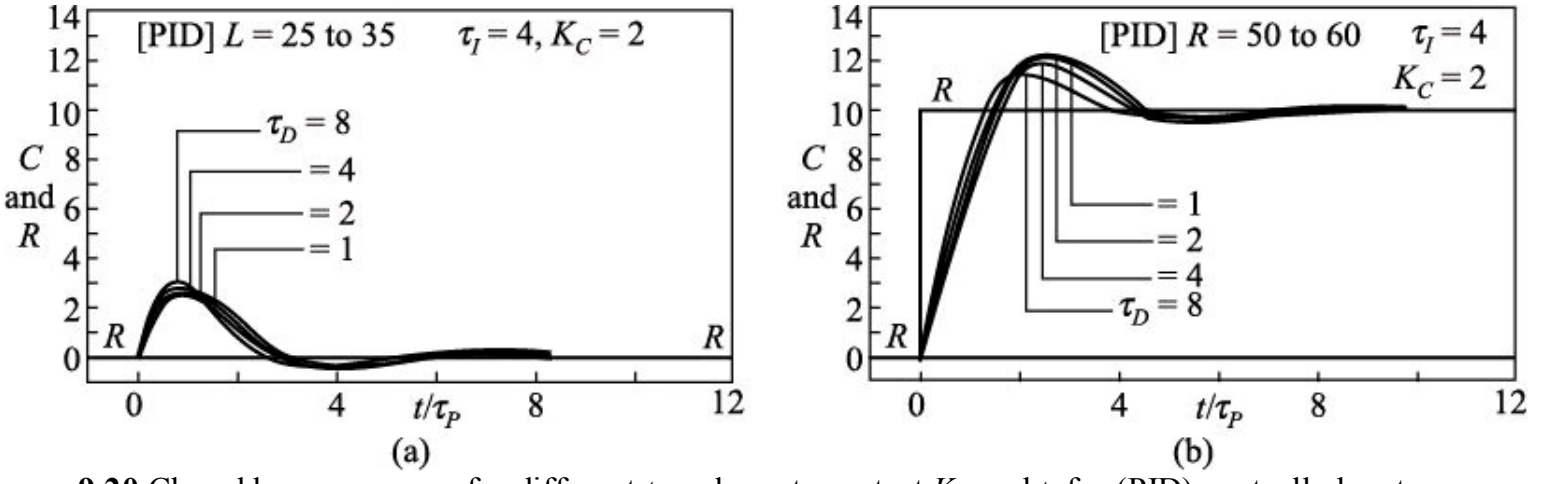

Figure 9.20 Closed loop responses for different t_D values at constant K_C and t_I for (PID) controlled system.

Comparing Figure 9.18 with 9.16, a reduction of overshoot as well as response time is observed that may be attributed to the effect of adding (D) action to a (PI) controller.

Again in Figure 9.18(b), the high value gain for K_C = 8 where the overshoot

increase has been observed with an insignificant change in w_{NCL}, as we did in Figure 9.17(b).

The comments made within the brackets ( ) for (PI) controller regarding the closed loop performance measure, in terms of decay ratio and response time, may be also applied for (PID) controller. Since the numbers of controller parameters are three, an implicit procedure would be required to find a solution with lesser number of relations. Let us do some examples to apply the relations that we have developed for quantifying the properties of closed loop response.

EXAMPLE 9.8 The set point of the control system illustrated in Figure 9.21 is given a step change of 0.1 unit. Determine: (a) the maximum value of C and the time at which it occurs; (b) the offset; (c) the period of oscillation.

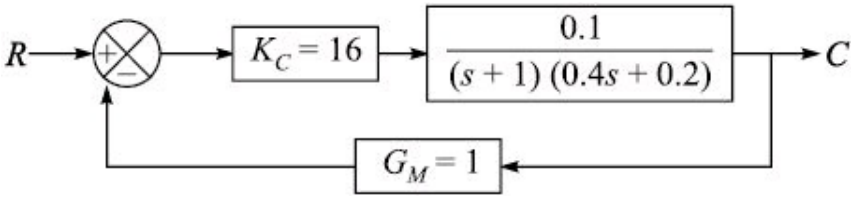

Figure 9.21 A second order process with a (P) controller.

Solution: The servo TF for this system is:

$$\frac{C}{R} = \frac{16 \times 0.1}{(s+1)(0.4s+0.2)} \Big/ 1 + \frac{16 \times 0.1}{(s+1)(0.4s+0.2)} = \frac{1.6}{0.4s^2 + 1.2s + 1.8} \sqrt{b^2 - 4ac}\, \frac{n!}{r!(n-r)!}\, \infty$$

$$= \frac{8}{9} \cdot \frac{9/2}{s^2 + (3/2)s + (9/2)}$$

The closed loop TF is that of a second order system. The parameters may be found as:

$$\omega_{NCL} = 3/\sqrt{2} = 2.12; \quad \zeta = 1/(2\sqrt{2}) = 0.354; \quad \zeta\omega_N = 0.75; \quad \sqrt{1-\zeta^2} = 0.9352$$

For $$R(s) = \frac{0.1}{s}\ ; \quad C(t) = \frac{0.8}{9}\left[1 - \frac{e^{-\zeta\omega_N t}}{\sqrt{1-\zeta^2}} \sin\left(\omega_N \sqrt{1-\zeta^2}\, t + \tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta}\right)\right]$$

or, $$C(t) = 0.089\left[1 - \frac{e^{-0.75t}}{0.935} \sin\left(1.983t + \tan^{-1} 2.642\right)\right]$$

Maximum value of $C(t)$ may be defined by the overshoot of the second order under damped response,

$$OS = e^{-\pi\zeta/\sqrt{1-\zeta^2}} = 0.3044; \quad \therefore \quad C(t)\,|_{\max} = 0.089(1 + 0.3044) = 0.1161$$

$$\text{Offset} = R - C(t)\,|_{t=\infty} = 0.1 - 0.089 = 0.011$$

Time period of oscillation, $TP = 2p/w_N = 2.96$ time units.

Frequency of oscillation, $\omega = \omega_N\sqrt{1-\zeta^2} = 1.983$ radians/time units.

Time to reach maximum value of response, T_M, would be half of time period,

$$T_M = \frac{T_P}{2} = 1.48 \text{ time units}$$

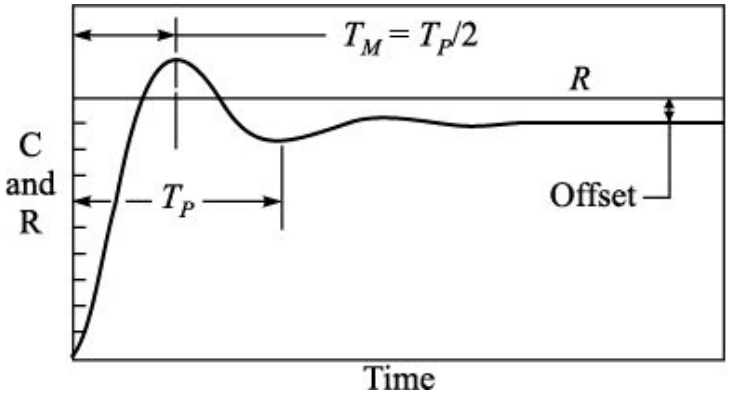

Figure 9.22 The response.

EXAMPLE 9.9 A liquid level control system is shown in Figure 9.23. As seen from this figure, the tanks are combined in a non-interacting fashion. The following data may be used for analyzing the system: (A) Resistances on the tank exit lines are linear. They have values

$R_1 = R_2 = 5 \times 10^{-4}$ cm/(cm^3/min); (B) Internal cross-sectional area of each tank is 2000 cm^2; (C) The level controller is a pneumatic (P) controller with only adjustable parameter K_C; (D) The control valve may be treated as a gain element with $K_V = 1000$ cc/psi.

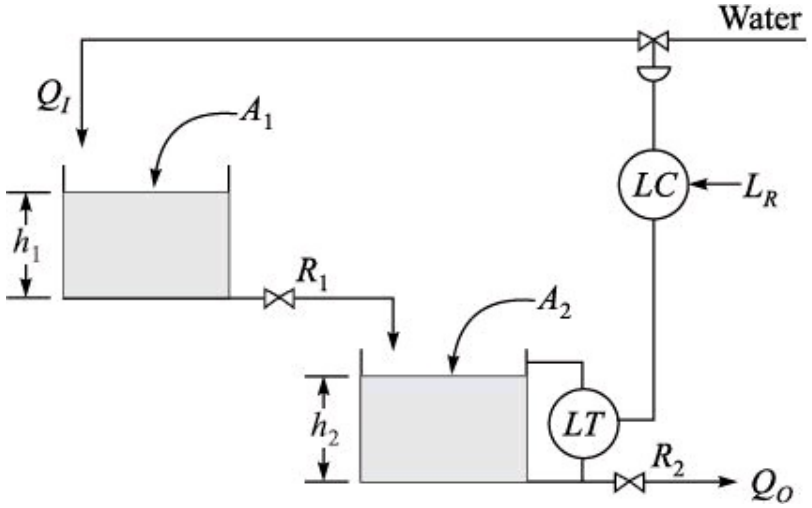

Figure 9.23 The liquid level control system.

Answer the following: (a) Draw a quantitative block diagram of this control system; (b) Calculate the value of K_C, so that the close loop response become critically

damped;

(c) If the tanks were connected in interacting fashion, calculate the value of K_C, so that the close loop response still remain critically damped; (d) using 1.5 times the value of K_C determined in part (c), find the response of the level in tank 2 to a step change in set point of 1 cm of level.

Solution: The parameter values of the system may be determined from the problem data:

$$A_1 = A_2 = 2000 \text{ cm}^2; \qquad R_1 = R_2 = 5 \times 10^{-4} \text{ cm/(cc/min)};$$

$$t_1 = t_2 = A_1R_1 = A_2R_2 = 1 \text{ min}, \qquad K_M = 1$$

For the valve $G_V = K_V = 1000$ cc/psi

and the process TF is, $G_P(s) = 5 \times 10^{-4}/[(s+1)^2]$ cm/(cc/min)

(a) For the non-interacting combination. Note that according to the pictorial description, there is load variable to the control system. Now we may draw the block diagram:

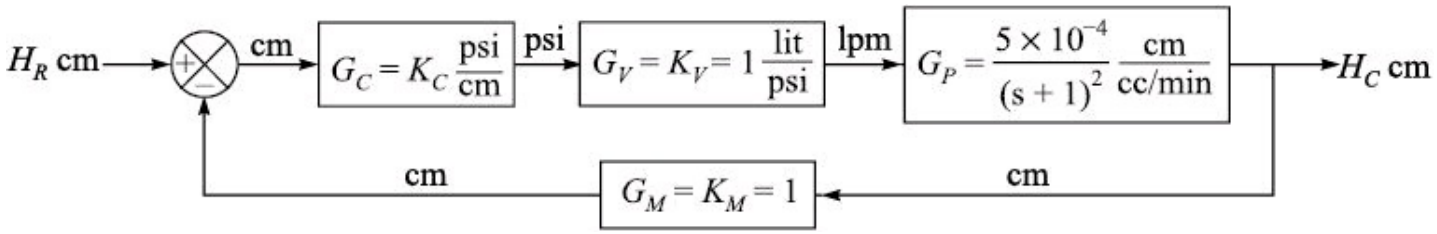

Figure 9.24 Block diagram of the level control system.

(b) The closed loop TF:

$$\frac{H_C}{H_R} = \left[\frac{5 \times 10^{-4} \times 1000K_C}{(s+1)^2}\right] \Big/ \left[1 + \frac{5 \times 10^{-4} \times 1000K_C}{(s+1)^2}\right] = \frac{0.5K_C}{s^2 + 2s + 0.5K_C + 1}$$

The roots of the characteristic equation $s_{1,2} = \{-2 \pm \sqrt{4 - 4(1 + 0.5K_C)}\}/2 = 1 \pm i\sqrt{0.5K_C}$

To render the closed loop response critically damped, we have to reduce K_C to zero value. This is equivalent to state that the system has no control, i.e. only an open loop situation may produce a critically damped response of $H_C(t)$ for a change in $H_R(t)$.

(c) In interacting combination, the process TF is $G_P(s) = 0.5/(s^2 + 3s + 1)$ ft/cfm

$$\text{The closed loop TF: } \frac{H_C}{H_R} = \frac{0.5K_C}{s^2 + 3s + 1 + 0.5K_C} \frac{\text{cm}}{\text{cc/min}}$$

To have a whole square (repeated (equal) roots of the characteristic equation) in

the denominator:

$s^2 + 3s + 1 + 0.5K_C = s^2 + 2 \times 1.5 \times s + (1.5)^2$; or $1 + 0.5K_C = 1.5^2$; or $K_C = 2.5$

(d) The required value of $K_C = 1.5 \times 2.5 = 3.75$; $H_R(s) = \frac{1}{s}$

$$H_C(s) = \left(\frac{1}{s}\right)\left(\frac{0.5 \times 3.75}{s^2 + 3s + 1 + 0.5 \times 3.75}\right) = \frac{0.6522}{s}\left(\frac{1.7^2}{s^2 + 2 \times 0.88 \times 1.7s + 1.7^2}\right)$$

The evaluated parameters are:

$$t = 1/1.7 = 0.59 \text{ min.}, z = 0.88; K_P = 0.6522; M = 1$$

The under damped response equation:

$$H_C(t) = MK_P\left[1 - \frac{e^{-t\zeta/\tau}}{\sqrt{1-\zeta^2}}\sin\left(\sqrt{1-\zeta^2}\,\frac{t}{\tau} + \tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta}\right)\right]$$

Putting the parameter values into the above equation, the response may be obtained.

EXAMPLE 9.10 A first order system with a (PID) controller is shown in Figure 9.25. Answer the following items: (a) For the closed loop system, develop formulae for the t and z in terms of K_C, t_I, t_D, and t_P.

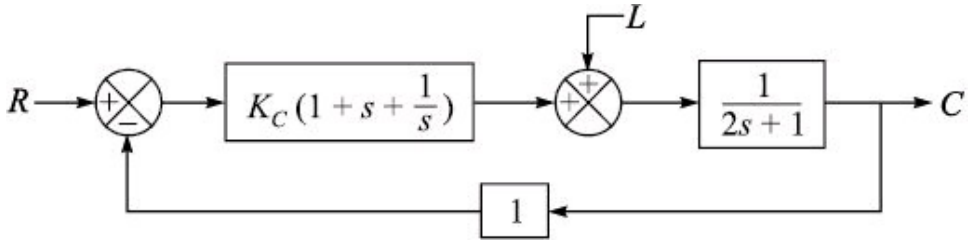

Figure 9.25 A first order system controlled by a (PID) controller.

Answer the following items: (a) For (i) $K_C = 0.5$ and (ii) $K_C = 2$, calculate the time constant and damping coefficient values exhibited by the closed loop system; (b) By increasing K_C, do the values of time constant and damping coefficient approach some limit? If so, find them; (c) For $K_C = 2$, calculate the offset when a unit step change in load is brought; (d) For $K_C = 2$ and 4, and unit step change in load, determine the maximum value of C and the time at which it occurs.

Solution: The general form of a (PID) controller and a unity gain first order system are:

$$G_C(s) = K_C\left[1 + \frac{1}{\tau_I s} + \tau_D s\right], \text{ and } G_P(s) = \frac{1}{\tau_P s + 1}$$

The closed loop TF with a unity feedback path will be:

$$\frac{C}{R} = \left[\frac{K_C(\tau_I\tau_D s^2 + \tau_I s + 1)}{\tau_I s(\tau_P s + 1)}\right] \Big/ \left[1 + \frac{K_C(\tau_I\tau_D s^2 + \tau_I s + 1)}{\tau_I s(\tau_P s + 1)}\right]$$

$$= \frac{K_C(\tau_I\tau_D s^2 + \tau_I s + 1)}{(\tau_I\tau_P + K_C\tau_I\tau_D)s^2 + (\tau_I + K_C\tau_I)s + K_C}$$

From the denominator,

$$\omega_N = \sqrt{\frac{K_C}{\tau_I(\tau_P + K_C\tau_D)}}$$

hence,

$$\tau = \frac{1}{\omega_N} = \sqrt{\frac{\tau_I(\tau_P + K_C\tau_D)}{K_C}}$$

and

$$\zeta = \frac{\tau_I(1 + K_C)}{2\tau_I(\tau_P + K_C\tau_D)}\sqrt{\frac{\tau_I(\tau_P + K_C\tau_D)}{K_C}} = \frac{(1 + K_C)}{2(\tau_P + K_C\tau_D)}\sqrt{\frac{\tau_I(\tau_P + K_C\tau_D)}{K_C}}$$

$$= \frac{(1 + K_C)}{2}\sqrt{\frac{\tau_I}{(\tau_P + K_C\tau_D)K_C}}$$

For $t_I = 1$; $t_D = 1$; $t_P = 2$; the expressions for t and z are:

$$\tau = \sqrt{\frac{2 + K_C}{K_C}}; \quad \zeta = \frac{(1 + K_C)}{\sqrt{(2 + K_C)K_C}}$$

Thus as K_C □ □; t □ 1 and z □ 0.5

Again for $t_I = t_D = 1$; $t_P = 2$; and $K_C = 0.5$; $t = 2.2361$; $z = 0.67082$

and for $t_I = t_D = 1$; $t_P = 2$; and $K_C = 2.0$; $t = 1.4142$; $z = 0.53033$

To evaluate the offset, let us first find the closed loop TF for load change.

$$\frac{C}{L} = \left(\frac{1}{\tau_P s + 1}\right) \Big/ \left(1 + \frac{K_C(\tau_I\tau_D s^2 + \tau_I s + 1)}{\tau_I s(\tau_P s + 1)}\right) = \frac{\tau_I s}{(\tau_I\tau_P + \tau_I\tau_D K_C)s^2 + (\tau_I + \tau_I K_C)s + K_C}$$

For, $L = 1/s$; $t_I = t_D = 1$; $t_P = 2$ and $K_C = 2$;

$$C(s) = \frac{1}{4s^2 + 3s + 2} = \frac{1}{2} \cdot \frac{1}{(\sqrt{2}s)^2 + 2\sqrt{2}\,\frac{3}{4\sqrt{2}}s + 1}$$

Thus, $C(t)$ is an Impulse response of a second order under damped system. At t =

∞, $C(t) = 0$. For $K_C = 0.5$, we may say that offset $= R(\infty) - C(t)|_{T\square\infty} = 0$.

Repeating the same analysis for any other finite values of K_C, it can be shown that zero offset values are zero.

To find the maxima let us write the load response equation,

$$C(t) = \frac{K_P}{\tau\sqrt{1-\zeta^2}} e^{-\zeta t/\tau} \sin\left(\sqrt{1-\zeta^2}\,\frac{t}{\tau}\right)$$

and differentiating,

$$\frac{dC(t)}{dt} = -\frac{K_P\zeta}{\tau^2\sqrt{1-\zeta^2}} e^{-t\zeta/\tau} \sin\left(\frac{\sqrt{1-\zeta^2}}{\tau}t\right) + \frac{K_P}{\tau^2} e^{-t\zeta/\tau} \cos\left(\frac{\sqrt{1-\zeta^2}}{\tau}t\right)$$

$$\left[p\cos\theta = -\frac{K_P\zeta}{\tau^2\sqrt{1-\zeta^2}};\quad p\sin\theta = \frac{K_P}{\tau^2};\quad \tan\theta = -\frac{\sqrt{1-\zeta^2}}{\zeta};\quad p = \frac{K_P}{\tau^2}\frac{1}{\sqrt{1-\zeta^2}} \right]$$

$$\therefore \qquad \frac{dC(t)}{dt} = \frac{K_P e^{-t\zeta/\tau}}{\tau^2\sqrt{1-\zeta^2}} \sin\left\{\frac{\sqrt{1-\zeta^2}}{\tau}t - \tan^{-1}\left(\frac{\sqrt{1-\zeta^2}}{\zeta}\right)\right\}$$

For maxima and minima

$$\frac{dC(t)}{dt} = 0$$

$$\therefore \qquad \sin\left\{\frac{\sqrt{1-\zeta^2}}{\tau}t - \tan^{-1}\left(\frac{\sqrt{1-\zeta^2}}{\zeta}\right)\right\} = 0 \quad = n\pi$$

because the exponential function will produce a trivial solution. From the above trigonometric solution, putting all the parameter values from the above analysis,

For $t = 1.4142$, $z = 0.53033$, $K_C = 2$ and $n = 0$

$$t = t_{\max} = \left\{\tan^{-1}\left(\frac{\sqrt{1-\zeta^2}}{\zeta}\right)\right\}\frac{\tau}{\sqrt{1-\zeta^2}} = 1.19345$$

For $t_{\mathrm{MAX}} = 1.19345$, and $K_C = 2$; $C(t)|_{\mathrm{MAX}} = C(t)|_{T=1.19345}$

EXAMPLE 9.11 The two close loop systems shown in Figures 9.26(a) and (b) have identical forward path and through the loop TF, but differ only by the entry point of

load variable into the process.

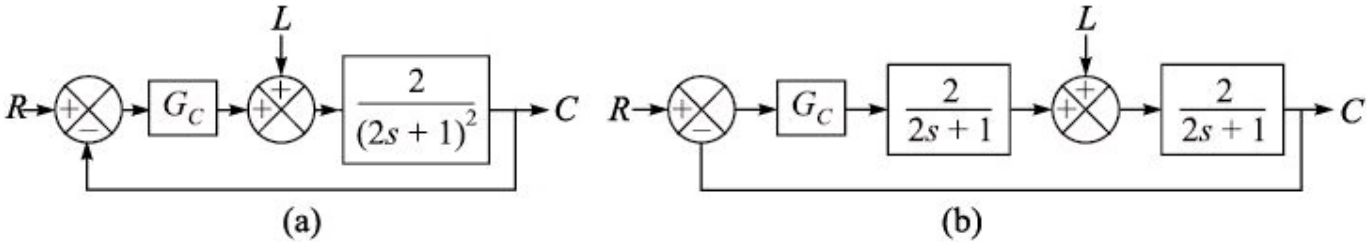

Figure 9.26 The block diagram of the system of example 9.11.

You have to check how this difference would effect the system's close loop performance by determining: (a) Offset; (b) Frequency of the transient response; (c) Transient response equations for a unit step of load as disturbance, with a (P) controller in the loop having a gain $K_C = 5$.

Solution: The closed loop TF for the systems in Figures 9.26(a) and (b) may be found as:

(a)

$$\frac{C(s)}{L_1(s)} = \frac{2}{4s^2 + 4s + 11};\ \tau = 0.603;\ \zeta = 0.3015;$$

$$= \frac{1}{11} \cdot \frac{2}{(4/11)s^2 + (4/11)s + 1}$$

for $L_1(s) = \frac{1}{s}$; $C(t)\,|_{t\to\infty} = \frac{2}{11}$

$$\text{Offset} = R(\infty) - C(t)\,|_{t\to\infty} = 0 - \frac{2}{11} = -\frac{2}{11}$$

(b)

$$\frac{C(s)}{L_2(s)} = \frac{2s+1}{4s^2 + 4s + 11};\ \tau = 0.603;\ \zeta = 0.3015$$

$$= \frac{1}{11} \cdot \frac{2s+1}{(4/11)s^2 + (4/11)s + 1}$$

for $L_2(s) = \frac{1}{s}$; $C(t)$ will have an impulse response part that will decay to 0 as t increases, and a step part that will become = K_P as t increases

$$C(t)\,|_{t\to\infty} = \frac{1}{11};\ \text{Offset} = -\frac{1}{11}$$

The response in both cases would have same frequency.

$$\omega = \omega_N\sqrt{1-\zeta^2} = \frac{\sqrt{1-\zeta^2}}{\tau} = 1.58114 \text{ rads/time}$$

The response equation for $L_1(s) = 1/s$, would be according to:

$$C(t) = K_P\left[1 - \frac{e^{-t\zeta/\tau}}{\sqrt{1-\zeta^2}}\sin\left(\sqrt{1-\zeta^2}\,\frac{t}{\tau} + \tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta}\right)\right]$$

The response equation for $L_2(s) = 1/s$, would be according to:

$$C(t) = K_{P1}\left[1 - \frac{e^{-t\zeta/\tau}}{\sqrt{1-\zeta^2}}\sin\left(\sqrt{1-\zeta^2}\,\frac{t}{\tau} + \tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta}\right)\right] + \frac{K_{P2}}{\tau\sqrt{1-\zeta^2}}e^{-\zeta t/\tau}\sin\left(\sqrt{1-\zeta^2}\cdot\frac{t}{\tau}\right)$$

The above step response equation describes the non zero part for the second type of input

For case (a) For case (b)

$\zeta = 0.3015;\quad \tau = 0.603;\quad K_P = \frac{2}{11}$ $\zeta = 0.3015;\quad \tau = 0.603;\quad K_{P1} = \frac{1}{11};\quad K_{P2} = \frac{2}{11}$

The responses may be plotted on a graph. The task is left to the reader.

EXAMPLE 9.12 A (PI) controller is in the closed loop with a first order process. If we increase the process time constant, the integral time and gain of the controller should be changed in what way, so that the close loop response may remain at the same specification?

Solution:

(a) Let us assume $G_M(s) = G_V(s) = 1$. The closed loop transfer function:

$$\frac{C}{R} = \frac{K_C\left[1+\frac{1}{\tau_I s}\right]\cdot\left[\frac{K_P}{\tau_P s+1}\right]}{1+K_C\left[1+\frac{1}{\tau_I s}\right]\cdot\left[\frac{K_P}{\tau_P s+1}\right]} = \frac{K_C K_P \tau_I s + K_C K_P}{\tau_I\tau_P s^2 + (\tau_I + K_C K_P \tau_I)s + K_C K_P}$$

this is a second order TF, with

$$\omega_N = \sqrt{\frac{K_C K_P}{\tau_I \tau_P}} \quad \text{(A)}$$

and $\zeta = \frac{(1+K_C K_P)}{2}\sqrt{\frac{\tau_I}{\tau_P K_C K_P}} = \frac{1}{2}\sqrt{\frac{\tau_I}{\tau_P}}\left[\frac{1+K_C K_P}{\sqrt{K_C K_p}}P\right]$ (B)

1. If t_P is □, to keep w_N constant, the t_I has to be □ and/or, K_C has to be □, from Equation A.
2. On the other hand, if t_P is □, to keep z constant, the t_I has to be □ and/or, K_C has to be □, from Equation B. Hence, the general conclusion is that, we cannot keep both w_N and z to their original value by adjusting t_I and K_C after an increase of t_P has occurred. The result for a given system with specific values of the parameter may show a different solution. However, if the overshoot characteristic is more important for a specific control loop, we should go for

the damping coefficient, z, but if the settling time is considered to be more significant, we have to keep the natural frequency, w_N, constant.

EXAMPLE 9.13 It is desired to control the exit, C_C, of the perfectly stirred constant holdup liquid blending system shown in Figure 9.27. The inlet stream 1 of a weak solution, whose concentration may vary from 0.005 to 0.01 g/l may be considered as the load variable. It is mixed with a concentrated solution stream 2 of same solute, to maintain the exit stream 3 concentration at the desired value of 0.006 g/l. Stream 1 flow rate q is constant at 10 l/min (l/m). Stream 2 flow rate is manipulated by control action, while its concentration is constant at 50 g/l.

The following information is available: (a) The blender tank volume, V = 500 l; (b) The control valve may be taken as a pure gain element, K_V = 0.6 l/m/psi; (c) All the streams and liquid inside the blender have equal and time invariant densities, and the amount of total flow through the blender may be taken equal to 10 l/m; (d) The transmitted output signal varies linearly from 4 to 20 mA as C_C varies from 0 to 0.02 g/l; (e) The current to pressure transducer has negligible dynamics and has a gain of 0.75 psi/mA; (f) Controller mode is (P) only.

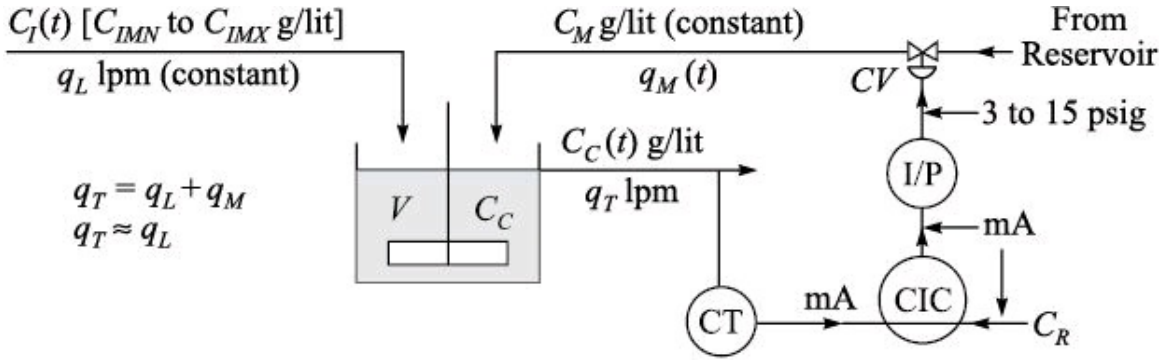

Figure 9.27 The concentration control system of a constant holdup blender.

Based upon this information, do the following: (a) Draw a block diagram for the composition control scheme, using the symbols in the above figure as much as possible; (b) Derive expressions for each TF in the block diagram and substitute numerical values for the parameters; (c) Derive an expression for overall TF of the control loop; (d) Find the closed loop response equation as $C_C(t)$, and the offset for a step change in C_R from 0.006 to 0.009 g/l.

Solution:

$$K_R = K_M = 16/0.02 = 800\ \frac{\text{mA}}{\text{g/l}};\ K_{IP} = 0.75\ \frac{\text{psi}}{\text{mA}};\ K_V = 0.6\ \frac{\text{l/m}}{\text{psi}} = 600\ \frac{\text{cc/min}}{\text{psi}}$$

An extra gain element in the load variable path K_L is needed to convert C_I g/l to g/m, and a gain element K_{MV} is needed after K_V to convert MV in l/m to MV in g/m.

These would be:

$K_L = q = 10$ l/m; $K_{MV} = 50$ g/l $= C_M$; also, $K_P = 1/q = 0.1$ min/l; $t_P = V/q = 50$ min;

$C_R = 0.006$ g/l; $C_{IMN} = 0.005$ gpl and $C_{IMX} = 0.01$ gpl.

The block diagram of this control system in Figure 9.28 becomes:

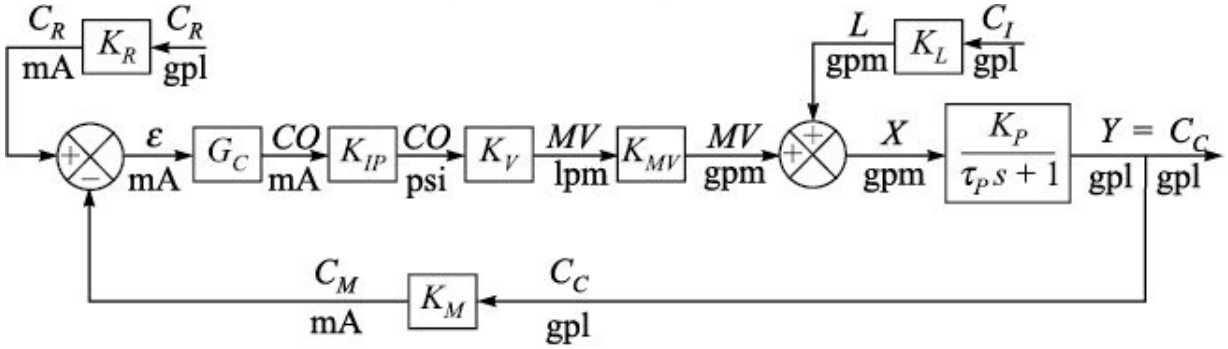

Figure 9.28 Block diagram of the concentration control system.

The closed loop TF for set point change,

$$\frac{C_C}{C_R} = \frac{(K_R K_C K_{IP} K_V K_{MV} K_P)/(\tau_P s + 1)}{1 + (K_C K_{IP} K_V K_{MV} K_M K_P)/(\tau_P s + 1)} = \frac{1800K_C/(\tau_p s + 1)}{1 + 1800K_C/(\tau_p s + 1)} = \frac{1800K_C/(1 + 1800K_C)}{[\tau_P/(1 + 1800K_C)]s + 1}$$

The input in set point is, $C_R(s) = \dfrac{0.003}{s}$.

$$C_C(s) = \frac{0.003}{s} \cdot \frac{1800K_C/(1 + 1800K_C)}{[50/(1 + 1800K_C)]s + 1}$$

$$\therefore \qquad K_{PCL} = \frac{1800K_C}{1 + 1800K_C} \quad \text{and} \quad \tau_{PCL} = \frac{50}{1 + 1800K_C}$$

At large values of time,

$$C_C(t)\,|_{t\to\infty} = \frac{0.003 \times 1800K_C}{1 + 1800K_C} \approx 0.003 \quad \left[\because \quad \frac{1800K_C}{1 + 1800K_C} \approx 1.0\right]$$

Hence, $\text{Offset} = R - C(t)\,|_{t\to\infty} = 0.003 - 0.003 = 0$

EXAMPLE 9.14 The concentration control system described in the above example is at zero error steady state for $C_R = 0.0075$ g/l, with a $C_I = 0.005$ g/l. A pulse type disturbance in C_I enters into the system (pulse height = 0.002 g/l, pulse width = 5 min). Find the response in $C_C(t)$ for (a) During duration of the pulse; (b) After the pulse has ended.

Solution: The closed loop TF for load change,

$$\frac{C_C}{C_L} = \frac{(K_L K_P)/(\tau_P s + 1)}{1 + (K_C K_{IP} K_V K_{MV} K_M K_P)/(\tau_P s + 1)} = \frac{K_C/(50s + 1)}{1 + 1800K_C/(50s + 1)} = \frac{K_C}{[50/(1 + 1800K_C)]s + 1}$$

$$K_{PCL} = K_C \quad \text{and} \quad \tau_{PCL} = \frac{50}{1 + 1800K_C}$$

and the pulse magnitude $M = 0.02$ g/l

During the continuation of the pulse, the +ve step response would be according to:

$$C_C(t)|_{0 \to T_1} = 0.02 \times K_{PCL}[1 - e^{-t/\tau_{PCL}}] = 0.02K_C[1 - e^{-t/\tau_{PCL}}]$$

At the end of pulse, C_C would attain a value, $C_C^* = 0.02K_C[1 - e^{-5/\tau_{PCL}}]$; and after the pulse has ended, $C_C(t)$ would show a response, $C_C(t)|_{T_1 \to \infty} = C_C^* e^{-(t-T_1)/\tau_{PCL}}$. Where, T_1 = 5 min.

EXAMPLE 9.15 To heat a liquid stream, the stream is cascaded through two tanks fitted with an electrical immersion heater and a mechanical agitator. As both tanks are situated at the same level, there is a pump which transfers the liquid from the first to the second tank. As liquid from the tanks exits as overflows, the liquid holdups in the tanks remain constant during operation. The flow rate of liquid, Q, is essentially constant through the system, while the inlet liquid stream temperature, T_I, may vary. The pump with its suction and delivery pipe constitutes a delay or dead time in the dynamics. Electrical heaters with thyristor drives are installed with both tanks. The thyristors may receive 4–20 mA as controller output, or as a constant current between the range. Heat outputs from the heaters vary linearly corresponding to the 4–20 mA signal. A Pt-100 temperature sensor-transmitter (Input: 0–200°C; output: 4–20 mA) situated in the 2nd tank supplies the process output to the controller. A schematic of the system is shown in Figure 9.29.

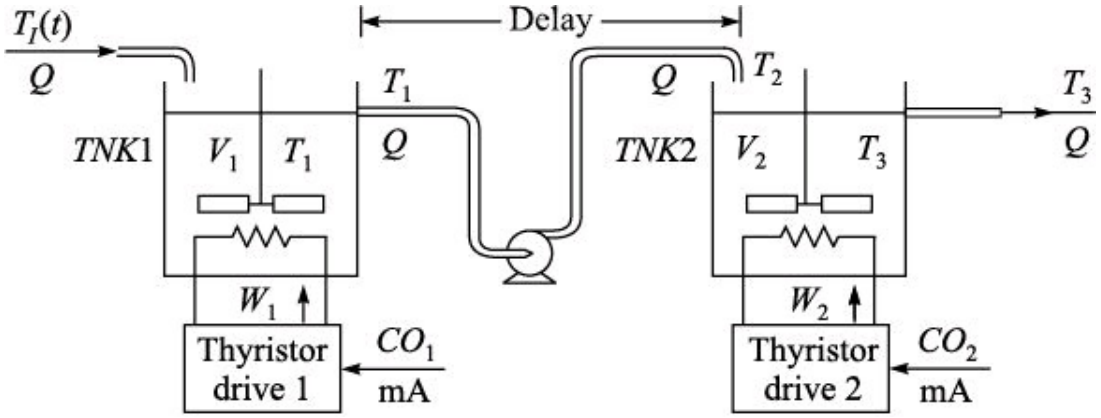

Figure 9.29 The electrically heated stirred liquid heater.

It is required to control the exit liquid temperature from the second tank, but as a single loop controller is selected for this duty, only one of the heaters will be

controller driven, and the remaining heater will produce a constant heat output corresponding to a constant value of current signal within the range 4–20 mA to the thyristor drive. Assuming the tanks are perfectly stirred, the bulk and exit liquid temperatures from these two tanks are identical. Further assuming that there is no heat transfer between the system and surrounding, liquid temperatures at the entry and exit ends of the delay line will be same at steady state condition, but they will be different in transient state.

Draw quantitative block diagrams for (a) Heater of the 2nd tank is controller driven, (b) Heater of the 2nd tank is controller driven. Also derive the CLTF for regulatory and servo type disturbances in both the cases. Available data for the system are: V_1 = 40 l,
V_2 = 30 l, Q = 2 l/m, Dead time, t_{DL} = 1min, the liquid properties are, r = 0.8948 g/cc, C_P = 0.95 cal/(g°C). The undisturbed steady values of temperatures are T_I = 40°C, T_1 = 65°C,
T_2 = 65°C, T_3 = 85°C. FCEs (heaters with thyristor actuators) may be taken as pure gain elements, their gain values are: K_{V1} = 5 kcal/mA, K_{V2} = 4 kcal/mA. While the first tank heater is controller driven, heater of the second tank receives a constant signal of CO_{2S} = 8.5 mA, similarly in case of second, is under control action, the first tank heater drive receives a constant signal CO_{1S} = 8.5 mA. These input signals are sufficient to maintain exit temperatures at their undisturbed normal steady state values mentioned above.

Solution: We can derive the TFs of the individual blocks of the system from the given information. The time constants of the tanks are $t_1 = V_1/Q$ = 20 min, $t_2 = V_2/Q$ = 15 min, and the delay time, t_{DL} = 1 min. The process TFs for the load path may be readily derived as:

$$G_{PL1}(s) = \frac{Q\rho C_p}{(V_1/Q)s+1} = \frac{1.7}{20s+1}\frac{\text{kcal/min}}{\text{°C}}; \; G_{PL2}(s) = \frac{1/(Q\rho C_p)}{(V_2/Q)s+1} = \frac{0.588}{15s+1}\frac{\text{°C}}{\text{kcal/min}}$$

The process TFs on the manipulated variable path for the first tank may be obtained as associated with the TF of FCE. For the second tank, as the summer operates with temperatures, hence its output has to be multiplied with process gain K_{PL2} Thus,

$$G_{PM1} = \frac{K_{V1}}{\tau_1 s+1} = \frac{5}{20s+1}\frac{\text{kcal/min}}{\text{mA}}; \; G_{PM2} = \frac{K_{V2}K_{PL2}}{\tau_2 s+1} = \frac{4\times 0.588}{15s+1} = \frac{2.352}{15s+1}\frac{\text{°C}}{\text{mA}}$$

For the same reason, the dimension of the output variable has to be changed from kcal/min to °C, by putting the process gain block. As this gain is a common feature for

both of the tanks, let this be denoted by $K_P = 1/(QrC_P) = 0.588$ °C/(kcal/min). The delay block and the measuring element has been already defined in the problem statement, thus, $G_{DL(S)} = e^{-S}$, and $G_M(s) = K_M = (20 - 4)/200 = 0.08$ mA/°C. The block diagram for Part 1 now may be proposed as shown in Figure 9.30.

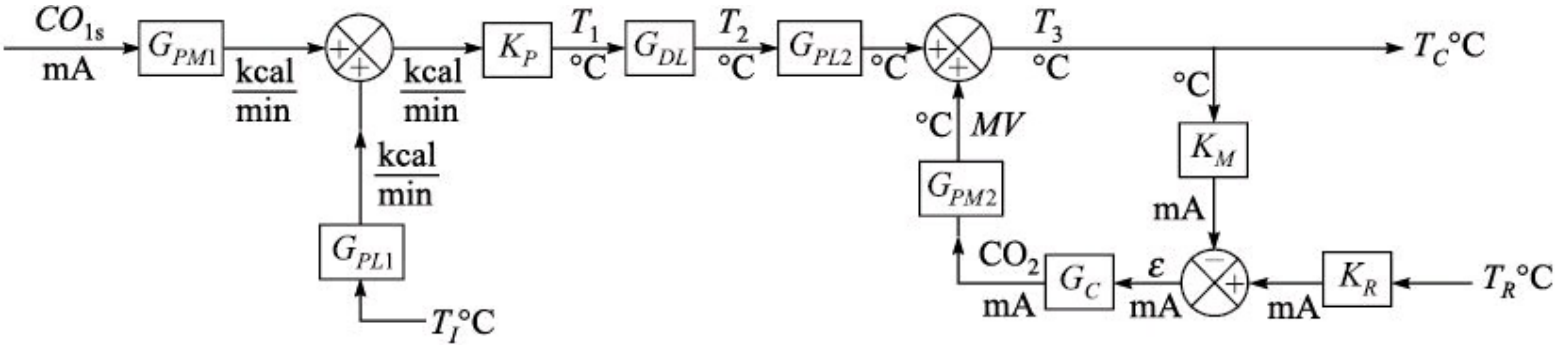

Figure 9.30 Block diagram for part 1.

One thing should be noted that as CO_{1S} is a constant quantity and is equivalent to a temperature rise of 25°C, it cannot enter into the block diagram analysis, and for its disappearance, the concerned temperatures should be redefined as:

$$T\square_I = T_I - 25°\text{C}; T\square_R = T_R - 25°\text{C}; T\square_C = T_C - 25°\text{C}$$

The close loop TFs may be easily determined from the block diagram:

$$\frac{T_C'(s)}{T_I'(s)} = \left[\frac{e^{-s}}{(20s+1)(15s+1)}\right] \Big/ \left[1 + \frac{0.188G_C}{(15s+1)}\right];$$

and,

$$\frac{T_C'(s)}{T_R'(s)} = \left[\frac{0.1882G_C}{15s+1}\right] \Big/ \left[1 + \frac{0.1882G_C}{(15s+1)}\right]$$

For Part 2, as CO_{2S} is a constant quantity and is equivalent to a temperature rise of 20°C, a new set of deviation variables in temperature may be defined as:

$$T\square\square_I = T_I - 20°\text{C}; T\square\square_R = T_R - 20°\text{C}; T\square\square_C = T_C - 20°\text{C}$$

Now the block diagram for Part 2 now may be proposed as shown in Figure 9.31.

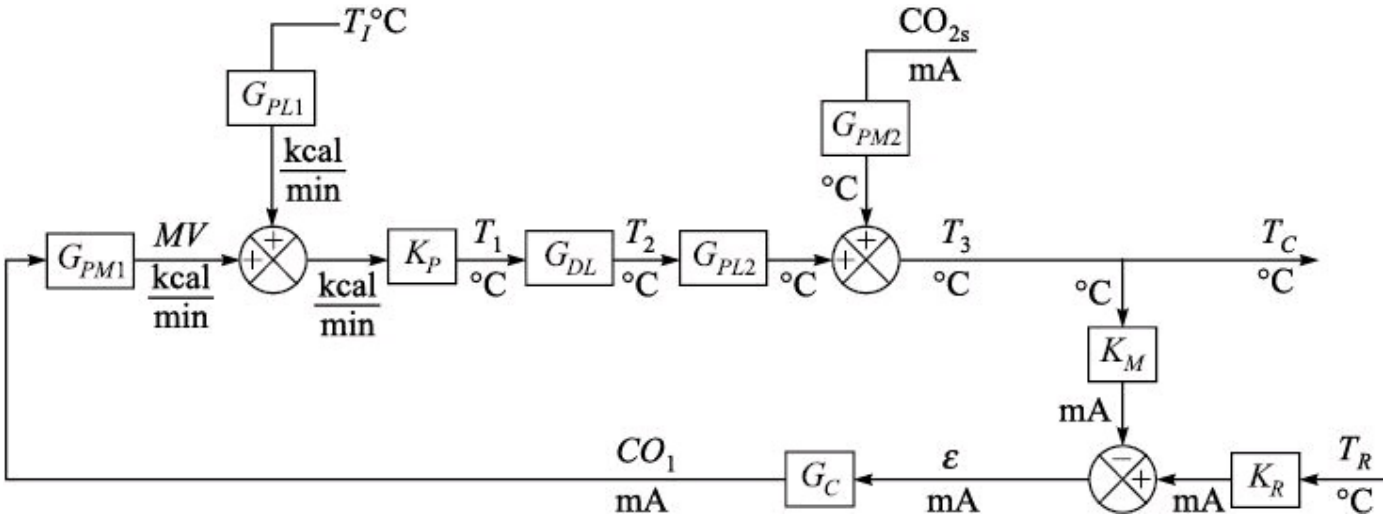

Figure 9.31 Block diagram for part 2.

From Figure 9.31, using the individual block TFs proposed above, the CLTF for regulatory and servo type disturbances may be found as:

$$\frac{T_C''(s)}{T_I''(s)} = \left[\frac{e^{-s}}{(20s+1)(15s+1)}\right] \Big/ \left[1 + \frac{0.2352G_C}{(20s+1)(15s+1)}\right]$$

and,

$$\frac{T_C''(s)}{T_R''(s)} = \left[\frac{0.1882G_C}{15s+1}\right] \Big/ \left[1 + \frac{0.1882G_C}{(15s+1)}\right]$$

During time domain analysis, the results in terms of deviation variables, the conversion in absolute temperatures may be obtained by the following simple addition:

$$T_I = T\square_I + 25°\text{C};\ T_R = T\square_R + 25°\text{C};\ T_C = T\square_C + 25°\text{C}$$

$$T_I = T\square\square + 20°\text{C};\ T_R = T\square\square + 20°\text{C};\ T_C = T\square\square_C + 20°\text{C}$$

9.4 Special Cases of Closed Loop Response

We all know that a (P) controlled system always shows offset in its closed loop response. But there is a possibility in case of processes containing an integrator in its TF that a zero offset response may be obtained with only a (P) controller in the loop under certain kind of disturbances. Let us demonstrate this behaviour by an example.

EXAMPLE 9.16: An interesting fact about the closed loop response property of processes that contain an integrator in their TF, or a process that is a pure integrator, is that their response may be of zero offset quality with only a (P) controller for certain kinds of disturbance. Consider the liquid level system illustrated in Figure 9.32, in which a constant throughput pump delivers a liquid to a downstream process the inflow or supply, q_I is variable due to which, the tank level may fall to endanger the

pump. The manipulated stream flow, q_M, is used keep up the level to its set value. Determine for which of the step disturbances, the response would be offset free with only (P) controller, in (a) set point; (b) load.

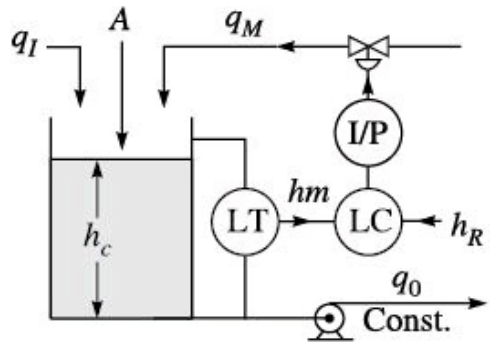

Figure 9.32 The level control system.

Solution: It has been established in Chapter 4 that an empty tank with the closed bottom discharge line, or a tank with a constant throughput discharge pump having suction connected to the tank bottom, dynamically represents an integrating element. Thus, the level of the liquid reservoir, H_C, in this problem has TFs with respect to Q_M and Q_I,

$$G_{PL}(s) = \frac{H_C(s)}{Q_I(s)} = \frac{1}{As} \text{ and } G_{PM}(s) = \frac{H_C(s)}{Q_M(s)} = \frac{1}{As}$$

The block diagram of this control system may be constructed as:

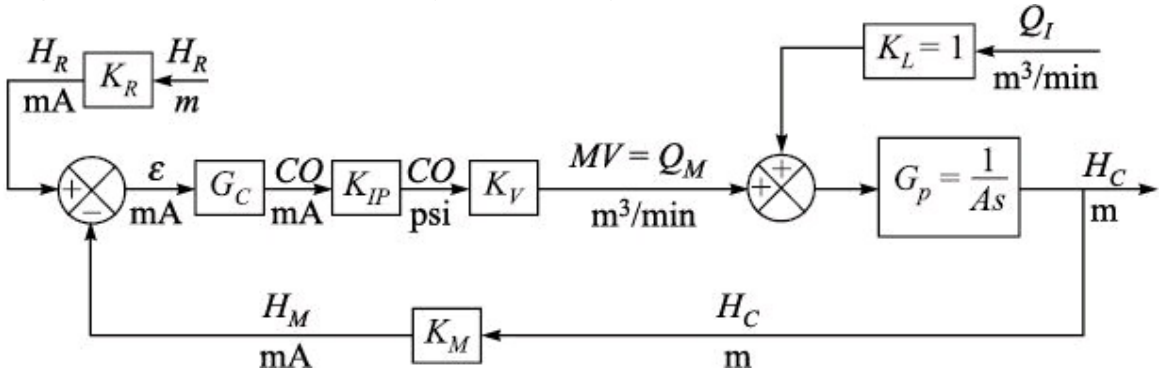

Figure 9.33 Block diagram of the level control system.

(a) For a (P) controller $G_C = K_C$. The closed loop TF for a set point change,

$$\frac{H_C}{H_R} = \frac{(K_R K_C K_{IP} K_V)/(As)}{1 + (K_C K_{IP} K_V K_M)/(As)} = \frac{K^*}{As} \cdot \frac{As}{As + K^*} \quad [\text{Assuming } K_R = K_M]$$

□ $$\frac{H_C}{H_R} = \frac{K^*}{As + K^*} = \frac{K^*/K^*}{(A/K^*)s + 1} = \frac{1}{\tau s + 1}$$

As the system gain is unity, for set point step change of magnitude M, the ultimate response value also would be M. Thus, offset value is zero.

(b) For a (P) controller $G_C = K_C$. The closed loop TF for a load change,

$$\frac{H_C}{Q_I} = \frac{1/(As)}{1+(K_C K_{IP} K_V K_M)/(As)} = \frac{1}{As} \cdot \frac{As}{As+K^*} = \frac{1/K^*}{\tau s+1}$$ [the close loop gain is $1/K^*$]

Thus for a unit step in load, the offset would be: $1 - 1/K^* = (K^* - 1)/K^*$.

As of now, we were mainly dealing with response to step disturbance. The following example demonstrates a special response property under ramp disturbance of set point of a control system. Such disturbances are common in programmed control application and often become a challenging exercise for a designer.

EXAMPLE 9.17 In the liquid level control system shown in Figure 9.34, there is a (PI) controller in the loop. The set point, H_R, follows a linear time dependent rise from an initial value, h_S, according to: $H_R = h_S + Mt$. For such a ramp input in set point, the response after the initial transients should show a steady linear increase. The set point and uncontrolled response should be parallel if the forward path gain is unity, and vertical distance between them is the steady *error* or offset at large values of time. The control action may reduce, or even eliminate this offset. Derive the relation between system parameters required for a zero-offset condition for such a close loop system.

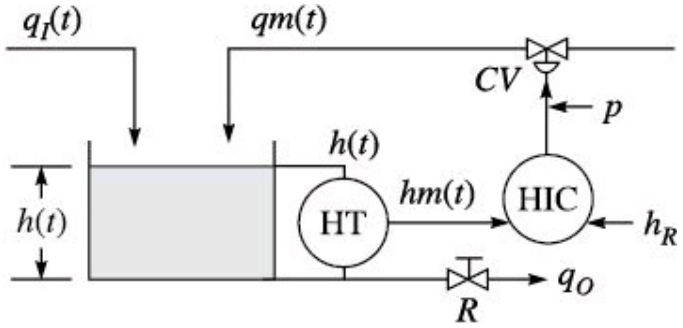

Figure 9.34 The level control system of example 9.17.

Solution: According to the description of the problem, the following block diagram as shown in Figure 9.35 may be drawn:

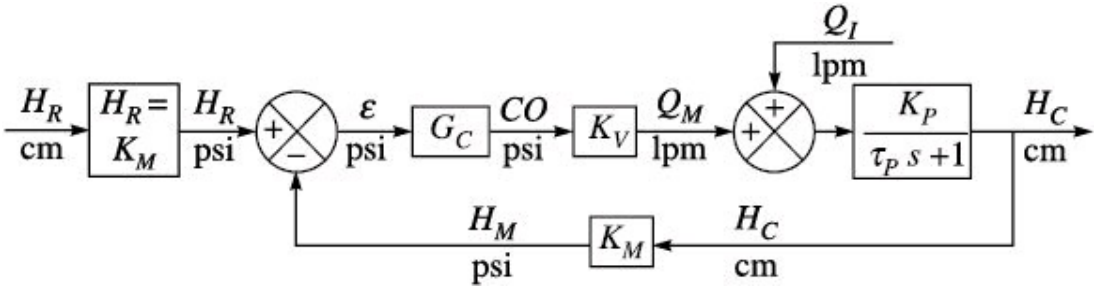

Figure 9.35 The block diagram of the level control system of example 9.17.

The TFs of the blocks in the above figure may be written as:

$$G_C = K_C\left[1+\frac{1}{\tau_I s}\right];\ G_P = \frac{K_P}{\tau_P s+1};\ K_P = R;\ K_R = K_M$$

Now the regulatory and servo closed loop TFs may be developed as:

$$\frac{H_C}{H_R} = \frac{K_R K_C[1+1/(\tau_I s)]K_V[K_P/(\tau_P s+1)]}{1+K_C[1+1/(\tau_I s)]K_V K_m[K_P/(\tau_P s+1)]}$$
$$= \frac{K_R K_C(\tau_I s+1)K_V K_P}{\tau_I\tau_P s^2 + \tau_I(1+K_C K_V K_P K_m)s + K_C K_V K_P K_m}$$

The above result could be splitted into two standard quadratic forms (second order),

$$\frac{H_C}{H_R} = \underbrace{\frac{K_1 s}{As^2+Bs+C}}_{\text{PART I}} + \underbrace{\frac{K_2}{As^2+Bs+C}}_{\text{PART II}}$$
$$= \underbrace{\frac{K_1}{C}\cdot\frac{(C/A)s}{s^2+(B/A)s+(C/A)}}_{\text{PART I}} + \underbrace{\frac{K_2}{C}\cdot\frac{(C/A)}{s^2+(B/A)s+(C/A)}}_{\text{PART II}}$$
$$= \underbrace{K_1^*\frac{\omega_N^2 s}{s^2+2\zeta\omega_N s+\omega_N^2}}_{\text{PART I}} + \underbrace{K_2^*\frac{\omega_N^2}{s^2+2\zeta\omega_N s+\omega_N^2}}_{\text{PART II}}$$

From the last expression of the two parts of the closed loop TF, we may write the closed loop second order system parameters as:

$$\therefore \quad \omega_{NCL} = \sqrt{\frac{C}{A}} = \sqrt{\frac{K_C K_V K_P K_m}{\tau_I\tau_P}}\ ;\ \zeta_{CL} = \left(\frac{B}{A}\right)\frac{1}{2\omega_{NCL}} = \frac{1+K_C K_V K_P K_m}{2\tau_P}\sqrt{\frac{\tau_I\tau_P}{K_C K_V K_P K_m}}$$

$$K_1^* = K_{1CL} = \frac{K_R K_C K_V K_P}{K_C K_V K_P K_m}\tau_I = \tau_I\ ;\ K_2^* = K_{2CL} = \frac{K_R K_C K_V K_P}{K_C K_V K_P K_m} = 1$$

For the ramp input in set point having a slope of M, $H_R(s) = M/s^2$, Part I of the TF would produce a step response and Part II of the TF would produce a ramp response.

The ultimate value of the step response and ultimate or stable part of ramp responses are:

$$\text{Response}_{\text{PT I}} = MK_{1CL} = Mt_I \text{ and } \text{Response}_{\text{PT II}} = MK_{2CL} = M$$

Also from the property of second order ramp response equations of Chapter 6, we may write the stable part of the ramp response

$$Y(t)\,|_{t\to\infty} = M\left[t - \frac{2\zeta}{\omega_N}\right]$$

from which the constant value of error (for a unity gain system) of the ramp response

$= 2Mz/w_N$.

Hence, for zero offset $M\tau_I = \dfrac{2M\zeta_{CL}}{\omega_{NCL}}$ or $t_I = 2z_{CL}t_{PCL}$

9.5 Effect of Measurement Lag on the Closed Loop Response

In all of the above analysis, the measuring element has been assumed to be a pure gain element. For a large number of applications this is satisfactory. However, for systems with higher response speed this assumption needs to be modified. At the beginning of Chapter 7, the second example that describes a continuous molasses sterilization system where the process response speed is comparable with the measuring element, the TF of the $G_M(s)$ should be considered during determination of the closed loop TF. A block diagram has been shown in Figure 9.36 of such a system, assuming a first order dynamics for the block, G_M.

The statements made in concern to the closed loop system of Figure 9.12 are all true for Figure 9.36, excepting that the measuring block contain a first order TF with unity gain. The responses are obtained by writing a computer programme that digitally simulates the control loop.

The closed loop TFs for the system in Figure 9.22 are:

For regulatory disturbance, $$\frac{C}{L} = \frac{1/(\tau_P s + 1)}{1 + 0.5G_C/[(\tau_P s + 1)(\tau_M s + 1)]} \qquad (9.16)$$

and for servo disturbance, $$\frac{C}{R} = \frac{0.5G_C/(\tau_P s + 1)}{1 + 0.5G_C/[(\tau_P s + 1)(\tau_M s + 1)]} \qquad (9.17)$$

As the TFs contain two distinct poles or a second order lag, the dynamic significance is that keeping the other parameters as they are, the dynamic lag would produce a more sluggish response, and we shall learn in the next chapter that such increase of lag, move the system toward more unstable conditions.

To check this statement, the responses of the system in Figure 9.36 with a (P) controller have been obtained. Figure 9.37 shows the response character for variation o f K_C at constant t_M, and Figure 9.38 for variation of t_M at constant K_C. The responses due to load change are in part (a) and that for set point change are in part (b) of each of these figures. Considering the on set of oscillatory response, it may be concluded that increase of either K_C or t_M begins and then pronounces the oscillation.

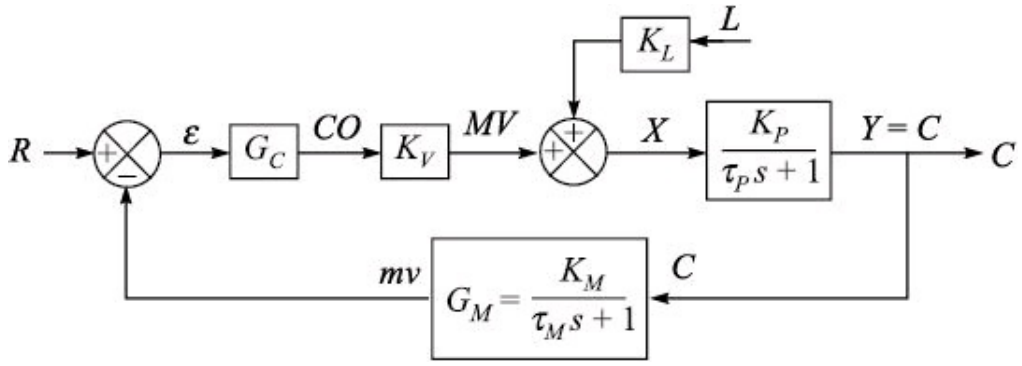

Figure 9.36 Control system containing measurement lag in the feedback path.

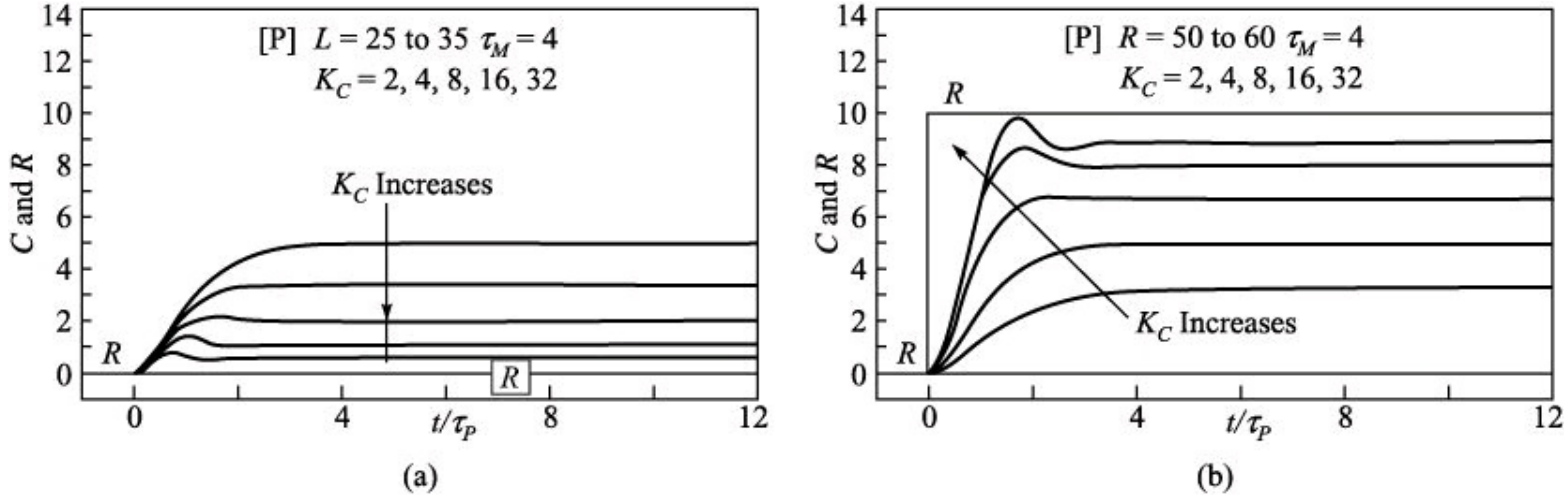

Figure 9.37 Closed loop responses for different K_C values at constant t_M for (P) controlled system.

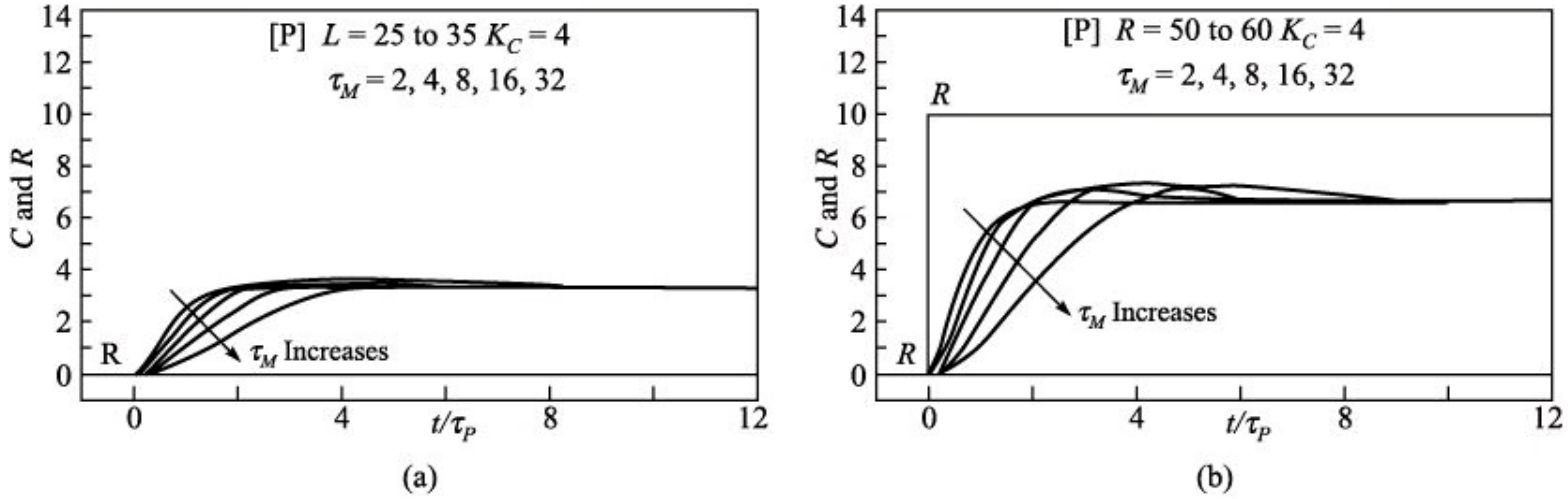

Figure 9.38 Closed loop responses for different t_{DL} values at constant K_C for (P) controlled system.

Coughanowr and Koppell have indicated an interesting result contrary to the conventional property of a (PD) controlled first order process in presence of a measurement lag in the loop through the following example.

EXAMPLE 9.18 We have already commented by adding derivative action to a (P) controller does not change the magnitude of offset, only increases the speed of response. Such a notion may not always be true if the feedback path of the loop contains RC-type lags. In this example, this possibility is investigated with respect to the closed loop response of the system depicted in Figure 9.41. Let us take the following TFs of blocks of the loop:

$$G_C(s) = K_C(1 + t_D s),\ G_P(s) = 1/(t_P s + 1),\ G_M(s) = 1/(t_M s + 1)$$

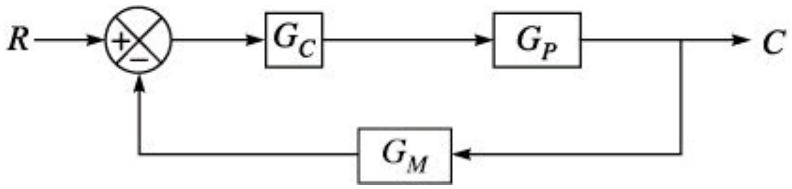

Figure 9.39 The closed loop response of a system.

With the following parameter values: $t_P = 60$ time unit, $t_M = 10$ time unit, determine the offset value for: (a) $t_D = 0$; and (b) $t_D = 3$ time unit, if the system is disturbed by a unit step in set point. Adjust the controller gain to obtain a slightly under-damped response, say for $z = 0.8$.

Solution: The closed loop TF for the above system is:

$$\frac{C}{R} = \left[\frac{K_C(1+\tau_D s)}{(\tau_P s+1)}\right] \Big/ \left[1 + \frac{K_C(1+\tau_D s)}{(\tau_P s+1)(\tau_M s+1)}\right] = \frac{K_C[\tau_D\tau_M s^2 + (\tau_D+\tau_M)s + 1}{\tau_P\tau_M s^2 + (\tau_P+\tau_M+\tau_D K_C)s + 1 + K_C}$$

or, $$\frac{C}{R} = \left(\frac{K_C}{1+K_C}C\right)\frac{\tau_D\tau_M s^2 + (\tau_D+\tau_M)s + 1}{[(\tau_P\tau_M)/(1+K_C)]s^2 + [(\tau_P+\tau_M+\tau_D K_C)/(1+K_C)s + 1]}$$

From the last equation, expressions for the close loop response parameters are:

$$\tau = \sqrt{\frac{\tau_P\tau_M}{1+Kc}},\ \zeta = \frac{\tau_P+\tau_M+\tau_D K_C}{2\tau_P\tau_M}\sqrt{\frac{\tau_P\tau_M}{1+K_C}} = \frac{\tau_P+\tau_M+\tau_D K_C}{2\sqrt{\tau_P\tau_M(1+K_C)}}$$

For a unit step in R, the ultimate response value may be found by applying final value theorem: $C(t)|_{T\to\infty} = K_C/(1 + K_C)$. Now the required value of K_C may be found from the derived equation for z. For $t_D = 0$

Thus $$0.7 = (60+10)/[2\sqrt{600(1+K_C)}] \text{ or } K_C = 3.167.$$

Hence, $$\text{Offset} = R - C(t)|_{T\to\infty} = 1 - K_C/(1+K_C) = 1 - 0.76 = 0.24$$

In case (b), for $t_D = 3$ time unit, the value of K_C is evaluated from the equation for z as,

$$0.7 = \frac{70 + 2K_C}{2\sqrt{600(1+K_C)}} \text{ or } 4K_C^2 + 896K_C + 3724 = 0$$

from which, $K_C = 4.25$ and 219.75

Taking the lower value of K_C, offset $= R - C(t)|_{T\to\infty} = 1 - K_C/(1+K_C) = 1 - 0.81 = 0.19$.

Taking the lower value of K_C, offset $= R - C(t)|_{T\to\infty} = 1 - K_C/(1+K_C) = 1 - 0.99 =$

0.01.

With (D) mode 'offset' reduces, T_P and w are increased. However, by addition of (D) mode in a (P) controller, this improvement of control quality is not a general phenomenon, needs evaluation for processes with other individual dynamic character.

One analysis about the use of (PD) controller may be forwarded in the following way: the analysis assumes a constant set point situation when the process has been disturbed from the steady state by a step change in load. It has been further assumed that the process variable would show a first order or an approximately first order dynamic response. If uncontrolled, the error, $e(t)$, will grow to a steady value, e_M, following a first order dynamics according to:

$$\varepsilon(t) = \varepsilon_M(1 - e^{-t/\tau_P}) \quad \text{or} \quad \tau_P \frac{d\varepsilon(t)}{dt} + \varepsilon(t) = \varepsilon_M \qquad (9.18)$$

If only a small magnitude of error appears at the beginning of this response, the output of (P) and (I) actions are hardly significant. But the first order nature of the error dynamics would show an initial slope of e_M/t_P, within this small magnitude, and the corrective action may be expressed as:

$$MV = K_C\varepsilon_M = K_C\left[\varepsilon(t) + \tau_P \frac{d\varepsilon(t)}{dt}\right] \qquad (9.19)$$

Equation 9.19 exactly expresses the dynamic equation of a (PD) controller. Thus, if the above assumptions are correct for a process, a (PD) controller would be adequate.

In the next section we shall consider the situation when there is a delay or dead time element in the loop. Irrespective of the dead time on the forward or feedback path of the loop, the dynamic significance will be identical. We have assumed that the dead time is associated with the first order process block, so that the process model may be expressed by an FOPDT (First Order Plus Dead Time) structure. This assumption would not be unrealistic, as in a large volume of process control, analysis and design literature have been built on the identical assumption, and if the dead time part is small enough the model reduces to first order.

9.6 Effect of Dead Time on Closed Loop Response

So far the effect of controller gain, K_C, and lag elements in the loop on the response have been studied mainly for absence or beginning and increment of oscillation due to a particular parameter change. Also, as a bi-product of the study, the 'settling time', which is a measure of the time required for driving away the effect of disturbance from the system, may be observed. These two measures are the fundamental guide line for controller tuning and optimization. In Chapter 13, we shall talk about this

topic. In studying the effect of 'lag', only the transfer lag has been considered so far. The process in Figures 9.12 and 9.36 contains a first order TF. Due to the presence of measurement lag, the closed loop TF for the system in Figure 9.36 has second order lag. Recalling Figure 8.12 in Section 8.3, the closed loop dynamics for a third order process with a (P) controller has been shown for different values of K_C. All these results confirm that an increase of number of poles increases the oscillatory character of the response for an identical value of K_C. But is this true only for transfer type of lag? What about the existence of a 'dead time' or 'transportation lag' in the process? Such dead times realistically appear in applications where the measuring element is installed at a physical distance from the process equipment exit point or there are inter-stage delays within a process equipment. Also, delay element commonly occurs in mixing systems in concentration or temperature control of a fluid stream.

To address this problem, a dead time element has been added to the process TF of Figure 9.12 to obtain the block diagram of Figure 9.40. Thus,

$$G_P(s) = \frac{K_P e^{-\tau_{DL} s}}{\tau_P s + 1}$$

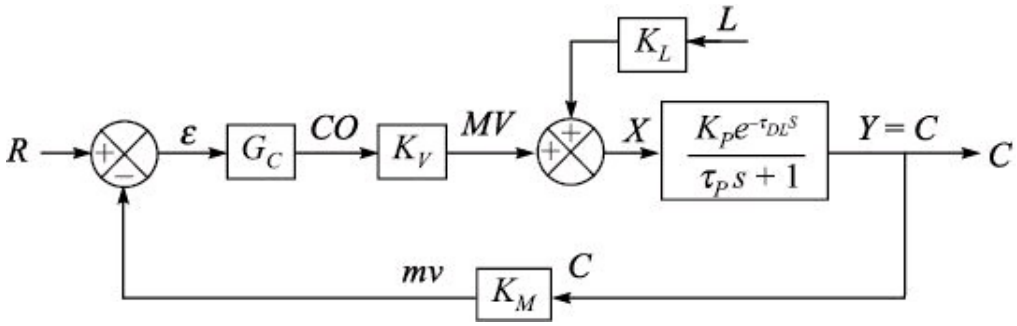

Figure 9.40 Control system—the process block contains a first order plus dead time element.

The statements made in concern to the closed loop system of Figure 9.12 are all true for Figure 9.40. Again, the responses are obtained by writing a computer programme that digitally simulates the control loop. Figure 9.41 shows the response character for variation of K_C at constant, t_{DL}, and Figure 9.42 for variation of t_{DL} at constant, K_C. The responses due to load change are in part (a) and that for set point change are in part (b) of each of these figures.

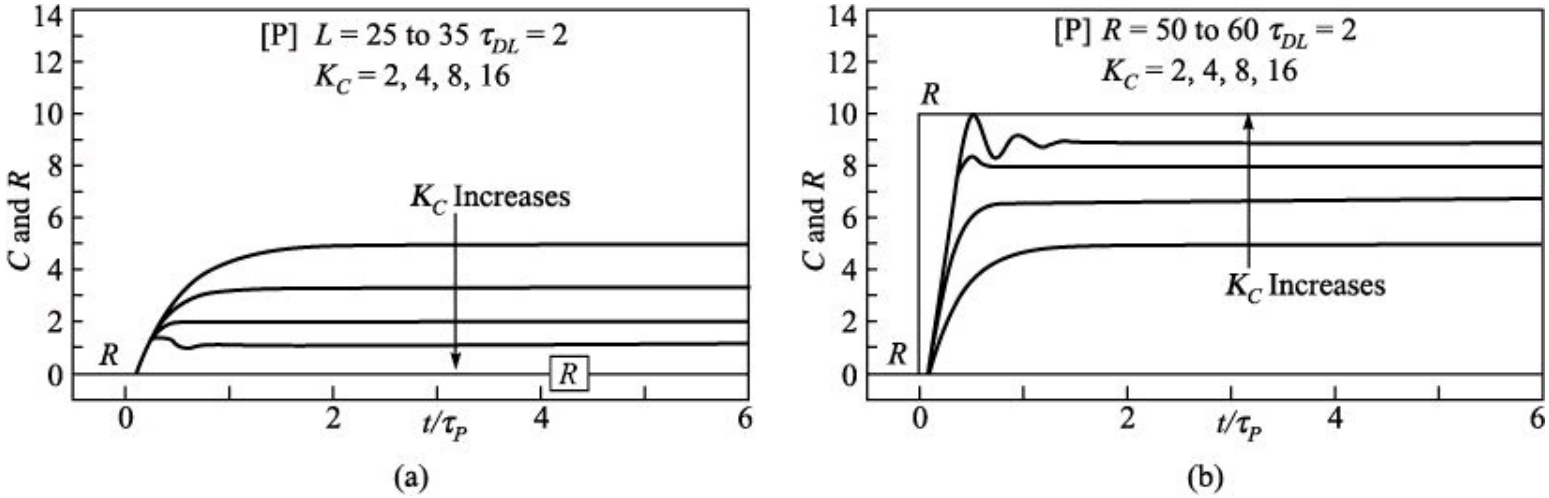

Figure 9.41 Closed loop responses for different K_C values at constant t_{DL} for (P) controlled system.

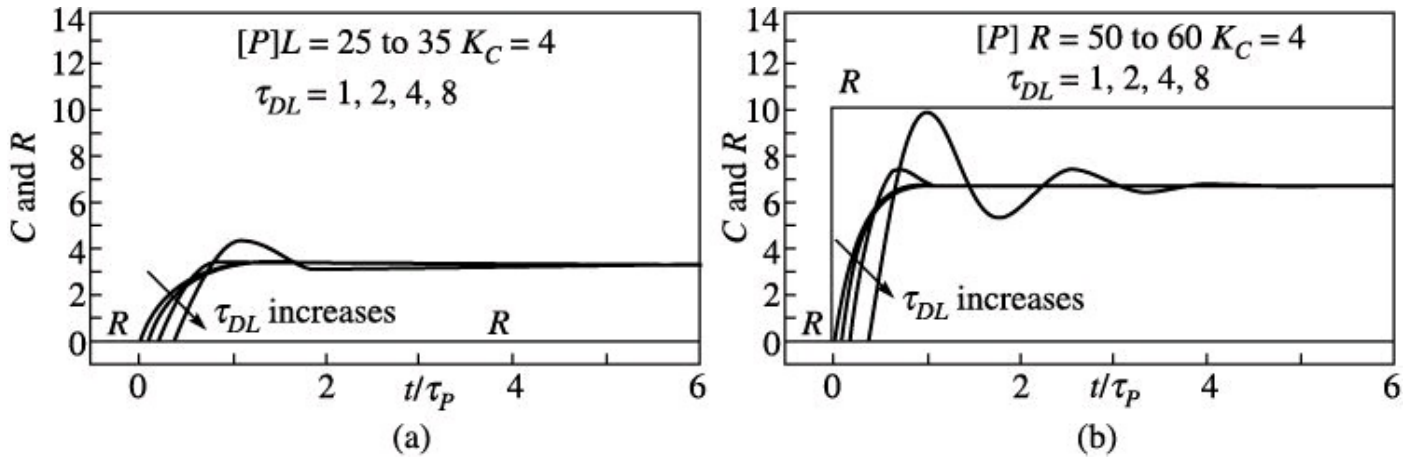

Figure 9.42 Closed loop responses for different t_{DL} values at constant K_C for (P) controlled system.

Hence, the conclusion made about effect of transfer lag on response character is also true for adding and increasing of dead time in the process. Not only that, the effect of dead time seems to be more pronounced toward producing oscillatory response. These are simulation results, however, for analytical treatment the dead time may be represented by the Pade's approximate TF. Already it has been mentioned in Chapter 6 that Pade's first order TF has the following form:

$$e^{-\tau_{DL}s} = \frac{1-\tau_{DL}s/2}{1+\tau_{DL}s/2} = \frac{2-\tau_{DL}s}{2+\tau_{DL}s}$$

Note that the introduction of the above equation will introduce a first order zero in the right half of s-plane (complex plane).

The Pade's and other approximation approaches have been explained in Chapter 6. The reader should be aware about the amount of error that is introduced in the resulting model by such an approximation. This has been demonstrated in the following example.

EXAMPLE 9.19 In a control system, the process block has a first order plus dead time
TF and the controller has (P) mode. The process and controller parameter values are:

$K_L = K_M = K_P = 1.0$; $K_C = 4$, $K_V = 0.5$; $x_P = 20$; $x_{DL} = 8$. At $t = 0$, a set point change equal to 10 units is introduced into the closed loop system. Determine the closed loop response by representing the dead time by Pade's first order formula, and compare this response with that of the simulated response.

Solution: The closed loop response for set point change would be:

$$\frac{C}{R} = \frac{K_C K_V K_P e^{-\tau_{DL}s}/(\tau_P s + 1)}{1 + K_C K_V K_P e^{-\tau_{DL}s}/(\tau_P s + 1)} = \frac{4 \times 0.5(2 - 8s)/[(2 + 8s)(20s + 1)]}{1 + 4 \times 0.5(2 - 8s)/[(2 + 8s)(20s + 1)]}$$

or,
$$\frac{C}{R} = \underbrace{\frac{2}{80s^2 + 16s + 3}}_{\text{Part I}} - \underbrace{\frac{8s}{80s^2 + 16s + 3}}_{\text{Part II}}$$

For a step change of R, Part I would produce a step and Part II, an impulse response of a second order system. The total response is the sum of these two. The parameters of the second order system, that represents a closed loop, may be determined by the method already shown. These are:

$$K_{1CL} = 2/3; K_{2CL} = 8/3; w_{NCL} = 0.19365; z_{CL} = 0.5164$$

The closed loop response for Part I of the response (Equation 6.80):

$$Y_I(t) = MK_{1CL}\left[1 - e^{-\omega_{NCL}\zeta_{CL}t}\left(\cos\omega_{NCL}\sqrt{1-\zeta_{CL}^2}\,t + \frac{\zeta_{CL}}{\sqrt{1-\zeta_{CL}{}^2}}\sin\omega_{NCL}\sqrt{1-\zeta_{CL}^2}\,t\right)\right]$$

and for the Part II of the response (Equation 6.94):

$$Y_2(t) = MK_{2CL}\frac{\omega_{NCL}}{\sqrt{1-\zeta_{CL}^2}}e^{-\omega_{NCL}\zeta_{CL}t}\{\sin(\omega_{NCL}\sqrt{1-\zeta_{CL}^2}\,t)\}$$

The analytical response, $Y(t) = Y_1(t) + Y_2(t)$. The analytical and simulated responses are drawn on a common time base in Figure 9.43.

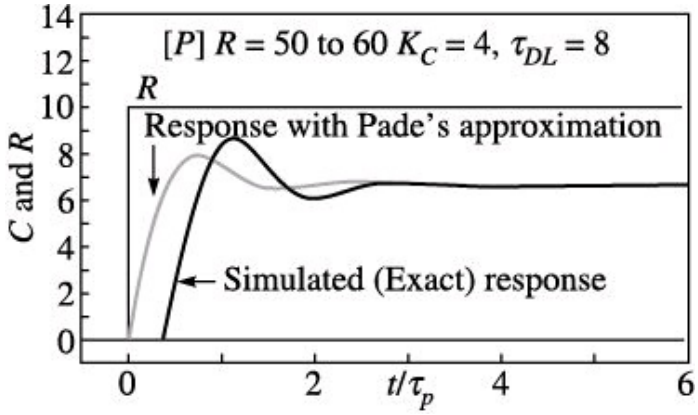

Figure 9.43 The error due to Pade's approximation in an FOPDT process.

Note that:

1. The approximate response has started from the instant of disturbance introduction, whereas the simulated response begins 8 time units after, which is correct.
2. The approximate response has zero initial derivative which is a property of second order process, but the simulation shows the initial response with a finite slope (in fact the maximum) as it should.
3. The overshoot is greater in case of simulation than the approximation result.
4. At high values of time, there is no difference between the response values obtained by the two methods of analysis. It may be thus concluded that, in case of property consideration at small time values, e.g. overshoot, decay ratio, etc., the simulation method should be trusted but for properties that need to be evaluated at large time values, both the methods are equally correct.

Effect of the lag or delay element brings the phenomenon of the controller remaining always unaware of the controlled variable for a certain time (equal to t_{DL} plus the effect of any transfer lag present in the process). This causes the controller to take action, which should have been adequate if taken t_{DL} time units before, and brings the poor quality of control.

Talking so much about the effect of lag, we must not forget about the other parameter that also may cause the process to become unstable. To emphasize the effect of that parameter only, a (P) controller has been chosen for the last closed loop systems of Figure 9.21 and Figure 9.24, that is the gain of the control loop. The K stands for the loop gain and it is also an important parameter that determines the stability and response speed of a control system. We shall deal with this topic in Chapter 10 on stability in complex domain and Chapter 11 on stability in frequency domain.

9.7 Closed Loop Response of a PWM (Pulse Width Modulation) Mode System

In Chapter 7 we introduced pulse width modulation as one of the special techniques of control system implementation, where a ON-OFF type switching device is the final control element instead of a conventional FCE, which throttles the manipulated variable in proportion to the controller output. We also mentioned the ‘ripple’ small amplitude oscillation in controlled variable resulting from the rapid switching of FCE. This is a limitation of PWM mode. However, the ripple amplitude could be reduced by optimally choosing window time, W, and the width of smallest pulse, DT. As an example, closed loop response for step change in load under PWM control with a

(PID) controller and an FOPDT process has been shown in Figure 9.44. The traces are drawn for two values of W and same DT. The block elements of the loop have following TFs:

$$G_C(s) = 1.5\left[1 + \frac{1}{8s} + s\right];\ G_P(s) = \frac{e^{-s}}{50s+1};\ K_M = 1;\ K_V = 0.5$$

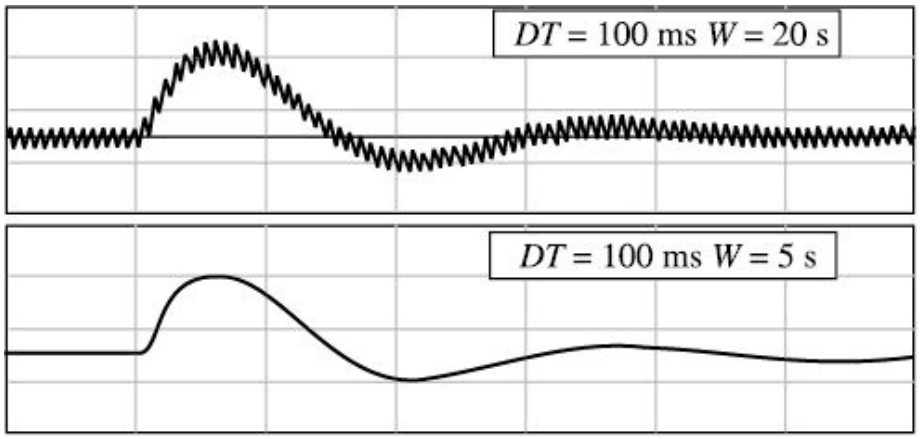

Figure 9.44 Closed loop response of PWM control of an FOPDT process for a step change in load.

There are two important stream configurations used in plant operation around process equipment for improving control quality, material utilization efficiency and ease of process operation. These configurations are:

1. Recycle and
2. By-pass streams.

Open and close loop dynamics of a process with and without these stream routings should be carefully attended to have a clearer grasp of the advantages they offer to a designer. As this part of the study does not directly come within the scope of close loop response analysis, the subject has been addressed in Appendix-III B.

EXERCISE PROBLEMS

9.1 Determine transfer functions *C*/*R*, and *C*/*L* for the following control system shown in Figure P9.1:

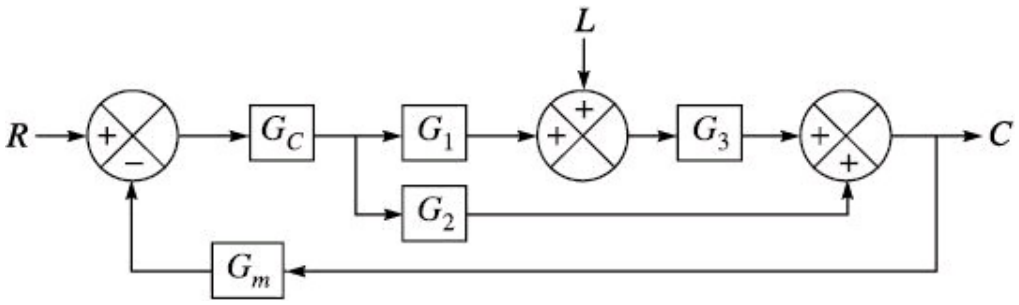

Figure P9.1

9.2 Determine transfer functions, *C*/*R* and *C*/*L* for the system shown in Figure P9.2:

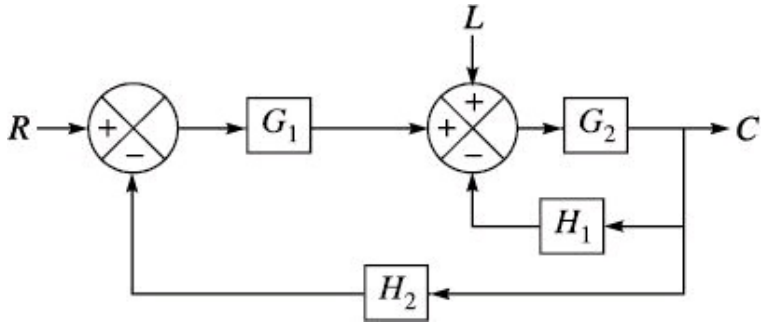

Figure P9.2

9.3 Determine transfer function, *C*/*R*, for the control system shown in Figure P9.3:

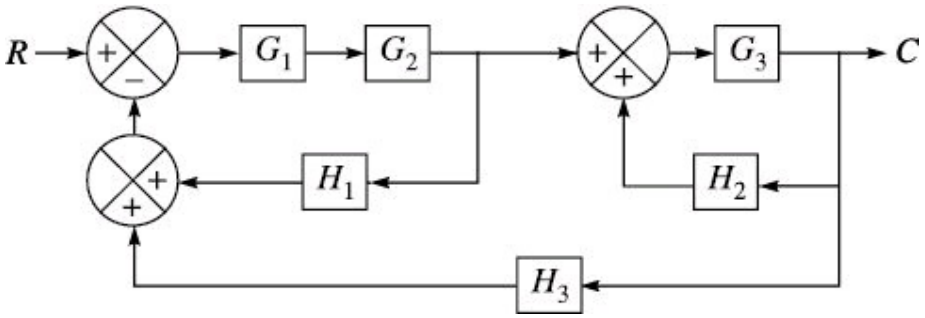

Figure P9.3.

9.4 Determine transfer function, *C*/*R*, for the position control system shown in Figure P9.4 of a Robot arm having three degrees of freedom:

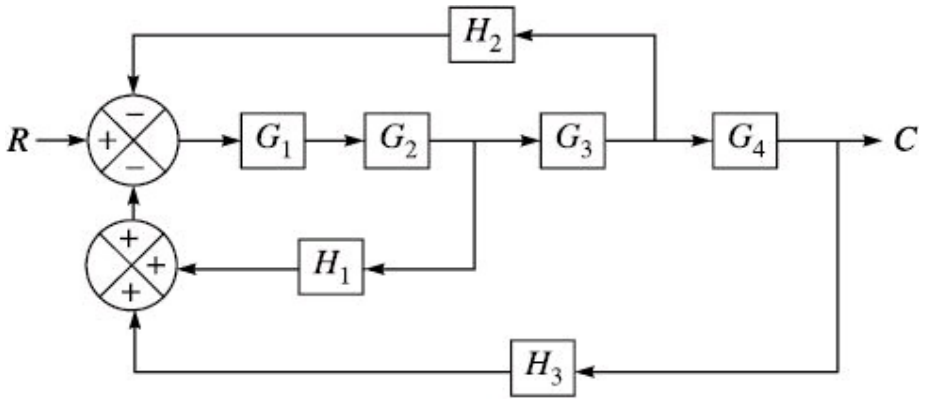

Figure P9.4.

9.5 Determine transfer function, *C*/*R*, for the position control system shown in Figure P9.5:

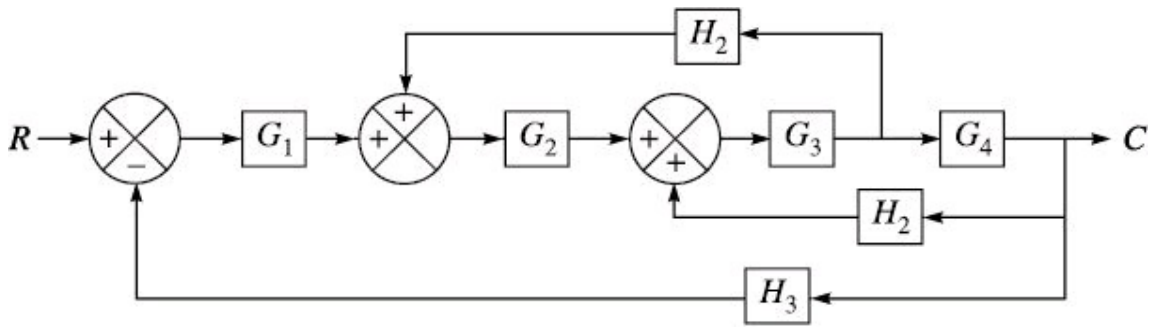

Figure P9.5.

9.6 The set point of the control system illustrated in Figure P9.6 is given a step change of 0.1 unit. Determine: (a) the maximum value of *C* and the time at which it

occurs; (b) the offset; (c) the period of oscillation.

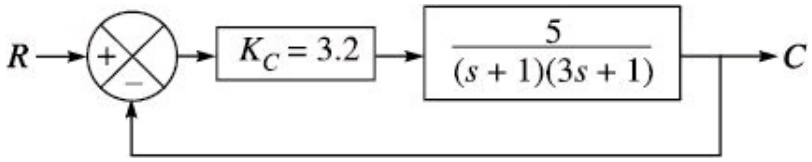

Figure P9.6.

9.7 A first order process is in a unity feedback loop with a (PI) controller. The process gain and time constants are K_P and t_P, and the controller parameters are K_C, and t_I. If $t_P = t_I$: (a) Show that the close loop system is always over or critically damped; (b) Determine the effective time constant for the system; (c) For a unit step change in set point, find the expression of $C(t)$; (d) For a unit step change in load, find the expression of $C(t)$.

9.8 Block diagram of a close loop process is shown in Figure P9.7. Answer the following questions:

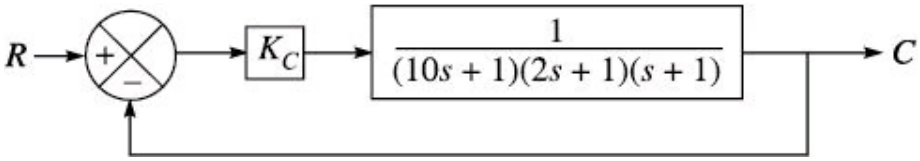

Figure P9.7 The system in problem 9.8.

(a) Find the value K_C that will produce a response with sustained oscillation in $C(t)$;

(b) If half of this value of K_C is used what would be the offset for a unit step in set point?

(c) For a unit step in set point what should be the value of K_C to obtain an offset of 15%?

(d) For the controller gain in part (c), what is the resulting damping coefficient?

9.9 A first order process is in a unity feedback loop with a (PI) controller. The process gain and time constants are $K_P = 2.0$ and $t_P = 5$ time units. Find the controller parameter values for which the loop will show a quarter decay type response.

9.10 A liquid stream is heated in an ideally stirred tank by an immersion type electrical heater. To improve the dynamic response of the system, the exit liquid goes through another stirred tank. The exit liquid temperature is measured by a sensor-transmitter having a measuring range of 0–100°C. A (PID) controller regulates the heater power through a thyristor type power regulator. The liquid properties are $C_{PL} = 0.87$ cal/°Cgm, $r_L = 0.9$ gm/cm^3. The desired exit liquid temperature is 80°C. The inlet liquid flow rate is constant and has a value of 800 cc/s, the inlet liquid temperature (load variable) may vary from 10 to 30°C. The

tank volumes are 16 and 24 l. Calculate the thyristor type power regulator with a factor of safety = 2. The integral and derivative times of the controller are, K_C = 2.5, t_I = 80 s, and t_D = 15 s. Assume that G_V and G_M are pure gain elements.

Draw the quantitative block diagram for the control system. Find the response equation for $C(t)$ for a 5°C step in: (a) set point; (b) load.

9.11 Consider the liquid level (PI) control system shown in Figure P9.8. The system has the following parameter values: A = 3000 sq.cm, $R = 1 \times 10^{-3}$ min/sq.cm, K_V = 180 cc/psi, K_M = 0.057 psi/cm, K_C = 4, and t_I = 3 min. The system was initially at steady state with a liquid level of 600 cm. If the set point is suddenly raised from 600 cm to 900 cm, how long will the liquid level in the tank, take to reach: (a) 750 cm, (b) 800 cm?

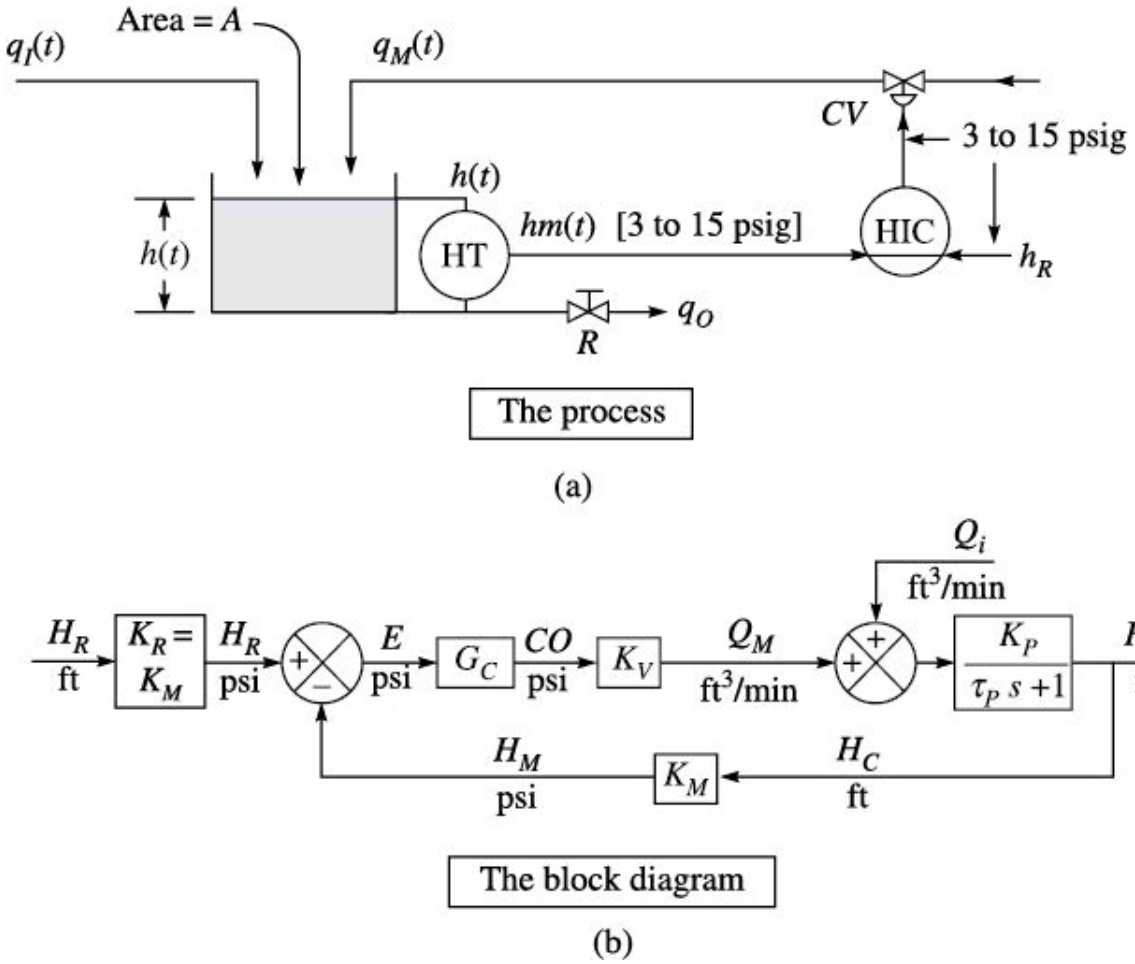

Figure P9.8 (a) The process flow sheet. (b) The block diagram of a level control system.

9.12 Consider the stirred liquid heating system described in Example 9.15. If a (PI)-controller is employed for maintaining the exit liquid temperature from the second tank with

t_I = 5 min, find the value of K_C to obtain a slightly under-damped response (z = 0.7) for the cases 1 and 2 described in the example. Compare the values of K_C in both cases and comment on the merits of the two arrangements on the basis of the

results.

REFERENCES

[1] Coughanowr, D.R. and Koppel, L.B., *Process System Analysis and Control*, Exercise Problem 16.2, McGraw-Hill Inc, 1965.

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10

Stability Analysis of Closed Loop Systems

10.1 Concept and Measure of Stability

The concept of stability was introduced by mathematicians long back in the first half of 18th century in concern of the solution of differential equations and particularly when the solution is represented by a combination of infinite series. The main objective of this exercise was to find whether the solution of a certain class of differential equations yielded a finite convergent or infinite divergent solution. Many of the concepts and results came into excellent use with the later development of dynamic system analysis and control system design. Examples of such usage are Routh-Hurwitz criterion and Liapunov's theorem.

The definition of stability for dynamic system is that: ***from steady state operational condition if a system is disturbed by a finite or bounded input disturbance, and the resulting response is also finite or bounded, the system is said to be stable. If under same circumstance the response is unbounded or infinite, the system is unstable.***

The finite input is an input that stays between an upper and lower limit at all values of time. As an example, we may cite the step and sinusoidal function of finite magnitude and amplitude respectively, whereas, a ramp input or an input of the type, $x(t) = te^t$, will always grow with time, hence it is unbounded and cannot be used for stability check of a system. Pure gain element, first-order-lag, are examples of stable system, which after being disturbed with a bounded input, show a bounded or finite response. But a pure integrating system, e.g. a closed cylindrical tank with a constant inflow would be an open loop unstable process, as the inside liquid level as output will constantly grow as time progresses.

If only a process without any control system is studied to reveal its dynamic qualities like stability, the result is commented as '**open loop**' stable or unstable, and when the process together with its control system becomes the subject of the same study, we would pronounce the result as '**closed loop**' stable or unstable.

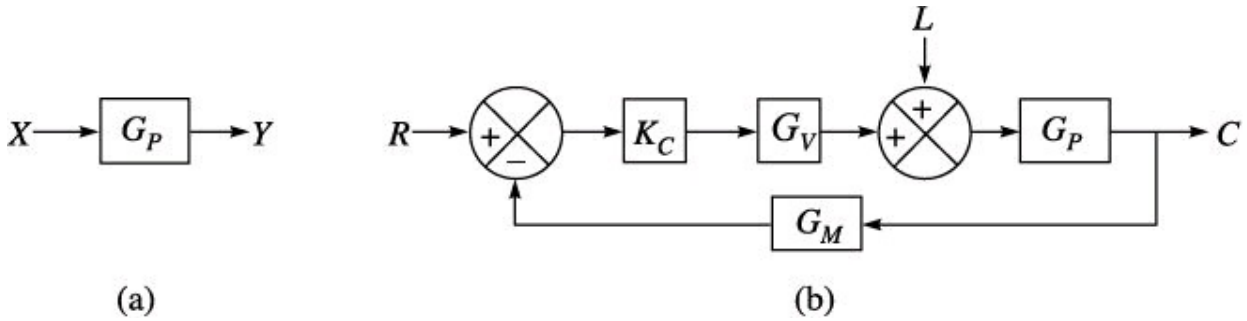


Figure 10.1 (a) The open loop process. (b) The closed loop process control system.

For stability analysis of a system the corresponding transfer function between input and output variables is first obtained. The denominator polynomial of the transfer function in terms of s when equated to zero, constitutes the characteristic equation of the system.

As an example, for a third order process, let the transfer function be:

$$\frac{Y(s)}{X(s)} = G_P(s) = \frac{K_P}{(\tau_1 s + 1)(\tau_2 s + 1)(\tau_3 s + 1)} \qquad (10.1)$$

and the system's characteristic equation is:

$$\tau_1\tau_2\tau_3 s^3 + (\tau_1\tau_2 + \tau_2\tau_3 + \tau_3\tau_1)s^2 + (\tau_1 + \tau_2 + \tau_3)s + 1 = 0 \qquad (10.2)$$

For the system shown in Figure 10.1, we know from the discussion of the previous chapter, that the closed loop transfer functions (CLTF) for set point and load changes contain similar expressions in the denominator.

$$\frac{C(s)}{R(s)} = \frac{G_C(s)G_V(s)G_P(s)}{1 + G_C(s)G_V(s)G_P(s)G_M(s)} \qquad (10.3)$$

and

$$\frac{C(s)}{L(s)} = \frac{G_P(s)}{1 + G_C(s)G_V(s)G_P(s)G_M(s)} \qquad (10.4)$$

If we generalize the transfer functions by writing it as a ratio of two polynomials of s,

$$\text{CLTF} = \frac{N(s)}{D(s)} \qquad (10.5)$$

where $N(s)$ and $D(s)$ are the polynomials in the numerator and denominator of the closed loop transfer function. As this point it is sufficient to say that the stability of a system, whether open or in close loop configuration, has a close relation with the roots of the characteristic equation obtained by equating the denominator of Equation 10.5 to zero, or, $D(s) = 0$. This topic will be emphasized further after the introductory concept.

We would like to introduce this concept of stability from the consideration of

dynamic response of a second order under damped system. We already know that a character transition of response occurs from monotonic to oscillatory when the damping coefficient z become less than unity. By decreasing z further, the response becomes more oscillatory. This transition and further increase of oscillatory character can be explained by the root location of the system's characteristic equation on the complex s-plane. Further development of this concept may become easy by considering the classical example of damped vibrator introduced in the discussion of second order system, redrawn in Figure 10.2.

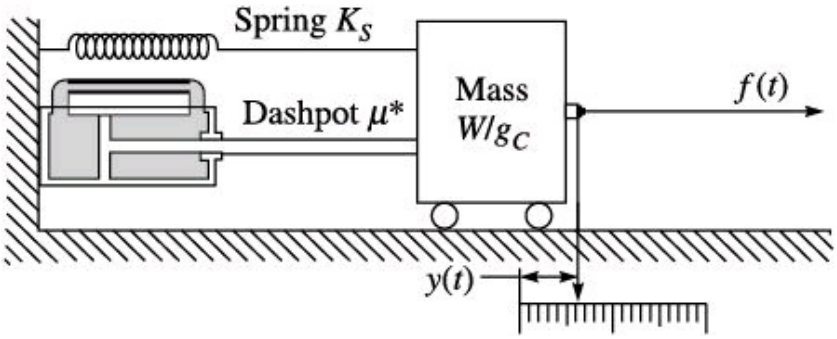


Figure 10.2 The damped vibrator.

Recalling the dynamic force balance equation of the damped vibrator,

$$\frac{W}{g_C K_S}\frac{d^2Y(t)}{dt^2}+\frac{\mu^*}{K_S}\frac{dY(t)}{dt}+Y(t)=\frac{F(t)}{K_S} \qquad (10.6)$$

and the corresponding transfer function,

$$\frac{Y(s)}{F(s)/K_S}=\frac{1}{\tau^2 s^2+2\zeta\tau s+1} \qquad (10.7),$$

we may write the characteristic equation as:

$$\tau^2 s^2+2\zeta\tau s+1=0 \qquad (10.8)$$

Equation 10.8 has the following roots in the complex plane:

$$s_1=-\frac{\zeta}{\tau}+i\frac{1}{\tau}\sqrt{(1-\zeta^2)} \qquad (10.9a)$$

$$s_2=-\frac{\zeta}{\tau}-i\frac{1}{\tau}\sqrt{(1-\zeta^2)} \qquad (10.9b)$$

If we plot the roots s_1, and s_2 on a complex plane, the dynamic parameters will have the following relations with the geometric dimensions, as shown in Figure 10.3.

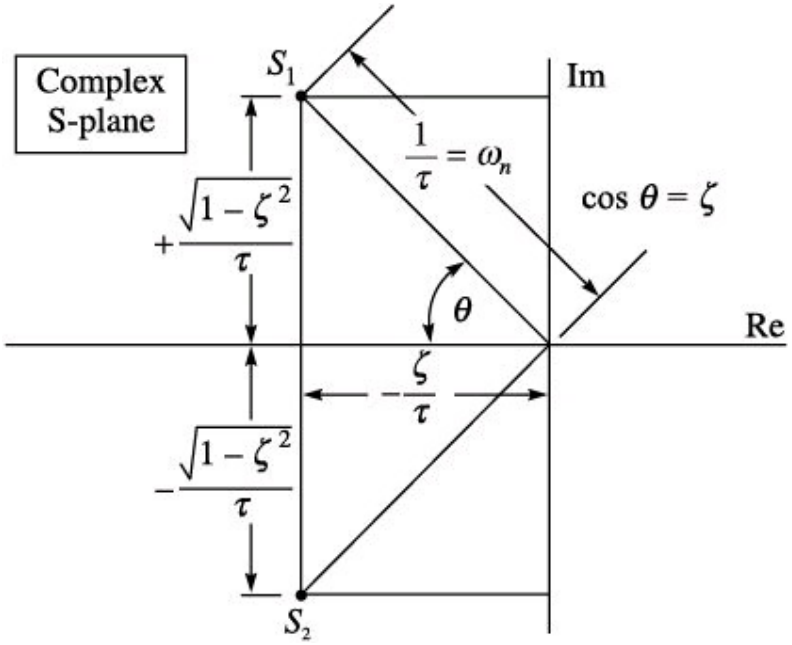


Figure 10.3 The location of roots and parameter relations on the complex plane.

Again, to establish the relations between dynamic and physical parameters of the system, we rewrite the substitutions made to obtain Equation 10.7 from Equation 10.6,

$$\tau = \sqrt{\frac{W}{g_C K_S}} \qquad (10.10)$$

and

$$\zeta = \frac{\mu^*}{2K_S}\sqrt{\frac{g_C K_S}{W}} = \frac{\mu^*}{2}\sqrt{\frac{g_C}{K_S W}} \qquad (10.11)$$

To study the root location on the complex plane as a manifestation of dynamic behaviour, we intend to change a physical parameter of the system and observe its effect on the damping coefficient, *z*. This approach will be more understandable from a practical viewpoint. Now let us take that the mass *W* and spring constant K_S are the two invariable system parameters, which is understandable from Equation 10.10 and Equation 10.11, because they will effect both *z* and *t* and it would not be possible to single out the effect of our target parameter *z*. Hence, we have chosen the dashpot constant, *m** (proportional to dashpot fluid viscosity), that will only bring change in *z* without affecting the time constant *t* of the system.

Let us start with the following values of the parameters *W* = 10 lbm; K_S= 20.0 lbf/ft, and we shall gradually decrease the value of the dashpot constant *m**, starting from *m** = 4.9, up to *m** = 0.1(lbf.s)/ft, calculate the values of *t*, and *z* and find the location of the roots s_1 and s_2 on the complex plane by using equations 10.6 through 10.9 given above. We have compiled the results in Table 10.1, and plotted s_1 and s_2 on the complex *s*-plane in Figure 10.4.

Table 10.1 Calculated parameter values and the Root Location as a function of m^*

n*	x	g	Coordinate of s_1 and s_2 in S-plane	
			Real part	Imaginary part
4.9	0.557	0.98	−1.76	±0.32
4.5	—	0.90	−1.62	±0.77
4.1	—	0.82	−1.47	±1.02
3.7	—	0.74	−1.33	±1.20
3.3	—	0.66	−1.19	±1.34
2.9	—	0.58	−1.04	±1.46
2.5	—	0.50	−0.9	±1.55
2.1	—	0.42	−0.76	±1.63
1.7	—	0.34	−0.61	±1.69
1.3	—	0.26	−0.47	±1.73
0.9	—	0.18	−0.32	±1.76
0.5	—	0.1	−0.18	±1.78
0.1	—	0.02	−0.036	±1.79

The following discussion on the system dynamics has been carried out for step disturbance, although the conclusions are equally valid for other bounded input functions. Verification of this notion is left for the reader as an exercise. Let us rewrite the step response equation for an

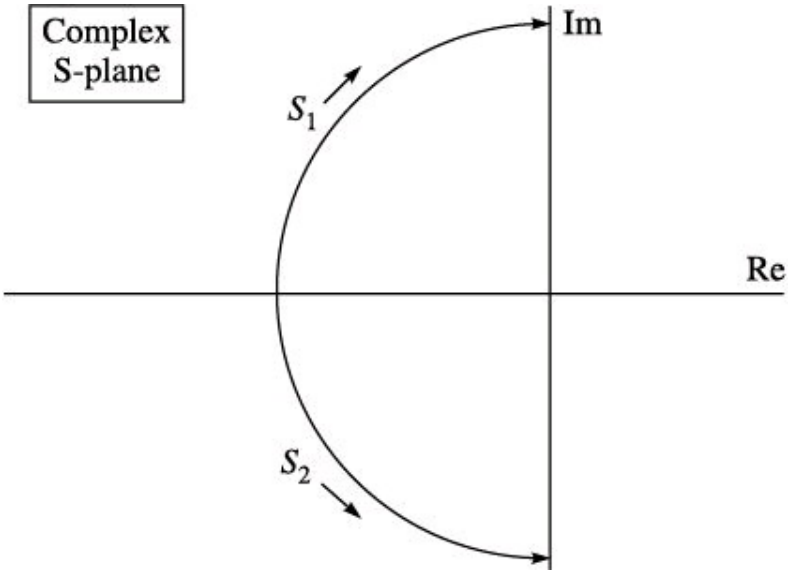


Figure 10.4 s_1 and s_2 goes closer to imaginary axis as the dashpot constant m^* decreases.

under damped second order system and examine the effect of this parameter change interpreted as the root movement in complex plane.

$$Y(t) = 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta t/\tau} \sin\left[(\sqrt{1-\zeta^2}) \frac{t}{\tau} + \tan^{-1}\left(\frac{\sqrt{1-\zeta^2}}{\zeta} \right) \right] \qquad (10.12)$$

The trigonometric sine function causes the oscillation in response and the coefficient term, that contains an exponential –ve time function, is in fact the amplitude of the sine function, which gradually reduces with progressive values of time. As z

decreases, surely the amplitude as well as the time for which the oscillatory part remains will increase. Thus, in the vicinity of very low value m^* we shall obtain very low values of z, as in Table 10.1, and very pronounced oscillatory character of the response. This may be generalized with the location of roots as – *"nearer the complex conjugate roots go to the imaginary axis, more oscillatory becomes the system response"* (overshoot increases), and more time is required for the oscillations to die down. If there is any doubt regarding the last statement due to the fact that response frequency w decreases with the decrease of damping coefficient z, according to $w = w_N\sqrt{(1 - z^2)}$, you may check, by evaluating the response time of the response for the parameter values of each row of Table 10.1. Let us continue this line of discussion by assuming that we can find a theoretical fluid having a zero viscosity and use that in the dashpot of our system, then m^* will become zero.

(With the volume of research going on Super-conductivity and related areas, in the near future, such a fluid may not remain as an utopia at all.)

In such a situation, the damping coefficient will be zero, that is, the real part of the roots will vanish and they become imaginary conjugates. The roots will be located on the imaginary axis. Thus, in the response, there is no existence of $\exp(-zt/t)$ term, so there will be no gradual decrease of the oscillation amplitude, because the trigonometric sine-function will be still there due to, $\exp[\pm i\sqrt{(1 - z^2)}/t] = \exp[\pm i/t]$ term. Thus, as the roots move and are located on the imaginary axis, the response is a sustained oscillation of undiminished amplitude, and in terms of dynamics, such a condition is called **Limiting Stability**. In control system application, this quality of closed loop response is also called **Continuous Cycling.**

Now we like to think up an imaginary situation where the damping coefficient has become –ve. It is not possible with the damped vibrator system by only changing m^*, but in certain electro-mechanical, and closed loop control system, we may face an equivalent dynamic condition. Later we shall see that many higher order systems that may be fairly approximated to a second order dynamic model may have such parameter condition.

In such a condition of damping coefficient becoming –ve, the real part of the roots will be +ve, and they will again become complex conjugates. The roots will be located on the +ve half of the complex plane. Thus in the response, due to the existence of $\exp(+zt/t)$ term, there will be gradual increase of the oscillation amplitude, because the trigonometric sine-function will be still present due to, $\exp[\pm i\sqrt{(1 - z^2)}/t]$ term. Thus, as the root moves into the +ve half of the complex plane, the response is oscillatory with a gradual increase of amplitude, and we have an **Unstable System**, because as time progresses, the increasing magnitude of the process output variable will

eventually cross the physical limits of the equipment system in which the process is operating. To have such a condition of instability, it is sufficient that a root of the characteristic equation lies on the +ve half of the *s*-plane, thus a +ve real root is sufficient for unstable dynamic condition. We have summarized the different categories of dynamic response corresponding to the root locations on the complex plane of second order system in Figure 10.5.

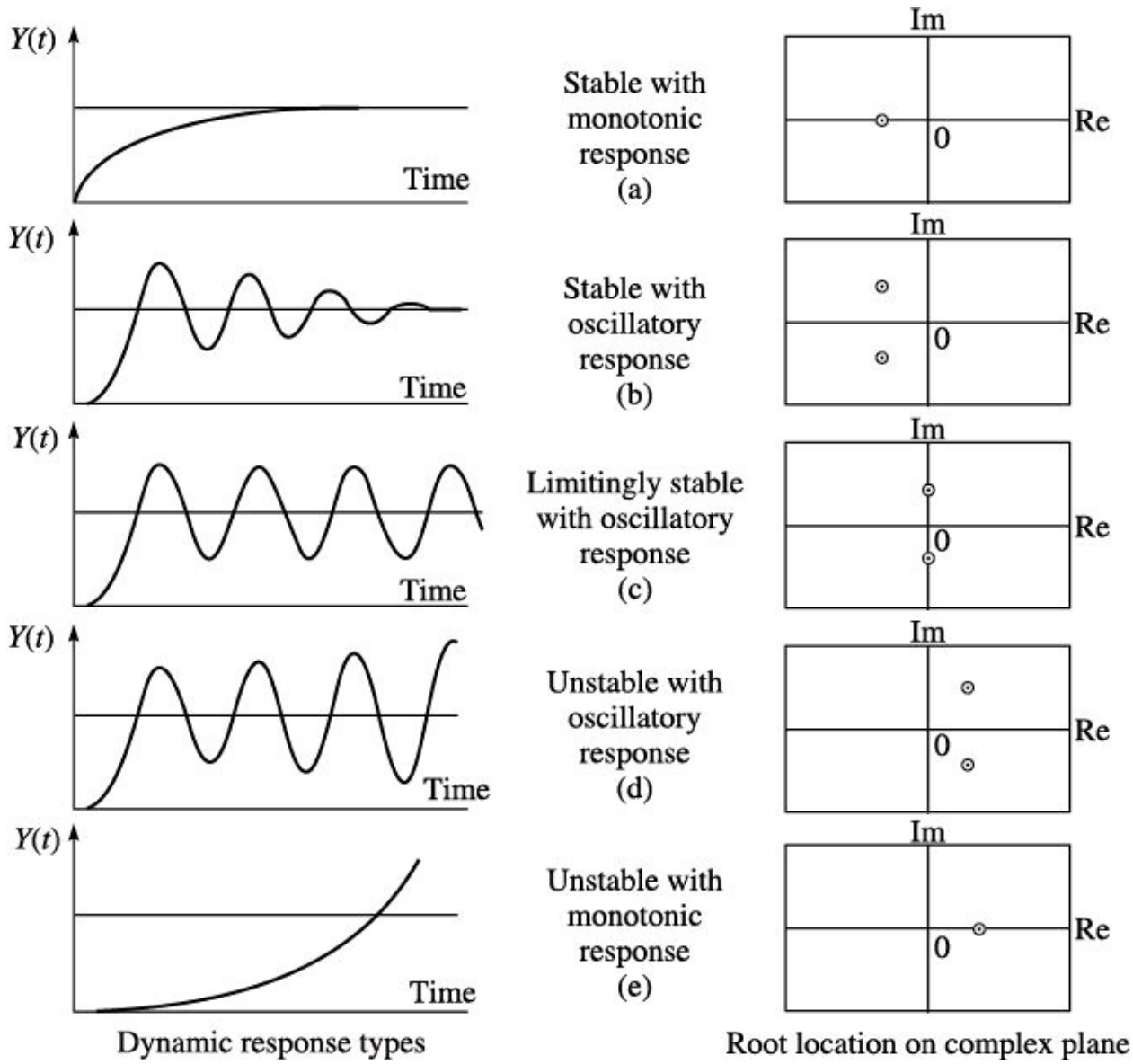


Figure 10.5 The type of dynamic response and the location of roots on the complex plane.

In control system operation, generally a slightly under-damped response which rapidly decays with time [Figure 10(b)] is desirable, but responses of the type given as in Figures 10.5(c), (d) and (e) are seriously undesirable dynamic conditions that should be avoided by proper choice of control loop parameters.

We may now propose another form of stability theorem more specific to the closed loop system treated in Laplace domain: ***A feedback control system is stable if and only if all roots of the characteristic equation are negative or have negative real parts. Otherwise the system is unstable.***

If we recall the expression for the Rise time of an under-damped second order

system and its dependency with z, another conclusion may be drawn, that as a system moves closer to being unstable, its response speed increases in terms of the time to reach a particular output value. Thus, in general, stability and response speed of a system are complementary to each other. It should be mentioned that there are few exceptions, but for majority of the practical control system applications, the above notion may be accepted. In fact, the controller optimization (tuning) procedures are largely based on this fundamental premise. We shall discuss this topic in the next chapter.

In a control loop as the process is given, so its dynamics in terms of transfer function $GP(s)$, is considered to be unchangeable. The analysis of the closed loop stability often becomes the study of stability limit as a function of controller parameters. The simplest situation would be to assume a proportional only controller with a single parameter K_C. Thus, we would like to emphasize on this point of assertion of stability of a closed loop system, by inspection of the characteristic equation root movement on complex plane as a function of the controller gain, K_C of a (P) controller in the loop. This has been a recognized practice in control system design literature not only due to relative simplicity of the procedure, but also for the fact that a wealth of literature has been accumulated on the subsequent control system optimization procedure that use this consideration as starting point. Thus, for a given closed loop system, to choose controller gain K_C as the variable controller parameter assuming a (P) only controller in the loop, and study the locus of the roots for increasing values of K_C starting from an arbitrary small value, will be our first address to stability analysis. This procedure of stability analysis is adequately called the **Root Locus** method.

10.2 Root-Locus Method

The root locus method starts with first constructing the characteristic equation of the given closed loop system. Then for a selected range of K_C, the roots of the characteristic equation are found and plotted on the complex s-plane, and the required locus is drawn through the root locations. Examination of the locus or simply the root coordinates, produces the information about system stability as has been demonstrated with the spring-dashpot system. Let us present a few examples to explain the procedure.

EXAMPLE 10.1 It is proposed to find the limiting value of K_C for the following closed loop system to remain stable.

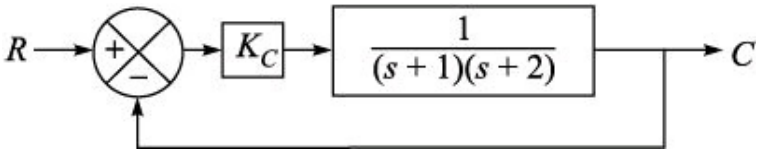


Figure 10.6 The closed loop system for example 10.1.

Solution: We first find the closed loop transfer function $C(s)/R(s)$ for the system to be:

$$\frac{C(s)}{R(s)} = \frac{K_C}{s^2 + 3s + 2 + K_C} \qquad (10.13)$$

and the characteristic equation is: $s_2 + 3s + 2 + K_C = 0$, with the roots r_1, r_2 expressed as:

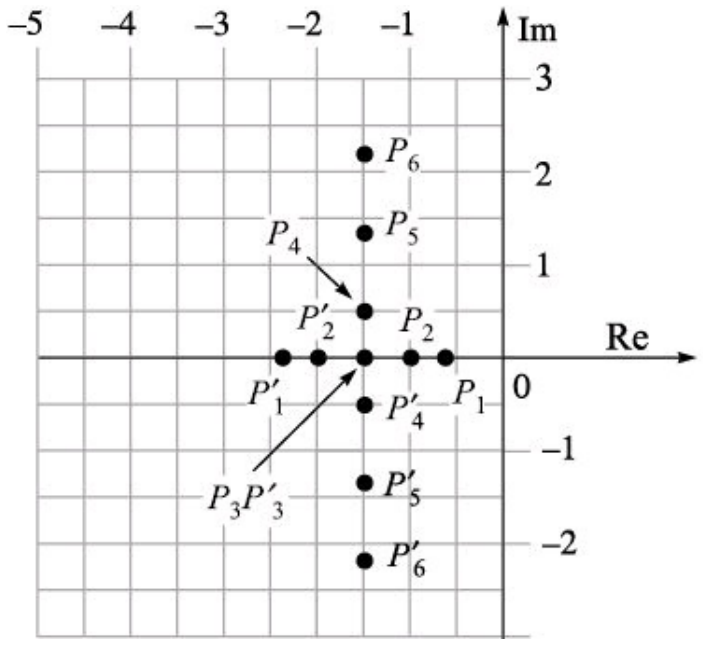
$$r_1 = -\frac{3}{2} + \frac{\sqrt{1 - 4K_C}}{2} \qquad (10.14a)$$

$$r_2 = -\frac{3}{2} - \frac{\sqrt{1 - 4K_C}}{2} \qquad (10.14b)$$

Putting gradually increasing values of K_C from 0 □ onward, the coordinate positions of the roots have been listed in Table 10.2 and plotted in Figure 10.7.

Table 10.2 The Root Locations of Example 1 for different values of K_C

K_C	r_1	r_2	Position on the Locus
0	–0.634	–2.366	P_1, P□1
0.5	–1.0	–2.0	P_2, P□2
0.75	–1.5	–1.5	P_3, P□3
1.0	–1.5 + 0.5*i*	–1.5–0.5*i*	P_4, P□4
2.0	–1.5 + 1.32*i*	–1.5–1.32*i*	P_5, P□5
5.0	–1.5 + 2.18*i*	–1.5–2.18*i*	P_6, P□6



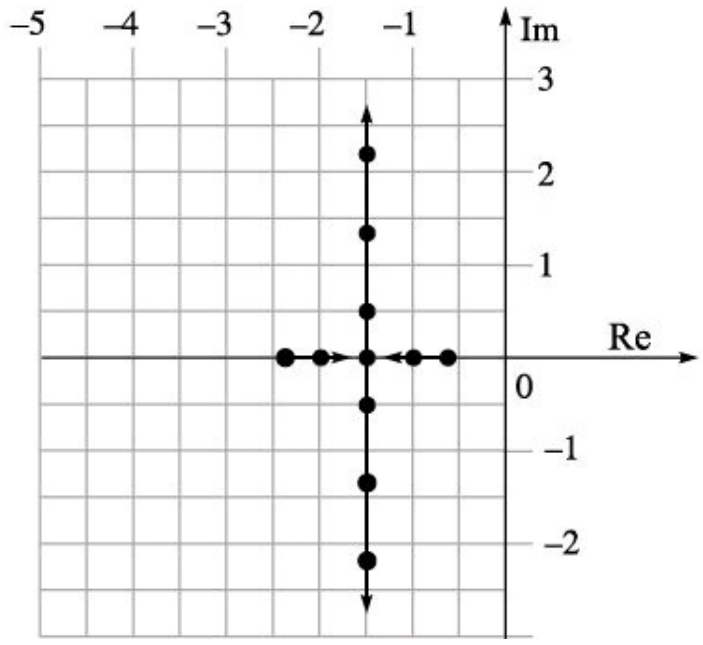


Figure 10.7 (a) The location of the roots. (b) The root locus for the system of example 10.2.

It is not even necessary to draw the locus. Looking at the values of the roots, we may conclude that by increasing K_C, the system could not be made unstable, because the real part of the roots are not a function of K_C and will always have a constant value of –1.5. However, as K_C increases, the z of the closed loop second order system will decrease, rendering the response oscillatory. Let us locate the various positions of the roots on the complex plane as given in Table 10.2.

Note that though the locus runs parallel to the imaginary axis at higher values of K_C , the damping coefficient still decreases, due to the increase of angle q, as has been depicted in Figure 10.2. We may conclude that by increasing controller gain, the closed loop system given in Figure 10.2 could be made more oscillatory but not unstable. This is a general behaviour of closed loop systems with second order characteristic equation of which the real part of the roots are not affected by any change of controller gain, K_C. This condition is true when process transfer function $G_P(s)$ is of second order, and other blocks in the loop are pure gain elements.

For an over-damped system (real unequal roots), as we increase K_C from the initial zero value, first the locus travels horizontally along the real axis until they meet, where the system become critically damped, then they separate vertically as complex conjugates resulting an under-damped condition of the dynamics.

For a critically-damped system at start, we will obtain only the vertical part of the root locus, starting from the real axis, and for a closed loop system which is under-damped at start, two conjugate half of the root locus will appear from two points, slightly above and below of the real axis.

However, for systems having higher order characteristic equation, we may obtain more complex root locus.

EXAMPLE 10.2 A third order characteristic equation may result either from a third order $G_P(s)$, or a second order $G_P(s)$ and a first order $G_M(s)$. If the parameters of the blocks in the two cases are similar, the resulting root locus will also be similar, as for the systems given in Figures 10.8(a) and (b).

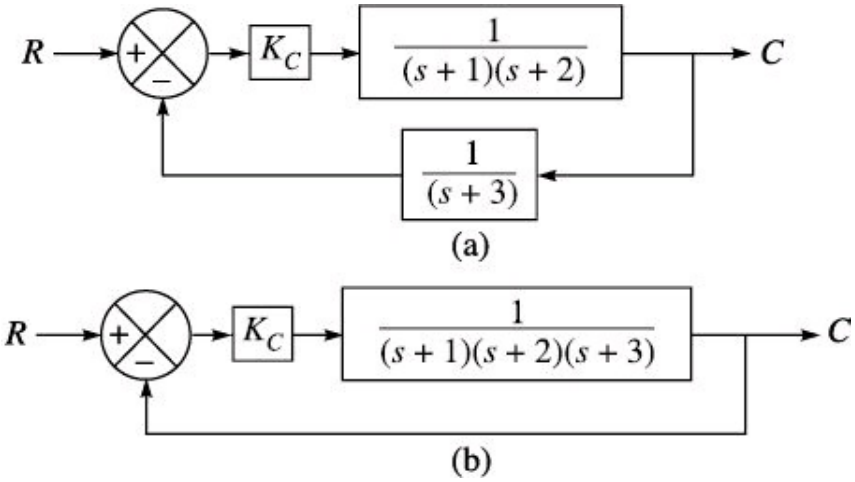


Figure 10.8 The two loops will present similar root locus as their characteristic equations are similar.

Solution: From the closed loop transfer function, we obtain the characteristic equation as:

$$s^3 + 6s^2 + 11s + 6 + K_C = 0 \qquad (10.15)$$

with known coefficient values, such a third degree equation may be solved for roots by the method discussed in higher algebra texts. For increasing values of K_C, the three roots of the above equation have been found out and listed in Table 10.3.

Table 10.3 The Root Locations of Example 2 for different values of K_C

K_C	r_1	r_2	r_3	Position on the Locus
0	–3	–2.0	– 1.0	P_1
0.23	–3.1	–1.11	– 1.75	P_2
0.39	–3.16	–1.42	–1.42	P_3
1.58	–3.45	$-1.28 + 0.76i$	$-1.28-0.76i$	P_4
6.6	–4.11	$-0.95 + 1.5i$	$-0.95-1.5i$	P_5
26.5	–5.10	$-0.45 + 2.5i$	$-0.45-2.5i$	P_6
60.0	–6.0	$0.0 + 3.32i$	$0.0-3.32i$	P_7
100.0	–6.72	$0.35 + 4.0i$	$0.35-4.0i$	P_8

We have plotted the locus of the roots corresponding to the values in the table, in Figure 10.8. The root r_1 has always real –ve values and moves further away from imaginary axis as K_C increases, hence, it does not bother us with any problem regarding stability. However, with the increment of K_C, r_2 and r_3 move toward each other on the real axis until they become equal at $K_C = 0.39$ (point P_3), after which they move apart vertically to become complex conjugates. As we further increase K_C, the locus curves towards the imaginary axis until they cross it at a controller gain value $K_C = 60$ (point P_7). To confirm the continuity of the locus beyond the crossing point, we have drawn the locus for $K_C = 100$ (point P_8). Thus, the closed loop system has a limiting value of $K_C = 60$. We may call this gain value as the **ultimate gain, K_U**.

Thus, for the given systems of Figure 10.7, $K_U = 60$. Any further increase of controller gain will push it to unstable condition. Any practical controller design will apply a K_C that is lower than K_U.

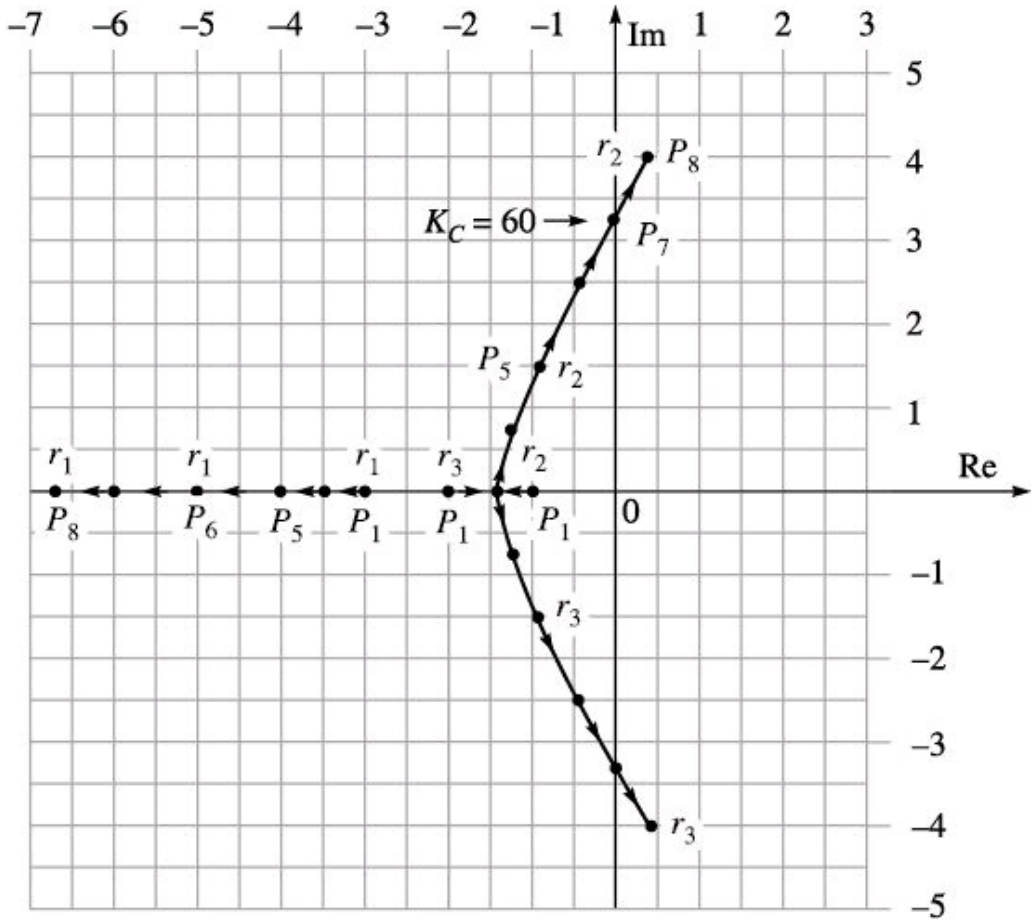


Figure 10.9 The root locus for example 10.2 shows an ultimate gain $K_U = 60$.

The root locus concept has been introduced to show the significance of location of characteristic equation roots on the complex plane, and recognize the +ve half of the complex plane as the forbidden zone, with respect to the stability of a dynamic system. To draw the root locus for complex systems having more number of poles and zeros in the transfer function, a method that uses partly graphical and analytical techniques has been given in detail in many control texts. An interested reader may go through any of the standard texts to get the detail of the procedure. This method gives result within acceptable accuracy.

In literature computer programmes are available for finding real and complex roots of higher degree characteristic equations based on some or other root solving algorithms (ref: Luyben, Smith & Corripio, Coughanowr). We have presented a computer programme POLRT written in QBASIC that may be used for drawing of root locus for a given control loop.

10.3 Programme C10-1A.BAS (POLRT)

This is a root finding program written in QBASIC for polynomial equations based on

Lin and Bairstow's method. We have adopted the basic structure of the programME presented by Coughanowr. In the present form, the input data are: degree of the polynomial, N, the initial coefficient array A1(N), A1(N – 1), …… A1(0), initial value of K_C = KCI, final value of K_C = KCF, increment of K_C = DKC and a scale factor = FCTR to be used for graphical plotting of root locus. The array A1 is converted to array A in the programme to include the dependency of K_C on one or more terms of the characteristic equation. The procedure is explained by the following examples:

(1) Let the closed loop system under consideration be expressed by the following block diagram (Figure 10.10):

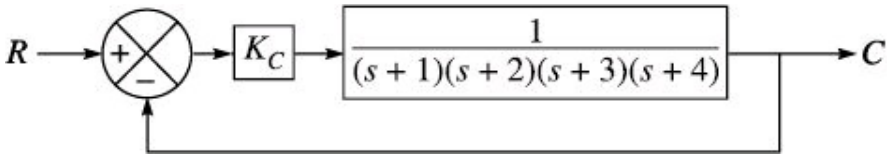


Figure 10.10

Which yields the following characteristic equation:

$$s^5 + 15s^4 + 85s^3 + 225s^2 + 294s + 120 + K_C = 0$$

Then the input data to be entered between programme lines 150 and 200, would be:

```
REM N=5: A1(5)=1: A1(4)=15: A1(3)=85: A1(2)=225: A1(1)=294: A1(0)=120
REM KCI=0: KCF=100: DKC=.1: KC=KCI: FCTR=20
```

and between lines 1050 and 1100, would be:

```
REM FOR I=N TO 1 STEP -1: A(I)=A1(I): NEXT I: A(0)=A1(0)+Kc
```

(2) A closed loop system containing zeros, as in the next block diagram (Figure 10.11):

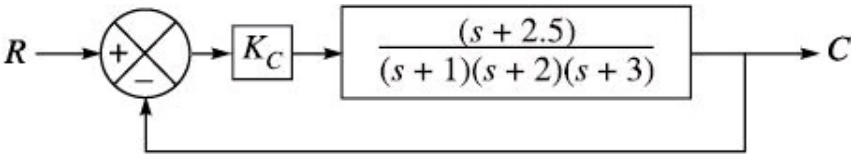


Figure 10.11

will yield a characteristic equation that has K_C dependency in more than one of its terms,

$$s^3 + 6s^2 + (11 + K_C)s + (6 + 2.5\,K_C S) = 0$$

Then the input data to be entered between programme lines 150 and 200, would be:

```
REM N=3: A1(3)=1: A1(2)=6: A1(1)=11: A1(0)=6
REM KCI=0: KCF=100: DKC=.1: KC=KCI: FCTR=20
```

and between lines 1050 and 1100, would be:

```
REM FOR I=N TO 2 STEP -1: A(I)=A1(I): NEXT I: A(1)=A1(1)+KC: A(0)=A1(0)+2.5*Kc
```

Selection of KCI, DKC, KCF and FCTR may require a few trials to ascertain the

determination of K_U and acceptable quality of graphic output. While determining K_U, and finding the value of K_C for which the locus goes through a transition (point at which two real roots joins together to become complex conjugates or vice versa), we recommend a higher printing lag by increasing the counting number, NCNT, of the delay loop at line no. 1700. This will help in observing the real and imaginary parts of the roots. Using a too low value of DKC may untimely hang-up the programme execution, may be due to numerical oscillation and failure of root finding for certain values of KC = KC*. In such cases a larger DKC may circumvent the problem, or the solution may be obtained in two parts, (1) KC = KCI to KC* – DKC and (2) KC = KC* + DKC to KCF.

In the following we present the program listing written in QBASIC, and solution of a few example problems.

```
50 REM ************************ PROGRAM NAME -> C10-1A.BAS
   ******************************
   REM *** THIS PROG. TRACES THE ROOT-LOCUS FOR A GIVEN CH.EQN. ***
   REM *** REQD. DATA : ***
   REM *** DEGREE OF THE CH. EQN. : 'N' ***
   REM *** COEFFS. OF THE POLYNOMIAL A1(I) IN THE CH. EQN. IN TERMS OF 's' ***
   REM *** AS: A1(N)s^N + A1(N-1)s^N-1 + … + A1(2)s^2 + A1(1)s + A1(0) = 0 ***
   REM *** THE A1 ARRAY ELEMENTS INCLUDING THEIR Kc DEPENDENCY & THE ***
   REM *** RANGE OF Kc THROUGH WHICH THE LOCUS HAS TO BE PLOTTED ARE ***
   REM *** WRITTEN BETWEEN LINES 150 & 200 AND 1150 &1200 AS SHOWN FOR ***
   REM *** THE TWO EXAMPLES: ***
   REM *** EX-1. ***
   REM *** between lines 150 & 200: ***
   REM *** N=5: A1(5)=1: A1(4)=15: A1(3)=85: A1(2)=225: A1(1)=294: A1(0)=120 ***
   REM *** KCI=0: KCF=100: DKC=.1: KC=KCI: FCTR=20 ***
   REM *** between lines 1150 & 1200: ***
   REM *** FOR I=N TO 1 STEP -1: A(I)=A1(I): NEXT I: A(0)=A1(0)+Kc ***
   REM *** THE ARRAY 'A1' IS CONVERTED TO ARRAY 'A' TO INCLUDE THE ***
   REM *** DEPENDENCY OF Kc. FCTR IS THE SCALE FACTOR FOR GRAPHIC ***
   REM *** DISPLAY, DENOTING THE CONSTANT THAT IS MULTIPLIED WITH THE ***
   REM *** REAL AND IMAG. PARTS OF THE ROOTS FOR PLOTTING. IF Kc IS ***
   REM *** INVOLVED IN MORE THAN ONE TERM AS IN SYSTEMS CONTAINING ***
   REM *** ZEROS IN THEIR TRANSFER FUNCTION: ***
   REM *** EX-2. ***
   REM *** between lines 150 & 200: ***
   REM *** N=4: A1(4)=5: A1(3)=36: A1(2)=87: A1(1)=76: A1(0)=0 ***
   REM *** KCI=0: KCF=100: DKC=.1: KC=KCI: FCTR=20 ***
   REM *** between lines 1150 & 1200: ***
   REM *** FOR I=N TO 1 STEP -1: A(I)=A1(I): NEXT I ***
   REM *** A(1)=A1(1)-KC: A(0)=A1(0)+2*KC ***
   REM *** AT THE TERMINATION OF THE PROGRAM USED VALUES OF KCI, KCF, ***
   REM *** KCF & FCTR ARE REPORTED TO FACILITATE LABELING OF GRAPH. ***
   REM
   ********************************************************************************
```

```
   DIM A1(20), A(20), B(20), C(20), D(20), RX(20), RY(20)
   CLS
   SCREEN 12
   XS = 320: YS = 230
   FOR I=1 TO 17: I-1=I1:LINE(XS+160,YS-160+I1*20)-(XS-160,YS-160+I1*20), 8: NEXT I
   FOR I=1 TO 17: I-1=I1:LINE(XS-160+I1*20,YS-160)-(XS-160+I1*20,YS+160), 8: NEXT I
   LINE(XS+160,YS)-(XS-160,YS):LINE(XS,YS-160)-(XS,YS+160)
1                                                                                5 0 REM
   ********************************************************************************
   N=5: A1(5)=1: A1(4)=15: A1(3)=85: A1(2)=225: A1(1)=294: A1(0)=120
   KCI=0: KCF=100: DKC=.1: KC=KCI: FCTR=20: NCNT=50
2                                                                                0 0 REM
   ********************************************************************************
   FOR I=1 TO 20: A1(I)=0: A(I)=0: B(I)=0: C(I)=0: D(I)=0: RX(I)=0: RY(I)=0: NEXT I
1000 IF KC>KCF THEN GOTO 1800
1                                                                  0 5 0 REM
   ********************************************************************************
   FOR I=N TO 1 STEP -1: A(I)=A1(I): NEXT I: A(0)=A1(0)+Kc
1                                                                  1 0 0 REM
   ********************************************************************************
1200 R=A(1)/A(2): S=A(0)/A(1)
1220 B(N)=A(N): C(N)=0: D(N)=0
   B(N-1)=A(N-1)-R*B(N): C(N-1)=-B(N): D(N-1)=0
1280 FOR I=2 TO N-2
   B(N-I)=A(N-I)-R*B(N-I+1)-S*B(N-I+2)
   C(N-I)=-B(N-I+1)-R*C(N-I+1)-S*B(N-I+2)
   D(N-I)=-B(N-I+2)-S*D(N-I+2)-R*D(N-I+1)
   NEXT I
   R1=A(1)-R*B(2)-S*B(3): S1=A(0)-S*B(2)
   T=-B(2)-R*C(2)-S*C(3): U=-B(3)-S*D(3)-R*D(2)
   V=-S*C(2): W=-B(2)-S*D(2)
   R2=(-R1*W+S1*U)/(T*W-U*V)
   S2=(-T*S1+V*R1)/(T*W-U*V): S=S+S2:R=R+R2
   IF ABS(R2)<.00001 THEN GOTO 1450
   GOTO 1220
1450 G=R*R-4*S
   IF G<0 THEN GOTO 1490
1470 RX(NR)=-R/2+SQR(G)/2: RY(NR)=0: NR=NR+1
   RX(NR)=-R/2-SQR(G)/2: RY(NR)=0: NR=NR+1
   GOTO 1500
1490 RX(NR)=-R/2+SQR(G)/2: RY(NR)=SQR(G)/2: NR=NR+1
   RX(NR)=-R/2+SQR(G)/2: RY(NR)=-SQR(-G)/2: NR=NR+1
   NR1=NR-1
   LOCATE 3,4
1494 PRINT "KC"; KC; "Real part"; RX(NR1); "Imaginary part"; RY(NR1)
1500 N=N-2
   IF N=0 THEN GOTO 1630
   FOR I=N TO 0 STEP -1: A(I)=B(I+2): NEXT I
   IF N>2 THEN GOTO 1200
   IF N<2 THEN GOTO 1610
   R=A(N-1)/A(N): S=A(N-2)/A(N)
   GOTO 1450
```

```
1610 PRINT "REAL ROOT"; -A(N-1)/A(N)
   RX(NR)=-A(N-1)/A(N): RY(NR)=0: NR=NR+1
1650 FOR I=N TO 0 STEP -1
   PSET(XS+RX(I)*FCTR, YS-RY(I)*FCTR)
   NEXT I
1700 NCNT=50: FOR I=1 TO NCNT: AI=I*I/I+I-I: NEXT I
   KC=KC+DKC
   GOTO 1000
1800 PRINT "KCI"; KCI; "KCF"; KCF; "DKC"; DKC; "FCTR"; FCTR
   END
```

Solution of the following examples has been obtained by drawing the root locus by using C10-1A.BAS. By the procedure discussed above, the limiting and transitional values of gain are determined.

EXAMPLE 10.3 A fourth order process of four distinct poles under proportional controller represented by the block diagram (Figure 10.12) obtains the characteristic equation given below.

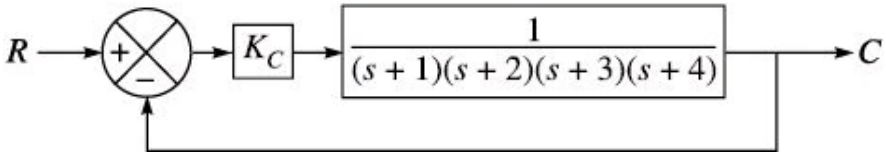


Figure 10.12

$$s^4 + 10s^3 + 35s^2 + 50s + 24 + K_C = 0$$

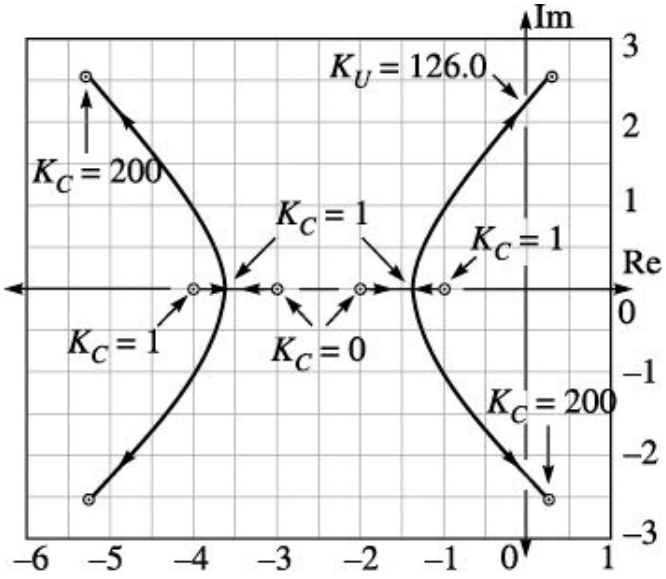


Figure 10.13 Four distinct pole process with a (P) controller.

EXAMPLE 10.4 There is a measurement lag (t_{DL} = 1) in the feedback loop of the following control system (Figure 10.14):

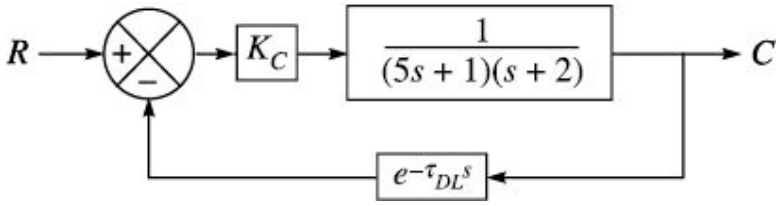


Figure 10.14

If the lag is expressed by first order Pade's approximation, the characteristic equation becomes:

$$5s^3 + 24s^2 + (24 - K_C)s + (4 + 2K_C) = 0$$

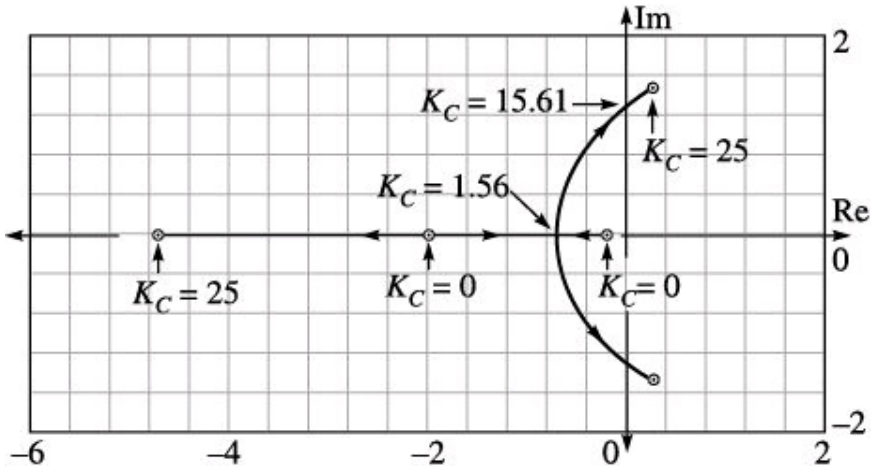


Figure 10.15 Second order process and a delay in the feedback path with a (P) controller.

EXAMPLE 10.5 A fourth order process of four repeated poles under proportional controller represented by the block diagram (Figure 10.16) obtains the characteristic equation given below.

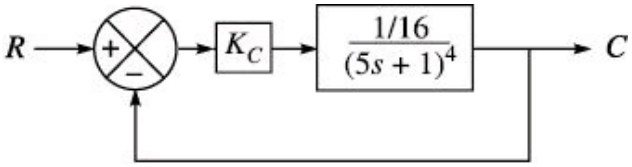


Figure 10.16

$$10000s^4 + 8000s^3 + 2400s^2 + 320s + 16 + K_C = 0$$

EXAMPLE 10.6 The same fourth order process of the last example with a proportional-derivative controller ($t_D = 5$) represented by the block diagram (Figure 10.17) obtains the characteristic equation given below.

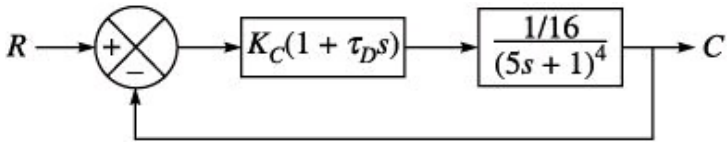


Figure 10.17

$$10000s^4 + 8000s^3 + 2400s^2 + (320 + 5K_C)s + (16 + K_C) = 0$$

or,
$$2000s^3 + 1200s^2 + 240s + 16 + K_C = 0$$

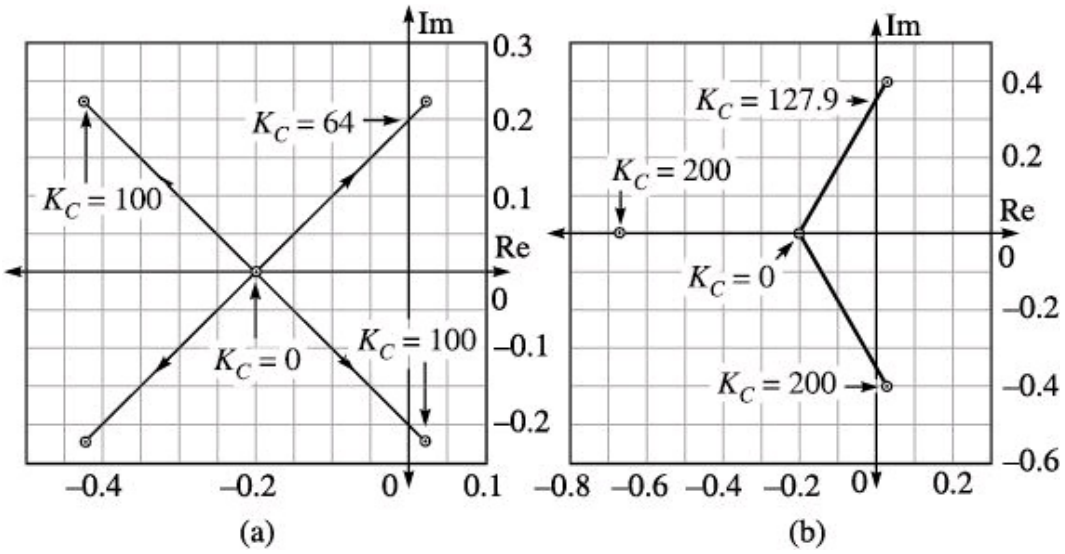


Figure 10.18 (a) Fourth order process having repeated roots with a (P) controller. (b) The same process with a (PD) controller.

EXAMPLE 10.7 A third order process with a proportional-integral controller ($t_I = 2$), represented by the block diagram (Figure 10.19), obtains the characteristic equation given below Figure 10.19: (a) Fourth order process having repeated roots with a (P) controller.

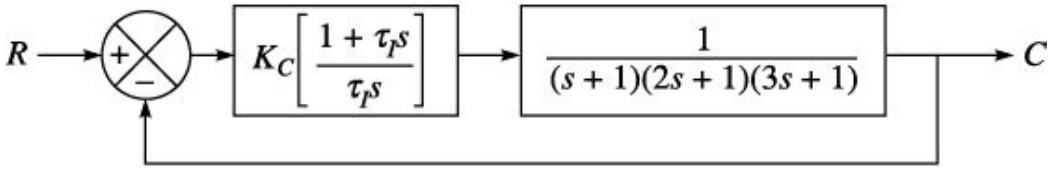


Figure 10.19

$$6s^3 + 8s^2 + 2s + K_C = 0$$

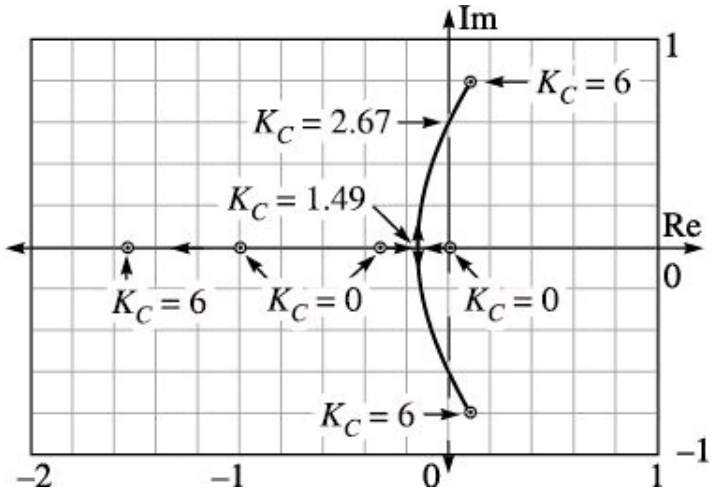


Figure 10.20 A third order process with a (PI) controller.

The reader is urged to check the result (K_U values) obtained in the above examples by the Routh and Hurwitz procedure, an algebraic method of stability determination that we shall discuss in the next section. As a stability analysis technique, the root locus analysis has certain limitations, i.e. if the system contains nonlinear gain character or dead time element, this method would be inapplicable. However, for the

determination of transitional gain values and finding the measure of stability, i.e. complex roots are how far away from the imaginary axis for a given value of K_C, root locus method perform commendably. We shall now go over to another method called Routh-Hurwitz stability criterion that is an algebraic procedure, developed during mid-eighteenth century.

10.4 Routh–Hurwitz Stability Criterion

This procedure, executed on the characteristic equation of a system, gives us the number of roots (if there are any) on the +ve half of the *s*-plane. In this text we are not presenting the complete analysis developed by Routh and Hurwitz, but only the result as an algebraic exercise of treating the characteristic equation to obtain the answer that whether there is any root/s in the forbidden zone of *s*-plane, and conclude about the system's stability condition. To explain the method, let us start with the following general form of the characteristic equation:

$$D(s) = a_N s^N + a_{N-1} s^{N-1} + a_{N-2} s^{N-2} + a_{N-3} s^{N-3} + \cdots + a_1 s + a_0 = 0 \qquad (10.16)$$

To ascertain the stability of a system, it is necessary to determine whether any one of the roots of $D(s)$ lies in the right half of the *s*-plane. Equation 10.16 may be written in the factored form as:

$$D(s) = a_N (r_1 - s)(r_2 - s)(r_3 - s), \ldots (r_I - s) \ldots (r_{N-1} - s)(r_N - s) = 0 \qquad (10.17)$$

where r_I is the ith root of the characteristic equation. Multiplying the factors in the above equation:

$$D(s) = a_N s^N - a_N(r_1 + r_2 + \cdots + r_N)s^{N-1} + a_N(r_1 r_2 + r_2 r_3 + r_1 r_3 + \cdots)s^{N-2}$$
$$- a_N(r_1 r_2 r_3 + r_1 r_2 r_4 + \cdots)s^{N-3} + \cdots a_N(-1)^N r_1 r_2 r_3 \cdots r_N = 0 \qquad (10.18)$$

Inspection of the above equation reveals that for an odd number –ve roots of characteristic equation, all the terms of Equation 10.18 will be –ve (which then may be multiplied by –1, to make them +ve), and for an even number of –ve roots, all terms will be +ve. Also by the same token of logic, there will be no zero coefficients.

Hence, for a stable system, the polynomial at L.H.S. of the characteristic equation must have non zero +ve coefficients. This is a necessary but not the sufficient condition for stability. That is, if we find a characteristic equation with one or more (but not all) –ve or zero coefficients, the system may be written off as an unstable system. But if the condition of +ve and non-zero coefficients is satisfied, our conclusion would be that the system may, or may not be stable, and further investigation should be made by using the Routh-Hurwitz stability criterion to ascertain its stability.

The Routh-Hurwitz criterion is carried out by first writing the characteristic equation in sequential order of decreasing power of *s* as:

$$a_N s^N + a_{N-1} s^{N-1} + a_{N-2} s^{N-2} + a_{N-3} s^{N-3} + \ldots + a_1 s + a_0 = 0 \qquad (10.19)$$

into an array form known as Routh's array, as given in Figure 10.21.

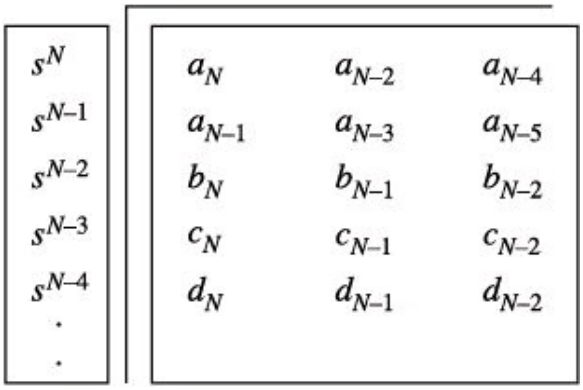


Figure 10.21 Construction of the Routh's Array.

The first two row elements are constructed by using the coefficients from the characteristic equation, and the following row elements are calculated by using a cross-multiplication type formula shown below:

$$b_N = \frac{a_{N-1}a_{N-2} - a_{N-3}a}{a_{N-1}}; \quad b_{N-1} = \frac{a_{N-1}a_{N-4} - a_{N-5}a}{a_{N-1}}; \quad b_{N-2} = \frac{a_{N-1}a_{N-6} - a_{N-7}a_N}{a_{N-1}}$$

$$c_N = \frac{b_N a_{N-3} - b_{N-1}a_{N-}}{b_N}; \quad c_{N-1} = \frac{b_N a_{N-5} - b_{N-2}a_{N-1}}{b_N}; \quad c_{N-2} = \frac{b_N a_{N-7} - b_{N-3}a_{N-1}}{b_N}$$

$$d_N = \frac{c_N b_{N-1} - c_{N-1}b}{c_N}; \quad d_{N-1} = \frac{c_N b_{N-2} - c_{N-2}b}{c_N}; \quad d_{N-2} = \frac{c_N b_{N-3} - c_{N-3}b_N}{c_N} \qquad (10.20)$$

This procedure is continued until a row is found with all zero elements. After constructing this Routh's array, the first column is examined. According to Routh-Hurwitz criterion, if all the first column elements are +ve, we can conclude that the system is stable. If there is any –ve elements in the first column, the system is unstable, and the number of sign changes equals the number of roots on the right half of *s*-plane.

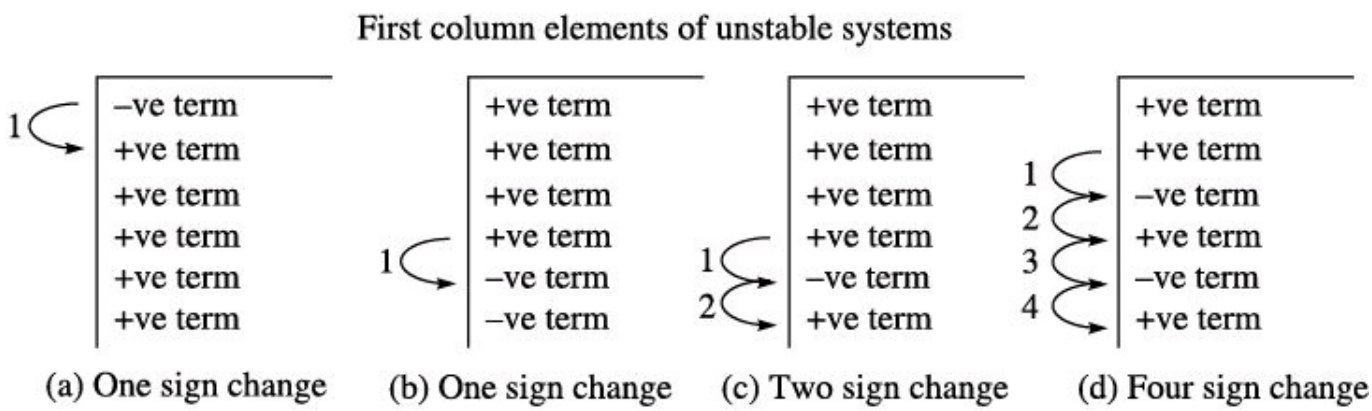


Figure 10.22 Demonstration of the rule for counting the number of sign changes in the first column of Routh's Array.

Let us work out a few examples to get familiar with this method and the associated types of problems.

EXAMPLE 10.8 It is required to ascertain the stability of a system represented by the following characteristic equation:

$$s^4 + 11s^3 + 41s^2 + 61s + 30 = 0 \quad (10.21)$$

Solution: To check the stability of the system, we construct the Routh's array as follows:

1 41 30
11 61 0
35.4545 30 0
51.6923 0 0
30 0 0
0 0 0

Elements of the third, fourth, fifth and sixth rows have been calculated according to Equation 10.20. By inspecting the first column of the array, we find that all elements have a +ve sign, and our conclusion is that the system is stable.

EXAMPLE 10.9 Let us redo the Example 10.2 to find the range of values of K_C within which the closed loop system remain stable.

Solution: We start by writing the characteristic equation of the system;

$$s^3 + 6s^2 + 11s + 6 + K_C = 0 \quad (10.15)$$

and construct the Routh's array as follows;

1 11
6 $6 + K_C$
$(60 - K_C)/6$ 0
$6 + K_C$ 0
0 0

By inspecting the first column of the array, we find two conditions have to be satisfied to make all the first column elements +ve, and ensure system stability.

Condition (1) : $(60 - K_C)/6 > 0$ or $K_C < 60$ (10.22)

and

Condition (2) : $6 + K_C > 0$ or $K_C > -6$ (10.23)

From Equations 10.22 and 10.23, the complete range of K_C is $-6 < K_C < 60$. Though this range has been found analytically, no practical controller can have +ve and –ve gain values simultaneously, thus, the practical range of K_C will be $0 < K_C < 60$ to ensure stability.

Note that upper limit value that was found by Root Locus method has been repeated by Routh-Hurwitz analysis.

10.4.1 Zero Elements in the First Column of Routh's Array

We have not covered the possibility of zero elements occurring in the first column of the array in our foregoing discussion. Usually this condition indicates that a pair of roots may be on the imaginary axes, which in turn spells a limitingly stable or unstable condition of the system. If a zero is found in the first column elements, the zero is replaced by a very small +ve number e that is allowed to approach zero after completing the array. There are two distinct types of this category encountered in Routh array construction.

(1) Zero in the first column but all other elements of the same row are non-zero. The associated systems are usually unstable. Let us take up an example:

EXAMPLE 10.10 It is required to ascertain the stability of a system represented by the following characteristic equation:

$$s^5 + 2s^4 + 2s^3 + 4s^2 + 11s + 10 = 0 \qquad (10.24)$$

Solution: To check the stability of the system we construct the Routh's array as follows:

1	2	11
2	4	10
ε	6	0
$c_1 = (4\varepsilon - 12)/\varepsilon$	10	0
$d_1 = (6c_1 - 10\varepsilon)/c_1$	0	0
10	0	0

Now as $e \to 0$; $c_1 = (4e - 12)/e = -12/e \to$ A large –ve number, and $d_1 = (6c_1 - 10e)/c_1 \to 6$.

Hence, there will be two sign changes in the first column and the system is unstable.

(2) Zero in the first column but all other elements of the same row are also zeroes. This occurs when two roots of the characteristic equation are imaginary conjugates, i.e. they are placed on the imaginary axes. .

EXAMPLE 10.11 It is required to ascertain the stability of a system represented by the following characteristic equation:

$$D(s) = s^3 + 2s^2 + 4s + 8 = 0 \qquad (10.25)$$

Solution: To check the stability of the system, we construct the Routh's array as follows:

s^3	1	4
s^2	2	8
s^1	ε	0
s^0	8	0

The imaginary conjugate roots in this case may be found by considering the auxiliary polynomial $A(s)$. The auxiliary polynomial is constructed by array elements of the row just preceding the row with all zeros. Thus,

$$A(s) = 2s^2 + 8 = 2(s + 2i)(s - 2i) \qquad (10.26)$$

That our original polynomial $D(s)$ contains $A(s)$ may be confirmed by dividing $D(s)$ by $A(s)$. The result is $0.5s + 1$, and the complete $D(s)$ may be written in the factored form as

$$D(s) = (s + 2)(s + 2i)(s + 2i) \qquad (10.27)$$

As the system contains roots on the imaginary axis that produce sustained oscillation with undiminished amplitude, we may conclude that the system is in limitingly stable condition which for practical purposes may be called unstable.

10.5 Method of Direct Substitution

Recall the root locus curves, specifically the point at which the locus crosses the imaginary axis. At this point the characteristic equation must be satisfied for $s = iw$. By this substitution, the stability condition may be found for an unknown parameter (say the controller gain K_C) of a closed loop system. Let us present some examples to explain this method.

EXAMPLE 10.12 Redo Example 10.2 by direct substitution method to find the range of value of K_C within which the closed loop system remain stable.

Solution: We start by writing the characteristic equation of the system;

$$s^3 + 6s^2 + 11s + 6 + K_C = 0 \qquad (10.15)$$

Putting $s = iw$ in the above equation, we obtain,

$$-iw^3 - 6w^2 + 11iw + 6 + K_C = 0 \qquad (10.28)$$

By equating the imaginary part of Equation 10.28,

$$-iw^3 + 11iw = 0 \text{ or } w = \sqrt{11} = 3.3166225 \text{ rads/time}$$

Putting this value of w in the real part of Equation 10.28, we obtain, $K_C = 60$.

The result tallies with the values of K_C found by root-locus and Routh's array methods. The value of $K_C = 60$ if put as the gain to the controller of closed loop system in Figure 10.5, the response will be a sustained oscillation of undiminished amplitude. This limiting value of K_C is called the Ultimate Gain, K_U, and the angular frequency of this oscillatory response is termed as w_U. The ultimate period, P_U, of this response is related to w_U, as $P_U = w_U/2\pi$.

For practical design of controllers the gain value used is less than K_U, but often the design procedure starts from the determination of K_U, and P_U, as we shall see in Chapter 13.

EXAMPLE 10.13 It is required to ascertain the stability of a system represented by the following characteristic equation:

$$s^4 + 11s^3 + 41s^2 + 61s + 30 + K_C = 0 \qquad (10.29)$$

Solution: We start by putting $s = iw$ in the above equation, and obtain,

$$w^4 - 11iw^3 - 41w^2 + 61iw + 30 + K_C = 0 \qquad (10.30)$$

By equating the imaginary parts of Equation 10.30,

$$-11iw^3 + 61iw = 0 \text{ or } w^2 = 61/11 \text{ or } w_U = 2.3549 \text{ rads/time}$$

By equating the real parts of Equation 10.30,

$$K_U = 41(2.3549)^2 - (2.3549)^4 - 30 = 227.3677 - 30.7532 - 30 = 166.6$$

Originally, this method was only used for linear systems. But dead time element, if present, could be incorporated into the analysis by writing linear approximate transfer function following the Pade's formula, or Euler's identity. Let us workout some examples to demonstrate this extension of Routh-Hurwitz procedure.

EXAMPLE 10.14 For a control system the following individual block transfer functions are given:

$G_P(s) = K_P/(t_P s + 1)$; $G_C(s) = K_C$; $G_M(s) = [\exp(-t_{DM} s)]/(t_M s + 1)$ and $G_V(s) = 1$

If the dead time is expressed by Pade's first order approximation formula, prove that the following relation $K_C > -(1/K_P)$ is necessary between controller and process gain to maintain stability of the control system.

Solution: We start by writing the Pade's formula for the dead time:

$$e^{-\tau_{DM}s} = \frac{1-(\tau_{DM}s)/2}{1+(\tau_{DM}s)/2} = \frac{1-\eta s}{1+\eta s}$$

Now we may write the CLTF as,

$$\frac{C}{R} = \frac{G_C G_P}{1+G_C G_P G_M} = \frac{(K_C K_P)/(\tau_P s+1)}{1+\frac{K_C K_P}{\tau_P s+1}\cdot\frac{1-\eta s}{1+\eta s}\cdot\frac{1}{\tau_M s+1}}$$

From above the required characteristic equation is:

$$\tau_P\tau_M\eta s^3 + (\tau_P\tau_M + \eta\tau_P + \eta\tau_M)s^2 + (\tau_P + \tau_M + \eta - K_C K_P \eta)s + 1 + K_C K_P = 0 \qquad (10.31)$$

Instead of constructing the Routh's array, we directly use the condition of stability, that all the coefficient terms of characteristic equation must be +ve to ensure system stability.

By examining the coefficient of s^0, we find:

$1 + K_C K_P > 0$ or $K_C > -(1/K_P)$

which may be taken as the required relation.

EXAMPLE 10.15 By means of the R.H. test, determine the stability of the system shown in the following figure, when $K_C = 2$.

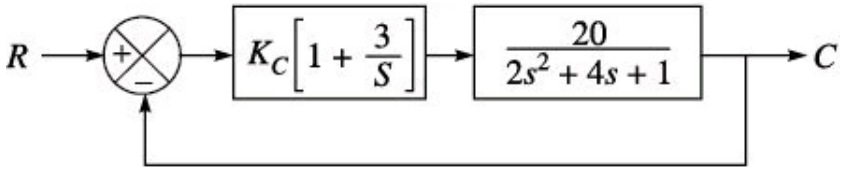


Figure 10.23 The control system of example 10.15.

Solution:

$$\frac{C}{R} = \frac{K_C\left[1+\frac{3}{s}\right]\left[\frac{20}{2s^2+4s+10}\right]}{1+2K_C\left[1+\frac{3}{s}\right]\left[\frac{20}{20s^2+4s+10}\right]} = \frac{20K_C(s+3)}{2s^3+4s^2+10(2K_C+1)s+60K_C}$$

□ The Ch. Eqn becomes: $s^3 + 2s^2 + 5(2K_C+1)s + 30K_C = 0$

By Routh's method: By direct substitution method:

$$-iw^3 - 2w^2 + 5iw(2K_C + 1) + 30K_C = 0$$

1. $5 + 10K_C$ Equating imaginary parts:

2. $30K_C - w^3 + 5w(2K_C + 1) = 0$ or $w_U = \sqrt{[10K_C + 5]}$

$5(1 - K_C)$ 0 Equating real parts:

$$30K_C\, 0 - 2w^2 + 30K_C = 0 \text{ or } K_U = 1$$

(i) Third row: $1 - K_C > 0$; or, $K_C < 1$

(ii) Fourth row: $K_C > 0$ Thus, the two results agree. The value of

Hence the range of stability:
$0 < K_C < 1.0$

$K_C = 2$ would render the system unstable because this value is greater than K_U.

EXAMPLE 10.16 In the following Figure 10.24, block diagram of a second order system with an integral controller is shown.

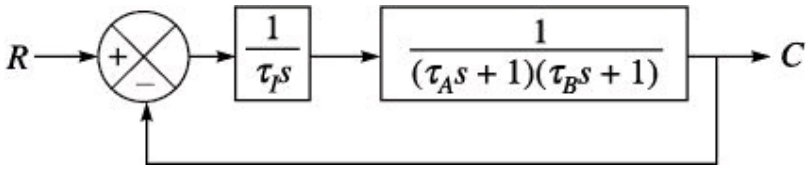


Figure 10.24 The control system of example 10.16.

To keep up closed loop stability it is necessary that the following relation:

$$\tau_I = \frac{\tau_1 \tau_2}{\tau_1 + \tau_2}$$

should exist between controller and process parameters.

Solution: The Ch. Eqn. of the closed loop system may be found as:

$$t_I t_1 t_2\, s^3 + (t_I t_1 + t_I t_2)s^2 + t_I s + 1 = 0$$

The Routh's array is constructed

$\tau_I \tau_1 \tau_2$	τ_I	From the first element of the third row, we
$\tau_I \tau_1 + \tau_I \tau_2$	1	obtain the required relation between τ_I
$\tau_I - [\tau_1 \tau_2/(\tau_1 + \tau_2)]$	0	and other time constants τ_A, and τ_B of
1	0	the system.
0	0	

$$\tau_I - \frac{\tau_1 \tau_2}{\tau_1 + \tau_2} > 0 \quad \text{or} \quad \tau_I > \frac{\tau_1 \tau_2}{\tau_1 + \tau_2} \tag{10.32}$$

EXAMPLE 10.17 Production of a special lubricant is done by heating and mixing a waxy solid with certain liquid hydrocarbons. The control system illustrated in Figure

10.25 is controlled by a pneumatic (PD) controller. Assume that there is heat flow between the surrounding and the process. Available data regarding the system:

$$q = 300 \text{ l/m}; V1 = 450 \text{ l}; V2 = 375 \text{ l}; V3 = 300 \text{ l}$$

mean heat capacity and density of the mixture are $C_P = 0.925$ kcal/(kg/°C), and $r = 0.9$ kg/l.

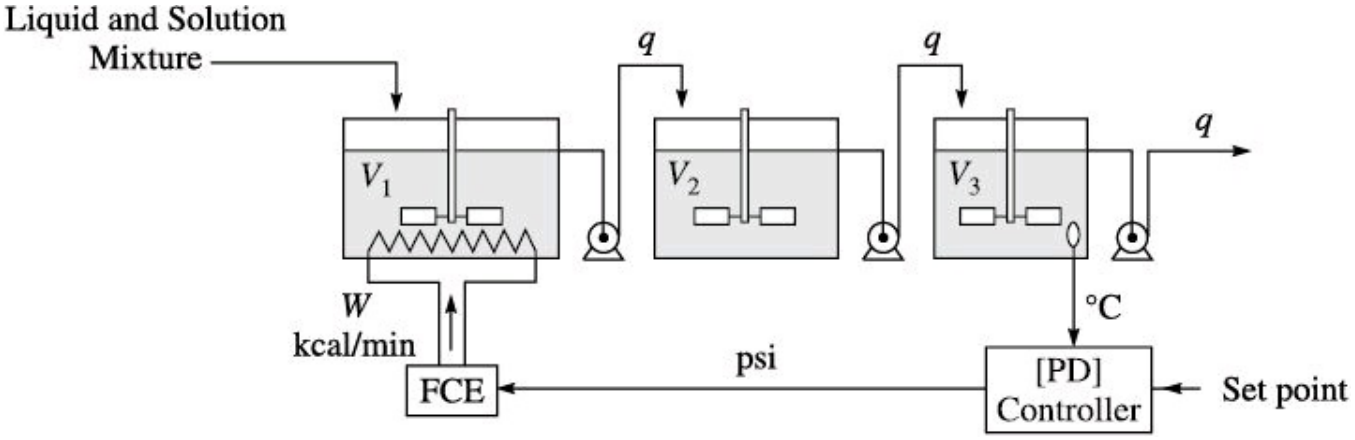


Figure 10.25 Temperature control of a 3-tank stirred heater.

Determine the value of gain in (psi/°C) which may just cause the system to become unstable if (a) $t_D = 0.25$ min. (b) $t_D = 0.5$ min.

Solution: Let us evaluate the gains and dynamic parameters of the system:

The time constants of the tanks are:

$$\tau_1 = \frac{450}{300} = 1.5 \text{ min}$$

similarly, $\tau_2 = 1.25$ min and $\tau_3 = 1$ min

The gains, $K_P = \dfrac{1}{(qC_P)} = 0.004$ (min/°C/kcal); $K_V = 500$ (kcal/min/°C);

The block diagram of the control system would assume as illustrated in Figure 10.26:

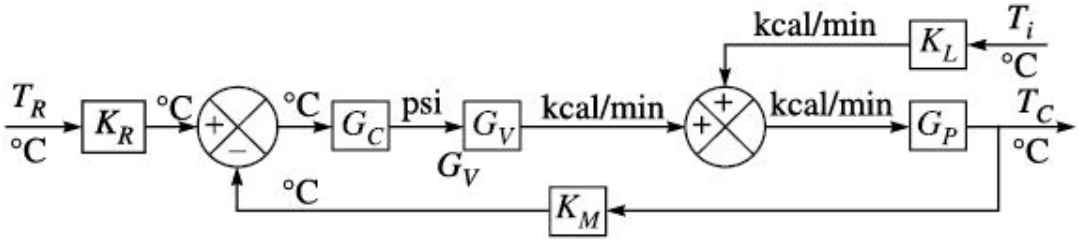


Figure 10.26 The block diagram of the control system in Figure 10.17.

The CLTF: $$\frac{C}{R} = \left[\frac{K_R G_C K_V K_P}{(s+1)(1.25s+1)(1.5s+1)}\right] \Big/ \left[1 + \frac{G_C K_V K_P K_M}{(s+1)(1.25s+1)(1.5s+1)}\right]$$

$$= \frac{2K_C(1+\tau_D s)}{1.875s^3 + 4.625s^2 + 3.75s + 1 + 2K_C(1+\tau_D s)}$$

Case I. For $\tau_D = 0.25$ min the Ch. Eqn. is:

$$1.875s^3 + 4.625s^2 + (3.75 + 0.5K_C)s + 2K_C = 0$$

The Routh's Array:		
	1.875	$3.75 + 0.5K_C$
	4.625	$2K_C$
	$3.75 - 0.311K_C$	$0 \rightarrow K_C < 12.065$

$\therefore$ The stability range:

$2K_C \quad 0 \rightarrow K_C > 0$

$0 < K_C < 12.065$ and $K_U = 12.065$

Case II. For $\tau_D = 0.4$ min the Ch. Eqn. is:

$$1.875s^3 + 4.625s^2 + (3.75 + 0.8K_C)s + 2K_C = 0$$

The Routh's Array:		
	1.875	$3.75 + 0.8K_C$
	4.625	$2K_C$
	$3.75 - 0.015K_C$	$0 \rightarrow K_C < 250.0$

$\therefore$ The stability range:

$2K_C \quad 0 \rightarrow K_C > 0$

$0 < K_C < 250.0$ and $K_U = 250$

The result shows that by using a slight increase of derivative time, the close loop stability has improved to a great extent. A much higher value of controller gain may be used to obtain a much faster response speed to drive away any disturbances and still remain within stability limit.

EXAMPLE 10.18 For a closed loop system, find the value of controller K_C needed for continuous cycling, if $G_C = K_C$, $G_V = G_M = 1$, and $G_P = 2/(4s+1)^N$, if (a) $N = 2$; (b) $N = 3$.

Solution: (a) For $N = 2$, the $G_P(s)$ is a second order process with equal roots at – 0.5. We know that such a process has a phase locus which asymptotically approaches – 180º. Hence, the crossover frequency would have no finite value, i.e. even for an infinitely large value of K_C, the system could not be brought to limitingly stable condition.

(b) The CLTF: $$\frac{C}{R} = \left[\frac{2K_C}{(4s+1)^3}\right] \Big/ \left[1 + \frac{2K_C}{(4s+1)^3}\right] = \frac{2K_C}{64s^3 + 48s^2 + 12s + 1 + 2K_C}$$

Hence, the Ch. Eqn. is: $64s^3 + 48s^2 + 12s + 1 + 2K_C = 0$

Routh's array:

64	12
48	1+2
$10.67 - 2.67K_C$	$0 \rightarrow 4 - K_C > 0$
$1 + 2K_C$	0

From first element of third row:

$4 - K_C > 0$ or $K_C < 4$

From first element of fourth row:

$$1 + 2K_C > 0 \quad \text{or} \quad K_C > -0.5$$

Range of stability: $0 < K_C < 4$

Applying direct substitution:

$-8i\omega^3 - 12\omega^2 + 6i\omega + 1 + 2K_C = 0$

Equating imaginary parts:

$K_C - 8i\omega^3 + 6i\omega = 0$; or, $\omega_U = 0.867$

Equating real parts:

$$-12\omega^2 + 1 + 2K_C = 0$$

or, $-12 \times 0.75 + 1 + 2K_C = 0$

or, $K_U = 4$

EXAMPLE 10.19 The exit tube liquid temperature T_2 of the shell and tube heat exchanger (shown in Figure 10.27) is to be controlled by manipulating the steam mass flow rate Ms. In this system the tube inlet liquid temperature T_1 is load variable. Dynamic model of the heat exchanger has been determined by step testing, and the results are interpreted as FOPDT models as:

$$\frac{T_2(s)}{Ms(s)} = \frac{3e^{-s}}{10s+1} \text{ °C/(kg/s)}; \quad \frac{T_2(s)}{T_1(s)} = \frac{0.9e^{-2s}}{5s+1} \text{ dimensionless}$$

where, the time constants and time delays have units of seconds. Steady state character of the control valve may be expressed as: $Ms = 0.273\sqrt{(c-4)}$ kg/(s/mA), where c is the controller output in mA. At steady operating condition $c = 12$ mA, and its dynamic response has a time constant, $t_V = 4$ s. The Temperature transmitter output varies linearly from 4 to 20 mA, as T_2 goes from 49 to 71°C.

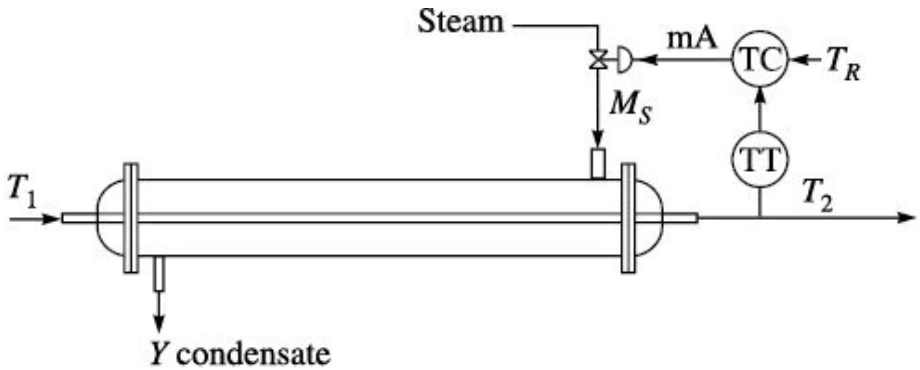


Figure 10.27 Temperature control of the exit tube liquid by manipulating shell steam flow.

Using the above information do the following: (a) If a (P) controller is used find the values of K_U and P_U by using direct substitution and Euler's identity to represent delay element; (b) Estimate K_U by R.H. criteria and first order Pade's formula to

represent the delay element.

Solution: We first draw the block diagram of the heat exchanger control system as shown in Figure 10.28:

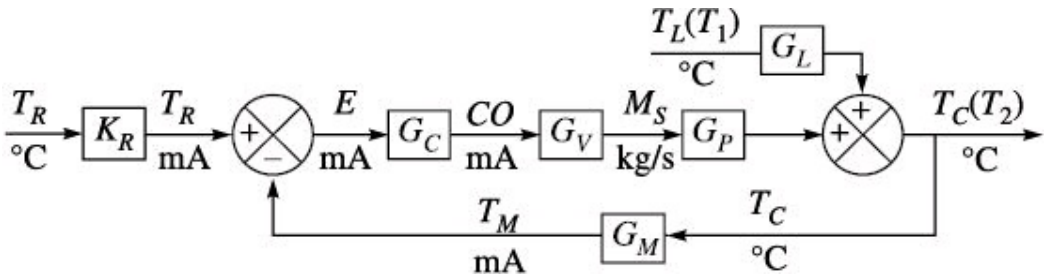


Figure 10.28 Block diagram of the heat exchanger control.

Then we write transfer functions of individual elements of the loop:

$$G_P(s) = \frac{T_2(s)}{W(s)} = \frac{3e^{-s}}{10s+1} \ \text{°C/(kg/s)}$$

and

$$G_L(s) = \frac{T_2(s)}{T_1(s)} = \frac{0.9e^{-s}}{5s+1} \ \text{dimensionless}$$

$$G_M = K_M = K_R = 16/40 = 0.72 \ \text{mA/°F};$$

$$G_V = \frac{K_V}{\tau_V s + 1}$$

Now as given in the problem: $5t_V = 20$ s, or, $t_V = 4$ s, and for the valve gain, K_V could be expressed by linearizing the relation, $M_S = 0.273\sqrt{(c-4)}$, around $c_S = 12$ mA. Thus,

$$K_V = \frac{\Delta y}{\Delta x} = \frac{M_S - Ms_S}{c - c_S} = \left[\frac{dMs}{dc}\right]_{S.S.} = \frac{0.273}{2\sqrt{12-4}} = 0.04826 \ \text{(kg/s)/mA};$$

Hence, the complete valve dynamics is

$$G_V = \frac{0.04826}{4s+1} \ \text{(kg/s)/mA}$$

We may now write the characteristic equation of the closed loop system as:

$$1 + G_C G_V G_P G_M = 0 \text{ or } 1 + \frac{K_C \times 0.04826 \times 3 \times 0.72 \times e^{-s}}{(4s+1)(10s+1)} = 0;$$

Putting $e^{-s} = \frac{2-s}{2+s}$ in the last equation, we obtain, after simplification,

$$40s^3 + 94s^2 + (29 - 0.1042K_C)s + 2 + 0.2084K_C = 0$$

(b) Now, the Routh's array may be constructed as,

40	$29 - 0.1042K_C$	Thus, condition (1) for stability is
94	$2 + 0.2048K_C$	$28.147 - 0.191K_C > 0$
$28.147 - 0.191K_C$	0	or, $K_C > 147.4$, or $K_U = 147.4$
$2 + 0.2048K_C$	0	and condition (2) is
0	0	$K_C > -9.766$

Practical range of K_C from this analysis: $0 < K_C < 147.4$

(a) Next we apply Euler's identity to represent the delay followed by direct substitution for stability check. Rewriting the denominator of the CLTF as,

$$1 + \frac{K_C \times 0.1042 \times e^{-s}}{(4s+1)(10s+1)} \text{ or } 40s^2 + 14s + 1 + K_C\, 0.1042e^{-s} = 0,$$

Putting $s = iw$, and using the Euler's identity: $e^{iw} = \cos w - i \sin w$, in the equation becomes,

$$-40w^2 + 14iw + 1 + 0.1042K_C \cos w - 0.1042iK_C \sin w = 0$$

Equating the imaginary parts of the above equation:

$$14\omega - 0.1042K_C \sin\omega = 0$$

or, $$K_C = 14\omega/(0.1042 \sin\omega) \quad (10.33$$

Equating the real part of the above equation:

$$-40\omega^2 + 1 + 0.1042K_C \cos\omega = 0$$

or, $$K_C = (40\omega^2 - 1)/(0.1042 \cos\omega) \quad (10.34)$$

From the above two expressions of K_C,

$$\tan\omega = \frac{14\omega}{40\omega^2 - 1} \quad (10.35)$$

To find the value of $w = w_U$ that will satisfy the tan w equation above, we apply an iterative procedure of 'trial and error'. The results are compiled in Table 10.4.

Table 10.4 Results of the Trial Error Procedure

~	tan ~	$14\sim/(40\sim^2 - 1)$	Remark
0.7	0.84	0.526	
0.6	0.684	0.6268	
0.59	0.6695	0.639	
0.58	0.6552	0.652	
0.57	0.641	0.6652	
0.575	0.648	0.6585	
0.579	0.653	0.653	End of search

So from above, we obtain $w_U = 0.579$. Putting this value of w_U in Equation 10.33, we obtain $K_U = 139.75$, and from Equation 10.34, we obtain $K_U = 139.86$. We take an average value
$K_U = 139.81$.

Hence, taking a more exact expression of the delay term, we obtain a more orthodox value of $K_U = 139.81$, compared to the analysis made by taking 1/1 Pade's approximation by which the value obtained was $K_U = 147.4$.

As the second method is more exact and devoid of any truncation error associated with 1/1 Pade's formula, The second result of K_U and w_U values should be used for control system design.

10.6 Stability Measure by Routh's Array

A criticism against Routh-Hurwitz criterion was that the result is only qualitative, i.e. the procedure can only answer in a 'yes-or-no'verdict about whether a system is *stable* or *unstable*. No quantitative information about the measure of stability could be obtained. This limitation may be addressed by using the property of imaginary conjugate roots, or pair of roots lying on the imaginary axis. This has been indicated earlier that such pair produces a row of zero elements in the Routh's array.

For the measure of stability of a system it is sufficient to determine the distance of the nearest roots of the characteristic equation from the imaginary axis, especially for complex conjugate roots because they will specify the amount of oscillation and overshoot in the response. Hence, for a stable system, if we shift the imaginary axis to a point so that all the elements of a row of Routh's array become zero, the distance of shift may be taken as the stability measure of the system. This analysis may be extended to controller gain design for tailoring of closed loop response to the desired value by shifting the nearest root to the imaginary axis to a new position. The procedure has been explained in the following example by considering the characteristic equation of a system in factored form.

EXAMPLE 10.20 The characteristic equation of a system have given by:

$$D(s) = (s + 1 + i)(s + 1 - i)(s + 2) = s^3 + 4s^2 + 6s + 4 = 0 \qquad (10.36)$$

Solution: The two complex conjugate pair are at distance 1 from the imaginary axis. The shift of axis may be performed by putting $s = s - 1$. By this substitution, the transformed characteristic equation becomes:

$$s^3 + s^2 + s + 1 = 0 \qquad (10.37)$$

The Routh's array for the above Equation 10.37

1	1	Thus, we have established the distance of the
1	1	nearest root from the imaginary axis to be = 1,
ε	0	and that may be taken as the stability measure
1	0	of the system.

For a real root being the nearest to imaginary axis or origin, a similar relocation will produce system transfer function with an integrator, thus in the characteristic equation there will be a missing term of a particular power of s. The power of s of the missing term will be less than the degree of the polynomial.

Let $$D(s) = (s + 1)(s + 2)(s + 3) = s^3 + 6s^2 + 11s + 6 = 0 \qquad (10.38)$$

As we know that the nearest root to the imaginary axis is at –1 on the real axis, we put

$s = s - 1$, and transform the characteristic equation to become,

$$s^3 + 3s^2 + s = 0 \qquad (10.39)$$

In the transformed characteristic Equation 10.39, s^0 term is missing. At this point, we are unable to say whether the system is limitingly stable or it has become unstable, because in this form the characteristic equation is not satisfying the conditions of Routh's analysis stated earlier. Thus, this procedure of ascertaining the stability measure fails if the system contains a real root that is closest to the imaginary axis.

EXERCISE PROBLEMS

10.1 Given the characteristic equation,

$$s^4 + 3s^3 + 5s^2 + 4s + 2 = 0$$

Determine the stability by R.H. method.

10.2 Find whether the systems having the following closed loop transfer function (CLTF) are stable or not.

$$\text{CLTF} = G(s)/[1 + G(s)]$$

(a) Where $G(s) = 8/[(3s + 1)^2 (s + 1)]$

(b) Where $G(s) = 10/[(10s + 1) (5s + 1)(s + 1)]$

(c) Where $G(s) = (2.5s + 1)/[5s(80s^3 + 60s^2 + 15s + 1)]$

10.3 For the closed loop control system in Figure P10.1, taking $t_1 = 1$, $t_2 = 2$, and $t_3 = 3$, determine the range of K_C for which the system is stable.

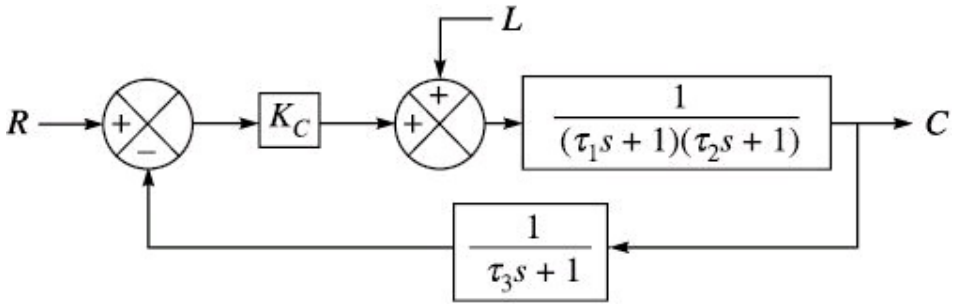


Figure P10.1.

10.4 Write the characteristic equation and construct the R.H. array for the control systen shown in Figure P10.2. Is the system stable for (a) $K_C = 9.5$; (b) $K_C = 11$; (c) $K_C = 12$.

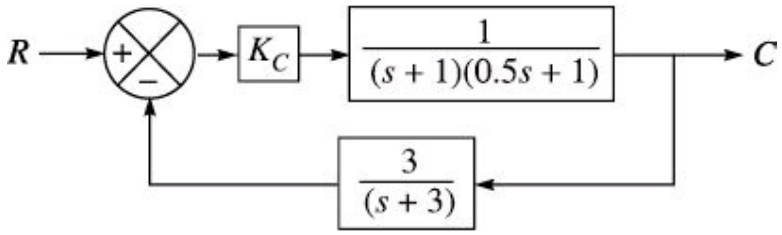


Figure P10.2.

10.5 By means of the R.H. test, determine the stability of the system shown in Figure P10.3, when $K_C = 2$.

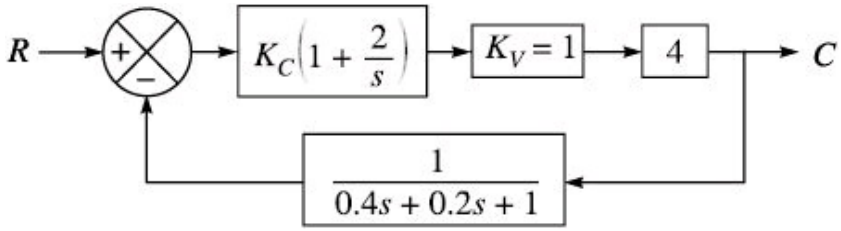


Figure P10.3.

10.6 For a control system the following individual block transfer functions are given:

$$G_P(s) = 1/(2s^2 + 4s + 4);\ G_C(s) = K_C[1 + 1/(t_I s)];\ \text{and}\ G_M(s) = Gv(s) = 1$$

Find a relation between K_C and t_I, so that the system stability is maintained.

10.7 For a control system the following individual block transfer functions are given:

$G_P(s) = K_P/(t_P s + 1)$; $G_C(s) = K_C$; $G_M(s) = [\exp(-t_{DM} s)]/(t_M s + 1)$ and $G_V(s) = 1$

Find a relation between K_C and K_P, so that the system stability is maintained. Express the delay term by Pade's first order approximation formula.

10.8 For a control system the following individual block transfer functions are given:

$$G_P(s) = 1/(s^2 + 2s + 2);\ G_C(s) = K_C[1 + 1/(t_I s)];\ \text{and}\ G_M(s) = G_V(s) = 1$$

Establish the condition of closed loop stability of the system, if $t_I = 0.1$.

10.9 For a control system the following individual block transfer functions are given:

$G_P(s) = 1/(s^2 + 2s + 3)$; $G_C(s) = K_C[1 + 1/(t_I s)]$; and $G_M(s) = G_V(s) = 1$

Find a relation between K_C and t_I, so that the system stability is maintained.

10.10 For a closed loop system:

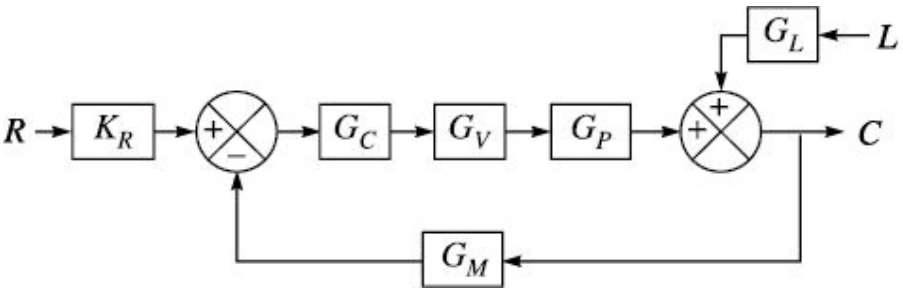
$$1 + G_p(s)G_M(s) = \frac{K_C(4 - s)}{[\tau_P s^2 + (2\tau_P + 1)s + K_C(4 - s) + 2]}$$

The most probable value of t_P is 2, but it may be anywhere between 1.8 and 2.1 and $K_C = 5$.

(a) Determine whether the system will be stable for $t_P = 2$, and $K_C = 5$.

(b) Determine whether the system will be stable for all values of t_P.

10.11 Consider the following block diagram with a (P) controller, $G_C(s) = K_C$, and the associated transfer functions of the individual blocks:



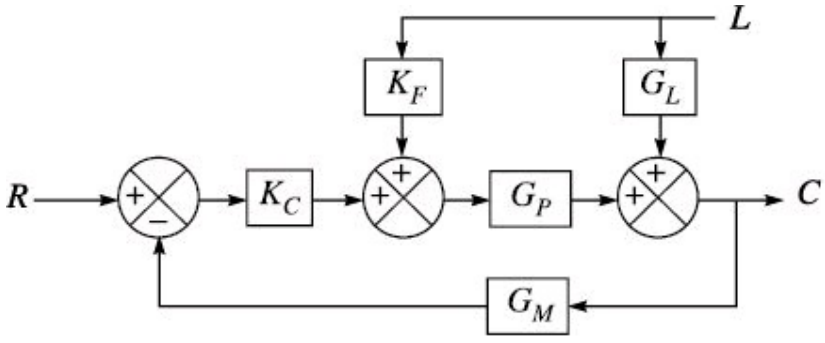
$$G_V(s) = 0.8 \quad G_p(s) = \frac{5e^{-2S}}{10s + 1}$$

$$G_I(s) = \frac{4e^{-S}}{15s + 1} \quad G_M(s) = \frac{1.2}{2s + 1}$$

Figure P10.4.

Which of the following changes would have more significant effect on the closed loop stability limit? (a) The process dead time is reduced from 2 to 1;(b) The measurement time constant is reduced from 2 to 1.

10.12 A block diagram of a feed forward—feedback control system is shown below. For what values of K_C and K_F , is the closed loop system stable?



$$G_P(s) = \frac{(s - 2)}{(s + 10)(s + 5)}$$

$$G_I(s) = \frac{1}{s + 1} \quad G_M(s) = \frac{1}{s + 1}$$

Figure P10.5.

10.13 For the following transfer functions of $G_P(s)$ and $G_C(s) = K_C$, find the value of K_U by Routh–Hurwitz criteria and Direct substitution method, assuming $G_V(s) = G_M(s) = 1$.

(a) $G_P(s) = \dfrac{1}{(s+1)}$

(b) $G_P(s) = \dfrac{1}{(3s+1)(2s+1)(s+1)}$

(c) $G_P(s) = \dfrac{0.5s+1}{(3s+1)(2s+1)(s+1)}$

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11

Frequency Response Analysis

11.1 Introduction

In this chapter we shall develop a new point of view toward dynamic systems that is how they behave while disturbed by sinusoidal input. In Chapter 5, we derived the sinusoidal response of first order systems. So the time domain solution from a given transfer function by partial fraction expansion is there for obtaining sinusoidal response, but the algebra become very lengthy and tedious for higher order systems. Here we shall propose a very simple method of converting Laplace domain transfer function into the required frequency response expression.

After going through this chapter, it should be realized that the dynamic information content about a system given by its frequency response is more than in its transient response traces. To emphasize this point we may cite a second order system. Its TF contains two time constants, one is much bigger than the other ($t_1 >> t_2$). If we consider the system's step response, it can be easily shown that the smaller time constant could be hardly recognizable or measured from the response trace. This fact is also true for any other type of non-periodic input functions, e.g. ramp or impulse.

Well, frequency domain description of a process could bring out such 'insignificant' poles or zeroes to surface for better identification, observation and analysis. Not only can this technique 'fish-out' subdued dynamic qualities, it moreover has the capability of stability analysis of systems with nonlinear elements and synthesis of optimum controller for a given process.

To understand a process from its frequency domain description, it is not always required to know a lot of complex variable algebra or trigonometry, because examples of this discipline are around us in our every day life.

Let us introduce the frequency domain concept by citing two such common examples, in which a couple of potential application of frequency response property of a system has been discussed.

11.1.1 A Little Photographic Exposure

Today photography is largely done with coloured films, but black and white photography is still considered to be the exclusive medium for creative work by a number of high-level professionals in the field.

In black and white photography coloured glasses are often used before the lens to

increase contrast between different shades of a principle colour. An assortment of objects, say a bed of red, purple and orange flowers under sunlight will emit maximum amount of light energy of frequency corresponding to red (the principle hue) plus relatively small amounts of energy of green, blue, indigo, even violet colours [Figure 11.1(a)]. If the maximum difference of emitted energy (other than red) corresponds to blue colour, we may use a blue glass between the object and the camera lens, and increase the difference of relative intensity in the blue region of the transmitted light [Figure 11.1(c)]. The blue glass allows the visible frequencies closer to the blue part of the *vibgyor* to go through, but suppresses those equal and near to red. Hence, the frequencies near and corresponding the red are filtered out, and we obtain a higher contrast between red, purple and orange flowers. It is not surprising that in photographic terminology, such coloured glasses are called **Optical Filters** or simply **Filters**.

As we know that the power or intensity of any type of energy communicated by sinusoidal motion of the medium may be expressed by the amplitude of the sinusoid, provided the transmission path resistance remains constant, the light energy transmission characteristic of a coloured glass at a particular frequency may be expressed by the amplitude ratio of the incident and transmitted sinusoids. Hence, a graph drawn for the full range of visible frequency as the abscissa and the corresponding amplitude ratio values plotted as ordinate may be called the **Frequency Response** of the optical filter [Figure 11.1(b)].

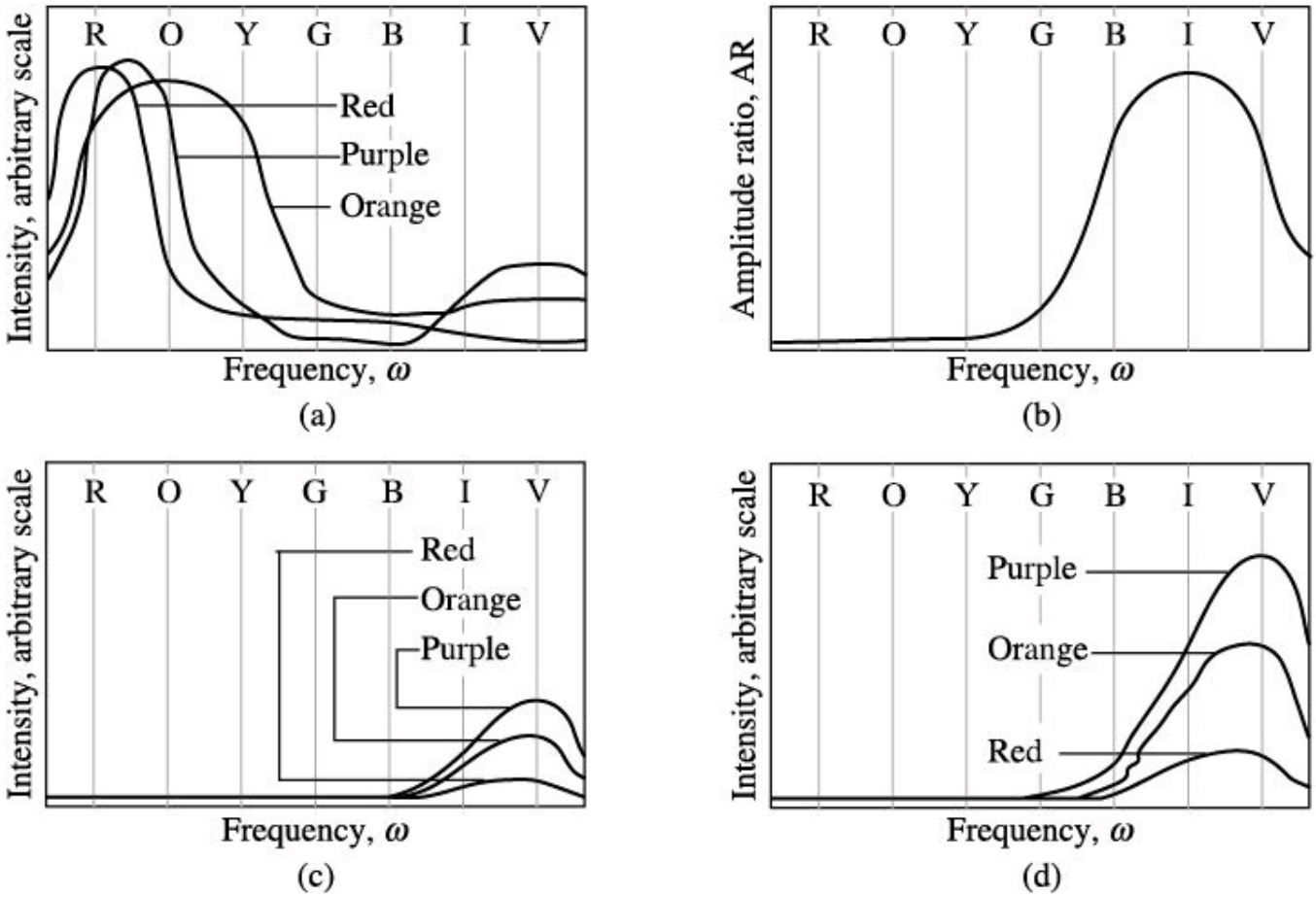


Figure 11.1 (a) The emission spectra of red, orange and purple flowers under sunlight. (b) The frequency response or transmittance of blue filter in the visible range. (c) Transmitted spectra of red, orange and purple flowers through blue filter before amplification. (d) Transmitted spectra of red, orange and purple flowers through blue filter after amplification.

It should be noted that any kind of filtering action is associated with loss of energy of the overall signal. We may have increased the contrast but reduced the overall light value by doing so. In photography the necessary amplification may be done by increasing the exposure value, either by using lower Shutter Speed, or by increasing the Aperture [Figure 11.1(d)]. Note that by this amplification, the contrast has further increased.

The optical filtering and amplification operation may be appreciated from Figure 11.1.

By the same line reasoning, a formation of flying cranes on the background of a cloudy sky if photographed through a light red filter, would produce a good contrast between sky, cloud and cranes.

11.1.2 Similar Frequency Domain Consideration about Sound Reproduction

Acoustic energy communication is also associated with sinusoidal motion of air particles within a vast range of frequency. Reproduction of audible sound from electronic recording devices occurs within a definite frequency range called Sonic range. Let us investigate a special sound reproduction problem. Suppose you have a recording of a solo presentation of the Bengali singing maestro Sri Hemanta Mukherjee. According to your opinion the supporting instrument artists did grievous injustice by being larger in volume than the singer's voice. You want to suppress the instrument sound and bring up the human voice. In Figure 11.2(a) the emission frequency ranges of common instruments and the baritone voice of Sri Mukherjee have been shown, just like the emission spectra of the flowers in Figure 11.1(a).

Modern sound reproduction equipment use three-speaker systems to cover the sonic frequency range. The largest diameter speaker called 'Woofer' can convert electronic signal to sound in the lowest frequency range, the medium diameter speaker called 'Normal' may do the conversion in the mid-frequency range and the smallest diameter speaker called 'Tweeter' operates in the highest frequency range of audible sound. So these speakers have a comparable frequency filtering capacity like coloured glass. The frequency response of these speakers as amplitude ratio against the sonic range of frequencies has been shown in Figure 11.2(b). May we call them audio-filters?

A solution of your problem may be that you switch off the woofer and tweeter of the audio system and run the recording only through the normal speaker. But this is not an adequate solution, as a part of the baritone voice would be lost due to the low

amplitude ratio response at the low and high frequency regions of the normal speaker [Figure 11.2(b)].

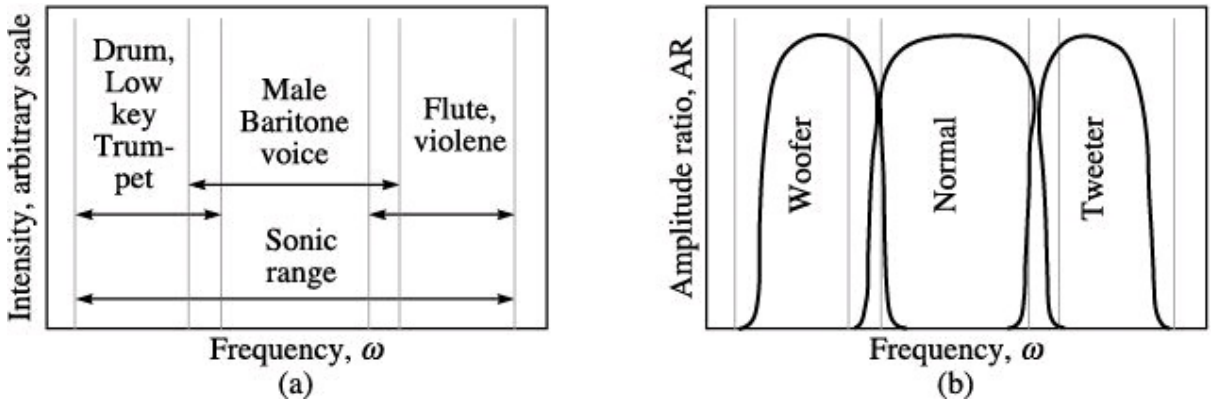


Figure 11.2 (a) The frequency ranges of instruments and male baritone voice. (b) The frequency response or transmittance of the three speakers.

It will be more elegant to address the problem by designing an electronic circuit that will filter out the sound of unwanted frequency range and allow the desired results. This may be a part of the electronic amplifier system so that the volume of the selected frequency audio output could also be adjusted. We may design such a filter, at least at the transfer function level. But before starting this interesting task we have to equip ourselves with a little bit of analytical techniques of the frequency domain.

11.2 Frequency Response of a Dynamic System

The stable part (i.e. after the transient part that is associated with the exponential –ve time function, $e^{-t/t}$, as in Equation 5.28, has decayed away) of the sinusoidal response of a process is called its **frequency response**. In such an expression, the independent variable is the angular frequency w rads/time, of the driving sinusoid because the stable part of response oscillates also with the same frequency = w. The frequency response information at a given w, has to contain:

1. The **Amplitude Ratio** (AR) of output to the input sinusoid, and
2. **Phase Difference** (□) between the output and input sinusoids in terms of angle.

Frequency Description or Frequency Response Description of a process is the AR and □ locus as a function of w within the given range of w. There are some alternative ways of expressing the same information. These will be revealed very shortly.

11.2.1 The Fundamental Substitution Theorem

This may be proved that in the transfer function $GP(s)$, of a process, by substituting iw in place of s the Laplace operator, the required frequency response may be obtained. Before trying to establish this, let us apply the theorem for a first order TF.

The first order TF: $G_P(s) = \dfrac{1}{1+\tau s}$ may be translated to the frequency domain as:

$$G_P(i\omega) = G_P(s)\,|_{S=I\omega} = \frac{1}{1+i\omega\tau} \qquad (11.1)$$

This is the fundamental frequency domain description of a first order TF. This may be rationalized into real and imaginary parts:

$$G_P(i\omega) = \frac{1-i\omega\tau}{1+\omega^2\tau^2} = \frac{1}{1+\omega^2\tau^2} - i\frac{\omega\tau}{1+\omega^2\tau^2} = \mathrm{Re}[G_P(i\omega)] + \mathrm{Im}[G_P(i\omega)] \qquad (11.2)$$

The real and imaginary components are just like the 'in-phase' and 'out-of-phase' currents in the vector diagram of an AC circuit. By analogy, the impressed cyclic voltage is the input sinusoid to the process. If we consider that input voltage to be the vector lying on the real axis and has an amplitude of A_I, the in-phase part of the resultant vector is the real component of output Re[$G_P(iw)$], the out-of-phase part of the resultant vector is the imaginary component of output Im[$G_P(iw)$], the vector magnitude or output amplitude, A_O, and phase angle, □, may be obtained by the following equations. If we assume that the input sinusoid amplitude is 1.0, then for a unity gain process the output amplitude is also the amplitude ratio, AR, thus,

$$A_O = AR = \sqrt{\left(\frac{1}{1+\omega^2\tau^2}\right)^2 + \left(\frac{\omega\tau}{1+\omega^2\tau^2}\right)^2} = \frac{1}{\sqrt{1+\omega^2\tau^2}} \qquad (11.3)$$

and

$$\Phi = \tan^{-1}\left(-\frac{\omega\tau/(1+\omega^2\tau^2)}{1/(1+\omega^2\tau^2)}\right) = -\tan^{-1}(\omega\tau) \qquad (11.4)$$

Another way of addressing $G_P(iw)$ output amplitude and phase is by expressing it as a resultant vector of magnitude, M, and argument □, so that,

$$G_P(i\omega) = |M|\angle\Phi \qquad (11.5)$$

By this definition the magnitude M becomes equal to AR, and argument is the phase angle. The analysis has been shown as a vector diagram in Figure 11.3. The above Equations 11.3 and 11.4 are similar to the Equations 5.31 and 5.34, we obtained as sinusoidal response of first, order systems.

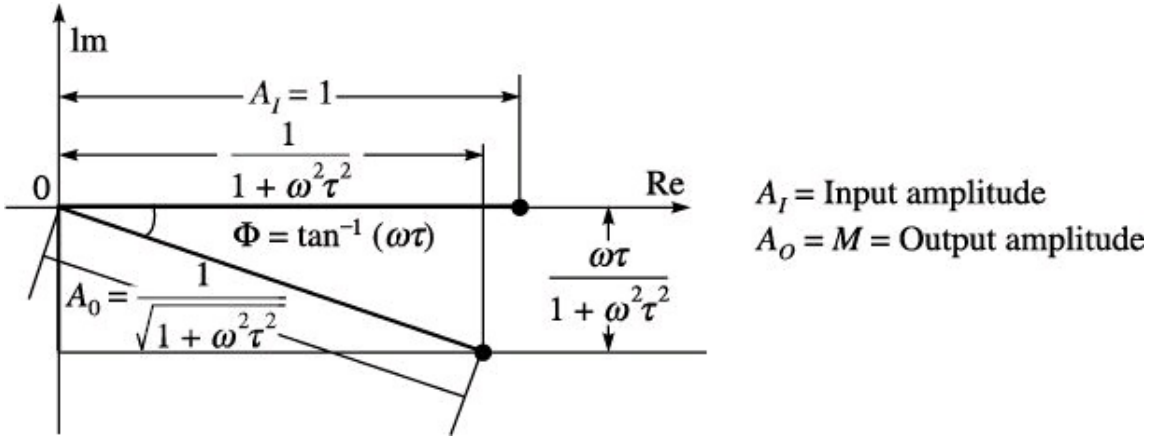


Figure 11.3 The vector representation of the input and response sinusoids on a complex plane.

This extremely useful tool of frequency response determination could be analytically proved as a complex variable procedure. We shall also do so. But before that, an explanation based on our past experience in process dynamics should be more interesting.

11.2.2 Dynamics in Complex s-Plane

A process could be identified by the location or position of the roots of its characteristic equation on complex *s*-plane. The root location describes the undisturbed dynamic character of the process. To make the following analysis simpler, let us consider a first order process.

When such a process is disturbed by say a step input, it is equivalent of adding an *s* in the denominator of TF. On the *s*-plane, the disturbance would be positioned at $s = 0$, the origin. In the mean time, in time domain the response, after showing an initial transient, which becomes less than 1% of its value after only five times of the process time constant, ultimately settles to a straight line parallel to the time axis, just like the depiction of a step function. Off course, the magnitude of this response step would depend on magnitude of the input step and gain of the process. As if in complex plane, the characteristic root representing the undisturbed process situated at a distance $-1/t_P$ from the origin, is dragged under influence of the disturbance toward the location where the disturbance is situated ($s = 0$) and within a short time, the moving root superposes on the input function, on complex domain.

The same rationale could be also true for ramp input, where the first order process response also becomes ramp like (linear function of time) after the initial transients are over, and the characteristic root of the process is dragged down to the input location at the *s*-plane origin. This time for the ramp there is a second degree *s* at the origin. Now let us try the pulse input. Remembering that this input is of a temporary category, we can say that the movement of the characteristic root towards origin continued until the +ve step part of the input was over. After which, under the influence of the –ve step of equal magnitude, the root again moves away to its original

undisturbed location at $-1/tP$ on the s-plane.

The case of impulse input is interesting, because it is a one-time function that has happened once at a particular value of time (for an infinitesimally small time instant). The process has been definitely knocked away from its location on s-plane (i.e. the roots) by the impact of input. But, as the input no longer exits at any +ve values of time, the process would gradually return to its steady state, and the roots eventually return to their original location on s-plane.

Consideration for sinusoidal input should first start by defining the imaginary axis in its polar form, i.e. from origin, the axes are expressed by $s = \pm iw$, so that the location of input function is on the imaginary axis at two points equidistant from origin, because the transformed input may be expressed as:

$$X(s) = \frac{M\omega}{s^2 + \omega^2} = \frac{M\omega}{(s + i\omega)(s - i\omega)} \tag{11.6}$$

The disturbed process after showing transient behaviour would be similarly dragged on the imaginary axis at the locations of the input function, to show its stable part of response. This new location of characteristic equation roots may be obtained by putting $s = \pm iw$ in the TF of the process. In fact, the stable part of the sinusoidal response, more formally known as the Frequency Response of the process, may be obtained by treating the TF in which the substitution $s = \pm iw$ has been made. The advantage of the new definition of the imaginary axis is that we may vary the value of w to obtain the complete frequency response of the process over a desired range of frequency.

11.2.3 Analytical Proof

The TF of a linear and stable system may be given as:

$$G_P(s) = \frac{Y(s)}{X(s)} = \frac{Z(s)}{P(s)}$$

where, the order, m, of the numerator polynomial, $Z(s)$, must be less than n the denominator polynomial $P(s)$. For a sinusoidal input:

$X(t) = M \sin wt$, that has the Laplace transform $X(s) = \dfrac{M\omega}{s^2 + \omega^2}$

The response would have two parts:

1. Due to process character called, Complementary Function,
2. Due to the response character, called Particular Integral.

The total solution may be written in terms of partial fraction as:

$$Y(s) = \left[\frac{k_1}{s-p_1} + \frac{k_2}{s-p_2} + \cdots + \frac{k_n}{s-p_n}\right] + \frac{c_1}{s+i\omega} + \frac{c_2}{s-i\omega}$$

(1) Complementary Function (2) Particular Integral

In time domain, Part (1) would produce exponential terms:

$$e^{p_1 t},\ e^{p_2 t},\ \ldots,\ e^{p_n t}$$

As we are concerned about stable systems, the poles will contain –ve real parts and hence the exponential functions would vanish as time progresses. Therefore, the stable part of the response due to part (2) would take the following form:

$$Y(t) = k_1 e^{-i\omega t} + k_2 e^{i\omega t}$$

The constants k_1 and k_2 may be found by the usual procedure of evaluating partial fraction constants as:

$$k_1 = \frac{[MG_P(-i\omega)]}{2i} \quad \text{and} \quad k_2 = \frac{[MG_P(i\omega)]}{2i}$$

Thus, the time domain response comes as:

$$Y(t) = \frac{MG_P(-i\omega)}{2i} e^{-i\omega t} + \frac{MG_P(i\omega)}{2i} e^{i\omega t}$$

By using the property of complex algebra, the complex variables $G_P(iw)$ and $G_P(-iw)$ may be expressed in polar form as:

$$G_P(i\omega) = |G_P(-i\omega)|e^{-i\phi} = |G_P(i\omega)|e^{-i\phi} \quad \text{and} \quad G_P(i\omega) = |G_P(i\omega)|e^{i\phi}$$

where, □ is the phase angle or argument of $G_P(iw)$. Substituting the above results in the last response equation:

$$Y(t) = \frac{M\,|G_P(i\omega)|}{2i} e^{-i(\omega t+\phi)} + \frac{M\,|G_P(i\omega)|}{2i} e^{i(\omega t+\phi)}$$

$$= M\,|G_P(i\omega)|\,\frac{e^{i(\omega t+\phi)} - e^{-i(\omega t+\phi)}}{2i}$$

Expressing the result in trigonometric form:

$$Y(t) = M\,|G_P(i\omega)|\,\sin(\omega t + \phi)$$

Thus it has been proved that:

1. For a sinusoidal input the stable part of the response is also sinusoidal.
2. The ratio of output to input amplitude may be expressed by putting $s = iw$ in

the TF of the system as the modulus of $GP(iw)$, and hence is a function of the input sinusoid of frequency, w.

3. The response sinusoid is shifted by a phase angle □ from the input sinusoid. □ is also a function of the input sinusoid frequency, w.

11.2.4 Different Descriptions of Frequency Response

By replacing s by iw in the TF of a process, we obtain the fundamental expression of its frequency response. If required, this expression is treated to obtain in a form having separate real and imaginary parts.

$$G_P(i\omega) = \mathrm{Re}[G_P(i\omega)] + \mathrm{Im}[G_P(i\omega)] \tag{11.6}$$

Following the analysis that we made through Equation 11.2 through 11.4, the amplitude ratio, AR, and phase angle, □, may be obtained by Equations 11.7 and 11.8.

$$AR = \sqrt{[\mathrm{Re}\{G_P(i\omega)\}]^2 + [\mathrm{Im}\{G_P(i\omega)\}]^2} \tag{11.7}$$

and

$$\Phi = \tan^{-1}\left[\frac{\mathrm{Im}\{G_P(i\omega)\}}{\mathrm{Re}\{G_P(i\omega)\}}\right] \tag{11.8}$$

This has been already done in case of first order system. Using the above equation, the frequency description of a process is expressed in three alternative ways:

(1) Bode Plot: This is a two graph representation of the frequency response data. One is w vs. AR on a log-log graph, and the other is the w vs. □ on a semi-log graph. In both graphs, w is plotted in a log scale abscissa. In the later period, another term Decibel (dB), coined from electronic amplifier practice, is used to express amplitude ratio.

$$\mathrm{dB} = 20 \log_{10}(AR) \quad (11.9)$$

as dB is logarithm of AR, hence w vs. dB is plotted on a semi-log graph. Hence, the complete Bode Plot may be drawn on a single semi-log paper with two ordinates on the left and right side of the graphs marked in units of dB and □ respectively, and a common log scale abscissa for w. Such a single plot expression has some advantage during post operations with the graph. Bode Plot renders the way toward frequency domain response and stability analysis.

(2) Nyquist Plot: In frequency domain, the response may be also expressed in polar (vector) form. The response vector is actually the $GP(iw)$ that is expressed in polar form corresponding to the input sinusoid with an angular frequency w:

$$G_P(i\omega) \equiv re^{i\phi} = r\cos\phi - ir\sin\phi \tag{11.10}$$

where, the vector magnitude, *r*, corresponding to a unit amplitude of input sinusoid is actually the amplitude ratio, AR, and □ is the phase angle. Hence, the above relation could be extended as:

$$G_P(i\omega) \equiv |AR| \angle\Phi = |AR|\cos\Phi - i|AR|\sin\Phi \tag{11.11}$$

Nyquist Plot to express the frequency response of a process is actually the locus of the end of the response vector as a function of frequency on the complex plane. This could be achieved by first finding the complex form of $GP(iw)$ containing 'real' and 'imaginary' parts as functions of *w*. For a given frequency, the coordinates of the vector head is obtained from real part of $GP(iw)$ as the *abscissa*, and imaginary part of $GP(iw)$ as the *ordinate*. The points are plotted on the complex plane for a desired frequency range. The chosen frequency values within the range should be close enough so that a smooth locus may be drawn through them to have good description of the system. Nyquist Plot is also called the polar plot of a process.

In certain problems, the AR and □ may be directly obtained from already developed relations for drawing the Bode plot. From these values the corresponding real and imaginary parts may be evaluated by using the above considerations as:

$$\text{Real Part} = \text{AR}\cos\Phi \tag{11.12a}$$

and

$$\text{Imaginary Part} = \text{AR}\sin\Phi \tag{11.12b}$$

The plot may be obtained for $w = 0$ to $+\infty$, and is actually half of the complete Nyquist diagram. The other half of the plot for $w = 0$ to $-\infty$ is, in fact, the mirror image of the first half. The frequency becomes a parameter along the curve in a Nyquist plot. This graph constitutes a convenient way of stability check of closed loop system in frequency domain.

(3) Nichol's Plot: This is a linear plot of dB vs. □. The □ is plotted on the abscissa and dB on the ordinate. Again the frequency becomes a parameter along the curve in such a plot. Nichol's Plot provides a method for analysis and design of closed loop response in frequency domain.

In the following section, examples of conversion of simple TFs into their frequency response plots have been presented. This would demonstrate certain frequency response properties of TF units and provide the basis for handling of more complex systems with multiple poles, zeros, dead time, integrators, etc.

EXAMPLE 11.1 First Order Element: The frequency response equations are

already derived. These are Equations 11.3 and 11.4. We shall use them to draw the different alternative descriptions of frequency response data. For the unity gain system putting $w = 0$, the AR = 1, or dB = 0, but for higher frequencies the AR and dB values have no lower limit.

For drawing the Bode Plot of a first order system with a gain of K_P, we will use the following relations:

$$AR = \frac{K_P}{\sqrt{1+\tau_P^2\omega^2}}\ ;\ \text{dB} = 20\log_{10}\left[\frac{K_P}{\sqrt{1+\tau_P^2\omega^2}}\right] \text{ and } \square = \tan^{-1}(-wt_P)$$

Putting $w = 0$ in the above equations we can check that at low frequencies the dB and □ locus starts from dB = $20\log_{10}(K_P)$, and □ = 0°. Similarly putting $w = \infty$ in the above equation, it can be confirmed that dB locus has a value of $-\infty$, but □ locus will converge to a value of –90°.

The character of dB and □ locus changes before and after the frequency value $w = 1/t_P$, or, in the reduced frequency scale $wt_P = 1$. This frequency value is called **natural** or **characteristic frequency** (w_N) of the system. w_N is also equal to the reciprocal of the time constant of the system. Both the locus goes downward, incase of dB locus, the –ve slope sharply changes after this point in frequency value. The curve in the low and high frequency regions relative to
$w = 1/t_P$, behaves as asymptotes to two straight lines:

1. *In the low frequency region ($w < 1/t_P$), it is the horizontal straight line with a decibel value of,*

$$\text{dB} = 20\log_{10}\left[\frac{K_P}{\sqrt{1+\omega^2\tau^2}}\right]$$

called the **low frequency asymptote (LFA)**, starting from $w = 1/t_P$

. In the high frequency region ($w > 1/t_P$), it is the straight line starting from
$w = 1/t_P$, goes down with –ve slope of –20dB/decade or one cycle. This is called the **high frequency asymptote (HFA).**

Whereas, the □ locus also goes down as w increases, with increasing –ve slope through the low frequency region and has maximum value of –ve slope at $w = 1/t_P$, after which the slope gradually reduces. In fact, in low and high frequency regions, the □ locus also behaves as two asymptotes to two straight lines; 0° and –90° horizontals.

The high and low frequency asymptotes of dB locus meet at $w = 1/t_P$, to make a

corner on the dB plot, hence this frequency is also called as the **Corner Frequency** (w_C). Thus,

$w = 1/tP = w_C = w_N$. At $w = w_N$, the AR, dB and □ values for a first order element may be found by using the above equations as:

$$AR = \frac{K_P}{\sqrt{2}},\ \text{dB} = 20\log_{10}\left[\frac{K_P}{\sqrt{2}}\right] \text{ and } \square = \tan^{-1}[-1] = -45°$$

The Bode Plot for three gain values have been drawn on a reduced frequency scale (wt) in Figure 11.4, in which the above results could be confirmed. The value of the three unknown values of gain could be found by already given equations:

Thus, K_{P1} = Anti $\log_{10}[10/20] = 3.1623$, and complete TF is $G_p(s) = \dfrac{3.1623}{\tau_p s + 1}$

and, K_{P2} = Anti $\log_{10}[1/20] = 1$, and the complete TF is $G_p(s) = \dfrac{1}{\tau_p s + 1}$

and, K_{P3} = Anti $\log_{10}[-10/20] = 0.3162$, and the complete TF is $G_p(s) = \dfrac{0.31623}{\tau_p s + 1}$

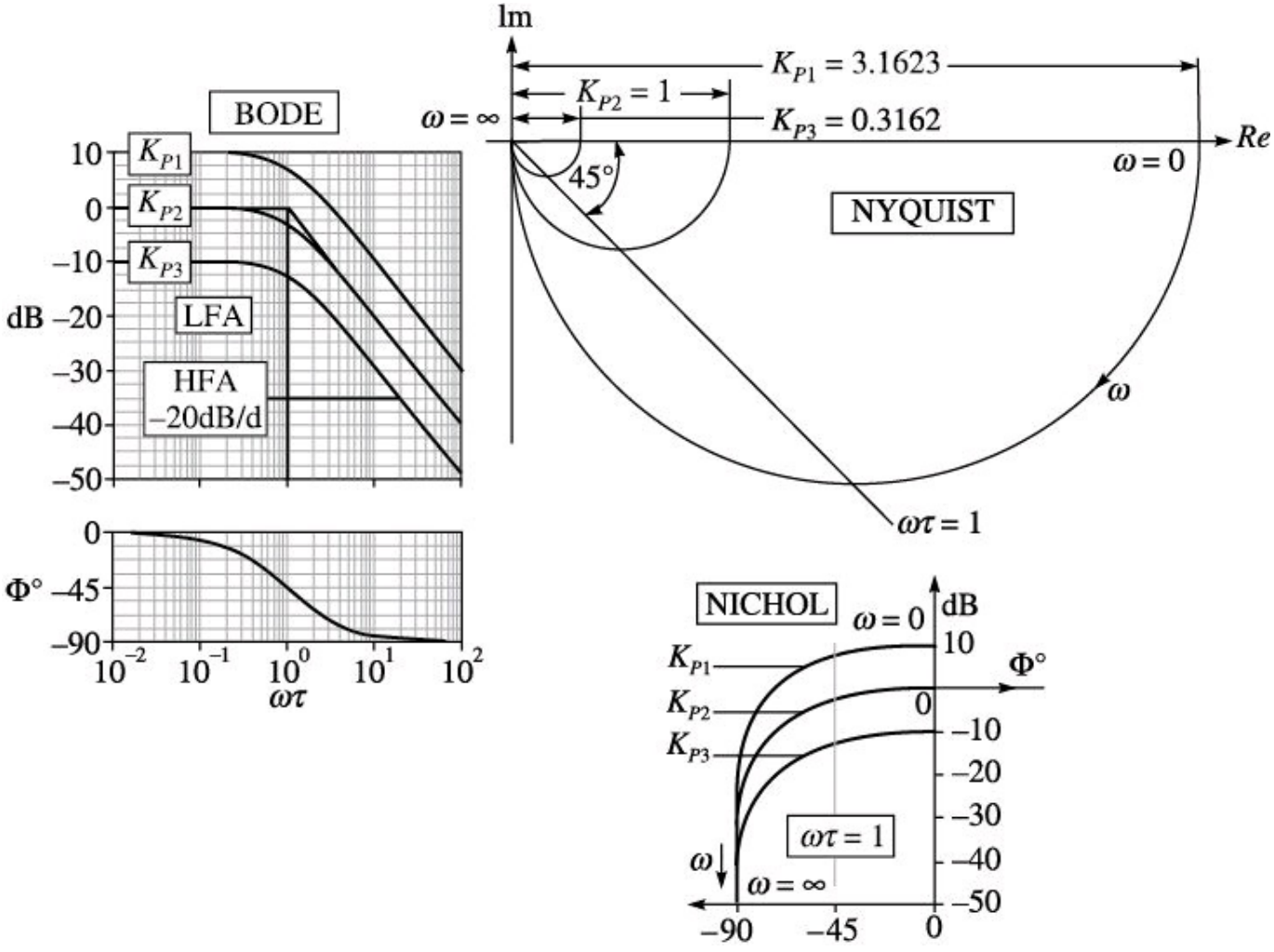


Figure 11.4 Different descriptions of the frequency response of a first order system.

It seems that the gain only lifts or depresses the dB curve at all frequencies than the unity gain system by an amount of 20 $\log_{10}[K_P]$.

The Nyquist Plot for a first order element, with gain value, may be drawn corresponding to a unit input vector lying on the +ve real axis either by:

1. Finding the vector magnitude (AR) and its angle □ by using the relations:

$$AR = \frac{K_P}{\sqrt{1+\omega^2\tau^2}} \text{ and } \square = -\tan^{-1}(wt)$$

or,

2. Determining the coordinates of the vector head (x, y), which may be found by using Equations 11.12a and 11.12b as:

$$x = AR \cos \square \text{ and } y = AR \sin \square$$

The plots starts from the +ve real axis at a distance K_P from the origin, because the output vector has a magnitude K_P and makes angle 0° with the real axis at $w = 0$. As w increases, the vector head will describe a semi-circle with decrease of □ and vector magnitude until at high frequencies the magnitude limitingly approaches zero value and □ approaches –90°. The point where $w = 1/t_P$, lies on the locus where the magnitude is $K_P/(\sqrt{2})$ and □ = 45°. Hence, for three such systems with different K_P values the three locus starts from three starting points, but all converge into the origin of the complex plane.

The Nichol's Plot drawn for the same systems should be understandable from the discussions of Bode and Nyquist Plots where the dB and □ are plotted on a linear grid with dB on y and □ on x-axis. On the plot the □ = 45° vertical is shown to mark the dB values for different system gains.

EXAMPLE 11.2 Inherently Second Order Element (Complex Conjugate Poles/Roots): The TF of such a unity gain second order system in terms of its natural frequency is:

$$G_P(s) = \frac{\omega_N^2}{s^2 + 2\zeta\omega_N s + \omega_N^2}$$

by substituting iw for s in the TF, hence, the $G_P(iw)$ may be found as:

$$G_P(i\omega) = \frac{\omega_N^2}{-\omega^2 + 2i\zeta\omega_N\omega + \omega_N^2} = \frac{\omega_N^2[(\omega_N^2 - \omega^2) - i2\zeta\omega_N\omega]}{[(\omega_N^2 - \omega^2) + i2\zeta\omega_N\omega][(\omega_N^2 - \omega^2) - i2\zeta\omega_N\omega]}$$

$$= \underbrace{\frac{\omega_N^2(\omega_N^2 - \omega^2)}{(\omega_N^2 - \omega^2)^2 + (2\zeta\omega_N\omega)^2}}_{\mathrm{Re}[G_P(i\omega)]} - i\underbrace{\frac{2\zeta\omega_N^3\omega}{(\omega_N^2 - \omega^2)^2 + (2\zeta\omega_N\omega)^2}}_{\mathrm{Im}[G_P(i\omega)]}$$

$$= \mathrm{Re}\,[G_P(i\omega)] + \mathrm{Im}\,[G_P(i\omega)] \tag{11.13}$$

From Equation 11.13 expressions for AR, dB and Φ may be found as:

$$AR = \frac{\omega_N^2}{\sqrt{(\omega_N^2 - \omega^2)^2 + (2\zeta\omega_N\omega)^2}} = \frac{1}{\sqrt{(1 - \omega^2\tau^2)^2 + (2\zeta\omega\tau)^2}} \tag{11.14}$$

$$\mathrm{dB} = 20\log_{10}(AR) = 20\log_{10}\left[\frac{\omega_N^2}{\sqrt{(\omega_N^2 - \omega^2)^2 + (2\zeta\omega_N\omega)^2}}\right] \tag{11.15a}$$

or,

$$\mathrm{dB} = 20\log_{10}(AR) = 20\log_{10}\left[\frac{1}{\sqrt{(1 - \omega^2\tau^2)^2 + (2\zeta\omega\tau)^2}}\right] \tag{11.15b}$$

and

$$\Phi = \tan^{-1}\left(-\frac{2\zeta\omega_N\omega}{\omega_N^2 - \omega^2}\right) = \tan^{-1}\left(-\frac{2\zeta\omega\tau}{1 - \omega^2\tau^2}\right) \tag{11.16}$$

In the frequency response plots to be drawn by using Equations 11.13 through 11.16, the damping coefficient, *z*, will be treated as a parameter. The Bode Plots show similar low and high frequency asymptotic character for all damping conditions and the HFA dips with a slope of –40 dB/decade or one cycle. Except for under damped conditions, the dB plot shows 'humps' over the $w = 1/t$. Again the natural or characteristic frequency and corner frequency of the system are all equal to $1/t$. The humps over $wt = 1$, are the manifestation of overshoot that increases with decreasing value of *z*, as we observed in time responses. The □ locus shows a similar low frequency asymptote as in a first order system but for a second order system, the

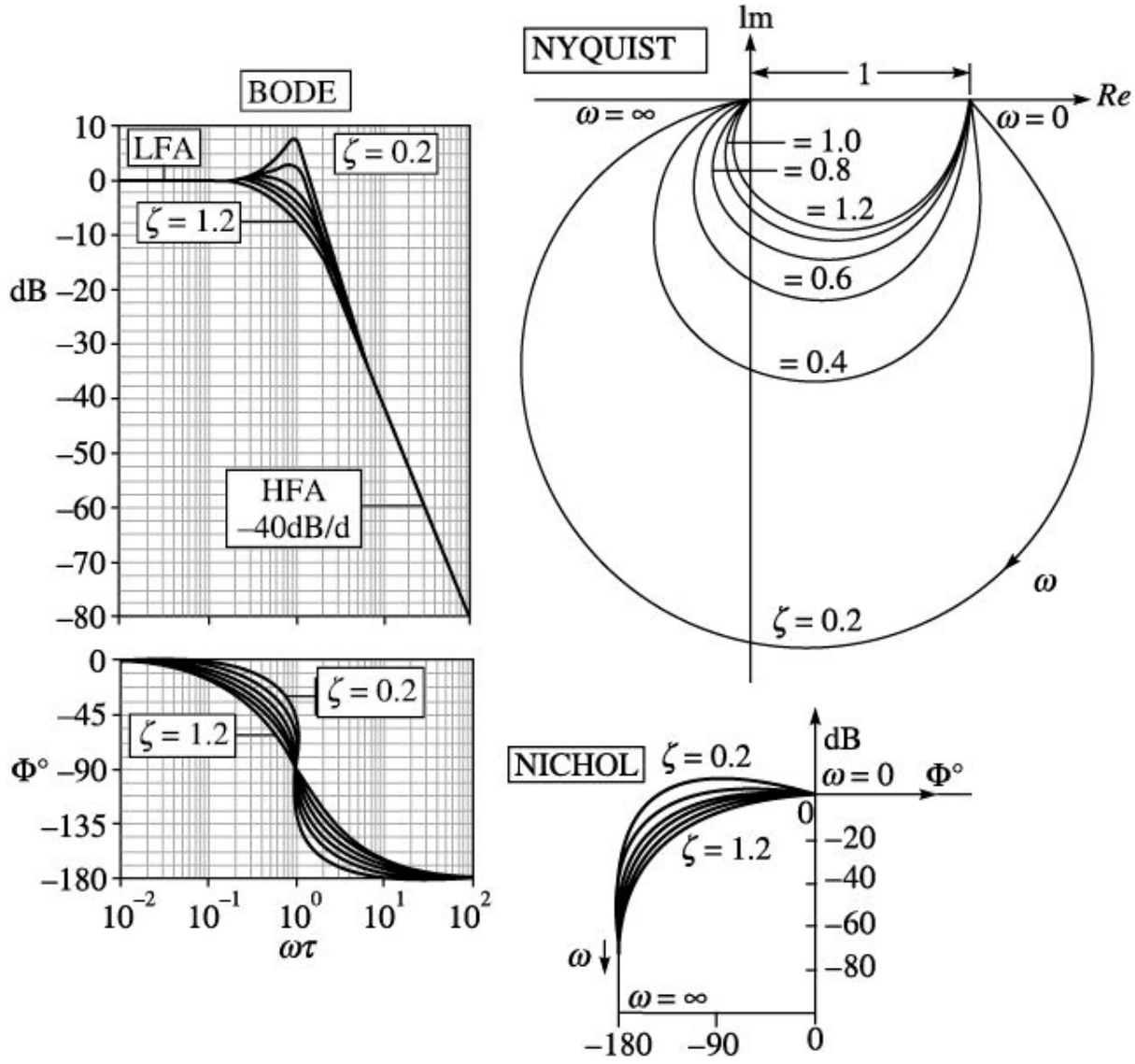


Figure 11.5 Different descriptions of the frequency response of an inherent second order system (complex conjugate poles).

high frequency locus is an asymptote to the □ = –180° horizontal. The maximum –ve slope of phase locus occurs at *wt* = 1, for □ = –90°. (It should be noted that as *wt* increases above 1.0, tan □ starts to approach zero but always remains +ve. This means that □ varies from –90 to –180 as *wt* goes from 1 to ∞.) At *w* = 1/*t*, the AR and dB values may be obtained from Equation 11.14. The value will depend on the damping condition of the system unlike that we had for a first order system.

The plots will depict the locus of the vector head starting from the point on the real axis

at unit distance from the origin. As *w* increases, all the locus crosses the imaginary axis for □ = –90°, irrespective of the value of *z*. The vector starts along the +ve real axis measuring a unit length from the origin for a system with unity gain, and gradually rotates as □ decreases for higher values of frequency with decreasing magnitude, ultimately for high values frequency, limitingly the magnitude, AR, approaches zero value and phase angle, □, approaches –180°.

The Nichol's Plot is drawn for all values damping coefficient (*z* = 0.2, 0.4, 0.6, 0.8,

1.0 and 1.2) by using Equations 11.15 and 11.16. The curves show the bulge above 0 dB horizontal for
$z < 1$, signifying the overshoot. Also, every locus converge on vertical line on which □ is –180°.

EXAMPLE 11.3 Dead Time Element:

$G_P(s) = e^{-\tau_{DL}s}$; by substituting *iw* for *s* in the TF,

$$G_P(i\omega) = e^{-i\omega\tau_{DL}} = \underset{\mathrm{Re}[G_P(i\omega)]}{\cos\omega\tau_{DL}} - \underset{\mathrm{Im}[G_P(i\omega)]}{i\sin\omega\tau_{DL}}$$

$$\therefore \quad AR = \sqrt{\cos^2\omega\tau_{DL} + \sin^2\omega\tau_{DL}} = 1 \tag{11.17}$$

$$\Phi = \frac{\tan^{-1}(-\sin\omega\tau_{DL})}{(\cos\omega\tau_{DL})} = -\omega\tau_{DL} \tag{11.18}$$

Equation (11.17) shows that AR is always 1.0 and dB always 0 for all values of frequency. However, the phase angle □ will linearly decrease as frequency is increased. Thus, in Bode Plot the dB is a straight line on the 0-dB horizontal and phase angle demonstrates an endlessly dipping locus starting from 0° with increasing *w*. If a system or a part thereof is a dead time element, the phase angle would be continuously decreasing with increasing frequency without converging to a finite minimum (as it happens for single or multiple poles). Such systems are termed as '**non-minimum phase systems**'.

As the response vector has a unit length independent of the frequency value, the Nyquist plot becomes a circle with unit radius, i.e. the vector rotates with undiminished magnitude as the phase lag increases with increasing frequency. The frequency value increases by $p/2t_{DL}$ as the vector passes through each quadrant of the complex plane.

The Nichol's plot is a straight line on the 0-dB horizontal extending from origin to –∞ because dB is always zero and □ is always decreasing with increasing frequency.

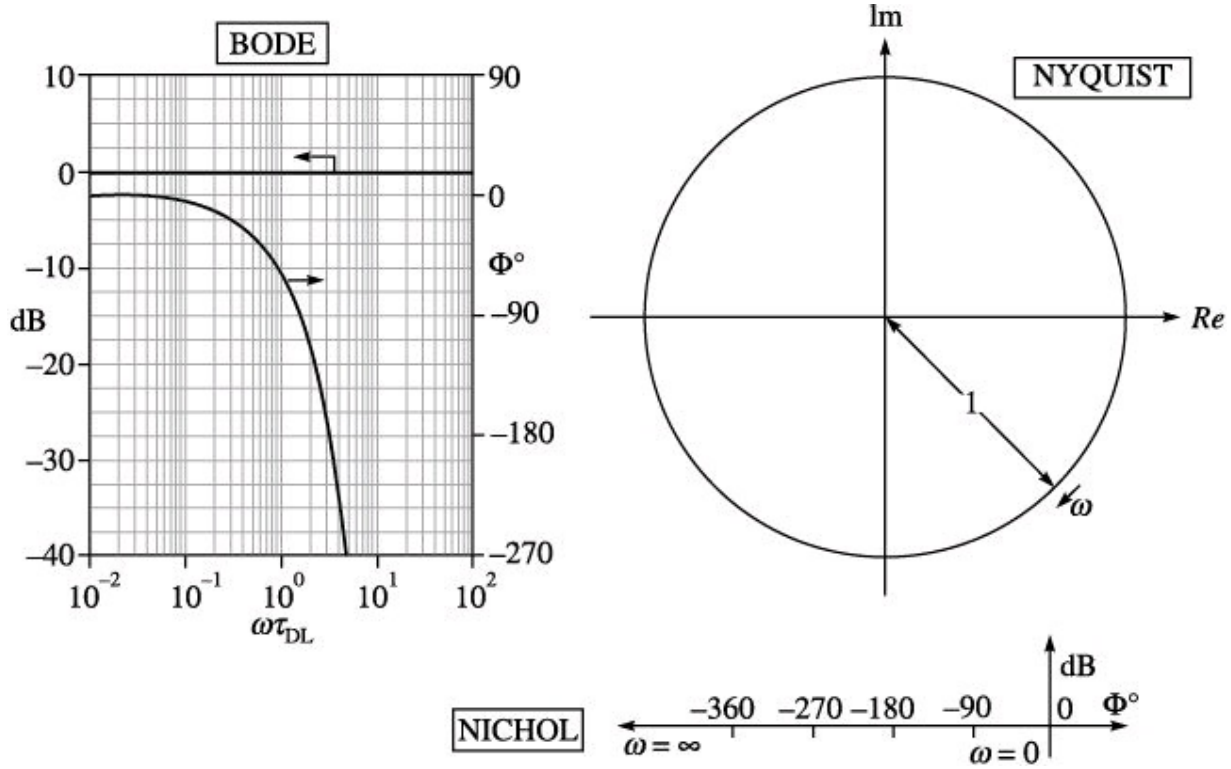


Figure 11.6 Different descriptions of the frequency response of a dead time element.

11.2.5 Frequency Response of Block Elements in Series

Let a process contains two dynamic elements in series, so that its TF may be written as:

$$G_P(s) = G_{P_1}(s)G_{P_2}(s)$$

By substituting *iw* for *s* in the TF of the individual elements and writing them in vector form,

$$G_{P_1}(i\omega) \equiv r_1 e^{i\phi_1} = |AR_1| \angle \phi_1 \quad \text{and} \quad G_{P_2}(i\omega) \equiv r_2 e^{i\phi_2} = |AR_2| \angle \phi_2$$

Applying the rule of vector multiplication:

$$G_{P_1}(i\omega)G_{P_2}(i\omega) = r_1 r_2 e^{i(\phi_1+\phi_2)} = |AR_1| \cdot |AR_2| \angle(\phi_1 + \phi_2) = |AR| \angle \phi$$

where, $|AR| = |AR_1| \cdot |AR_2|$ and $\angle\phi = \angle(\phi_1 + \phi_2)$

In terms of dB the above expression becomes:

$$\text{dB} = 20 \log_{10}(AR) = 20 \log_{10}(AR_1) + 20 \log_{10}(AR_2) = \text{dB}_1 + \text{dB}_2 \qquad (11.19)$$

and,
$$\Phi = \phi_1 + \phi_2 \qquad (11.20)$$

Thus, the general rule for compound TFs has been established as:

$$G_P(i\omega) = G_{P_1}(i\omega)G_{P_2}(i\omega)$$

If we use the values of AR and □ for constructing the Bode Plot one 'Log-Log' graph for AR vs. *w* and a 'Semi-Log' graph for □ vs. *w* has to be used, but if *AR* is expressed as dB, a single 'Semi-Log' graph may contain both of the plots. This single graph representation is convenient for dynamic and stability analysis in frequency

domain. In this text we shall use this single Semi-Log graph to draw Bode Plot.

This approach may be extended for constructing the Bode Plot for processes with more than one elements in its TF. Instead of evaluating the overall dB for the $GP(iw)$ dB values for individual units may be found and graphically added to evaluate the total dB with less computational effort.

EXAMPLE 11.4 Two first Order Elements in Series:

$$G_P(s) = \frac{1}{(\tau_{P1}s + 1)(\tau_{P2}s + 1)}$$

$GP(s)$ may be thought as comprising of two first order blocks in series and their frequency response description is already known. Applying the above approach the frequency response of $GP(s)$ may be written as:

$$G_P(i\omega) = \frac{1}{(1-\omega^2\tau_{P1}\tau_{P2}) + i\omega(\tau_{P1}+\tau_{P2})} = \frac{(1-\omega^2\tau_{P1}\tau_{P2}) - i\omega(\tau_{P1}+\tau_{P2})}{(1-\omega^2\tau_{P1}\tau_{P2})^2 + \omega^2(\tau_{P1}+\tau_{P2})^2}$$

$$= \underbrace{\frac{(1-\omega^2\tau_{P1}\tau_{P2})}{(1-\omega^2\tau_{P1}\tau_{P2})^2 + \omega^2(\tau_{P1}+\tau_{P2})^2}}_{\mathrm{Re}[G_P(i\omega)]} - i\underbrace{\frac{\omega(\tau_{P1}+\tau_{P2})}{(1-\omega^2\tau_{P1}\tau_{P2})^2 + \omega^2(\tau_{P1}+\tau_{P2})^2}}_{\mathrm{Im}[G_P(i\omega)]}$$

The expression for AR and □ may be derived from the above equation or directly written as the product of *AR*s or sum of dBs and □s for two first order elements:

$$|AR| = \frac{1}{\sqrt{1+\omega^2\tau_{P1}^2}} \cdot \frac{1}{\sqrt{1+\omega^2\tau_{P2}^2}} = \frac{1}{\sqrt{(1+\omega^2\tau_{P1}^2)(1+\omega^2\tau_{P2}^2)}} \tag{11.21}$$

$$\mathrm{dB} = \mathrm{dB}_1 + \mathrm{dB}_2 = 20\log_{10}\left(\frac{1}{\sqrt{1+\omega^2\tau_{P1}^2}}\right) + 20\log_{10}\left(\frac{1}{\sqrt{1+\omega^2\tau_{P2}^2}}\right) \tag{11.22}$$

and,

$$\Phi = \phi_1 + \phi_2 = -\tan^{-1}(\omega\tau_{P1}) - \tan^{-1}(\omega\tau_{P2}) \tag{11.23}$$

To identify two HFA in the Bode plot and an undulation in Nichol's plot, as in Figure 11.7, it is required that the two characteristic frequencies ($1/tP1$ and $1/tP2$) should be far apart on the frequency scale. The Nyquist plot has only small bulge in the fourth quadrant because systems of this category can only posses an over- or critically damped character.

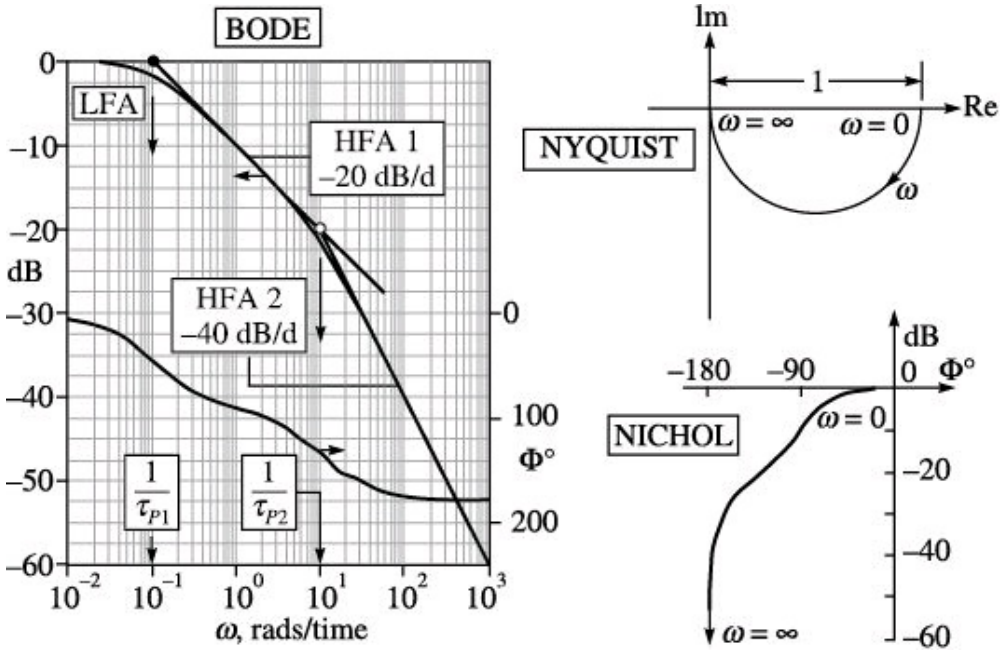


Figure 11.7 Different descriptions of the frequency response of a second order system. (two distinct poles)

EXAMPLE 11.5 A First Order plus Dead Time Element: The TF of this element may be splitted as:

$$G_P(s) = \frac{K_P e^{-\tau_{DL}s}}{1+\tau_P s} = G_{P_1}(s)G_{P_2}(s)G_{P_3}(s) = K_P \frac{1}{1+\tau_P s} e^{-\tau_{DL}s}$$

The three TFs in frequency domain would yield the following AR, dB and □ expressions:

$$AR_1 = K_P;\ \mathrm{dB}_1 = 20\log_{10} K_P;\ f_1 = 0;$$

$$AR_2 = \frac{1}{\sqrt{1+\omega^2\tau_P^2}}\quad \mathrm{dB}_2 = 20\log_{10}\frac{1}{\sqrt{1+\omega^2\tau_P^2}};\quad \phi_2 = -\tan^{-1}(\omega\tau_P)\ ;\ \square$$

$$AR_3 = 1;\ \mathrm{dB}_3 = 0;\ f_3 = -wt_{DL}$$

The total AR, dB and □ may be obtained by multiplying the ARs and adding the dB and □ of individual elements:

$$AR = AR_1 AR_2 AR_3 \tag{11.24}$$

$$\mathrm{dB} = \mathrm{dB}_1 + \mathrm{dB}_2 + \mathrm{dB}_3 \tag{11.25}$$

$$\Phi = \phi_1 + \phi_2 + \phi_3 \tag{11.26}$$

The real and imaginary parts of $G_P(iw)$ for Nyquist Plot is obtained as:

$$\mathrm{Re}[G_P(iw)] = AR\cos\square \text{ and } \mathrm{Im}[G_P(iw)] = AR\sin\square$$

Figure 11.8 shows the three kinds of frequency responses for this system. In the Bode Plot, the frequency scale is marked with $1/t_P$ and $1/t_{DL}$. The delay element does not contribute

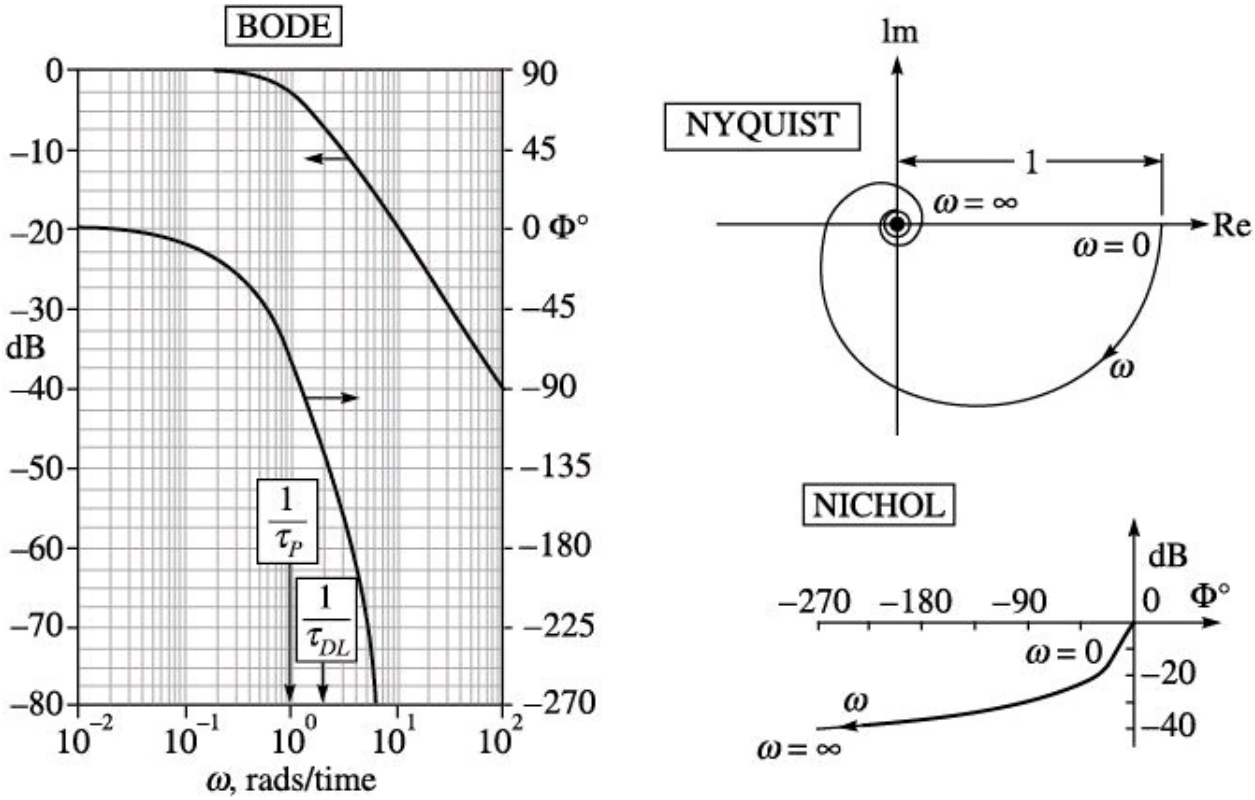


Figure 11.8 Different descriptions of the frequency response of a first order + a dead time element.

to the dB curve, hence it is possible to mark $1/t_P$, by joining the LFA and HFA at the corner frequency. However, to mark $1/t_{DL}$, it is required to use the phase property. At $w = 1/t_{DL}$ the total phase angle is:

$$\Phi = \phi_2 + \phi_3 = \tan^{-1}\left(\frac{\tau_P}{\tau_{DL}}\right) + \frac{\tau_P}{\tau_{DL}}$$

Thus, unknown parameters of such a system could be estimated from its frequency response.

The spiral shape of Nyquist Plot appears as the response vector looses its magnitude under influence of the first order pole. The Nichol's Plot also shows the influence of delay element on the first order pole, as the locus moves much faster to the left than it dips down.

EXAMPLE 11.6 The Integrator and Differentiator: An integrator, $1/s$, or a differentiator, s, appearing in the TF may be separately analyzed in the following way:

$$G_P(s) = \frac{1}{s}; \quad \therefore \; G_P(i\omega) = \frac{1}{i\omega} = \frac{1}{i\omega}\cdot\frac{i}{i} = -i\frac{1}{\omega} \qquad (11.27)$$

Hence, there is no real part. The response vector will always lie on the –ve part imaginary axis. Its magnitude, $1/w$, will increase from 0 to $-\infty$ as w increases from 0 to ∞. Thus, the expression of dB and □ would be:

$$\text{dB} = 20\log_{10}[1/w] = -20\log_{10}[w] \text{ and } \square = -90° \qquad (11.28)$$

The dB locus would show a slope of –20 dB/decade, and the □ would remain

constant at –90º at all frequencies.

The only significant frequency response description for this family of systems is Bode Plot. These are shown in Figure 11.9.

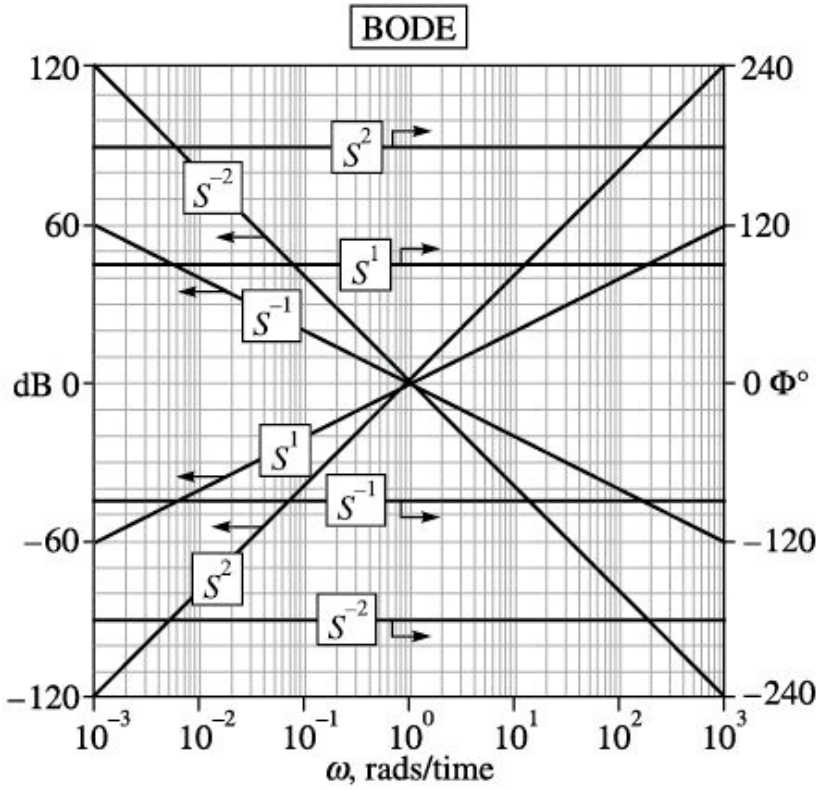


Figure 11.9 Frequency response of integrator (first and second order) and differentiator (first and second order).

By a similar analysis it may be shown that for a differentiator, *s*. The response vector will always lie on the +ve part imaginary axis. Its magnitude, *w*, will increase from 0 to + ∞ as *w* increases from 0 to ∞. Thus, the expression of dB and □ would be:

$$\text{dB} = 20 \log_{10}[w] \text{ and } \square = +90°$$

The dB locus would show a slope of +20 dB/decade, and the □ would remain constant at +90º at all frequencies.

By generalization for a dynamic element TF ($s \pm N$):

B locus is a straight line with a constant slope of = ± N × 20 dB/decade

he □ locus is a horizontal line expressing constant phase angle, □ = ±N × 90°.

In the following three examples, frequency domain expressions of three ideal controllers (PD), (PI) and (PID) have been derived. (P) controller has been left out because a proportional controller is in fact a gain element and we already know that for a gain element, the dB remains constant at 20 $\log_{10}$ (K_C) and the phase remains 0 at all frequencies.

EXAMPLE 11.7 A First Order Zero or (PD) Controller: Due to the following dynamic similarity between the two elements, their frequency description would be

identical.

$$G_P(s) = K_P[1 + \tau_Z s] \equiv G_C(s) = K_C[1 + \tau_D s$$

Starting with the first order zero with unity gain, the substitution of $s = iw$ splits the result into real and imaginary parts:

$$G_P(iw) = 1 + iwt_Z$$

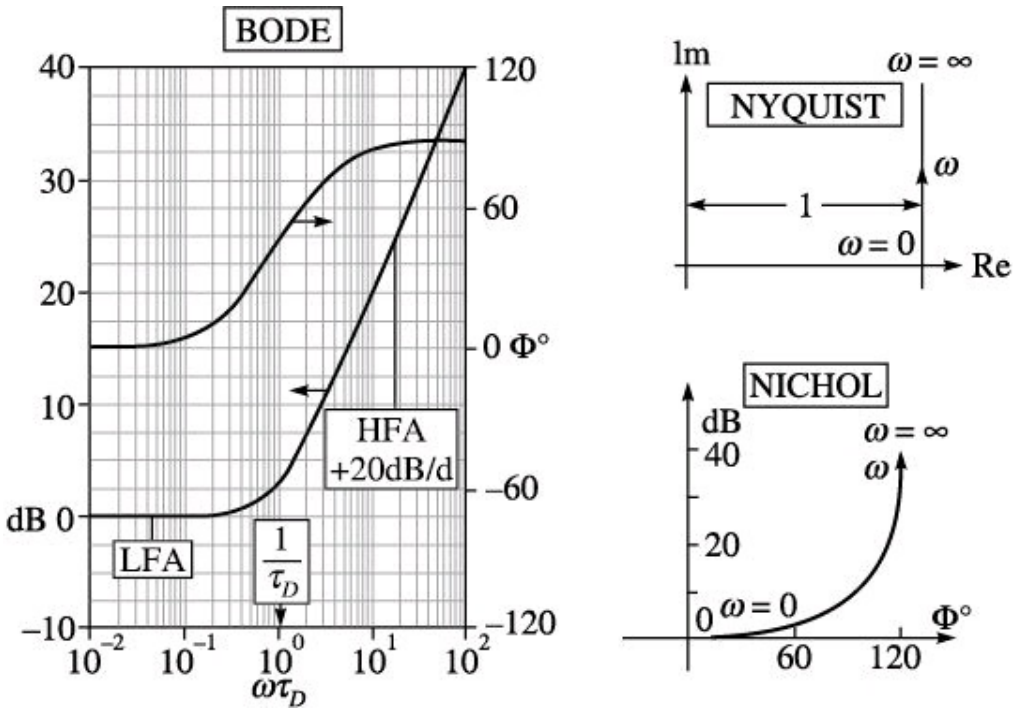


Figure 11.10 Different descriptions of the frequency response of a first order zero element or a (PD) controller.

The Nyquist Plot becomes a vertical straight line (parallel to the imaginary axis at a distance = 1) starting from the real axis at $w = 0$, moving to $+\infty$ as w □ $+\infty$. The dB and □ expressions are:

$$AR = \sqrt{1 + \omega^2 \tau_Z^2} \tag{11.29}$$

$$\text{dB} = 20 \log_{10} \sqrt{1 + \omega^2 \tau_Z^2} \tag{11.30}$$

$$\Phi = \tan^{-1}(\omega \tau_Z) \tag{11.31}$$

EXAMPLE 11.8 A (PI) Controller: The TF of a (PI) controller indicates that this is a combination of an integrator and a first order zero element. However, we shall analyze the TF in its combined form:

$$G_C(s) = K_C\left[1 + \frac{1}{\tau_I s}\right] = \frac{K_C}{\tau_I}\left[\frac{1 + \tau_I s}{s}\right]$$

$$G_C(i\omega) = K_C\left[1 + \frac{1}{i\omega\tau_I}\right] = \left[K_C - i\frac{K_C}{\omega\tau_I}\right]$$

$\therefore$ Real part = K_C; Imaginary part $= -\dfrac{K_C}{\omega\tau_I}$

The frequency plots are prepared assuming $K_C = 1$ and $t_I = 10$. The expressions for *AR*, dB and □ are found as:

$$AR = \sqrt{1 + \frac{1}{\omega^2\tau_I^2}} \tag{11.32}$$

$$\text{dB} = 20\log_{10}(AR) = \frac{\sqrt{1 + \omega^2\tau_I^2}}{\omega\tau_I} \tag{11.33}$$

and

$$\Phi = -\tan^{-1}\left(\frac{1}{\omega\tau_I}\right) \tag{11.34}$$

The dB and □ plots check with the superimposed plots of an integrator + a first order zero. The Nyquist Plot is a vertical straight line parallel to imaginary axis at a distance of 1.0. It starts from $-\infty$ for $w = 0$ and tends toward 0 as w □∞. The Nichol's Plot starts for low frequency ($w = 0.001$) at dB = 40 and □ = –90°, and ends for high frequency ($w = 10$) at dB = 0 and □ = 0°. The results are shown Figure 11.11.

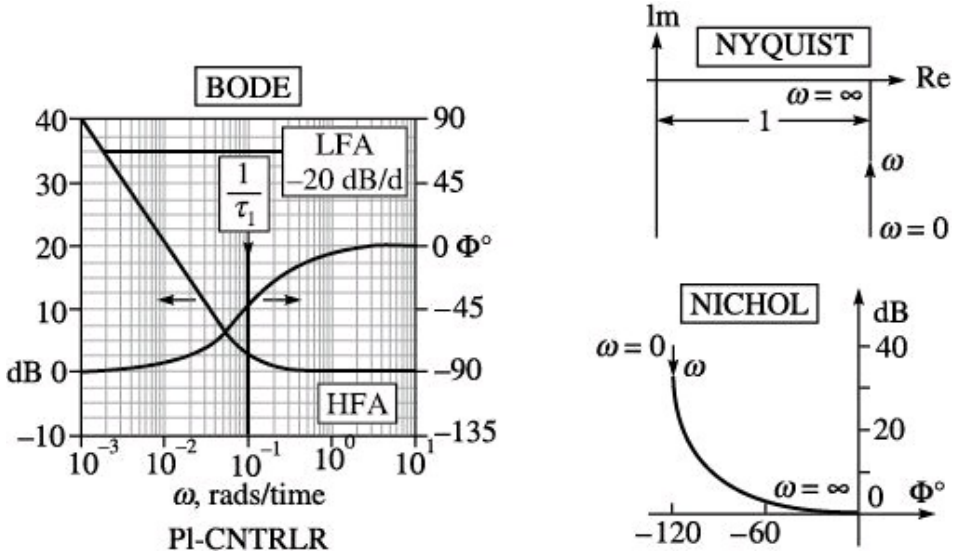


Figure 11.11 Different descriptions of the frequency response of a (PI) Controller.

EXAMPLE 11.9 A (PID) Controller:

$$G_C(s) = K_C\left[1 + \frac{1}{\tau_I s} + \tau_D s\right]$$

$$\therefore \quad G_C(i\omega) = K_C + \frac{K_C}{i\omega\tau_I} + i\omega K_C\tau_D = K_C + iK_C\left[\omega\tau_D - \frac{1}{\omega\tau_I}\right]$$

Thus, $\quad \mathrm{Re}[G_P(i\omega)] = K_C \quad \text{and} \quad \mathrm{Im}[G_P(i\omega)] = K_C\left[\omega\tau_D - \frac{1}{\omega\tau_I}\right]$

From the above results, the expressions for *AR*, dB and □ may be found as:

$$AR = K_C\sqrt{\left(\omega\tau_D - \frac{1}{\omega\tau_I}\right)^2 + 1} \tag{11.35}$$

$$\mathrm{dB} = 20\log_{10}(AR) \tag{11.36}$$

and

$$\Phi = \tan^{-1}\left[\omega\tau_D - \frac{1}{\omega\tau_I}\right] \tag{11.37}$$

The frequency responses for the (PID) controller are prepared assuming a unity value of controller gain. Also, the value of the ratio t_I/t_D was assumed as 100. Such a high value of this ratio helps to show the three asymptotes with the following slopes:

1. –20 dB/decade,
2. 0 dB/decade,
3. + 2 0 dB/decade, and the phase plot to recognize that total TF has the contribution of an integrator and a second order zero. Again the Nyquist Plot is a vertical straight line parallel to the imaginary axis at a distance of 1.0. It starts from –∞ (because
   $-1/wt_I = -\infty$ and $wt_D = 0$) for $w = 0$ and tends toward +∞ (because wt_D □ +∞ and
   $1/wt_I$ □ 0) as w □ ∞. The Nichol's Plot starts for low frequency ($w = 0.01$) at dB = 40 and □ = –90°, touches dB = 0, □ = 0° in the mid-frequency region, where the response is in phase with the input vector and ends for high frequency ($w = 10000$) at dB = 0 and □ = 0°. The results are shown in Figure 11.12.

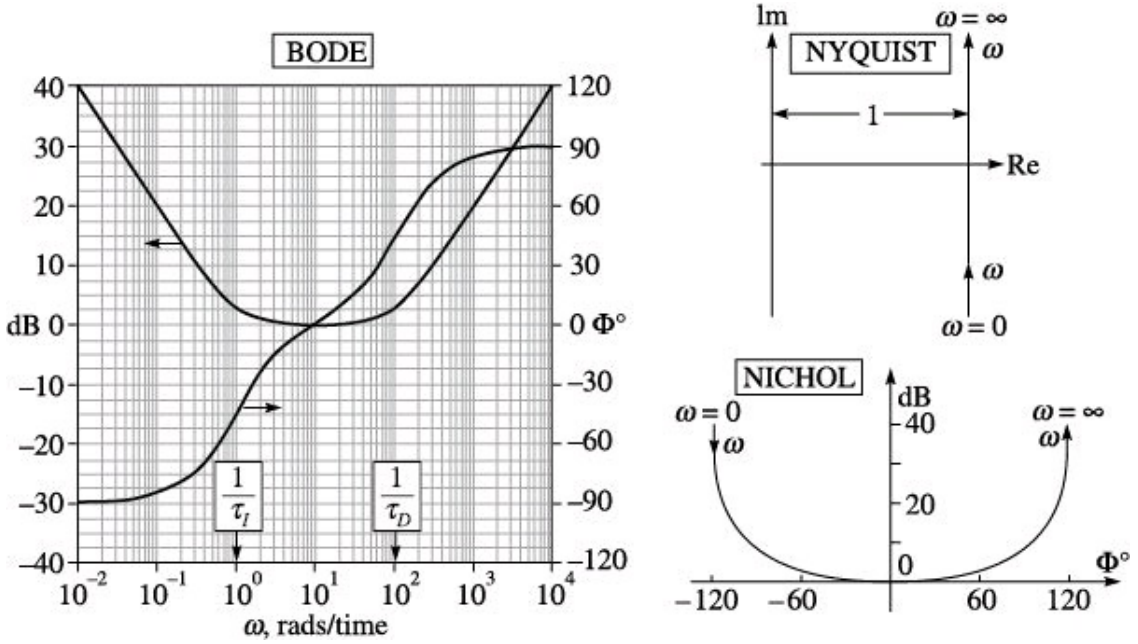


Figure 11.12 Different descriptions of the frequency response of a (PID) controller.

Phase lead or lag depend on t_I, t_D and w, I hope you remember such a result was obtained in Chapter 8 while discussing the sinusoidal response of (PID) controller.

EXAMPLE 11.10 A Lead-Lag Element: A lead-lag element is dynamically significant as we have seen its application in Chapter 8 and will again observe in a later part of the text while discussing the 'feed-forward' control.

$$G_P(s) = \frac{1+\tau_Z s}{1+\tau_P s} = (1+\tau_Z s)\frac{1}{1+\tau_P s}$$

$$G_P(i\omega) = \frac{1+i\omega\tau_Z}{1+i\omega\tau_P} = \frac{(1+i\omega\tau_Z)(1-i\omega\tau_P)}{(1+i\omega\tau_P)(1-i\omega\tau_P)} = \frac{1+\omega^2\tau_Z\tau_P}{1+\omega^2\tau_P^2} + i\frac{\omega(\tau_Z-\tau_P)}{1+\omega^2\tau_P^2} \tag{11.38}$$

$$AR = \sqrt{(\mathrm{Re})^2 + (\mathrm{Im})^2} = \frac{\sqrt{1+2\omega^2\tau_P\tau_Z+\omega^4\tau_P^2\tau_Z^2+\omega^2\tau_Z^2-2\omega^2\tau_P\tau_Z+\omega^2\tau_P^2}}{1+\omega^2\tau_P^2} \tag{11.39}$$

and,
$$\Phi = \tan^{-1}\left[\frac{(\tau_Z-\tau_P)\omega}{(1+\omega^2\tau_Z\tau_P)}\right] \tag{11.40}$$

These expressions have different configurations depending upon the relative values of t_Z and t_P. For a relatively bigger value of t_Z, the lead-lag element behaves like a differentiator, a property used to simulate the derivative mode in a practical (PID) controller as we have discussed in Chapter 8. On the other hand, if t_P is considerably larger than t_Z, the element dynamically resembles an integrator. To check these properties in the frequency domain, the element has been examined for the following

two combination of time constant values.

Case I, $t_Z = 10.0$ and $t_P = 0.1$ Case II, $t_Z = 0.1$ and $t_P = 10.0$

$$AR = \sqrt{1+\omega^2\tau_Z^2} \quad AR = \frac{1}{\sqrt{1+\omega^2\tau_P^2}}$$

and, □ = $\tan^{-1}(wt_Z)$ and □ = $-\tan^{-1}(wt_P)$

(The above expressions have been obtained by neglecting the negligibly small values of parameter combinations that would appear in the general expression of dB and □ developed above.)

Corresponding to Case I, the approximated expression resembles that of a first order zero and for Case II, we obtain that of a first order pole, and the Bode and Nyquist Plots are expected to resemble the shapes of these two dynamic elements. If the characteristic frequencies are far separated or the investigated frequency range do not include them, the predicted shapes would have been satisfied. But in the present case, neither of these conditions have been satisfied, hence, the two forms of frequency response depict the real picture of 'lead-lag' element, while it behaves like a Differentiator (Case I), and while it behaves like an Integrator (Case II). The results are shown in Figure 11.13.

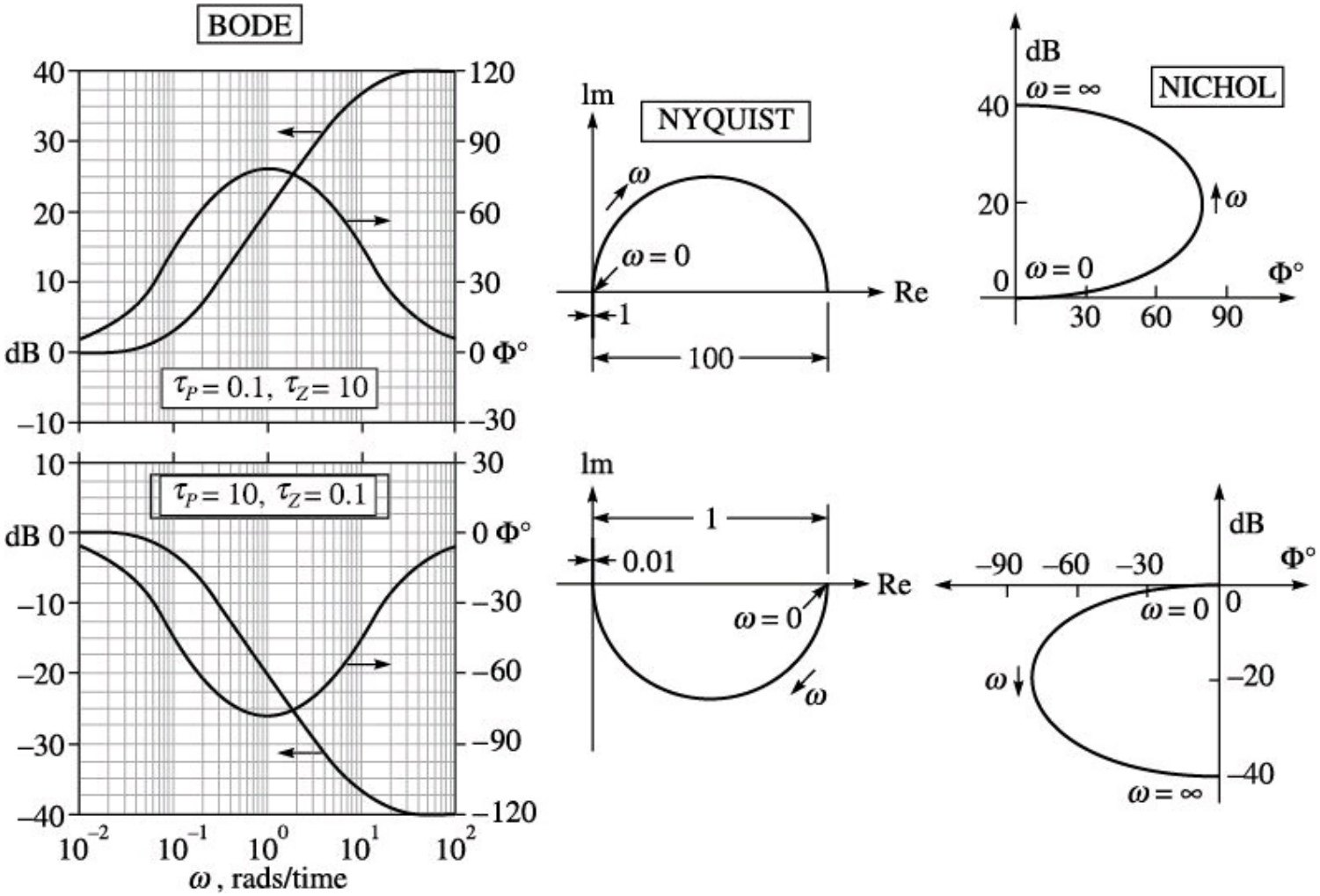


Figure 11.13 Different descriptions of the frequency response of a lead-lag element.

Let us make a property list of the Bode Plot that we have learnt from the above examples. This will be useful in diagnosing the frequency response of an unknown system and may also be a guideline for predicting the response shape of a system of known dynamics. The process transfer function is postulated using the following frequency response properties:

11.2.6 Properties of Bode Plot of Individual Dynamic Blocks

1. For each pole in the transfer function, the high frequency asymptote has a slope = – 20 dB/decade. The high and low frequency asymptotes intersect at corner frequency, w_C.
2. For each zero in the transfer function, the high frequency asymptote has a slope = + 20 dB/decade. The high and low frequency asymptotes intersect at corner frequency, w_C.
3. For each under and critically damped second order pole in the transfer function, the high frequency asymptote has a slope = – 20 dB/decade. The high and low frequency asymptotes intersect at corner frequency, w_C.
4. For each under and critically damped second order zero in the transfer function, the high frequency asymptote has a slope = + 20 dB/decade. The high and low frequency asymptotes intersect at corner frequency, w_C.
5. For each order of integration in the transfer function, the dB locus has a slope = – 20 dB/decade, and it passes through 0 dB horizontal at $w = 1$.
6. For each order of differentiation in the transfer function, the dB locus has a slope =
   +20 dB/decade, and it passes through 0 dB horizontal at $w = 1$.
7. By multiplying the transfer function by a constant K, which is equivalent of changing the gain, the dB plot at all frequency is shifted vertically by 20. $\log_{10}(K)$.
8. The phase plot has a maximum lag of 90º per pole in the transfer function, and has a maximum slope at the corner frequency, $w = 1/t$, where the phase lag is 45º.
9. The phase plot has a maximum lead of 90º per pole in the transfer function, and has a maximum slope at the corner frequency, $w = 1/t$, where the phase lead is 45º.
10. The phase plot has a maximum lag of 180º for each under or critically damped second order pole in the transfer function, and has a maximum slope at the corner frequency, $w = 1/t$, where the phase lag is 90º.

11. The phase plot has a maximum lead of 180º for each under or critically damped second order pole in the transfer function, and has a maximum slope at the corner frequency, $w = 1/t$, where the phase lag is 90º.
12. The phase plot has a constant lag of 90º at all frequencies for each order of integration in the transfer function.
13. The phase plot has a constant lead of 90º at all frequencies for each order of differentiation in the transfer function.
14. A dead time element (e^{-tDLS}) in the transfer function only contributes to the phase plot as a linear function of frequency ($= wt_{DL}$), but does not affect the dB plot. Thus a dead time element has a phase lag value of 1.0 radian at $w = 1/t_{DL}$.

To develop the transfer function it is recommended to draw the dB and □ locus on a single semi-logarithm graph paper so that value or slope comparison of dB or phase angle at any frequency becomes more accurate. Methods of this category differ only by the way the frequency response is obtained, not by the way it is treated.

In the following section, we shall first consider certain examples of drawing frequency response for systems having complex transfer functions involving multiple poles, zeroes dead time, integrator, etc.

The next category of examples would be considered to demonstrate the use of the above listed properties of frequency response, for synthesizing TF of a given process, or some frequency related properties, from the frequency response.

EXAMPLE 11.11 Find the frequency response of the following process TF. Construct the Bode Plot in terms of dB and □ for the individual dynamic elements of the TF, and the overall plot as summation of the individual traces. Also show the low and high frequency asymptotes to indicate the corner frequencies as intersections for the single zero and the two poles of the TF.

$$G_P(s) = \frac{(3s+1)e^{-0.05s}}{s(50s+1)(0.1s+1)}$$

Solution: The frequency range should cover the characteristic frequencies 1/0.05 = 20, 1/50 = 0.02, 1/3 = 0.3333, 1/0.1 = 10. We have chosen a six-cycle frequency range 10^{-3}, 10^{-2}, 10^{-1}, 10^{0}, 10^{1}, 10^{2}, 10^{3}.

The Bode Plot for individual dynamic elements and the overall TF has been shown in

Figure 11.14. Part(a) contains the dB plots and part(b) shows the phase angle □, traces. The corner frequencies for the single zero and two poles have been identified

as the intersection points of low and high frequency asymptotes of the respective dynamic elements.

Note that for one zero and two poles, the overall dB plot would have an ultimate slope of –20 dB/decade at high frequencies. But due to the presence of the integrator, $1/s$, in the TF, the slope has become –40 dB/decade.

Also due to the presence of a relatively small amount of dead time (ratio of $t_{DL}/t_{P\text{MAX}} = 0.05/50 = 0.0001$), the phase shows a sharp decrease at high frequencies. Whereas, in absence of the dead time the phase angle would have an ultimate value of –180º (because +90º for the pole, –180º for two poles, and –90º for the integrator adds up to –180º)

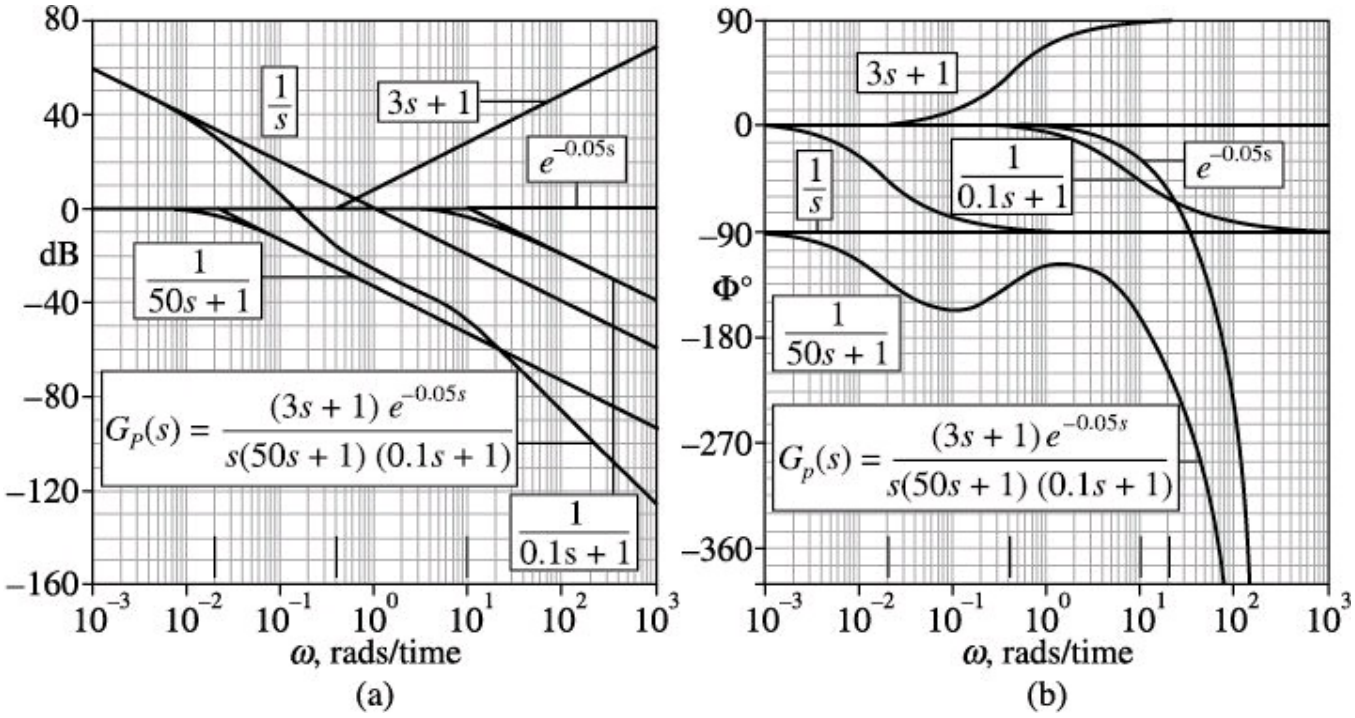


Figure 11.14 (a) Log-modulus. (b) Plots for the system in Example 11.11.

While manually solving such problems, after deciding about the frequency range, you may select a number of frequencies. These frequency values should be close enough so that a smooth curve could be drawn through them to reflect the system characteristics, as well as their number should be small enough so that the complete solution may be obtained within a reasonable amount of workout time.

Example 11.12 The overall TF of a control loop is:

$$G_{FP}(s) = G_C(s)G_V(s)G_P(s) = \frac{K_C K_P(\tau_Z s + 1)e^{-\tau_{DL}s}}{(\tau_{P1}s + 1)(\tau_{P2}s + 1)(\tau_{P3}s + 1)}$$

The TF parameters have the following numerical values:

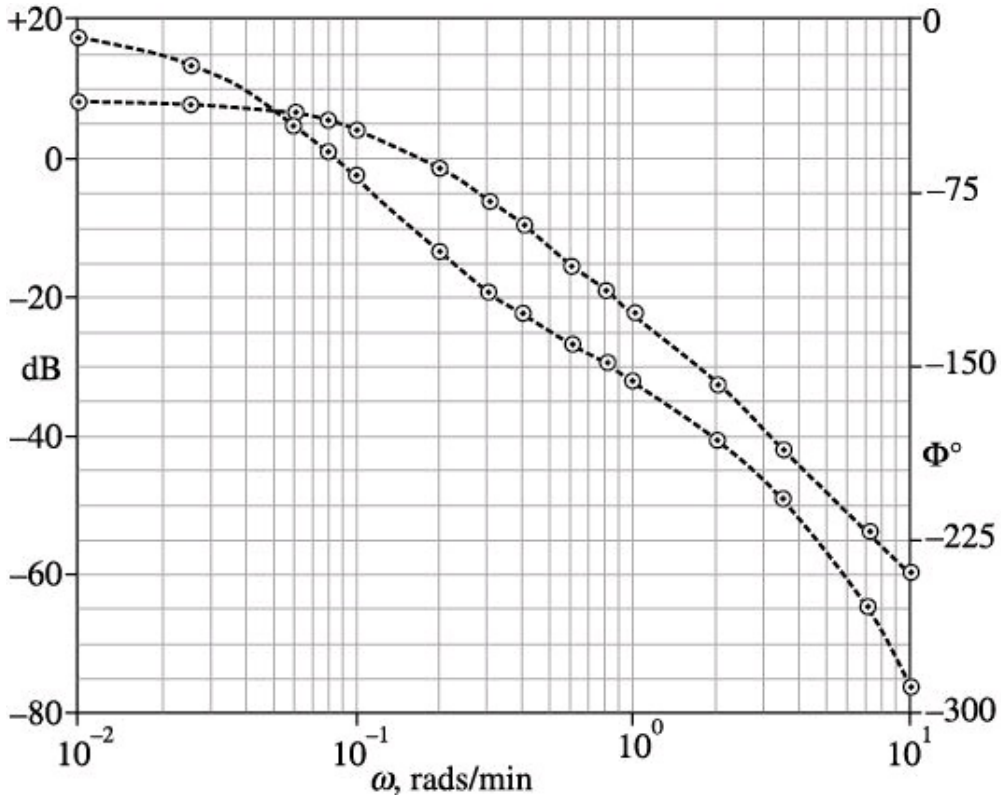
$$K_C K_P = 2.5;\ t_Z = 2.0;\ t_{DL} = 0.2;\ t_{P1} = 1;\ t_{P2} = 5;\ t_{P3} = 10$$

All time constants are in min.

Draw a Bode Plot for the above TF using the following frequency values in rads/min on a suitably marked semi-log graph. (w = 0.01, 0.025, 0.06, 0.08, 0.1, 0.2, 0.3, 0.4, 0.6, 0.8, 1.0, 2.0, 3.5, 7.0, 10.0)

Solution: The expressions for individual dB and □ in the TF and their sum total are:

Items	Decibel (dB)	Phase Angle □,°
Gain	$dB_K = 20\log(2.5)$	$z_K = 0$
Zero	$dB_Z = 20\log\sqrt{1+(2\omega)^2}$	$z_Z = \tan^{-1}(2\sim)$
Delay	$dB_{DL} = 0$	$z_{DL} = -0.2\sim$
Pole-1	$dB_{P1} = 20\log[1/\sqrt{1+\omega^2}]$	$z_{P1} = \tan^{-1}(\sim)$
Pole-2	$dB_{P2} = 20\log[1/\sqrt{1+25\omega^2}]$	$z_{P2} = -\tan^{-1}(5\sim)$
Pole-3	$dB_{P3} = 20\log[1/\sqrt{1+100\omega^2}]$	$z_{P3} = -\tan^{-1}(10\sim)$
Total	$dB = dB_K + dB_Z + dB_{DL} + dB_{P1} + dB_{P2} + dB_{P3}$	□ $= z_K + z_Z + z_{DL} + z_{P1} + z_{P2} + z_{P3}$



ω, $\frac{\text{rads}}{\text{min}}$	dB	Φ°
0.01	7.90	–8.11
0.025	7.635	–20.0
0.06	6.29	–44.9
0.08	5.247	–56.9
0.1	4.1	–67.1
0.2	–1.57	–100.2
0.3	–6.2	–117.0
0.4	–9.83	–127.1
0.6	–15.18	–139.75
0.8	–19.1	–148.67
1.0	–22.25	–156.0
2.0	–32.8	–181.82
3.5	–42.03	–207.4
7.0	–53.86	–253.7
10.0	–60.02	–290.0

Figure 11.15 The Bode Plot for the system of example 11.12.

The dB and □ values have been calculated and are plotted in Figure 11.15. The calculation results are given in the included Table.

EXAMPLE 11.13 The asymptotic Bode Plot as dB vs. w for a system is given in Figure 11.16. Find the actual AR, dB and □ at w = 0.01, 0.1, 1.0, 10 and 100. The process TF is:

$$G_p(s) = \frac{1}{(0.1s+1)(s+1)(10s+1)}$$

where, the time constants are in min.

Solution: As $K_P = 1$, the low frequency asymptote would coincide with

$$\text{dB} = 20 \times \log_{10}(1) = 0$$

The first high frequency asymptote (I) would start from very low values of ~ up to the smallest corner frequency $w_{N1} = 1/10 = 0.1$ rads/min with a zero slope. The second high frequency asymptote (II) starts from w_{N1} and ends at $w_{N2} = 1$ rads/min, with a slope of – 20 dB/decade. The third high frequency asymptote (III) starts from w_{N2} and ends at $w_{N3} = 10$ rads/min with a slope of – 40 dB/decade. Finally, the fourth high frequency asymptote (IV) starts from w_{N3} and extends up to infinite values of w.rads/min with a slope of – 60 dB/decade. The result is shown in Figure 11.16.

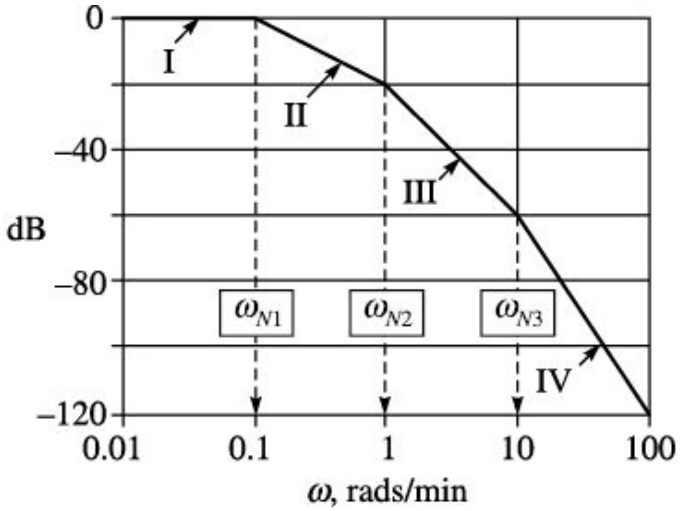


Figure 11.16 The asymptotic Bode' Plot.

The dB and □, at $w = 0.01$, 0.1, 1.0, 10 and 100 has been given in the following table.

~ **rads/min**	$\text{dB} = \text{dB}_1 + \text{dB}_2 + \text{dB}_3 + \text{dB}_4$ $= 0 + 20 \times \log_{10}(\frac{1}{\sqrt{1+0.01w^2}}) + 20 \times (\frac{1}{\sqrt{1+w^2}}) + 20 \times (\frac{1}{\sqrt{1+100w^2}})$	$□° = □_1 + □_2 + □_3 + □_4$ $= 0 + \tan^{-1}(0.1w) + \tan^{-1}(w) + \tan^{-1}(10w)$
0.01	-4.36×10^{-4}	−6.34
0.1	−3.053	−51.28
1.0	−23.09	−135.0
10.0	−63.04	−218.7
100.0	−120.02	−263.66

EXAMPLE 11.14 Prove that for an under damped second order system under sinusoidal disturbance, the normalized amplitude ratio, AR_N , will show a maximum and a resonance frequency only if $0 < z < 0.707$. Also, prove that the expression for maximum value of normalized amplitude ratio, AR_N, is given by,

$$[AR_N]_{\max} = \left[\frac{AR}{K_P}\right]_{\max} = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$$

and the corresponding resonance frequency w_R is given by:

$$\omega_R = \frac{\sqrt{1-2\zeta^2}}{\tau}$$

Solution: Following from Equation 11.14, the normalized expression for the AR_N may be written as:

$$AR_N = \frac{AR}{K_P} = \frac{1}{\sqrt{[1-(\omega/\omega_N)^2 + \{(2\zeta\omega)/\omega_N\}^2}}$$

To find the maximum we differentiate AR_N with respect to w,

$$\frac{d(AR_N)}{d\omega} = \frac{-4\omega\tau^2 + 4\omega^3\tau^4 + 8\omega\tau^2\zeta^2}{-2[(1-\omega^2\tau^2)^2 + (2\omega\tau\zeta)^2]^{1.5}}$$

To render this derivative to have zero value at the maximum, equate the numerator to zero. Thus,

$$4w^3t^4 + 8wt^2z^2 = 4wt^2 \text{ or } 1 - w^2t^2 - 2z^2 = 0 \text{ or } \omega_R = \frac{\sqrt{1-2\zeta^2}}{\tau}$$

The above solution for w_R is real only if the square rooted term remains +ve. That is z may have values greater than zero up to $2z^2 \leq 1$. Hence,

$$0 < \zeta < \frac{1}{\sqrt{2}} \text{ or } 0 < z < 0.707$$

Now by putting the expression of w_R in the equation of AR_N , we may obtain the solution for $[AR_N]_{\text{MAX}}$,

$$[AR_N]_{\max} = [(1-\omega_R^2\tau^2)^2 + (2\omega_R\tau\zeta)^2]^{-0.5} = [4\zeta^4 + 4\zeta^2 - 8\zeta^4]^{-0.5} = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$$

EXAMPLE 11.15 The block diagram of a control system is given in Figure 11.17. (a) Plot the dB vs. w of $D(s)/e(s)$ as an asymptotic Bode Plot; (b) If by increasing K_C, the system oscillates continuously with a frequency, w = 1.5 rads/min. Find the value of the dead time t_{DL}.

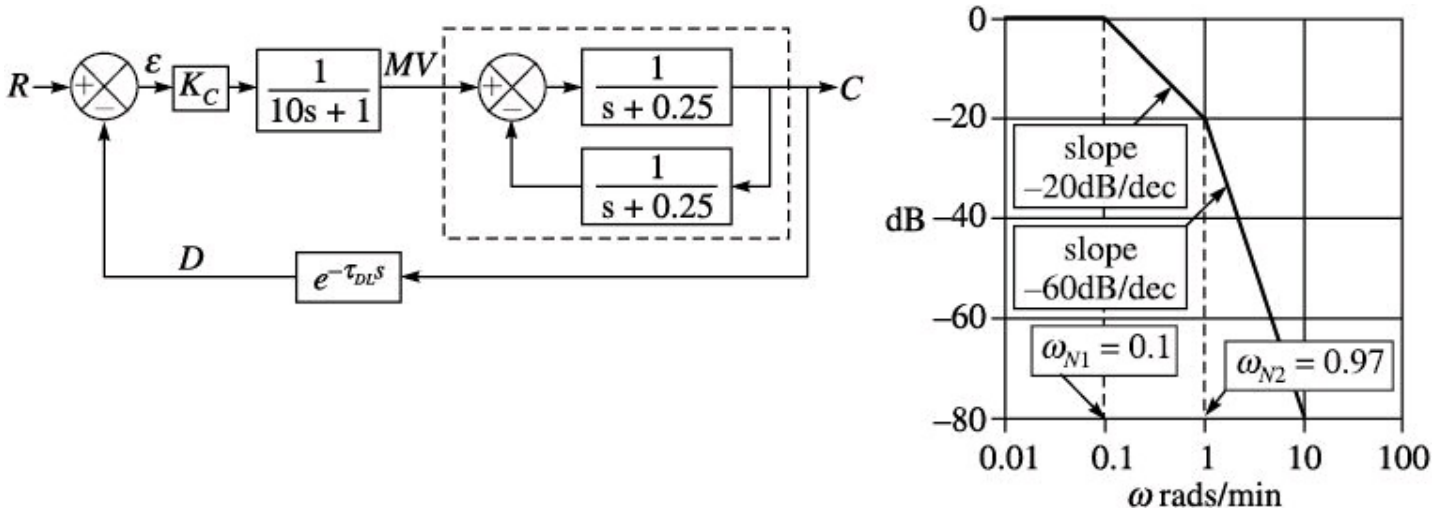


Figure 11.17 The control system, and the Bode Plot.

Solution: The nested loop has the following CLTF:

$$\frac{C(s)}{MV(s)} = \frac{1}{(s+0.25)^2} \Bigg/ \left\{1 + \frac{1}{(s+0.25)^2}\right\} = \frac{1}{s^2 + 0.5s + 1.0625} = \frac{1}{(s+0.25+i)(s+0.25-i)}$$

$$\omega_N = 0.97 \ \frac{\text{rads}}{\text{min}}, \zeta = 0.2577 \text{ and } \frac{D(s)}{\varepsilon(s)} = \frac{K_c e^{-\tau_{DL}s}}{(10s+1)(s+0.25+i)(s+0.25-i)}$$

The pole at $w_{P1} = 0.1$ for $t_{P1} = 10$ is of first order, but the repeated poles at $w_{P2} = 0.97$ for $t_{P2} = 1.031$ are of second order. We have drawn the asymptotic Bode Plot as a part of
Figure 11.17. Up to $w = 0.1$, the dB locus runs along the 0 dB parallel. From $w = 0.1$ to $w = 0.97$, it will have a slope of –20 dB/decade (cycle). Beyond $w = 0.97$, the dB graph will have a slope = (–20 dB/decade –40 dB/decade) = – 60 dB/decade.

From the data: Total □ = – 180º at $w = 1.5$ rads/min.

$$-180° = \frac{-1.5 \times \tau_{DL} \times 180}{\pi} - \tan^{-1}(1.5 \times 10) - 2 \times \tan^{-1}(1.5 \times 0.97)$$

or, $$\tau_{DL} = \frac{(180 - 119.8084 - 23.553)\pi}{180} = 0.63946 \text{ min} = 38.4 \text{ s}$$

EXAMPLE 11.16 Two thermometers are placed close to each other, as shown in Figure 11.18, in an air oven whose temperature could be driven according to sinusoidal function of variable frequency. The thermometers posses first order dynamics with unity gain, but different time constants. At different frequencies, the difference of phase angle of these thermometers is shown in the following table. This data is obtained from the stable part of the response. Time constant, t, of the first thermometer is 9s. Using the data, find the time constant of the second thermometer.

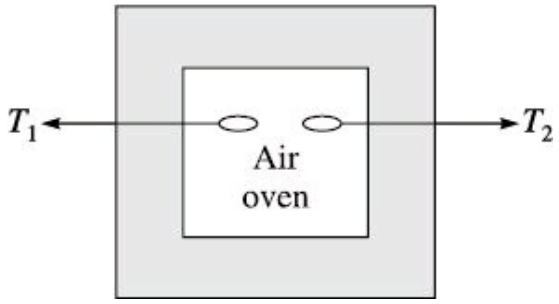


Figure 11.18 Two thermometers in an air oven to measure oscillating air temperature.

~, rads/min	0.3	0.6	1.2	2.5	4.8	6.0	9.0	12.0
□□ = |□1 – □2|°	4.266	8.358	15.437	24.444	26.734	25.393	21.005	17.287

Solution: Let us define the sinusoidally varying air temperature and the two TC outputs as the input and two responses respectively expressed as:

$$X(t) = M \sin wt;\ Y_1 = \frac{M}{\sqrt{1+\omega^2\tau_1^2}} \sin(\omega t - \Phi_1) \text{ and } Y_2 = \frac{M}{\sqrt{1+\omega^2\tau_2^2}} \sin(\omega t - \Phi_2)$$

□2 may be obtained from the difference of phase data and calculated at different frequencies. We present the result in the following table:

w rads/min	□1 = –[tan (*wt*1)]°	□□ from above table = |□1 – □2|°	□2 = (□1 – □□)°
0.3	–2.5766	4.2662	–6.8428
0.6	–5.1428	8.353	–13.500
1.2	–10.204	15.437	–25.641
2.5	–20.556	24.444	–45.000
4.8	–35.754	26.734	–62.488
6.0	–41.987	25.393	–67.380
9.0	–53.471	21.005	–74.476
12.0	–60.945	17.287	–78.232

Corresponding to $w = 2.5$ rads/min; □2 = – 45°. Applying the property, that at $wt_2 = 1.0$, □2 = – 45°.

□ $t_2 = 1/2.5 = 0.4$ min = 24 s

the TF of the second thermometer: $G_{M2}(s) = \dfrac{1}{(0.4s+1)}$.

(The evaluation of time constant could have been done by using the data corresponding to
any of the frequencies given in the above table, but we have chosen the frequency against □2 = –45°, because the phase locus of a first order system shows highest gradient at –45°, hence the phase angle difference data should be of highest accuracy at this frequency, even if determined manually.)

EXAMPLE 11.17 A thermocouple is placed in an oven for a long time. The oven

temperature is oscillating according to:

$$x(t) = x(s) + M \sin wt$$

The thermocouple has a first order dynamics. Its gain is 1.0 and the time constant is t. Find the maximum error, $E(t)$, between the true temperature of the oven and the output temperature, $Y(t)$, registered by the thermocouple.

Solution: The stable part of the sinusoidal response of a first order system and the input both in deviation variable form may be written as:

$$Y(t) = \frac{M}{1+\omega^2\tau^2}\sin\omega t - \frac{M\omega\tau}{1+\omega^2\tau^2}\cos\omega t \text{ and } x(t) = M \sin wt$$

Hence, the error may be defined as: $E(t) = X(t) - Y(t)$

Or, $$E(t) = M\left\{1 - \frac{1}{1+\omega^2\tau^2}\right\}\sin\omega t + \frac{M\omega\tau}{1+\omega^2\tau^2}\cos\omega t = \frac{M\omega^2\tau^2}{1+\omega^2\tau^2}\sin\omega t + \frac{M\omega\tau}{1+\omega^2\tau^2}\cos\omega\tau$$

Applying the trigonometry theorem for condensing two trigonometric variables into a single one,

$$p \sin(A + B) = p \sin A \cos B + p \cos A \sin B$$

For this problem, we may identify the terms as:

$$\sin A \equiv \sin\omega t; \quad p\cos B \equiv \frac{M\omega^2\tau^2}{1+\omega^2\tau^2}; \quad \cos A \equiv \cos\omega t\ ; \quad p\sin B \equiv \frac{M\omega\tau}{1+\omega^2\tau^2}$$

Thus, $$p = \sqrt{p^2\cos^2 B + p^2\sin^2 B} = \frac{M\omega\tau}{\sqrt{1+\omega^2\tau^2}} \quad \text{and} \quad B = \tan^{-1}\left(\frac{p\sin B}{p\cos B}\right) = \tan^{-1}\left(\frac{1}{\omega\tau}\right)$$

Applying the above result the error function becomes:

$$E(t) = \frac{M\omega\tau}{\sqrt{1+\omega^2\tau^2}}\sin\left\{\omega t + \tan^{-1}\left(\frac{1}{\omega\tau}\right)\right\}$$

As the error also varies sinusoidally, its maximum value will be the amplitude of the function [because $E(t)$ is the difference of two sine waves having the same average value], thus,

$$E_{\max} = \frac{M\omega\tau}{\sqrt{1+\omega^2\tau^2}}$$

EXAMPLE 11.18 In this example, the damping performance of input flow fluctuation is compared of two configurations of surge tank systems. The fluctuations are assumed as sinusoidal. The two kinds of surge tank arrangements are: (A) Two tanks of 90 cm diameter, 150 cm height connected in non-interacting fashion; (B) A single tank of 120 cm diameter, 168.75 cm height. These arrangements along with

their dimension and parameter values are shown in Figure 11.19. The threshold flow rate through the tanks is 400 l/min around which the oscillations occur. The valves on the exit lines are so adjusted that the tanks are half full when q_I = 400 l/min. Compare the damping characteristic of the two configurations for sinusoidal disturbances in flow rate, q_I, as function of frequency, w, of the disturbance.

Solution:

$$q_{IS} = 400 \text{ l/m} = 4 \times 10^5 \text{ cm}^3/\text{min}; \; h_S = \frac{q_{IS}}{R}; \; \frac{H(s)}{Q_I(s)} = \frac{R}{(ARs + 1)}$$

$$q_I(t) = q_{IS} + M \sin wt; \; h(t) = h_S + \frac{MR}{\sqrt{1 + \omega^2\tau^2}} \sin(\omega t - \varphi)$$

We only need to show the relative value of AR of the two configurations, as ratio of two ARs, as a function of inflow sinusoid frequency, w.

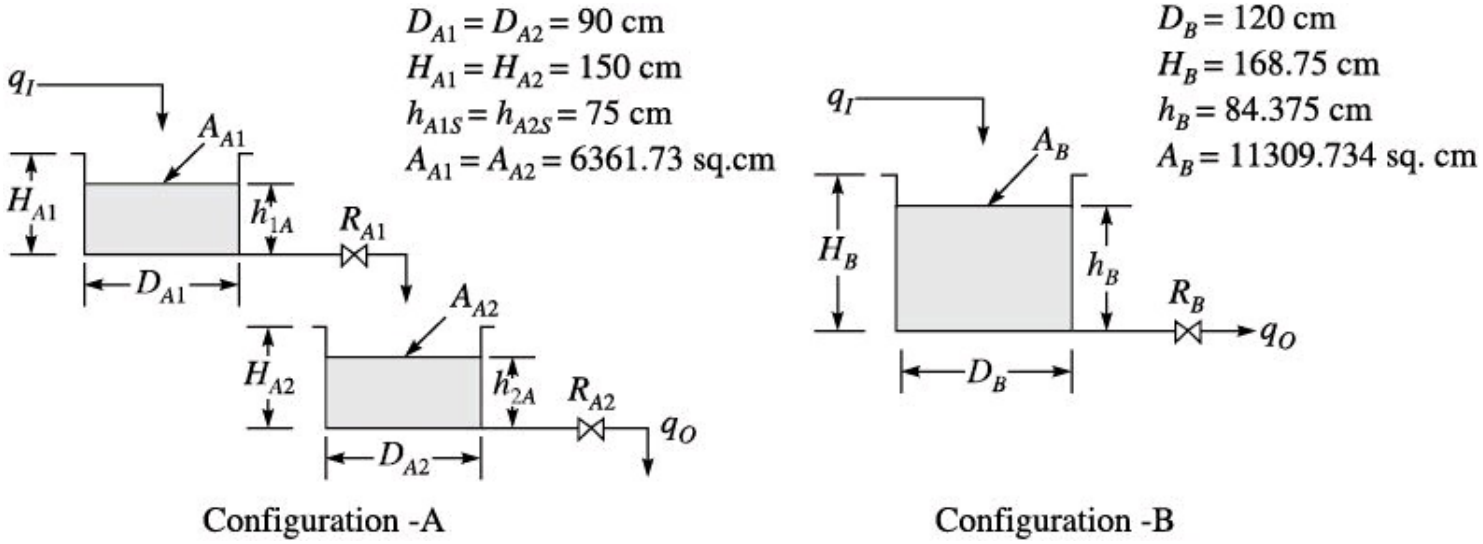


Figure 11.19 The competitive tank configurations for flow fluctuation damping.

The two tank arrangement (A)

$$h_{1AS} = h_{2AS} = \frac{H_{A1}}{2} = 75 \text{ cm};$$

$$R_{A1} = R_{A2} = \frac{h_{2AS}}{q_{IS}} = \frac{75}{400}$$

$$= 1.875 \times 10^{-4} \text{ cm/(cc/min)}$$

$$A_{A1} = A_{A2} = 11309.7336 \text{ cm}^2$$

$$\tau_{A1} = \tau_{A2} = 1.1928 \text{ min} = 71.57 \text{ s}$$

The *AR* of this system's frequency response will be:

$$AR_A = \frac{M \times 1.875 \times 10^{-4}}{\left(\sqrt{1 + (\omega \times 71.57)^2}\right)^2}$$

The single tank arrangement (B)

$$H_B \times \pi \times \left(\frac{D_B}{2}\right)^2 = 2 \times H_{A1} \times \pi \times \left(\frac{D_{A1}}{2}\right)^2$$

or, $H_B = 168.75$ cm

$$\therefore \quad h_{BS} = \frac{H_B}{2} = 84.375 \text{ cm}$$

$$\therefore \quad R_B = \frac{h_{BS}}{q_{IS}} = 2.11 \times 10^{-4} \text{ cm/(cc/min)}$$

$$A_B = \pi \times \left(\frac{D_B}{2}\right)^2 = 12.566 \text{ ft}^2$$

$$\tau_B = 2.386 \text{ min} = 143.14 \text{ s}$$

$$\therefore \quad AR_B = \frac{M \times 2.11 \times 10^{-4}}{\sqrt{1 + (\omega \times 143.14)^2}}$$

The relative value of the amplitude ratios may be expressed as their ratio, *R*, thus,

$$R_E = \frac{AR_A}{AR_B} = \frac{R_{A1}}{R_B} \frac{\sqrt{1 + (143.14 \times \omega)^2}}{[\sqrt{1 + (71.57 \times \omega)^2}\,]^2} \qquad \text{(A)}$$

R_E as a function of *w* are found and listed in the following table:

w rad/s	1.0	2.0	3.0	4.0	5.0	6.0	7.0	8.0	9.0	10.0
R_E	1.15	0.614	0.459	0.355	0.288	0.245	0.21	0.185	0.165	0.148

We observe that except in the low frequency region with increase of frequency, system-A's performance becomes steadily better.

EXAMPLE 11.19 Concentration C_I of a stream of solution is varying periodically due to an ON-OFF control action in an upstream equipment. This periodicity may be described as a sine function as:

$$C_I(t) = C_S + M_I \sin(wt)$$

The stream is proposed to be passed through four (4) equal volume, well stirred mixers so that the oscillation amplitude become 1/6th of that in the inlet stream. Find the required relation between the stream flow rate, *Q*, the mixer volume of a single

stage, V, and the angular frequency of oscillation, w.

Solution: If these quantities (Q, V and w) are expressed in consistent engineering units, then the time constant of each mixer will be $t = V/Q$, and the expression of amplitude reduction in 4 consecutive stages is:

$$M_4 = \frac{M_I}{[\sqrt{1+\omega^2\tau^2}]^4}$$

By the condition of the problem $M_4 = M_I/6$. Solving for wt, we obtain:

$$(1 + w^2t^2)^2 = 6 \text{ or } wt = 1.204$$

as $t = V/Q$, hence, the required relation is:

$$V = \frac{(1.204Q)}{\omega}$$

This relation is consistent with our notion that by increasing the oscillation frequency, the amplitude will be further reduced for given flow rate and mixer volume.

EXAMPLE 11.20 A circular pipe carries a liquid whose temperature is varying sinusoidally according to: $T(t) = T_S + M \sin[(2pt)/T_P]$, where, $T(t)$ = Temperature of the liquid as a function of time; T_S = The threshold temperature of the liquid around which the oscillations occur = 90°C; M = Amplitude of oscillation = 45°C; T_P = Time period of oscillation =
1.0472 min.

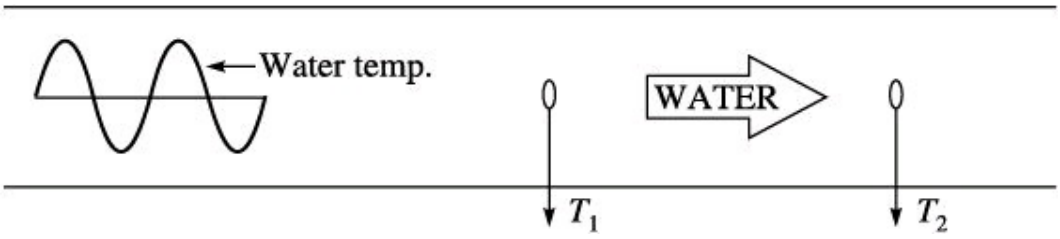


Figure 11.20 Schematic of the system.

A schematic of the system is shown in Figure 11.20. Two temperature sensors having unity gain first order dynamics (t_1 = 3 s, t_2 = 5 s) are inserted at a longitudinal distance D in the pipe for temperature measurement flowing liquid. The head of these sensors are located on the axis of the pipe. Measurements are made after the initial transients are eliminated. It is observed that the oscillation peak of the downstream sensor 2 occurs 10.522 s after the upstream sensor 1. If the volumetric flow rate of liquid, Q = 25 l/m, and the cross sectional area of the pipe, A = 10000 sq.cm, from these information calculate the distance D. Assume a flat velocity profile inside the pipe.

Solution: We need to run the analysis by using the phase lag relation of a first order system.
The radian frequency *w* may be found as

$$\omega = \frac{2\pi}{T_P} = 6 \text{ rads/min} = 0.1 \text{rads/s}$$

The phase lag of the two sensors are:

$$\phi_1 = -\tan^{-1}(wt_1) = -16.7° \text{ and } \phi_2 = -\tan^{-1}(wt_2) = -26.565°$$

$\phi_2 - \phi_1 = -9.87°$, which is equivalent to 1.722 s. As the total measurement lag between two temperature sensors are 10.522 sec, the dead time or transportation lag will be, $t_{DL} = 10.522 - 1.722 = 8.8$ s. Hence, the distance D = (25000/10000) × 8.8 = 22 cm.

EXAMPLE 11.21 The frequency response of a first order process is to be determined experimentally. How many cycles are required before the *AR* is within 5% of the stable value? How many minutes are required if the process time constant is 10 min?

Solution: For a first order system:
If the input $X(t) = M \sin wt$, the response is,

$$Y(t) = \frac{M\omega\tau}{1+\tau^2\omega^2} e^{-t/\tau} + \frac{M}{\sqrt{1+\tau^2\omega^2}} \sin(\omega t + \phi)$$

Clearly the first term in the RHS of the response equation $[(Mwt)/(1 + t^2w^2)]e^{-t/t}$ is the decaying function of time that determines the difference between *transient* and *stable* state of the response. In fact this may be called as the envelope function of the sinusoidal oscillations of the response. The variation of magnitude of this function as time increases, indicates that how close the response has come to its stable value. To derive time for 95% recovery or, 5% deviation,

$$e^{-t/\tau} = 0.05; \quad \text{or, } \frac{-t}{\tau} = \ln 0.05 = -3; \quad \text{if } \tau = 10 \text{ min}; \quad t = 30 \text{ min}.$$

Number of cycles, *N*, within this time *t* = 30 min may be determined only if we know the value of the *w* frequency of the input sinusoid. The time period *TP* may be expressed as:

$$T_P = \frac{2\pi}{\omega}; \quad \therefore N = \frac{30}{T_P} = \frac{30\omega}{2\pi} = \frac{15\omega}{\pi} = 4.775\omega$$

EXAMPLE 11.22 Show that in the phase part of the Bode plot of a process TF containing two zeros, four poles and an integrator, if the characteristic frequencies are

properly chosen, there will be more one frequencies at which the lag will be 180º.

Solution: The TF of such a system will be:

$$G_P(s) = \left(\frac{K_P(\tau_{Z1}s + 1)(\tau_{Z2}s + 1)}{\tau_I s(\tau_1 s + 1)(\tau_2 s + 1)(\tau_3 s + 1)(\tau_4 s + 1)} \right)$$

In the phase plot of □ vs. *w* the two 'zeros' demonstrate phase leads, each of which asymptotically approach + 90º. The integrator has a constant lag of –90º at all frequency, and all the four poles have phase lags, each of which asymptotically approach –90º. If these zeros, poles and integrator are placed in the □ vs. *w* plot in the following way: two poles in the low frequency followed by two zeros in the mid-frequency and then two poles in the high frequency region.

Due to the presence of the integrator (1/*s*) term in the TF, the □ curve will start from:
–90º and gradually go down by the influence of two low frequency poles. In the vicinity of
$w = 1/t_{P1}$ it will first go below –180º due to the more significant contribution of the poles
$[1/(t_{P1}s + 1)]$ and $[1/(t_{P2}s + 1)]$ and less heavy contribution of the two zeros. As *w* increases beyond $1/t_{P2}$ in the mid-frequency region, the □ will go up and become higher than –180º due to the heavy contributions of two zeros and very small contribution of the remaining poles
$[1/(t_{P3}s + 1)]$ and $[1/(t_{P4}s + 1)]$.

However, at still higher frequency, the phase contributions of these remaining two poles will become significant and □ curve will again dip beyond –180º. Ultimately the phase curve would asymptotically approach a value of (–5 × 90º + 2 × 90º =) –270º.

The placement, individual and overall contribution of poles and zeros, are shown in Figure 11.21. The parameter values used for obtaining the response have been:

$$K_P = 1.0;\ t_{P1} = 0.01;\ t_{P2} = 0.05;\ t_{Z1} = 0.001;\ t_{Z2} = 0.005;\ t_{P3} = 0.00001;\ t_{P4} = 0.00006$$

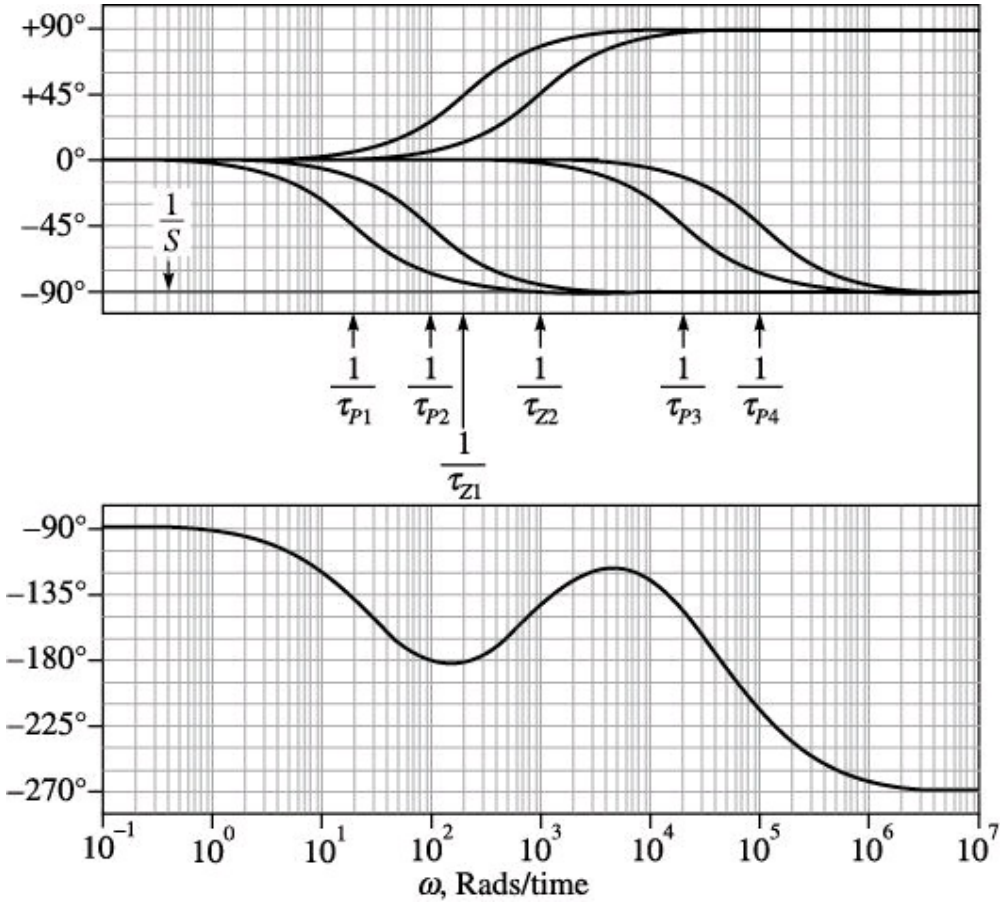


Figure 11.21 The frequency response as solution of example 11.21.

11.3 Stability Analysis in Frequency Domain

We shall consider two procedures of stability analysis in frequency domain, e.g., Bode and Nyuuist stability criterion.

11.3.1 Bode Stability Criterion

We shall begin this topic by considering a dynamic system having the following open loop block diagram shown in Figure 11.22, and its input output variables denoted as X and Y respectively. Note that the variability of the system gain has been recognized by labeling the gain element as K_C, as if a (P) controller is included in the system while the other blocks constitute a unity gain.

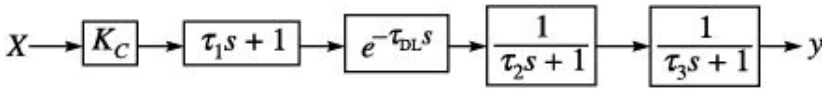


Figure 11.22 An open loop process with a (P) controller.

This block diagram may be reconfigured to represent a control system as in the Figure 11.23. If the loop is severed (opened) at point B, the dynamics between input, X, and output, Y, remain identical. This statement is also true irrespective of the fact that whether the dead-time element is placed either on the forward or on the feedback path.

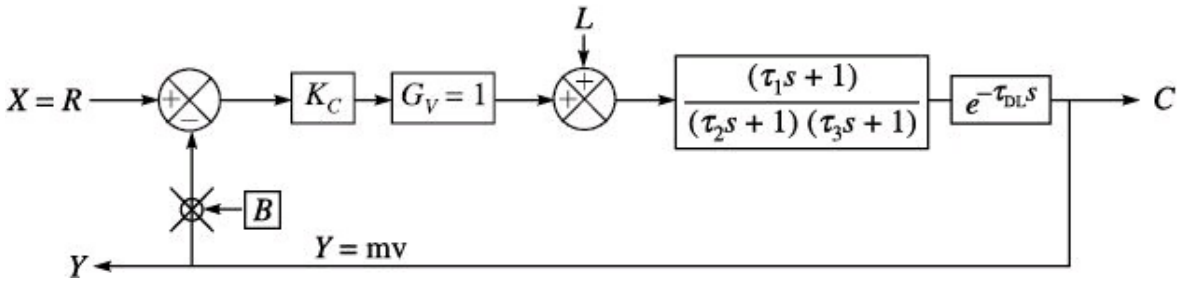


Figure 11.23 The closed loop block system corresponding to Figure 11.22.

That is, if this reconfigured loop is put under servo type of disturbance through set point change, $R(t) = X$ while it has been severed at B, similar response in C would be obtained as Y for the open loop system of Figure 11.22.

Let R be disturbed in a sinusoidal manner according to:

$$R = M \sin (wt)$$

If the system parameters (K_C and t_{DL}) are adjusted in a manner that the overall gain and phase lag values between X and Y are 1.0 and 180º, the output Y will be a sinusoid of identical amplitude M, as the input X, but with a 180º phase lag, i.e.

$$Y = -M \sin (wt)$$

This condition of operation has been shown in Figure 11.24(a). Now two changes are made simultaneously:

1. R is turned to zero, and
2. The loop is rejoined at point B as in Figure 11.24(b).

What would be the result after these changes? In absence of any input ($R = 0$) and the fact that when Y goes through the comparator-block its sign changes from +1 to –1, causes the sinusoidal variation that is already continuing through the loop to become self-sustained. This means that the signal cycles through the loop as a sustained constant amplitude sine wave.

In such a closed loop system, three possible categories of stability conditions may be realized by applying following of parameter values:

1. If the overall gain < 1.0, the sine wave will gradually diminish in amplitude and the system is stable.
2. If the overall gain = 1.0, the sine wave will sustain with a constant amplitude and the system is limitingly stable.
3. If the overall gain become > 1.0, the sine wave will gradually increase in amplitude and the system is unstable.

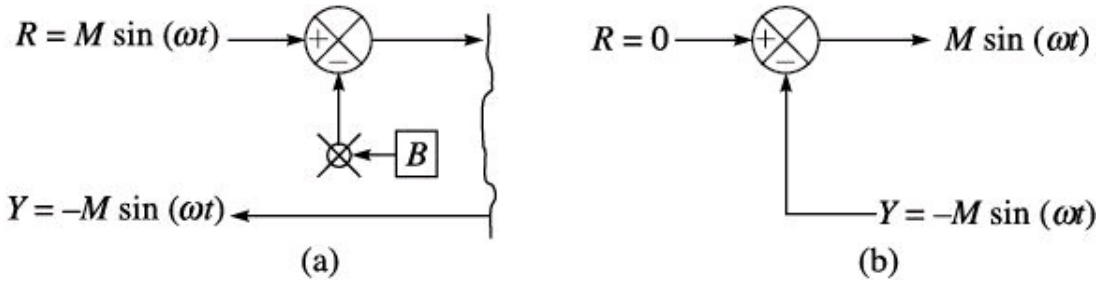


Figure 11.24 Synthetically producing the limitingly stable condition in a closed loop system.

The notions expressed above are the fundamental considerations of **Bode Stability Criterion**. Based on the discussion that we just had, the Bode Stability Criterion under frequency domain may be stated as:

"**In order for a control system to remain stable the amplitude ratio must be less than 1.0 (dB < 0) when the phase angle □ = –180° (–*p* radians)**." (The frequency at which the phase angle □ = –180° is called the Cross-Over Frequency w_{CO}.)

To determine system stability by this approach, we require to determine frequency response of the control system in a 'straitened form of the loop', determine the frequency at which the phase angle □ = –180° and check whether at this frequency the amplitude ratio is equal to or more than 1.0 (dB = 0 or < 0). The term 'straightened form of the loop' signifies that if there is any dynamic block in the feedback path of the block diagram, it may be included in the forward path between *X* and *Y*, as shown in Figure 11.25. Thus, frequency response of the forward path is required to be determined for this method of stability analysis, considering the control system to be of unity feedback configuration.

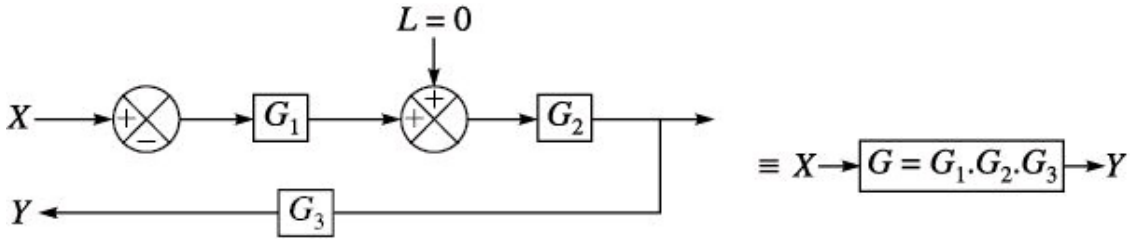


Figure 11.25 To configure the forward path of a closed loop system.

For feedback loop with non-unity gain in the feedback path, the block diagram may be modified in the manner shown in Figure 11.26 to reconfigure in the unity feedback form. In following discussion the forward path will be considered with a K_C representing a (P) controller and the process TF $G_P(s)$ that include $G_V(s)$, $G_P(s)$, and $G_M(s)$, if any.

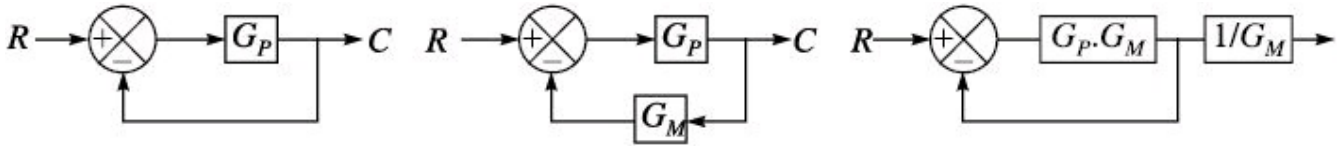


Figure 11.26 Change from non-unity to unity feedback loop.

11.3.2 Measure of Stability

A system may be stable, but how much is its measure of stability, or how far away it is from the stability limit? Similarly for an unstable system – what is its separation or distance from the stable operation condition? The more important question is whether these measures could be interpreted in terms of parameter values, i.e. by suitably adjusting the controller parameter values, could an unstable system be made stable? In frequency domain, these queries are answerable. From these considerations it is possible that a given closed loop system be designed to obtain response of a specified speed and stability measure.

Let the closed loop systems frequency response for a given set of parameters (K_C and t_{DL}) is given as in the Figure 11.27(a).

The Bode Plot in Figure 11.27(a) shows that at the cross-over frequency, w_{CO}, the dB value is –ve, hence, the system is stable. The system is at a safe distance from stability limit with respect to gain by an amount ΔdB. A term known as the **Gain Margin** (GM) has been defined as the multiplier or gain value that may be applied (say by increasing the controller gain K_C) to bring the system to its stability limit. The GM may be calculated from the ΔdB value as:

The AR corresponding to this ΔdB is: AR = Anti-log(ΔdB/20)]

Hence, $GM = 1/AR$.

Thus, K_C may be increased by the amount of GM to reach stability limit.

Hence, $GM = K_{\text{ULTIMATE}}$ (K_U).

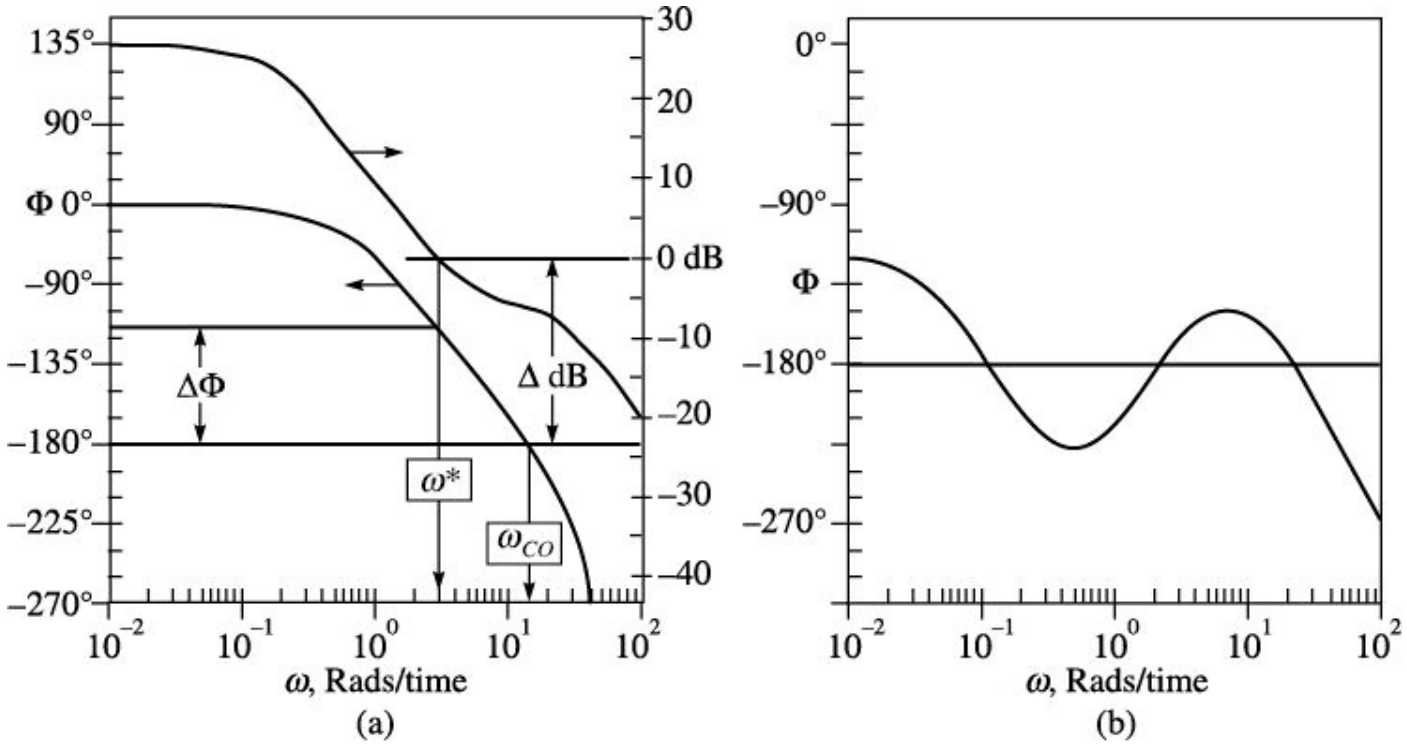


Figure 11.27 (a) The pictorial definition of phase and gain margin. (b) Complex systems for which w_{CO} has multiple

values.

The system could be also away from stability limit with respect to its phase angle by an amount $\Delta\phi$, known as the Phase Margin (PM). $\Delta\phi$ is determined as the difference of phase angle between –180° and the phase value for the frequency at which dB = 0. Let us denote this frequency as w^*. To express this measure of stability with respect to phase, a term **Phase Margin (PM)** is defined as the difference in phase angle value at w^* and –180°.

Thus, from Figure 11.27(a): $PM = |-180° - (-114°)| = 66°$.

From the value of w_{CO}, we may also find the period of oscillation P_{ULTIMATE} (P_U) with which the system would oscillate in this limitingly stable condition. These two parameters are the starting point of exercising Ziegler-Nichol's Tuning formulae for the design of control system in frequency domain. We shall discuss this topic in Chapter 13.

A limitation of applying Bode Stability criterion is that it fails to analyze systems that demonstrates a ϕ curve cutting the –180° horizontal at more than one frequency or, the systems having multiple values of w_{CO} within the range of investigated frequency. The ϕ curve of such a system has been shown in Figure 11.27(b). We shall talk more about the systems of this category in the next section to find any practical detour possible for their analysis. Recall Example 11.21 where it has been shown that a process contains four poles with a (PID) controller may show multiple values of w_{CO}.

Let us work out some examples to strengthen our assimilation of Bode Stability criterion.

EXAMPLE 11.23 In Example 11.11, we have derived the frequency response as Bode Plot of the following closed loop TF:

$$G_P G_M(s) = \frac{K_C K_P (3s+1)e^{-0.05s}}{s(50s+1)(0.1s+1)}$$

The plots for individual dynamic blocks and the total plot as summation of the individuals are shown in Figure 11.14. Here we use this overall Bode Plot to find w_{CO}, GM, PM and K_U for the process.

Solution: The system features one zero, two poles, a dead time element, an integrator, a unity process gain ($K_P = 1$) and a (P) controller having unity gain ($K_C = 1$). The overall dB vs. w and ϕ vs. w plots are reproduced from Figure 11.14 in part (a) and (b) of Figure 11.28, respectively. The crossover frequency, w_{CO}, is found to be 11.7 rads/time from the intersection of ϕ locus with the –180° horizontal in part (b).

Corresponding to this frequency, the dB value is –53, obtained from part (a). Hence, gain margin, may be evaluated in the way as already discussed.

$$\square \text{dB} = -53$$

The *AR* corresponding to dB = –53 would be:

$$AR = \text{Anti-log}\left[\frac{-53}{20}\right] = 0.0022387$$

Thus, value of *GM* would be reciprocal of this *AR*.

$$GM = KL = \frac{1}{0.0022387} = 446.684$$

From Figure (a) the frequency at which the dB value becomes zero is denoted as w^*, and magnitude is found to be 0.128 rads/time. From Figure (b), the phase angle at $w^* = 0.128$ rads/time is $\square = -154°$. Hence, $PM = \square\square = 180 - 154 = 26°$.

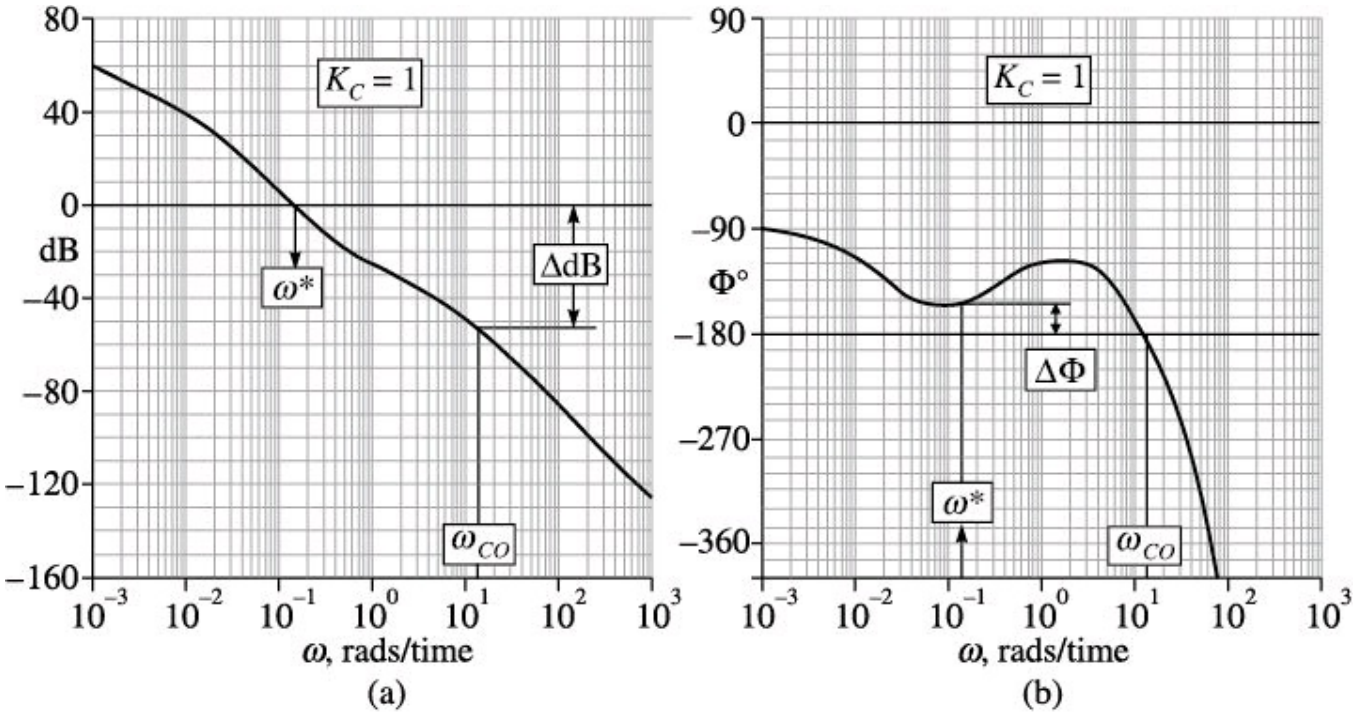


Figure 11.28 Determination of stability in frequency domain.

EXAMPLE 11.24 A closed loop process has been shown in Figure 11.29, in block diagram form. Time constants and dead time values are in minutes. At the frequencies $w = 1.07$ and 1.1 rads/min, the value of phase difference $\square = -178.4°$ and $-181.1°$ respectively. Assuming that w versus $\square$ has a linear relation in the above range of frequency, find K_U, P_U, *GM*, and *PM* of this control system.

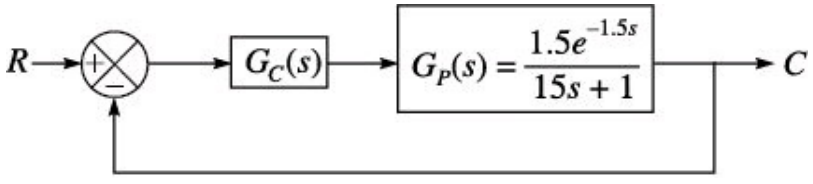


Figure 11.29 An FOPDT process with a (P) controller.

Solution: From the data and problem statement, slope S of w vs. □ curve may be found as:

$$S = \frac{\Phi_2 - \Phi_1}{\omega_2 - \omega_1} = \frac{-181.1 - (-178.4)}{1.1 - 1.07} = -90 \ \frac{\text{deg}}{\text{rads/min}}$$

To find w_{CO}, we may write,

$$S = \frac{\Phi_{CO} - \Phi_1}{\omega_{CO} - \omega_1} \quad \text{or} \quad \omega_{CO} = \omega_1 + \frac{\Phi_{CO} - \Phi_1}{S} = 1.6 + \left[\frac{\{-180 - (-178.1)\}}{-59}\right] = 1.6322 \ \frac{\text{rads}}{\text{min}}$$

$$\therefore \quad P_U = \frac{2\pi}{\omega_{CO}} = 5.55 \,\text{min}$$

At Cross-Over frequency w_{CO}, the amplitude ratio,

$$AR = \frac{1.5}{\sqrt{1 + (1.1322 \times 15)^2}} = 0.0882$$

$$\therefore \quad GM = \frac{1}{0.0882} = 11.34$$

$$\therefore \quad K_U = 11.34$$

To find PM we first evaluate the frequency, ω, at which $AR = 1$, thus,

$$\frac{1.5}{\sqrt{1 + (15 \times \omega)^2}} = 1.0 \quad \text{or} \quad (1.5)^2 = 1 + (15 \times \omega)^2 \quad \text{or} \quad \omega = 0.0745 \text{ rads/min.}$$

At this frequency the phase lag,

$$\Phi = -\omega\tau_D - \tan^{-1}(\omega\tau_P) = -0.0745\frac{180}{\pi} - \tan^{-1}\left[15 \times 0.0745\right] = -4.27 - 48.18 \cong -52.45^\circ$$

Hence, $PM = 180 - 52.45 = 127.55^\circ$.

We have first mentioned in Chapter 1 that a closed loop process stability mainly depends on the 'lag' and 'gain' of the loop. The lag may comprise of transfer as well as transportation-lag elements in the loop and they principally constitute the 'phase angle' of the loop dynamics, and determines the frequency value at which phase angle, □, becomes −180°. On the other hand, the overall loop gain, which is the product of gains of individual blocks, defines the frequency value (w^*) at which dB becomes zero ($AR = 1$). Hence, closed stability may be improved by decreasing either the lag or the gain of the loop.

These conclusions are demonstrated in the following example.

EXAMPLE 11.25 Analyze the stability of the closed loop system given in Figure 11.30 in the form of block diagram, to establish the validity of the notion that "A stable feedback control system may be made unstable either by increasing the forward path gain at constant value of dead time or by increasing the dead time situated on the forward path". Select two sets of K_C and t_{DL} values for this purpose by a 'trial and error' procedure, if required.

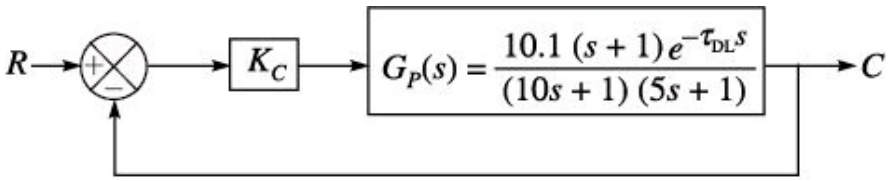


Figure 11.30 The block diagram of a closed loop system.

Solution: The process contains a gain, two poles, one zero and a dead time element. The constant parameters of the process are listed as:

$$K_P = 10.1;\ t_{P1} = 10;\ t_{P2} = 5;\ t_Z = 1.$$

The remaining process parameter t_{DL} and (P) controller gain K_C are adjustable. Two set of values of K_C and t_{DL} have been selected after a few trials.

K_C, dimensionless	0.2	0.8	3.2	12.8	51.2
x_{DL} , Arbitrary. time units	0.1	0.6	1.1	1.6	2.2

Selection of the two parameter values is such that the combination ($K_C = 3.2$, $t_{DL} = 1.1$) renders a limitingly stable condition of the system. Two sets of Bode Plots have been drawn in Figure a, for $K_C = 3.2$, and $t_{DL} = 0.1, 0.6, 1.1, 1.6, 2.2$, and in Figure b, for $t_{DL} = 1.1$ and $K_C = 0.2, 0.8, 3.2, 12.8, 51.2$. As the process gain, K_P, has a value of 10.1, it should be remembered that effective forward path become $10.1 \times K_C$ for any K_C setting.

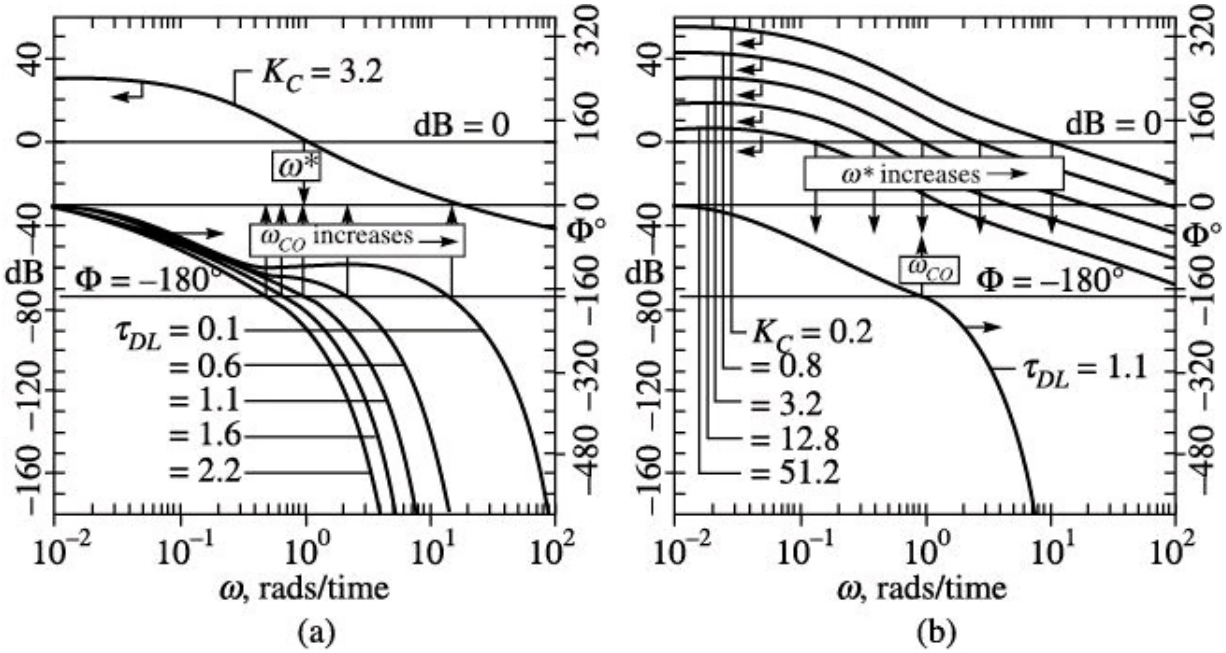


Figure 11.31 Two sets of Bode Plot for the system shown in Figure 11.30.

In both figures, increase of one parameter has gradually changed the stability measure from stable to limitingly stable and ultimately to unstable condition. This change of stability measure has been reflected in the shifting of relative position of w_{CO} with respect to w^* in Figure a, and that of w^* with respect to w_{CO} in Figure b. Thus, we have analytically established the aforesaid notion.

Have you noticed that from the definition of *GM* and *PM* and by working out the last example, we have set up an alternative definition of Bode Stability Criterion in terms of the two characteristic frequencies, w^* and w_{CO}.

It may be stated that: The closed loop system,

1. would be Stable if $w^* < w_{CO}$,
2. would be Limitingly Stable if $w^* = w_{CO}$,
3. would be Unstable if $w^* > w_{CO}$.

11.3.3 Nyquist Stability Criterion

To understand the fundamental considerations about this procedure of stability analysis, we have to refresh ourselves with certain topics that are part of the complex variable theory.

Mapping

If $f(x)$ be a function of x, then the graph $f(x)$ vs. x is said to be the mapping of x in $f(x)$. Each point $x_1, x_2, \ldots\ldots$ maps to corresponding values of $f(x_1), f(x_2), \ldots\ldots$, etc. This idea may be clearer from Figure 11.32.

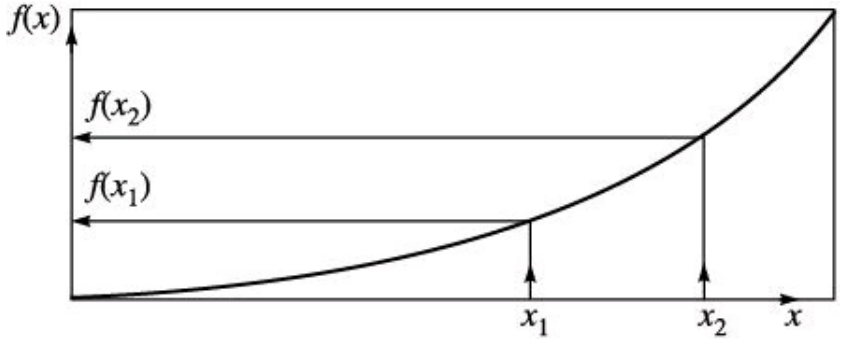


Figure 11.32 Mapping of a single dimension function.

This single dimensional translation becomes two dimensional if the function $f(u, v)$ has two dimensions. The independent variables and the dependent function require individual planes to describe any of their variations. This happens in case of complex variables. If we consider s to be the independent variable having real and imaginary parts, and $GP(s)$ be the dependent variable also complex, then any point in s-plane will have a corresponding point in $GP(s)$ plane depending upon the function $GP(s)$.

As an example let the complex function be:

$$G_P(s) = \frac{1}{2s+1}$$

and we take some points on the imaginary axis. Among the points A, B, C, O, D, E and F, O is at origin, A, B, and C are on the +ve part and D, E, and F are on the –ve part. Their coordinates are listed in the following table:

A	B	C	O	D	E	F
$0+i$	$0+0.5i$	$0+0.25i$	0	$0-0.25i$	$0-0.5i$	$0-i$

The translation of the first point A on the s-plane may be carried out to find the coordinate of the corresponding point A' on the $GP(s)$ plane as:

$$A' = \frac{1}{2(0+i)+1} = \frac{1}{1+2i} = \frac{1-2i}{(1+2i)(1-2i)} = \frac{1}{5} - i\frac{2}{5}$$

By similar procedure, the translated coordinates of other points have been found and listed in the following table:

A'	B'	C'	O'	D'	E'	F'
$0.2-0.4i$	$0.5-0.5i$	$0.8-0.4i$	1.0	$0.8+0.4i$	$0.5+0.5i$	$0.2+0.4i$

The s-plane points and their mapped points on $GP(s)$ plane have been shown in Figures 11.33(a) and (b).

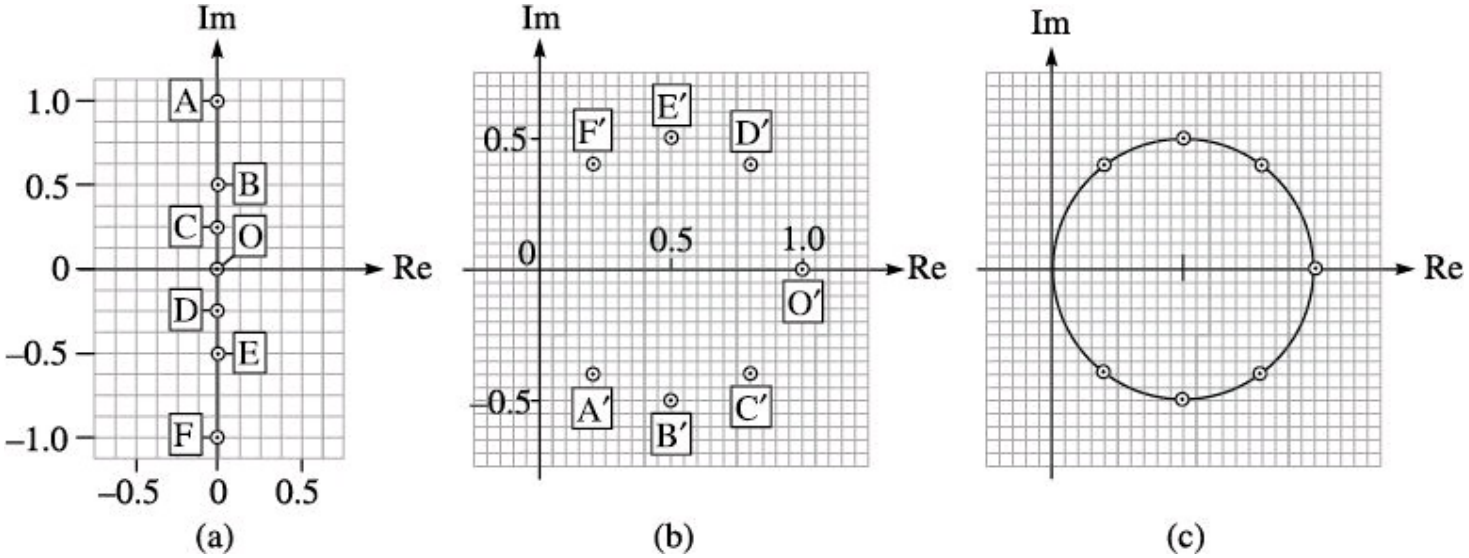


Figure 11.33 Mapping *s*-plane points on to $G_P(s)$ plane.

In Figure 11.33 (b), the *s*-plane points on the +ve part of the imaginary axis are mapped to obtain –ve imaginary values on the $G_P(s)$ plane, and the points lying on the –ve part of the imaginary axis of *s*-plane are translated to obtain +ve imaginary values on the $G_P(s)$ plane. That the points *A*□, *B*□, *C*□, *O*□, *D*□, *E*□ and *F*□ lie on a circle as shown in Figure 11.33(c) must have been appreciated as a foregone result of the earlier Example 11.1 of this chapter.

The Nyquist Plots given in examples 11.1 through 11.10 are actually the translation or mapping of the +ve part of imaginary axis through the functions $G_P(s)$ of the respective example. The difference is that the imaginary axis has been defined as *iw* instead of *iy* and we have used only the +ve part of the imaginary axis for drawing of Nyquist Plots. This has been done to obtain the response for +ve part of frequency variation from 0 to +∞. However, any of the translated points would lie on the circle of radius 0.5 and centre at 0.5 may be proved in an alternative way:

The case of points lying on the imaginary axis mapping into a part of circle of radius = ½ is of special interest to us. Let us prove that every point on the imaginary axis will map into such a circle by the function, $G_P(s) = 1/(2s + 1)$.

Equation of a circle of radius *R* and centre at origin, is $x^2 + y^2 = R^2$ under Cartesian coordinates. With centre at ½ and radius = ½, the equation of the circle is:

$$\left[\mathrm{Re}\{G_P(i\omega)\} - \frac{1}{2}\right]^2 + [\mathrm{Im}\{G_P(i\omega)\}]^2 = \left(\frac{1}{2}\right)^2 = \frac{1}{4} \qquad (11.41)$$

Representing the points lying on the imaginary axis by $s = iw$, where, *w* takes the values from –∞ to +∞. Using this representation $G_P(iw)$ becomes:

$$G_P(i\omega) = \frac{1}{1+2i\omega} = \frac{1-2i\omega}{1+4\omega^2} = \frac{1}{1+4\omega^2} - i\frac{2\omega}{1+4\omega^2} \qquad (11.42)$$

Substituting this result in Equation 11.41:

$$\left[\frac{1}{1+4\omega^2} - \frac{1}{2}\right]^2 + \left[\frac{-2\omega}{1+4\omega^2}\right]^2 = \left[\frac{1-4\omega^2}{2(1+4\omega^2)}\right] - \frac{4\omega^2}{(1+4\omega^2)^2} = \frac{1+8\omega^2+16\omega^4}{4(1+4\omega^2)^2} = \frac{1}{4} \qquad (11.43)$$

Hence, we have proved the statement. Thus, variation of w from 0 to ∞ in s-plane draws the lower half of the circle in $G_P(iw)$ plane by the function $G_P(s) = 1/(2s + 1)$ and this in fact is the Nyquist Plot for frequency response description of $G_P(s)$ that we obtain by replacing s by iw. Figure 11.34 is such a plot for any first order system. To bring the generality, wt values instead of w values have been labeled at various points on the locus, and the diameter of the semi-circle has been labeled as K_P the numerator of the first order mapping function instead of the numeric value of unity which is the specific value for the present example.

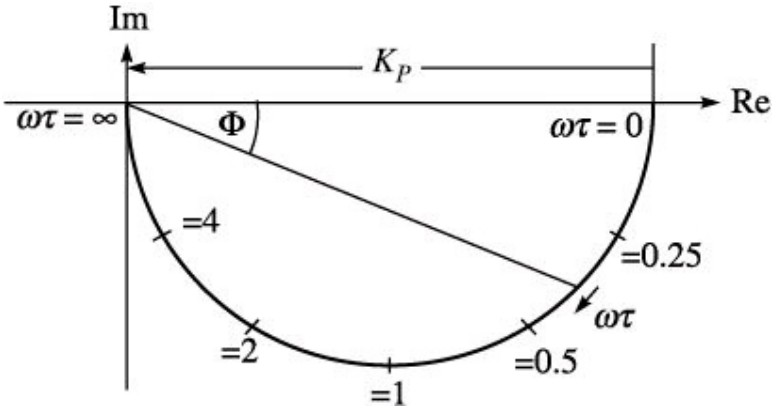


Figure 11.34 Mapping of +ve imaginary axis by a first order transfer function.

Before proposing the Nyquist Criterion, we need to know another property of complex variables called ***N = Z – P* Theorem**. Another consideration about the notation of the mapping function is that the forward path TF of a closed loop system is denoted by $G_P(s)$ and that of the feedback path is $G_M(s)$, so that the overall TF round the loop is $G_P(s)G_M(s)$, which in short form becomes $G_PG_M(s)$.

N = Z – P Theorem

Consider a function $G_PG_M(s)$ of complex variable having two zeros z_1 and z_2, and three poles p_1, p_2 and p_3, so that $G_PG_M(s)$ may be written as:

$$G_P G_M(s) = \frac{(s - z_1)(s - z_2)}{(s - p_1)(s - p_2)(s - p_3)} \tag{11.44}$$

The phase angle or argument of $G_P G_M(s)$ is:

$$\angle[G_P G_M(s)] = \angle\left[\frac{(s - z_1)(s - z_2)}{(s - p_1)(s - p_2)(s - p_3)}\right]$$

$$= \angle(s - z_1) + \angle(s - z_2) - \angle(s - p_1) - \angle(s - p_2) - \angle(s - p_3) \tag{11.45}$$

These zeros and poles are called **singular points** or **singularities** of the function $G_P G_M(s)$. They are located in Figure 11.35 inside a contour C that encircles all the singularities. Let c be a point on the contour C. The angle between the line drawn from z_1 to c' and the horizontal line is the argument of $(s - z_1)$. If c moves completely around the contour, q_{Z1} will increase by $2p$ radians. Hence, arg[$G_P G_M$(s)] will increase by $2p$ radians per each zero and reduce $2p$ radians per each pole. Thus a plot of $G_P G_M(s)$ in the $G_P G_M$-plane must encircle the origin once as c goes completely around the contour, because there are two zeros and three poles in the TF of Equation 11.44. The result in abbreviated form is the theorem: $\boldsymbol{N = Z - P}$, where, $\boldsymbol{N}$ = Number of Encirclement of the Origin;
$\boldsymbol{Z}$ = Number of Zeros; $\boldsymbol{P}$ = Number of Poles.

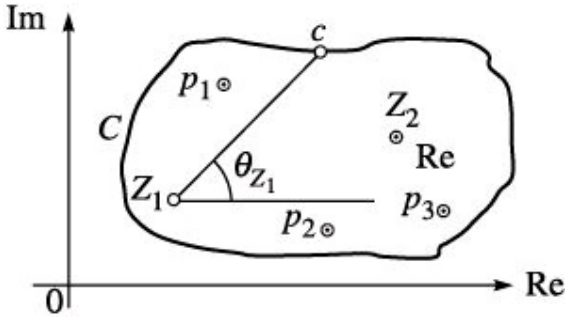


Figure 11.35 Contour in s-plane encircling all singularities of $G_P G_M(s)$.

The point that is encircled is the value of the complex function at zeros. Obviously this value is located at the origin of the $G_P G_M(s)$ plane. Now let a unity feedback path be built to render the open loop TF to a closed loop system (From the treatment that has been described in concern with Bode stability criterion, we may conclude that non-unity feedback systems may also be converted to unity feedback configurations.). Thus, the general expression of open loop TF is $G_P G_M(s)$, and the corresponding characteristic equation is:

$$G_P G_M(s) + 1 = 0 \tag{11.46}$$

or,
$$G_P G_M(s) = -1 \tag{11.47}$$

Now we shall apply the above results to find whether there are any roots of Equation 11.46 in the RHS of the *s*-plane. To do this, the following steps may be followed:

1. Define a contour in *s*-plane that completely encircles the RHS of *s*-plane as shown in Figure 11.36. The imaginary axis is described in polar form as $s = iw$, where w goes from $-\infty$ to $+\infty$ to describe the imaginary axis and an infinitely large radius, R that describes a half circle to completely encircle the right half of *s*-plane.

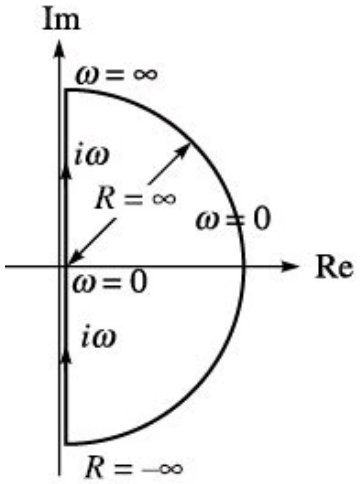


Figure 11.36 The contour encircling all singularities of the right-half of *s*-plane.

Choose the mapping function to be $G_P G_M(s)$, i.e. the open loop TF of the given control system.

For one complete travel of the *s*-plane contour from $w = 0$ and back, if there is N number of encirclement of origin in $1 + G_P G_M(s)$ plane, or if there is N number of encirclement of $(-1, 0)$ point in the $G_P G_M(s)$ plane according to Equation 11.47, the closed loop system have N roots in RHS of *s*-plane.

The different configurations of encirclement have been explained in Figure 11.38.

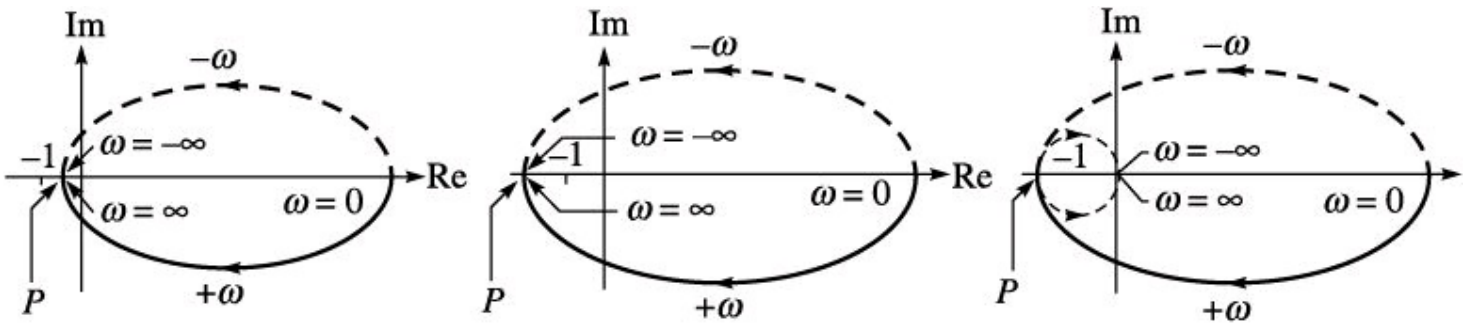


Figure 11.37 (a) Closed loop stable. (b) Closed loop unstable-one root in RHS. (c) Closed loop unstable-two roots in the RHS.

In Figure 11.37(a), the Nyquist or polar Plot goes without encircling the point –1, 0, or the intersection point P of the polar plot and the real axis is at right of point –1, 0 and we may conclude the system is closed loop stable.

In Figure 11.37(b), the polar plot encircles the point –1, 0 once, i.e. point P is at left of –1, 0 and we may conclude that the system is closed loop unstable and the characteristic equation of the system has one root in the RHS of *s*-plane.

In Figure 11.37(c), the polar plot encircles the point –1, 0 twice, again point P is at left of –1, 0 and we may conclude that the system is closed loop unstable and the characteristic equation of the system has two roots in the RHS of *s*-plane.

Now the **Nyquist Stability Criterion** may be proposed as:

If the Polar plot of the open loop transfer function $G_PG_M(s)$ of a given system encircles – 1, 0 point in the $G_PG_M(s)$ plane in a clockwise fashion, as the angular frequency w goes from zero to infinity, system is closed loop unstable. Also the number of such encirclements is equal to the roots of the characteristic equation in the right half of *s*-plane.

A corollary of this definition is that a counter-clockwise encirclement of – 1, 0 point indicates the presence of a +ve zero in the process TF.

In the following examples, we shall show that:

1. As w varies from 0 to $+\infty$ or from 0 to $-\infty$, the solid line part and the dotted line part of the map are drawn respectively as in Figure 11.37 and that these two parts are in fact 'mirror-image' to each other. The solid line part has been already introduced as the Nyquist Plot as a description of frequency response of dynamic systems.
2. As a semi-circle of $R = \infty$ is traveled in *s*-plane, it maps into the point P on the $G_PG_M(s)$ plane.
3. Encirclement of (–1, 0) point is easily determined by counting the net number of rotations made by the vector from the (–1, 0) point to the Nyquist Plot as the head of the vector traces the entire contour.

EXAMPLE 11.26 Determine the value of K_C for which the control system will remain stable. Its open loop TF is given by:

$$GH(s) = \frac{K_C}{(s+1)(s/2+1)(s/3+1)}$$

Solution: The Nyquist diagram for $G_PG_M(s)$ with $K_C = 1$ has been drawn in Figure

11.38 as the solid line curve cuts the –ve part of the real axis at –0.1. Hence, instability will first occur if K_C is raised to 1/0.1 = 10. This is true because raising K_C to 10 is equivalent to multiplying all magnitudes by 10, which would shift the left portion of the contour to the dashed curve that just encircles the point (–1, 0). Hence for stable operation, K_C must be less than 10.0.

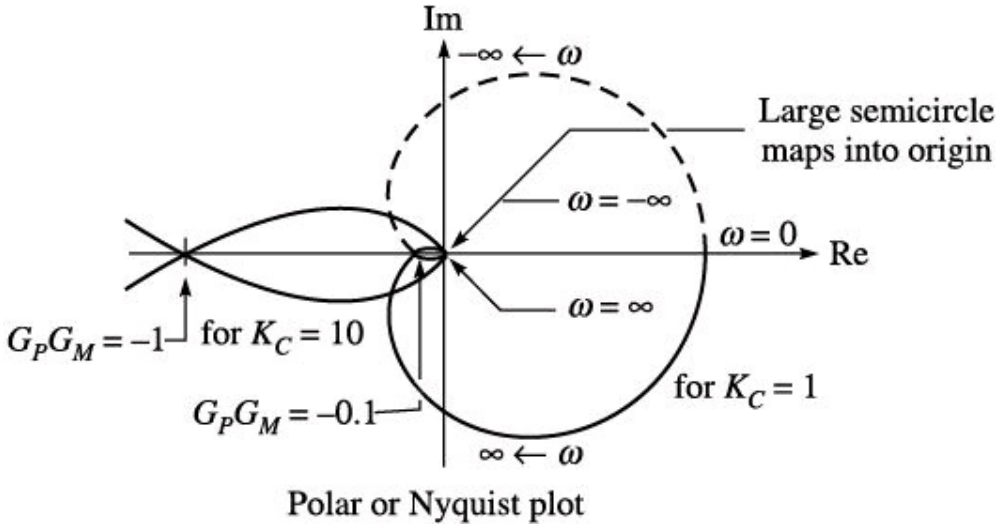


Figure 11.38 Stability analysis and determination of K_U by Nyquist method.

Large Semi-Circle in *s*-Plane Maps into the Origin of $G_P G_M(s)$ Plane: On the large semi-circle of radius $R = \infty$; $s = Re^{iq}$. R will go to ∞ and q takes on values from $+p/2$ through 0 to $-p/2$. Substituting into $G_P G_M(s)$:

$$G_P G_M(s) = \frac{K_C}{(Re^{i\theta} + 1)(Re^{i\theta/2} + 1)(Re^{i\theta/3} + 1)} \tag{11.48}$$

as R goes to ∞ $G_P G_M(s)$ □ 0 as the semicircle part of the contour is described.

This procedure is applicable for systems open loop TF, which contain greater number of poles than zeros. Such a condition is satisfied in practically all control system applications.

Systems with Integrator in Their Open Loop TF

Systems with an S (or any higher powers of s) in the denominator of open loop TF require that the s-plane contour to be modified a little, because if the contour passes through origin of s-plane, at origin the mapping function $G_P G_M(s)$ will have an infinite value due to presence of the integrator. The modified contour is shown in Figure 11.39. To avoid the origin, there is a small semicircular detour around the origin of the s-plane.

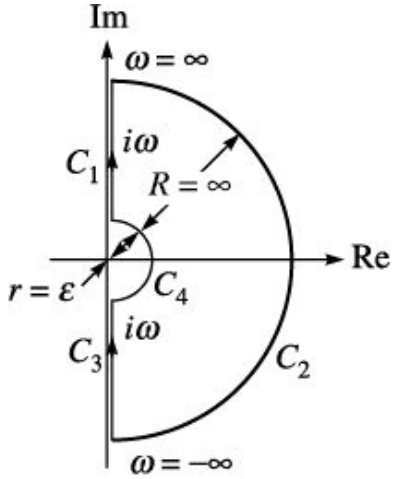


Figure 11.39 Modified contour on the *s*-plane for systems having integrator in their TF.

The traverse may begin as w goes from r to $+\infty$ (contour C_1), then along the large semicircle of infinite radius $R = \infty$ (contour C_2) as q goes from $+p/2$ to $-p/2$. Next is the path from $w = -\infty$ to r (contour C_3) and lastly the small semicircular part is traversed as q goes from $-p/2$ to $+p/2$ radians (contour C_3). If the radius r of the small semicircle approaches an arbitrarily very small value e, the total contour effectively envelops the entire right half of *s*-plane.

The treatment for evaluating the mapping of a complex function for the modified contour has been presented in the next example.

EXAMPLE 11.27 A unity gain process contains two poles and an integrator in its TF, with the following time constant values: $t_{P1} = 20$, $t_{P2} = 40$ time units. Construct the Nyquist Plot for the system. By trial, find the controller gain K_C for limiting stability condition. Luyben has analyzed the system considering a general form of the TF of this system given by Equation 11.49

$$G_P G_M(s) = \frac{K_C}{s(T_1 s + 1)(T_2 s + 1)} \tag{11.49}$$

He has proposed the following relations to define the system's limiting stability condition:

$$K_U = -\frac{T_1 + T_2}{T_1 T_2} \tag{11.50a}$$

and

$$\omega_U = \frac{1}{\sqrt{T_1 T_2}} \tag{11.50b}$$

Check the validity of Equations 11.50(a) and 11.50(b) in context to this problem using the time constants stated in this problem.

Solution: The straitened forward path TF of the system is:

$$G_P G_M(s) = \frac{K_C}{s(20s+1)(40s+1)}$$

The TF has a pole at the *s*-plane origin, hence the contour has to be modified by adding a small semicircle of radius *r* at the origin. As the contour is traced, the origin is avoided by riding on the added semicircular part. The corresponding mapping is obtained for the small semicircle part by making $r \to 0$, and the large semicircle approaches to cover the total RHS of the *s*-plane as $R \to \infty$.

We shall examine the $G_P G_M(s)$ plane mapping as different parts of the *s*-plane contour are traversed.

1. C_1 contour, $s = iw$ as w goes from r to R, as $r \to 0$ and $R \to \infty$.

$$G_P G_M(i\omega) = \frac{K_C}{i\omega(1+i\omega T_1)(1+i\omega T_2)} = \frac{60K_C\omega - iK_C(1-800\omega^2)}{3600\omega^3 + \omega(1-800\omega^2)^2}$$

which happens to be the Nyquist Plot as shown in Figure 11.40.

2. C_2 contour, $s = Re^{iq}$, as R goes from $0 \to \infty$,

$$G_P G_M(i\omega) = \frac{K_C}{Re^{i\theta}(1+20Re^{i\theta})(1+40Re^{i\theta})} \quad \text{or} \quad \lim_{R\to\infty}[G_P G_M(s)] = \lim_{R\to\infty}\left[\frac{K_C}{800R^3}e^{i\theta}\right] = 0$$

That is, C_2 maps into the origin of the s-plane.

3. C_3 contour, $s = iw$, where w goes from $-R$ to $-r$, as $r \to 0$ and $R \to \infty$,

The polar plot of $G_P G_M(s)$ is obtained as the mirror image of that obtained for C_1.

4. C_4 contour, as r goes to 0 and q goes from $-p/2$ to $+p/2$ radians and $G_P G_M(s)$ becomes:

$$G_P G_M(i\omega) = \frac{K_C}{re^{i\theta}(1+20re^{i\theta})(1+40re^{i\theta})} \quad \text{or} \quad \lim_{r\to 0}[G_P G_M(s)] = \lim_{r\to 0}\left[\frac{K_C}{r}e^{-i\theta}\right] = \infty$$

Thus, C_4 contour maps into a semicircle in $G_P G_M(s)$ plane that has a radius which goes to infinity as $r \to 0$, and a phase angle that goes from $+p/2$ through to $-p/2$ as shown in the figure.

Few trials are made to find the limiting stability condition. Results of the last three trials are shown in Figure 11.41. The limiting value of K_C and w are found as, $K_U = 0.075$ and $wU = 0.03535$ rads/time. By using Equations 11.50(a) and 11.50(b), value of these parameters are found as:

$$K_U = \frac{40+20}{40\times 20} = 0.075 \text{ and } \omega_U = \frac{1}{\sqrt{40\times 20}} = 0.035355339 \text{ rads/time.}$$

Hence, the results corroborate the proposed relations.

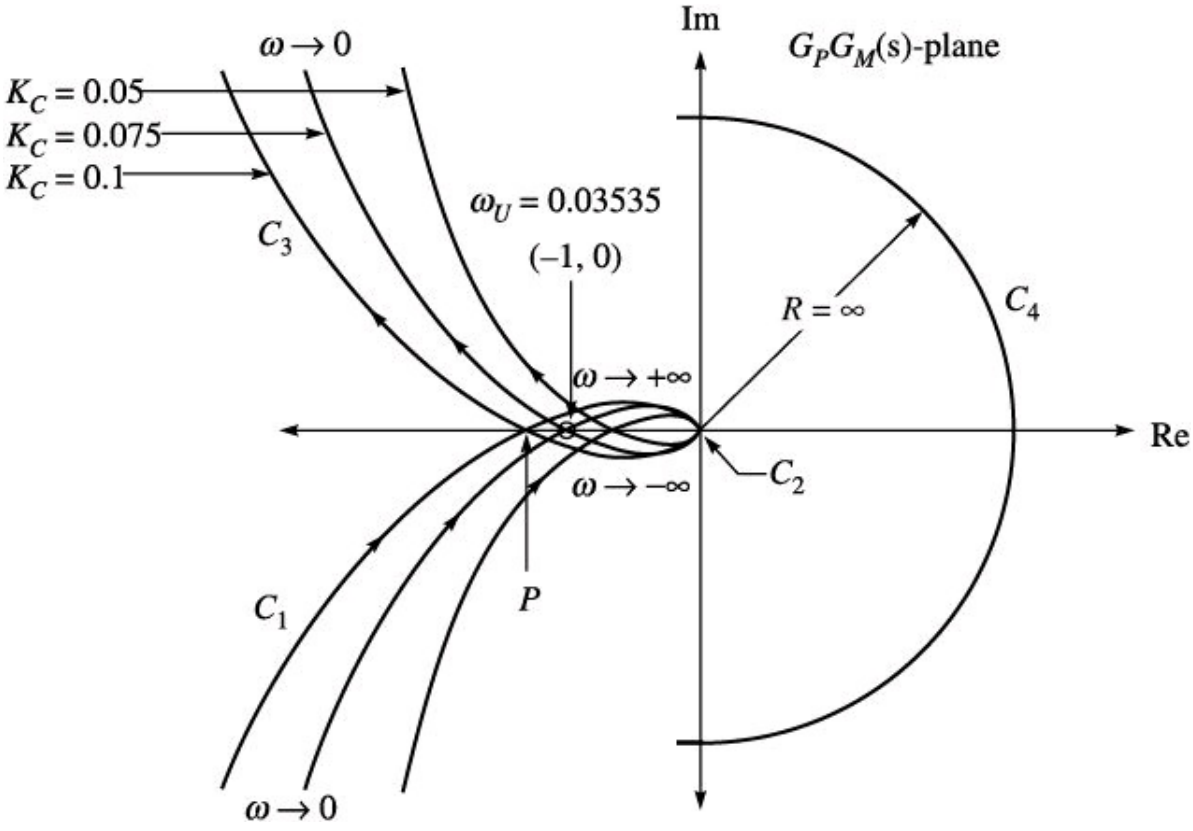


Figure 11.40 Nyquist Plot for a system with two poles and an integrator.

In the next example, Nyquist Plot of a system, having an unstable zero and two poles, has been analyzed. As we see that the polar plot for *w* variation from 0 to – ∞ is the reflection of the plot for *w* variation from 0 to + ∞, from now on we shall be using only the 0 to + ∞ part or Nyquist Plot for stability analysis.

EXAMPLE 11.28 The open loop and closed loop TFs of a system are given as:

$$G_P(s) = \frac{K_P[s/3-1]}{(10s+1)(2s+1)} \quad \text{and} \quad G_PG_M(s) = \frac{K_CK_P[s/3-1]}{(10s+1)(2s+1)}$$

By drawing the Nyquist Plot and few trials establish the value of ultimate gain for the closed loop system.

Solution: Starting with an initial value of $K_CK_P = 1$ after a few trials the Nyquist Plot that exactly rides over the –1, 0 point has been obtained for a K_CK_P value of 3.6. Thus, this is the ultimate gain value. As this is the product of process and controller gains, to obtain the value of K_U, K_P must be known. The result of the analysis has been illustrated in Figure 11.41.

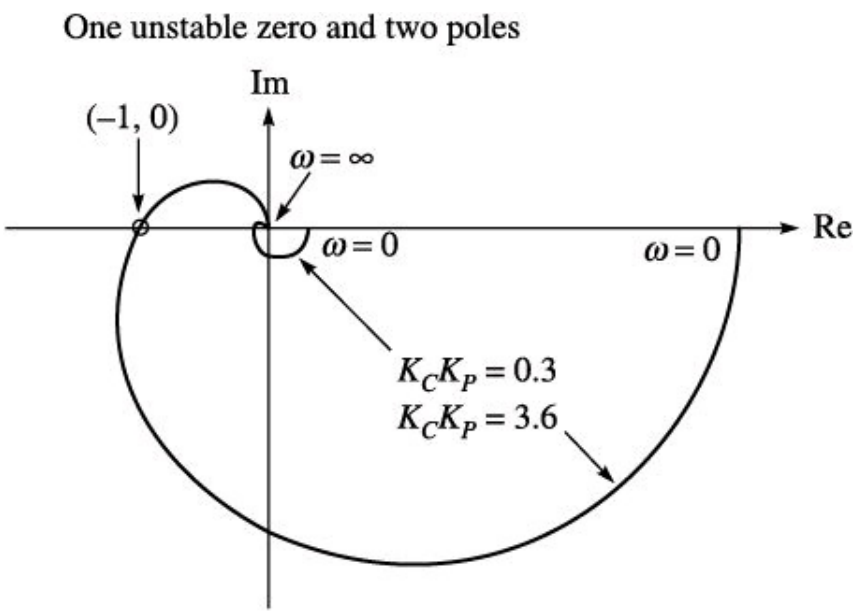


Figure 11.41 Nyquist Plot for a system with an unstable zero.

Probably from the introduction of lead-lag element we are commenting that existence of a differentiator, or lead or zero in the TF improves the stability character of a system. But what is the quantitative estimate of this improvement, and whether this amount depends on the placement of lead or zero element relative to the other elements in the TF with respect to their characteristic frequencies? To address these issues, analysis of a system containing one zero and four poles has been done.

EXAMPLE 11.29 The open loop TF of the process contains four poles only:

$$G_P(s) = \frac{1}{(\tau_{P1}s+1)(\tau_{P2}s+1)(\tau_{P3}s+1)(\tau_{P4}s+1)}$$

A (PD) controller has been added to this process so that the closed loop TF becomes:

$$G_P G_M(s) = \frac{K_C(\tau_Z s+1)}{(\tau_{P1}s+1)(\tau_{P2}s+1)(\tau_{P3}s+1)(\tau_{P4}s+1)}$$

The time constant of the lead or zero element t_Z is equated to the derivative time t_D of the controller to impart more generality of the treatment. The question is, could the derivative time of the controller be adjusted to produce the best stability character of the closed loop system.

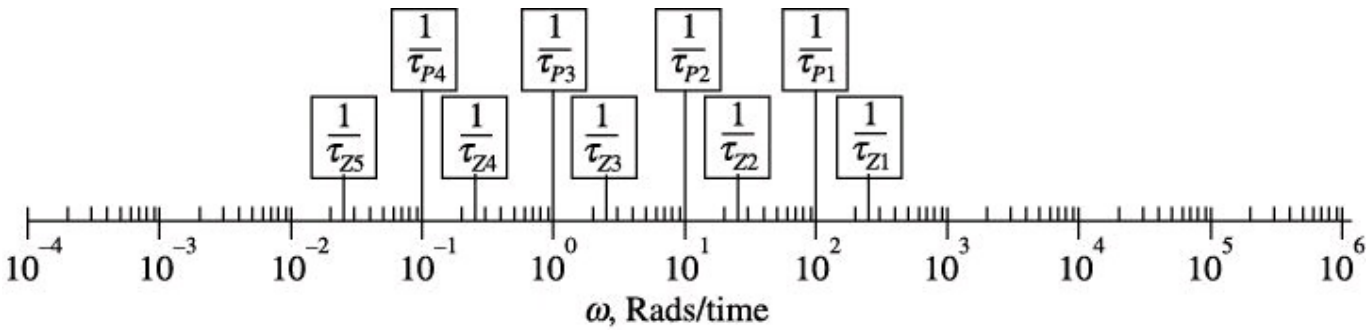


Figure 11.42 Alternative placement of the zero on the frequency scale.

The time constants of the four poles are be taken as: $t_{P1} = 0.01$, $t_{P2} = 0.1$, $t_{P3} = 1$, $t_{P4} = 10$. To examine the effect of placement of zero, five alternative values of t_Z starting before the largest t_P up to and beyond the smallest t_P with three in between values of each t_P have been applied. These values are $t_Z = 0.004, 0.04, 0.4, 4, 40$. In frequency scale the placement of characteristic frequencies of the four poles and the five zeros is in Figure 11.42.

Solution: The Nyquist Plots of the open loop system and five values of t_Z are drawn, and with a few trials, the K_U for the open loop situation and each value of t_Z has been established. To confirm the value of K_U, plots with slightly higher and lower values of K_C than K_U are also drawn for each of the six cases. The enlarged locus near the –1, 0 points are shown in Figure 11.43.

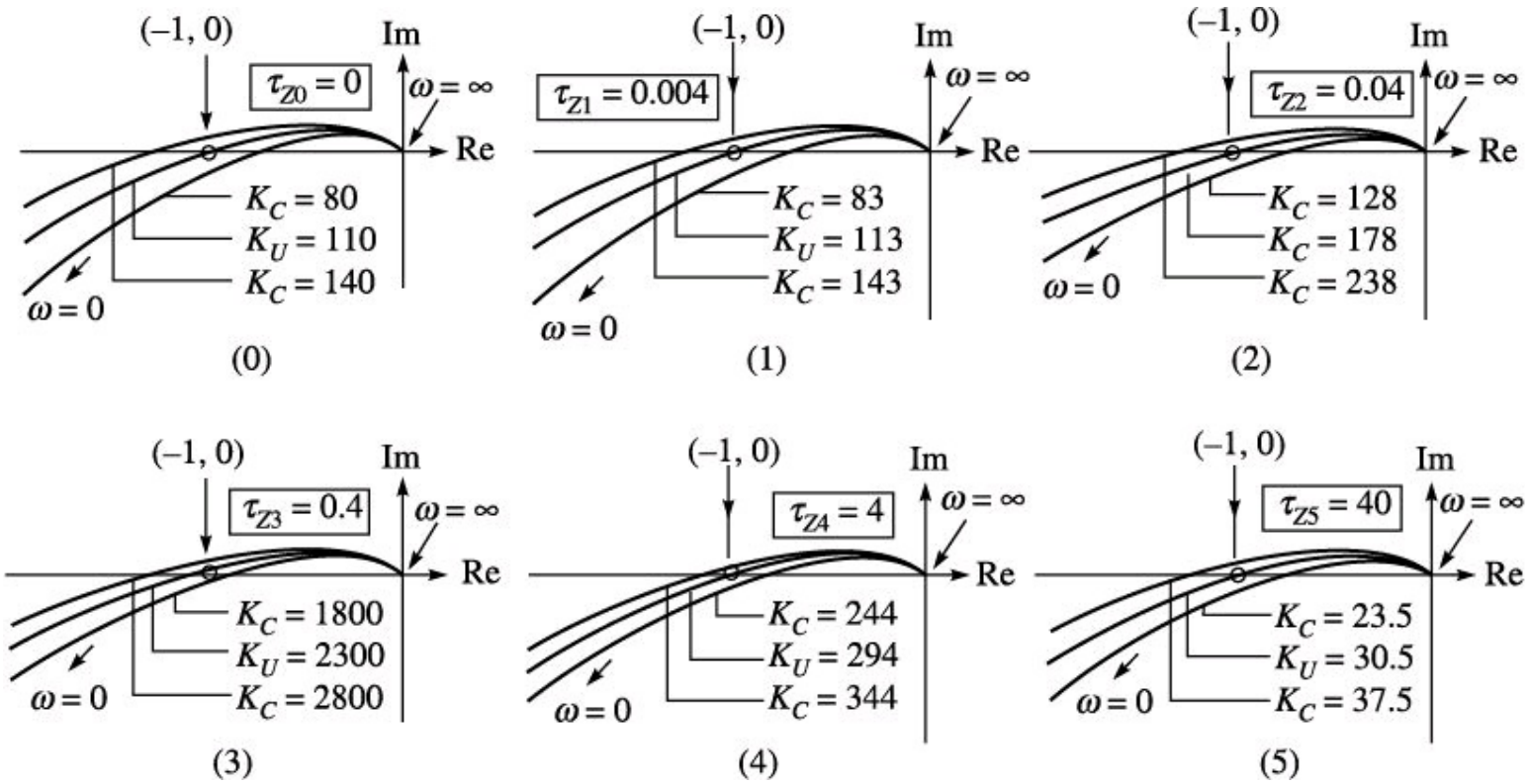


Figure 11.43 Shift of K_U depending on the derivative time value.

Let us prepare a list of K_U vales corresponding to the time constants of the zero element or derivative time:

Figure No.	B(0)	B(1)	B(2)	B(3)	B(4)	B(5)
$x_Z =$	0	0.004	0.04	0.4	4	40
$K_U =$	110	113	178	2300	294	30.5

The results show that the relative improvement increases as the t_Z is progressively made higher to occupy location between pairs of higher t_Ps, up to a particular pair of t_P. After that in B(4) for $t_Z = 4$, the relative improvement decreases, but K_U is still

higher than that obtained in B(0) and B(5) for $t_Z = 40$, this decrease is so much that K_U becomes lower than the K_U obtained in B(0). To address this problem we have resorted to tracing the Bode Plots for the open loop and five closed loop cases.

You should note that the label *GM* has been put in a slightly wrong way that differs from the definition of *GM*. However, this has been done to express the measure of how far a closed loop system is from the stability limit in terms of dB.

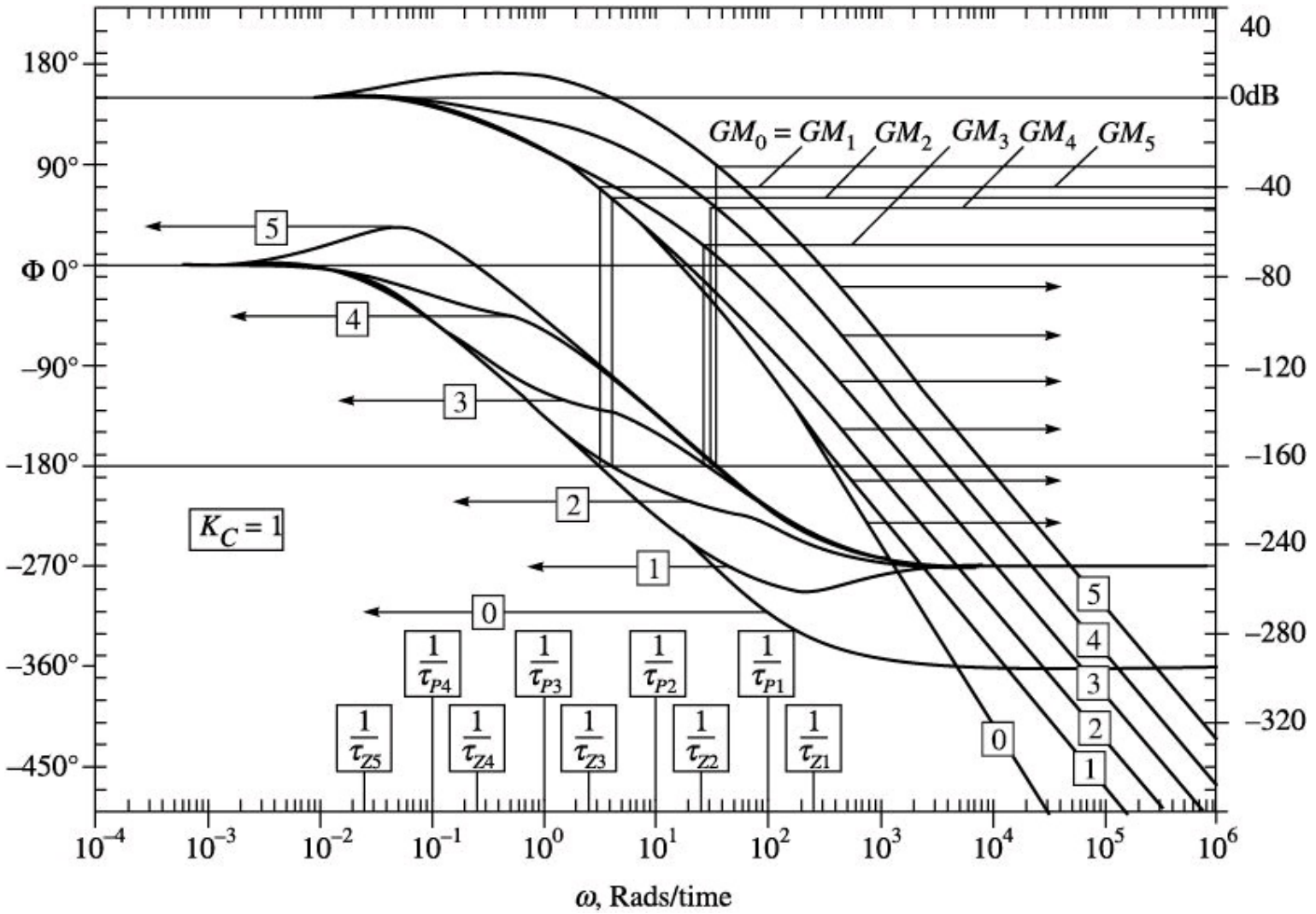


Figure 11.44 Complete frequency response of the system.

The 'bulge' that appears in the □ curves due the presence of the zero in the closed loop TF recedes with the increase of t_Z. This affects in increasing the w_{CO} but in an irregular manner because the effect is maximum when the 'bulge' near the – 180° horizontal. When this effect goes so far behind that there is a progressively lower effect on the value of w_{CO}, the improvement on stability character decreases, as in case (4).

Whereas, considering the dB curves presence and increase of t_Z all the plots starts from 0 dB except for case (5) hence, for cases (1) to (4) there is no increase of w^* as all other curves starts from 0 dB value. This increase of w^* deteriorates the stability measure so much that it decreases below that obtained in case (0) i.e. without the (PD)

control action.

While discussing about the limitations of Bode stability criterion it has been mentioned that the method fails for systems that gives a □ curve with multiple intersections with the –180° horizontal. In Example 11.21, the phase plot of such a system was demonstrated. The next example has been designed to check whether Nyquist method of analysis may answer such a problem. The example also exposes an interesting phenomenon in concern with such systems.

EXAMPLE 11.30 The open loop process contains four poles and the closed loop system contains four poles, two zeros and an integrator as if by putting a (PID) controller in the loop.

$$G_P(s) = \frac{K_P}{(\tau_{P1}s+1)(\tau_{P2}s+1)(\tau_{P3}s+1)(\tau_{P4}s+1)}$$

and,
$$G_P G_M(s) = \frac{K_C K_P(\tau_{Z1}s+1)(\tau_{Z2}s+1)}{s(\tau_{P1}s+1)(\tau_{P2}s+1)(\tau_{P3}s+1)(\tau_{P4}s+1)}$$

The following parameter values are used for analysis:

$$K_P = 1,\ t_{P1} = 0.9,\ t_{P2} = 1.1,\ t_{P3} = 90,\ t_{P4} = 110,\ t_{Z1} = 9,\ t_{Z2} = 11$$

Solution: At first the Bode plot of the closed loop system has been constructed and it is shown in Figure 11.45.

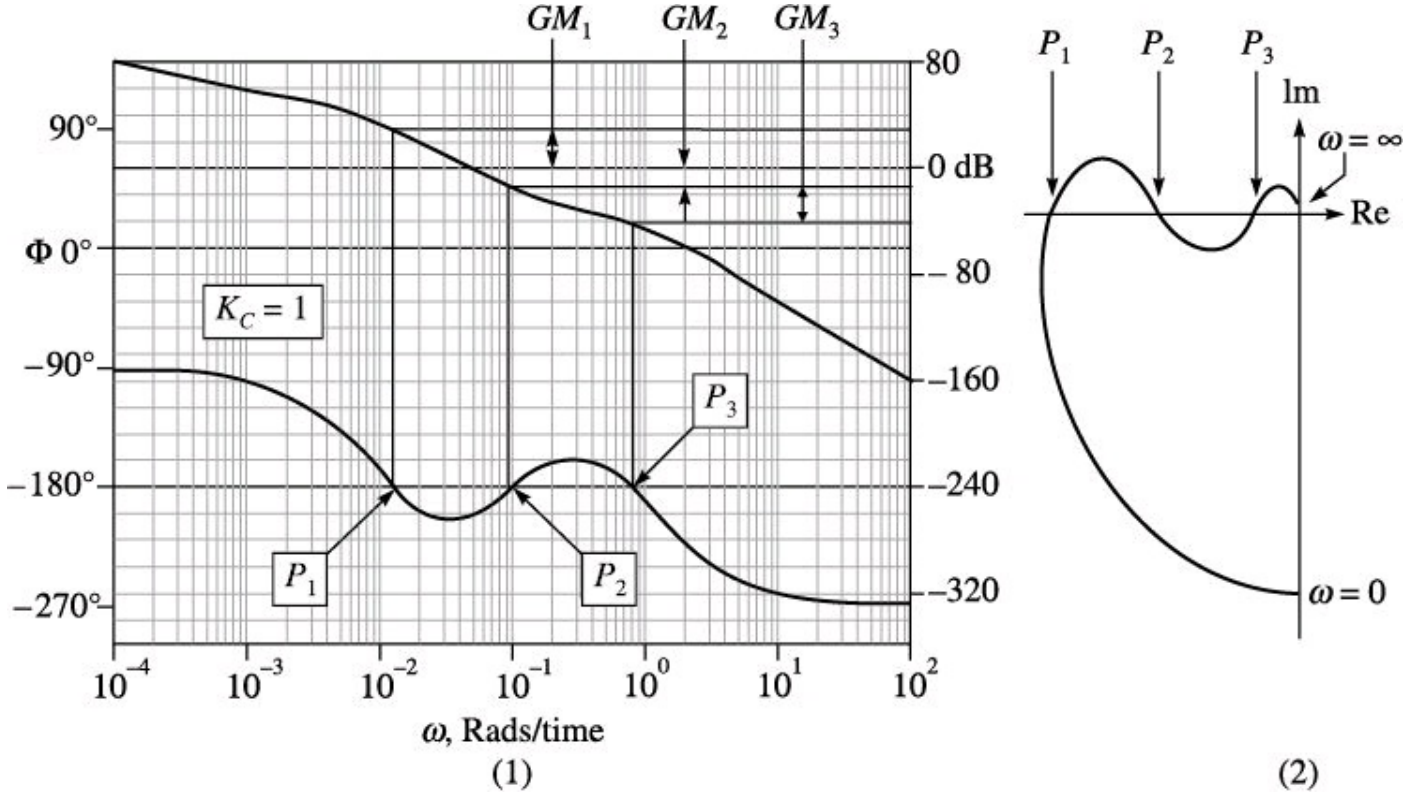


Figure 11.45 Bode Plot of a conditionally stable system.

To investigate the stability aspect corresponding to the three cutting points P1, P2, P3 of the □ locus with the –180° horizontal, the Nyquist Plot (not to scale) has been

shown in
Figure 11.45. The points P1, P2 and P3 in Figures 11.45 and 11.46 have identical significance.

That the curve starts at very low frequencies with a – 90° phase angle (magnitude vector direction), and ends up with – 270° phase angle (magnitude vector direction) is confirmed from the closed loop TF. This would be subsequently verified that the qualitative shape tallies with the actual plot, but Figure 11.45 has been drawn not according to scale because the vector magnitude obtained at low frequencies are so huge compared to that at high frequencies that it is impossible to depict them with respect to a common scale. To establish the stability measure for the points P1, P2 and P3, Nyquist Plots have been drawn in Figure 11.46 to find the K_C values required to shift the points P1, P2, P3 to the location of –1, 0 point on the real axis respectively in (1), (2) and (3). The obtained K_C values are listed below.

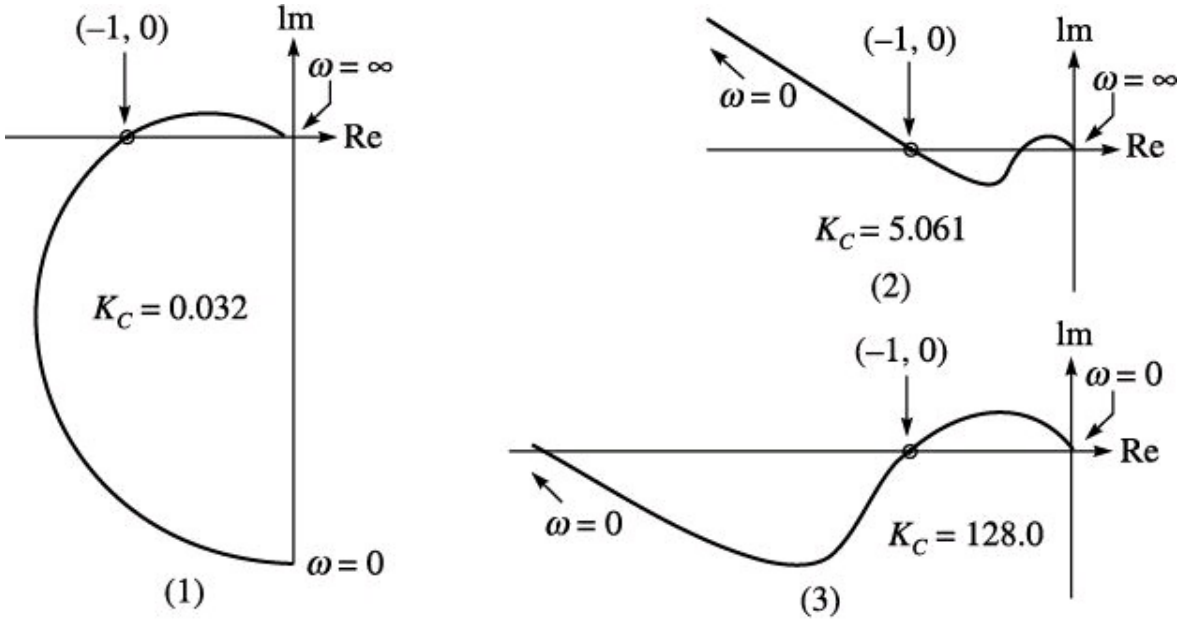


Figure 11.46 Nyquist Plot of a conditionally stable system.

For point P1	For point P2	For point P3
$K_C = 0.032$	$K_C = 5.061$	$K_C = 128$

From the analysis it appears that the closed loop system is stable if $K_C < 0.032$, it is unstable if $0.032 < K_C < 5.061$, again the system remains stable if $5.061 < K_C < 128$, and finally it becomes unstable if $K_C > 128$. This is a very interesting result in the fact that the system becomes unstable at lower K_C, again becomes stable within an intermediate range and becomes stable again at a higher value of controller gain. You must have appreciated that this happens due to the multiple intersections of the curve with the –180° horizontal.

This phenomenon is known as **Conditional Stability,** and the systems that show this property are called **Conditionally Stable Systems**.

In this chapter we have learnt about obtaining the dynamics and analyzing the stability of a process in frequency domain. The next step would be to know, how a control system may be designed for a given process also through frequency response. This topic is discussed in Chapter 8 of *Advanced Process Dynamics and Control*.

(*Frequency response plots in this chapter largely are obtained by executing computer programmes described in the later part of Chapter 14. The reader should also try this route for problem solving.)*

EXERCISE PROBLEMS

11.1 Prepare Bode, Nyquist and Nichol's Plot for the open loop TF:

$$G_P G_M(s) = K_C \left(1 + \frac{1}{s}\right)\left[\frac{4s+1}{(2s+1)(5s+1)}\right]$$

(a) Find the ultimate gain and frequency

(b) From value of K_C from Nyquist stability criterion for which the closed loop system will be limitingly stable.

11.2 Prepare a Nyquist Plot for the open loop TF:

$$G_P G_M(s) = \frac{K_C}{s(s+1)}$$

and find whether the system will be closed loop stable for $K_C = 1$, and for $K_C = 10$.

11.3 For each of the following transfer functions, sketch the dB vs. w as an asymptotic Bode Plot. For each case find the actual AR, dB and □ at $w = 10$.

(a) $\dfrac{80}{(8s+1)(2s+1)}$ (b) $\dfrac{12s}{(s+1)(0.5s+1)}$

(c) $\dfrac{(s+1)}{(0.5s+2)(s+1)^2}$ (d) $\dfrac{(s-1)}{(0.2s+1)(2s+1)}$

11.4 For each of the following transfer functions sketch the dB vs. w, and □ vs. w as an exact Bode Plot. (Choose the range of w to cover all the characteristic frequencies of the system, and use 12 to 15 frequency values within the range for each plot.)

(a) $\dfrac{6.5}{2s+1}$ (b) $e^{-2.5s}$ (c) $5\left[1+\dfrac{2}{s}\right]$

(d) $\dfrac{2.5}{(s+2)^2}$ (e) $\dfrac{5}{s^2+s+2}$ (f) $2.8\left[1+\dfrac{1}{4s}+s\right]$

11.5 In Examples 11.8, 9 and 10, the frequency response plots are obtained from composite form of $GP(iw)$, though the TFs contain more than one dynamic blocks. Draw the same figures considering the series combination of individual blocks as illustrated in Example 11.11.

11.6 The asymptotic dB vs. w plot as a part of the Bode diagram for a process has been given in Figure P11.1. The process has been identified as a minimum phase system. Find its transfer function.

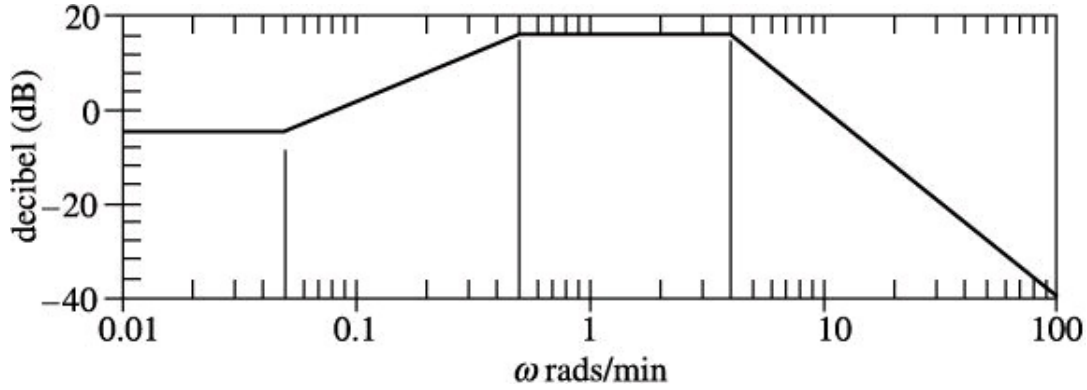


Figure P11.1

11.7 The amplitude ratio AR vs. w, for a process has been shown in Figure P11.2 as the asymptotic plot. The phase angle plot shows that the system is a non-minimum phase type. At, w = 1.1 rad/min., the phase angle □ was equal to – 250 degrees. Find the process TF containing gain, time constants, and dead time, if any.

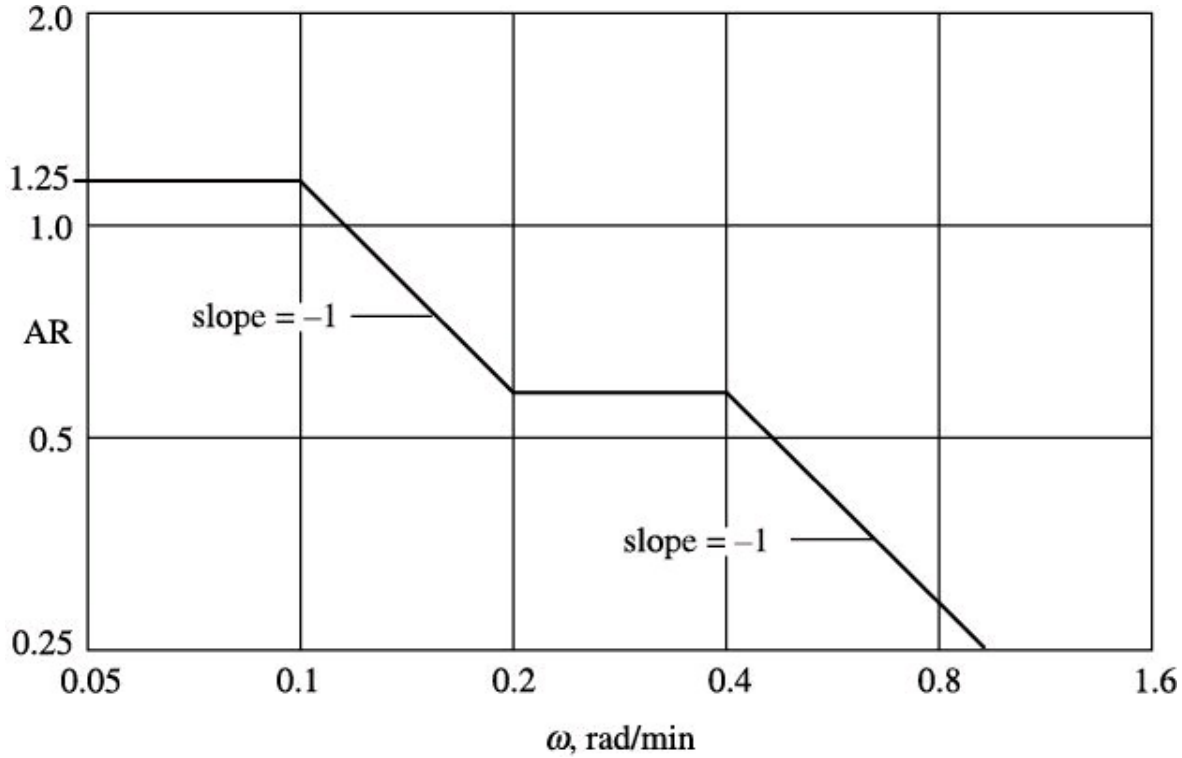


Figure P11.2

11.8 The asymptotic dB vs. w plot as a part of the Bode diagram for a process has been given in Figure P11.3. The process has been identified as a minimum phase system. Find its transfer function.

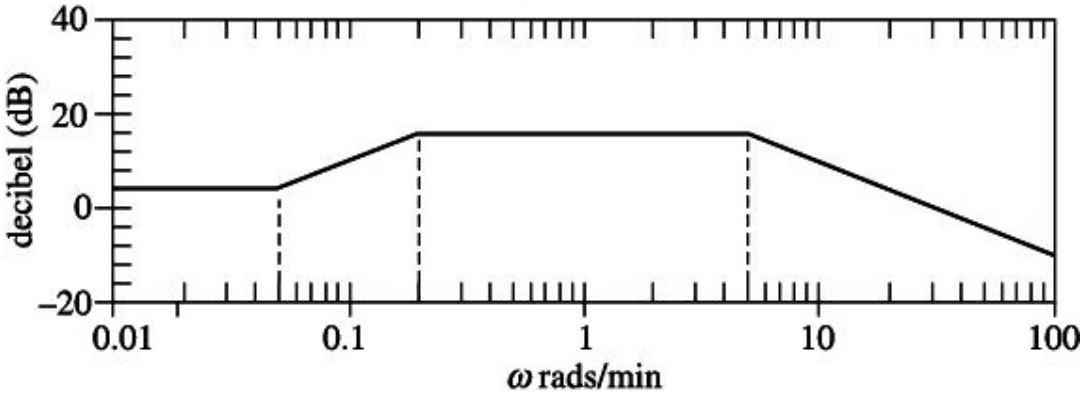


Figure P11.3

11.9 As we had promised in the introductory part of this chapter, let us try to design the transfer function of an acoustic system (very justifiably electronic engineers call them audio filters), that will subdue the instrumental interference and unmask the golden voice of Sri Hemanta Mukherjee. The asymptotic *w* vs. dB specification of the filter is as follows:

A. From very low frequency to 316 Hz: dB = – 20.

B. From 316 Hz to 3160 Hz: dB increases from – 20 to + 20.

C. From 3160 Hz to 10000 Hz: dB is constant at +20.

D. From 10000 Hz to 20000 Hz: dB decreases from + 20 dB to – 20.

E. Beyond 20000 Hz, up to 35000 Hz (the end of audio range) dB remains constant at –20.

11.10 The frequency response of a first order process is to be determined experimentally. How many cycles are required before the *AR* is within 5% of the stable value? How many minutes are required if the process time constant is 10 min?

11.11 A control loop has four first order systems in series with time constant $t_1 = 8$, $t_2 = 4$, $t_3 = 2$, $t_4 = 0.1$ min. Calculate *w* and K_U. If controller gain K_C is set to $K_C = 0.45\ K_U$, find the phase margin.

11.12 A unity gain process contains three poles in its TF, with the following time constant values: $t_{P1} = 0.75$ s, $t_{P2} = 12$ s and $t_{P3} = 0.25$ s. The process is controlled by a (PI) controller with an integral time value, $t_I = 3$ s. Analyze the stability condition of the closed loop system for controller gain values $K_C = 0.25$, 0.5 and 0.75.

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Engineers, McGraw-Hill, pp. 401, 1973.

[2] Coughanowr, D.R. and Koppel, L.B., *Process System Analysis and Control*, McGraw-Hill Inc., 1965.

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12

Process Identification (Experimental Methods of Dynamic Model Development)

In Chapter 2, we have discussed about the analytical way of process model development that is the fundamental exercise to develop an unsteady state principal entity balance by using only ideal physical and chemical laws and equations. This procedure gave forth a single or a set of differential equations as the dynamic characteristic of the process. In this chapter we would be exposing the experimental way of dynamic model building. These methods are useful where the analytical method shows its limitation due to certain dynamic complexity of the process or existence of more than one sub-process within a single system boundary. These experimental methods are extensively used in designing control systems and, in fact, many of these techniques have come up in conjunction with some particular tuning procedure.

All the methods have some common feature. Applying a chosen type of disturbance function in the form of real process variable, the given process is disturbed from its steady state. The input disturbance, along with the response or output from the process, is recorded until the process attains the next steady state. The recorded data is mathematically processed through a developed algorithm to produce the required process dynamic model, generally as the transfer function.

12.1 On-Line Methods

These methods commonly reside as a part of the control algorithm. While the process is running in closed loop with the control system, the control software at a pre-selected time interval runs the process identification part of the control algorithm. These procedures either use the regular kind of disturbances as the load varies during plant operation or creates some particular disturbance function through manipulated variable as the dynamic test input to the process. The response is recorded along with the input, and the model building technique is implemented. Generally, such identification techniques are implemented with Adaptive Control Strategies for processes that feature dynamic anomalies, e.g. poorly defined processes requiring self-learning type control schemes, or variable process parameters requiring interim evaluation and updating the controller tuning on the current dynamic status of the

process. We have described the Auto Tuning procedure, an example of on-line technique in the chapter on Controller Tuning.

12.2 Off-Line Methods

The given process is treated in an open loop fashion, i.e. there is no control loop associated with the process while it is dynamically tested. An algorithm treats the recorded input and response data to obtain the process model. Usually, a computer programme implements the procedure. Accuracy of a derived model is found by calculating an error criterion between the experimental (Y_E) and model (Y_M) response. The Standard Deviation (SD) has been extensively used for this purpose. If we have the response data as N number of data points, the standard deviation may be defined as:

$$\text{SD} = \sqrt{\frac{\sum_{I=1}^{N}(Y_E - Y_M)^2}{N}} \qquad (12.1)$$

If there is more than one model that has the possibility of fitting the experimental data, the model that offers the least SD value is chosen.

The off-line methods are extensively used in practical process modeling, optimization and controller tuning. They are classified according to the type of analysis procedure that is used for their development.

12.3 Open Loop Methods Based on Lumped Parameter Model Structures

This method of model building fundamentally uses certain types of transfer functions to approximately express the dynamics of the given process. These transfer functions are the dynamic description of certain simple lumped parameter model structures. The method uses the Step Response Locus or Process Reaction Curve (PRC) of the process together with the magnitude of the input step. There are four types of model structures corresponding to the following transfer functions:

*1. First Order (***FO***) Model, expressed by the transfer function,*

$$\frac{Y(s)}{X(s)} = \frac{K_P}{\tau_P s + 1}\text{; Model parameters: } K_P, t_P. \qquad (12.2)$$

*2. First Order Plus Dead Time (***FOPDT***) Model, expressed by the transfer function,*

$$\frac{Y(s)}{X(s)} = \frac{K_P e^{-\tau_{DL} s}}{\tau_P s + 1}\text{; Model parameters: } K_P, t_P, t_{DL} \qquad (12.3)$$

*3. Second Order (***SO***) Model, expressed by the transfer function,*

$$\frac{Y(s)}{X(s)} = \frac{K_P}{(\tau_{P1}s+1)(\tau_{P2}s+1)}$$; Model parameters: K_P, t_{P1}, t_{P2} (12.4)

*4. Second Order Plus Dead Time (***SOPDT***) Model, expressed by the transfer function,*

$$\frac{Y(s)}{X(s)} = \frac{K_P e^{-\tau_{DL}s}}{(\tau_{P1}s+1)(\tau_{P2}s+1)}$$; Model parameters: K_P, t_{P1}, t_{P2}, t_{DL} (12.5)

The objective of the algorithm is to determine the parameter value of a chosen model structure from the step response of the process. An approximate choice or rather the elimination of certain type of models from the above list, may be done by examining the process response shape. If we recall the step response shapes of the first and second order system, particularly their initial part—that only first order system shows a finite +ve initial value (in fact the maximum value) of the slope of response, and any higher order system shows an initial zero value of the response slope that is characterized by the occurrence of 'foot' in the response.

Thus, existence of a foot or initial zero value of the response slope will discard the FO model from the choice list. Again, if the foot or initial zero part is too long or pronounced, one can discard the SO type, because we know that second order processes, even with large difference between two time constants, show a small initial zero part of the response (recall the discussion regarding 'Transfer Lag'). Hence, our choice narrows down to the models that include dead time in their structures, i.e. FOPDT and SOPDT. In fact, only the comparison of some error measure, e.g. Standard Deviation (SD), determined between actual and model response obtained for these model types, may find the best-suited model for the process.

The procedures adopted for the determination of the model parameter values should start with process gain K_P, because it is determined for all the model types from its definition as:

$$\text{Process Gain} = K_P = \frac{\Delta \text{ Output}}{\Delta \text{ Input}} = \frac{Y_{2S} - Y_{1S}}{X_{2S} - X_{1S}} = \frac{Y_M}{X_M} \quad (12.6)$$

where, Y_{1S} and Y_{2S} are steady state response values before and after the input change has occurred from X_{1S} to X_{2S}, and applying the definition of deviation variable:

$X_{2S} - X_{1S} = X_M$ = Magnitude of the input step

$Y_{2S} - Y_{1S} = Y_M$ = Response change from initial to final steady state

The remaining parameters, i.e. time constants and dead time determination procedures are different for the four model types.

12.4 Determination of FO Model Parameters

The time constant of the FO model may be done by using the property of a first order system step response, that is, its initial slope or the tangent drawn on the response locus drawn at
$t = 0$, crosses the total response magnitude within a time equal to the time constant of the system. The required graphical construction has been shown in Figure 12.1.

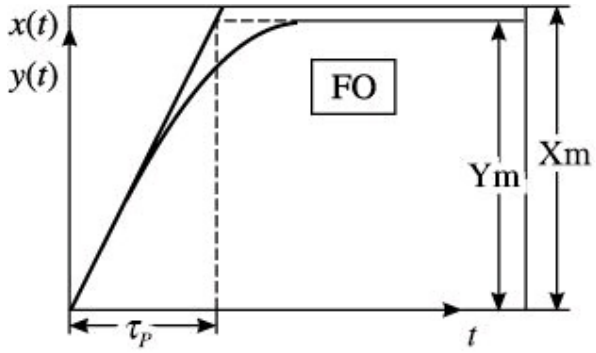

Figure 12.1 Determination of time constant of an FO model.

12.5 Determination of FOPDT Model Parameters

Probably the FOPDT is the type of model that is most widely used for dynamic representation of processes. This is because:

1. Numbers of parameters to be evaluated are practically only two (t_P and t_{DL}), that do not require a very large amount of computational effort, and
2. The obtained FOPDT model has a reasonable accuracy even when used for higher order or distributed parameter systems if the parameters are evaluated with an adequate algorithm. A large amount of literature exits in this area.

(A) **Method 1** The oldest approach is an extension of the method of first order time constant determination given in Figure 12.1. For responses with 'foot', the point of tangent construction shifts to the point of 'inflection' (point at which the response has maximum slope) on the response locus. The tangent is drawn for the total response magnitude. If the right angle triangle formed by the tangent, the perpendicular from the top intersection of the tangent with the ultimate response value to the abscissa, and the abscissa are considered, the base of the triangle = t_P, and the length of the foot of the triangle from the origin = t_{DL}. These ideas have been expressed in Figure 12.2.

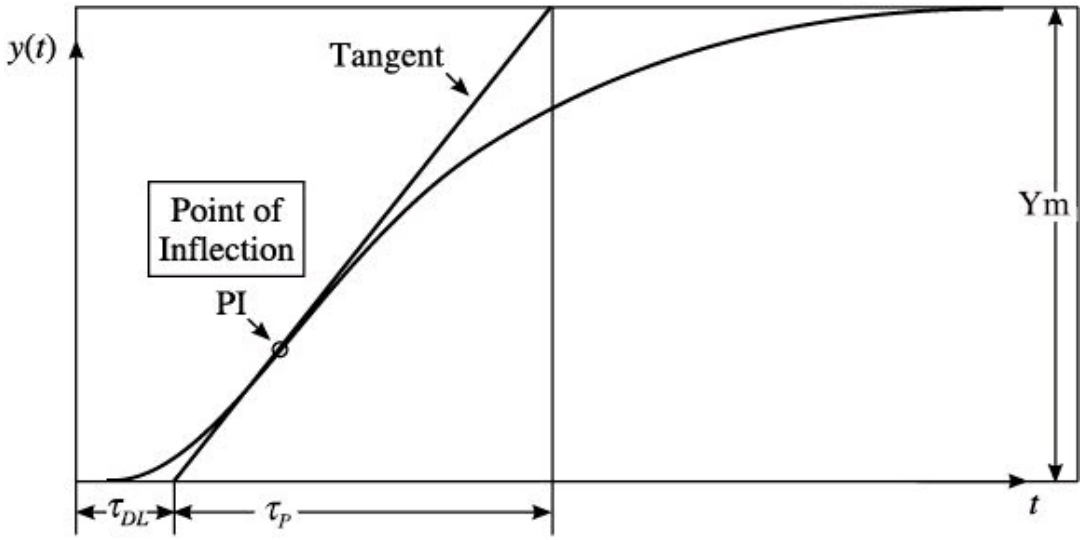

Figure 12.2 Determination of FOPDT model parameters from PRC.

(B) **Method 2** A modification of this method determines the dead-time x_{DL} in the same way as shown in Figure 12.2, but t_P is measured from the foot of the tangent up to where the response has reached 63.21% of the response magnitude. Usually the t_P found by this approach has lower values than by Method 1. Also the accuracy obtained is better.

(C) **Method 3** It was later established by Smith that model accuracy may be improved by taking two points on the PRC corresponding to 28.35% and 63.21% recovery of the response. This is based on the following analysis:

The unit step response eqn. of first order + Dead time model for any finite +ve response value may be written as:

$$y(t) = K_P[1 - e^{-(t-\tau_{DL})/\tau_P}]; \quad \text{for } t > \tau_{DL} \tag{12.7}$$

where, M = Input step magnitude.

Thus, if the FOPDT response $y(t)$ is considered, for $t = t_1 = t_{DL} + t_P/3$ and $t = t_2 = t_{DL} + t_P$, the response will be:

$$y(t_1) = 0.2835Y_M \quad \text{and} \quad y(t_2) = 0.6321Y_M$$

hence,

$$t_1 = \tau_{DL} + \tau_P/3 \tag{12.8}$$

$$t_2 = \tau_{DL} + \tau_P \tag{12.9}$$

From Equations 12.8 and 12.9

$$\tau_P = 1.5(t_2 - t_1) \tag{12.10}$$

$$\tau_{DL} = 0.5(3t_1 - t_2) \tag{12.11}$$

From the given step response locus two times t_1 and t_2 are found corresponding to 28.35% and 63.21% of maximum response magnitude. This has been shown in Figure 12.3.

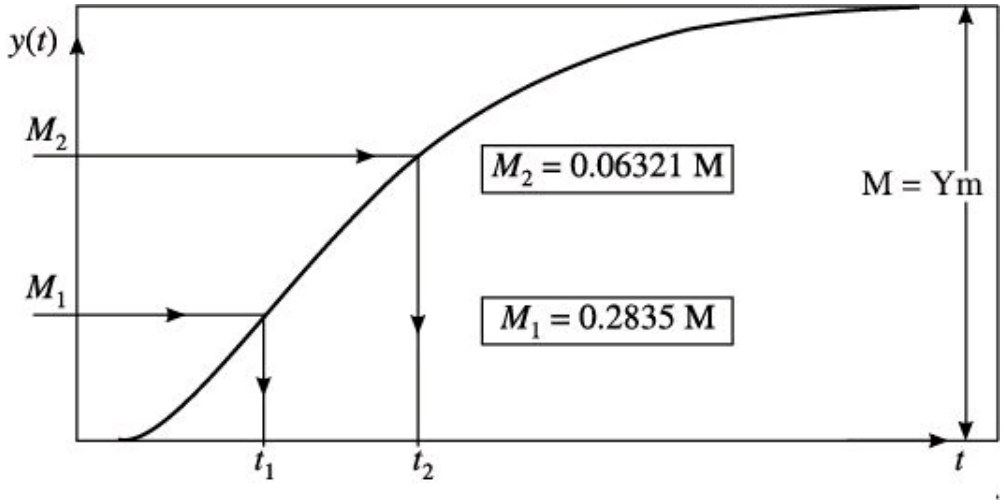

Figure 12.3 Smith's method of FOPDT model parameter estimation.

(D) **Method 4 (A Modification of Method 3)** A later modification of this procedure assumes that the two fractions of maximum response magnitude may not be fixed but depend on certain shape characteristics of the given PRC. It has been recognized that the shape of a given PRC depends on the location of the inflection point (PI) and the slope of the tangent, as in Figure 12.2. A perpendicular from the point PI renders two triangles instead of one as in Figure 12.2. This is illustrated in Figure 12.4, together with the geometrical definition of time parameters TA, TB and TC. In developing the method, the dimensionless time parameter, TBACA = TBA/TCA = (TB–TA)/(TC – TA) has been used to characterize a PRC. This has been shown in Figure 12.4. For the TBA/TCA value corresponding to the given PRC, the fractional response recovery M_1 and M_2 have been found by an optimum search programme that obtains minimum standard deviation between the given and model response. This procedure has been repeated for more than 50 different TBA/TCA values within the range 0.1 to 0.5. Figure 12.5 depicts the dependence of M_1 and M_2 against TBA/TCA.

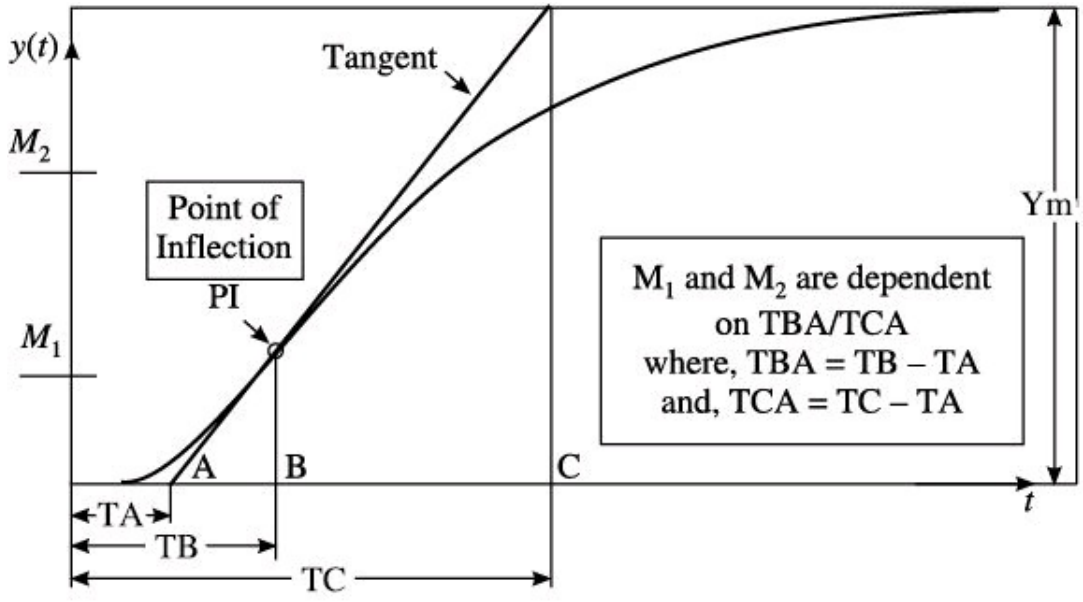

Figure 12.4 Modification of Smith's FOPDT model parameter estimation.

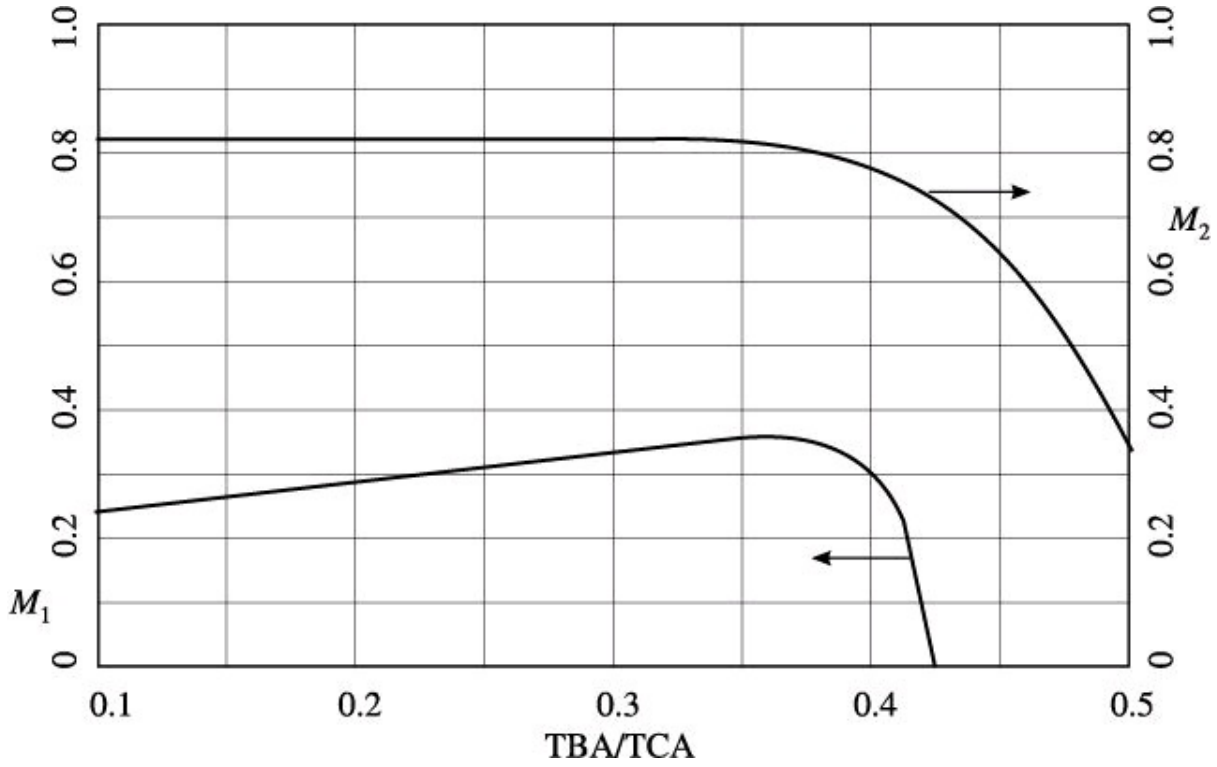

Figure 12.5 Dependence of M_1 and M_2 with TBA/TCA with reference to Figure 12.4.

For the given step response of a process as in Figure 12.4, after locating the PI and drawing the tangent, the time parameters TA, TB, and TC are found and reduced time parameter TBA/TCA is calculated. From Figure 12.5, the corresponding values of M_1 and M_2 are read out. The following equations may be applied to find the required time constants t_D and t_P of the FOPDT model.

$$\tau_P = \frac{t_2 - t_1}{[\ln(1 - M_1) - \ln(1 - M_2)]} \tag{12.12}$$

$$\tau_{DL} = t_1 + \tau_P \ln(1 - M_1); \quad \text{or} \quad \tau_{DL} = t_2 + \tau_P \ln(1 - M_2) \tag{12.13}$$

Equations 12.12 and 12.13 may be easily derived from the FOPDT response Equation 12.7.

To facilitate the use of this modified method without Figure 12.5, as in a computer programme, following empirical relations between TBA/TCA – M_1 and TBA/TCA – M_2 have been developed:

$$M_1 = \frac{A_1 + B_1 X}{1 + C_1 X + D_1 X^2} \tag{12.14}$$

and

$$M_2 = A_2 - B_2 \exp(-C_2 X^{D_2}) \tag{12.15}$$

where, X = TBA/TCA, and the constants of the equation have the following values:

$A_1 = 0.2169$ $\quad A_2 = 0.8231$

$B_1 = -0.5111$ $\quad B_2 = 67.5527$

$C_1 = -3.6489 \qquad C_2 = 1.4374$

$D_1 = 3.1268 \qquad D_2 = -1.7807$

These relations are in good agreement with the obtained graphs of Figure 12.5.
You should note a couple of things about using this method:

1. The accuracy is better for responses that show the TBACA value larger than 0.34.
2. Accuracy also improves, if the model parameters are determined discarding the initial 10% of the response.

Incidentally a process under dynamic conditions of closed loop operation largely resides in the latter part of the response where the model accuracy is better.

A typical example of the FOPDT model parameter estimation is certain optimum controller setting determination procedures described in Chapter 13.

EXAMPLE 12.1 It has been established that in biological waste digestion, to produce bio-gas, the fermentation reactions fall under two principal categories:

1. Acedogenic, and
2. Methanogenic reactions.

The raw material used in an experimental digester is the fruit peel-offs and compacted pulps after juice extractions from a food processing unit. These are dried and shredded to 2 to 3 mm size and put into the digestion system together with water, as slurry. The system comprise of two reactors:

1. Acedogenic fermentor, vol: 6.0 m^3,
2. Methanogenic fermentor, vol: 9.0 m^3, operating in series.

The input slurry, fed at a rate of 5 kg/h, contains 10% solid waste, has an average density of 1kg/l. The low pH of the effluent from the first reactor is raised to 6.3 by adding slurry of lime at a rate of 100 g/h. Biogas obtained from both reactors is collected in a holder. A small amount of generated gas is recycled into the reactors to facilitate agitation. Under these operating conditions the gas generation rate is steady at 1.5 m^3/h.

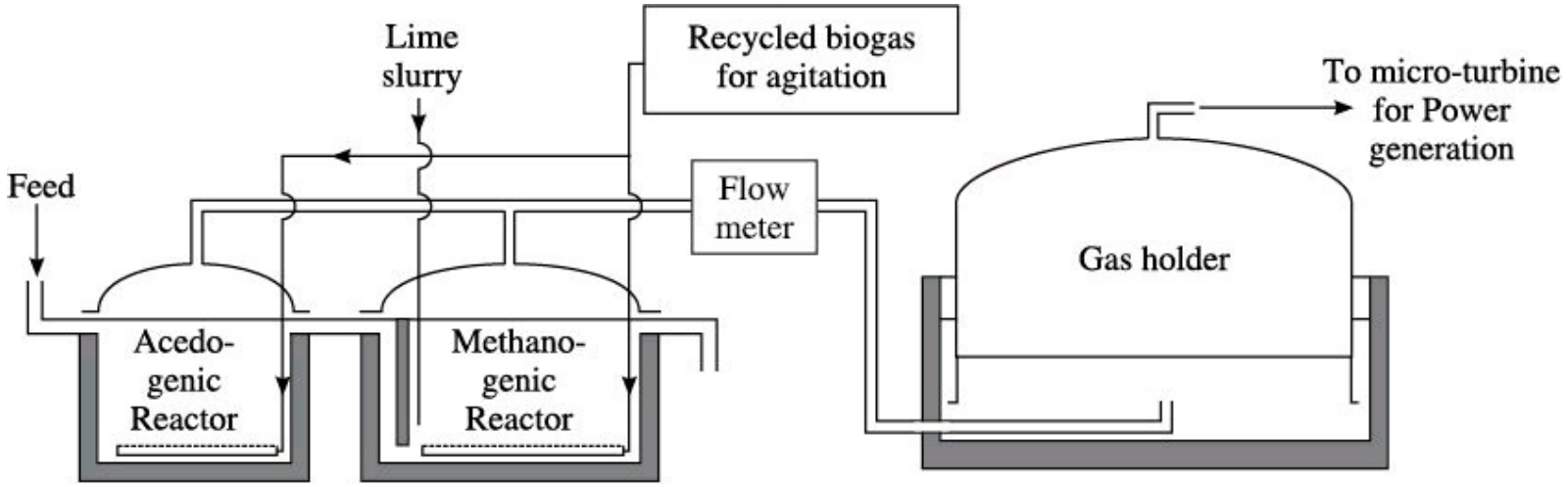

Figure 12.6 A schematic of the biogas generation process.

In Figure 12.6, a flow sheet of the gas generation process is illustrated. To design a proposed system for controlling gas-generation rate, where manipulated variable is the rate of lime slurry, an experiment is carried by increasing the lime flow rate to 130 g/h according to a step change, and recording the increment in gas flow rate from the flowmeter against time to establish a process model for the methanogenic reactor. The recorded data is given in the following table:

Time, min	0	0.01	25	50	75	125	175
ΔQ, m^3/h	0	0	0.0029	0.0193	0.0541	0.1744	0.337
Time, min	250	325	400	375	550	625	700
ΔQ, m^3/h	0.597	0.822	0.99	1.107	1.183	1.23	1.259
Time, min	775	850	925	1000	1075	1175	1200
ΔQ, m^3/h	1.277	1.286	1.292	1.295	1.297	1.298	1.3

Find out the FOPDT model parameters for the process by using the first three methods given above and compare their performance in terms of standard deviation from the experimental data.

Solution: The recorded data is plotted on a graph and a smooth curve is drawn through them. This is shown in Figure 12.7.

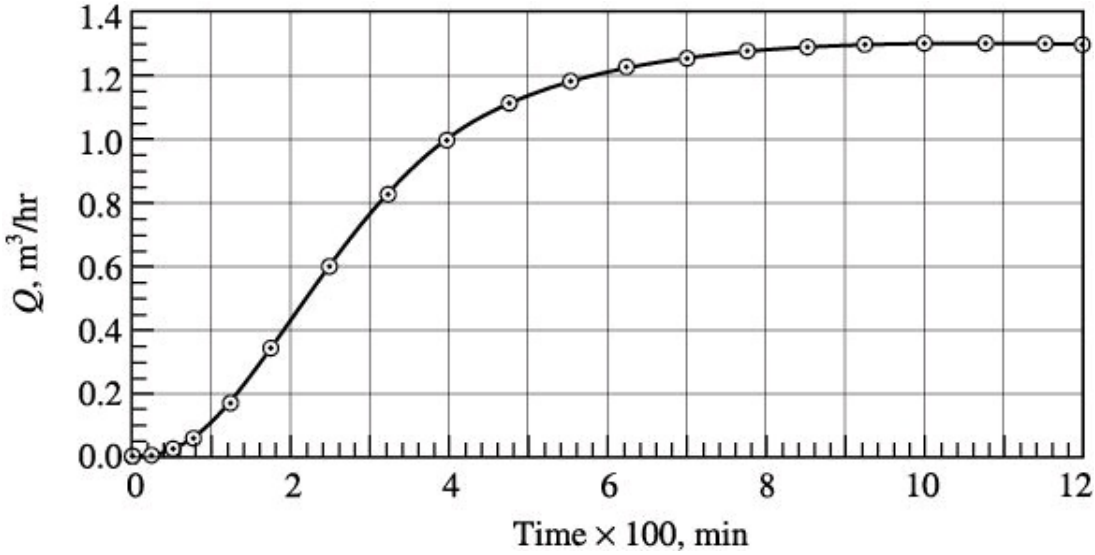

Figure 12.7 Plot of tabulated data.

The data is expressed as a smooth curve by plotting normalized response ($Y_N = Q/Q_{MAX}$) against time up to 1000 min in Figures 12.8(a) and (b). We may readily find the process gain, K_P as:

$$K_P = \frac{\Delta \text{ Output}}{\Delta \text{ Input}} = \frac{1.3 \text{ m}^3/\text{h}}{30 \text{ g/h}} = 0.0433 \text{ m}^3/\text{g}$$

In Figure 12.8(a), the graphical construction is made to find TA = 79.15 min, TB = 192.23 min, and TC = 448.17 min.

Thus, the model parameters found by Method 1 are, t_P = TC – TA = 448.17– 79.15 = 369.02 min, and t_{DL} = TA = 79.15 min.

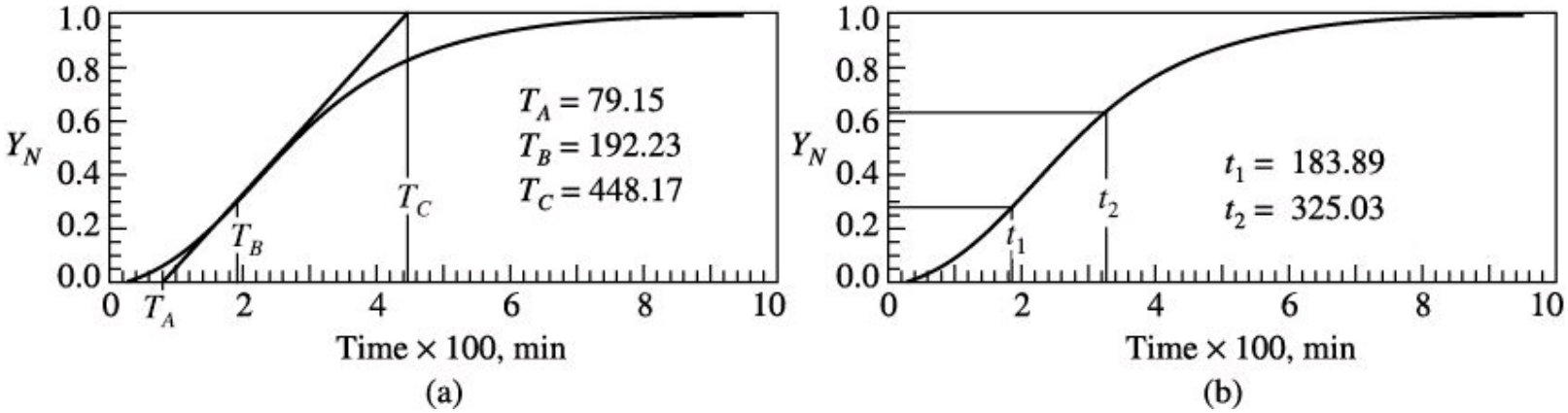

Figure 12.8 (a) Graphical construction to find TA, TB and TC. (b) Determination of t_1 and t_2 corresponding to step recovery of 28.35% and 63.21%.

Method 2: As discussed in the text, the model parameters are, $t_P = t_2$ = 325.03 min [the time required for 63.21% step recovery as in Figure 12.8(b)], and $t_{DL} = T_A$ [as in Figure 12.8(a)] = 79.15 min.

Method 3: To find the model parameters, we apply Smith's equations (Equations 12.8 through 12.11), thus, t_P = 1.5 (325.03 – 183.89) = 211.71 min, and t_{DL} = 0.5(3 × 183.89 – 325.03) = 113.32 min.

The three model responses (Y_{M1}, Y_{M2}, Y_{M3}), are drawn along with the experimental data (Y_E) in Figure 12.9.

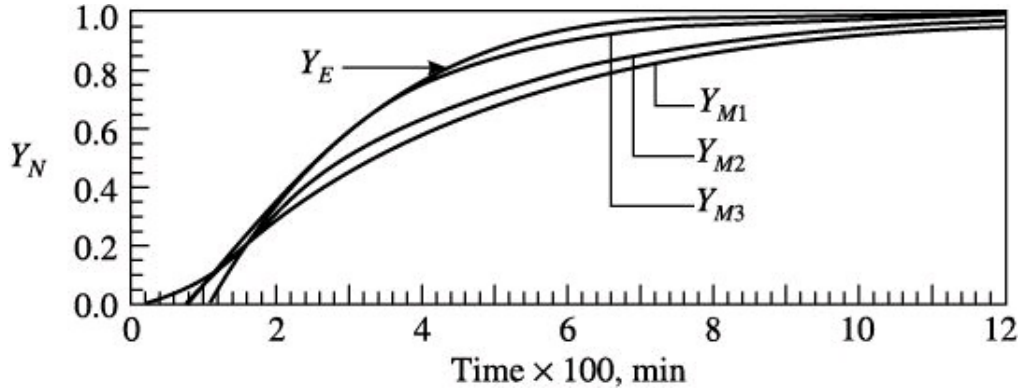

Figure 12.9 Comparison of the experimental with the model responses.

Standard deviations of the model responses with respect to the experimental data are calculated and reported in the following table:

Model number	1	2	3
Standard deviation	0.1046	0.0718	0.0237

The process TF is formulated by the parameters of the most accurate model.

$$G_p(s) = \frac{K_p e^{-\tau_{DL} s}}{\tau_p s + 1} = \frac{0.0433 e^{-113.32 s}}{211.7 s + 1} \frac{\text{m}^3}{\text{g}}$$

Data recording from dynamic experiments or plant are performed by procedures that depend on the response speed of the process and available hardware. Generally, manual recording of data is practical for a slow process. If the process is fast enough, some kind of recording instrument is used. A potentiometric single or multi-channel recorder would be suitable for a medium speed process. A digital data logger, instrument tape recorder, storage oscilloscope, and specially built movie cameras may be used as recording devices for high speed process. From these instruments, the data is available as a hardcopy trace of the variable preferably on a graph, or a file from a storage device. Using a hardcopy, the data set may be prepared by putting required number points on the trace and manually reading the variable value. Such a data set may be used for parameter search of a chosen model configuration.

EXAMPLE 12.2 A catalytic Plug Flow Reactor (PFR) is used for cracking of hydrocarbon in gas phase. Reaction temperature is 375–400°C. In the following, the step response of molar % of C_3 molecule in the product corresponding to a 7.3% increase of input flow rate is shown in Figure 12.10. The C_3 fraction is easily separated from the product because of its boiling point difference. This fraction is mixed with a predetermined amount of air and sent to the detector. The detector system has a catalytic converter that totally burns the product fraction and a thermal conductivity detector measures the amount of CO_2 present and computes the % C_3 in the product.

It can be assumed that there is no change of number of moles before and after the reaction so that it is a constant density process occurring under isothermal condition.

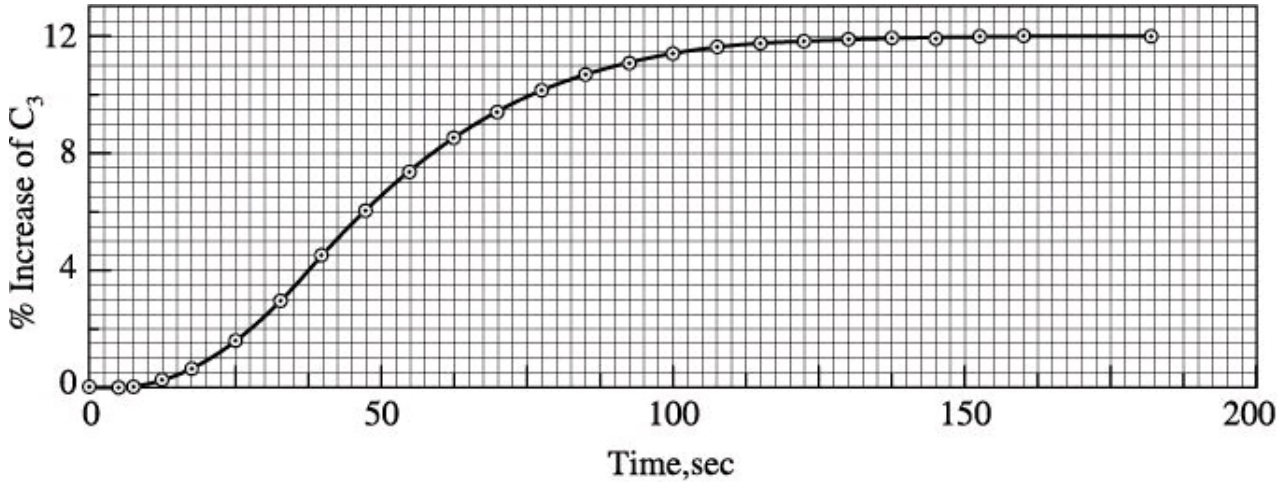

Figure 12.10 The step response in the product concentration to a feed rate change.

Develop an FOPDT model for the reactor using Method 3 and 4.

Solution: On the response plot, data points are manually located in sufficiently close intervals at regions where slope change is maximum, and they are at a wider gap where the curve approaches saturation. The data point locations are chosen to accurately reflect the dynamics of the process. The coordinates of the points are given in the following table:

Time, s	0	2.5	5	7.5	12.5
$\Delta\%\ C_3$	0	0	0.009	0.04	0.21
Time, s	17.5	25	32.5	40	47.5
$\Delta\%\ C_3$	0.59	1.59	2.96	4.49	5.99
Time, s	55	62.5	70	77.5	85.1
$\Delta\%\ C_3$	7.34	8.49	9.41	10.13	10.67
Time, s	92.6	100	107.6	115.1	122.6
$\Delta\%\ C_3$	11.06	11.35	11.55	11.69	11.79
Time, s	130.1	137.6	145.1	160.1	180.1
$\Delta\%\ C_3$	11.86	11.9	11.93	12	12

The tabulated data is expressed as a smooth curve by plotting normalized response ($Y_N = \Delta\%C_3/\Delta\%C_{3MAX}$) against time up to 180 s. in Figure 12.10(a) and Figure 12.11.

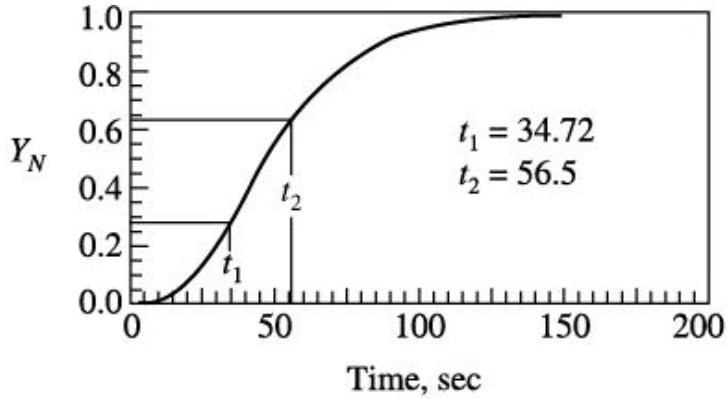

Figure 12.10(a) Determination of t_1 and t_2 in method 3.

On Figure 12.10(a), the time, t_1 needed for 28.35%, and time t_2 for 63.21% step

recovery are obtained as the starting point of method-3 implementation. As shown in the previous example, the model parameters are evaluated as:

$$K_P = \frac{12\%}{7.3\%} = 1.644\ \%/\%;\ tP = 32.67\ \text{s};\ t_D = 23.83\ \text{s}$$

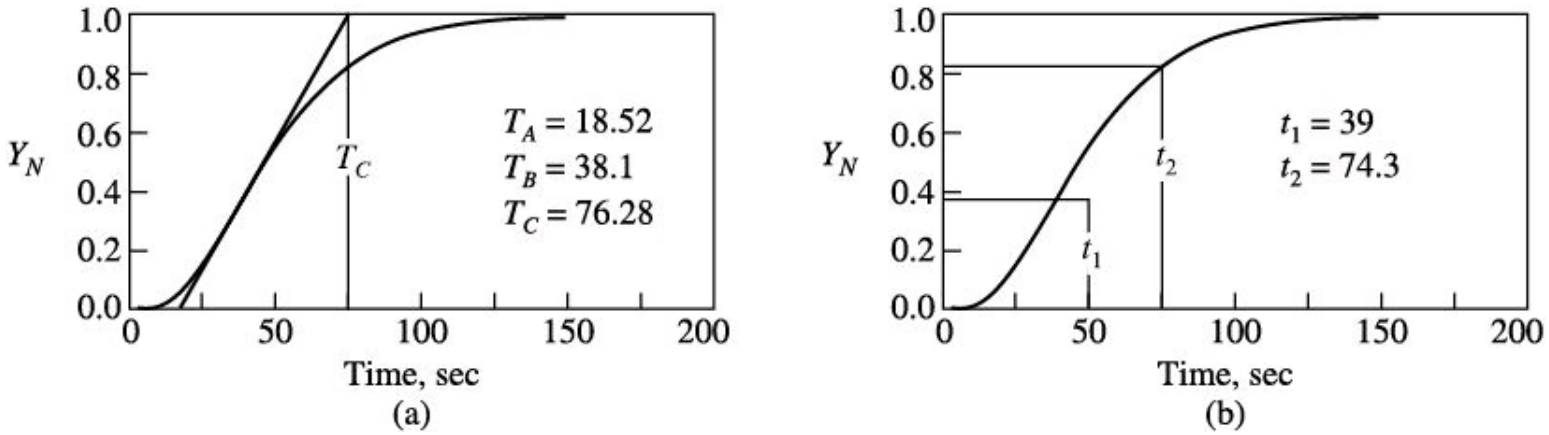

Figure 12.11 The graphical construction and determination of t_1 and t_2 in method 4.

For Method 4, on the curve of Figure 12.10(a), the graphical construction for determining TA, TB, and TC is done in Figure 12.11(a). Using Figure 12.5, the step recovery magnitudes are found to be $M_1 = 0.356$ and $M_2 = 0.82$. The times corresponding to these recovery magnitudes, constructed in Figure 12.11(b), are found as $t_1 = 39$ s and $t_2 = 74.3$ s. By Equations 12.12 and 12.13, the model time constants are found as:

$$tP = 27.72\ \text{s};\ t_D = 26.8\ \text{s}$$

The process gain value will be the same as found above. Next, to compare the calculated model configurations, the normalized responses of the experimental and the two models are shown in Figure 12.12.

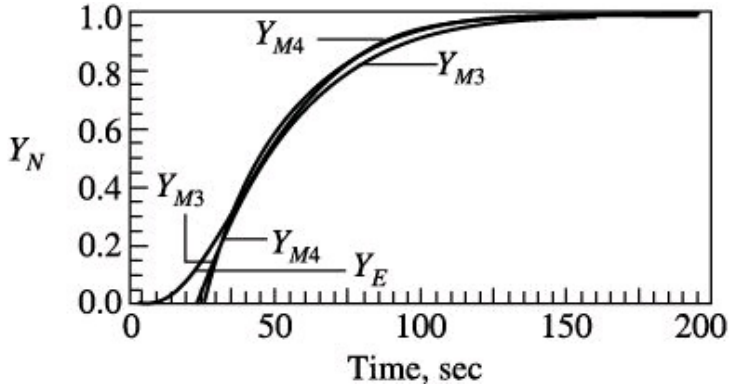

Figure 12.12 Performance comparison the model 3 and model 4.

Standard deviations for the model responses are found to be, $SD_{M3} = 0.057$ and $SD_{M4} = 0.057$. But if we consider the responses leaving their initial 10%, standard deviation values become, $SD_{M3} = 0.0541$ and $SD_{M4} = 0.0498$. By observing the

model responses, it may be said that model 3 deviates less than model 4 at initial parts, whereas, model 4 comes closer to the actual response for the latter part. Thus, the result supports this visual observation following the comments made at the end of Method 4.

12.6 Determination of SO Model Parameters

Corresponding to Equation 12.4, a typical trace of the step response of an over damped second order process is shown in the normalized form in Figure 12.13. The normalization is done by expressing the response value as $y(t) = Y(t)/(MK_P)$, where, M = step magnitude, K_P = steady state gain, i.e. $y(t)$ can have values only between 0 and 1.0. At present, we shall confine the discussion for over damped responses without or with dead time because such responses are most widely found in process application. The case of under damped second order response will be taken up later.

The response starts with a zero slope but eventually a time comes when its slope becomes maximum, after which the slope gradually reduces to zero at the steady state condition where response attains its maximum value $Y_M = 1$. The point of maximum slope has been identified as the point of inflection (PI), and the corresponding time as TB. A tangent drawn through the point PI and the triangles ABPI and ACD, renders the three time parameters of the response TA, TB and TC that have been extensively used in the model parameter analysis.

In case of over damped response, there is no direct or explicit route for model parameter estimation by analytical method. Determination of the two-time constants t_{P1} and t_{P2} can be only done by some implicit procedure or a graphical construction to circumvent the implicit step. This problem has attracted a lot of attention. Quite a number of approaches and their results have been reported. These methods contain easy-to-use guide plots for finding SO model parameters from the experimentally obtained step response data. A class of these methods, require the determination of point of inflection at which the response attains maximum slope and the certain time parameters (point PI and the time values of TA, TB, TC) shown in Figure 12.13. These values could be easily obtained by graphical construction over the experimental data as shown in Figure 12.13. Two methods are worth mentioning in this class are:

1. Oldenbourg-Sartorius, and
2. Perlmutter.

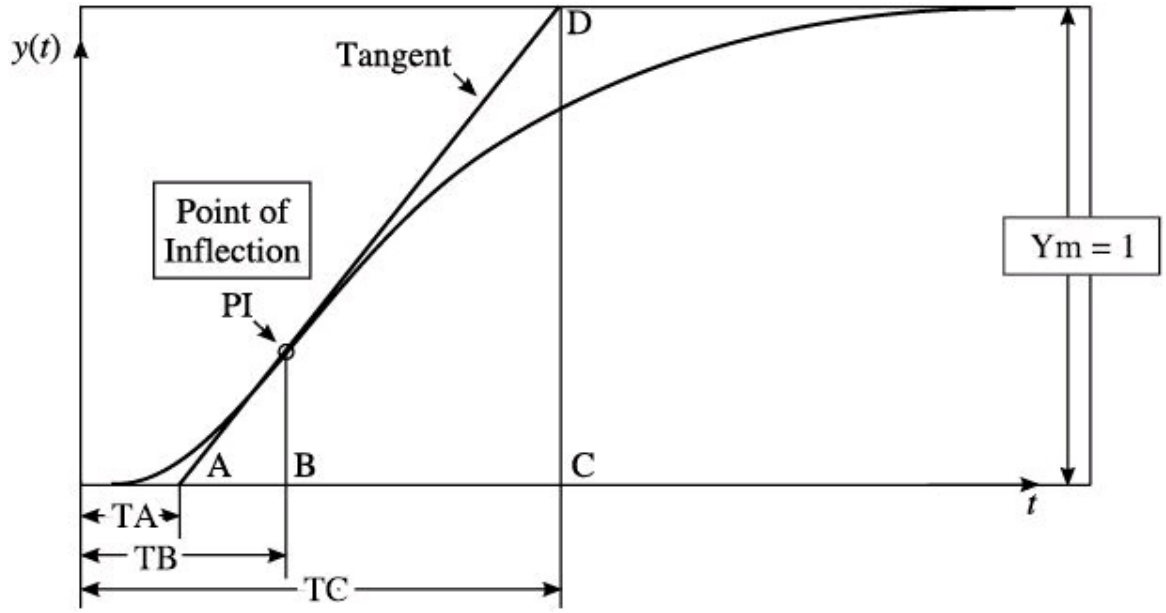

Figure 12.13 Characterization of the step response of an over damped SO process model.

We have not included the detail of methods by Oldenbourg-Sartorius and Perlmutter because it has been found that the accuracy of the predicted models by these methods is somewhat limited. Interested readers may find the details from the mentioned references. However, in the following section, we present an indigenously developed procedure that may be called a modification of the previous two methods and offer a better accuracy.

One thing should be noted that all these methods loose their accuracy when the solution approaches a critically damped condition. This is understandable because all the analysis starts from assuming two real unequal roots of the second order characteristic equation.

Method 3 A Modification of Methods 1 and 2.

In the following method, the analysis starts from the unity gain second order over damped transfer function,

$$\frac{Y(s)}{X(s)} = \frac{1}{(\tau_1 s + 1)(\tau_2 s + 1)} \qquad (12.16)$$

With two distinct –ve real roots of the characteristic equation, the corresponding unit step response would be:

$$y(t) = 1 - \frac{\tau_1 \tau_2}{\tau_1 - \tau_2}\left(\frac{1}{\tau_2} e^{-t/\tau_1} - \frac{1}{\tau_1} e^{-t/\tau_2}\right) \qquad (12.17)$$

Differentiating Equation 12.17 with respect to time,

$$\frac{dy(t)}{dt} = \frac{1}{\tau_1 - \tau_2}(e^{-t/\tau_1} - e^{-t/\tau_2}) \qquad (12.18)$$

Again differentiating Equation 12.18,

$$\frac{d^2y(t)}{dt^2} = -\frac{1}{(\tau_1 - \tau_2)\tau_1}e^{-t/\tau_1} + \frac{1}{(\tau_1 - \tau_2)\tau_2}e^{-t/\tau_2} \qquad (12.19)$$

According to the property of Inflection Point, at t = TB, the second derivative would be zero. Solving for TB,

$$\text{TB} = \frac{[\ln(\tau_1/\tau_2)]\tau_1\tau_2}{(\tau_1 - \tau_2)} \qquad (12.20)$$

Putting the value of t = TB in Equation 12.17, the response value YB = $y(t_{TB})$ may be found. Also, putting t = TB in Equation 12.18, the maximum value of the slope of the tangent in Figure 12.6, could be found. As the height of the triangle ACD is 1.0 (due to unit magnitude of the step input), the base of the triangle TCA = TC – TA is found as:

$$\text{TCA} = \frac{1}{dy/dt\big|_{t=TB}} = \frac{\tau_1 - \tau_2}{e^{-TB/\tau_1} - e^{-TB/\tau_2}} \qquad (12.21)$$

From the two equivalent triangles, ABPI and ACD,

$$\text{TA} - \text{TB} = \text{TAB} = \text{YB}\square\text{TAC} \qquad (12.22)$$

For different ratios of t_1/t_2 ranging from 0.01 to 0.99, the reduced time parameter TBACA = TBA/TCA, T1CA = t_1/TCA and T2CA = t_2/TCA have been found by using Equations 12.17 through 12.22. The results are plotted in Figure 12.14, with TBACA in abscissa and T1CA and T2CA on the two ordinates.

From a given response, after the geometrical construction indicated in Figure 12.6, TBACA, T1CA and T2CA may be found. Applying these values in Figure 12.14, the required time constants, t_1 and t_2 may be obtained.

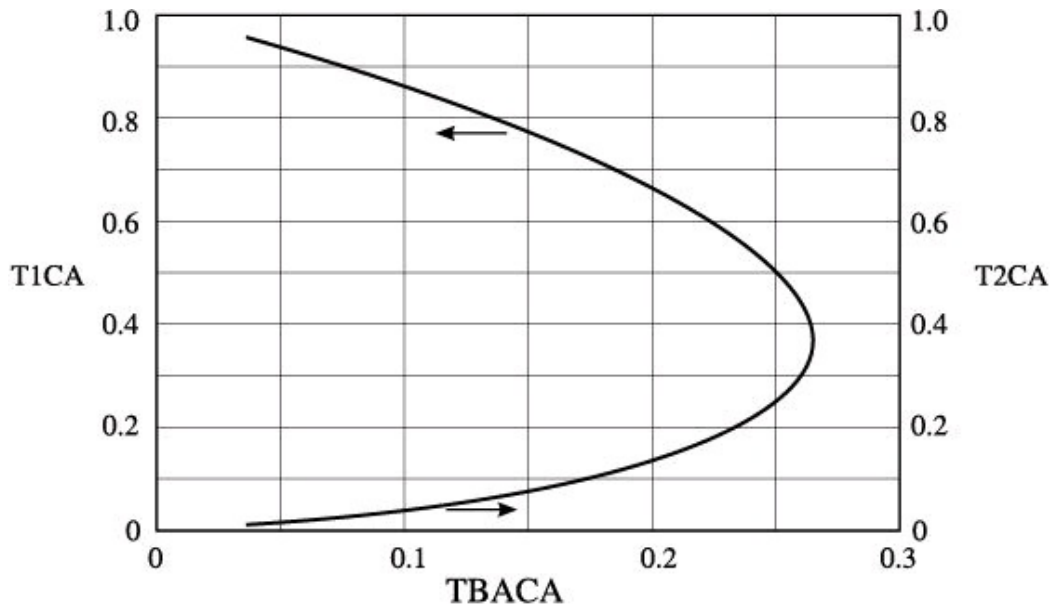

Figure 12.14 Guide chart for SO model parameter estimation.

To facilitate the use of this method without Figure 12.14, as in a computer

programme, the following empirical relations between TBACA vs. T1CA and TBACA vs. T2CA have been found:

$$Y_1 = \frac{A_1 + B_1 X}{1 + C_1 X + D_1 X^2} \qquad (12.23)$$

and

$$Y_2 = A_2 + B_2 X + C_2 X^2 + D_2 X^3 + E_2 X^4 + F_2 X^5 \qquad (12.24)$$

where, X = TBACA = TBA/TCA, Y_1 = T1CA, Y_2 = T2CA, and the constants of the equation have the following values:

$A_1 = 1.03285$ $A_2 = -0.14893$

$B_1 = -3.62063$ $B_2 = 8.3757$

$C_1 = -1.57403$ $C_2 = -153.5975$

$D_1 = -5.57877$ $D_2 = 1322.1118$

$E_2 = -5139.0107$

$F_2 = 7501.7005$

These relations are in good agreement (SD ≈ 0.0075) with the obtained graphs of Figure 12.16. To substantiate the statements made let us do an example.

EXAMPLE 12.3 The NH_3 concentration response in ppm in the exit gas from a two-packed chamber ammonia scrubbing system acting in series, is recorded after suddenly
increasing water (absorbent) flow rate from 100 to 125 l/min. The absorber is employed for removing NH_3 from an Air + Ammonia mixture. The recorded data is given in the following table. Derive the parameters of SO model structure according to the three methods discussed in the text.

Time, s	0	0	25	50
C_{NH3}, ppm	108	108	107.68	106.92
Time, s	75	125	174	250
C_{NH3}, ppm	105.95	103.94	102.2	100.35
Time, s	325	400	475	550
C_{NH3}, ppm	99.25	98.65	98.33	98.16
Time, s	625	700	775	850
C_{NH3}, ppm	98.08	98.04	98.02	98.0

Solution: The recorded data is plotted on a graph as shown in Figure 12.15, and the points are joined by a smooth curve to represent the step dynamics of the process.

The Process Gain may be found as, $K_P = \frac{\Delta y}{\Delta x} = \frac{98 - 108 \text{ ppm}}{125 - 100 \text{ l/min}} = -0.4 \ \frac{\text{ppm}}{\text{l/min}}$

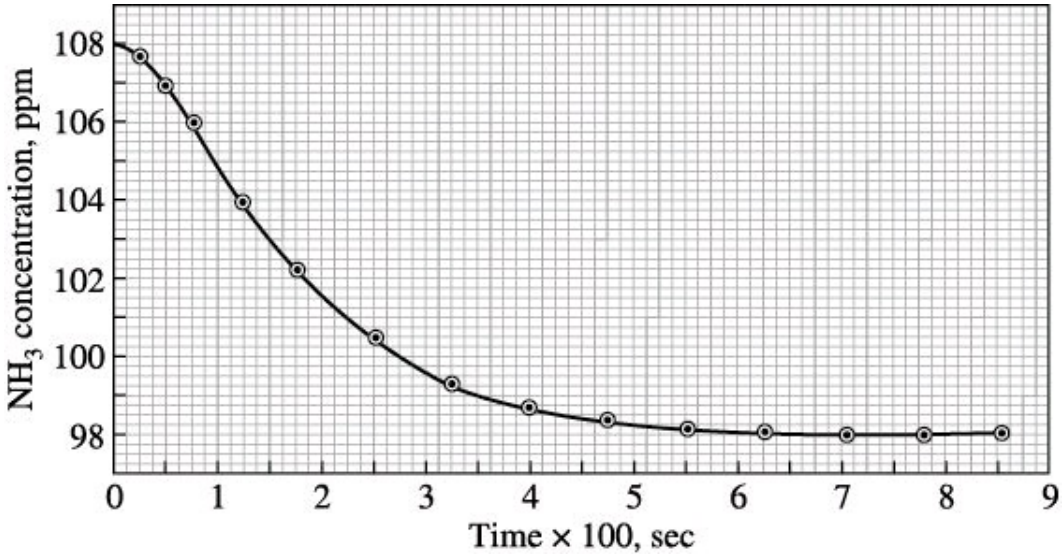

Figure 12.15 NH_3 concentration response for a step increase in absorbent flow rate.

The dynamics is redrawn by converting the response variable to its normalized form in Figure 12.16, and the construction for determining TA, TB, TC AND TBA/TCA is done. The values obtained are:

TA = 25.11, TB = 86.95, TC = 268.95, TBA/TCA = 0.2537.

Model 1 (Sartorius): From the above values, TCB/TCA = 0.7463. Now using the method described in ref: 4, model time constants are found as:

t_1 = 12.029 s and t_2 = 6.215 s

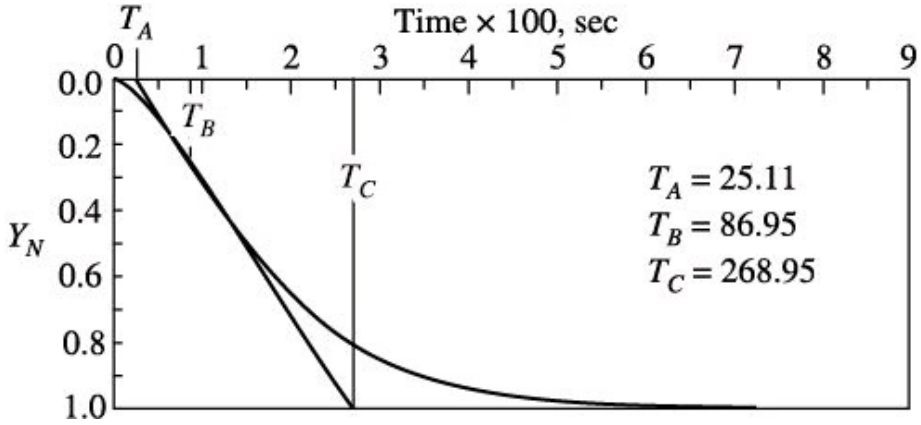

Figure 12.16 The dynamic response in normalized form.

Model 2 (Perlmutter): From the above values, TCA/TCB = 1.34. Now using the method described in ref: 5, model time constants are found as:

t_1 = 12 s and t_2 = 6.6 s

Model 3 (Modified Model): From Figure 12.9, corresponding to TBA/TCA = 0.2537, the parameters t_1/TCA = 0.482 and t_2/TCA = 0.26 are obtained. Putting the value of TCA in these expressions the model time constants are found as:

t_1 = 11.75 s and t_2 = 6.34 s

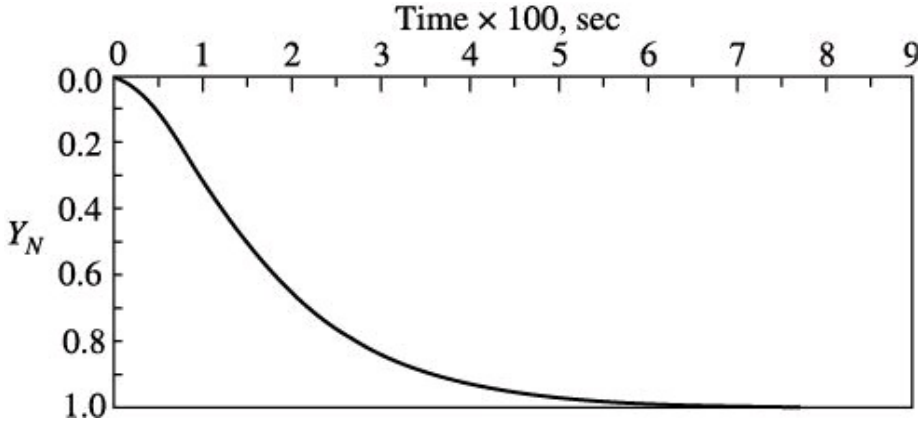

Figure 12.17 Performance comparison of the SO models.

The model responses are drawn together with the experimental one in Figure 12.17. The responses are so close to the experimental data that it is hardly possible to label them as we did in the previous examples. However, a comparison of performance may be done by calculating standard deviations. The values obtained are:

$$SD_{M1} = 0.008172,\ SD_{M2} = 0.0138 \text{ and } SD_{M3} = 0.006737 \text{ (best fit)}.$$

We may write the process TF on the most accurate model parameters:

$$G_P(s) = \frac{-0.4}{(11.75s + 1)(6.34s + 1)} \frac{\text{ppm}}{\text{l/min}}$$

The main criticism that was raised for the above class of models is that their predictability largely depends on accuracy of determining the point of inflection (PI). This accuracy becomes questionable in case the response contain slightest amount of noise. A recommendation made for increasing the accuracy: if a data set with increased number data points around the suspected location of PI is available or could be generated, a polynomial model may be obtained by any of the standard software (e.g. CurVe-EXPert). By differentiating this model and finding the location of maximum derivative value will increase the accuracy.

Further along the line originated a second category of model prediction procedures, which either do not require PI determination or the evaluation may be done by response property analysis, or by computer simulation of the given process model. In the following section, we shall discuss the method due to Harriot. Another method due to Sundersen will be shown in concern to SOPDT model parameter estimation.

(*In the following couple of examples, computer programmes written in* QBASIC *are shown to compute the solutions. The reader is urged go through Chapter 14 to acquire the necessary exposure of program writing*.)

Method 4 (Harriot): This method is based on an interesting property of step response of over damped second order systems, that is, instead of absolute time, if the normalized response $Y_N = Y/(MK_P)$ is plotted against reduced time defined as $[t/(t_1 +$

t_2)] for different values of the ratio (t_1/t_2), then,

1. All the traces meet and cross each other at $y = 0.73$ and this crossing point occurs at $t/(t_1 + t_2) = 1.3$, and
2. All the traces have maximum spread or separation along vertical axis at $t/(t_1 + t_2) = 0.5$. To check this notion, Figure 12.18 has been prepared with the help of a computer programme.

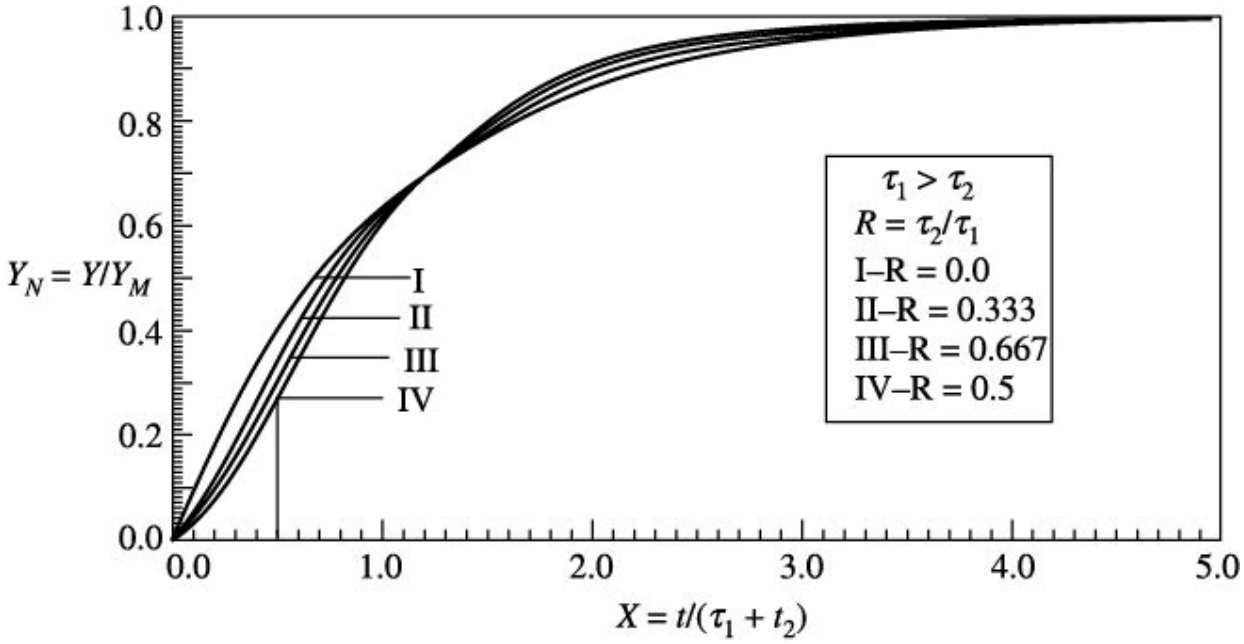

Figure 12.18 Normalized step response of over damped second order systems against reduced time.

Based on these observations for SO model parameter determination, at the value of $t/(t_1 + t_2) = 0.5$, the normalized response is plotted against $t_1/(t_1 + t_2)$ by programming the procedure on a computer. This plot is shown in Figure 12.19.

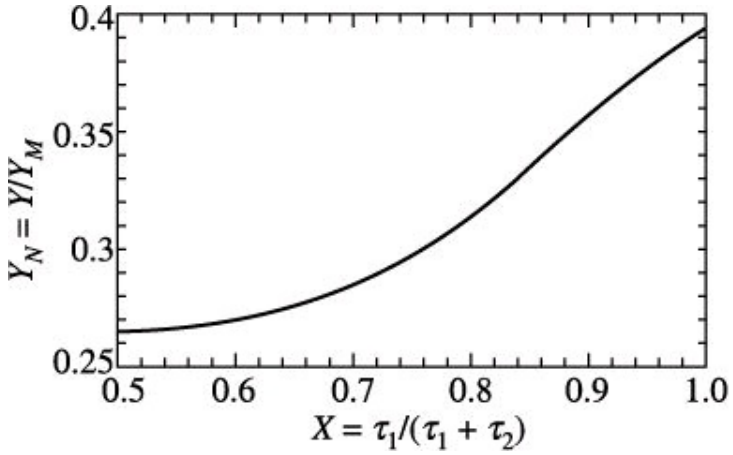

Figure 12.19 The SO model parameter estimation chart by Harriot's method.

Following steps are required to be completed to exercise this procedure.

1. From the experimental data a plot of Y_N vs. t is prepared. Using this plot, the

time required for $Y_N = 0.73$ is obtained as T_{73}, thus, $(t_1 + t_2) = T_{73}/1.3$ is noted,

2. Also, from the same plot, value of Y_N corresponding to $t/(t_1 + t_2) = 0.5$ is noted,
3. Denoting this time as T_{125}, the corresponding value of Y_N is read,
4. Putting the last obtained value of Y_N in Figure 12.19, the corresponding value of
   $t_1/(t_1 + t_2)$ is read,

is found by multiplying the ratio $t_1/(t_1 + t_2)$ by $(t_1 + t_2)$ obtained in step (1), then t_2 is found by subtracting t_1 from $(t_1 + t_2)$.

The described procedure could be implemented by writing a computer programme that will contain the experimental data set, and the trace in Figure 12.19 as a data set in another array. This procedure is demonstrated in next example.

EXAMPLE 12.4 The normalized response against time of step disturbed process is presented in the following table:

T sec	0	2	5	9	14	20	28	35
Y	0	0.0138	0.72	0.185	0.34	0.51	0.683	0.788

T sec	45	55	65	75	90	105	120
Y	0.8835	0.937	0.966	0.982	0.993	0.997	1.0

Determine the SO model parameter values using Method 4.

Solution: A computer programme in QBASIC is written for solving this problem in which the response data is stored in the arrays: AT (= time, s) and AYN (= Y_N). Also, using the procedure discussed above, the trace in Figure 12.19, is stored as a data set in the arrays XB
[= $t_1/(t_1 + t_2)$] and ZB [= Y_N].

XB	0.998	0.9597	0.924	0.891	0.861	0.832	0.805	0.78	0.7565
ZB	0.393	0.38	0.366	0.351	0.338	0.326	0.316	0.307	0.299

XB	0.737	0.713	0.694	0.675	0.657	0.64	0.624	0.609
ZB	0.293	0.288	0.284	0.28	0.277	0.274	0.272	0.27

XB	0.595	0.581	0.568	0.555	0.553	0.531	0.52	0.51

ZB	0.269	0.267	0.266	0.2657	0.265	0.2647	0.2644	0.2643

Reading of intermediate dependent variable values corresponding to a given value of independent variable occur at three stages of calculation. These are performed by linear interpolation method, because the inter-distance between data points are reasonably close. The programme is shown in the following.

```
4900    REM ************ THE PROGRAM FOR SOLVING EXAMPLE – 12.4 *********************
    DIM TB(50), YB(50), YM(50)
    DIM XB(50), ZB(50)
    SCREEN 12: XS = 250: YS = 400
    TB(1) = 0: TB(2) = 2: BTB(3) = 5: TB(4) = 9: TB(5) = 14
    YB(1) = 0: YB(2) =.0138: YB(3) = .072: YB(4) =.185: YB(5) = .34
    TB(6) = 20: TB(7) = 28: TB(8) = 35: TB(9) = 45: TB(10) = 55
    YB(6) = .51: YB(7) = .683: YB(8) = .788: YB(9) = .8835: YB(10) = .937
    TB(11) = 65: TB(12) = 75: TB(13) = 90: TB(14) = 105: TB(15) = 120
    YB(11) =.966: YB(12) = .982: YB(13) = .993: YB(14) = .997: YB(15) = 1
    ND1 = 15
    XB(1) = .998: XB(2) = .9597: XB(3) = .924: XB(4) = .891: XB(5) = .861
    ZB(1) = .393: ZB(2) = .38: ZB(3) = .366: ZB(4) = .351: ZB(5) = .338
    XB(6) = .832: XB(7) = .805: XB(8) = .78: XB(9) = .7565: XB(10) = .734
    ZB(6) = .326: ZB(7) = .316: ZB(8) = .307: ZB(9) = .299: ZB(10) = .293
    XB(11) =.713: XB(12) = .694: XB(13) = .675: XB(14) = .675: XB(15) = .64
    ZB(11) = .288: ZB(12) = .284: ZB(13) = .28: ZB(14) = .277: ZB(15) = .274
    XB(16) = .624: XB(17) = .609: XB(18) = .595: XB(19) = .581: XB(20) = .568
    ZB(16) = .272: ZB(17) = .27: ZB(18) = .269: ZB(19) = .267: ZB(20) = .266
    XB(21) = .555: XB(22) = .543: XB(23) = .531: XB(24) = .52: XB(25) = .51
    ZB(21) = .2657: ZB(22) = .265: ZB(23) = .2647: ZB(24) = .2644: ZB(25) = .2643
    ND2 = 25
    Y73 = .73
    FOR I = 1 TO ND1
    IF Y73 <= YB(I + 1) THEN GOTO 4912
    NEXT I
4912    SPL1 = (TB(I + 1) – TB(I)) / (YB(I + 1) – YB(I))
    T73 = TB(I) + SLP1 * (Y73 – YB(I))
    T12 = T73 / 1.3: T125 = T12 * .5
    FOR I = 1 TO ND1
    IF T125 <= TB(I + 1) THEN GOTO 4914
    NEXT I
4914    SLP2 = (YB(I + 1) – YB(I)) / (TB(I + 1) – TB(I))
    Y125 = YB(I) + SLP2 * (T125 – TB(I))
    FOR I = 1 TO ND2
    IF Y125 <= ZB(I + 1) THEN GOTO 4916
    NEXT I
4916    SLP3 = (XB(I + 1) – XB(I)) / (ZB(I + 1) – ZB(I))
    TRAT = XB(I) + SLP3 * (Y125 – ZB(I))
    T1 = TRAT * T12: T2 = T12 – T1
    PRINT "T1"; T1; "T2"; T2
    XS = 50: YS = 350: LINE (XS, YS) – (XS + 250, YS – 150), , B
    M = 1: ER = 0
    T = 0: DT = .01: TF = 125
4920    IF T > TF THEN GOTO 4930
```

```
      Y1 = M * (1 – (T1 / (T1 – T2)) * EXP( –T / T1)))
      Y2 = M * ((T2 / (T1 – T2)) * (EXP(–T / T2)))
      YN = Y1 + Y2
      PSET (XS + T * 2, YS – YN * 150)
      T = T + DT
      GOTO 4920
4930      FOR I = 1 TO ND1
      CIRCLE (XS + TB(I) * 2, YS – YB(I) * 150), 2
      Y11 = M * (1 – (T1 / (T1 – T2)) * EXP( –TB(I) / T1)))
      Y12 = M * ((T2 / (T1 – T2)) * (EXP(–TB(I) / T2)))
      YM(I) = Y11 + Y12
      ER = ER + (YB(I) – YM(I)) ^ 2
      NEXT I
      SD = (ER ^ .5) / ND1
      PRINT "SD"; SD
```

The result after execution of the programme is shown in Figure 12.20.

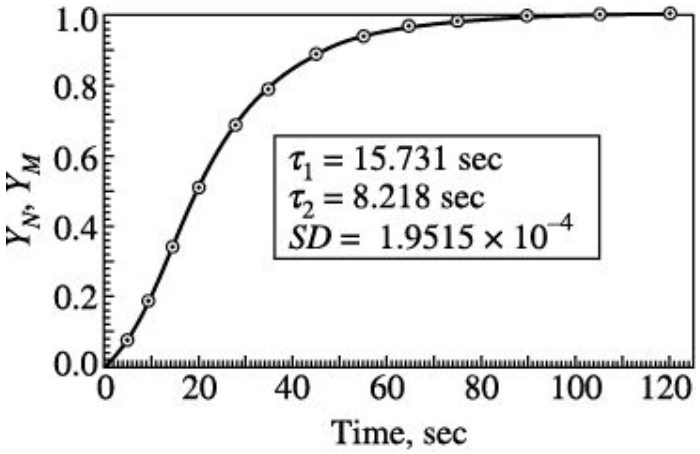

Figure 12.20 Experimental and model responses with model parameters with error.

12.7 Determination of SOPDT Model Parameters

A recognizable 'foot' with an inflection point in the response trace, indicates that a dead time added to the over damped second order model, would increase its accuracy. In the four methods discussed above, a dead time may be added by measuring the time through which the response remains at its zero value in the initial part. The SO model determination may be applied by shifting the time axis origin to the time after which the response shows first finite value.

Sundersen, et al. have presented a more accurate procedure for SO and SOPDT model parameter determination for over damped, as well as for under damped second order processes. We shall only discuss the method for over-damped process. A distinct advantage of this method is that the procedure does not require the graphical way of inflection point determination.

Sundersen Method

The over damped SOPDT transfer function is:

$$G_P(s) = \frac{Y(s)}{X(s)} = \frac{e^{-\tau_{DL}s}}{(\tau_1 s + 1)(\tau_2 s + 1)} \tag{12.25}$$

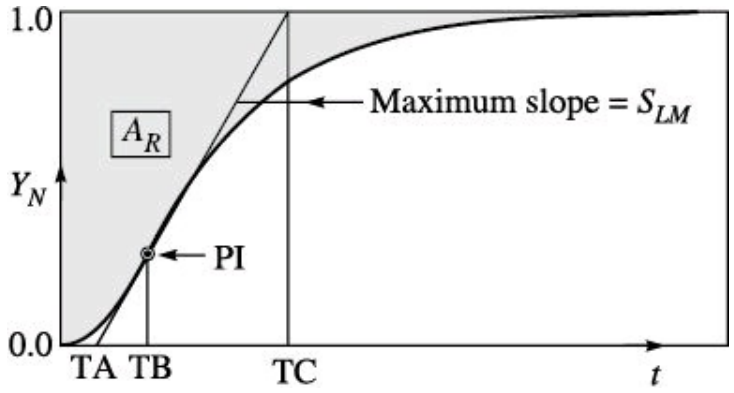

Figure 12.21 Over damped second order response.

Figure 12.21 is a redrawing of Figure 12.13 with the one exception that the response shape resembles certain distribution functions well known in statistical analysis, and the shaded area, A_R, over the trace up to the ultimate value of the process output, measures the exact value of function's first moment. Thus, first moment of the response function $Y(t)$ is,

$$A_R = \int_0^\infty [1 - Y(t)]dt \tag{12.26}$$

The expression of this first moment could also be found by applying final value theorem to derivative of TF with respect to s. This yields to,

$$A_R = -\left.\frac{dG_P(s)}{d_S}\right|_{s=0} = \tau_{DL} + \tau_1 + \tau_2 \tag{12.27}$$

We shall use the equality of above two equations for parameter estimation.

With a dead time in the dynamic model, the step response equation may be written as,

$$y(t) = 1 - \frac{\tau_1\tau_2}{\tau_1 - \tau_2}\left(\frac{1}{\tau_2}e^{-(t-\tau_{DL})/\tau_1} - \frac{1}{\tau_1}e^{-(t-\tau_{DL})/\tau_2}\right)u(t - \tau_{DL}) \tag{12.28}$$

By twice differentiating Equation 12.28 and putting $t = 0$, as we did with Equation 12.17, the expression for TB, resulting in Equation 12.20. But the existence of dead time will add up in this expression to obtain:

$$\text{TB} = \tau_{DL} + \frac{[\ln(\tau_1/\tau_2)]\tau_1\tau_2}{(\tau_1 - \tau_2)} \tag{12.29}$$

By introducing, $a = \dfrac{\tau_1\tau_2}{\tau_1 - \tau_2}$, and $b = \dfrac{\tau_1}{\tau_2}$ in Equation 12.29, we obtain,

$$\text{TB} = t_{DL} + a \ln b \tag{12.30}$$

Putting this value of TB in the first derivative equation, the expression for maximum slope, S_{LM} may be found as,

$$S_{LM} = \frac{b^{1/(1-b)}}{(b-1)a} \tag{12.31}$$

This is the slope of a tangent on the response curve that passes through PI and reaches the ultimate value of response at t = TC. The value of TC could found from the slope value as,

$$\text{TC} = \tau_{DL} + a\left[\ln b + \frac{b^2 - 1}{b}\right] \tag{12.32}$$

Equations 12.27, 12.31 and 12.32 may be combined to obtain:

$$(\text{TC} - A_R)S_{LM} = \frac{b^{1/(1-b)}}{(b-1)} \ln b \tag{12.33}$$

Putting □ = (TC – A_R)S_{LM}, Equation 12.35 reduces to:

$$\Lambda = \frac{b^{1/(1-b)}}{(b-1)} \ln b \tag{12.34}$$

For a given process TF, the response plot value of □ could be found by first determining the values of TC, A_R. and S_{LM}. Now the corresponding value of b may be found either by an implicit convergence procedure or from a plot of □ vs. b. This plot generated by a computer programme is shown in Figure 12.22.

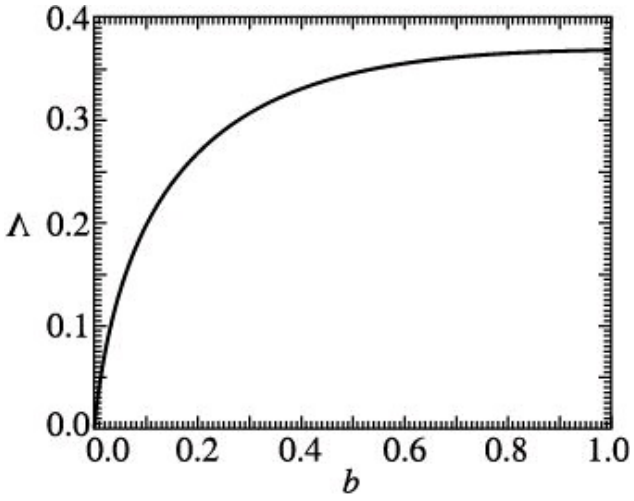

Figure 12.22 The plot of □ vs. b.

By examining Equation 12.34 we see that for critically damped condition, b = 1 and □ has its maximum value at □ = e^{-1}. Subsequently, Equation 12.30, and definition equations of a and b may be solved together to establish the following relations for determining the model parameter values:

$$\tau_1 = \frac{b^{b/(1-b)}}{S_{LM}} \tag{12.35}$$

$$\tau_2 = \frac{b^{1/(1-b)}}{S_{LM}} \tag{12.36}$$

$$\tau_{DL} = A_R - \left(\frac{b^{b/(1-b)}}{S_{LM}}\right)\left(\frac{b+1}{b}\right) \tag{12.37}$$

In case when the system is critically damped, $b = 1$. The above three equations become:

$$\tau_1 = \tau_2 = \frac{1}{S_{LM}e} \tag{12.38}$$

and

$$\tau_{DL} = A_R - \frac{2}{S_{LM}e} \tag{12.39}$$

The procedure has been demonstrated by the following example.

EXAMPLE 12.5 The TF of a third order process is given as follows:

$$G_P(s) = \frac{1}{(0.4s+1)(s+1)(2.2s+1)}$$

The time constants in this TF are in minutes. Find the parameters of an SOPDT model for this process by Sundarsen method.

Solution: As the TF of the given process is available, it will be better to write a computer programme for simulating the process. The step response generation, finding the values of A_R, S_{LM}, TA, TB and TC could be done with good accuracy by employing a small enough DT (time step of integration). After this, the value of □ could be calculated by using the RHS expression of Equation 12.35. For finding the corresponding value of b, instead of using Figure 12.21, we shall employ an implicit convergence procedure called 'Decimal Section Rejection', described in Chapter 14. As the value of b obtained is less than e^{-1}, the model parameter values are obtained by employing Equations 12.35, 12.36 and 12.37.

The computer programme written for the solution of the problem is as follows:

```
100   REM ********* PROGRAM FOR SOLVING EXAMPLE – 12.5 ***************
      DIM AX(30), ER(30)
      CLS: SCREEN 12
      XS = 50: YS = 300: LINE (XS, YS) – (XS + 300), YS – 150), , B
      XF = 20: YF = 150
      KP = 1: TP1 = .4: TP2 = 1: TP3 = 2.2
```

```
    X = 0: Y1O = 0: Y2O = 0: Y3O = 0
    T = 0: DT = .01: TF = 15
120     IF T > TF THEN GOTO 130
    X=1
    DYDT1 = X * (KP / TP1) – Y1O / TP1
    Y1N = Y1O + DY1DT * DT
    DYDT2 = Y1O / TP2 – Y2O / TP2
    Y2N = Y2O + DYDT2 * DT
    DYDT3 = Y2O / TP3 – Y3O / TP3
    Y3N = Y3O + DYDT3 * DT
    YN = Y3N: YO = Y3O
    SL = (YN – YO) / DT
    IF SL > SLM THEN GOTO 122
    GOTO 124
122     SLM = SL: TB = T: YI = YN
124     PSET (XS + T * XF, YS – YO * YF)
    AR = AR + ((1 – YO) + (1 – YN)) * (DT / 2)
    Y1O = Y1N: Y2O = Y2N: Y3O = Y3N
    T = T + DT
    GOTO 120
130     AB = YI / SLM: TA = TB – AB: AC = 1 / SLM: TC = TA + AC
    LMD = (TC – AR) * SLM
    PRINT "AR"; AR; "SLM"; SLM; "TA"; TA; "TB"; TB; "TC"; TC; "LMD"; LMD
    YG = LMD: BI = .001: BF = .999
    GOSUB 240
    TAU1 = (X ^ (X / (1 – X))) / SLM
    TAU2 = (X ^ (1 / (1 – X))) / SLM
    TAUDL = AR – (TAU2) * (((X + 1) / X)
    PRINT "TAU1"; TAU1; "TAU2"; TAU2; "TAUDL"; TAUDL
    M = 1: T1 = TAU1: T2 = TAU2
    T = 0: DT = .01: TF = 15
160     IF T > TF THEN GOTO 170
    IF T > TAUDL THEN GOTO 162
    YN = 0
    GOTO 164
162     Y1 = M * (1 – (T1 / (T1 – T2)) * (EXP( – (T – TAUDL) / T1)))
    Y2 = M * ((T2 / (T1 – T2)) * (EXP(– (T – TAUDL) / T2)))
    YN = Y1 + Y2
164     PSET (XS + T * XF, YS – YN * YF)
    T = T + DT
    GOTO 160
170     GOTO 5000

240     REM ********** SUBROUTINE DCSCRJ **********
    XI = BI: XF = BF: DX = (XF – XI) / 10: ERF = .001: NI = 1
250   FOR I = 2 TO 12
    AX(I – 1) = XI + DX * (I – 1)
    X = AX(I – 1): GOSUB 300: YC = Y
    ER(I – 1) = YG – YC
    IF ABS(ER(I – 1)) <= ERF THEN GOTO 280
    IF I > 2 THEN GOTO 255
    GOTO 260
```

```
255    IF (ER(I – 1) * ER(I – 2)) < 0 THEN GOTO 270
260    NEXT I
270    XI = AX(I – 2): XF = AX(I – 1): DX = (XF – XI) / 10: NI = NI + 1
   GOTO 250
280    RETURN
   END

300    REM *************** SUBROUTINE FX **********************
   B = X
   FX = ((B ^ (1 / (1 – B))) / (B – 1)) * (LOG(B))
   Y = FX
   RETURN
   END

5000   END
```

By executing the programme, the following results are obtained, Area A_R = 3.589707 min, Slope of the tangent through inflection point, S_{LM} = 0.225924 min^{-1}, TA = 0.732 min, TB = 1.94 min, TC = 5.1581 min. Using the RHS of Equation 13.35, □ = .35434. By applying the convergence procedure, value of b is 0.56986. Finally applying Equations 12.37, 12.38, and 12.39, model parameter values are calculated as:

$$t_1 = 2.1012 \text{ min}, \; t_2 = 1.1974 \text{ min}, \; t_{DL} = 0.2911 \text{ min}$$

Thus, the model for the process is found to be:

$$G_P(s) = \frac{e^{-0.2911s}}{(2.1012s + 1)(1.1974s + 1)}$$

The performance of the model is tested by drawing the model response along with the process response in Figure 12.23.

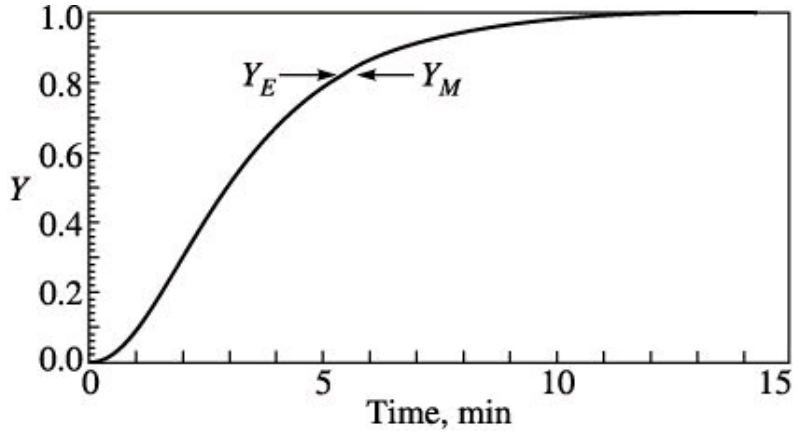

Figure 12.23 The third order process (Y_E) and SOPDT model (Y_M) responses.

Except for, very near to the initial part, experimental and model responses practically retrace each other indicating a recommendable performance of this procedure. Task of evaluating standard deviation between these traces is left to the reader.

12.8 Second Order Under-Damped Plus Dead Time Model

Generally, chemical processes do not show oscillatory response which is characteristic of under-damped second order systems. Due to the mechanism of transfer processes and hydro-dynamics that are embodied to realize the material conversion inside a process, inductance as a dynamic element is rarely evidenced in a process model. However, in Chapter 9, we have seen how even a first order system may show an oscillatory response under closed loop condition. Such responses have a good visual resemblance with those of second order under-damped systems. These approximate second order under-damped models, with or without a dead time element, serve in certain applications in controller design. Sundersen and Smith has proposed procedures for parameter estimation of an SOPDT model from an oscillatory step response data. Accuracy found by these procedures is limited, and especially Sundersen's method is only suitable for a class of response shapes, and the amount of algebra involved is also rather tedious.

In Chapter 6, we followed a direct and simple method of under-damped SO model by a simple graphical construction on the oscillatory step response. The same procedure will be followed only with a modification to include a dead time element. This will be done by noting the length of the 'foot' of response. Let us try an example.

EXAMPLE 12.6 The same mixer-heater system that we are following up from Chapter 1 is presented with a (PI) controller in its loop for controlling the exit liquid temperature. The closed loop system is shown by the following block diagram in Figure 12.24. The process features an FOPDT model. The time constants in the block elements are in seconds.

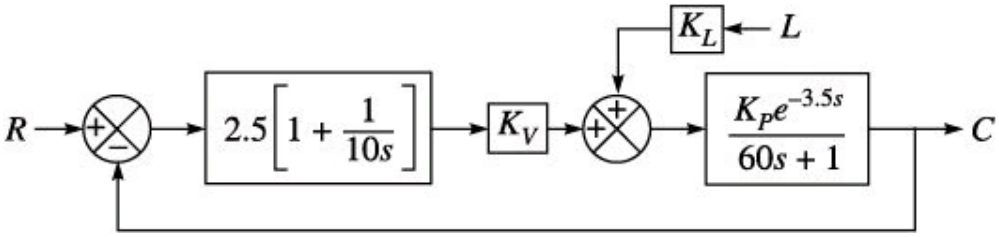

Figure 12.24 Block diagram of an FOPDT process controlled by a (PI) controller.

The system is running at steady state with a set point, $R = 50°C$, when R is suddenly raised to 51°C according to step change. The exit temperature trace is shown in Figure 12.25.

Determine the parameters of an under-damped second order model.

Solution: The required construction is made on the closed loop response. Following the proposed guide lines, the t_{DL} is found from the length of response foot as, $t_{DL} =$ 3.6 s. The damping coefficient is found from the value of overshoot, *OS*. *OS* is found from the dimensioned lengths of the response. Thus,

$$OS = \text{B/A} = 0.3665848$$

Applying the equation,

$$OS = e^{-\zeta\pi/\sqrt{1-\zeta^2}}$$

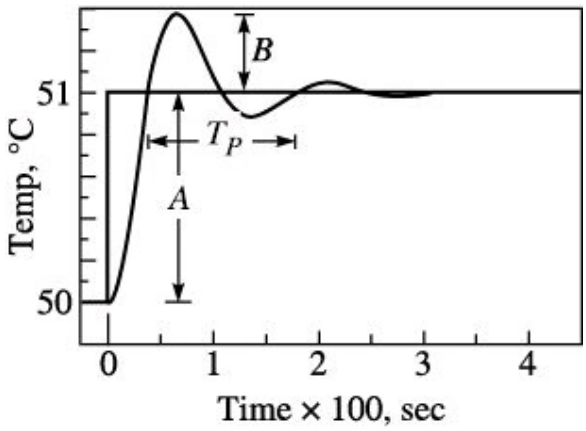

Figure 12.25 Servo response of the mixer-responses.

The damping coefficient is determined as, $z = 0.3053579$. Then, from the first three crossing points of response with the ultimate response value the time period of oscillation, $TP = 187.2 - 45.8 = 141.4$ s. Hence the corresponding radian frequency, w, is:

$$\omega = \frac{2\pi}{T_P} = 0.0443477 \text{ rads/s}$$

Using the relation between radian frequency and natural frequency (w_N) of the system, and knowing that $tP = 1/w_N$, the time constant of the model TF is calculated as:

$$\omega_N = \frac{\omega}{\sqrt{1-\zeta^2}} = 0.0466635 \text{ rads/s.}$$

Thus,

$$\tau_P = \frac{1}{\omega_N} = 21.43 \text{ s}$$

The system gain KP, is unity because the ultimate change in exit temperature is equal to the step magnitude. The model TF is:

$$G_P(s) = \frac{K_P e^{-\tau_{DL}s}}{\tau_P^2 s^2 + 2\zeta\tau_P s + 1}$$

where, the parameter values are given above.

To check model accuracy the experimental, Y_E and model, Y_M, responses are simultaneously drawn in Figure 12.26, and standard deviation (SD) between the responses have been found to be, $SD = 0.08225$.

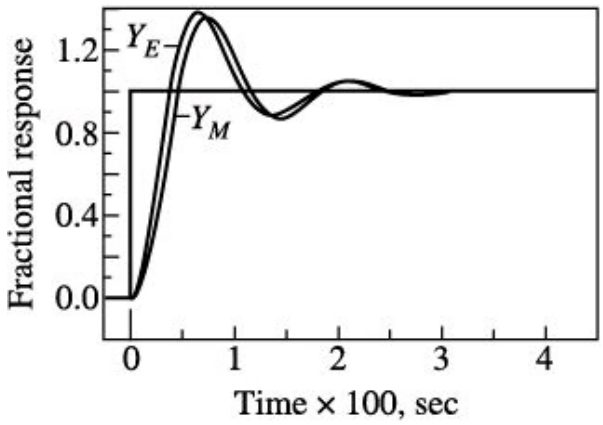

Figure 12.26 The experimental and model heater system.

12.9 Approximation of SO Response by FOPDT Model

Probably to demonstrate the application potential of FOPDT model, some workers have tried to fit SO response data by an FOPDT model. In this area, Martin's work is worth mentioning. Assuming unity gain, if the two time constants t_1 and t_2 of the second order system are given, the required t_P and t_{DL} of the FOPDT model can be expressed by the following two empirical equations:

$$\frac{\tau_P}{\tau_1} = 0.828 + 0.812\frac{\tau_2}{\tau_1} + 0.172e^{-6.9\frac{\tau_2}{\tau_1}} \qquad (12.40)$$

$$\frac{\tau_{DL}}{\tau_1} = \frac{1.116.\tau_2}{\tau_1 + 1.208.\tau_2} \qquad (12.41)$$

Along with these relations, Martin has also presented a graphical guide chart to find t_P and t_{DL} from the SO time constants. The FOPDT parameter prediction performance of the above two equations are quite acceptable compared to the guide chart, hence we refrain to add it in this text. To prove this, the following example is presented.

EXAMPLE 12.7 The TF of the given second order system is,

$$G_P(s) = \frac{1}{(8s+1)(5s+1)}$$

Find the equivalent FOPDT model parameters by Martin's method. The time constants are in minutes.

Solution: Applying Equations 12.42 and 12.43, the required FOPDT parameters are

found. They are, t_P = 10.7024 min, and t_{DL} = 3.1795 min. To compare between the true SO and modeled FOPDT responses, a simulation programme is written. We are not including the programme, the resulting true and model responses together with the calculated standard deviation between them are shown in Figure 12.27.

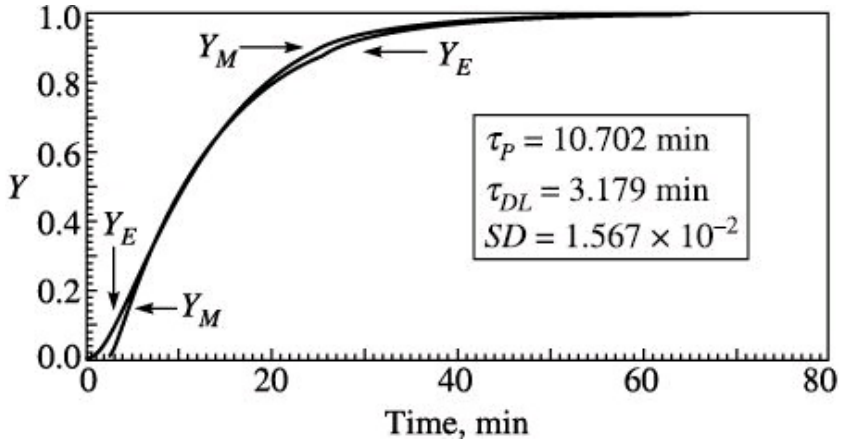

Figure 12.27 True SO and modeled FOPDT responses.

Fitting of higher order (order > 1) system's responses with FOPDT model faces the primary difficulty of adjusting the rounded portion with a 'foot' (for order > 2) or without a 'foot' (for order = 2), corresponding to the 'knee' (at the end of dead time and beginning of first order part of the FOPDT response. A large part of the error usually comes from the mismatch in this part. Martin's algorithm seems to have worked reasonably well in this aspect because, the knee and the initial rounded part are fairly close, and the responses have crossed each other three times within the time to reach steady state

12.10 Frequency Domain Methods

There are several ways by which, from the frequency response data of an unknown process, its dynamic model may be obtained in the transfer function form. Usually, the frequency response data is expressed in the form of a Bode's plot for this purpose. The process transfer function is postulated using the following frequency response properties:

1. For each pole in the transfer function, the high frequency asymptote has a slope =
   –2 0 *dB*/decade. The high and low frequency asymptotes intersect at corner frequency, w_C.
2. For each zero in the transfer function, the high frequency asymptote has a slope =
   + 20 *dB*/decade. The high and low frequency asymptotes intersect at corner

frequency, w_C.

3. For each under and critically damped second order pole in the transfer function, the high frequency asymptote has a slope = – 20 *dB*/decade. The high and low frequency asymptotes intersect at corner frequency, w_C.
4. For each under and critically damped second order zero in the transfer function, the high frequency asymptote has a slope = + 20 *dB*/decade. The high and low frequency asymptotes intersect at corner frequency, w_C.
5. For each order of integration in the transfer function, the dB locus has a slope = –20 *dB*/decade, and it passes through 0 dB horizontal at $w = 1$.
6. For each order of differentiation in the transfer function, the dB locus has a slope = + 20 *dB*/decade, and it passes through 0 dB horizontal at $w = 1$.
7. By multiplying the transfer function by a constant K, which is equivalent of changing the Gain, the *dB* plot at all frequency is shifted vertically by 20. $\log_{10}$(K).
8. The phase plot has a maximum lag of 90º per pole in the transfer function, and has a maximum slope at the corner frequency. $w = 1/t$, where the phase lag is 45º.
9. The phase plot has a maximum lead of 90º per pole in the transfer function, and has a maximum slope at the corner frequency. $w = 1/t$, where the phase lead is 45º.
10. The phase plot has a maximum lag of 180º for each under or critically damped second order pole in the transfer function, and has a maximum slope at the corner frequency, $w = 1/t$, where the phase lag is 90º.
11. The phase plot has a maximum lead of 180º for each under or critically damped second order pole in the transfer function, and has a maximum slope at the corner frequency, $w = 1/t$, where the phase lag is 90º.
12. The phase plot has a constant lag of 90º at all frequencies for each order of integration in the transfer function.
13. The phase plot has a constant lead of 90º at all frequencies for each order of differentiation in the transfer function.
14. A dead time element ($e^{-t_{DLS}}$) in the transfer function only contributes to the phase plot as a linear function of frequency ($= wt_{DL}$), but does not affect the dB plot. Thus a dead time element has a phase lag value of 1.0 radian at $w = 1/t_{DL}$.

To develop the transfer function, it is recommended to draw the dB and □ locus on

a single semi-log graph paper so that value or slope comparison of dB or phase at any frequency becomes more accurate. Methods of this category differ only by the way the frequency response is obtained; not by the way it is treated.

12.10.1 Direct Sine Wave Testing

The disturbance or input to any process plant is usually effected either by changing the Control Valve position, or the Set point to the Controller in a sinusoidal manner. This can be accomplished in a number of ways, but commercially available mechanical, electronic or pneumatic sine wave generators are usually employed.

After waiting till all transients have died out and a steady oscillation in the output has been established, the amplitude ratio as dB and phase angle, ϕ, are found by recording the input and output sine waves. Thus, any of the standard frequency response representations may be used to express the data at as many frequency values as required by repeating the above procedure. Direct sine wave testing is an extremely useful way of obtaining precise dynamic information of a process. Order of a system, damping character, time constants, all can be accurately determined.

The main disadvantage of direct sine wave testing is that it may be very time consuming when applied to large scale process equipment and plants, because steady oscillatory conditions must be established at each frequency value that may take many minutes or even hours. Also, there are operational hazards of running a plant to its stability limit by bringing an oscillation of untoward amplitude. Furthermore, a running plant may not be allowed to operate in an oscillatory manner for a long time, producing a large amount of out of quality product, incurring a huge economic loss. Hence, the next development in this area has been by the way of using a transient or non-oscillatory input for frequency response generation. Pulse type of input has been used for this purpose with greater degree of success.

How to treat frequency response data to obtain the TF of a process with unknown dynamics has been discussed to some extent with worked out examples in Chapter 11. Example 11.13 should be revisited for this purpose.

In this chapter, the experimental methods of transfer function postulation of an unknown process have been surveyed. The discussion is restricted within bounds of deterministic signal processing and applications are for control system design of a variety of processes. If we consider the parent subject ‘Process System Analysis’ of which this chapter title is only a part, and be aware that there always has been an exchange of information between these two disciplines of applied science at various levels, then an interested reader may want to know about the state of affairs in the larger scenario of the parent subject. For example,

1. Flow modeling of non-ideally mixed systems,
2. Processing of noisy signal for TF determination,
3. Use of synthetic noisy input for model building, etc.

We have cited a few literature references that cover these topics. In Chapter 14, two example programmes are included to show the technique of model building of non-ideally mixed systems.

EXERCISE PROBLEMS

12.1 The % of exit tube temperature change due to a 155% step change of tube inlet temperature of a small laboratory concentric tube heat exchanger is recorded in the following. Find the SO model parameters by Harriot's, and the Indigenous method. Also find the Standard Deviation between observed and model response obtained by these methods.

t, s	0	2	5	9	14	20	28	35
Y%	0	2.4	9.8	22.9	38.7	54.2	69.2	78.3

t, s	45	55	65	75	90	105	120
Y%	86.8	92	95.2	97	98.6	99.3	100

12.2 Generate a set of at least 20 number of data for the unit step response of the following second order transfer function:

$$\frac{Y(s)}{X(s)} = \frac{1}{(5s+1)(2s+1)}$$

Use at least two approximations for SO model parameters and one for FOPDT model parameters. Compare their performance by calculating SD.

12.3 A two stirred tank constant holdup mixing system is processing a solution of an organic acid. The system is at steady state with an inlet and exit concentrations of 0.15 g/l, when a step increase of 0.15 g/l is made at the inlet. Assume that the solution density is independent of concentration. If the solution flow rate is steady at 12 l/min, find the tank volumes from the SOPDT model parameters using following step response data of the exit response.

t, s	0	10	12.5	16.25	21.25	27.5	35	45	53.75
C_O g/l	0.15	0.15	0.152	0.16	0.175	0.195	0.22	0.24	0.254

t, s	66.25	78.75	91.25	103.75	122.5	141.25	160
C_O g/l	0.269	0.28	0.286	0.29	0.295	0.297	0.3

12.4 The inlet air temperature of a tunnel type continuous drier is raised from 70°C to 82.5°C, according to step change. The rise in exit temperature air is recorded as ΔT°C against time in minute as in the following. Find the FOPDT and SOPDT model parameters from this data by as many methods as you can and compare their performances.

t, min	0	2	2.5	3.25	4.25	5.5	7.0	9
ΔT, °C	0	0	0.06	0.317	0.838	1.57	2.4	3.3

t, min	10.75	13.25	15.75	18.25	20.75	24.5	28.25	32
ΔT, °C	3.85	4.36	4.66	4.82	4.9	4.96	4.99	5.0

12.5 In a pilot scale biogas generator, the produced gas is accumulated in a gas holder. The following data shows the integrated gas volume in cubic meter against time in hours. Assuming this is the step response data of an equivalent continuous system that provides a residence time of 120 hours, derive the system model by using frequency response technique.

t, h	0	2	5	9	14	20	28	35
Q_G m^3	0	0.03	0.16	0.42	0.76	1.2	1.64	1.92

t, h	45	55	65	75	90	105	120
Q_G m^3	2.18	2.33	2.4	2.45	2.48	2.49	3.0

12.6 For the following fourth order process TF, find the parameters of an SOPDT model by Sundarsen's method and evaluate the model performance by calculating standard deviation.

$$G_P(s) = \frac{1}{(s+1)(s+4)(s+4.5)(s+5)}$$

12.7 Determine the FOPDT model parameters by Martin's method for a unity gain over-damped second order system having two constants, t_1 = 3.5 min and t_2 = 5 min.

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13

Controller Tuning

13.1 Controller Tuning

Uptil now, in our discussion, we have learnt to analyze the stability of a linear dynamic system, whether it is an open loop process or a close loop control system. If such a system contains entities like delay or dead time, or if any of the loop elements have a nonlinear gain character, we know how to linearize them and evaluate the system stability, at least in the vicinity of its operating region. Also, we know from the discussion regarding root-locus analysis that by decreasing the controller gain K_C the stability measure of a close loop system with a (P) controller, may be increased. The above notion is true if we exclude systems that are conditionally stable or possess other complexities in their dynamic character, e.g. inverse response processes, non-stationary processes, etc.

But how far we should go in increasing stability? Is there any other kind of adverse dynamic feature that may crop up in operating a control system at very high stability conditions and decrease its performance? If so, then what should be the optimum stability condition for running a controlled process? These are important considerations, addressed by the optimum controller parameter design exercise or more commonly known as controller tuning.

We know from our close loop dynamic response studies that by increasing stability, we loose speed of the response of the system. That is, the controller requires more time to drive out a quantum of bad quality product that has entered into the process by way of disturbance. This feature will definitely deteriorate the close loop performance. Hence, controller tuning in principle is a 'trade-off' between speed and stability of a control system. In case of a multiple mode controller application for a given process, the loop stability becomes a function of all the parameters present in the controller, and the tuning of such systems involves a complex adjustment of all these parameters (e.g. K_C, t_I, and t_D). The optimization exercise is carried with respect to a definite objective function or criterion that is usually a close loop performance specification. Majority of these analyses involve a multiple parameter search procedure to seek and find the optimum point in terms of the given performance specification.

In the past few decades, a number of analytical controller tuning approaches have been developed both from the industry and academic field. Practically every tuning

method has two parts in its exercise:

(A) To ascertain the dynamic characteristics of the process loop (process + measuring element + final control element), except the controller;

(B) To formulate the controller characteristics in a way so that it matches the process character to produce an output of desired quality.

The development of these formulae proceeds as: For a given process, the controller parameters are gradually adjusted until the desired objective function value is reached by a search procedure. A large number of process conditions are used, and the resulting controller parameter values are combined to form a set of equations or charts from which the optimum value of controller may be found as a function of process parameters.

Before discussing such analytically developed methods, we feel that a 'trial-and-error' based tuning procedure should be presented that an instrument mechanic follows to tune a (PI) controller already installed on line. This method has come up following a long legacy of experience gathered and descended by generations of instrumentation professionals. This method has been presented not only for an historical interest, but as you shall see in the later sections, that how close the results of this method are to those obtained from some of the analytical methods developed later.

13.1.1 On line Trial and Error Tuning Method

Dynamic testing on the given process is performed by executing the following stepwise procedure to find the optimum controller parameter values.

Step 1. Eliminate integral and derivative action by setting t_I to its maximum value and t_D to its minimum value.

Step 2. Set K_C at a low value (say K_C = 0.2 to 0.5), put the controller on automatic and apply a small step in load or set point.

Step 3. Increase the K_C by small increments, and each increment is followed by a small set point or load change, until sustained oscillation with constant amplitude (continuous cycling) of the measured variable is observed.

Step 4. Reduce K_C by a factor of 2.

Step 5. Decrease t_I in small increments, and each increment is followed by a small set point or load change, until a sustained oscillation with a constant amplitude (continuous cycling) of the measured variable is observed. Set t_I equal to three times of this value.

Step 6. Increase t_D in small increments, and each increment is followed by a small set

point or load change, until a sustained oscillation with a constant amplitude (continuous cycling) of the measured variable is observed. Set t_D equal to one third of this value.

The value of K_C, that results in continuous cycling, is termed as the ultimate gain and is denoted by K_U. We already know the term K_U from the studies on stability analysis by direct substitution method.

Due to the presence of saturation in the process and/or controller characteristics, a type of sustained oscillation may occur for $K_C > K_U$, that should not be considered as true continuous cycling and may lead to an inaccurate estimate of controller setting value. Practically all industrial controllers posses this saturation feature and during the testing, care must be exercised to differentiate between curves presented in the following two types of cycling shown in Figure 13.1.

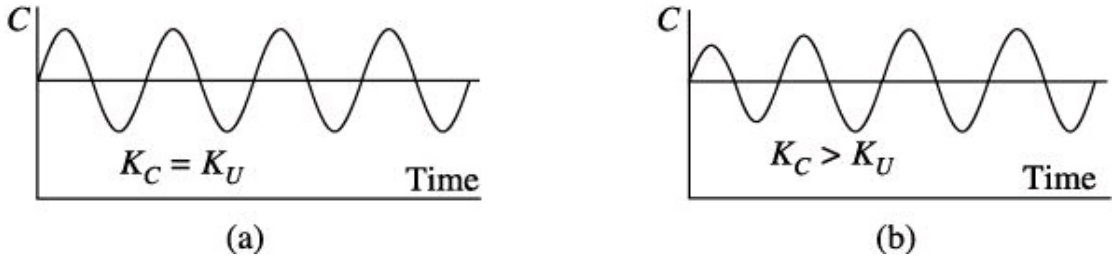

Figure 13.1 (a) Proper continuous cycling at limiting stability. (b) Fictitious cycling at unstable condition due to saturation.

For an available process model K_U may be calculated by applying the direct substitution method.

Disadvantages of trial and error procedure

1. In case of a slow process, this would be a time-consuming procedure due to the requirement of large number of trials, and this feature may also lead to formation of a large amount of out of specification product.
2. May lead the process operation to hazardous conditions, as it is pushed to the state of limiting stability condition.
3. Not applicable to open loop unstable processes, as they are unstable at high or low values of K_C, but may be stable for intermediate values of K_C.
4. Certain true lumped parameter systems (e.g. processes having FO or SO model configuration without dead time), do not possess an ultimate gain character, and hence this method is inapplicable for such systems.

13.1.2 Ziegler–Nichol's Continuous Cycling Method

Ziegler and Nichol in their famous publication in 1942, presented the controller setting

formulae for (P), (PI), and (PID) controllers based on continuous cycling experiment carried on the close loop process. The sustained oscillatory condition is to be obtained by the way as described in the previous section after having the controller turned to manual with only the proportional mode remaining active. Their objective function was that the tuned close loop system would produce an under damped oscillatory response in the disturbed condition, with quarter decay ratio
($DR = 0.25$) character. The Z–N method consists of the following implementation steps:

1. The close loop system is turned to manual control, and the integral and derivative actions, if present, are nulled by setting t_I to its maximum value and t_D to its minimum value.
2. From an initial small value, K_C is increased in small steps, and for each K_C setting, a small step change in set point is made. This procedure is continued until we obtain continuous cycling of the measured variable. We already know that this value of K_C is the ultimate gain K_U, and the period of oscillatory response is the ultimate period P_U.

Using the values of K_U and P_U, Ziegler and Nichol recommended the following controller parameter setting formulae:

Table 13.1 Ziegler and Nichol's Controller Setting Formulae

Controller Mode	K_C	x_I	x_D
(P), $G_C(s) = K_c$	$0.5K_U$	—	—
(PI), $G_C(s) = K_C[1 + 1/(x_I s)]$	$0.45K_U$	$P_U/1.2$	—
(PID), $G_C(s) = K_C[1 + x_D s + 1/(x_I s)]$	$0.6K_U$	$P_U/2$	$P_U/8$

In the above table, it is observed that by including (I) mode for a (PI) controller, the recommended gain is lower than for a (P) only controller, and further addition of a (D) mode for a (PID) controller, the recommended gain is higher than for a (P) only controller. This observation tallies with the fact that integration contributes a LAG in the close loop dynamics by moving the system closer to the stability limit, while a (D) action adds a LEAD, which further stabilizes the system.

Disadvantages of Z-N Controller Setting Formulae based on Continuous cycling

1. As in the trial and error procedure, uneconomic and hazardous plant operations may result due to limiting the stability conditions during the testing.

2. No knowledge of the close loop dynamic model could be obtained from such testing, hence, for any change of any of the process parameters, the whole testing sequence has to be repeated.
3. To obtain $DR = 0.25$, it is required that the overshoot should be 50%. Such an initial overshoot may be unacceptable in many process operations, especially in case of set point disturbance.
4. The parameter combination, found by Z-N formula for quarter decay ratio type response is not unique, except for a (P) controller. For (PI) and (PID) controlled loop, many other parameter combinations could be found that also produce quarter decay ratio type response, and there are possibilities that some of them may offer close loop response with more desirable response characteristics. One of the most important of such characteristics is the 'settling time' or 'response time' that determines how long the error stays in the process.

In the 1950s, the quarter decay ratio characteristics were criticized as being too oscillatory and certain modification of original Z-N setting formulae appeared in literature. One of them claims to offer a more orthodox close loop dynamic response. They are presented only for PID tuning in Table 13.2.

Table 13.2 Original and Modified Z-N Setting Formulae for a PID Controller

Feature	K_C	x_I	x_D
Original ($DR = 0.25$)	$0.6\,K_U$	$P_U/2$	$P_U/8$
Some Overshoot	$0.33\,K_U$	$P_U/2$	$P_U/3$
Zero Overshoot	$0.2\,K_U$	$P_U/2$	$P_U/3$

Another modification forwarded by Tyreus and Luyben appeared in the text in 1997. This approach also uses the continuous cycling to determine K_U and P_U, and determines the controller setting for (PI) and (PID) controllers from the formulae given in Table 13.3.

Table 13.3 Tyreus-Luyben Modification of Z-N Settings

Controller Mode	K_C	x_I	x_D
(PI)	$K_U/3.2$	$2.2P_U$	–
(PID)	$K_U/2.2$	$2.2P_U$	$P_U/6.3$

An important criticism against continuous cycling method is that such estimation does not offer any information about the process transfer function (Item 1. in the above disadvantage list). This issue has been addressed by the development of estimation methods based on process reaction curve. But before discussing them, we should review a more recent method based on forced continuous cycling of processes.

13.1.3 Auto-Tuning of Controllers by Forced Cycling

Astrom and Haglund presented a procedure of automatic tuning (auto-tuning) of controllers. This method has been originally designed as an on-line controller-tuning algorithm as a part of an Adaptive Control programme. Later it was found that the method may be also applied for the determination of K_U and P_U in an off-line fashion, after which the Z-N formulae may be applied to find the controller parameter values. The method has the following features:

1. The analog controller present in the loop could be bypassed by an ON-OFF controller with a differential. The maximum and minimum value of the manipulated variable issued from the FCE due to control action can be adjusted. Usually, the difference between these values is only a few percent of the full range of MV, and it is adjusted in such a way that the controlled variable can play above and below the set point with approximately equal magnitude having an approximately sinusoidal shape due to the presence of RC-lags in the process loop. In case of a noisy signal, this range is reasonably high than the noise threshold. The time period of the relay action is also adjustable.
2. Usually a single experiment is sufficient for the estimation of K_U and P_U.

The FCE action and CV response is illustrated with the required construction required in this method has been illustrated in Figure 13.2.

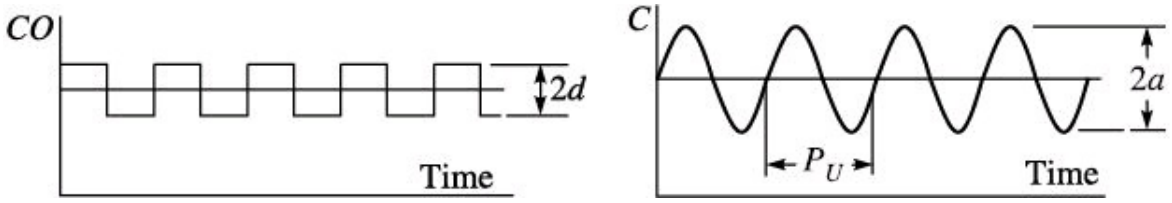

Figure 13.2 Auto-tuning using relay controller.

The ultimate period P_U is found by measuring the period of process variable oscillation. The ultimate gain is found as:

$$K_U = \frac{4d}{\pi a} \qquad (13.1)$$

The controller setting may be found by applying Z-N formulae given in Table 13.1.

To address the criticism against continuous cycling that the technique conveys no information about the process dynamic model, a number of methods were developed based on the Process Reaction Curve (PRC) for determination of the process model. The first one of them was again published by Ziegler and Nichol in 1942. Before discussing them, let us understand what PRC is, and how it is experimentally obtained

and processed to obtain the process dynamic model.

13.1.4 Process Reaction Curve (PRC)

The PRC of a process is obtained by putting a step input to the process under an 'off-line' condition and recording the response data. Thus, PRC may be alternatively called as Step Response Curve (SRC). For further processing of the result, input magnitude of the step is also required.

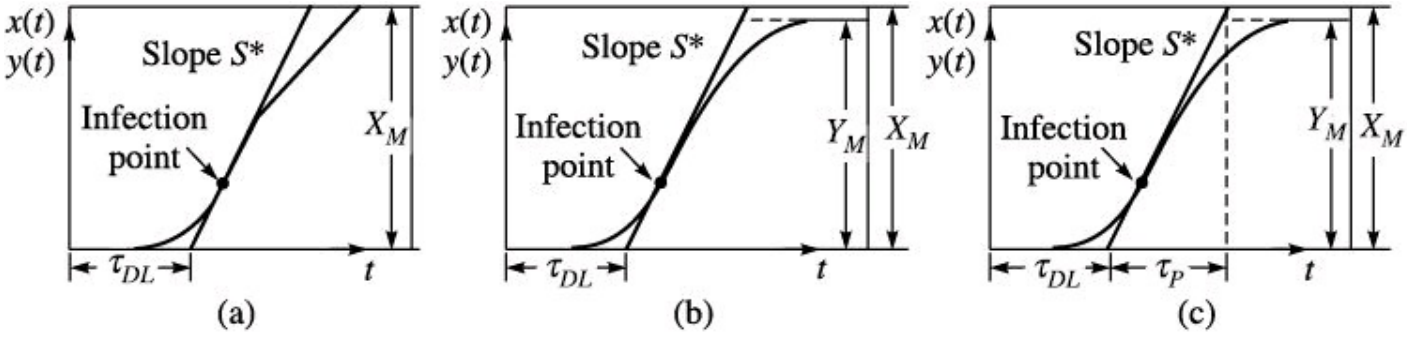

Figure 13.3 The different graphical construction on PRC trace used for process parameter estimation. (a) PRC for unstable process, only t_{DL} and S^* are used for characterization. (b) PRC for stable processes, only t_{DL} and S^* are used for characterization. (c) PRC for stable processes, all FOPDT parameters are used for characterization.

Two fundamental types of PRC are obtained:

1. For open loop stable processes, where the output shows a finite ultimate value, and
2. For open loop unstable processes, where the output grows out of bounds, and there is no finite ultimate value.

A major number of industrial processes show open loop stable type PRC, and the most popular transfer function configuration recommended for them is First Order Plus Dead Time (FOPDT) model. The transfer function of an FOPDT model may be expressed as:

$$G_P(s) = \frac{K_P e^{-\tau_{DL} s}}{\tau_P s + 1} \qquad (13.2)$$

The model parameters, K_P, t_P, and t_{DL} may be evaluated from the PRC. The t_P, and t_{DL} from the graphical construction and K_P by using the definition of process gain:

$$K_P = \frac{Y_M}{X_M} \qquad (13.3)$$

With the tuning formulae based on PRC, we should either use the FOPDT model, or

use the graphical parameters, t_{DL}, and S^*, where S^* is slope value of the tangent through inflection point or the maximum slope point on the step response locus. In absence of an ultimate value for open loop unstable processes, use of t_{DL}, and S^* has been made to encompass stable and unstable process under the same framework of tuning formulae. In the following we shall discuss such a method also due Ziegler and Nichol.

13.1.5 Ziegler–Nichol's Formulae Based on PRC

In the same paper published in 1942, Ziegler and Nichol presented a method of optimum controller tuning based on the PRC or step testing of the process. They used the same quarter decay ratio of the close loop response as their criterion of optimization. This method uses t_{DL}, and S^*, two parameters of the PRC locus, for unstable or stable response types. The controller parameters may be estimated by using the Table 13.4.

Table 13.4 Ziegler and Nichol's Formulae based on PRC Parameters

Controller Mode	K_C	x_I	x_D
(P)	$1.0/(x_{DL}\,S^*)$	–	–
(PI)	$0.9/(x_{DL}\,S^*)$	$3.33\,x_{DL}$	–
(PID)	$1.2/(x_{DL}\,S^*)$	$2\,x_{DL}$	$0.5\,x_{DL}$

Smith and Corripio have forwarded a modification of these formulae that uses the FOPDT model parameters determined by Method 1 described in Chapter 12, or from PRC for stable processes as given in Figure 13.3(c). The formulae are listed in Table 13.5.

Table 13.5 Ziegler and Nichol's Formulae based on FOPDT Parameters

Controller Mode	K_C	x_I	x_D
(P)	$(x_P/K_P x_{DL})$	—	—
(PI)	$(0.9x_P/K_P x_{DL})$	$3.33x_{DL}$	—
(PID)	$(1.2x_P/K_P x_{DL})$	$2.0x_{DL}$	$0.5x_{DL}$

The relations given in Table 13.5 will be used to find controller settings in this chapter and they will be denoted as Z-N* to represent the modified Ziegler-Nichol formulae. They fall under the same framework of process parameter evaluation as other tuning formulae based on PRC as due to Cohen-Coon and Lopez discussed in the following sections.

13.1.6 Cohen and Coon Formulae Based on PRC

Cohen and Coon, in 1953, published a set of controller tuning formula based on the

FOPDT model parameter estimation from PRC as indicated in the extreme left diagram in Figure 13.3. Like Z-N the Cohen, Coon (C-C) formulae are also developed considering the quarter decay ratio of the close loop response as the criterion of optimization. After finding the model parameters, K_P, t_P, and t_{DL} from the indicated graphical construction on PRC, controller parameters K_C, t_I, and t_D may be evaluated by using Table 13.6.

Table 13.6 Cohen and Coon Formulae based on FOPDT Parameters

Controller Mode	K_C	x_I	x_D
(P)		–	–
(PI)			–
(PID)			
(PD)		—	

A word about the process TFs that are treated for finding optimum controller setting exercise is that such a TF should contain all the dynamic blocks in the loop because the controller is going to appear as the dynamic compensator of the total loop. Hence, such a TF will be represented as $G(s)$ instead of $G_P(s)$, and $G(s) = G_V(s)G_P(s)G_M(s)$. In special cases where the feedback path has a unity TF $G(s) = G_V(s)G_P(s)$, and if $G_V(s) = G_M(s) = 1$, then $G(s) = G_P(s)$. In the following few examples, the controller setting determination techniques, and performance of closed loop performance using some of the setting formulae given above, are compared.

EXAMPLE 13.1 The system of four reactor CSTR battery handling a first order irreversible reaction is the subject in this example. It is required to find the controller setting values by the different formulae that have been introduced.

Solution: The process TF between exit and inlet concentrations, is found as:

$$G_P(s) = \frac{C_O(s)}{C_I(s)} = \frac{1/16}{(5s+1)^4}$$

1. A control system is simulated with a (P) controller in which the other block elements have unity TF. The closed loop response with respect to a step change in set point is studied for increasing values of K_C. With a first set of K_C values of 30, 40, 50, 60 and 70, it was found that the limiting value lies between 60 and 70. Further trials with gradually smaller increments in K_C ultimately ascertained the limiting K_C value to be 63.8.

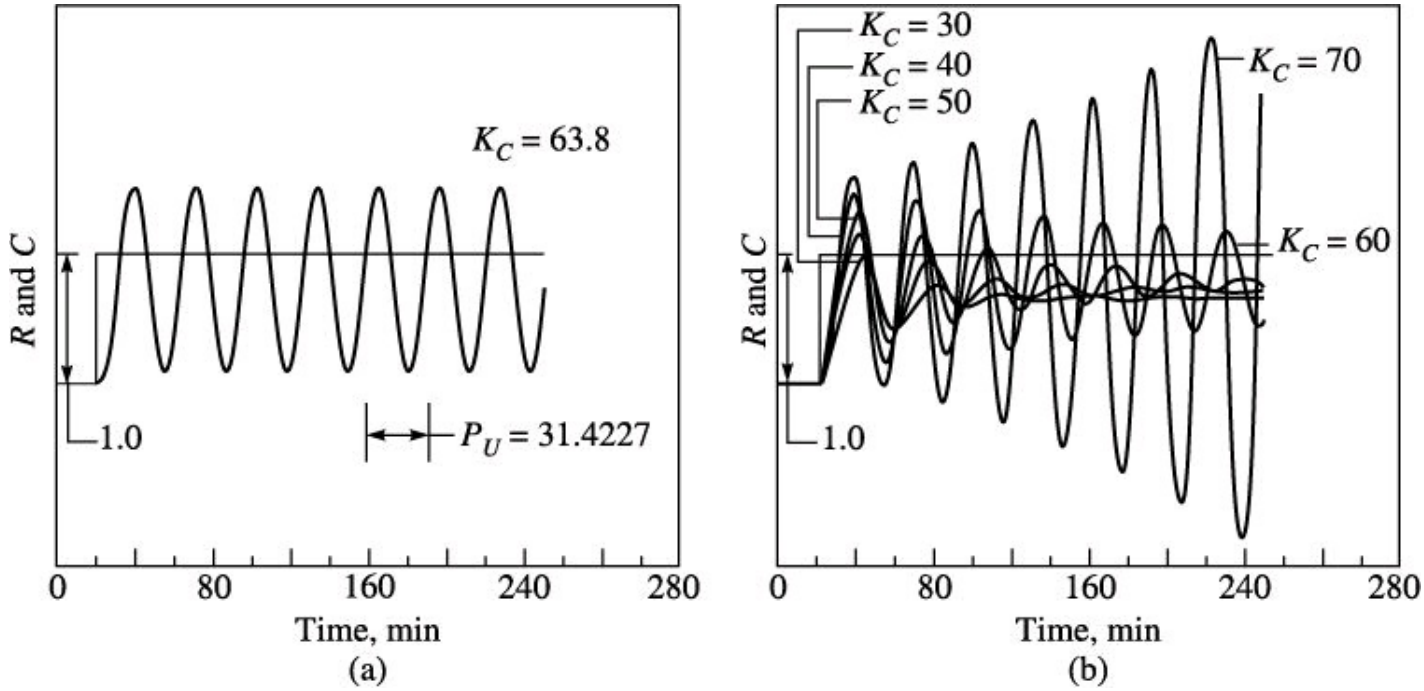

Figure 13.4 The trials with K_C to determine K_U and P_U.

The slight discrepancy with the earlier found value of 64 with the same system in Chapter 10 may be due to the inherent error associated with the integration algorithm used for solving differential equations (in this system they are four in series). The results are shown in Figures 13.4(a) and (b). In Figure 13.4(b), from the sustained oscillation the time period has been found. The results are,

$$K_U = 63.8,\ P_U = 31.4227 \text{ min},\ w_U = 0.199956888 \text{ rads/min}.$$

2. The open loop system is simulated. A step response or process reaction curve (PRC) is generated with the model. The construction shown in Figure 13.3(c) is repeated on the response in Figure 13.5.

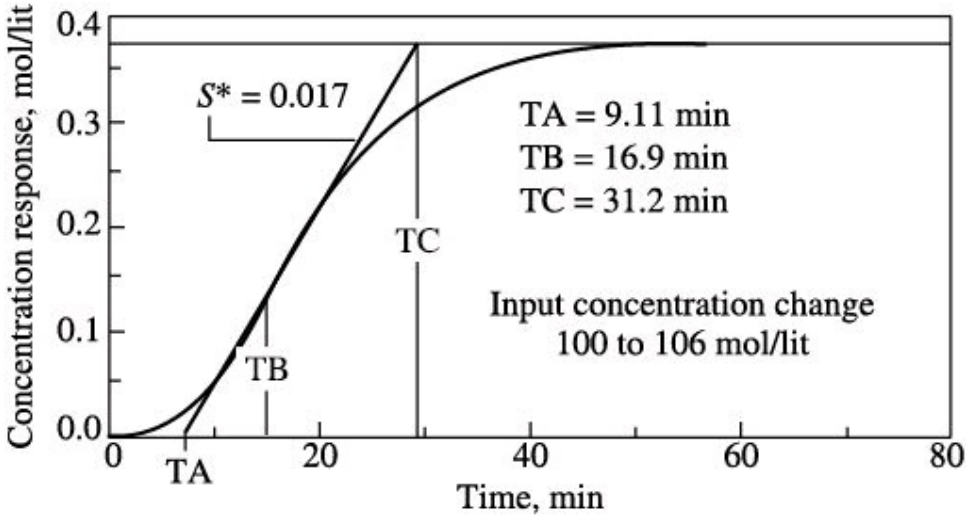

Figure 13.5 The construction on the PRC.

The FOPDT model time parameters found from Figure 13.5 and process TF are:

$$K_P = 1/16 = 0.0625;\ t_P = \text{TC} - \text{TA} = 22.09 \text{ min};\ t_{DL} = \text{TA} = 9.11 \text{ min}.$$

Based on these findings, the controller parameter values are found according to several guidelines that have been discussed so far.

Controller settings according to Z-N formulae (Continuous Cycling).

Controller Mode	K_C	x_I, min	x_D, min
(P)	31.9	–	–
(PI)	28.71	26	–
(PID)	38.28	15.6	3.9

Controller settings according to modified Z-N formulae.

Controller Mode	K_C	x_I, min	x_D, min
(P)	49.725	–	–
(PI)	44.75	23.67	–
(PID)	19.2	14.21	3.55

Controller settings according to Z-N* formulae (PRC based).

Controller Mode	K_C	x_I, min	x_D, min
(P)	8.29	–	–
(PI)	7.46	23.67	–
(PID)	9.95	14.21	3.55

Controller settings according to C-C Formulae (FOPDT based).

Controller Mode	K_C	x_I, min	x_D, min
(P)	55.06	–	–
(PI)	46.09	14.26	–
(PID)	70.3	15.48	2.44

From the parameter values given in the above tables we are able make a few comments:

1. K_C for a (PI) controller always falls between that for a (P) and (PID) in any set of formulae. The reason being that to compensate for the lag effect (which pushes the system closer to its stability limit) introduced by the integrator in $G_C(s)$, K_C has to be reduced for a (PI), but when a differentiator does the compensation in a (PID) controller, K_C may have a more liberal value,
2. Among these guidelines, the C-C seems to be the most adventurous and PRC based Z-N, the most orthodox with respect to K_C value. Thus, with a same quarter decay ratio feature, a faster response is obtained with C-C settings. This feature results due to the knowledge of process TF put into the determination of controller parameters. The presence of a first order and a dead time part in the approximate TF should be more appropriate for systems containing an

actual delay, and also for systems of higher order.

3. Practically all the formulae claim a quarter decay ratio feature of loop performance, hence, it seems that for a given process parameter values, the solution for controller parameter values offering $DR = 0.25$, is not unique. It is required to have an independent scale of measurement that could find the best among the betters. Integral error criterion is a measure of loop performance which may be used for this purpose. We shall discuss this topic in the next section.

Besides the continuous cycling, frequency response techniques may be used to determine K_U and P_U for the application of Z-N settings provided we have a prior knowledge of the process TF. However, the Z-N method was originally developed as a direct plant testing procedure for controller performance optimization in time domain, where the designer has no information about the dynamics of the process.

EXAMPLE 13.2 For the following two processes, described by their loop TFs:

1. $$G(s) = \frac{1}{(s+1)(s+2)(s+3)(s+4)} = \frac{0.041667}{(s+1)(0.5s+1)(0.3333s+1)(0.25s+1)}$$

2. $$G(s) = \frac{2e^{-0.5s}}{5s+1}$$

It is required to find Z-N settings for the closed loop system around these processes using frequency domain techniques.

Solution: The basic method is to draw the Bode's Plot, determine the w_{CO} and GM (gain margin) from the plot. Now, $w_U = w_{CO}$ and P_U may be found as $P_U = 2\pi/w_U$. From the AR value the corresponding to the GM may be found as $GM = 1/AR$. Now K_U is found as:

$$K_U = \frac{1}{AR|_{\omega=\omega_{co}}}$$

The problem reduces to find the frequency (w) for which the $\angle G_P(iw) = -180°$. This can be done definitely from an open loop Bodes' Plot, but there is a computational method following which the graph plotting exercise may be avoided. The expression for AR and $\angle$ are written for the given TF. By using an implicit (trial-n-error) procedure, the value of w_U ($= w_{CO}$) is found for which $\angle = 180°$. (This method has been described as the 'decimal section rejection' in

Chapter 14). Putting the value of w_U in the expression for AR the ultimate gain $K_U(=1/AR)$ is found.

The expressions for the first problem are:

$$\square = -\tan^{-1}(w) - \tan^{-1}(0.5w) - \tan^{-1}(0.3333w) - \tan^{-1}(0.25w)$$

and,

$$AR = \frac{K_P}{\sqrt{1+\omega^2}\sqrt{1+(0.5\omega)^2}\sqrt{1+(0.3333\omega)^2}\sqrt{1+(0.25\omega)^2}}$$

The expressions for the second problem are:

$$\square = -\tan^{-1}(5w) - 0.5w \text{ and } AR = \frac{K_P}{\sqrt{1+(5\omega)^2}}$$

The results obtained by following the implicit procedure are:

For the first problem: $K_U = 126.002$; $w_U = 2.2361$ rads/min; $P_U = 2.81$ min.

For the second problem: $K_U = 8.1753$; $w_U = 3.264$ rads/min; $P_U = 1.925$ min.

These results could be obtained by another way, that is, by applying direct substitution method discussed in Chapter 10. Let apply this for the system in problem 1. The characteristic equation for a unity feedback loop around this process with a (P) controller is:

$$s^4 + 10s^3 + 35s^2 + 50s + 24 + K_C = 0$$

By putting $s = iw$ into this equation,

$$(iw)^4 + 10(iw)^3 + 35(iw)^2 + 50iw + 24 + K_C = 0$$

Equating the imaginary parts, $w_U = 2.236$ rads/min.

Equating real parts and putting the value of w_U into the resulting equation, $K_U = 126$.

This method gives poor results for systems containing dead time or any nonlinear elements due to the in-accuracy in the linear model constructed for the system. For the system in this example, the ultimate gain value obtained by the two methods are identical. This result was also obtained in Chapter 10 by applying root locus method to the same system. Now the ultimate gain and period values are applied to find the Z-N settings. The results are listed in the following table.

Controller Mode	Problem 1			Problem 2		
(P)	$K_C = 63$			$K_C = 4.09$		
(PI)	$K_C = 56.7$	$x_I = 2.34$ min		$K_C = 3.68$	$x_I = 1.6$ min	

(PID)	$K_C = 75.6$	$x_I = 1.4$ min	$x_D = 0.35$ min	$K_C = 4.91$	$x_I = 0.96$ min	$x_D = 0.24$ min

An interesting possibility crops up that if the FOPDT model becomes more accurate by applying Smith's equations described in Chapter 12, should the parameter values produce a better closed loop performance? Let us try this thought in the next example. You should note that for such a demonstration we should not choose a pure FOPDT process for obvious reasons. We shall denote the PRC based Z-N formula as Z-N* in the following analysis.

EXAMPLE 13.3 Find the (PID) controller parameters for the process TFs given for the following two processes by using the FOPDT model parameters estimated by Method 1 and Method 4 (Smith's equation) described in Chapter 12. Compare the responses for (a) Load, and (b) Set point change for controller parameters calculated by Z-N* and C-C formulae.

1. Process 1, a SOPDT process, $G_P(s) = \dfrac{e^{-4s}}{(10s+1)(16s+1)}$;
2. Process 2, a fourth order process with repeated poles, $G_P(s) = \dfrac{1/16}{(5s+1)^4}$

Solution: In case of a (PID) controller, any change in model parameter values will affect maximum number of controller parameters.
For process 1, the FOPDT model using Method 1 and 4 are respectively evaluated as:

$$G_P(s) = \frac{1e^{-7.516587s}}{34.99621s+1} \text{ (by Method 1, } a = 0.215);$$

$$G_P(s) = \frac{1e^{-10.31514s}}{21.37549s+1} \text{ (by Method 4, } a = 0.4826)$$

Where, a is the so-called 'controllability ratio' defined in context to an FOPDT type process model as, $a = t_{DL}/t_P$.

The resulting controller parameters according to modified Z-N based on PRC and C-C formulae are listed below.

FOPDT parameters	Z-N* settings based on PRC			C-C settings		
	K_C	x_I	x_D	K_C	x_I	x_D
Method 1	5.587038	15.03317	3.758293	6.457819	17.00047	2.630576
Method 4	2.486693	20.63028	5.157571	3.012992	21.34872	3.448399

The corresponding load change (regulator) and set point change (servo) responses are shown in Figure 13.6.

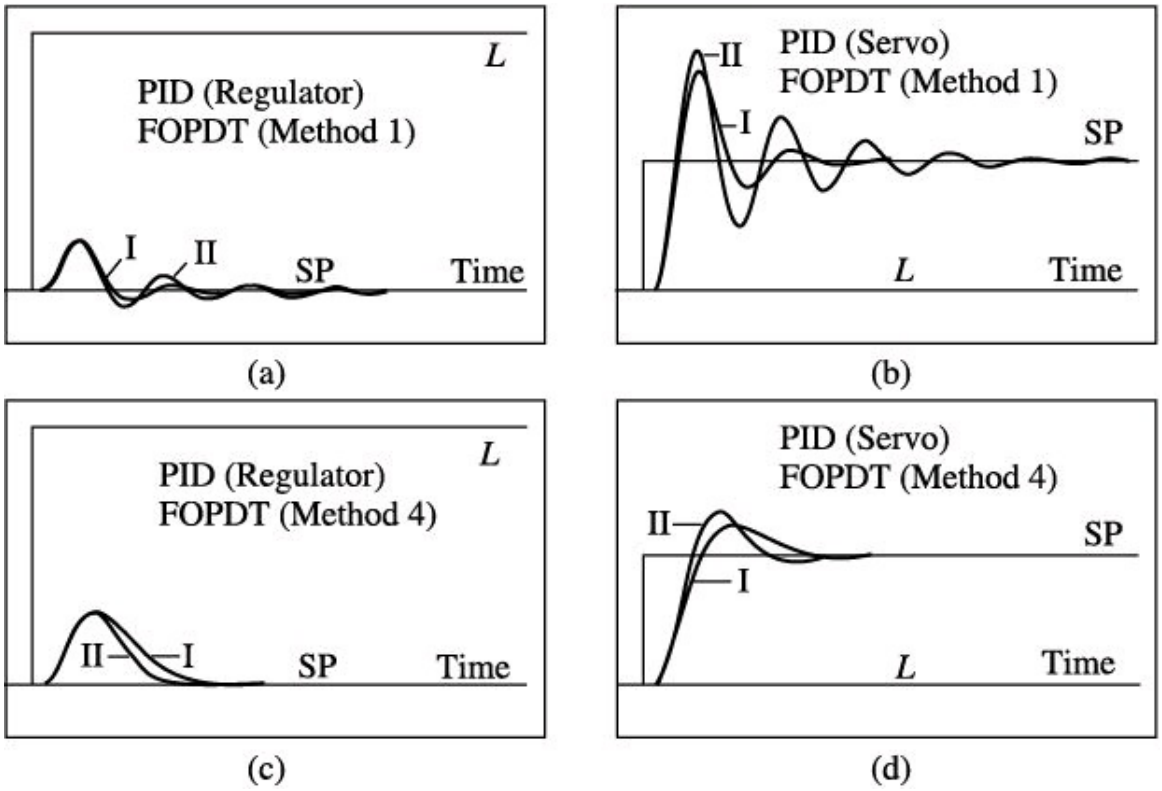

Figure 13.6 (a) and (c) Load change responses. (b) and (d) Set point change responses of a (PID) controlled loop for process 1. The traces marked by I are for Z-N* and those marked by II are for C-C settings.

For process 2, the FOPDT model using Method 1 and 4 are respectively evaluated as:

$$G_P(s) = \frac{0.0625e^{-7.128973s}}{22.2995s + 1} \text{ (by Method 1, } a = 0.32)$$

$$G_P(s) = \frac{0.0625e^{-9.284319s}}{12.47721s + 1} \text{ (by Method 4, } a = 0.74)$$

The resulting controller parameters, according to modified Z-N* based on PRC and C-C formulae, are listed below

FOPDT parameters	Z-N* settings based on PRC			C-C settings		
	K_C	x_I	x_D	K_C	x_I	x_D
Method 1	60.0578	14.25795	3.564487	70.73088	15.54241	2.449949
Method 4	25.80292	18.56864	4.642159	32.66991	17.86273	2.973788

The corresponding load change (regulator) and set point change (servo) responses are shown in Figure 13.7.

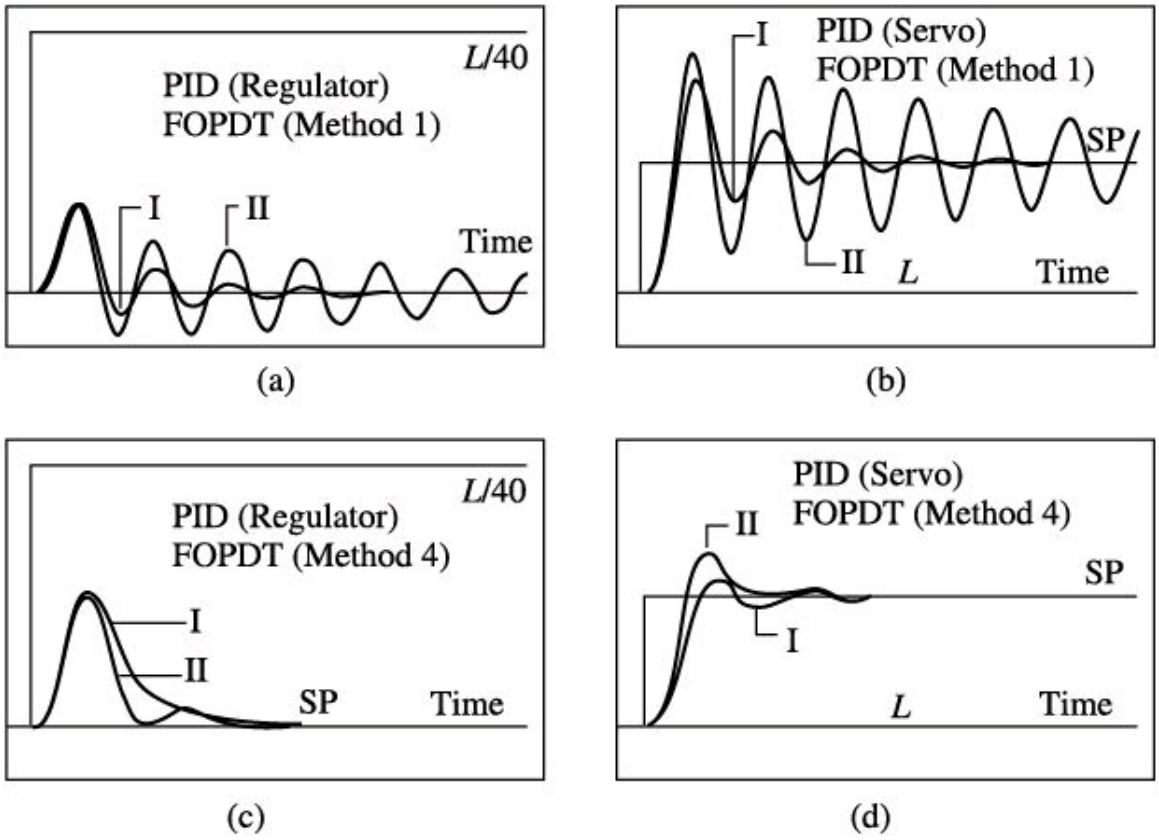

Figure 13.7 (a) and (c) Load change responses. (b) and (d) Set point change responses of a (PID) controlled loop for process 1. The traces marked by I are for Z-N* and those marked by II are for C-C settings.

[Due to the low value of process gain (K_P = 0.0625), the load variable in Figures 13.7(a) and (c) is plotted with a reduced scale factor (SF = 1/40) to keep it in the same frame with controlled variable.]

The improvement in response quality by following Method 4 for FOPDT model calculation is appreciable for example processes that have different dynamic models. An interesting observation about the 'controllability ratio' (a) values calculated for the FOPDT models obtained for the two processes, is that the models determined by Method 4 show a higher value of a than for the models determined by Method 1. That is, the model that gives better loop response has an inferior value of a.

If we look at the responses obtained by applying the two set of formulae, at a first glance one would say that the Z-N* is the better choice, but in certain cases this choice may lead to over-damped or near over-damped type of response. Such response quality often leads to higher settling time, although the overshoot may have a low value. This type of loop response is also seen for other processes featuring different dynamic character types. If we examine the calculated controller parameters by formula sets forwarded so far, the values of t_I and t_D are approximately equal whereas that of K_C is appreciably low for Z-N* compared to Z-N and C-C settings. This over compensation of controller gain is probably responsible for such a character of loop response.

Though claimed by the authors, many of the responses do not show the 'quarter

decay ratio' feature, still Z-N and to a lower extent C-C settings, are set as a standard for comparison for evaluation for any controller tuning or synthesis efforts. The next example examines the performance of the principal setting formulae for certain processes in terms of closed loop response. Henceforth, we shall use Method 4 for PRC based FOPDT model parameter evaluation.

EXAMPLE 13.4 Compare the performance of two processes:

(a) Process 1. $G(s) = \dfrac{1e^{-8s}}{20s+1}$;

(b) Process 2. $G(s) = \dfrac{1}{(5.5s+1)(6s+1)(9.5s+1)(10s+1)}$.

by using: (a) Z-N, (b) Z-N*, and (d) Cohen-Coon guidelines for load (Regulator) and set point (Servo) disturbances with (P), (PI) and (PID) controllers.

Solution: For process 1, the FOPDT model need not be calculated; it will be identical to the process TF:

$$G(s) = \frac{1e^{-8s}}{20s+1} \quad (a = 0.4)$$

The resulting controller parameters according to modified Z-N, Z-N* and C-C formulae are listed below:

Cntrlr Mode	Z-N settings			Z-N* settings			C-C settings		
	K_C	x_I	x_D	K_C	x_I	x_D	K_C	x_I	x_D
(P)	2.2934	–	–	2.501	–	–	2.8343	–	–
(PI)	2.064	23.394	–	2.251	26.624	–	2.334	14.676	–
(PID)	2.752	14.036	3.509	3.001	15.99	3.998	3.5846	16.978	2.7103

The closed loop responses with (P), (PI) and (PID) controller modes for regulator and servo disturbance situations are found by a simulation programme. The responses are shown in Figure 13.8. By examination, the C-C responses show a better quality with respect to 'quarter decay ratio' and 'settling time' over that of Z-N and Z-N* in a major number of cases. The (PI) and (PID) servo responses are probably the only exceptions from this general trend. Again, the over compensation of K_C value in Z-N* of (PID) settings seems to be reason of such a feature.

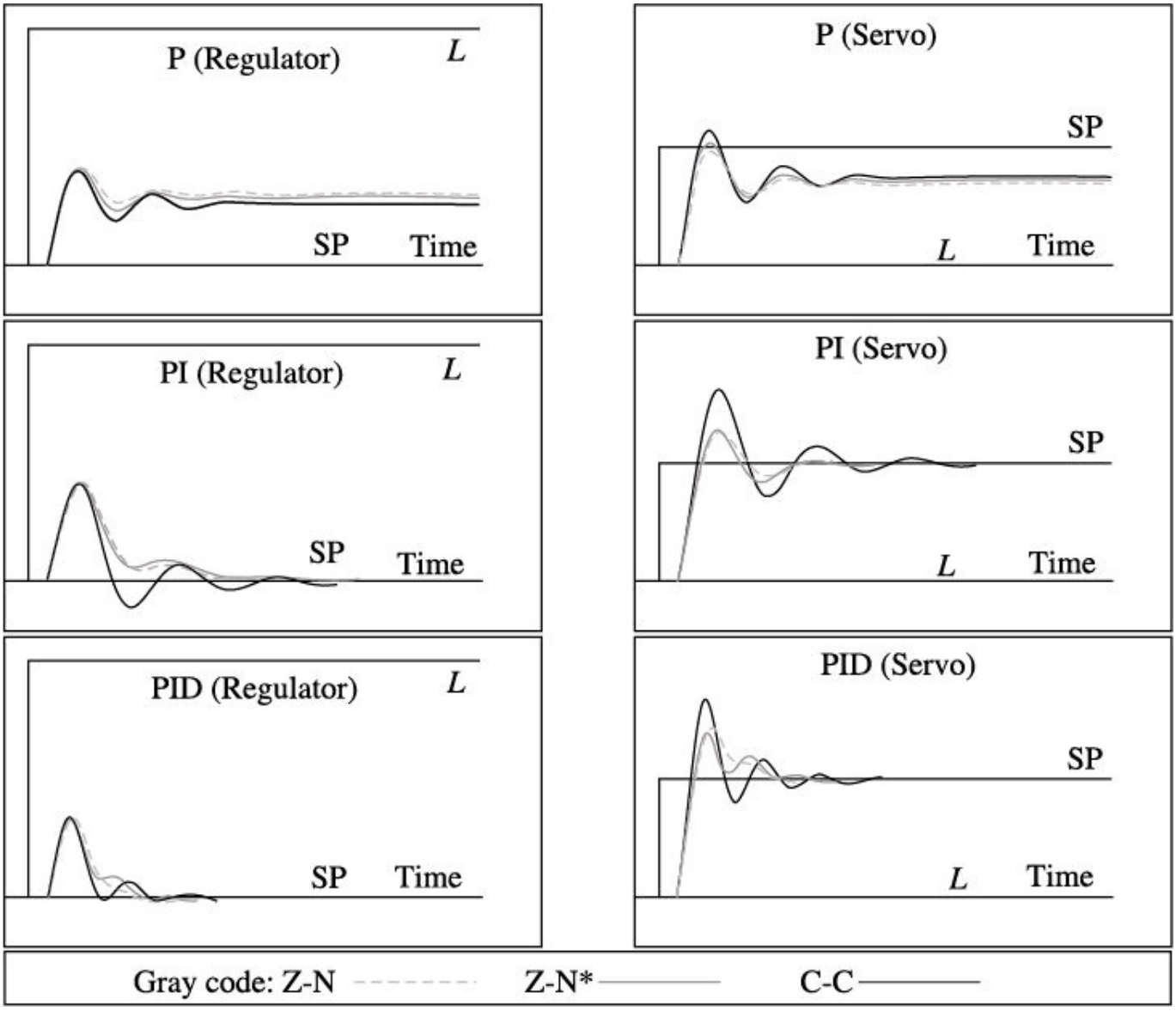

Figure 13.8 Loop responses of the FOPDT process.

For process 2, the FOPDT model using Method 4 is evaluated as:

$$G(s) = \frac{0.0625.e^{-13.9722s}}{19.623s + 1} \quad (a = 0.712)$$

The resulting controller parameters according to modified Z-N, Z-N* and C-C formulae are listed below.

Cntrlr Mode	Z-N settings			Z-N* settings			C-C settings		
	K_C	x_I	x_D	K_C	x_I	x_D	K_C	x_I	x_D
(P)	2.1452	–	–	1.4044	–	–	1.7377	–	–
(PI)	1.9307	39.183	–	1.264	46.527	–	1.3473	19.32	–
(PID)	2.5743	23.51	5.877	1.6853	27.944	6.986	2.1226	27.107	4.498

The closed loop responses with (P), (PI) and (PID) controller modes for regulator and servo disturbance situations are found by a simulation programme. The responses are shown in Figure 13.9. By examination of responses it may be found that in major number of cases the Z-N responses show a better quality with respect to ‘quarter

decay ratio' and 'settling time' over that of Z-N* and C-C. The (PID) servo response probably is the only exception from this general trend. Again, the over compensation of K_C value in Z-N* of (PID) settings seems to be reason of such a feature.

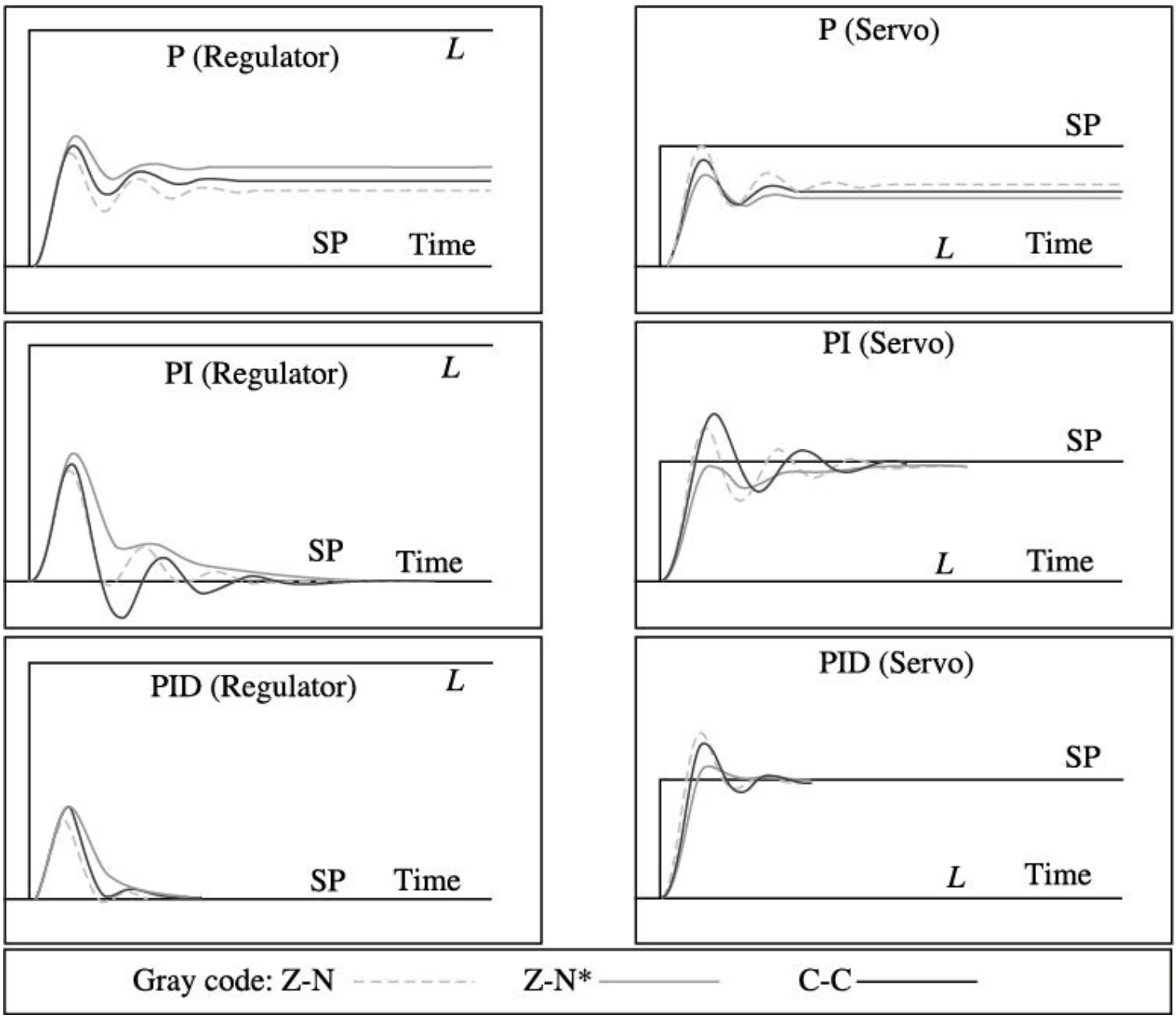

Figure 13.9 Loop responses of the fourth order process.

The responses corresponding to Z-N* settings, in general, seem to show less oscillatory character that hardly follow the ¼ *DR* property. It has been observed that for many processes, the Z-N* settings produce over-damped responses with longer settling time which are unacceptable where tighter loop performance is demanded. A general comment made in many literatures is that the C-C settings perform better for a process containing a dead time, and Z-N settings for a process where R-C lag is prevailing. The results of the last example more or less support this comment.

In the area of control technology, this was the state of affairs up to late 1950s, when the primary inspection of control quality depended on the visual quality of the response traces made over a strip chart recorder or may be on a computer monitor.

Though properties like ¼ *DR* and settling time still carried some significance, it was difficult to choose between a response that offered large overshoot with small settling time, and another that took a longer time to settle with small amplitude oscillation. Furthermore, the disadvantage of quarter decay ratio response still remained with the application of these formulae.

Also, with the growing awareness about limitations of resources in terms of energy and material and increasing competition among manufacturers, the old era of high profitability was coming to an end and demand towards tighter control of plant to accrue higher profit through efficiency escalation was continually increasing. Thus, the drive was for 'Survival through more sophistication of Technology'.

It was gradually being felt that to satisfy newer demands from the industry, some quantitative estimate of response quality was needed. The basic idea from which the development of a new generation of response quality measurement began is that *in a disturbed process, the travel of process variable above or below the set point until it again settles back to set point, may be a measure of the amount of waste or out of quality material* (refer to Figure 13.10).

These are the background of the outcome of **Integral Error Criterions**, a set of definitions to quantify the loop response for a given controller settings. The reasons for this development could be summarized as:

1. The non-uniqueness of ¼ *DR* criterion.
2. The criteria have no account of the time duration required to drive away an amount of disturbance/error that has entered into the process.
3. Increasing demand of controller optimization procedures that may offer lesser amount of out of specification/defective product in transient state.
4. Increasing demand of controller optimization procedures that may offer highest possible material transfer and energy utilization efficiency on a plant wide basis.

13.1.7 Integral Error Criterions

In the later part of 1960s, a number of publications appeared where the FOPDT model parameter estimation from the PRC locus was used to characterize the process. The objective was to minimize the total error or deviation of the controlled variable from the set point during the transient period or as long as the error stays in the process. To find the total error magnitude, the error values at each time instant has to be summed together for the whole duration of the transient period. This is, by definition, the integral of error with respect to time or the shaded area in the response graphs illustrated in Figure 13.10.

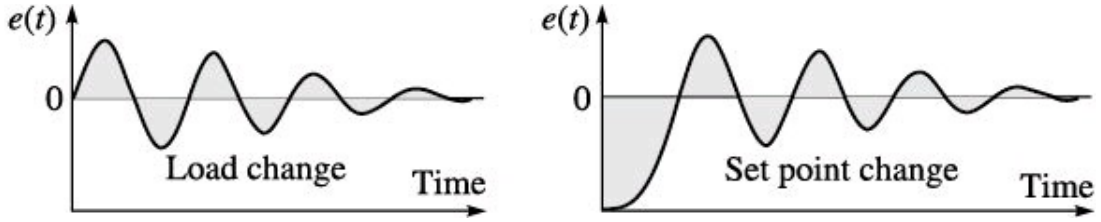

Figure 13.10 Definition of the error Integral for load and set point change.

To prevent that –ve and +ve error values may not cancel each other during integration and lead to a false estimate, the error magnitude with their algebraic sign are either absoluted or squared. Thus, two fundamental error integral criterion formulae are forwarded as:

$$\text{Integral Absolute Error (IAE)} = \int_0^\infty |e(t)|\,dt \qquad (13.4)$$

$$\text{Integral Squared Error (ISE)} = \int_0^\infty e^2(t)dt \qquad (13.5)$$

It is obvious that by minimizing ISE, we put more emphasis on the overshoots than by minimizing IAE because higher error values produce comparatively large squared values in the resulting integral. By the same line of reasoning, ISE puts less weightage on smaller errors occurring at the end point of transient response. Hence, a minimum IAE response may show a highly oscillatory nature with a large *DR* value with error oscillating around zero for a relatively long time. This phenomenon suggests the performance criteria should contain a penalty for the time elapsed from the start of response. The following error integrals contain such a penalty by including a weight for the elapsed time.

$$\text{Integral Time-Absolute Error (ITAE)} = \int_0^\infty t\cdot|e(t)|dt \qquad (13.6)$$

$$\text{Integral Time-Squared Error (ITSE)} = \int_0^\infty te^2(t)dt \qquad (13.7)$$

The above four equations constitute the four basic error integrals that can be minimized for a given loop by adjusting the controller parameters. Unfortunately, the optimum parameter values are not only a function of the type of criterion chosen, but also of the type of input disturbances.

However, Lopez et al. in 1967, developed tuning formulae for minimum error integral criteria for step changes in set point and load. The process loop under their consideration has the feature that the process transfer function with respect to load and set point changes are same as illustrated in Figure 13.11.

A process model with FOPDT structure is obtained from PRC locus (Figure 13.3),

or preferably by Method 4 for FOPDT parameter estimation described in Chapter 12, in terms of K_P, t_P, t_{DL}. In the next step, the controller parameters K_C, t_I, t_D may be found from the relations listed in Tables 13.7 and 13.8.

Table 13.7 Minimum Error Integral Tuning Formulae for Set point Change

Controller Mode	For IAE	For ITAE
(PI) Controller $K_C = \frac{a_1}{K_P}\left(\frac{t_{DL}}{t_P}\right)^{b_1}$ $\tau_I = \frac{t_P}{[a_2 + b_2(t_{DL}/t_P)]}$	$a_1 = 0.758$ $b_1 = -0.861$ $a_2 = 1.02$ $b_2 = -0.323$	$a_1 = 0.586$ $b_1 = -0.916$ $a_2 = 1.03$ $b_2 = -0.165$
(PID) Controller $K_C = \frac{a_1}{K_P}\left(\frac{t_{DL}}{t_P}\right)^{b_1}$ $\tau_I = \frac{t_P}{[a_2 + b_2(t_{DL}/t_P)]}$ $\tau_D = a_3 t_P\left(\frac{t_{DL}}{t_P}\right)^{b_3}$	$a_1 = 1.086$ $b_1 = -0.869$ $a_2 = 0.740$ $b_2 = -1.30$ $a_3 = 0.348$ $b_3 = -0.914$	$a_1 = 0.965$ $b_1 = -0.855$ $a_2 = 0.796$ $b_2 = -0.147$ $a_3 = 0.308$ $b_3 = -0.9292$

Table 13.8 Minimum Error Integral Tuning Formulae for Load Change

Controller Mode	ISE	IAE	ITAE
(P) Controller $K_C = \frac{a_1}{K_P}\left(\frac{t_{DL}}{t_P}\right)^{b_1}$	$a_1 = 1.411$ $b_1 = -0.917$	$a_1 = 1.902$ $b_1 = -0.985$	$a_1 = 1.490$ $b_1 = 10.84$
(PI) Controller $K_C = \frac{a_1}{K_P}\left(\frac{t_{DL}}{t_P}\right)^{b_1}$ $\tau_I = \frac{t_P}{a_2}\left(\frac{t_{DL}}{t_P}\right)^{b_2}$	$a_1 = 1.305$ $b_1 = -0.959$ $a_2 = 0.492$ $b_2 = 0.739$	$a_1 = 0.984$ $b_1 = -0.986$ $a_2 = 0.608$ $b_2 = 0.707$	$a_1 = 0.859$ $b_1 = -0.977$ $a_2 = 0.674$ $b_2 = 0.680$
(PID) Controller $K_C = \frac{a_1}{K_P}\left(\frac{t_{DL}}{t_P}\right)^{b_1}$ $\tau_I = \frac{t_P}{a_2}\left(\frac{t_{DL}}{t_P}\right)^{b_2}$ $\tau_D = a_3 t_P\left(\frac{t_{DL}}{t_P}\right)^{b_3}$	$a_1 = 1.495$ $b_1 = -0.945$ $a_2 = 1.101$ $b_2 = 0.771$ $a_3 = 0.560$ $b_3 = 1.006$	$a_1 = 1.435$ $b_1 = -0.921$ $a_2 = 0.878$ $b_2 = 0.749$ $a_3 = 0.482$ $b_3 = 1.137$	$a_1 = 1.357$ $b_1 = -0.947$ $a_2 = 0.842$ $b_2 = 0.738$ $a_3 = 0.381$ $b_3 = 0.995$

The close loop model used for the optimization study is illustrated in Figure 13.11.

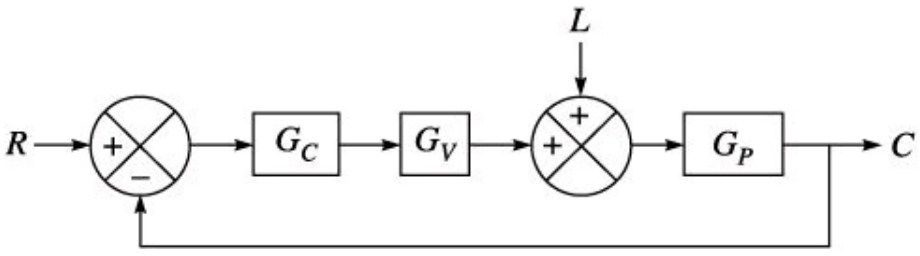

Figure 13.11 The configuration of closed loop system used by Lopez, et al. for their controller tuning studies.

In the past, the IAE and ISE were popular in controller tuning in practice and research areas, but gradually the significance of ITAE was appreciated and its popularity has increased rapidly. In recent times, practically the result of any innovations in this line of investigation is expressed in terms of ITAE criterion. We shall also use this measurement standard in the following example.

To express the result in a more general form, the following definition of ITAE is used in response quality calculation.

$$\text{ITAE} = \frac{\sum_{I=1}^{N} (SP - Y_I) T_I}{\sum_{I=1}^{N} T_I} \qquad (13.8)$$

EXAMPLE 13.5 Compare the performance of Lopez formulae with Z-N, Z-N* and C-C guidelines for the two processes given in Example 13.3. The comparison will be done by drawing the closed loop responses and calculating the ITAE values for all responses obtained with the different controller settings by Lopez formulae.

Solution: For process 1, the controller parameters are calculated corresponding to the process TF, using the Lopez formulae given in Tables 13.7 and 13.8. The results are listed below. The loop responses are shown for Lopez and other settings in Figure 13.12.

Controller Mode	Load (Regulatory) disturbance			Set point (Servo) disturbance		
	K_C	x_I	x_D	K_C	x_I	x_D
(P)	1.280599	—	—	—	—	—
(PI)	2.103511	15.90589	—	1.356952	20.74163	—
(PID)	3.23287	12.07301	3.060122	2.11305	27.12268	2.627605

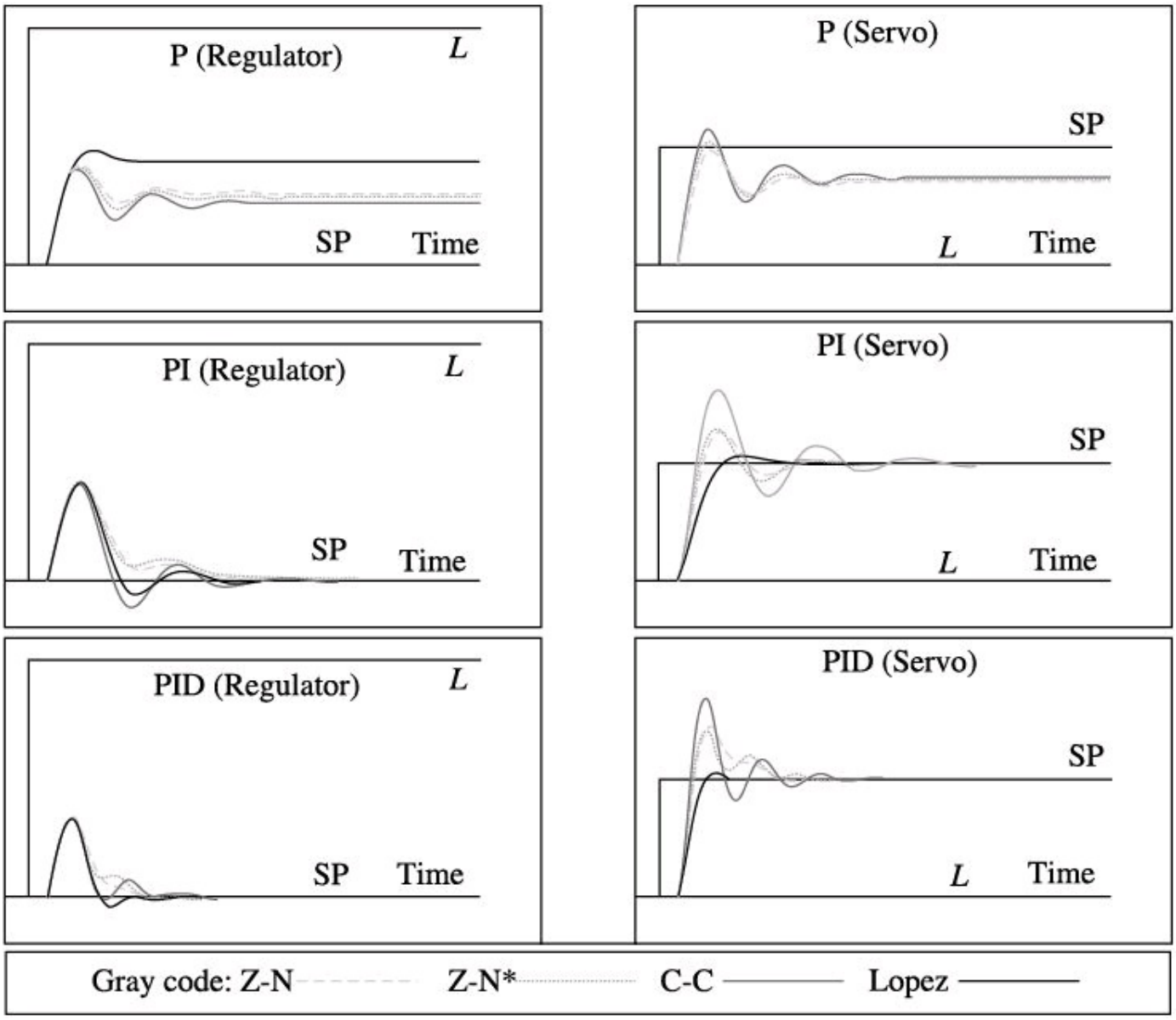

Figure 13.12 Lopez response along with others for process 1.

The ITAE values are calculated for the responses shown in Figure 13.12. These values are listed in the following table.

Cntrlr Mode	Z-N		Z-N*		C-C		Lopez	
	L	SP	*L*	SP	*L*	SP	*L*	SP
(P)	57.656	115.6	54.214	108.753	49.5314	99.8573	83.0333	–
(PI)	4.1237	5.633	4.7249	6.5245	3.6347	15.8787	2.7867	3.8116
(PID)	1.3649	5.603	1.3873	5.4015	1.3081	7.5572	1.0197	2.1968

For process 2, the controller parameters are calculated corresponding to the process TF, using the Lopez formulae given in Tables 13.7 and 13.8. The settings are listed below:

Controller Mode	Load (Regulatory) disturbance			Set point (Servo) disturbance		
	K_C	x_I	x_D	K_C	x_I	x_D
(P)	0.6994721	–	–	–	–	–
(PI)	1.197003	23.11008	–	0.7998391	21.50397	–
(PID)	1.187179	18.13822	5.332435	1.290135	28.38392	4.408151

The closed loop responses are shown in Figure 13.13.

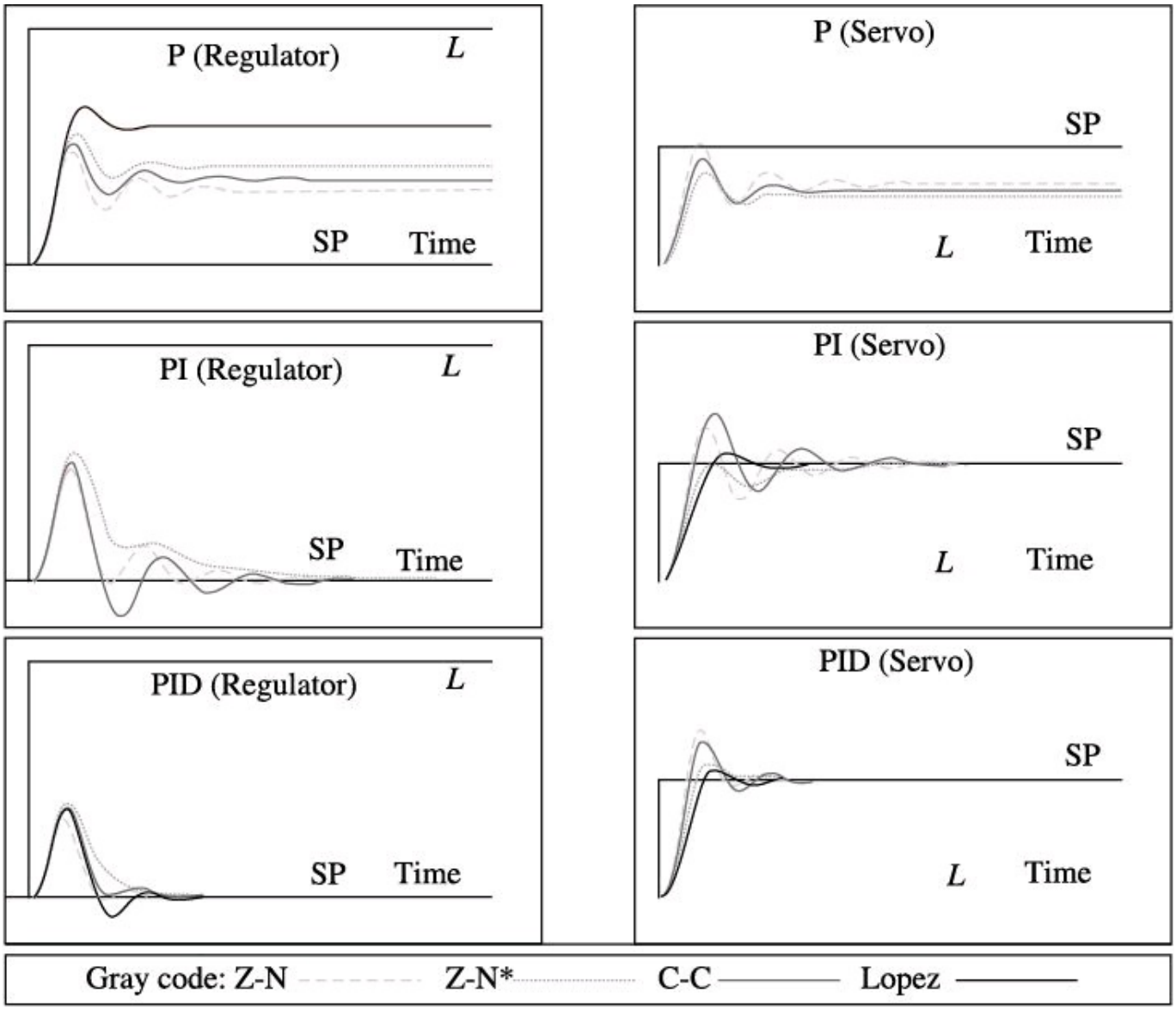

Figure 13.13 Lopez responses along with others for process 2.

The ITAE values are calculated for the responses shown in Figure 13.13. These values are listed in the following table.

Cntrlr Mode	Z-N		Z-N*		C-C		Lopez	
	L	SP	L	SP	L	SP	L	SP
(P)	124.05	247.9	162.135	324.526	142.382	284.94	229.1	–
(PI)	6.6	18.39	15.376	19.282	7.9	21.134	5.634	6.959
(PID)	1.666	5.939	3.801	4.283	2.644	6.484	2.769	4.611

Now we have a yard-stick or quantitative method of judging a loop response, which possesses a dynamic significance about the performance of the applied controller in the loop. Curiously, Lopez setting has a poor performance with (P) controller for both of the processes that fall into two representative dynamic categories. Probably this is the reason that Lopez, et al. did not propose (P) controller setting for servo disturbances. In the result table of process 2, the ITAE value is slightly bigger for

(PID) load and (PID) set point than the obtained minimum, but for process 1, Lopez results are blemish-less. Again we come to the notion that as the basic process model on which the Lopez formulae was built contains dead time element, they show better performance for process containing dead tine than that with R-C lags. In actual implementation in an off or on-line situation, there is a little difficulty associated due to the formula difference in regulatory and servo type of disturbances. In many process plants, usual disturbances are regulatory, but occasionally the operator may change the set point for various reasons. By simulation we may test the performance of a formula when applied for a wrong kind of disturbance.

EXERCISE PROBLEMS

13.1 The block elements in a feedback control loop have the following TFs:

$$G_V(s) = 3.2; \quad G_P(s) = \frac{e^{-2.5s}}{(12s+1)(3s+1)}; \quad G_M = 1$$

Calculate the values of K_U and P_U by the method of:

(a) Frequency response of the open loop TF,

(b) Routh-Hurwitz criteria after representing the dead time by 1–1 Pade's approximation.

(c) Calculate the values of (P), (PI) and (PID) controller settings by Z-N formulae from the two sets of values of K_U and P_U obtained in part (a) and (b).

(d) Comment on the quality of responses that may be obtained with (PI) and (PID) controllers with the parameter values calculated in part (c).

13.2 By applying Martin's method described in Chapter 12, find the FOPDT equivalent of:

$$G_P(s) = \frac{e^{-2.5s}}{(12s+1)(3s+1)}.$$

Using the FOPDT parameters find (PI) and (PID) controller parameters according to Z-N* and C-C formulae and compare with that obtained in part (c) of problem 1.

13.3 A shell-n-tube heat exchanger is proposed for the service of feed pre-heater of a distillation column. The feed, a liquid hydrocarbon mixture flowing at rate of 450 kg/min is heated in the tubes of the exchanger by employing mixture of hot and cold Thermic-fluid on the shell side. To have a precision control performance, manipulation is to be done by proportioning the flows of hot and cold thermic fluid mixture prepared by a 3-way mixing type control valve. To design a (PID) controller for this service, the exchanger is dynamically tested by making a step

change of 12ºC of the shell fluid. Record of the fluid temperature is shown in Figure 13.P1.

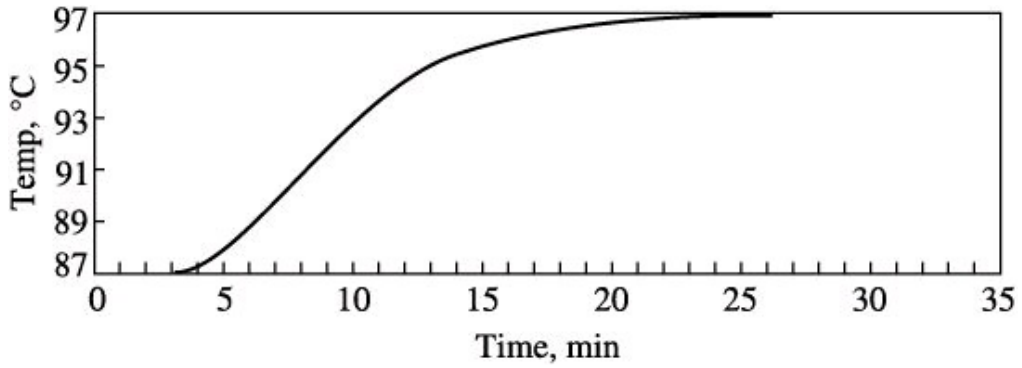

Figure 13.P1 Step response in the tube fluid temperature.

Derive the FOPDT model between shell and tube fluid of the exchanger. Then find (PID) parameters according to Z-N*, C-C and Lopez formulae. Write a computer programme to obtain close loop responses for regulator and servo disturbances. Also calculate ITAE values of the responses. Comment on the response qualities.

13.4 For the following open loop process TFs, find the K_U and P_U by (a) Direct substitution method, and (b) frequency response technique:

(i) $G(s) = \dfrac{1/6}{(s+1)(s+0.5)(s+0.3333)}$

(ii) $G(s) = \dfrac{1}{(s+5)^4}$

(iii) $G(s) = \dfrac{(0.5s+1)}{(3s+1)(2s+1)(s+1)}$

(iv) $G(s) = \dfrac{1}{(s+1)(2s+1)(3s+1)(4s+1)}$

Calculate the parameters of (P), (PI) and (PID) controllers according to Z-N formulae for the above processes.

13.5 In an air conditioning system for a small auditorium, the inlet air temperature is made by mixing atmospheric air with cold air from a refrigeration system. In normal operation, the two air streams are mixed in 1:1 ratio to keep the room temperature at 78ºF. To dynamically test the system in open loop condition, the ratio of cold to atmospheric temperature is suddenly raised from 0.65:0.35 to 0.67:0.33. The step response in the room temperature is recorded in following table.

Time, s	0	0.1	2.5	5	7.5	12.5	17.5	25
Temp,ºF	54	54	53.84	53.46	52.98	51.97	51.1	50.18
Time, s	32.5	40	47.5	55	62.5	70	77.5	85
Temp,ºF	49.63	49.33	49.17	49.08	49.04	49.02	49.01	49.0

Prepare an FOPDT model for the system and find (PID) parameters according to Z-N*, C-C and Lopez formulae. Write a computer programme to obtain close loop responses for regulator and servo disturbances. Also calculate ITAE values of the responses. Comment on the response qualities.

13.6 The TF of a process is expressed as: $G_P = 1.8e^{-0.2s}/(s + 1)$. Compare the (PI) and (PID) controller settings that are obtained by the following methods:

(a) Cohen-Coon formulae

(b) Lopez formulae for load change

(c) Lopez formulae for set point change

(d) Direct synthesis with $t_C = 0.3$ and 1.1.

Which of the methods give the most conservative estimate of the parameters and which of them obtains the least conservative values of the parameters? (i.e. the most over-damped or under-damped responses.)

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14

Digital Simulation of Processes and Control Systems

Fundamentally, simulation is the exercise of creating a replica or facsimile or a model of a real physical system. The simulated model may not be identical or contain all the features and properties of the actual system, but should have enough similarity so that it's response for any given input disturbance becomes comparable and remains within a required accuracy limit to the system's response.

A simulation program also may be used to train a person or system by creating an artificial environment that has enough similarity to the actual one. A 'Flight Simulator' trains a pilot without endangering a novice to the real hazards of aircraft navigation. A special software generates an artificial flight deck with controls, a dynamically changing outside view during the flight—every visual details on the monitor of a computer. The trainee may control this simulated aircraft from the computer keyboard.

A person trains his watchdog by fixing the dog's food at the throat of a man-size figurine during the dog's feeding time. This exercise uses a simulated human being to train the dog for negotiating prowlers in the premises.

We have not mentioned one important thing about simulated models that they should be easy to construct and handle, cheap enough and involve very less hazards during operation, compared to the real thing.

Digital simulation of process models and perfecting them till they meet the accuracy demands and operability of actual equipment is an excellent breakthrough in the design and optimization in engineering science. Selection of the appropriate control scheme for the process may be also done by adding simulated control loops to the simulated process. By using the resulting model, the controller parameters may be tuned for an optimum closed loop performance without using or endangering any of the costly hardware to any operational difficulties. Also, raw material is not wasted during any trial or testing. These features of digital simulation have made this activity so popular in the field of process dynamics and control.

14.1 Historical

Analog computers were probably the pioneers in this field. They embody configurable

electronic circuits to represent dynamic systems. The most important unit in an analog computer is the Operation Amplifier (OP-AMP). An OP-AMP may be configured in a number of ways to serve various mathematical operations, e.g. an adder, subtractor, constant multiplier, constant divider, integrator, differentiator, comparator and signal inverter. Additional hardware, like voltage dividers, is also provided for fraction multiplication.

The programmer develops the dynamic model in the form of a set of differential and algebraic equations (if there is any). The model is ultimately programmed as an analog circuit that is realized by proper wiring network on the computer console to represent the dynamic system. A calibrated DC voltage may be introduced as input to the circuit and the voltage as a function of time that comes at the output end represents the response of the modeled system.

During late 1950s or early 1960s, analog computers were phased away by digital computers primarily due to the gradual development, which became an explosion in ’80s of the later variety and partially due to certain inherent limitations of the former one.

The digital simulators came into view probably in ’70s. The forerunner software were FORTRAN based CSMP, MIMIC, LEANS, etc. A little later, DYNSYS, PRODYC and REMUS were introduced. In the ’80s, we saw the earliest versions of ASPEN, MATLAB, and LABVIEW. Generally, the software within the first two groups was developed on the language base of FORTRAN. They offered an environment to the user to develop their own program by providing certain built-in procedures to ease out the intellectual effort.

Then awareness for the extensive graphic facilities for input and response function display gradually increased. As FORTRAN had certain limitation in this area, a third group was designed either by using newer languages having graphic features (e.g. the visual versions of BASIC or C++), or resorting to the application level of software development. In this group, LABVIEW should be mentioned first due to its novel graphic programming feature, and probably the first control system simulator that could be hooked up with a real time process through the computer I/O ports. The simulated instrumentation panel may be displayed on the computer monitor that reacts in real time with the process.

We have heard in the recent years about intelligent simulators that automatically correct the user error up to a certain level and have certain built in deterministic and stochastic data processing features. These features provide the capability to realize fuzzy logic, pattern recognition, statistical process control (SPC) systems and ultimately the Virtual Image environment.

14.2 Process Control System Analyzer (PCSA)

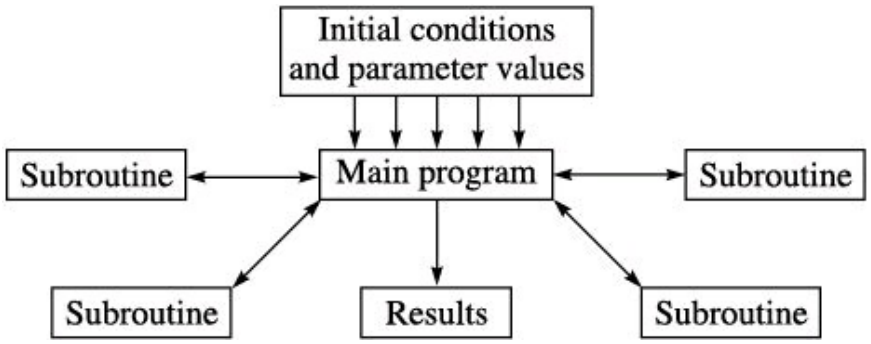
In this chapter, we shall discuss a procedure or platform for digital simulation of dynamic processes with or without a control system. The basic philosophy is to show the guideline for developing a main program that contains the dynamic character of the process and the call sequence to a number of subroutines. These subroutines act somewhat like utilities that provide certain essential computational and signal processing facilities. For computer implementation, Q-Basic has been used as the language base for writing the programs. Choice of this language base is primarily for the following reasons:

1. The program writing rules are very simple, simpler than in FORTRAN and C++. One feels that the language being used is very close to English.
2. The subroutine usage do not require an argument list, hence the associated problem of maintaining a correct sequence of parameter and variables in the argument list completely disappears. You need not use a COMMON (which is again very stringent about this sequence), because in a dynamic program, all the updated variable values are available to the concerned subroutines from the main program or any of the subroutines (however, use of COMMON as a declaration is allowed in QBASIC but only for special applications).
3. The Q-Basic has special facilities that also allow writing structured programs so that a user can start by writing the simplest version of programs. Later, as his experience and skill increases, he may attempt to write more compact and sophisticated versions.
4. The graphic features are extensive and easy to use, thus, tracing of dynamic variables and mimic display of system configuration are possible.

PCSA probably owes its conceptual legacy to the FORTRAN based simulator DYFLO, explained by R.G.E. Franks in his text.

(It has been assumed that the reader has a preliminary training in program building in GWBASIC or BASIC, because these languages are taught in school level, as program development tools. The fundamental language rules are same for BASIC and QBASIC, however, QBASIC provides a cluster of facilities that are extra to BASIC or BASICA. The reader may refer to any of the texts written on this subject. We have tried to keep the example programs as simple as possible, so that a working knowledge of BASIC is sufficient to try them. We have included QBASIC source reference to obtain interpreter in the accompanying CD. Refer to the preface of this book for including this facility in your computer so that you can write the example programs and programs for solving the problems offered in this chapter.)

We shall first introduce certain utility programs that act as service subroutines performing simple numerical procedures that are required during execution of simulation programs. The basic structure of the mathematical model for the given process is generally spelled in the main program segment. The parameter values, initial conditions and input functions, if any, are either supplied by the user or available in the main program. From the main program the required subroutines are called in proper sequence for execution of the simulation procedure. The procedure of program operation may be appreciated from the following block diagram.



To demonstrate the usage of these subroutines, simple and short main programs will be shown to gradually familiarize the reader in the art of writing more complex simulation programs.

The placement of subroutines will be in a nested fashion within the main program.

14.3 Subroutine for One Dimensional Linear Interpolation (LNNT1A), (LNNT1B), (LNNT1C)

```
1200 REM ********** SUBROUTINE LNNT1A ********************
1202 FOR I = 1 TO ND - 1
1204 IF X <= AX(I + 1) GOTO 1208
1206 NEXT I
1208 Y = AY(I) + ((AY(I + 1) - AY(I)) / (AX(I + 1) - AX(I))) * (X - AX(I))
1210 RETURN
1212 END
```

The operation of the of SUBROUTINE LNNT1A.BAS happens by first finding the two data points that are closest to the independent variable for which the dependent variable value is called from the main program. This happens in the FOR loop written in line numbers 1202 to 1205. When the comparison is successful in line number 1204, the execution is shifted to line number 1208, where the linear interpolation is executed to find the required dependent variable value. Principle of this interpolation is shown in Figure 14.1.

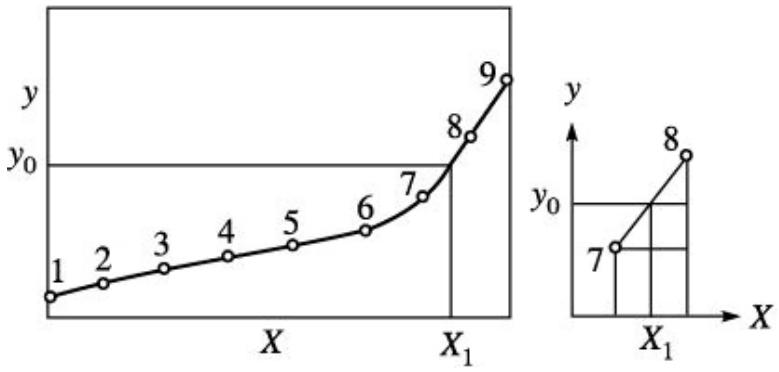


Figure 14.1 Principle of interpolation subroutines.

To use this subroutine, the main programe has to contain the data set containing independent and corresponding dependent variable values, respectively written as array entries AX(1), AX(2), …, AX(ND) and AY(1), AY(2), …, AY(ND). Where, ND is the number of data in the set. Hence, at the beginning of the main program, there should be a dimension declaration to provide adequate space for these arrays. Calling of any subroutine requires a single statement:

GOSUB 1200

where the number after GOSUB is the starting line number of the subroutine. The variables that are used in the subroutine should be available before the GOSUB statement. In the present case, it is the independent variable value X for which the dependent variable value is required. We shall cite an example to explain the use of LNNT1A.

EXAMPLE 14.1 In an oil fired furnace the optimum value of % of excess air, y, is determined as a function of the heat load, x, because fuel flow rate may be considered as proportional to heat load. In the following data set, the heat load is reported as the % heat load relative to normal or rated output of the furnace.

x	80	85	90	95	100	105	110	115	120	125
y	20	20.12	20.48	21.08	21.92	23	24.32	25.88	27.68	29.7

Such a data set may be used by a fuel/air ratio controller. The controller will calculate the set point to the fuel flow controller from the heat demand information obtained from the process side (where the furnace heat is utilized), and calculate the gain of the ratio station, K_R, by using the % heat load as x, to call the subroutine LNNT1A.BAS and obtain the % of excess air as y. From the fuel flow set point, and K_R, the value of the set point to the air flow controller may be calculated. Write a program to execute this duty.

Solution: The main program and the nested subroutine for this task will appear as:

```
10      REM ********** PROGRAM NAME-> C14-01.BAS ******************
15      DIM AX(50), AY(50)
20      AX(1)=80: AY(1)=20: AX(2)=85: AY(2)=20.12: AX(3)=90: AY(3)=20.48: AX(4)=95: AY(4)=21.08
25      AX(5)=100: AY(5)=21.92: AX(6)=105: AY(6)=23: AX(7)=110: AY(7)=24.32: AX(8)=115: AY(8)=25.88
30      AX(9)=120: AY(9)=27.68: AX(10)=125: AY(10)=29.7
35      ND=10
40      INPUT "% HEAT LOAD="; PHL
45      X=PHL: GOSUB 1200: PEA=Y
50      PRINT "% HEAT LOAD="; PHL; "% EXCESS AIR="; PEA
60      GOTO 5000

1200    REM ********** SUBROUTINE LNNT1A ******************
1202    FOR I = 1 TO ND - 1
1204    IF X <= AX(I + 1) GOTO 1208
1206    NEXT I
1208    Y = AY(I) + ((AY(I + 1) - AY(I)) / (AX(I + 1) - AX(I))) * (X - AX(I))
1210    RETURN
1212    END

5000    END
```

The main program starts with DIMension declaration to allocate 50 data spaces for the arrays AX(50) and AY(50), followed by the actually writing the values of AX(1), AY(1), …… AX(10), AY(10) on lines 20, 25 and 30. Number of data ND = 10 is given after that on line number 35. When this will be used as a part of furnace control software, the input '% Heat Load' will come as the process side Load. As shown in this demonstration program, the nested configuration of subroutines will be assumed in the latter example programs if not specifically mentioned otherwise.

Execution of any program written in QBASIC may be started by pressing the F5 key (or by first pressing ALT then repeatedly pressing (□) cursor key to bring the selector on RUN menu. Pressing (□) cursor twice to select START option, and finally pressing the ENTER key). The output screen will look like:

% HEAT LOAD = ?

With a blipping cursor at the right side of '?', say, we enter a value of 112 for PHL and press ENTER, the output screen will appear as:

% HEAT LOAD = ? 112
% HEAT LOAD = 112 % EXCESS AIR = 24.944

Where, the value of %EXCESS AIR is the desired result.

```
% HEAT LOAD=? 112
% HEAT LOAD= 112 % EXCESS AIR= 24.944
```

Note:

1. The number of data points is finite;
2. The number of data points in the set may not be equidistant with respect to the

independent variable;

3. In certain applications, the data may be generated during execution of the program, the generated data has to be stored in the pre-dimensioned arrays assigned for such duty and subsequently used in the fashion indicated in the previous example;
4. The independent variable values have to be stored in the array in ascending order;
5. The number of data points ND is to be included after filling up the arrays.

Problem 14.1 The $\sin^{-1}$ function is not available in QBASIC library. Write a program that will first generate and store q and $\text{Sin}^{-1}\, q$, when q varies from 0º to 90º with a step of 1º, and is stored in the arrays ATH(), ASTH() respectively. Then calculate: (1) q, for $\sin^{-1}(.20)$; (2) q, for $\sin^{-1}(.35)$; (3) q, for $\sin^{-1}(.55)$; and (4) q, for $\sin^{-1}(.75)$, by using SUBROUTINE LNNT1A.BAS. Report the accuracy of the results at these four points.

The linear interpolation can be used to generate a time dependent variable, where the time values are stored as the independent variable in the data array. The progress of time is implemented by a **'Time Loop'** written within the main program. The time loop starts and finishes with following program steps:

```
      T = *: DT = **: TF = **
50      IF T > TF THEN GOTO 5000
      {Executable statements in each time step e.g.,
      GOSUB statement to the interpolation
      Subroutine}
      T = T + DT
      GOTO 50
5000  END
```

The general structure of such a time loop is shown in the above program segment. The first line sets the starting value of time by T = ** (usually this value is 0), the incremental value of T, as DT = **, and the finishing value of T, as TF = ** (the value of T up to which the program has to run). In the next line, the comparison statement terminates (ends) the program when T becomes greater than TF. A program segment follows the comparison statement, which contains all the statements needed to be executed in each time step. After completion of these steps, the time value is progressed by the amount DT and the program execution is returned to line number 50.

We should present another form of linear interpolation procedure to address a relevant query: what value of the dependent variable, y, the subroutine should return, if the independent variable value, x, goes beyond the range given in the data set, i.e.

Y = ?; if X < AX(1), and Y = ?; if X > AX(ND)

In practical application of dynamic simulation, it is advantageous to answer this problem as:

Y = AY(1); if X < AX(1), and Y = AY(ND); if X > AX(ND)

SUBROUTINE LNNT1B shown in the following has these facilities.

```
1230    REM ********** SUBROUTINE LNNT1B ********************
1232    IF X <= AX(1) GOTO 1246
1234    IF X >= AX(ND) GOTO 1250
1236    FOR I = 1 TO ND - 1
1238    IF X <= AX(I + 1) GOTO 1242
1240    NEXT I
1242    Y = AY(I) + ((AY(I + 1) - AY(I)) / (AX(I + 1) - AX(I))) * (X - AX(I))
1244    RETURN
1246    Y = AY(1)
1248    RETURN
1250    Y = AY(ND)
1252    RETURN
1254    END
```

This 'beyond the range' feature in subroutine LNNT1B is realized in program lines 1232, 1246 and 1234, 1250. To explain how these features add advantage when used in dynamic modeling duties, the following examples may be considered:

In the earlier chapters it was said that dynamic response analysis pre-assumes that a process remains at steady state before it is disturbed. The instant a disturbance or excitation enters into a process, it is denoted by the zero value in time scale. If any variable value is required before the disturbance has entered, it will be sufficient to return the value that was recorded at $t = 0$.

Suppose a programmed control system is employed for a batch process which itself calculates the set point and maintains the process from start to the finishing point of a batch. It should be an essential feature of such a controller to bring the process in a 'fail-safe' state at the end of run. Hence, the set point value at the end of batch time may be used for time values longer than the reaction time until the finished batch is discharged from the processing equipment.

Let us demonstrate the procedure by the help of the following example:

EXAMPLE 14.2 The programmed temperature control system of a temperature scheduled batch reactor should follow a time-set point temperature history expressed by the following data set:

Time, h	0	0.4	1.05	1.65	2.0	3.5	3.95
Temp.°C	30	95	165	185	195	195	30

Write a program to generate set point temperature at a time interval of (a) 0.25 h; (b) 0.01 h. For (a) to show the output as a data of time and temperature; for part (b), to

show the output as a graph of temperature versus time with a scale factor of 10 for time and 1 for temperature. The origin of the graph should be at 50 px, 300 px of the monitor and it will be displayed inside a box of 300 × 200.

Solution: Part (a): The time loop parameters are: T = 0: DT = 0.25: TF = 5. The main program may be written as follows:

```
10        REM ********* PROGRAM NAME -> C14-02A.BAS ********************
DIM AX(20), AY(20)
AX(1) = 0: AY(1) =30: AX(2) = .4: AY(2) = 95
AX(3) = 1.05: AY(3) = 165: AX(4) = 1.65: AY(4) = 185
AX(5) = 2: AY(5) = 195: AX(6) = 3.5: AY(6) = 195
AX(7) = 3.95: AY(7) = 30: ND = 7
T = 0: DT = 0.25: TF = 5
50        IF T > TF THEN GOTO 5000
```

The executable statements for each time step would be a call to LNNT1B.BAS follwed by the output print statement, thus,

```
X = T: GOSUB 1230: TEMP = Y
REM PRINT "TIME ="; T; "SET POINT ="; TEMP
REM PRINT TAB(3); "T="; TAB(17); "TEMP="; TEMP
```

And finally the loop will be terminated by the following statements:

```
T = T + DT
GOTO 50
5000 END
```

Two print statements are written within the loop. Removing the REMark from the first PRINT statement and executing the program, results will appear as in the box (A) of

Figure 14.1(a). If you want to print in an organized manner so that data in each row has to start from a particular number of column value, install the REM of the first and remove the REM from the second PRINT. By execution, the time values will be written from the third column, and temperature values will written from the seventeenth column as shown in box(B) of

Figure 14.1(a): TAB(NC) declaration containing the column number NC within parentheses preceding a PRINT statement, will cause the printing to start from column no. NC.

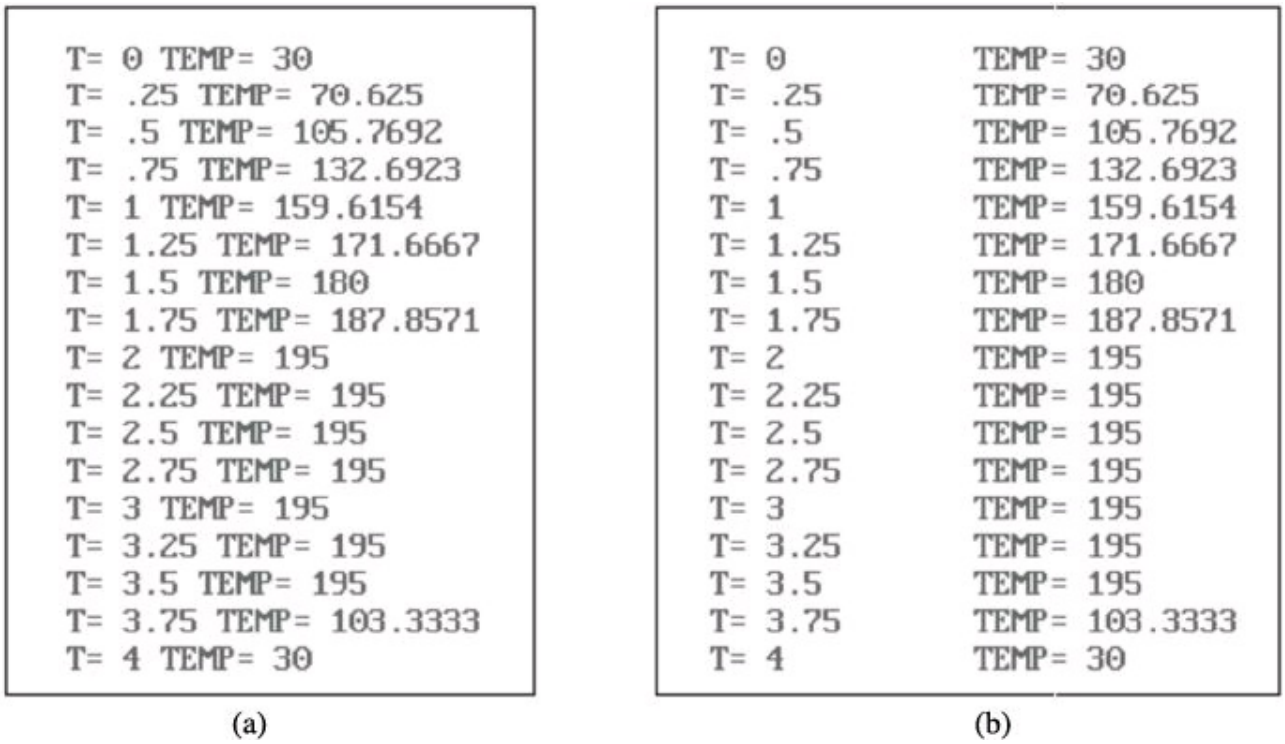


Figure 14.1(a) Results of example 14.2.

14.4 Introducing the Graphics

Part (b) of example 14.2. In such a reactor temperature control application, set point information after each 0.25 h (15 min) is too long an interval so that while it should follow a 'ramp', in fact, the set point profile will be presented to the controller as a staircase function. This may produce unsatisfactory reactor performance. In this part, we shall use a much smaller time interval, $DT = 0.01$ h (0.6 min = 36s). Rather than to use PRINT as the output statement, we shall use a graphic output statement to generate the set point profile against time.

To obtain graphic output, a screen number declaration should be at the beginning of the main program. In QBASIC, there are a number of screens available. We shall use SCREEN 12 in majority of the programs discussed in this text, which has a pixel size: 640 pixel □*x*-axis; 480 pixel□*y*-axis. Origin of every graphic screen is situated at the top left hand corner of the screen. After the screen number declaration, it is necessary to specify the coordinate of the origin from which the time versus set point profile will be drawn. According to the specification given in the problem, the profile has to be drawn within a box measuring 300 pixel x 200 pixel with its origin at (50, 300). This point is 50 pixels away from the left hand edge, and 300 pixels below from the top edge of the screen. This is done by:

```
SCREEN 12
XS = 50: YS = 300
LINE (XS, YS) – (XS + 300, YS – 300), , B
```

The LINE statement is usually applied for drawing a straight line between points (X1, Y1) and (X2, Y2). After the first "," a number from 0 to 15 may be put to colour

the line. After the second "," the letter 'B' will draw a rectangular box with (X1, Y1) and (X2, Y2) as its two opposite corners, and a 'BF' will draw a box filled with the colour as per the number put between the two commas. A blank space or 15 within the commas will produce white colour. Thus, the above LINE statement will produce a box of white colour.

To obtain a graphic output, we can use PSET statement as:

```
PSET(XS + T * XF. YS – Y * YF), 12
```

Where, XF and YF are scale factors in X and Y directions respectively, and T and Y are the time and output variables of a given process. The number 12 after the comma is the colour number for Red, hence, if XF = YF = 1, then the above statement will produce a red coloured point (single pixel) at XS + T and YS – Y. With colour number 10 (green), the same PSET statement will produce a green point at the same location. Thus, executing the time loop,, corresponding to will be depicted as a series of points by using the subroutine LNNT1B. In PSET statement, if there is no comma and colour number, the point will be drawn in white. If *DT* is taken sufficiently small, the output will be a continuous trace without any break.

If only the supplied data points are required to be plotted, we can use a CIRCLE statement within a loop as:

```
CIRCLE(X(I), Y(I)), 2
```

where, I increase from 1 to ND, and the number after the comma is pixel value of radius of the drawn circle. If the data points are to be joined by straight line segments, we can use LINE statement within a loop in the following fashion.

```
LINE(X(I), Y(I)) – (X(I + 1), Y(I + 1))
```

where, I increase from 1 to ND – 1. With such knowledge, we can write the main program for Part (b) as:

```
10      REM ******************* PROGRAM NAME -> C14-02B.BAS *********************
    DIM AX(20), AY(20)
    SCREEN 12
    XS = 50: YS = 300: XF = 100: YF = 1
    LINE (XS, YS) – (XS + 300, YS – 200), , B
    AX(1) = 0: AY(1) =30: AX(2) = .4: AY(2) = 95
    AX(3) = 1.05: AY(3) = 165: AX(4) = 1.65: AY(4) = 185
    AX(5) = 2: AY(5) = 195: AX(6) = 3.5: AY(6) = 195
    AX(7) = 3.95: AY(7) = 30: ND = 7
    T = 0: DT = 0.01: TF = 4
50      IF T > TF THEN GOTO 5000
    X = T: GOSUB 1230: TEMP = Y
    PSET(XS + T * XF. YS – Y * YF), 12
    T = T + DT
    GOTO 50
5000    END
```

Readers are asked to execute this program and observe the graphic result. We shall

execute the same program after some modifications are introduced.

Problem 14.2 Modify the program written for Example 14.2 to produce: (a) the data points in the same graphic box by circles of radius of 2 pixels; (b) the data points in the same graphic box joined by straight line segments.

14.5 Scale Divisions and Captions on the Graphic Trace of A Variable

Let us consider that the box in which the result of part (b) is displayed, has to have proper scale marking on the ordinate and abscissa and captioned in terms of Temp. (ºC) and Time(h). The scale marks can be put by LINE statements operating in two loops for temperature and time. If we choose 20ºC and 0.25 h as the minimum scale divisions for temperature and time, the required program lines to be put just below the drawing of box will be:

```
XS = 50: YS = 300: XF = 100: YF = 1
LINE (XS, YS) – (XS + 300, YS – 200), , B
FOR I = 1 TO 9
LINE (XS, YS – I * 20) – (XS + 5, YS – I * 20)
NEXT I
FOR J = 1 TO 15
LINE (XS + I * 25, YS) – (XS + I * 25, YS – 5)
```

We have assumed that same scale factors will be used for drawing variable trace. The scale division marks will be 5 pixel in length. For captions we can use PRINT statement. To print a letter anywhere on the screen, the LOCATE Y, X statement may be used before the print, which commands the cursor to write what ever text is in PRINT statement from row number Y, and column number X of the screen. For SCREEN 12, the available rows are 28, and available columns are 80. Each character is 9 pixels high and 6 pixels wide. The vertical distance between two characters is 7 pixels, and horizontal distance between two characters is 2 pixels. It is advantageous to place the caption statements before the dynamic calculations. The caption writing statements just below the scale division writing statements will be:

```
LOCATE 7, 4: PRINT "200"
LOCATE 12, 2: PRINT "T"
LOCATE 13, 2: PRINT "e 100"
LOCATE 14, 2: PRINT "m"
LOCATE 15, 2: PRINT "p"
LOCATE 17, 2: PRINT "C"
LOCATE 19, 6: PRINT "0"
LOCATE 20, 7: PRINT "0 1 2 3 4"
LOCATE 21, 29: PRINT "Time, HR"
```

Now the complete program may be written as:

```
10      REM ******************* PROGRAM NAME -> C14-02B.BAS ********************
DIM AX(20), AY(20)
SCREEN 12
```

```
    XS = 50: YS = 300: XF = 100: YF = 1
    LINE (XS, YS) – (XS + 300, YS – 200), , B
    FOR I = 1 TO 9
    LINE (XS, YS – I * 20) – (XS + 5, YS – I * 20)
    NEXT I
    FOR J = 1 TO 15
    LINE (XS + J * 25, YS) – (XS + J * 25, YS – 5)
    NEXT J
    LOCATE 7, 4: PRINT "200"
    LOCATE 12, 2: PRINT "T"
    LOCATE 13, 2: PRINT "e 100"
    LOCATE 14, 2: PRINT "m"
    LOCATE 15, 2: PRINT "p"
    LOCATE 17, 2: PRINT "C"
    LOCATE 19, 6: PRINT "0"
    LOCATE 20, 7: PRINT "0 1 2 3 4"
    LOCATE 21, 29: PRINT "Time, HR"
    AX(1) = 0: AY(1) =30: AX(2) = .4: AY(2) = 95
    AX(3) = 1.05: AY(3) = 165: AX(4) = 1.65: AY(4) = 185
    AX(5) = 2: AY(5) = 195: AX(6) = 3.5: AY(6) = 195
    AX(7) = 3.95: AY(7) = 30: ND = 7
    T = 0: DT = 0.002: TF = 4
50      IF T > TF THEN GOTO 5000
    X = T: GOSUB 1230: TEMP = Y
    PSET(XS + T * XF. YS – TEMP * YF)
    T = T + DT
    GOTO 50
5000    END
```

The graphic result is shown in Figure 14.2.

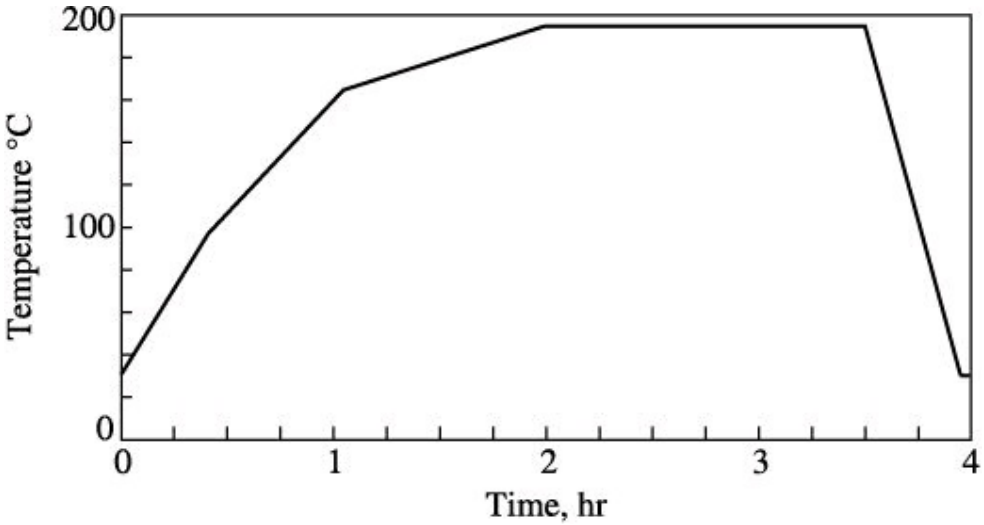


Figure 14.2 Time-Temperature schedule of the batch reactor of Example 14.2.

Though it has been shown in this example how the scale divisions and captions could be added by programming, however, in the later examples of this chapters and also in many other figures of this text that have been generated by computer programming, these additions are made by first running the QBASIC program through a special software DOS-BOX and then bringing the figures to MS-PAINT by

copy-pasting. This method is preferable due to several reasons:

1. Reduction of programming effort,
2. There are many limitations of text writing in QBASIC, e.g. single font style and size is allowed, no rotation facility of figure or text, etc. For the details of program running in DOS-BOX, see Annexure-IV.

Ideal and complex input functions (in which more than one ideal function sequentially applied, as discussed in Chapter 5) can be introduced into dynamic model by LNNT1B or LNNT1A.

To show another advantage of LNNT1B over LNNT1A in introducing input functions, consider a 'ramp' function that starts at a time value of 50 units from 0 value of the function and ends at a time value of 100 with a function value of 50. If the function is to be traced from $t = 0$ to $t = 400$, the data set required if LNNT1A is applied will be:

```
AX(1) = 0: AY(1) = 0: AX(2) = 50: AY(2) = 0: AX(3) = 100: AY(3) = 50: AX(4) = 400: AY(4) = 50: ND =4
```

whereas, if LNNT1B is applied the required data set will be:

```
AX(1) = 50: AY(1) = 0: AX(2) = 100: AY(2) = 50: ND =2
```

14.6 Introducing The Branching Statement ON (NPS) GOTO A, B, C

During execution, the control may be diverted to program line numbers A, B or C, depending on the value of path selector number NPS. NPS is an integer that may have values 1, 2, 3, … etc.

If NPS = 1, the control goes to line number A
If NPS = 2, the control goes to line number B
If NPS = 3, the control goes to line number C
And so on …

The next example demonstrates the use of ON() GOTO statement.

EXAMPLE 14.3 Write a main program to generate the ideal input functions: Step, Ramp and Impulse, using LNNT1B.

Solution: The main program may be written following what we did in Example 14.2, part (b). Note that we shall denote the output from subroutine as XIN because that is the nomenclature followed in this text, for the input or disturbance variable to a dynamic process.

```
10      REM ******************** PROGRAM NAME -> C14-03.BAS ********************
    DIM AX(20), AY(20)
    SCREEN 12
    XS = 50: YS = 300: XF = 50: YF = 10
    LINE (XS, YS) – (XS + 300, YS – 200), , B
```

```
      PRINT "ENTER VALUE OF YOUR INPUT CHOICE (INC)"
      PRINT "FOR STEP, INC=1; FOR RAMP, INC=2; FOR IMPULSE, INC=3"
      INPUT "INC="; INC
      ON (INC) GOTO 20, 25, 30
20    AX(1) = 1: AY(1) = 1: AX(2) = 1.1: AY(2) = 11: ND = 2
      GOTO 40
25    AX(1) = 1: AY(1) = 1: AX(2) = 4: AY(2) = 11: ND = 2
      GOTO 40
30    AX(1) = .9: AY(1) = 1: AX(2) = 1: AY(2) = 11
      AX(3) = 1.1: AY(3) = 1: ND = 3
40    T = 0: DT = .0001: TF = 3
50    IF T > TF THEN GOTO 5000
      X = T: GOSUB 1230: XIN = Y
      PSET (XS + T * XF, YS – XIN * YF)
      T = T + DT
      GOTO 50
5000 END
```

Following points about the main program C14-3.BAS should be noted:

1. Though the time count starts from 0, the input function occurs at a finite +ve value of time. There is no information of the function value at $t = 0$ in the data array. Due to the '*beyond the range*' feature of LNNT1B, it returns the function value corresponding to the data point at minimum value of time. The same could be said for the function returned during time interval between AX(ND) and TF;
2. The highest limit of argument list A, B, C … of ON () GOTO is 20 in QBASIC;
3. Ideal step or impulse functions are not realizable through a digital program. We have approximated the step by a ramp of extremely high slope and ideal unit impulse by a Dirac delta function also of very high slope edges having unit enclosed area. The results of the program are shown in Figures 14.3(a), (b), (c).

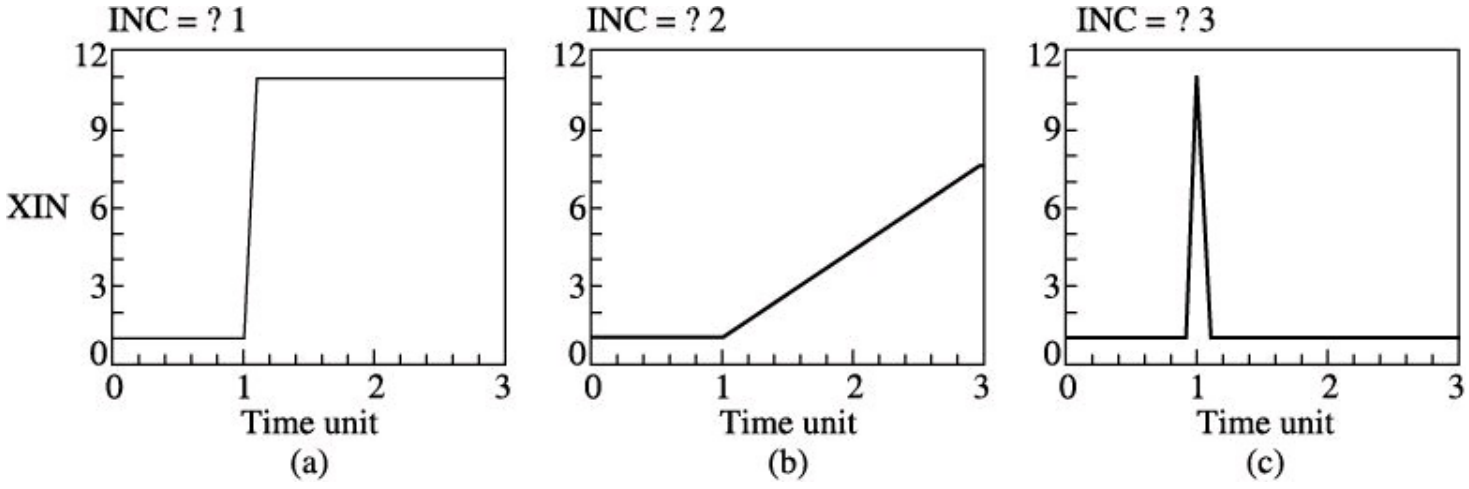


Figure 14.3 Digital simulation of (a) Step. (b) Ramp. (c) Impulse disturbance functions.

Problem 14.3 With the graphic parameters presented in Example 14.3, draw the

sinusoidal input function by using the SIN function. No data array is necessary because you are not reading the function from a data set but creating it from built in facility of QBASIC. Use the following statements associated with the time loop:

```
      A = 50: TP = 50: PI = 3.14159
      T = 0: DT = .01: TF = 200: TINP = 0
50     IF T > TF THEN GOTO 1000
      IF T > TINP THEN XIN = A * SIN((2 * PI)/TP) * T)
      PSET (XS + T * XF, YS – XIN * YF)
```

(In drawing a sine function, it is always convenient to express the radian frequency w in terms of time period TP, because in that way you know how many oscillations will show up in the result display box.)

In many applications, more than one dependent variables are related to a single independent variable, as $y_1 = f_1(x), y_2 = f_2(x), \ldots y_N = f_N(x)$. The relations could be arbitrary in nature and can be expressed only by the data set of following nature,

X	y_1	y_2	…	y_N
… … … … …	… … … … …	… … … … …	… … … … …	… … … … …
NC = 1	NC = 2	NC = 3	……	NC = N

The NCs written below each column of the above table are the column numbers of the data set. These data are to be stored in a two dimensional array, where the first dimension is the column number and entries in each column equals to ND. That is, number of data of independent and all the dependent variables are equal to ND. After the necessary DIMension declaration, the filled up array will look as follows:

```
DIM AR(..., ...)
AR(1, 1) = ... : AR(2, 1) = ... : AR(3, 1) = ... : ..... AR(N, 1) = ...
AR(1, 2) = ... : AR(2, 2) = ... : AR(3, 2) = ... : ..... AR(N, 2) = ...
.                  .              .              .....              .
.                  .              .              .....              .
AR(1, ND) = ... : AR(2, ND) = ... : AR(3, ND)  = ... : ..... AR(N, ND) = ...
```

An interpolation subroutine to handle such a data set to find a specified dependent variable, say $y3$, corresponding a given value of the independent variable, x, must be associated with the column number, NC = 4, of the data array containing the values of $y3$. But if we think about putting an extra facility of applying the same interpolation procedure for finding x values corresponding to a given value of any one of the dependent variables, it will be more versatile in use, e.g. for prediction of a number of properties of a material at a given condition of temperature, pressure or any other physical variables. Also, finding the condition at which a given property is shown by

the material would be possible by such a subroutine. That is, we can select the input (independent) and dependent by mentioning the column number NC in the GOSUB call to the subroutine. This activity is equivalent to interchanging of axes in Figure 14.1. One protection has to be taken in case the data values in any of the columns are in descending instead of ascending order. SUBROUTINE LNNT1C.BAS has been designed with all the requirements discussed above.

By examining the listing of SUBROUTINE LNNT1C.BAS, it can be seen that the 'check', to see whether the input column store data in descending order, happens in line number 1274, and rearrangement of these data in ascending order happens in the FOR loop in line numbers 1288, 1290, 1292 and 1294. In the selected segment, at first the slope, SL, of the dependent variable, *y*, with respect to the independent variable, *x*, is determined in line numbers 1282 or 1296, then the required value of *y* is calculated in lines 1284, or 1298.

```
1270 REM ********** SUBROUTINE LNNT1C.BAS **********
1272 REM ********** X, NC1, NC2, ND ---> |S.R.| ---> Y **********
1274 IF AR(NC1, 1) > AR(NC1, ND) THEN GOTO 1288
1276 FOR I = 1, ND
1278 IF (AR(NC1, I) >= X THEN GOTO 1282
1280 NEXT I
1282 SL = (AR(NC2, I) – AR(NC2, I – 1)) / (AR(NC1, I) – AR(NC1, I – 1))
1284 Y = AR(NC2, I – 1) + (X – AR(NC1, I – 1)) * SL
1286 RETURN
1288 FOR I = 1 TO ND – 1
1290 I1 = ND – I: I2 = ND – I + 1
1292 IF AR(NC1, I1) >= X THEN GOTO 1296
1294 NEXT I
1296 SL = (AR(NC2, I1) – AR(NC2, I2)) / (AR(NC1, I1) – AR(NC1, I2))
1298 Y = AR(NC2, I2) + (X – AR(NC1, I2)) * SL
1300 RETURN
1302 END
```

EXAMPLE 14.4 A process is tested with sinusoidal input of different radian frequencies, *w*. For each frequency the Log modulus of amplitude ratio, dB, and phase angle, □, was recorded only after a stable oscillation was obtained. The following data set contains the dB and □ against the input frequency, *w*.

~ rads/s	0.001	0.002	0.003	0.004	0.006	0.008	0.01	0.02
dB	23.1	17.4	12.8	9.2	3.8	–0.1	–3.3	–13.8
□°	–102	–135	–152	–162	–175	–184	–191	–207

Write a program that determines the values of dB and *w*, for □ = –180º.

Solution: The main program is written in a similar way that has been followed in earlier examples but without any graphic facility because only numerical output result print is required.

```
10    REM ************* PROGRAM NAME --> C14-04.BAS ***************
      DIM AR(5, 30)
      AR(1, 1) = 0.001: AR(2, 1) = 23.1: AR(3, 1) = –102
      AR(1, 2) = 0.002: AR(2, 2) = 17.4: AR(3, 2) = –135
      AR(1, 3) = 0.003: AR(2, 3) = 12.8: AR(3, 3) = –152
      AR(1, 4) = 0.004: AR(2, 4) = 9.2: AR(3, 4) = –162
      AR(1, 5) = 0.006: AR(2, 5) = 3.8: AR(3, 5) = –175
      AR(1, 6) = 0.008: AR(2, 6) = –0.1: AR(3, 6) = –184
      AR(1, 7) = 0.01: AR(2, 7) = –3.3: AR(3, 7) = –191
      AR(1, 8) = 0.012: AR(2, 8) = –13.8: AR(3, 8) = –207
      ND = 8
      PHI = –180
      X = PHI: NC1 = 3: NC2 = 1: GOSUB 1270: OMG = Y
      PRINT "PHI ="; PHI; "OMG ="; OMG
      X = OMG: NC1 = 1: NC2 = 2: GOSUB 1270: DB = Y
      PRINT "PHI ="; PHI; "OMG ="; OMG; "DB ="; DB
      GOTO 5000
1270 REM ************* SUBROUTINE LNNT1C.BAS **************
5000 END
```

The result after execution of the program comes as:

```
PHI=-180 OMG= 7.111111E-03
PHI=-180 OMG= 7.111111E-03 DB= 1.633334
```

14.7 Linear Interpolation in Two Dimensions

In certain cases of variable description, the dependent variable, y, depends on the values of two independent variables, x_1 and x_2. Such a relation may be expressed as:

$$y = f(x_1, x_2) \tag{14.1}$$

As an example, we may consider the problem of estimating melt viscosity of a polymer from the melt temperature and average molecular weight of the polymer. Any analytical model is hardly available for the system. However, experimental value of melt viscosity is available for different molecular weights at various temperatures. Dependence of slurry viscosity of a solid upon solid/liquid ratio and average particle size of the solid in the slurry is another such example.

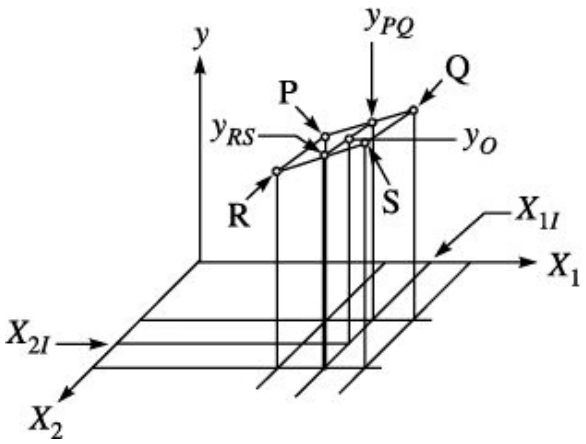


Figure 14.4 The principle of two dimensional interpolation.

The principle of such interpolation has been shown in Figure 14.4. Corresponding to the inputs x_{1I} and x_{2I}, the couple of two adjacent values in x_1 and x_2 directions are found from the data set. The y-values at these four points are P, Q, R and S. Applying linear interpolation respectively for P, Q and R, S, the two y-values y_{PQ} and y_{RS} are determined. Finally, again using linear interpolation between y_{PQ} and y_{RS}, the required output y_O is found.

```
1320 REM ********** SUBROUTINE LNNT2 **********
1322 FOR I = 1 TO ND1
1324 IF X1 < AX1(I) THEN GOTO 1328
1326 NEXT I
1328 IP = I – 1
1330 FOR J = 1 TO ND2
1332 IF X2 < AX2(J) THEN GOTO 1336
1334 NEXT J
1336 JP = J – 1
1338 SL1 = (AY1(IP + 1, JP) – AY1(IP, JP)) / (AX1(IP + 1) – AX1(IP))
1340 YPQ = AY1(IP, JP) + SL1 * (X1 – AX1(IP))
1342 SL2 = (AY1(IP + 1, JP + 1) – AY1(IP, JP + 1)) / (AX1(IP + 1) – AX1(IP))
1344 YRS = AY1(IP, JP + 1) + SL2 * (X1 – AX1(IP))
1346 SL3 = (YRS – YPQ) / (AX2(JP + 1) – AX2(JP))
1348 Y = YPQ + SL3 * (X2 – AX2(JP))
1350 RETURN
1352 END
```

To implement this subroutine, the data set has to be stored in arrays AX1(ND1), AX2(ND2) and AY(ND1, ND2). Let us do an example.

EXAMPLE 14.5 Determination of optimum amount of % excess air in a liquid fuel fired furnace depending only on % heat demand (as discussed in Example 14.1) can be further improved by considering the % of CO detected in the flue gas of the furnace. The following data is obtained from a test furnace using furnace oil as fuel.

x_1 □	85	90	95	100	105	110	115	120	125	130
□	□ Values of the dependent variable y as a function of x_1 and x_2 □									
$x_2 = 1$	20.17	20.58	21.23	22.12	23.25	24.62	26.23	28.08	30.17	32.5
= 2	20.22	20.68	21.38	22.32	23.5	24.92	26.58	28.48	30.62	33.0
= 3	20.27	20.78	21.53	22.52	23.75	25.22	26.93	28.88	31.07	33.5
= 4	20.32	20.88	21.68	22.72	24.0	25.52	27.28	29.28	31.52	34.0
= 5	20.37	20.98	21.83	22.92	24.25	25.82	27.63	29.68	31.97	34.5
= 6	20.42	21.08	21.98	23.12	24.5	26.12	27.98	30.08	32.42	35.0
= 7	20.47	21.18	22.13	23.32	24.75	26.42	28.33	30.48	32.87	35.5
= 8	20.52	21.28	22.28	23.52	25.0	26.72	28.68	30.88	33.32	36.0

= 9	20.57	21.38	22.43	23.72	25.25	27.02	29.03	31.28	33.77	36.5
= 10	20.62	21.48	22.58	23.92	25.5	27.32	29.38	31.68	34.22	37.0

Write a main program that will receive inputs of % heat demand (x_1) and % CO in flue gas (x_2) by calling SUBROUTINE LNNT2 and will calculate and report % excess air (y). Find y for x_1 = 112%, and x_2 = 7.8%.

Solution: The main program is styled in a manner as in Example 14.1, the only difference being that in the present program, there will be two inputs X1 and X2.

```
10   REM ************ PROGRAM NAME --> C14-05.BAS ***************
     DIM AX1(20), AX2(20), AY(20, 20)
     AX1(1)=85: AY(1,1)=20.17: AY(1,2)=20.58: AY(1,3)=21.23: AY(1,4)=22.12: AY(1,5)=23.25
     AX1(2)=90: AY(1,6)=24.62: AY(1,7)=26.23: AY(1,8)=28.08: AY(1,9)=30.17: AY(1,10)=32.5
     AX1(3)=95: AY(2,1)=20.22: AY(2,2)=20.68: AY(2,3)=21.38: AY(2,4)=22.32: AY(2,5)=23.5
     AX1(4)=100: AY(2,6)=24.92: AY(2,7)=26.58: AY(2,8)=28.48: AY(2,9)=30.62: AY(2,10)=33.0
     AX1(5)=105: AY(3,1)=20.27: AY(3,2)=20.78: AY(3,3)=21.53: AY(3,4)=22.52: AY(3,5)=23.75
     AX1(6)=110: AY(3,6)=25.22: AY(3,7)=26.93: AY(3,8)=28.88: AY(3,9)=31.07: AY(3,10)=33.5
     AX1(7)=115: AY(4,1)=20.32: AY(4,2)=20.88: AY(4,3)=21.68: AY(4,4)=22.72: AY(4,5)=24.0
     AX1(8)=120: AY(4,6)=25.52: AY(4,7)=27.28: AY(4,8)=29.28: AY(4,9)=31.52: AY(4,10)=34.0
     AX1(9)=125: AY(5,1)=20.37: AY(5,2)=20.98: AY(5,3)=21.83: AY(5,4)=22.92: AY(5,5)=24.25
     AX1(10)=130: AY(5,6)=25.82: AY(5,7)=27.63: AY(5,8)=29.68: AY(5,9)=31.97: AY(5,10)=34.5
     AX2(1)=1: AY(6,1)=20.42: AY(6,2)=21.08: AY(6,3)=21.98: AY(6,4)=23.12: AY(6,5)=24.5
     AX2(2)=2: AY(6,6)=26.12: AY(6,7)=27.98: AY(6,8)=30.08: AY(6,9)=32.42: AY(6,10)=35.0
     AX2(3)=3: AY(7,1)=20.47: AY(7,2)=21.18: AY(7,3)=22.13: AY(7,4)=23.32: AY(7,5)=24.75
     AX2(4)=4: AY(7,6)=26.42: AY(7,7)=28.33: AY(7,8)=30.48: AY(7,9)=32.87: AY(7,10)=35.5
     AX2(5)=5: AY(8,1)=20.52: AY(8,2)=21.28: AY(8,3)=22.28: AY(8,4)=23.52: AY(8,5)=25.0
     AX2(6)=6: AY(8,6)=26.72: AY(8,7)=28.68: AY(8,8)=30.88: AY(8,9)=33.32: AY(8,10)=36.0
     AX2(7)=7: AY(9,1)=20.57: AY(9,2)=21.38: AY(9,3)=22.43: AY(9,4)=23.72: AY(9,5)=25.25
     AX2(8)=8: AY(9,6)=27.02: AY(9,7)=29.03: AY(9,8)=31.28: AY(9,9)=33.77: AY(8,10)=36.5
     AX2(9)=9: AY(10,1)=20.62: AY(10,2)=21.48: AY(10,3)=22.58: AY(10,4)=23.92: AY(10,5)=25.5
     AX2(10)=10:AY(10,6)=27.32: AY(10,7)=29.38: AY(10,8)=31.68: AY(10,9)=34.22: AY(10,10)=37.0
     ND1 = 10: ND2 = 10
     X1 = 112: X2 = 7.8
     GOSUB 1320
     PRINT "FOR"; X1; "% HEAT LOAD and"; X2; "% CO in FLUE"
     PRINT "OPTIMUM % EXCESS AIR is"; Y
     GOTO 5000
1320 REM ************ SUBROUTINE LNNT2 ***************
5000 END
```

Result after execution of the program will appear as:

```
% EXCESS LOAD= 112.5 % CO IN FLUE= 7.8
OPTIMUM % EXCESS AIR FOR COMBUSTION= 27.635
```

14.8 Dead Time or Delay Element (Subroutine DLAY1 and DLAY2)

Dead time elements are abundant in chemical process systems. A subroutine to represent such a dynamic element is an important addition to the armory of utilities of

a simulation software.

14.8.1 DLAY1

Operation of this subroutine may be appreciated from Figure 14.5. For each increment of time step, DT, subroutine receives a call from the main program. The main segment contains the single dimension array ADL1. The effective dead time, t_{DL}, is equal to the product NDL × DT time units, where, NDL = the number of delay units given in main program.

For each call from the main program, the subroutine performs three sequential operations:

1. Assigning the value stored in ADL1 (NDL) to the subroutine output YDL;
2. Sequential shuffling of the content of ADL1 (I) to ADL1 (I + 1) progressing in the reverse direction as I decreases from NDL – 1 to I = 1;
3. Assigning the input variable value XDL to the array location ADL1 (1).

The subroutine output YDL is returned to the main program.

```
1370 REM ****** SUBROUTINE DLAY1 ********
1372 YDL = ADL1(NDL)
1374 FOR I = 1 TO NDL - 1
1376 II = NDL - I
1378 ADL1(II + 1) = ADL1(II)
1380 NEXT I
1382 ADL1(1) = XDL
1384 RETURN
1386 END
```

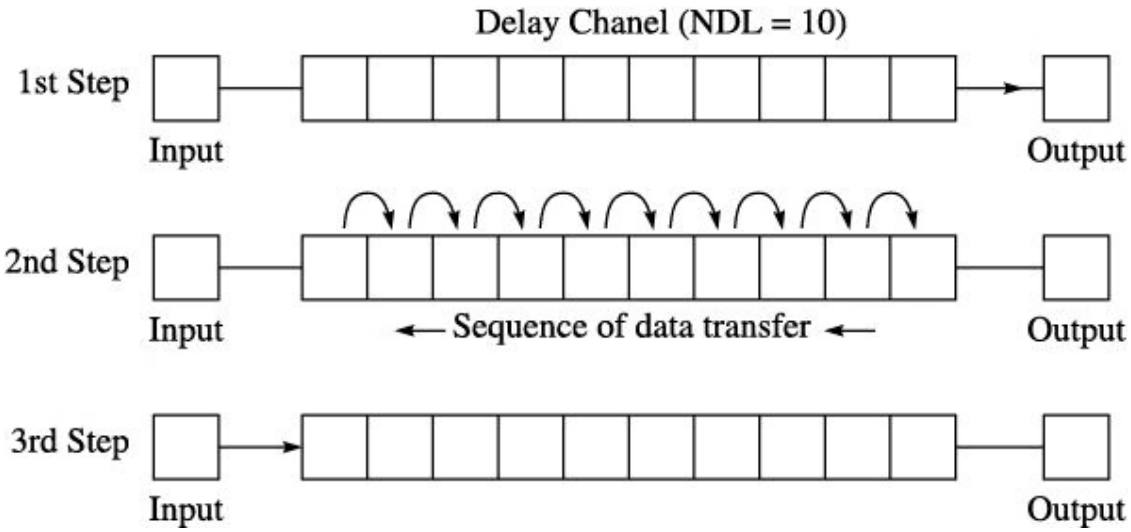


Figure 14.5 The operation of delay element simulator.

14.8.2 DLAY02

This subroutine may be used to represent multiple delay units in larger processes. This

could be easily done by adding an extra parameter, NC, the channel number for each delay line. Hence, the array ADL2 is two-dimensional. The calls are sequenced from the main program by putting proper NC values to them. If the number of delay units, NDL is not equal for all channels, then NDL should also posses a single dimension, or they may have different names, e.g., NDL1, NDL2 … etc.

```
1400 REM ****** SUBROUTINE DLAY2 ********
1402 YDL = ADL2(NC, NDL(NC))
1404 FOR I = 1 TO NDL(NC) - 1
1406 II = NDL(NC) - I
1408 ADL2(NC, II + 1) = ADL2(NC, II)
1410 NEXT I
1412 ADL2(NC, 1) = XDL
1414 RETURN
1416 END
```

To implement any of the DLAY subroutines, a time loop is necessary in the main program, because manifestation of a dead time happens only in dynamic or time variant situations. As we already said, the main program will also contain array/s containing the delay channel/s, number of delay units in the channels, initialization or putting initial values (values at $t = 0$) of the variable in all locations of the channel/s and value of the time increment step, DT, before the loop operation begins.

EXAMPLE 14.6 In Example 14.3, formation of step, ramp, and impulse input functions was shown. Write a program to delay these functions by 200 time increment units, i.e. NDL = 2000, and $t_{DL} = 2000 \times$ DT.

Solution: The main program retains the style in which it was written for Example 14.3. The array ADL1() is included in the DIMension declaration, and initialization of this array in a loop from 1 up to NDL locations happens before the beginning of time loop. Two boxes are drawn by two LINE statements to display the functions entering and leaving the delay element.

```
10    REM ************ PROGRAM NAME --> C14-6.BAS ******************
      DIM AX(20), AY(20), ADL1(5000)
      SCREEN 12
      XS1 = 50: XS2 = 50: YS1 = 400: YS2 = 200: XF = 10: YF = 1: NDL = 2000
      INC = 1
15    IF INC > 3 THEN GOTO 5000
      ON (INC) GOTO 20, 25, 30
20    AX(1) = 1: AY(1) = 10: AX(2) = 1.01: AY(2) = 110: ND = 2
      GOTO 40
25    AX(1) = 1: AY(1) = 10: AX(2) = 11: AY(2) = 110: ND = 2
      GOTO 40
30    AX(1) = .99: AY(1) = 10: AX(2) = 1: AY(2) = 110
      AX(3) = 1.01: AY(3) = 10: ND = 3
40    LINE (XS1, YS1) – (XS1 + 200, YS1 – 120), ,B
      LINE (XS2, YS2) – (XS2 + 200, YS2 – 120), ,B
```

```
      FOR I = 1 TO NDL
      ADL1(I) = AY(1)
      NEXT I
      T = 0: DT = .001: TF = 20
50    IF T > TF THEN GOTO 70
      X = T: GOSUB 1230: XIN = Y
      XDL = XIN: GOSUB 1370: XDLO = YDL
      PSET (XS + T * XF, YS2 – XIN * YF)
      PSET (XS + T * XF, YS1 – XDLO * YF)
      T = T + DT
      GOTO 50
70    REM ******************
      DO
      LOOP WHILE INKEY$ = “”
      REM ******************
      CLS
      INC = INC + 1
      GOTO 15
5000 END
```

Besides fulfilling the requirements asked in the statement of Example 14.6, the program shows few new programming techniques and a new statement CLS. Let us explain them:

1. There is an outer loop in which INP increases, and an inner loop, the time loop that we already know;
2. The two statement sequence between two REMs:

```
70    REM ******************
      DO
      LOOP WHILE INKEY$ = “”
      REM ******************
```

will temporarily pause the program execution at that stage of the program, until any of the keys are depressed;

. The CLS statement after the pause clears the screen for the display of the next input type. We may put an interim PRINT statement, as shown in the following, which will be displayed during the pause, and will vanish after the ENTER button is pressed to start execution.

```
70    REM ******************
      DO
      LOCATE 28, 16: PRINT “Press any KEY”
      LOOP WHILE INKEY$ = “”
      REM ******************
```

The end result of the program is displayed in Figure 14.6.

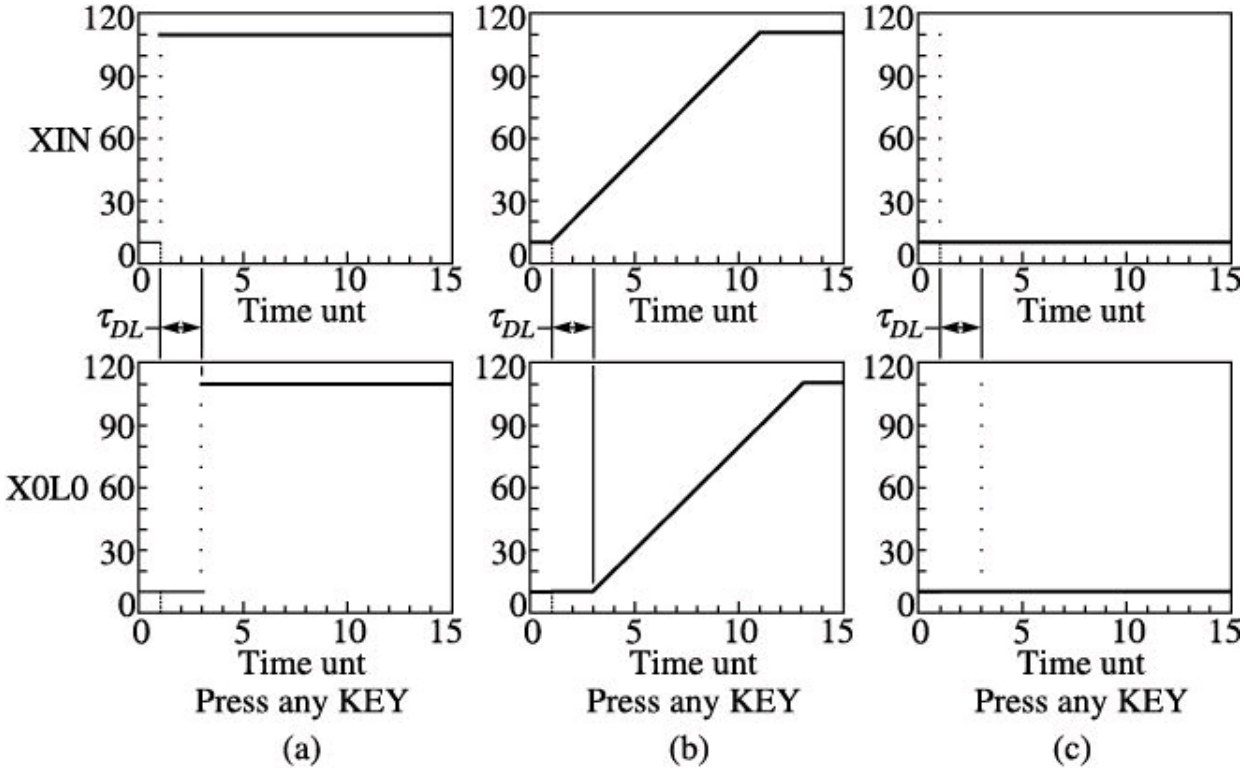


Figure 14.6 The (a) Step, (b) Ramp, and (c) Impulse inputs are delayed by subroutine DLAY1.

Usage of DLAY2 will be discussed after we introduce the solution of differential equations.

14.9 Integration Algorithms

Practically all the integration algorithms have a common feature, i.e. they apply the same calculation scheme for the output magnitude at Nth time instant, y_N, with the knowledge of the value of output at the N – 1 time instant, y_{N-1}, and derivative or slope of the function evaluated in a way so that the calculated y_N is obtained as close to its true value as possible.

$$y_N = y_{N-1} + \left(\frac{dy}{dt}\right)^* \Delta t \qquad (14.2)$$

We have put an asterix (*) with the derivative expression to express that the way of calculating this derivative is where the algorithms differ. We shall discuss the following ones and compare their performance with respect to the solution of a First order dynamic equation:

$$\tau \frac{dy}{dt} + y = K_P x \quad \text{or} \quad \frac{dy}{dt} = \frac{K_P}{\tau} x - \frac{1}{\tau} y \qquad (14.3)$$

For a step input of magnitude M, $x = M$ for all +ve values of t.

Euler First Order

The expression of derivative is calculated at $t = t_{N-1}$, as $[dy/dt]_{N-1}$ or $[dy/dt]_O$ (these

two expressions of derivative mean that they are evaluated at previous or 'old' time instant). Next, the new value of the output is obtained as:

$$y_N = y_O + \left[\frac{dy}{dt}\right]_O \Delta t \qquad (14.4)$$

To obtain the current value of y as y_N, the necessary equations to be solved for each time step are:

$$\left[\frac{dy}{dt}\right]_O = A_1 x_O - A_2 y_O \quad \text{followed by Equation 14.4} \qquad (14.5)$$

where $A_1 = K_P/t$, and $A_2 = 1/t$, $y = y_O$ at $t = t_{N-1}$, and $y = y_N$ at $t = t_N$.

As this is an explicit procedure involving only a two step calculation, Euler first order algorithm may be used as embedded in the main program instead of as a subroutine.

Euler Second Order

This algorithm involves an iterative procedure and derivative evaluation at mid-point of the time step. The calculation of y_N proceeds through the following steps:

Step 1 The following values of input are calculated by functional relation [$x = f(t)$], or by using the interpolation routine, $x_O = x$ at $t = t_{N-1}$, $x_N = x$ at $t = t_N$, where $t_N = t_{N-1} + \Delta t$.

Step 2 $(dy/dt)_1 = A_1 x_O - A_2 y_O$

Step 3 $y_{N1} = y_O + (dy/dt)_1 \Delta t$

Step 4 $(dy/dt)_2 = A_1 x_N - A_2 y_{N1}$

Step 5 $y_{N2} = y_O + [(dy/dt)_1 + (dy/dt)_2]/2\ \Delta t$

Step 6 If the difference between y_{N1} and y_{N2} is less than an arbitrarily small number, the result is evaluated in the next step. If not, y_{N1} is made equal to y_{N2} and the procedure is looped back to Step 4.

Step 7 $y_N = y_{N2}$.

Runge–Kutta Third Order

This is an explicit method that proceeds according to the following sequence:

Step 1 The following values of input are calculated by functional relation [$x = f(t)$], or by using the interpolation routine, $x_O = x$ at $t = t_{N-1}$, $x_{N2} = x$ at $t = t_N$, and, $x_{N1} = x$ at $t = t_{N-1}$,+ $\Delta t/2$.

Step 2 $(dy/dt)_1 = A_1 . x_O - A_2 . y_O$

Step 3 $y_{N1} = y_O + (dy/dt)_1 . (\Delta t/2)$

Step 4 $y_{N2} = y_O + (dy/dt)_1 . \Delta t$

Step 5 $(dy/dt)_2 = A_1 . x_{N2} - A_2 . y_{N2}$

Step 6 $y_{N3} = y_O + (dy/dt)_2 . \Delta t$

Step 7 $(dy/dt)_3 = A_1 . x_{N1} - A_2 . y_{N2}$

Step 8 $(dy/dt)_4 = A_1 . x_{N2} - A_2 . y_{N3}$

Step 9 $y_N = y_O + (1/6).[(dy/dt)_1 + 4.(dy/dt)_3 + (dy/dt)_4]. \Delta t$

Runge–Kutta Fourth Order

This is an explicit method that proceeds according to the following sequence:

Step 1 The following values of input are calculated by functional relation $x = [f(t)]$, or by using the interpolation routine, $x_O = x$ at $t = t_{N-1}$, $x_{N2} = x$ at $t = t_N$, and, $x_{N1} = x$ at
$t = t_{N-1}, + \Delta t/2$.

Step 2 $(dy/dt)_1 = A_1 x_O - A_2 y_O$

Step 3 $y_{N1} = y_O + (dy/dt)_1 (\Delta t/2)$

Step 4 $(dy/dt)_2 = A_1 x_{N1} - A_2 y_{N1}$

Step 5 $y_{N2} = y_O + (dy/dt)_2 (\Delta t/2)$

Step 6 $(dy/dt)_3 = A_1 x_{N1} - A_2 y_{N2}$

Step 7 $y_{N3} = y_O + (dy/dt)_3 \Delta t$

Step 8 $(dy/dt)_4 = A_1 x_{N2} - A_2 y_{N3}$

Step 9 $y_N = y_O + (1/6)[(dy/dt)_1 + 2(dy/dt)_2 + 2(dy/dt)_3 + (dy/dt)_4] \Delta t$

The result of performance comparison is shown in Figure 14.7. It is evident from Figure 14.7 that with a slightly smaller step size the simplest algorithm Euler first order performs with a comparable accuracy than the more accurate ones. The ODEs that make up the models of most practical chemical engineering systems usually represent a mixture of fast and slow dynamics. For example, in a plate type absorption column, the liquid flow or hydrodynamic response occurs fairly rapidly, of the order of few seconds per tray. But the liquid composition dynamics is fairly slow– minutes or even hours for big columns. Systems with such mixtures of fast and slow ODEs are called *Stiff* Systems.

If we want to integrate all ODEs with high accuracy, the fast ones will require a small step size, much smaller than would be required for the slower ones. Therefore, it is often quite acceptable to sacrifice accuracy on the fast ODEs and run at a step size for which the fast ODEs are still stable and the slower ones are quite accurate. Based on these considerations, Luyben states that "therefore my experience has been that for most of the complex systems that chemical engineers have to live with, a simple, easily programmed Euler integration is just as good as the more complex fourth order Runge-Kutta, and very often is even better".

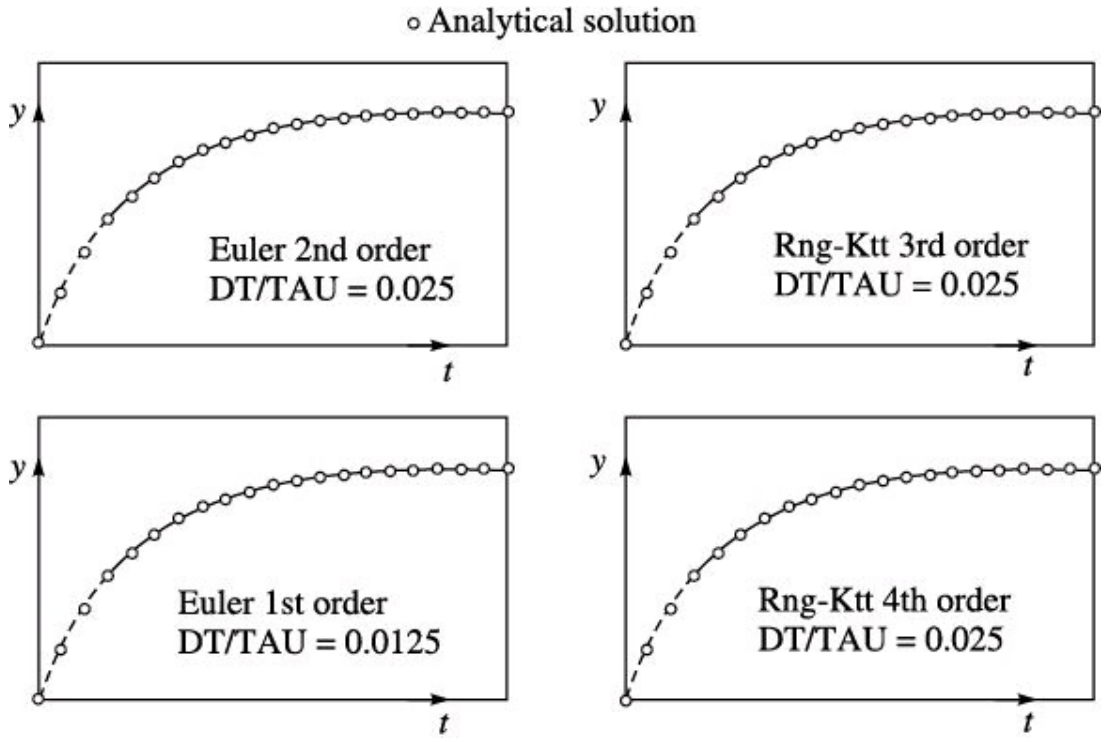


Figure 14.7 Performance comparison of integration algorithms for step response of a first order process.

Thus, in the simulation examples in this chapter, we have used Euler first order algorithm for integration of ODEs.

Until now we are discussing about first order ODEs. In case of a second or higher order system, the dynamic model has to be broken down to a set of first order ODEs just as we do for the derivation of state model of a system. The solution sequence of such a set may follow the path of signal flow; for a system comprising of serially connected first order blocks, the integration sequence will be along the direction of material or energy flow.

Problem 14.4 From the steps given for the four algorithms, the reader is asked to write the programs in subroutine form for them and then try to draw the traces in Figure 14.7.

14.9.1 First and Second Order Dynamic Blocks (SUBROUTINE FOTF, SOTFA)

Usage of these subroutines are described in the following:

FOTF

Equation 14.5 may be used for finding the derivative expression, and Equation 14.4 for y_N. The following parameters and initial value are to be supplied from the main program to the subroutine:

KP = Process Gain; TAUP = Process time constant; Initial value: YO. Also, the input variable, X, is supplied from the main program to the subroutine for all values of time.

```
1500 REM ****** SUBROUTINE FOTF (FIRST ORDER TRANSFER FUNCTION) ********
1502 DYDT = (1 / TAU) * (KP * X - YO)
1504 YN = YO + DYDT * DT
1506 RETURN
1508 END
```

Before the time increment statement within the loop, the main program must contain the statement for assigning a new value of the dynamic variable to its old one's name as:

```
YO = YN
```

SOTFA

We start with the *t-z* form of second order dynamics

$$\tau^2 \frac{d^2y}{dt^2} + 2\zeta\tau \frac{dy}{dt} + y = K_P x(t) \qquad (14.6)$$

and convert this equation into two first order ODE as:

$$\frac{dy}{dt} = y_1 \qquad (14.7a)$$

and

$$\frac{dy_1}{dt} = \frac{K_P}{\tau^2} x(t) - \frac{2\zeta}{\tau} y_1 - \frac{1}{\tau^2} y \qquad (14.7b)$$

Equations 14.7a and 14.7b may be programmed to obtain the second order system response. The following parameters and initial values are to be supplied from the main program to the subroutine: KP = Process Gain; TP = Process time constant; ETA = Damping coefficient. Initial values: YO and DYDTO. Also, the input variable X is supplied from the main program to the subroutine for all values of time.

```
1510 REM ****** SUBROUTINE SOTFA (SECOND ORDER TRANSFER FUNCTION) ********
1512 DYDT2 = (1 / (TP * TP)) * (KP * X – YO) – ((2 * ETA) / TP) * (DYDT1O)
1514 DYDT1N = DYDT1O + DYDT2 * DT
1516 YN = YO + DYDT1N * DT
1518 DYDT1O = DYDT1N
1520 RETURN
1522 END
```

Before the time increment statement within the loop, the main program must contain the statement for assigning a new value of the dynamic variable to its old one's name as:

```
YO = YN
```

For introducing input disturbance, we have shown the way to do it by using linear interpolation for which a data set is required. A simpler way to do this is shown in the following example. The input is introduced by writing the function describing statement/s after the first statement of the time loop. The statement would appear as an IF statement, containing the new input function value and the time value at which the change is desired. The 'IF' will compare the current time with the desired time and the function, written as argument of the statement, would be introduced when the comparison becomes true.

Second order TF may have another configuration, i.e. t_1, t_2 form:

$$\frac{Y(s)}{X(s)} = \frac{K_P}{(\tau_1 s + 1)(\tau_2 s + 1)} \qquad (14.8)$$

A subroutine written from the time domain differential equation corresponding to the above TF may be presented as:

```
1528 REM ************************* SUBROUTINE SOTFB ******************************
1530 DY1DT = X * (KP / TP1) – Y1O * (1 / TP1)
1532 Y1N = Y1O + DY1DT * DT
1534 DY2DT = Y1O * (KP / TP2) – Y2O * (1 / TP2)
1536 Y2N = Y2O + DY2DT * DT
1538 RETURN
1540 END
```

The statement for assigning new value of the dynamic variables to its old ones will appear before the time increment step as:

```
Y1O = Y1N: Y2O = Y2N
```

In many modeling problems, the first and/or second order TF subroutines need to be called more than once. For this purpose, subscripted version of subroutines FOTF, SOTFA, SOTFB could be written as FOTF2, SOTFA2, and SOTFB2. SOTFB2 will be presented in a later section associated with a relevant example. Development of the subroutines FOTF2 and SOTFA2 are left to the reader.

EXAMPLE 14.7 Write programs that would call subroutines FOTF and SOTF to obtain: (a) Responses for step, ramp and sinusoidal inputs with system parameters, $K_P = 1$, $t_P = 20$, and (b) Step response for a second order system with system parameters, $K_P = 1$, $t_P = 30$; and $z = 0.2$, 0.4, 0.6, 0.8, 1.0, 1.2 and 1.4.

Solution: Part (a): The main program is written following the style we followed in solving Example 15.6. Two boxes are drawn to display the inputs and responses. To

examine them with your knowledge of Chapter 5, these figures will remain in the screen until you press any key to break the 'pause'.

```
10    REM ************ PROGRAM NAME --> C14-07.BAS ******************
      SCREEN 12
      XS1 = 50: XS2 = 50: YS1 = 400: YS2 = 200: XF = 1: YF = 1
      KP = 1: TAU = 20: A = 40: TP = 30: PI = 3.14159
      INC = 1
15    IF INC > 3 THEN GOTO 5000
      LINE (XS1, YS1) – (XS1 + 200, YS1 – 120), ,B
      LINE (XS2, YS2) – (XS2 + 200, YS2 – 120), ,B
      X = 50: YO = 50
      T = 0: DT = .01: TF = 200: TINP1 = 10: TINP2 = 110
20    IF T > TF THEN GOTO 70
      ON (INC) GOTO 25, 30, 35
25    IF T > TINP1 THE X = 100
      GOTO 40
30    IF T > TINP1 AND T < TINP2 THEN X = 50 + .5 * (T – TINP1)
      GOTO 40
35    IF T > TINP1 THEN X = A * SIN(((2 * PI) / TP) * (T – TINP1))
40    GOSUB 1500
      PSET (XS + T * XF, YS – X * YF)
      PSET (XS + T * XF, YS – YN * YF)
      YO = YN
      T = T + DT
      GOTO 20
70    DO
      LOCATE 28, 16: PRINT "Press any KEY"
      LOOP WHILE INKEY$ = ""
      CLS
      INC = INC + 1
      GOTO 15
5000 END
```

Display of the last result is shown in Figures 14.8(a), (b) and (c)

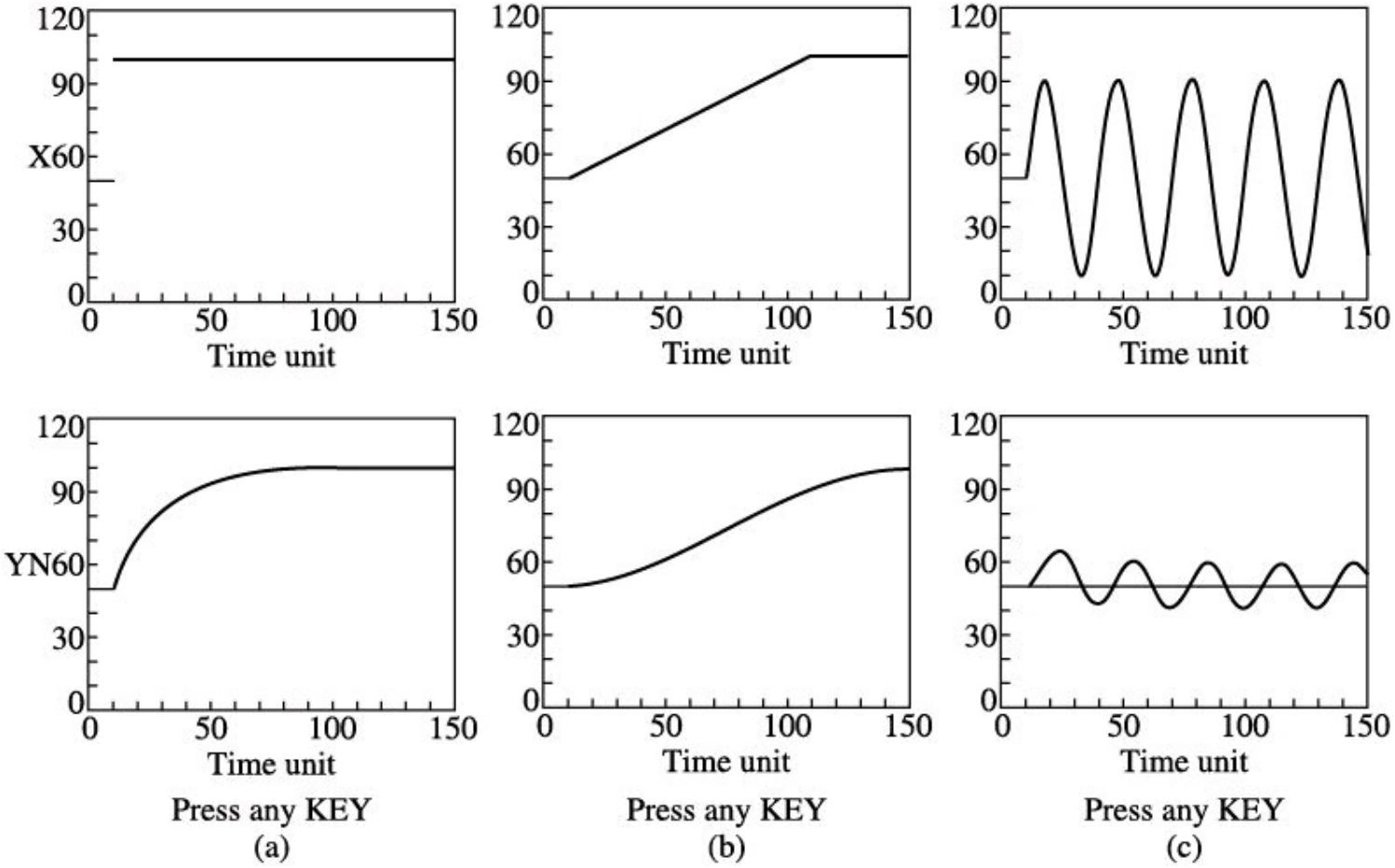


Figure 14.8 Simulation of first order system responses of ideal inputs.

Solution: Part (b):

```
10    REM ************ PROGRAM NAME --> C14-7B.BAS ******************
      SCREEN 12
      XS = 50: YS = 300: XF = .5: YF = 100
      LINE (XS, YS) – (XS1 + 200, YS1 – 160), ,B
      KP = 1: TAU = 20
      ETA = .2
15    IF ETA > 1.41 THEN GOTO 5000
      X = 0: YO = 0: DYDT1O = 0
      T = 0: DT = .01: TF = 400: TINP = 10
20    IF T > TF THEN GOTO 70
      IF T > TINP THEN X = 1
      GOSUB 1520
      PSET (XS + T * XF, YS – X * YF)
      PSET (XS + T * XF, YS – YN * YF)
      YO = YN
      T = T + DT
      GOTO 20
      ETA = ETA + .2
      GOTO 15
5000 END
```

In this program, we have not used the pause and screen clear statements, thus all the responses for ascending values of damping coefficient will be drawn in the same box together with the step input to obtain Figure 14.9. The result must make you recall the

step response shapes we had seen in Chapter 6.

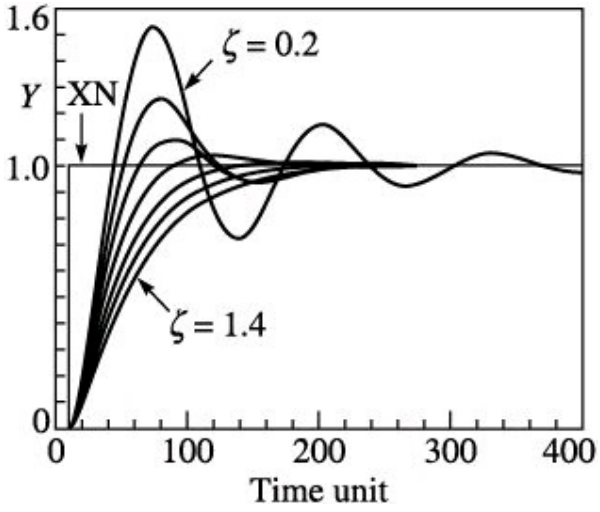


Figure 14.9 Step response of a second order system for different values of *z*.

Problem 14.5 Use PROGRAM C14-07B.BAS to trace the impulse responses of an under-, critically- and over-damped second order system.

Problem 14.6 Write a program to simulate dynamic liquid level in two interacting tanks in series. The parameter values are: A_1 = 100 sq.cm, R_1 = 0.1 cm/(cm^3/s), A_2 = 50 sq.cm, R_2 = 0.2 cm/(cm^3/s). The system is operating with a steady in flow Q_I = 50 cm^3/s. To study the model dynamics, use a pulse input according to:

1. For 0 < t < 20 s, Q_I = 50 cm^3/s;
2. For 20 s < t < 30 s, Q_I = 100 cm^3/s;
3. For 30 s < t, Q_I = 50 cm^3/s.

(You may use SUBROUTINE SOTFB for solving this problem after ascertaining the parameter values.)

In many systems, the dynamic model is obtained as differential equations, and it becomes convenient to directly program those equations without deriving their transfer functions. Let us do an example.

14.10 Modeling of Non-linear Systems

It has been mentioned during transfer function analysis of dynamic systems that the TF of a non-linear system is its linearized approximation. Digital or computer simulation is a recognized technique to obtain the exact version of dynamics of a non-linear system. Off course, such an analysis would be always subject to the truncation error inherent to any digital calculation procedure. The topic is treated by citing the

following examples.

EXAMPLE 14.8 A liquid stream is heated in an ideally stirred jacketed mixing tank of 10000 cm^3 capacity. The normal liquid flow rate is Q = 1000 cm^3/min. The liquid is heated by transferring heat from the jacket fluid through the common wall between the tank and jacket with an effective heat transfer area, A = 2000 sq.cm. The jacket fluid is saturated steam. If steam pressure remains constant, there is no appreciable change in jacket temperature during heat transfer. Jacket temperature is changed by varying the inlet steam pressure.

Liquid properties are: r = 1.0 g/cm^3, C_P = 1.0 cal/(g°C). The overall heat transfer coefficient may be taken as, U = 1.5 cal/(min/sq.cm°C). The recognizable input variables are inlet liquid temperature, T_I°C, and jacket fluid temperature, T_J°C, while the output variable is exit liquid temperature, T_O°C. The system was running under steady state conditions, T_I = 30°C, Q = 1000 cm^3/min, T_J = 110°C and T_O = 90°C. Assuming that there is no heat transfer between surrounding and the liquid heating system, develop a lumped parameter heat transfer model for the system. Write a program for simulating the model and obtain the response traces of liquid exit temperature for: (a) Inlet liquid temperature changes from 30 to 45°C, keeping liquid inflow and jacket temperature constant; (b) Inlet flow rate changes from 1000 to 1200 cm^3/min, keeping T_I and T_J constant; (c) Jacket temperature changes from 110 to 115°C, keeping T_I and Q constant.

Solution: The process model for the stirred liquid heater may be written as:

$$Q\rho C_P T_I - Q\rho C_P T_O + UA(T_J - T_O) = V\rho C_P \frac{dT_O}{dt}$$

or

$$\frac{dT_O}{dt} = \frac{Q}{V}T_I - \frac{Q}{V}T_O + \frac{UA}{V\rho C_P}(T_J - T_O) \qquad (14.9)$$

The above first order differential equation describes the dynamic model of the stirred tank jacketed heater. Program C14-08.BAS is written to obtain the response in the exit temperature T_O for step changes in T_I (for INC = 1), Q (for INC = 2) and T_J (for INC = 3).

```
10    REM ************ PROGRAM NAME --> C14-08.BAS ******************
      SCREEN 12
      XS = 50: YS1 = 420: YS2 = 270: YS3 = 120: XF = 5: YF = 10
      LINE (XS, YS1 + 40) – (XS + 180, YS1 – 60), ,B
      LINE (XS, YS2 + 40) – (XS + 180, YS2 – 60), ,B
```

```
      LINE (XS, YS3 + 40) – (XS + 180, YS3 – 60), ,B
      INC = 1
20    IF INC > 3 THEN GOTO 5000
      CP = 1: RO = 1: V = 10000: U = 1.5: A = 2000
      Q = 1000: TI = 30: TJ = 110: TOO = 90
      T = 0: DT = .01: TF = 36: TINP = 2
50    IF T > TF THEN GOTO 80
      ON (INC) GOTO 55, 60, 65
55    IF T > TINP THEN TI = 45
      YS = 420
      GOTO 70
60    IF T > TINP THEN Q = 1200
      YS = 270
      GOTO 70
65    IF T > TINP THEN TJ = 115
      YS = 120
70    DTODT = (TI – TOO) * (Q / V) + (TJ – TOO) * (U * A) / (V * CP * RO)
      TON = TOO + DTODT * DT
      PSET(XS + T * XF, YS – (TON – 90) * YF)
      TOO = TON
      T = T + DT
      GOTO 50
80    INC = INC + 1
      GOTO 20
5000 END
```

Results of this program are shown in Figures 14.10(a), (b) and (c).

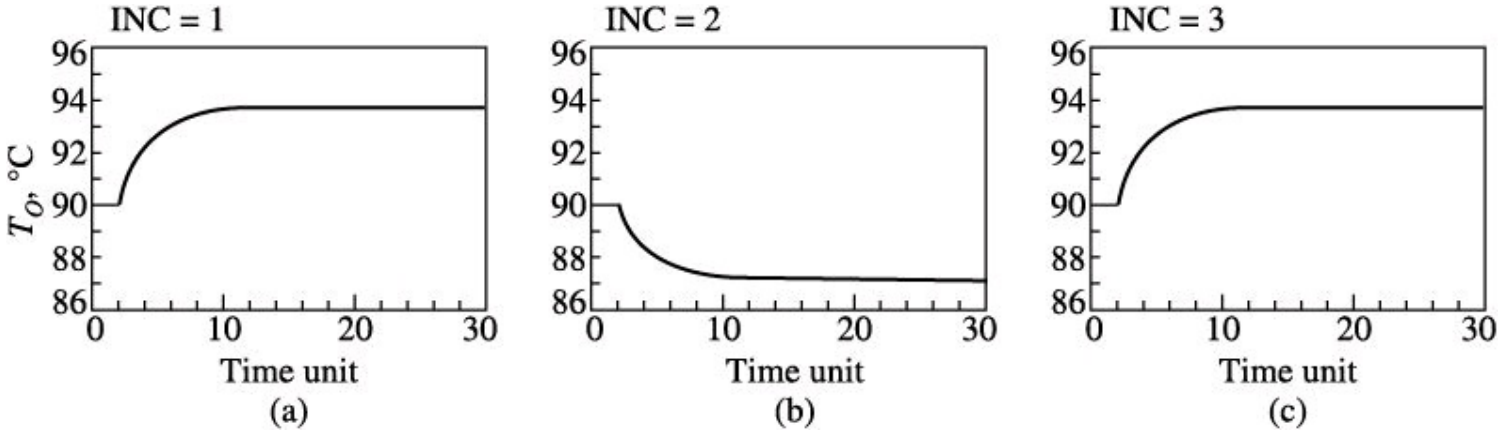


Figure 14.10 Model responses of T_O for the stirred liquid heater of example 14.7.

In the above example, it is shown how a dynamic system may be modeled without going for its transfer function. The reasons are:

1. Determination of TFs for the three inputs: $T_O(s)/T_I(s)$, $T_O(s)/Q(s)$ and $T_O(s)/T_J(s)$ are possible only with the linearized form of the model Equation 14.9. That is, C14-8.BAS represents the exact model of the system;
2. A small but not insignificant advantage is that no subroutines are required to be called in this program.

While showing its transient behaviour, the output variable T_O starts from its steady state value of 90ºC and may move toward either higher or lower values while showing the dynamic response. Hence, 90ºC should be at a suitable height from the bottom of the scale. This has been accomplished by the graphic statement:

```
PSET(XS + T * XF, YS – (TON – 90) * YF)
```

Also, to obtain a good resolution of the output trace, a scale factor YF = 10 has been chosen.

The initial value of exit temperature, T_O = 90ºC, has been found by steady state heat balance of the system. For multi-stage lumped parameter system such a balance would be very difficult. An alternative method is to run the program with inputs set at their initial values and a trial value of the output variable. If the system has sufficient 'self regulation' the system will stabilize after a finite run time giving the desired initial values of all outputs of each stage. The reader should check the accuracy of the obtained steady states values of T_O from the dynamic trace by independently calculating these values by applying heat balance for the three input changes.

In the earlier part of the text, while presenting linear analysis of open and closed loop processes, it had been mentioned a few times that for exact modeling of such systems, keeping their non-linear features intact, digital simulation is an excellent tool. We have already demonstrated this in Example 14.8, where the nonlinearity is due to the terms on the right hand side of Equation 14.9 containing product of variables. In the following two examples, we will treat systems that contain dynamic elements with nonlinear characteristics.

EXAMPLE 14.9 A single tank liquid level system has one inflow line and one outflow line with a valve. For a given value of valve opening the head versus exit flow data at different steady state conditions of the system, is shown in the following table:

h, cm	10	20	30	40	50
q_O, cm^3/min	316.22	447.21	547.72	632.45	707.1

h, cm	60	70	80	90	100
q_O, cm^3/min	774.59	856.66	894.42	948.68	1000.0

The internal cross sectional area of the tank is 200 sq.cm, and it is operating at steady state with an inflow rate q_I = 400 cm^3/min. If q_I is suddenly increased to 500 cm^3/min, find the new steady state level with the dynamic trace of level response by using: (a) The exact nonlinear model, and (b) An approximate linear model.

Solution: From a Log-log plot of the above data, the following relation is obtained between outflow rate and level of the tank:

$$q_O = 100\sqrt{h} \qquad (14.10)$$

Equation 14.10 tallies with the general form of relation that describes head versus liquid flow through an orifice, which is:

$$q_O = C_D\sqrt{h} \qquad (14.11)$$

The exact model in terms of absolute values of the variables is:

$$A\frac{dh}{dt} = q_I - C_D\sqrt{h} \quad \text{or} \quad \frac{dh}{dt} = \frac{q_I}{A} - \frac{C_D\sqrt{h}}{A} \qquad (14.12)$$

The linear model, in terms of deviation variable would be,

$$\frac{dH}{dt} = \frac{Q_I}{A} - \frac{C_D}{2A\sqrt{h_S}}H \qquad (14.13)$$

From Equation 14.11 the value of the discharge coefficient is $C_D = 100$. Note that in the exact model, the variables are in absolute values but as the linear model is valid for only within a small excursion from its steady state hence, the variables are in their deviation form from the steady state. The parameter values of the model are:

$$C_D = 100,\ A = 200 \text{ sq.cm},\ h_S = (400/100)^2 = 16 \text{ cm}$$

We may now write the program:

```
10    REM ************ PROGRAM NAME --> C14-09.BAS ******************
      SCREEN 12
      XS = 50: YS = 300: XF = 1: YF = 10
      LINE (XS, YS + 10) – (XS + 200, YS – 100), ,B
      A =200: CD = 100: HNLO = 0: QI = 400: QO =400
      HO = 16: HS = 16: QIL = 0: QOL = 0
      T = 0: DT = .01: TF = 200: TINP = 5
      REM ******** The Time Loop ********************
50    IF T > TF THEN GOTO 5000
      IF T > TINP THEN QI = 500
      IF T > TINP THEN QIL = 100
      REM ************** Non Linear Model **************
      DHDT = QI / A – CD * (HO ^ .5) * (1 / A)
      HN =HO + DHDT * DT
      REM ************** Linear Model ******************
      DHNLDT = QIL / A – (CD * HNLO) / (2 * (HS ^ .5) * A)
      HNLN = HNLO + DHNLDT * DT
      REM ************** Output Section ****************
      REM PRINT "T"; T; "HN"; HN; "HNLN"; HNLN
      PSET (XS + T * XF, YS – (HN – HS) * YF)
      PSET (XS + T * XF, YS – HNLN * YF)
      REM ************** New to Old Variable Value Transfer ***********
      HNLO = HNLN: HO = HN
```

```
      T = T + DT
      GOTO 50
5000 END
```

The results of this program are shown in 14.11.

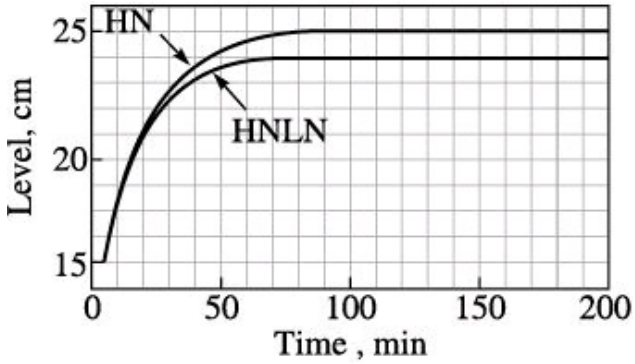


Figure 14.11 Comparison of response dynamics of linear and nonlinear model of a process.

Accuracy of the exact or nonlinear model can be confirmed by putting q = 500 cm^3/min into Equation 14.13 to obtain the new steady state value of level h_S = $(500/100)^2$ = 25 cm. This result is in absolute agreement with steady state value of the level obtained from the dynamic trace. As the output variable starts its dynamic excursion from the previous steady state
(h_S = 16 cm), the output graphic statement is written in the same form as we did in Example 14.8.

EXAMPLE 14.10 A CSTR with working volume, V = 10000 l, is handling an isothermal constant density irreversible second order reaction with zero heat effect. The reaction proceeds according to:

2A □ Product; with the kinetic model: $\frac{dC_A}{dt} = kC_A^2$, where, k = 0.1 l/s.

The reactor is operating at a steady state with a reactant concentration in the inlet solution, C_{AI} = 1.0 mole/l. The solution flows into the reactor at a constant rate, Q = 1000 cm^3/s. Write the exact and linear model for this system. If the inlet concentration of the solution is suddenly raised to 1.1 moles/l, determine the response in the exit concentration obtained from the two model configurations by a computer program.

Solution: The unsteady state balance of the reacting component around the reactor is:

$$QC_{AI} - QC_{AO} - kVC_{AO}^2 = V\frac{dC_{AO}}{dt} \qquad (14.14)$$

From Equation 14.14, the nonlinear model will be:

$$\frac{dC_{AO}}{dt} = \frac{Q}{V}(C_{AI} - C_{AO}) - kC_{AO}^2 \qquad (14.15)$$

and the linear model will be:

$$\frac{d\bar{C}_{AO}}{dt} = \frac{Q}{V}(\bar{C}_{AI} - \bar{C}_{AO}) - 2k\bar{C}_{AO}C_{AOS} \qquad (14.16)$$

where, $\bar{C}_{AI}$, $\bar{C}_{AO}$ are the deviation forms of the corresponding variables. The deviation is measured from their steady state values, C_{AIS}, C_{AOS}.

To determine C_{AOS}, a program is written only with nonlinear model. The program was run putting $C_{AOO} = C_{AI} = 1.0$. Running the program for a sufficient long time, the exit concentration C_{AO} became steady at the value of 0.8284292 mol/l. The complete program as given in following is written with $C_{AOS} = C_{AOO} = 0.8284292$ mol/l.

```
10    REM ************ PROGRAM NAME --> C14-10.BAS ******************
      SCREEN 12
      XS = 50: YS = 300: XF = 2: YF = 200
      LINE (XS, YS + 20) – (XS + 200, YS – 90), , B
      Q = 1000: V = 10000: K = .025
      CAI = 1: CAOO = .8284292
      CAIL = 0: CAOL = 0: CAOS = .8284292
      T = 0: DT = .01: TF = 100: TINP = 5
50    IF T > TF THEN GOTO 5000
      IF T > TINP THEN CAI = 1.5
      IF T > TINP THEN CAIL = .5
      DCADT = (Q / V) * (CAI – CAOO) – K * (CAOO ^ 2)
      CAON = CAOO + DCADT * DT
      DCALDT = (Q / V) * (CAIL – CAOL) – (2 * K * CAOS * CAOL)
      CANL = CAOL + DCALDT * DT
      REM PRINT "T"; T; "CAON"; CAON; "CANL"; CANL
      PSET(XS + T * XF, YS – (CAON – CAOS) * YF)
      PSET(XS + T * XF, YS – CANL * YF)
      CAOO = CAON: CAOL = CANL
      T = T + DT
      GOTO 50
5000 END
```

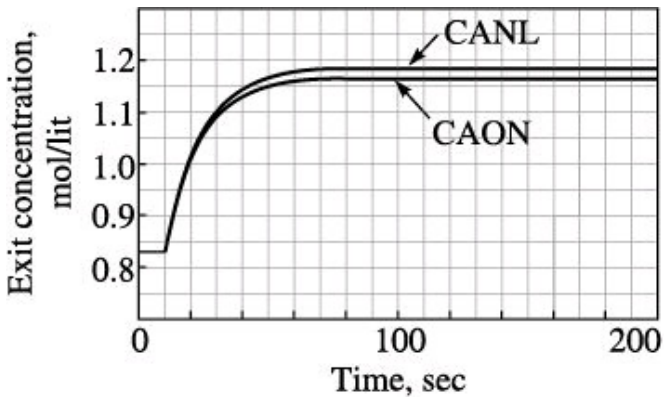


Figure 14.12 Comparison of response dynamics of linear and nonlinear model for a second order reaction in a CSTR.

Problem 14.7 Develop first order models from the set of step responses in Figure

14.11 and 14.12, by using the methods discussed in Chapter 12.

Problem 14.8 Why is the final steady state value obtained from the nonlinear model is *smaller* in case of the reactor but larger for liquid level system than the approximate value obtained from the linear model?

14.11 Lead-Lag Element

Lead-lag elements are important dynamic blocks in control system synthesis that we have seen during the discussion of practical realization of Derivative action in a (PID) controller, and also in concern of feed forward controller design. We shall attempt to write a program segment that will simulate the dynamic action of a lead-lag element by assuming the following TF of the lead-lag block:

$$\frac{Y(s)}{X(s)} = \frac{1+\tau_{LD}s}{1+\tau_{LG}s} \qquad (14.17)$$

For $t_{LG} < t_{LD}$, the block behaves like a differentiator (lead) and for $t_{LG} > t_{LD}$, the block will represent an integrator (lag). Equation 14.17 in time domain becomes:

$$\tau_{LG}\frac{dy}{dt} + y = \tau_{LD}\frac{dx}{dt} + x \quad \text{or} \quad \frac{dy}{dt} = \frac{\tau_{LD}}{\tau_{LG}}\frac{dx}{dt} + \frac{x}{\tau_{LG}} - \frac{y}{\tau_{LG}} \qquad (14.18)$$

In the following program C14-11.BAS Equation 14.18 is simulated to obtain unit step response of a lead-lag element for $t_{LD}/t_{LG} = 10$, and $t_{LD}/t_{LG} = 0.1$.

```
10    REM ************ PROGRAM NAME --> C14-11.BAS ******************
      SCREEN 12
      XS = 50: YS = 300: XF = 1: YF = 10
      LINE (XS, YS + 10) – (XS + 200, YS1 – 100), ,B
      TLD = 40: TLG = 4
      REM TLD = 4: TLG = 40
      A = TLD / TLG: B = 1 / TLG
      XN = 0: XO = 0: YO = 0
      T = 0: DT = .01: TF = 200: TINP = 5
      50 IF T > TF THEN GOTO 5000
      IF T > TINP THEN XN = 1
      DYDT = A * ((XN – XO) / DT) + B * (XO – YO)
      YN = YO + DYDT * DT
      REM **********OUTPUT SECTION *********************
      PSET (XS + T * XF, YS – YN * YF)
      REM ******************************************************
      YO = YN: XN = XO
      T = T + DT
      GOTO 50
5000 END
```

The reader should modify this program to obtain responses for different values of *a,* and to obtain a figure like that was obtained in Chapter 8, lead-lag section.

14.12 Implicit Convergence (Subroutines INTHAV, NEWRAF and DCSCRJ)

In a relation of the following type,

$$y = f(x) \tag{14.19}$$

if the argument contains non-linear terms, the calculation of x_S corresponding to a given y_I requires some implicit (trial-error) procedure that converges to the solution. In implementing these methods, there is the question of numerical stability, that is, the x values obtained in successive iterations should not oscillate around or diverge away from the true solution. A little less significant aspect due to the lightning speed of modern computers is the calculation speed of a given method that may be expressed as the number of iteration steps required to arrive within the desired error limit from the solution. There are many such algorithms available in the literature for implicit way of solution seeking procedures. In the following section three of the more important types have been discussed.

14.12.1 INTHAV

Subroutine INTHAV starts with a given trial value and step size of x and progresses in the direction in which the error is reducing. When there is a change of algebraic sign of error, the search is continued in reverse direction with half step size. The solution is obtained after a comparatively large number of steps but the method has impeccable stability feature. The following Figure 14.13(a) displays the interval halving technique.

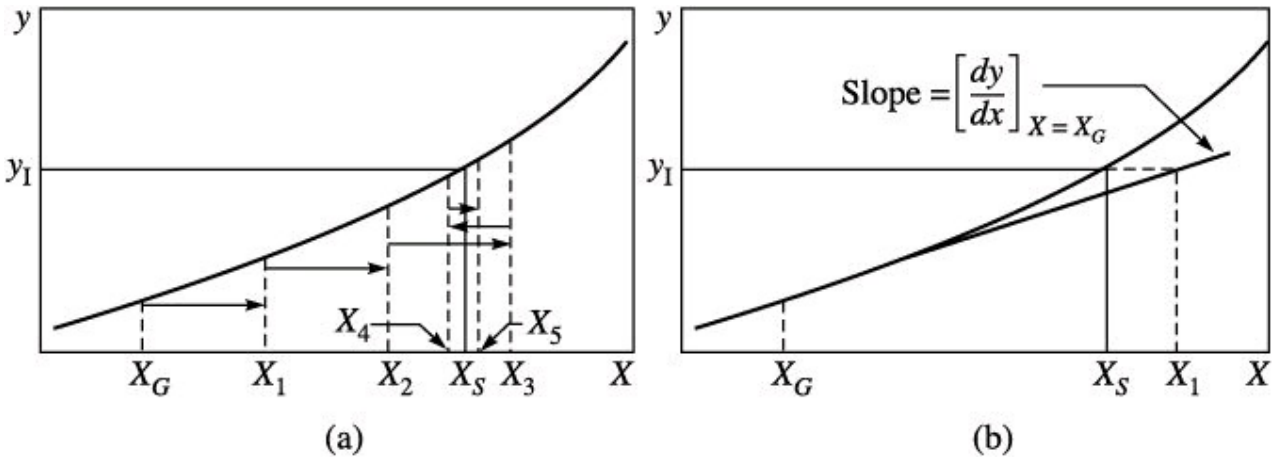


Figure 14.13 Implicit convergence by: (a) Interval halving, (b) Newton-Raphson techinques.

```
1550 REM ********** SUBROUTINE INTHAV **********
1552 GOSUB 1580
1554 IF ABS(YS – YC) < ER THEN GOTO 1576
1556 IF YC < YS THEN GOTO 1560
1558 IF YC > YS THEN GOTO 1564
1560 NJ = 1: X = X + DX * NJ: NKO = NKN
1562 GOTO 1552
1564 NJ = –1: NKN = NKO * –1
```

```
1566 IF (NKN * NKO) < 0 AND NJ = –1 THEN GOTO 1570
1568 GOTO 1572
1570 DX = DX / 2
1572 X = X + DX * NJ: NKO = NKN
1574 GOTO 1552
1576 RETURN
1578 END
```

14.12.2 NEWRAF

Subroutine NEWRAF employs Newton-Raphson algorithm which is more fast and requires lesser numbers of iteration steps, but if the function has pronounced non-linear features (too much curvature around the solution), stability problem may arise. A little experience about the function locus obtained by a few trials will guide the user to select a proper first guess value to eliminate this problem.

Redefining Equation 14.19 as:

$$f^*(x) = f(x - x_S) \tag{14.20}$$

The problem renders to that we want to find the value of x that makes $f^*(x) = 0$. [i.e. we want to find the root of $f^*(x)$]. We make a guess of $x = x_G$, and we evaluate the function:

$$f^*(x)|_{X=X_G}$$

Next we evaluate the slope of the function at $x = x_G$, as $\left[\frac{df^*(x)}{dx}\right]_{x=x_G}$, then from Figure 14.13(b),

$$\left[\frac{df^*(x_G)}{dx}\right] = f'^*(x_G) = \frac{-f^*(x_G)}{x_1 - x_G} \quad \text{or} \quad x_1 = x_G - \frac{f^*(x_G)}{f'^*(x_G)} \tag{14.21}$$

Equation 14.21 is used as the recursive solution for finding new trial values until the required amount of convergence is obtained. Progress of the procedure is shown in Figure 14.13(b).

```
1610 REM ********************** SUBROUTINE NEWRAF ***************************
1614 TK = X: GOSUB 1630
1616 IF ABS(YS – YC) < ER THEN GOTO 1622
1618 RE = YC – YS: X = X – RE / DYDX
1620 GOTO 1614
1622 XS = X
1624 RETURN
1626 END
```

14.12.3 Decimal Section Rejection (DCSCRJ)

Users of this procedure should have an estimate about the range of independent variable, x, within which the solution in terms of the dependent variable, y, exists.

This range is divided into ten (10) equal length sections. Values of y are found at the 11 junction points of these sections. The section that contains the solution is found by serially checking for which two points, the error (= YS – YC, where, YS is required solution, and YC the calculated value of y) changes its algebraic sign. The selected interval is again subdivided into ten sub-sections and they are searched to find the solution containing section. This process is continued until the absolute value of error becomes smaller than an arbitrarily small number ER.

```
1660 REM ********** SUBROUTINE DCSCRJ **********
1662 DX = (XF – XI) / 10
1664 XO = XI: X = XO: GOSUB 1700: YCO = Y: ERO = YS – YCO
1666 XN = XI + DX
1668 X = XN: GOSUB 1700: YCN = Y
1670 ERN = YS – YCN
1672 IF ABS(ERN) < ER THEN GOTO 1686
1674 IF ERN * ERO < 0 THEN GOTO 1682
1676 XO = XN: ERO = ERN
1678 XN = XN + DX
1680 GOTO 1668
1682 XI = XO: XF = XN: DX = (XF – XI) / 10
1682 GOTO 1666
1684 XS = XN
1686 RETURN
1690 END
```

Before discussing the use of these subroutines, it should be mentioned that in all these implicit procedures, the function $y = f(x)$ has to be evaluated for each value of x as the iteration is progressed. In the Newton-Raphson method, the derivative, dy/dx, is also needed along with the function value. This function evaluation is specific to a given problem, hence, may be contained in an associated subroutine that will run along the implicit subroutines. Following such a procedure, the implicit procedure subroutines could be presented in a general form. A main program is always required to run these subroutines that will merely contain the function value YS (solution) for which the independent variable value XS is required, and parameter values for function evaluation. In the following section the implicit procedures are discussed with the main and function evaluation programs (FXs) through three specific examples.

EXAMPLE 14.11 A liquid mixture contains benzene (B) and toluene (T) at different proportions given by the mole fraction of the more volatile component, x_B. (a) Write a program to find the boiling temperature of the mixture, T_B°C, and vapour composition, y_B, in equilibrium with the boiling liquid by using interval halving procedure, for x_B = 0, 0.25, 0.5, 0.75 and 1; (b) Write a program to draw X-Y and T-X-Y type equilibrium diagram for the system using the same subroutines. The

prevailing pressure is 1 atm = 760 mm Hg.

The temperature dependency of vapour pressures for pure benzene and toluene are given by the following relations:

$\text{Log}_{10}(P_{BENZENE}) = 7.84135 - 1750/T$, and $\text{Log}_{10}(P_{TOLUENE}) = 8.0884 - 1985/T$

where, temperature T is in ºK.

Solution: Part (a) for use in the program, these relations are simplified as:

For benzene, $P_B = 10^{7.84135-1750/T}$, and for toluene, $P_T = 10^{8.0884-1985/T}$

The temperature of the boiling system is the independent variable, x, and the dependent variable, y, is the total pressure exerted by the liquid mixture. As the first guess value of the independent variable, X = TK = 337°K is used, which is below the atmospheric boiling point of the more volatile component. Now we may write the main program C15-12.BAS and place subroutines INTHAV and FX1, sequentially in the following manner.

```
10    REM ********** PROGRAM NAME -> C14-12A.BAS **********
      XB = 0: XT = 1 – XB: YS = 760
      TK = 337: TC = TK – 273: X = TK: DX = 10
      ER = .001: NKO = 1: NKN = 1: NJ = 1
      GOSUB 1550
      PPB = XB * VPB: PPT = XT * VPT: TOTP = PPB + PPT
      YB = PPB / TOTP: YT = 1 – YB: BPT = X – 273
      PRINT "BACKCHECK, SUPPLD TOT PR =": YS; "CALCD TOT PR=": YC
      PRINT "RESULT, LIQ.CMPSN=": XB; "VP.CMPSN=": YB; "B.PT=": BPT
      GOTO 5000
1550 REM ********** SUBROUTINE INTHAV **********
1580 REM ********** SUBROUTINE FX1 (VP FROM TEMP) **********
1582 AB = 7.48135: BB = 1750: AT = 8.0884: BT = 1985
1584 VPB = 10 ^ (AB – BB / X): VPT = 10 ^ (AT – BT / X)
1586 YC = VPB * XB + VPT * (1 – XB)
1588 RETURN
1590 END
5000 END
```

The result is shown for XB = 0.5.

```
BACKCHECK, SUPPLD TOT PR= 760 CALCD TOT PR= 760.0004
RESULT, LIQ.CMPSN= .5 VP.CMPSN= .7144387 B.PT= 91.16293
```

The results corresponding to other values of XB are found and given in the following table.

X_B mole fraction	0.00	0.25	0.50	0.75	1.00
Y_B mole fraction	0.00	0.4473	0.7144	0.8851	1.00
T_B ºC	108.175	98.714	91.163	84.974	79.784

Part (b) The main program written for part (a) could be used for solution of this part

with a little modifications, e.g.

1. The graphic screen declaration because we want to draw the equilibrium diagrams;
2. DIMension declaration of arrays in which the equilibrium compositions and boiling temperatures will be stored;
3. A loop in which XB will be increased by 0.1% from 0 to 100%, and for each liquid composition the corresponding vapour composition and boiling temperatures are stored in the arrays; and
4. Drawing the two equilibrium diagrams by the PSET statements within a FOR loop. An *x-y* diagonal has been drawn to render a familiar look to the X-Y plot.

```
20    REM ***************** PROGRAM NAME -> C14-12B.BAS ************
      DIM AXB(2000), AYB(2000), ATC(2000)
      SCREEN 12
      XBI = 0: XBF = 1: DXB = .001: XB = XBI: NP = 0
30    IF XB > XBF THEN GOTO 560
      XT = 1 – XB: YS = 760: X = TK: NP = NP + 1
      TK = 337: TC = TK – 273: X = TK: DX = 10
      ER = .001: NKO = 1: NKN = 1: NJ = 1
      GOSUB 1550
      PPB = XB * VPB: PPT = XT * VPT: TOTP = PPB + PPT
      YB = PPB / TOTP: YT = 1 – YB: BPT = X – 273
      AXB(NP) = XB: AYB(NP) = YB: ATC(NP) = BPT
      XB = XB + DXB
      GOTO 530
560   XS1 = 50: YS = 300: XS2 = 300
      LINE (XS1, YS) – (XS1 + 200, YS – 200), , B
      LINE (XS1, YS) – (XS1 + 200, YS – 200)
      LINE (XS1, YS) – (XS1 + 200, YS – 200), , B
      FOR I = 1 TO 1001
      PSET(XS1 + AXB(I) * 200, YS – AYB(I) * 200)
      PSET(XS1 + AXB(I) * 200, YS – ATC(I) * 5)
      PSET(XS1 + AYB(I) * 200, YS – ATC(I) * 5)
      NEXT I
      GOTO 5000
```

This main program can operate with SUBROUTINE INTHAV and SUBROUTINE FX1 to obtain the following results.

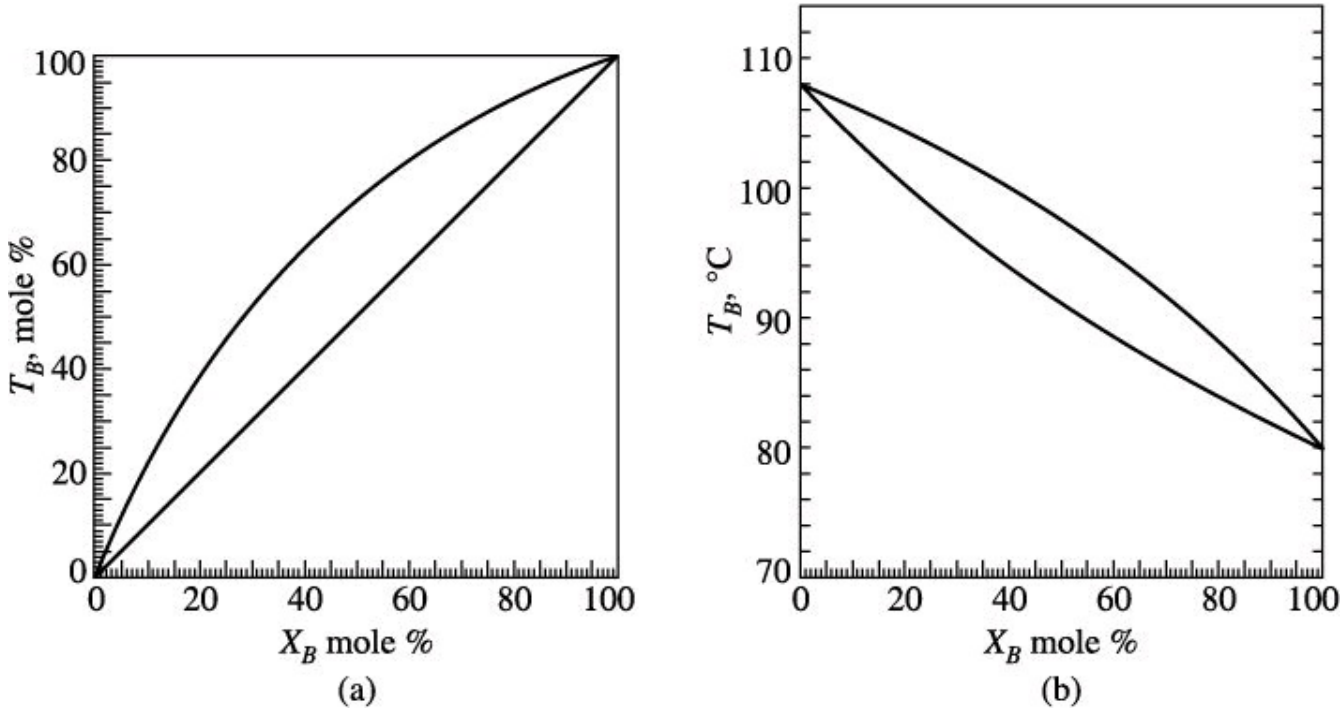


Figure 14.14 Vapour-Liquid equilibrium for benzene-toluene mixture. (a) X-Y plot. (b) T-X-Y plot.

EXAMPLE 14.12 An organic reacting species, A, undergoes two parallel irreversible reactions giving B and C, according to: (1) A □ B, (2) A □ C, where B is the desired product but C is an unwanted waste. The reactions will be conducted in a CSTR. The temperature dependency of the rate constants of the two reactions are expressed by the following Arrhenius type relations:

$$k_1 = k_{O1} e^{-B1/T} \text{ and } k_2 = k_{O2} e^{-B2/T}$$

where, $k_{O1} = 0.52 \text{ min}^{-1}$, $k_{O2} = 0.09 \text{ min}^{-1}$, B1 = 400ºK, B2 = 300ºK and T in ºK.

To increase the relative production of B, it has been proposed to operate the reactor at an elevated temperature so that the ratio k_1/k_2 becomes 4.5. Find the temperature by Newton-Raphson method.

```
10    REM ************ PROGRAM NAME -> C14-13.BAS ************
      KO1 = .52: B1 = 400: KO2 = .09: B2 = 300: RKO = KO1 / KO2
      YS = 4.5: X = 200: ER = .001
      GOSUB 1610
      PRINT "YS"; YS; "YC"; YC; "XS"; XS
      PRINT "TO OBTAIN A k1/k2 RATIO="; YS
      PRINT "REACTOR TEMP="; XS – 273; "deg C"
      GOTO 5000
1610 REM ********************** SUBROUTINE NEWRAF ***************************
1630 REM ***************** SUBROUTINE FX2 & DFXDX ********************
1632 YC = RKO * EXP((B2 – B1) / TK)
1634 DYDX = RKO * ((B1 – B2) / (TK * TK)) * EXP((B2 – B1) / TK)
1636 RETURN
1638 END
5000 END
```

Result after executing the program is obtained as:

```
YS 4.5 YC 4.499998 XS 400.0927
TO OBTAIN A k1/k2 RATIO= 4.5
REACTOR TEMP= 127.0927 deg C
```

EXAMPLE 14.13 The open loop transfer function of a process is given in the following:

$$G_P(s) = \frac{2.5(2.5s+1)e^{-2s}}{(s+1)(5s+1)(10s+1)}$$

All time constants in the transfer function are in minutes. Find the cross over frequency at which the phase of the system will be –180º, by decimal section rejection method. Also find K_U, P_U values and parameter values of (P), (PI), and (PID) controllers for the process, according to Ziegler-Nichol's formulae.

```
10   REM ************ PROGRAM NAME -> C14-14.BAS ************
     YS = –180: XI = .01: XF = 10: ER = .0001
     KP = 2.5: TZ1 = 2: TP1 = 1: TP2 = 5: TP3 = 10: TD = .2
     PI = 3.14159: FA = 180 / PI
     GOSUB 1660
     X = XS
     PHP = 0
     PHZ1 = (ATN( X * TZ1)) * FA
     PHP1 = (ATN( – X * TP1)) * FA
     PHP2 = (ATN( – X * TP2)) * FA
     PHP3 = (ATN( – X * TP3)) * FA
     PHDT = – (TD * X * FA)
     YC = PHP + PHZ1 + PHP1 + PHP2 + PHP3 + PHDT
     PRINT "YS="; YS; "YC="; YC; "XS="; XS
     PRINT
     AK = KP
     AZ1 = SQR(1 + TZ1 * TZ1 * X * X)
     AP1 = 1 / (SQR(1 + TP1 * TP1 * X * X)
     AP2 = 1 / (SQR(1 + TP2 * TP2 * X * X)
     AP3 = 1 / (SQR(1 + TP3 * TP3 * X * X)
     ADT = 1
     AR = AK * AZ1 * AP1 * AP2 * AP3 * ADT
     Wco = XS: Pu = (2 * PI) / Wu: Ku = 1 / AR
     PRINT "Wco"; Wco; "Pu"; Pu; "Ku"; Ku
     PRINT "CONTROLLER SETTINGS ACCORDING TO Z-N FORMULAE"
     KC = Ku * .5
     PRINT "[P]-CONTROLLER"; TAB(19); "KC="; KC
     KC = Ku * .45: TAUI = Pu / 1.2
     PRINT "[PI]-CONTROLLER"; TAB(19); "KC="; KC; "TAUI="; TAUI
     KC = Ku * .6: TAUI = Pu / 2: TAUD = Pu / 8
     PRINT "[PID]-CONTROLLER"; TAB(19); "KC="; KC; "TAUI="; TAUI; "TAUD="; TAUD
     GOTO 5000
1660 REM ************** SUBROUTINE DCSCRJ ******************
```

```
1700 REM ***************** SUBROUTINE FX3 ******************
      PHP = 0
      PHZ1 = ATN(X * TZ1) * FA
      PHP1 = (ATN( –X * TP1)) * FA
      PHP2 = (ATN( –X * TP2)) * FA
      PHP3 = (ATN( –X * TP3)) * FA
      PHDT = –(X * TD * FA)
      Y = PHP + PHZ1 + PHP1 + PHP2 + PHP3 + PHDT
1706 RETURN
1708 END
5000 END
```

The program results are shown below. Such a procedure should have a potential use in controller parameter design applications.

```
YS=-180 YCALC=-180 XS= .46564

Wu .46564 Pu 13.49364 AR .2566289 Ku 3.896678
CONTROLLER SETTINGS ACCORDING TO Z-N FORMULAE
[P]-CONTROLLER     KC= 1.948339
[PI]-CONTROLLER    KC= 1.753505 TAUI= 11.2447
[PID]-CONTROLLER   KC= 2.338007 TAUI= 6.746822 TAUD= 1.686705
```

The use of TAB(--) declaration in a PRINT statement is used to control the print duty after TAB(A) to start from column number A. This is a means to print a series of data with their left hand edge arranged in vertical line.

14.13 Modeling for Non-Ideal Mixing

In many process models, more than one dynamic blocks are identified to operate simultaneously. These dynamic units are either first or second order type lumped parameter blocks or dead time elements. Within a process, the intra-connection configuration between these blocks may be a series or parallel or series-parallel network. Such a network may be recognized as bypass or recycle path within the process. While writing a program for simulating a process if the dynamic block subroutines are to be called more than once, it is better to use their subscripted version. The subscript is usually the stream or channel number of the stream identified in the model configuration. We have already presented the subscripted version of dead time as DLAY2 and first and second order TFs. In the following two examples, we shall show the use of these subroutines for two process models that comprise of parallel signal processing and recirculation paths to simulate non-ideal mixing phenomena in processing equipment.

Tracer technique is a popular method of process testing, in which an inert but easily detectable material is injected as a narrow width pulse in a vehicle fluid stream into the process. The vehicle fluid is essentially an inert material that passes through the process equipment at a constant rate and experiences all the hydrodynamic excursions

through the equipment volume identical to the actual process fluid. The tracer concentration is detected by a suitable detector or sensor having a linear characteristic. Among others, optical, electrolytic, and radio-active tracers are commonly used in industry and R&D studies.

In the first of the following two examples, a 'defective' packed bed is tracer tested to identify the nature and quantum of its defect. In the second example, the non-ideal mixing model of a stirred liquid mixing vessel is tracer tested to determine the dynamic concentration traces of the tracer and thereby determine the mixing time of the mixer.

EXAMPLE 14.14 From the unsatisfactory performance of an up-flow ion-exchange column, it is suspected that the bed has leakage channel/s due to defective packing of the bed. It is proposed that the leakage paths, as well as the normally packed portion of the bed may be represented by separate parallel branches having SOPDT model structure. The bed is tracer tested with a small pulse of an optical tracer material (say, a non-polar dye solution). The detector output is plotted on a potentiometric recorder. The dynamic trace shows two earlier sharp 'humps' for at least two leakage channels, followed by a regular pulse response for the normally packed portion of the bed.

A schematic of the defective bed together with the tracer pulse injector on the inlet and its detector on the outlet is shown in Figure 14.15(a). In Figure 14.15(b), the three parallel channels model configuration has been shown in which the channel numbers are denoted as NC. Write a program to represent the model. With suitable values of the model parameters show that the pulse response qualitatively produces the shape of the tracer response.

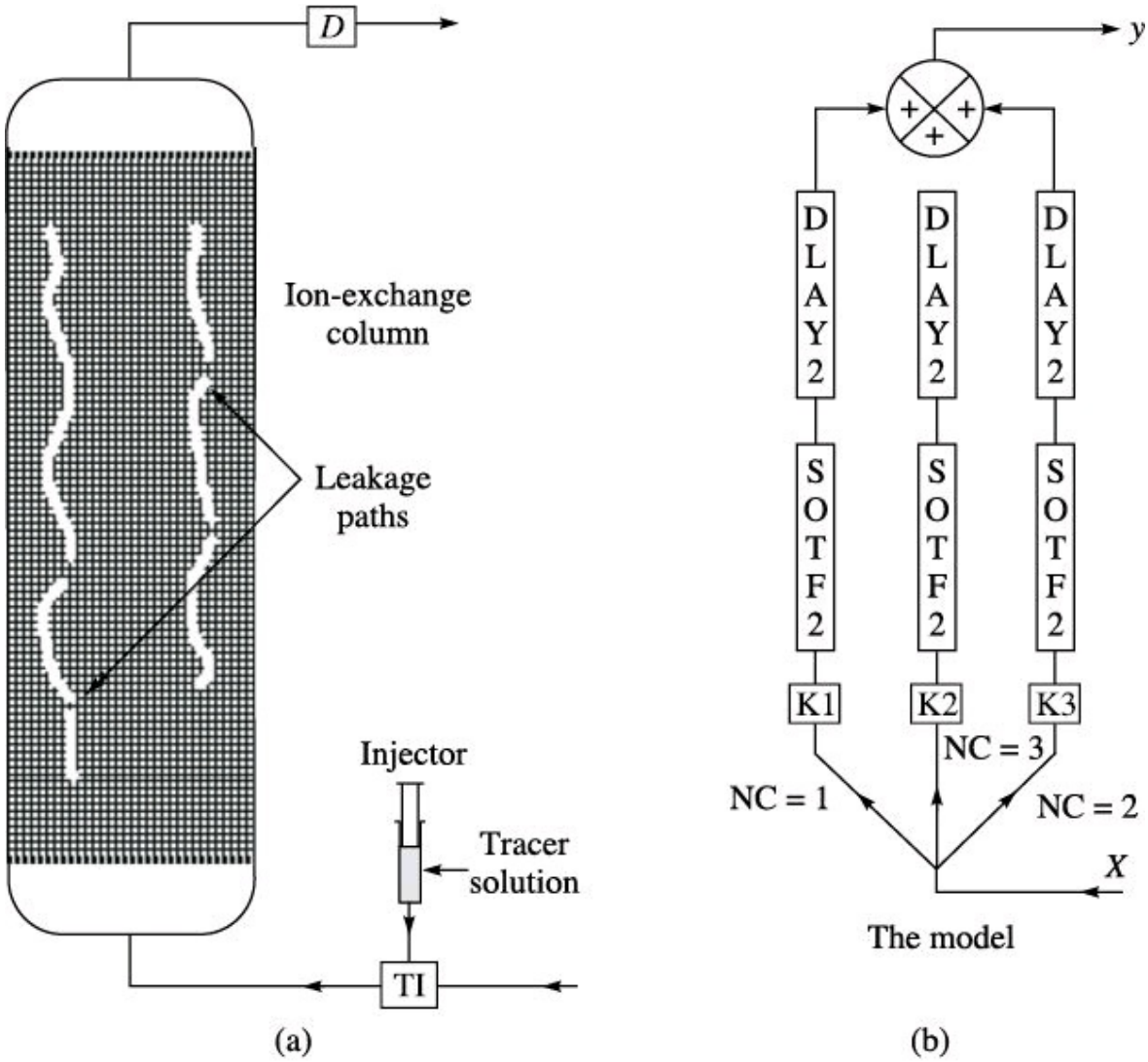


Figure 14.15 (a) The schematic and (b) the model of an ion-exchange column with defective packing.

Solution: The main program C14-15.BAS is prepared to simulate the process, which calls upon subroutine SOTFB2 and DLAY2 for NC = 1, 2 and 3 to act as the subscripts in the respective subroutines. The process parameters as given in the main program are so chosen that the two leakage channels and an adequately packed zone are represented. In the initialization section, adequate number of locations of ADEL2 array is filled up with outlet variable values of second order blocks. Carefully go through the main program, subroutines and their call sequence, which starts by mentioning the channel number.

```
10    REM ************* PROGRAM NAME -> C14-15.BAS *************
      DIM ADL2(4, 2900), NDL(4), KP(4), TP(4), DY1DT(4), DY2DT(4)
      DIM Y1O(4), Y1N(4), Y2O(4), Y2N(4)
      CLS
      SCREEN 12
      XS = 50: YS = 400: XF = 2: YF = 2
      LINE(XS, YS) – (XS + 400, YS – 120), , B
      REM ************* PARAMETER VALUES **********************
      K1 = .1: K2 = .1: K3 = .8: KP(1) = 1: KP(2) = 1: KP(3) = 1
      TP1(1) = 2.5: TP2(1) = 2.5: TP1(2) = 2: TP2(2) = 2
      TP1(3) = 15: TP2(3) = 20
```

```
      NDL(1) = 200: NDL(2) = 750: NDL(3) = 2900
      REM ***************** INITIAL VALUES *************************
      X1 = 0: Y1O(1) = 0: Y2O(1) = 0: Y1O(2) = 0: Y2O(2) =0: Y1O(3) = 0: Y2O(3) = 0
      FOR I = 1 TO NDL(1): ADL2(1, I) = Y2O(1): NEXT I
      FOR I = 1 TO NDL(2): ADL2(2, I) = Y2O(2): NEXT I
      FOR I = 1 TO NDL(3): ADL2(3, I) = Y2O(3): NEXT I
      REM ****************** THE TIME LOOP ************************
      T = 0: DT = .01: TF =200
50    IF T > TF THEN GOTO 70
      IF T > 1 THEN X1 = 2000
      IF T > 2 THEN X1 = 0
      NC = 1: X = K1 * X1: GOSUB 1540: XDL = Y2N(NC)
      GOSUB 1400: YDL1 = YDL
      NC = 2: X = K2 * X1: GOSUB 1540: XDL = Y2N(NC)
      GOSUB 1400: YDL2 = YDL
      NC = 3: X = K1 * X3: GOSUB 1540: XDL = Y2N(NC)
      GOSUB 1400: YDL3 = YDL
      Y = YDL1 + YDL2 + YDL3
      PSET (XS + T * XF, YS – Y * YF)
      FOR I = 1 TO 3:Y10(I) = Y1N(I):Y2O(I) = Y2N(I): NEXT I
      T = T + DT
      GOTO 50
70    GOTO 5000
1540 ************************* SUBROUTINE SOTFB2 ******************************
      DY1DT(NC) = X * (KP(NC) / TP1(NC)) – Y1O(NC) * (1 / TP1(NC))
      Y1N(NC) = Y1O(NC) + DY1DT(NC) * DT
      DY2DT(NC) = Y1O(NC) * (KP(NC) / TP2(NC)) – Y2O(NC) * (1 / TP2(NC))
      Y2N(NC) = Y2O(NC) + DY2DT(NC) * DT
      RETURN
      END
1400 REM ********************** SUBROUTINE DLAY2 ************************
5000 END
```

Result of the program is shown in Figure 14.16. In the early part of the resulting trace, two sharp peaks can be identified as the output of the leakage paths followed by the pulse response of the adequately packed portion of the bed. The short residence times offered in the leakage paths are the cause of reduction in bed performance.

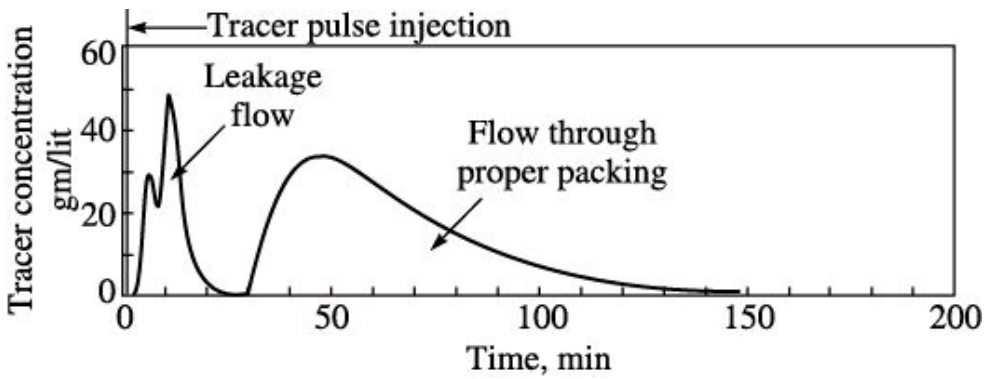


Figure 14.16 Simulation of diagnostic trace test of a defective packed bed.

Problem 14.9 By using program number C14-15.BAS, can you decide which of the

two leakage paths in the model produces the first peak?

Problem 14.10 By changing the parameters of program number C14-15.BAS, can you separate the two early peaks of the response?

EXAMPLE 14.15 Ideal stirring condition or perfect mixing in a stirred mixer is also called 'back-mixing'. This is a very common assumption in process analysis, and the conceptual foundation of 'lumping of parameters' in a blender or reactor is dependent on this assumption. However, in reality there exists no such mixer that may be called ideal or perfect in which the bulk fluid concentration is identical everywhere in the mixer volume, and exit stream concentration is always equal to that of the bulk fluid. In practical mixers, concentration gradient exists within the bulk fluid, and in a semi-batch type of blender or reactor, after adding a new slug into the stirred volume of liquid contained in the mixer, a finite time is always required to obtain homogeneity. Determining the 'mixing time' of a blending system is important in process applications to assure a definite product quality.

Non-ideal mixing models are important branches of chemical engineering studies that can predict fluctuations of concentration within the bulk of fluid and the time required to attain homogeneity after a disturbance. This area has attracted a large number of finest workers and over several decades of their effort has resulted in a wealth of literature on the subject. These models are categorized into several classes in which 'compartmented' models are an important class. Models of this class are constructed by combining blocks of FOTF, SOTFA and DLAY in different combinations of series and/or parallel network.

A particular configuration of this class has been considered in the next example. The streamlines existing in a stirred pot of liquid is modeled by 4 numbers of dead time elements representing the recirculation paths and a single first order element for the centrally located well-stirred zone around the moving impeller. The stirred vessel is provided with a tracer introduction tube. The laser LED as the source, S, and an optical detector, D, are placed outside of the transparent vessel at a level below the impeller, so that the laser beam goes unhindered through the stirred portion. These arrangements are shown in Figure 14.17(a).

The model structure is depicted in Figure 14.17(b). The model structure provides an inlet stream for tracer injection and an exit stream considering the system to be a constant holdup equipment. The tracer injection tube is physically removed after the introduction of the tracer solution pulse to eliminate any flow disturbance.

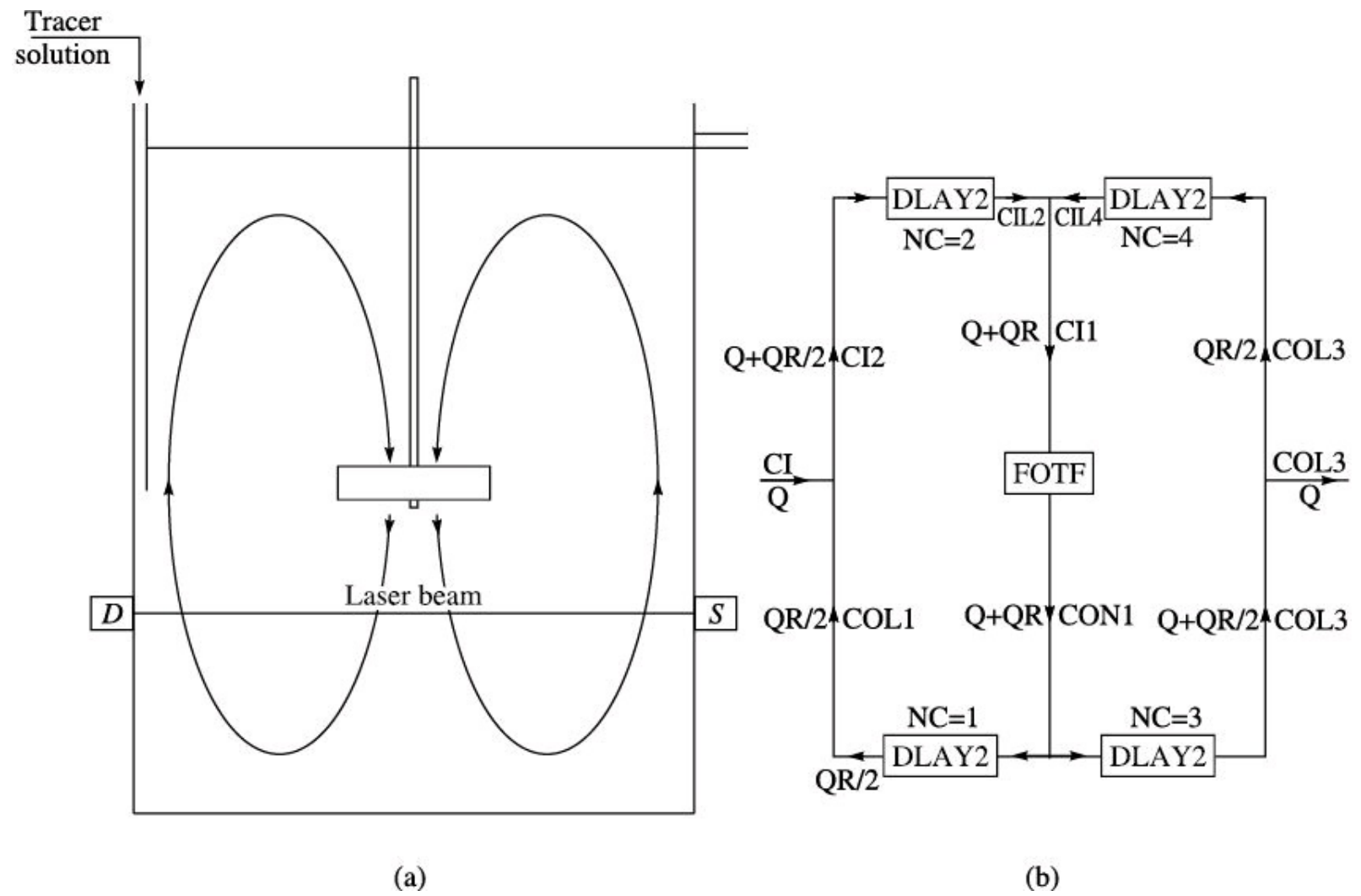


Figure 14.17 (a) The streamlines in a stirred vessel with tracer injection and detection system. (b) The structure of the compartmental model for the stirred vessel.

EXAMPLE 14.16 Write a program to simulate the mixing model for a stirred vessel that will produce the response to a pulse of tracer as a dynamic trace and it will be possible to determine the mixing or homogenization time from the dynamic trace. The following parameter values may be used for simulation.

Inlet and exit flow, Q, happens only during tracer entry, Q = 100 cm^3/s. Recirculation flow rate, QR = 2000 cm^3/s. Total mixer volume, V = 10000 cm^3. Volume of ideally mixed zone, $VM = 0.2.V$ cm^3. Total volume of the delay lines, $VDL = 0.8.V$ cm^3. The time constants are calculated neglecting the inlet flow which is applied only during the pulse injection time.

Time constant of the well mixed zone, $TP = VM/QR$ = 1 s. Time constant of each of four dead time elements, $TDL = (VDL/4)/QR$ = 2 s.

Solution: Using subroutines DLAY2 and FOTF, the following main program is prepared for simulating the mixing model:

```
10    EM ********** PROGRAM NAME -> C14-16.BAS **********
      DEFINT N
      CLS
```

```
      SCREEN 12
      XS = 50: YS = 400: XF = 4: YF = 10
      LINE (XS, YS) – (XS + 400, YS – 150), , B
      REM ***************** PARAMETER VALUES ***************************
      Q = 0: QR = 2000: V = 10000: VDL = .8 * V: VM = ,2 * V
      TDL(1) = 2: TDL(2) = 2: TDL(3) = 2: TDL(4) = 2: TP = 1
      NDL(1) = 200: NDL(2) = 200: NDL(3) = 200: NDL(4) = 200
      REM ****************** INITIAL VALUES ************************
      CI =0: CI2 = 0: CI1 = 0: CON1 = 0
      FOR I = 1 TO NDL(1): ADL2(1, I) = CON1: NEXT I
      FOR I = 1 TO NDL(2): ADL2(2, I) = CI2: NEXT I
      FOR I = 1 TO NDL(3): ADL2(3, I) = CON1: NEXT I
      FOR I = 1 TO NDL(4): ADL2(4, I) = COL3: NEXT I
      REM ******************* THE TIME LOOP ******************
      T = 0: DT = .01: TF = 100
50    IF T > TF THEN GOTO 70
      IF T > 1 THEN CI = 500
      IF T > 1 THEN Q = 100
      IF T > 2 THEN CI = 0
      IF T > 2 THEN Q = 0
      CI2 = (CI * Q + COL1 * .5 * QR) / (Q + .5 * QR)
      CI1 = (CIL2 * (Q + .5 * QR) + CIL4 * .5 * QR) / (Q + .5 * QR)
      NC = 1: XDL = CON1: GOSUB 1400: COL1 = YDL
      NC = 2: XDL = CI2: GOSUB 1400: CIL2 = YDL
      NC = 3: XDL = CON1: GOSUB 1400: COL3 = YDL
      NC = 4: XDL = COL3: GOSUB 1400: CIL4 = YDL
      X = CI1: GOSUB 1500: CON1 = YN
      PSET (XS + T * XF, YS – COL3 * YF)
      YO = YN
      T = T + DT
      GOTO 50
70    GOTO 5000
1400 REM ************ SUBROUTINE DLAY2 *****************
1500 REM ************ SUBROUTINE FOTF *****************
5000 END
```

The result of the above program is shown in Figure 14.18. The tracer slug after entering into liquid bulk, which is already in motion due to impeller action, gradually disperses as it moves in a circular motion along with the liquid. This dispersing dye solution repeatedly cuts the laser beam and ultimately the liquid bulk becomes homogeneous with respect to the tracer concentration. These are reasons for which an oscillation is obtained in the output and the initial amplitude of oscillation gradually diminishes into a steady horizontal trace. The time, from the instant of tracer introduction up to when the steady horizontal response is obtained, is the ‘mixing time’ of the mixing system.

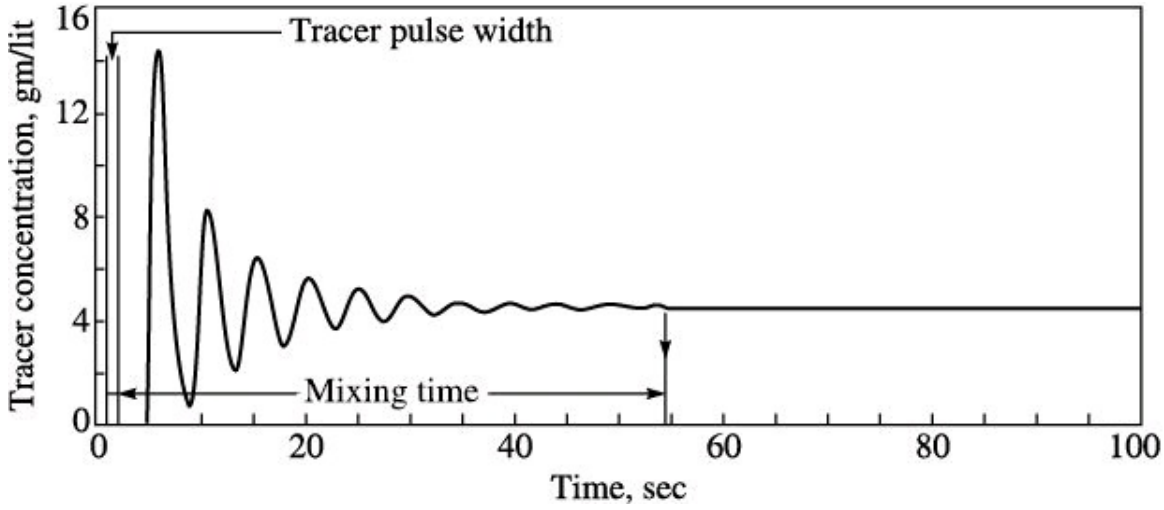


Figure 14.18 The tracer concentration dynamics in a mixing vessel.

Problem 14.11 Write a program to simulate a process with an inverse response. The block diagram of such a process is shown in Figure 14.19(a). Following parameter values may be assumed for the system: $KP1 = 1$, $t_1 = 10$ min, $KP2 = -0.25$, $t_2 = 2$ min. Draw the unit step response for the system by executing the program.

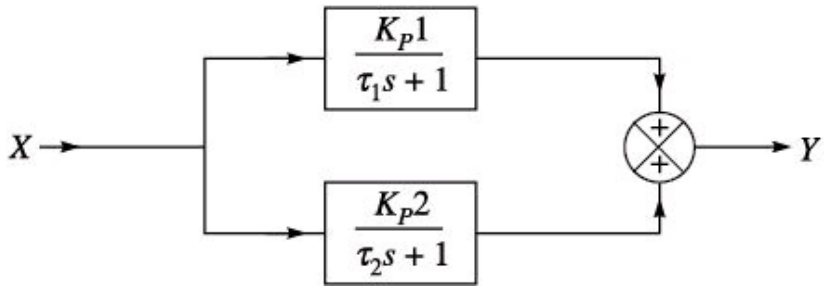


Figure 14.19 Block diagram of a system having inverse response.

With this amount of preliminary exposure of model development of simple systems and translating the models into computer programs, we should move on to the task of model building for more complex ones. There is practically no difference in the principle of model making that is required for the so called complex systems. The term complex arises from the existence of:

1. A large number of differential equations in the model;
2. Complex configuration of signal transfer paths, i.e. the input-output variable paths situated between these equations;
3. Appearance of integral equations, along with differential equations in the model;
4. Non-linear or time dependency of gain and dynamic parameters of the system.

As we did in Chapter 3, the following system examples are divided into three categories:

1. Lumped parameter systems,
2. Multiple lumped parameter stage systems, and
3. Distributed parameter systems.

See Chapter 3 for the fundamental considerations of model building and their classification.

14.14 Lumped Parameter Systems

The model building procedure could be explained by citing some representative examples.

14.14.1 Air-Water Vapour Equilibrium System—Psychometric Variables in Drying

In the analysis of drying systems it is often required to calculate certain air-water vapour equilibrium properties or psychometric variables. In a computer program that simulates the dynamic state, such properties may require to be calculated many times during execution (usually for all time values within the time loop). This can be done from a data set relating basic thermodynamic properties of water vapour against temperature and calculating any required property for a given temperature by using a suitable linear interpolation subroutine. This data set will be always used within its minimum and maximum values, hence no 'beyond-the-range' facility is required and we can use subroutine LNNT1A.

For implementing a simple model for drying system, only the vapour pressure of water against temperature may be sufficient. In literature, data set of several thermodynamic properties of water vapour at close variation of prevailing temperatures are available. A part of the data set, i.e. vapour pressure of water against temperature, is presented in the following table:

```
REM ********** (REF: PERY, 7TH ED. P12-8) ***********
REM TEMP F VP IN HG TEMP F VP IN HG
AT(1) = 30: AP(1) = 0.1661: AT(2) = 32: AP(2) = 0.1803
AT(3) = 34: AP(3) = 0.1955: AT(4) = 36: AP(4) = 0.2118
AT(5) = 38: AP(5) = 0.2292: AT(6) = 40: AP(6) = 0.2478
AT(7) = 42: AP(7) = 0.2677: AT(8) = 44: AP(8) = 0.2891
AT(9) = 46: AP(9) = 0.3120: AT(10) = 48: AP(10) = 0.3364
AT(11) = 50: AP(11) = 0.3626: AT(12) = 52: AP(12) = 0.3906
AT(13) = 54: AP(13) = 0.4203: AT(14) = 56: AP(14) = 0.4520
AT(15) = 58: AP(15) = 0.4858: AT(16) = 60: AP(16) = 0.5218
AT(17) = 62: AP(17) = 0.5601: AT(18) = 64: AP(18) = 0.6000
AT(19) = 66: AP(19) = 0.6422: AT(20) = 68: AP(20) = 0.6903
AT(21) = 70: AP(21) = 0.7392: AT(22) = 72: AP(22) = 0.7912
AT(23) = 74: AP(23) = 0.8462: AT(24) = 76: AP(24) = 0.9046
```

AT(25) = 78: AP(25) = 0.9666: AT(26) = 80: AP(26) = 1.0321
AT(27) = 82: AP(27) = 1.1016: AT(28) = 84: AP(28) = 1.1750
AT(29) = 86: AP(29) = 1.2527: AT(30) = 88: AP(30) = 1.3347
AT(31) = 90: AP(31) = 1.4215: AT(32) = 92: AP(32) = 1.5131
AT(33) = 94: AP(33) = 1.6097: AT(34) = 96: AP(34) = 1.7117
AT(35) = 98: AP(35) = 1.8192: AT(36) = 100: AP(36) = 1.9325
AT(37) = 102: AP(37) = 2.0519: AT(38) = 104: AP(38) = 2.1775
AT(39) = 106: AP(39) = 2.3099: AT(40) = 108: AP(40) = 2.4491
AT(41) = 110: AP(41) = 2.5955: AT(42) = 112: AP(42) = 2.7494
AT(43) = 114: AP(43) = 2.9111: AT(44) = 116: AP(44) = 3.0806
AT(45) = 118: AP(45) = 3.2589: AT(46) = 120: AP(46) = 3.4458
AT(47) = 122: AP(47) = 3.6420: AT(48) = 124: AP(48) = 3.8475
AT(49) = 126: AP(49) = 4.0629: AT(50) = 128: AP(50) = 4.2887
AT(51) = 130: AP(51) = 4.5251: AT(52) = 132: AP(52) = 4.7725
AT(53) = 134: AP(53) = 5.0314: AT(54) = 136: AP(54) = 5.3022
AT(55) = 138: AP(55) = 5.5852: AT(56) = 140: AP(56) = 5.8812
AT(57) = 142: AP(57) = 6.1903: AT(58) = 144: AP(58) = 6.5132
AT(59) = 146: AP(59) = 6.8500: AT(60) = 148: AP(60) = 7.2020
AT(61) = 150: AP(61) = 7.5690: AT(62) = 152: AP(62) = 7.9520
AT(63) = 154: AP(63) = 8.3510: AT(64) = 156: AP(64) = 8.7670
AT(65) = 158: AP(65) = 9.2000: AT(66) = 160: AP(66) = 9.6520
AT(67) = 162: AP(67) = 10.122: AT(68) = 164: AP(68) = 10.611
AT(69) = 166: AP(69) = 11.120: AT(70) = 168: AP(70) = 11.649
AT(71) = 170: AP(71) = 12.199: AT(72) = 172: AP(72) = 12.772
AT(73) = 174: AP(73) = 13.366: AT(74) = 176: AP(74) = 13.983
AT(75) = 178: AP(75) = 14.625: AT(76) = 180: AP(76) = 15.291
AT(77) = 182: AP(77) = 15.982: AT(78) = 184: AP(78) = 16.699
AT(79) = 186: AP(79) = 17.443: AT(80) = 188: AP(80) = 18.214
AT(81) = 190: AP(81) = 19.014: AT(82) = 192: AP(82) = 19.843
AT(83) = 194: AP(83) = 20.703: AT(84) = 196: AP(84) = 21.593
AT(85) = 198: AP(85) = 22.515: AT(86) = 200: AP(86) = 23.467
AT(87) = 202: AP(87) = 24.455: AT(88) = 204: AP(88) = 25.475
AT(89) = 206: AP(89) = 26.531: AT(90) = 208: AP(90) = 27.625
AT(91) = 210: AP(91) = 28.755: AT(92) = 212: AP(92) = 29.922
ND = 92

If we assume that air-water vapour mixture obeys ideal gas law, some of the basic psychometric variables can be calculated by the following procedures:

Procedure 1 If D_{BT} = Dry bulb temperature and D_{PT} = Dew point are known, to calculate AH and PRH. Let V_{P1} and V_{P2} be the respective values of vapour pressures of water found by interpolation corresponding to these two temperatures.

Mole fraction of water vapour in air = Molar humidity

$$MH = \frac{V_{P2}}{29.922 - V_{P2}}$$

Thus, Absolute humidity, $AH = MH \times \frac{18}{28.84}$.

As the saturation vapour pressure at D_{BT} is V_{P1}, the percent relative humidity, PRH will be,

$$PRH = \frac{V_{P1}}{V_{P2}} \times 100$$

Procedure 2 Let D_{BT} and AH are known, then PRH may be calculated as:

At D_{BT}, let V_{P1} be the vapour pressure of water found by interpolation. Corresponding to the given value of AH, the molar humidity is found as:

$$MH = AH \times \frac{28.84}{18}$$

If we define V_{P2} as the partial pressure of water vapour in the unsaturated mixture, then

$$V_{P2} = \frac{MH \times 29.922}{1 + MH}$$

Thus,

$$PRH = \frac{V_{P2}}{V_{P1}} \times 100$$

Procedure 3 If PRH and D_{BT} are known, then AH may be calculated as:

$$RH = \frac{PRH}{100}$$

Now V_{P1} is the vapour pressure of water at D_{BT}, then in the air-water vapour mixture, partial pressure of water vapour, $P_{PW} = V_{P1} \times RH$, and partial pressure of air,

$$P_{PA} = 29.922 - P_{PW}$$

Thus,

$$AH = \frac{P_{PW}}{P_{PA}} \times \frac{18}{28.84}$$

This procedure can be repeatedly done by implementing two loops, the outer loop changes PRH by steps of 10% and the inner or nested loop changes the D_{BT} by steps of say 1°C. The results may be graphically displayed to show the fundamental traces of the psychometric chart, as in Figure 14.20.

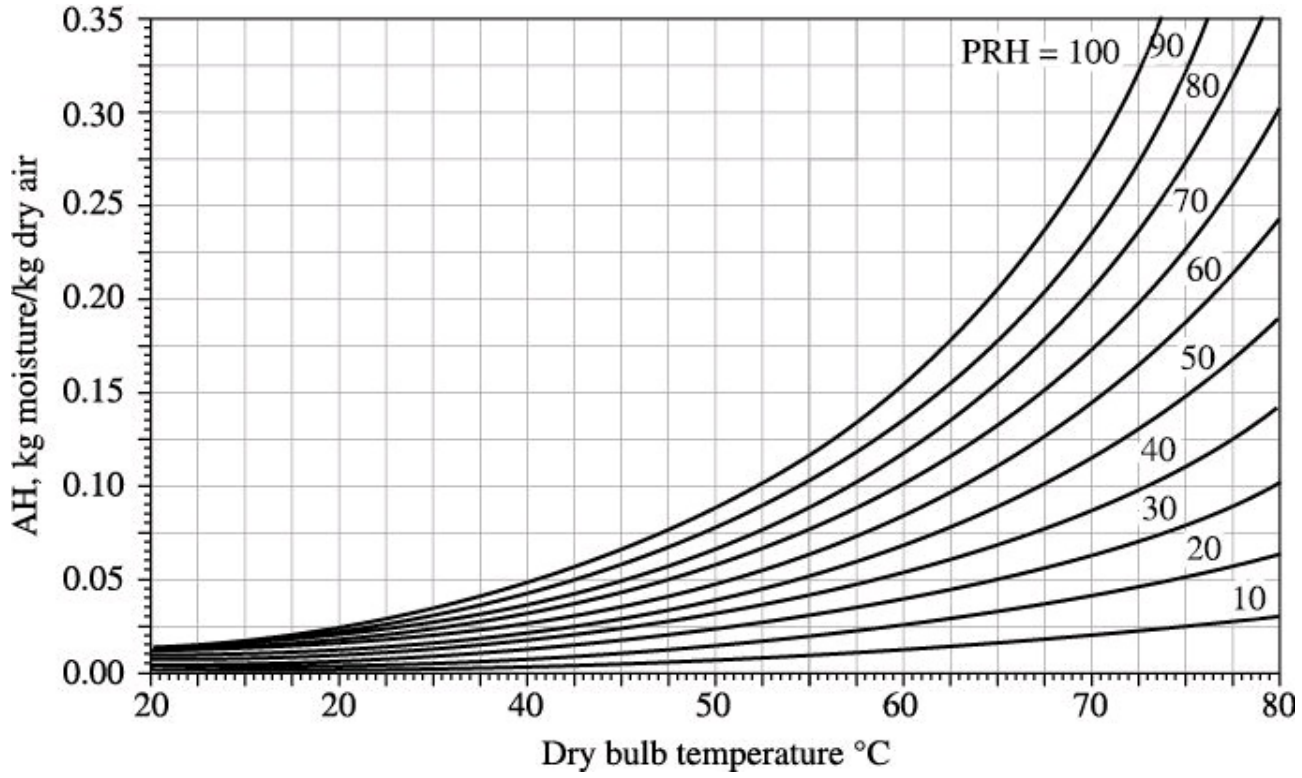


Figure 14.20 Drawing of a simplified version of psychometric chart.

EXAMPLE 14.17 (Simulation of a Drying System) The objective of the following effort is to simulate a drying system. Schematic of a batch type dryer is shown in Figure 14.21(a), in which a number of wet solid cakes kept on stainless steel meshes are subjected to the flowing air stream through the dryer chamber. The cakes are arranged in an orderly manner with similar gaps between them so that all the six sides of each cake are evenly exposed to the air stream.

The drying air is prepared by drawing atmospheric air through a filter by a blower, then processing the air through an air conditioning system. The conditioning system comprises of a refrigeration unit in the form of a finned-tube heat exchanger and a steam heater, which is also a finned tube heat exchanger. The refrigerated air cooler reduces air temperature below the dew point [Path from point A to point C through point B in Figure 14.21(b)], so that the leaving air is in fully saturated condition and its temperature is equal to the dew point or wet bulb temperature. In the heater, the air temperature is raised [Path from point C to point D in Figure 14.21(b)], so that exit temperature of the flowing air becomes its dry bulb temperature.

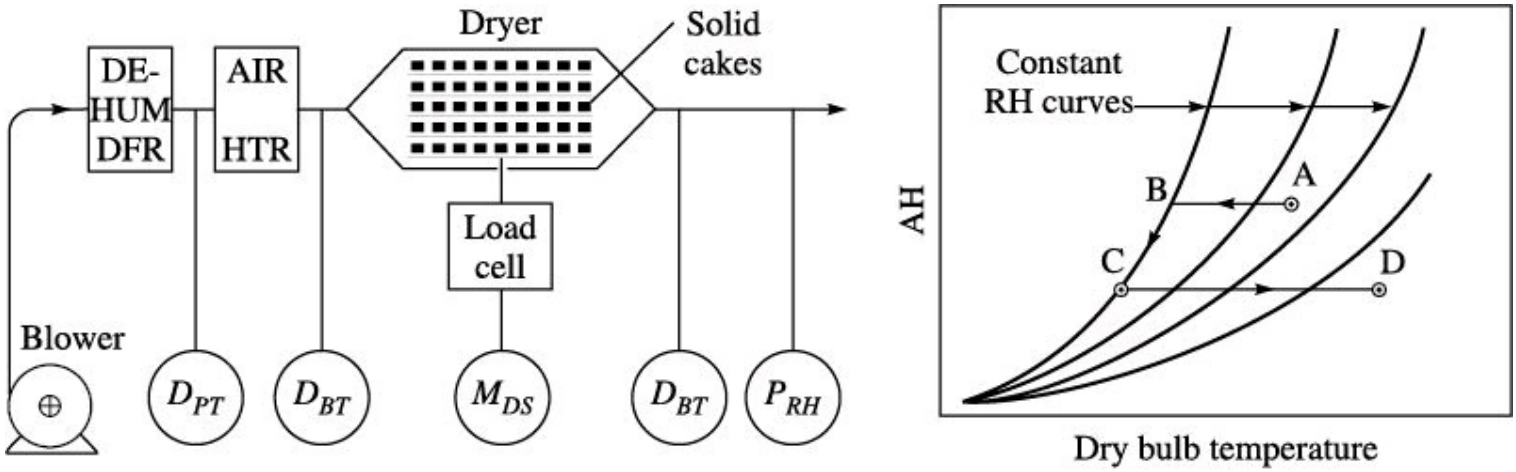


Figure 14.21 (a) Batch dryer. (b) The psychometric changes in the air conditioning system.

Thus, after going through this conditioning system, the air enters the dryer chamber at controlled temperature and humidity. The air entering into the dryer chamber is measured for its dry bulb temperature and its dew point is measured at the exit of the air cooler. Also, the total weight of the wet solid is measured by a load cell to continuously measure the extent of drying. The exit air from the dryer is measured for dry bulb temperature, D_{BT}, and percent relative humidity, P_{RH}.

The model for the drying system has been developed from the following considerations. Lumped parameter condition is assumed for the air inside the dryer chamber and the wet solid cakes. This enables us to assume homogeneous moisture concentration and temperature throughout the solid mass and dryer chamber.

It is also assumed in the enthalpy balance that enthalpy change of the air is appreciably larger to that of the solid cakes and hence, enthalpy of solid may be neglected. In this regard, Shinsky has commented that in practical applications, the minimum temperature the drying solid can assume is its dew point under the prevailing condition of the dryer, hence, enthalpy variation of the solid will be restricted and that too within the initial stages of drying.

Further, due to the small concentration range of water vapour in the air stream, all properties of pure air are assumed in the calculation. Lastly, volume and surface area of the solid cakes are assumed to remain constant during drying operation.

In the analysis, moisture concentration in the wet solid is denoted by X and that of air by C. The first suffix D or W on these letters signify whether the concentration is expressed on 'dry'or 'wet' basis. The second suffix I, C and F on X are used to denote *initial*, *critical* and *final* states of drying of the solid. The second suffix I and O on C refer to the *inlet* and *outlet* condition of air in the dryer chamber. It is easy to show that moisture concentration of any solid or air sample on dry and wet basis are related by the following equations:

$$X_{DI} = \frac{X_{WI}}{1 - X_{WI}} \qquad (14.22a)$$

$$C_{DO} = \frac{C_{WO}}{1 - C_{WO}} \qquad (14.22c)$$

$$X_{WI} = \frac{X_{DI}}{1 + X_{DI}} \qquad (14.22b)$$

$$C_{WO} = \frac{C_{DO}}{1 + C_{DO}} \qquad (14.22d)$$

The model equations comprise of:

1. Mass balance of moisture in solid;
2. Mass balance of moisture in air;
3. Heat balance on air for (a) constant rate, and (b) falling rate period of drying.

The following parameter values are required for construction of the model:

Dimension of the dryer chamber, A_D m (Length) $\times B_D$ m (Breadth) $\times C_D$ m (Height),

Volume of the dryer, $V = A_D B_D C_D$ m^3,

Air flow rate, Q_A m^3/min,

Air density, r_A kg/m^3,

Air heat capacity, Cp_A kcal/(kg/°C),

Latent heat vapourization of water, l kcal/kg,

Dimension of each solid cake, A_C m (Length) $\times B_C$ m (Breadth) $\times C_C$ m (Height),

Volume of each solid cake, $V_{SC} = A_C B_C C_C$ m^3,

External surface area of each solid cake, $s_{SA} = 2(A_C B_C + B_C C_C + A_C C_C)$m^2,

Dry solid density, r_S kg/m^3,

Mass of dry solid in each cake, $m_{DS} = V_{SC} r_S$ kg,

Number of cakes in the stack in the dryer chamber = A_{NC}.

External surface area of all solid cakes, $S_{SA} = A_{NC} s_{SA}$ m^2,

The drying rate constant during constant rate period, K_{DC} kg water/(m^2/min),

The drying rate constant during falling rate period, K_{DF} kg water/(m^2/min/Δc),

Mass of dry solid in all cakes, $M_{DS} = A_{NC} m_{DS}$ kg,

As M_{DS} remain unchanged during drying, this will be 'tie' material in our calculation. The following air properties and wet solid properties are required to implement the model:

Entering air dry bulb temperature, D_{BTI} °C,

Entering air dew point, D_{PTI}°C,

Initial moisture in solid in wet basis, X_{WI} kg water/kg wet solid,

Critical moisture in solid in wet basis, X_{WC} kg water/kg wet solid,

Final moisture in solid in wet basis, X_{WF} kg water/kg wet solid,

The following variable values will be calculated in the model during the simulation run:

Mass of wet solid in all the cakes, M_{WS} kg,

Percent relative humidity of exit air, R_{HO},

Dry bulb temperature of exit air, D_{BTO}°C,

As shown in Figure 14.21(a) D_{BTI}, D_{PTI}, M_{WS}, R_{HO} and D_{BTO} will be continuously measured in the dryer setup. Such an instrumentation scheme is sufficient for monitoring and controlling the drying process. Thus,

Initial mass of wet solid in all the cakes, $M_{SWCI} = M_{SDC}(1 + X_{DI})$ kg.

Variable mass of wet solid in all the cakes, $M_{SWC} = M_{SDC}(1 + X_D)$ kg.

Also the AH and PRH of the entering air is found from the measured D_{BT}, D_{PT} values by the method described as procedure 1. We may now formulate the model equations applicable during constant and falling rate periods:

During constant rate period mass balance on solid:

1. Rate of moisture going from a cake to air, $D_{WSA} = K_{DC}.\ S_{SA}$,
2. Amount of moisture going from all cakes to air in time DT = $K_{DC}S_{SA}$DT, = D_{WSA}.DT

Thus, moisture balance on solid:

1. $M_{SDCN} = M_{SDCO} - D_{WSA}$DT (14.23)

The moisture balance on air: As same amount of moisture goes into air that has come from the solid cakes,

2. Amount of moisture coming into air from all cakes in time DT = $K_{DC}.\ S_{SA}$.DT,

Thus, $$Q_A\rho_A C_{WI} + D_{MSDC} - Q_A\rho_A C_{WO} = V_D\rho_A \frac{dC_{WO}}{dt}$$

or, $$\frac{dC_{WO}}{dt} = \frac{Q_A\rho_A C_{WI} + D_{MSDC} - Q_A\rho_A C_{WO}}{V_D\rho_A} \quad (14.24)$$

The enthalpy balance on air:

1. The entering air temperature, T_I = Dry bulb temperature of entering air, D_{BTI},
2. Rate of heat loss, $D_{HDT} = lD_{WSA}$,
3. Heat removal for water evaporation in time DT, D_{HDT}DT = lD_{WSA}DT

$$Q_A\rho_A C_{p_A} T_I - D_{HDT} - Q_A\rho_A C_{p_A} T_O = V_D\rho_A C_{p_A} \frac{dT_O}{dt}$$

or, $$\frac{dT_O}{dt} = \frac{(Q_A\rho_A C_{p_A} T_I - D_{HDT} - Q_A\rho_A C_{p_A} T_O)}{V_D\rho_A C_{p_A}} \quad (14.25)$$

During falling rate period:

Mass balance on solid:

1. Rate of moisture going from a cake to air, $D_{WSA} = K_{DF}S_{SA}(C_{DO} - X_{DO})$,
2. Amount of moisture going from all cakes to air in time DT

$$= K_{DF}S_{SA}(C_{DO} - X_{DO})\text{DT} = D_{WSA}\text{DT}$$

Thus, moisture balance on solid:

1. $M_{SDCN} = M_{SDCO} - D_{WSA}$DT (14.26)

The moisture balance on air: As same amount of moisture goes into air that has come from the solid cakes,

2. Amount of moisture coming into air from all cakes in time DT = $K_{DC}S_{SA}$DT,

Thus,

$$Q_A\rho_A C_{WI} + D_{MSDC} - Q_A\rho_A C_{WO} = V_D\rho_A \frac{dC_{WO}}{dt}$$

or,

$$\frac{dC_{WO}}{dt} = \frac{Q_A\rho_A C_{WI} + D_{MSDC} - Q_A\rho_A C_{WO}}{V_D\rho_A} \quad (14.27)$$

The enthalpy balance on air:

1. The entering air temperature, T_I = Dry bulb temperature of entering air, D_{BTI},
2. Rate of heat loss, $D_{HDT} = lD_{WSA}$,
3. Heat removal for water evaporation in time DT, D_{HDT}DT = lD_{WSA}DT

$$Q_A\rho_A C_{p_A} T_I - D_{HDT} - Q_A\rho_A C_{p_A} T_O = V_D\rho_A C_{p_A}\frac{dT_O}{dt}$$

or

$$\frac{dT_O}{dt} = \frac{Q_A\rho_A C_{p_A} T_I - D_{HDT} - Q_A\rho_A C_{p_A} T_O}{V_D\rho_A C_{p_A}} \qquad (14.28)$$

In the time cycle, during the constant rate period, Equations14.23, 14.24, 14.25 are integrated, and while in the falling rate period, Equations 14.26, 14.27, 14.28 are integrated. Irrespective of these drying conditions, just before the time increment statement, exit air P_{RH} is found from its dry bulb temperature D_{BTI} and C_{WO} by the procedure 2.

The following program has been written for simulation of the batch dryer:

```
10    REM ***************** PROGRAM NAME -> C14-17.BAS ********************
      DIM AX(200), AY(200)
      CLS: SCREEN 12
      XS1 = 50: YS1 = 450: XS2 = 50: YS2 = 300: XS3 = 50: YS3 = 150
      LINE (XS1, YS1) – (XS1 + 500), YS1 – 120), , B
      LINE (XS2, YS2) – (XS2 + 250), YS2 – 120), , B
      LINE (XS3, YS3) – (XS3 + 250), YS3 – 120), , B
      REM ********* TEMP VS H2O VP DATA (REF: PERRY) **************
      ND = 92
      REM **** PARAMETER VALUES & INITIAL CONDITIONS ****
      AD = 2: BD = .6: CD = .6: VD = .72: ROA = .9: CPA = .2402
      AC = .015: BC = .01: CC = .005: VSC = .00000075: ANC = 3600
      ssaa = .00055: SSA = 1.98: ROS = 5600: LMD = 540
      mdss = .0042: MDS = 15.12
      QA = 1.35: DBTI = 60: DBT = DBTI: DPTI = 20: DPT = DPTI
      X = 32 + 1.8 * DBT: GOSUB 1200: VPW1 = Y
      X = 32 + 1.8 * DPT: GOSUB 1200: VPW2 = Y
      MH = VPW2 / (29.922 – VPW2): AH = MH * (18 / 28.84)
      RH = VPW2 / VPW1: PRH = RH * 100
      AHI = AH: CDI = AHI: CWI = CDI / (1 + CDI)
      RHI = RH: PRHI = PRH: CWOO = CWI: TOO = DBTI
      XWI = .25: XWC = .15: XWF = .02: KDC = .02: KDF = .15
      XDI = XWI / (1 – XWI): XDC = XWC / (1 – XWC): XDF = XWF / (1 – XWF)
      REM ** INITIAL MASS OF WET SOLID IN 3600 CAKES = MWSI
      MWSI = MDS * (1 + XDI)
      REM ** INITIAL MASS OF MOISTURE IN 3600 CAKES = MMI
```

```
    MMI = MDS * XDI
    XWO = XWI: XDO = XDI: MWSO = MWSI: MMO = MMI
    XWN = XWI: XDN = XDI: MWSN = MWSI: MMN = MMI
    REM ********** TIME LOOP PARAMETERS ***************
    T = 0: DT = .1: TF = 250
    REM ************** THE TIME LOOP ********************
50  IF T > TF THEN GOTO 80
    IF XWN < XWF THEN GOTO 90
    IF XWN < XWC THEN GOTO 60
    REM ******** DURING THE CONST RATE PERIOD ********
    REM ************** Moisture balance on cakes ***************
    DWSA = KDC * SSA: MMN = MMO – DWSA * DT: XDN = XDO – ((DWSA * DT) / MDS)
    XWO = XDO / (1 + XDO): XWN = XDN / (1 + XDN): MWSN = MWSO – DWSA * DT
    REM ***************** Moisture balance on air *****************
    DCWODT = (QA * ROA * CWI + DWSA – QA * ROA * CWOO) / (VD * ROA)
    CWON = CWOO + DCWODT * DT: CDON = CWON / (1 – CWON)
    REM ***************** Enthalpy balance on air *****************
    DHDT = LMD * DWSA: TI = DBTI
    DTODT = (QA * ROA * CPA * TI – DHDT – QA * ROA * CPA * TOO) / (VD * ROA * CPA)
    TON = TOO + DTODT * DT: DBTO = TON
    GOTO 70
60  REM ******** DURING THE FALLING RATE PERIOD ********
    REM ************** Moisture balance on cakes ***************
    DWSA = KDF * SSA * (XWO – CWOO): MMN = MMO – DWSA * DT
    XDN = XDO – ((DWSA * DT) / MDS)
    XWO = XDO / (1 + XDO): XWN = XDN / (1 + XDN): MWSN = MWSO – DWSA * DT
    REM ***************** Moisture balance on air *****************
    DCWODT = (QA * ROA * CWI + DWSA – QA * ROA * CWOO) / (VD * ROA)
    CWON = CWOO + DCWODT * DT: CDON = CWON / (1 – CWON)
    REM ***************** Enthalpy balance on air *****************
    DHDT = LMD * DWSA: TI = DBTI
    DTODT = (QA * ROA * CPA * TI – DHDT – QA * ROA * CPA * TOO) / (VD * ROA * CPA)
    TON = TOO + DTODT * DT: DBTO = TON
70  REM ********* DETERMINATION OF PRHO ***********
    AHO = CDON
    X = 32 + 1.8 * DBTO: GOSUB 1200: VPW = Y
    MH = AHO * (28.84 / 18): PPW = (MH * 29.922) / (1 + MH)
    PRHO = (PPW / VPW) * 100
    REM ***************** OUTPUT SECTION ******************
    PSET (XS3 + T, YS3 + 500 – TON * 10)
    PSET (XS3 + T, YS3 + 500 – TI * 10)
    PSET (XS2 + T, YS2 + 300 – MWSN * 200)
    PSET (XS2 + T, YS2 – PRHO * 5)
    PSET (XS1 + XWN * 2000, YS1 – DWSA * 20000)
    REM ************ OLD <– NEW TRANSFER ***************
    MMO =MMN: XDO = XDN: XWO = XWN: MWSO = MWSN: CWOO = CWON: TOO = TON
    REM ********************************************************
    T = T + DT
    GOTO 50
80  PRINT "INSUFFICIENT DRYING TIME"
90  LINE (XS1 + XWC * 2000, YS1) – (XS1 + XWC * 2000, YS1 – 90)
    LINE (XS1 + XWF * 2000, YS1) – (XS1 + XWF * 2000, YS1 – 60)
```

```
      GOTO 5000
1200 REM ********** SUBROUTINE LNNT1A ******************
      RETURN
      END
5000 END
```

Program results are shown in Figures 14.22(a), (b), and (c). The figure in part (c) is the drying characteristic curve that we have seen in unit operation texts. The horizontal part signifies the constant rate period of drying. This horizontal part is also evidenced in the TO and RHO plots in part (a) and (b) of Figure 14.22 in constant rate period, because the moisture transfer rate remains constant in that range of moisture variation.

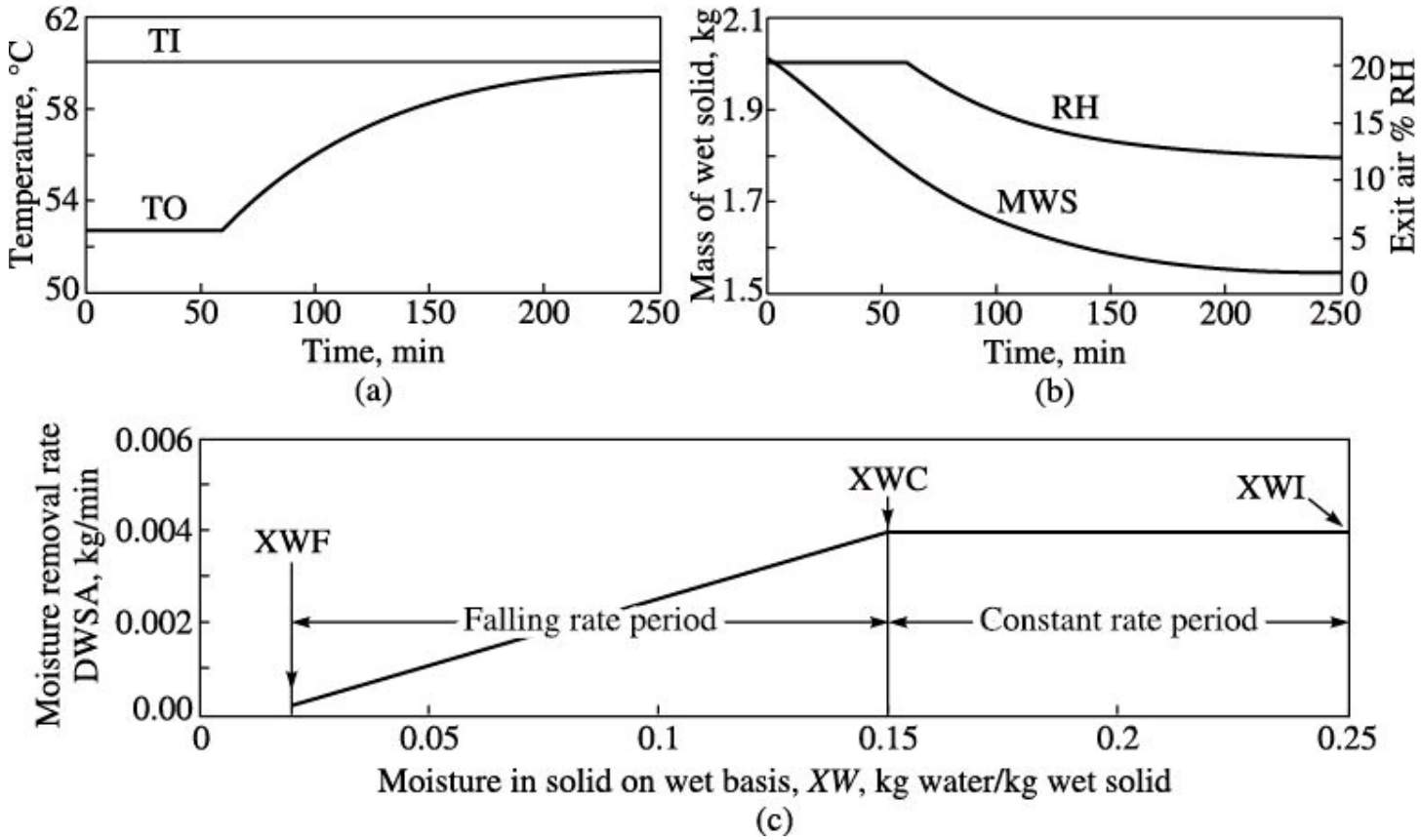


Figure 14.22 Response of the model during a batch run of drying.

In part (b) of Figure 14.22, the exit PRH starts at value around 20 and comes down to 12. One can get the idea that a larger portion of drying potential of the conditioned air (an expensive commodity) remains unutilized.

Problem 14.12 Setup a model for the dryer with a recirculation path for the air around the dryer to maximize the utilization of the drying potential of the conditioned air. Use all the pertinent data used in PROGRAM NAME -> C14.17.BAS.

There exists a class of systems in which one or more integral equation has to be solved along with a set of differential equations. This class of systems features what is called Integro-differential model. As in a bio-chemical fermentor, the kinetic model will consist of differential equations that describe the concentration dynamics of the

biochemical species and the equations that express the rate of biogas generation. The gas generation has to be integrated to determine the cumulative amount of gas produced at a given time instant, and the rest of the differential equations have to be treated in the conventional manner as we have done in so many previous examples.

14.14.2 Bio-Chemical Process

Among many categories we discuss the following example having ample application potential.

EXAMPLE 14.18 (A Batch Fermentor for Bio-Gas Production) An experimental batch type anaerobic fermentor is used for biogas production with solid city waste material as raw feed. A five stage kinetic model for the overall process is available in the literature together with the parameter values. These are presented as follows:

$$\frac{dCs_O}{dt} = -\beta Cx_{1O} Cs_O \tag{14.29}$$

$$\frac{dCx_1}{dt} = (\mu_1 - k_1)Cx_1 \tag{14.30}$$

$$\frac{dCs_1}{dt} = \beta Cx_1 Cs_O - \frac{\mu_1 Cx_1}{y_1} \tag{14.31}$$

$$\frac{dCx_2}{dt} = (\mu_2 - k_2)Cx_2 \tag{14.32}$$

$$\frac{dCs_2}{dt} = y_B \mu_1 Cx_1 - \frac{\mu_2 Cx_2}{y_2} \tag{14.33}$$

The specific growth rates are found by the following algebraic relations:

$$\mu_1 = \frac{\mu_{1MAX} Cs_1}{Ks_1 + Cs_1} \tag{14.34}$$

$$\mu_2 = \frac{\mu_{2MAX} Cs_2}{Ks_2 + Cs_2 + 1 + (Cs_2/K_I)} \tag{14.35}$$

$$Q(t) = y_G \mu_2 Cx_2 \tag{14.36}$$

The last equation gives the gas evolution rate. The following parameter values are used in the simulation:

$$\mu_{1\text{MAX}} = 0.4;\ \mu_{2\text{MAX}} = 0.4;\ Ks_1 = 160;\ Ks_2 = 0.82;\ k_1 = 0.004;\ k_2 = 0.004$$

$$y_1 = 0.0264;\ y_2 = 0.0242;\ \beta = 0.5;\ y_B = 45.51;\ y_G = 74.54;\ K_I = 3432$$

A computer program C14-18.BAS is written with graphic display facility. To calculate the amount of the reacting species present at any given time, the model equations are treated as we have done before, except the gas generation Equation

14.36, which is solved to obtain generated gas in each DT and added with its previous value. The program listing is shown below. The gas generation data against time, from an experimental biogas generator, is given in the data set at the beginning of the program. The simulated gas generation will be compared with the experimental data.

```
10    REM ********************* PROGRAM NAME -> C14-18.BAS ****************************
      REM *** THIS PROGRAM DEMONSTRATES THE OPENLOOP DYNAMICS OF AN ***
      REM ****************************** ANAEROBIC DIGESTOR
      *********************************
      DIM AX(50), AY(50)
      CLS : SCREEN 12
      AX(1) = 0: AY(1) = 0: AX(2) = 12: AY(2) = 6.23
      AX(3) = 24: AY(3) = 14.7: AX(4) = 36: AY(4) = 22.27
      AX(5) = 48: AY(5) = 26.7: AX(6) = 60: AY(6) = 38.57
      AX(7) = 72: AY(7) = 40.77: AX(8) = 84: AY(8) = 43.3
      AX(9) = 96: AY(9) = 46.8: AX(10) = 108: AY(10) = 48.3
      AX(11) = 120: AY(11) = 49.27: AX(12) = 132: AY(12) = 50.97
      AX(13) = 144: AY(13) = 51.5: AX(14) = 156: AY(14) = 51.93
      AX(15) = 168: AY(15) = 52: AX(16) = 180: AY(16) = 52.38
      AX(17) = 192: AY(17) = 52.4: AX(18) = 200: AY(18) = 52.42
      ND = 18
      XS = 50: YS = 400: DX = 500: DY = 300
      LINE (XS, YS) – (XS + DX, YS – DY), , B
      YF1 = 20: YF2 = 3
      CSOO = 14.24: CX1O = .1: CS1O = 7: CX2O = .01: CS2O = 3
      MU1MX = .4: KS1 = 160: Y1 = .0264: K1 = .004
      MU2MX = .4: KS2 = .82: Y2 = .0242: K2 = .004
      BETA = .5: YB = 45.51: YG = 74.54: KI = 3432
      QT = 0
      T = 0: DT = .1: TF = 250
      REM ***************** THE TIME LOOP ********************
50    IF T > TFIN THEN GOTO 70
      DCSODT = (-1) * BETA * CX1O * CSOO
      DCX1DT = (MU1 - K1) * CX1O
      DCS1DT = BETA * CX1O * CSOO - (MU1 * CX1O) / Y1
      DCX2DT = (MU2 - K2) * CX2O
      DCS2DT = YB * MU1 * CX1O - (MU2 * CX2O) / Y2
      CSON = CSOO + DCSODT * DT
      CX1N = CX1O + DCX1DT * DT
      CS1N = CS1O + DCS1DT * DT
      CX2N = CX2O + DCX2DT * DT
      CS2N = CS2O + DCS2DT * DT
      MU1 = (MU1MX * CS1N) / (KS1 + CS1N)
      MU2 = (MU2MX * CS2N) / ((KS2 + CS2N) + (1 + (CS2N / KI)))
      Q = YG * MU2 * CX2N: QT = QT + Q * DT
      REM ***************** GRAPHIC OUTPUT ************************
      PSET (XS + T * 2, YS - CSON * YF1)
      PSET (XS + T * 2, YS - CX1N * YF1)
      PSET (XS + T * 2, YS - CS1N * YF1)
      PSET (XS + T * 2, YS - CX2N * YF1)
      PSET (XS + T * 2, YS - CS2N * YF1)
```

```
      PSET (XS + T * 2, YS - QT * YF2)
      REM *************** OLD <- NEW TRANSFER *****************
      CSOO = CSON: CX1O = CX1N: CS1O = CS1N: CX2O = CX2N: CS2O = CS2N
      T = T + DT
      GOTO 50
70    FOR I = 1 TO ND
      CIRCLE (XS + AX(I) * 2, YS - AY(I) * YF2), 2
      NEXT I
      CIRCLE (215, 150), 2: LOCATE 10, 29: PRINT "Experimental data"
      LINE (204, 142) – (365, 159), , B
5000 END
```

Experimental gas generation data is plotted on the gas generation locus obtained from the model. Results of the analysis are shown in Figure 14.23. The initial increase followed by the dip in the concentration dynamics of Cs_2 confirms the notion that biochemical reactions retain some of the features of an 'auto-catalytic' reaction.

14.15 Systems Featuring Lumped Parameter Units in Series

We shall treat similar examples under this heading as we did in Chapter 3.

14.15.1 CSTR Battery

The first example of this system category features three CSTRs operating in series. The first order reaction, which is carried out in this reactor system, has rate constant that is dependent on reactor temperature but the heat of reaction is negligible. To add a little twist it has been assumed that reactors are kept at different temperatures.

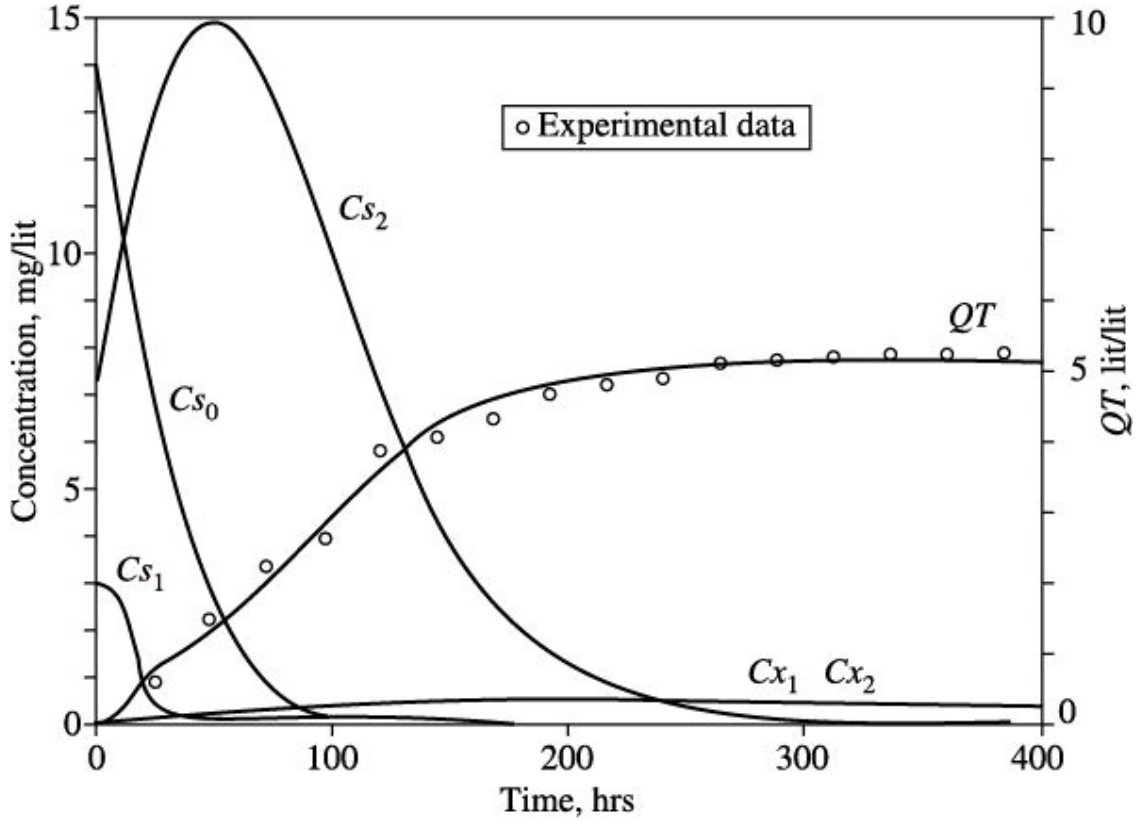


Figure 14.23 Open loop dynamics of an anaerobic fermentor.

EXAMPLE 14.19 Let us consider a species A that undergoes a first order irreversible reaction in a three tank CSTR battery. The rate constant k varies with reaction temperature T°C according to:

$$k_T = 0.01 + 0.001 \times T°C$$

Such a linear description of temperature dependency of reaction rate may be valid for a small range of temperature, say 10 – 80°C. The reaction has a negligible ΔH_R. The reactor volumes are: V_1 = 15 l, V_2 = 20 l, V_3 = 25 l. Reactant A, in the form of a solution made with an inert solvent, is fed to the first reactor at a constant flow rate, Q = 0.25 l/min, with the inlet concentration C_I = 1.0 mole/l. The three reactors are kept at constant temperatures of 20, 40 and 60°C. C_{A1}, C_{A2}, and C_{A3} are the exit concentrations from the three reactors. Write a program for simulating the reactors and obtain response of the exit concentrations from them if C_I changes to 1.2 mole/l according to a step.

Solution: The model for the reactor battery may be obtained from the reacting component balance around them:

For the first reactor: $\frac{dC_{A1}}{dt} = \frac{q}{V_1}(C_{AI} - C_{A1}) - k_{T1}C_{A1}$

(14.37)

For the second reactor: $\frac{dC_{A2}}{dt} = \frac{Q}{V_2}(C_{A1} - C_{A2}) - k_{T2}C_{A2}$

(14.38)

For the 3rd reactor: $\frac{dC_{A3}}{dt} = \frac{q}{V_3}(C_{A2} - C_{A3}) - k_{T3}C_{A3}$

(14.39)

A program C14-19.BAS has been written to represent the system with the following parameter values.

Q = 0.25 l/min; V_1 = 10 l; V_2 = 15 l; V_3 = 20 l;

k_{T1} = 0.012 min^{-1}, k_{T2} = 0.014 min^{-1}, k_{T3} = 0.016 min^{-1}.

To find the initial values of exit concentration from the three reactors, we run the program with zero initial inlet and exit concentration values, make an inlet concentration change to 1.0 according to:

$C_{A1} = C_{A1}$, $C_{A2} = C_{A2} = 0$ for $t < 0$, and $C_{A1} = 1.0$ for $t > 0$

The program is run for sufficient time until a steady state is reached and exit

concentrations from the reactors are printed. Following steady state values in g/l are obtained:

$$C_{A1} = 0.6755952,\ C_{A2} = 0.3671227,\ C_{A3} = 0.1609926$$

For the required dynamic run, these initial values are used.

The PROGRAM NAME C14-19.BAS, written for the dynamic response of the reactor system, is given below:

```
10    REM ************** PROGRAM NAME -> C14-19.BAS ***************
      REM ****** DYNAMIC RESPONSE OF THREE CSTR HANDLING *******
      REM ***** 1ST ORD. REACN. THE REACTORS ARE KEPT UNDER *****
      REM ************ DIFFERENT TEMPERATURE CONDITIONS ************
      SCREEN 12
      XS = 50: YS = 350: XF = .5: YF = 100
      LINE (XS, YS) – (XS + 250, YS – 140), , B
      REM ************* PARAMETER VALUES **************
      Q = .25: V1 = 10: V2 = 15: V3 = 20
      KT1 = .012: KT2 = .014: KT3 = .016
      REM ************** INITIAL CONDITIONS **************
      CAI = 1
      CA1O = .6755952: CA2O = .3671227: CA3O = .16709926
      T = 0: DT = .01: TF = 500
      REM ***************** THE TIME LOOP *****************
50    IF T > TF THEN GOTO 70
      IF T > 10 THEN CAI = 1.2
      DCA1DT = (Q / V1) * (CAI – CA1O) – KT1 * CA1O
      DCA2DT = (Q / V2) * (CA1O – CA2O) – KT2 * CA2O
      DCA3DT = (Q / V3) * (CA2O – CA3O) – KT3 * CA3O
      CA1N = CA1O + DCA1DT * DT
      CA2N = CA2O + DCA2DT * DT
      CA3N = CA3O + DCA3DT * DT
      PSET (XS + T * XF, YS – CAI * YF)
      PSET (XS + T * XF, YS – CA1N * YF)
      PSET (XS + T * XF, YS – CA2N * YF)
      PSET (XS + T * XF, YS – CA3N * YF)
      CA1O = CA1N: CA2O = CA2N: CA3O = CA3N
      T = T + DT
      GOTO 50
70    GOTO 5000
5000 END
```

The result after execution of the program is shown in Figure 14.24. We have not labeled the responses from first, second and third reactor because the obvious shapes and gradually increasing lags in the responses speak for them.

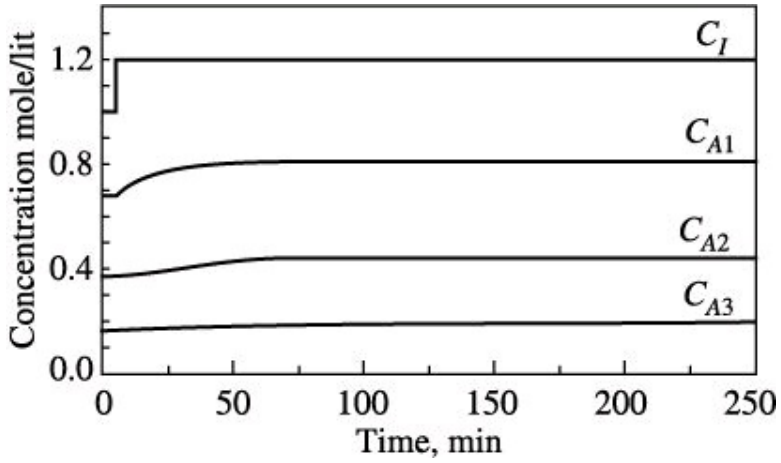


Figure 14.24 3-CSTR dynamics for an inlet concentration step change.

Problem 14.13 In many reaction engineering texts, steady state performance of serially operating CSTRs have been graphically expressed as in Figure 14.25. The lines emerging from origin are rate of reaction against concentration in a particular stage. In Example 14.19, as the rates are different as the operating temperatures are dissimilar, there are three such lines with slope proportional to rate constant k_T. The lines with –ve slope start from the x-axis at inlet concentration for each of the tanks, the slope being equal to $-(1/t_K)$, where t_K is the residence time of the k-th tank. By running PROGRAM NAME -> C14-19.BAS, the following steady state concentration values are obtained:

$$C_{A1} = 1.2,\ C_{A1} = 0.8107300,\ C_{A2} = 0.4405657,\ C_{A3} = 0.1932044$$

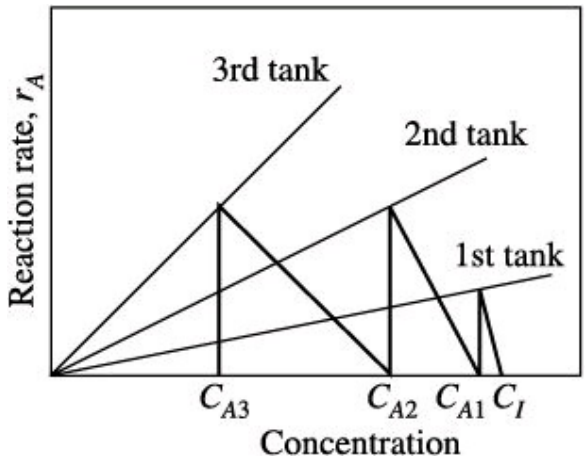


Figure 14.25 Steady state performance of 3-CSTRs in series.

From these data and model parameters, check whether the graphical representation of steady state characteristic tallies with the program results.

14.15.2 Plate Type Gas Absorber

A simple model of plate type absorber may be constructed by assuming that the gas and liquid on a plate are perfectly mixed and hence, may be represented by a lumped parameter stage. Thus, the lumped parameter stages in such an absorber are serially

connected by gas and liquid streams going in a counter current way to each other.

EXAMPLE 14.20 The model for an Absorber, Extraction systems, Rectifying and Stripping section of a distillation column are fundamentally similar. Two immiscible phases containing some transferable component are brought into intimate contact, usually by dispersing one of the phases in the form of small droplets or bubbles. The concentration difference of the transferable component reduces and moves towards equilibrium in the liquid held up over a plate. The reduction of concentration difference depends on the residence time and mixing efficiency available in the hold up volume over a plate. In Chapter 3 we had presented the fundamental equations of a simple model for a plate type absorber containing ideal plates. In the following section, certain aspects of the model are discussed that are required for a simulation effort. We start from the equilibrium relation.

The equilibrium relation may be expressed as:

$$y_N = mx_N \tag{14.40}$$

where, m is a constant, and y and x are mole fractions of the transferable component in gas and liquid phase respectively. The suffix N refers to the plate number. To further simplify the modeling effort it has been assumed that:

1. molal flow changes of gas G and liquid L streams are negligible through the column;
2. the molar liquid hold up H on each plate are constant. Gas and liquid streams contacting on the I-th plate may be visualized as in Figure 14.26.

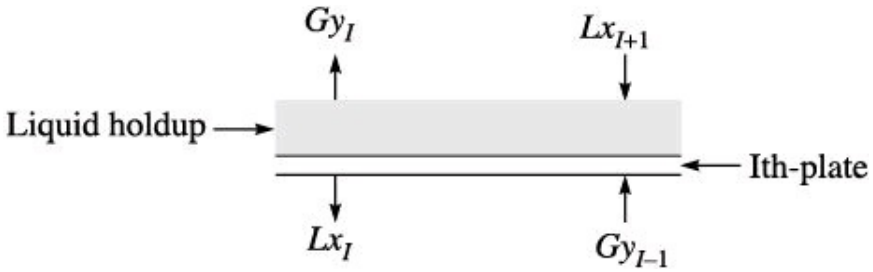


Figure 14.26 The liquid and gas phases contacting on the I-th plate.

The dynamic component balance for I-th plate of the column may be written from Figure 14.26 as:

$$H\frac{dx_I}{dt} = Gy_{I-1} - Gy_I + Lx_{I+1} - Lx_I \tag{14.41}$$

Now we introduce the Plate Efficiency (EP) to find an independent relation between entering-leaving stream compositions. For 100% efficient plates (EP = 1, the streams will be bound by Equation 14.40 as shown in Figure 14.27(a)).

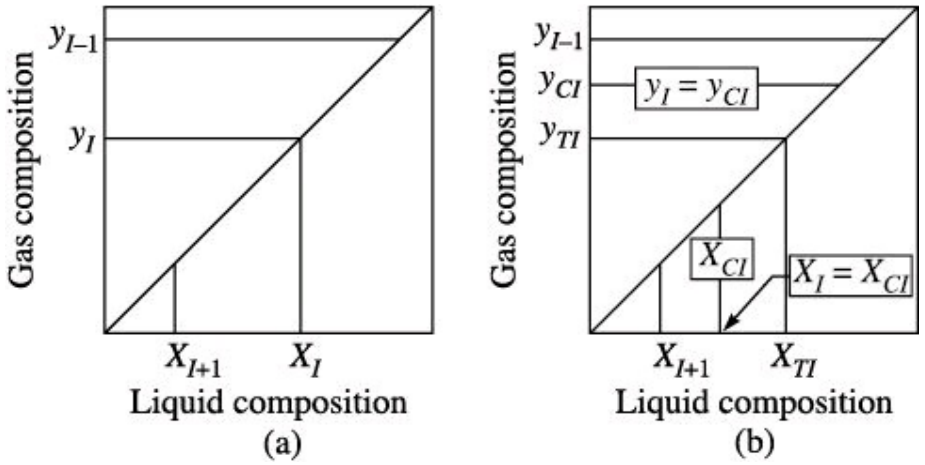


Figure 14.27 The gas, liquid compositions on (a) Ideal. (b) Non-Ideal plate.

In the last equation, the gas composition may be expressed in terms of liquid composition by using the equilibrium relation given by Equation 14.40

$$H\frac{dx_I}{dt} = mLx_{I+1} - (L + mG)x_I + mGx_{I-1} \qquad (14.42)$$

In case of a non-ideal plate, we first define the plate efficiency as:

$$EP = \frac{xc_I - x_{I+1}}{xt_I - x_{I+1}} \quad \text{or} \quad xc_I = x_{I+1} + EP(xt_I - x_{I+1}) \qquad (14.43)$$

where, xt_I is in equilibrium with yt_I. The idea has been illustrated in Figure 14.27(b).

Considering a five plate column, we may write the dynamic balance equations. Assuming that the pure liquid absorbent is coming at the top plate, $x_6 = 0$, the gas composition from the top plate, y_6, is the output variable, and the gas composition, y_0, coming from below the bottom plate, is the input variable to the system. We write the component balance of the absorbed material, for the general I-th plate and write the dynamic balance as the following set of differential equations:

For a general (I-th) plate (I from 2 to NS – 1):

$$\frac{dx_I}{dt} = C_1[xy_{I-1} - xy_I] + C_2[x_{I+1} - x_I] \qquad (14.44)$$

In forming the above equation, the old values of all the variables have been used. Also, the gas composition's *y*s have been converted to the corresponding equilibrium liquid composition's *x*s by applying Equation 14.40 and they are written as *xy*s. By integrating Equation 14.44, the current value of *x*s for EP = 1 is found as *xt*s. Next, the corrected value liquid composition *xcs* are found for any value of EP (= 0 to 1) by using Equation 15.43 and the idea is expressed in Figure 14.27(b).

$$xc_I = EP[xt_I - x_{I+1}] + x_{I+1} \qquad (14.45)$$

The corresponding gas compositions are found by applying Equation 14.46.

$$xyc_I = xy_{I-1} - C_3[xc_I - x_{I+1}] \qquad (14.46)$$

The constants used in the above equations have the following definitions:

where,

$$C_1 = \frac{Vm}{H}; \quad C_2 = \frac{L}{H}; \quad C_3 = \frac{L}{Vm}$$

A program C14-20.BAS is written for the simulation of a 5-plate absorption column. The program has the facility for demonstrating the dynamic response in exit liquid and gas composition corresponding to the two pertinent input variables, viz., inlet gas composition, xy_O, and flow rate of the absorbent liquid, L. A time list for a given sequence of several input changes have been inbuilt in the program just after line number 50. These input changes and their corresponding response have been recorded respectively in the top and bottom part of Figure 14.28. Note that at certain sequences of disturbance, both the inputs are acting simultaneously on the system. Hence, the model as such can handle variable product type non-linearity.

```
10    REM ***************** PROGRAM NAME -> C14-20.BAS **********************
      REM ******* SIMULATION OF A SEVEN PLATE GAS ABSORBER *********
      REM ************ Parameter Values & Variable Initialization ************
      SCREEN 12
      XS = 50: YS = 400: XF = 1
      LINE (XS, YS + 10) – (XS + 400, YS – 160), , B
      LINE (XS, YS – 170) – (XS + 400, YS – 350), , B
      EP = .8: L = 2.5: V = .25: H = 6: M = 3.5: B = .01: NS = 5
      FOR I = 1 TO NS + 1
      XO(I) = 0: XYO(I) = 0: XNT(I) = 0: XNC(I) = 0: XYNC(I) = 0
      NEXT I
      B = 0: YO = 0: XYOO = (YO - B) / M
      T = 0: DT = .01: TFIN = 400
      REM ********************** The Time Loop ************************
50    IF T > TFIN GOTO 70
      IF T > 15 THEN XYOO = .05
      IF T > 75 THEN L = 4
      IF T > 130 THEN L = 5
      IF T > 170 THEN L = 3
      IF T > 225 THEN XYOO = .025
      IF T > 250 THEN L = 4
      IF T > 300 THEN XYOO = .04
      IF T > 325 THEN L = 3
      C1 = (V * M) / H: C2 = L / H: C3 = L / (V * M)
      REM ******* For Five Plates Considering Plate Efficiency (EP) ******
      REM ***************** The First (BOTTOM) Plate *********************
      DXDT(1) = C1 * (XYOO – XYO(1)) + C2 * (XO(2) – XO(1))
      XNT(1) = XO(1) + DXDT(1) * DT
      XNC(1) = EP * (XNT(1) – XO(2)) + XO(2)
      XYNC(1) = XYOO – C3 * (XNC(1) – XO(2))
      REM ************ The General (ith,i=2 to NS-1) Plates *************
      FOR I = 2 TO NS – 1
```

```
    DXDT(I) = C1 * (XYO(I – 1) – XYO(I)) + C2 * (XO(I + 1) – XO(I))
    XNT(I) = XO(I) + DXDT(I) * DT
    XNC(I) = EP * (XNT(I) – XO(I + 1)) + XO(I + 1)
    XYNC(I) = XYO(I – 1) – C3 * (XNC(I) – XO(I + 1))
    NEXT I
    REM ******************** The Top (NSth) Plate ***********************
    DXDT(NS) = C1 * (XYO(NS – 1) – XYO(NS)) – C2 * XO(NS)
    XNT(NS) = XO(NS) + DXDT(NS) * DT
    XNC(NS) = EP * (XNT(NS) – XO(NS + 1))
    XYNC(NS) = XYO(NS – 1) – C3 * XNC(NS)
    YOUT = XYNC(NS) * M
    REM ****************** GRAPHIC OUTPUT SUBROUTINE ********************
    PSET (XS + T * XFCTR * .5, YS – XNC(1) * 5000) 'exit liq.
    PSET (XS + T * XFCTR * .5, YS – YOUT * 500) 'exit gas
    PSET (XS + T * XFCTR * .5, YS – 180 – XYOO * 500) 'inlet gas
    PSET (XS + T * XFCTR * .5, YS – 210 – L * 10) 'liq rate
    REM *********** TRANSFER OF NEW VALUE--->TO--->OLD VALUE ************
    FOR I = 1 TO NS + 1
    XO(I) = XNT(I)
    XYO(I) = XNT(I) * M
    NEXT I
    T = T + DT
    GOTO 50
70    GOTO 5000
5000 END
```

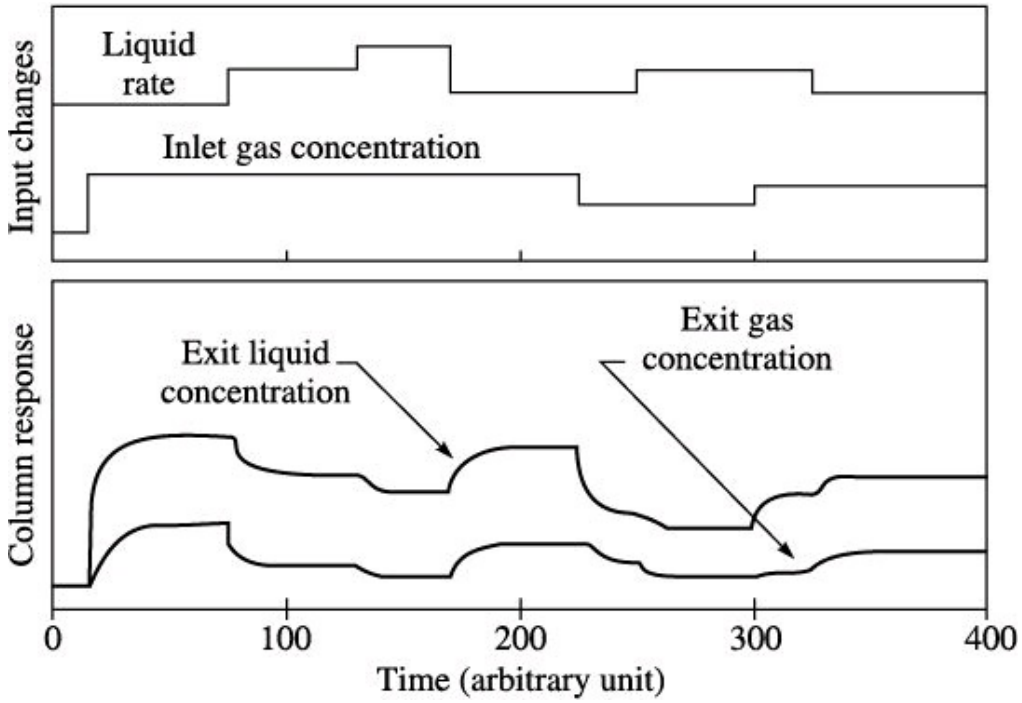


Figure 14.28 The absorption column response for the two kinds of disturbance.

14.16 Distributed Parameter Systems

It was discussed in Chapter 3 that a way of modeling distributed parameter systems is to split them into a number of lumped parameter sub-systems that are connected in series. By such representation, the models of co-current and counter-current double

pipe heat exchangers are proposed. Without the detail formulation (which are given in Chapter 3), the model equations will be presented in the state form and an effort will be made for writing programs for simulation of the models.

14.16.1 Co-Current Double Pipe Heat Exchanger (without considering metal wall capacity)

The schematic of the model to represent a double pipe co-current heat exchanger is shown in Figure 14.29.

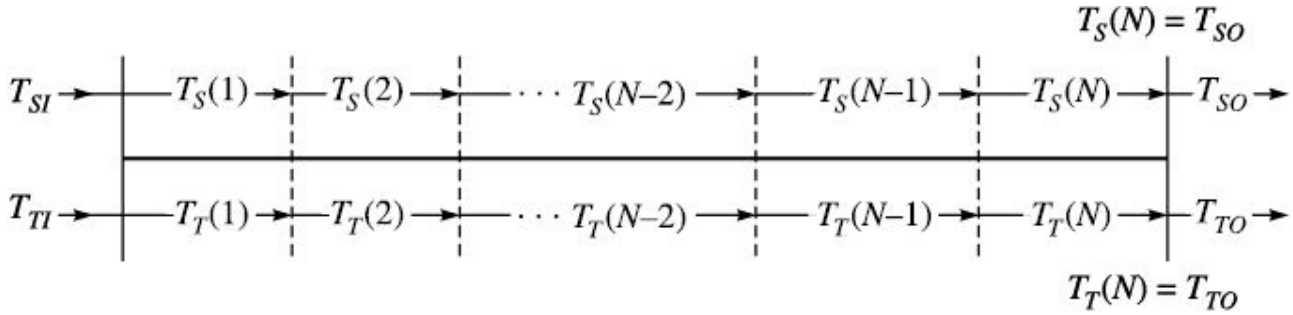


Figure 14.29 The multi-stage lumped parameter model for a co-current heat exchanger.

The following parameter values are relevant for construction of the model:

QS —Flow rate of shell fluid, cm^3/s^{-1}; QT—Flow rate of tube fluid, cm^3/s^{-1};

CPS —Ht. Cap. of shell fluid, $cal/g^{-1}/^oC^{-1}$; CPT—Ht. Cap. of shell fluid, $cal/g^{-1}/^oC^{-1}$;

rS—Shell fluid density, g/cm^{-3}; rT—Tube fluid density, g/cm^{-3};

N—Number of segments; L—Length of exchanger, cm;

VS—Shell side volume, cm^3; VT—Tube volume, cm^3;

AS—Cross sectional area of shell, cm^2; AT—Cross sectional area of tube, cm^2;

A—Heat transfer area, cm^2; U—Heat transfer coefficient, $cal/s^{-1}/cm^{-2}/^oC^{-1}$;

$$VSN = \frac{VS}{N}; \quad VTN = \frac{VT}{N}; \quad AN = \frac{A}{N}$$

Suffix S refers to shell side; Suffix T refers to tube side.
The model equations in state variable are given as:
At the entry segment:

for shell, $$\frac{d}{dt}T_S(1) = BS[T_{SI} - T_S(1)] - CS[T_S(1) - T_T(1)] \tag{14.47}$$

for tube, $$\frac{d}{dt}T_T(1) = BT[T_{TI} - T_T(1)] + CT[T_S(1) - T_T(1)] \quad (14.48)$$

At a general Ith segment:

for shell, $$\frac{d}{dt}T_S(I) = BS[T_S(I-1) - T_S(I)] - CS[T_S(I) - T_T(I)] \quad (14.49)$$

for tube, $$\frac{d}{dt}T_T(I) = BT[T_T(I-1) - T_T(I)] + CT[T_S(I) - T_T(I)] \quad (14.50)$$

At the exit segment:

for shell, $$\frac{d}{dt}T_S(N) = BS[T_S(N-1) - T_S(N)] - CS[T_S(N) - T_T(N)] \quad (14.51)$$

for tube, $$\frac{d}{dt}T_T(N) = BT[T_T(N-1) - T_T(N)] + CT[T_S(N) - T_T(N)] \quad (14.52)$$

where the groups in the above model equations are the following combination of system parameters.

$$BS = \frac{Q_S}{V_{SN}}; \quad BT = \frac{Q_T}{V_{TN}}; \quad CS = \frac{UA_N}{C_{p_S}\rho_S V_{SN}}; \quad CT = \frac{UA_N}{C_{p_T}\rho_T V_{TN}}$$

Now the developed equations may be put in the form of a computer program to simulate the co-current heat exchanger. System parameter values with consistent engineering units and initial values of the variables are required to run the program before putting any disturbance to the system. We have shown how by running such a program, representing a multiple lumped parameter stage system, with starting variable value for long enough time (long enough to settle all the variables to their steady state), the steady state values may be printed for subsequent use as initial conditions before the system is subjected to a disturbance. This has been done for the three reactor CSTR battery. But this method of finding steady state values would be tedious and impractical if we divide the heat exchanger into, say 40 lumped parameter stages. We have to print 80 variable values (40 for shell side and 40 for the tube side) then apply them. Instead, after determining the required settling time and calling it TINP2, the disturbance may be applied from the value of T when T > TINP2.

EXAMPLE 14.21 Write a program to simulate a double pipe co-current heat exchanger assuming the following parameter values:

$Q_S = 0.015$ l/s^{-1}; $Q_T = 0.015$ l/s^{-1}; $C_{PS} = 0.9$ cal/g^{-1}/°C^{-1};

C_{PT} = 0.95 cal/g^{-1}/°C^{-1}; r_S = 0.9 g/cm^{-3}; r_T = 0.95 g/cm^{-3};

N = 40; L = 1000 cm; D_S, Shell ID = 60 mm;

D_{TT}, Tube ID = 30 mm; U = 0.02 cal/s^{-1}/cm^{-2}/°C^{-1}.

Let us assume that the exchanger is running at steady state with a tube inlet temperature of 20°C and shell inlet temperature of 100°C, when,

1. Tube inlet temperature is raised to 30°C keeping shell inlet temperature constant and then this step response dynamics of tube exit temperature is recorded;
2. Shell inlet temperature is raised to 110°C keeping tube inlet temperature constant at 20°C, and the resulting step response dynamics in tube exit temperature is recorded.

From these dynamic responses, find the parameters of FOPDT model structures for the two input disturbances.

Solution: Program C14-21.BAS, based on the model structure in Figure 14.29, and Equations 14.47 through 14.52, has been prepared for this purpose. The program starts with the initial value of 20°C for all variables. At T = TINP1, the shell inlet temperature is raised to 100°C, and the system is allowed to come to steady state with this value of shell inlet temperature, until at T = TINP2, either a step of 10°C is introduced in tube inlet temperature or in shell inlet temperature.

By trial, TINP2 is found when the variable values are showing a constant magnitude at least for five time unit (i.e. for 100 integration steps). In the third and fourth line after line number 50, facilities for introducing disturbances are provided. The PSET statement records the response with time values starting from TINP2 and only the incremental value of temperature from its steady state value TST is recorded.

Also, as a check of validity of simulation performance of the program, the steady values at T = TINP2 are recorded as a temperature profile along the shell and tube side of the exchanger are recorded. These profiles for co-current as well as counter-current exchangers are shown in Figures 14.30(a) and (b).

```
10    REM ******* PROGRAM NAME -> C14-21.BAS ***********
      REM *** RESPONSE OF A CO-CURRENT SHELL & TUBE HT.EXCHGR **********
      DIM TSO(50), TTO(50), TSN(50), TTN(50), DERS(50), DERT(50)
      SCREEN 12
      XS = 50: YS2 = 280: YS3 = 430: XF = 1: YF = 20
      LINE (XS, YS2) – (XS + 240, YS2 – 120), , B
      LINE (XS, YS3) – (XS + 240, YS3 – 120), , B
      REM *** PHYSICAL PARAMETERS ********
      QS = .015: QT = .01: PI = 3.14159: N = 40
      L = 1000: DS = 60: DTT = 30: A = (PI * DTT * L) / 100
```

```
    VT = (PI * DTT * DTT * L) / 4000: VS = ((PI * DS * DS * L) / 4000) – VT
    AN = A / N: VSN = VS / N: VTN = VT / N: U = .02
    CPS = .9: CPT = .95: ROS = .9: ROT = .95
    BS = (QS * 1000) / VSN: CS = (U * AN) / (VSN * CPS * ROS)
    BT = (QT * 1000) / VTN: CT = (U * AN) / (VTN * CPT * ROT)
    REM ****** INITIAL VALUES OF THE VARIABLES ******
    TSI = 20: TTI = 20: TSOO = 20: TTOO = 20
    FOR I = 1 TO N – 1
    TSO(I) = 20: TTO(I) = 20
    NEXT I
    REM ****** TIME LOOP PARAMETERS *******
    T = 0: DT = .005: TF = 480: TINP1 = 20: TINP2 = 240
    REM ********* THE TIME LOOP ************
50   IF T > TF THEN GOTO 70
    IF T > TINP1 THEN TSI = 100
    REM IF T > TINP2 THEN TTI = 30 'For input in TTI
    REM IF T > TINP2 THEN TSI = 110 'For input in TSI
    DERS(N) = BS * (TSO(N – 1) – TSOO) – CS * (TSOO – TTOO)
    DERT(N) = BT * (TTO(N – 1) – TTOO) + CT * (TSOO – TTOO)
    TSON = TSOO + DERS(N) * DT: TTON = TTOO + DERT(N) * DT
    FOR I = 2 TO N – 1
    J = N – I: J1 = J + 1
    DERS(J1) = BS * (TSO(J) – TSO(J1)) – CS * (TSO(J1) – TTO(J1))
    DERT(J1) = BT * (TTO(J) – TTO(J1)) + CT * (TSO(J1) – TTO(J1))
    TSN(J1) = TSO(J1) + DERS(J1) * DT
    TTN(J1) = TTO(J1) + DERT(J1) * DT
    NEXT I
    DERS(1) = BS * (TSI – TSO(1)) – CS * (TSI – TTO(1))
    DERT(1) = BT * (TTI – TTO(1)) + CT * (TSI – TTO(1))
    TSN(1) = TSO(1) + DERS(1) * DT: TTN(1) = TTO(1) + DERT(1) * DT
    REM ***** GRAPHICAL OUTPUT **********
    IF T < TINP2 – DT THEN GOTO 60
    IF T > TINP2 – 2 * DT AND T < TINP2 – DT THEN GOTO 54
    IF T > TINP2 THEN GOTO 58
54   TST = TTON
    CIRCLE (XS + 20, YS2 – TSI), 1
    CIRCLE (XS + 20, YS2 – TTI), 1
    FOR I = 1 TO N – 1
    CIRCLE (XS + 20 + 5 * I, YS2 – TSN(I)), 1
    CIRCLE (XS + 20 + 5 * I, YS2 – TTN(I)), 1
    NEXT I
    CIRCLE (XS + 20 + 5 * I, YS2 – TSON), 1
    CIRCLE (XS + 20 + 5 * I, YS2 – TTON), 1
    GOTO 60
58   PSET (XS + (T – TINP2) * XF, YS3 – (TTON – TST) * YF)\
60   REM *** NEW VALUES TRANSFERED TO OLD VALUES *****
    TSOO = TSON: TTOO = TTON
    FOR I = 1 TO N – 1
    TSO(I) = TSN(I): TTO(I) = TTN(I)
    NEXT I
    T = T + DT
    GOTO 50
```

```
70      GOTO 5000
5000 END
```

The program results are shown in Figures 14.30(a) and (b).

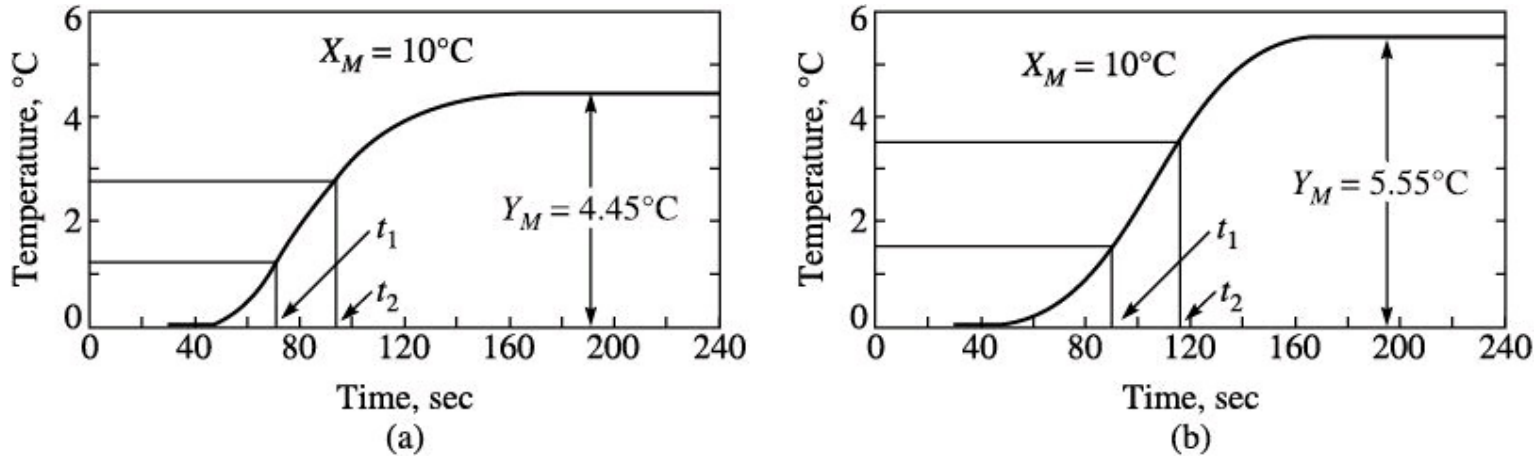


Figure 14.30 Tube exit temperature response for: (a) Step input of tube inlet temperature. (b) Step input at shell inlet temperature.

For input in tube inlet temperature:

t_1 = 71 s, t_2 = 94 s (from response trace)

Applying Equations 12.10 and 12.11:

$t_{PT} = 34.5$ s; $t_{DLT} = 59.5$ s

$$K_{PT} = \frac{Y_M}{X_M} = 0.445$$

Thus,

$$G_{PT}(s) = \frac{0.445e^{-59.5s}}{34.5s + 1} \qquad (14.53)$$

For input in shell inlet temperature:

t_1 = 91s, t_2 = 116 s (from response trace)

Applying Equations 12.10 and 12.11:

$t_{PS} = 37.5$ s; $t_{DLS} = 78.5$ s

$$K_{PT} = \frac{Y_M}{X_M} = 0.555$$

$$G_{PS}(s) = \frac{0.555.e^{-78.5s}}{37.5s + 1} \qquad (14.54)$$

In the program C14-21.BAS, note for the mid-sections, how the new values, old values and derivatives are calculated in the FOR loops and stored as subscripted variables in the respective array locations. The graphic output also uses a FOR loop for depicting the steady state variable values shown later in Figure 14.33(a).

14.16.2 Sensible Heat Transfer in a Counter Current Exchanger

The schematic of the model to represent a double pipe counter-current heat exchanger is shown in Figure 14.31.

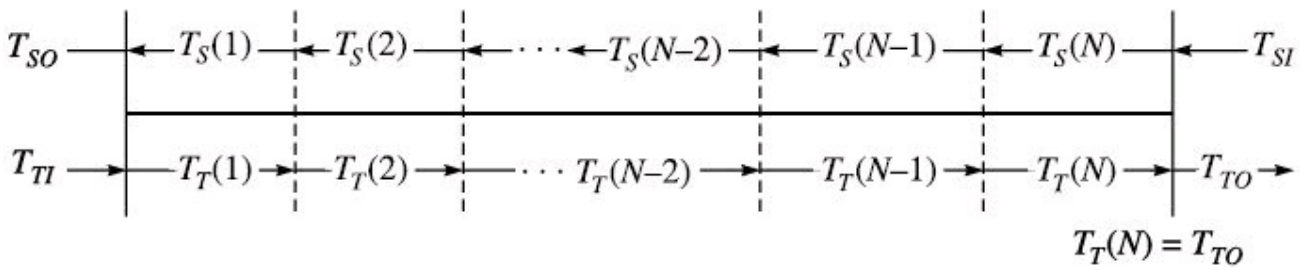


Figure 14.31 The multi-stage lumped parameter model for a counter-current heat exchanger.

The nomenclature for system parameters that has been proposed in the co-current situation will also hold through this analysis. After writing the dynamic heat balance equations, the identical group of parameters have been applied to simplify them. These state equations are presented as follows:

At the tube and shell entry segment:

for shell, $$\frac{d}{dt}T_S(1) = BS[T_{SI} - T_S(1)] - CS[T_S(1) - T_T(1)] \quad (14.55)$$

for tube, $$\frac{d}{dt}T_T(N) = BT[T_T(N-1) - T_T(N)] + CT[T_S(N) - T_T(N)] \quad (14.56)$$

At a general Ith segment: [in shell side|2 □ I □ (N–2); in tube side|2 □ I □ (N–2)]

for shell, $$\frac{d}{dt}T_S(I) = BS[T_S(I+1) - T_S(I)] - CS[T_S(I) - T_T(I)] \quad (14.57)$$

for tube, $$\frac{d}{dt}T_T(I) = BT[T_T(I+1) - T_T(I)] + CT[T_S(I) - T_T(I)] \quad (14.58)$$

At the tube and shell exit segment:

for shell, $$\frac{d}{dt}T_S(N) = BS[T_{SI} - T_S(N)] - CS[T_S(N) - T_{TO}] \quad (14.59)$$

for tube, $$\frac{d}{dt}T_T(1) = BT[T_{TI} - T_T(1)] + CT[T_{SO} - T_T(1)] \quad (14.60)$$

Again the developed equations may be put in the form of a computer program to simulate the counter-current heat exchanger. System parameter values with consistent engineering units and initial values of the variables are again required to run the program. The procedure shown in the previous example may be followed for setting the variables to their steady state values before introducing any kind of disturbance.

EXAMPLE 14.22 In this example, we shall write a program to simulate a double pipe heat exchanger with identical parameter values and operational data, as that in the last example. The only difference being that the stream arrangements in this exchanger are counter-current fashion.

Assuming the same initial operating conditions and two types of input disturbance sequence, record the responses from a simulation program and find the parameters of FOPDT, or any other suitable model structures for the two input disturbances.

Solution: The developed program is named as C14.22.BAS. By trial, TINP2 is found when the variable values are showing a constant magnitude at least for five time unit (i.e. for 1000 integration steps). The value of TINP2 is much larger than that for the co-current configuration which is definitely a dynamic feature of counter-current configuration. Also, the ultimate response values have so much difference that two scale factors are used for recording responses with respect to inputs in tube temperature (YF1) and shell temperatures (YF2).

```
10    REM ******* PROGRAM NAME -> C14-22.BAS ***********
      REM *** RESPONSE OF A COUNTER-CURRENT SHELL & TUBE HT.EXCHGR **********
      DIM TSO(50), TTO(50), TSN(50), TTN(50), DERS(50), DERT(50)
      SCREEN 12
      XS = 50: YS2 = 280: YS3 = 430: XF = 1: YF1 = 40: YF2 = 10
      LINE (XS, YS2) – (XS + 240, YS2 – 120), , B
      LINE (XS, YS3) – (XS + 240, YS3 – 120), , B
      REM *** PHYSICAL PARAMETERS ********
      QS = .015: QT = .01: PI = 3.14159: N = 40
      L = 1000: DS = 60: DTT = 30: A = (PI * DTT * L) / 100
      VT = (PI * DTT * DTT * L) / 4000: VS = ((PI * DS * DS * L) / 4000) – VT
      AN = A / N: VSN = VS / N: VTN = VT / N: U = .02
      CPS = .9: CPT = .95: ROS = .9: ROT = .95
      BS = (QS * 1000) / VSN: CS = (U * AN) / (VSN * CPS * ROS)
      BT = (QT * 1000) / VTN: CT = (U * AN) / (VTN * CPT * ROT)
      REM ****** INITIAL VALUES OF THE VARIABLES ******
      TSI = 20: TTI = 20: TSOO = 20: TTOO = 20
      FOR I = 1 TO N
      TSO(I) = 20: TTO(I) = 20
      NEXT I
      REM ****** TIME LOOP PARAMETERS *******
      T = 0: DT = .005: TF = 1095: TINP1 = 20: TINP2 = 615
      REM ********* THE TIME LOOP *************
50    IF T > TF THEN GOTO 70
      IF T > TINP1 THEN TSI = 100
      IF T > TINP2 THEN TTI = 30 'For input in TTI
      REM IF T > TINP2 THEN TSI = 110 'For input in TSI
      DERS(1) = BS * (TSO(2) – TSO(1)) – CS * (TSO(1) – TTO(1))
      DERT(1) = BT * (TTI – TTO(1)) + CT * (TSO(1) – TTO(1))
      TSN(1) = TSO(1) + DERS(1) * DT: TTN(1) = TTO(1) + DERT(1) * DT
      TSON = TSN(1)
      FOR I = 2 TO N – 1
      I1 = I – 1: I2 = I + 1
      DERS(I) = BS * (TSO(I2) – TSO(I)) – CS * (TSO(I) – TTO(I))
      DERT(I) = BT * (TTO(I1) – TTO(I)) – CT * (TSO(I) – TTO(I))
      TSN(I) = TSO(I) + DERS(I) * DT: TTN(I) = TTO(I) + DERT(I) * DT
      NEXT I
      DERS(N) = BS * (TSI – TSO(N)) – CS * (TSO(N) – TTO(N))
      DERT(N) = BT * (TTO(N – 1) – TTO(N)) + CT * (TSO(N) – TTO(N))
      TSN(N) = TSO(N) + DERS(N) * DT: TTN(N) = TTO(N) + DERT(N) * DT
      TTON = TTN(N)
      REM ***** GRAPHICAL OUTPUT **********
```

```
      IF T < TINP2 – DT THEN GOTO 60
      IF T > TINP2 – 2 * DT AND T < TINP2 – DT THEN GOTO 54
      IF T > TINP2 THEN GOTO 54
54     TST = TTON
      CIRCLE (XS + 20, YS2 – TSON), 1
      CIRCLE (XS + 20, YS2 – TTI), 1
      FOR I = 1 TO N – 1
      CIRCLE (XS + 20 + 5 * I, YS2 – TSN(I)), 1
      CIRCLE (XS + 20 + 5 * I, YS2 – TTN(I)), 1
      NEXT I
      CIRCLE (XS + 20 + 5 * I, YS2 – TSI), 1
      CIRCLE (XS + 20 + 5 * I, YS2 – TTON), 1
      GOTO 60
58     PSET (XS + (T – TINP2) * XF, YS3 – (TTON – TST) * YF1)
60     REM *** NEW VALUES TRANSFERED TO OLD VALUES *****
      TSOO = TSON: TTOO = TTON
      FOR I = 2 TO N – 1
      TSO(I) = TSN(I): TTO(I) = TTN(I)
      NEXT I
      TSO(N) = TSN(N): TTO(N) = TTN(N)
      T = T + DT
      GOTO 50
70     GOTO 5000
5000 END
```

The program results are shown in Figures 14.32(a) and (b).

For input in tube inlet temperature:

t_1 = 68 s, t_2 = 94 s (from response trace)

Applying Equations 12.10 and 12.11:

$$\tau_{PT} = 45 \text{ s}; \ \tau_{DLT} = 53 \text{ s}$$

$$K_{PT} = Y_M/X_M = 0.27$$

Thus,

$$G_{PT}(s) = \frac{0.27e^{-53s}}{45s+1} \qquad (14.61)$$

For input in shell inlet temperature:

t_1 = 26 s, t_2 = 86 s (from response trace)

Applying Equations 12.10 and 12.11:

$$\tau_{PS} = 90 \text{ s}; \ \tau_{DLS} = -4 \text{ s}$$

$$K_{PT} = Y_M/X_M = 0.72$$

Due to a finite value of initial slope and –ve value of the dead time in FOPDT model, we propose an FO model structure as:

$$G_{PS}(s) = \frac{0.72}{90s+1} \qquad (14.62)$$

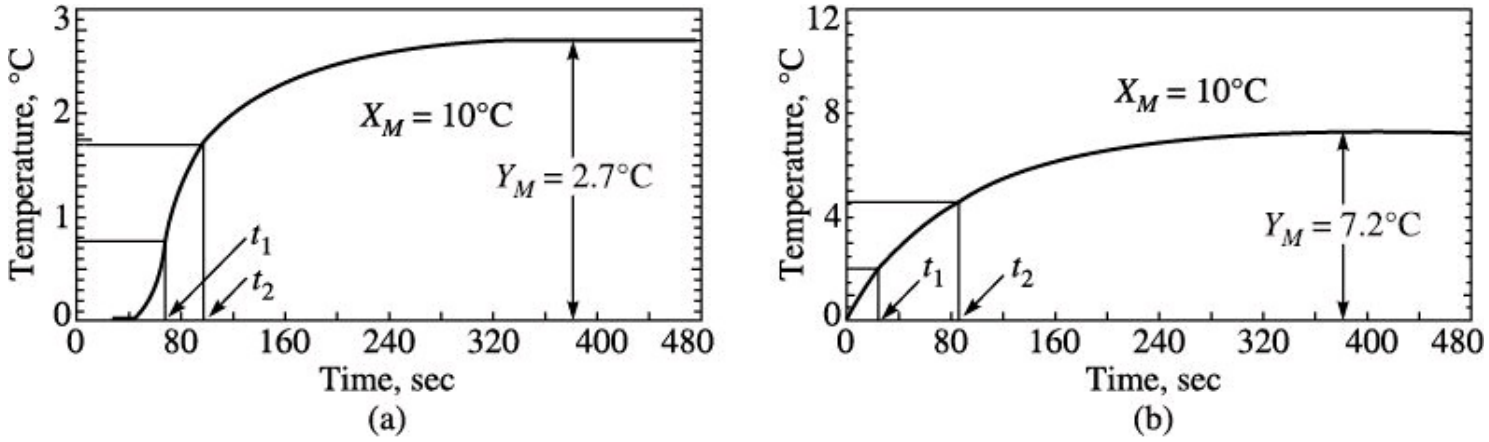


Figure 14.32 Tube exit temperature response for: (a) Step input of tube inlet temperature. (b) Step input at shell inlet temperature.

The non-zero initial slope of response for shell inlet temperature change is understandable because temperature change has occurred at the stage of shell which is nearest to the tube exit. The low value of ultimate change (2.7°C), in case of disturbing the tube liquid temperature, indicates that much of the heat transfer potential remain unexploited, a fact also supported by the steady state temperature profile plots for co-current and counter-current arrangement. These are shown in Figures 14.33(a) and (b). These figures, by their shapes, also confirm the validity of these models because they are comparable with the steady state profiles given in any heat transfer text books. However, the relatively large difference between terminal temperatures in case of counter-current case explains the smaller value of gain in the transfer function $G_{PT}(s)$.

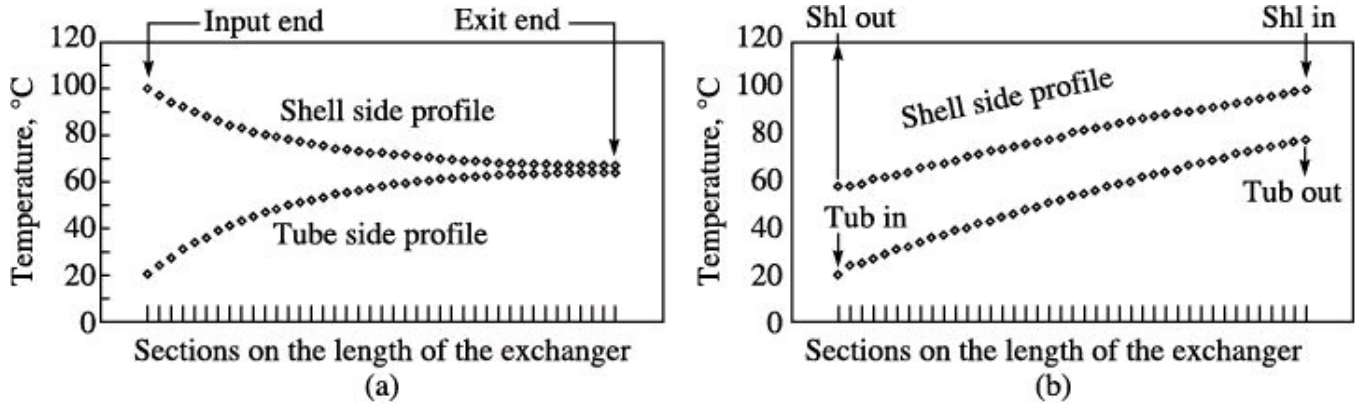


Figure 14.33 Steady state temperature profiles in: (a) Co-Current (b) Counter current double pipe heat exchangers.

Problem 14.14 The next topic of interest will be a heat transfer system where one fluid is changing phase for rejection of heat while the other is receiving only sensible heat. If the fluid that is supplying heat is condensing under saturated condition, its flow rate may be considered negligibly small as compared to the liquid that is heated by sensible heat only. Thus, we may assume that the total volume of the section where vapour is condensing (let us say the shell side) will be under constant temperature

condition. Shell temperature may be changed by regulating the pressure of the saturated vapour (let us say by a control valve on the shell inlet line). Dynamically speaking, in such systems a distributed parameter part and a lumped parameter part coexists. In Chapter 3, Example 3.14 discusses a system where saturated steam is condensing on the shell side and tube fluid is heated only by sensible heat. The developed model is expressed by Equations 3.93 through 3.97. Assume all parameter and operational conditions identical to the values given for the tube side as in the last example. For shell side, assume a linear relation between saturation pressure and temperature given as:

$$T_S = (100 + P_S)\ ^\circ\text{C}$$

where, T_S and P_S are the saturation temperature in °C and pressure in psig. Assume that the relation is valid for the pressure range 0 to 20 psig. Initially, the heat exchanger is running with tube inlet temperature of 20°C and shell inlet pressure of 2 psig. Then it is disturbed either by:

1. Increasing the tube inlet temperature to 30°C, or by
2. Increasing the shell pressure to 10 psig. Write a program to simulate the system and the responses.

We have discussed about modeling and simulation for a number of processes of assorted dynamic varieties through the programs corresponding to these models. The processes demonstrated their dynamic character subject to initial conditions, parameter values and inputs or disturbances. The models and the generated dynamics are called Open Loop in control engineering terminology. The counter parts of these models are Closed Loop models that will produce closed loop or controlled process response when a simulated controller in the form of a computer program is added to the open loop process program. It is time to present the controller programs in the form of subroutines.

14.17 Subroutines for Control Hardware

The control hardware, as a group of subroutines for performing their task within a closed loop system, are listed as follows:

1. **Measuring Element** [SUBROUTINE MEL1 (for single loop) and SUBROUTINE MEL2 (for multiple loop)].
2. **Comparator** [SUBROUTINE CMP1(for single loop) and SUBROUTINE CMP2 (for multiple loop)].
3. **Controller** [SUBROUTINE CNTR1(for single loop) and SUBROUTINE

CNTR2 (for multiple loop)].

4. **Final Control Element** [SUBROUTINE FCE1(for single loop) and SUBROUTINE FCE2 (for multiple loop)].

14.17.1 Measuring Element

Output signal from any instrument is proportionally converted within a numerical range, encompassing all necessary calculations for realizing the control algorithms under different modes. The measuring range of a measuring element is defined by maximum and minimum values of the measured variable, MVCL and MVZR, and the numerical range of control mode calculation is performed within the range defined by maximum and minimum values, CL and ZR. In the measuring element subroutine MEL1, the signal values are proportionally converted within the numerical range between CL and ZR, by the statements of the form:

```
CVN = (CVNN – MVZR) * ((CL – ZR)/(MVCL – MVZR)
```

In the above step, CVNN is the current value of controlled variable from the measuring element with proper engineering unit and CVN is its proportionally converted value within the numerical range of CL and ZR. As an example, we may consider a temperature sensor with measuring range of 0 to 250°C. For CL = 100 and ZR = 0, if the process temperature, CVNN = 50°C, the corresponding CVN will be (50 – 0) × [(100 – 0)/(250 – 0)] = 20. Similarly, for a pressure sensor with the working range of 3 to 15 psig, a measured signal of 9 psig will obtain CVN = 50. The above relation is also valid for ‘suppressed’ and ‘elevated’ zero instrument.

The old value of controlled variable CVON and set point *SP* are similarly processed in MEL1 to obtain CVO and SPN. Generally used values for CL and ZR are respectively 100 and 0.

```
2000 REM ***************** SUBROUTINE MEL1 *****************
CVN = (CVNN – MVZR) * ((CL – ZR) / (MVCL – MVZR)
CVO = (CVON – MVZR) * ((CL – ZR) / (MVCL – MVZR)
SPN = (SP – MVZR) * ((CL – ZR) / (MVCL – MVZR)
RETURN
END
```

14.17.2 Comparator

The subroutine CMP1 calculates the current and old values of error as ERN and ERO, to be sent to the controller subroutine CNTR1.

```
2020 REM ***************** SUBROUTINE CMP1 *****************
ERN = SPN – CVN: ERO = SPN – CVO
RETURN
END
```

14.17.3 Controller

In the second line of subroutine CNTR1, the ON (NMD) GOTO statement sends the command to any of four modes of control, depending on the mode identifier NMD [for NMD = 1, to (P) control action; for NMD = 2, to (PI) control action; for NMD = 3, to (PD) control action; for NMD = 4, to (PID) control action]. Beside NMD, the following parameter values are also received by the CNTR1 from main program: ACN = 1 (for direct action), ACN = –1 (for reverse action), OPB = % output bias, K_C = Gain of the controller, TAUI = Integral time, TAUD = Derivative time. Significance of these parameters have been already discussed in Chapter 8.

```
2040 REM *************** SUBROUTINE CNTR1 ***************
     ON (NMD) GOTO 2044, 2048, 2054, 2060 ‘ Mode selection
2044 REM *************** [P] CNTRLR ********************
     CO = OPB + ACN * KC * ERN
     IF CO > CL THEN CO = CL
     IF CO < ZR THEN CO = ZR
     GOTO 2070
2048 REM *************** [PI] CNTRLR *******************
     IF T = 0 THEN GOTO 2050
     COP = ACN * KC * ERN
     COI = COI + ACN * (KC / TAUI) * (ERN + ERO) * (DT / 2)
     CO = OPB + COP + COI
     GOTO 2052
2050 CO = OPB
2052 IF CO > CL THEN CO = CL
     IF CO < ZR THEN CO = ZR
     GOTO 2070
2054 REM *************** [PD] CNTRLR *****************
     COP = ACN * KC * ERN
     COD = ACN * KC * TAUD * ((ERN – ERO) / DT)
     CO = OPB + COP + COD
     IF CO > CL THEN CO = CL
     IF CO < ZR THEN CO = ZR
     GOTO 2070
2060 REM ************* [PID] CNTRLR **************
     IF T = 0 THEN GOTO 2062
     COP = ACN * KC * ERN
     COI = COI + ACN * (KC / TAUI) * (ERN + ERO) * (DT / 2)
     COD = ACN * KC * TAUD * ((ERN – ERO) / DT)
     CO = OPB + COP + COI + COD
     GOTO 2064
2062 CO = OPB
2064 IF CO > CL THEN CO = CL
     IF CO < ZR THEN CO = ZR
2070 RETURN
     END
```

From the program steps of subroutine CNTR1, it should be noted that the maximum and the minimum numerical value of control action calculation, CL and ZR

respectively, are used as the 'limiter', which is important in preventing the 'reset-windup' problems. Integration and differentiation of error terms are done by the Trapezoidal rule and first order difference formulae respectively. In some literature, more rigorous approach, e.g. Newton's formula and second order difference equations are also suggested. It has been found in a number of simulation exercises that by choosing a small enough time step value, the difference in the results is so little that it may as well be considered insignificant. Moreover, as the error terms resulting from dynamic response of the process are calculated by first order Euler algorithm, it is a bit superfluous to apply a higher degree approximation for a part of the simulated closed loop system. However, to check this statement we present the following example.

EXAMPLE 14.23 Four modes of control are tested with four ideal input types as error functions to the controllers. The controller parameters are: $K_C = 0.5$, TAUI = 20, TAUD = 8, OPB = 50, ACN = 1, CL = 100, ZR = 0. The input functions have been defined as:

1. Step: ERN = 0 for $t < 10$, ERN = 10 for $t > 10$;
2. Ramp: ERN = 0 for $t < 10$, ERN = $0.8(t - 10)$;
3. Pulse: ERN = 0 for $t < 10$, ERN = 10 for $10 < t < 50$, ERN = 0 for $t > 50$;
4. Sinusoidal: ERN = 0 for $t < 10$, ERN = $20\ \mathrm{Sin}[0.1(t - 10)]$.

A program, C14-23.BAS, is written using CNTR1 to obtain controller responses corresponding to these inputs. To compare the responses, controller outputs are also calculated by time domain controller response equations developed in Chapter 8. The input functions as well as the simulated and calculated controller responses are shown in Figure 14.34.

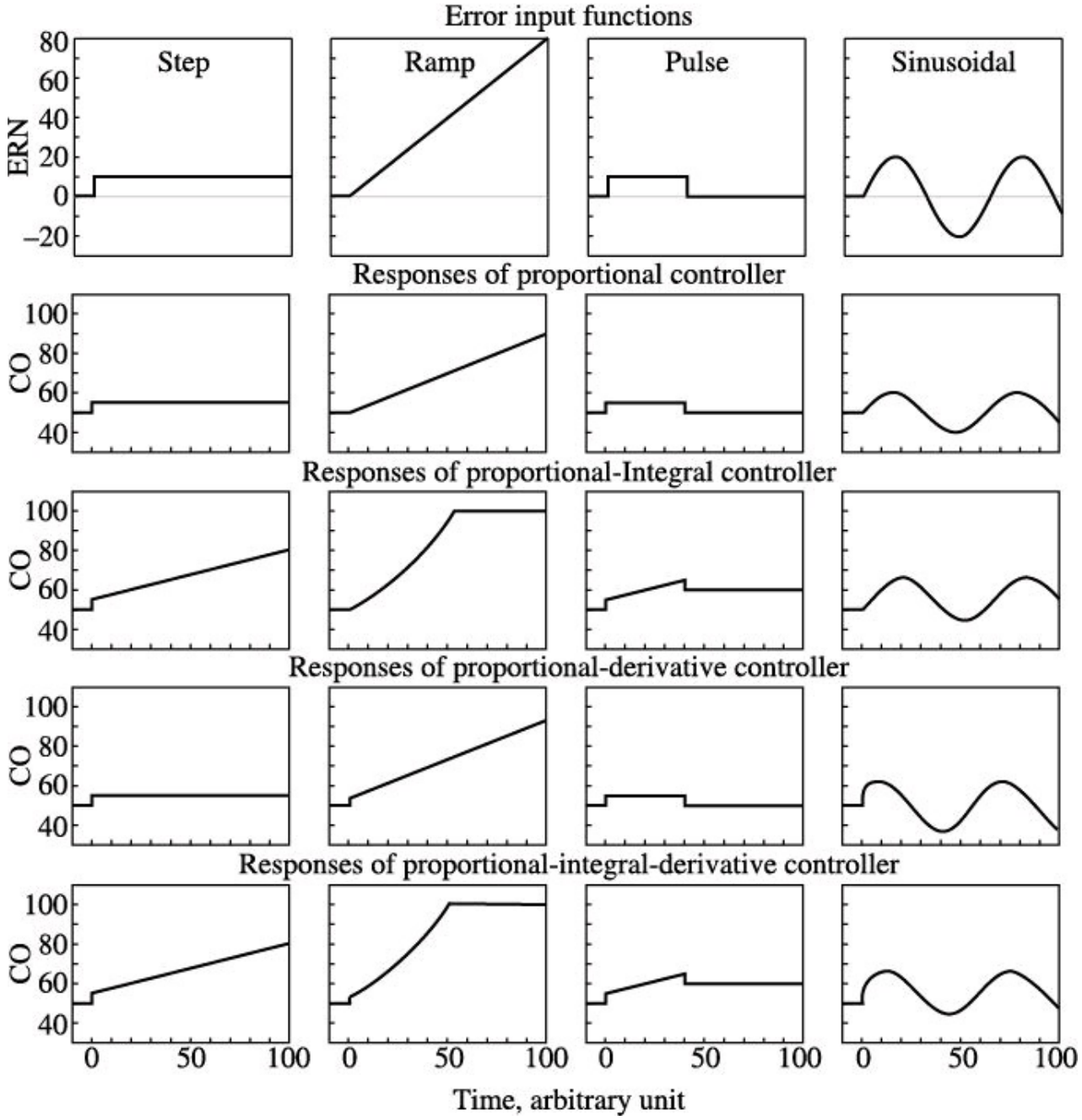


Figure 14.34 Simulation results of controller responses by using CNTR1.

The calculated responses are obtained by using Equations 8.8, 8.12, 8.19 and 8.27. From Figure 14.34, it can be seen that the simulated and calculated responses are so close that they have superimposed on each other. The response shapes in Figure 14.34 should be compared with those shown in Figures 8.7, 8.9 and 8.11 as a second check of their validity. We have not included the computer program in the text (see Appendix-IV).

Problem 14.15 Modify subroutine CNTR1 to represent the (D) action by a lead-lag element. The program segment for simulating a lead-lag has been described in an earlier example in this chapter.

14.17.4 Final Control Element

The processing on the signal from measuring element that is done in subroutine

MEL1, has to be reversed to convert the controller output CO, (a numerical value within the range of CL to ZR), to a proportional magnitude of the manipulated variable within the range MAVCL and MAVZR. In fact, MAVCL and MAVZR define the capacity measure of a final control element. A linear characteristic has been assumed for the FCE.

```
2180 REM ********** SUBROUTINE FCE1 **********
     MAV = MAVZR + (CO – ZR) * ((MAVCL – MAVZR) / (CL – ZR)
     RETURN
     END
```

Finally, the FCE1 output MAV goes to the process as the dynamic compensation. For example, if the control valve, as the final control element of a control system, has a flow range of 0 to 300 l/m of water, corresponding to, CO = 30, the subroutine output will be MAV = 90 l/m.

EXAMPLE 14.24 Write a program for the study of controlled variable dynamics of a liquid level control system comprising of 4-cylindrical tanks in non-interacting cascade combination. The first tank has one line for inflow, Q_I, with the control valve at the top and a line for exit flow, Q_O, with a valve, at the bottom of the fourth tank. The exit line valve resistance,
R_4, may be considered as the load to the closed loop system. The A liquid level sensor-transmitter having the measuring range of 0 to 100 cm of liquid height is used as the measuring element. The control valve has a linear characteristic and it is designed to handle a maximum flow of 200 cc/s of liquid. The internal areas of the tanks are: $A_1 = A_1 = A_1 = A_1 = 100$ cm^2 and the valve resistances are $R_1 = R_2 = R_3 = R_4 = 1$cm/(cm^3/s). The set point for tank level is
50 cm. Initially, the system is running at error free steady state for $Q_I = Q_O = 100$ cm^3/s and $R_4 = 1$ cm/(cm^3/s). Apply a step disturbance by increasing R_O to 1.25, at t = 50 s. Draw the resulting response traces for control modes (P), (PI), (PD) and (PID). Two sets of controller and graphic output parameters are used, one for (P) and (PD) control mode, and the other for (PI) and (PID) control mode. These are given below:

```
REM KC = 2.5: TAUD = 30: TF = 10000: XF = .02 ‘ for NMD=1&3
REM KC = .08: TAUI = 20: TAUD = 100: TF = 20000: XF = .01 ‘ for NMD=2&4
```

Other parameters used are common to all modes of control, and are: OPB = 50, ACN = 1, CL = 100, ZR = 0, MVCL = 100, MVZR = 0, MAVCL = 100, MAVZR = 50.

The program C14-24.BAS has been written as a solution of this example. Thus, assigning the process output as the controlled variable (CVNN = HN) followed by the GOSUB call sequence to the control hardware subroutines, and finally defining the

control action as manipulation to be included to the process, QI = MAV, has appeared as the new feature of a closed loop extra to that of an open loop simulation program. A few trials have been made with each of the control modes to bring up the demonstrable properties of the close loop response, hence it is not surprising that we have ended up with different sets of controller settings for (P)-(PD) and (PI)-(PID) mode.

```
10    REM ****************** PROGRAM NAME -> C14-24.BAS ******************
      REM **** LIQ LEVEL CNTRL OF 4-CYLN TNKS IN NON-INTRCTNG CASCADE *****
      REM ** EXIT VALVE RESIST=LOAD, ENTRY LINE FLOW=MANPLTD VARIABLE **
      SCREEN 12
      A1 = 100: A2 = 100: A3 = 100: A4 = 100
      R1 = 1: R2 = 1: R3 = 1: R4 = 1
      NMD = 1
20    IF NMD > 4 THEN GOTO 70
      ON (NMD) 22, 24, 26, 28
22    XS = 50: YS = 200: XF = .02: YF = 4: TF = 10000
      KC = 2.5
      LOCATE 7, 22: PRINT "NMD = 1"
      GOTO 30
24    XS = 300: YS = 200: XF = .01: YF = 4: TF = 20000
      KC = .08: TAUI = 20
      LOCATE 7, 53: PRINT "NMD = 2"
      GOTO 30
26    XS = 50: YS = 400: XF = .02: YF = 4: TF = 10000
      KC = 2.5: TAUD = 30
      LOCATE 20, 22: PRINT "NMD = 3"
      GOTO 30
28    XS = 50: YS = 200: XF = .02: YF = 4: TF = 10000
      KC = .08: TAUI = 20: TAUD = 100
      LOCATE 20, 53: PRINT "NMD = 4"
30    LINE (XS, YS + 20) – (XS + 200, YS – 120), , B
      CL =100: ZR = 0: MVCL = 100: MVZR = 0: MAVCL = 100: MAVZR = 0
      COP = 0: COI = 0: COD = 0: CO = 0
      SP = 50: OPB = 50: ACN = 1
      H1O = 50: H2O = 50: H3O = 50: H4O = 50
      Q1 = H1O / R1: Q2 = H2O / R2: Q3 = H3O / R3: Q4 = H4O / R4
      MAV = 50
      QI = MAV
      T = 0: DT = .1
50    IF T > TF THEN GOTO 60
      IF T > 50 THEN R4 = 1.25
      QI = MAV
      DH1DT = (QI – Q1) * (1 / A1)
      H1N = H1O + DH1DT * DT: Q1 = H1O / R1
      DH2DT = (Q1 – Q2) * (1 / A2)
      H2N = H2O + DH2DT * DT: Q2 = H2O / R2
      DH3DT = (Q2 – Q3) * (1 / A3)
      H3N = H3O + DH3DT * DT: Q3 = H3O / R3
      DH4DT = (Q3 – QO) * (1 / A4)
```

```
      H4N = H4O + DH4DT * DT: QO = H4O / R4
      CVNN = H4N: CVON = H4O
      GOSUB = 2000: GOSUB 2020: GOSUB 2040: GOSUB 2180
      PSET (XS + T * XF, YS1 – 50 – (H4N – 50) * YF)
      PSET (XS + T * XF, YS1 – SP * 1)
      H1O = H1N: H2O = H2N: H3O = H3N: H4O = H4N
      T = T + DT
      GOTO 50
60     NMD = NMD + 1
      GOTO 20
70     GOTO 5000

2000 REM ***************** SUBROUTINE MEL1 *****************
2020 REM ***************** SUBROUTINE CMP1 *****************
2040 REM ***************** SUBROUTINE CNTR1 ****************
2180 REM ***************** SUBROUTINE FCE1 *****************

5000 END
```

The results are shown in Figure 14.35. The advantage of adding (D) action to either (P) or (PI) modes has been shown by the reduction of overshoot and settling time. It is felt that the scope of further improvement of the close loop responses still remains. It is left to the reader as an exercise to tune the loop with each control mode by few more trials.

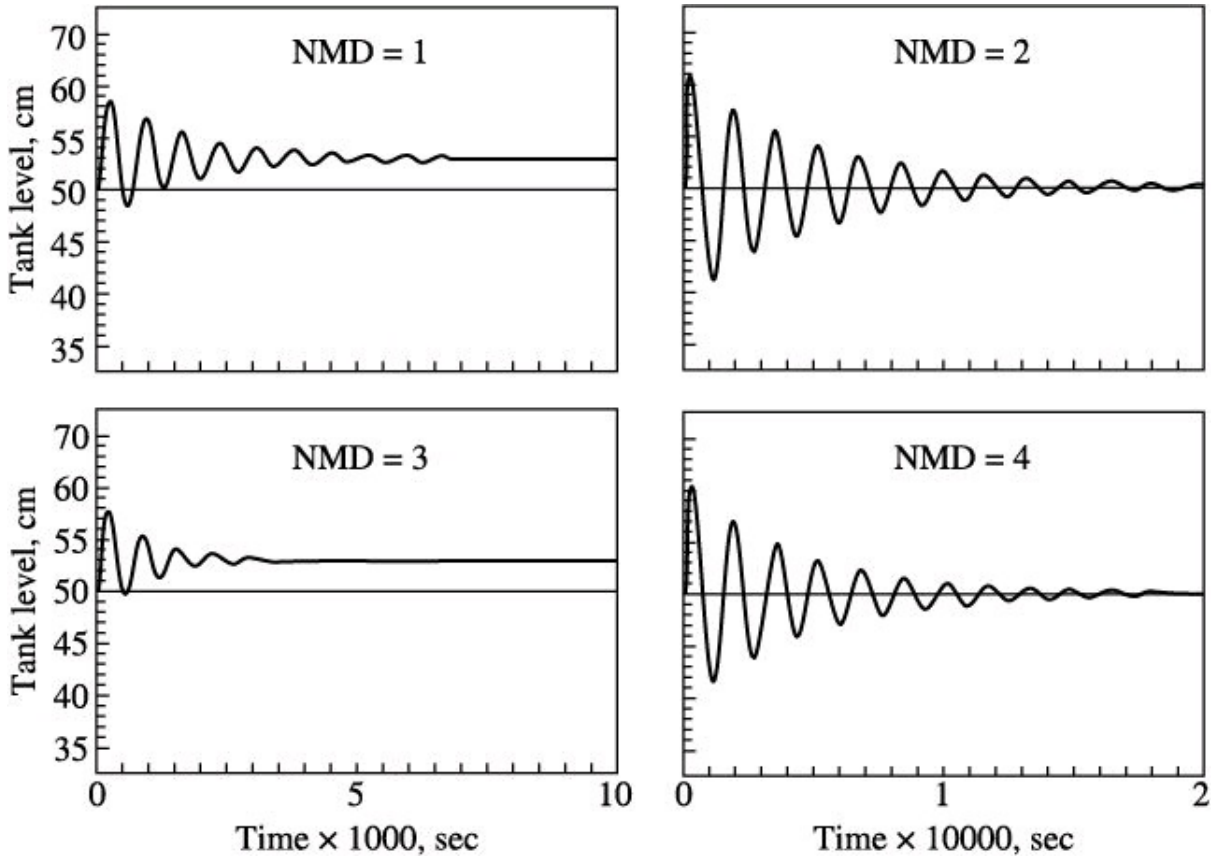


Figure 14.35 Closed loop level dynamics of the 4-noninteracting tanks in series for 4-modes of control.

An off-set appears and shows a constant value for (P) and (PD) modes, only the response settles more quickly by addition of D-action. The off-set value is around 3 cm for a load change of 25%, hence, if the control specification for such a liquid

reservoir is to keep the tank level within high- and low-level limits which are more than the possible off-set value for normal loading of the tank in service, one should select a (P) controller rather than go for more complex and costlier modes. The response settles more quickly by adding (D) action to a (PI) controller can be seen from the output of NMD = 2 and NMD = 4. This observation is also consistent with results we have obtained in Chapters 8 and 9.

14.17.5 Tuning of Controller

Tuning methods of feedback controllers have been discussed in Chapter 13. In Chapter 15, we would introduce a term **control difficulty**, and discuss the significance of this term with respect to the limitation of these tuning procedures for systems that possess certain kinds of these properties. Existence of a large dead time element has proved to have detrimental effect to the control quality of a system. In the next example, we shall show the limitation of using a tuned controller for a system having a large dead time element.

EXAMPLE 14.25 In Example 14.21, we presented the open loop simulation program for a double pipe co-current heat exchanger. Let us include a tuned (PID) controller for controlling the tube exit temperature, T_{TO}, of the heat exchanger by manipulating the shell inlet temperature, T_{SI}. In this problem, the tube entry temperature, T_{TI}, is considered to be the load variable. The heat exchanger has been simulated keeping all parameter values as they are in Example 14.21, except the overall heat transfer coefficient, U, for which a slightly lower value, $U = 0.016818$ has been assumed so that with $T_{TI} = 30°C$ and $T_{SI} = 100°C$, the tube exit temperature, T_{TO} settles to a steady state value of 68°C. To find the optimum controller settings, the step response in T_{TO} is obtained, as T_{TI} is changed from 20 to 30°C with $T_{SI} = 100°C$. The response trace, the construction on the trace and the FOPDT model parameters are shown in Figure 14.36.

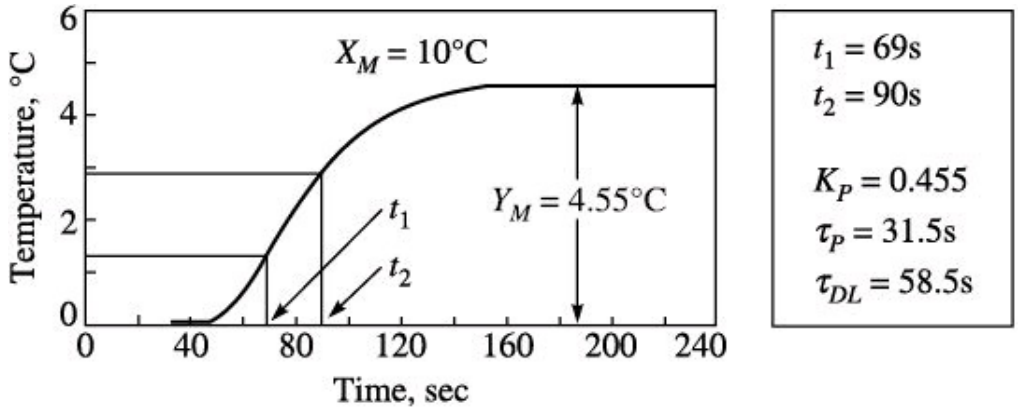


Figure 14.36 The FOPDT model parameters from the step response.

From this data the optimum controller settings, obtained accordingly to Cohen-Coon

formulae for a (PID) controller are listed as:

Cohen-Coon settings: $K_C = 2.13$, $t_I = 90.6$ s, $t_D = 15.9$ s.

The control system with the heat exchanger is shown in Figure 14.37(a).

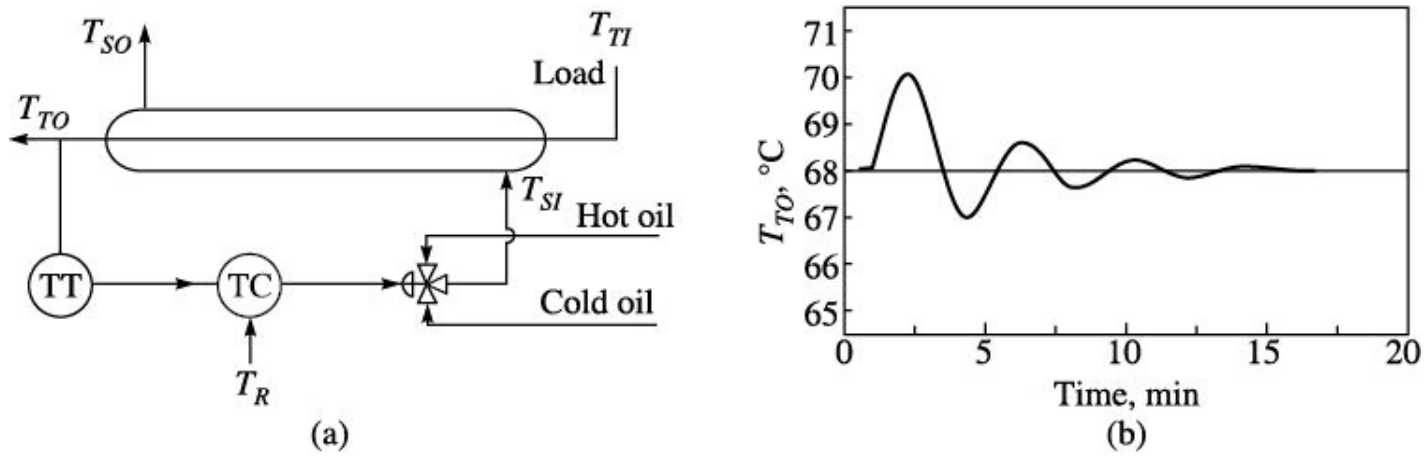


Figure 14.37 (a) The control system, and (b) The closed loop response of T_{TO} when T_{TI} is raised from 30°C to 35°C according to a step.

The closed loop simulation program has been written by the name of C14-25.BAS. The closed loop section has been written following the open loop dynamic section as in C14-21.BAS. The close loop part of the listing that starts from the last statement of the open loop part is shown as follows:

```
70    PRINT "T"; T; "TTI"; TTI; "TTON"; TTON; "TSI"; TSI; "TSON"; TSON
      CLS
      LINE(XS, YS2) – (XS + 240, YS2 – 140), , B
      REM ********** VARIABLE INITIAL VALUES **********
      TSI = 100: TTI = 30: TSOO = 71.6383: TTOO = 68
      REM ********** CONTROLLER PARAMETERS **********
      SP = 68: OPB = 50: ACN = 1: NMD = 4
      KC = 2.13: TAUI = 90.6: TAUD = 15.9
      CL = 100: ZR = 0: MVCL = 100: MVZR = 0: MAVCL = 170: MAVZR = 30
      MAV = 100
      REM ********** TIME LOOP PARAMETERS **********
      T = 0: DT = .05: TF = 1200: TINP = 10
150   IF T > TF THEN GOTO 170
      IF T > TINP THEN TTI = 35
      DERS(N) = BS * (TSO(N – 1) – TSOO) – CS * (TSOO – TTOO)
      DERT(N) = BT * (TTO(N – 1) – TTOO) + CT * (TSOO – TTOO)
      TSON = TSOO + DERS(N) * DT: TTON = TTOO + DERT(N) * DT
      FOR I = 2 TO N – 1
      J = N – I: J1 = J + 1
      DERS(J1) = BS * (TSO(J) – TSO(J1)) – CS * (TSO(J1) – TTO(J1))
      DERT(J1) = BT * (TTO(J) – TTO(J1)) + CT * (TSO(J1) – TTO(J1))
      TSN(J1) = TSO(J1) + DERS(J1) * DT
      TTN(J1) = TTO(J1) + DERT(J1) * DT
      NEXT I
      DERS(1) = BS * (TSI – TSO(1)) – CS * (TSI – TTO(1))
      DERT(1) = BT * (TTI – TTO(1)) + CT * (TSI – TTO(1))
```

```
      TSN(1) = TSO(1) + DERS(1) * DT: TTN(1) = TTO(1) + DERT(1) * DT
      CVNN = TTON: CVON = TTOO
      GOSUB 2000: GOSUB 2020: GOSUB 2040: GOSUB 2180: TSI = MAV
      REM ************ GRAPHIC OUTPUT **************
      PSET (XS + T * .2, YS2 – 70 – (TTON – SP) * 20)
      PSET (XS + T * .2, YS2 – 70 – (SP – 68) * 20)
      REM **** NEW VARIABLE VALUES TRANSFEREF TO OLD ****
      TSOO = TSON: TTOO = TTON
      FOR I = 1 TO N – 1
      TSO(I) = TSN(I): TTO(I) = TTN(I)
      NEXT I
      T = T + DT
      GOTO 150
170   GOTO 5000
2000 REM ****************** SUBROUTINE MEL1 ********************
2020 REM ****************** SUBROUTINE CMP1 ********************
2040 REM ****************** SUBROUTINE CNTR1 ********************
2180 REM ****************** SUBROUTINE FCE1 ********************
5000 END
```

At the end of the operation of open loop part of the program, the ultimate values of T_{TO}, T_{SO} and [$T_{TO(I)}$, $T_{SO(I)}$, I = 1 TO N – 1] automatically become the initial values for executing the closed loop part.

The close loop response shown in Figure 14.37(b), obtained with Cohen-Coon settings, is barely satisfactory. An optimum response is expected to show lesser decay ratio and settling time. This deterioration of control quality may be attributed to the presence of the large dead time element in the system dynamics.

Besides the selection of controller mode and determination of its parameter values, the tuning procedure of a specific control loop includes capacity design of the selected final control element. The importance of this exercise may be demonstrated by selecting a few pairs of MAVCL and MAVZR, keeping the OPB = 50 and the initial value of T_{SI} = 100°C. The reader is urged to run the program with the following pairs:

```
MAVCL = 150: MAVZR = 50
MAVCL = 160: MAVZR = 40
MAVCL = 170: MAVZR = 30
MAVCL = 180: MAVZR = 20
MAVCL = 190: MAVZR = 10
```

Observe how the response goes more under-damped and ultimately goes unstable as the span of FCE output (MAVCL – MAVZR) is increased. However, by reducing the FCE action span, we shall also reduce the maximum magnitude of load or disturbance that the control system could handle. By using a smaller span of FCE action, we are in fact applying the ‘load splitting’ technique described in Chapter 8 in the context of

ON-OFF controllers.

Problem 14.16 Examine the close loop response with the Zeigler-Nichol's settings shown in the above by using the same program. Try the MAVCL and MAVZR values mentioned above and observe the closed loop response quality.

14.18 Systems with Multiple Control Loops

There are numerous examples to be found in the process industry where more than one variable is required to be controlled by employing that many number of control loops. To fulfill such control duties, subscripted versions of the subroutines for measuring element, comparator, controller and final control element have been written. These subroutines are presented in the following section. The subscript name NL stands for 'number of loop', and acts as the loop identifier.

```
2010 REM **************SUBROUTINE MEL2 ******************
     SLP1(NL) = ((CL(NL) – ZR(NL)) / (MVCL(NL) – MVZR(NL)))
     CVN(NL) = ((CVNN(NL) – MVZR(NL)) * SLP1(NL)
     CVO(NL) = ((CVON(NL) – MVZR(NL)) * SLP1(NL)
     SPN(NL) = ((SP(NL) – MVZR(NL)) * SLP1(NL)
     RETURN
     END
2030 REM ******************** SUBROUTINE CMP2 **********************
     ERN(NL) = SPN(NL) – CVN(NL): ERO(NL) = SPN(NL) – CVO(NL)
     RETURN
     END
2140 REM ******************** SUBROUTINE CNTR2 ************************
     REM ************* CONTROLLER MODE SELECTION ************
     ON (NMD(NL)) GOTO 2144, 2148, 2156, 2160
2144 REM ********************** [P] CONTROLLER **********************
     CO(NL) = OPB(NL) + ACN(NL) * KC(NL) * ERN(NL)
     IF CO(NL) > CL(NL) THEN CO(NL) = CL(NL)
     IF CO(NL) < ZR(NL) THEN CO(NL) = ZR(NL)
     GOTO 2170
2148 REM ********************** [PI] CONTROLLER ************************
     IF T = 0 THEN GOTO 2150
     COP(NL) = ACN(NL) * KC(NL) * ERN(NL)
     TI = (KC(NL) / TAUI(NL)) * (ERN(NL) + ERO(NL)) * (DT / 2)
     COI(NL) = COI(NL) + ACN(NL) * TI(NL)
     CO(NL) = OPB(NL) + COP(NL) + COI(NL)
     GOTO 2152
2150 CO(NL) = OPB(NL)
2152 IF CO(NL) > CL(NL) THEN CO(NL) = CL(NL)
     IF CO(NL) < ZR(NL) THEN CO(NL) = ZR(NL)
     GOTO 2170
2156 REM ******************** [PD] CONTROLLER ********************
     COP(NL) = ACN(NL) * KC(NL) * ERN(NL)
     COD(NL) = ACN(NL) * KC(NL) * TAUD(NL) * ((ERN(NL) – ERO(NL)) / DT)
     CO(NL) = OPB(NL) + COP(NL) + COD(NL)
```

```
      IF CO(NL) > CL(NL) THEN CO(NL) = CL(NL)
      IF CO(NL) < ZR(NL) THEN CO(NL) = ZR(NL)
      GOTO 2170
2160 REM ********************* [PID] CONTROLLER ***************
      IF T = 0 THEN GOTO 2162
      COP(NL) = ACN(NL) * KC(NL) * ERN(NL)
      TI = (KC(NL) / TAUI(NL)) * (ERN(NL) + ERO(NL)) * (DT / 2)
      COI(NL) = COI(NL) + ACN(NL) * TI(NL)
      COD(NL) = ACN(NL) * KC(NL) * TAUD(NL) * ((ERN(NL) – ERO(NL)) / DT)
      CO(NL) = OPB(NL) + COP(NL) + COI(NL) + COD(NL)
      GOTO 2164
2162 CO(NL) = OPB(NL)
2164 IF CO(NL) > CL(NL) THEN CO(NL) = CL(NL)
      IF CO(NL) < ZR(NL) THEN CO(NL) = ZR(NL)
      GOTO 2170
2170 RETURN
      END
2190 REM ********************* SUBROUTINE FCE2 ********************
      SLP2(NL) = ((MAVCL(NL) – MAVZR(NL)) / (CL(NL) – ZR(NL)))
      MAV(NL) = MAVZR(NL) + (CO(NL) – ZR(NL)) * SLP2(NL)
      RETURN
      END
```

To implement these subroutines, the main program should contain DIMENSION declarations for all the subscripted variables and subscripted constants required for controller operation. These are given as follows:

```
DIM CVNN(10), CVON(10), CVN(10), CVO(10), ERN(10), ERO(10), SP(10)
DIM NMD(8), ACN(10), OPB(10), KC(10), TAUI(10), TAUD(10), SPN(10)
DIM CL(10), ZR(10), MVCL(10), MVZR(10), MAVCL(10), MAVZR(10), MAV(10)
DIM CO(10), COP(10), COI(10), COD(10), TI(10), SLP1(10), SLP2(10)
```

Usage of these subroutines is demonstrated by considering the following examples.

EXAMPLE 14.26 Pre-processing of a liquid mixture of aromatic hydrocarbons, before the stream enters into a continuous reactor requires that the stream should be at a slightly higher pressure (30 mm H_2O column) and temperature (60°C). This slightly higher temperature is required to supply the activation energy required to commence the reaction, and as the formed products are near to saturated condition, the reactor operates at a +ve pressure. Hence, the reactant stream is injected at a pressure that is slightly above the reactor operating pressure. The pre-processing is done in a vertically situated cylindrical vessel with hemispherical ends which has a jacket on the lower half for heat transfer from the hot thermic-liquid, flowing through the jacket. Constant air pressure is applied to the vessel from an air manifold. Two control loops are associated with the pre-processing vessel:

1. A level loop to keep the vessel liquid at a desired value (210 cm from the bottom of the tank). This is done for keeping the liquid above the jacket level,

also so that the exit flow, Q_O, only depends on the exit line resistance, R_O. In fact R_O is the final control element of a flow loop of the reactor. Normal flow rates of the process and jacket liquids are Q_I = 240 l/m (4000 cm^3/s); Q_J = 120 l/m (2000 cm/s); R_O = 0.06 cm/(cm^3/s); the constant pressure head in the tank, H_{EPO} = 30 cm;

2. A temperature loop to keep the exit liquid temperature at 60°C. The tank dimensions and control systems are shown in Figure 14.38. Assuming the following dimensions, parameters and properties, write a program for simulating the control systems associated with the pre-processing tank:

Q_I = 4000 cm/s; Q_J = 2000 cm/s; L = 300 cm; L_J = 150 cm; D = 100 cm; D_J = 110 cm; r_L = 0.9 g/cm^3; Cp_L = 0.85 cal/(g/°C); r_J = 0.9 g/cm^3; Cp_J = 0.95 cal/(g/°C); U = 0.234 cal/(cm^2/°C/s). Temperatures of hot and cold liquid to the 3-way control valve are 180°C and 60°C respectively. The level and temperature loops will operate with set points, L_R = 210 cm; T_R = 60°C.

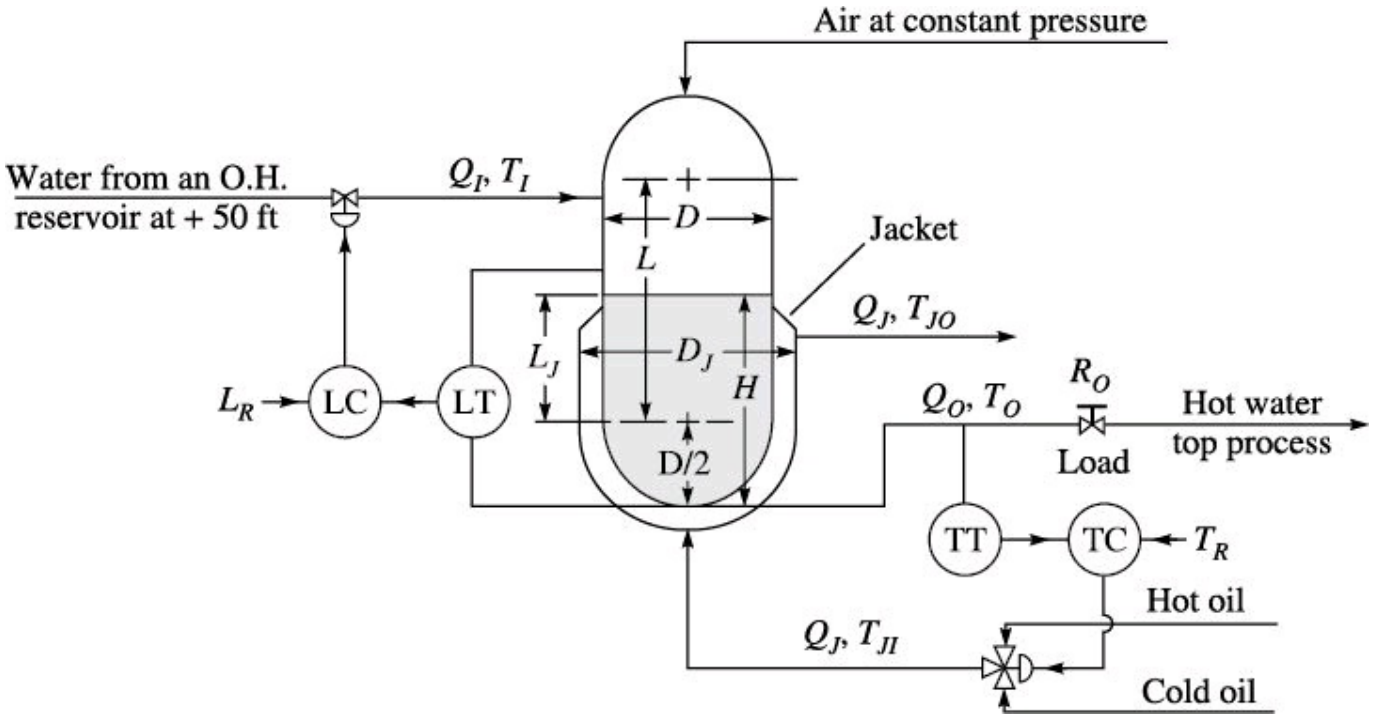


Figure 14.38 Dimensions and control loops of the pre-processing tank.

Solution: The following parameter values are calculated from the given dimensions:

$$V_T = \left(\frac{\pi}{6}\right)D^3 + \left(\frac{\pi L_J}{4}\right)D^2 = 1701686 \text{ cm}^3;$$

$$V_J = \left(\frac{\pi}{12}\right)(D_J^3 - D^3) + \frac{\pi L_J}{4}\cdot(D_J^2 - D^2) = 334055.8 \text{ cm}^3; R_O = 0.06 \text{ cm/(cm}^3\text{/s)}$$

$$A_T = \left(\frac{\pi}{4}\right)D^2 = 7853.975\text{cm}^2; A_H = \pi D L_J + \left(\frac{\pi}{2}\right)D^2 = 62831.8\ \text{cm}^2; H_{EPO} = 30\ \text{cm};$$

Assuming that during the transient period, the tank level will never reach the hemispherical bottom part of the tank and there is zero heat transfer between the system and surrounding, the lumped parameter model for the system may be derived from the following unsteady state principle entity balances given by the following set of equations:

Mass balance:

$$Q_O = \frac{H}{R_O};\ \rho_L Q_I - \rho_L Q_O = \rho_L A_T \frac{dH}{dt}$$

or

$$\frac{dH}{dt} = \frac{Q_I}{A_T} - \frac{Q_O}{A_T} \qquad (14.63)$$

Heat balance on the Tank:

$$Q_I \rho_L C p_L T_I - Q_O \rho_L C_{PL} T_O + U A_H (T_{JO} - T_O) = \rho_L C_{PL} \frac{d(V_L T_O)}{dt}$$

Simplifying, we obtain:

$$\frac{dT_O}{dt} = \frac{1}{A_T}(Q_I T_I - Q_O T_O) + \frac{U A_H}{\rho_L C_{PL} A_T H}(T_{JO} - T_O) - \left[\frac{Q_I - Q_O}{A_T H}\right] T_O \qquad (14.64)$$

Heat balance on the Jacket:

$$Q_J \rho_J C_{PJ} T_{JI} - Q_J \rho_J C_{PJ} T_{JO} - U A_H (T_{JO} - T_O) = \rho_J C_{PJ} V_J \frac{dT_{OJ}}{dt}$$

or,

$$\frac{dT_{OJ}}{dt} = \frac{Q_J}{V_J}(T_{JI} - T_{JO}) - \frac{U A_H}{V_J \rho_J C_{PJ}}(T_{JO} - T_O) \qquad (14.65)$$

Let us discuss the construction of the simulation program. At first, we assign loop number NL = 1 for the level loop, and NL = 2 for the temperature control loop. As there is always a constant pressure head of 30 mm of water column above the liquid, we shall use MVZR(1) = 30 mm and MVCL(1) = 430 mm water column. Also, for the temperature control, as the cold liquid and hot liquid temperatures are 180°C and 60°C, we shall use MAVZR(2) = 60°C and MAVCL(2) = 180°C. Thus, in this example, we demonstrate, how ‘zero-suppression’ feature of a measuring element and threshold value of a final control element can be incorporated in a simulation

program. Program C14-26.BAS is written for simulating the two loop control system where both controller modes are (PID) and the controller parameters given in the program listing are chosen with few trials. It should be recognized that the load and manipulated variables for the level loop are R_O and Q_I and that for temperature loop are T_I and T_{JI} respectively.

The system runs at zero error steady state with Q_I = 4000 cm^3/s, R_O = 0.06 $cm/(cm^3/s)$,
Q_J = 2000 cm^3/s, T_I = 30°C, T_{JI} = 120°C with the set points, L_R = 210 cm; T_R = 60°C. It can be seen that the closed loop response is obtained for both kinds of load changes:

1. by increasing R_O from 0.06 to 0.07 $cm/(cm^3/s)$, at t = 100 s and after initial transients when the response has settled,
2. T_I is changed from 30 to 40°C at t = 4000 s.

```
10    REM ***************** PROGRAM NAME -> C14-26.BAS *********************
      REM ***** CLOSED LOOP DYNAMICS OF A LIQUID HEATER UNDER PRESSURE *****
      DIM CVNN(8), CVON(8), CVN(8), CVO(8), ERN(8), ERO(8), SP(8)
      DIM NMD(8), ACN(8), OPB(8), KC(8), TAUI(8), TAUD(8), SPN(10)
      DIM CL(8), ZR(8), MVCL(8), MVZR(8), MAVCL(8), MAVZR(8), MAV(8)
      DIM CO(8), COP(8), COI(8), COD(8), TI(8), SLP1(8), SLP2(8)
      SCREEN 12
      XS = 50: YS2 = 350: YS3 = 200
      LINE (XS, YS2 + 10) – (XS + 400, YS2 – 110), , B
      LINE (XS, YS3 + 10) – (XS + 400, YS3 – 110), , B
      REM *************** PROCESS & OPERATIONAL PARAMETERS ************
      QI = 4000: QJ = 2000: D = 100: L = 300: LJ = 150: DJ = 110: HO = 210: AT = 7853.975
      AH = 62831.8: VT = 1701695: VJ = 334055.8: VL = 1518435: AE = 1701695
      HEPO = 30: RO = .06
      REM *************** INITIAL VALUES ******************
      HO = 210: TOO = 60: TJOO = 66.34
      VL = 1518435: AE = 1701695
      QI = 4000: QJ = 2000: D = 100: L = 300: LJ = 150: DJ = 110: HO = 210: AT = 7853.975
      AH = 62831.8: VT = 1701695: VJ = 334055.8: VL = 1518435: AE = 1701695
      HJ = 200
      REM ****************** INITIAL VALUES ******************
      TOO = 60: HO = 210: TJOO = 66.34: PI = 3.14159
      VL = (PI / 4) * D * D * (HO – D / 2) + (PI / 12) * D * D * D: AE = VL / HO
      REM ****************** CONTROLLER PARAMETERS *********************
      ACN(1) = 1: OPB(1) = 50: KC(1) = 1: TAUI(1) = 8: TAUD(1) = 2
      CL(1) = 100: ZR(1) = 0: MVCL(1) =430: MVZR(1) =30: NMD(1) =4
      SP(1) = 210: MAVCL(1) = 8000: MAVZR(1) = 0: MAV(1) = 4000
      ACN(2) = 1: OPB(2) = 50: KC(2) = 1: TAUI(2) = 8: TAUD(2) = 2
      CL(2) = 100: ZR(2) = 0: MVCL(2) =120: MVZR(2) =0: NMD(2) =4
      SP(2) = 60: MAVCL(2) = 180: MAVZR(2) = 60: MAV(2) = 120
      QI = MAV(1): TJI = MAV(2): TI = 30
```

```
      REM ************ TIME LOOP PARAMETERS *****************
      T = 0: DT = .1: TF = 8000
      REM ************ THE TIME LOOP ********************
50    IF T > TF THEN GOTO 80
      IF T > 200 THEN TI = 40
      IF T > 4000 THEN RO = .07
      QO = (HO + HEPO) / RO
      DHDT = QI / AT – QO / AT
      HN = HO + DHDT * DT
      C1 = 1 / (AT * HO)
      C2 = (U * AH) / (AT * HO * ROL * CPL)
      C3 = (QI – QO) / (AT * HO)
      DTODT = C1 * (QI * TI – QO * TOO) + C2 * (TJOO – TOO) – C3 * TOO
      TON = TOO + DTODT * DT
      C4 = QJ / VJ: C5 = (U * AH) / (VJ * ROJ * CPJ)
      DTJODT = C4 * (TJI – TJOO) – C5 * (TJOO – TOO)
      TJON = TJOO + DTJODT * DT
      CVNN(1) = HN: CVON = HO
      NL = 1: GOSUB 2010: GOSUB 2030: GOSUB 2140: GOSUB 2190: QI = MAV(1)
      CVNN(2) = TON: CVON(2) = TOO
      NL = 2: GOSUB 2010: GOSUB 2030: GOSUB 2140: GOSUB 2190: TJI = MAV(2)
      REM ***************** OUTPUT GENERATION ***********************
      PSET(XS + T * .05, YS3 – 50 – (SP(1) – 210) * 10)
      PSET(XS + T * .05, YS3 – 50 – (HN – 210) * 10)
      PSET(XS + T * .05, YS3 – (RO – .06) * 1000)
      PSET(XS + T * .05, YS2 – 50 – (SP(2) – 60) * 10)
      PSET(XS + T * .05, YS2 – 50 – (TON – 60) * 10)
      PSET(XS + T * .05, YS2 – (TI – 30) * 1)
      REM ***************** OLD <- NEW VALUE TRANSFER ***************
      HO = HN: TJOO = TJON
      T = T + DT
      GOTO 50
80    GOTO 5000
2010 REM ***************** SUBROUTINE MEL2 ********************
2030 REM ***************** SUBROUTINE CMP2 ********************
2040 REM ***************** SUBROUTINE CNTR2 ********************
2190 REM ***************** SUBROUTINE FCE2 ********************
5000 END
```

Results after executing the program are recorded in Figure 15.39. The lower part of the figure shows the effect of load change of inlet liquid temperature at t = 100 s, followed by

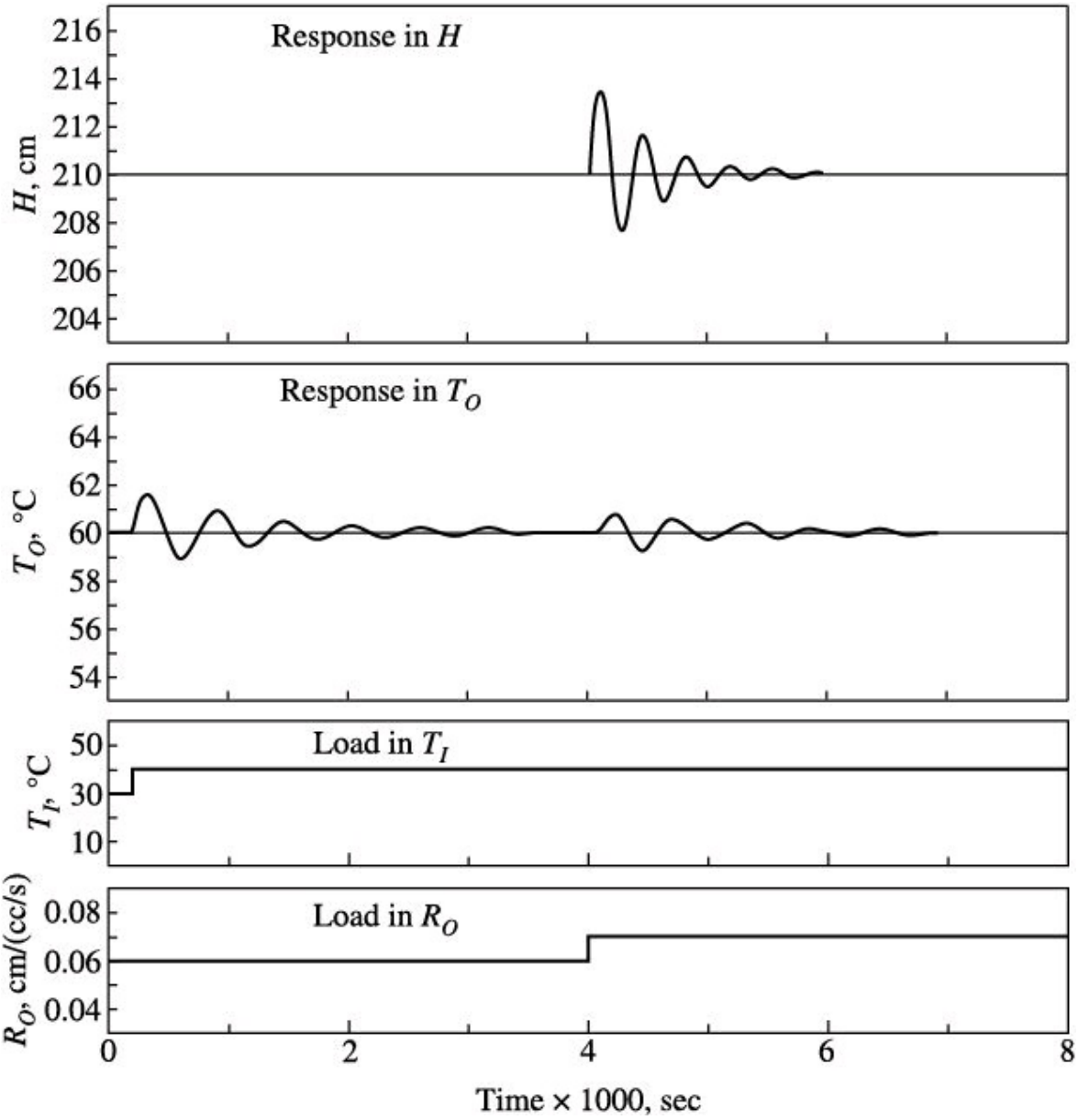


Figure 14.39 The load changes and corresponding close loop responses of the multi-loop control system.

change in the exit resistance at t = 4000 s. Corresponding responses in tank level and exit liquid temperature are shown in the top two traces. A noteworthy feature of these responses is that for the inlet liquid temperature change, understandably the exit liquid temperature changes, but there is no change in level response. But, on the other hand, corresponding to an exit flow resistance change, the level as well as exit liquid temperature shows under-damped response before settling. We can very easily put a physical explanation of this phenomenon by mentioning that by oscillations in level, the total thermal capacity of the tank liquid varies which causes this exit liquid temperature fluctuation. However, in control vocabulary, in a multiple loop system when the control action in one of the loops (due to a disturbance) affects the controlled variable/s of the other loop/s, the phenomenon is called 'Interaction'.

We have many more things to talk about this property of multi-variable control system known as interaction, and how to remove or at least reduce its detrimental influence on control quality. These discussions are included in Chapter 16.

14.19 Control Valve

In the examples that are discussed so far, the flow manipulation situations are modeled by assuming a linear relation between controller output and flow rate obtained from the FCE. For more critical applications, an exact form of such relations are presented in the following section. The flow, Q, through the valve is expressed as:

$$Q = F_O C_V f(\Delta P) \tag{14.66}$$

The constant, C_V, is called the valve coefficient, and refers to the flow rate through the valve in fully open condition for 1psi pressure drop across the valve; F_O is the fractional opening of the valve; and $f(\Delta P)$ is a pressure function, its exact form depends on phase of the fluid and sometimes on the up and downstream pressure levels. Certain typical forms of the above equation are given below:

$$Q = F_O C_V \sqrt{P_1 - P_2} \quad \text{(for liquids)} \tag{14.67}$$

$$Q = F_O C_V \sqrt{P_M (P_1 - P_2)} \quad \text{(for gases under sub-critical flow)} \tag{14.68}$$

$$Q = 0.85 F_O (C_V / \sqrt{2}) P_1 \quad \text{(for gases under critical flow)} \tag{14.69}$$

where, P_1 = Upstream pressure, P_2 = Downstream pressure, $P_M = (P_1 + P_2)/2$, the average pressure. The critical flow condition is defined as, when $P_2 \leq 0.53 P_1$.

It only remains to relate fractional opening, F_O, to the controller output, CO. This relation is the so-called valve characteristic equation. Such relations are available for:

1. Linear;
2. Equal percent; and
3. Parabolic valves as given in Appendix-III.

The total calculations can be used in the form of a subroutine that can work along with other control hardware subroutines.

Problem 14.17 Write a subroutine for simulating a control valve having an equal percent characteristic. (See Chapter 8 and Appendix-III.)

14.20 Frequency Response Plotting

From a given transfer function, the frequency response plots (Bode's, Nyquist and Nichols) can be obtained as graphic output from a program. For Bode plotting, the log-log or semi-log graphs have to be drawn as the background grid. This facility should be within the program and has to be executed before drawing the frequency response trace. First, we shall discuss about drawing of log-log and semi-log graph,

then we shall introduce Bode, Nyquist and Nichols Plots drawing programs.

Log-log plots are required when frequency, *w*, versus amplitude ratio, AR, traces are drawn. In that case, for plotting *w* versus phase angle, □, a separate semi-log plot is necessary. But by converting *AR* to dB, a single semi-log plot may contain both dB and □ locus with the common *w* values on the log-scale as abscissa. Hence, semi-log graphs are most widely used for expressing and analyzing frequency response information. However, beside semi-log graph, we are presenting the program for drawing of log-log graph, just in case, if it is required for some special problem.

14.20.1 Program for Drawing of a Log-log Graph

The parameters to be specified in the program are:

1. CYLN = the length of one cycle or decade in pixel value;
2. NXF, NYF = these are the number of cycles that are required to be in the graph in X and Y direction respectively. If SCREEN 12 is used, at least a CYNL = 60 should be used to obtain a reasonable accuracy of the grid. These two values will be reported at the top of the graph to facilitate labeling during post editing of the graph.

A program for drawing log-log graph is as follows:

```
10    REM ********** PROGRAM NAME -> C14-F1A.BAS ***********
      SCREEN 12
      XS = 50: YS = 400: CYLN = 60
      NXI = 0: NXF = 3: NX = NXI
      NYI = 0: NYF = 3: NY = NYI
      15 IF NX > NXF THEN GOTO 50
      X = 1
20    IF X > 10 THEN GOTO 40
      DXT = CYLN * (NX + (LOG(X) / 2.303))
      LINE (XS + DXT, YS) – (XS + DXT, YS – CYLN * (NYF + 1)), 8
      X = X + 1
      GOTO 20
40    NX = NX + 1
      GOTO 15
50    IF NY > NYF THEN GOTO 70
      Y = 1
55    IF Y > 10 THEN GOTO 60
      DYT = CYLN *(NY + (LOG(Y) / 2.303))
      LINE (XS, YS – DYT) – (XS + CYLN * (NYF + 1), YS – DYT), 8
      Y = Y + 1
      GOTO 55
60    NY = NY + 1
      GOTO 50
      70 LINE(XS, YS) – (XS + CYLN * (NXF + 1), YS – CYLN * (NYF + 1)), B
5000 END
```

The result of the above program after execution is shown in Figure 14.40(a).

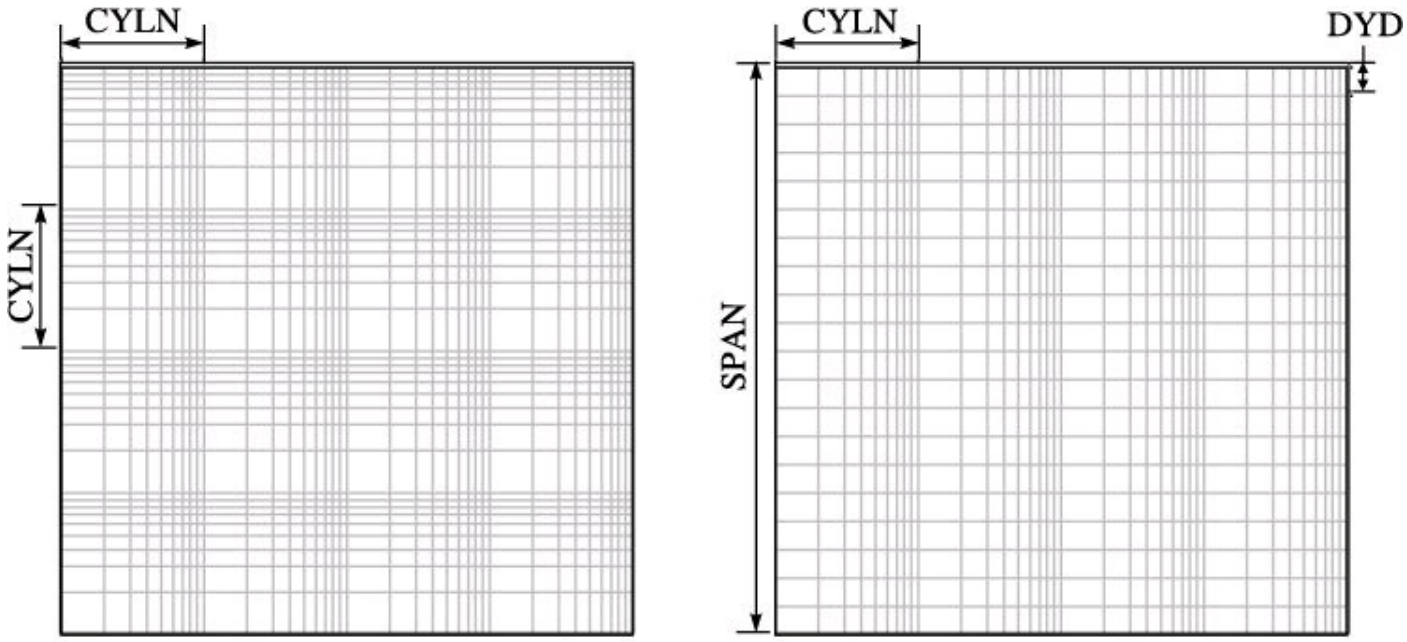


Figure 14.40 Sample of (a) Log-Log graphs. (b) Semi-log graphs.

14.20.2 Program for Drawing of a Semi-log Graph

The parameters CYLN and NXF are again required carrying an identical significance as in case of log-log graph. For drawing the linear scale for depicting of dB and □ on opposite sides of *y*-scale, following parameters are required:

1. SPAN – Total length of y-axis in pixel value;
2. NDIV – Number of divisions in the y-axis.

A program for drawing semi-log graph is given below:

```
110  REM ********** PROGRAM NAME -> C14-F1B.BAS ***********
     SCREEN 12
     XS = 350: YS = 400: CYLN = 60
     NXI = 0: NXF = 3: NX = NXI
     SPAN = 240: NDIV = 20: DYD = SPAN / NDIV
115   IF NX > NXF THEN GOTO 140
     X = 1
120   IF X > 10 THEN GOTO 130
     DXT = CYLN * (NX + (LOG(X) / 2.303))
     LINE (XS + DXT, YS) – (XS + DXT, YS – SPAN), 8
     X = X + 1
     GOTO 120
130  NX = NX + 1
     GOTO 115
140  FOR Y = 0 TO SPAN STEP DYD
     LINE (XS, YS – Y) – (XS + CYLN * (NXF + 1), YS – Y), 8
     NEXT Y
     LINE (XS, YS) – (XS + CYLN * (NXF + 1), Y – SPAN), B
5000 END
```

The result of the above program after execution is shown in Figure 14.40(b).

14.20.3 Drawing the Bode Plot on a Semi-log Graph

Beside the three CYLN, SPAN and NDIV, the following parameters are also required for this graphic output:

3. DBI – initial value of dB at the top of *y*-scale (say, on the left hand side of the graph);
4. DBF – end value of dB at the bottom of *y*-scale;
5. PHII – initial value of □ at the top of *y*-scale (say, on the right hand side of the graph);
6. PHIF – end value of □ at the bottom of *y*-scale.

The reader should select the values of SPAN, ΔdB (= dBF – dBI) and Δ□ (= PHIF – PHII) in a way so that SDD (dB per division on *y*-scale = ΔdB/NDIV) and SDP (□ per division of *y*-scale = Δ□/NDIV) are two integer numbers.

It is also convenient to draw on the graph the horizontals corresponding to DB = 0 and □ = –180°, as we know for further information processing. By the following program lines for executing the two LINE statements, the horizontals may be drawn in 'blue' and 'red' respectively.

```
YDO = (SPAN / (DBI – DBF)) * (0 – DBF)
YPPI = (SPAN / (PHII – PHIF)) * (– 180 – PHIF)
LINE (XS, YS – YDO) – (XS + NX * CYCL, YS – YDO), 11
LINE (XS, YS – YPPI) – (XS + NX * CYCL, YS – YPPI), 12
```

The Frequency Loop

Just as in for plotting of time domain response we used a time loop, here for frequency response, we shall apply a frequency loop. To have equal spacing between plotted data on a log scale, the frequency will be increased according to a Geometric Progression (GP) instead of Arithmetic Progression (AP) as we did in a time loop. Let the number of points to be plotted per cycle or decade be NP, then the increment ratio per loop traverse would be:

```
DMG = 1/(10 ^ NP)
```

The frequency loop starts as:

```
    OMGI = ……: DMG = ……: OMGF = ……: W = OMGI
50 IF W > OMGF THEN GOTO 100
    |-----------------------------------------------|
    | Body of the program containing |
    | all the executable statements for |
    | each frequency value |
    |-----------------------------------------------|
    W = W * DMG
    GOTO 50
100 END
```

In the body of the program there will be statements for calculating *AR*, *DB* and □

for each value of w (W) dictated in the frequency loop, followed by the graphic plotting statement (PSET). The *AR*, *DB* and □ are calculated for each type of dynamic blocks identified in the given transfer function. The possible varieties and the calculation statements are given in the following list.

Type	TF	Statements
1. Gain	K_P	AK = KP: DBK = FD * (LOG(KP)) PHK = 0
2. First order pole	$1/(T_P s + 1)$	ARP = 1/(SQR(1 + (TP * TP * W * W)) DBP = FD * (LOG(ARP)) PHP = FA * (ATN(– TP * W))
3. First order zero	$(T_Z s + 1)$	ARZ = SQR(1 + (TZ * TZ * W * W)) DBZ = FD * (LOG(ARZ)) PHZ = FA * (ATN(TZ * W))
4. Two complex poles	$1/(T^2 s^2 + 2zTs + 1)$	T1 = (1 – TPCM * TPCM * W * W) ^ 2 T2 = (2 * ETA * TPCM * W) ^ 2 APCM = 1/(SQR(T1 + T2) DPCM = FD * (LOG(APCM)) ANUM = (– 2 * ETA * TPCM * W) ADNM = (1 – TPCM * TPCM * W * W) PHPCM = FA * (ATN(ANUM/ADNM)

[The phase angle of poles or zeros are calculated by ATN [arctan()] or $\tan^{-1}()$ function. If the argument of this function varies from 0 to $-\infty$ when w increases from 0 to ∞ as in the case of a first order pole, the resulting □ goes 0 to – 90°. But in case of an inherently second order system involving two complex poles, when the argument after reaching $-\infty$, progressively becomes slightly less than $-\infty$ (in the third quadrant), the resulting □ becomes +ve, that increases to +90° as the argument goes $-\infty$ to 0. This discontinuity is a property of the $\tan^{-1}()$ function. This discontinuity is not consistent with the physical behaviour of an inherently second order pole that shows □ to go from 0 to –180° as w increases from 0 to ∞. To avoid this problem, the following program segment (in bold) has to be used:

```
PHPCM = FA * (ATN(…))
IF PHPCM > 0 THEN PHPCM = – 90 – PHPCM
```

Note that this problem does not exist in case of a second order system comprising of two first order systems in series.]

5. Dead time	e^{-TD}s	ADT = 1: DBDT = 0 PHDT = – (TDL * FA * W)
6. Integrator (first order)	$1/s$	API = 1/W DPI = FD * (LOG(API)) PHPI = –90
7. Differentiator (first order)	s	APD = W DPD = FD * (LOG(APD)) PHPD = 90
** In the above list FD = 20/2.303, and FA = 180/3.14159.		

After the required calculation statements, the next step are to calculate total or

overall values of *AR*, *DB* and PHI for the given TF, say for a process TF,

$$G_P(s) = \frac{K_P e^{-TDs}}{(TPs + 1)}$$

The calculation steps are:

```
AR = AK * ADT * ARP
DB = DBK + DBT + DBP
PHI = PHK + PHDT + PHP
```

Let us consider the case of depicting *DB* and PHI on the left and right hand sides on *y*-axis with W (*w*) on the *x*-axis as the independent variable. To display these values as points, first the location of zeros are calculated as:

```
YDO = (SPAN / (DBI – DBF)) * (0 – DBF)
YPPO = (SPAN / (PHII – PHIF)) * (0 – PHIF)
```

Next, the scale factors for plotting *DB* and PHI are calculated as:

```
YFDB = SPAN / ABS (DBF – DBI)
YFPH = SPAN / ABS(PHIF – PHII)
```

For the frequency scale the scale factor per CYLN will be:

```
XDS = (LOG(W / OGI)) / 2.303
```

The complete graphic plotting statements are:

```
XDS = (LOG(W / OGI)) / 2.303
PSET (XS + CYLN * XDS, YS – YDO – DB * YFDB), 11
PSET (XS + CYLN * XDS, YS – YPO – PHI * YFPH), 12
```

Let us solve an example to demonstrate the procedure that we have presented for drawing the Bode Plot.

EXAMPLE 14.27 The transfer function of a process is expressed as:

$$G_P(s) = \frac{K_P(\tau_Z s + 1)e^{-\tau_{DL}s}}{(\tau_{P1}s + 1)(\tau_{P2}s + 1)}$$

where, $K_P = 5$, $t_Z = 1$, $t_{P1} = 0.1$, $t_{P2} = 10$, $t_{DL} = 0.05$

Write a program for construction of Bode Plot for the process.

Solution: The program C14-F2.BAS has been written following the above discussion. The maximum and minimum value of *w* scale is chosen so that all the characteristic frequencies are included. The value of SPAN and maximum, minimum values of *DB* and □ scale is chosen from the knowledge of contribution of *DB* and □ of the individual units in the $G_P(s)$.

```
10    REM ************** PROGRAM NAME -> C14-F2.BAS ************
      SCREEN 12
      XS = 50: YS = 400: CYLN =80
      NXI = 0: NXF = 3: NX = NXI
      SPAN = 240: NDIV = 20: DYD = SPAN / NDIV
```

```
    DBI = 20: DBF = – 80: PHII = 30: PHIF = – 270
    YFDB = SPAN / ABS (DBF – DBI): YFPH = SPAN / ABS(PHIF – PHII)
    REM ********** DRAWING THE SEMI-LOG GRAPH **********
    LINE (XS – 10, YS + 10) – (XS + 330, YS – 250), 15, BF
15   IF NX > NXF THEN GOTO 40
    X = 1
20   IF X > 10 THEN GOTO 30
    DXT = CYLN * (NX + (LOG(X) / 2.303))
    LINE (XS, DXT, YS) – (XS + DXT, YS – SPAN), 8
    X = X + 1
    GOTO 20
30   NX = NX + 1
    GOTO 15
40   FOR Y = 0 TO SPAN STEP DYD
    LINE (XS, YS – Y) – (XS + CYLN * (NXF + 1), YS – Y), 8
    NEXT Y
    LINE (XS, YS) – (XS + CYLN * (NXF + 1), YS – SPAN), 0, B
    REM ********** DRAWING THE DB=0 & PHI= –180 HORIZONTALS *********
    YDO = (SPAN / (DBI – DBF)) * (0 – DBF)
    YPPI = (SPAN / (PHII – PHIF)) * (– 180 – PHIF)
    LINE (XS, YS – YDO) – (XS + CYLN * (NXF + 1), YS – YDO), 1
    LINE (XS, YS – YPPI) – (XS + CYLN * (NXF + 1), YS – YPPI), 4
    REM ************ PROCESS PARAMETERS **********
    KP = 5: TZ = 1: TP1 = .1: TP2 = 10: TD = .05
    FD = 20 / 2.303: FA = 180 / 3.14159
    OGI = .01: NP = 500: DMG = 10 ^ (1 / NP): OGF = 10: W = OGI
    YFDB = SPAN / ABS (DBF – DBI): YFPH = SPAN / ABS(PHIF – PHII)
    REM ************ THE FRQUENCY LOOP **********
50   IF W > OGF THEN GOTO 5000
    REM ************ CALCULATION OF AR, DB & PHI **********
    ARK = KP: DBK = FD * (LOG(ARK)): PHK = 0
    ARZ = SQR(1 + TZ * TZ * W * W): DZ = FD * (LOG(ARZ))
    PHZ = FA * (ATN(W * TZ))
    ARP1 = 1 / (SQR(1 + TP1 * TP1 * W * W)): DP1 = FD * (LOG(ARP1))
    PHP1 = FA * (ATN(– TP1 * W))
    ARP2 = 1 / (SQR(1 + TP2 * TP2 * W * W)): DP2 = FD * (LOG(ARP2))
    PHP2 = FA * (ATN(– TP2 * W))
    ARDT = 1: DBDT = 0: PHDT = – (FA * TD * W)
    REM ********** TOTAL DB & PHI **********
    DB = DBK + DZ + DP1 + DP2 + DBDT
    PHI = PHK + PHZ + PHP1 + PHP2 + PHDT
    REM ********** PLOTTING THE RESULT **********
    XDS = (LOG(W / OGI)) / 2.303
    PSET (XS + CYLN * XDS, YS – YDO – DB * YFDB), 1
    PSET (XS + CYLN * XDS, YS – YPO – PHI * YFPH), 4
    W = W * DMG
    GOTO 50
5000 END
```

After running the program, the graphic result is depicted in Figure 14.41.

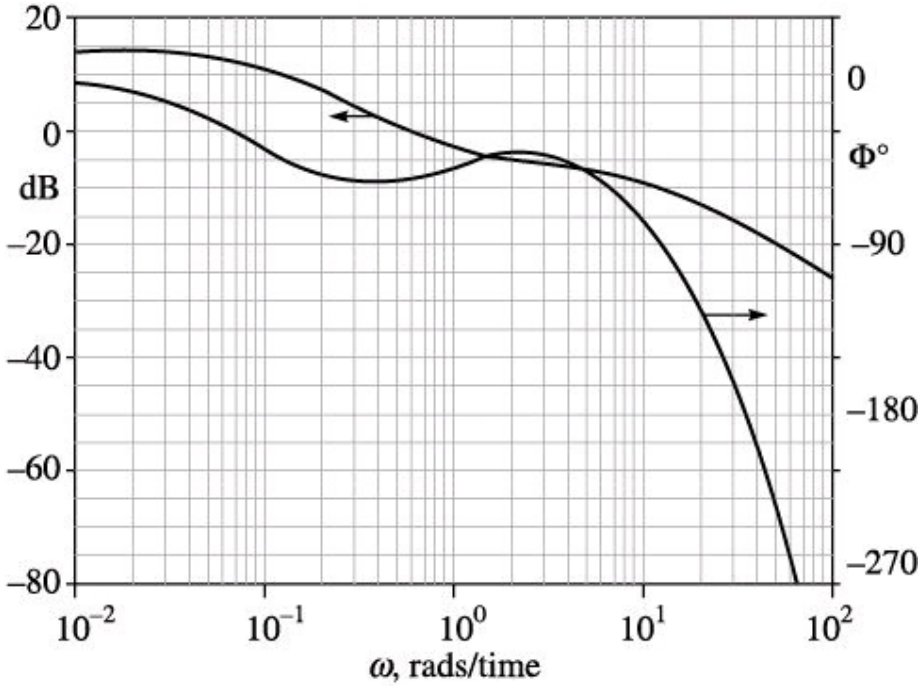


Figure 14.41 The Bode plot.

Problem 14.18 Determine the values of w_{CO}, K_U and P_U from the Bode Plot of Figure 14.41.

Problem 14.19 Use program C14-F2.BAS to draw the Bode Plot for the process represented by the following TF:

$$G_P(s) = \frac{2.5e^{-0.5s}}{10s + 1}$$

14.20.4 Drawing of Nyquist Plot

For the Nyquist Plot, a linear graph showing all four quadrants of the complex *s*-plane is prepared by two straight lines, one horizontal and one vertical to represent real and imaginary axes respectively. But as we have seen in Chapter 11, the maximum manifestation of Nyquist plots occurs in the third and fourth quadrants, it is justified to leave a larger area for these two quadrants in the graph. The frequency loop and calculation steps for finding the overall *AR* and PHI are identical to those in Bode Plotting program. After that, the coordinates of the end point of the response vector, corresponding to a unit input vector is found by the following statements:

```
X = AR * COS(PHI / FA): Y = AR * SIN(PHI / FA)
```

and the plotting statement will be:

```
PSET(XS + X * XF, YS – Y * YF)
```

where, XS and YS are coordinates of the origin of complex plane, and XF, YF are the scale factors in *x*- and *y*-direction respectively. The complete locus is obtained as W (*w*) varies from OGI (w_I) to OGF (w_F).

EXAMPLE 14.28 Write a program to obtain the Nyquist Plot of a process expressed

by the following TF:

$$G_P(s) = \frac{K_P}{(\tau_{P1}s + 1)(\tau_{P2}s + 1)(\tau_{P3}s + 1)}$$

where, $K_P = 5$, $t_{P1} = 1$, $t_{P2} = 2$, $t_{P3} = 3$

Solution: The program for drawing the Nyquist Plot is named as C15-F3.BAS. The listing of the program is shown below. Locate the –1, 0 point on the –ve real axis in the resulting plot.

```
10    REM *************** PROGRAM NAME -> C14-F3.BAS *************
      SCREEN 12
      XS = 220: YS = 200: XF = 25: YF = 25
      LINE (XS, YS + 100) – (XS, YS – 100)
      LINE (XS + 200, YS) – (XS – 200, YS)
      CIRCLE (XS – 1 * XF, YS), 2
      REM ************ PROCESS PARAMETERS ********************
      KP = 5: TP1 = 1: TP2 = 2: TP3 = 3
      FD = 20 / 2.303: FA = 180 / 3.14159
      OGI = .001: NP = 500: DMG = 10 ^ (1 / NP): OGF = 100: W = OGI
      REM ************* THE FRQUENCY LOOP ************************
50    IF W > OGF THEN GOTO 5000
      REM ************** CALCULATION OF AR & PHI ****************
      ARK = KP: PHK = 0
      ARP1 = 1 / (SQR(1 + TP1 * TP1 * W * W)): PHP1 = FA * (ATN(– TP1 * W))
      ARP2 = 1 / (SQR(1 + TP2 * TP2 * W * W)): PHP2 = FA * (ATN(– TP2 * W))
      ARP3 = 1 / (SQR(1 + TP3 * TP3 * W * W)): PHP3 = FA * (ATN(– TP3 * W))
      REM *************** TOTAL AR & PHI *******************
      AR = ARK + ARP1 + ARP2 + ARP3
      PHI = PHK + PHP1 + PHP2 + PHP3
      REM ************** PLOTTING THE RESULT *************
      XDS = AR * COS(PHI / FA): YDS = AR * SIN(PHI / FA)
      PSET (XS + XDS * XF, YS – YDS * YF)
      W = W * DMG
      GOTO 50
5000 END
```

The result of the program is shown in Figure 14.42.

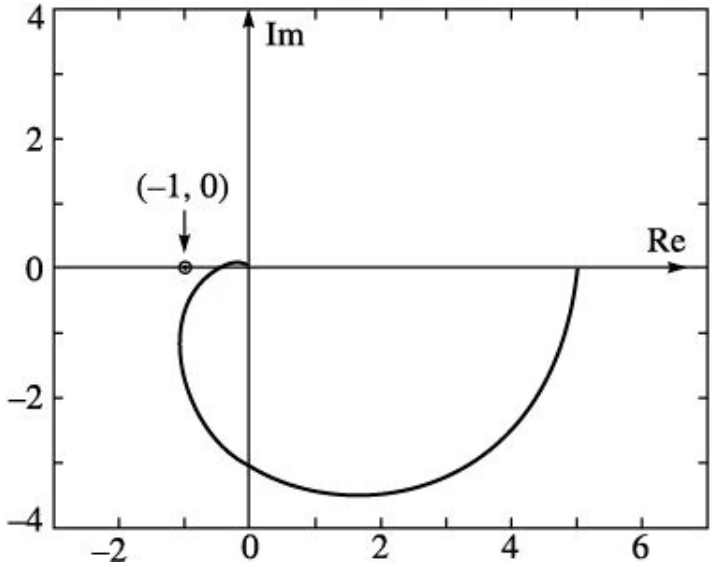


Figure 14.42 The Nyquist plot.

Problem 14.20 Use program C14-F3.BAS to find the value of w_{CO} and K_U for the above system of Example 14.28.

14.20.5 Drawing of Nichol's Plot

Nichol's Plot is also drawn on a linear graph. The program steps for initial settings and the structure of frequency loop and the calculations of *DB* and PHI are identical to those in Bode Plotting program. After finding *DB* and PHI, the plotting statement is:

```
PSET(XS + PHI * XF, YS – DB * YF)
```

EXAMPLE 14.29 Write a program for drawing of Nichol's Plot for the following second order system represented by the TF,

$$G_P(s) = \frac{K_P}{\tau^2 s^2 + 2\zeta\tau s + 1}$$

Plot for three values of the damping coefficient, z = 0.2, 1.0 and 1.8.

Solution: The program C14-F4.BAS is given in the following listing:

```
10    REM ************** PROGRAM NAME -> C14-F4.BAS ************
      SCREEN 12
      XS = 220: YS = 200: XF = 25: YF = 25
      LINE (XS, YS – 40) – (XS – 200, YS + 200), , B
      ETA = .2
20    IF ETA > 1.8 THEN GOTO 5000
      REM ************ PROCESS PARAMETERS ********************
      KP = 1: TPCM = 1
      FD = 20 / 2.303: FA = 180 / 3.14159
      OGI = .01: NP = 500: DMG = 10 ^ (1 / NP): OGF = 200: W = OGI
      REM ************* THE FRQUENCY LOOP ***********************
50    IF W > OGF THEN GOTO 70
      REM ************** CALCULATION OF AR & PHI ***************
      T1 = (1 – TPCM * TPCM * W * W) ^ 2
```

```
      T2 = (2 * ETA * TPCM * W) ^ 2
      APCM = 1 / (SQR(T1 + T2))
      DPCM = FD * (LOG(APCM))
      NUM = – 2 * ETA * TPCM * W
      DNM = 1 – TPCM * TPCM * W * W
      PHPCM = FA * (ATN(NUM / DNM))
      IF PHPCM > 0 THEN PHPCM = – 180 + PHPCM
      DB = DBK + DPCM
      PHI = PHPCM
      PSET (XS + PHI * XF, YS – DB * YF)
      W = W * DMG
      GOTO 50
70    ETA = ETA + .8
      GOTO 20
5000 END
```

Result of the program after execution is shown in Figure 14.43.

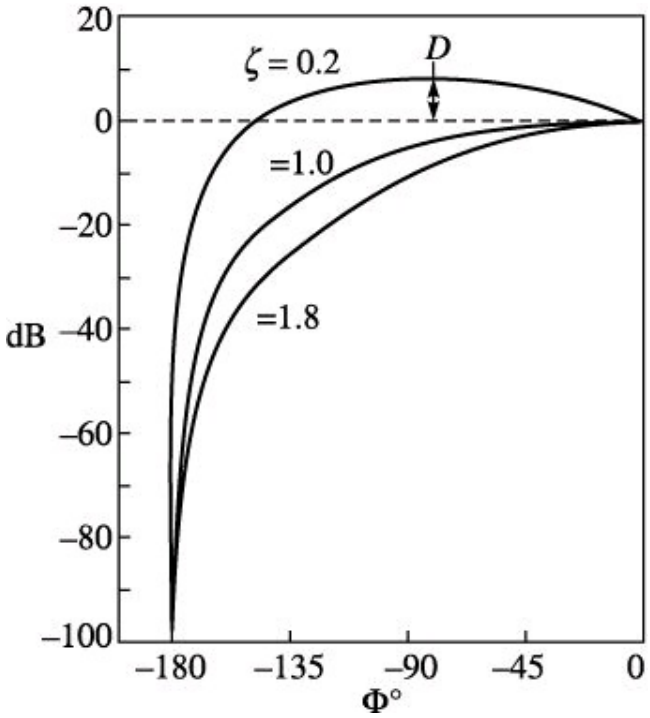


Figure 14.43 The Nichol's plot of a second order system under three values of damping coefficient.

Problem 14.21 In the above figure, the 'bulge' of *DB* value over 0-*DB*, measures 'D'. Determine the overshoot from this measure and check this value corresponding to that obtained from the equation of overshoot, given in Chapter 6, for $z = 0.2$.

The programs shown in this chapter are contained in the CD accompanying the text. The reader should run these programs, and try to modify them and attempt to solve problems presented in this chapter by writing new ones. With such knowledge acquired, the reader may try to build up programs for solving the problems given in chapters 9 through 13.

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15

Advanced Control Strategies

It has been recognized that feedback control for many processes may produce poor performance due to certain specialties in their dynamic character, or unpredictable quality variation of support systems necessary to maintain the control action. The feedback control fails due to existence of anomalies in the fundamental premises on which the feedback logic is built up, that is on the backbone of linear analysis. These special features appear as an unpleasant surprise to the designer, because feedback control offers a poor performance for such applications with respect to either load or set point change. To cater for such needs, certain control strategies have come up through long years of research and practice. According to their time of development and launching in industrial practice, some of them are relatively older than the others, but all of them have appeared after feedback control. In the present chapter we shall discuss some of the more important categories of this class of control strategies. But before that, it would be justified to know about the type of specialties in the process dynamics and/or the utilities that are used in implementing the control action for which the feedback control fails because of which such advanced strategies are required to be applied.

15.1 Dynamic Specialities Causing Control Difficulties

A list of some of the more important classes of these dynamic specialities or anomalies have been presented as follows:

1. Large values of Transfer or Transportation Lag
2. Non-linear character of the process
3. Difficulty in manipulating the process variable
4. Unpredictable potential fluctuation of utilities that have to be used as the manipulated variable source
5. Unpredictable or time dependent variation of process parameters
6. Integration in the process characteristics.
7. Inverse response characteristics of the process
8. Difficulty in measuring the controlled and other variables of the process
9. Non-stationary, time scheduled set point tracking or programmed control of process

10. Limited knowledge about the dynamic model of the process

In this chapter we shall discuss some of the more important categories of advanced control strategy that are theoretically established as answers to the above dynamic difficulties, and are successfully used in industrial practice. It should be noted that the list of the topics discussed here is not complete by any means, but will increase with the growing social and economic demands to the control system.

15.2 Feed Forward Control

Erlier we have discussed the analysis and design of feedback control, and treated the subject up to a definite depth. Feed forward strategy embodies a control philosophy that may be thought as the first new viewpoint or approach for dynamically compensating a process, which is categorically different from the feedback scheme. In a nut-shell, the essence of this strategy may be expressed by the famous phrase:

"Prevention is better than cure."

It is always better to take anticipatory corrective action as soon as or before the disaster strikes than to wait until it has affected the total system and the symptoms have become detectable. The feed forward action may be equated to the preventive measure, and the cure becomes parallel to feedback strategy. Historically, this kind of activity is nothing new, to store or hoard foodstuff or crops as a safety against a drought, or inoculation against possible epidemics are the preventive actions that definitely have an essence of feed forward philosophy.

However, there is no anticipatory part in control system application. Feed forward action can appear only when a disturbance is detected at the entry of a process as a change in load variable. Obviously, the corresponding control action will enter through regulation of the manipulated variable.

The next question is what are the categories of process for which, feed forward control would produce a better closed loop performance than feed back control. We may state that a process with a large capacity or connected multiple capacities or any other form of *lag*, if disturbed, requires a considerable amount of time before the sensor situated at the process output would be aware of the disturbance. By the time the controller takes action, the whole process may be *contaminated* (affected). Thus, under feedback control, the output from such a process may remain 'off-quality' for a long time. In many applications this condition is highly undesirable.

(*If we consider the steady state operation of a control system, say the feed forward temperature control of the exit liquid flowing at a constant rate from an ideally stirred mixer-heater, the compensation required for a sudden change of inlet liquid temperature may be calculated by a simple heat balance around the mixer. The feed*

forward compensation would finally reach this value if the inlet liquid temperature remains at its new value long enough. By performing such a steady state calculation for maximum possible disturbance magnitude, we may have a good estimate of FCE sizing and the gain value of the final control element. In certain literatures such an analysis has been called the design of feed forward controller under steady state condition.)

In feed forward control compensation for a disturbance is produced as it enters the process. The disturbances are measured at the entry point of the process by the measuring element G_{MFF} and sent to the **Feed Forward Controller** G_{CFF}. This controller calculates the required value of manipulation to maintain the controlled variable at its set point. A feed forward control loop is shown in Figure 15.1. The control action is done by the FCE of the loop. It is obvious that such a controller requires a prior knowledge about the dynamics of the process in the form of a process model to compute the compensation with respect to a disturbance. The process model information is put into the controller as a part of the control algorithm.

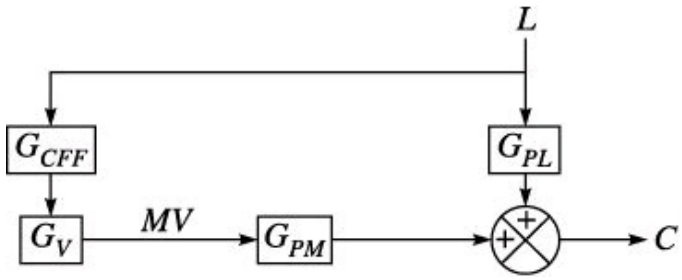


Figure 15.1 The feed forward control scheme.

We should clarify some important aspect of the system depicted in Figure 15.1. In the first example of Chapter 7, we had a preliminary discussion about the temperature control of a heat exchanger. There we introduced two forms of process model, instead of a single one. These two forms are, one for load variation, G_{PL}, and the other for the variation of manipulation, G_{PM}, respectively. This in fact is the reality that the internal dynamic processing inside a system may follow two different pathways in terms of hydrodynamics as well as energy and/or mass transfer mechanisms. Thus, the summer block that up till now was located before the process and formed the input of the process, now should be shifted after the process TFs, and their outputs are joined in the summer to produce the total process output, controlled variable C.

This difference of dynamic processing may be cited by Example 1.5, the process of ideally stirred mixer-heater, the enthalpy flux through load (inlet liquid) is physically carried into the system but the manipulated variable (heat flux through the electric heater) enters by heat transfer.

15.2.1 Design of a Feed Forward Controller

Let us assume that the given process may be represented by FOPDT models with respect to load as well as manipulated variable. To approach this problem we refer to the block diagram of Figure 15.1. The two process TFs are recognized as G_{PL} relating C/L and G_{PM} relating

C/MV. If the load and manipulation entry points are not too far located and the variables enjoy approximately similar kind of dynamic processing, the two TFs may be written as:

$$G_{PL}(s) = \frac{K_{PL}e^{-\tau_{DL}s}}{\tau_{PL}s + 1} \qquad (15.1)$$

and

$$G_{PM}(s) = \frac{K_{PM}e^{-\tau_{DM}s}}{\tau_{PM}s + 1} \qquad (15.2)$$

From the above Figure 15.1, the synthesis equation for G_{CFF} may be developed as:

$$C = LG_{PL} + LG_{CFF}G_VG_{PM}$$

For perfect control $C = 0$. Putting this into the above equation,

$$G_{CFF} = -\frac{G_{PL}}{G_VG_{PM}} \quad (15.3)$$

Equation 15.3 is the basic design equation of feed forward controller synthesis. Assuming $G_V = 1$ and putting the expressions for G_{PL} and G_{PM} from Equations 15.1 and 15.2 into the last equation,

$$G_{CFF} = -\left(\frac{K_{PL}}{K_{PM}}\right)\left(\frac{\tau_{PM}s + 1}{\tau_{PL}s + 1}\right)e^{s(\tau_{DM} - \tau_{DL})} = -K_{CF}\left(\frac{\tau_{PM}s + 1}{\tau_{PL}s + 1}\right)e^{s(\tau_{DM} - \tau_{DL})} \quad (15.4)$$

Term 1 Term II Term III

Thus, synthesized TF of a feed forward controller given by Equation 15.4 contains three Terms. These are characteristic dynamic elements and should be commented.

Term I is the controller gain. The –ve sign has a definite dynamic significance in the compensation produced by feed forward action.

Term II is a Lead-Lag element. For $t_{PL} > t_{PM}$, this would act like an integrator, and if $t_{PL} < t_{PM}$, the element would be a differentiator.

Term III For $t_{DM} > t_{DL}$, this becomes a pure predictor (ability to predict what will happen in future!!), hence, unrealizable. For $t_{DL} > t_{DM}$, the term is a dead time element that is physically realizable. However, for many process applications it has

been found that $t_{DL} = t_{DM}$. Thus, two possible configurations of G_{CFF} are:

$$G_{CFF} = -K_{CFF}\left(\frac{\tau_{PM}s+1}{\tau_{PL}s+1}\right)e^{s(\tau_{DM}-\tau_{DL})} \quad (15.5)$$

and $$G_{CFF} = -K_{CFF}\left(\frac{\tau_{PM}s+1}{\tau_{PL}s+1}\right) \quad (15.6)$$

15.2.2 Combining Feedback with Feedforward Action

The success of a feed forward control largely depends on the accuracy of the process model. In practical applications, the model building is done by process identification methods. We have already discussed in Chapter 12 that the process model developed by these experimental methods always contains an amount of error. Hence, generally in all such control system designs, the feed forward scheme is used together with feedback control. The basic notion is that the feedback control would take care of the error produced in feed forward compensation and obtain an error free response. A feed forward-feedback control loop has been shown in Figure 15.2. It can be established that this combined control scheme is preferable to any of the individual schemes singly applied.

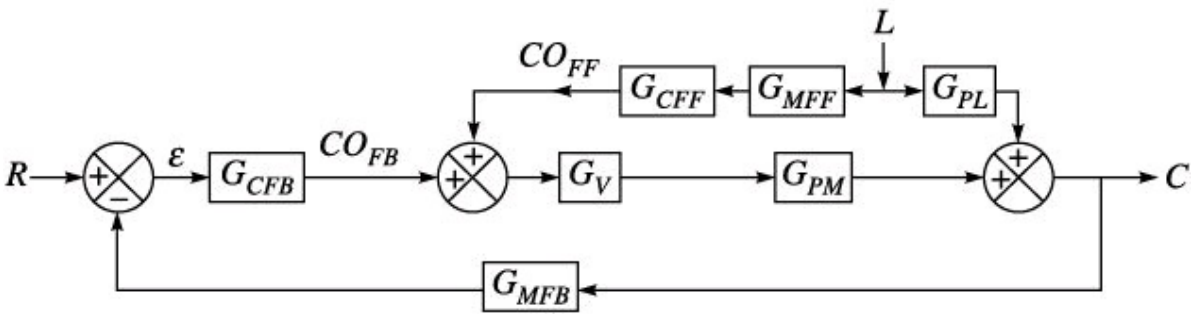


Figure 15.2 Combined FF and FB Control of a process.

(*The advantage of using such a mixed control strategy of feed forward-feedback control may become more apparent from an example out of a totally different application area. Say in a military action, an infantry gunner is trying to hit an enemy position. He adjusts the gun point by knowing the geographical position of his enemy and his own location, makes compensation for relative elevation of the target, air speed, air density, etc., and fires the gun. Uptill this point he is resorting only to feed forward strategy. But as soon as he lifts his binoculars, views where the actual hit has taken place and readjusts the gun barrel so that the next shot becomes more accurate, he is using the feedback strategy.*)

Another dynamic aspect about FF+FB control strategy implementation is the possibility of controller output saturation if FF and FB controllers act simultaneously. In fact, the FF control action will appear within a short time after the entry of

disturbance, whereas FB controller action will be delayed amounting to the dynamic lag of the process. Thus, we need not worry about summation of the outputs from the two controllers resulting in a possibility of output saturation. On the contrary, there is a possibility that the –ve sign before FF action may drive the total output toward lower saturation limit (toward ZR of subroutine CNTRL1, as described in Chapter 14).

In certain applications an alternative solution of this problem has been suggested by introducing a weight or confidence factor, a, in the output equation before FF action. The value of a may range from 0 to 1, and its complementary value is put before FB action.

15.2.3 An Industrial Example of Feed Forward-Feedback Control Application

The control configuration is shown in Figure 15.3. Controlled variable is the exit tube liquid temperature, load is inlet liquid temperature and manipulation is the steam flow rate.

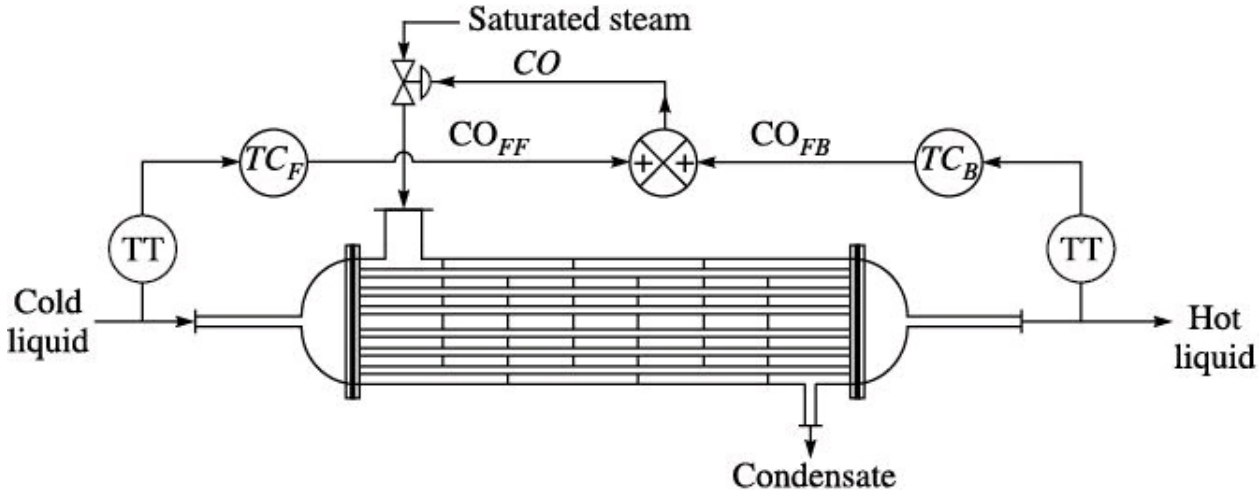


Figure 15.3 Feed forward-feedback control scheme of a heat exchanger.

Thus the two TFs of the process would have different forms. It should be remembered that we introduced a feedback control system of a shell and tube heat exchanger in
Example 7.1 and first talked about the process TF with respect to load and manipulation change. Let us rewrite those TFs:

$$G_{P_L}(s) = \frac{Tc(s)}{T_L(s)} = \frac{0.43.e^{-2.8\min s}}{12.5\min s + 1};\ G_{P_M}(s) = \frac{Tc(s)}{Q_M(s)} = \frac{0.3^{o}C/l/me^{-2.1\min s}}{10.8\min s + 1} \quad (7.1)$$

Using Equation 7.1, the feed forward controller TF becomes:

$$G_{CFF} = (-1.433)\left(\frac{10.8s + 1}{12.5s + 1}\right)e^{-0.7s}\ \frac{\text{lpm}}{^{\circ}\text{C}} \quad (15.7)$$

You should not be confused with the dimension of G_{CFF}, because in fact the TF of

the control valve G_V as a gain element K_V is imbedded with the TF of the controller.

15.2.4 Stability in Presence of Feed Forward Controller in an FF-FB Loop

We may derive the regulator and servo TFs for the FF-FB loop of Figure 15.2 from the procedure developed in Chapter 9.

$$\frac{C}{L} = \frac{G_{PL} + G_{MFF} G_{CFF} G_V G_{PM}}{1 + G_{MFB} G_{CFB} G_V G_{PM}} \qquad (15.8a)$$

and

$$\frac{C}{R} = \frac{G_{CFB} G_V G_{PM}}{1 + G_{MFB} G_{CFB} G_V G_{PM}} \qquad (15.8b)$$

Examination of the above expressions reveals that the characteristic equation of the closed loop system would not be influenced by the feed forward controller, because the denominators do not contain the feed forward controller TF, G_{CFF}. Hence, presence of the FF controller does not affect the stability of an FF-FB loop.

The above statement follows the cardinal rule of stability evaluation, but during analysis if an approximate model is used, namely to represent a distributed parameter process, exception of this notion may be evidenced. We shall workout some examples to examine this aspect and the effect of adding feed forward control in a feedback control system on the overall closed loop performance.

15.2.5 Tuning of an FF-FB Loop

In control system applications the determination of process model may often become too difficult or impractical. Tuning an FF-FB loop in such cases could be a time consuming procedure that may lead the running plant to remain under disturbed conditions and produce too much of out of quality products. Coughanowr has presented a tuning guide line that partially shortens the process modeling exercise.

The FF-FB control is considered without the feedback loop. A simulation exercise may be used for certain process systems for which the TF with respect to load change, G_{PL}, could be known, but there is an uncertainty about G_{PM}. The tuning procedure, to some extent, removes the effort for determining G_{PM}. This procedure determines the feed forward controller TF, G_{CFF} that contains only the –ve gain $-K_{CFF}$ and the Lead-Lag element, $[(t_{LD}s + 1)/(t_{LG}.s + 1)]$ without considering any delay element that may result from a modeling effort of the process. For this objective, the FF-FB loop is reconfigured in the following way:

1. The feedback action is eliminated from the loop;
2. The dynamic part of the G_{CFF} is also omitted (In a computer based controller, this change may be easily done by slightly changing the controller algorithm. For a modular controller, this change could be achieved by making the lead and lag time constants identical and equal to some small value.); and
3. $-K_{CFF}$ is adjusted so that the forward path gain from L to C through G_{CFF} becomes –1. The control loop in this form appears as in Figure 15.4.

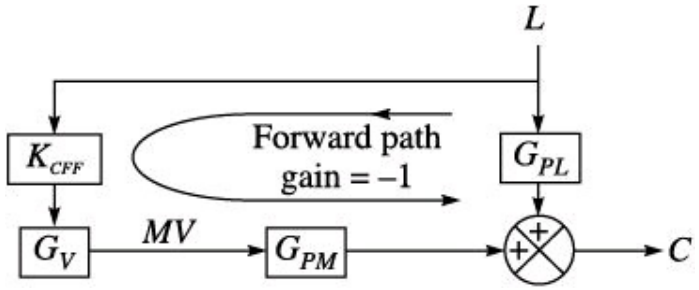


Figure 15.4 The reconfigured control loop.

This system is experimentally tested by introducing a step in load of small magnitude. The corresponding open loop process response resembles any of the two types of curve shown in Figure 15.5.

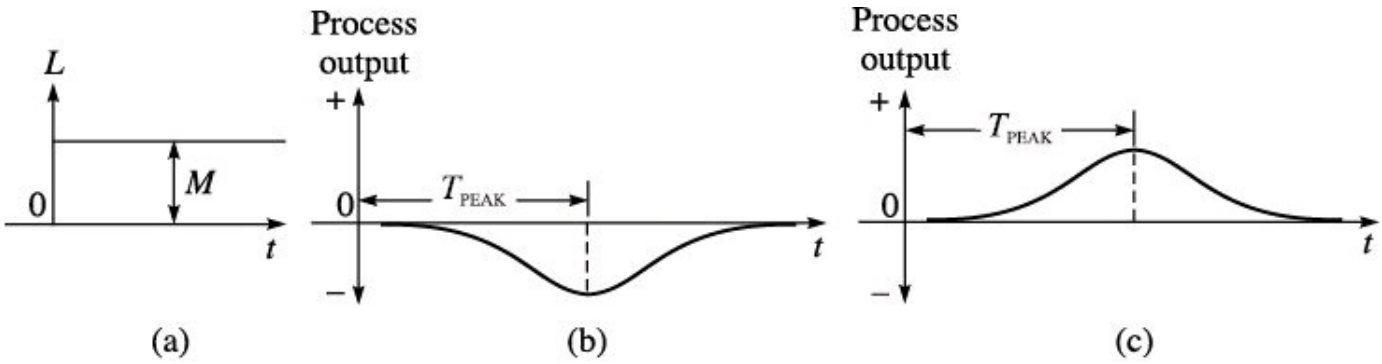


Figure 15.5 Two categories of process response of the system of Figure 15.4.

From the obtained curve the time elapsed between the step introduction and occurrence of either maxima or minima of the curve is determined by graphical construction. This time value is termed as T_{PEAK}. The G_{CFF} may be determined by the guide relations given in Table 15.1.

Table 15.1 The Tuning Relations

Type of the Curve	Figure No.	Lag Time Constant x_{LG}, in the Lead-Lag	Lead Time Constant x_{LD}, in the Lead-Lag
Minima	15.5(b)	1.5 T_{PEAK}	0.7T_{PEAK}
Maxima	15.5(c)	0.7T_{PEAK}	1.5T_{PEAK}

To test the performance of this tuning procedure, we shall workout the following example.

EXAMPLE 15.1 The temperature control of a process is considered where the TFs,

G_{PL} and G_{PM} are known. These are given as:

$$G_{PL} = \frac{1}{(20s+1)^4} \text{ and } G_{PM} = \frac{1}{(20s+1)^3}$$

For the sake of calculation simplicity, let us assume that $G_{MFB} = G_{MFF} = 1$.

Solution: With the given dynamic descriptions of the elements in the FF-FB loop, the system block diagram takes the form as shown in Figure 15.6.

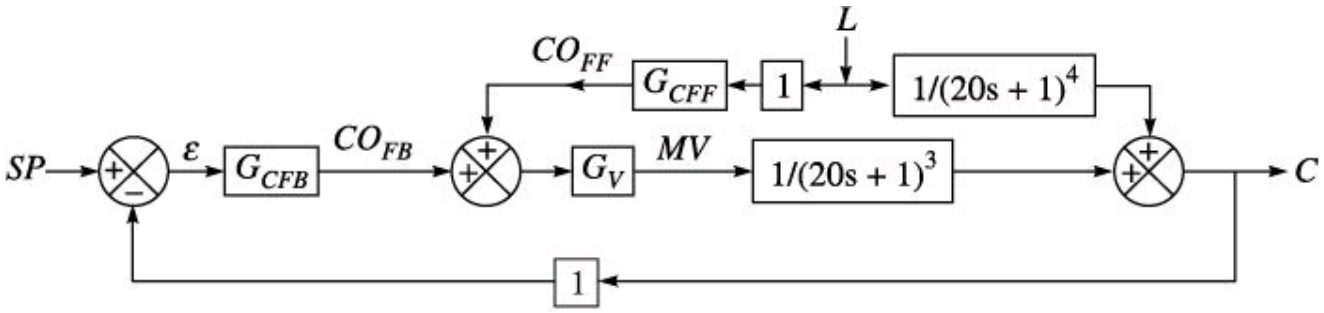


Figure 15.6 The block diagram of the control system for example 15.1.

We should mention another aspect of the unbiased FF-FB loop given in Figure 15.2. The CO_{FF} always comes with a –ve sign for +ve disturbances but CO_{FB} is always +ve if the feedback controller features direct action. The *OPB* is adjusted to such a value so that for normal disturbances the CO_{FB} would not reach the lower saturation. But due to the presence of –ve valued CO_{FF}, the total controller output CO may reach the lower saturation of the FCE for a moderate value of +ve disturbance, and a large part of the feed forward control action might be lost. This problem is addressed by introducing another constant, *MVO*, in the relation between controller output, CO, and manipulated variable, MV. Now the modified relation becomes:

$$MV = (MVO + CO)K_V \tag{15.9}$$

In a simulation exercise, the higher and lower saturation values of the feedback controller and FCE may be respectively introduced as *CL*, *ZR* and *MVCL*, *MVZR*. By incorporating these constants Equation 15.9 becomes:

$$MV = (MVO + CO)\frac{(MVCL - MVZR)}{(CL - ZR)} \tag{15.10}$$

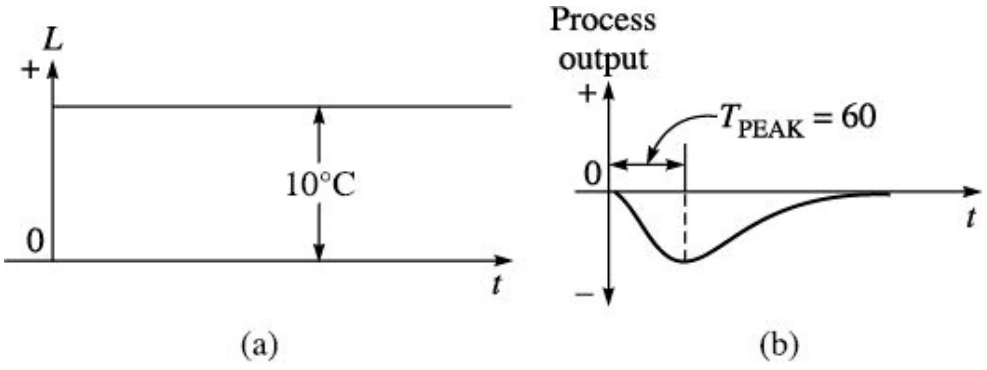


Figure 15.7 The open loop step test of the process of example 15.1.

A simulation program is prepared for the step test. Figure 15.7 shows the result that features a minima. The K_{CFF} has been found to be –1, and the response shows a minima with T_{PEAK} = 60 s. Using the relations of Table 15.1, the feed forward controller TF, according to Coughanowr's method, is found as:

$$G_{CFF} = -\frac{42s+1}{75s+1} \qquad (15.11)$$

The parameters of the (PID) controller in the feedback loop are determined by the following method. The feedback controller output will go through only the load part of process. The step response of G_{PL} is obtained for an arbitrarily small step magnitude. From the response curve 28.2% and 63.2% step recovery times, T1 and T2 are found. By using the Equations 12.10 and 12.11 developed in Chapter 12, parameters of the FOPDT model for G^*_{PL} as an approximation for G_{PL} have been found. From the FOPDT model parameters, by applying frequency response technique the crossover frequency, w_{CO}, is found, followed by the determination of ultimate gain, K_U, and period, P_U. Finally applying the Ziegler–Nichol's formulae the controller parameters of a (PID) controller have been obtained. This procedure has been already explained by citing an example in Chapter 13. For the present problem, the procedure has been pictorially presented in Figure 15.8.

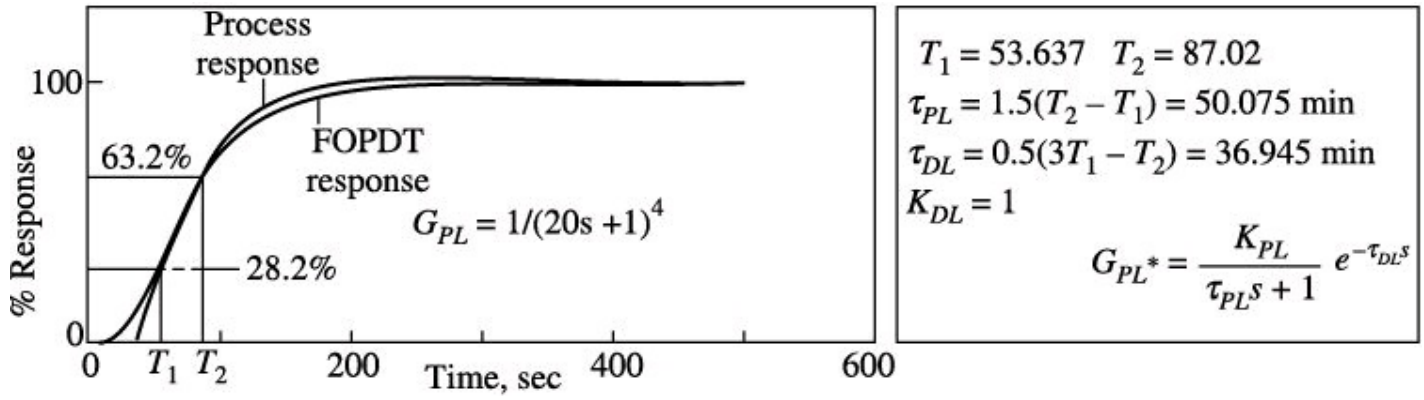


Figure 15.8 Determination of the FOPDT model parameters from step response of G_{PL}.

The obtained values from the frequency response method are:

$w_{CO} = 0.0405146$

The phase angle calculated at this frequency, $f = -180.0002°$. From this, $K_U = 2.261827$, and
$P_U = 155.0844$ s. The (PID) controller values according to Ziegler–Nichol's formulae: $K_C = 1.3571$; $t_I = 77.5422$ s; and $t_D = 19.386$ s.

Following the same procedure, the FOPDT model G^*_{PL}, as the approximation for G^*_{PM}, is obtained as:

$$G^*_{PM} = \frac{K_{PM} e^{-\tau_{DM} s}}{\tau_{PM} s + 1} = \frac{1 e^{-22.827 s}}{42.309 s + 1} \qquad (15.12)$$

From the calculated FOPDT model parameters of G^*_{PL} in Figure 15.8, G^*_{PM} as given in the above Equation 15.12, the feed forward controller TF, according to Equation 15.5 becomes:

$$G_{CFF} = -\frac{42.309 s + 1}{50.075 s + 1} e^{-14.118 s} \qquad (15.13)$$

A simulation program is written for observing the closed loop response of the designed FF-FB combined controller that has the facility for testing the performance of three types of controller, e.g.

1. FF-FB control where the FF controller has been designed according to Equation 15.13;
2. FF-FB control where the FF controller is designed by Coughanowr's method using Equation 15.11; and
3. only an FB controller is employed for the control action.

The feedback controller parameters are designed by Ziegler–Nichol's formulae as given above.

The following values of Controller and FCE parameters have been used in the simulation: $CL = MVCL = 100$, the upper saturation values, $ZR = MVZR = 0$, the lower saturation values, and $OPB = 20$, $MVO = 25$.

The loop is tested for:

1. A step input of 10ºC in load, with Set point, SP = 35ºC and initial value of the load = 25ºC; and
2. A pulse input of 20ºC with 100 s duration, with Set point, SP = 35ºC and initial value of the load = 15ºC. The step and pulse responses for three types of

control have been respectively shown in Figure 15.9 and Figure 15.10. The ITAE values are calculated for each of the responses and compiled in Table 15.2.

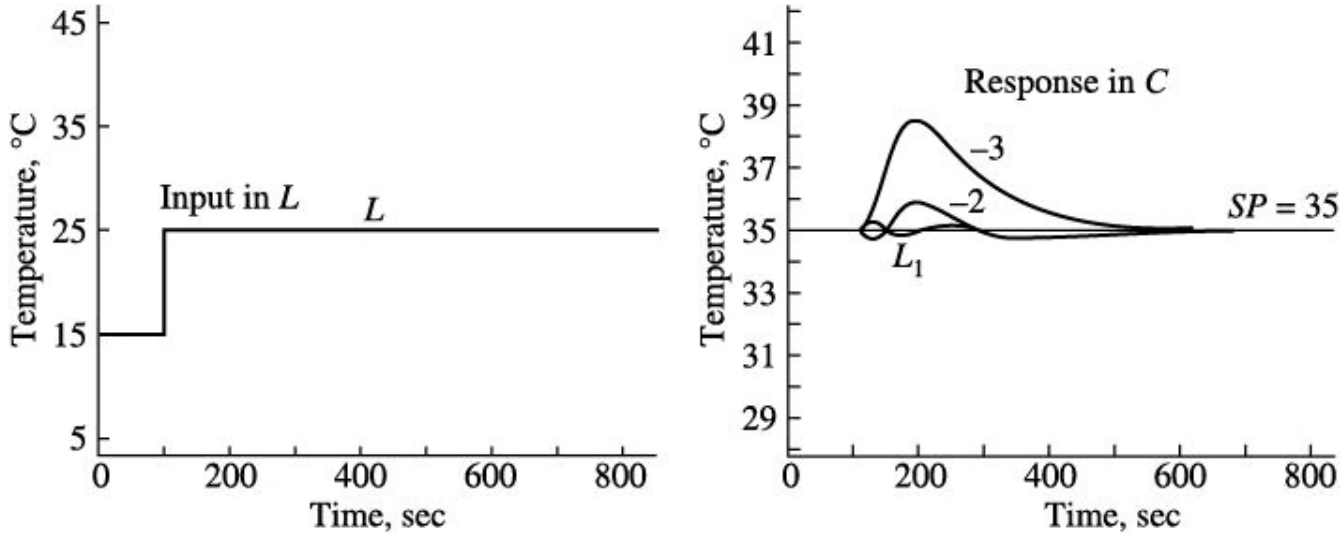


Figure 15.9 Step response of the controlled system under three types of control.

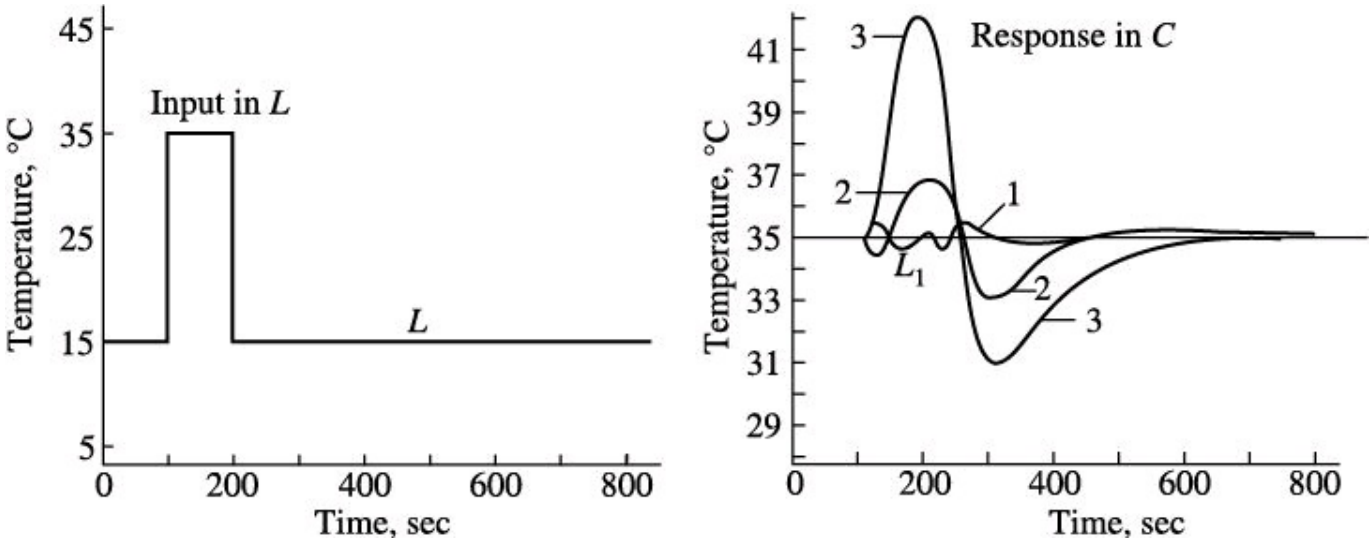


Figure 15.10 Pulse response of the controlled system under three types of control.

Table 15.2 Performance of Three Control Types for Step and Pulse Input

Step Input in Load			**Pulse Input in Load**		
Curve No.	**Type of Control**	**ITAE**	**Curve No.**	**Type of Control**	**ITAE**
1	FB + FF (Equation 15.11)	2.737	1	FB + FF (Equation 15.14)	9.491
2	FB + FF (Equation 15.13)	28.69	2	FB + FF Equation 15.12)	74.848
3	FB (only)	81.64	3	FB (only)	208.18

Comments on the simulation results:

1. *The improvement in response for both step and pulse inputs, with an FF controller designed by the FOPDT models for G_{PL} and G_{PM} combined with a tuned FB controller, has been evidenced quantitatively (ITAE values) as well*

as qualitatively (response shapes). The performance is satisfactorily above that of the FB + FF combination when the G_{CFF} is tuned by Coughanowr's method (Equation 15.12) and way above when only feedback control is applied. But it is felt that the difference in performance between curve 1 and 2 would not be so dramatic if the system featured in this example had not got such an amount of dead-time difference (14.118 s) between load and manipulated variable response. This notion is supported by the small initial undershoot for both kind response in curve 2.

However, we should not overlook the advantage of using Coughanowr's method that only a relatively simple step test is required to determine the T_{PEAK}.

2. *The role of –ve sign of the G_{CFF} is to reduce the over- and undershoots produced by the G_{CFB} and to bring the overall response quickly closer to the set point. Absence of this correction has been shown by the curve 3.*
3. *In running a simulation program, in which there are many intermediate variables, often the selection of initial values, so that from beginning of run time an error free response is obtained, becomes a tedious and time-consuming exercise. This may be avoided by putting as many of the terminal variables values as possible (obtained by say, steady state energy and mass balances) and allowing the program to run without any disturbance for sufficiently long time. From the graphic output ensuring that the initial transients have died down, all the required variable values could be printed and then used in the closed loop test run.*

As we shall see in our future discussions that feed forward and a mixture of feed forward-feed back schemes have appeared as the back bone of many advanced control strategies. Probably the principle of these basic control techniques has stimulated the imaginations of workers in this field to such an extent that many beautiful research efforts keep appearing, even in the recent years.

15.2.6 An Alternative Configuration of FF+FB Loop

At the end of Section 15.2.2, we mentioned about an alternative configuration of implementing FF+FB control by introducing 'bias' or 'confidence factor', a, where a may have any value between 0 and 1. Here, a is a multiplier by which the controller output of feed forward controller, CO_{FF}, is multiplied before it goes to a summer block, and the output from feedback controller, CO_{FB}, is multiplied by $(1 - a)$ before the same summer block. Figure 15.11 shows the block diagram of this configuration.

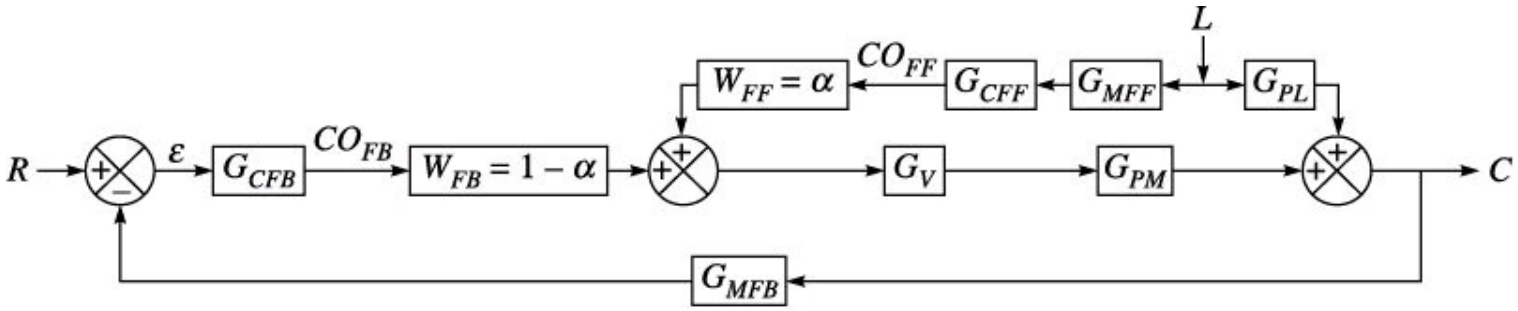


Figure 15.11 The combined FF-FB loop with bias on two control actions.

The output from the summer is the controller output of the combined control loop that is used for actuation of FCE. This complementary biasing may be expressed in the following equation,

$$CO = aCO_{FF} + (1 - a)CO_{FB} \tag{15.14}$$

By putting $a = 0$, the control system would become solely a feedback loop, and by putting $a = 0.5$, we would be introducing FF and FB action with equal bias.

In time domain, the FF action appears swiftly following a disturbance, but FB compensation becomes effective after some time, and the time depends on the amount of lag of the process. It is felt that biasing the FF and FB action by using Equation 5.14 should be used for systems having lower values of response lag. Let us examine this method with a first order system.

EXAMPLE 15.2 A feedback-feedforward control loop has been implemented for a stirred liquid heater. The heating tank has a working volume, V = 100 l. The liquid enters at a constant flow rate, q = 10 l/m, and temperature, T_I that may vary from $T_{I\text{MIN}}$ = 15ºC to $T_{I\text{MAX}}$ = 35ºC. The desired exit temperature, $T_R = T_{\text{SET}}$ = 90ºC. Liquid density and heat capacity are respectively, r = 0.95 g/cm^3, and Cp = 0.9 cal/(g, °C). The exit liquid temperature is measured by a sensor inserted in the pipeline at a finite distance from the actual tank exit. A (P) controller is used in the feedback loop. An adjustable confidence factor, $W_{FF} = a$, is put on the controller output, CO_{FF}, from the feed forward, and the complementary confidence factor, $W_{FB} = 1 - a$, is put on the controller output, CO_{FB}, from feedback controller. The schematic diagram of the heating system control is shown in Figure 15.12. The final control element may be considered as pure gain. The temperature sensors have equal gains, $K_{MB} = K_{MF}$ = 0.16 mA/ºC. Estimate the %OPB of the feedback controller assuming a linear process characteristic.

(a) Find the ultimate gain of the feedback controller, K_U for (1) a = 0.5 and (2) a = 0.

(b) With K_C = 2 for the feedback controller, find the value of 'offset' for (1) a = 0.5 and (2) a = 0 for a unit Load disturbance.

Solution: The block diagram in Figure 15.13 may be used to represent the system in this example. To find the immersion heater capacity, FCE, let us perform a heat balance for minimum and maximum values of the inlet liquid temperature,

Maximum heating rate = $q \times r \times Cp \times (T_R - T_{IMIN}) = 641.25$ kcal/min.

Minimum heating rate = $q \times r \times Cp \times (T_R - T_{IMAX}) = 470.25$ kcal/min.

Average heating rate = (641.25 + 470.25)/2 = 555.75 kcal/min.

Assuming a linear process characteristic, the %OPB of the feedback controller may be estimated as, OPB = (555.7/641.25) × 100 = 86.7%.

The final control element gain, $K_V = 641.25/16 = 40.078$ (kcal/min)/mA.

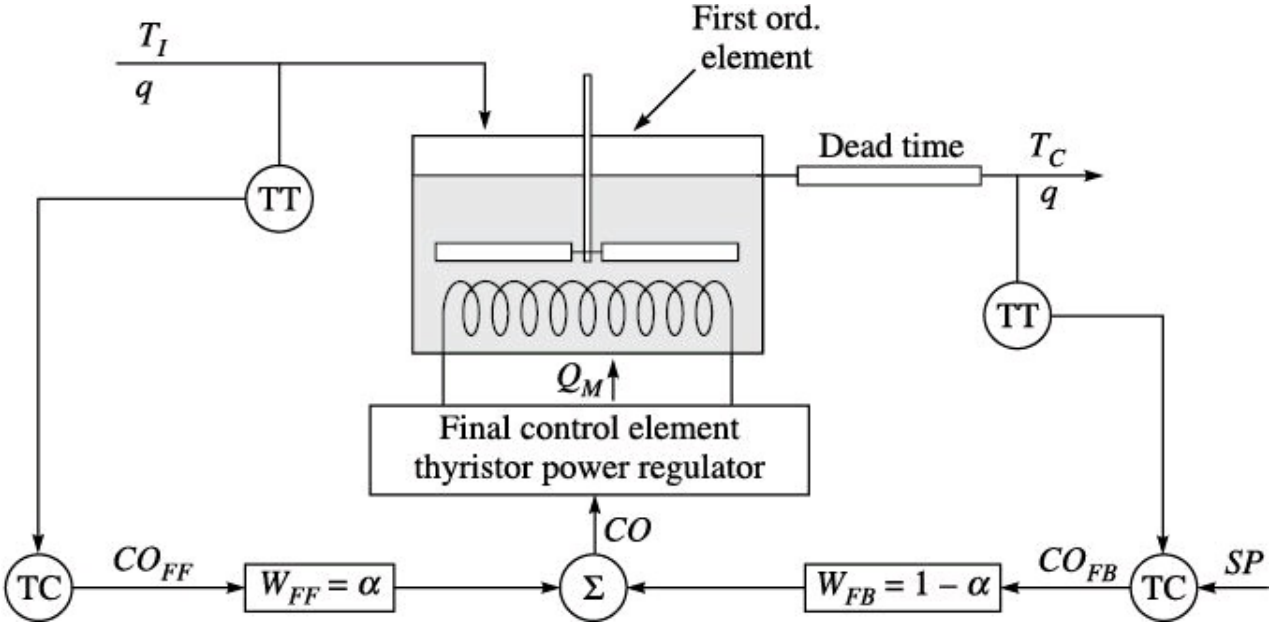


Figure 15.12 Feedforward-Feedback control of the stirred liquid heater.

To obtain a clear idea of the gains situated in the path from load entry point, and from manipulation entry point, to the summer that results in the controlled variable, let us draw the signal paths.

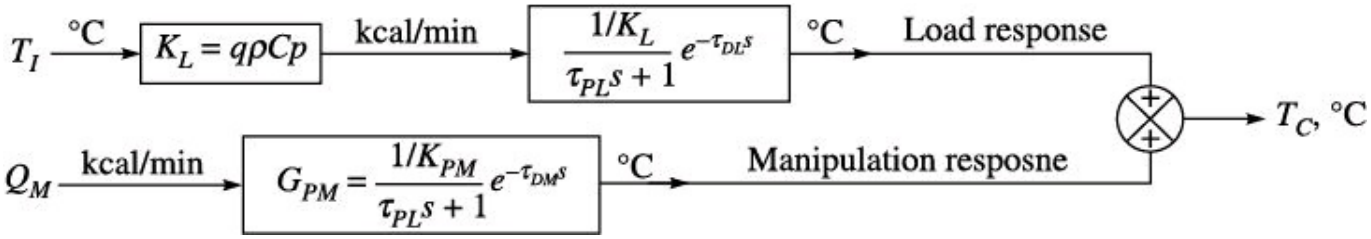


Figure 15.13 Block diagram of the control System of Figure 15.12.

The gain terms K_{PL}, K_{PM} may be evaluated as:

$$K_L = q\rho Cp = 8.55 \text{ kcal/min}; \quad K_{PL} = K_L\left(\frac{1}{K_L}\right) = 1; \quad K_{PM} = \frac{1}{K_L} = \frac{1}{8.55}(\text{kcal/min})^{-1}$$

The time constants, t_{PL} and t_{PM} are numerically identical because both the load

and the manipulation have to go through similar hydrodynamic processing for the two paths. Also, as the process reaction, whether due to load or manipulation change, has to go through the same length of pipeline before it comes to the measurement point; hence, the dead times t_{DL} and t_{DM} are also identical. The parameter values are evaluated as:

$$\tau_{PL} = \tau_{PM} = \frac{V}{q} = 10 \text{ min}; \; \tau_{DL} = \tau_{DM} = \frac{V_L}{q} = 1 \text{ min}.$$

The process transfer functions are:

$$G_{PL}(s) = \frac{e^{-s}}{10s+1} \quad \text{and} \quad G_{PM}(s) = \frac{(1/8.55)e^{-s}}{10s+1} \text{ kcal/min}^{-1}$$

The two measuring elements are pure element $K_{MFB} = K_{MFF} = 0.16$ mA/°C

The feed forward controller transfer function may be found now:

$$G_{CFF}(s) = -\frac{K_{PL}}{K_{PM}K_V K_{MF}} = -\frac{855}{641.25}$$

The feed forward controller is a gain element.

The closed loop TF with respect to load change is:

$$\frac{C}{L} = \frac{G_{PL} + G_{MF}G_{CFF}G_V G_{PM}G_{PM}\alpha}{1 + G_{MB}G_{CFB}G_V G_{PM}(1-\alpha)}$$

Putting the values of individual TFs in the above equation and approximating the dead time by Pade's first order equation, we get:

$$\frac{C}{L} = \frac{\dfrac{e^{-s}}{(10s+1)} - \dfrac{16}{100}\cdot\dfrac{855}{641.25}\cdot\dfrac{641.25}{16}\cdot\dfrac{e^{-s}}{8.55(10s+1)}\cdot\alpha}{1 + K_C\cdot\dfrac{641.25}{16}\cdot\dfrac{e^{-s}}{8.55(10s+1)}\cdot\dfrac{16}{100}(1-\alpha)} = \frac{\left(\dfrac{1-\alpha}{10s+1}\right)\left(\dfrac{2-s}{2+s}\right)}{1 + \dfrac{0.75K_C(1-\alpha)}{(10s+1)}\left(\dfrac{2-s}{2+s}\right)}$$

$$= \frac{(2-s)(1-\alpha)}{10s^2 + 21s + 2 + 0.75K_C(2-s)(1-\alpha)}$$

For $\alpha = 0$ (only FB control):

Characteristic Equation:

$$10s^2 + (21 - 0.75K_C)s + 2 + 1.5K_C = 0$$

Applying R.H. Criteria: $K_U = 28$.
For $K_C = 2$, the regulatory TF:

$$\frac{C}{L} = \frac{2 - s}{10s^2 + 19.5s + 5}$$

If a unit step disturbance is made in Load, applying final value theorem,

$$C(t)\Big|_{t=\infty} = \underset{s\to 0}{\text{Lt}}[sC(s)] = \frac{2}{5} = 0.4$$

$$\text{Offset} = R - C(t)\Big|_{t=\infty} = 0 - 0.4 = -0.4$$

For $\alpha = 0.5$ (Equal confidence to FF and FB control):

Characteristic Equation:

$$10s^2 + (21 - 0.375K_C)s + 2 + 0.75K_C = 0$$

Applying R.H. Criteria: $K_U = 56$.
For $K_C = 2$, the regulatory TF:

$$\frac{C}{L} = \frac{1 - 0.5s}{10s^2 + 20.25s + 5}$$

If a unit step disturbance is made in Load, applying final value theorem

$$C(t)\Big|_{t=\infty} = \underset{s\to 0}{\text{Lt}}[sC(s)] = \frac{1}{5} = 0.2$$

$$\text{Offset} = R - C(t)\Big|_{t=\infty} = 0 - 0.2 = -0.2$$

The results show an improvement in K_U and offset by putting feed forward loop along with a feedback control. The offset decrease is not unique for $K_C = 2$. The reader should check the trend by putting other +ve values for K_C. For such an implementing strategy of an FF-FB loop that contains the bias factor a, the tuning problem to find the optimum value for a that obtains the best closed loop performance with respect to any error criterion, say ITAE, could be addressed only by the digital simulation procedure.

Robustness of a control strategy is an important item for investigation, particularly when any model based algorithm is a part of the control strategy. A question may be raised–to what extent the performance improvement would suffer, if the designed control algorithm is based on a process model that may contain an amount of error? You must agree that Feed forward control is a suitable candidate for such investigation.

EXAMPLE 15.3 Consider the system of the last example of the stirred liquid heater. All the variable and parameter values are identical. The only difference is that the first order time constants, when determined (say by the process reaction curve method), have shown a slight difference from their actual values in the process. These estimated TFs are:

$$G_{PL}^* = \frac{e^{-s}}{11s + 1} \quad \text{and} \quad G_{PM}^* = \frac{(1/8.55)e^{-s}}{9s + 1}$$

They will be used for synthesizing G_{CFF}. Thus,

$$G_{CFF} = -\frac{K^*_{PL}}{K^*_{PM} K_V K_{MF}} = -\frac{e^{-s}/(11s+1)}{e^{-s}/(9s+1)} \cdot \frac{8.55 \times 16}{641.25} \cdot \frac{100}{16} = -1.333 \frac{(9s+1)}{(11s+1)}$$

All other block TFs remain same as have been used in the previous example. These TFs are rewritten for convenience, as:

$$G_{CFB} = K_C; \quad G_V = 40.1 \ \frac{\text{kcal/min}}{\text{mA}}; \quad G_{PL} = \frac{e^{-s}}{10s+1}; \quad G_{PM} = \frac{(1/8.55)e^{-s}}{(10s+1)} \frac{\text{kcal/min}}{\text{mA}}$$

and $K_{MB} = K_{MF} = 0.16$ mA/°C

Putting these TF expressions in the regulatory closed loop TF equation developed in the previous example,

$$\frac{C}{L} = \frac{G_{PL} + G_{MF} G_{CFF} G_V G_{PM} G_{PM} \alpha}{1 + G_{MB} G_{CFB} G_V G_{PM} (1-\alpha)} = \frac{\frac{e^{-s}}{10s+1} - 0.16 \times 1.333 \times \left(\frac{9s+1}{11s+1}\right) \cdot \frac{40.1}{8.55} \cdot \frac{e^{-s}\alpha}{(10s+1)}}{1 + K_C \times 0.16 \times 40.1 \times \frac{(1/8.55)e^{-s}(1-\alpha)}{(10s+1)}}$$

$$= \frac{[11s+1-(9s+1)\alpha](2-s)}{(11s+1)[9(10s+1)(s+2) + 0.75K_C(1-\alpha)(2-s)]}$$

For $\alpha = 0$ (only FB control):

Characteristic Equation:

$$110s^3 + (241 - 8.25K_C)s^2 + (43 + 15.75K_C)s + 1.5K_C + 2 = 0$$

Applying R. H. Criteria two solutions of K_C are obtained $K_C = 28$ and -2.786
Taking the highest +ve value for K_U,

$K_U = 28$.

Applying Final Value theorem to the Closed loop regulatory TF, for $K_C = 2$:

$$C(t)\big|_{t=\infty} = \underset{s\to 0}{\text{Lt}}[sC(s)] = \frac{1}{12} = 0.0833$$

$$\text{Offset} = R - C(t)\big|_{t=\infty} = -0.08333$$

For $\alpha = 0.5$ (Equal confidence to FF and FB control):

Characteristic Equation:

$$110s^3 + (241 - 4.125K_C)s^2 + (43 + 7.875K_C)s + 0.75K_C + 2 = 0$$

Applying R. H. Criteria two solutions of K_C are obtained $K_C = 42.38$ and 7.37
Taking the highest +ve value for K_U,

$K_U = 42.38$

Applying Final Value theorem to the Closed loop regulatory TF, $K_C = 2$:

$$C(t)\big|_{t=\infty} = \underset{s\to 0}{\text{Lt}}[sC(s)] = \frac{1}{21} = 0.0476$$

$$\text{Offset} = R - C(t)\big|_{t=\infty} = -0.04762$$

Thus, the results show an improvement in K_U and offset by putting feed forward

loop along with a feedback control even when an erroneous process model contaminates the controller TF.

15.3 Ratio Control

Ratio Control is a special application of feed forward strategy that has been discussed in
Section 15.2. Application examples of ratio control are numerous:

1. Ratio control of raw materials into a Cement kiln.
2. Air/fuel ratio control to the burner of a furnace in face of changing heat demand.
3. Ratio control of feed flow to the column and steam to the re-boiler of a distillation column.
4. Ratio control of added chemicals to waste water in water treatment and pollution control applications.

To understand this strategy, we consider the blending operation of two streams of pure A and pure B. The control task is to obtain a mixture in which A and B are present at a desired ratio, R_D, in terms of volume. The ratio scheme implements the flow control of two streams to the blender in correct proportion so that the product stream is obtained at the desired ratio. A simple feedback system may perform this task by employing two flow loops for the individual streams. Each of the loops have flow measuring elements, final control elements and flow controllers, which act according to their set points, R_{FA} and R_{FB}, so that the streams go to and exit from the mixer at the desired ratio, R_D. The control system is shown in Figure 15.14.

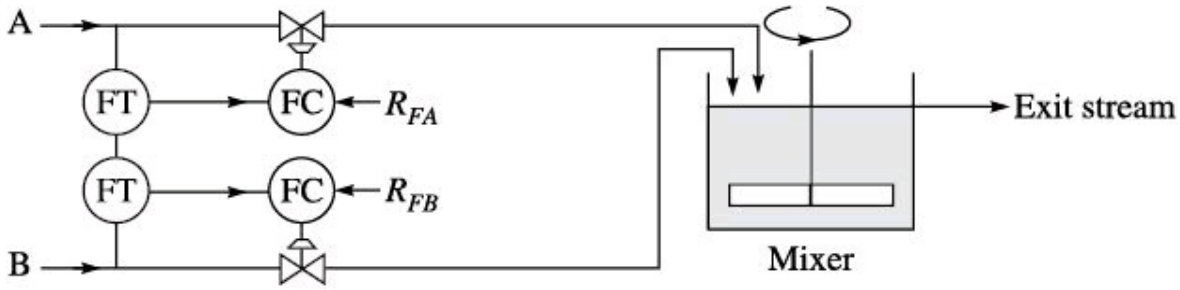


Figure 15.14 Ratio control of two streams by feedback loops.

Ratio control in the true sense has certain advantages over the simple feedback scheme. Realization of ratio control has been possible only after the development of pneumatic and electronic modules that had special computing powers during 1940s. The modules operate within the standard variable range of their actuation, (pneumatic: 3–15 psig; electronic: 4–20 mA). Two such hardwares with 'dividing' and

'multiplying' facility are of special interest to us. The divider as well as the multiplier module features two inputs, X_1 and X_2, and one output, Y. The signal processing action is expressed in Equations 15.15 and 15.16 and Figure 15.15 depicts their equivalent block diagram form.

$$Y = \frac{X_2}{X_1} \tag{15.15}$$

$$Y = X_1 X_2 \tag{15.16}$$

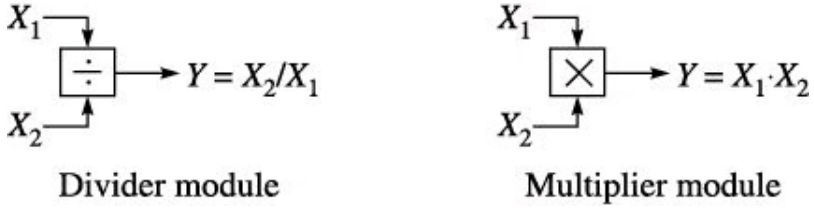


Figure 15.15 Operation of a divider and a multiplier module.

In ratio control application, the input variables are the flow rates to any of the modules. The divider module accepts the stream flow rates from the flow sensors as the two inputs, X_1 and X_2, and produces the flow ratio as its output, Y. This ratio may be accepted by a feedback controller where it is compared with the desired ratio as the set point and produces necessary control action to regulate the flow of a chosen stream called manipulated stream. The other stream is called the load stream.

In the practice of ratio control, the multiplier module is called **Ratio Station** (**RS**). This is used in the following manner: Flow rate of one stream (called the load stream) goes to the module as input X_1, the other input X_2 to the module is the desired ratio R_D. Thus, the output Y, from the module, is the required flow rate of the manipulated stream. The input X_2 is an external input for fixed ratio applications; it is set by the operator from the instrument front panel or '*facia*' by turning a knob on a scale. In case the ratio is variable, in variable ratio applications, X_2 may be a time dependent signal. In ratio control practice, X_2 is called the gain of the ratio station K_R. The term '**ratio**' is generally understood as the ratio of manipulated to the load variable.

Use of these modules in ratio control would produce two different configurations of the loop. The loops are shown in Figures 15.16(a) and (b) with the necessary changes of variable name conforming to control system nomenclature.

In the block diagrams of Figure 15.16, the notation R_R and K_R have been used instead of R_D, though they have similar significance. During practical implementation of ratio control, one has to consider the span of the flow sensors of the streams that are going to measure the flow rates and send either electronic (4–20 mA) or pneumatic (3–15 psig) signal to the ratio

station.

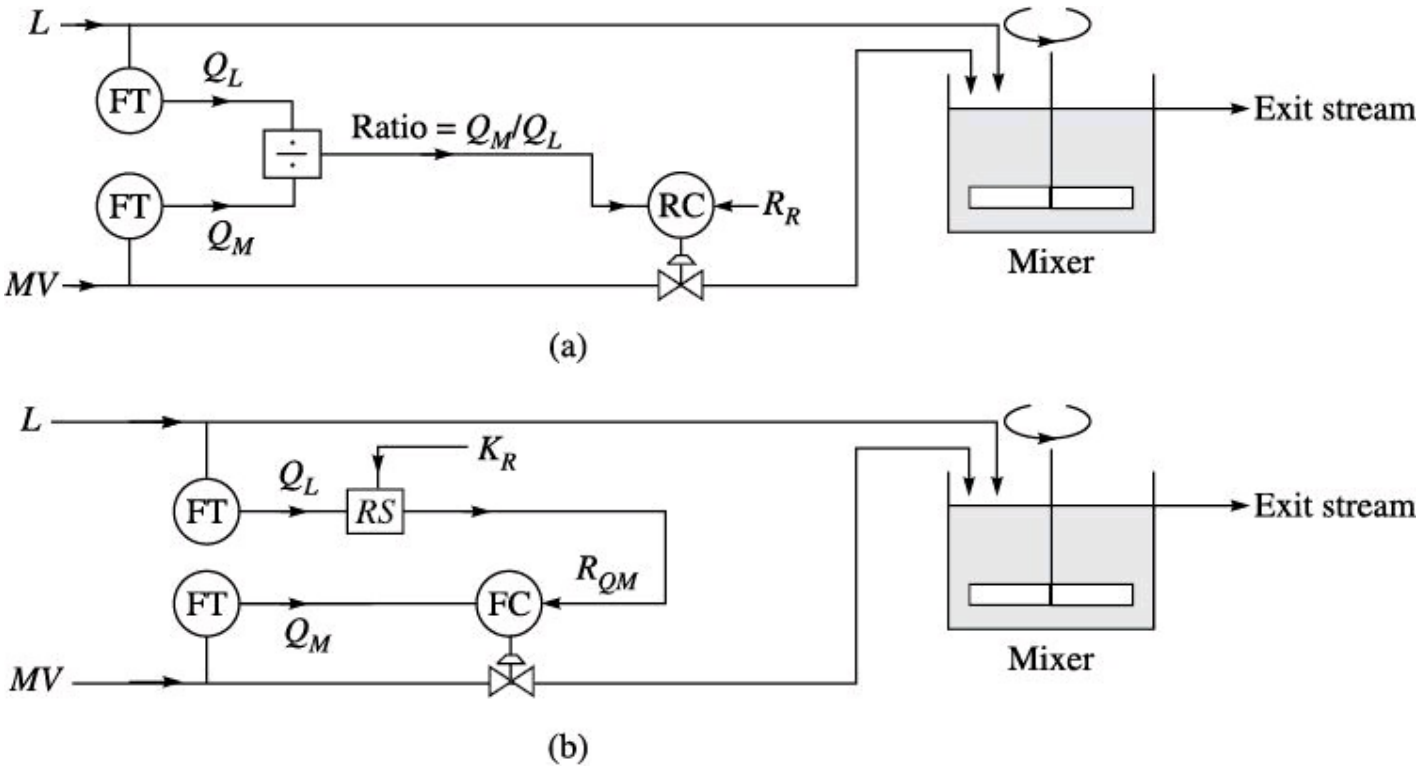


Figure 15.16 Ratio control configurations using: (a) Divider. (b) Ratio station module.

In an application, let the design flow rates for streams L and MV respectively be Q_L and Q_M, in any consistent engineering unit. For the selected instruments, if the ranges of flow sensors for the two streams are respectively $Q_{L\text{MIN}}$ to $Q_{L\text{MAX}}$ and $Q_{M\text{MIN}}$ to $Q_{M\text{MAX}}$, then spans of these instruments would be $S_L = Q_{L\text{MAX}} - Q_{L\text{MIN}}$ and $S_M = Q_{M\text{MAX}} - Q_{M\text{MIN}}$.

Also the desired flow ratio for MV with respect to L is $R_D = Q_M/Q_L$. For an electronic sensor transmitter only considering the variable part of the output:

For load (L) stream the output signal would be

$$m_L = \frac{Q_L}{S_L} \times 16 \text{ mA}$$

and for manipulated variable, MV, the sensor output would be

$$m_M = \frac{Q_M}{S_M} \times 16 \text{ mA}$$

Thus, the ratio station gain, K_R which when multiplied by the output from stream L sensor would produce an electronic signal proportional to the desired flow of MV may be found as:

$$m_L K_R = m_M \text{ or } K_R = \frac{m_M}{m_L} = \frac{Q_M}{Q_L} \cdot \frac{S_L}{S_M} = R_D \cdot \frac{S_L}{S_M} \qquad (15.17)$$

where, the desired flow ratio for *MV* stream with respect to *L* stream is R_D

In both of the ratio control schemes of Figure 15.16, the load variable is measured and the information sent forward to realize the control action. This is the reason why such a scheme may be considered as a class of feed forward control application. Such a control strategy may be successful if the down stream system, into which the streams enter, do not involve any kinetic rate process. In certain applications, a correction of the R_D (ratio of the reactant stream flow rates) may become necessary, depending on the conversion efficiency. The measurement of conversion may be done by a suitable composition analyzer situated at the process exit, and the information sent back to the control system. This modification brings feedback strategy into a feed forward control loop.

Between the two streams, selection of load variable is easy if the flow rate of a particular stream is uncontrollable at its source. In applications where both streams are product of certain upstream processes, this selection depends on certain other aspects.

1. *In ammonia synthesis, where gaseous Hydrogen and Nitrogen streams are ratio controlled before entry to the catalytic converter, the Hydrogen stream flow rate may be selected as the load variable.*
2. *In Ostwald's synthesis of HNO_3, where Air and NH_3 are the feed streams, the air flow rate may be the load variable.*
   The load variable selection in the above two cases may be because relatively small flow rate streams are to be manipulated.
3. *However, in Air/fuel ratio control of a furnace, the fuel flow rate depends on the thermal energy demand, because energy production comes by the combustion of fuel, where air flow loop is said to 'follow' the fuel control loop. Hence, fuel flow is selected as the load and air flow becomes the manipulated variable.*

In Figure 16.16, the second control scheme (b) has some advantage over the first (a) one. To demonstrate this, a gain analysis of the two loops is in order. The system in Figure 16.16(b) solves the equation: $Q_M = R_D Q_L$. The gain of this arrangement, that is, the amount of output change per unit change in Q_L is given by:

$$\frac{\partial Q_M}{\partial Q_L} = R_D \qquad (15.18)$$

This is a constant. The first scheme (A) effects the relation: $R_D = Q_M/Q_L$. The gain of this system is given as:

$$\frac{\partial R_D}{\partial Q_L} = -\frac{Q_M}{Q_L^2} \qquad (15.19)$$

The –ve sign recognizes that an increment of Q_L will reduce R_D. Equation 15.19 shows that the first control scheme (a) brings up a non-linear character of the loop. This is the demerit of scheme (a).

In industrial applications, where the flow rates have moderate to high values and the entering streams to the process may be under undesirable load-fluctuations, the hardware eliminations as indicated in the above ratio control schemes would not be judicious. If the system has a specified processing capacity in particular, then a too high value of total feed would deteriorate the conversion efficiency, correspondingly too small value of the total feed would reduce the capacity utilization of the system. Hence, all the streams are dealt with flow control loops and a ratio station is implemented between them to obtain a tighter control of total feed rate as well as the ratio of the materials entering into the processing system. A ratio control scheme with these features is shown in Figure 15.17.

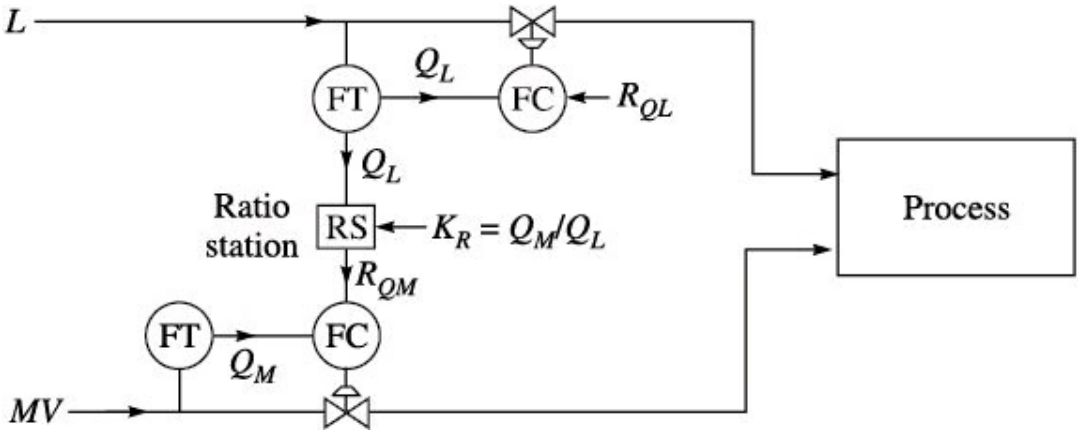


Figure 15.17 Ratio control with manipulation in both loops.

An important feature of the control scheme of Figure 15.17 is that for changing the amount of total throughput of the mixed feed to the process, one has to only adjust the set point R_{QL} to the load variable flow-loop.

EXAMPLE 15.4 In this example, we discuss the ratio control of the feed streams in the Ostwald's method of Nitric Acid synthesis by high temperature gas phase reaction between NH_3 and aerial O_2. The reaction happens in three steps:

Step 1 $4NH_3 + 5O_2 = 6H_2O + 4NO$

Step 2 $2NO + O_2 = 2NO_2$

Step 3 $3NO_2 + H_2O = 2HNO_3 + NO$

The first step reaction is carried over a catalyst at 700ºC. The product of this reaction is processed in a tower type converter to complete the oxidation (second

step). A part of the exit stream from the tower is recycled back to convert the NO obtained from the third step reaction into NO_2. Air, as the source of O_2, is used in 20% excess of the stoichiometric requirement. A pilot reactor has been set up with a capacity to process 17 kg of NH_3 per min (i.e. 1 kg-mole). The feed-streams, NH_3 and air at the proper mass ratio, enter into the pre-heater unit at 20ºC and 755 mm Hg. A ratio control scheme is used for this purpose. Find the Gain of the Ratio Station, K_R. The control scheme is shown in Figure 15.18.

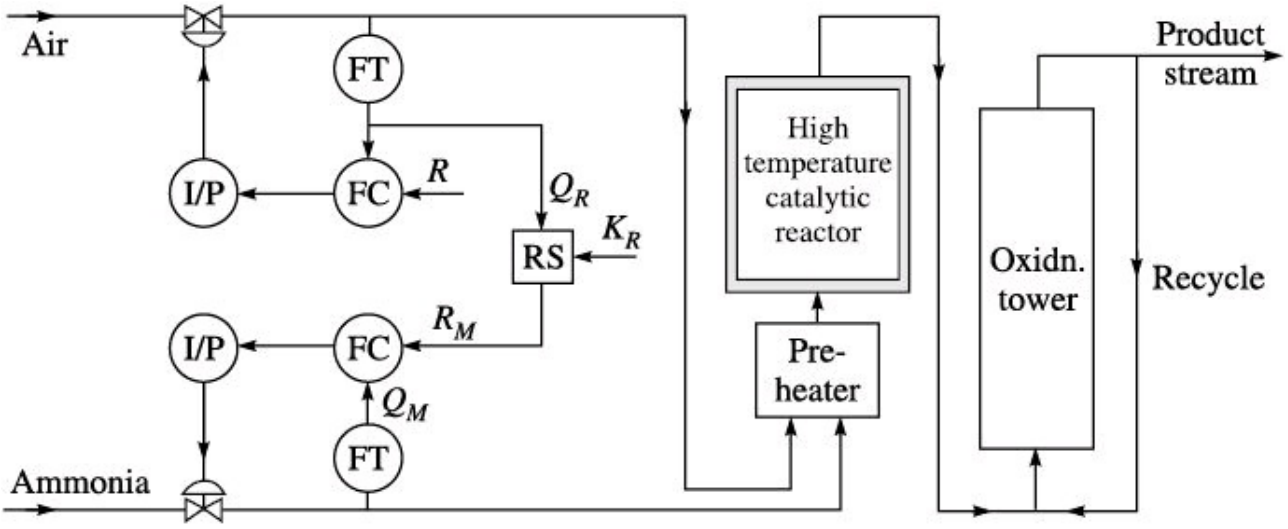


Figure 15.18 The ratio control of the feed streams for nitric acid synthesis.

Data: The flow sensor transmitters for air has a range of 0 to 40 m^3/min, and output 4–20 mA, and for NH_3 has a range of 0 to 40 m^3/min, and output 4–20 mA. The flow transmitters have individual '*square-root extractors*'. The Electro-pneumatic-converter (EPC) has a gain K_{IP} = 0.75 psi/mA. The control valves have linear characteristics.

Solution: To find the overall stoichiometry, multiply step 1 reaction by 2, step 2 reaction by 6 and step 3 reaction by 4, then add them together and finally divide the result by 8, to obtain:

$$NH_3 + 2O_2 = HNO_3 + H_2O$$

Stoichiometric Basis: 1 kg-mole NH_3.

For 1 kg-mole NH_3, the stochiometric requirement of O_2 is 2 kg-mole.

For 1 kg-mole NH_3, the amount of oxygen supplied is $2 \times 1.2 = 2.4$ kg-mole.

2.4 kg-mole of O_2 is available from 2.4/0.21 = 11.4286 kg-mole air.

2.4 kg-mole of O_2 at 20ºC and 755 mm Hg will occupy a volume

$$= 2.4 \times 22.414 \times \frac{293 \times 760}{273 \times 755} = 58.12 \text{ m}^3$$

and, 11.4286 kg-mole of Air at 20ºC and 755 mm Hg will occupy a volume

$$= \left(\frac{11.43}{2.4}\right) \times 58.12 = 276.75\,\text{m}^3$$

The electronic flow transmitters for NH_3 has a span, S_M = 40 m^3/min, and for air has a span, S_R = 350 m^3/min. The air flow rate, Q_R, may be considered as the load variable for the purpose of ratio control. The input to the ratio station is Q_R. The output signal, R_M, serves as the set point for the NH_3 flow rate controller. R_M may be found as:

$$R_M = K_R Q_M$$

From the information that 20% excess air than the theoretically required amount is used, the desired ratio is:

$$R_D = \frac{Q_M}{Q_R} = \frac{58.12}{276.75} = 0.21$$

Because the transmitter gains are inversely proportional to their spans, the ratio station gain following Equation 15.17 may be found as:

$$K_R = R_D \frac{S_R}{S_M} = 0.21 \times \frac{350}{75} = 0.98$$

The next example is about the air/fuel ratio control of the furnace of a mini-boiler installation. The required heat throughput and its fluctuation for downstream load variation are small enough so that a simple ratio scheme with fixed K_R is all that is necessary.

15.3.1 Air-Fuel Ratio Control of Combustion Systems

Analysis of this important application of ratio control may be appreciated by considering the following examples.

EXAMPLE 15.5 (The Air/Fuel Ratio Control in The Furnace of a Mini-Boiler)
Air/Fuel ratio control in a mini-boiler furnace using gaseous or liquid fuel is a common application area of ratio control. A boiler for steam generation employs a furnace as the heat source. The furnace burner uses a liquid fuel (say furnace oil FO) for combustion. The generated steam goes to the steam header connected to the loads or systems that use the steam from the header by individual supply lines. The increase or decrease in steam demand is reflected in the variation of header pressure.

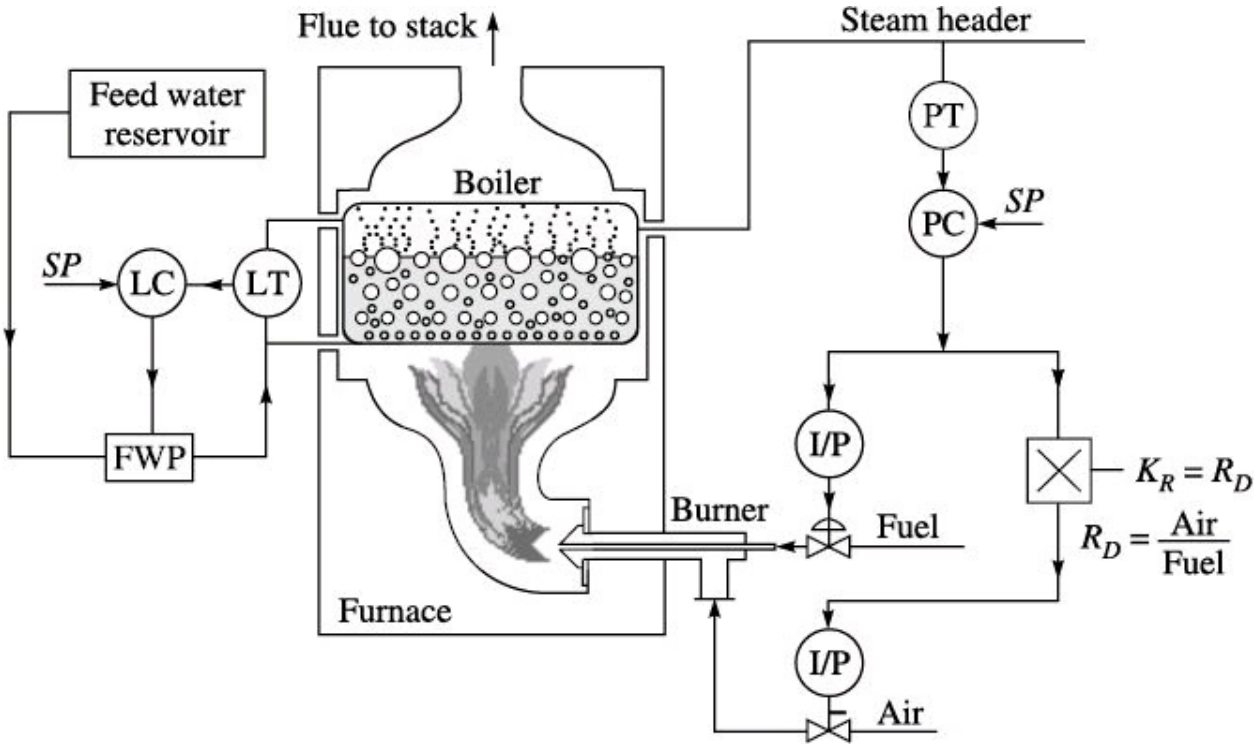


Figure 15.19 The air/fuel ratio control of a mini-boiler

The steam header pressure is sensed and sent to a pressure controller. The output of this electronic controller is converted by an EPC (I/P) to a proportional pneumatic signal that actuates a pneumatic control valve to regulate the fuel flow rate. The same controller output in a parallel path comes to the ratio station where it is multiplied by the ratio station gain, K_R, that has been set equal to the desired ratio, R_D, of 'Air flow' to the 'Fuel flow rate'. The output signal from the ratio station, i.e. the desired Air flow rate goes to the Butterfly type control valve for air flow manipulation. The ratio control features the second configuration of ratio station in Figure 15.16(b), in which it is used as multiplier.

An auxiliary level control loop that is an ON-OFF controller with a wide differential between HI and LO set points always keep the water level in between them. The manipulation being the start and stop of the feed water pump (FWP).

Note that there is no control loop with respect to fuel or air flow. The pressure controller output is implemented through the two control valves. In contrast to a large capacity system, such a mini-boiler set-up features relatively small individual stream flow rates and their variations are also small. For such low capacity systems, a simple ratio scheme with fixed K_R is all that is necessary, a more detail instrumentation may not be economically justified.

However, for moderate to large capacity installations, as we have said before, control loops are necessary for individual streams to obtain highest product quality with maximum capacity utilization. Also, in large capacity systems where the limiting reactant flow rate has to be varied according to the demand of downstream load, the desired ratio, R_D, of the streams has to be varied.

A relatively large furnace is an example of a similar situation where for an increased heat load, higher amount of fuel is required to be burned within the same space or combustion volume of the furnace interior. A detailed analysis of combustion and hydrodynamic models for a given furnace indicates that for higher values of fuel flow rate, the amount of excess air should be increased to keep the combustion efficiency constant. Such a correction is often called '**Oxygen Trim**'. Therefore, the ratio station gain, $K_R = R_D$, should have the facility of variation. In the next example, a simplified version of such a control scheme has been discussed.

In the previous example, the amount of heat load or demand has been recognized as pressure variation in the steam header. In other similar applications, this manifestation may be reflected in other variables of the system that constitutes the load. As in a Thermic-Oil heating furnace that supplies hot thermic-oil for process heating applications, the heat demand may be obtained as a function of the heating fluid flow rate (the manipulated variable of the process temperature control loop).

EXAMPLE 15.6 A thermic-fluid (SERVO-Therm) heating system is used to supply hot oil as the heat transfer fluid in the tank-coil of an endothermic reactor. The heating system is a furnace that burns a liquid fuel with excess air. The reactor temperature control loop uses the flow forcing of the thermic-fluid as the manipulation. The heat load is sensed by a flow sensor of the thermic-fluid flow rate.

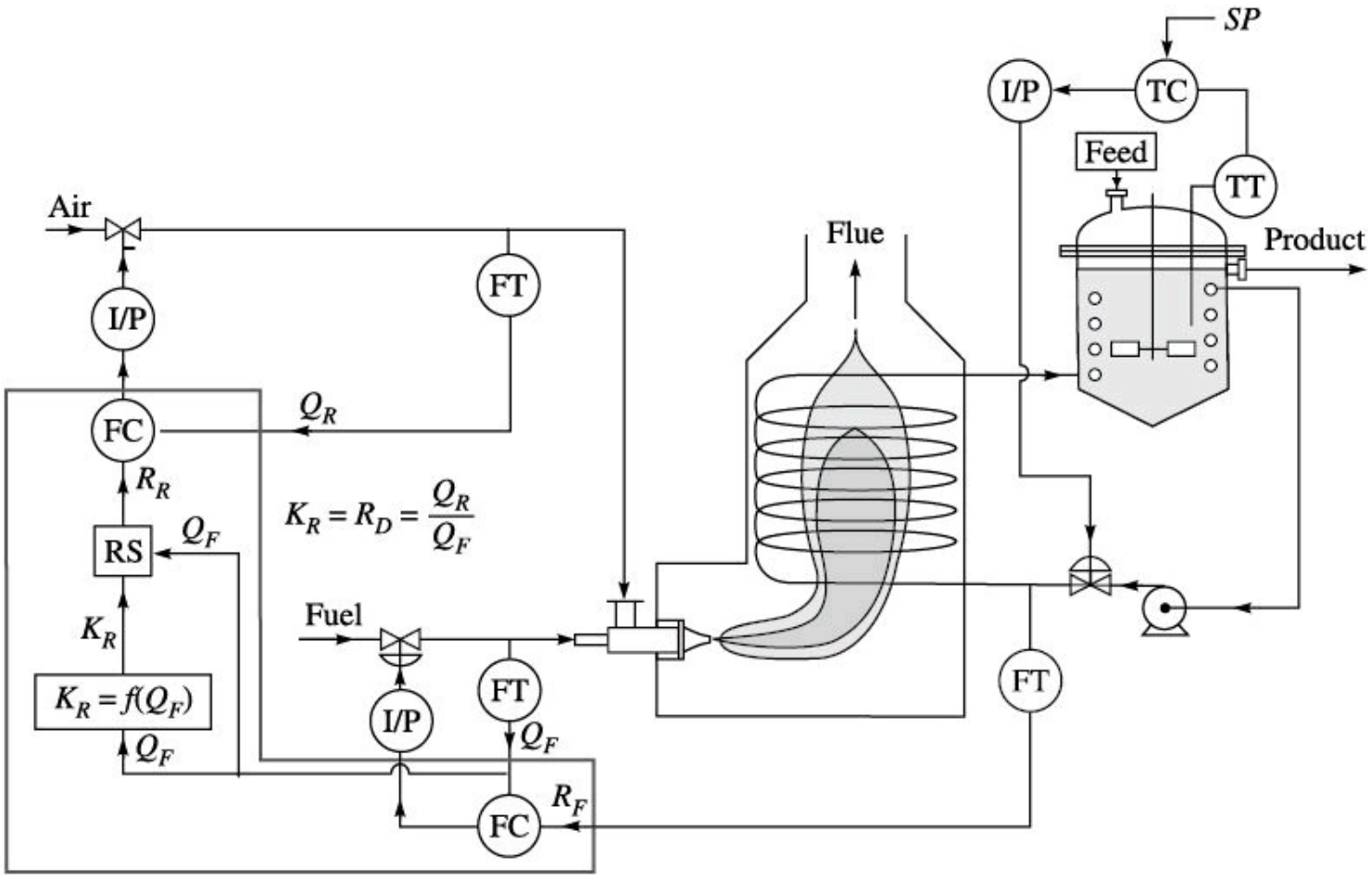


Figure 15.20 The air/fuel ratio control of a large capacity furnace with O_2 trim.

The fuel and air flow have individual control loops. The thermic-fluid flow signal may be used as the set point R_F, to the fuel flow controller. The fuel flow rate signal goes to the fuel flow controller as well as to the computing block where current value of ratio station gain, K_R, is calculated according to a function of fuel flow rate, Q_F, as $[K_R = f(Q_F)]$. This function constitutes the oxygen trim. In the simplest form, this may be a linear increase of K_R with respect to Q_F. In this application the ratio station gain K_R is effectively equal to the desired ratio
R_D = (Air flow)/(Fuel flow) = Q_R/Q_F. Q_F also goes to the ratio station where it is multiplied by the corrected value of K_R to obtain the desired value of air flow Q_R that is used as the set point R_R in the air flow control loop.

Data: A medium capacity furnace is used for thermic-liquid heating facility. The hot oil is used in the tank-coil of a continuous reactor handling an endothermic reaction. To negotiate the minimum load and maximum possible load variation, it has been estimated that the required minimum and maximum flow of a liquid fuel, FO, to be burned are respectively 80 and 100 l/h. Fuel and air intake occurs at 30ºC and 760 mm Hg. It has been also estimated that for liquid fuel flow rate, Q_F = 80 l/h. The excess air requirement is 25% and that for Q_F = 100 l/h is 45%. The liquid fuel FO has been analyzed as:

H–13.7%; C–77.3%; N–2%; O–6.0% and S–1.0%; FO density r = 0.95 g/cm^3

The flow sensor transmitters for air has a range of 0 to 500 l/min, and output 4–20 mA, and for the fuel has a range of 0 to 2.0 l/min, and output 4–20 mA. The flow transmitters have individual 'square-root' extractors. The Electro-pneumatic-converters (I/P) have gain K_{IP} = 0.75 psi/mA. The control valves have linear characteristics.

Assuming a linear relation between the 'Excess Air' and 'Fuel Burned', prepare two graphs:

1. %Excess Air (PEA) vs. Q_F (l/h.);
2. The Ratio Station Gain K_R vs. Q_F (l/min.). Thus, find the required function:

$$K_R = f(Q_F)$$

Solution: Calculating the theoretical Air requirement (Basis: 1 kg Fuel)
H – 0.137 kg = 0.137/2 kg-mole, requires + 0.137/4 kg-mole O_2 for burning

C – 0.773 kg = 0.773/12 kg-mole, requires + 0.773/12 kg-mole O_2 for burning

S – 0.01 kg = 0.01/32 kg-mole, requires + 0.01/32 kg-mole O_2 for burning

O_2 available is 0.06 kg = – 0.06/32 kg-mole,

Theoretical Oxygen requirement for burning of 1 kg fuel:

= 0.137/4 + 0.773/12 + 0.01/32 – 0.06/32 = 0.0971045 kg-mole O_2, and that would be available in 0.0971045 × (100/21) = 0.46240 kg-mole air. At 30ºC and 760 mm Hg this amount of air will occupy 0.46240 × 22.414 × (303/273) = 11.50322 m^3.

Theoretical air requirement is 11.50322 m^3 for 1 kg fuel, or 1/0.95 = 1.0526 l fuel.

Thus, theoretical air requirement for 1 l fuel is 11.50322/1.0526 = 10.928 m^3.

For minimum fuel flow condition	For maximum fuel flow condition
Fuel flow = 80 l/h = 1.333 l/min Air flow (with 25% excess) = 303.56 l/min Desired ratio of $Q_R/Q_F = R_D$ = 303.56/1.333 = 227.6757 Minimum value of K_R $K_R = R_D \frac{SPAN_{FUEL}}{SPAN_{AIR}} = 227.6757 \frac{2}{500} = 0.9107$	Fuel flow = 100 l/h = 1.667 l/min Air flow (with 45% excess) = 440.158 l/min Desired ratio of $Q_R/Q_F = R_D$ = 440.158/1.667 = 264.0895 Maximum value of K_R, $K_R = R_D \frac{SPAN_{FUEL}}{SPAN_{AIR}} = 264.0895 \frac{2}{500} = 1.05636$

Now the relation between K_R and Q_F may be graphically shown as:

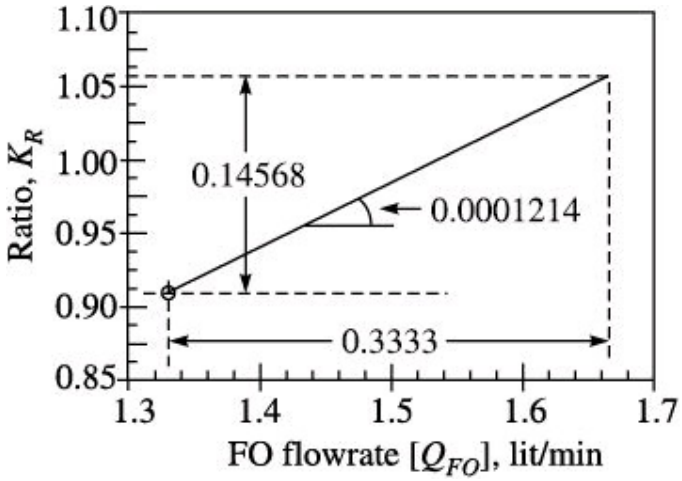


Figure 15.21 Graphical development of equation 15.20.

and the linear relation between K_R and Q_F may be developed as:

$$K_R = 0.9107 + 0.0001214 \times (Q_F - 1.333) \qquad (15.20)$$

Preparation of the first graph is left to the reader as an exercise. Computation and implementation of Equation 15.21 requires arithmetic power that may be realized by Special Computing Blocks or Digital Control System, indicated by the gray boundary in Figure 15.20.

The oxygen trim strategy that realizes the variation of ratio station gain, K_R, as a function of changing oxygen demand for variable fuel flow rate, Q_F, has been presented in an over simplified manner in the above Example 15.6. The development of Equation 15.20 requires an appreciable amount of experimental data on a given furnace with same fuel quality, and expecting the result would be a linear relation is an over simplification.

This method is sort of an explicit approach that lacks any correction of K_R based on measurement of degree of completion of combustion process in the furnace. The combustion efficiency may be estimated by implementing a Flue Gas Analyzer to measure the amounts of O_2, CO_2 and CO present in the flue and send the information to a computation block that calculates the combustion efficiency and finds out the extra amount of O_2 in terms of air to be added to obtain the target combustion efficiency. The result of this calculation comes as the ΔK_R that has to be added to the minimum K_R required for minimum fuel flow condition. The control of air-fuel ratio for a large capacity furnace with oxygen trim, implementing the feedback information from a 'Flue-Gas-Analyzer', has been shown in the next example.

EXAMPLE 15.7 The control scheme of Figure 15.22 differs from that in Figure 15.20 only by the fact that variation of K_R is calculated as ΔK_R to be added to the base or initial value of K_R*. ΔK_R is calculated with respect to the current information of composition of O_2, CO_2 and CO present in the flue obtained from the Flue Gas Analyzer. Other features of the system are same as in the previous example.

In Examples 15.4 and 15.5 the implementation of ratio control remained true to our earlier statement that compensation depended only on feed forward strategy. But Examples 15.6 and 15.7 feature a feedback along with feed forward action in the form of using the information regarding the process performance to fine tune the ratio of reactants in the feed. Application of such a mixed strategy is required when process gain shows non-linearity with respect to load changes. In the industry, we should remember that there are many furnaces still run on modular instruments with limited computing power, also the fuel used may be of solid or gaseous variety. What should be the ratio control configuration for such cases? This topic has been discussed in Chapter 13 of *Advanced Process Dynamics and Control*, P.K. Sarkar (PHI Learning).

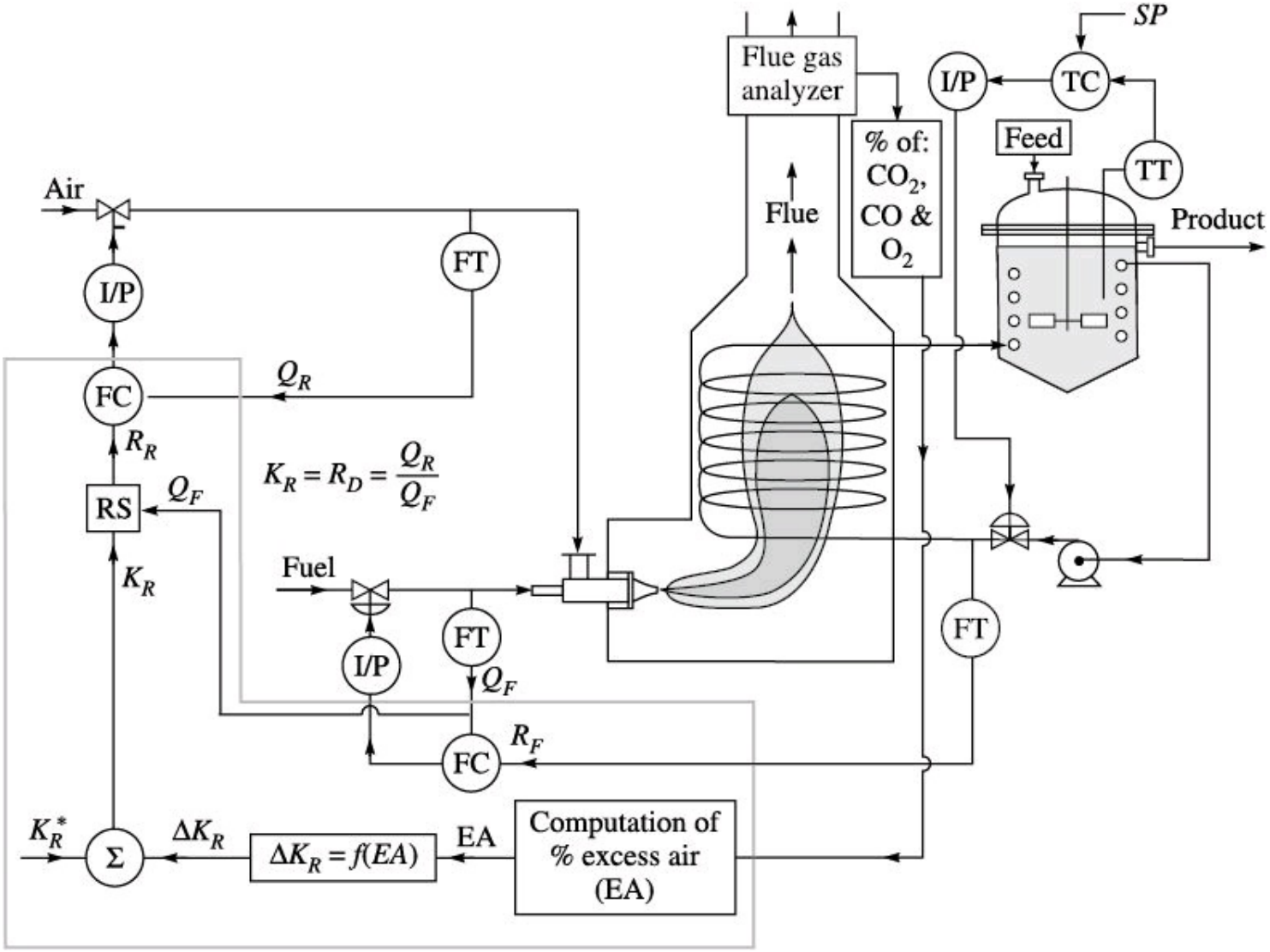


Figure 15.22 Air-Fuel ratio control of a furnace with O_2 trim, using the feedback information from a 'Flue-Gas-Analyzer'.

15.4 Cascade Control

Two application categories of this control strategy are commonly seen in process industries.

1. In a process comprising of serially connected dynamic elements, the manipulation goes to the first of these elements but the controlled variable is available from the last or any of the intermediate stages. The control difficulty is that we cannot directly put the required manipulation into the actual problem area, and that the manipulation has to travel through the intermediate stages and obviously be influenced by their dynamics.
2. The manipulated variable is drawn from a suitable utility or energy source to the control loop, as in the case of steam heated heat transfer equipment. The manipulation is effected by throttling the steam flow through a control valve available from an adjacent steam header. Ideally, the header pressure should remain constant. But if steam is supplied to other equipment from the same

steam supply, the header pressure would fluctuate quite arbitrarily, depending on the steam requirement of these parallely connected loads. This demand pattern is unpredictable on which there is no regulation. Control quality of any of the loads would become poorer because of the uncontrolled fluctuation of manipulated variable potential at its source.

To handle these problem, two measurements are used, one for the controlled variable G_{MM}, and the other, G_{MS}, for the manipulated variable management in a way to bring up the necessary correction for the reasons we have already stated. There are two controllers, G_{CM} and G_{CS}, associated with the measuring elements. G_{CM} has its external set point, but G_{CS} does not have external set point; the output of G_{CM} goes to G_{CS} as its set point. This internal set point may be considered as the *corrected set point* to negotiate the dynamic difficulty of the process. This fashion of information flow, i.e. the first controller's output becoming the set point of the second controller, is called *cascading of information*, which has been recognized in its nomenclature: **Cascade Control**.

The first controller, G_{CM}, that has an independent set point facility and the associated measuring element to sense the variable whose maintenance is the main objective of the loop, is said to be part of the **Master Control Loop**. The controller of this loop is called the **Master Controller** that produces the set point to the second controller. The second controller, G_{CS}, and the associated measuring element, is said to be part of the **Slave Control Loop**. The controller of this loop is called the **Slave Controller**.

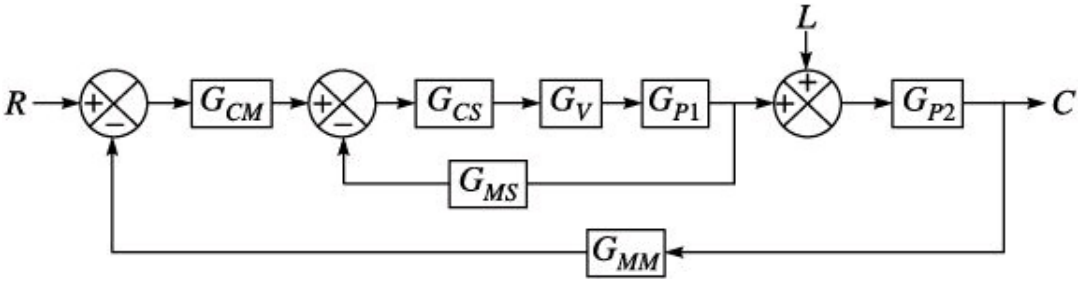


Figure 15.23 Configuration of a cascade control loop.

The general configuration of a cascade control loop has been shown in Figure 15.23. Block diagram wise, as the master controller resides on the outer loop and the slave controller on the inner loop, they are also called **Outer** and **Inner** loop controllers respectively. In this figure, the process is represented by two parts—G_{P1} and G_{P2}. In such a system, G_{P1} represents the external system characteristics, and the interaction between G_{P1} and G_{P2} brings up the control difficulty. Figure 15.23

reveals that the load enters the outer or master control loop, but it has no manipulation facility. Whereas, the inner or slave control loop is devoid of load entry, but has manipulation facility. To expose the interrelation between the master (outer) loop and slave (inner/nested) loop let us discuss a couple of examples.

15.4.1 Example of Cascade Control Application

This example is about the temperature control of a tray in the bottom part of a distillation column by regulating the vaporization rate in the re-boiler. The actual manipulated variable is the steam flow rate to the re-boiler throttled by a control valve, the FCE of the loop. The single feedback loop control suffers due to frequent variation of steam pressure in the steam header
(supply line). During the steady state condition of operation, a sudden decrease of steam pressure may happen as a result of, say starting of another load connected in parallel fashion to the steam header. This will decrease the steam flow through the valve than the required rate. However, this undesirable decrease will affect the vapour generation rate that in turn will decrease the tray temperature and ultimately the feedback loop will take action by further opening the valve to drive away the disturbance. Think about the length of response time of the loop shown in Figure 15.24(a) as the disturbance finally produces a control action.

The cascade configuration for the same control duty has been shown in Figure 15.24(b). The tray temperature control is the main objective of this loop. The temperature measuring element and the temperature controller constitute the master loop. The master controller output goes to the steam flow rate controller as its set point. The steam flow rate measuring element and the flow controller make up the slave loop of the system. The variation of steam flow rate due to pressure fluctuation will be sensed by the flow sensor and will be communicated to the flow controller, but its set point will come from the temperature controller, which will act following any discrepancy in tray temperature due to steam pressure variation, and produce a controller output that will be received by the flow controller as its set point and drive away the disturbance.

This cascade control scheme is expected to act with more speed and precision than the feedback loop shown in Figure 15.24(a).

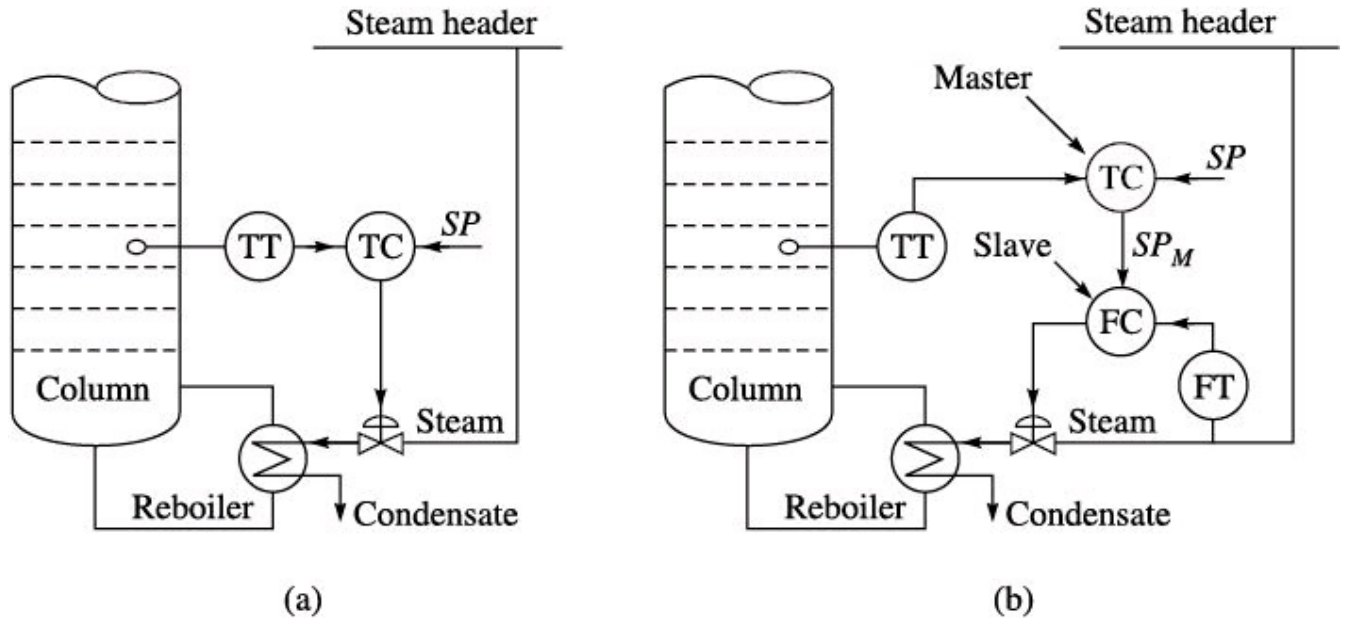


Figure 15.24 Comparison of cascade with a feedback loop applied to the temperature control of stripping section of a distillation column.

15.4.2 Another Example of Cascade Control Application

In this example, we shall introduce the cascade control of the temperature of an exothermic CSTR. The control difficulty ensues from the fact that though the temperature of the reactor could be measured, but manipulation resides only on the jacket (a) fluid flow rate, or (b) temperature.

[The flow manipulation or 'flow forcing' gives a less accurate control quality than temperature manipulation or 'temperature forcing'. The hardware requirement for the second option is costlier than the first choice. The control schemes for two configurations are shown in Figures 15.25(a) and (b). For more critical applications where tolerance from the set point is less than the conventional reactor control, the second choice is preferred.]

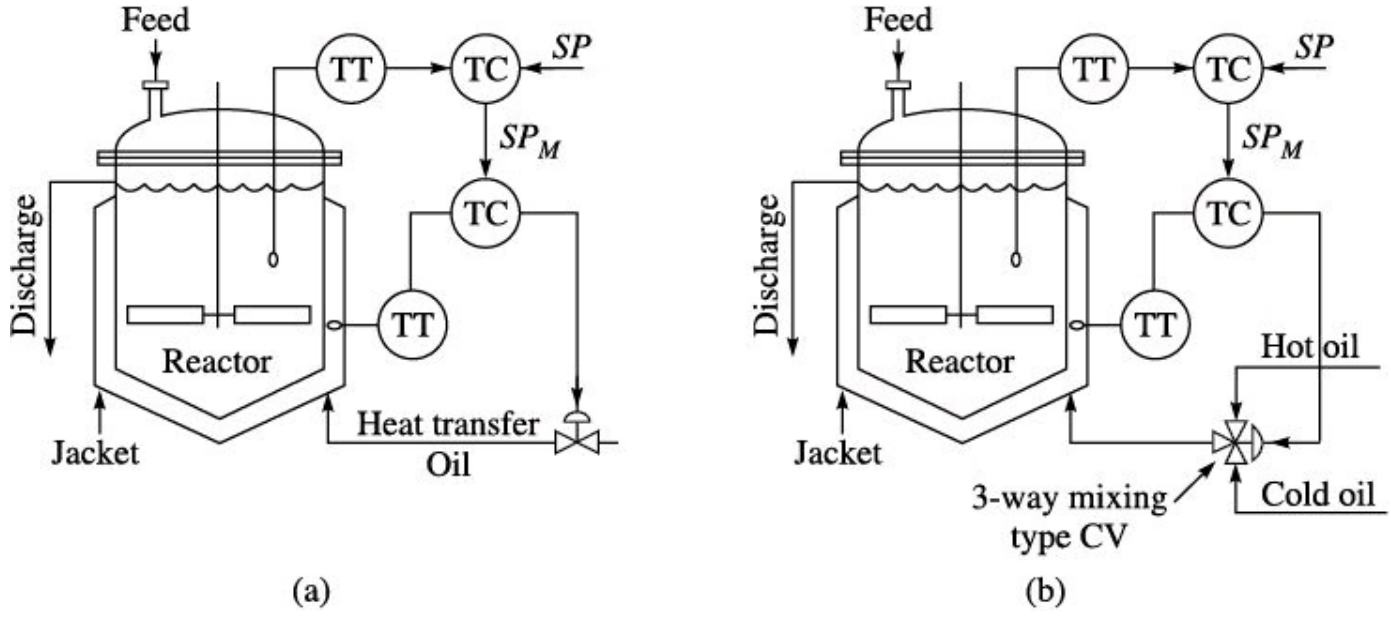


Figure 15.25 Cascade control of a CSTR implementing (a) Flow forcing, (b) Temperature forcing as the manipulation.

The rest of the cascade loops in Figure 15.25 are identical in configuration. The reactor temperature is the target variable to be controlled. The reactor temperature controller and reactor temperature measuring element constitute the master loop. Set point to the jacket liquid temperature controller comes as the output from the master controller. The source of control difficulty is the sluggish response speed due to overall dynamics of two serially connected *RC* type elements (the jacket and the reactor) in an interactive fashion.

15.4.3 Analysis of Cascade Control System

For analysis of a cascade loop, a more general block diagram should be considered in which load may enter in the inner as well outer loop of the control system. To examine this possibility two application examples have been considered.

In the example of tray temperature control in the bottom part of the distillation column, use of cascade loop has been cited, that covers up any fluctuation of the steam header pressure. This steam pressure change enters the outer loop and necessary correction is made by implementing the cascade algorithm to maintain a constant tray temperature. But there is a possibility that another load, in the form of temperature change of the feed, may enter into the inner loop. The cascade control has the capability to deal with such kind of disturbance and still maintain the tray temperature; hence, a more generalized block diagram of cascade control loop should also contain this feature that provide entry facility to both kind of disturbances into the loop.

Another example of this situation may arise in a jacketed semi-batch reactor that is handling an endothermic reaction and heat is added by the steam flowing through the jacket of the reactor. The steam comes from a header to the jacket entry through a control valve that is the FCE of the loop. There is an addition of a particular reactant to the reaction mixture that causes the progress of the reaction. This may be considered as the load to the inner loop. Any variation of steam header pressure may be considered as the load variation in the outer loop.

Thus, the generalized block diagram of a cascade loop should appear as in Figure 15.26.

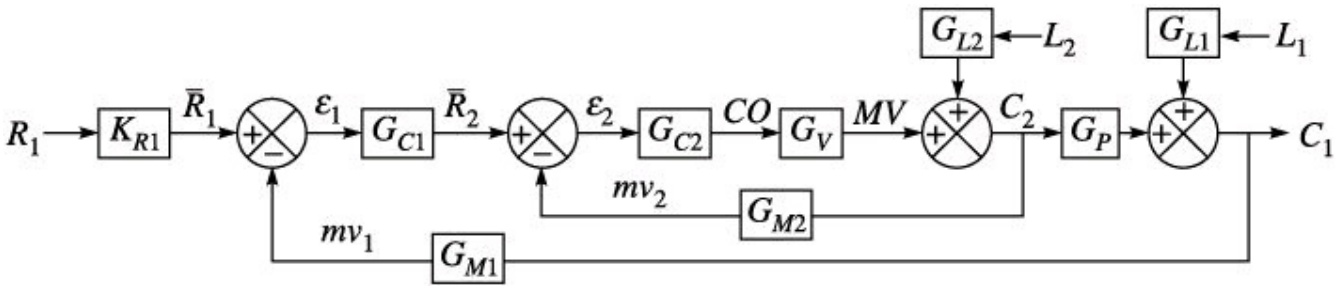


Figure 15.26 Generalized block diagram of cascade control loop.

The closed loop transfer functions may be obtained as:

$$\frac{C_1}{L_1} = \frac{G_{L1}(1 + G_{C2}G_V G_{M2})}{1 + G_{C2}G_V G_{M2} + G_{C1}G_{C2}G_V G_P G_{M1}}; \frac{C_1}{R_1} = \frac{G_{C1}G_{C2}G_V G_P K_{R1}}{1 + G_{C2}G_V G_{M2} + G_{C_1}G_{C2}G_V G_P G_{M1}}$$

$$\frac{C_1}{L_2} = \frac{G_P G_{L2}}{1 + G_{C2}G_V G_{M2} + G_{C1}G_{C2}G_V G_P G_{M1}}; \frac{C_2}{R_2} = \frac{G_{C2}G_V}{1 + G_{C2}G_V G_{M2}}$$

The characteristic equation of the cascade loop would be:

$$1 + G_{C2}G_V G_{M2} + G_{C1}G_{C2}G_V G_P G_{M1} = 0 \qquad (15.21)$$

If the inner loop is removed from the system (i.e. $G_{C2} = 1$, and $G_{M2} = 0$), the characteristic equation would be identical for a conventional feedback control system:

$$1 + G_{C1}G_V G_P G_{M1} = 0 \qquad (15.22)$$

EXAMPLE 15.8 For a cascade loop with reference to the generalized block diagram of
Figure 15.19, the individual blocks contain the following TFs:

$$G_V = \frac{5}{s+1}; \; G_P = \frac{4}{(2s+1)(4s+1)}; \; G_{L2} = 1 \; ; \; G_{M1} = 0.2; \; G_{L1} = \frac{1}{3s+1} \; ; \text{and, } G_{M1} = 0.05$$

where, the time constants are in minute, and gains have consistent units.

Determine the stability limits of a conventional feedback control with a (P) controller, as well as for a cascade control consisting of two (P) controllers. Assume $G_{C2} = K_{C2} = 4$, for the inner (slave) loop. Calculate the resulting 'offset' for a unit step change in L_2.

Solution: In the cascade arrangement for the inner loop, we substitute values in the closed loop TF, C_2/R_2, and obtain the following:

$$\frac{C_2}{R_2} = \frac{4\,[5/(s+1)]}{1 + 4 \times 0.2 \times [5/(s+1)]} = \frac{4}{0.2s+1}$$

Hence, from the above equation, the closed loop time constant of the inner loop is 0.2 min. But the feedback control system has a time constant of 1 min, since in this case,

$$\frac{C_2}{R_2} = G_V = \frac{5}{s+1}$$

Thus, cascade control significantly speeds up the response of C_2. For a (P) controller
($G_{C1} = K_{C1}$) in the outer loop, the characteristic equation is:

$$1+4\times\left(\frac{5}{s+1}\right)\times 0.2+K_{C1}\times 4\times\left(\frac{5}{s+1}\right)\left(\frac{4}{(2s+1)(4s+1)}\right)\times 0.05=0$$

or,

$$8s^3+46s^2+31s+5+4K_{C1}=0 \qquad (15.23)$$

By Routh-Hurwitz's method, for stability limit K_U = 43.3; For $G_{C1} = K_{C1}$; For conventional feedback loop with (P) controller, the characteristic equation is:

$$8s^3+14s^2+7s+1+K_{C1}=0 \qquad (15.24)$$

Again by Routh-Hurwitz's method, for stability limit K_U = 11.25; For $G_{C1} = K_{C1}$. Therefore, the cascade configuration has increased the ultimate gain by nearly a factor of 4.

The offset for C_1 for a unit step change in L_2 may be obtained by setting $s = 0$ in the right hand side of the closed loop TF, C_1/L_2, obtained above (using the final value theorem)

Thus,

$$e_1(t\to\infty)=R_1-C_1(t\to\infty)=\frac{-2}{5+8K_{C1}}$$

For the conventional feedback control, the offset would be,

$$e_1(t\to\infty)=\frac{-2}{1+2K_{C1}}$$

By comparing the above expressions, it may be concluded that by applying cascade control, the offset has been reduced for similar value of K_{C1}.

EXAMPLE 15.9 A cascade loop has been proposed for maintaining the level in a two tank liquid level system that is under non-interacting combination. The controlled variable is the level of the second tank. The level control system is shown in Figure 15.27(b). We shall compare the performance of this cascade control system with that of a feedback loop. The same level control with a feedback loop has been shown in Figure 15.27(a). Disturbance comes as an additional

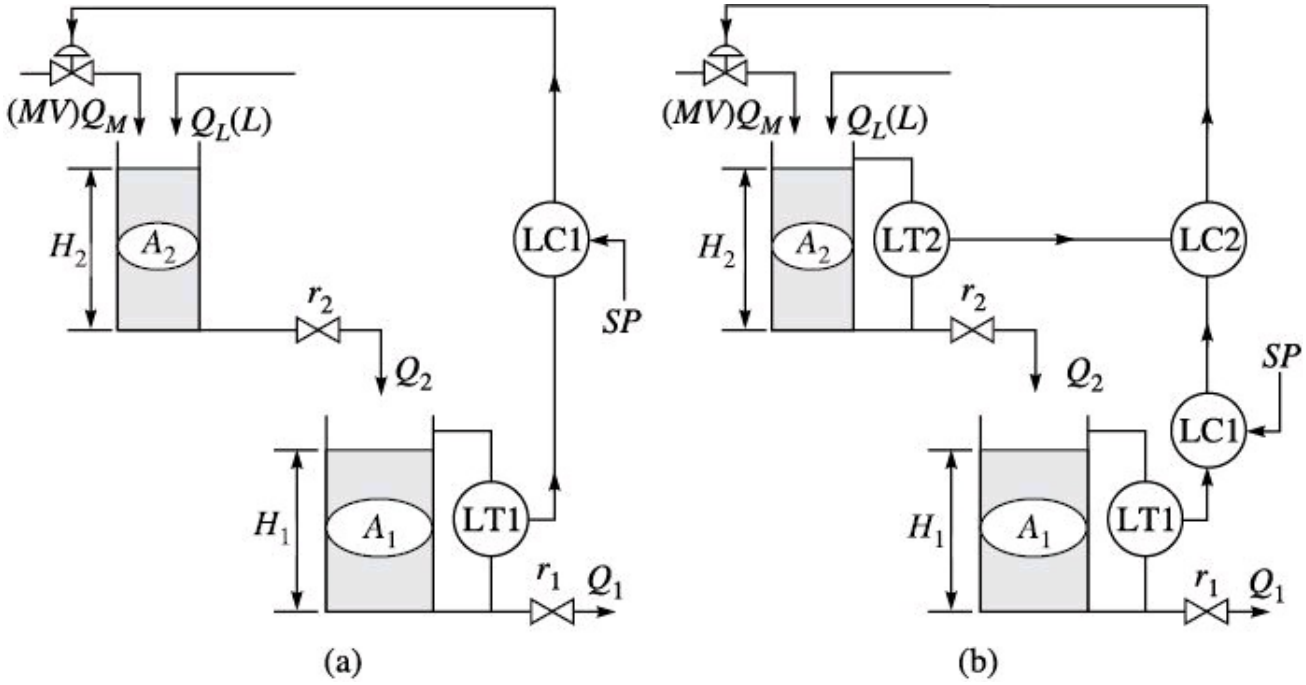


Figure 15.27 The two configurations of level control loop for a non-interacting, two-tank system: (a) Feedback. (b) Cascade.

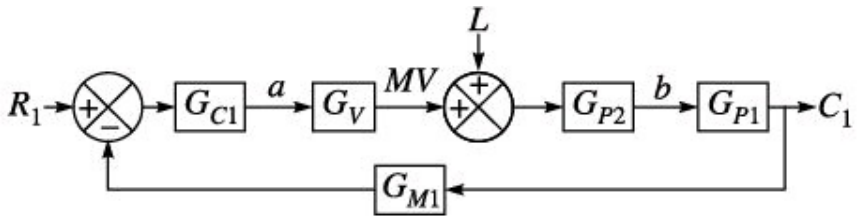
stream into the top tank that is negotiated by the exit flow manipulation through the control valve. It has been assumed that there is a constant head supply at the entry of the control valve which has linear characteristics. Also, the instruments used are all electronic so that no electro-pneumatic signal conversion is required. The system parameter values are presented as follows:

Tank 1: Area, A_1 = 600 cm^2, Exit line resistance, r_1 = 1 cm/(cm^3/s), t_1 = 600 s = 10 min.

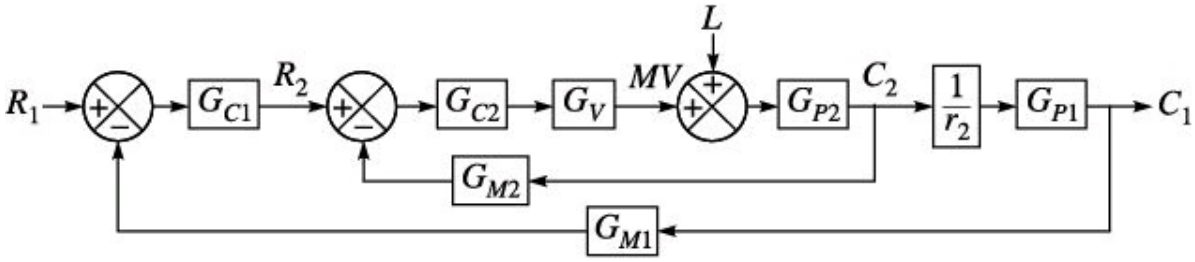
Tank 2: Area, A_2 = 120 cm^2, Exit line resistance, r_2 = 1 cm/(cm^3/s), t_2 = 120 s = 2 min.

$$G_V = \frac{10}{s+1}; \quad G_{P1} = \frac{1}{10s+1}; \quad G_{P2} = \frac{1}{2s+1}; \quad G_{M1} = G_{M2} = K_M = 0.16\frac{\text{mA}}{\text{cm}}$$

The corresponding block diagrams of above control systems are given in Figure 15.28. You should note that in Figure 15.28(b) a block containing the TF: $1/r_2$, has been placed between G_{P2} and G_{P1}. The output variable from G_{P2} is C_2, the level of the top tank that is sensed by G_{M2}. But the input variable to the bottom tank is the flow rate Q_2. Hence, the intermediate block
$1/r_2$ has been introduced to keep the dimensional consistency.



(a)



(b)

Figure 15.28 Block diagrams of the control loops given in Figure 15.27: (a) Feedback. (b) Cascade.

First, the closed loop regulatory TF for the feedback loop may be found as,

$$\frac{C_1}{L} = \frac{G_{P2}G_{P1}}{1 + G_{M1}G_{C1}G_V G_{P2}G_{P1}}$$

Putting the TFs of the individual blocks into the last equation and simplifying, the closed loop TF finally reduces to:

$$\frac{C_1}{L} = \frac{s+1}{20s^3 + 32s^2 + 13s + 1 + 10K_{C1}}$$

From the above result the characteristic equation is, $20s^3 + 32s^2 + 13s + 1 + 10K_{C1} = 0$. By applying the Routh-Hurwitz criteria, the ultimate gain value, $K_U = 1.98$ is obtained. Taking $K_{C1} = 1$, and using the final value theorem on the closed loop TF, the offset is evaluated as, OS = –0.091.

Now we may develop the closed loop regulatory TF of the cascade loop from Figure 15.27(b):

$$\frac{C_1}{L} = \frac{G_{P2}[1/r_2]G_{P1}}{1 + G_{M1}G_{C1}G_{C2}G_V G_{P2}[1/r_2]G_{P1} + G_{M2}G_{C2}G_V G_{P2}}$$

Putting the TFs of the individual blocks into the last equation and simplifying the closed loop finally reduces to:

$$\frac{C_1}{L} = \frac{s+1}{20s^3 + 32s^2 + (13s + 16K_{C2})s + 1 + 1.6K_{C2}(1 + K_{C1})}$$

Taking $K_{C2} = 1$, the characteristic equation becomes:

$$20s^3 + 32s^2 + 29s + 2.6 + 1.6K_{C1} = 0$$

By applying Routh-Hurwitz criteria, the ultimate gain value, $K_U = 27.375$ is obtained.

The offset may be obtained if the final value theorem is put on to the closed loop TF for a unit step change in load. Thus, the following expression is obtained for offset:

$$\text{OS} = 0 - \frac{1}{1 + 1.6K_{C2}(1 + K_{C1})}$$; for $K_{C2} = 1$, and $K_{C1} = 6$ the $OS = -0.082$

Note that as we have obtained a relatively high value of the ultimate gain, still higher values of K_{C1} is permitted to be used.

Finally, the effect of introducing the cascade over the feedback loop, on relative speed of response, may be calculated. For this exercise we shall find the TF between output and input variables identified at the points between which the inner loop of the cascade configuration is situated, because this inner loop is the sole manifestation of cascade control over feedback strategy. For the cascade scheme, this TF would be between C_2 and R_2 in the Figure 15.28(b). The equivalent points are b and a for feedback loop in Figure 15.28(a).

For feedback loop the TF between variables b and a:

$$\frac{b(s)}{a(s)} = \frac{10}{(s+1)(2s+1)} = \frac{10}{2s^2 + 3s + 1} = \frac{10}{1} \cdot \frac{(1/2)}{s^2 + (3/2)s + (1/2)}$$

From this second order TF, the parameters may be evaluated as:

$$K_P = 10,\ \tau = \sqrt{\frac{1}{\omega_N^2}} = \sqrt{\frac{1}{(1/2)}} = 1.414 \text{ and } \zeta = \frac{(3/2)}{2.\omega_N} = 1.061$$

For cascade loop, the TF between variables C_2 and R_2 is:

$$\frac{C_2}{R_2} = \frac{G_{C2}G_V G_{P2}}{1 + G_{M2}G_{C2}G_V G_{P2}}$$

putting block TF values into this expression and simplifying,

$$\frac{C_2}{R_2} = \frac{10}{2s^2 + 3s + 2.6}$$ from which, $K_P = 10$, t = 0.88, and $z = 0.658$.

The above results have been tabulated to facilitate performance comparison of cascade relative to the feedback loop.

Properties	Feedback Control	Cascade Control

K_U	1.98	27.375
OS	– 0.091	– 0.082
K_P	10	10
T	1.414	0.88
Z	1.061	0.658

15.4.4 Tuning of Cascade Control System

Discussion on this topic will follow the approach mentioned by Harriot and later treated by Deshpande and Ash. The tuning of a given cascade system given by its block diagram (say Figure 15.29), may be achieved by performing the calculation steps according to the following sequence:

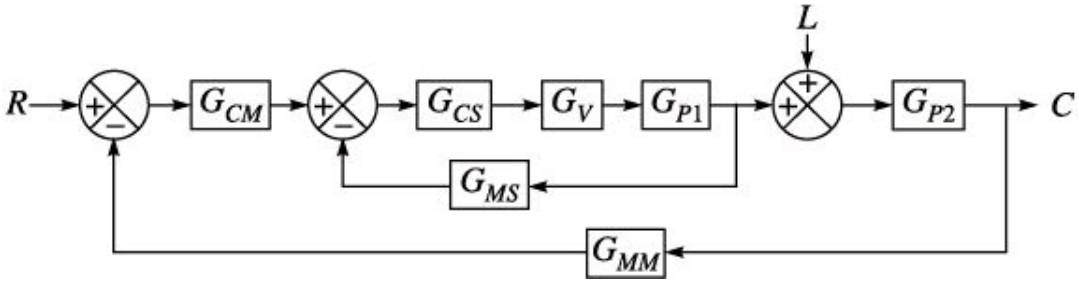


Figure 15.29 Configuration of a cascade control loop.

1. Assuming that G_{CS} be a proportional controller of unity gain ($K_{CS} = 1$), open loop Bode' plot for the inner loop is drawn for the TF:

$$G_{OLI} = K_{CS}G_VG_{P1} = G_VG_{P1}$$

From this first plot w_{CO}, and then K_U, and P_U are evaluated, from which the controller settings according to Zeigler-Nichol's formulae are calculated.

2. The unity feedback closed loop Bode's plot is drawn for the closed loop TF:

$$G_{CLI} = \frac{G_{CS}G_VG_{P1}}{1 + G_{CS}G_VG_{P1}}$$

3. Assuming that G_{CM} be a proportional controller, open loop Bode's plot for the outer loop is drawn for the TF:

$$G_{OLO} = G_{CM}G_{CLI}G_{P2}$$

From this plot, the controller settings for the outer loop controller are calculated by the steps described in step 1.

In many applications, a (P) or (PI) controller for the outer loop and a (PI) controller for the inner loop has given satisfactory performance. (D) mode is rarely used in cascade control due to its limitation in face of input fluctuations that may happen for the inner loop. We shall demonstrate the tuning procedure by the following example. The solution would require a numerical simulation method.

EXAMPLE 15.10 A cascade loop has been proposed for maintaining the temperature of an exothermic continuous flow reactor. The heat transfer fluid is put in the reactor jacket. An evaluation program for the loop has been designed to operate the reactor as a simple stirred liquid heater with the existing feedback loop and also with the proposed cascade loop.

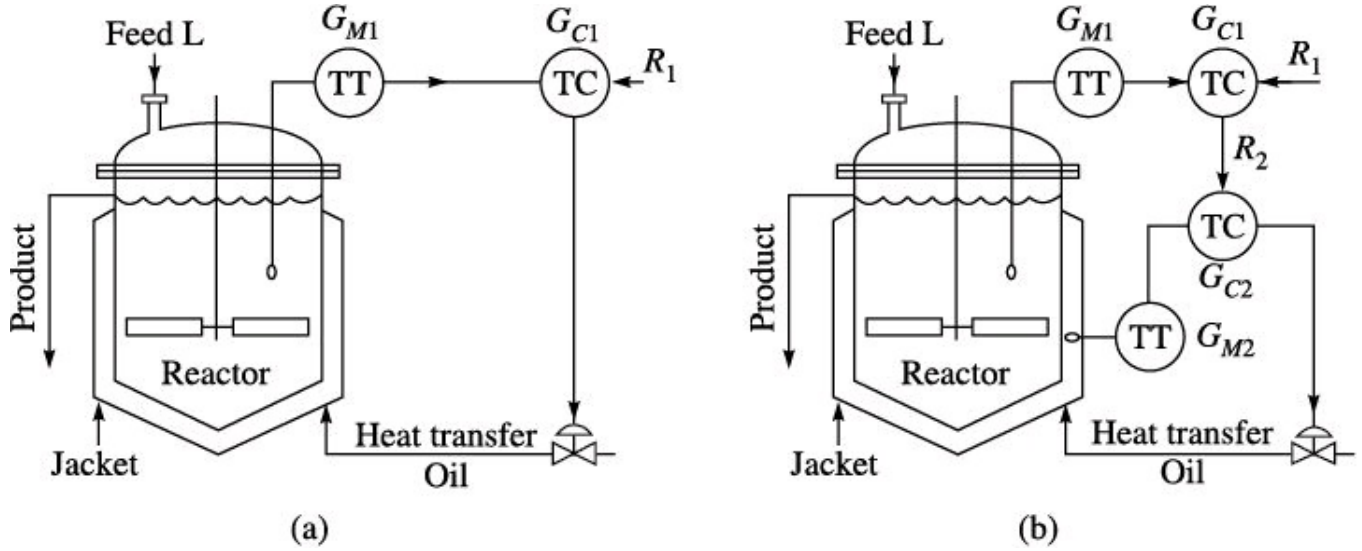


Figure 15.30 The two configurations of temperature control loop for an exothermic CSTR: (a) Feedback. (b) Cascade.

The block diagrams of the two systems are shown in Figures 15.30(a) Feedback. (b) Cascade. The notations associated with instrument symbols have been used to express their location in the corresponding block diagrams shown in the Figure 15.31.

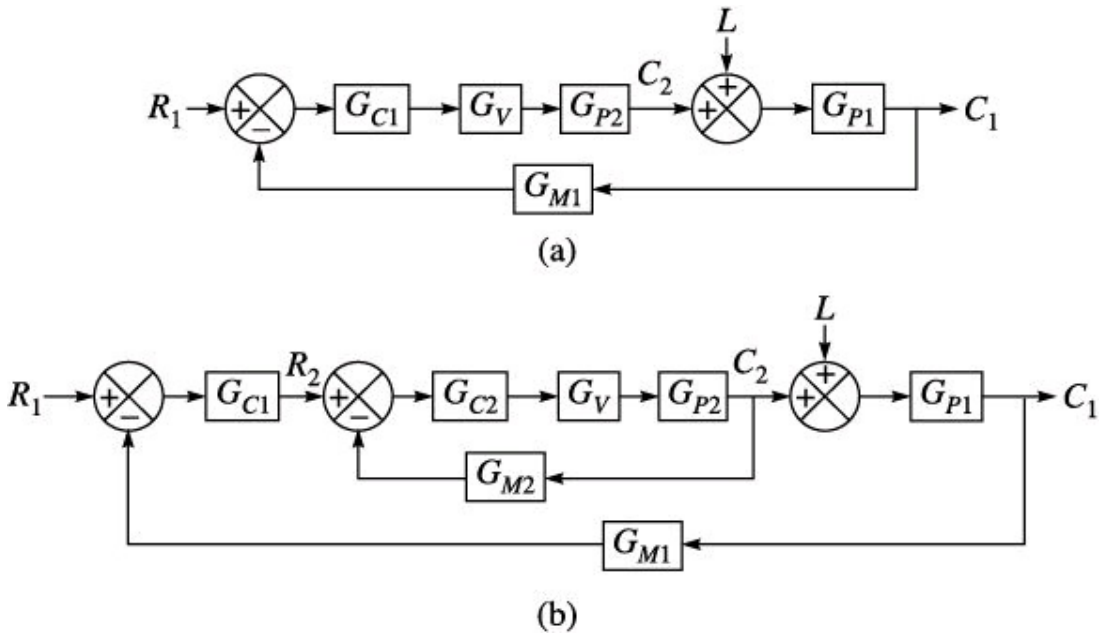


Figure 15.31 Block diagrams of the control loops of Figure 15.24. (a) Feedback. (b) Cascade.

The dynamics of individual blocks has been given in following. The TFs are a bit simplified to reduce the arithmetic retaining all the dynamic character of the loop.

$$G_{P1} = \frac{0.82}{(30s+1)(10s+1)}; \quad G_{P2} = \frac{0.735}{(60s+1)(20s+1)(10s+1)}; \quad G_V = 1; \quad K_{M1} = 1; \quad K_{M2} = 1$$

Tuning of the Feedback loop: The tuning of the feedback controller is carried through the frequency response approach. The forward path TF of the FB loop with a unity gain (P) controller is given as:

$$\frac{K_{CFB} \times 0.82 \times 0.735}{(30s+1)(10s+1)(60s+1)(20s+1)(10s+1)} = \frac{0.6027}{(60s+1)(30s+1)(20s+1)(10s+1)^2}$$

From the Bode's plot shown in Figure 15.32, of the above TF, the w_{CO}, K_U, and P_U values are found, and from these values the controller settings according to Zeigler-Nichol's formulae are found.

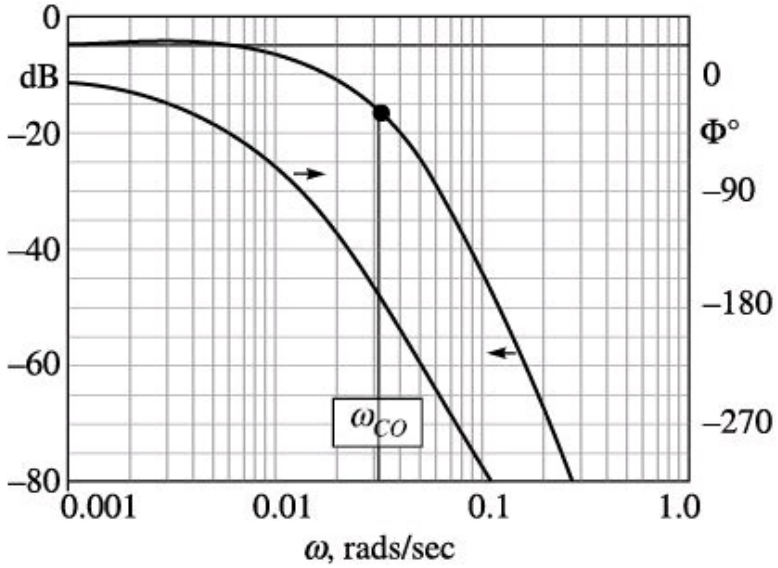


Figure 15.32 Bode's plot for the feedback loop.

(P) Controller	(PI) Controller		(PID) Controller		
K_C	K_C	x_I	K_C	x_I	x_D
1.7464	1.572	155.593	2.0956	93.355	23.339

Tuning of the Cascade loop:

1. Tuning of the inner loop is worked out as given for the feedback loop above. The forward path TF of the inner loop with a unity gain (P) controller is given as:

$$\frac{0.735}{(60s+1)(20s+1)(10s+1)}$$

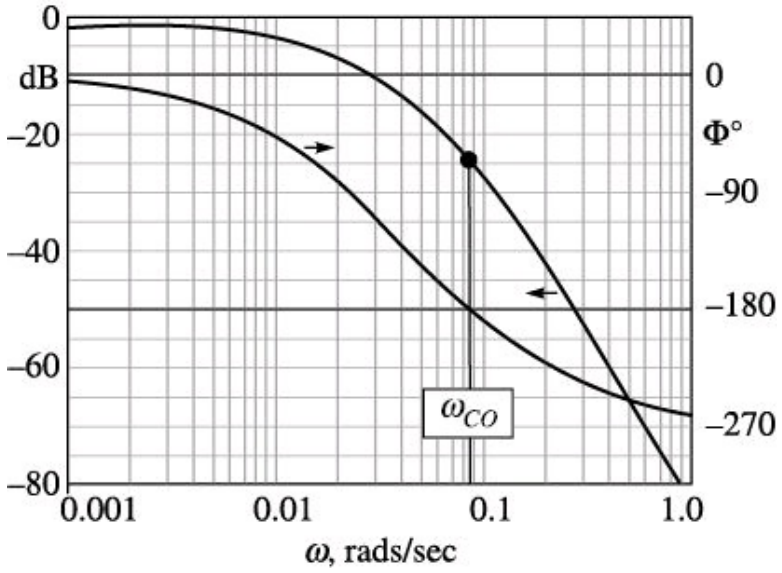


Figure 15.33 Bode's plot for inner loop.

From the Bode's plot of the above TF, the w_{CO}, K_U, and P_U values are found, and from these values the controller settings according to Zeigler-Nichol's formulae are calculated.

Controller settings for inner loop controller					
(P) Controller	(PI) Controller		(PID) Controller		
K_C	K_C	x_I	K_C	x_I	x_D
8.565	7.71	60.365	10.278	36.22	9.055

2. The equations for calculating unity feedback closed loop amplitude ratio, AR_{CL} and closed loop phase angle ϕ_{CL} from the open loop amplitude ratio AR and open loop phase angle ϕ, could be calculated from the following relation (see Chapter 8 of *Advanced Process Dynamics and Control*).

$$AR_{CL} = \frac{1}{\sqrt{[1 + (\cos\phi/AR)]^2 + (\sin\phi/AR)^2}} \text{ and } \phi_{CL} = \tan^{-1}\left[\frac{\sin\phi/AR}{1 + (\cos\phi/A)}\right]$$

3. For tuning of the outer loop controller of the cascade loop, the required Bode's Plot is prepared for the forward path comprising of G_{P1}, and closed inner loop. This is shown in Figure 15.34. Then the controller settings are found by the similar method as in
step 1 and step 2.

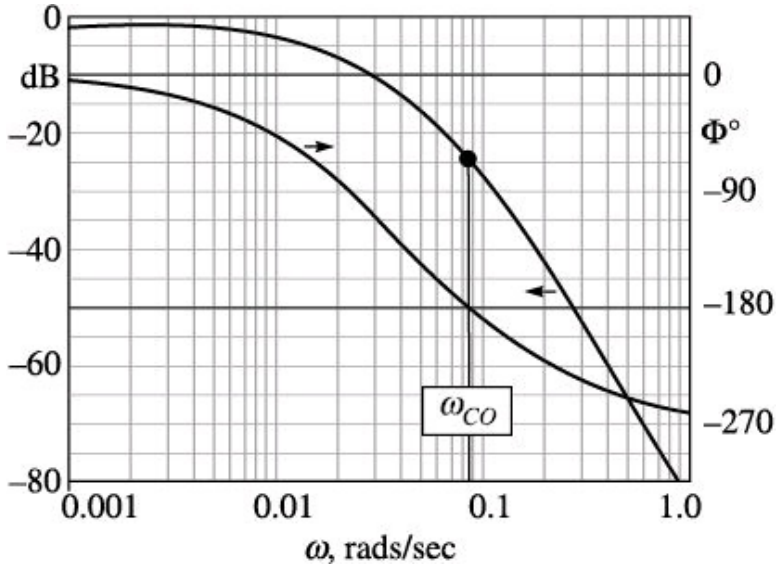


Figure 15.34 Bode's plot for the cascade control system.

Controller settings for outer loop controller					
(P) Controller	(PI) Controller		(PID) Controller		
K_C	K_C	x_I	K_C	x_I	x_D
3.91	3.52	139.95	4.69	83.97	21

Simulation programs are prepared for evaluating closed loop performance in terms of ITAE for the process:

1. With a feedback;
2. With a cascade controller.

After a few trials, (PI) mode has been selected for both inner and outer loop controller for cascade controller. To make the performances comparable, the feedback controller mode is selected as (PID). The loops are tested for step and pulse input in load because, in such a process, load variation is the most common type of disturbance. Beside the controller settings, values of the other parameters used for these tests are listed below:

For Feedback Loop:
NMD = 2, SP = 50, OPB = 50, CL = 100, ZR = 0, MAVCL = 100, MAVZR = 0

For Cascade Loop:
In outer loop SP = 50, OPB = 50, CL = 100, ZR = 0
In inner loop OPB = 20, CL = 100, ZR = 0, MAVCL = 100, MAVZR = 0

The evaluated ITAE values listed in Table 15.3 show improvement with cascade over feedback control. But we have tried a few trials for further adjustment of controller settings to seek further betterment, if any, of loop performance. It is found that by slightly reducing the integral time (t_I = 100) of the outer loop controller, the

ITAE could be reduced to an appreciable extent. The results are listed in Table 15.4 and shown in Figure 15.35.

Table 15.3 ITAE Values with Tuned Controller Parameters

	Step in Load	Pulse in Load
FB Control	74.84139	105.3099
Cascade	70.56	69.87

Table 15.4 ITAE Values with Manually Adjusted Integral Time of Outer Loop

	Step in Load	Pulse in Load
FB Control	74.84139	105.3099
Cascade Control	37.16	59.5213

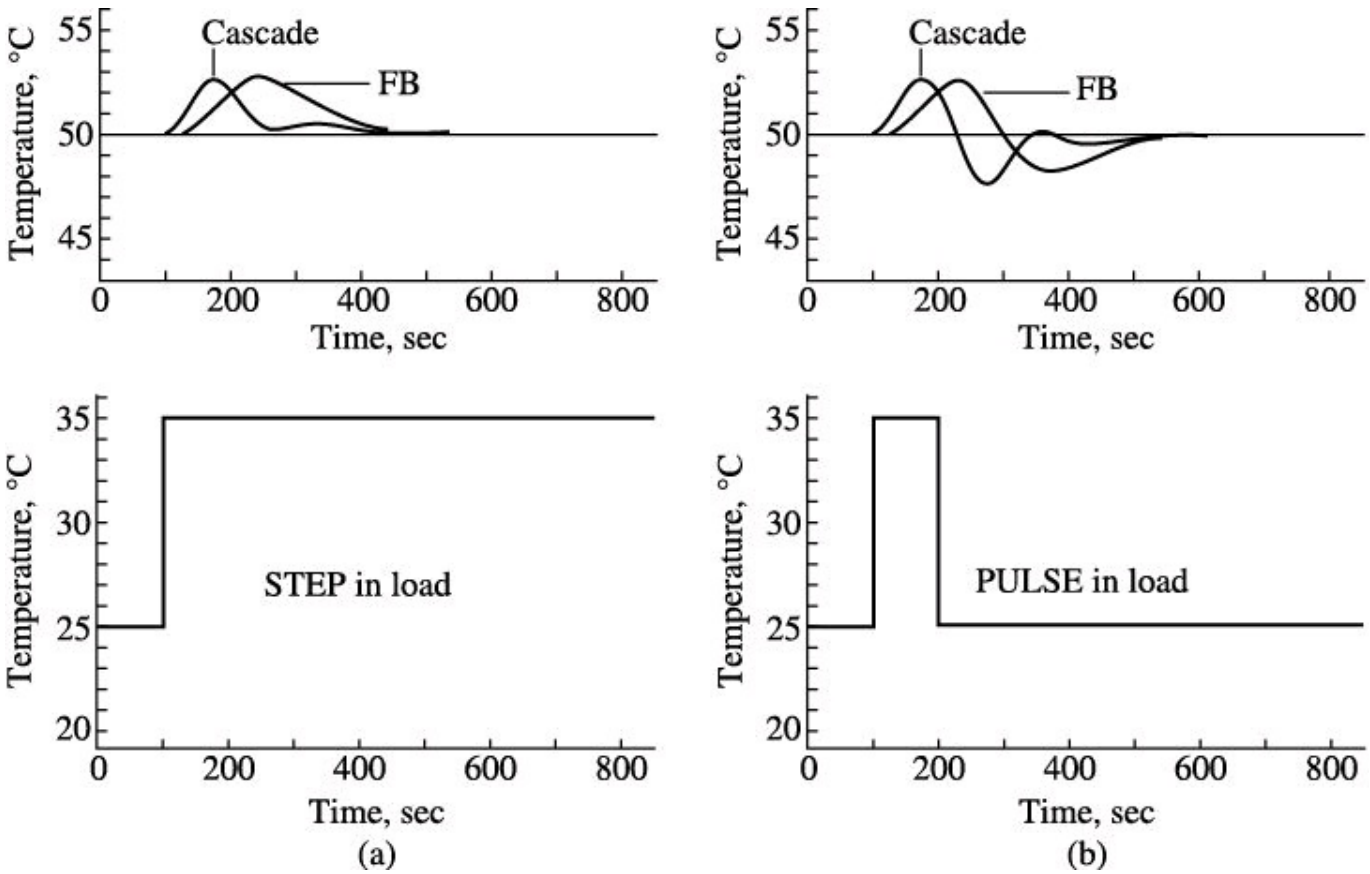


Figure 15.35 Comparison of responses to (a) Step in Load, (b) Pulse in load, for feedback and cascade control of a fifth order process.

Improvement of ITAE by manual trial shows the limitation of Z-N settings for higher order processes. Also, there is a possibility of a numerical error in the tuning procedure by frequency response method, especially in case of outer loop where the result comes from the evaluation through more than one Bode's plots.

15.5 Smith's Predictor Algorithm

This is a model based approach for better control of systems with large dead times.

The existence of dead time may be due to multiple lumped parameters or distributed parameter character of the process. Assuming that such a process may be represented by an FOPDT model, a simple feedback control system for the process has been shown in Figure 15.36(a). Presence of a relatively large amount of delay will definitely reduce the closed loop performance. Let us conceptually split the dead time from the *R-C* lag part of the process, the result may be shown as in Figure 15.36(b).

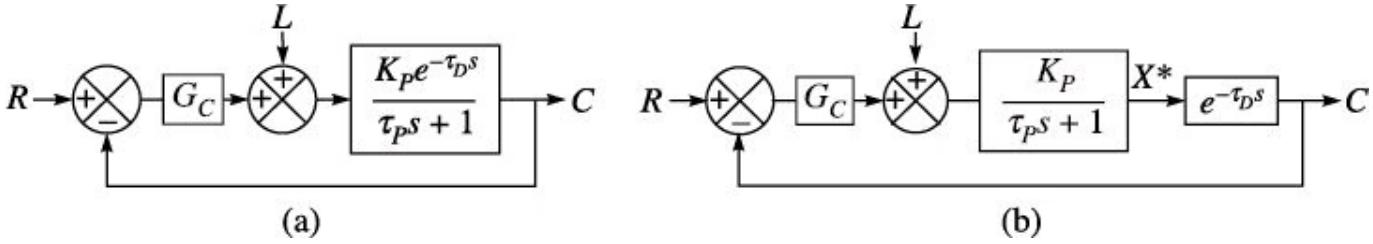


Figure 15.36 (a) Control applied to an FOPDT process. (b) Conceptually splitting of process into FO and DT parts.

To eliminate the effect of delay, it would be convenient if the control action be realized on the measurement of intermediate variable X^*. However, this control scheme, shown in
Figure 15.37, is unrealizable because the intermediate variable X^* cannot be observed.

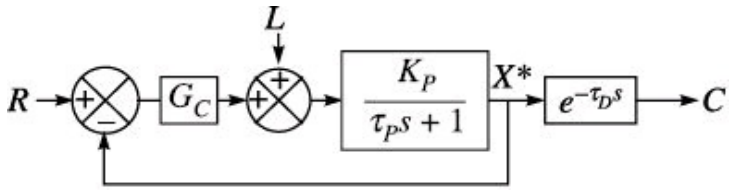


Figure 15.37 A very convenient but impossible control scheme.

But X^* can be obtained with a simulated model. When employing a computer controller, the simulated model may be a part of the control algorithm, and control performance could be improved by using X^* as the feedback signal, i.e. the control loop in Figure 15.38 is realizable. Note that the process, with all its dynamic complexities, is denoted by the TF $G_P(s)$ and the simulated FOPDT model produces the intermediate variable X^*.

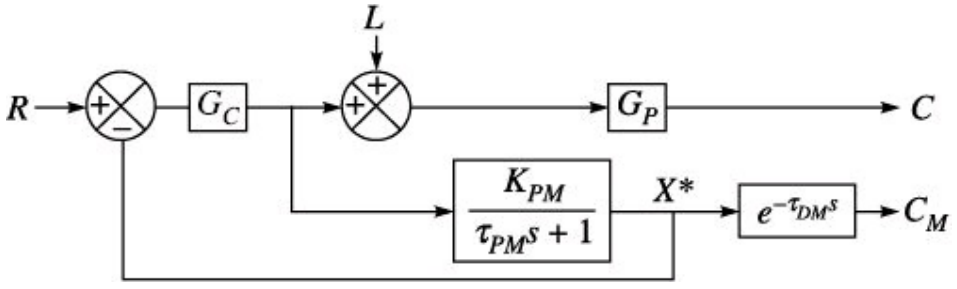


Figure 15.38 A realizable control scheme by using a computer controller.

Success of the control scheme of Figure 15.32 largely depends on the accuracy of the process model. Performance could be further improved by adding an extra feedback loop to the existing one, where the error $e^* = R - e_M$ would be used as an

extra compensation or correction. This modified loop is called **Smith's Predictor Algorithm,** and it has been shown in
Figure 15.39.

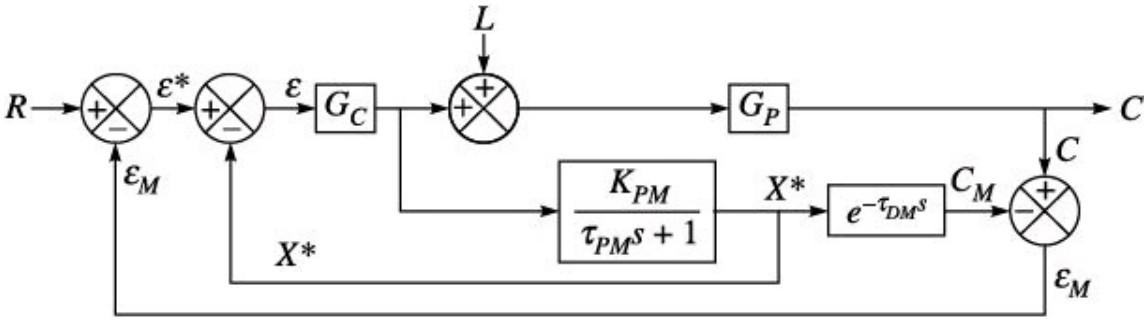


Figure 15.39 The Smith's predictor control scheme.

(*You should compare this modified scheme of Figure 15.39 with a Feed forward-Feedback loop. Because the system of Figure 15.7 the 'one-shot' feed forward scheme that relies only on the model. As soon as there is a doubt about that, one feels about the necessity of looking back and correcting the 'future action' based on feedback philosophy, as it has been done by implementing the second feedback path.)*

15.5.1 Analysis of Smith's Predictor Algorithm Block Diagram

The following nomenclature of blocks, shown in Figure 15.33, have been used in developing the closed loop TF of the system:

$$G_{PM}(s) = \frac{K_{PM}}{\tau_{PM}s+1};\ G_{DM} = e^{-\tau_{DM}s}$$

The closed loop servo TF may be derived by the method shown below:

$$C = \varepsilon G_C G_P = RG_C G_P - CG_C G_P + G_C G_P C_M(1 - G_{DM}) \qquad (15.25)$$

Using the following relation:

$$C_M = \frac{CG_{PM}G_{DM}}{G_P} \qquad (15.26)$$

The unwanted variable in eqn.(15.25) may be eliminated to obtain:

$$\frac{C}{R} = \frac{G_C G_P}{1 + G_C G_P + G_C G_{PM}(1 - G_{DM})} \qquad (15.27)$$

EXAMPLE 15.11 The relative stability of closed loop system for a third order process is evaluated in this example for: (a) Feedback controller, and (b) Smith's predictor control algorithm. The process TF is given as:

$$G_P(s) = \frac{1}{(s+1)(2s+1)(4s+1)}$$

Solution: As a first step, the FOPDT model for $G_P(s)$ is found by frequency response method as has been shown in examples of previous sections of this chapter. Thus, the model TFs are evaluated as:

$$G_{PM}(s) = \frac{1}{5.16s + 1} \text{ and } G_{DM}(s) = e^{-2.27s}$$

(a) Feedback control: Assuming $G_V = G_M = 1$, and a proportional controller ($G_C = K_C$) in the loop, if only a feedback control is applied, the closed loop servo TF becomes:

$$\frac{C}{R} = \frac{K_C G_P}{1 + K_C G_P} = \frac{K_C}{8s^3 + 14s^2 + 7s + 1 + K_C}$$

Performing the Rooth's test on the characteristic equation:

$$8s^2 + 14s^2 + 7s + 1 + K_C = 0$$

The range of K_C for stable loop operation is obtained as: $0 > K_C > 11.25$. Thus $K_U = 11.25$.

(b) Smith's Predictor Control: With the similar assumption that $G_V = G_M = 1$, and a proportional controller ($G_C = K_C$) in the loop, if the Smith's Predictor control algorithm is applied, the closed loop servo TF becomes:

Taking Pade's first order approximate TF for the dead time term and simplifying, the characteristic equation may be obtained as:

$$\frac{C}{R} = \frac{G_C G_P}{1 + G_C G_P + G_C G_{PM}(1 - G_{DM})}$$
$$= \frac{K_C/[(s+1)(2s+1)(4s+1)]}{1 + K_C/[(s+1)(2s+1)(4s+1)] + K_C/(5.16s+1)[1 - e^{-2.27s}]}$$

Taking Pade's first order approximate TF for the dead time term and simplifying, the characteristic equation may be obtained as:

$$93.71s^5 + (264.7 + 36.32K_C)s^4 + (274.25 + 63.56K_C)s^3 + (127.9 + 43.48K_C)s^2$$
$$+ (26.6 + 17.14K_C)s + 2 + 2K_C = 0$$

By applying Routh's test for determination of K_U, the algebra becomes too laborious as the evaluation of K_C from different first column terms of Routh's array may involve solutions of higher degree equations in terms of K_C. Alternatively, we have applied the root finding program C10-11A.BAS for higher degree equations that

has been described in Chapter 10. It would be sufficient to establish that even for a very high value of K_C, the real part of the roots of the fifth degree characteristic equation remain –ve to prove that Smith's Predictor offers a better stability character than only feedback control. The values of the roots are obtained for K_C = 1 to 1000. Observing values of all the five roots (3 real and 2 complex conjugates), a concised number of results are listed in Table 15.5 from which trend is easily observed. A root-locus plot could also be drawn from these data. Except for the third real root that goes to higher –ve values, the other roots converge to definite –ve values as K_C increases. This feature may be attributed to the fact that under SP control, the dead time is eliminated from the inner loop where the feedback control action actually takes place, hence only a first order lag element G_{PM}, having a relatively simple dynamics is to be controlled, thus reducing the complexity of control. It may be recalled that convergence of roots toward a limiting value for higher controller gains have been observed in Chapter 10 for lower order systems.

Table 15.5 Roots of Fifth order Characteristic Equation in Example 15.11

K_C	Real Roots			Complex Roots
1	−0.181	−0.941	−1.463	−0.314 ± i0.27
2	−0.1799	-0.944	−1.873	−0.301 ± i0.332
3	−0.1795	−0.945	−2.267	−0.2977 ± i0.386
4	−0.1793	−0.9458	−2.656	−0.2971 ± i0.0.3857
5	−0.1792	−0.9461	−3.04	−0.2968 ± i0.4
6	−0.1791	−0.9463	−3.43	−0.2973 ± i0.4105
7	−0.179	−0.9465	−3.81	−0.298 ± i0.4185
8	−0.179	−0.9466	−4.20	−0.2987 ± i0.4248
9	−0.1789	−0.9467	−4.54	−0.2994 ± i0.4299
10	−0.1789	−0.9467	−4.97	−0.3 ± i0.4341
20	−0.1788	−0.9471	−8.84	−0.3044 ± i0.4545
30	−0.1787	−0.9472	−12.71	−0.3064 ± i0.4619
40	−0.1787	−0.9472	−16.58	−0.3076 ± i0.4657
50	−0.1787	−0.9472	−20.46	−0.3084 ± i0.468
60	−0.1787	−0.9473	−24.33	−0.3089 ± i0.4695
70	−0.1787	−0.9473	−28.21	−0.3093 ± i0.4706
80	−0.1787	−0.9473	−32.08	−0.3096 ± i0.4715
90	−0.1787	−0.9473	−35.96	−0.3099 ± i0.4721
100	−0.1787	−0.9473	−39.83	−0.3101 ± i0.4726
200	−0.1786	−0.9474	−78.59	−0.311 ± i0.475
300	−0.1786	−0.9474	−117.35	−0.3113 ± i0.4758
400	−0.1786	−0.9474	−156.86	−0.3114 ± i0.4762
500	−0.1786	−0.9474	−194.86	−0.3115 ± i0.4765
600	−0.1786	−0.9474	−233.62	−0.3116 ± i0.4766
700	−0.1786	−0.9474	−272.38	−0.3116 ± i0.4767
800	−0.1786	−0.9474	−311.13	−0.3117 ± i0.4768
900	−0.1786	−0.9474	−349.89	−0.3117 ± i0.4769
1000	−0.1786	−0.9474	−388.65	−0.3117 ± i0.477

The above are results obtained by using the program C10-1A.BAS for each single value of K_C. It is left to the reader to obtain the corresponding root-locus using the same program.

The last example shows that (Smith's Predictor) SP control has a higher stability character than the FB control. But to obtain the closed loop dynamics for visual examination and quantitatively express the closed loop performance quality by ITAE, the next example has been solved by using simulation programs for feedback and SP control.

EXAMPLE 15.12 The process that is put under the two kinds of control is of sixth order with non-unity gain. Such a higher order process has been chosen to emphasize the lag in process dynamics.

$$G_P(s) = \frac{0.735}{(20s+1)^6}$$

For the simplicity of calculation $G_V = G_M = 1$.

Solution: A (PID) controller is to be used by both the loops. The controller is tunned according Ziegler-Nichol's formulae. For this purpose, the FOPDT model $G_M(s)$ is determined from the process TF $G_P(s)$, and by applying frequency response technique, the controller settings have been found. The results are shown in the following:

$$G_M(s) = \frac{0.735e^{-33.63s}}{33.63s+1}$$

(PID) controller settings:

$$K_C = 1.85;\ t_I = 48.7;\ t_D = 12.17$$

The simulated closed loop systems are run for identical time for set point and load inputs. Responses are obtained for step and pulse type of disturbances. Initial values for load and set point are, $SP = 40$ and $L = 25$. For step SP is raised to 50 and L is raised to 35. For pulse SP is raised to 50 and brought back to 40 after 100 s, and L is raised to 35 and then brought back to 25 after 100 s.

Figure 15.40 shows the closed loop responses for SP (Smith's predictor) and FB (feed back) control for step and pulse disturbances in set point. The ITAE values are compiled as follows:

Disturbance Type	ITAE for *SP* Control	ITAE for FB Control
Step in *SP*	58.79	1376.722
Pulse in *SP*	113.18	1811.607

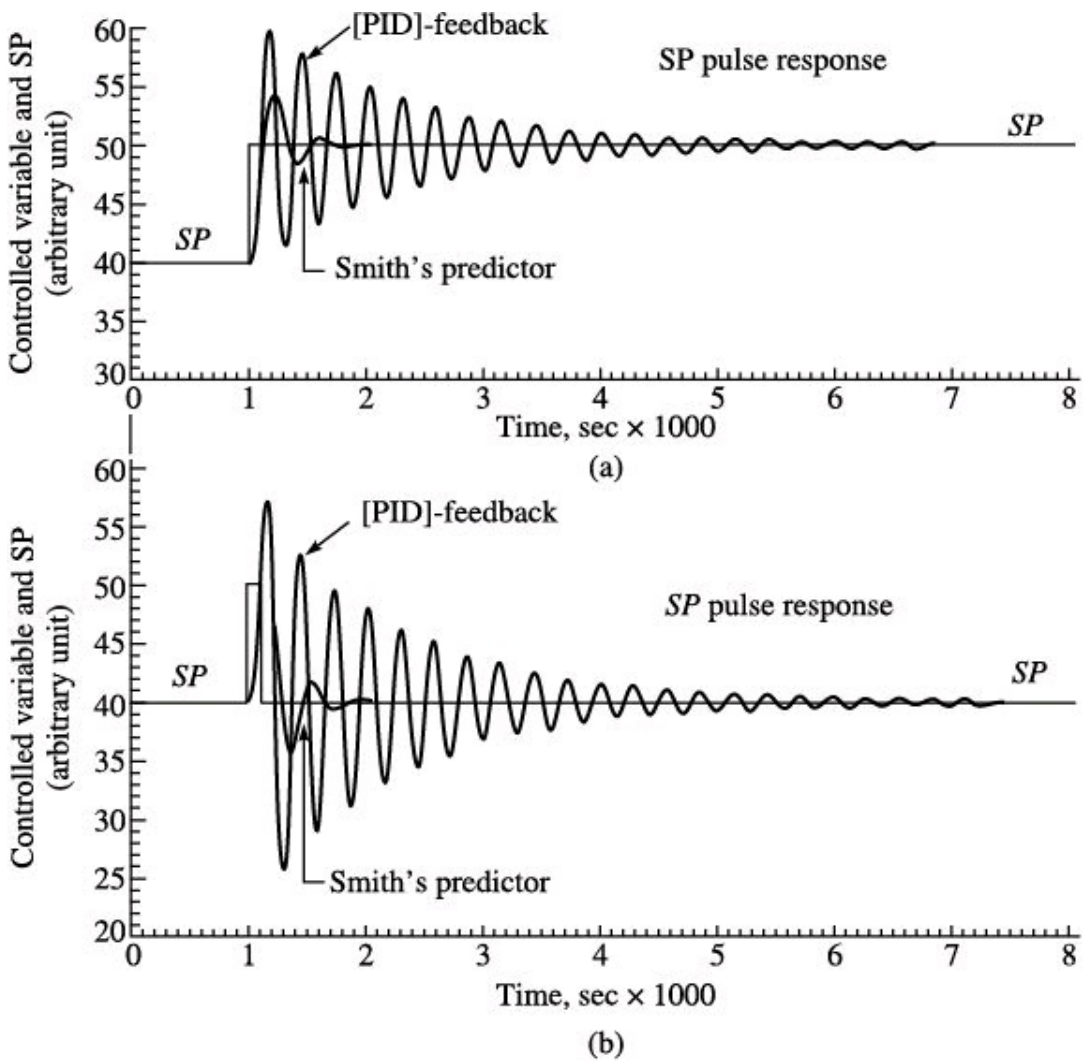


Figure 15.40 Responses of *SP* and FB control for set point change. (a) Step input. (b) Pulse input.

Figure 15.41 shows the closed loop responses for *SP* and FB control for step and pulse disturbances in load. The ITAE values are compiled as follows:

Disturbance Type	ITAE for *SP* Control	ITAE for FB Control
Step in *L*	26.539	444.69
Pulse in *L*	42.843	830.66

Coughanowr has demonstrated these improvements with a unity gain fourth order process, where it is less marked than in the present example. This fact may confirm that *SP* control shows its strength for systems where FB control quality gradually deteriorates as the amount of transfer lag or dead time increases in the process dynamics.

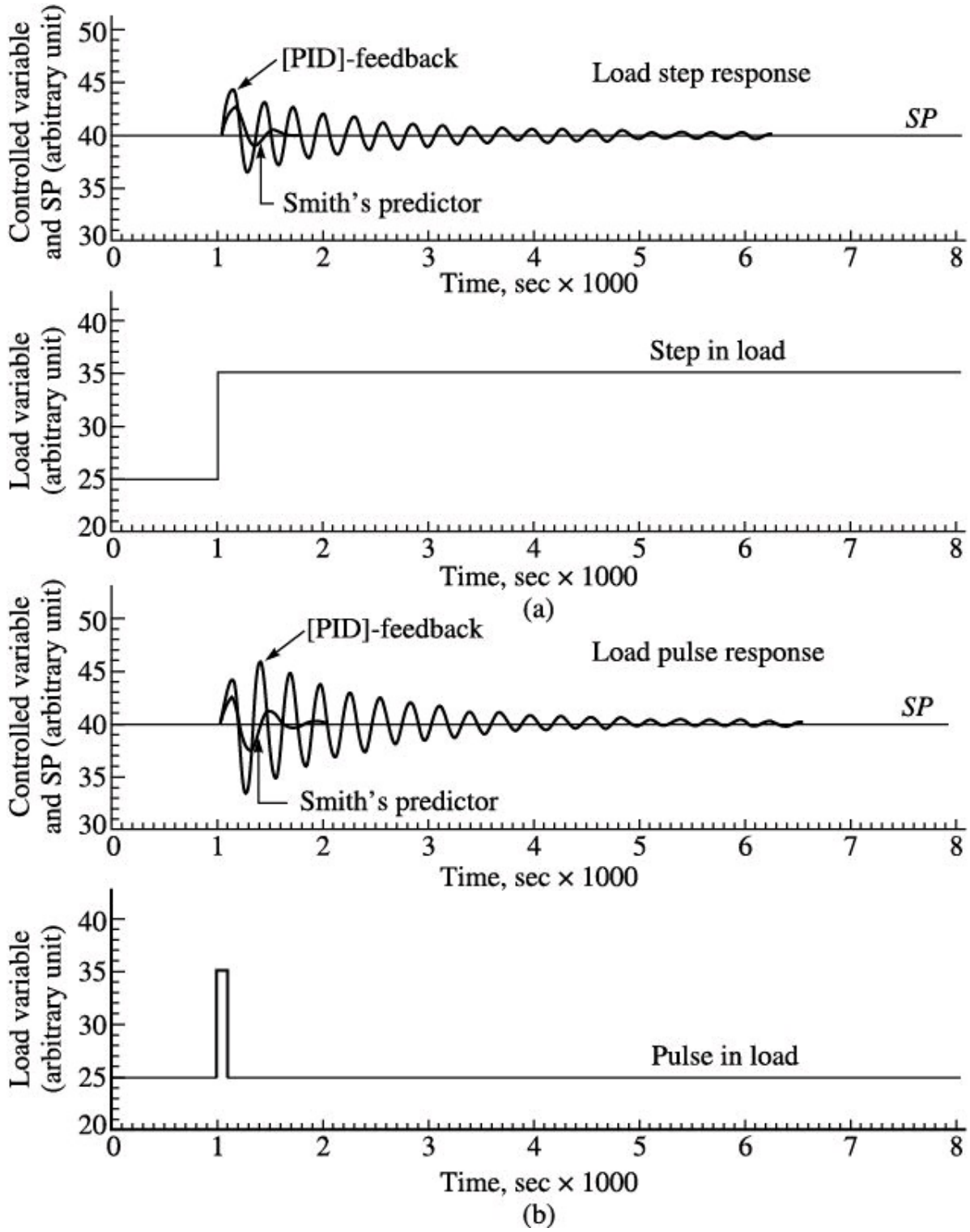


Figure 15.41 Responses of *SP* and FB Control for load change. (a) Step input. (b) Pulse input.

15.6 Process with Inverse Response

An interesting type of dynamic difficulty may appear in the response of certain processes where a +ve input change may produce a temporary initial response in the –ve direction. As time progresses, the response gradually attains zero value and then becomes +ve. Think about a feedback controller that is managing the process. At a certain instance it takes a +ve manipulation as the remedy for an error magnitude but the process variable moves in a direction that further increases the error. Generally the process remains in such a state for a relatively short time, but consider the ambiguous

situation this would put the controller into. Such an anomalous dynamic behaviour is called 'Inverse Response Character'.

What kind of dynamic processing may happen inside a process that would manifest in such an anomalous behaviour? It has been recognized that inverse response processes contain two dynamic processing paths inside their system boundary, and generally in both paths there are two simple dynamic elements that simultaneously process the input and the final output is the linear addition or superimposition of the outputs of the two parallel branches. The types of TF of the dynamic elements on these paths are simple lumped models, e.g. first order, second order, Integrator, FOPDT and SOPDT elements. The common feature is that one of the branches has a slow dynamic response and large +ve gain (larger $+K_P$, and larger t_P), while the second branch has a fast dynamic response with a small –ve gain (smaller $-K_P$, and smaller t_P). We will discuss two process examples that demonstrate an inverse response but contain different kinds of dynamic elements within them.

15.6.1 An Example

In a distillation column, the response of bottom composition corresponding to a change in vapour boil up can show an inverse behaviour. In a binary distillation column we know that an increase in vapour rate must drive more low boiling compound component up the column, therefore x_B (mole% of more volatile in the bottoms) is expected to decrease. However, some unexpected results may appear due to effect of tray hydraulics.

Depending on the tray design, as in the case of sieve trays, more amount of vapour coming up tends to 'seal off' the liquid falling down through the tray, as there is insufficient liquid head on the tray in the initial stage. This would result in an increase of liquid falling down through the down-comer, which contains more amounts of more volatiles. Thus, initially x_B will increase. Eventually the liquid head over the tray would build up and a dynamic equilibrium would be established between up-coming vapour and down- coming liquid through the tray. This happens due to better mixing between liquid and vapour to ensure a rated amount of mass transfer between them. Thus, x_B will be reduced to its proper value.

The tray hydraulics in the initial stage after the input change has just occurred and subsequently when the normal operation is gradually taking hold, it may be thought as two parrallely acting first order processes. The first one is due to the –ve response; it is fast and has a small –ve gain value because the second process overcomes and drives away the first. Hence, it has a higher +ve gain and also it is slower than the first, because it takes more time to produce the full effect. Thus, the hydrodynamics of

these two effects (initially up-going vapour putting resistance to downward liquid flow and then gradual on set of normal liquid down flow) are like first order systems because both involve separate pair of R-C couples.

The model configuration of this process and its response to a +ve step input has been shown in Figure 15.42.

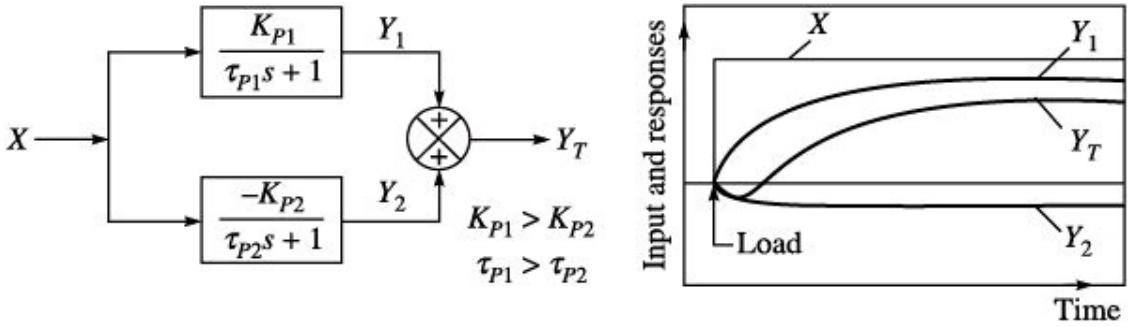


Figure 15.42 Conceptual block diagram and step response of an inverse response process and its individual blocks.

15.6.2 Another Example

The vaporizer is a boiler in which the water is kept boiling to produce steam of proper quality and pressure. A level sensor is employed to measure the boiling water level inside the vaporizer and when the level comes down to the lowest allowable value so that the flue bearing tubes may not come above the water surface, the feed water pump is automatically turned-on. The pump delivers feed water into the vaporizer drum so as to increase the water level. This is what is desirable. But to understand what really happens, we should be aware that a level sensor when measuring the level of a boiling water mass that has a frothing surface, actually measures the level where the two phases, vapour and liquid, separate. That is, it includes the froth or foams into the liquid level. What happens when the colder feed water runs into the drum? The temperature drops, foams subside and the level sensor indicates a –ve response in level due to +ve manipulation. Gradually the temperature of the liquid mass increases and level response rises to its proper value.

To analyze this process, with respect to feed water, the process behaves like an integrator because the mass rate of water is far grater than its rate of evaporation and it has a +ve gain. But this is a slower phenomenon than the collapsing of foams which occurs at a much faster pace and the heat transfer from the body of foam to the liquid bulk may be assumed to be of first order dynamics (involving a lumped parameter R-C couple). The model configuration of this process and its response to a +ve step input has been shown in Figure 15.43.

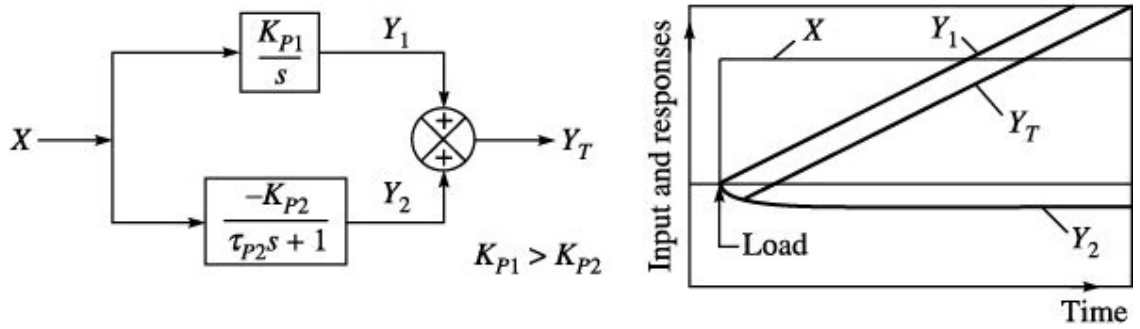


Figure 15.43 Conceptual block diagram of a simple inverse response process, and the step response of the overall process as well as its individual blocks.

A list of different dynamic system configurations with inverse response has been presented as follows:

1. Pure Capacitance minus first Order element:

$$G_P(s) = \frac{K_1}{s} - \frac{K_2}{\tau_2 s + 1} = \frac{(K_1\tau_2 - K_2)s + K_1}{s(\tau_2 s + 1)} \qquad (15.28)$$

Necessary condition for inverse response and +ve zero:

$K_1 t_2 < K_2$, so that, $z = -\frac{K_1}{K_1\tau_2 - K_2} > 0$, is the location of zero.

2. Difference between two first Order elements:

$$G_P(s) = \frac{K_1}{\tau_1 s + 1} - \frac{K_2}{\tau_2 s + 1} = \frac{(K_1\tau_2 - K_2\tau_1)s + (K_1 - K_2)}{(\tau_1 s + 1)(\tau_2 s + 1)} \qquad (15.29)$$

Necessary condition for inverse response and +ve zero:

$\frac{\tau_1}{\tau_2} > \frac{K_1}{K_2} > 1$, so that, $z = -\frac{K_1 - K_2}{K_1\tau_2 - K_2\tau_1} > 0$, is the location of zero.

3. Difference between two FOPDT elements:

$$G_P(s) = \frac{K_1 e^{-\tau_{DL1}s}}{\tau_1 s + 1} - \frac{K_2 e^{-\tau_{DL2}s}}{\tau_2 s + 1} \qquad (15.30)$$

Necessary condition for inverse response and +ve zero:

$K_1 > K_2$ and $t_{DL1} > t_{DL2}$ □ 0

4. Difference between a second order and a first Order element:

$$G_P(s) = \frac{K_1}{\tau_1^2 s^2 + 2\zeta\tau_1 s + 1} - \frac{K_2}{\tau_2 s + 1} \qquad (15.31)$$

Necessary condition for inverse response and +ve zero: $K_1 > K_2$

5. Difference between two second order elements:

$$G_P(s) = \frac{K_1}{\tau_1^2 s^2 + 2\zeta_1\tau_1 s + 1} - \frac{K_2}{\tau_2^2 s^2 + 2\zeta_2\tau_2 s + 1} \qquad (15.32)$$

Necessary condition for inverse response and +ve zero: $\frac{\tau_1^2}{\tau_2^2} > \frac{K_1}{K_2} > 1$

6. Difference between two SOPDT elements:

$$G_P(s) = \frac{K_1 e^{-\tau_{DL1}s}}{\tau_1^2 s^2 + 2\zeta_1\tau_1 s + 1} - \frac{K_2 e^{-\tau_{DL2}s}}{\tau_2^2 s^2 + 2\zeta_2\tau_2 s + 1} \qquad (15.33)$$

Necessary condition for inverse response and +ve zero: $K_1 > K_2$ and $t_{DL1} > t_{DL2}$ □ 0

15.6.3 Control System Design for Inverse Response Processes

There are two approaches for this exercise.

Method I. Starting with a process with two parallel acting combinations of first order elements, the closed loop block diagram may be represented as in Figure 15.44.

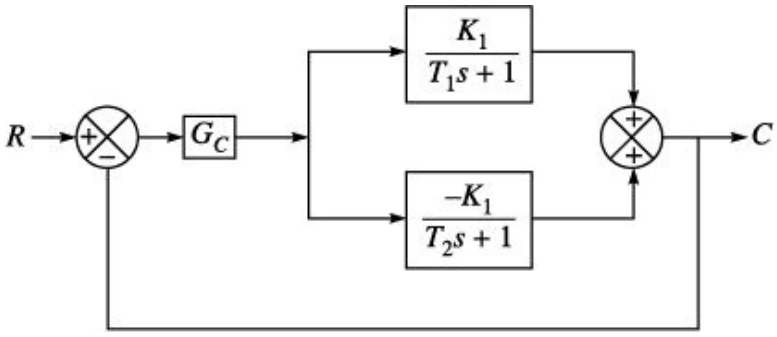


Figure 15.44 Block diagram of an inverse response process with feedback controller.

The overall TF of the forward path of this block diagram would be:

$$C(s) = G_C(s)\frac{(K_1T_2 - K_2T_1)s + (K_1 - K_2)}{(T_1 s + 1)(T_2 s + 1)}R(s) \qquad (15.34)$$

To eliminate the inverse response character of the system it is sufficient to remove the +ve zero from the above TF. This may be done by adding a dynamic compensator $D_C(s)$ in the control loop. The $D_C(s)$ may be given as:

$$D_C(s) = G_C(s)P\left(\frac{1}{T_2 s + 1} - \frac{1}{T_1 s + 1}\right)R(s) \qquad (15.35)$$

where $P \geq \frac{K_2T_1 - K_1T_2}{T_1 - T_2}$

Thus, after addition of the compensator, overall TF of the forward path becomes:

$$C^*(s) = C(s) + D_C(s) = G_C(s)\frac{[(K_1T_2 - K_2T_1) + P(T_1 - T_2)]s + (K_1 - K_2)}{(T_1s + 1)(T_2s + 1)}R(s) \qquad (15.36)$$

For the above value of P, the zero of the resulting TF becomes non +ve,

$$z = \frac{K_1 - K_2}{(K_1T_2 - K_2T_1) + P(T_1 - T_2)} \leq 0 \qquad (15.37)$$

The compensated signal $C^*(s)$ is formed by adding $D_C(s)$ to $C(s)$. That is how the dynamic compensation is implemented to reduce the inverse character of the closed loop response. The block diagram of the compensated loop has been shown in the Figure 15.45.

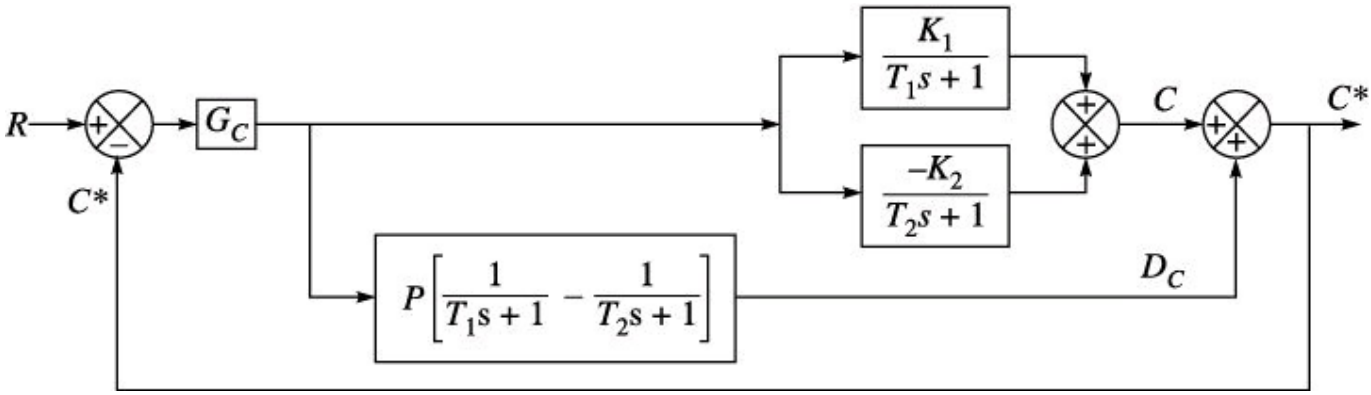


Figure 15.45 Close loop block diagram of an inverse response process with dynamic compensator.

It should be noted that the controlled variable in the compensated loop has become $C^*(s)$ instead of $C(s)$, because in fact, the controller acts on to reduce e^* not e, that is computed as, $e^* = R - C^*$. If the control system requirement is such that the management of the modified controlled variable, C^*, instead of uncompensated controlled C is acceptable, the control scheme of Figure 15.45 has satisfactory performance standards. The dynamic compensator $D_C(s)$ has an adjustable parameter, P given by Equation 15.35, has an inequality sign that may be expressed as:

$$P = C_K\frac{K_2T_1 - K_1T_2}{T_1 - T_2}$$

where C_K □ 1 is an adjustable constant.

Method II. A feedback loop with a (PID) mode controller may be used without the dynamic compensator. This controller should be tuned by some standard tuning method. Such a tuned feedback system may produce an acceptable closed loop performance if the requirements are not too stiff. The (D)mode by its nature would anticipate the 'wrong' direction of process response and provide the proper corrective action to limit (not eliminate) the inverse shoot.

The following example demonstrates the performance of the two strategies.

EXAMPLE 15.13 The process features two parallely acting first order blocks. For calculation simplicity, instrumentation hardware is assumed to have unity gain dynamics. TF of the individual blocks in the loop, where the time constants are in seconds, may be given as:

$$G_{P1} = \frac{1}{60s+1}; \quad G_{P2} = \frac{-0.4}{20s+1}; \quad G_V = G_M = 1 \text{ and } D_C = \left(\frac{1}{20s+1} - \frac{1}{60s+1}\right)(0.1C_K)$$

It has been proposed to compare the performance of: (a) a feedback loop with (b) a compensated feedback loop. A tuned (PID) controller has to be used for both cases.

Solution: The feedback control loop in block diagram form is given in Figure 15.46.

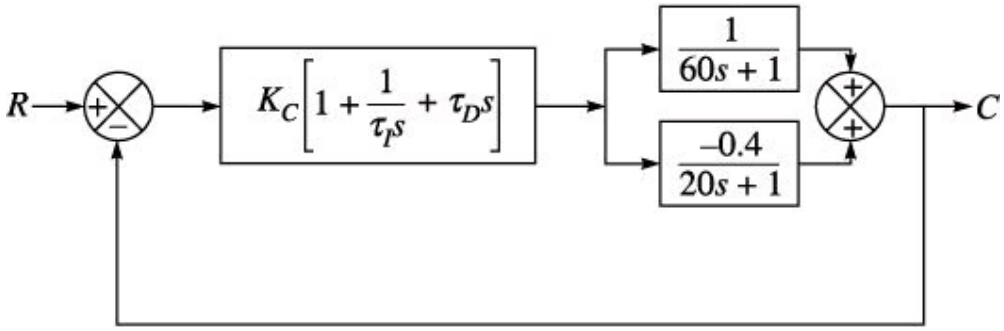


Figure 15.46 The process of example 16.13 with a (PID) controller.

The TF of the dynamic compensator $D_C(s)$ is obtained as:

$$D_C(s) = \left(\frac{1}{20s+1} - \frac{1}{60s+1}\right)(0.1C_K)$$

where, $P = 0.1$, and C_K □ 1, and adjustable parameter.

The feedback loop with the dynamic compensator is shown in Figure 15.47. For controller parameter optimization of the feedback loop, a simulation program is prepared that uses a multi-parameter gradient search optimization procedure, where the objective function is the integral error criterion ITAE obtained for load or set point disturbance. This objective function has been minimized by adjusting the controller parameter values (K_C, t_I, and t_D) within a 3-D parameter space. The optimum parameter values obtained are: K_C= 2.1, t_I = 80 s, t_D =20 s.

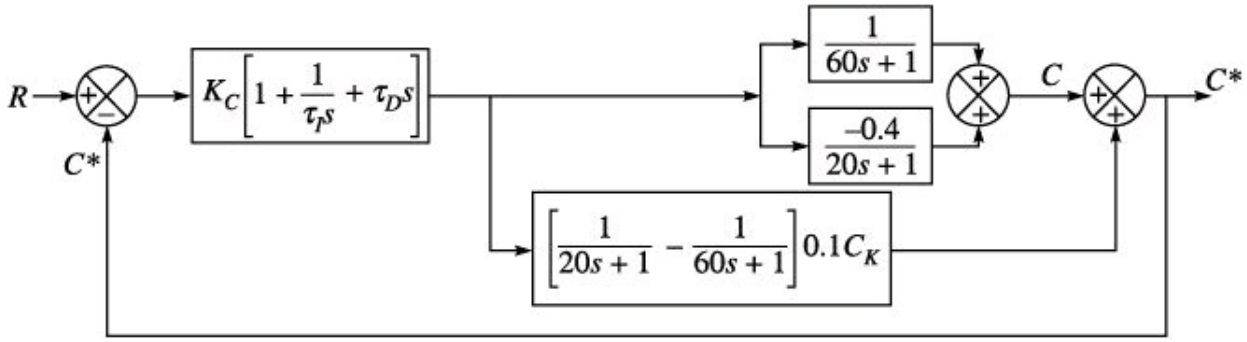


Figure 15.47 The closed loop system of Figure 15.35 with dynamic compensator.

Following the block diagram, two simulation programs are prepared for the study of the feedback and dynamically compensated feedback control. Identical (PID) parameter values are used in both programs. The initial steady state variable values used are $L = 25$ and $SP = 40$. Step (magnitude = 10 units) and pulse (magnitude = 10, duration = 100 s) disturbances are used in *Load* and *Set Point* to check the closed loop response.

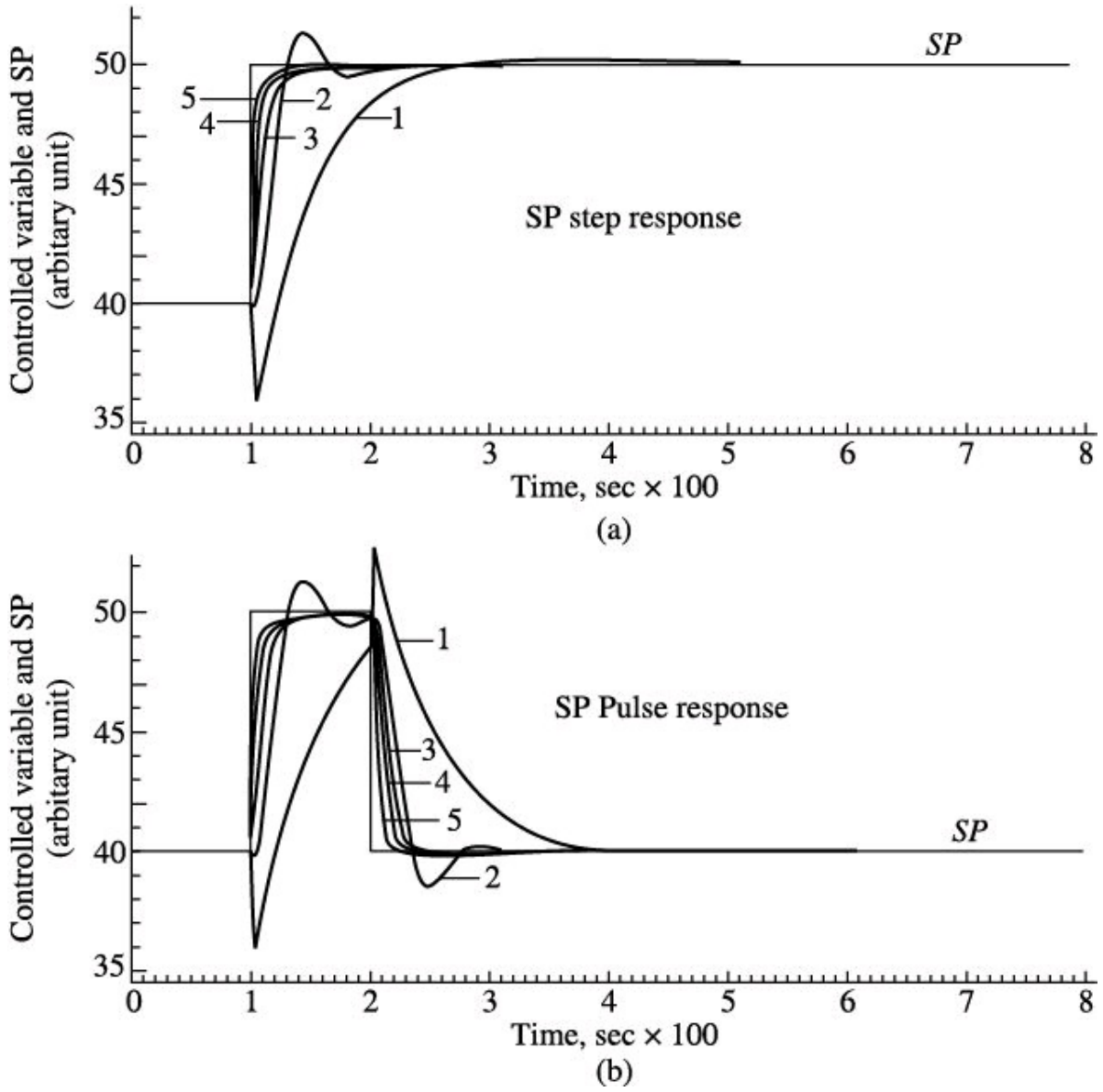


Figure 15.48 Close loop response for: (a) Step in set point. (b) Pulse in set point.

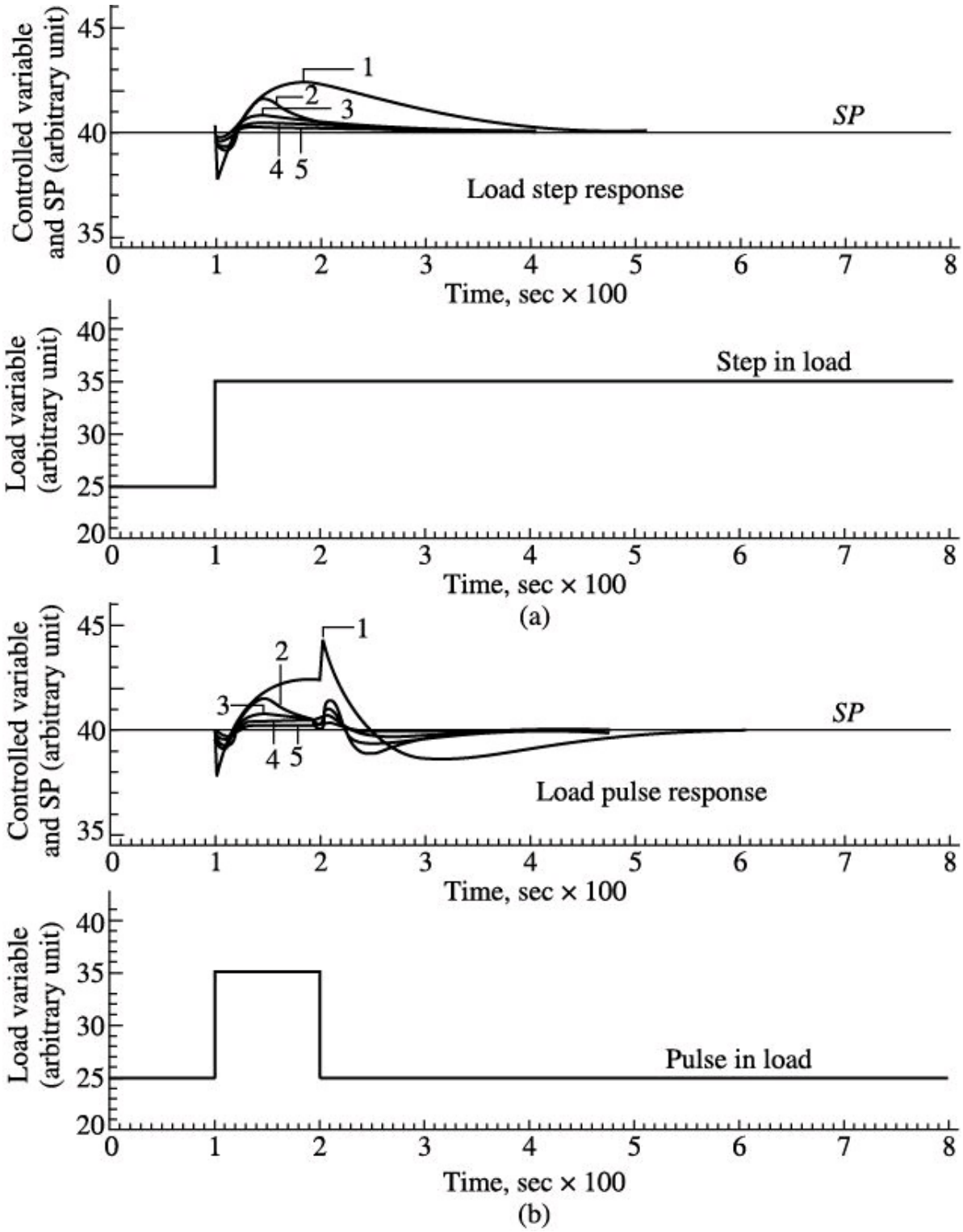


Figure 15.49 Close loop response for: (a) Step in load. (b) Pulse in load.

Figure 15.48 depicts the step and pulse response for set point change and Figure 15.49 shows the step and pulse response for load change. The nomenclature of response traces and corresponding ITAE values are given in the following table.

As we observe from the response traces as well as from the tabulated ITAE values, the improvement in response character is obvious by the application of the dynamic compensator in a feedback loop. Also by increasing C_K, which we may now recognize as a tunable controller parameter, the response dynamics could be categorically modified; as an example in the step and load response for set point

disturbance, the inverse shoot in the response is completely removed and the control approaches towards the ideal response character at higher values of C_K. However, use of too high a value of C_K may lead the loop to unstable condition.

Comparison of the close lopp performance

Input in	Disturbance	Fig. No.	Trace No.	Type of Control	ITAE	C_K
Set point	Step	Figure 15.48(a)	1	FB	38.54	–
			2	FB + Dc	9.72	1
			3	FB + Dc	6.46	2
			4	FB + Dc	5.39	4
			5	FB + Dc	4.69	8
	Pulse	Figure 15.48(b)	1	FB	125.8	–
			2	FB + Dc	35.47	1
			3	FB + Dc	21.28	2
			4	FB + Dc	16.22	4
			5	FB + Dc	14.52	8
Load	Step	Figure 15.49(a)	1	FB	55.96	–
			2	FB + Dc	15.11	1
			3	FB + Dc	10.66	2
			4	FB + Dc	8.1	4
			5	FB + Dc	5.64	8
	Pulse	Figure 15.49(b)	1	FB	78.69	–
			2	FB + Dc	26.04	1
			3	FB + Dc	17.81	2
			4	FB + Dc	11.93	4
			5	FB + Dc	8.0	8

15.7 Inferential Control

In certain processes the disturbance L, manipulated variable MV, and controlled variable C, all are un-measurable, but an associated or connected variable and its manipulation may be measurable. Based on these measurements, an estimation may be developed in the following manner for performing the control duty:

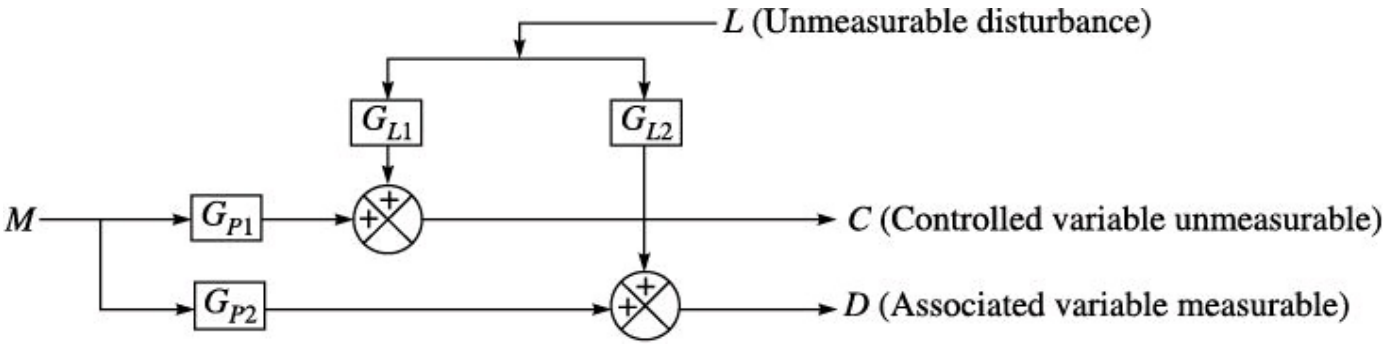


Figure 15.50 Variable stream arrangement for implementing inferential control.

From the description of the process in Figure 15.50, the following relations are developed:

$$C = G_{P1}M + G_{L1}L \tag{15.38}$$

and

$$D = G_{P2}M + G_{L2}L \tag{15.39}$$

From Equation (15.39)

$$L = D\frac{1}{G_{L2}} - M\frac{G_{P2}}{G_{L2}} \tag{15.40}$$

Putting (15.40) in (15.38), we obtain:

$$C = G_{P1}M + G_{L1}\left(\frac{D}{G_{L2}} - \frac{MG_{P2}}{G_{L2}}\right) = M\left(G_{P1} - \frac{G_{L1}G_{P2}}{G_{L2}}\right) + D\frac{G_{L1}}{G_{L2}} \tag{15.41}$$

In Figure 15.51, a control scheme for implementing the estimator and the associated loop is demonstrated.

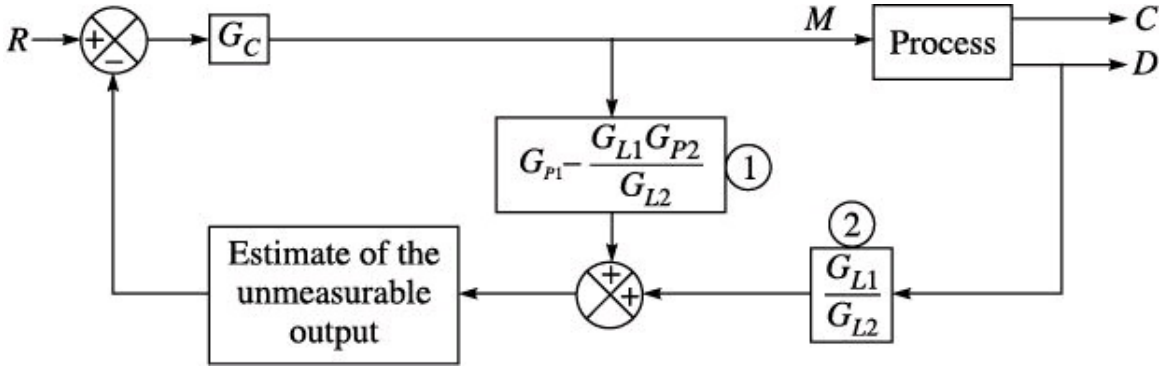


Figure 15.51 Inferential control scheme of a process with dynamic estimator.

Obviously the success of the control scheme depends largely on the accuracy of the TFs of the blocks (1) and (2).

EXAMPLE 15.14 For the preparation of a commercial adhesive, a linear chain polymeric material has to be diluted with a solvent. The feed is available at a high concentration which is mixed with the solvent in a stirred mixer. The exit solution concentration is controlled. The problem is that there is no composition sensor available for the polymer in the solution system. However, the solution viscosity has a reliable dependence to the concentration. The following data is obtained by determining viscosity with an Instron viscometer, in centipoises for solutions of different polymer concentration in g/10 cm^3 or g/centi l at a constant temperature.

μ, cp	3.61	3.996	4.435	4.935	5.503	6.149	6.884	7.72	8.67	9.753	11
C, g/cl	0	0.025	0.05	0.075	0.1	0.125	0.15	0.175	0.2	0.225	0.25
μ, cp	12.38	13.97	15.78	17.84	20.18	22.85	25.88	29.32	33.24	37.7	–
C, g/cl	0.275	0.3	0.325	0.35	0.375	0.4	0.425	0.45	0.475	0.5	–

A relation between solution viscosity, μ and solution concentration, C has been found as:

$$m = m_O + Ae^{BC} \tag{15.42}$$

where, $m_O = 0.8$, $A = 2.81$, $B = 5.15$

The above relation holds if the molecular weight of the polymer remains constant in the feed.

Steady state conditions:

Feed solution flow rate, $Q_L = 50$ l/min (constant).

Feed solution concentration, $C_L = 0.27$ g/cl (with possible variation between 0.23–0.32 g/cl).

Desirable concentration of the exit solution (set point), $C_R = 0.08$ g/cl. This is the set point with respect to concentration of the exit solution from the mixer. As the actual controlled variable would be viscosity, the corresponding set point in terms of viscosity would be,

$m_R = m_O + Ae^{BC_R} = 5.0426$ cp.

Fresh solvent ($C_M = 0$) as the manipulated variable is added to the perfectly stirred mixer.

Measuring element is a continuous type capillary viscometer with electronic transmitter manufactured by Taylor Instruments. The selected measuring range is 2.0 to 50 cp and produces a proportional output of 4–20 mA. Thus, in terms of viscosity,

$$G_M = K_M = \frac{48}{16} = 3 \text{ cp/mA}$$

The controller is an electronic PID controller that receives and generates 4–20 mA signal.

The final control element is a motorized control valve that accepts 4–20 mA.

By making a polymeric component balance for maximum load condition,

$$0.32 \text{ g/cl} \times 50 \times 100 \text{ cl} + 0 \times Q_M = 0.08 \times (50 \times 100 + Q_M)$$

Thus,

$$Q_M = \frac{1200}{0.08} = \frac{120000}{8} = 15000 \text{ cl/min} = 150 \text{ l/m}.$$

The G_V may be taken as a pure gain element. The maximum capacity may be calculated by taking a design factor of 1.5, so that,

$$G_V = K_V = \frac{150 \times 1.5}{16} = \frac{225}{16} = 14.0625 \text{ l/m/mA} = 1406.25 \text{ (cl/min)/mA}$$

From the above information, we may propose the control block diagram for the dilution control system that is depicted in Figure 15.52.

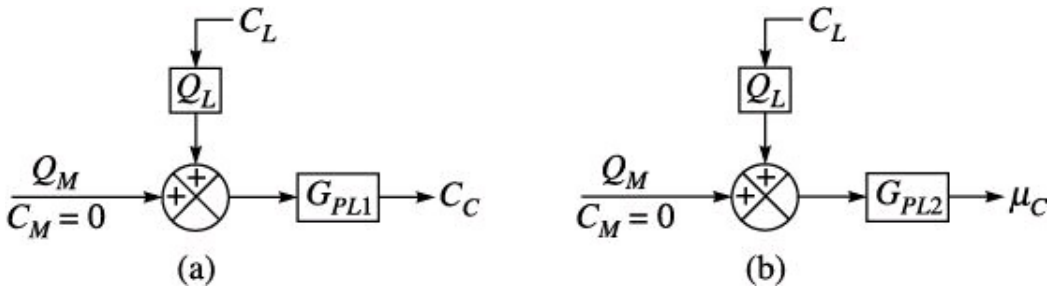


Figure 15.52 The node points of the inferential control system.

The inputs to G_{PL1} and G_{PL2}, both have dimension of g/min. The gain term in the numerator of G_{PL2} would contain the relation between viscosity and concentration. This is a non-linear function, hence the linear equivalent evaluated at steady state condition would be:

$$\left.\frac{d\mu}{dC}\right|_{C=C_S} = ABe^{BC_S} \qquad (15.43)$$

and

$$K_{PL2} = \frac{ABe^{BC_S}}{Q_{MS} + Q_{LS}} \qquad (15.44)$$

But the gain term of G_{PL1} may be simply given as, $K_{PL1} = 1/(Q_{MS} + Q_L)$. Thus,

$$G_{PL1} = \frac{1/(Q_{MS} + Q_L)}{[V/(Q_{MS} + Q_L)]s + 1} \qquad (15.45)$$

and

$$G_{PL2} = \frac{ABe^{BC_S}/(Q_{MS} + Q_L)}{[V/(Q_{MS} + Q_L)]s + 1} \qquad (15.46)$$

Note that steady state value of manipulated flow has been used also for G_{PL1}, because the two TFs operate under identical variable description.

Following the principle shown earlier, the estimator formulation may be proceeded as,

$$C_C = C_L Q_L G_{PL1} + C_M Q_M G_{PL1} \qquad (15.47)$$

and

$$\mu_C = C_L Q_L G_{PL2} + C_M Q_M G_{PL2} \qquad (15.48)$$

with a little manipulation, the estimator block may be found as,

$$\frac{C_C}{\mu_C} = \frac{G_{PL1}}{G_{PL2}} = \frac{1}{ABe^{BC_S}} \qquad (15.49)$$

Equation 15.49 is nothing but the concentration to viscosity conversion at steady state condition. We have prepared a simulation program for studying the closed loop dynamics of this control system. The process has two types of non-linearity:

1. The concentration-viscosity relation;
2. Variability of time constant of the mixer when solvent flow rate (manipulated variable) changes due to control action.

In such a study it is not needed to use the linear estimator, and the process has been simulated with its non-linear characteristics as such.

The control action is based on viscosity measurement; however, in a simulation program, the dynamics of concentration change may be computed. The closed loop dynamics has been displayed for both viscosity and concentration. Such a system would normally face disturbances in load. Initial values of load (concentration of the thick polymer solution) and Set Point (desirable exit solution concentration) are, C_L = 0.27 g/cl, and C_R = 0.08 g/cl. A pulse in load of 100 min duration is introduced by increasing C_L to 0.32 g/cl. A step in load is introduced by increasing C_L to 0.32 g/cl. The pulse and step responses in load are shown respectively in Figures 15.53 (a) and (b). We have also used set point change to examine the performance of the controller in face of the system's non-linear features. A pulse in set point of 100 min duration is introduced by increasing C_R to 0.1 g/cl. A step in load is introduced by increasing C_R to 0.1 g/cl. The pulse and step responses in load are shown respectively in Figures 15.53 (a) and (b). The pulse and step responses in set point are shown respectively in Figures 15.54 (a) and (b). The integral error measure ITAE has been calculated with respect to the concentration response for all four kinds of responses. These values indicate the amount of out-of-quality product that would result from a disturbed plant.

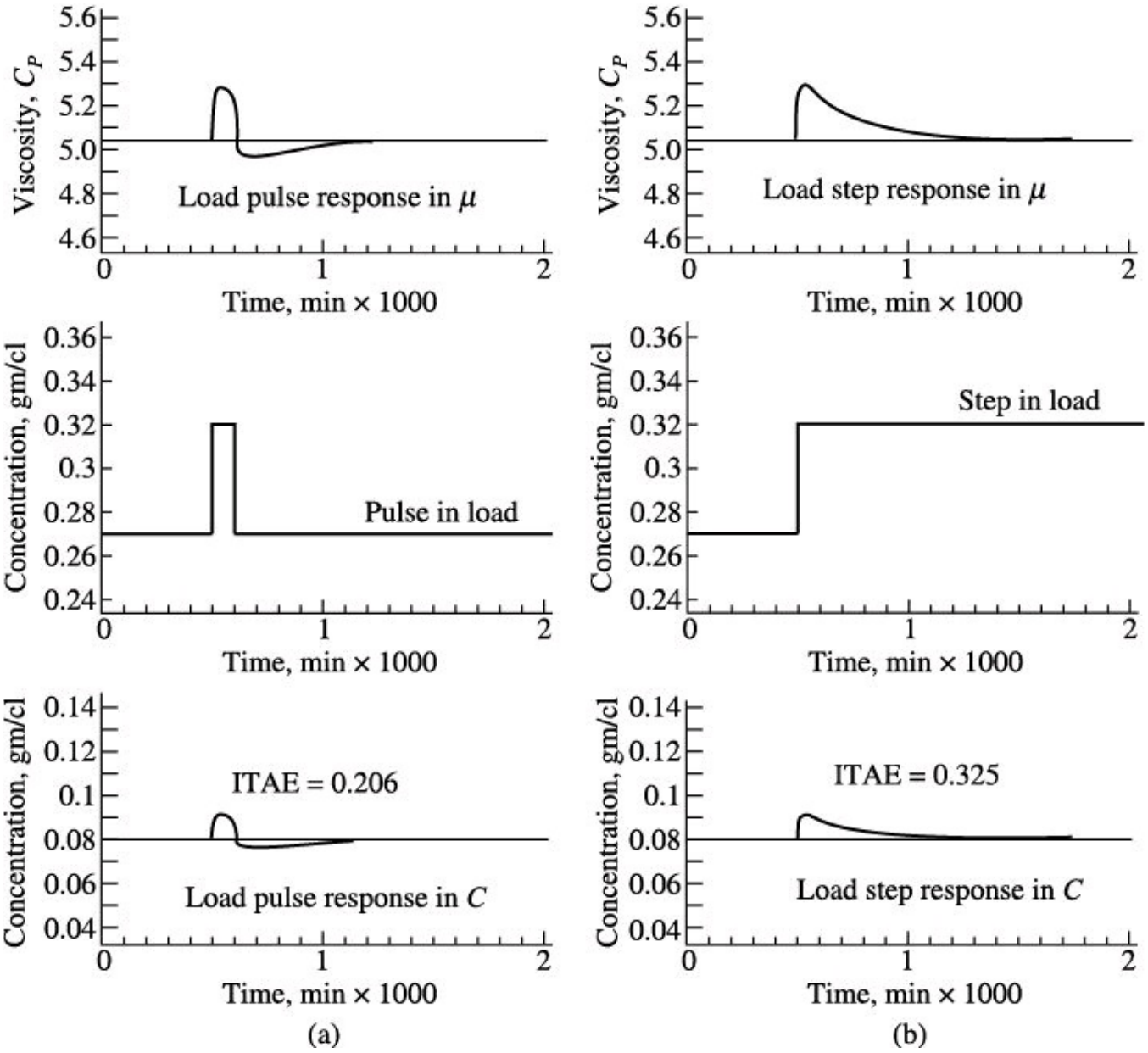


Figure 15.53 Pulse and step response in load.

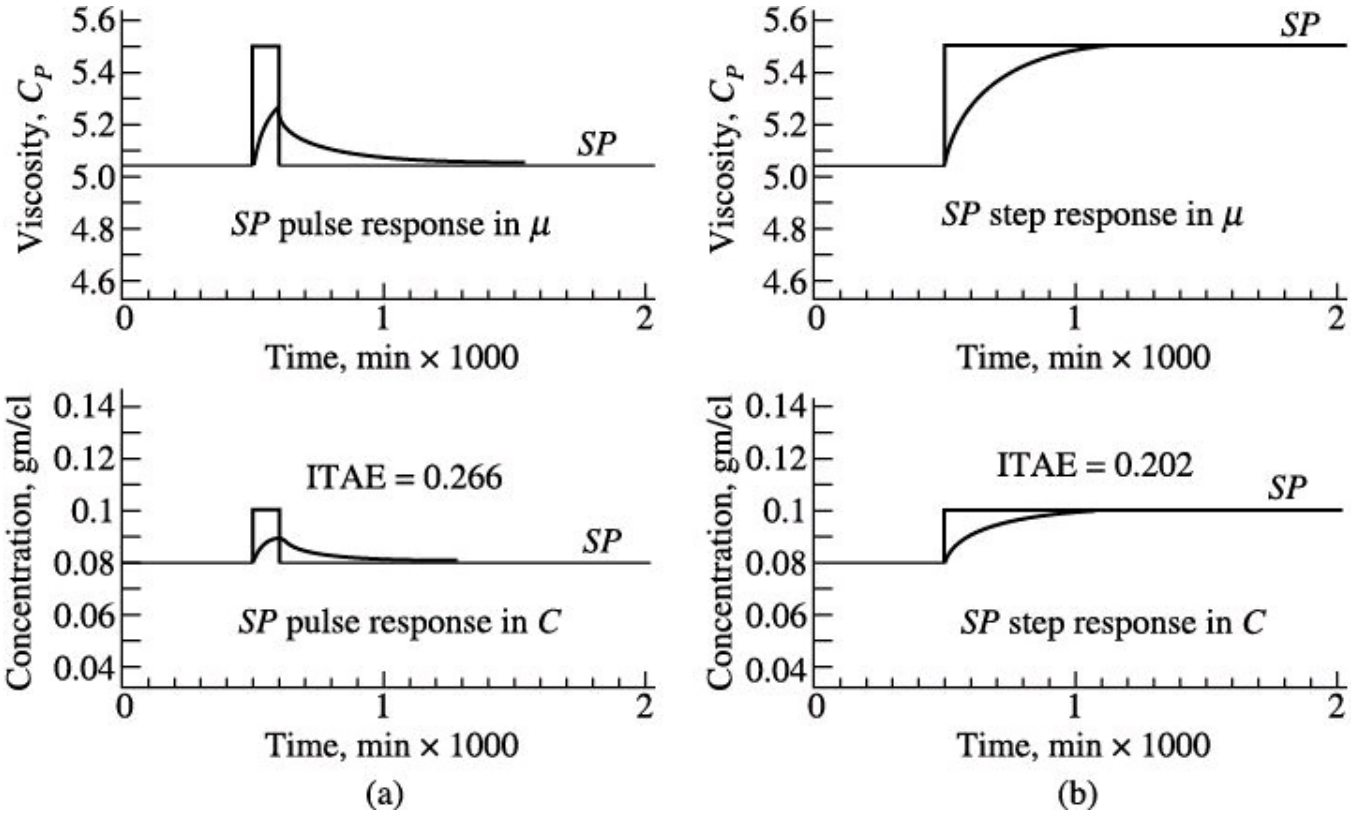


Figure 15.54 Pulse and step Response for set point change of the process under inferential control.

15.8 Selective and Override Control

If there are more controlled variables than manipulated variables in a process to be controlled, a common solution to this problem is to use a **selector** to choose the appropriate process variable from among a number of available measurements. Selectors can be based on multiple measurement points, multiple FCE or multiple controllers. Selectors are used to improve the control system performance as well as to protect equipment from unsafe operating conditions.

One type of selector module chooses the highest (or lowest) of two or more inputs and sends them as its output to the controller. This operation is also called **Auctioneering** in certain literature. For example, a high selector may be used to determine the 'hot spot'in a fixed catalytic bed tubular reactor in which an exothermic reaction is taking place. The process may run away due to disturbance or other changes in the reactor. Because a hot spot may develop at one of several possible locations in the reactor, multiple sensing points should be provided. Such an arrangement enables a high speed control action for safe operation. The control system has been shown in Figure 15.55.

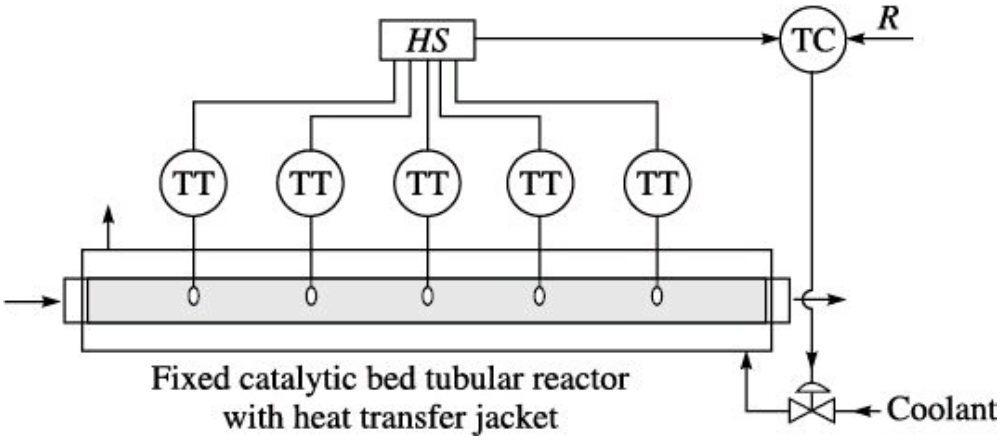


Figure 15.55 Selective zone temperature control of a tubular reactor.

In an **Override Control** the selected control action depends on whether the process variable has reached any of the two pre-chosen limits called hi- or lo-limits. The feature of 'anti-reset-windup' in feedback controller is a category of override.

Consider the process in Figure 15.56(a), where a hot saturated liquid pours in surge tank (capacity element to reduce flow fluctuation) and from there is pumped under flow control to some down stream process. Under normal operation, the controller tries to keep the level at H_1. If, for some reason, the level drops down to H_2, the liquid will not have enough net +ve suction head (NPSH) and the pump may get vapour bound which is a highly undesirable condition in pump operation. Therefore, it is necessary to introduce a control scheme to avoid this situation. This new scheme has been shown in Figure 15.56(b).

The level in the tank is now measured and controlled. The FCE in the flow loop is a variable speed pump that pumps more liquid for an increase of controller output. Hence, the flow controller (FC) must have a reverse action, while the level controller (LC) is a direct acting controller. The output of each controller goes to Low Selector (LS). The low selector's output goes to the pump. Under normal operating condition with level at H_1 (which is above the set point), the LC would try to speed-up the pump by increasing its output to saturation (say to 20 mA), while the output from FC might be 10–12 mA. The low selector (LS) will allow only the FC output to manipulate pump speed. This is the desired operating condition.

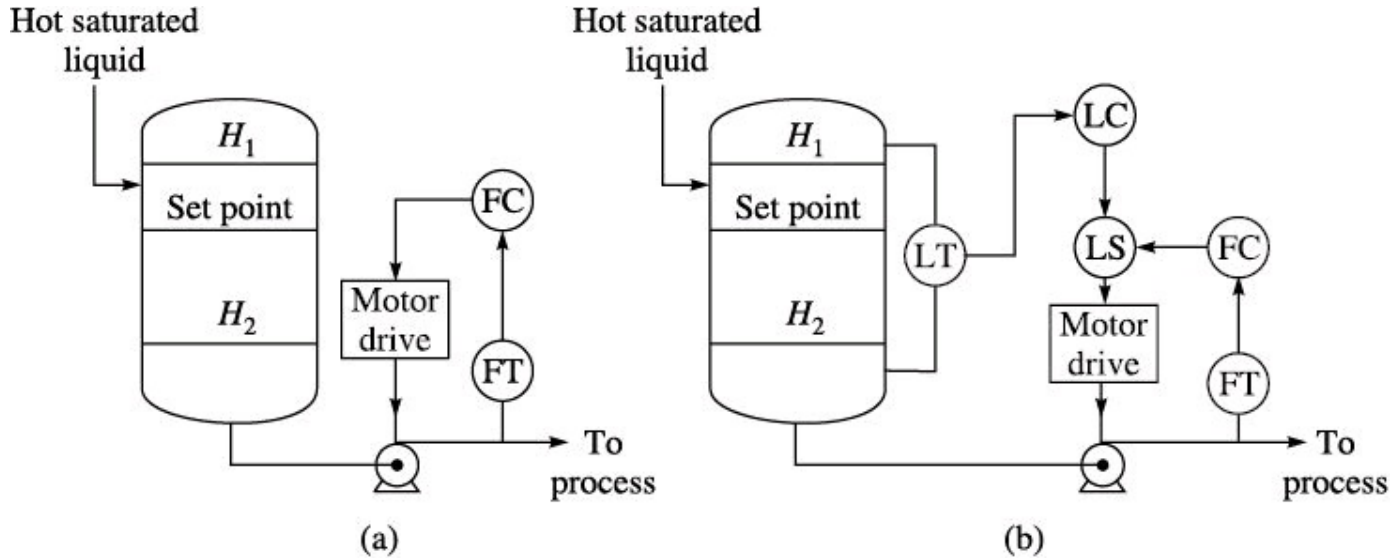


Figure 15.56 Tank level control. (a) Conventional. (b) Control with override.

Another kind of contingency may appear if the in-flow rate of the hot liquid decreases so that tank level continues to drop down. As the level drops below set point, the LC will decrease its output to slow down the pump speed. If LC output goes below that of FC, the LS will again allow only the LC output to manipulate pump speed thus avoiding any undesirable low level value inside the tank. It can be said that the LC overrides the FC.

15.9 Adaptive Control Schemes

There are many processes that face untoward external disturbances and/or internal parameter changes when they are operating under industrial environment. In such cases, the controllers inevitably need retuning of its parameters as and when necessary, to obtain satisfactory closed loop performance. In face of a significant change in process characteristics, operating conditions, or in the environment, the controller may have to be retuned. If these changes occur frequently, an adequate solution would be to use adaptive control schemes. An adaptive system is one in which the controller parameters are adjusted automatically to compensate for changing process conditions.

An adaptive control system constitutes a controller with adjustable parameters and a computing module containing the algorithm for adjusting them. Principally, the controller is non-linear in nature because of the parameter adjustment facility. The block diagrams of such systems are categorically different than any of the schemes that we have seen earlier. An adaptive control system can be thought of as having two loops. The **internal** loop may be a conventional feedback loop while the **external** loop contains mechanisms or algorithm blocks that execute the controller parameter calculation based on information from process input as well as its response. The path carrying currently calculated parameter values to the controller block completes the

outer loop. A block diagram of an adaptive scheme has been shown in Figure 15.57. The parameter adjustment loop is generally slower than the normal feedback loop, or acts intermittently.

A number of adaptive control schemes have been developed during recent years mainly for systems which possess the 'dynamic specialties', and also for systems whose dynamics is poorly defined. Adaptive control algorithms are designed not just to keep the process variable at set

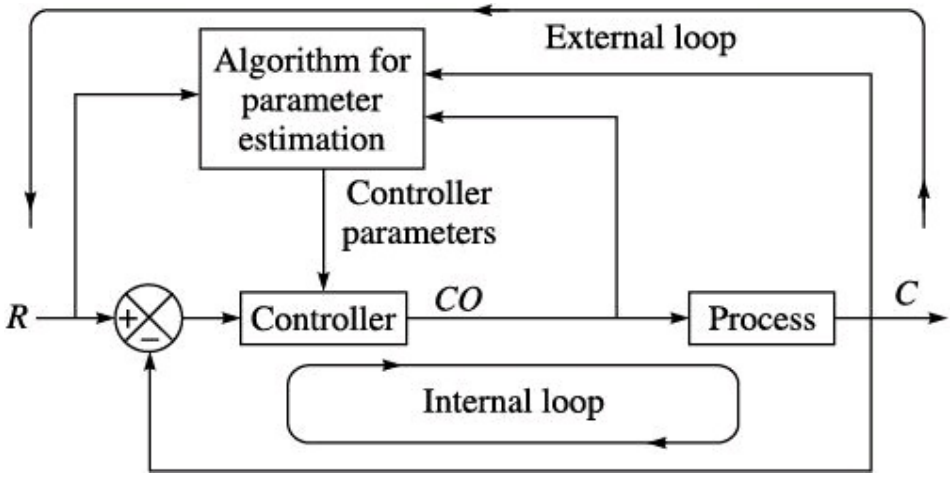


Figure 15.57 The general block diagram of an adaptive control scheme.

point (this could be done by feedback or any other control schemes discussed in earlier sections of this chapter), but may also have to fulfill a pre-selected closed loop response criterion. These may be:

1. Quarter Decay Ratio;
2. Minimum ITAE or ISE;
3. GM and PM specification.

We shall discuss some of the more important schemes of this class.

15.9.1 Gain Scheduling Type Adaptation

Many industrially important processes possess non-linear gain characteristics. The non-linearity may be in the mass or energy transfer mechanism, or it could come from the flow-head relation of a valve or characteristics of the sensor that is employed for process variable measurement. A feedback controller usually can control such a process if the amount of non-linearity is moderate. To get an idea about the measure of non-linearity for any input-output relation, (x □ y), the one can check the derivative (dy/dx) through the range of input variable change. Example 15.14 may be cited in this context where a feedback loop with a (PID) controller demonstrated a fair closed loop performance because the non-linearity in the concentration-viscosity relation that affects the 'dynamic compensator block' is not very much. To

demonstrate this fact, the μ vs. C and $d\mu/dC$ vs. C, for the process of Example 15.14, has been drawn in Figure 15.58(a) and 15.58(b) respectively.

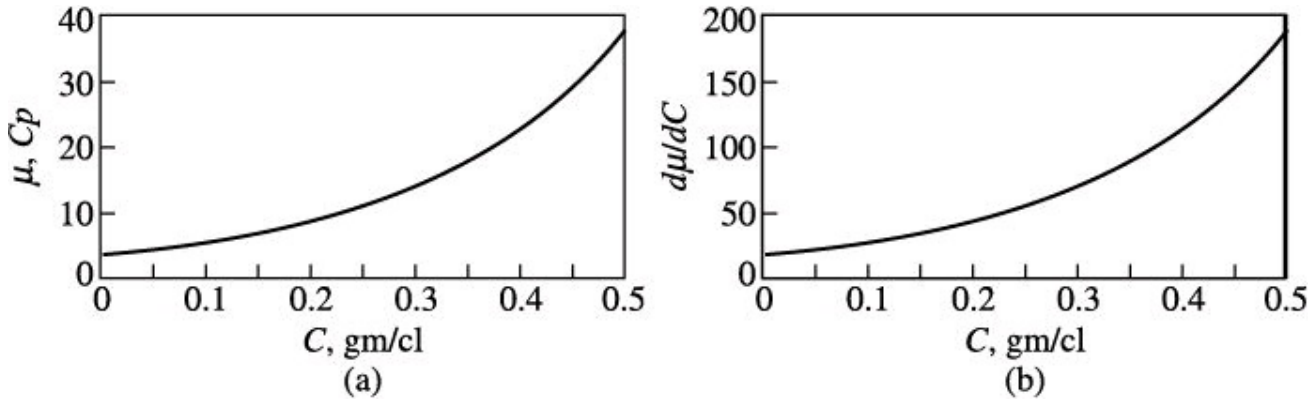


Figure 15.58 (a) Process with mild non-linearity. (b) Process with heavy non-linearity.

But there are processes that possess a more severe non-linear character, or in which the process gain changes drastically during the dynamic operation. As the controller gain K_C is within the forward path along with the process gain K_P, the controller gain needs to be evaluated for any process input change. This calculation is done on the external loop and results are sent to the controller for parameter readjustment. The calculations may be executed by a relation of output to gain. Alternatively, the relation may be expressed as a number of data set or schedules covering the range of process operation.

This approach for adaptation is called gain scheduling because the scheme was originally used to measure the process gain and then change controller gain. A block diagram of a control system with gain scheduling is shown in Figure 15.59, in the internal and external loops are also cited for identification.

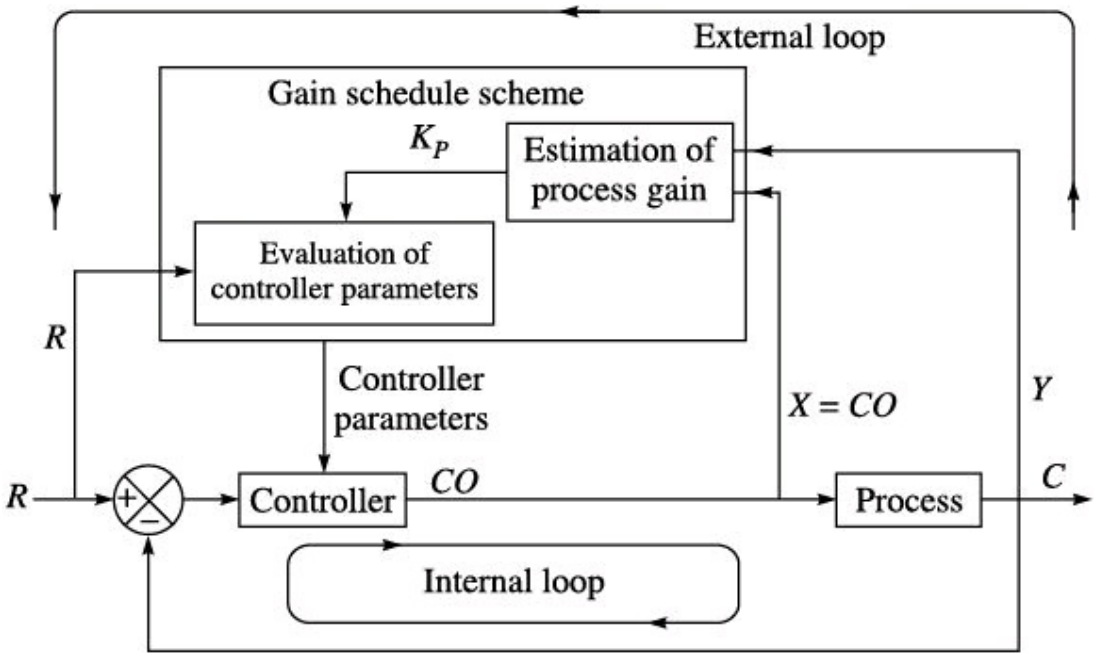


Figure 15.59 Block diagram of a control system with gain scheduling.

A typical application example of gain scheduling is pH control in a neutralization

process. Astrom and Wittenmark have analyzed this process in CSTR for a strong acid alkali pair.

EXAMPLE 15.15 Control of pH (the –ve $\log_{10}$ of H^+) is a well-known control problem that presents difficulties due to large variations in process gain in terms of static nonlinearity between pH and acid-alkali mole ratio exiting in the reaction mass as shown in Figure 16.60(a). pH is defined as:

$$pH = -\log_{10} [H^+]$$

where (H^+) denotes the hydrogen ion concentration. Water molecules are dissociated according to,

$$H_2O \rightleftharpoons H^+ + OH^-$$

The ionic equilibrium between (H^+) and (OH^-) is expressed by:

$$\frac{(H^+)(OH^-)}{(H_2O)} = \text{Constant} \qquad (15.50)$$

As only a small fraction of water is dissociated and activity coefficient of water is practically unity,

$$(H^+)(OH^-) = K_W \qquad (15.51)$$

where the equilibrium constant K_W value is 10^{-14} $(\text{mol/l})^2$ at 25°C.

Consider neutralization of m_{AC} mole of HCl by m_{AL} mole of NaOH in an aqueous solution according to:

$$HCl + NaOH \rightleftharpoons H^+ + Cl^- + Na^+ + OH^-$$

If the reaction is considered to be a constant density process happening in a total solution volume of V litres, the concentration of Cl^- ions is then,

$$(Cl^-) = x_{AC} = m_{AC}/V$$

and concentration of Na^+ ions would be,

$$(Na^+) = x_{AL} = m_{AL}/V$$

Due to dissociation, the of +ve and –ve ions are equal, hence,

$$x_{AC} + (OH^-) = x_{AL} + (H^+)$$

The concentration of (OH^-) ions may be related to the (H^+) ion concentration by Equation 15.51,

$$x = x_{AL} - x_{AC} = \frac{K_W}{(H^+)} - (H^+) = 10^{pH-14} - 10^{pH} \qquad (15.52)$$

Solving for (H^+) obtains,

$$(H^+) = \sqrt{\frac{x^2}{4 + K_W}} - \frac{x}{2}$$

$$(OH^-) = \sqrt{\frac{x^2}{4 + K_W}} + \frac{x}{2}$$

Finally,

$$pH = f(x) = -\log(\sqrt{\frac{x^2}{4} + K_W} - \frac{x}{2} \qquad (15.53)$$

The plot of function $f(x)$, shown in Figure 15.60(a), describes the fundamental nonlinearity of neutralization process. A considerable amount of slope change of the plot occurs if we move either from very low or high pH to the neutral pH of 7.0. As a quantitative check, the derivative of $f(x)$ is given by:

$$f'(x) = \frac{\log_{10} e}{2\sqrt{x^2/4 + K_W}} = \frac{\log_{10} e}{10^{pH-14} + 10^{-pH}} \qquad (15.54)$$

The derivative has its largest value 3.6×10^6 at pH = 7. It decreases rapidly for larger and smaller values of pH. At pH = 4 and 10, the derivative value is 4.3×10^3. To show these features, pH vs. x and dpH/dx vs. x, is drawn in Figures 15.60(a) and (b), respectively.

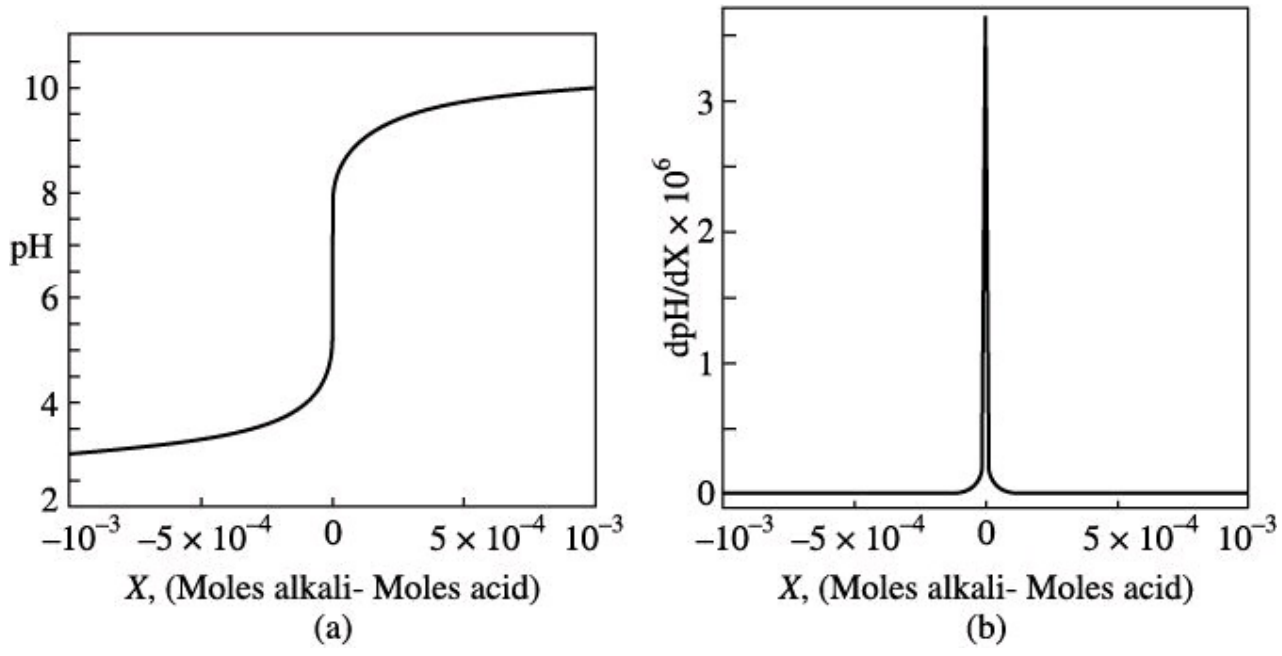


Figure 15.60 (a) The nonlinear characteristics of the pH vs alkali-acid system. (b) Derivative of the characteristic.

In an effluent treatment plant, a CSTR (V_R = 2000 l) is employed to control the pH of the incoming acidic effluent (Q_L = 50 l/min, C_{CL} = 0.08 mol/l) is treated with an alkali solution ($Q_M(t)$ = 0 to 4 l/min, C_{CM} = 10 mol/l). The control scheme is shown in Figure 15.61, in which the gain scheduling block effects the K_C evaluation and sends it to the controller.

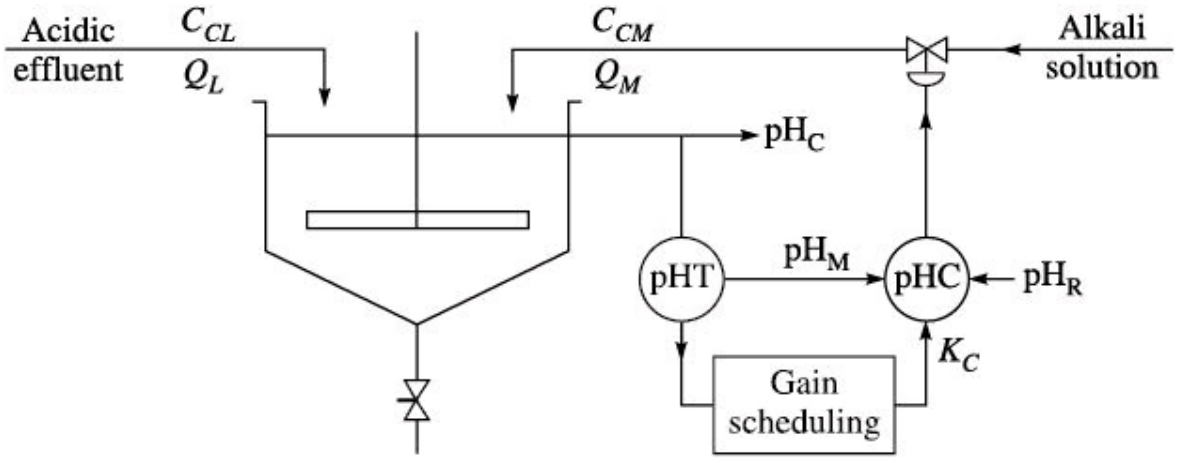


Figure 15.61 pH control of an acidic effluent by gain scheduling type adaptation.

Here the gain scheduling is done by a simple algorithm, i.e. $K_C = K_{CO}/K_P$, where K_C is the process gain evaluated by Equation 15.54, and K_{CO} is a tunable parameter. The value of
K_{CO} = 800000 has been found to obtain a good closed loop performance.

A simulation program is prepared to simulate the control system either with only a (PI) controller or a (PI) controller with gain scheduling. The (PI) controller parameters are tuned by trial-error procedure. The process is stabilized for a set point of pH_R = 4, then it is disturbed by step in the set point to pH_R = 7, 8, 9. The simulation results are shown in Figures 15.62(a) and (b).

The results show the elimination of oscillatory instability at pH = 7. But the loss of response speed is marked for all of the three pH values under gain scheduling type adaptation. The reason may be the use of a constant value for the tunable parameter, K_{CO} that effects the adaptation. A more accurate adaptation may be where information of process output as well as set point is used to evaluate the controller gain.

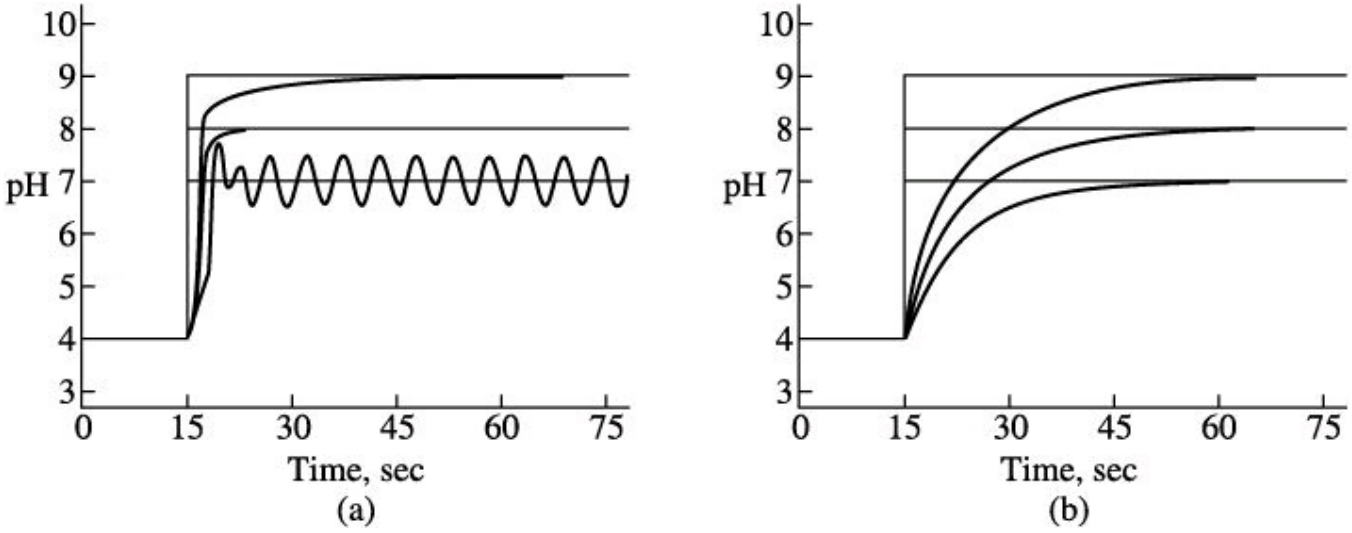


Figure 15.62 Response of pH control with, (a) (PI) controller. (b) (PI) controller with gain scheduling.

Application of this adaptive strategy for pH control in a batch reactor has been shown in *Advanced Process Dynamics and Control.*

15.9.2 Reference Model Adaptive Control (RMAC)

The model reference adaptive control systems are an important adaptive strategy. It may be regarded as an adaptive servo system in which the desired performance specification is expressed in terms of a reference model that produces the satisfactory response to a given disturbance. This is a convenient way for specifying a servo control problem. A block diagram of the system is shown in Figure 15.63.

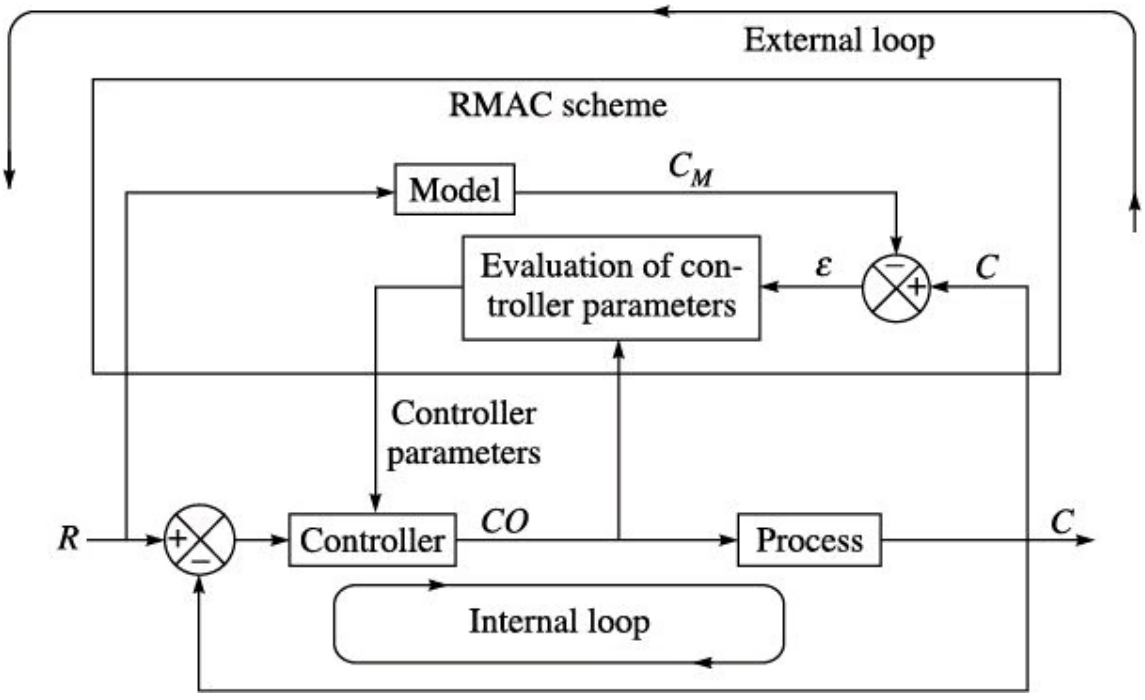


Figure 15.63 Adaptive strategy of reference model class.

The control system has a conventional feedback loop composed of the process and the controller and another feedback loop that changes the controller parameters. As we have said earlier, these are respectively called the 'Internal' and 'External' loops.

In the external loop the controller parameters are changed on the basis of error feedback. The error is the difference between process output C, and reference model output C_M. The algorithm for adjusting the controller parameters in such an adaptive system may differ in principle of implementation and style depending on the application, for which it is intended. Two popular methods of performing this task are either by using a gradient method or by applying stability theory.

In the MRAC system, the desired behaviour of the system is specified by a model, and the parameters of the controller are adjusted on the basis of the error that is obtained as the difference between the process and model responses ($C - C_M$). MRAC systems were first introduced for continuous time deterministic processes. The developed version for discrete time and stochastic system, came later.

Application of this adaptive strategy for non-stationary set point tracking in a heat treatment furnace has been shown *Advanced Process Dynamics and Control*.

15.9.3 Self Tuning Regulators (STR)

In the above two strategies, the adaptive compensation act is a sort of an explicit manner, i.e. the controller parameter determination steps always move forward, whereas, if the changes in process characteristics cannot be measured or predicted, the process estimation must follow an implicit (trial-n-error) or recursive routine. After this step is complete, the process model information may go to the controller parameter estimation algorithm, where the close loop performance may be available as a specification. Finally, the evaluated controller parameters from this module are communicated to the controller module. Such an adaptive scheme is called Self Tuning Regulator (STR). An STR tries to embody the tasks as described and perform them automatically. Figure 15.64 shows the action principle of an STR in block diagram form.

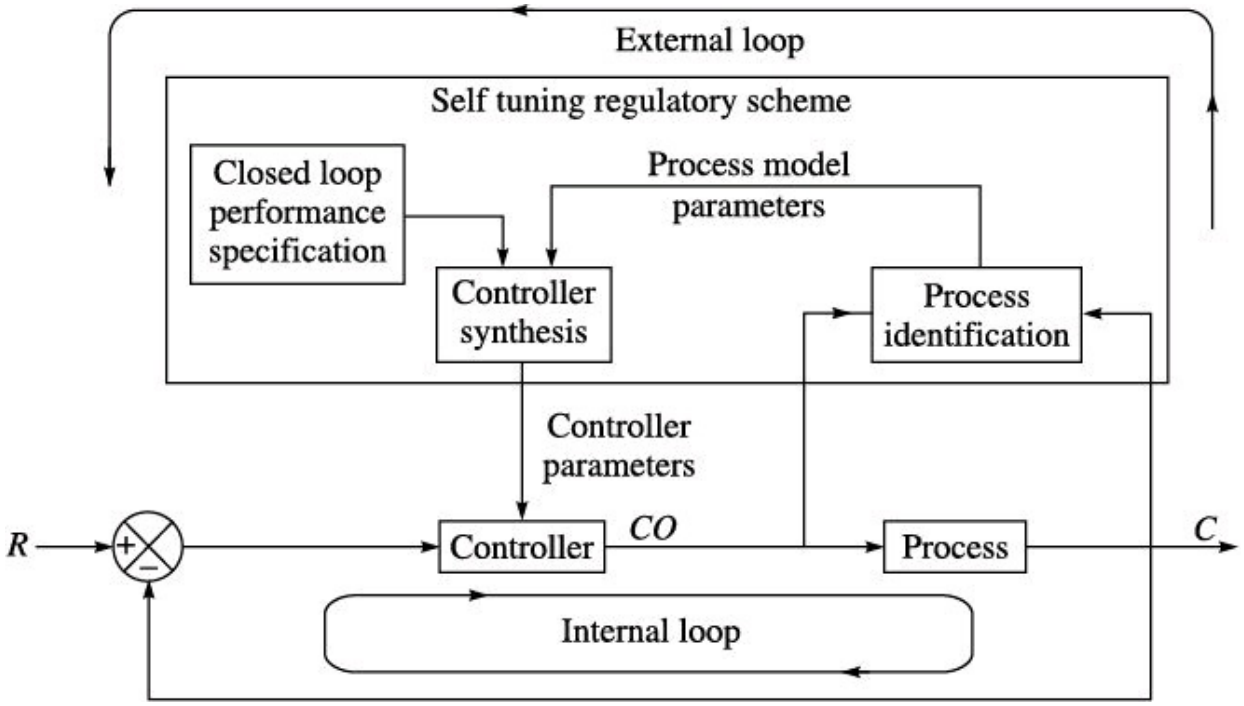


Figure 15.64 Block diagram of a self tuning regulator.

It is assumed that though the process model structure is known, the process parameters are non-stationary. Parameters of the model are estimated on-line based upon the information received of the input (manipulated variable) and process response (controlled variable). This is done in the block labeled Process Identification. Such an on-line process identification procedure has been discussed in Chapter 12. The block labeled Controller Synthesis contains procedures that are required to perform the design of a controller with a selected method and a few design parameters that can be chosen externally. In the block labeled controller, the designed parameters are implemented on the selected controller structure.

The main reason for using an adaptive controller is that the process or its environment is changing continuously. The term Self Tuning has been used to express the property that the controller parameters converge to the controller configuration that would have been designed if the process was known.

The tasks shown in the block diagram may be realized in many different methods. There are many possible choices of model and controller configurations. Identification can be performed continuously or intermittently. In digital implementations, different sampling rates can be used for the identifier, controller synthesis action and the controller. It is also possible to use a hybrid scheme in which control is done continuously and the identification-synthesis-and-parameter updating is done discretely.

EXAMPLE 15.16 As an application example of this control strategy, consider a fixed bed catalytic tubular reactor that is handling a gas phase exothermic reaction. To obtain higher conversion it was proposed to run the reactor with cyclic variation of temperature to increase the average production because the catalyst activity increases

exponentially with temperature.

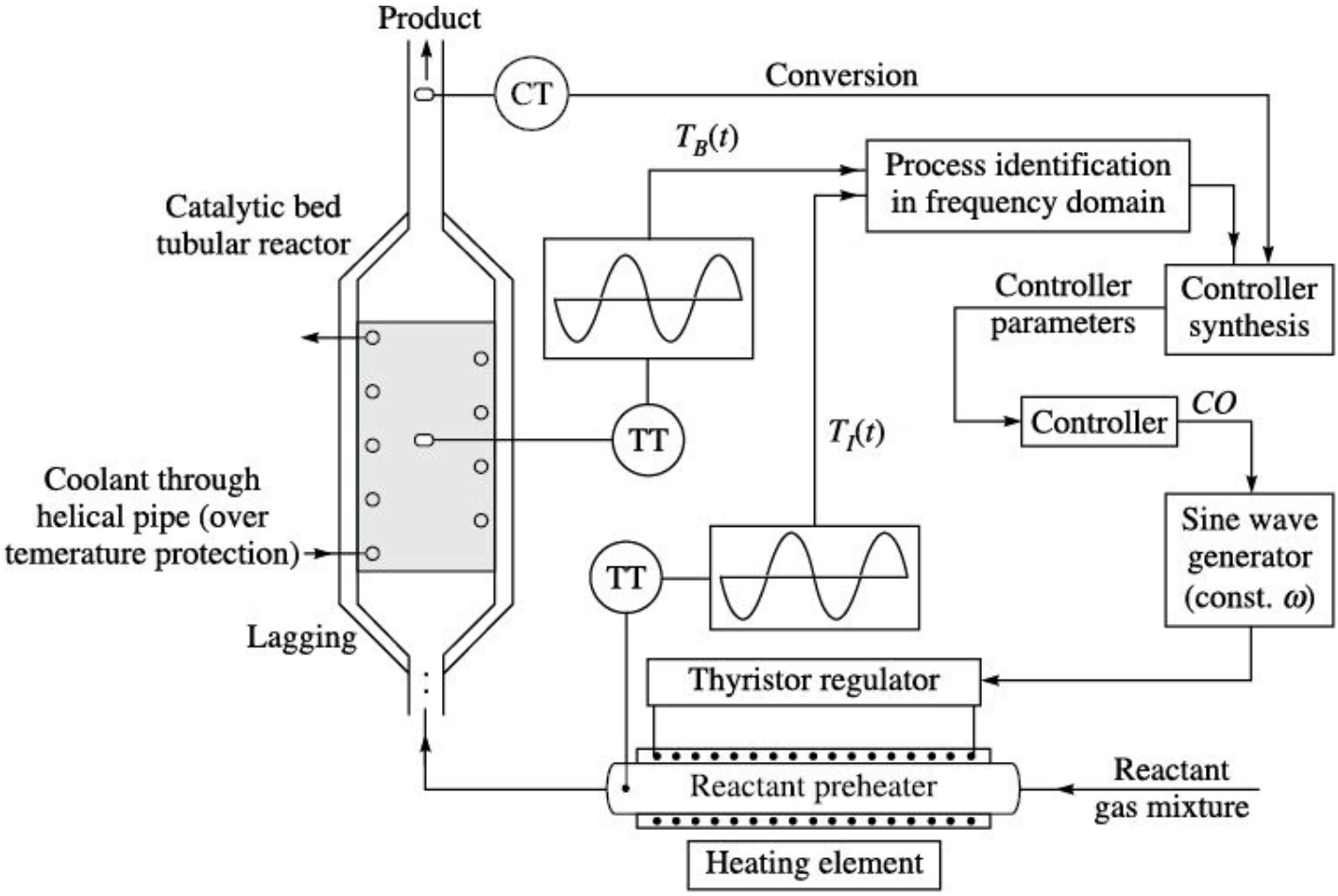


Figure 15.65 Operation and control of a PFR under forced temperature cycling.

However, there is an upper limit of catalyst temperature, T_M, that is, if the catalyst bed is subjected to a temperature, $T_C > T_M$, for more than a limiting time, t_M, it may be irreversibly damaged.

A simplified flow diagram of the reactor and its control system is shown in Figure 15.65. If we consider the reactor as the process, and the same is modeled conforming to an FOPDT structure, then the model parameters will vary during a reactor run mainly due to the poisoning of catalyst. The reactor is kept under cyclic temperature variation. It is achieved by a reactant preheater, through which the inlet gases gain heat from an electric band heater installed on the outer surface of the preheater tube, and the heating element receives a sinusoidally varying current from the thyristor regulator. The thyristor is in fact the FCE of control system. From the reactor inlet temperature, T_I, and catalyst bed temperature, T_B, the model parameters (in the Example 6.16, such an analysis has already been indicated) may be evaluated by using the frequency domain process identification procedure.

A suitable composition sensor may be applied to the reactor outlet gases for measuring the amount of conversion. This information is a measure of closed loop performance of the reactor system. If the conversion becomes lower than the desired

specification, which would gradually happen as the catalyst gets poisoned, it will be required to increase the amplitude of the temperature sinusoids of the entering gas to keep it up. The process model parameters and conversion as the closed loop performance measure, enters the controller synthesis block where the required analysis is done to obtain the controller parameters [for a (P) controller, the K_C values]. This gain value may be used to calculate the amplitude of current sinusoid to be applied to the electric heater of the preheater.

15.9.4 Programmed or Scheduled Adaptive Control

The process model structure (generally FOPDT is selected, but one may also select SO or SOPDT for more complex process or higher model accuracy requirement) is known and there is an auxiliary variable, by measuring which the model parameters may be determined. In such a system, a program or schedule may be resident in the control algorithm that continuously receives the value of the auxiliary variable from an auxiliary measurement point and adequately readjusts the process model parameter values.

Using these updated process parameter values the optimum, controller parameters are computed conforming to any pre-selected optimum response criterion and the parameters of the controller on the feedback loop are adequately adjusted. Hence, the external loop is the parameter estimator and the internal loop is the process controller.

Figure 15.66 shows the operation of a programmed or scheduled adaptive control scheme in block diagram form.

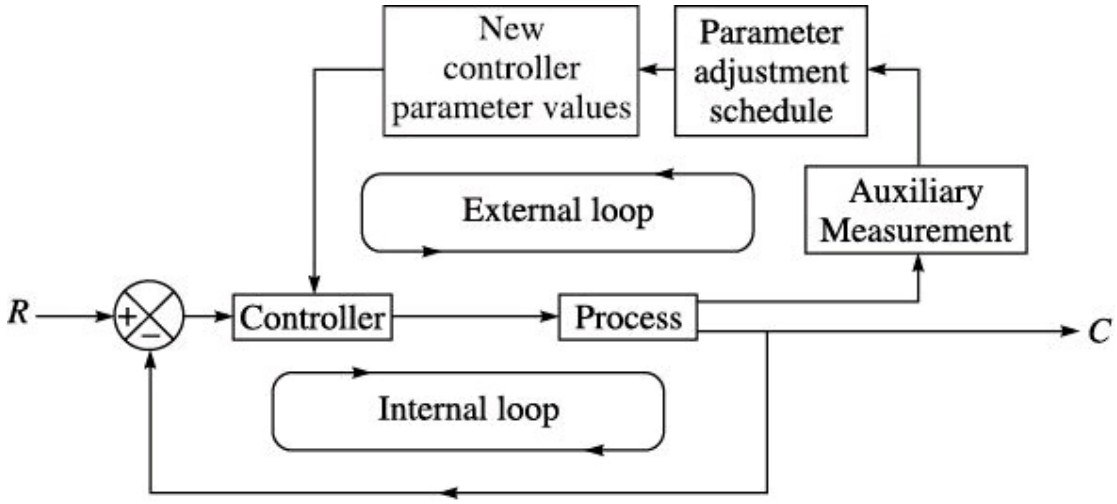


Figure 15.66 Block diagram of a programmed or scheduled adaptive control system.

There are few other variations of the same strategy available in literature. They only differ in the technique of obtaining and implementing the auxiliary variable for finding the current process parameter values.

EXERCISE PROBLEMS

15.1 The following block TFs are found for an FF + FB control system:

$$G_{PL}(s) = \frac{1.5}{(s+1)(4s+1)(0.25s+1)};\ G_{PM}(s) = \frac{1}{(s+1)(4s+1)}$$

$$G_{ML}(s) = G_{MM}(s) = G_V(s) = 1$$

Determine the feedforward controller TF, $G_{CFF}(s)$, to keep the controlled variable at its set point, for a step change in load.

15.2 A feedforward controller has to be designed on the basis of following TFs of the elements of the loop:

$$G_{PL}(s) = \frac{0.5}{(2s+1)(4s+1)(6s+1)^2};\ G_{PM}(s) = \frac{0.5s+1}{(2s+1)(4s+1)};\ G_V(s) = \frac{2}{0.1s+1}$$

The measuring elements in the loop may be assumed to have unity gain. Design the feedforward controller TF. If a feedback controller with (P) mode is coupled to this FF controller, *Ku*, *Pu* and off-set for a load change of unit step. Evaluate these quantities for (a) Only FB control, and for (b) FB + FF control.

15.3 The TFs, G_{PL} and G_{PM} of a concentration control system are known. These are given as,

$$G_{PL}(s) = \frac{1.4}{(2s+1)^4} \text{ and } G_{PM} = \frac{0.7}{(2s+1)^3}$$

The other elements of the FF + FB control system may be assumed to have unity gain TFs. Design a feedforward controller by the method discussed in Section 15.1. Prepare programs for the simulation of the closed loop system with: (a) FB; (b) FF + FB control. The program should have the facility of obtaining step response for load change.

15.4 Schematic diagram of mixing system of two streams (A) and (B) at different flow rates and concentrations are joined in a mixer to constitute a mixed stream. As shown
Figure P15.1.

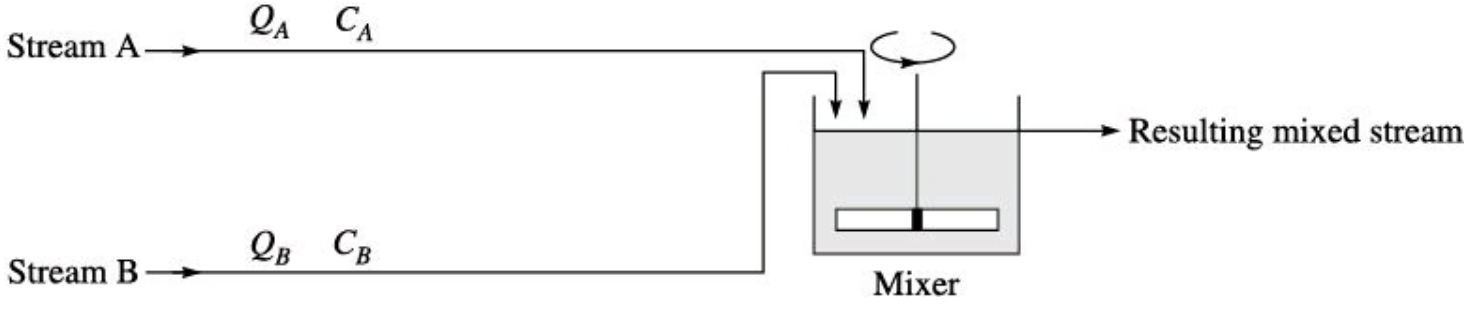


Figure P15.1 Schematic diagram of a mixing system.

Prepare a ratio control scheme to maintain the concentration of exit stream by considering stream A to be wild, however, its concentration remains constant, while Q_B is uncontrollable, but C_B is also constant and Q_B can be regulated. Draw the control scheme in terms of block diagram. Assume that the exit concentration sensor, inlet flow sensors and the control valve are pure gain elements.

15.5 The raw materials N_2 and H_2 for ammonia synthesis by Haber's process is obtained from Producer gas (mainly a mixture of N_2, CO_2, CO and O_2) and Water gas (mainly a mixture of H_2, N_2, CO_2, CO and O_2) respectively. Oxygen is removed by a thermodynamic cycle. CO is converted to CO by catalytic conversion called Shift's reaction. The purified producer gas contains only N_2, and that from water gas contain H_2 – 87% and N_2 – 13% by volume. The following reaction absorbs heat and there is reduction of 1 mole per 2 moles of the reactant converted.

$$N_2 + 3H_2 \rightleftharpoons 2NH_3, -\Delta H_R.$$

Hence, the reaction is carried at high temperature (450°C) and pressure (680 psig) by contacting the reactant gases with a special catalyst in a Catalytic tubular reactor. After separating ammonia from the exit gas steam, the unconverted N_2 and H_2 are recycled into the reactor with fresh feed. The fresh gas streams used are 10% excess than their stoichiometric requirement.

For production of 10000 kg/h of ammonia, calculate the amount of purified producer and water gas in m^3/h at the reaction temperature and pressure. Area type flow metres with square root extractor are to be used for measuring the flow of two gases. Specify the range of these flow metres assuming that the normal flow rate falls at 65% of their span. If the flow rate of water gas is considered to be the wild variable, design the K_R of a ratio controller to manipulate the producer gas stream at the reactor inlet.

15.6 In the following Figures P15.2(a) and (b), feedback and cascade control schemes for a process are demonstrated respectively.

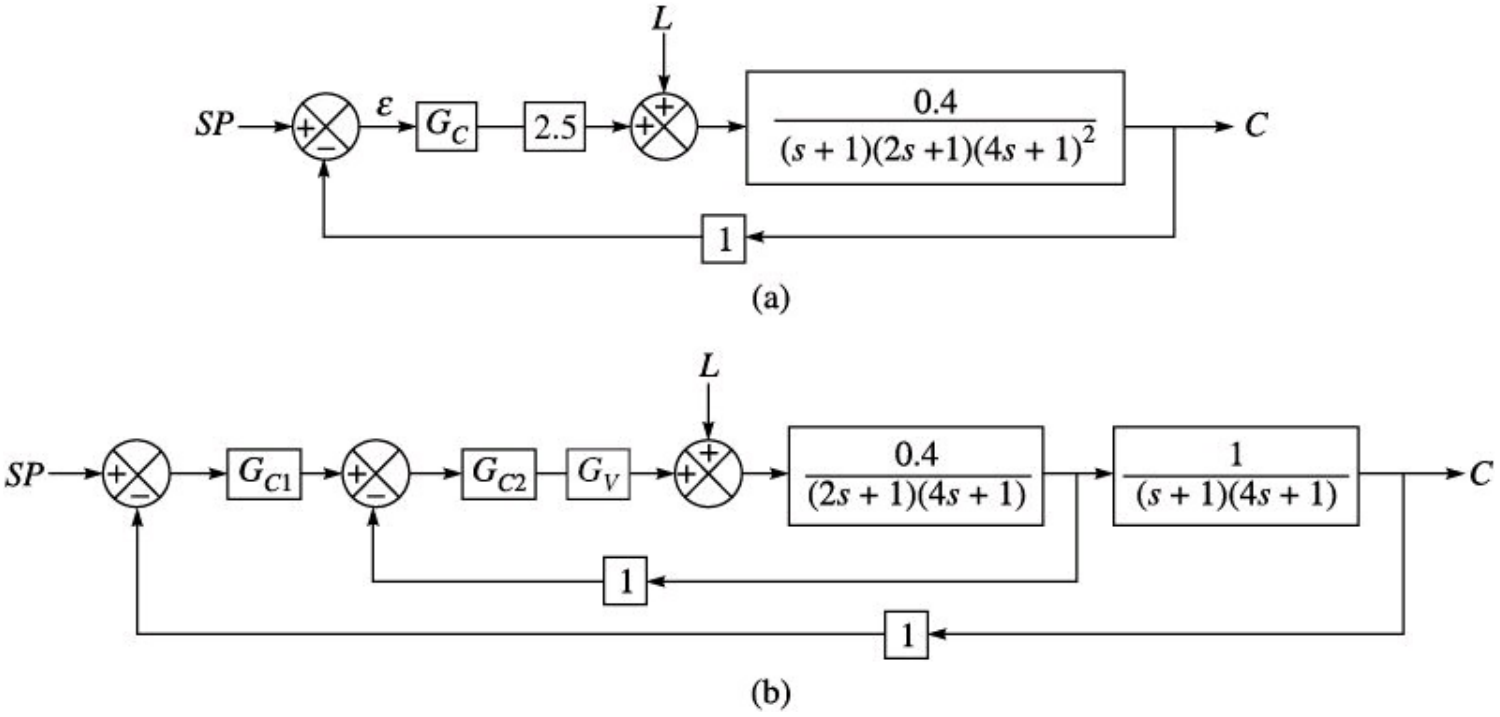


Figure P15.2 (a) Feedback. (b) Cascade control of a fourth order process.

Design the cascade controller according to method shown Section 15.3. Prepare programs for the simulation of the closed loop system with: (a) FB, (b) Cascade control. The program should have the facility of obtaining step response for load and set point change.

15.7 A dynamics of a process is expressed by a fifth order TF as:

$$G_P(s) = \frac{1.5}{(s+1)(2s+1)(s+4)(s+5)(5s+1)}$$

The process is to be controlled by: (a) Feedback controller, (b) A controller designed according to Smith's Predictor Algorithm. Other elements of the loop may be assumed to be of unity TF. Write simulation programs to study and compare the performances under control schemes (a) and (b).

15.8 The dynamics of an inverse response process is expressed by the following block diagram in P15.3.

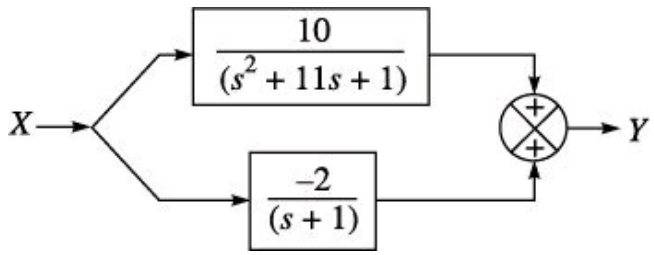


Figure P15.3 The dynamic model of an inverse response process.

The process has to be controlled by a (PID) controller. Derive the location of 'zero' in the process model and the expression for dynamic compensator, $D_C(s)$, in the control loop. Obtain the closed response with: (a) only tuned controller, and

(b) tuned controller + the dynamic compensator by means of simulation programs.

15.9 A liquid phase combination type reaction between two organic chemical substances A and B, happens in a CSTR to form an organic product C according to:

$$A + B \square C$$

Rate of the reaction, $r = kC_AC_B$, and the specific rate constant, $k = 0.1$ mole/(l/s).

Two solution streams containing A and B enters from top of the continuously stirred constant holdup reactor. The exit stream concentration has to be controlled. The control scheme is shown in Figure P15.4.

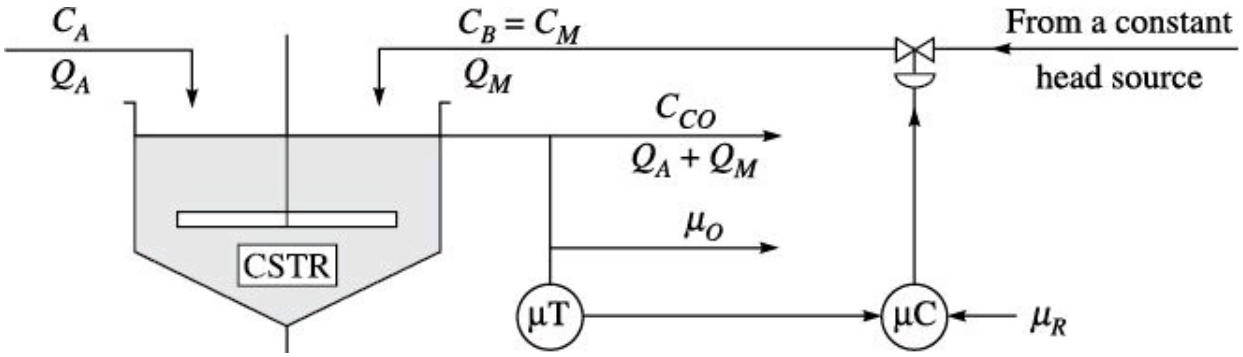


Figure P15.4 Exit concentration control in a CSTR.

The difficulty in this control is that there is no sensing device available for the direct measurement of product concentration, C_A. However, the refractive index of the exit solution bears the following relation with C_C, but is independent of concentration of A and B. The relation may be expressed as:

$$m_C = 0.15 + 0.2C_C$$

Design the reactor volume, V_R assuming 80% conversion, with normal variable values given below. Derive the dynamic estimator for the system for designing an inferential control scheme. Prepare a simulation program to study the close loop step response in load and set point: (a) with FB, and (b) with FB and dynamic estimator.

Data: $Q_A = 40$ l/min (constant), $C_A = 17.55 \pm 2.45$ g/l (load),

$Q_B = Q_M$ l/min (manipulation), $C_B = 13.2$ g/l (constant),

$Q_C = Q_A + Q_B$ l/min. C_R is equivalent to 80% conversion.

Molecular wt. of A, $M_A = 117$, Molecular wt. of B, $M_B = 88$, Molecular wt. of C, $M_C = 205$.

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16

State Variables and Multiple Input-Output Systems

Part 1: STATE VARIABLE ANALYSIS

16.1 State Variable Analysis

In Chapter 3 the state variable model has been introduced in a preliminary way. At least you know how to formulate the matrix representation of a dynamic system from the fundamental differential equations that are derived following the unsteady state balance of principal entity. Such matrix representation is called Input-Output model description. It is obvious that matrix representation is possible only when the number of input or output variables is more than one and, by multiplicity they may be called **vectors**. For single input single output systems, the variables are called **scalar**, and they are better treated by the way of transfer function technique.

The number of **states** is defined by the number of first order ODEs that may be written to describe the dynamics of a given system. At any time instant, t, to describe the state of a system having N-number of state variables, it is sufficient to locate the point in the N-dimensional space with coordinates $(y_1, y_2, \ldots, y_{N-1}, y_N)$, where, y_1, y_2, …, y_{N-1}, y_N, are the magnitude of the state variables at time t. For a given time span, the dynamics of such a system may be described by the trajectory obtained by joining individual points within such an N-dimensional space.

Before beginning any serious treatment, we shall present some of the preliminary matrix algebra and subsequently start from assuming that from the process description, the reader knows how to present the dynamics in the matrix form. The notations used in this chapter would be a carry over from Chapter 3.

16.1.1 Matrix Algebra

Rules of matrix algebra in a concised form are given in the following:

Matrix Addition/Subtraction

If **[A]** and **[B]** are two square matrices, their addition/subtraction is effected in the following way:

$$[\mathbf{A}] \pm [\mathbf{B}] = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \pm \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11} \pm b_{11} & a_{12} \pm b_{12} \\ a_{21} \pm b_{21} & a_{22} \pm b_{22} \end{bmatrix} \quad (16.1)$$

Matrix Multiplication

Multiplication of two matrices, **[A]** and **[B],** is possible if the number of rows of the first matrix, **[A],** equals the number of columns of the second matrix, **[B]**.

If **[A]** = $[a_{11} \quad a_{12}]$, and **[B]** = $\begin{bmatrix} b_{11} \\ b_{21} \end{bmatrix}$, then **[AB]** = $[a_{11}b_{11} + a_{12}b_{21}] = a_{11}b_{11} + a_{12}b_{21}$. This is simply a scalar. Thus, a matrix with a single element is considered as a scalar.

The product, **[AB],** of two square matrices, **[A]** and **[B],** is expressed in the following way:

$$[\mathbf{AB}] = [\mathbf{A}] \times [\mathbf{B}] = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \times \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{12}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{bmatrix} \quad (16.2)$$

Generally, **[AB]** ≠ **[BA]**, because **[BA]** is given by:

$$[\mathbf{BA}] = [\mathbf{B}] \times [\mathbf{A}] = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \times \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} + a_{21}b_{12} & a_{12}b_{11} + a_{22}b_{12} \\ a_{11}b_{21} + a_{21}b_{22} & a_{12}b_{21} + a_{22}b_{22} \end{bmatrix} \quad (16.3)$$

If the equality **[AB] = [BA]** holds, the matrices are said to be *Commutable*.

Transpose of a Matrix

The transpose $[\mathbf{A}]_\mathbf{T}$ of a square matrix **[A]** is found by interchanging the row elements by column elements in the following way:

If

$$[\mathbf{A}] = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \text{ then } [\mathbf{A}]_\mathbf{T} = \begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{bmatrix} \quad (16.4)$$

Identity or Unit Matrix, [I]

It is a square matrix whose diagonal terms are 1 and all other terms being zero. A 2 × 2 unit matrix, **[I],** may be given as: $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

Null Matrix

If all the elements of a matrix are zero, the matrix is said to be a Null Matrix.

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Determinant of a Matrix

The determinant of a matrix, **[A]**, is a scalar quantity denoted by |A| and is obtained

by:

If

$$\mathbf{[A]} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \text{ then } |\mathrm{A}| = a_{11}a_{22} - a_{12}a_{21} \qquad (16.5)$$

If the determinant of a matrix is zero (0), the matrix is called a **Singular Matrix.**

Adjoint of a Matrix

The adjoint of a matrix **[A]** is found by exercising the following sequence of operations. First a matrix is obtained by evaluating the co-factor of each element of the original matrix. The co-factors are the determinants formed by eliminating the row and column in which each element lies, using the remaining N – 1 rows and columns, and multiplying this value by $(-1)^{i+j}$, where, i is the row and j is the column in which the element is located. The transpose of this matrix is called the adjoint and denoted as Adj**[A]**.

For

$$[\mathbf{A}] = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \text{ the co-factor matrix is } = \begin{bmatrix} A_{11} & -A_{12} & A_{13} \\ -A_{21} & A_{22} & -A_{23} \\ A_{31} & -A_{32} & A_{33} \end{bmatrix}$$

Where the co-factors are:

$$A_{11} = a_{22}a_{33} - a_{32}a_{23},\ A_{12} = a_{21}a_{33} - a_{31}a_{23},\ A_{22} = a_{11}a_{33} - a_{31}a_{13}, \ldots, A_{33} = a_{11}a_{22} - a_{21}a_{12}$$

The transpose of the co-factor matrix is called the adjoint matrix of **[A]**. Thus,

$$\mathrm{Adj}[\mathbf{A}] = \begin{bmatrix} A_{11} & -A_{21} & A_{31} \\ -A_{12} & A_{22} & -A_{32} \\ A_{13} & -A_{23} & A_{33} \end{bmatrix} \qquad (16.6)$$

Rank of a Matrix

If we select any r rows and r columns from the elements of a matrix **[A]**, then the determinant formed by these $r \times r$ elements is called the **minor** of **[A]** of order r. Obviously, there will be a number of different minors of the same order.

A matrix is said to be of rank r, when it has at least one non-zero minor of order r, and all minor of order greater than r are zero.

Matrix Inversion

The inverse of a matrix $\mathbf{[A]}^{-1}$ is a matrix which, when multiplied by the original matrix **[A]**, gives the Identity Matrix, thus, $[\mathbf{A}][\mathbf{A}]^{-1} = [\mathbf{I}]$. Inverse of matrix **[A]** may be calculated by the following equation:

$$[\mathbf{A}]^{-1} = \frac{\text{Adj}[\mathbf{A}]}{\text{Det } A} = \frac{[\text{co-factor } \mathbf{A}]^{\mathrm{T}}}{|A|} \qquad (16.7)$$

16.1.2 State Variable Representation of Dynamic Processes

In Chapter 3, it has been shown that the dynamic character of a process may be developed from the unsteady state principal entity balance and can be expressed in the form of an *n*-th order differential equation, where, *n* is order of the process.

$$a_N \frac{d^n y}{dt^n} + a_{N-1} \frac{d^{n-1} y}{dt^{n-1}} + \cdots + a_1 \frac{dy}{dt} + a_0 y = b_0 x(t) \qquad (3.7)$$

From this fundamental description, the corresponding input-output type state variable model can be written. For convenience, we are rewriting Equations 3.8, 3.9 and 3.10 and renumbering them:

$$\left.\begin{aligned} dy/dt &= y_1 \\ dy_1/dt &= y_2 \\ &\vdots \\ dy_{N-2}/dt &= y_{N-1} \\ dy_{N-1}/dt &= (b_0/a_N)x(t) - (a_{N-1}/a_N)y_{N-1} - \cdots - (a_1/a_N)y_1 - (a_0/a_N)y \end{aligned}\right\} \qquad (16.8)$$

and expressing the set of equations in the following matrix form:

$$[\dot{\mathbf{x}}] = [\mathbf{A}][\mathbf{x}] + [\mathbf{B}]\mathbf{u} \qquad (16.9)$$

$$\mathbf{y} = [\mathbf{C}][\mathbf{x}] \qquad (16.10)$$

where,

$\dot{x} = \begin{bmatrix} \dot{y} \\ \dot{y}_1 \\ \cdot \\ \cdot \\ \dot{y}_{N-2} \\ \dot{y}_{N-1} \end{bmatrix}$ is the vector representing the derivative of state variable vector,

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & . & . & 0 & 0 \\ 0 & 0 & 1 & . & 0 & 0 \\ . & . & . & . & . & . \\ . & . & . & . & . & . \\ 0 & 0 & . & . & 1 & 0 \\ -\frac{a_0}{a_N} & -\frac{a_1}{a_N} & . & . & -\frac{a_{N-1}}{a_N} & 0 \end{bmatrix}$$ is the process characteristic matrix,

$$\mathrm{B} = \begin{bmatrix} 0 \\ 0 \\ . \\ . \\ 0 \\ \frac{b_0}{a_N} \end{bmatrix}$$ is the input vector,

$\mathbf{u} = [0 \quad 0 \quad . \quad . \quad 0 \quad 1]$ is the perturbation or input disturbance,

$\mathbf{C} = [1 \quad 0 \quad . \quad . \quad 0 \quad 0]$ is the output matrix, and

$$\boldsymbol{x} = \begin{bmatrix} y \\ y_1 \\ . \\ . \\ y_{N-2} \\ y_{N-1} \end{bmatrix}$$ is the state vector.

Some examples of converting process models from differential equation to the corresponding state variable form has been discussed in Chapter 3, however, for the sake of continuity, we shall present a couple of examples to demonstrate this procedure.

EXAMPLE 16.1 An interacting two tank liquid level dynamic system is shown in Figure 16.1, in which the process parameters and variables are labeled. Derive the state variable model.

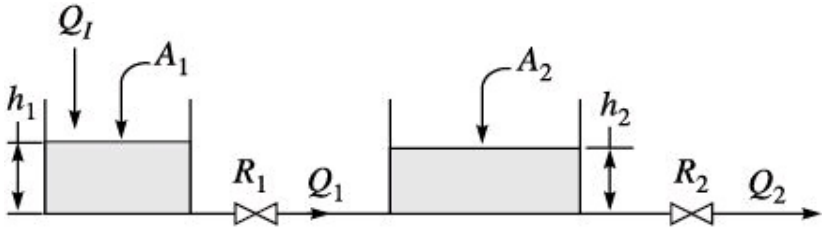

Figure 16.1 The two tank interacting system.

Solution: Recognizing Q_I as the input variable, the unsteady state mass balance around Tank 1 and 2, produces the following fundamental differential equations:

$$Q_I - Q_1 = A_1 \frac{dh_1}{dt} \tag{16.11}$$

$$Q_1 - Q_2 = A_2 \frac{dh_2}{dt} \tag{16.12}$$

and the flow-head relations:

$$h_1 - h_2 = R_1 Q_1 \tag{16.13}$$

$$h_2 = R_2 Q_2 \tag{16.14}$$

Using these relations, the intermediate variable, Q_1, may be eliminated to obtain the two differential equations in the desired form:

$$\frac{dh_1}{dt} = \frac{Q_I}{A_1} - \frac{h_1}{A_1 R_1} - \frac{h_2}{A_1 R_2} \tag{16.15}$$

$$\frac{dh_2}{dt} = \frac{h_1}{A_2 R_1} - \frac{h_2}{A_2 R_1} - \frac{h_2}{A_2 R_2} \tag{16.16}$$

From the above equations, the corresponding input-output form of the state variable model may be written as:

$$\begin{bmatrix} \dot{h}_1 \\ \dot{h}_2 \end{bmatrix} = \begin{bmatrix} -1/A_1 R_1 & 1/A_1 R_1 \\ 1/A_2 R_1 & -(1/A_2 R_1 + 1/A_2 R_2) \end{bmatrix} \begin{bmatrix} h_1 \\ h_2 \end{bmatrix} + \begin{bmatrix} 1/A_1 \\ 0 \end{bmatrix} Q_I \tag{16.17}$$

If we choose Q_2 as the output variable, the output relation would be:

$$Q_2 = \begin{bmatrix} 0 & 1/R_2 \end{bmatrix} \begin{bmatrix} h_1 \\ h_2 \end{bmatrix} \tag{16.18}$$

EXAMPLE 16.2 An electrically heated furnace is shown in Figure 16.2. The resistive heating element is embedded inside the refractory wall of the furnace. The object to be heated perfectly fits the inside space of the refractory. The temperature of the wall, T_1, and that of the object, T_2 are uniform through the corresponding masses, m_1 and m_2, respectively. The corresponding heat capacities are C_1 and C_2. Heat transfer area between the refractory wall and the object is A_H and the overall heat transfer

coefficient is U. The amount of heat that dissipates through the wall to the atmosphere is $U_O A_O(T_1 - T_A)$, where, U_O = Heat transfer coefficient between furnace body and atmosphere, A_O = Outside surface area of the furnace, T_A = Atmospheric temperature. Assuming T_A = 0ºC, write the heat balance equation and derive the state variable model for the system with heat input W, and object temperature T_2, as the input and output variables.

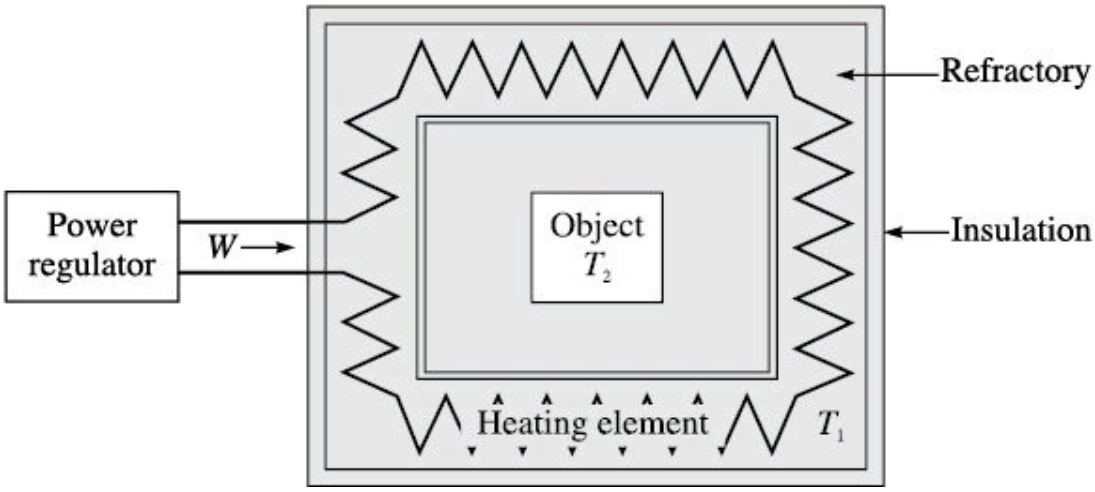

Figure 16.2 Thermal dynamics in an electrically heated furnace.

Solution: Thermal balance around the refractory wall and the object results in the following two differential equations:

$$W - A_H U(T_1 - T_2) - A_O U_O T_1 = m_1 C_1 \frac{dT_1}{dt}$$

$$A_H U(T_1 - T_2) = m_2 C_2 \frac{dT_2}{dt}$$

Introducing, $A = 1/(m_1 C_1)$; $B = (A_H U)/(m_1 C_1)$; $C = (A_H U)/(m_2 C_2)$; and $D = (A_O U_O)/(m_1 C_1)$, the above two equations are converted to the required form as:

$$\frac{dT_1}{dt} = AW - T_1(B + D) + BT_2 \qquad (16.19)$$

$$\frac{dT_2}{dt} = CT_1 - CT_2 \qquad (16.20)$$

from which the required state variable model may be written as:

$$\begin{bmatrix} \dot{T}_1 \\ \dot{T}_2 \end{bmatrix} = \begin{bmatrix} -(B+D) & B \\ C & -C \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \end{bmatrix} + \begin{bmatrix} A \\ 0 \end{bmatrix} W \qquad (16.21)$$

and

$$T_2 = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \end{bmatrix} \qquad (16.22)$$

In the above two examples, we have obtained the input-output type state variable models starting from fundamental dynamic balance of principle entity. The obtained models have the following general form of matrix expression:

$$[\dot{\mathbf{x}}] = [\mathbf{A}][\mathbf{x}] + [\mathbf{B}]\mathbf{u} \tag{16.9}$$

and the output, $\mathbf{y} = [\mathbf{C}][\mathbf{x}]$ (16.10)

For a system if there is no input or perturbation, the first Equation, 16.9, can be written without the input part **[B]u** as:

$$[\dot{\mathbf{x}}] = [\mathbf{A}][\mathbf{x}] \tag{16.23}$$

The above equation represents the dynamics of an unperturbed system, but the output is still expressed by Equation 16.10. Before showing how these equations can be solved to obtain the time domain response, certain properties of the characteristic matrix, **(A),** needs to be discussed. This will establish an important conclusion regarding the states of a given process, and also will help in solving Equations 16.9 and 16.10, or, 16.23 and 16.10.

16.1.3 Characteristic Equation and Eigenvalues of [A]

The characteristic equation of **[A]** is found by equating the polynomial generated by the determinant $|l\mathbf{I} - \mathbf{A}|$ to zero, thus,

$$\det|l\mathbf{I} - \mathbf{A}| = 0 \tag{16.24}$$

The eigenvalues, l,s, for a given **[A]** are found by determining the roots of Equation 16.24.

EXAMPLE 16.3 Let us find the eigenvalues by considering the system described by the following third order ordinary differential equation:

$$\frac{d^3y}{dt^3} + 9\frac{d^2y}{dt^2} + 23\frac{dy}{dt} + 15y = 2u \tag{16.25}$$

To convert Equation 16.25 into state variable form, define the following state variables as:

$$x_1 = y; \quad x_2 = \frac{dy}{dt}; \quad x_3 = \frac{d^2y}{dt^2}$$

By introducing the above variable definitions, Equation 16.25 becomes:

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = x_3$$
$$\dot{x}_3 = -15x_1 - 23x_2 - 9x_3 + 2u$$

The above set of first order ordinary differential equations can be put into standard state variable form as:

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix}=\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -15 & -23 & -9\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}+\begin{bmatrix}0\\ 0\\ 2\end{bmatrix}u \text{ and } y=\begin{bmatrix}1 & 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

The above two equations can be put into the standard form of

$$[\dot{\mathbf{x}}]=[\mathbf{A}][\mathbf{x}]+[\mathbf{B}]\mathbf{u} \text{ and } \mathbf{y}=[\mathbf{C}][\mathbf{x}]$$

where $[\mathbf{A}]=\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -15 & -23 & -9\end{bmatrix}$, $[\mathbf{B}]=\begin{bmatrix}0\\ 0\\ 2\end{bmatrix}$ and $[\mathbf{C}]=\begin{bmatrix}1 & 0 & 0\end{bmatrix}$

Starting with the matrix **[A]** for the given system, the characteristic equation and its roots may be found as:

$$\det|l\mathbf{I}-\mathbf{A}|=\begin{bmatrix}\lambda & -1 & 0\\ 0 & \lambda & -1\\ 15 & 23 & \lambda+9\end{bmatrix}=\lambda^3+9\lambda^2+23\lambda+15=0; \text{ or, } (l-1)(l-3)(l-5)=0$$

Then the required eigenvalues are: $l_1=-1$, $l_2=-3$, and $l_3=-5$, and they are *distinct*.

To describe a dynamic system, the choice of state variables is not unique. In transfer function analysis, we have found that depending on the convenience of a measuring device, a liquid level dynamic system can be described either in terms of tank level or exit flow. This notion is true for a multiple state system also. To prove this, we shall use what is called as the **Linear Transformation** of matrix **[A]**.

16.1.4 Invariance of Eigenvalues by Linear Transformation

It frequently happens that the state variables used in the original formulation of the dynamics of the system are not as convenient as another set of state variable that may be found by a simple matrix operation without loosing any of dynamic properties of the system. If we consider the original set of matrix differential equations as:

$$[\dot{\mathbf{x}}]=[\mathbf{A}][\mathbf{x}] \qquad (16.23)$$

and introduce the linear transformation,

$$\mathbf{[x]=[P][y]} \qquad (16.26)$$

into the above equation, then Equation 16.26 becomes:

$$[\mathbf{P}][\dot{\mathbf{y}}]=[\mathbf{A}][\mathbf{P}][\mathbf{y}] \text{ or, } [\dot{\mathbf{y}}]=[\mathbf{P}]^{-1}[\mathbf{A}][\mathbf{P}][\mathbf{y}] \qquad (16.27)$$

To establish the invariance of the eigenvalues and the dynamic character of the process as well, under linear transformation, we have to show that the characteristic polynomials

$|l\mathbf{I} - \mathbf{A}|$ and $|l\mathbf{I} - \mathbf{P}^{-1}\mathbf{AP}|$ are identical. Since the determinant of a product is the product of the determinants, we obtain,

$$|l\mathbf{I} - \mathbf{P}^{-1}\mathbf{AP}| = |l\mathbf{P}^{-1}\mathbf{P} - \mathbf{P}^{-1}\mathbf{AP}| = |\mathbf{P}^{-1}(l\mathbf{I} - \mathbf{A})\mathbf{P}| = |\mathbf{P}^{-1}||l\mathbf{I} - \mathbf{A}||\mathbf{P}| = |\mathbf{P}^{-1}||\mathbf{P}||l\mathbf{I} - \mathbf{A}| = |l\mathbf{I} - \mathbf{A}|$$

The above identity holds because the product of the determinants $|\mathbf{P}^{-1}||\mathbf{P}|$ is determinant of the product $|\mathbf{P}^{-1}\mathbf{P}| = 1$. A very useful corollary of linear transform is that by choosing a proper transformation matrix, **[P]**, the matrix $\mathbf{[P^{-1}AP]}$ could be obtained in a diagonal form. As we shall see that by obtaining the matrix **[A]** in diagonal form, considerably reduces certain procedural effort in state variable analysis.

16.1.5 Diagonalization of a Square Matrix with Distinct Eigenvalues

Such an *n*-th order matrix with distinct eigenvalues could be given as:

$$\mathbf{[A]} = \begin{bmatrix} 0 & 1 & . & . & 0 & 0 \\ 0 & 0 & 1 & . & 0 & 0 \\ . & . & . & . & . & . \\ . & . & . & . & . & . \\ 0 & 0 & . & . & 1 & 0 \\ -\frac{a_0}{a_N} & -\frac{a_1}{a_N} & . & . & -\frac{a_{N-1}}{a_N} & 0 \end{bmatrix}$$

The transformation **[x]** = **[P][z]**, where,

$$\mathbf{[P]} = \begin{bmatrix} 1 & 1 & . & . & 1 & 1 \\ \lambda_1 & \lambda_2 & \lambda_3 & . & \lambda_{N-1} & \lambda_N \\ \lambda_1^2 & \lambda_2^2 & . & . & \lambda_{N-1}^2 & \lambda_N^2 \\ . & . & . & . & . & . \\ \lambda_1^{n-1} & \lambda_2^{n-1} & . & . & \lambda_{N-1}^{N-1} & \lambda_N^{N-1} \\ \lambda_1^N & \lambda_2^N & . & . & \lambda_{N-1}^N & \lambda_N^N \end{bmatrix}$$

where, l_1, l_2, ..., l_N are distinct eigenvalues of **[A]** will transform into a diagonal matrix

or

$$[\mathbf{P}^{-1}\mathbf{AP}] = [\Lambda] = \begin{bmatrix} \lambda_1 & 0 & . & . & 0 & 0 \\ 0 & \lambda_2 & 0 & . & 0 & 0 \\ 0 & 0 & \lambda_3 & . & 0 & 0 \\ . & . & . & . & . & . \\ 0 & 0 & . & . & \lambda_{N-1} & 0 \\ 0 & 0 & . & . & 0 & \lambda_N \end{bmatrix} \tag{16.28}$$

EXAMPLE 16.4 Let us consider the system equation given in Example 16.3 and reformulate the state equation by linear transforming by the matrix **[P],** where **[P]** is found for diagonalization by the procedure as indicated above:

$$[\mathbf{P}] = \begin{bmatrix} 1 & 1 & 1 \\ \lambda_1 & \lambda_1 & \lambda_3 \\ \lambda_1^2 & \lambda_2^2 & \lambda_3^2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -3 & -5 \\ 1 & 9 & 25 \end{bmatrix} \tag{16.29}$$

We define a set of new state variables z_1, z_2, z_3 by the transformation:

$$\mathbf{[x] = [P][z]} \tag{16.30}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -3 & -5 \\ 1 & 9 & 25 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$$

Substituting Equation 16.30 into 16.9,

$$[\mathbf{P}][\dot{\mathbf{z}}] = [\mathbf{A}][\mathbf{P}][\mathbf{z}] + [\mathbf{B}]u$$

Premultiplying both sides of the last equation by **[P]**, we obtain,

$$[\dot{\mathbf{z}}] = [\mathbf{P}]^{-1}[\mathbf{A}][\mathbf{P}][\mathbf{z}] + [\mathbf{P}]^{-1}[\mathbf{B}]u \tag{16.31}$$

Putting the element values in each of the matrices in Equation 16.31,

$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \\ \dot{z}_3 \end{bmatrix} = \begin{bmatrix} 1.875 & 1 & 0.125 \\ -1.25 & -1.5 & -0.25 \\ 0.375 & 0.5 & 0.125 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -15 & -23 & -9 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ -1 & -3 & -5 \\ 1 & 9 & 25 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} +$$

$$\begin{bmatrix} 1.875 & 1 & 0.125 \\ -1.25 & -1.5 & -0.25 \\ 0.375 & 0.5 & 0.125 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} u$$

or

$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \\ \dot{z}_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} + \begin{bmatrix} 0.25 \\ -0.5 \\ 0.25 \end{bmatrix} u \qquad (16.32)$$

Equation 16.32 is also a state equation that describes the same system as defined by the Equation 16.25. Similarly, the output equation is transformed to:

$$y = \mathbf{([C])([P])[(z])} \qquad (16.33)$$

or

$$y = [1 \quad 0 \quad 0] \begin{bmatrix} 1 & 1 & 1 \\ -1 & -3 & -5 \\ 1 & 9 & 9 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = [1 \quad 1 \quad 1] \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$$

If the characteristic matrix **[A]** has repeated eigenvalues, there are some ways of finding the transformation matrix **[P]**. This discussion is included in *Advanced Process Dynamics and Control*, in the section regarding analysis of complete controllability.

There are certain important results that could be derived from Equation 16.23, the state equation of an unperturbed system, but before that discussion, we shall show how a time domain solution of the equation with perturbation or input disturbance is possible, provided the initial value of the states, **[x(0)]** is available.

16.1.6 Response Dynamics of State Variable Models

Let us first consider the equivalent scalar equation and its solution corresponding to an input change while all the initial conditions are known.

$$\dot{x} = ax + bu \quad \text{or} \quad \dot{x} - ax = bu \qquad (16.34)$$

Multiplying by, e^{-at}, $e^{-at}[\dot{x}(t) - ax(t)] = e^{-at}bu(t)$

or

$$\frac{d}{dt}[e^{-at}x(t)] = e^{-at}bu(t) \qquad (16.35)$$

Integrating Equation 16.35 between 0 and t obtains,

$$e^{-at}x(t) = x(0) + \int_0^t e^{-at}bu(t)dt$$

or

$$x(t) = e^{at}x(0) + e^{at}\int_0^t e^{-at}bu(t)dt \qquad (16.36)$$

Total Solution = Homogeneous Solution + Particular Solution

Remarks made under terms of Equation 16.36 might remind you the course on differential equation solution taught in your school days. The starting equation without the $bu(t)$ term, that is, the input disturbance term, is called a homogeneous relation that depends on the process character and the initial conditions. Thus, the solution only due to this term is called the homogeneous solution and the corresponding solution that appears due to a given or particular input function is called the particular solution. The total or complete solution is obtained by summing up these two parts.

The same procedure may be followed for the solution of a state equation:

or

$$[\dot{\mathbf{x}}] = [\mathbf{A}][\mathbf{x}] + [\mathbf{B}]\mathbf{u} \quad \text{or} \quad [\dot{\mathbf{x}}] - [\mathbf{A}][\mathbf{x}] = [\mathbf{B}]\mathbf{u} \tag{16.37}$$

multiplying by $e^{-[\mathbf{A}]t}$, and proceeding as we have done for the scalar equation, the solution is obtained as:

$$[x(t)] = e^{[\mathbf{A}]t}[x_0] + \int_0^t e^{-[\mathbf{A}]\tau}[\mathbf{B}][u(\tau)]d\tau \tag{16.38}$$

EXAMPLE 16.5 Calculate the time domain response for the system,

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}0 & 1\\ -2 & -3\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}0\\ 1\end{bmatrix}[u] \text{ with initial conditions, } [x(0)] = \begin{bmatrix}0\\ 0\end{bmatrix} \tag{16.39}$$

The input, $u(t)$ is given as,

$$\begin{bmatrix}u(t) = 0, & \text{for } t < 0\\ u(t) = 1, & \text{for } t \geq 0\end{bmatrix} \tag{16.40}$$

Solution:

$$[\mathbf{A}] = \begin{bmatrix}0 & 1\\ -2 & -3\end{bmatrix}, \quad [\mathbf{A}]^2 = \begin{bmatrix}-2 & -3\\ 6 & 7\end{bmatrix}$$

The matrix exponentials may be calculated as:

$$e^{[\mathbf{A}]t} = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} + \begin{bmatrix}0 & 1\\ -2 & -3\end{bmatrix}t + \frac{1}{2}\begin{bmatrix}-2 & -3\\ 6 & 7\end{bmatrix}t^2 = \begin{bmatrix}1 - t^2 + \cdots + t - (3/2)t^2 + \cdots\\ -2t + 3t^2 + \cdots + 1 - 3t + (7/2)t^2 + \cdots\end{bmatrix}$$

$$e^{-[\mathbf{A}]t} = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} - \begin{bmatrix}0 & 1\\ -2 & -3\end{bmatrix}t + \frac{1}{2}\begin{bmatrix}-2 & -3\\ 6 & 7\end{bmatrix}t^2 = \begin{bmatrix}1 - t^2 + \cdots + -t - (3/2)t^2 + \cdots\\ -2t + 3t^2 + \cdots + 1 - 3t + (7/2)t^2 + \cdots\end{bmatrix}$$

$$\int_0^t e^{-[\mathbf{A}]\tau}[\mathbf{B}]u(\tau)d\tau = \int_0^t \begin{bmatrix} 1-\tau^2+\cdots+-\tau-(3/2)\tau^2+\cdots \\ 2\tau+3\tau^2+\cdots+1+3\tau+(7/2)\tau^2+\cdots \end{bmatrix}\begin{bmatrix}0\\1\end{bmatrix}1d\tau$$

$$= \int_0^t \begin{bmatrix} -\tau-(3/2)\tau^2+\cdots \\ 1+3\tau+(7/2)\tau^2+\cdots \end{bmatrix} d\tau = \begin{bmatrix} -(1/2)t^2-(1/2)t^3+\cdots \\ t+(3/2)t^2+(7/6)t^3+\cdots \end{bmatrix} \qquad (16.41)$$

Now the complete solution may be obtained by Equation 16.38, remembering that $[x_0] = \begin{bmatrix}0\\0\end{bmatrix}$

Thus,
$$[x(t)] = e^{[\mathbf{A}]t}[x_0] + \int_0^t e^{-[\mathbf{A}]t}[\mathbf{B}][u(t)]dt$$

$$\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} 1+t^2+\cdots+t-(3/2)t^2+\cdots \\ -2t+3t^2+\cdots+1-3t-(7/2)t^2+\cdots \end{bmatrix}\begin{bmatrix}0\\0\end{bmatrix} + \begin{bmatrix} -(1/2)t^2-(1/2)t^3+\cdots \\ t+(3/2)t^2-(7/6)t^3+\cdots \end{bmatrix}$$

$$= \begin{bmatrix} -(1/2)t^2-(1/2)t^3+\cdots \\ t+(3/2)t^2-(7/6)t^3+\cdots \end{bmatrix} \qquad (16.42)$$

EXAMPLE 16.6 Let us find the response in the level of the second tank, h_2, corresponding to a unit flow change in the inflow, Q_I, in the system described in Example 16.1. We further assume that the tanks were originally empty and there was no inflow to the first tank. Following parameter values may be used during solution: $A_1 = 1; R_1 = 1; A_2 = 1; R_2 = 0.5$.

Solution: Putting the parameter values in the derived state variable model, the matrices in Equations 16.17 and 16.18 become:

$$[\dot{\mathbf{x}}] = \begin{bmatrix}\dot{h}_1\\ \dot{h}_2\end{bmatrix};\ [\mathbf{A}] = \begin{bmatrix}-1 & 1\\ 1 & -3\end{bmatrix};\ [\mathbf{x}] = \begin{bmatrix}h_1\\ h_2\end{bmatrix};\ [\mathbf{B}] = \begin{bmatrix}1\\0\end{bmatrix};\ [\mathbf{x(0)}] = \begin{bmatrix}0\\0\end{bmatrix}$$

and the input $u(t) = Q_I = 0$ for $t < 0$; $u(t) = Q_I = 1$ for $t \geq 0$

From the above results, we may derive the following:

$$[\mathbf{A}] = \begin{bmatrix}-1 & 1\\ 1 & -3\end{bmatrix};\ [\mathbf{A}]^2 = \begin{bmatrix}2 & -4\\ -4 & 10\end{bmatrix};\ [\mathbf{A}]^3 = \begin{bmatrix}-6 & 14\\ 14 & -34\end{bmatrix}$$

$$e^{[\mathbf{A}]t} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} -1 & 1 \\ 1 & -3 \end{bmatrix} t + 1/2 \begin{bmatrix} 2 & -4 \\ -4 & 10 \end{bmatrix} t^2 + 1/6 \begin{bmatrix} -6 & 14 \\ 14 & -34 \end{bmatrix} t^3$$

$$= \begin{bmatrix} 1 - t + t^2 - t^3 \ldots & t - 2t^2 + (14/6)t^3 \ldots \\ t - 2t^2 + (14/6)t^3 \ldots & 1 - 3t + 5t^2 - (34/6)t^3 \ldots \end{bmatrix}$$

$$e^{-[\mathbf{A}]t} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} -1 & 1 \\ 1 & -3 \end{bmatrix} t + 1/2 \begin{bmatrix} 2 & -4 \\ -4 & 10 \end{bmatrix} t^2 - 1/6 \begin{bmatrix} -6 & 14 \\ 14 & -34 \end{bmatrix} t^3$$

$$= \begin{bmatrix} 1 - t + t^2 - t^3 \ldots & t - 2t^2 - (14/6)t^3 \ldots \\ -t - 2t^2 - (14/6)t^3 \ldots & 1 + 3t + 5t^2 + (34/6)t^3 \ldots \end{bmatrix}$$

Now the response equation may be derived:

$$\begin{bmatrix} h_1(t) \\ h_2(t) \end{bmatrix} = e^{[\mathbf{A}]t} \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \int_0^t e^{-[\mathbf{A}]\tau} [\mathbf{B}] u(\tau) d\tau$$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \int_0^t \begin{bmatrix} 1 - \tau + \tau^2 + \tau^3 \ldots & \tau - 2\tau^2 - (14/6)\tau^3 \ldots \\ -\tau - \tau^2 - (14/6)\tau^3 \ldots & 1 + 3\tau + 5\tau^2 + (34/6)\tau^3 \ldots \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} d\tau$$

or, $$\begin{bmatrix} h_1(t) \\ h_2(t) \end{bmatrix} = \begin{bmatrix} t - t^2/2 + t^3/3 + t^4/4 \ldots \\ -t^2/2 - t^3/3 - (14/6)t^4 \ldots \end{bmatrix}$$

The required response of the second tank level is:

$$h_2(t) = -\frac{t^2}{2} - \frac{t^3}{3} - \left(\frac{14}{24}\right) t^4$$

The evaluation of $e^{[\mathbf{A}]t}$ may become considerably tedious if **[A]** has order more than three. This effort could be simplified by linearly transforming **[A]** into a diagonal form as shown above. Another way to avoid the lengthy arithmetic is by the way of applying the *Sylvester's theorem*.

16.1.7 Sylvester's Theorem

If the eigenvalues l_I (I = 1, 2, … N) of an N × N matrix **[A]** are distinct, and if □(**A**) is any polynomial in **[A],** then,

$$\square(\mathbf{A}) = \sum_{R=1}^{N} \left[\Phi(\lambda_R) \prod_{\substack{R=1 \\ K \neq R}}^{N} \left(\frac{\lambda_K \mathbf{I} - \mathbf{A}}{\lambda_R - \lambda_K} \right) \right] \qquad (16.43)$$

EXAMPLE 16.7 The characteristic matrix [A] of a system is: [A] = $\begin{bmatrix} -1 & 0 \\ 1 & -2 \end{bmatrix}$, find the value of $e^{[A]t}$.

Solution: The eigenvalues of system may be found as:

$$\det |l\mathbf{I} - \mathbf{A}| = 0 \text{ or } (l+1)(l+2) = 0 \text{ or } l_1 = -1,\ l_2 = -2$$

$$e^{[A]t} = e^{-t}\frac{[-2\mathbf{I} - \mathbf{A}]}{[-1-(-2)]} + e^{-2t}\frac{[-1\mathbf{I} - \mathbf{A}]}{[-2-(-1)]} = e^{-t}\begin{bmatrix} -1 & 0 \\ -1 & 2 \end{bmatrix} + e^{-2t}(-1)\begin{bmatrix} 0 & 0 \\ -1 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} -e^{-t} & 0 \\ -e^{-t} + e^{-2t} & 2e^{-t} + 2e^{-2t} \end{bmatrix}$$

16.1.8 Conversion of State Variable Model to Transfer Function

State variable models and their treatment is not a separate analysis domain by any means, they may be converted to equivalent transfer functions and subsequent processing may follow the procedures established in the earlier chapters. This selection entirely depends on the problem statement and the designer's choice. Let us rewrite the input-output model in state variable form introduced in Chapter 3 as:

$$\left.\begin{aligned} \dot{\mathbf{x}} &= [\mathbf{A}]\mathbf{x} + [\mathbf{B}]\mathbf{u} \\ \mathbf{y} &= [\mathbf{C}]\mathbf{x} + [\mathbf{D}]\mathbf{u} \end{aligned}\right\} \qquad (16.44)$$

Taking LT of the first state equation,

$$s\mathbf{X}(s) - \mathbf{x}(0) = \mathbf{A}\mathbf{X}(s) + \mathbf{B}\mathbf{u}(s)$$

or

$$s\mathbf{X}(s) - \mathbf{A}\mathbf{X}(s) = \mathbf{x}(0) + \mathbf{B}\mathbf{u}(s)$$

Solving for $\mathbf{X}(s)$,

$$(s\mathbf{I} - \mathbf{A})\mathbf{X}(s) = \mathbf{x}(0) + \mathbf{B}\mathbf{u}(s) \qquad (16.45)$$

To obtain an expression for $\mathbf{X}(s)$, we multiply Equation 16.45 by $(s\mathbf{I} - \mathbf{A})^{-1}$;

$$(s\mathbf{I} - \mathbf{A})^{-1}(s\mathbf{I} - \mathbf{A})\mathbf{X}(s) = (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{x}(0) + (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}\mathbf{u}(s)$$

This equation may be simplified to:

$$\mathbf{X}(s) = (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{x}(0) + (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}\mathbf{u}(s) \qquad (16.46)$$

To obtain the time solution, $\mathbf{x}(t)$, the above equation needs to be inverted,

$$\mathbf{x}(t) = \mathrm{L}^{-1}[(s\mathbf{I} - \mathbf{A})^{-1}\,\mathbf{x}(0)] + \mathrm{L}^{-1}[(s\mathbf{I} - \mathbf{A})^{-1}\,\mathbf{B}\mathbf{u}(s)] \qquad (16.47)$$

But we already have the time domain solution:

$$\mathbf{x}(t) = e^{[\mathbf{A}]t}[x_0] + \int_0^t e^{-[\mathbf{A}]\tau}[\mathbf{B}][u(\tau)]d\tau \qquad (16.48)$$

By comparing Equation 16.47 with 16.48, we can see that,

$$\mathbf{e}^{\mathbf{A}t} = \mathrm{L}^{-1}[(s\mathbf{I} - \mathbf{A})^{-1}] \qquad (16.49)$$

This exponential time function $\mathbf{e}^{\mathbf{A}t}$, of characteristic matrix **[A]**, is also a matrix having identical dimension that of **[A]**, and is called the **state transition matrix**. To qualify this statement, we expand $\mathbf{e}^{\mathbf{A}t}$ as an exponential series to as many terms as we like, evaluate all the terms from the lowest through to the highest order, then add up to obtain the resulting matrix. The state transition matrix is also denoted as □(t). □(t) and its inverse $□^{-1}(t) = \mathbf{e}^{-\mathbf{A}t}$ has significant importance in state variable analysis. Thus,

$$□(t) = \mathbf{e}^{\mathbf{A}t} = \mathrm{L}^{-1}[(s\mathbf{I} - \mathbf{A})^{-1}] \qquad (16.50)$$

and, $□^{-1}(t) = \mathbf{e}^{-\mathbf{A}t}$ (16.51)

By using the developed expression for the state transition matrix, the solution of the state variable model can be obtained to have the following form:

$$\int_0^t e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau)d\tau = \mathrm{L}^{-1}[(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}\mathbf{u}(s)] \qquad (16.52)$$

Time domain solution of matrix differential equation is also possible by detouring the procedure through evaluation of the state transition matrix, □(*t*).

EXAMPLE 16.8 Calculate the time domain response by evaluating the state transition matrix for the system,

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}0 & 1\\ -2 & -3\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}0\\ 1\end{bmatrix}[u] \text{ with initial conditions, } [x(0)] = \begin{bmatrix}0\\ 0\end{bmatrix} \qquad (16.53)$$

The input, $u(t)$ is given as:

$$\begin{bmatrix}u(t) = 0, \text{ for } t < 0\\ u(t) = 1, \text{ for } t \geq 0\end{bmatrix} (16.54)$$

Solution: For this system:

$$\mathbf{[A]} = \begin{bmatrix}0 & 1\\ -2 & -3\end{bmatrix} \text{ and } [\mathbf{B}] = \begin{bmatrix}0\\ 1\end{bmatrix}$$

To evaluate the state transition matrix for this system, we first determine,

$$[s\mathbf{I}-\mathbf{A}]=\begin{bmatrix}s & 0\\ 0 & s\end{bmatrix}-\begin{bmatrix}0 & 1\\ -2 & -3\end{bmatrix}=\begin{bmatrix}s & -1\\ 2 & s+3\end{bmatrix}$$

In the next step, let us determine $[s\mathbf{I}-\mathbf{A}]^{-1}$,

$$[s\mathbf{I}-\mathbf{A}]^{-1}=\frac{1}{(s+1)(s+2)}\begin{bmatrix}s+3 & 1\\ -2 & s\end{bmatrix}=\begin{bmatrix}\dfrac{s+3}{(s+1)(s+2)} & \dfrac{1}{(s+1)(s+2)}\\ \dfrac{-2}{(s+1)(s+2)} & \dfrac{s}{(s+1)(s+2)}\end{bmatrix}$$

Hence, $$\Phi(t)=\mathrm{L}^{-1}[(s\mathbf{I}-\mathbf{A})^{-1}]=\begin{bmatrix}2e^{t}-e^{2t} & e^{t}-e^{2t}\\ -2e^{t}+2e^{2t} & -e^{t}+2e^{2t}\end{bmatrix}$$

The response to the unit step input could be obtained as:

$$\mathbf{x}(t)=e^{At}\mathbf{x}(0)+\int_0^t\begin{bmatrix}2e^{-(t-\tau)}-e^{-2(t-\tau)} & e^{-(t-\tau)}-e^{-2(t-\tau)}\\ -2e^{-(t-\tau)}+2e^{-2(t-\tau)} & -e^{-(t-\tau)}+2e^{-2(t-\tau)}\end{bmatrix}\begin{bmatrix}0\\ 1\end{bmatrix}[1]\,d\tau$$

or,

$$\begin{bmatrix}x_1(t)\\ x_2(t)\end{bmatrix}=\begin{bmatrix}2e^{-t}-e^{-2t} & e^{-t}-e^{-2t}\\ -2e^{-t}+2e^{-2t} & -e^{-t}+2e^{-2t}\end{bmatrix}\begin{bmatrix}x_1(0)\\ x_2(0)\end{bmatrix}+\begin{bmatrix}1/2-e^{-t}+1/2e^{-2t}\\ e^{-t}-e^{-2t}\end{bmatrix}$$

As the initial conditions are zero, or $\mathbf{x}(0)=0$, the solution simplifies to:

$$\begin{bmatrix}x_1(t)\\ x_2(t)\end{bmatrix}=\begin{bmatrix}1/2-e^{-t}+1/2e^{-2t}\\ e^{-t}-e^{-2t}\end{bmatrix}$$

16.1.9 Transfer Function Matrix

A term transfer function matrix could be defined from its state variable model which has a similar significance as we had with transfer function in the complex domain analysis.

For an unperturbed system setting, **[D]** = 0 in output part of Equation 17.44, and $\mathbf{X}(s)$ as derived by Equation 16.46, the transformed output $\mathbf{y}(s)$ becomes:

$$\mathbf{y}(s)=\mathbf{[C]}\Phi(s)\mathbf{[B]u}(s) \tag{16.55}$$

In the case where **u** and **y** are scalar quantities, $\mathbf{[C]}\Phi(s)\mathbf{[B]}$ must be scalar. A little generalization with the treatment on Laplace domain dynamics that we had in the earlier chapters should reveal that $\mathbf{[C]}\Phi(s)\mathbf{[B]}$ must be the transfer function of the system. Thus, we can rewrite Equation 16.55 as:

$$y(s)=G_P(s)u(s) \tag{16.56}$$

where, $G_P(s) = \mathbf{[C]}\Phi(s)\mathbf{[B]}$ (16.57)

Now $\Phi(s)$ may be also expressed as, $\Phi(s) = (s\mathbf{I} - \mathbf{A})^{-1} = \dfrac{\text{adj}(s\mathbf{I} - \mathbf{A})}{\det(s\mathbf{I} - \mathbf{A})}$ (16.58)

Putting this result in Equation 16.57,

$$G_P(s) = \mathbf{[C]}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{[B]} = \frac{\mathbf{[C]}[\text{adj}(s\mathbf{I} - \mathbf{A})]\mathbf{[B]}}{\det(s\mathbf{I} - \mathbf{A})} \qquad (16.59)$$

EXAMPLE 16.9 A battery of three perfectly stirred reactors is used to carry out an isothermal irreversible first order reaction. The only input variable is the inlet concentration C_I (mole/l). The constant reactant solution flow rate is Q (l/m), reactor volumes are, V_1, V_2, and V_3 (litres), and specific reaction rate is k (min^{-1}). The reaction occurs at constant density. The reaction and its kinetics may be given by:

Reactant → Product; and, rate of reaction, $r = -\dfrac{dC}{dt} = -kC$, where C is the reactant concentration present in a reactor.

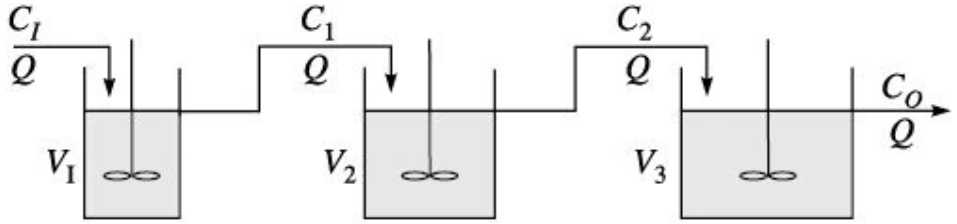

Figure 16.3 The reactor and stream arrangement in the 3-CSTR battery.

Find the expression for TF that relates outlet concentration C_O to the inlet concentration C_I.

Solution: The reactant mass balance relations for the three reactors are:

$$\frac{dC_1}{dt} = \frac{Q}{V_1}C_I - \frac{Q}{V_1}C_1 - kC_1 \qquad (16.60a)$$

$$\frac{dC_2}{dt} = \frac{Q}{V_2}C_1 - \frac{Q}{V_2}C_2 - kC_2 \qquad (16.60b)$$

$$\frac{dC_O}{dt} = \frac{Q}{V_3}C_2 - \frac{Q}{V_3}C_O - kC_O \qquad (16.60c)$$

and the corresponding state equations are:

$$\begin{bmatrix}\dot{C}_1\\ \dot{C}_2\\ \dot{C}_O\end{bmatrix}=\begin{bmatrix}-(Q/V_1+k) & 0 & 0\\ Q/V_2 & -(Q/V_2+k) & 0\\ 0 & Q/V_3 & -(Q/V_3+k)\end{bmatrix}\begin{bmatrix}C_1\\ C_2\\ C_O\end{bmatrix}+\begin{bmatrix}Q/V_1\\ 0\\ 0\end{bmatrix}C_I \qquad (16.61)$$

$$[\dot{\mathbf{x}}]=[\mathbf{A}][\mathbf{x}]+[\mathbf{B}]\mathbf{u}$$

and the output equation is:

$$C_O=\begin{bmatrix}0 & 0 & 1\end{bmatrix}\begin{bmatrix}C_1\\ C_2\\ C_O\end{bmatrix} \qquad (16.62)$$

$$\mathbf{y}=[\mathbf{C}][\mathbf{x}]$$

$$[s\mathbf{I}-\mathbf{A}]=\begin{bmatrix}s+Q/V_1+k & 0 & 0\\ -Q/V_2 & s+Q/V_2+k & 0\\ 0 & -Q/V_3 & s+Q/V_3+k\end{bmatrix}$$

$$[s\mathbf{I}-\mathbf{A}]^{-1}=1/\begin{bmatrix}(s+Q/V_1+k)\\ (s+Q/V_2+k)\\ (s+Q/V_3+k)\end{bmatrix}\begin{bmatrix}s+Q/V_1+k & 0 & 0\\ (Q/V_2)(s+Q/V_3+k) & s+Q/V_2+k & 0\\ 0 & -Q/V_3 & s+Q/V_3+k\end{bmatrix}$$

$$[\Phi(s)]=[s\mathbf{I}-\mathbf{A}]^{-1}=1/[(s+Q/V_1+k)(s+Q/V_2+k)(s+Q/V_3+k)]\,x$$

$$\begin{bmatrix}(s+Q/V_2+k)(s+Q/V_3+k) & 0 & 0\\ (Q/V_2)(s+Q/V_3+k) & (s+Q/V_1+k)(s+Q/V_3+k) & 0\\ (Q/V_2)(Q/V_3) & (s+Q/V_1+k)(s+Q/V_3+k) & (s+Q/V_1+k)(s+Q/V_2+k)\end{bmatrix}$$

Hence, $G_P(s)=[\mathbf{C}][\Phi(s)][\mathbf{B}]=\dfrac{(Q/V_1)(Q/V_2)(Q/V_3)}{(s+Q/V_1+k)(s+Q/V_2+k)(s+Q/V_3+k)}$

After simplification, $G_P(s)=\dfrac{\left(\dfrac{Q}{Q+kV_1}\right)\left(\dfrac{Q}{Q+kV_2}\right)\left(\dfrac{Q}{Q+kV_3}\right)}{\left[\left(\dfrac{V_1}{Q+kV_1}\right)s+1\right]\left[\left(\dfrac{V_2}{Q+kV_2}\right)s+1\right]\left[\left(\dfrac{V_3}{Q+kV_3}\right)s+1\right]}$

(16.63)

The eigenvalues of this system may be found as:

$$[\lambda \mathbf{I} - \mathbf{A}] = \begin{bmatrix} \lambda + Q/V_1 + k & 0 & 0 \\ -Q/V_2 & \lambda + Q/V_2 + k & 0 \\ 0 & -Q/V_3 & \lambda + Q/V_3 + k \end{bmatrix}$$

The characteristic equation of the system can be obtained by equating the determinant of the [*l***I** – **A]** matrix to zero. Thus, the characteristic equation is obtained as:

$$\left(\lambda + \frac{Q}{V_1} + k\right)\left(\lambda + \frac{Q}{V_2} + k\right)\left(\lambda + \frac{Q}{V_3} + k\right) = 0 \qquad (16.64)$$

As the equation is already factored into its roots, hence the roots or eigenvalues are:

$$\lambda_1 = -\frac{Q}{V_1} - k; \quad \lambda_2 = -\frac{Q}{V_2} - k; \quad \lambda_3 = -\frac{Q}{V_3} - k \qquad (16.65)$$

These roots would always have –ve real values for any possible variation of system parameters, hence the stability of the system is guaranteed.

EXAMPLE 16.10 A battery of two perfectly stirred reactors is used to carry out an isothermal irreversible first order reaction. The reactant solution flow rate Q (l/m), as well as its inlet concentration C_I (mole/l) are variable. Reactor volumes are, V_1, V_2, and V_3 (litres), and specific reaction rate is k (min^{-1}). The reaction occurs at constant density. Find the expression for TFs that relates: (a) Outlet concentration C_O to the inlet concentration C_I, and (b) Outlet concentration C_O to the reactant flow rate Q.

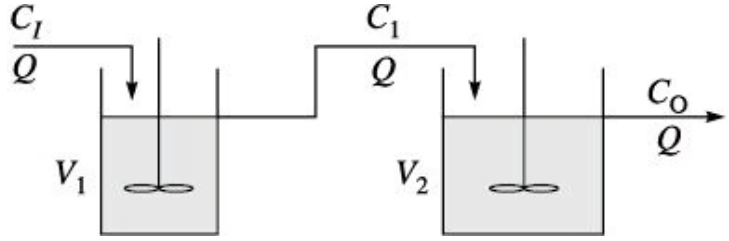

Figure 16.4 The reactor and stream arrangement in the 2-CSTR battery.

The reaction and its kinetics may be given by,

Reactant □ Product; and, rate of reaction, $r = -\frac{dC}{dt} = -kC$, where C is the reactant concentration present in a reactor.

Solution: The linearized form of reactant mass balance relations for the two reactors is:

$$\frac{dC_1}{dt} = -\left(\frac{Q_S}{V_1} + k\right)C_1 + \frac{Q_S}{V_1}C_I + \left(\frac{C_{IS} - C_{1S}}{V_1}\right)Q \qquad (16.66a)$$

$$\frac{dC_O}{dt} = \frac{Q_S}{V_2}C_1 - \left(\frac{Q_S}{V_2} + k\right)C_O + \left(\frac{C_{1S} - C_{OS}}{V_2}\right)Q \qquad (16.66b)$$

where suffix S is used to indicate the steady state value of the respective variable. From Equations 16.66(a) and 16.66(b),

$$\begin{bmatrix}\dot{C}_1 \\ \dot{C}_O\end{bmatrix} = \begin{bmatrix}-(Q_S/V_1 + k) & 0 \\ Q_S/V_2 & -(Q_S/V_2 + k)\end{bmatrix}\begin{bmatrix}C_1 \\ C_O\end{bmatrix} + \begin{bmatrix}Q_S/V_1 & (C_{IS} - C_{1S})/V_1 \\ 0 & (C_{1S} - C_{OS})/V_2\end{bmatrix}\begin{bmatrix}C_I \\ Q\end{bmatrix}$$

$$[\dot{\mathbf{x}}] = [\mathbf{A}][\mathbf{x}] + [\mathbf{B}]\mathbf{u}$$

$$\begin{bmatrix}C_1 \\ C_O\end{bmatrix} = \begin{bmatrix}1 & 0 \\ 0 & 1\end{bmatrix}\begin{bmatrix}C_1 \\ C_O\end{bmatrix}$$

y = [C][x] for an unperturbed system.

Now, $[s\mathbf{I} - \mathbf{A}] = \begin{bmatrix}s + Q_S/V_1 + k & 0 \\ -Q_S/V_2 & s + Q_S/V_2 + k\end{bmatrix}$

$$[\Phi(s)] = [s\mathbf{I} - \mathbf{A}]^{-1} = \frac{1}{(s + Q_S/V_1 + k)(s + Q_S/V_2 + k)}\begin{bmatrix}(s + Q_S/V_2 + k) & 0 \\ Q_S/V_2 & (s + Q_S/V_1 + k)\end{bmatrix}$$

$$= \begin{bmatrix}1/(s + Q_S/V_1 + k) & 0 \\ \dfrac{Q_S/V_2}{(s + Q_S/V_1 + k)(s + Q_S/V_2 + k)} & 1/(s + Q_S/V_2 + k)\end{bmatrix} \qquad (16.67)$$

$$G_P(s) = [\mathbf{C}][\Phi(s)][\mathbf{B}] = \begin{bmatrix}1 & 0 \\ 0 & 1\end{bmatrix}\begin{bmatrix}1/(s + Q_S/V_1 + k) & 0 \\ \dfrac{Q_S/V_2}{(s + Q_S/V_1 + k)(s + Q_S/V_2 + k)} & 1/(s + Q_S/V_2 + k)\end{bmatrix}$$

$$\begin{bmatrix}Q_S/V_1 & (C_{IS} - C_{1S})/V_1 \\ 0 & (C_{1S} - C_{OS})/V_2\end{bmatrix}$$

By abbreviating as: $C_1 = (C_{IS} - C_{1S})/V_1$ and $C_2 = (C_{IS} - C_{2S})/V_2$, the result becomes:

$$G_P(s) = \begin{bmatrix}\dfrac{Q_S/V_1}{s + Q_S/V_1 + k} & \dfrac{C1}{s + Q_S/V_1 + k} \\ \dfrac{(Q_S/V_1)(Q_S/V_2)}{(s + Q_S/V_1 + k)(s + Q_S/V_2 + k)} & \dfrac{(Q_S/V_2)C1 + (s + Q_S/V_1 + k)C_2}{(s + Q_S/V_1 + k)(s + Q_S/V_2 + k)}\end{bmatrix} \qquad (16.68)$$

To express the result in block diagram form it will be convenient to represent $G_P(s)$

as:

$$G_P(s) = \begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix}$$

Thus,

$$\begin{bmatrix} C_1(s) \\ C_O(s) \end{bmatrix} = \begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} C_I(s) \\ Q(s) \end{bmatrix} \qquad (16.69)$$

and the block diagram form of input-output relation may be expressed in Figure 16.5.

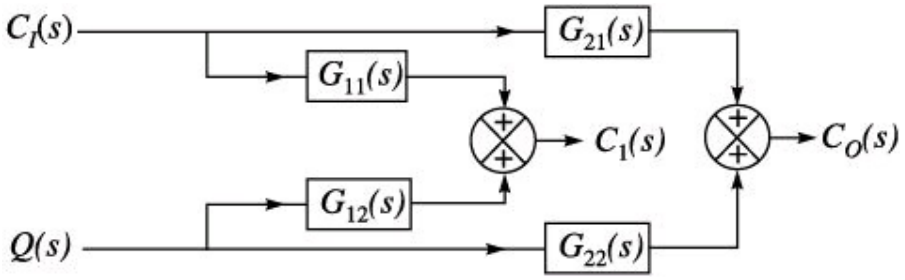

Figure 16.5 Block diagram of the 2-CSTR battery.

EXAMPLE 16.11 Two tanks are connected in a non-interacting fashion. The inflow Q_I enters the first tank. The exit flows and liquid levels of the first and second tanks are Q_1, h_1, and Q_2, h_2 respectively. The physical arrangement and parameters are shown in Figure 16.6. The following parameter values will be used during analysis: $A_1 = 2$; $A_2 = 2$; $R_1 = 0.5$; $R_2 = 1$.

If Q_I changes according to a unit step at $t = 0$, find the dynamic responses of h_1 and h_2.

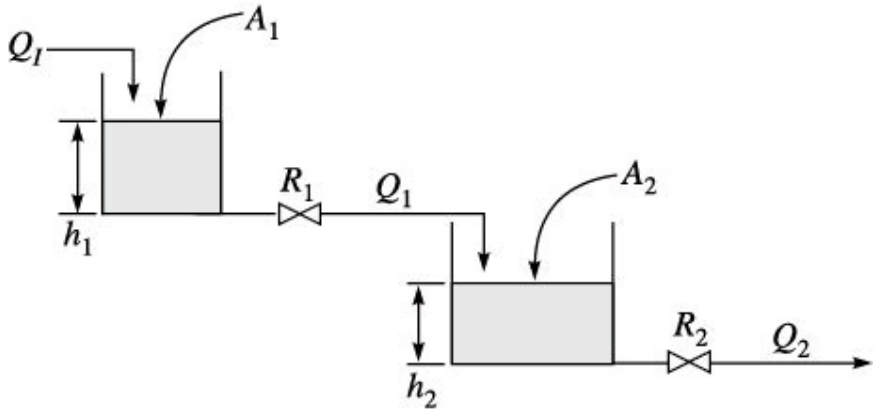

Figure 16.6 Non-Interacting two tank liquid level dynamic system.

Solution: The dynamic mass balance for the two tanks are:

$$Q_I - Q_1 = A_1 \frac{dh_1}{dt} \qquad (16.70a)$$

and

$$Q_1 - Q_2 = A_2 \frac{dh_2}{dt} \qquad (16.70b)$$

from which we can obtain:

$$\frac{dh_1}{dt} = \frac{Q_I}{A_1} - \frac{h_1}{A_1 R_1} \qquad (16.71a)$$

and

$$\frac{dh_2}{dt} = \frac{h_1}{A_2 R_1} - \frac{h_2}{A_2 R_2} \qquad (16.71b)$$

The last two differential equations can be represented by the following state equations:

$$\begin{bmatrix} \dot{h}_1 \\ \dot{h}_2 \end{bmatrix} = \begin{bmatrix} -1/A_1R_1 & 0 \\ 1/A_2R_1 & -1/A_2R_2 \end{bmatrix} \begin{bmatrix} h_1 \\ h_2 \end{bmatrix} + \begin{bmatrix} 1/A_1 \\ 0 \end{bmatrix} Q_I \qquad (16.72)$$

Putting the parameter values, the above state equation becomes:

$$\begin{bmatrix} \dot{h}_1 \\ \dot{h}_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} h_1 \\ h_2 \end{bmatrix} + \begin{bmatrix} 0.5 \\ 0 \end{bmatrix} 1$$

Hence $\mathbf{[sI - A]} = \begin{bmatrix} s+1 & 0 \\ -2 & s+2 \end{bmatrix}$; Det $|\mathbf{A}| = (s+1)(s+2)$

And co-factor of $\mathbf{[sI - A]} = \begin{bmatrix} s+2 & 2 \\ 0 & s+1 \end{bmatrix}$; Adj $\mathbf{[sI - A]} = \begin{bmatrix} s+2 & 0 \\ 2 & s+1 \end{bmatrix}$

Finally, $G_P(s) = \mathbf{[C][sI - A]^{-1}[B]} = \dfrac{\begin{bmatrix} s+2 & 0 \\ 2 & s+1 \end{bmatrix} \begin{bmatrix} 0.5 \\ 0 \end{bmatrix}}{(s+1)(s+2)} = \begin{bmatrix} \dfrac{1}{s+1} & 0 \\ \dfrac{2}{(s+1)(s+2)} & \dfrac{1}{s+2} \end{bmatrix} \begin{bmatrix} 0.5 \\ 0 \end{bmatrix}$

$$= \begin{bmatrix} \dfrac{0.5}{s+1} \\ \dfrac{1}{(s+1)(s+2)} \end{bmatrix}$$

The response in Laplace domain is obtained corresponding to $Q_I(s) = 1/s$, as:

$$H_1(s) = \frac{0.5}{s(s+1)} \text{ and } H_2(s) = \frac{1}{s(s+1)(s+2)}$$

The time responses may be found by inverting the above expressions:

$$h_1(t) = h_{1S} + 0.5(1 - e^{-t}) \text{ and } h_2(t) = h_{2S} + 0.5\,[1 - (2e^{-t} - e^{-2t})]$$

That the same result could be obtained by evaluating the state transition matrix $\Phi(t) = \mathbf{e}^{\mathbf{A}t} = \mathrm{L}^{-1}[(s\mathbf{I} - \mathbf{A})^{-1}]$ and following the procedure discussed in Example 16.5. This exercise is left to the reader.

16.1.10 State Variable Model from the Transfer Function

This conversion is very straight forward. We can introduce the approach by considering an example. But by effecting such a conversion, we may arrive at more than a single solution. Off course, the obtained solution must fulfill the input-output variable need of the given process application. This fact again confirms the classical comment that "*for a system a given set of state variables is not unique*".

EXAMPLE 16.12 Consider the system defined by:

$$\dddot{y} + 10\ddot{y} + 31\dot{y} + 30y = 30u$$

where, y □ system output, u □ system input

Solution: (A) By taking Laplace transform of the system equation, and assuming $\ddot{y}(0) = \dot{y}(0) = y(0) = 0$, we obtain,

$$\frac{y(s)}{u(s)} = \frac{30}{s^3 + 10s^2 + 31s + 30} = \frac{30}{(s+2)(s+3)(s+5)} \qquad (16.73)$$

By partial fraction expansion, the above TF may be expressed as,

$$\frac{y(s)}{u(s)} = \frac{10}{s+2} + \frac{-15}{s+3} + \frac{5}{s+5} \text{ or } y(s) = \frac{10}{s+2}u(s) + \frac{-15}{s+3}u(s) + \frac{5}{s+5}u(s) \qquad (16.74)$$

Let us define: $x_1(s) = \frac{10}{s+2}u(s)$, $x_2(s) = \frac{-15}{s+3}u(s)$, $x_3(s) = \frac{5}{s+5}u(s)$

and $y(s) = x_1(s) + x_2(s) + x_3(s)$

Inverting the above transforms,

$$\dot{x}_1 = -2x_1 + 10u, \quad \dot{x}_2 = -3x_2 - 15u, \quad \dot{x}_3 = -5x_3 + 5u \qquad (16.75)$$

and

$$y = x_1 + x_2 + x_3 \qquad (16.76)$$

Now we may write the matrix equivalent for the equations in time domain as:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -2 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 10 \\ -15 \\ 5 \end{bmatrix} u \text{ and } y = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} u \qquad (16.77)$$

$$[\dot{\mathbf{x}}] = [\mathbf{A}][\mathbf{x}] + [\mathbf{B}]\mathbf{u} \qquad \mathbf{y} = [\mathbf{C}][\mathbf{x}]$$

A noteworthy thing about this conversion is that the characteristic matrix **[A]** has become a diagonal matrix whose elements are the eigenvalues of the system. This feature is absent in the conventional method that has been shown in the following solution.

Solution: (B) Let the state variables be: $x_1 = y; x_2 = \dot{y}; x_3 = \ddot{y}$.

Converting the third order system differential equation into three first order differential equations, we obtain:

$$\dot{x}_1 = x_2; \quad \dot{x}_2 = x_3; \quad \dot{x}_3 = -30x_1 - 31x_2 - 10x_3 + 30u$$

In the vector matrix notation, the above set of equations appear as:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -30 & -31 & -10 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 30 \end{bmatrix} [u], \text{ and the output equation, } y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

(16.78)

$$[\dot{\mathbf{x}}] = [\mathbf{A}][\mathbf{x}] + [\mathbf{B}]\mathbf{u} \qquad \mathbf{y} = [\mathbf{C}][\mathbf{x}]$$

Thus, from the same system equation, two different configurations of state variable models are obtained. The internal signal transfer arrangements in these two models are obviously different and this in turn indicates that significance of intermediate variables between stages (if we consider each integrator as stages) are different in these two descriptions.

Let us now try to analyze a closed loop system through the state variable technique. This may be done by citing a well-known example discussed elsewhere in this text.

EXAMPLE 16.13 Construct the state equations for the control system illustrated in Figure 16.7. Also determine range of K_C for which the system would remain stable by eigenvalue technique.

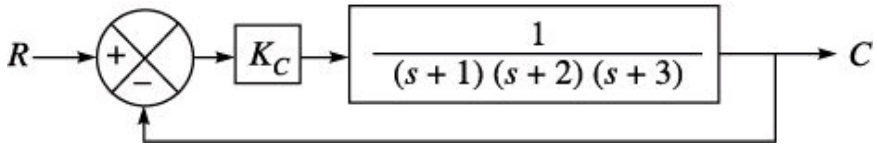

Figure 16.7 Closed loop system of example 16.13.

Solution: For analysis, we split up the closed loop system by defining certain internal variables defined in the Figure 16.8.

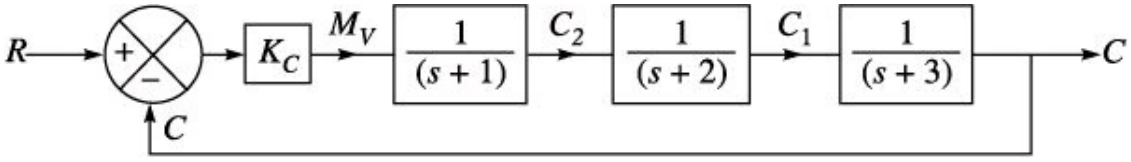

Figure 16.8 Reconfiguration of the loop.

The input/output relations for all blocks of the above figure are first written as algebraic relations and then converted to their equivalent differential form:

$$\frac{C_2}{M_V} = \frac{1}{s+1} \tag{16.79}$$

$$M_V = (R - C)K_C \tag{16.80}$$

$$\frac{C_1}{C_2} = \frac{1}{s+2} \tag{16.81}$$

$$\frac{C}{C_1} = \frac{1}{s+3} \tag{16.82}$$

From (16.79) and (16.80): From (16.81): From (16.82):

$$sC_2 + C_2 = (R - C)K_C \tag{16.83}$$

$$sC_1 + 2C_1 = C_2 \tag{16.84}$$

$$sC + 3C = C_1 \tag{16.85}$$

From (16.83): From (16.84): From (16.85):

$$\frac{dC_2}{dt} = RK_C - C_2 - CK_C \tag{16.86}$$

$$\frac{dC_1}{dt} = C_2 - 2C_1 \tag{16.87}$$

$$\frac{dC}{dt} = C_1 - 3C \tag{16.88}$$

Now we can write the state equations as:

$$\begin{bmatrix} \dot{C}_2 \\ \dot{C}_1 \\ C \end{bmatrix} = \begin{bmatrix} -1 & 0 & -K_C \\ 1 & -2 & 0 \\ 0 & 1 & -3 \end{bmatrix} \begin{bmatrix} C_2 \\ C_1 \\ C \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ K_C \end{bmatrix} \tag{16.89}$$

and

$$C = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} C_2 \\ C_1 \\ C \end{bmatrix} \tag{16.90}$$

Equations 16.89 and 16.90 are the required state variable models.

To obtain the characteristic equation of the system, let us find $[l\mathbf{I} - \mathbf{A}]$.

$$[l\mathbf{I} - \mathbf{A}] = \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} - \begin{bmatrix} -1 & 0 & -K_C \\ 1 & -2 & 0 \\ 0 & 1 & -3 \end{bmatrix} = \begin{bmatrix} \lambda + 1 & 0 & K_C \\ 1 & \lambda + 2 & 0 \\ 0 & 1 & \lambda + 1 \end{bmatrix}$$

Now the characteristic equation may be formed as $\det|l\mathbf{I} - \mathbf{A}| = 0$

$$\det|l\mathbf{I} - \mathbf{A}| = (l+1)(l+2)(l+3) + K_C = l^3 + 6l^2 + 11l + 6 + K_C = 0 \qquad (16.91)$$

By replacing l with the Laplace operator s in Equation 16.91, we obtain our known form of characteristic equation with which we can apply Routh-Hurwitz criteria to determine the range of K_C for stable operation of the closed loop system. In Chapter 10 we have already done it, let us mention the result as: $60 > K_C > 0$.

EXAMPLE 16.14 It is required to find the time response of a first order process under a pure integral controller facing a unit step change of set point. A state variable analysis to the problem has to be formulated. The parameters of the closed loop system are shown in Figure 16.9.

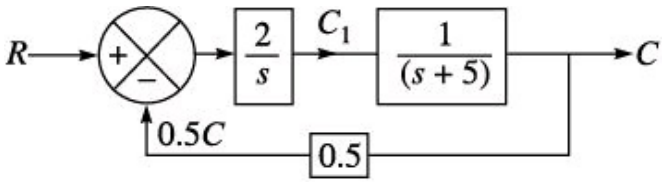

Figure 16.9 A first order process under an integral controller.

Solution: As in the last example, the input/output relations for all blocks of the above figure are first written as algebraic relations and then converted to their equivalent differential form:

$$sC_1 = 2R - C \text{ and } sC = C_1 - 5C \text{ from which } \frac{dC_1}{dt} = 2R - C \text{ and } \frac{dC}{dt} = C_1 - 5C$$

From the last two first order differential equations, the state variable model for the closed loop process may be proposed as:

$$\begin{bmatrix} \dot{C}_1 \\ C \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & -5 \end{bmatrix} \begin{bmatrix} C_1 \\ C \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \end{bmatrix} R \text{ and the output, } \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} C_1 \\ C \end{bmatrix}$$

$$[s\mathbf{I} - \mathbf{A}] = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & -1 \\ 1 & -5 \end{bmatrix} = \begin{bmatrix} s & 1 \\ -1 & s+5 \end{bmatrix}; \det |\, s\mathbf{I} - \mathbf{A} \,| = s(s+5) + 1 = s^2 + 5s + 1$$

$$\text{Cofactor } [s\mathbf{I} - \mathbf{A}] = \begin{bmatrix} s+5 & -1 \\ 1 & s \end{bmatrix}; \text{ Adj}[s\mathbf{I} - \mathbf{A}] = \begin{bmatrix} s+5 & 1 \\ -1 & s \end{bmatrix}$$

$$[s\mathbf{I} - \mathbf{A}]^{-1} = \begin{bmatrix} \dfrac{s+5}{s^2+5s+1} & \dfrac{1}{s^2+5s+1} \\ \dfrac{-1}{s^2+5s+1} & \dfrac{s}{s^2+5s+1} \end{bmatrix}$$

□

and $GP(s) = \frac{C(s)}{R(s)} = \mathbf{[C][sI - A]^{-1}[B]} = [0\ 1]\begin{bmatrix} \frac{s+5}{s^2+5s+1} & \frac{1}{s^2+5s+1} \\ \frac{-1}{s^2+5s+1} & \frac{s}{s^2+5s+1} \end{bmatrix}\begin{bmatrix} 2 \\ 0 \end{bmatrix} = \frac{2}{s^2+5s+1}$

For $$R(s) = \frac{1}{s},\ C(s) = \frac{2}{s(s^2+5s+1)}$$

Thus the response will be the step response of a second order system. The parameter values may be found from the general constants ($KP = 2$, $A = 1$, $B = 5$, $C = 1$) of a second order system as shown in Chapter 6. The calculated values are: $KP = 2$; $w_N = 1$; $z = 2.5$. It is left to the reader to select the step response equation of an over-damped second order system and putting the calculated parameter values to arrive at the final result.

16.1.11 Eigenvalues and Eigenvectors of a State Variable Model

The state variable model for a dynamic system without an input or perturbation may be written as:

$$[\dot{\mathbf{x}}] = [\mathbf{A}][\mathbf{x}] \qquad (16.92)$$

This representation has some useful properties that need to be discussed because an alternative solution procedure for such a model through Laplace transformation stems from this approach.

It has been already shown that a set of differential equations expressing the dynamic model of a system may be presented in matrix notation as:

$$\begin{aligned} \dot{x}_1 &= a_{11}x_1 + a_{12}x_2 + \ldots + a_{1N}x_N \\ \dot{x}_2 &= a_{21}x_1 + a_{22}x_2 + \ldots + a_{2N}x_N \\ &\cdot \\ &\cdot \\ \dot{x}_N &= a_{N1}x_1 + a_{N2}x_2 + \cdots + a_{NN}x_N \end{aligned} \quad \Rightarrow [\dot{\mathbf{x}}] = [\mathbf{A}][\mathbf{x}] \qquad (16.93)$$

Note that we are only considering the unperturbed system, hence the input vector part, i.e. the **[B]u** term in the input-output state model configuration has been omitted. By denoting the time derivative d/dt as Δ, and rearranging, the above set of equations may be given as:

$$
\begin{aligned}
(\Delta - a_{11})x_1 - a_{12}x_2 - \cdots - a_{1N}x_N &= 0 \\
-a_{21}x_1 + (\Delta - a_{22})x_2 - \cdots - a_{2N}x_N &= 0 \\
&\cdot \\
&\cdot \\
-a_{N1}x_1 - a_{N2}x_2 - \cdots (\Delta - a_{NN})x_N &= 0
\end{aligned}
$$

or
$$
\begin{bmatrix}
\Delta - a_{11} & -a_{12} & -\cdots - a_{1N} \\
-a_{21} & \Delta - a_{22} & -\cdots - a_{2N} \\
\cdots & & \\
-a_{N1} & -a_{N2} & -\cdots \Delta - a_{NN}
\end{bmatrix}
\begin{bmatrix} x_1 \\ x_2 \\ \cdots \\ x_N \end{bmatrix} = 0
$$
or **[ΔI – A]x** = 0 (16.94)

By taking Laplace transform of Equation 16.94,

$$(s\mathbf{I} - \mathbf{A})\mathbf{x}(s) = \mathbf{x}(0) \tag{16.95}$$

where $\mathbf{x}(0)$ is LT of initial conditions. In computing the inverse of $(s\mathbf{I} - \mathbf{A})$, the following condition about the given determinant has to be true:

$$|l\mathbf{I} - \mathbf{A}| = 0 \tag{16.96}$$

We already know that if the determinant is expanded, a polynomial in l will result,

$$b_N l^n + b_{N\text{-}1} l^{n-1} + \ldots + b_1 l + b_0 = 0 \tag{16.97}$$

and Equation 16.97 is the *characteristic equation* of the system. From Equation 16.97, n-roots of the form, l_N, l_{N-1}, ..., l_2, l_1 are obtainable, provided all roots of Equation 16.97 are distinct. These roots are called the *eigenvalues* of the given system from which **[A]**, the characteristic matrix has been obtained. They spell the same significance about stability measure, response shape and speed of a given system as we have experienced in the transfer function analysis.

The solution of the set of differential equations may be given in terms of these eigenvalues in the following equation form:

$$
\begin{aligned}
x_1 &= c_{11}e^{\lambda_1 t} + c_{12}e^{\lambda_2 t} + \cdots + c_{1N}e^{\lambda_N t} \\
x_2 &= c_{21}e^{\lambda_1 t} + c_{22}e^{\lambda_2 t} + \cdots + c_{2N}e^{\lambda_N t} \\
&\cdot \\
&\cdot \\
x_N &= c_{N1}e^{\lambda_1 t} + c_{N2}e^{\lambda_2 t} + \cdots + c_{NN}e^{\lambda_N t}
\end{aligned}
\tag{16.98}
$$

This may be written in the equivalent matrix notation:

$$\begin{bmatrix} x_1 \\ x_2 \\ . \\ . \\ x_N \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} & . & . & c_{1N} \\ c_{21} & c_{22} & . & . & c_{2N} \\ . & . & . & . & . \\ . & . & . & . & . \\ c_{N1} & c_{N2} & . & . & c_{NN} \end{bmatrix} \begin{bmatrix} e^{\lambda_1 t} \\ e^{\lambda_2 t} \\ . \\ . \\ e^{\lambda_N t} \end{bmatrix} \qquad (16.99)$$

or $\mathbf{x} = \mathbf{C}e^{-lt}$ (16.100)

The **[C]** matrix groups $n \times n$ constants. Let n linearly independent group of constants are chosen within them and each group lie on each column of **[C]** matrix, then these constants in the form of column vectors are called characteristic vectors or **eigenvectors**. As each characteristic vector corresponds to one of the eigenvalues, the *j*-th eigenvector is:

$$\mathbf{c_J} = \begin{bmatrix} c_{1J} \\ c_{2J} \\ . \\ . \\ c_{NJ} \end{bmatrix} \qquad (16.101)$$

and with such a form, the solution may be written as:

$$x = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t} + \cdots + c_N e^{\lambda_N t} \qquad (16.102)$$

This is a linear combination of linearly independent eigenvectors, the exponential time functions are scalar weighting factors. The individual terms in the left hand side of
Equation 16.102) are called the natural modes of the system, i.e. the term,

$$c_J e^{\lambda_J t} \qquad (16.103)$$

is the *j*-th natural mode associated with the *j*-th eigenvalue.

Because each of exponential time functions must separately satisfy the system equation, putting Equation 16.103 into 16.93

$$\lambda_J c_J e^{\lambda_J t} = \mathbf{A} c_J e^{\lambda_J t} \quad \text{or} \quad \lambda_J c_J = \mathbf{A} c_J \qquad (16.104)$$

These n-number of equations are homogeneous but not linearly independent, hence only $n(n-1)$ unknown constants could be found by them. For the remaining n constants, we shall use n-number of initial conditions.

Solution by the procedure that we have discussed above is a bit lengthy and may often become tedious. Douglas has applied this method to find the dynamic response

of an inherently second order system. However, through the determination of eigenvalues and eigenvector, there is an alternate way of diagonalization of the characteristic matrix **[A]**. This diagonalization procedure is helpful is the analysis of controllability and observability of a system.

16.1.12 Linear Independence of Vectors

The vectors, $\mathbf{x}_1, \mathbf{x}_2, \ldots \mathbf{x}_N$ are said to be linearly independent if,

$$c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + \ldots + c_N\mathbf{x}_N = 0 \qquad (16.105)$$

where $c_1, c_2, \ldots, c_N$ are constants and this condition implies that $c_1 = c_2 = \ldots = c_N = 0$.

Whereas, the vectors $\mathbf{x}_1, \mathbf{x}_2, \ldots \mathbf{x}_N$ are said to be linearly dependent if $\mathbf{x}_I$ can be expressed as a linear combination of $\mathbf{x}_J$ ($J = 1, 2, \ldots N; J \neq I$), or

$$x_I = \sum_{\substack{J=1 \\ J \neq I}}^{N} c_J \mathbf{x}_J \qquad (16.106)$$

is true for some set of constants c_J. This means that if x_I can be expressed as a linear combination of the other vectors in the set, it is linearly dependent upon them or it is not an independent member of the set.

A practical test of linear independence for a dynamic process may be done by testing whether its characteristic matrix **[A]** is singular [det|A| = 0] or non-singular [det|A| ≠ 0]. Non-singularity of an $n \times n$ square matrix proves that n-column (or row) vectors are linearly independent, and matrix has a rank of n. Similarly, singularity of an $n \times n$ square matrix proves that n-column (or row) vectors are linearly dependent, and the matrix may have a rank less than n.

16.1.13 Canonical Transformation

Let us consider Equation 16.104 and expand it for a few eigenvalues:

$$\lambda_1 c_1 = \mathbf{A}c_1, \ \lambda_2 c_2 = \mathbf{A}c_2, \ldots, \lambda_N c_N = \mathbf{A}c_N \qquad (16.107)$$

Expanding in matrix form, Equation 16.107 becomes:

$$\lambda_1 \begin{bmatrix} c_{11} \\ c_{21} \\ . \\ c_{N1} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & . & a_{1N} \\ a_{21} & a_{22} & . & a_{2N} \\ . & . & . & . \\ a_{N1} & a_{N2} & . & a_{NN} \end{bmatrix} \begin{bmatrix} c_{11} \\ c_{21} \\ . \\ c_{N1} \end{bmatrix}, \quad \lambda_2 \begin{bmatrix} c_{12} \\ c_{22} \\ . \\ c_{N2} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & . & a_{1N} \\ a_{21} & a_{22} & . & a_{2N} \\ . & . & . & . \\ a_{N1} & a_{N2} & . & a_{NN} \end{bmatrix} \begin{bmatrix} c_{12} \\ c_{22} \\ . \\ c_{N2} \end{bmatrix},$$

$$\lambda_N \begin{bmatrix} c_{1N} \\ c_{2N} \\ . \\ c_{NN} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & . & a_{1N} \\ a_{21} & a_{22} & . & a_{2N} \\ . & . & . & . \\ a_{N1} & a_{N2} & . & a_{NN} \end{bmatrix} \begin{bmatrix} c_{1N} \\ c_{2N} \\ . \\ c_{NN} \end{bmatrix} \qquad (16.108)$$

These are a set of vector equations. If the column vectors (eigenvectors) are collected into a matrix and factor out the **[A]** matrix from the right hand side of the equation,

$$\begin{bmatrix} \lambda_1 c_{11} & \lambda_2 c_{12} & . & \lambda_N c_{1N} \\ \lambda_1 c_{21} & \lambda_2 c_{22} & . & \lambda_N c_{2N} \\ . & . & . & . \\ \lambda_1 c_{N1} & \lambda_2 c_{2N} & . & \lambda_N c_{NN} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & . & a_{1N} \\ a_{21} & a_{22} & . & a_{2N} \\ . & . & . & . \\ a_{N1} & a_{N2} & . & a_{NN} \end{bmatrix} \begin{bmatrix} c_{11} & c_{12} & . & c_{1N} \\ c_{21} & c_{22} & . & c_{2N} \\ . & . & . & . \\ c_{N1} & c_{2N} & . & c_{NN} \end{bmatrix} \qquad (16.109)$$

Noting that the matrix on the left hand side of the above equation can be expressed as the product of two matrices, one of which is a diagonal matrix,

$$\begin{bmatrix} \lambda_1 c_{11} & \lambda_2 c_{12} & . & \lambda_N c_N \\ \lambda_1 c_{21} & \lambda_2 c_{22} & . & \lambda_N c_{2N} \\ . & . & . & . \\ \lambda_1 c_{N1} & \lambda_2 c_{2N} & . & \lambda_N c_{NN} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & . & a_{1N} \\ a_{21} & a_{22} & . & a_{2N} \\ . & . & . & . \\ a_{N1} & a_{N2} & . & a_{NN} \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & . & 0 \\ 0 & \lambda_2 & . & 0 \\ . & . & . & . \\ 0 & 0 & . & \lambda_N \end{bmatrix} \qquad (16.110)$$

Calling the diagonal matrix as **[□]**, to denote that the diagonal elements are the eigenvalues of the system and combining Equations 16.109 and 16.110 obtains the result:

$$\mathbf{[M][\square] = [A][M]} \qquad (16.111)$$

or

$$\mathbf{[\square] = [M]^{-1}[A][M]} \qquad (16.112)$$

The transformation matrix is denoted by **[M]** because it is sometimes called *Modal Matrix*. Now returning to the discussion on linear transformation, it becomes apparent that if we choose the transformation matrix, **[M],** as the matrix of eigenvectors, **[C]**, the equation,

$$[\dot{\mathbf{y}}] = [\mathbf{A}][\mathbf{y}], \text{ would become } [\dot{\mathbf{y}}] = [\Lambda][\mathbf{y}] \qquad (16.113)$$

The choice of **[□]** as diagonal matrix means that all the equations will be uncoupled,

$$\begin{bmatrix} \dot{y}_1 \\ \dot{y}_2 \\ . \\ \dot{y}_N \end{bmatrix} = \begin{bmatrix} \lambda_1 & 0 & . & 0 \\ 0 & \lambda_2 & . & 0 \\ . & . & . & . \\ 0 & 0 & . & \lambda_N \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ . \\ y_N \end{bmatrix} \qquad (16.114)$$

This transformation is called a *canonical transformation* and the coordinates in the y-domain are called *canonical coordinates*.

For an unperturbed system, i.e. **u** = 0, the canonical equations become:

$$\begin{aligned} dy_1/dt &= \lambda_1 y_1 \\ dy_2/dt &= \lambda_2 y_2 \\ &\cdot \\ &\cdot \\ dy_N/dt &= \lambda_N y_N \end{aligned} \quad \text{with the initial condition} \quad \begin{aligned} y_1(0) &= y_{1S} \\ y_2(0) &= y_{2S} \\ &\cdot \\ &\cdot \\ y_N(0) &= y_{NS} \end{aligned} \qquad (16.115)$$

It is possible to prove that n-eigenvector c_js are linearly independent. This fact guarantees that the solution of Equation 16.102,

$$\mathbf{x} = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t} + \cdots + c_N e^{\lambda_N t} \qquad (16.102)$$

is unique and that the matrix **[C]** is non-singular, and original linear transformation with **[M]** = **[C]** can be written in terms of the eigenvectors as:

$$\mathbf{x} = \mathbf{[M]y} = \mathbf{[C]y} = c_1 y_1 + c_2 y_2 + \ldots + c_N y_N \qquad (16.116)$$

The use of modal matrix **[M]** to transform the characteristic matrix **[A]** into the diagonal form **[□]** is a useful technique of analyzing controllability of a system.

In certain systems, the state variables are not described by first order ODEs that are serially connected. Instead, they operate in a parallel manner. Generally the inputs and outputs of the first order processes are different. Input to and/or output to any of processes could have influence on other processes through the argument of the describing differential equation. Such a process may be encountered while solving Problem P16.8 in the above problem set. In literature, these systems are justifiably called as multiple-input-multiple-output (MIMO) systems. The analysis of such systems resorts to many of the procedures that have been used in state variable analysis, e.g. the digonalization of characteristic matrix, by which control of a MIMO system becomes more easy and free of variable interaction. We shall discuss analysis of multiple-input-multiple-output systems in the next part of this chapter.

Part 2: MULTIPLE INPUT AND OUTPUT (MIMO) SYSTEMS

16.2 Multiple Input and Multiple Output (MIMO) Systems

We have introduced the multiple input-output kind of system by two examples in Chapters 3 and 4 [Examples 3.6 and 4.19]. In many industrial processes, we come across such systems that feature more than one input and output variables. The most simple of these systems are expressed as 2×2, i.e. systems featuring two inputs and two outputs. A few examples of this class of MIMO systems are shown in Figure 16.10.

The first example shows a mixing or blending process where the input variables are

the flow rates of two streams of pure A and pure B, and output variables are flow rate and composition in terms of either A or B, of the exit stream which is a mixture of A and B. The exothermic CSTR has been already introduced in Chapter 4 to define the input and output variables as of the feed and exit stream compositions and temperatures respectively, where the composition may be defined as the concentration of the reacting species. The third example shows a single effect evaporator operating under constant vacuum (–ve pressure head). The input variables are feed solution and steam flow rates, and output variables are composition of the exit stream and liquid level inside the evaporator body.

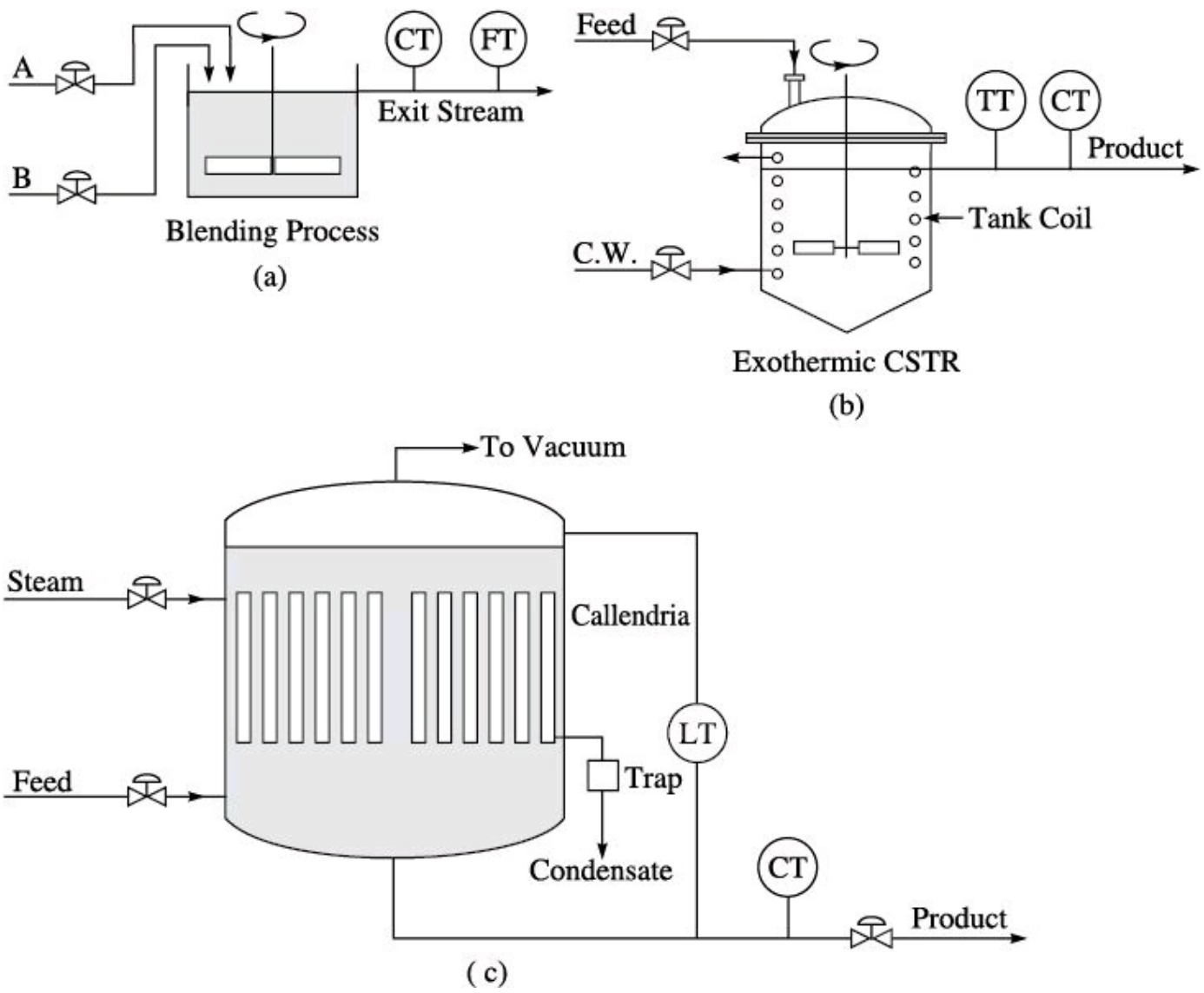

Figure 16.10 Examples of 2 × 2 MIMO systems.

16.2.1 Block Diagram Representation

In all the above examples, the output variables are related to the inputs and these relations are defined by physical laws and kinetic rate processes (in case of the reactor). To treat the subject from the consideration of designing a control system around such a process, it is convenient to describe the process dynamics by block diagrams. A linear model of an exothermic CSTR was introduced in Chapter 4

(Example 4.19). A block diagram of the model was also shown in Figure 4.19 that depicted the internal signal flows between the first order differential nodes of the process model. Hence, Figure 4.19 describes the linear model of an exothermic CSTR which is a 2 × 2 MIMO system, but this representation differs from the type of block diagrams used for MIMO system analysis in the literature. Let us examine what this difference amounts to.

The main problem to be addressed in designing the control system for a multiple input multiple output process is the dynamic feature called '**interaction**' which means that any one of the input change may affect more than one of the response variables. When such a process is considered within a closed loop situation, the inputs are the manipulated variables (MV) and responses are the controlled variables (C). The problem of controlling such a system is how the manipulation MV_1, targeted toward C_1 is going to influence C_2, beside C_1. This is also true for MV_2, C_2 and C_1.

In Example 4.29, the interaction was explained by the feedback paths exited to realize the integration of the two differential equations that comprise process model. Whereas, in a block diagram used conventionally to express or analyze a MIMO process, no feedback path is recognized. The signal streams always flow in forward direction, i.e. from input to the output through the TF blocks. Generally the TF of the blocks are determined experimentally by the methods described in Chapter 13. Hence, they should be obtained from the real process not from its linear approximate model. The same procedure may be followed for a simulated process model whenever that is used for deriving the block TFs. Usually an FOPDT type model structure is attempted with a step response data. The new kind of process model for such a block diagram construction may be given as:

$$C_1 = MV_1 \frac{k_{11} e^{-\tau_{DL11} s}}{\tau_{11} s + 1} + MV_2 \frac{k_{12} e^{-\tau_{DL12} s}}{\tau_{12} s + 1} \qquad (16.117a)$$

and,

$$C_2 = MV_1 \frac{k_{21} e^{-\tau_{DL21} s}}{\tau_{21} s + 1} + MV_2 \frac{k_{22} e^{-\tau_{DL22} s}}{\tau_{22} s + 1} \qquad (16.117b)$$

In a compact form, the above equations will assume the following form:

$$C_1 = MV_1 G_{11} + MV_2 G_{12} \text{ and } C_2 = MV_1 G_{21} + MV_2 G_{22} \qquad (16.118)$$

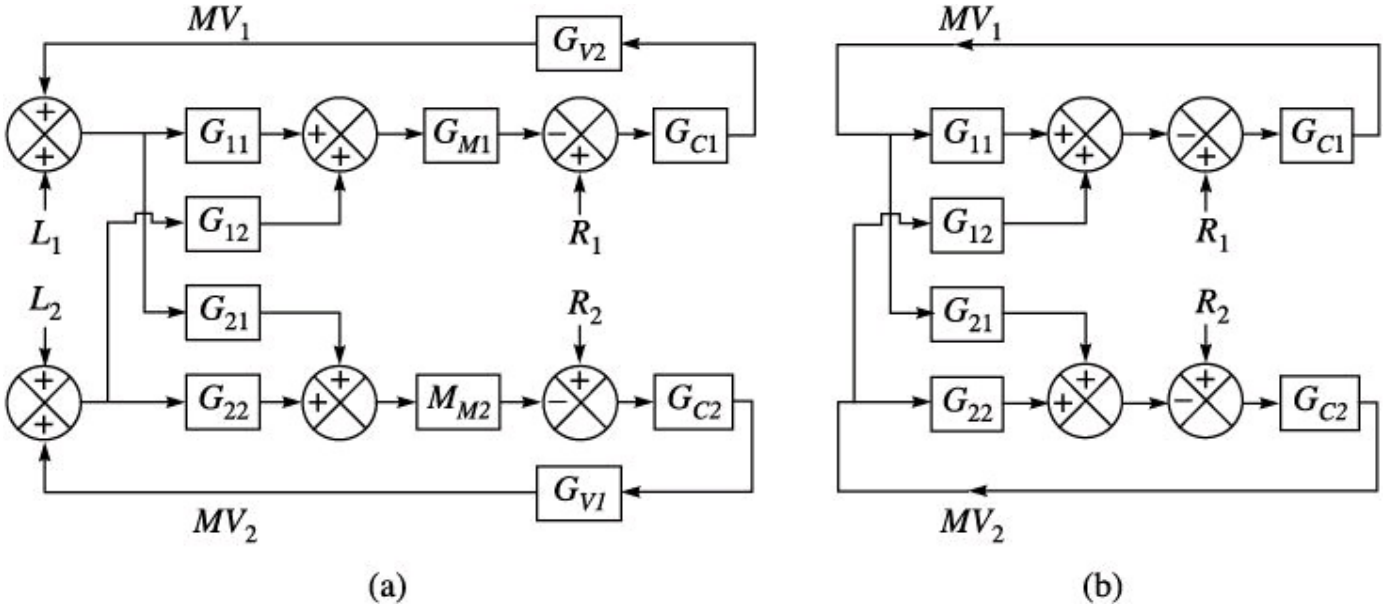

Figure 16.11 The block diagram of a closed loop MIMO process.

In standard matrix notation, Equations 16.118 takes the form:

$$\begin{bmatrix} C_1 \\ C_2 \end{bmatrix} = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix} \begin{bmatrix} MV_1 \\ MV_2 \end{bmatrix} \text{ or } [\mathbf{C}] = [\mathbf{G}]\,[\mathbf{MV}] \tag{16.119}$$

Matrix **[G]** is called the "**transfer matrix**" of the MIMO process.

Block diagrams in Figure 16.11 are constructed from Equations 16.118. In Figure 16.11(a), the process is drawn with the controllers and disturbance paths for load as well as set point are provided. In Figure 16.11(b), the control system can be disturbed only for set point disturbances. Generally the latter figure will be considered for analysis. Interaction in such a representation comes from the branches containing G_{21} and G_{12} due to which the control action affects the outputs other than the target controlled variable. The interaction of variables in a MIMO system often produces poor control quality, which may even lead to instability of the closed loop process. Now a question may arise that "is it possible to find the pairs of manipulated and controlled variables amongst all the possibilities, so that interaction may be reduced, if not removed in the resulting closed loop process?" In many such processes, it has been found that with manipulation, if employed for the variable apparently for which it is targeted, the resulting systems performance become poorer. This may be called as the incorrect pairing. The exercise of finding the correct input-output variable pair to reduce interaction is called '**pairing**' of variables.

By means of such an effort, if it is found that G_{C1} should send manipulation to process block G_{21}, G_{22}, and G_{C2} to G_{11}, G_{12}, the pairing configuration is called 1-2|2-1 pairing. Thus the system shown in Figure 16.12 is known to have a 1-1|2-2 pairing.

If it so happens that input-1 can only affect output-1, and input-2 has influence

solely on output-2, then the corresponding sets of I/O are said to be completely **paired**. Such processes are easily controllable because manipulation of a particular loop does not affect the other loop in any way. Figure 16.12 expresses the idea of complete pairing of variables.

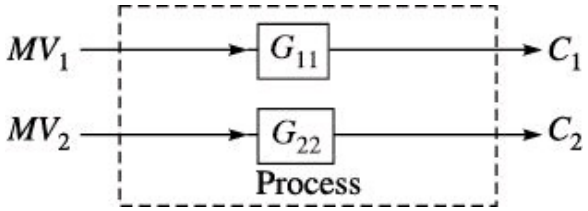

Figure 16.12 A completely paired MIMO process.

In matrix notation, a completely paired system is expressed as:

$$\begin{bmatrix} C_1 \\ C_2 \end{bmatrix} = \begin{bmatrix} G_{11} & 0 \\ 0 & G_{22} \end{bmatrix} \begin{bmatrix} MV_1 \\ MV_2 \end{bmatrix} \qquad (16.120)$$

To exercise an effort for finding the correct pairing for a given system, first we must have quantitative estimate of interaction that a particular pairing offers.

16.2.2 Measure of Interaction

There is a way to measure this interaction that has proven to be successful in numerous industrial processes. The following treatment is done for a 2×2 MIMO system, however, the result is applicable for systems of higher order also. To begin with, it is logical to control each controlled variable with manipulation that has the greatest 'influence' on it. In this context, influence and gain have same significance. To make a decision, let us find all the gain terms of the system. For the 2×2 system with interaction that is given in Figure 16.11(b), in which MV_1 affects C_1 and C_2, and MV_2 affects C_2 and C_1, there will be four steady state gains defined as:

$$\begin{aligned} k_{11} &= [\partial C_1/\partial M_1]_{M_2=\text{ Const.}}, & k_{21} &= [\partial C_2/\partial M_1]_{M_2=\text{ Const.}} \\ k_{12} &= [\partial C_1/\partial M_2]_{M_1=\text{ Const.}}, & k_{22} &= [\partial C_2/\partial M_2]_{M_1=\text{ Const.}} \end{aligned} \qquad (16.121)$$

These four gains may be arranged in the form of an array (Steady State Gain Array □ *SSGA*) or matrix (Steady State Gain Matrix □ *SSGM*) to obtain a more graphical description of their relationship to the controlled and manipulated variables.

$$SSGA = \begin{array}{c|cc} & MV_1 & MV \\ \hline C_1 & k_{11} & k_{12} \\ C_2 & k_{21} & k_{22} \end{array} \quad SSGM = \begin{array}{c|c} & \begin{matrix} MV_1 & MV_2 \end{matrix} \\ \hline \begin{matrix} C_1 \\ C_2 \end{matrix} & \begin{bmatrix} k_{11} & k_{12} \\ k_{21} & k_{22} \end{bmatrix} \end{array}$$

At a certain steady state condition, the following equations may be written to express the changes in the controlled variables:

$$\Delta C_1 = \left.\frac{\partial C_1}{\partial MV_1}\right|_{MV_2=\text{Const.}} \Delta MV_1 + \left.\frac{\partial C_1}{\partial MV_2}\right|_{MV_1=\text{Const.}} \Delta MV_2 = k_{11}\Delta MV_1 + k_{12}\Delta MV_2 \qquad (16.122)$$

$$\Delta C_2 = \left.\frac{\partial C_2}{\partial MV_1}\right|_{MV_2=\text{Const.}} \Delta MV_1 + \left.\frac{\partial C_2}{\partial MV_2}\right|_{MV_1=\text{Const.}} \Delta MV_2 = k_{21}\Delta MV_1 + k_{22}\Delta MV_2 \qquad (16.123)$$

From the *SSGA* or *SSGM*, we may choose the combination of *C* and *MV* that yields the largest absolute value in each row to obtain the right combinations. That is, if $|k_{12}|$ is larger than $|k_{11}|$, then MV_2 is chosen to control C_1. However, this method of decision making about pairing is not foolproof because the gain terms may contain different dimensions, consequently their comparison is impossible. A new kind of gain has been proposed by Bristol by which it is possible to derive what has been called **Relative Gain,** *l*, of the system. Relative gains are the normalized form of steady state gains, and free of their short comings. First we define the new gain term, b_{IJ}, by considering the case when instead of holding MV_2 constant (while finding the *k*s) during making the small change in MV_1, MV_2 is simultaneously manipulated to bring back C_2 to its original steady state value. From the result obtained, another kind of gain may be defined between C_1 and MV_1. Thus,

$$\begin{aligned} b_{11} &= [\partial C_1/\partial MV_1]_{C_2=\text{ Const.}}; \quad b_{12} = [\partial C_1/\partial MV_2]_{C_2=\text{ Const.}} \\ b_{21} &= [\partial C_2/\partial MV_1]_{C_1=\text{ Const.}}; \quad b_{22} = [\partial C_2/\partial MV_2]_{C_1=\text{ Const.}} \end{aligned} \qquad (16.124)$$

The ratio of *k*/*b* is called the **Relative Gain** and denoted as *l*s. Thus,

$$\lambda_{11} = \frac{k_{11}}{b_{11}}; \quad \lambda_{12} = \frac{k_{12}}{b_{12}}; \quad \lambda_{21} = \frac{k_{21}}{b_{21}}; \quad \lambda_{22} = \frac{k_{22}}{b_{22}} \qquad (16.125)$$

The b_{IJ} gains may also be computed from the k_{IJ} gains in following manner:

Equations 16.122 and 16.123 may be rewritten as:

$$\Delta C_1 = k_{11}\Delta M_1 + k_{12}\Delta M_2 \qquad (16.126)$$

$$\Delta C_2 = k_{21}\Delta M_1 + k_{22}\Delta M_2 \qquad (16.127)$$

The b_{11} by definition is $\Delta C_1/\Delta M_1$ with $\Delta C_2 = 0$.

Thus, from Equation 16.127

$$\Delta C_2 = 0 = k_{21}\Delta M_1 + k_{22}\Delta M_2$$

Solving for ΔM_2,

$$\Delta M_2 = -(k_{21}/k_{22})\Delta M_1 \qquad (16.128)$$

Substituting 16.128 into 16.126

$$\Delta C_1 = k_{11}\Delta M_1 - \frac{k_{12}k_{21}}{k_{22}}\Delta M_1 = \left(k_{11} - \frac{k_{12}k_{21}}{k_{22}}\right)\Delta M_1$$

Thus,

$$b_{11} = \left(\frac{\Delta C_1}{\Delta M_1}\right)_{\Delta C_2 = 0} = k_{11} - \frac{k_{12} - k_{21}}{k_{22}} = \frac{k_{11}k_{22} - k_{12}k_{21}}{k_{22}}$$

The relative gain terms *l*s are then:

$$\lambda_{11} = \frac{k_{11}}{b_{11}} = \frac{k_{11}k_{22}}{k_{11}k_{22} - k_{12}k_{21}};\ \lambda_{12} = \frac{k_{12}}{b_{12}} = \frac{k_{12}k_{21}}{k_{12}k_{21} - k_{11}k_{22}} \qquad (16.129)$$

$$\lambda_{21} = \frac{k_{21}}{b_{21}} = \frac{k_{12}k_{21}}{k_{12}k_{21} - k_{11}k_{22}};\ \lambda_{22} = \frac{k_{22}}{b_{22}} = \frac{k_{11}k_{22}}{k_{11}k_{22} - k_{12}k_{21}}$$

Like *SSGM*, the relative gains are also expressed in array or matrix form:

$$RGA = \begin{array}{c|cc} & MV_1 & MV_2 \\ \hline C_1 & \lambda_{11} & \lambda_{12} \\ C_2 & \lambda_{21} & \lambda_{22} \end{array} \quad RGM = \begin{array}{c|cc} & MV_1 & MV_2 \\ \hline C_1 & \lambda_{11} & \lambda_{12} \\ C_2 & \lambda_{21} & \lambda_{22} \end{array}$$

Bequette has shown that "**For an n × n system, the element l_{IJ} of the relative gain matrix may be obtained as the product of *ij* element of the steady state gain matrix with *ji* element of the inverse of steady state gain matrix**".

(The reader is asked to prepare a computer program that will receive the steady gain values as an input array, calculate and report the elements of relative gain matrix as the result.)

Relative gain matrix has a useful property that there may be a 'back-check' for eliminating arithmetic mistakes in calculating the array elements – i.e. the sum of diagonally, horizontally or vertically placed array element's is unity.

As we stated that the summation of all terms in each column and each row must equal 1.0, hence for a 2 × 2 system, only one term need to be calculated while the others may be obtained using this property. In a 3 × 3 system, which has nine terms in the matrix, only four independent terms need to be evaluated, the other five terms are then obtained by simple arithmetic.

In the following examples, problems have been solved to demonstrate the steady state and relative gain evaluation from the developed model of the system. Also, the pairing problem is discussed in the context of the results obtained. Later, we shall summarize certain rules of pairing as a generalization of these conclusions.

EXAMPLE 16.15 A stirred liquid heater with variable hold up where the heat

transfer occurs between jacket liquid and tank liquid through the tank wall. As the heat transfer happens only through the wetted part of the inside surface of the tank, the heat transfer area will be a function of the process liquid level in the tank. The input variables (manipulations) are the inlet liquid rate Q_I, to the tank and inlet temperature of the jacket liquid, T_{JI}. The output (controlled) variables will be process liquid level in the tank, H_T , and exit temperature of the process liquid, T_O. Other process parameters may be understood from the schematic of the process shown in Figure 16.13.

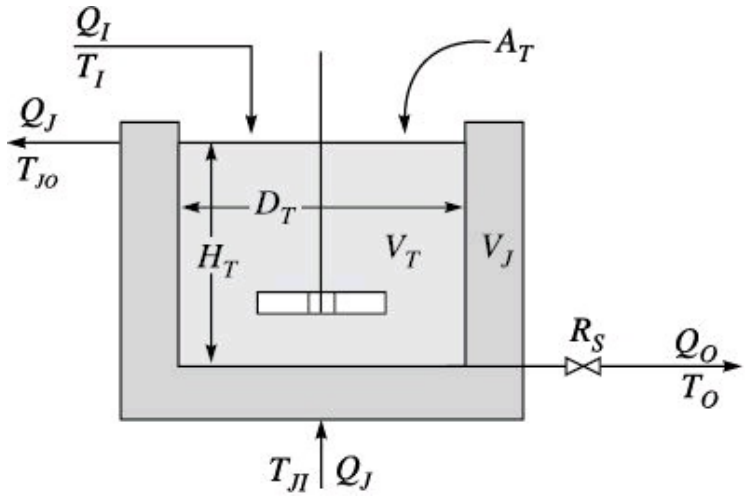

Figure 16.13 A variable volume stirred liquid heating process.

The lumped parameter model of the liquid heating system may be written as:

1. Tank Mass balance:

$$Q_I - \frac{H_T}{R_S} = A_T \frac{dH_T}{dt} \qquad (16.130)$$

2. Mixer Heat balance:

$$T_I Q_I \rho_L C_{PL} - T_O \frac{H_T}{R_S} \rho_L C_{PL} + UA_H (T_{JO} - T_O) = A_T H_T \rho_L C_{PL} \frac{dT_O}{dt} \qquad (16.131)$$

3. Jacket Heat balance:

$$T_{JI} Q_J \rho_J C_{PJ} - T_{JO} Q_J \rho_J C_{PJ} - UA_H (T_{JO} - T_O) = V_J \rho_J C_{PJ} \frac{dT_{JO}}{dt} \qquad (16.132)$$

where, heat transfer area, $A_H = \pi (D_T^2/4) H_T$.

The model is simulated with the following parameter values:

D_T = 25 cm; A_T = 490.87 cm^2; V_T = 196.35 l; V_J = 58.9 l; U = 1.625 cal/(min/cm^2/ºC);

r_L = 1 g/cm^3; C_{PL} = 1 cal/(g/ºC); r_J = 1 g/cm^3; C_{PJ} = 1 cal/(g/ºC); R_S = 0.1

cm/(cm^3/s).

For evaluation of four *SSG*s, first the system was run with following initial values of the variables, Q_I = 2.0 l/min, T_I = 30ºC, Q_{JI} = 1.0 l/min, and T_{JI} = 50ºC. After attainment of steady state, the system is disturbed by small step changes in Q_I (MV_1) and T_{JI} (MV_2) respectively. The resulting changes in H_T (C_1) and T_O (C_2) are noted.

The *SSG*s are determined as:

$$k_{11} = \left.\frac{\partial C_1}{\partial MV_1}\right|_{MV_2} = \frac{20}{0.2} = 100 \text{ cm/l/m}; \; k_{12} = \left.\frac{\partial C_1}{\partial MV_2}\right|_{MV_1} = \frac{200-200}{5} = 0 \text{ cm/ºC}$$

$$k_{21} = \left.\frac{\partial C_2}{\partial MV_1}\right|_{MV_2} = \frac{1.985}{0.2} = 9.925 \text{ ºC/l/m}; \; k_{22} = \left.\frac{\partial C_2}{\partial MV_2}\right|_{MV_1} = \frac{38.12-36.51}{5} = 0.322 \text{ ºC/ºC}$$

From the *SSGM* $\begin{bmatrix} 100 & 0 \\ 9.925 & 0.322 \end{bmatrix}$, the *RGM* is found as: $\lambda = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

The result tells us that C_1 (H_T) is solely influenced by MV_1 (Q_I) and C_2 (T_O) by MV_2 (T_J). This is an example where complete pairing exists between manipulated and controlled variables. Hence, the obvious choice of control configuration should be 1-1|2-2.

EXAMPLE 16.16 Two liquid streams at different temperature are mixed in a stirred tank having variable hold up. The two streams enter at constant temperature but their flow rates are variable. The inflow rates Q_{I1} and Q_{I2} are the manipulated variables (MVs) to the process. The tank level, H_T, and exit liquid temperature, T_O, are the controlled variables (Cs). The process is shown in Figure 16.14.

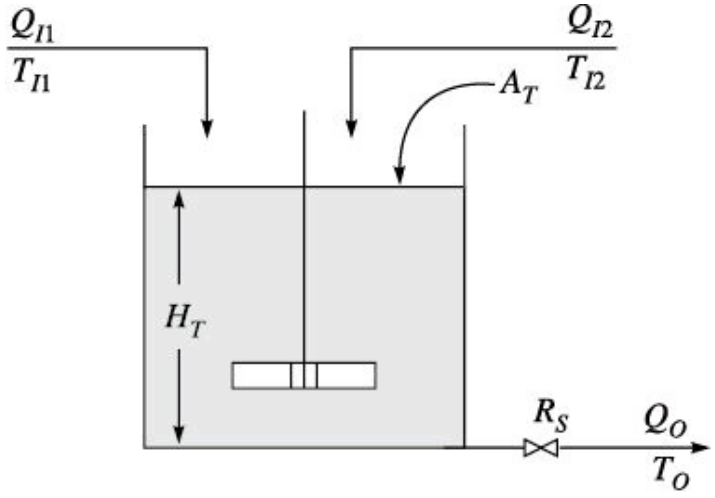

Figure 16.14 Mixing two liquid streams of different temperatures within a variable volume mixer.

If a diabatic condition is assumed, the model equations may be written as:

1. Tank Mass balance:

$$Q_{I1} + Q_{I2} - \frac{H_T}{R_S} = A_T \frac{dH_T}{dt} \qquad (16.133)$$

2. Mixer Heat balance:

$$T_{I1} Q_{I1} \rho_L C_{PL} + T_{I2} Q_{I2} \rho_L C_{PL} - T_O \frac{H_T}{R_S} = A_T \frac{dT_O}{dt} \qquad (16.134)$$

The model is simulated with the following parameter values:

A_T = 490.87 cm^2; V_T = 196.35 l; r_L = 1 g/cm^3; C_{PL} = 1 cal/(g/ºC), R_S = 0.1 cm/(cm^3/s).

For evaluation of four *SSG*s, first the system was run with following initial values of the variables, Q_{I1} = 1.0 l/min, T_{I1} = 30ºC, Q_{I2} = 1.0 l/min, and T_{I2} = 70ºC. After attainment of steady state, the system is disturbed by small step changes in Q_{I1} (MV_1) and Q_{I2} (MV_2) respectively. The resulting changes in H_T (C_1) and T_O (C_2) are noted. The *SSG*s are determined as:

$$k_{11} = \left.\frac{\partial C_1}{\partial MV_1}\right|_{MV_2} = \frac{220 - 200}{0.2} = 100 \text{ cm/l/m}; \; k_{12} = \left.\frac{\partial C_1}{\partial MV_2}\right|_{MV_1} = \frac{220 - 200}{0.2} = 100 \text{ cm/l/p}$$

$$k_{21} = \left.\frac{\partial C_2}{\partial MV_1}\right|_{MV_2} = \frac{48.2 - 50}{0.2} = -9 \text{ ºC/l/m}; \; k_{22} = \left.\frac{\partial C_2}{\partial MV_2}\right|_{MV_1} = \frac{51.83 - 50}{0.2} = 9.15 \text{ ºC/l/m}$$

From the *SSGM* $\begin{bmatrix} 100 & 100 \\ -9 & 9.15 \end{bmatrix}$, the *RGM* is found as: $\lambda = \begin{bmatrix} 0.504 & 0.496 \\ 0.496 & 0.504 \end{bmatrix}$

As the diagonal elements are still heavy, hence the 1-1|2-2 control configuration may be applied. That their differences with the non-diagonal elements are relatively small, follows from the values of k_{21} and k_{22}.

EXAMPLE 16.17 A two tank liquid level system with interaction is presented in Figure 16.15. The system has two inputs (*MV*s) as the two inlet liquid flow rates *Q*, and two outputs (*C*s) as the liquid levels *H*, in the two tanks. Find the relative gain matrix from a simulation run of the system model assuming a suitable set of parameter and variable values.

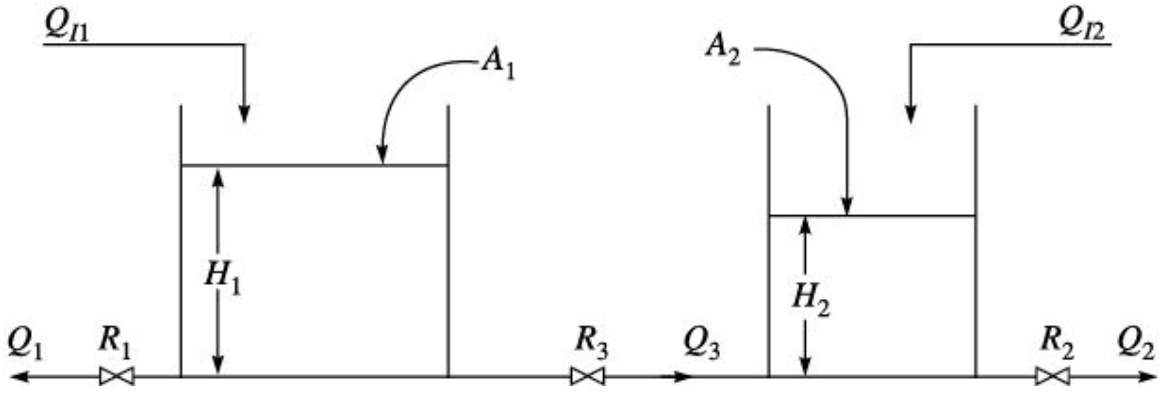

Figure 16.15 The interacting liquid levels in two tanks as a 2×2 MIMO process.

The model equations for the above process may be written as:

1. Mass balance for Tank 1:

$$Q_{I1} - \frac{H_1 - H_2}{R_3} - Q_1 = A_1 \frac{dH_1}{dt} \qquad (16.135)$$

2. Mass balance for Tank 2:

$$Q_{I2} + \frac{H_1 - H_2}{R_3} - Q_2 = A_2 \frac{dH_2}{dt} \qquad (16.136)$$

The model is simulated with the following parameter values:

$A_1 = 1\ \text{m}^2$, $A_2 = 0.5\ \text{m}^2$, $R_1 = 1\ \text{m/(m}^3\text{/h)}$, $R_2 = 2\ \text{m/(m}^3\text{/h)}$, $R_3 = 0.5\ \text{m/(m}^3\text{/h)}$.

For evaluation of four *SSG*s, first the system was run with following initial values of the variables, $Q_{I1} = 5.0\ \text{m}^3\text{/h}$, and $Q_{I2} = 5.0\ \text{m}^3\text{/h}$. After attainment of steady state, the system is disturbed by small step changes in Q_{I2} (MV_1) and Q_{I2} (MV_2) respectively. The resulting changes in H_1 (C_1) and H_2 (C_2) are noted.

The *SSG*s are determined as:

$$k_{11} = \left.\frac{\partial C_1}{\partial MV_1}\right|_{MV_2} = \frac{7.32 - 6.43}{1.25} = 0.712\ \text{m/m}^3;\ k_{12} = \left.\frac{\partial C_1}{\partial MV_2}\right|_{MV_1} = \frac{7.143 - 6.43}{1.25} = 0.57\ \text{m/m}^3$$

$$k_{21} = \left.\frac{\partial C_2}{\partial MV_1}\right|_{MV_2} = \frac{7.86 - 7.14}{1.25} = 0.571\ \text{m/m}^3;\ k_{22} = \left.\frac{\partial C_2}{\partial MV_2}\right|_{MV_1} = \frac{8.21 - 7.143}{1.25} = 0.857\ \text{m/m}^3$$

From the *SSGM* $\begin{bmatrix} 0.712 & 0.571 \\ 0.571 & 0.857 \end{bmatrix}$, the *RGM* is found as: $\lambda = \begin{bmatrix} 2.147 & -1.147 \\ -1.147 & 2.147 \end{bmatrix}$

If the same model is simulated with the following new set of parameter values:

$A_1 = 2\ \text{m}^2$, $A_2 = 0.5\ \text{m}^2$, $R_1 = 2\ \text{m/(m}^3\text{/h)}$, $R_2 = 1.5\ \text{m/(m}^3\text{/h)}$, $R_3 = 1.0\ \text{m/(m}^3\text{/h)}$

and similar step responses are obtained, the *SSGM* and *RGM* becomes:

$$SSGM = [k] = \begin{bmatrix} 1.391 & 0.557 \\ 0.556 & 0.973 \end{bmatrix} \text{ and } RGM = \lambda = \begin{bmatrix} 1.297 & -0.297 \\ -0.297 & 1.297 \end{bmatrix}$$

For the two sets of parameter values, the system relative gain matrix shows +ve values on the diagonal. The chosen variable pairing should be C_1 with MV_1 and C_2 with MV_2. We may now summarize the variable pairing rules depending on the element values of the *RGM*.

EXAMPLE 16.18 Two liquid hydrocarbon streams (1 and 2) are mixed in a blender to produce the feedstock for cracking process. Stream 1 contains a mixture of compounds, A and B, while stream 2 contains only B. Effectively, after blending the product is enriched by B compound than stream1.

Data: Stream 1: 40%(v) of A, 60%(v) of B; Stream 2: 100%(v) of B. Flow rate of :

Stream 1, Q_{I1} = 16 l/m; Flow rate of Stream 2, Q_{I2} = 4 l/m.

Desirable product stream: 30%(v) of A, 70%(v) of B.

Let C_{A1} be the volume fraction of A in stream 1 and C_{O1} the volume fraction of A in the exit stream. Since the composition of A in stream 2 is zero, the total and component mass balances are obtained as:

$$Q_O = Q_{I1} + Q_{I2} \tag{16.137}$$

and

$$Q_O C_{AO} = Q_{I1} C_{A1} \tag{16.138}$$

Solving the above two equations:

$$C_{AO} = \frac{Q_{I1} C_{A1}}{Q_{I1} + Q_{I2}} \tag{16.139}$$

To define a 2×2 MIMO process from this blending system, the controlled variables are identified as composition of A in the product stream C_{AO} as C_1 and product flow rate Q_O as C_2. The manipulated variables are flow rates of stream 1: Q_{I1} (MV_1) and stream 2: Q_{I1} (MV_2).

The steady state gains may be determined as:

$$k_{11} = \left.\frac{\partial C_{AO}}{\partial Q_{I1}}\right|_{Q_{I2}} = \frac{Q_{I2} C_{A1}}{(Q_{I1} + Q_{I2})^2} = \frac{4 \times 0.4}{20^2} = 0.004 \text{ mol fraction/l/m}$$

$$k_{12} = \left.\frac{\partial C_{AO}}{\partial Q_{I2}}\right|_{Q_{I1}} = \frac{-Q_{I1} C_{A1}}{(Q_{I1} + Q_{I2})^2} = \frac{-16 \times 0.4}{20^2} = -0.016 \text{ mol fraction/l/m}$$

$$k_{21} = \left.\frac{\partial Q_O}{\partial Q_{I1}}\right|_{Q_{I2}} = 1 \text{ lpm/lpm};\ k_{22} = \left.\frac{\partial Q_O}{\partial Q_{I2}}\right|_{Q_{I1}} = 1 \text{ lpm/lpm}$$

Thus *SSGM* is found as:

$$K = \begin{bmatrix} 0.004 & -0.016 \\ 1 & 1 \end{bmatrix}$$

The corresponding *RGM* is found as:

$$\lambda = \begin{bmatrix} 0.2 & 0.8 \\ 0.8 & 0.2 \end{bmatrix}$$

As the cross diagonal elements are heavier, so MV_1 should be paired with C_2 and MV_2 with C_1, i.e. the 1-2|2-1 control configuration is valid for the system.

Let us talk about a few more revelations about the relative gain term, l_{IJ}. First, considering the new gain term b_{IJ}, its significance may be stated as b_{IJ} is the gain of MV_J on C_I, when all other loops are closed and all other controllers have integral action, and therefore the controlled variables under them will return to set points (zero deviation condition). Thus, we can write,

$$\lambda_{IJ} = \frac{\text{Gain when the other loops are open}}{\text{Gain when the other loops are closed}} = \frac{k_{IJ}}{b_{IJ}} \qquad (16.140)$$

From this definition, we see that the relative gain term is a dimensionless quantity that can be used to measure the magnitude of interaction and decide on the correct pairing of controlled and manipulated variables. In fact, l_{IJ} is a measure of the effect of closing all other loops on the process gain for a given C and MV pair. Let us examine the interaction significance of certain extreme values of l_{IJ}.

Case I $l_{IJ} = 1$. This means that the process gain has same value with all other loops, either open or closed, or there is no interaction and the respective *C*s and *MV*s are completely paired.

Case II $l_{IJ} = 0$. Such a condition may arise out of two possible causes:

1. The open loop gain $|\Box C_1/\Box MV_J|_{MV}$ may be zero. Hence, C_I is not influenced by MV_J at least in terms of open loop condition;
2. The closed loop gain is so large that for a low enough value of the numerator, the ratio is negligibly small, meaning that in order to keep other controlled variables constant, the other loops are interacting significantly with the loop in

question.

For any of the two possibilities, the conclusion is that C_I should not be controlled by MV_J.

Case III $l_{IJ} = \infty$. Such a value of relative gain may be obtained when the closed loop gain is zero or very small, signifying that when the other loops are in automatic, the loop under consideration cannot be controlled, because MV_J affects C_I very little or not at all. So to control this loop, only manual option remains. This is an undesirable situation.

With this background, the rule of pairing of the controlled to manipulated variable may be stated as:

"Always pair on +ve RGM elements that are closest to 1.0. Check the resulting pairing for stability using Niederlinsky's theorem. If the pairings yield an unstable system, choose other +ve pairings with closest to 1.0. Avoid –ve pairings if possible."

This rule was first proposed by Bristol and later modified by Koppel.

The proposed pairing rule is easy to use as we need to know the steady state gain values only, which are generally available during the design stage of the process and consequently, it is not necessary to obtain any operational data from the plant.

Consider two possible *RGM*s:

(I)

$$\begin{array}{c|cc} & MV_1 & MV_2 \\ \hline C_1 & 0.25 & 0.75 \\ C_2 & 0.75 & 0.25 \end{array}$$

(II)

$$\begin{array}{c|cc} & MV_1 & MV_2 \\ \hline C_1 & 2.5 & -1.5 \\ C_2 & -1.5 & 2.5 \end{array}$$

(I) The terms $l_{11} = l_{22} = 0.25 = ¼$, mean that for this pairing, the gain of each loop increases by a factor of 4 when the other loop is closed. The terms $l_{12} = l_{21} = 0.75 = ¾$ indicates that for this pairing, the gain increases only for a factor of 1.333. Thus, we decide for the pairing of C_1 with MV_2 and C_2 with MV_1.

(II) The terms $l_{11} = l_{22} = 2.5 = 1/0.4$ indicate that the gain of each loop is affected by a factor of 0.4, when the other loop is closed. The terms $l_{12} = l_{21} = – 1.25$ indicate that the gain of each loop not only changes sign, but are also multiplied by a factor 1.25. Certainly, this last case is undesirable because it means that the action of the controller depends on whether the other loop is closed or open. Hence, the correct pairing should be $MV_1 – C_1$, and $MV_2 – C_2$.

EXAMPLE 16.19 The *SSGM* of a 3 × 3 MIMO process is given in the following:

$$K = \begin{bmatrix} 1 & 1 & 2.5 \\ 0.5 & 1 & 1 \\ 1 & 0.5 & 1 \end{bmatrix}$$

Determine the relative gain matrix and find correct pairing of variables

Solution: From the above *SSGM* the corresponding *RGM* is calculated as:

$$\lambda = \begin{bmatrix} -0.57143 & -0.57143 & 2.14286 \\ -0.14286 & 1.71429 & -0.57143 \\ 1.71429 & -0.14286 & -0.57143 \end{bmatrix}$$

The only +ve elements in the matrix are on the cross diagonal, hence the pairing should be:

MV_3 with C_1, MV_2 with C_2, MV_1 with C_3.

EXAMPLE 16.20 The *RGM* of a 3 × 3 MIMO process is given in the following:

$$\lambda = \begin{bmatrix} -1 & 1 & 1 \\ 2 & 3 & 4 \\ 0 & 4 & -3 \end{bmatrix}$$

How to choose manipulation-controlled variable pairing for the process?

Solution: Starting from the given *l*, the progressive sequence of the selection process of variable pairs is shown in the following:

$$\lambda = \begin{bmatrix} -1 & 1 & 1 \\ 2 & 3 & 4 \\ 0 & 4 & -3 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 1 & 1 \\ 2 & 3 & 4 \\ 0 & [4] & -3 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 1 & [1] \\ 2 & 3 & 4 \\ 0 & [4] & -3 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 1 & [1] \\ [2] & 3 & 4 \\ 0 & [4] & -3 \end{bmatrix}$$

I II III

Two guidelines should be followed during the selection:

1. Pairing not to be done on a –ve relative gain;
2. Also, a zero element should be avoided, because a zero value of l_{IJ} means MV_J has no influence on C_I.

In the third row that relates C_3 with all the *MV*s, the first and third elements are discarded as one of them is zero and the other being –ve, hence, the second element is selected for pairing and the choice has been registered by putting a bracket on the

element (I).

In the first row, both second and third column elements could have been selected but as the second column is already occupied, the choice goes to the third column and the element is similarly bracketed.

All the three elements in the second row are +ve but as the chosen elements of first and third row are situated in third and second columns, there is no other way but to select the first column element.

To express the chosen elements on the diagonal of the relative gain matrix, the controlled variables may be renamed as:

$$C_1^* = C_2;\ C_2^* = C_3;\ C_3^* = C_1$$

With such renamed variables, the *RGM* becomes:

$$\lambda^* = \begin{bmatrix} [2] & 3 & 4 \\ 0 & [4] & -3 \\ -1 & 1 & [1] \end{bmatrix}$$

To reduce interaction in a MIMO system, finding the right pairs of manipulated and controlled variable is not always going to obtain the right answer to the problem. Recall the resulting *RGM* obtained in Example 17.20:

$$\lambda = \begin{bmatrix} 0.504 & 0.496 \\ 0.496 & 0.504 \end{bmatrix}$$

Though we have decided to pair C_1 with MV_1 and C_2 with MV_2 because the diagonal elements are heavier, but a question may be asked as to how much heavier are they than the cross diagonal elements. In fact the difference is not much [{(0.504 – 0.496) × 100}/1.0 = 0.8%]. Hence, it is expected that a considerable amount of interaction should still remain even after the target pairing is achieved.

The next approach in removing or reducing interaction is to design and install dynamic compensators within the control loop of a MIMO system. This approach is called '**decoupling**', and the dynamic compensators are denoted as '**Decouplers**'. These topics are included in Chapter 11 of *Advanced Process Dynamics and Control*.

EXERCISE PROBLEMS—PART I

16.1 Find the eigenvalues of the following characteristic matrices:

(a) $\mathbf{[A]} = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$; (b) $\mathbf{[A]} = \begin{bmatrix} 1 & 4 \\ 3 & 2 \end{bmatrix}$; (c) $\mathbf{[A]} = \begin{bmatrix} 1 & 4 \\ 3 & 2 \end{bmatrix}$

(d) $\mathbf{[A]} = \begin{bmatrix} 2 & 3 & -2 \\ -2 & 1 & 1 \\ 1 & 0 & 2 \end{bmatrix}$; (e) $\mathbf{[A]} = \begin{bmatrix} 8 & -6 & 3 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$; (f) $\mathbf{[A]} = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$

16.2 Determine the transformation matrix **[P],** the matrix **[A]** given in parts (a), (b), (c), (d), (e) and (f) of problem 1 above. Also, find out the diagonal **[*L*]** for all of them.

16.3 A damped vibrator model is shown in Figure P16.1.

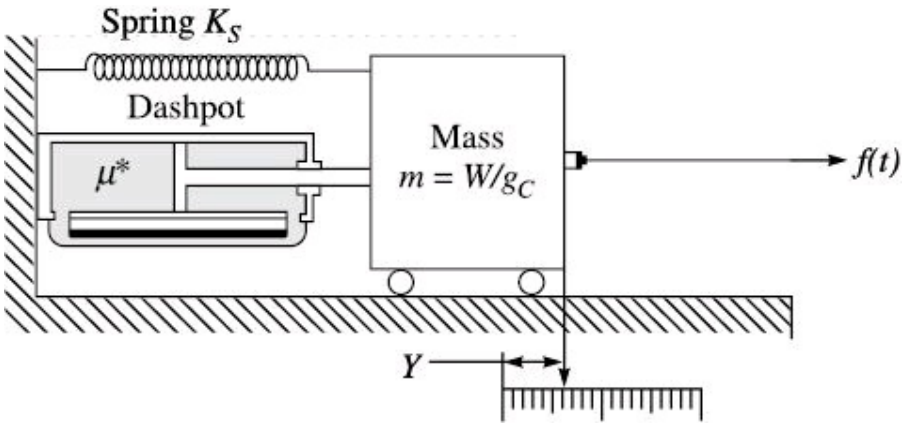

Figure P16.1 The damped vibrator.

Derive the state variable equations for the system considering the applied force $f(t)$ to be the input and the displacement as the output. For the following parameter values of the system:

$W = 154.56$ lbm, $g_C = 32.2$(lbm.ft)/(lbf.s^2), $K_S = 1.2$ lbf/ft, and $\mu^* = 6$(lbf)/(ft/s).

(a) Determine the characteristic matrix **[A]** and **[B]**. Also find the eigenvalues for the system.

(b) The system was initially at rest ($Y = 0$). If the force $f(t)$ is applied according to: $f(t) = 0$, for $t < 0$ and $f(t) = 1.2$ lbf, for $t \geq 0$, find the dynamic response as $Y(t)$ of the system.

16.4 Using following data of the electrically heated furnace of Example 16.2, obtain the dynamics of the object temperature assuming $T_1 = T_2 = T_A = 0$°C, if the heater power W changes from 0 to 10000 cal/min at $t = 0$.

Data: $m_1 = 4000$ g, $C_1 = 2.5$C/(g/cal), $m_2 = 2500$ g , $C_2 = 3$C/(g/cal)

$A_H = 5000$ cm^2, $U = 3$ cal/(cm^2/°C/min), $A_O = 12500$ cm^2

$U_O = 2$ cal/(cm^2/°C/min).

16.5 A non-interacting two tank liquid level dynamic system is as in Figure 16.6. The state variable model of this system is found to be:

$$\begin{bmatrix}\dot{h}_1\\ \dot{h}_2\end{bmatrix}=\begin{bmatrix}-4 & 0\\ 8 & -2\end{bmatrix}\begin{bmatrix}h_1\\ h_2\end{bmatrix}+\begin{bmatrix}0.5\\ 0\end{bmatrix}Q_I \text{ and } h_2=[0 \quad 1]\begin{bmatrix}h_1\\ h_2\end{bmatrix}$$

If $A_1/A_2 = 2$, find the values of A_1, A_2, R_1 and R_2 with reference to Figure 16.6. Also find state transition matrix $\Phi(t)$ of the system.

16.6 The transfer function between output $Y(s)$ and input $U(s)$ of a third order process is:

$$\frac{Y(s)}{U(s)}=\frac{3}{s^3+3.5s^2+3.5s+1}$$

Derive the state equations for the system.

16.7 An interacting 3-tank liquid level system is shown in Figure P16.2. Derive its state variable model.

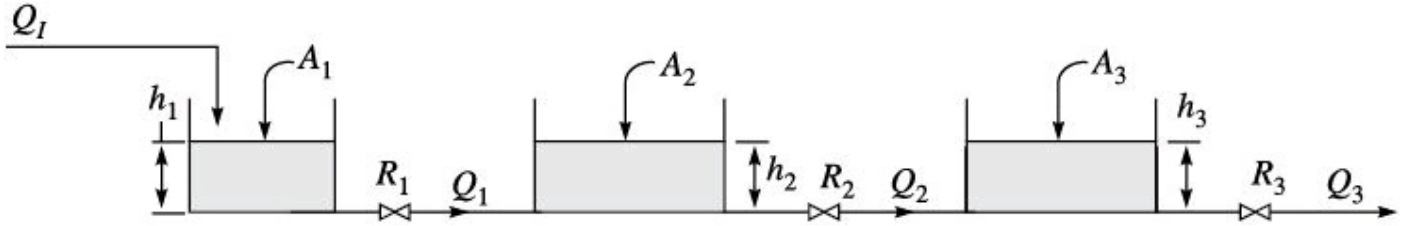

Figure P16.2 Three tanks in interacting series combination.

16.8 The stirred liquid heater uses a hot liquid passing through its jacket as the heating medium, shown in Figure P16.3. Both mixer and jacket liquids are perfectly mixed. The mixer and the jacket have effective volumes of V and V_J respectively. All the variables are illustrated in the following figure. Derive the state equation for the system.

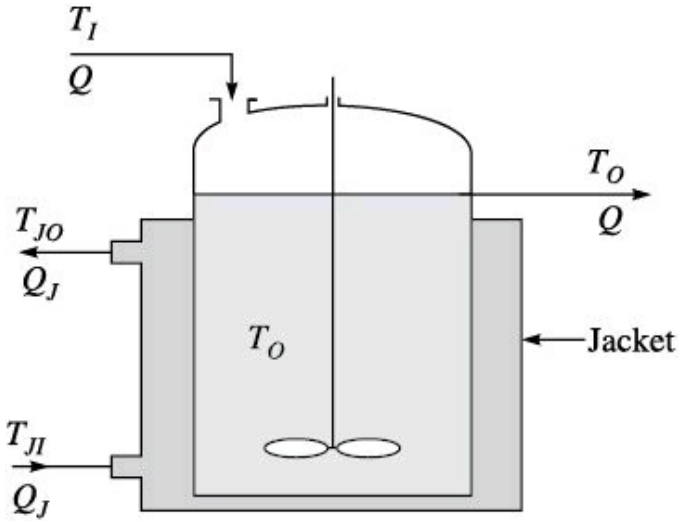

Figure P16.3 The jacketed stirred liquid heater.

16.9 A proportional controller is employed for a second order process and the feedback path contains a first order lag associated with the measuring element. The

closed loop system is shown in Figure P16.4 in the block diagram form. Derive the state equations for the system and find the response for: (a) A regulatory change (Load change only); (b) A servo change (Set point change only). The response in time domain should be determined by two methods: (i) By evaluating the state transition matrix; (ii) Transfer function matrix. Comment on the length of algebraic efforts encountered in case of these two methods for part (a) and (b).

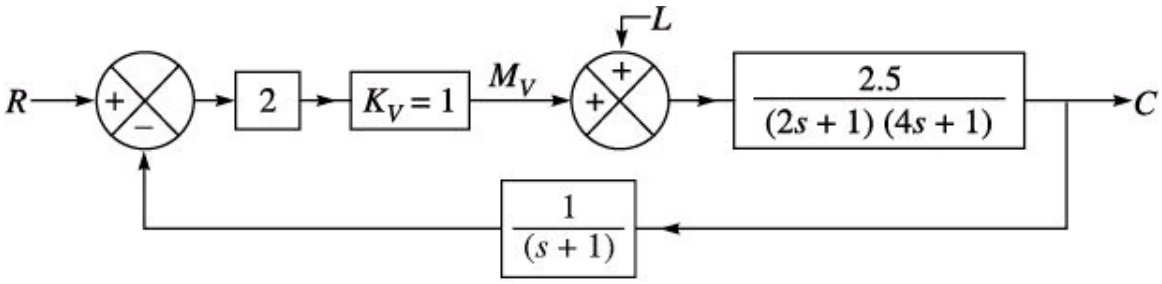

Figure P16.4 Closed loop block diagram of the system in problem 9.

16.10 Convert the closed loop for a second order system with a proportional-integral controller shown in Figure P16.5, to the state variable model. [Hint: To work with a closed loop T.F. with zero in its numerator, see reff. (Cowghanowr), (Ogata)].

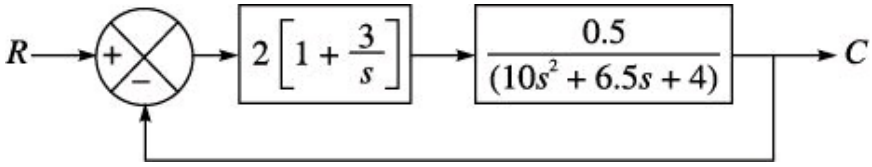

Figure P16.5 Closed loop block diagram of the system in problem 10.

EXERCISE PROBLEMS—PART II

16.11 The steady state gain values of a 2 -input, 2-output process is expressed as:

$$K_P = \begin{bmatrix} 2.5 & 5 \\ 1.25 & 2.55 \end{bmatrix}$$

Find the relative gain matrix [l] and decide on the correct pairing of variables.

16.12 A 3-input, 3-output process is tested for its steady state gain values. The following steady state gain array is obtained:

$$K_P = \begin{bmatrix} 2 & 2 & 3.333 \\ 0.667 & 2 & 2 \\ 2 & 0.667 & 2 \end{bmatrix}$$

Determine the relative gain matrix [l] and decide on the correct pairing of variables for controlling the process.

16.13 From the open loop testing of a pilot scale binary distillation column, the following individual dynamic models as TFs are obtained and arranged in an array,

$$\begin{bmatrix} x_D(s) \\ x_B(s) \end{bmatrix} = \begin{bmatrix} (0.07e^{-7s})/(12s+1) & (-0.05e^{-s})/(15s+1) \\ (0.1e^{-4s})/(11s+1) & (-0.1e^{-2s})/(10s+1) \end{bmatrix} \begin{bmatrix} L(s) \\ V(s) \end{bmatrix}$$

where x_D and x_B are the distillate and bottoms composition respectively, L and V are the reflux and boil-up rates respectively. Determine the RGA and decouplers for the system.

16.14 The steady state gain values are obtained to relate D (distillate rate) and V (boil-up rate) to x_D (distillate composition) and x_B (bottoms composition) of multiple plate distillation column handling a binary feed.

	D	V
x_D	-0.102885×10^{-1}	0.456211×10^{-4}
x_B	5.136286×10^{-3}	-0.322101×10^{-4}

Based on these data, decide on the correct pairing.

16.15 A two tank interacting type dynamic liquid level system with two input and two exit streams is shown in Figure P16.6. It is required to control the levels h_1 and h_2 in the two tanks by manipulating Q_{I1} and Q_{I2}. Flow rate Q_{I4} is the principal load variable. The flow-head relations applicable for the system are:

$$Q_1 = \frac{h_1}{R_1},\ Q_2 = \frac{h_2}{R_2},\ Q_3 = \frac{(h_1 - h_2)}{R_3}$$

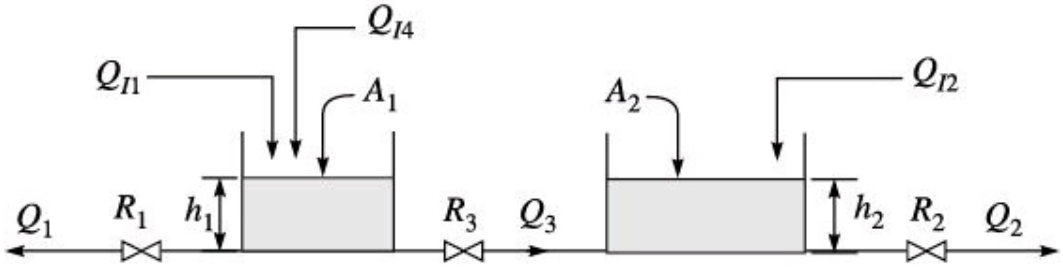

Figure P16.6 The two interacting tank system.

R_1, R_2 and R_3 are the flow resistances associated with the flow paths as shown in the figure. Following parameter values can assumed for analysis:

$R_1 = 0.005$ cm/l/m, $R_2 = 0.004$ cm/l/m, $R_3 = 0.0025$ cm/l/m, $A_1 = A_2 = 1200$ cm^2.

The steady state operating values of the levels in the two tanks are: $h_{S1} = 1200$ cm, $h_{S2} = 900$ cm.

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Appendix 1A

Common Unit Conversions and Physical Properties

No.	To convert	To	Multiply by
1.	Feet	Centimetre	304.8
2.	Kilometre	Mile	1/1.6
3.	Centimetre	Inches	0.3937
4.	Degrees	Radians	0.017453
5.	Square feet	Square metre	0.0929
6.	Square inches	Square centimetre	6.452
7.	Cubic inches	Cubic centimetres	16.3871
8.	Cubic feet	Cubic metres	0.028317
9.	Gram	Pounds	0.002204585
10.	Pounds	Kilogram	0.45359
11.	Calories	B.T.U.	0.003968
12.	Calories	Joules	4.1868
13.	Kilo calories	Kilowatt-hours	0.0011626
14.	Kilo calories/s	Kilowatt	4.185
15.	Kilowatts	Horsepower	1.341
16.	Foot pounds	B.T.U.	0.0012856
17.	Watts	B.T.U./h	3.413
18.	Watt-hour	Joule	3600
19.	Atmosphere	Millimetre Hg	760
20.	Atmosphere	psi	14.696
21.	Centi-poise	Lb/(ft/s)	0.000672
22.	Centi-poise	Lb/(ft/h)	2.4192
23.	Cal/(cm^2/s/°C/cm)	B.T.U./(ft^2/h/°F/ft)	242.131
24.	B.T.U./(ft^2/h/°F)	Calorie/(m^2/s/°C)	1.3562
25.	B.T.U./(Lb/°F)	Calorie/(g/°C)	1

Universal gas constant:

R = 1.987 B.T.U./(Lb-mole.°R) = 1.987 cal/(g-mole.°C) = 82.057 (l/atm)/(mol/°K).

Temperature conversion:

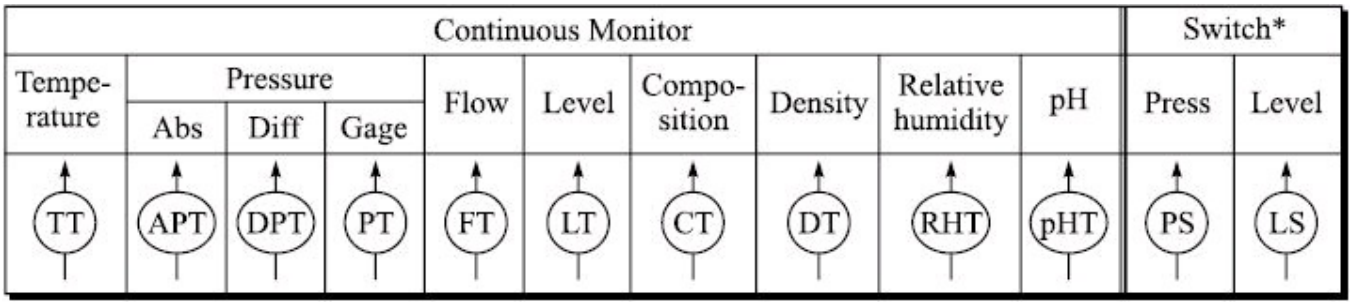
°F = °C × (5/9) + 32; °C = (°F – 32) × 5/9; °R = °F + 459.69; °K = °C + 273.15.

Appendix 1B

Instrument Symbols Used in This Book

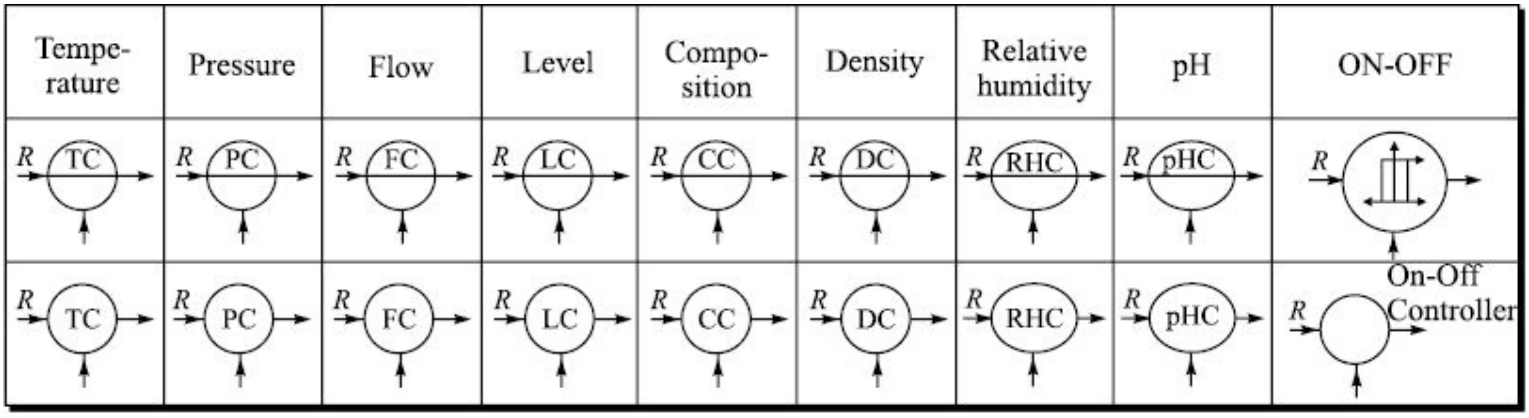
In the following tables, instrument symbols are equipped with outgoing/incoming-outgoing arrows. The outgoing arrows from any instrument indicate the transmitted signal from the hardware and an incoming arrow signifies an entering signal to the hardware from an upstream instrument in a control system. The arrow positions may vary in a P&I diagram due to different orientations of the process elements and instrument position in such a diagram.

Table A1B.1 Process Variable Sensor-Transmitter and Switches



* The switches operate as ON-OFF controllers. They are available with either variable Set point, and constant Differential, or variable Set point and variable Differential. Output from switches may be used to activate Solenoid Valves and/or Alarms.

Table A1B.2 Controllers



In Table A1B.1, the symbols in the upper tier are generally used in the relatively low density diagrams, and for high density, diagrams have been used in high density diagrams involving multiple loop systems.

Table A1B.3 Final Control Elements

Control Valves								Special Elements	
Actuators			Bodies		3-way Valves		Solenoid Valves	Var. Freq. Drives	Thyristors
Pneumtc.	Hydrlc.	Motorzd.	Hi Press	Lo Press	Mixing	Diverting			
	H	M		Butterfly			S	VFD	THY

Table A1B.4 Signal Transducers (Electro-Pneumatic, Analog-to-Digital and Digital-to-Analog), and Alarms

Signal Transducers			Alarms	
EPT	A to D	D to A	Audio	Visual
4–20 mA → I/P → 3–15 psig	4–20 mA → A/D → Digital pulses 4–20 mA → ADC → Digital pulses	Digital pulses → D/A → 4–20 mA Digital pulses → DAC → 4–20 mA		

Table A1B.5 Arithmatic/Algebraic Control Functions

Additn.	Subtrcn.	Multplcn.	Divn.	Lead-Lag	Square	Square root	Exptl.	Spl. funcn.
$Y = X_1 + X_2$	$Y = X_1 - X_2$	$Y = X_1.X_2$	$Y = X_1/X_2$		$Y = A.X^2$	$Y = A.\sqrt{X}$	$Y = A.e^{B.X}$	$Y = f(X)$

Table A1B.6 Logical Control Functions

Lo-selector	Hi-selector	Lo-limiter	Hi-limiter
If $X_L < X_H$, then $Y = X_L$	If $X_L < X_H$, then $Y = X_H$	If $X < X_L$, then $Y = X_L$	If $X > X_H$, then $Y = X_H$

Appendix 2

Laplace Transform

Laplace transform is an operational method of analysis. In process dynamic analysis, this transform has extensive applications, because by this procedure time domain functions are converted to simpler expressions and ordinary differential equations are reduced to algebraic equations. They may be handled more easily in this form. By this transformation, a time domain function or equation is translated into a complex or *s*-domain. Fundamentally the transform is executed by performing the following definition integral:

$$f(s) = L[f(t)] = \int_{-\infty}^{\infty} f(t)e^{-st}dt \qquad \text{(A2.1)}$$

However, in process control applications every variable is considered to have zero values by definition, or by converting the variables from absolute to their deviation form. We have dealt with this topic at the beginning of Chapter 2. Hence, by considering only +ve time part, the definition integral takes the following form:

$$f(s) = L[f(t)] = \int_{0}^{\infty} f(t)e^{-st}dt \qquad \text{(A2.2)}$$

A2.1 Properties of Laplace Transformation

Fundamentally the Laplace transformation of a function is a linear operation. That is, if K is a constant, it follows that:

$$L[Kf(t)] = Kf(s)$$

Also, due to linear property the distributive property is also true, i.e.

$$L[f_1(t) + f_2(t)] = L[f_1(t)] + L[f_2(t)] = f_1(s) + f_2(s)$$

A2.2 Transform of Ideal Time Functions

A2.2.1 Step Function

$f(t) = 0$, for $t < 0$ and $f(t) = M$ (a constant) for $t \geq 0$,

$$f(s) = L[f(t)] = \int_{0}^{\infty} Me^{-st}dt = -\left[\frac{1}{s}e^{-st}\right]_{0}^{\infty} = \frac{M}{s} \qquad \text{(A2.3)}$$

for a step of unit magnitude, i.e. if $M = 1$, $f(s) = 1/s$. It is conventional to denote a unit

step by $U(t)$ or by using lower case as $u(t)$, thus a step of magnitude M may be written as $MU(t)$ or $Mu(t)$.

A2.2.2 Ramp Function

$f(t) = 0$, for $t < 0$ and $f(t) = Mt$ (where M is the constant slope) for $t \geq 0$

$$f(s) = L[f(t)] = \int_0^\infty Mte^{-st}dt = -\left[\frac{1}{s}te^{-st}\right]_0^\infty - \left[1\frac{1}{s^2}e^{-st}\right]_0^\infty = \frac{M}{s^2} \qquad \text{(A2.3)}$$

for a ramp of unit slope, i.e. if $M = 1$, $f(s) = 1/s^2$.

A2.2.3 Sine Function

$f(t) = 0$, for $t < 0$ and $f(t) = M \sin(wt)$

$$f(s) = L[f(t)] = \int_0^\infty M\sin(\omega t)e^{-st}dt = -\frac{Me^{-st}}{s^2+\omega^2}[s\sin(\omega t) + \omega\cos(\omega t)]_0^\infty = \frac{M\omega}{s^2+\omega^2} \qquad \text{(A2.3)}$$

A2.2.4 Exponential Function

$f(t) = 0$, for $t < 0$ and $f(t) = e^{-at}$ for for $t \geq 0$,

$$f(s) = L[f(t)] = \int_0^\infty e^{-(s+a)t}dt = -\frac{1}{s+a}[e^{-(s+a)t}]_0^\infty = \frac{1}{s+a} \qquad \text{(A2.4)}$$

A2.2.5 Transform of Derivatives

First derivative

$$L\left[\frac{df(t)}{dt}\right] = \int_0^\infty \frac{df(t)}{dt}e^{-st}dt$$

to integrate by parts let $u = e^{-st}$ and $v = f(t)$

$$du = -se^{-st} \text{ and } \frac{dv}{dt} = \frac{df(t)}{dt} \quad \text{or} \quad dv = \left[\frac{df(t)}{dt}\right]dt$$

Since, $\int u\,dv = uv - \int v\,du$,

we have $$L\left[\frac{df(t)}{dt}\right] = \int_0^\infty \frac{df(t)}{dt}e^{-st}dt = [f(t)e^{-st}]_0^\infty + s\int_0^\infty f(t)e^{-st}dt = -f(0) + sf(s) \qquad \text{(A2.5)}$$

Second derivative

$$L\left[\frac{d^2 f(t)}{dt^2}\right] = L\left[\frac{d}{dt}\left\{\frac{df(t)}{dt}\right\}\right] = sL\left\{\frac{df(t)}{dt}\right\} - \frac{df(t)}{dt}\Big|_0^\infty = s^2 f(s) - sf(0) - f'(0) \qquad \text{(A2.6)}$$

Generalizing the results the nth derivative may be evaluated as:

n-th derivative

$$L\left[\frac{d^n f(t)}{dt^n}\right] = s^n f(s) - s^{n-1} f(0) - s^{n-2} f^{(1)}(0) - \cdots - sf^{(n-2)}(0) - f^{(n-1)}(0) \qquad \text{(A2.7)}$$

By definition, if the initial value (at $t = 0$) of the variable and all its derivatives have zero values, the transform of derivatives take the following form:

$$L\left[\frac{df(t)}{dt}\right] = sf(s); \quad L\left[\frac{d^2 f(t)}{dt^2}\right] = s^2 f(s); \quad L\left[\frac{d^n f(t)}{dt}\right] = s^n f(s) \qquad \text{(A2.8)}$$

A2.2.6 Transform of Integral

Evaluation of $L\left[\int_0^t f(t)dt\right]$ may be performing the integration of $\int_0^\infty \left(\int_0^t f(t)dt\right) e^{-st}dt$ by parts:

Let $u = e^{-st}$, and $v = \int_0^t f(t)dt$; then $du = -se^{-st}\,dt$ and $dv = f(t)dt$. Applying these results to the integration,

$$f(s) = L\left[\int_0^t f(t)dt\right] = e^{-st}\int_0^t f(t)dt\Big|_0^\infty + s\int_0^\infty \left[\int_0^t f(t)dt\right] e^{-st}dt$$

Term 1 Term 2

The first term vanishes for the upper limit due to the existence of e^{-st} function, also it vanishes for the lower limit by applying the initial zero value assumption. Thus,

$$f(s) = s\int_0^\infty \left[\int_0^t f(t)dt\right] e^{-st}dt = sL\left[\int_0^t f(t)dt\right] \quad \text{or} \quad L\left[\int_0^t f(t)dt\right] = \frac{f(s)}{s} \qquad \text{(A2.9)}$$

This is the required result. We may generalize from the results obtained for differentiation and integration that multiplying $f(s)$ by an s is equivalent of taking the first derivative of $f(t)$, and dividing $f(s)$ by an s is equivalent of single integration of $f(t)$. This has been already proven that the result holds for higher orders of derivatives and it can be also proved for higher orders of integration.

For some commonly encountered time functions, $[f(t)]$, in process dynamics and

control system analysis, and their corresponding transforms, [$f(s)$], have been compiled as a Laplace transform chart (short form). The chart is presented at the end of this chapter.

A2.3 Transfer Functions of a System

If the dynamic characteristics of a process is defined by the following differential equation between its output y and input x,

$$a_n \frac{d^n y}{dt^n} + a_{n-1} \frac{d^{n-1} y}{dt^{n-1}} + \cdots + a_1 \frac{dy}{dt} + a_0 y = b_m \frac{d^m x}{dt^m} + b_{m-1} \frac{d^{m-1} x}{dt^{m-1}} + \cdots + b_1 \frac{dx}{dt} + b_0 x \qquad \text{(A2.10)}$$

and if the initial zero value condition is applicable for the variables, then by transforming both sides of the above equation we obtain:

$$a_n s^n y(s) + a_{n-1} s^{n-1} y(s) + \cdots + a_1 s y(s) + a_0 y(s) = b_m s^m x(s) + b_{m-1} s^{m-1} x(s) + \cdots + b_1 s x(s) + b_0 x(s)$$

This may be expressed in ratio of transformed output to the input variable in the form,

$$\frac{y(s)}{x(s)} = \frac{b_m s^m + b_{m-1} s^{m-1} + \cdots + b_1 s + b_0}{a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0} \qquad \text{(A2.11)}$$

The Equation A2.11 is called the transfer function of the given system. In the following factored form,

$$\frac{y(s)}{x(s)} = \frac{(s - z_1)(s - z_2) \cdots (s - z_{m-1})(s - z_m)}{(s - p_1)(s - p_2) \cdots (s - p_{n-1})(s - p_n)} \qquad \text{(A2.12)}$$

the TF becomes more easily solvable by expanding Equation A2.12 in the Partial Fraction for the system's time domain response for $t \geq 0$, if the input function, $x(t)$, is known for $t \geq 0$. From the consideration of system stability, in Equation A2.12, n must be $\geq m$.

A 2 . 4 Solution of Ordinary Differential Equations by Laplace Transform

In all the following examples of solution of differential equations, the inversion of partial fraction expansion to time domain, we have used the Laplace Transform Chart provided at the end of this chapter.

The solution procedure may be demonstrated by considering few examples.

EXAMPLE A2.1 To find the solution of the ODE, $\frac{d^3 y}{dt^3} + 6\frac{d^2 y}{dt^2} + 11\frac{dy}{dt} + 6y = x$ for all +ve

values of time. The initial conditions are: $y(0) = y\square(0) = y\square\square(0) = 0$, The input $x = 0$,

for $t < 0$, and $x = 2$, for $t \geq 0$.

Step 1. Taking LT of the ODE we obtain,

$$s^3y(s) + 6s^2y(s) + 11sy(s) + 6y(s) = \frac{2}{s}$$

or

$$y(s) = \frac{2}{s(s^3 + 6s^2 + 11s + 6)} = \frac{2}{s(s+1)(s+2)(s+3)} \qquad \text{(A2.13)}$$

It is convenient to express the denominator of the $y(s)$ expression in factored form. If the polynomial in s is equated to zero to form an equation, the factors contain the roots of the equation. The next steps of solution slightly differ depending on the nature of these roots. However, let us proceed with the present given example in which one root is 0 (at the origin of the complex s-plane) and the rest of the roots are real and –ve (lye on the –ve part of the real axis of the complex s-plane).

Step 2. The factored form of y(s) expression in Equation (A2.13) is expanded in terms of partial fraction:

$$y(s) = \frac{2}{s(s+1)(s+2)(s+3)} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+2} + \frac{D}{s+3} \qquad \text{(A2.14)}$$

Inversion to time domain to get the solution is possible after the constants A, B and C are evaluated.

Step 3.

To find A, multiply $y(s)$ by s and put $s = 0$ in the resulting expression, thus, $A = 1/3$.

To find B, multiply $y(s)$ by $s + 1$, and put $s = -1$ in the resulting expression, thus, $B = -1$.

To find C, multiply $y(s)$ by $s + 2$, and put $s = -2$ in the resulting expression, thus, $C = 1$.

To find D, multiply $y(s)$ by $s + 3$, and put $s = -3$ in the resulting expression, thus, $D = -1/3$.

Step 4. Putting the values of these constants in the partial fraction expansion, $y(s)$ becomes,

$$y(s) = \frac{1/3}{s} - \frac{1}{s+1} + \frac{1}{s+2} - \frac{1/3}{s+3} \qquad \text{(A2.14)}$$

Inversion of the individual terms of Equation A2.14 is possible by consulting the Laplace transform table given at the end of this section. Thus the complete solution is obtained as:

$$y(t) = \frac{1}{3} - e^{-t} + e^{-2t} - \left(\frac{1}{3}\right)e^{-3t} \qquad \text{(A2.15)}$$

EXAMPLE A2.2 In this example, we consider the case in which at least one factor in Equation A2.14 has the power more than one, i.e. at least one root is real and repetitive. Let us proceed with the following ODE to find solution for all +ve values of time.

$$\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y = x$$

with initial conditions: $y(0) = y\square(0) = 0$. But the input in this case is different, i.e. $x = 0$, for $t < 0$, and $x = t$, for $t \geq 0$.

Solution: We proceed by following steps shown in the previous example.

Step 1. Taking LT of the ODE we obtain,

$$s^2y(s) + 5sy(s) + 6y(s) = \frac{1}{s^2}$$

or

$$y(s) = \frac{1}{s^2(s^2 + 5s + 6)} = \frac{1}{s^2(s+2)(s+3)} \qquad \text{(A2.16)}$$

Step 2.

$$y(s) = \frac{1}{s^2(s+2)(s+3)} = \frac{A}{s^2} + \frac{B}{s} + \frac{C}{s+2} + \frac{D}{s+3} \qquad \text{(A2.17)}$$

An extra term with constant B is needed in partial fraction expansion to recognize that the system has two roots at the origin of s-plane.

Step 3. To find A, multiply $y(s)$ by s^2 to obtain:

$$\frac{1}{s(s+2)(s+3)} = A + \frac{B}{s} + \frac{Cs}{s+2} + \frac{Ds}{s+3} \qquad \text{(A2.18)}$$

and, put $s = 0$ in Equation A2.18, thus, $A = 1/6$. But to find B, if we multiply $y(s)$ by s and put $s = 0$ in the resulting expression, value of B will appear as infinity, an unacceptable result. This is because there are two roots at the origin of s-plane. To find B, we apply L'Hospital's theorem. Differentiating both sides of Equation A2.18 with respect to s, we obtain:

$$-\frac{2s+5}{(s+2)^2(s+3)^2} = B + \frac{Cs(s+4)}{(s+2)^2} + \frac{Ds(s+6)}{(s+3)^2}$$

and putting $s = 0$ in the expression, the value of $B = -5/36$. Now C and D may be

found by using Equation A2.17 as has been done in the previous example.

To find C, multiply $y(s)$ by $s + 2$, and put $s = –2$ in the resulting expression, thus, $C = 1/4$.

To find D, multiply $y(s)$ by $s + 3$, and put $s = –3$ in the resulting expression, thus, $D = –1/9$.

Putting the values of these constants in the partial fraction expansion, $y(s)$ becomes,

$$y(s) = \frac{1/6}{s^2} - \frac{5/36}{s} + \frac{1/4}{s+2} - \frac{1/9}{s+3} \qquad \text{(A2.19)}$$

The time domain solution may be obtained by inverting Equation A2.19. This would be:

$$y(t) = \left(\frac{1}{6}\right)t - \left(\frac{5}{36}\right) + \left(\frac{1}{4}\right)e^{-2t} - \left(\frac{1}{9}\right)e^{-3t} \qquad \text{(A2.20)}$$

In this example, the repeated roots at $s = 0$ have appeared due to the time dependent input function, [$x = t$]. But such a situation may occur from the dynamic characteristics of the system itself, i.e. from the ODE representing the system.

EXAMPLE A2.3 To find the solution of the ODE: $\alpha^3 \frac{d^3y}{dt^3} + 3\alpha^2 \frac{d^2y}{dt^2} + 3\alpha \frac{dy}{dt} + y = x$, for all

+ve values of time. The initial conditions are: $y(0) = y\square(0) = y\square\square(0) = 0$. The input $x = 0$, for

$t < 0$, and $x = 1$, for $t \geq 0$.

Solution:

Taking LT of the ODE we obtain,

$$\alpha^3 s^3 y(s) + 3\alpha^2 s^2 y(s) + 3\alpha s y(s) + y(s) = \frac{1}{s}$$

or

$$y(s) = \frac{1}{s(\alpha^3 s^3 + 3\alpha^2 s^2 + 3\alpha s + 1)} = \frac{1}{s(s+\alpha)^3} \qquad \text{(A2.21)}$$

or

$$y(s) = \frac{1}{s(s+\alpha)^3} = \frac{A}{s} + \frac{B}{(s+\alpha)^3} + \frac{C}{(s+\alpha)^2} + \frac{D}{s+\alpha} \qquad \text{(A2.22)}$$

To find A, Equation A2.22) is multiplied by s, and $s = 0$ is put into the result thus, $A = 1/a^3$.

To find B, Equation A2.22 is multiplied by $(s + a)^3$ to obtain:

$$\frac{1}{s} = \frac{A(s+\alpha)^3}{s} + B + C(s+\alpha) + D(s+\alpha)^2 \qquad \text{(A2.23)}$$

Putting $s = -a$ in Equation A2.23, we get $B = -(1/a)$

To find C, Equation A2.23 is differentiated with respect to s to obtain:

$$-\frac{1}{s^2} = A\left[\frac{3(s+\alpha)^2}{s} - \frac{(s+\alpha)^3}{s^2}\right] + C + 2D(s+\alpha) \qquad \text{(A2.24)}$$

Putting $s = -a$ in Equation A2.24, we get $C = -(1/a^2)$

To find D, Equation A2.24 is differentiated with respect to s and putting $s = -a$ in the result, we obtain $D = -(1/a^3)$.

Putting the values of these constants in the partial fraction expansion, $y(s)$ becomes,

$$y(s) = \frac{1/\alpha^3}{s} - \frac{1/\alpha}{(s+\alpha)^3} - \frac{1/\alpha^2}{(s+\alpha)^2} - \frac{1/\alpha^3}{(s+\alpha)} \qquad \text{(A2.25)}$$

The time domain solution may be obtained by inverting Equation A2.19. This would be:

$$y(t) = \frac{1}{\alpha^3} - \frac{1}{2\alpha}t^2e^{-\alpha t} - \frac{1}{\alpha^2}te^{-\alpha t} - \frac{1}{\alpha^3}e^{-\alpha t} \qquad \text{(A2.26)}$$

The next categories of factors that may appear in the expression of $y(s)$, and require special treatment in finding the related constants, are Imaginary or Complex conjugates. As in the case of repetitive roots, they may appear from the input function or may be embedded in the dynamic characteristics of the system. We shall demonstrate the treatment of this kind of roots by considering the following two examples.

EXAMPLE A2.4 To find the solution of the ODE: $5\frac{dy}{dt} + y = x$, for all +ve values of time

The initial condition is: $y(0) = 0$. The input $x = 0$, for $t < 0$, and $x = M \sin(wt)$, for $t \geq 0$.

Solution: Taking LT of the ODE, we obtain,

$$5sy(s) + y(s) = x(s)$$

and taking LT of $x(t)$, we obtain: $x(s) = \frac{M\omega}{s^2 + \omega^2}$.

Thus,

$$y(s) = \frac{1}{5s+1} \cdot \frac{M\omega}{s^2 + \omega^2} = \frac{M\omega}{(5s+1)(s+i\omega)(s-i\omega)}$$

or

$$\frac{M\omega}{(5s+1)(s+i\omega)(s-i\omega)} = \frac{A}{5s+1} + \frac{B}{s+i\omega} + \frac{C}{s+i\omega} \qquad \text{(A2.27)}$$

To find *A*, multiply both sides of Equation A2.27 by $s + 1/5$, and put $s = -1/5$.

Thus, $$A = \frac{5M\omega}{25\omega^2 + 1}$$

To find *B*, multiply both sides of Equation A2.27 by $s + iw$, and put $s = -iw$.

Thus, $$B = -\frac{M}{2}\left(\frac{5\omega + i}{25\omega^2 + 1}\right)$$

To find *C*, multiply both sides of Equation A2.27 by $s - iw$, and put $s = iw$.

Thus, $$C = \frac{M}{2}\left(\frac{5\omega + i}{25\omega^2 + 1}\right)$$

Putting these in Equation A2.27,

$$y(s) = \frac{5M\omega}{25\omega^2 + 1} \cdot \frac{1}{s + 1/5} - \frac{M}{2} \cdot \frac{5\omega + i}{25\omega^2 + 1} \cdot \frac{1}{s + i\omega} + \frac{M}{2} \cdot \frac{5\omega + i}{25\omega^2 + 1} \cdot \frac{1}{s - i\omega} \qquad \text{(A2.28)}$$

The second and third terms of Equation A2.28 may be inverted to time domain by using the Inversion Theorem for Complex Roots:

$$\text{If } x(s) = \frac{u + iv}{s + j + ik} + \frac{u - iv}{s + j - ik}, \text{ then } y(t) = 2e^{-jt}\{u\cos(kt) + v\sin(kt)\} \qquad \text{(A2.29)}$$

Comparing Equation A2.28 with A2.29, the following values of the constants are assigned: $u = 5w$; $v = 1$; $j = 0$; $k = w$. Using the constant values in Equation A2.29, the complete time domain solution may be given as:

$$y(t) = \frac{5M\omega e^{-t/5}}{25\omega^2 + 1} - \frac{5M\omega}{25\omega^2 + 1}\cos(\omega t) + \frac{M}{25\omega^2 + 1}\sin(\omega t) \qquad \text{(A2.30)}$$

The complex roots may exist in the ODE representing the System characteristics.

EXAMPLE A2.5 To find the solution of the ODE: $\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 13y = x$, for all +ve values

of time. The initial condition is: $y'(0) = y(0) = 0$. The input $x = 0$, for $t < 0$, and $x = 1$, for $t \geq 0$.

Solution:

Taking LT of the ODE, we obtain, $s^2y(s) + 4sy(s) + 13y(s) = x(s)$

and taking LT of $x(t)$, we obtain: $x(s) = 1/s$.

Thus,

$$y(s) = \frac{1}{s(s^2 + 4s + 13)} = \frac{1}{s(s + 2 - 3i)(s + 2 + 3i)}$$

or

$$\frac{1}{s(s + 2 - 3i)(s + 2 + 3i)} = \frac{A}{s} + \frac{B}{s + 2 - 3i} + \frac{C}{s + 2 + 3i} \qquad \text{(A2.31)}$$

To find A, multiply both sides of Equation A2.31 by s, and put $s = 0$ to get, $A = 1/13$.

To find B, multiply both sides of Equation A2.31 by $s + 2 - 3i$, and put $s = -2 + 3i$ to get,

$$B = \left(\frac{-18 + 12i}{468}\right)$$

To find C, multiply both sides of Equation A2.31 by $s + 2 + 3i$, and put $s = -2 - 3i$ to get,

$$C = \left(\frac{-18 - 12i}{468}\right)$$

Putting these in Equation A2.31,

$$y(s) = \frac{1}{13s} + \frac{1}{468}\left(\frac{-18 + 12i}{s + 2 - 3i}\right) + \frac{1}{468}\left(\frac{-18 - 12i}{s + 2 + 3i}\right) \qquad \text{(A2.32)}$$

Following the procedure shown in the previous example, we assign the values of constants of the Inversion Theorem for Complex Roots for the second and third terms of Equation A2.32, $u = -18\ ; v = 12\ ; j = 2\ ; k = 3$

$y_{T2,T3}(t) = \frac{1}{234} e^{-2t}\{-18\cos(3t) + 12\sin(3t)\}$, and the complete solution becomes,

$$y(t) = \frac{1}{13} + e^{-2t}\left[\frac{1}{19.5}\sin(3t) - \frac{1}{13}\cos(3t)\right] \qquad \text{(A2.33)}$$

We may generalize as, whether a type of root appears due to input variable specification or from the process characteristics the solution procedure would only depend on the type of root.

A2.5 Time Translation Theorem

If a function of time $f(t)$ that may be a process variable, is displaced along the time axis then this displaced or translated function is *delayed* in time. As we have seen in

Chapter 6 that process time delays are caused by *transportation lag*. This phenomenon shown in Figure A2.1 is known as *dead time*. Because Laplace transform does not contain information about the original function for –ve time, the delayed function must be assumed to be zero for times less than the time delay. The theorem is stated as:

$$L[f(t-q)] = e^{-sq} f(s) \qquad \text{(A2.34)}$$

Proof: Using the definition of Laplace transform, we get,

$$L[f(t-\theta)] = \int_0^\infty f(t-\theta)e^{-st}dt \qquad \text{(A2.35)}$$

Let, $t = t - q$; or, $t = q + t$. Putting this result in Equation A2.35,

$$L[f(t-\theta)] = \int_{\tau=-\theta}^{\tau=\infty} f(\tau)e^{-s(\theta+\tau)}d(\theta+\tau) = \int_{\tau=0}^{\tau=\infty} f(\tau)e^{-s\theta}e^{-s\tau}d\tau$$

$$= e^{-s\theta}\int_0^\infty f(\tau)e^{-s\tau}d\tau = e^{-s\theta}f(s)$$

Hence, the theorem is proved. The proof has made use of the fact that $f(t) = 0$ for $t < 0$ $[t < q]$.

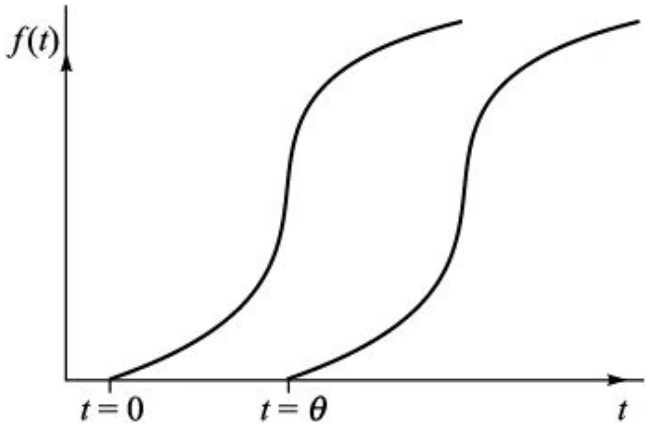


Figure A2.1 Time translation of $f(t)$.

A2.6 Transform of Unit Pulse Function

Find the Laplace transform of $f(t)$ having the following time domain specification:

$$f(t) = 0, \text{ for } t < 0; f(t) = H, \text{ for } 0 < t < \frac{1}{H}; f(t) = 0, \text{ for } t > \frac{1}{H}$$

The function called unit pulse may be represented by two successively occurring step functions. First, a +ve step magnitude H appearing at $t = 0$, the next, a –ve step of identical magnitude occurs at $t = 1/H$. According to the linear property of Laplace transform, these functions may be superimposed to obtain the unit pulse function

shown in Figure A2.2. Hence,

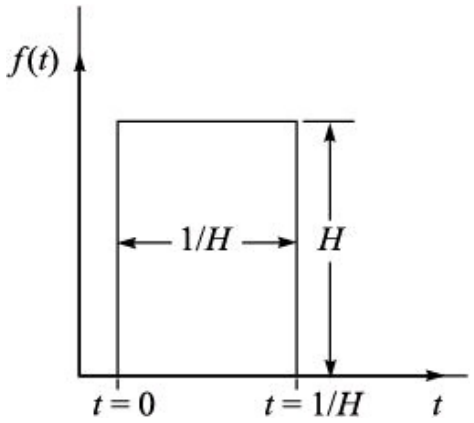


Figure A2.2 Unit pulse

$$f(t) = H[u(t) - u\left(t - \frac{1}{H}\right)$$

where, $u(t - 1/H)$ is the unit step translated $1/H$ units of time to the right. Now by using the linearity property of Laplace transform, we may write,

$$f(s) = H\left[\frac{1 - e^{-s/H}}{s}\right] \qquad \text{(A2.36)}$$

A2.7 Transform of Unit Impulse Function

Let the width of the pulse $(1/H)$ in Figure A2.2 be allowed to shrink while its height is increased so that the area under the function always remains equal to 1. As $1/H$ approaches zero, H will tend to infinity. In the limiting conditions, the function shown in Figure A2.3 is called unit impulse function and it is usually represented by $d(t)$. The Laplace transform of $d(t)$ may be obtained by letting $1/H$ go to zero in Equation A2.36. Applying L'Hospital's rule,

$$L[\delta(t)] = \underset{1/H \to 0}{\text{Lt}}\left[\frac{1 - e^{-s(1/H)}}{(1/H)s}\right] = \underset{1/H \to 0}{\text{Lt}}\left[\frac{se^{-s(1/H)}}{s}\right] = 1 \qquad \text{(A2.37)}$$

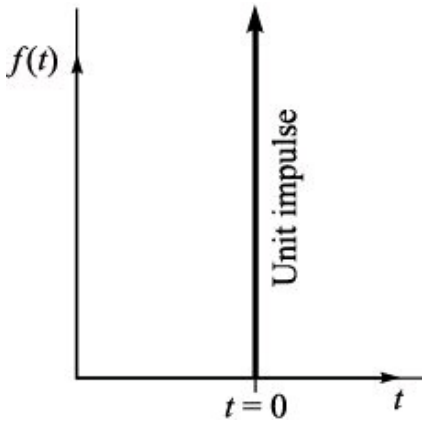


Figure A2.3 Unit impulse.

A2.8 Final Value Theorem

If $f(s)$ is the Laplace transform of $f(t)$, then,

$$\underset{t\to\infty}{\text{Lt}}\,[f(t)] = \underset{s\to 0}{\text{Lt}}\,[sf(s)] \qquad \text{(A2.38)}$$

provided $sf(s)$ does not become infinity for any value of s, satisfying $Re(s) \geq 0$. If this condition does not hold, $f(t)$ does not approach a limit as $t \to \infty$.

Proof: Let us rewrite the Laplace transformation of the first derivative of a function $f(t)$,

$$\int_0^\infty \frac{df(t)}{dt} e^{-st} dt = sf(s) - f(0)$$

Hence,

$$\underset{s\to 0}{\text{Lt}} \int_0^\infty \frac{df(t)}{dt} e^{-st} dt = \underset{s\to 0}{Lt}\,[sf(s)] - f(0)$$

It may be shown that the order of integration and limit operation on the left side of the above equation can be interchanged, provided the conditions of the theorem are applicable for the given system. Performing this operation,

$$\int_0^\infty \frac{df(t)}{dt} = \underset{s\to 0}{\text{Lt}}\,[sf(s)] - f(0)$$

Evaluating the integral, $\underset{t\to\infty}{\text{Lt}}\,[f(t)] - f(0) = \underset{s\to 0}{\text{Lt}}\,[sf(s)] - f(0)$ or, $\underset{t\to\infty}{\text{Lt}}\,[f(t)] = \underset{s\to 0}{\text{Lt}}\,[sf(s)]$.

Hence, the theorem is proved. The final value theorem allows us to compute the value that a function approaches as $t \to \infty$, when its Laplace transform is known.

EXAMPLE A2.6 Find the value of $f(t)$ as $t \to \infty$, when $f(s) = \dfrac{s+2}{s(s+1/3)(s+1/2)(s+1)}$.

Solution: Using the final value theorem,

$$\underset{t\to\infty}{\text{Lt}}\, f(t) = \underset{s\to 0}{\text{Lt}}\,[sf(s)] = \underset{s\to 0}{\text{Lt}}\left[\frac{s(s+2)}{s(s+1/3)(s+1/2)(s+1)}\right] = \frac{2}{(1/3)(1/2)(1)} = 12$$

A2.9 Initial Value Theorem

If $f(s)$ is the Laplace transform of $f(t)$, then,

$$\underset{t\to 0}{\text{Lt}}\,[f(t)] = \underset{s\to\infty}{\text{Lt}}\,[sf(s)] \qquad \text{(A2.39)}$$

The proof of this theorem is omitted because this may be performed in the same way as we have done for the final value theorem. The initial value theorem allows us

to compute the value that a function starts after being disturbed or approaches backward as $t \to 0$, when its Laplace transform is known.

EXAMPLE A2.7 Find the value of $f(t)$ at $t = 0$, when $f(s) = \dfrac{(s-1)(s+1)}{s(s+3)(s-4)}$.

Solution: Using the final value theorem,

$$\lim_{t\to 0} f(t) = \lim_{s\to\infty}[sf(s)] = \lim_{s\to\infty}\left[\frac{s(s-1)(s+1)}{s(s+3)(s-4)}\right] = \lim_{s\to\infty}\left[1+\frac{s+11}{s^2-s-12}\right] = 1$$

A2.10 Derivation of First Order Response Equations for Ideal Input Functions

The following first order Transfer Function has been used for deriving the following response equations:

$$\frac{Y(s)}{X(s)} = \frac{1}{\tau s+1} \qquad \text{(A2.40)}$$

A2.10.1 Step Input

Step input of magnitude M, i.e. $X(t) = M$. By taking Laplace transform, $X(s) = M/s$. Putting $X(s)$ in Equation A2.40:

$$Y(s) = \frac{M}{s(\tau s+1)} = \frac{M}{\tau}\left(\frac{1}{s(s+1/\tau)}\right) \qquad \text{(A2.41)}$$

Let us expand the above Equation A2.41 in terms partial fraction as:

$$Y(s) = \frac{M}{\tau}\left(\frac{1}{s(s+1/\tau)}\right) = \left(\frac{A}{s}+\frac{B}{s+1/\tau}\right) \qquad \text{(A2.42)}$$

To find A, multiply both sides of Equation A2.41 by s, and put $s = 0$. Thus $A = M$

To find B, multiply both sides of Equation A2.41 by $s + 1/t$, and put $s = -(1/t)$. Thus $B = -M$

Equation A2.41 becomes: $Y(s) = \dfrac{M}{s} - \dfrac{M}{s+1/\tau}$

(A2.43)

By inverting to time domain; $Y(t) = M - Me^{-t/t} = M(1 - e^{-t/t})$

(A2.44)

A2.10.2 Ramp Input

Ramp input of slope M, i.e. $X(t) = Mt$. By taking Laplace transform, $X(s) = M/s^2$. Putting $X(s)$ in Equation A2.40 and expanding by way of partial fractions,

$$Y(s) = \frac{M}{s^2(\tau s+1)} = \frac{M}{\tau}\left(\frac{1}{s^2(s+1/\tau)}\right) = \frac{A}{s^2} + \frac{B}{s} + \frac{C}{s+1/\tau} \qquad \text{(A2.45)}$$

To find A, multiply both sides of Equation A2.41 by s^2, and put $s = 0$. Thus, $A = 1$
To find B, we have to use L'Hospital's theorem. Differentiating Equation A2.45 with respect to s;

$$-\frac{M}{\tau}\cdot\frac{1}{(s+1/\tau)^2} = B + Cs\left[\frac{(s+1/\tau)-s}{(s+1/\tau)^2}\right]$$

Putting $s = 0$, we obtain, $B = -Mt$

To find C, multiply both sides of Equation 5 by $(s + 1/t)$, and put $s = -(1/t)$. Thus $C = Mt$

Equation A2.45 becomes:

$$Y(s) = \frac{M}{s^2} - \frac{M\tau}{s} + \frac{M\tau}{s+1/\tau} \qquad \text{(A2.46)}$$

By inverting to time domain:

$$Y(t) = Mt - M\tau + M\tau e^{-t/\tau} = M(t - \tau + \tau e^{-t/\tau}) \qquad \text{(A2.47)}$$

A2.10.3 Impulse Input

Impulse input of magnitude, M, i.e. $X(t) = Md(t)$. By taking Laplace transform, $X(s) = M$. Putting $X(s)$ in Equation A2.40:

$$Y(s) = \frac{M}{\tau s+1} = \frac{M}{\tau}\cdot\frac{1}{(s+1/\tau)} \qquad \text{(A2.48)}$$

Equation A2.48 may be straightway inverted to time domain by consulting the Laplace transform chart given above.

$$Y(t) = \frac{M}{\tau}e^{-t/\tau} \qquad \text{(A2.49)}$$

A2.10.4 Sinusoidal Input

The Sinusoidal input is given by: $X(t) = M\sin(wt)$. By taking Laplace transform,

$$X(s) = \frac{M\omega}{s^2+\omega^2}$$

Putting $X(s)$ in Equation A2.40:

$$Y(s) = \frac{M\omega}{s^2+\omega^2}\cdot\frac{1/\tau}{s+1/\tau} = \frac{M\omega}{\tau}\left[\frac{1}{(s+i\omega)(s-i\omega)(s+1/\tau)}\right] = \frac{A}{s+i\omega} + \frac{B}{s-i\omega} + \frac{C}{s+1/\tau} \qquad \text{(A2.50)}$$

To find A, multiply both sides of Equation A2.50 by $s + iw$, and put $s = -iw$.

Thus, $$A = -\frac{M}{2}\left(\frac{\tau\omega + i}{\tau^2\omega^2 + 1}\right)$$

To find *B*, multiply both sides of Equation A2.50 by *s – iw*, and put *s = iw*.

Thus, $$B = -\frac{M}{2}\left(\frac{\tau\omega + i}{\tau^2\omega^2 + 1}\right)$$

To find *C*, multiply both sides of Equation A2.50 by *s + 1/t*, and put *s = – 1/t*.

Thus, $$C = \frac{M\omega\tau}{\tau^2\omega^2 + 1}$$

Putting these in Equation A2.50,

$$Y(s) = -\frac{M}{2}\cdot\frac{\tau\omega + i}{\tau^2\omega^2 + 1}\cdot\frac{1}{s + i\omega} + \frac{M}{2}\cdot\frac{\tau\omega + i}{\tau^2\omega^2 + 1}\cdot\frac{1}{s - i\omega} + \frac{M\omega\tau}{\tau^2\omega^2 + 1}\cdot\frac{1}{s + 1/\tau} \qquad \text{(A2.51)}$$

The first two terms of the above expression may be inverted to time domain by using the Inversion Theorem for Complex Roots:

If $X(s) = \frac{a_1 + ib_1}{s + k_1 + ik_2} + \frac{a_1 - ib_1}{s + k_1 - ik_2}$; Then, $X(t) = 2e^{-k_1 t}\{a_1\cos(k_2 t) + b_2\sin(k_2 t)\}$

$$Y(t) = \frac{M\omega\tau e^{-t/\tau}}{\tau^2\omega^2 + 1} - \frac{M\omega\tau}{\tau^2\omega^2 + 1}\cos(\omega t) + \frac{M}{\tau^2\omega^2 + 1}\sin(\omega t) \qquad \text{(A2.52)}$$

we shall use the following trigonometric identity to condense the two periodic functions in Equation (A2.52)

$$p\cos A + q\sin A = r\sin(A + j)$$

where, $r = \sqrt{(p^2 + q^2)}$, and $\phi = \tan^{-1}\left(\frac{p}{q}\right)$

Applying the above identity, Equation A2.52 becomes,

$$Y(t) = \frac{M\omega\tau}{1 + \tau^2\omega^2}e^{-t/\tau} + \frac{M}{\sqrt{1 + \tau^2\omega^2}}\sin\left[\omega t - \tan^{-1}(\omega\tau)\right] \qquad \text{(A2.53)}$$

As in the Ramp response equation, first term of Equation A2.52 contains time decay factor ($e^{-t/t}$) that will become negligible at large values of time. Ignoring this transient term, we may have the stable part of the response, as:

$$Y(t) = \frac{M}{\sqrt{1 + \tau^2\omega^2}}\sin[\omega t - \tan^{-1}(\omega\tau)] \qquad \text{(A2.54)}$$

(***Note: In all of the above derivations, the inversion from Laplace to time domain***

have been effected by consulting the Transform Chart provided in the following.)

A2.11 Laplace Transform Chart (Short Form)

No	*f(t)*	*f(s)*		*f*(*t*)	*F*(*s*)
1	M	M/s	**8**	$M\cos(\sim t)$	$Ms/(s^2+\sim^2)$
2	Mt	M/s^2	**9**	$M\sinh(\sim t)$	$Mw/(s^2-\sim^2)$
3	Mt^n	$Mn!/s^{n+1}$	**10**	$M\cos(\sim t)$	$Ms/(s^2-\sim^2)$
4	$e^{-at}M$	$M/(s+a)$	**11**	$Me^{-at}\sin(\sim t)$	$M\sim/[(s+a)^2+\sim^2]$
5	$\mathrm{d}(t)$	1	**12**	$Me^{-at}\cos(\sim t)$	$[Ms+a]/[(s+a)^2+\sim^2]$
6	$M\mathrm{d}(t)$	M	**13**	$df(t)/dt$	$sf(s)-f(0)$
7	$M\sin(\sim t)$	$M\sim/(s^2+\sim^2)$	**14**	$\int f(t)\,dt$	$f(s)/s$

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[2] Coughanowr, D.R. and Kopel, L.B., *Process System Analysis and Control*, Ch-2, 3 & 4, McGraw-Hill Inc., 1965.

Appendix 3A

Control Valves, Their Steady State Characteristics and Sizing

In Chapter 7 we proposed the dynamic model of control valve with two types of actuators, pneumatic and electric motor driven. Further to this dynamic consideration, it is essential to have knowledge about steady state characteristic of control valves to understand their working as a final control element and selecting the correct valve for a given application.

A3A.1 Introduction

Control Valves are an important part of process control system. They are the most commonly found final control element (FCE) found in process industry. They are used to regulate flows of manipulated variable for keeping the controlled variable at its set point. Operationally they may be thought as power amplifier for increasing the strength of controller output signal, which is essentially low in power, up to an adequate level for moving heavy mechanism required for throttling of high energy fluid flow. A control valve, in effect, acts as a variable resistance in a process pipe. By changing its opening, it changes the resistance to flow and, thus, flow itself.

A3A.2 Construction of a Control Valve

There are so many varieties and classes of control valves available commercially, that it is difficult to present a general description. However, description of any of them will expose the basic function and interrelation between different parts. For this purpose we choose a Control Valve with Single Plug-seat assembly and Pneumatic actuator. It is probably the oldest and most widely used of its class. Figure A3A.1 represents such a design. The **actuator** is where power amplification of the controller output signal takes place. As for example, 15 psig air pressure signal on a 10-inch diameter diaphragm can produce force, to the tune of 1200 lb or half of a ton. The stem movement attached to the diaphragm occurs against the spring force until the balance point is reached. As air pressure is the source of energy, hence, the arrangement is called pneumatic actuator.

There are other modes of actuator used in control valves, e.g. **Hydraulic actuators** are used where extreme high force is required for sealing high pressure or large flow

rates or both. The basic unit comprises a piston-cylinder assembly with or without a servo action. A hydraulic fluid is used to communicate the pressure. Sealing force of the order of 20 ton and above may be generated in such systems. **Electric Motor driven actuators** are used for more fast and high precision response. High torque Servos, as well as Stepper motors are used for this purpose.

In many designs, the force balance spring is replaceable, so that the valve could be used in other pneumatic specification than 3–15 psig. The valve stem is exposed between the frame where a **pointer** on a **scale** is provided for indicating stem position or the valve opening.

Except for a very small size, the top portion comprising of the actuator and stem-position indicator, is either screwed, or flanged, or welded to the flow line assembly. Flow line assembly is actually the **valve body** containing the **valve seat**, fluid **inlet port** and **outlet port,** and the **bonnet** through which the stem enters into the valve body. At the stem entry point, a **seal or stuffing box** is provided against any leakage. A **lubrication port** is provided in large size valves for lubricant introduction to ensure smooth movement of the stem. At the end of the stem, the **valve plug** is welded.

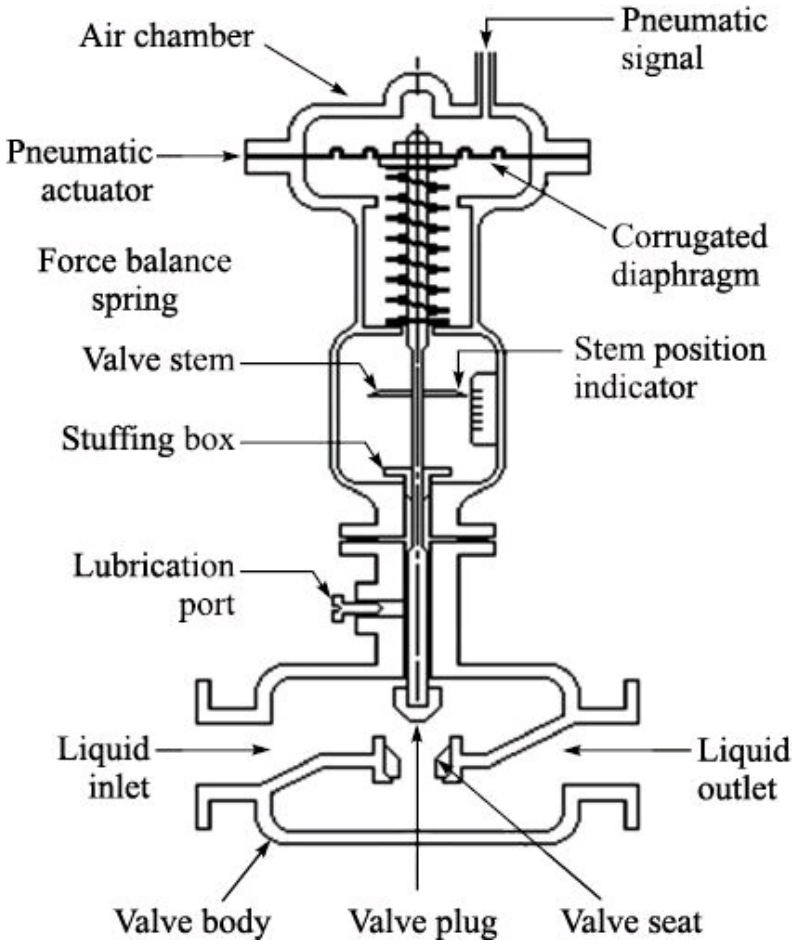


Figure A3A.1 A pneumatically actuated control valve.

The plug seats on the **valve seat** are attached to the body tightly enough at zero flow condition of the valve. There are designs where double seats and double plug assembly is used to obtain more sealing pressure. The profile of the seat and the plug

are carefully designed and matched to obtain required flow characteristic of the valve.

A3A.3 Flow Characteristics of Control Valves

The flow characteristic of a valve is defined by the flow behaviour of the valve as it is operated through its rated stem length. It is usually presented by a plot of valve opening or stem travel, x, in percent of its full value, against flow, Q, in percent of its maximum.

There are two kinds of characteristics that a valve possesses:

1. **Inherent characteristic** is obtained with a designated constant pressure drop (ΔP = 1psi) maintained across the valve body between inlet to outlet ports. This is the basic flow characteristic built into the valve by its manufacturer. Inherent characteristic is obtained considering the valve as the only resistive element in a flow-head system. The inherent characteristic can be generated on a testing setup by creating different values of port opening or stem position, and adjusting the flow to generate ΔP = 1 psi across the valve. A series of recorded values of flow through the valve and corresponding valve opening can be plotted to produce the characteristic curve. In this testing procedure, the effect of pressure drop due to the associated pipe line and other resistive elements on flow through the valve and variable pressure drop across the valve, which happens in a practical application are eliminated.

2. **Installed characteristic** of a valve is shown as a plot of flow versus valve opening relationship, when the valve is operating in a given flow system. In such a situation, the shape of its characteristic changes from the inherent curve given by the manufacturer. The most common reason for such a change is the variation in pressure drop across the valve as the flow changes in response to the demand of the process. This alteration of the inherent characteristic is often important and must be considered when analyzing a given control problem.

A3A.3.1 Inherent Characteristic of Control Valve

According to the relation between %Maximum Flow, Q, and %Stem Travel, x, or the inherent characteristic, valves fall into three main categories:

(1) **Linear characteristic:** The characteristic curve bears a linear relation between x and Q. Thus,

$$Q = Kx \tag{A3A.1}$$

where Q = % of maximum flow under constant pressure drop; x = % of maximum stem travel; K = A constant. The characteristic curve is given in Figure A3A.2(a).

(2) **Equal percent characteristic:** The equal percent character may be stated as: from an initial value of 100% opening, if 20% stem travel reduces the flow by 20% from

100%, then the next 20% travel will produce another 20% flow reduction of the remaining 80%, that is, after the second 20% stem travel, flow will be $80 - 80 \times 0.2 = 64\%$. In a similar way, it can be shown that after third, fourth and fifth travel, each of 20%, the corresponding stem position or port opening are 41.2%, 32.96% and 26.368% respectively. We can examine the results after each 20% travel in the following table.

x %	100%	80%	60%	40%	20%	0%
Q %	100%	80%	64%	41.2%	32.96%	26.368%

The objective of this analysis is two fold, one to show that even after 100% stem travel from fully open condition, the valve is still open to some extent, thus there will be a leakage flow even when the valve is apparently in fully closed condition, and such relation should remind us of the 'half-life' of a reacting species following an irreversible first order reaction, in which the starting amount of the reactant gets reduced by 50% during the half-life time period. In the next half-life period, the remaining 50% is reduced to 25% and so on. Thus, we should expect that a similar exponential relation between Q and x exists as it does between remaining amount of reactant and time, in case of a first order reaction. In fact the relation may be expressed as:

$$Q = be^{ax} \tag{A3A.2}$$

where b and a are constants. These constants can be evaluated considering the minimum or leakage flow, Q_0, at minimum (x = 0%) stem travel. Hence, $a = Q_0$. Constant b can be represented by introducing a ratio factor R. R is defined as the ratio of flow rates through the valve at its maximum and minimum opening, under constant ΔP. With these modifications Equation A3A.2 becomes,

$$Q = Q_0 e^{(\ln R)(x/X)} \tag{A3A.3}$$

where, X is maximum value of x (i.e., 100%). For example, for R = 50, i.e. 2% leakage flow, Equation A3A.3 becomes,

$$Q = 2e^{0.0391x} \tag{A3A.4}$$

Characteristic curves of equal % valve for R = 25, 50 and 100 corresponding to Q_O = 4, 2 and 1%, are shown in Figure A3A.2(b).

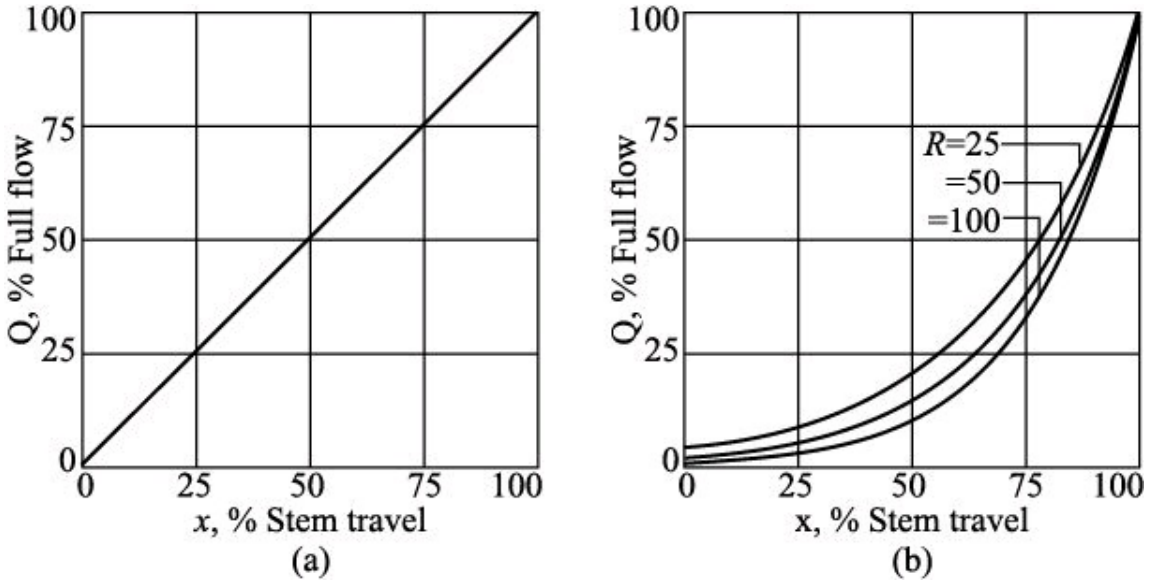


Figure A3A.2 Characteristic curves for (a) Linear, (b) Equal percent valves.

The second objective is that, if such a valve is properly designed and combined with a process having saturation type nonlinearity, the combination would show a much extended linear character.

The Valve Flow Coefficient C_V

Before proceeding, it is required to introduce the valve flow coefficient, C_V, a measure of flow through the fully open valve that occurs under a prescribed pressure drop condition. By definition C_V equals the number of US-gallons of water that flows through a fully open valve, in 1 min, under the pressure difference of 1 psi across the valve faces. In fact, C_V is a capacity index that is reported in valve manufacturer's literature against all available sizes and characteristics to help designers in choosing the correct size and characteristic of the valve in a control problem. Excerpts from such a literature has been shown in Table A3A.1.

A little later, we shall present quantitative relation for calculation of C_V for different types of fluids and thence to arrive at the right size of the valve for a given application. At present, it should be noted that such a flow coefficient may exist for any flow resistance in the form of an orifice, an aperture, or a lengthy pipeline as in the tube side of a multiple pass heat exchanger or a tank coil dipped in a reactor for heat transfer. For a flowing liquid, the general relation between such a flow coefficient and the flow parameters may be given as:

$$Q = C_F \sqrt{\frac{\Delta P}{\rho}} \qquad \text{(A3A.5)}$$

where, Q = Volumetric flow rate; C_F = General notation for the flow coefficient; ΔP = Pressure drop across the valve; r = Density of the liquid. With this much of pre-requisite, we may go for establishing the installed characteristics of control valves.

A3A.3.2 Installed Characteristics of Control Valve

Let us define a liquid flow system in which a control valve supplies a liquid stream at variable flow rate to a process for a given control objective. Such a system is illustrated in Figure A3A.3. The flow coefficients of valve and the process are given as C_V and C_P respectively. The flow regulation occurs by changing the port area, which results in a variation in flow resistance.

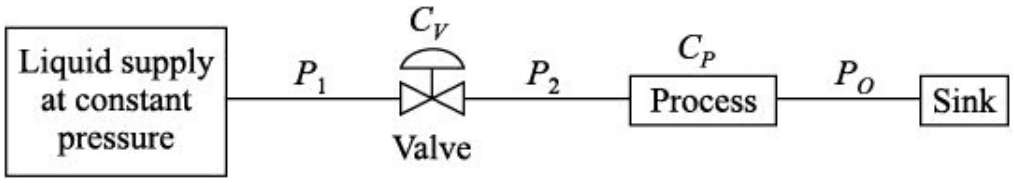


Figure A3A.3 A flow control scheme by a control valve.

The flow resistance of the process essentially remains constant while the valve resistance changes for flow regulation. If the flowing liquid is water with $r = 1$, Equation A3A.5 for maximum flow can be written as:

$$Q_{MAX}^2 = C_V^2(P_1 - P_2) = C_P^2(P_2 - P_O) \qquad \text{(A3A.6)}$$

Eliminating P_2 from Equation A3A.6,

$$Q_{MAX}^2 = \frac{P_1 - P_O}{1/C_V^2 + 1/C_P^2} \qquad \text{(A3A.7)}$$

For a linear valve, let C_V represent the maximum valve opening. Let for a fraction or percent Q, of maximum flow Q_{MAX}, the fraction or percent x, of the maximum stem travel X, then the fractional opening could be represented by $x.C_V$. Putting these quantities in Equation A3A.7,

$$(QQ_{MAX})^2 = \frac{P_1 - P_O}{1/(xC_V)^2 + 1/C_P^2} \qquad \text{(A3A.8)}$$

Relation between fractional flow stem can be obtained by dividing Equation A3A.8 by A3A.7 and taking square root of the result:

$$Q = \left[\frac{(C_V/C_P)^2 + 1}{(C_V/C_P)^2 + (1/x)^2}\right]^{0.5} \qquad \text{(A3A.9)}$$

Usually, the flow resistance due to process is equal or higher than the valve resistance so that the ratio C_V/C_P is equal to or greater than 1.0. We have plotted the result of Equation A3A.9 in Figure A3A.4(a) as the installed characteristic of a linear valve for $C_V/C_P = 1$, 2.5, 5 and 10.

In Equation A3A.8, the term xC_V actually represents fraction of full opening. It has been possible to write stem position in place of port opening due to the linear nature

of valve characteristic. For an equal percent valve, the fractional stem position, x, is related to the fraction of maximum flow or port opening by Equation A3A.3. To find the installed characteristics of an equal percent valve, we shall use Equations A3A.3 and A3A.9 in the following form:

$$y = Q_0 e^{(\ln R)(x/X)} \qquad \text{(A3A.10a)}$$

and

$$Q = \left[\frac{(C_V/C_P)^2 + 1}{(C_V/C_P)^2 + (1/y)^2} \right]^{0.5} \qquad \text{(A3A.10b)}$$

The fraction flow obtained from Equation A3A.10a is put into Equation A3A.10b. These equations have been used to construct the installed characteristics curves for C_V/C_P = 1, 2.5, 5 and 10, shown in Figure A3A.4(b).

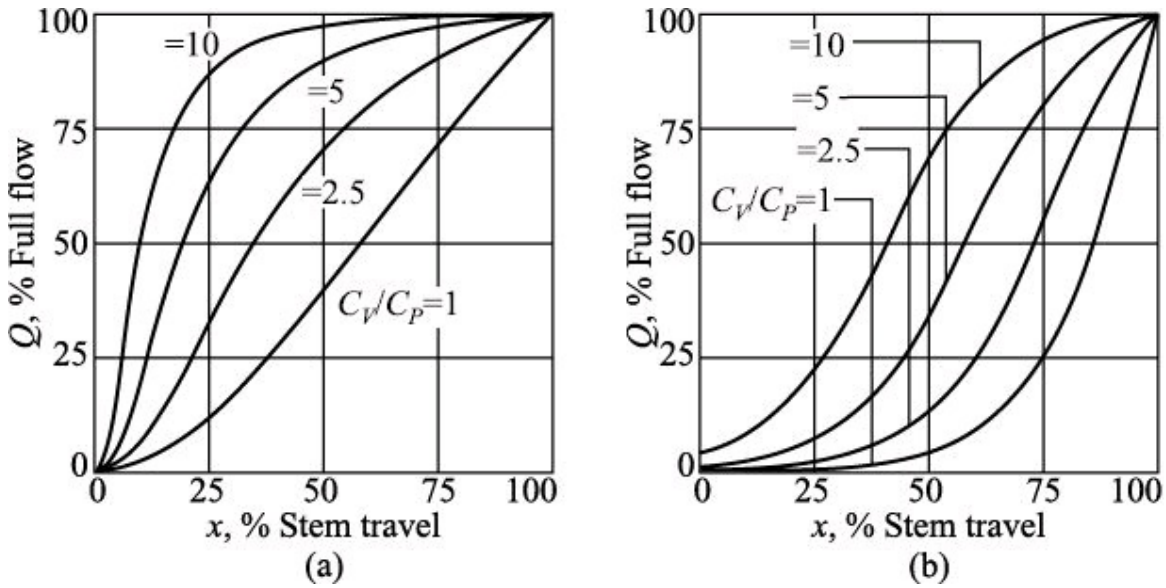


Figure A3A.4 Installed characteristics for (a) Linear, (b) Equal percent valves.

Note that the installed characteristics of an equal percent valve gradually approach that of a linear valve with increase of the ratio C_V/C_P and the same for a linear valve approach that of a quick opening valve. The characteristic of a quick opening valve is described in the following section and depicted in Figure A3A.5(a). For a designer these characteristic curves are important in finding the valve gain for analyzing the dynamic compensation the valve is going to contribute in operational condition.

(3) **Quick opening characteristic:** Quick opening characteristic is used where a large reduction of flow is required for a small change of controller output. In fact these valves can be used as flow switches. About 80–90% flow increase (or reduction) occurs for only a 15–20% stem travel. Their characteristic curves are arbitrarily designed and do not bear any mathematical relation.

(4) **Parabolic characteristics:** The characteristic curve of such a valve bears a

quadratic relation between x and Q. Thus,

$$Q = Kx^2 \qquad \text{(A3A.11)}$$

where, Q = % of maximum flow under constant pressure drop; x = % of maximum stem travel; K = A constant. The characteristic curve of quick opening and parabolic valves are shown in Figure A3A.5(b).

For processes with non-linear gain characteristic, equal % and parabolic valves are used to bring down the combined non-linearity of process + FCE, by properly designing the valve characteristic equation parameters.

(5) **Characteristic of a three-way valve:** Three-way control valves have three ports. These ports are located on the valve body in similar configuration to the branches of a Tee. These valves are available in two categories:

1. Mixing type;
2. Diverting type.

In a mixing type valve, two inlet streams enter the valve through the inlet ports, P1 and P2, which are opposite to each other, and the mixed stream exits through the outlet port, P3 that is in right angle to the inlet ports. In a diverting type valve, there is one inlet stream entering through the port, P3, that is similar in configuration to the outlet port of a mixing type valve and the entering stream is divided into two parts to leave the valve through two outlet ports, P1 and P2, similar to the inlet ports of a mixing type valve. In both of these types, the inherent characteristic spells that two streams either get mixed or divided according to a ratio that varies linearly with the stem position. To realize such a characteristic, it is necessary that the pressure drop through P1 and P2 are equal and constant, which is rarely the case in practical application. Figure A3A.5(b) shows the characteristics of 3-way control valve. The inset of the figure shows the stream arrangements of the two types.

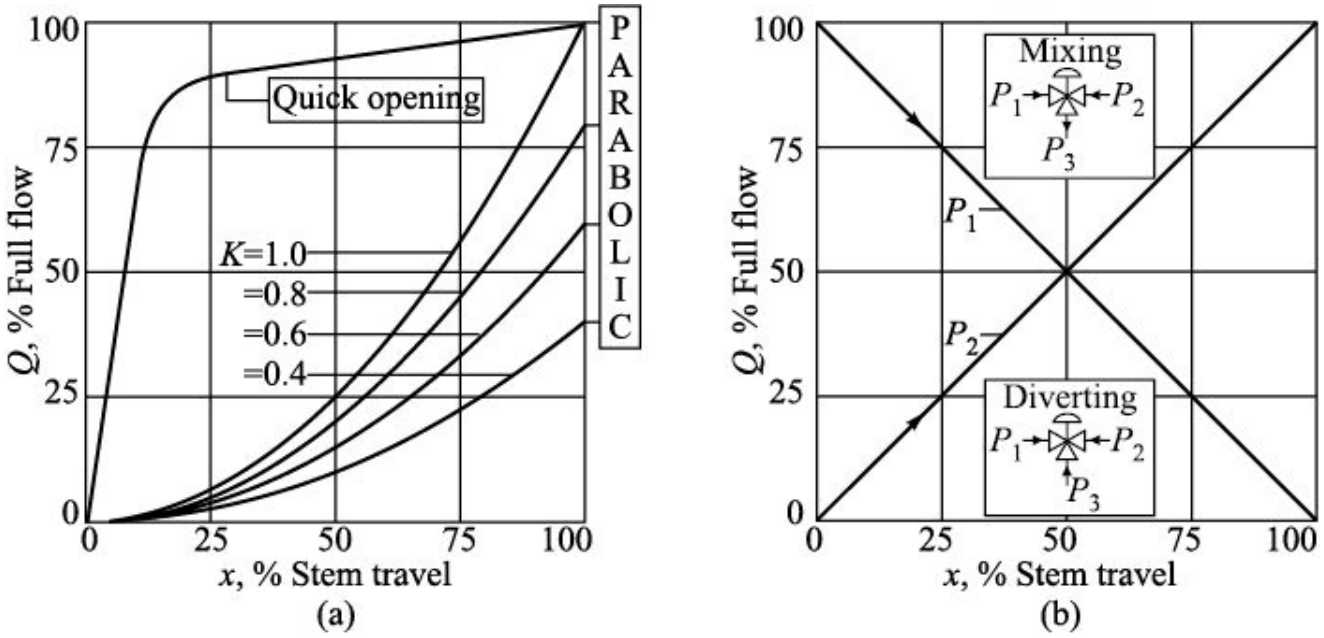


Figure A3A.5 Inherent characteristics of (a) Quick opening and parabolic (b) 3-way control valves.

A3A.4 Control Valve Action

The basic consideration while choosing the action of a control valve is that it should move to the safest position with respect to the process operation, in case of energy failure. We should mention that there are two possible 'action' configurations in which a control valve is available, similar in all other respect. A valve that moves from close to open position by supply of energy is called '**Air- to-open**'. The term 'Air' refers that energy in the form of air pressure is supplied to the valve top. Similarly, a valve which moves from open to close position by the supply of energy is called '**Air- to-close**'. Which of these two configurations should be chosen for given control duty? How to define the safest condition of a running plant? We seek the answer with reference to a real process plant.

EXAMPLE A3A.1 An industrial fermentor producing yeast from molasses has a temperature control system for controlling the fermenting broth temperature between 29 and 31^{O}C. This is achieved by a stream of cold water at 12^{O}C flowing through the jacket of the fermenter. A control valve regulates the flow rate of the cold water. What should be its action? In case of energy failure, air pressure falling to zero, the valve should move to 'fully-open' position, so that maximum cold water flow occurs lowering the broth temperature below the set point. This will definitely slow down the reaction rate but would not allow any rise of temperature above 41^{O}C, and degeneration of the broth. Degeneration of the fermenting broth will lead to loss of costly raw material and, create a serious pollution management problem during disposal of the contaminated broth. Thus we choose an air-to-close type control valve for the application.

EXAMPLE A3A.2 We may cite another example from the petroleum refining industry. In a certain fraction from the fractional distillation, a mixture of 'Naphtha' and 'Lube-oil' is obtained. To recover the naphtha, a solvent extraction is performed with a solvent MIBK (methyl-isobutyl-ketone). Water is added into the extractor after draining out the lube portion, the mass is separated into two layers:

1. Top layer containing naphtha;
2. Bottom water layer containing unwanted material.

The draining of the bottom layer is regulated by a control valve. A density sensor located nearer to the valve is employed to measure the density of the passing liquid and acts in conjunction with a level controller. As soon as the lighter naphtha layer is sensed, the controller closes the control valve which, in fact, is the bottom drain valve of the extractor. But what should be its action? In case of energy failure, the valve should remain closed for preventing the drainage loss of costly naphtha and solvent. Thus, the action chosen is 'Air-to-open'.

A3A.5 Control Valve Sizing and Valve Capacity

Control valve sizing is the procedure of calculating the valve flow coefficient, C_V. The 'C_V method' was introduced by Masoneilan Int. Corpn., in 1944. Once the required C_V is found and the valve type is selected, the valve size may be found from any of the manufacturer's catalogue. Four basic types of fluid have been recognized, and for services of these, different formulae are required for C_V determination. The fluid types are:

1. Liquids (Incompressible Fluids);
2. Pure Gases;
3. Steam;
4. Vapours other than Steam.

For the determination of C_V, we shall use certain physical variable values of the flow system. These variables will be defined against their notations as they appear in the following.

For Liquid Service: The basic equations for valve size and valve capacity for liquids are:

$$C_V = V\sqrt{\frac{G}{\Delta P}} \qquad \text{(A3A.12a)}$$

and

$$V = C_V\sqrt{\frac{\Delta P}{G}} \qquad \text{(A3A.12b)}$$

where, V = Liquid flow rate in US-GPM; P_1 = Upstream pressure at maximum flow in psia;
P_2 = Downstream pressure at maximum flow in psia; ΔP = Pressure drop at maximum flow in psi ($P_1 - P_2$); G = Specific gravity (water = 1.0); C_V = Valve flow coefficient.

(When flowing temperature of the liquid is above 200°F, use specific gravity and other quantities at flowing conditions. When viscosity of flowing liquid exceeds 100 SSU or 20 centi-stokes, apply viscosity correction.)

For Gas Service: The basic equations for valve size and valve capacity for liquids are:

$$C_V = \frac{Q\sqrt{GT_A}}{1360\sqrt{(\Delta P)P_2}} \qquad \text{(A3A.13a)}$$

and

$$Q = \frac{1360C_V\sqrt{(\Delta P)P_2}}{\sqrt{GT_A}} \qquad \text{(A3A.13b)}$$

where, Q = Gas flow rate in cu.ft/hr at 14.7psia and 60°F; P_1 = Upstream pressure at maximum flow in psia; P_2 =Downstream pressure at maximum flow in psia; ΔP = Pressure drop at maximum flow in psi ($P_1 - P_2$); G = Specific gravity (air = 1.0); T_A = Flowing temperature in °R (°R = °F + 460); C_V = Valve flow coefficient.

For Steam Service: The basic equations for valve size and valve capacity for liquids are:

$$C_V = \frac{WK}{3\sqrt{(\Delta P)P_2}} \qquad \text{(A3A.14a)}$$

and

$$W = \frac{3C_V\sqrt{(\Delta P)P_2}}{K} \qquad \text{(A3A.14b)}$$

where, W = Steam flow rate in lb/h; K = 1 + 0.0007 × (°F super heat); P_1 = Upstream pressure at maximum flow in psia; P_2 = Downstream pressure at maximum flow in psia; ΔP = Pressure drop at maximum flow in psi ($P_1 - P_2$); G = Specific gravity (air = 1.0); T_A = Flowing temperature in °R (°R = °F + 460); C_V = Valve flow coefficient.

For Vapour Service other than Steam: The basic equations for valve size and valve capacity for liquids are:

$$C_V = \frac{W}{63.4}\sqrt{\frac{v_2}{(\Delta P)}} \qquad \text{(A3A.15a)}$$

and

$$W = 63.4 C_V \sqrt{\frac{(\Delta P)}{v_2}} \qquad \text{(A3A.15b)}$$

where, W = Vapour flow rate in lb/h; v_2 = Specific volume of vapour at downstream pressure in cu.ft/lb; P_1 = Upstream pressure at maximum flow in psia; P_2 = Downstream pressure at maximum flow in psia; ΔP = Pressure drop at maximum flow in psi ($P_1 - P_2$); G = Specific gravity (air = 1.0); C_V = Valve flow coefficient. (When P_2 is less than $0.5P_1$, use the value of $0.5P_1$ in place of ΔP and use v_2 corresponding to $0.5P_1$.)

Viscosity Correction for Liquids: If viscosity of the flowing liquid is greater than 20 centi-stokes (or 100 T_{SU}) a correction factor is required to be determined with which the C_V calculated by Equation A3A.1 has to be multiplied to obtain the correct C_V. To implement this procedure, first determine the factor R by using either of the following equations. If T_{SU} and h are the liquid viscosities in SSU and centi-stokes at flowing temperature:

For viscosity $< 200T_{SU}$, $$R = \frac{10000 \times \text{Flow in GPM}}{\sqrt{C_V \eta}}$$

(A3A.16)

For viscosity $> 200T_{SU}$, $$R = \frac{46500 \times \text{Flow in GPM}}{\sqrt{C_V T_{SU}}}$$

(A3A.17)

[To convert viscosity from SSU units, T_{SU}, to stokes units, h, of kinematic viscosity use the following equation:

$$\eta = 0.0022T_{SU} - \frac{1.8}{T_{SU}} \qquad \text{(A3A.18)]}$$

Figure A3A.6 can be used to find the correction factor with the value of calculated R from above.

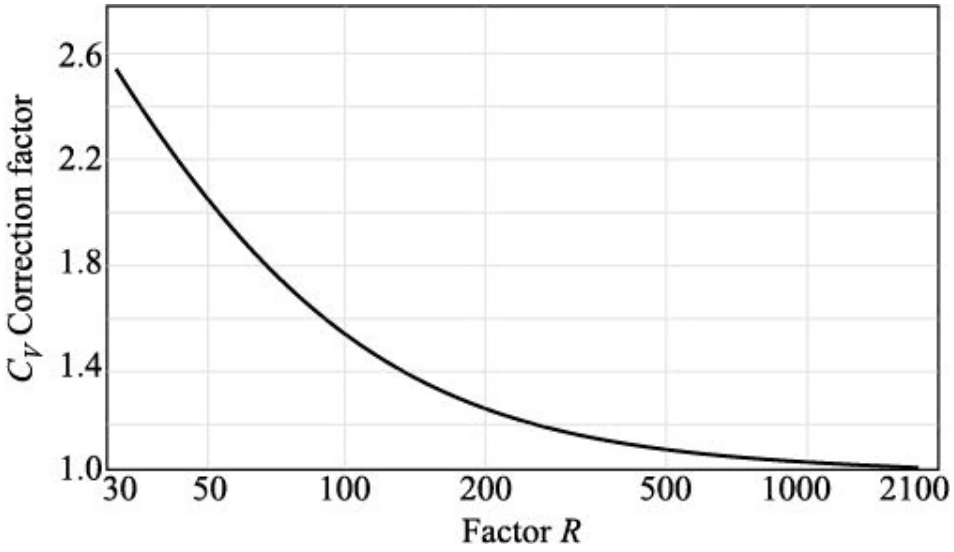


Figure A3A.6 The viscosity correction factor as a function of R.

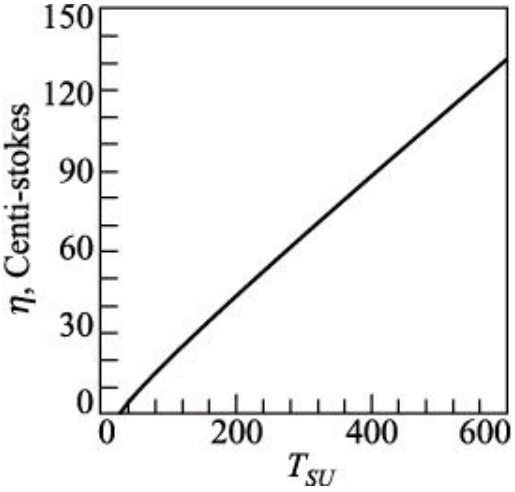


Figure A3A.7 Chart for h vs. T_{SU} conversion.

Table A3A.1 Sizing Coefficients of Single Ported Globe Styled Valve Bodies

Valve Size (inches)	Flow Characteristics	Port Dia. (inches)	Rated Travel (inches)	Flow Coeff. (C_V)
1/2	Equal %	0.38	½	2.41
3/4	Equal %	0.56	½	5.92
1	Linear	1–5/16	¾	20.6
1	Equal %	1–5/16	¾	17.2
1½	Linear	1–7/8	¾	39.2
1½	Equal %	1–7/8	¾	35.8
2	Linear	2–5/16	1–1/8	72.9
2	Equal %	2–5/16	1–1/8	59.7
3	Linear	3–7/16	1–1/2	148
3	Equal %	3–7/16	1–1/2	136
4	Linear	4–3/8	2	236
4	Equal %	4–3/8	2	224
6	Linear	7	2	433
6	Equal %	7	2	394
8	Linear	8	3	846
8	Equal %	8	3	818

EXAMPLE A3A.3 In the final viscosity polishing step of a special lubricant manufacturing plant, a petroleum derivative is blended with a binary mixture of linear chain butyl polymers. The exit stream from the blending vessel is viscosity controlled

by manipulating the butyl polymer stream input flow rate, which depends on the consistency of the petroleum stock, the load variable of the system. The controller obtains the measurement from a continuous capillary type viscometer on exit line from the blender and regulates the butyl polymer flow rate.

A plant study has revealed that maximum and minimum flow rate of the butyl polymer are 350 and 267 kg/min respectively. Also, the mixture viscosity varies linearly with amount of polymer in the mixture. A control valve with linear characteristic is recommended. The physical properties of the liquid butyl polymer are: r = 1.13 g/cm^3, μ = 65 centi-poise. Find the value of C_V and select the control valve size from Table A3A.1 to be installed on the butyl polymer inlet line. Also, it is recommended that a maximum pressure drop across the control valve may be taken as 6.0 psi with a down stream pressure P_2 = 2.5 psig. Select the valve size for the application.

Solution: Let the maximum flow rate of the manipulated stream is Q in GPM, as in Equation A3A.1. Knowing 1US-gallon = 3.7857 litre,

$$Q = \frac{350}{1.13 \times 3.7857} = 81.817 \text{ GPM ; sp.gr., } G = 1.13 \text{ ; } \Delta P = 6.0 \text{ psi.}$$

Applying Equation A3A.12a,

$$C_V = Q\sqrt{\frac{G}{\Delta P}} = 81.817\sqrt{\frac{1.13}{6}} = 35.5$$

Let us check whether viscosity correction is necessary. First we convert the dynamic viscosity, μ, to kinematic viscosity, h, as: $h = \mu/r$ = 65/1.14 = 57.5 centi-stokes.

Kinematic viscosity may be converted to Saybolt Seconds Universal (SSU) or T_{SU} viscosity by using Equation A3A.18 with trial-n-error procedure or by the help of the conversion chart given in Figure A3A.7. The result is corresponding to h = 57.5 centi-stokes, T_{SU} = 246.5. As this value is greater than 200, we apply Equation A3A.17 to find factor R,

$$R = \frac{46500 \times Q}{\sqrt{C_V} \times T_{SU}} = \frac{46500 \times 81.817}{\sqrt{35.5} \times 246.5} = 2590.386$$

From Figure A3A.6, the correction factor is 1.0

Hence, the corrected C_V = 35.5. As the chosen characteristic is linear and the nearest higher value of linear valve in Table A3A.1 is 39.2 for 1½ inch valve, we choose a 1½ inch Single Ported Globe Styled Valve.

EXAMPLE A3A.4 A (1–4) shell and tube heat exchanger is employed to heat a

petroleum derivative as the tube side liquid to a controlled temperature of 95°C, while the maximum and minimum entry temperatures are 15°C and 35°C respectively. The flow rate and other physical properties of the process liquid are: Q = 200 l/min, r = 0.95 g/cm^3, CP = 0.9 cal/(g/°C). Saturated steam (Ts = 110°C), applied on the shell side is the manipulated variable. The result of a study on the heat exchanger under several steady state conditions shows that the exchanger operates at an average heat transfer efficiency of 68%.

The total shell side pressure drop has been calculated to be 30 psi. It is recommended that maximum allowable pressure drop through the control valve should not exceed 20% of the total pressure drop of shell side and a factor of safety of 25% should be considered while calculating the heat duty of the exchanger. Calculate the C_V of the control valve for this application and select the valve from Table A3A.1.

Solution: Following are the thermodynamic properties of 110°C (= 230°F) saturated steam: P_S = 20.779 psig; v = 19.381 cu.ft/lb; H_L = 198.33 BTU/lb; H_V = 1157.1 BTU/lb.

The maximum heat load, w_{MAX}, of the exchanger considering 68% heat transfer efficiency and 25% factor of safety excess of theoretical requirement can be calculated as:

$$w_{MAX} = \frac{200 \times 0.95 \times 0.9 \times 80 \times 1.25 \times 3.9716}{0.68} = 99874.05882 \text{ BTU/min}$$

The corresponding steam requirement, W, in lb/h will be:

$$W = \frac{w_{MAX} \times 60}{(H_V - H_L)} = 6250.136664 \text{ lb/h}$$

The maximum allowable $\Delta P = 30 \times 0.2 = 6$ psi. Also, the entering steam, after going through this amount of pressure drop, should still possess sufficient pressure to remain in saturated condition so that an amount of heat = $(H_V - H_L)$, be available for heating of process liquid. Hence, the steam at the entry of the control valve should have a few degrees of super heat at least to remain in saturated condition after suffering the pressure drop through the valve. Let us assume the entering steam has 5°C (= 9°F) of super heat.

Thus, the factor K will have a value of, 1 + 0.0001 × (degree of super heat).

Now before we apply Equation A3A.14a to calculate C_V, let us arrange the item values required to do so, as follows:

W = 6250.137; K = 1.0063; downstream pressure P_2 = 20.779 psi (the saturation pressure); ΔP = 6 psi.

Thus, $$C_V = \frac{6250.137 \times 1.0063}{3 \times \sqrt{6 \times 20.779}} = 214.438$$

We choose equal % characteristic for the steam service valve, due to reasons described in Section 13.1 of P.K. Sarkar's, *Advanced Process Dynamic and Control*, PHI Learning. In Table A3A.1, the nearest higher value of an equal % valve is 224 against a valve size of 4 inch. Hence, the selected valve size is 4 inch Single Ported Globe Styled Valve having equal % characteristic.

The valve sizing procedure that we have described is simple and easy to use. The method offers good results provided the flow system is not associated with the following phenomena:

(1) **Choked flow correction:** The phenomenon choked flow occurs if a very high flow rate is warranted of a relatively small size valve by increasing the pressure drop across the valve. The relation Q vs. ΔP bears a non-linear relation that shows saturation characteristic beyond a limiting value of Q. That is, any further increase of ΔP has no effect on Q. This limiting flow rate is called the **Choked Flow**. Such a condition may occur for gases, as well as liquids. In cases where this possibility is pertinent, it is necessary to account for the occurrence of choked flow during the sizing process to ensure against undersizing a valve. Prediction must be made using a valve recovery coefficient, F_L, for liquids and X_T for gases.

(2) **Piping correction:** If there are fittings in pipeline attached to near vicinity of valve faces, e.g. reducers, elbows, tees, etc., in either upstream or downstream side, it is necessary to account for these additional pipe fitting losses in the form of a correction factor F_P to be added in the sizing formula.

(3) **Viscous flow correction:** In case the flow rate too small so that the flow occurs in laminar or viscous region, a correction factor F_R is to be added in the sizing formula. F_R is dependent on the Reynold number, N_{RE}, of flow system.

A more detailed approach to control valve sizing incorporating the correction factors may found in the concerned literature.

A3A.6 Cavitation in a Control Valve

Another phenomenon called **cavitation** may occur even in correctly designed valves. Cavitation is known as the growth and collapse of cavities in a flowing liquid. The cavities are essentially made of vapour, formed due to very low pressure regions where the pressure is very near to the vapour pressure of the flowing liquid, and exists in certain portions of the valve body. Usually local pressure in the downstream side of the valve causes the cavities to collapse. High velocities associated with the

collapse are the source of most of the problems. Cavitation can shorten the operating life of critical and expensive hardware and upset process control. To avoid this phenomenon, three remedies are recommended:

1. To lift the mean pressure profile across the valve so that the minimum pressure always remains above liquid vapour pressure, the valve is located in a region of pipeline that has a high overall pressure,
2. To put a fixed resistance in the form of an orifice or valve on the downstream side of the valve, so that the back pressure on the valve increases keeping the same inlet pressure. Usually the pressure drop across the valve decreases by this action and the minimum pressure point in pressure profile is lifted to a value greater than the liquid vapour pressure,
3. To put a stream of non-condensable gas through the valve along with the flowing liquid. A gas which would not condense under the prevailing downstream conditions is injected into the liquid near the *vena-contracta*. The continued presence of the gas during pressure recovery disrupts the cavity collapse process, thus limiting its detrimental effect.

Before closing this discussion, we shall present two more topics about control valve technology. The first one is about practical regulation capability of a valve, and the other is an accessory hardware used to improve the regulation accuracy of a control valve.

A3A.7 Rangeability of a Control Valve

Related to the capacity of a valve is the term rangeability. The rangeability, R, of a valve is defined as the ratio of maximum controllable flow to the minimum controllable flow:

$$R = \frac{Q_{MXC}}{Q_{MNC}} \qquad \text{(A3A.19)}$$

where, Q_{MXC} = Maximum controllable flow through the valve; Q_{MNC} = Minimum controllable flow through the valve.

The definition of maximum or minimum controllable flow is very subjective. Some people like to define a controllable flow between 10% and 90% of the valve opening, while others define between 5% and 95%. The valve must have sufficient rangeability to dominate the control action. A value of R from 12 to 18 is satisfactory for most of the applications.

A3A.8 Valve Positioner

A valve positioner is really a servo-system used to improve the regulation accuracy of a control valve. An electric motor driven valve uses electrical servo and a pneumatic valve needs a pneumatic servo as the positioner. The controller output is sent to the servo as the input signal if the output from the servo is unequal to the input, the difference in signals is sensed as the error. The hardware of the servo system moves in a direction to reduce the error to zero. The valve stem being attached to this moving part, is displaced proportionally to a position, which is accurate with respect to the controller output.

The action principle just described reveals that a positioner could be also called as a feedback controller situated locally to the control valve. A positioner is generally used if a high accuracy of valve regulation is needed in certain special loops, or due to long distance between control station and FCE there is a possibility of losing the quality of control signal.

The relation between instrument signal and valve stem movement is usually linear, but some models are available with a ‘cam’ and the feedback linkage that can be used to change the characteristics of the valve. This may be a useful tool in stabilizing complex control systems.

Beside the above topics, it has been discussed in Section 14.20, the procedure to represent a control valve of known size and characteristic under variable pressure drop condition within a control system.

Appendix 3B

Processes with Recycle Path

In process operation, often the optimum design of equipment are such that in a single pass of the inlet stream, the conversion potential is not fully utilized due to limitation in the equipment volume. This limitation comes from the consideration of various techno-economic criteria. Operational significance is that, in such equipment, the residence time required to obtain a satisfactory amount of conversion is too big to be provided in a single pass of the inlet stream. As a remedy to this problem, an amount of the exit stream is recycled back with the inlet stream to go again through the equipment.

Relative amount of recycled stream flow rate to the inlet or outlet streams depends on the amount of conversion achieved in the single pass, which depends on the available residence time in the equipment. There are numerous applications of recycling technique in the process industry. Three such examples are shown in Figure A3B.8.

Bequette and Marlin have shown that a process with recycle path could be represented in block diagram form by a loop around the process containing a positive feedback as shown in the Figure A3B.9. The existence of +ve feedback is justified by the stream configuration that the recycled stream joins with the input (feed) before entering the process.

The dynamics of the process and the recycle path are represented by the transfer functions $G_P(s)$ and $G_R(s)$ respectively. The closed loop TF of such a process with recycle would be:

$$\frac{Y(s)}{X(s)} = \frac{G_P(s)}{1 - G_R(s)G_P(s)} \qquad \text{(A3B.20)}$$

In a recycle system the gain K_R is defined as the ratio of the recycled stream flow to the output flow rate. The simplest description of such a loop would be that the process has a first order dynamics and the recycle path does not have any dynamic lag, so that it can be represented by a pure gain element having a finite gain value of K_R. Thus, $G_R(s) = K_R$. The gain of the recycle path may be determined from a material balance around the process, in case the process involves a conversion mechanism and the unconverted material is recycled with the feed after

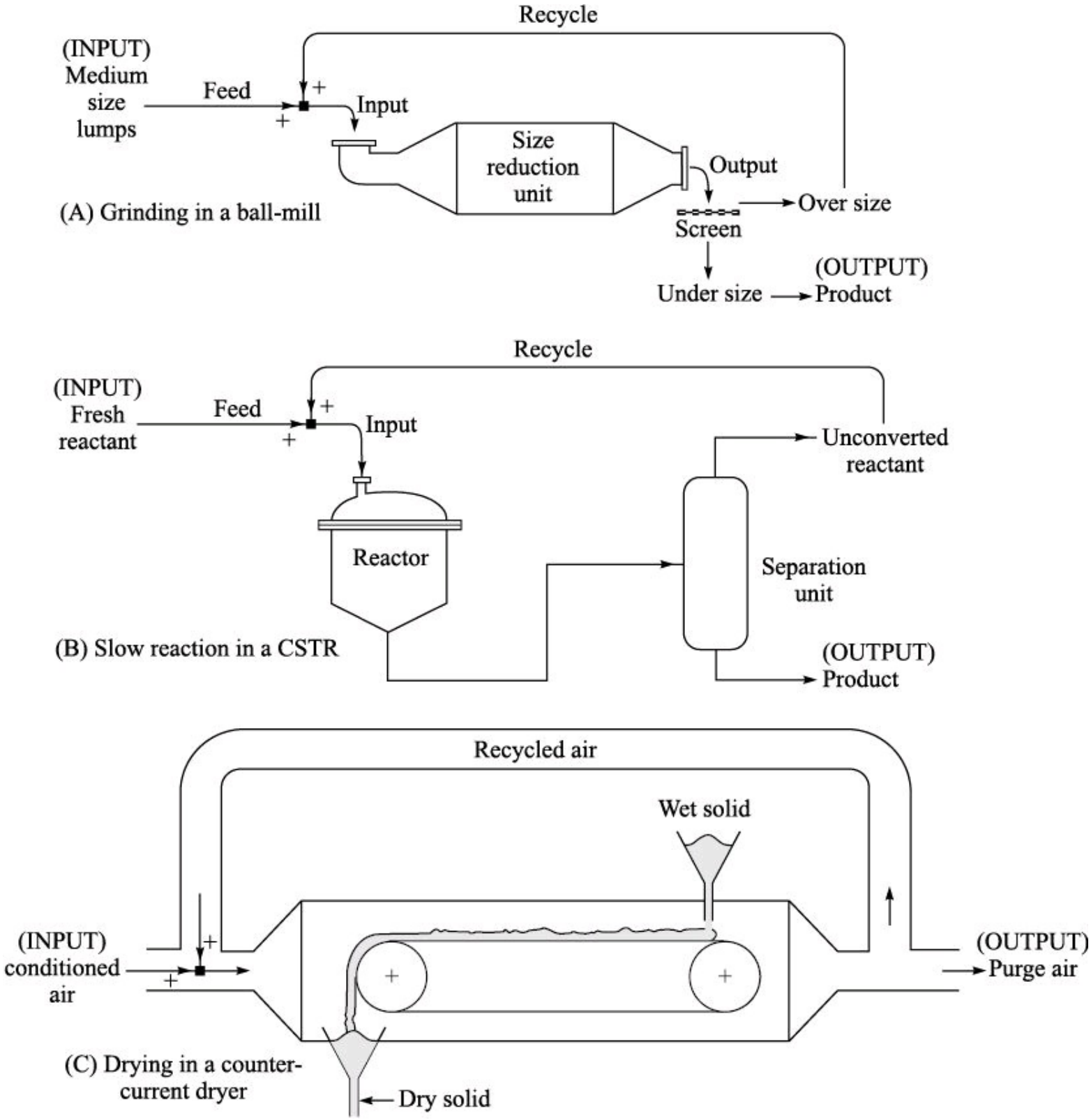


Figure A3B.8 Three example systems of recycling technique in process application.

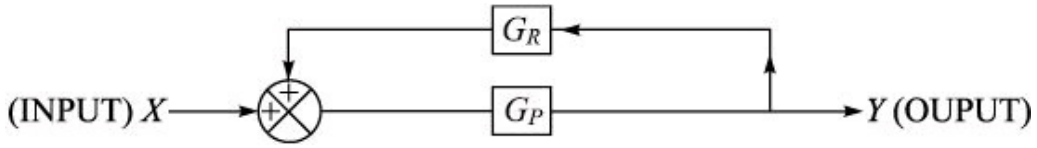


Figure A3B.9 Block diagram of a process with recycle.

being separated from the exit stream of the process. The close loop TF of such a first order process [$G_P(s) = K_P/(t_P s + 1)$] with finite gain recycle path shown in Figure A3B.10(a) is:

$$\frac{Y(s)}{X(s)} = \frac{K_P/(1-K_PK_R)}{[\tau_P/(1-K_PK_R)]s+1} \qquad \text{(A3B.21)}$$

Effect of joining a recycle path around a process is having a +ve feedback loop around it. This fact acts adversely to the stability of the loop and also slows down the dynamic response of the loop relative to the process alone. The effect of recycle also imparts nonlinear gain characteristics to the overall dynamics. This can be shown by:

1. Step response of the loop, as a function of K_R at constant K_P; and
2. the same, as a function of K_P at constant K_R.

For part 1, following values of the parameters have been chosen: K_P = 1; t_P = 10; K_R = 0, 0.2, 0.4, 0.6, 0.8. The result as the unit step response of the recycle system is shown in
Figure A3B.10(b), and for part 2, the chosen parameter values are: K_P = 0.8, 0.9, 1.0, 1.1, 1.2; t_P = 10; K_R = 0.65. The result as the unit step response of the recycle system is shown in Figure A3B.10(c).

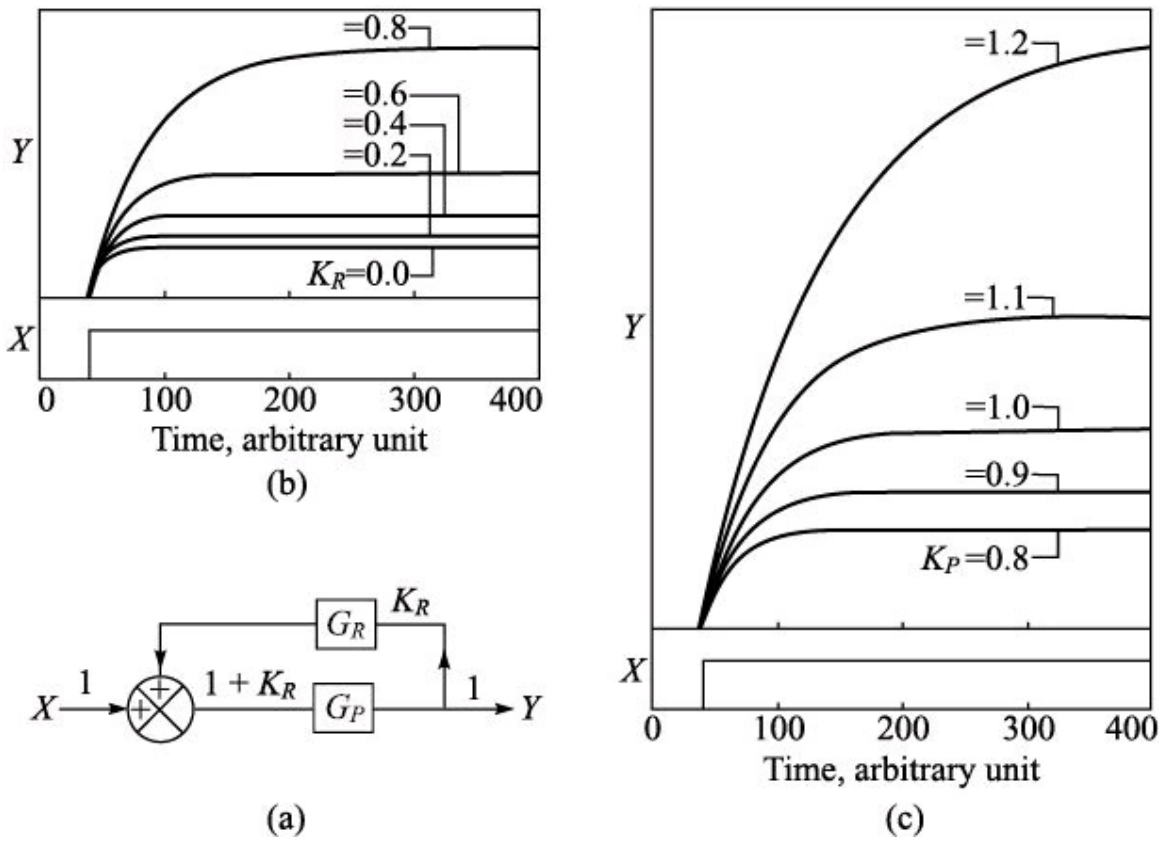


Figure A3B.10 (a) The recycle loop. (b) Step response for different K_R at constant value of K_P. (c) Step response for different K_P at constant value of K_R.

Examine Figure A3B.10(b) to note that with an increase of recycle ratio, the decrease of response speed and increase of overall gain of the recycled process happens rapidly in a non-linear fashion. To examine this feature quantitatively, let us write the TF of

Equation A3B.21 for $K_R = 0.9$:

$$\frac{Y(s)}{X(s)} = \frac{K_P/0.1}{[\tau_P/0.1]s+1} = \frac{10K_P}{10\tau_P s+1} \qquad \text{(A3B.22)}$$

Thus, the effective gain and time constant has increased tenfold by applying a recycle ratio of 0.9. Even for a moderate recycle ratio, $K_R = 0.65$, the effective gain increases even for low values of process gains around unity, in a non-linear fashion. These dynamic behaviours are shown in part (c) of Figure A3B.10. The reason of this dynamic quality is attributed to the effect of positive feedback of a recycle loop.

Another effect of +ve feedback is the reduction of stability measure of the overall system. This can be examined with a recycle element that has a dead time with a gain element. The gain element has a similar significance as that of recycle ratio, and the assumption of dead time in the recycle path is understandable because of the transportation lag required to bring the process output from exit end to its entry point through the recycle path. Example of such recycle path delay may be found in many industrial applications. With a moderate increase of process gain K_P, such processes may show unstable response even for a small pulse type disturbance.

To demonstrate this response property, a close loop process with following block TFs and recycle ratio is disturbed with a pulse input. The system and the response are shown in

Figures A3B.11(a) and (b) respectively. Following values of the system parameters have been assumed for the system of Figure A3B.11(b): $K_P = 1.0, 1.5, 2.0$; $t_P = 1$; $K_R = 0.6$; $t_{DR} = 8$.

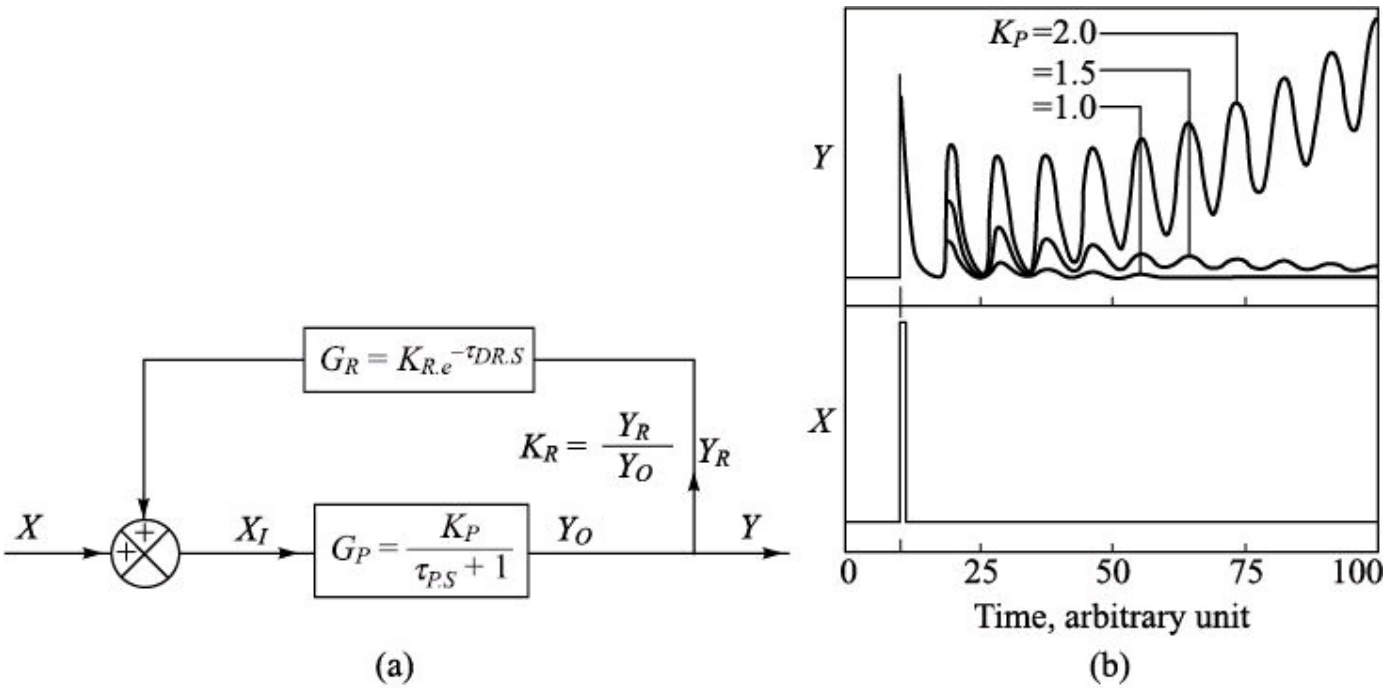


Figure A3B.11 Pulse response of a first order system with a dead time in the recycle path.

We had initially mentioned that if the conversion of an input stream through a process equipment is not enough during its passage, the unconverted part of the exit stream after separation may be recycled through the equipment. The recycled stream flow rate bears a ratio to the overall exit stream flow rate known as the recycle ratio. Up to this point, we have discussed about the effect of recycle ratio on the dynamic behaviour of the overall process with recycle, but now we ask the question that – whether the recycle ratio should be treated as an adjustable parameter or is there any way of fixing the value of this parameter for a given process? To answer this, we should recognize that the recycled amount depends on the degree of conversion achieved through the process, hence, if the internal conversion mechanism by which the input is converted to output in a process is known, the amount of unconverted material and thus, the recycle ratio could be predicted. In Figure A3B.8 for example systems in (a) and (c), the process models are not well defined, hence, recycle ratio for these systems remains an adjustable parameter. But for the CSTR of Figure A3B.8(b), the conversion is predictable, and that would determine the magnitude of recycle ratio, K_R.

EXAMPLE A3B.5 A first order, isothermal, irreversible reaction happens in a CSTR, to convert the reactant A to the product B, according to: A □ B. The exit stream, which is a mixture of A and B, is put into a distillation column to split the mixture into un-reacted A and the product B. The reactant A is recycled back and fedback into the reactor along with the fresh feed. The fundamental dynamic balance on the reactant for the CSTR would be:

$$QC_I - QC_O - kVC_O = V\frac{dC_O}{dt}$$

where, Q = Reactant solution flow rate, lit/min; C_I and C_O are respectively the inlet and exit concentrations of the reactant, mol/l/k = Specific reaction rate constant, min^{-1}; V = Reactor volume, litre.

By examining the above equation, it should be appreciated that the value of the product kV is the determining factor in measuring conversion through the reactor. Taking Q = 100 l/min, C_I = 1mol/l, and either (1) varying V from 1000 l to 5000 l with $k = 0.01\text{min}^{-1}$ in 40 steps, or (2) varying k from 0.005 min^{-1} to 0.0.25 min^{-1} with V = 2000 l, in 40 steps, similar data points are obtained to relate K_R with the product $k.V$. Figure A3B.12 shows this relation in which the product $k.V$ has been denoted by K_V.

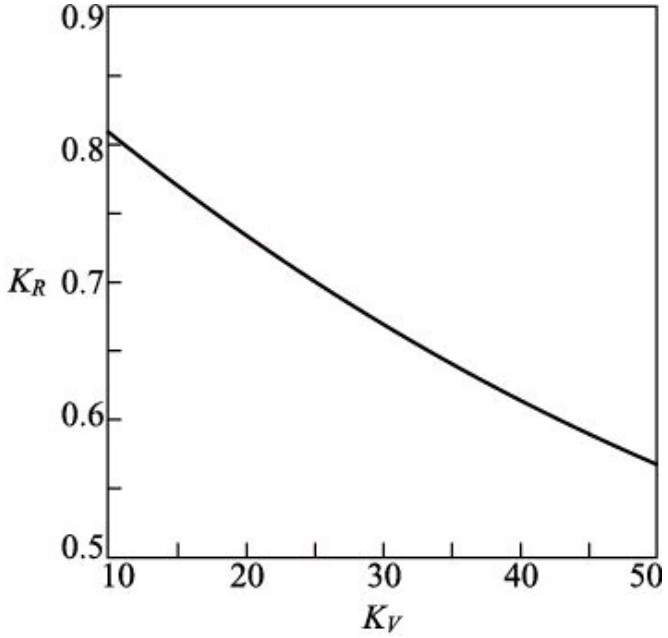


Figure A3B.12 Relation between K_V vs. K_R for CSTR handling an irreversible first order reaction.

The objective of this discussion is to make the designer aware of the type and magnitude of changes in the dynamics of a process when a recycle loop is joined around, and also to search for a suitable model of the given process so that a usable value of recycle ratio may be found without wasting a large amount of time and material by making a number of trials.

A3B.10 Processes with Parallel Streams or Bypass Connection

A bypass or parallel paths in or around a process are sometimes found as inherent dynamic properties of a process or in certain applications they are intentionally created to be a part of the inlet stream to move around the equipment until it joins with the exit stream from the process.

EXAMPLE A3B.6 An example of the first type would be the leakage paths in packed columns that appear as a type of packing defect. Such a flaw in packing may seriously affect the column performance. To detect the parallel paths, the commonly used method is the tracer injection technique. A vehicle fluid, which has approximately similar properties as that of the process fluid, is flown through the equipment. At the input end a relatively small amount of tracer is put into the vehicle fluid stream by a tracer injection device. Tracer is an easily detectable but otherwise inert material with respect to the vehicle fluid and wetted parts of the equipment. A detector measures tracer concentration in the flowing vehicle fluid at the exit end of the process equipment. The response to step and pulse input of tracer through such a defectively packed column with two parallel leakage paths are shown in Figure A3B.13.

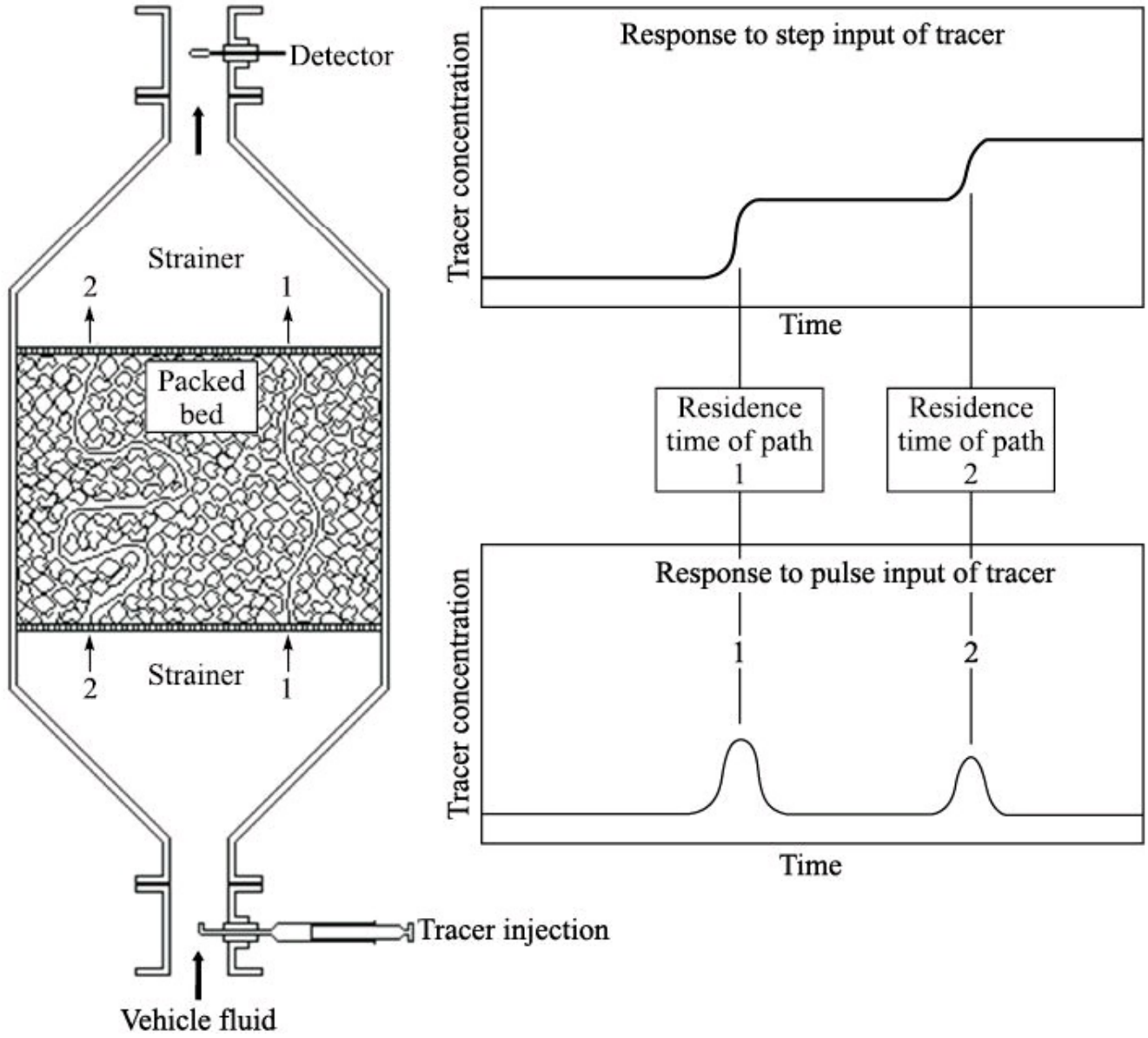
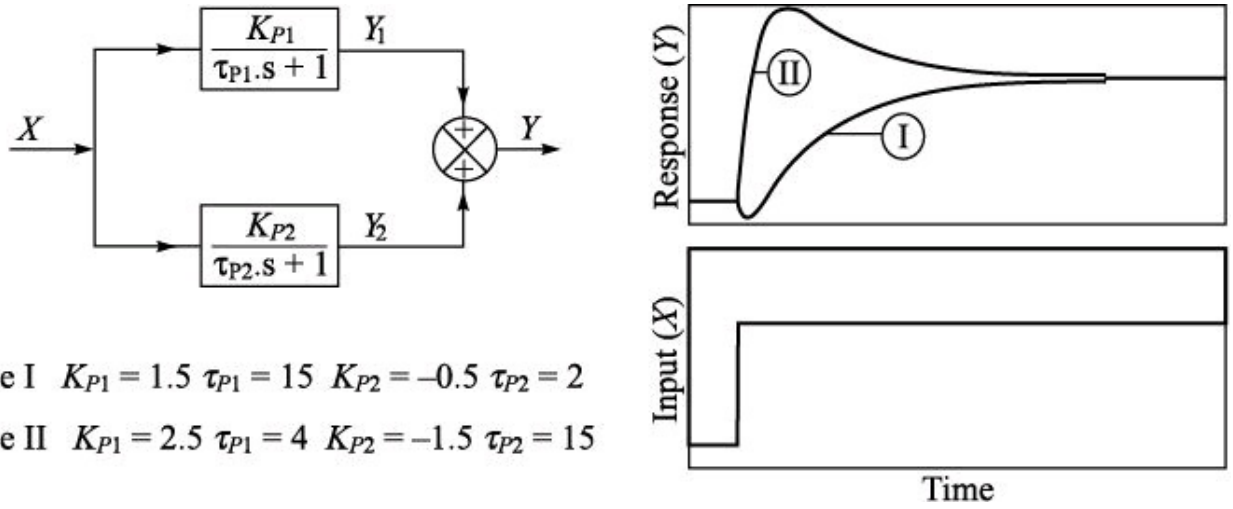


Figure A3B.13 Tracer injection technique for detecting parallel leakage paths in a packed column.

It is understandable that the lesser resistance, due to its shorter length, offered by path 1 will allow more of the tracer material in less time to reach the output end that will cause a higher height and magnitude of response step and pulse respectively, than for path 2.

EXAMPLE A3B.7 The second example of inherent parallel paths is evidenced in processes with inverse response. Two first order blocks may constitute such a process. One block responds much faster than the other, and also, it has a negative gain of smaller magnitude, than the first dynamic block. We have discussed in Chapter 16 about the application examples, dynamic features and design procedure of control system for an inverse response process. Here it only remains to show the effect of extreme parameter value conditions (i.e. if the faster system has a positive and higher magnitude gain than the first) on the response of the system. Figure A3B.14 shows the response of such a system with inherent parallel structure, along with the response of an inverse response process.



Case I $K_{P1} = 1.5$ $\tau_{P1} = 15$ $K_{P2} = -0.5$ $\tau_{P2} = 2$

Case II $K_{P1} = 2.5$ $\tau_{P1} = 4$ $K_{P2} = -1.5$ $\tau_{P2} = 15$

Figure A3B.14 Two categorical response types in processes having inherent parallel structure.

For Case I parameter settings, we obtain the conventional inverse response. The initial –ve dip is due to the –ve value of K_{P2} and it is short lived because $t_{P2} << t_{P1}$. The ultimate response value is equal to the input step magnitude because the overall gain of the system, $K_P = K_{P1} + K_{P2} = 1.5 + (-0.5) = 1.0$.

In Case II, we examine the situation if the settings are fixed the other way round, i.e. the slower sub-process has –ve and the faster sub-process has the +ve gain. Also, the overall gain value is still unity [$K_P = K_{P1} + K_{P2} = 2.5 + (-1.5) = 1.0$], hence ultimate response value is identical to the input step magnitude.

For the following two sets of parameter values, the response to a unit step will be identical to that of a unity gain first order system having a time constant of 10 time units.

Set A: $K_{P1} = 0.5$; $t_{P1} = 10$ time units; $K_{P2} = 0.5$; $t_{P2} = 10$ time units, and

Set B: $K_{P1} = 1.5$; $t_{P1} = 10$ time units; $K_{P2} = -0.5$; $t_{P2} = 10$ time units.

It is left for the reader to prove the above conclusion.

Another example of parallel operation may be found in the practice of continuous reactor operation. A parallel reactor system has an extra degree of freedom compared with a single or series system because, the total reactor volume and reactant flow rate can be arbitrarily divided between the parallel elements. When significant amount of heat has to be transferred to, or from the reaction mass, identical small volume reactors operating in parallel may be preferred because a larger heat transfer area would be available.

Difference between parallel structure and bypass system is that in the bypass line, no conversion or processing is done on the input stream, which is conveyed to the exit point as fresh feed to be mixed with the processed stream.

In certain applications, a bypass line is intentionally created by the designer around the main process to improve overall performance. In Chapter 13 of P.K. Sarkar's,

Advanced Process Dynamics and Control, PHI Learning, we have discussed the control of a heat exchanger with a bypass to carry one of the fluid streams around the exchanger. This was done to increase the response speed, which is usually very slow due to the distributed parameter character of the system. We shall examine the closed loop performance of this bypass based control scheme and that of the conventional feedback scheme, by applying these to a practical heat exchanger application.

EXAMPLE A3B.8 A shell and tube heat exchanger is applied to heat an aqueous solution in the tubes by using saturated steam on the shell side. From an experimental step dynamic test done on the heat exchanger, following TFs are found:

$$G_{PL}(s) = \frac{0.9e^{-4s}}{7.5s+1} \text{ dimensionless, and } G_{PM}(s) = \frac{6e^{-2s}}{15s+1} \text{ °C/(kg steam/s)}$$

where, $G_{PL}(s)$ is the process TF with respect to load variable (tube inlet temperature) change, $G_{PM}(s)$ is the process TF with respect to manipulated variable (steam flow rate) change. The tube fluid flow rate is 90 kg/s, and the fluid properties are r = 0.91g/cm^3, C_P = 1.1 cal/(g/°C). The minimum and maximum inlet temperatures of tube liquid, T_{TI}, are 20°C and 60°C. Assuming the latent heat of steam, l = 540 cal/g, from a steady state heat balance, the minimum and maximum steam requirement are 6 kg/s and 12 kg/s.

The two control schemes to be tested are shown in Figures A3B.15(a) and (b).

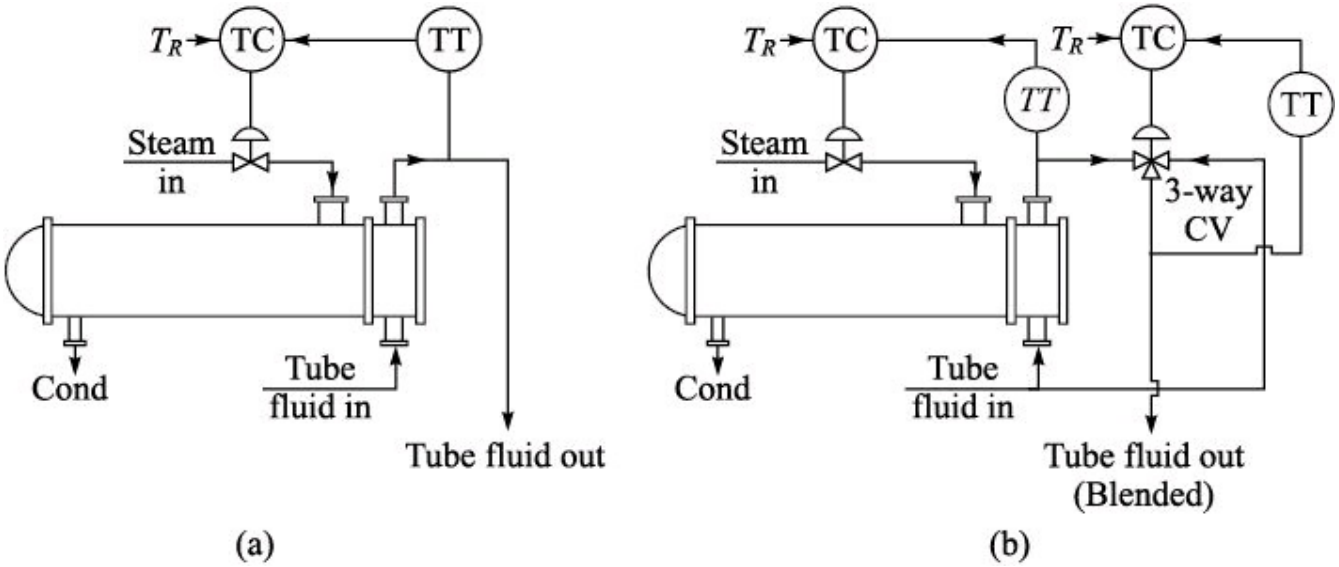


Figure A3B.15 (a) The feedback and (b) Feedback + tube liquid bypass based control of a heat exchanger.

The simulation programs are written for these two control schemes in which the process is represented by the above two TFs, $G_{PL}(s)$ and $G_{PM}(s)$. The simulated closed loop processes has been disturbed by a step change in tube inlet temperature, T_{TI}. The controller used for these two schemes are of (PI) mode. The closed loop

dynamic responses are shown in Figures A3B.16(a) and (b).

The comparison of response speeds of the two schemes may be done from the settling times shown in their response characteristics. The improvement in response speed in scheme B is marked by the reduction in settling time in Figure A3B.16(b) than in (a) for an identical magnitude of disturbance.

It is interesting to note the initial 'kick' in the response in scheme B just as the disturbance arrives in the system. The FCE used here is a 3-way control valve which was in an adequate value of stem position in the initial steady state. With the introduction of disturbance, an amount of warmer liquid goes through the valve before the controller reacts to change the stem position. This causes the initial rise in tube outlet temperature, T_{TO}.

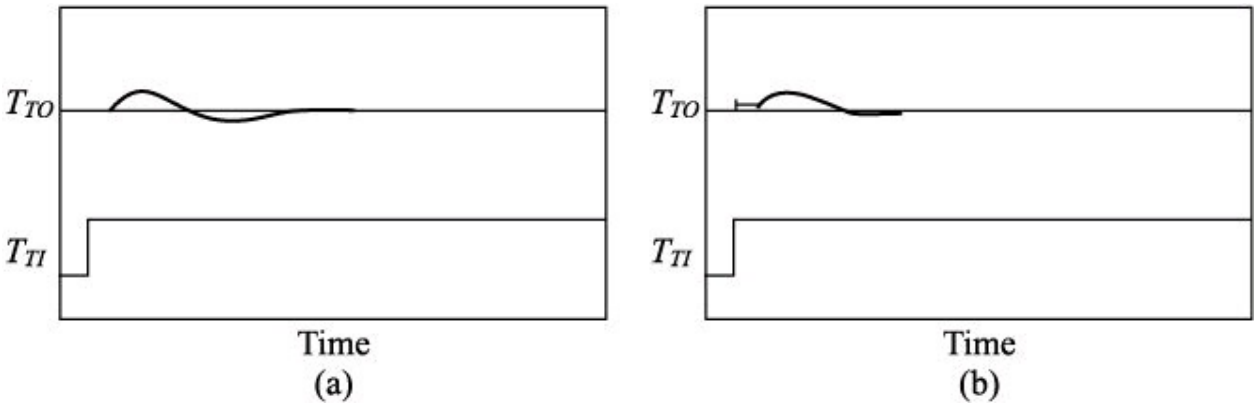


Figure A3B.16 The disturbance and response of (a) Feedback and (b) Feedback and tube liquid bypass control scheme.

A second example of this class may be cited as the introduction of Cl_2 gas into drinking water in a city water supply system. The actual chlorination is done in a bypass line through which a small amount of total water flow is diverted into the bypass line. The advantage of this procedure is that the relatively small amount of chlorine is better mixed in a smaller volume of water under a low Reynold Number condition than in a large body of water flowing at a highly turbulent condition.

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Appendix 4

Transform Chart, Response Equations and Controller Tuning Formulae

Laplace' Transform Chart (Short Form)

No	$f(t)$	$f(s)$		$f(t)$	$f(s)$
1	M	M/s	8	$M\cos(\sim t)$	$Ms/(s^2+\sim^2)$
2	Mt	M/s^2	9	$M\sinh(\sim t)$	$M\sim/(s^2-\sim^2)$
3	Mt^n	$Mn!/s^{n+1}$	10	$M\cos(\sim t)$	$Ms/(s^2-\sim^2)$
4	$e^{-at}M$	$M/(s+a)$	11	$Me^{-at}\sin(\sim t)$	$Mw/[(s+a)^2+\sim^2]$
5	$d(t)$	1	12	$Me^{-at}\cos(\sim t)$	$[Ms+a]/[(s+a)^2+\sim^2]$
6	$Md(t)$	M	13	$df(t)/dt$	$sf(s)-f(0)$
7	$M\sin(\sim t)$	$M\sim/(s^2+\sim^2)$	14	$\int f(t)dt$	$f(s)/s$

Response Equations of First Order System for Ideal Input Functions.

$$G_P(s)=\frac{Y(s)}{X(s)}=\frac{1}{\tau s+1}$$

1. Step Input:

$X(t) = M$, hence, $X(s) = M/s$; $\quad Y(t) = M - Me^{-t/t} = M(1 - e^{-t/t})$

2. Ramp Input:

$X(t) = Mt$, hence, $X(s) = M/s^2$; $\quad Y(t) = Mt - Mt + Mte^{-t/t} = M(t - t + te^{-t/t})$

3. Impulse Input:

$X(t) = Md(t)$, hence, $X(s) = M$; $\quad Y(t)=\left(\frac{M}{\tau}\right)e^{-t/\tau}$

4. Sinusoidal Input:

$X(t) = M\sin(wt)$, hence, $\quad X(s)=\frac{M\omega}{s^2+\omega^2}$

$$Y(t)=\frac{M\omega\tau e^{-t/\tau}}{\tau^2\omega^2+1}-\frac{M\omega\tau}{\tau^2\omega^2+1}\cos(\omega t)+\frac{M}{\tau^2\omega^2+1}\sin(\omega t)$$

also, $$Y(t) = \frac{M\omega\tau}{1+\tau^2\omega^2} e^{-t/\tau} + \frac{M}{\sqrt{1+\tau^2\omega^2}} \sin[\omega t - \tan^{-1}(\omega\tau)]$$

Response Equations of Second Order System for Ideal Input Functions.

$$G_P(s) = \frac{K}{As^2 + Bs + C} = \left(\frac{K}{C}\right)\frac{(C/A)}{s^2 + (B/A)s + (C/A)} = \left(\frac{K}{C}\right)\frac{\omega_N^2}{s^2 + 2\zeta\omega_N s + \omega_N^2}$$

1. Step Response for a step input of magnitude *M*: $X(t) = M$, hence, $X(s) = M/s$.

Case I: Under Damped condition ($z < 1$).

$$Y(t) = \frac{MK}{C}\left[1 - e^{-\omega_N\zeta t}\left(\cos\omega_N\sqrt{1-\zeta^2}\,t + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\omega_N\sqrt{1-\zeta^2}\,t\right)\right]$$

also, $$Y(t) = \frac{MK}{C}\left[1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\omega_N\zeta t}\sin\left(\omega_N\sqrt{1-\zeta^2}\,t + \tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta}\right)\right]$$

Case II: Critically damped condition ($z = 1$).

$$Y(t) = \left[\frac{MK}{C}\right][1 - e^{-\omega_N\zeta t}\sin(1 + \omega_N t)]$$

Case III: Over damped condition ($z > 1$).

$$Y(t) = \frac{MK}{C}\left[1 - e^{-\omega_N\zeta t}\left(\cosh\omega_N\sqrt{\zeta^2-1}\,t + \frac{\zeta}{\sqrt{\zeta^2-1}}\sinh\omega_N\sqrt{\zeta^2-1}\,t\right)\right]$$

2. Ramp Response for a ramp of slope *M*: $X(t) = Mt$, hence, $X(s) = M/s^2$.

Case I: Under Damped condition ($z < 1$).

$$Y(t) = \frac{MK}{C}\left[t - \frac{2\zeta}{\omega_N} + \frac{1}{\omega_N\sqrt{1-\zeta^2}} e^{-\omega_N\zeta t}\sin\left\{\omega_N\sqrt{1-\zeta^2}\,t + \tan^{-1}\left(\frac{\zeta\sqrt{1-\zeta^2}}{\zeta^2 - 0.5}\right)\right\}\right]$$

Case II: Critically Damped condition ($z = 1$).

$$Y(t) = \frac{MK}{C}\left[\left(t - \frac{2}{\omega_N}\right) + \left(t + \frac{2}{\omega_N}\right)e^{-\omega_N t}\right]$$

Case III: Over Damped condition ($z > 1$).

$$Y(t) = \frac{MK}{C}\left[t - \frac{2\zeta}{\omega_N} + e^{-\omega_N t}\left\{ 2\zeta \cosh\left(\omega_N\sqrt{\zeta^2 - 1}t\right) + \frac{2\zeta^2}{\sqrt{\zeta^2 - 1}} \sinh\left(\omega_N\sqrt{\zeta^2 - 1}t\right)\right\}\right]$$

3. Impulse Response or an Impulse input of Magnitude *M*:

$X(t) = Md(t)$, hence, $X(s) = M$; $\qquad Y(t) = \left(\frac{M}{\tau}\right)e^{-t/\tau}$

Case I: Under Damped condition ($z < 1$).

$$Y(t) = \left(\frac{MK}{C}\right)\frac{\omega_N}{\sqrt{1-\zeta^2}}e^{-\omega_N\zeta t}\{\sin(\omega_N\sqrt{1-\zeta^2}t)\}$$

Case II: Critically Damped condition ($z = 1$).

$$Y(t) = \left[\frac{MK}{C}\right]\omega_N^2 te^{-\omega_N t}$$

Case III: Over Damped condition ($z > 1$).

$$Y(t) = \left(\frac{MK}{C}\right)\frac{\omega_N}{\sqrt{\zeta^2-1}}e^{-\omega_N\zeta t}\sinh(\omega_N\sqrt{\zeta^2-1}t)$$

Zeigler-Nichol's Controller Tuning Formulae

Control Mode	$G_C(s)$	K_C	x_I	x_D
P	K_C	$0.5K_U$	–	–
PI	$K_C[1 + 1/(x_I s)]$	$0.45K_U$	$Pu/1.2$	–
PID	$K_C[1 + 1/(x_I s) + x_D s]$	$0.6K_U$	$P_U/2.0$	$P_U/8.0$

Cohen-Coon's Controller Tuning Formulae $[G_P(s) = (K_P e^{-t_D s})/t_P(s+1)]$

Control Mode	K_C	x_I	x_D
P	[illegible]	—	—
PI	[illegible]	[illegible]	—
PID	[illegible]	[illegible]	[illegible]
PD	[illegible]	—	[illegible]

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